nsh

process design review · 840 steps · generated by CarbonEDA v0 on 2026-11-09 19:58 EST

735.3

kgCO2e per wafer start

498.5
EPW energy · 68%

68.3
MPW materials · 9%

168.5
GPW gases · 23%

wafer 300 mm · yield 1 · carbon intensity 0.43 kgCO2e/kWh · facility overhead 0.4 x tool energy
EPW 498.5 = tool 356.1 + facility overhead 142.4 kgCO2e (828 + 331 kWh)

flow technologies/logic/nsh/nsh_final.csv
layout technologies/logic/nsh/nsh_layout.gds
structure images from the spx run of 2026-11-09 18:17:22

CarbonEDA parameters

config/carbon_config.yaml and materials.yaml as used for every number in this report

EPW energy

(tool + 0.4 x tool facility overhead) x 0.43 kgCO2e/kWh

process area	kWh / step	steps
RIE	1.33	158
Ion_implantation	1.33	7
Diffusion	1.33	37
Metallization	2	64
Wet_etch	0.28	4
Wafer_clean	0.28	156
Resist	0.28	70
CMP	0.28	46
CVD	1.09	224
EUV	3.25	29
ArF-i	0.36	28
ArF	0.33	8
KrF	0.33	3
Hg	0.33	6

GPW process gases

kg x (1 - utilization) x (1 - abatement) x GWP, plus byproducts kg/kg x (1 - abatement) x GWP; defaults: utilization 1.0, abatement 0.9

area	gas	g / step	upstream*	utilization	abatement	byproducts (kg per kg gas)
CVD	NF3	2.54	12.5	0.8	0.95	CF4 0.046, C2F6 0.045, CHF3 0.025, CH2F2 0.00086,...
RIE	C4F8	0.16	12.1	0.82	0.98	CF4 0.045, C2F6 0.031, C4F6 0.018, CHF3 0.012, CH...
RIE	SF6	0.439	125.6	0.71	0.95	CF4 0.04, C2F6 0.044, CHF3 0.0037, CH2F2 2.9e-05

* kgCO2e per kg of gas made, counted in MPW

GWP (kgCO2e per kg, 100-yr) and abatement of every gas emitted

gas	GWP	abatement
CF4	7390	0.89
C2F6	12200	0.95
C4F6	0	0.98
C4F8	10300	0.98
CHF3	14800	0.98

gas	GWP	abatement
CH2F2	675	0.98
CH3F	116	0.98
NF3	17200	0.95
SF6	22800	0.95

MPW materials

blanket film over a 300 mm wafer: thickness x area x density x GWP; the wafer itself 26000 gCO2e (300 mm); yield 1

material	g/cm³	gCO2e/g
TiN	5.21	43.98
W	19.3	75.81
TaN	13.7	25.82

material	g/cm³	gCO2e/g
Cu	8.96	6.09
Co	8.9	44.5

Step #20: Wafer

substrate · Wafer

26	52	3.5%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

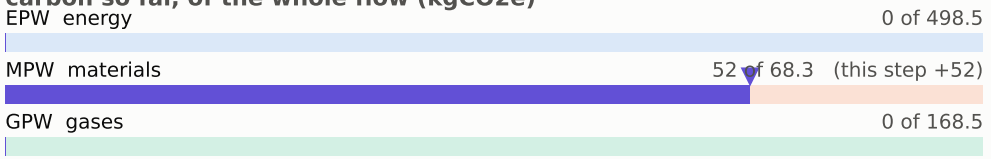
zoomed to fit

this step

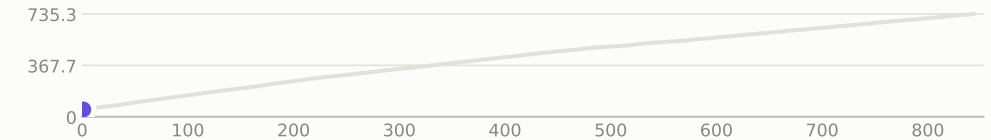
process area	Wafer	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	26 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**26 kgCO2e**

Wafer step: not counted as a process step (uncounted_process_areas)

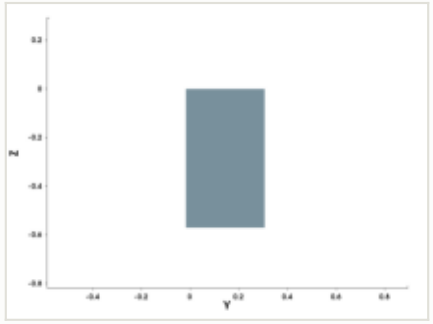
carbon so far, of the whole flow (kgCO2e)



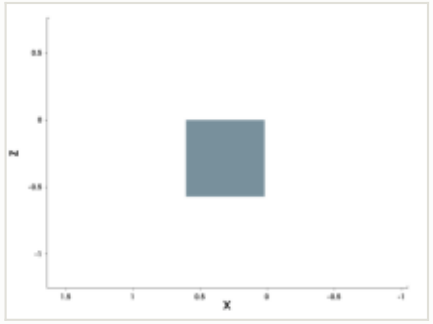
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

no mask for this step

steps by process area

materials deposited (g)

gases emitted (g)

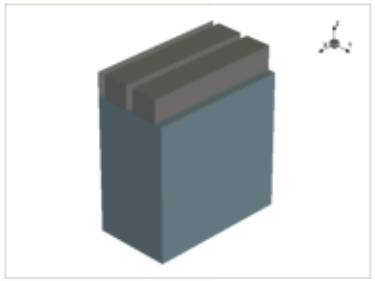
Step #40: KrF

litho_patterning · KrF

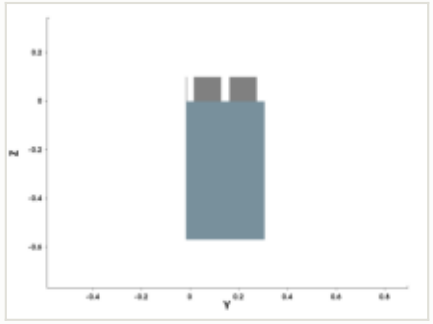
0.199	52.2	0.027%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

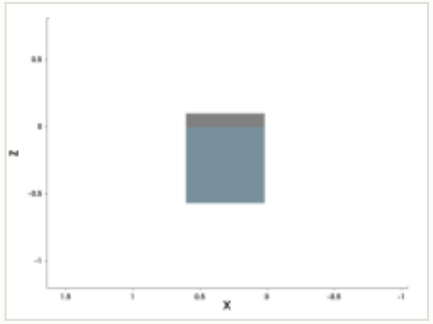
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm

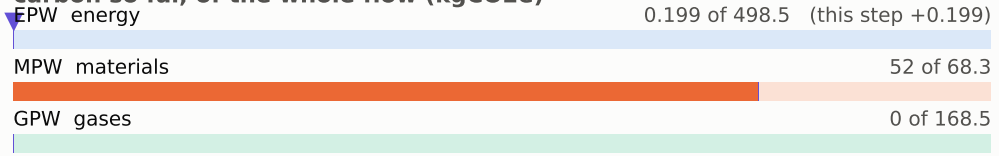


mask: layer 8:0 (GDSII)

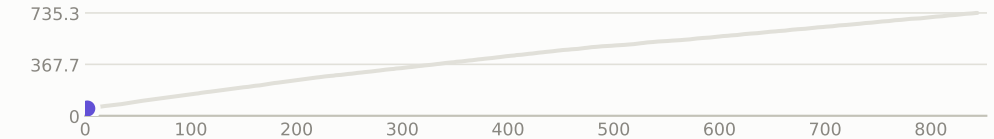
this step

process area	KrF	EPW	0.199 kgCO2e
material	—	EPW tool	0.142 kgCO2e
thickness	—	EPW overhead	0.0568 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.33 kWh + 0.132 overhead	embodied carbon	**0.199 kgCO2e**

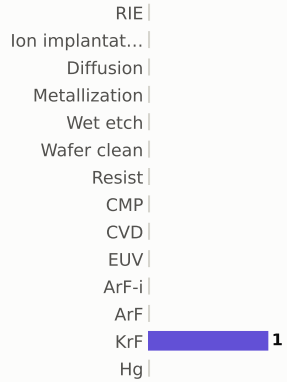
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)



steps by process area



materials deposited (g)

gases emitted (g)

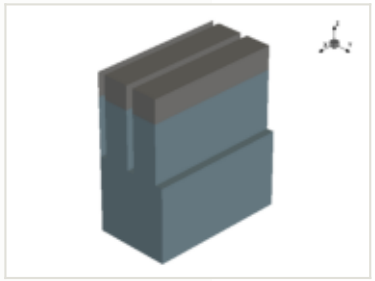
Step #50: RIE

etch_anisotropic · RIE

1.04	53.2	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

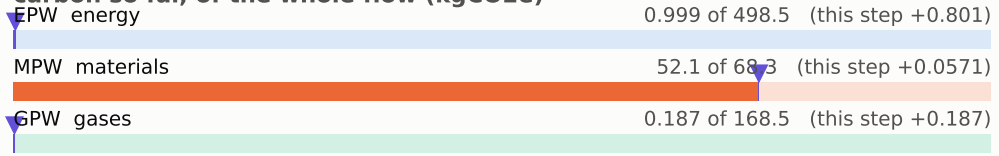


this step

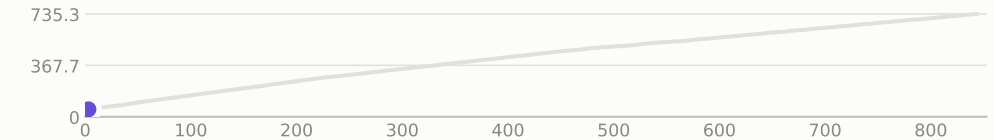
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

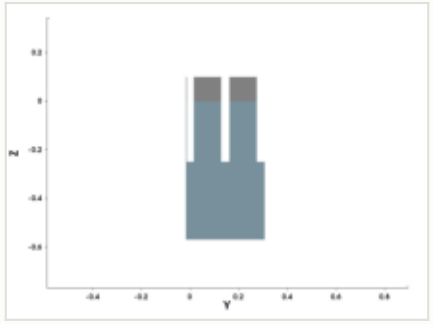
carbon so far, of the whole flow (kgCO2e)



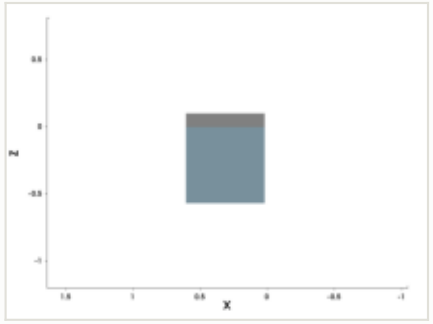
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



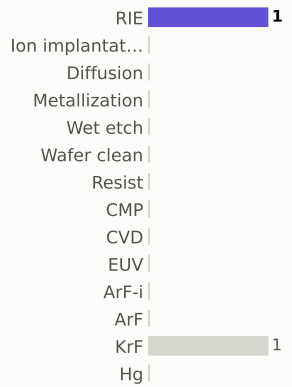
y cut at 0.184 µm



mask for this step

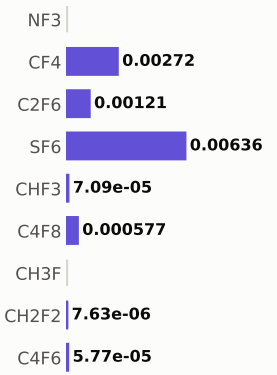
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



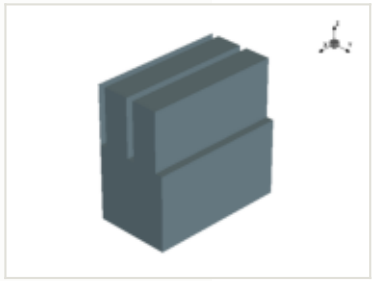
Step #60: Resist

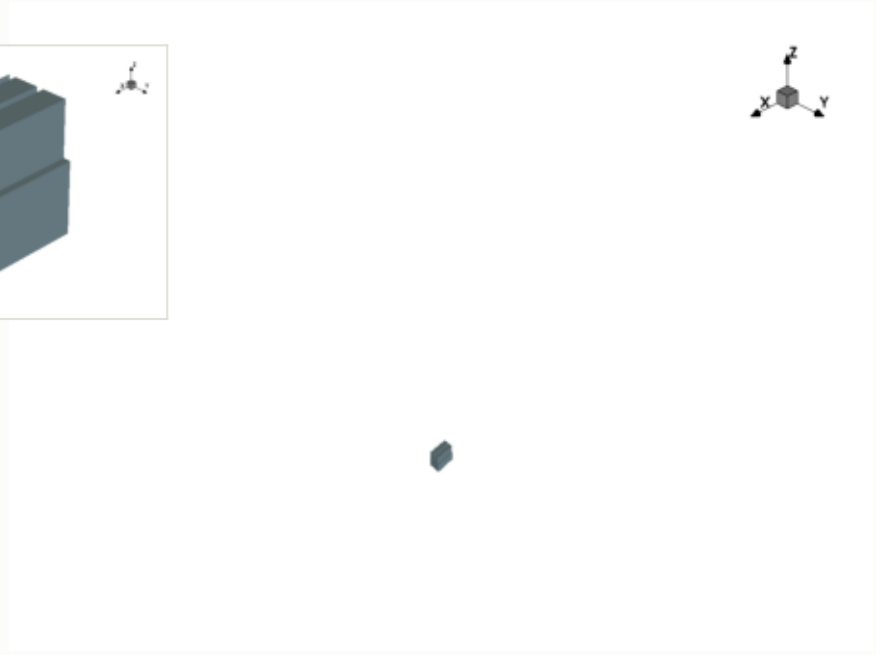
etch_strip · Resist

0.169	53.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



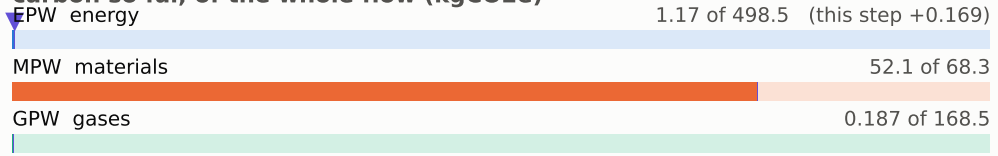


this step

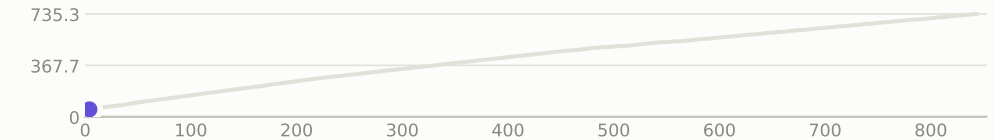
process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

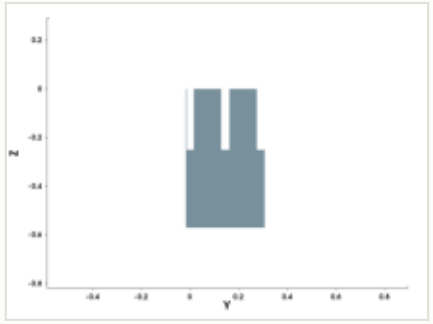
carbon so far, of the whole flow (kgCO2e)



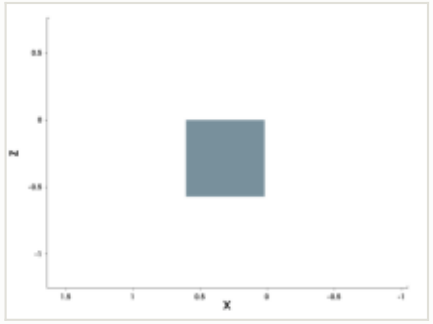
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



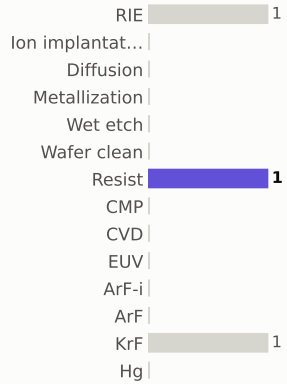
y cut at 0.184 µm



mask for this step

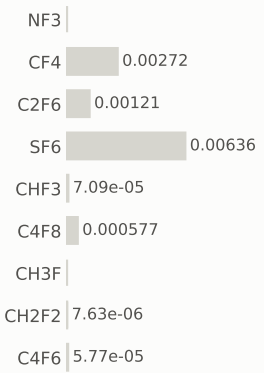
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



Step #70: CVD

depo_isotropic · CVD

1.31	54.7	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

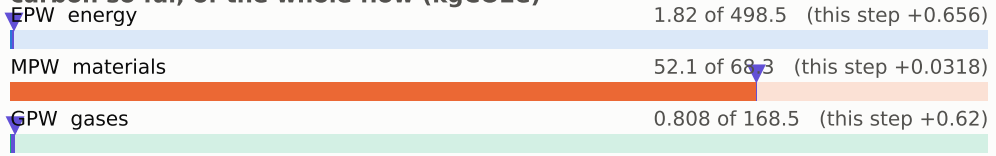
zoomed to fit

this step

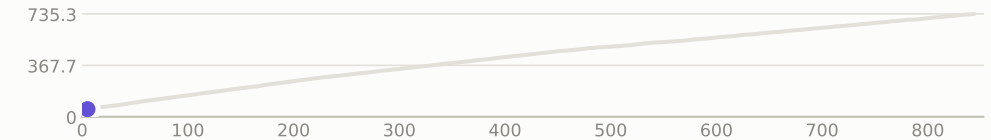
process area	CVD
material	—
thickness	0.005 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

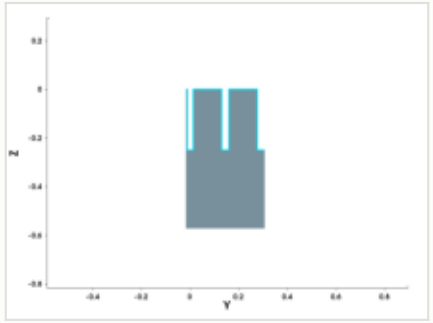
carbon so far, of the whole flow (kgCO2e)



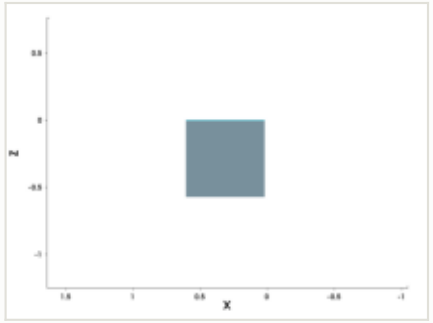
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



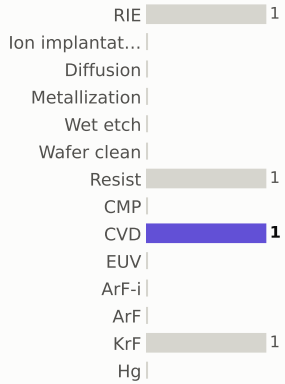
y cut at 0.184 µm



mask for this step

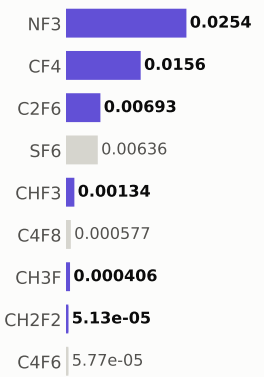
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



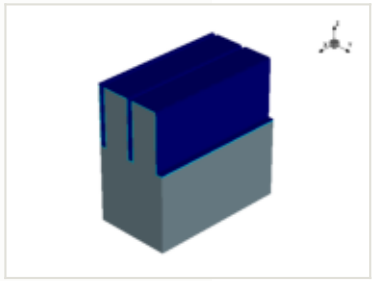
Step #80: Metallization

depo_isotropic · Metallization

1.2	55.9	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

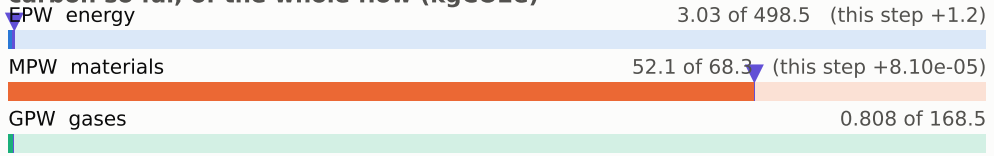


this step

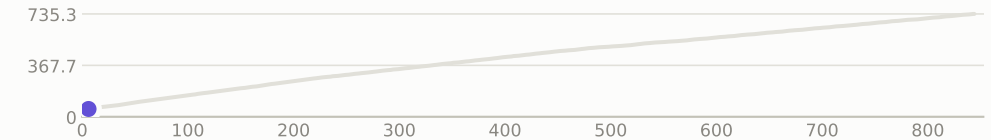
process area	Metallization
material	TiN
thickness	0.005 µm
material GWP	43.9778 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	8.10e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

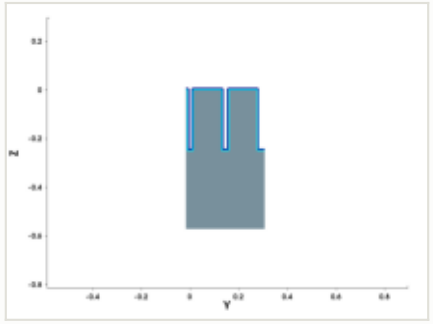
carbon so far, of the whole flow (kgCO2e)



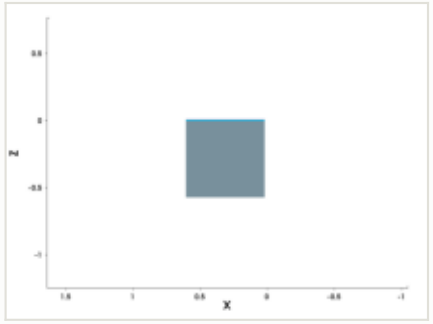
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



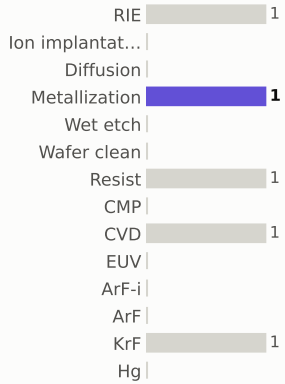
y cut at 0.184 µm



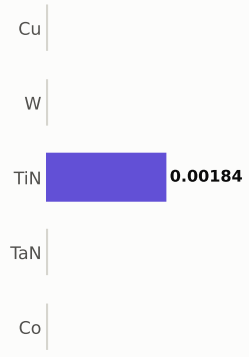
mask for this step

no mask for this step

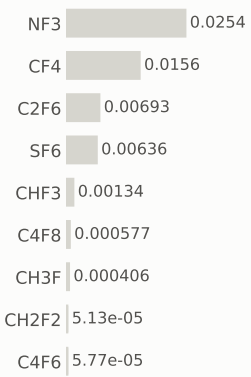
steps by process area



materials deposited (g)



gases emitted (g)



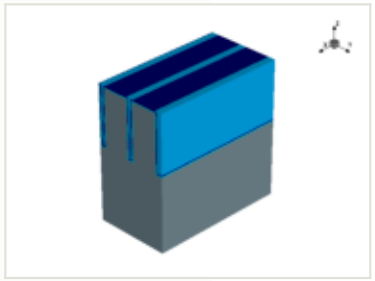
Step #90: Metallization

depo_fill · Metallization

1.2	57.1	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

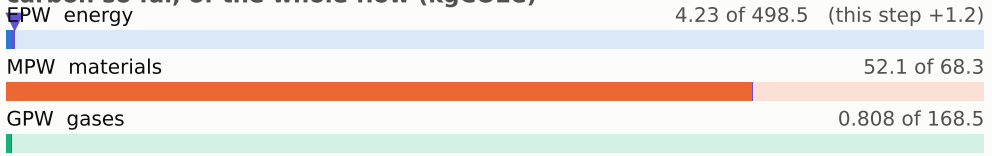


this step

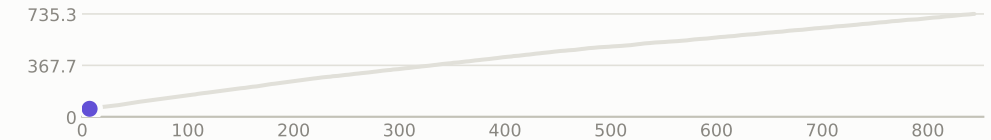
process area	Metallization
material	—
thickness	0 µm
material GWP	—
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

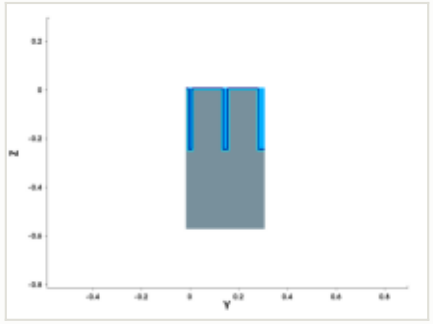
carbon so far, of the whole flow (kgCO2e)



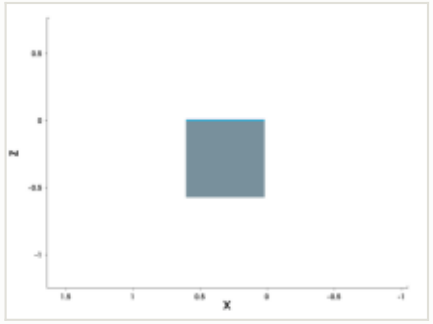
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



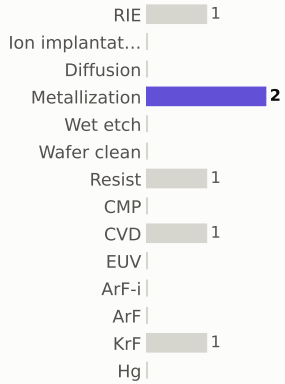
y cut at 0.184 µm



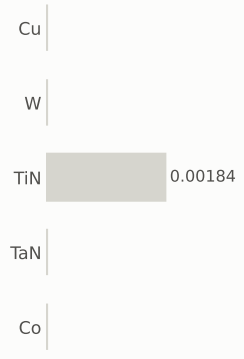
mask for this step

no mask for this step

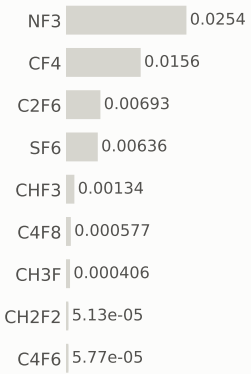
steps by process area



materials deposited (g)



gases emitted (g)



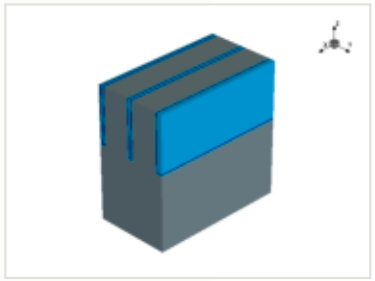
Step #100: CMP

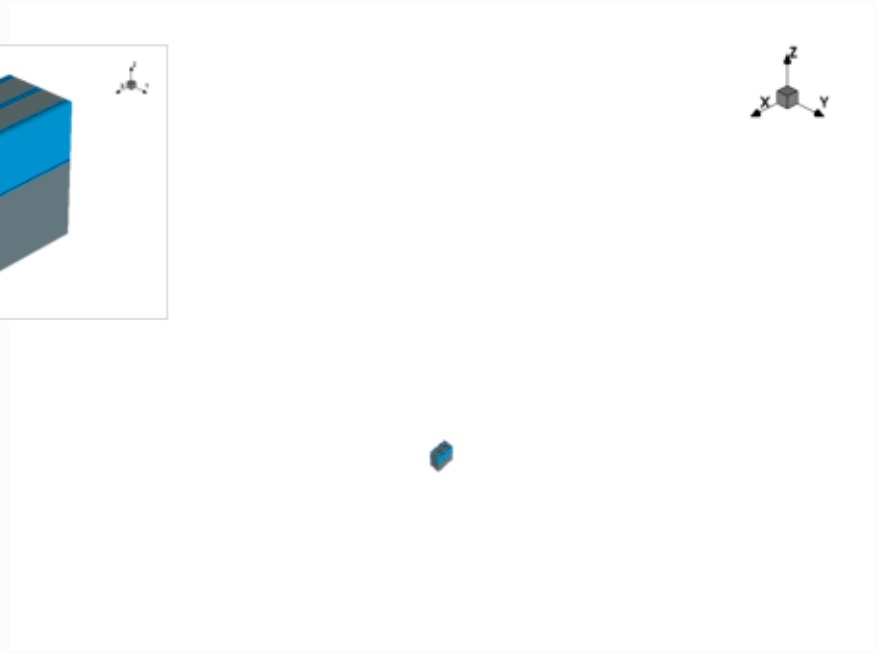
etch_cmp · CMP

0.169	57.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



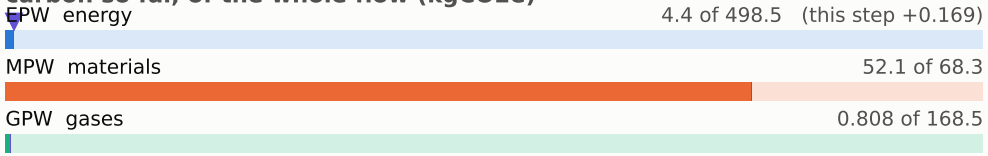


this step

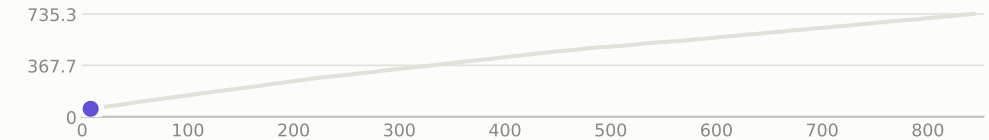
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

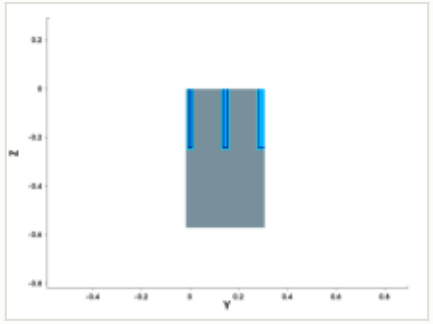
carbon so far, of the whole flow (kgCO2e)



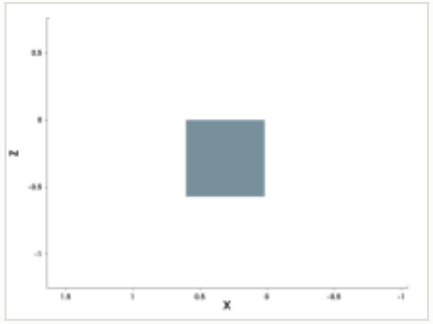
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



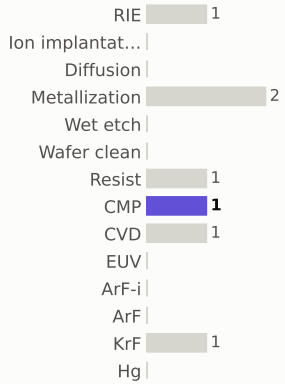
y cut at 0.184 µm



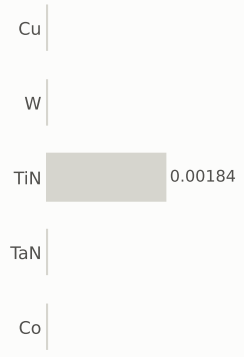
mask for this step

no mask for this step

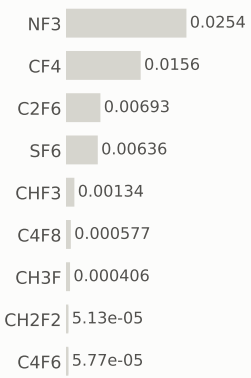
steps by process area



materials deposited (g)



gases emitted (g)



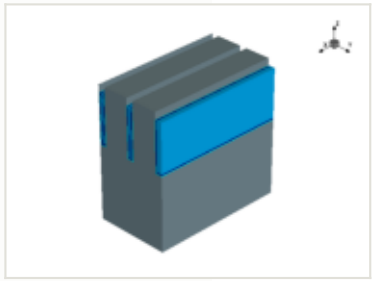
Step #110: RIE

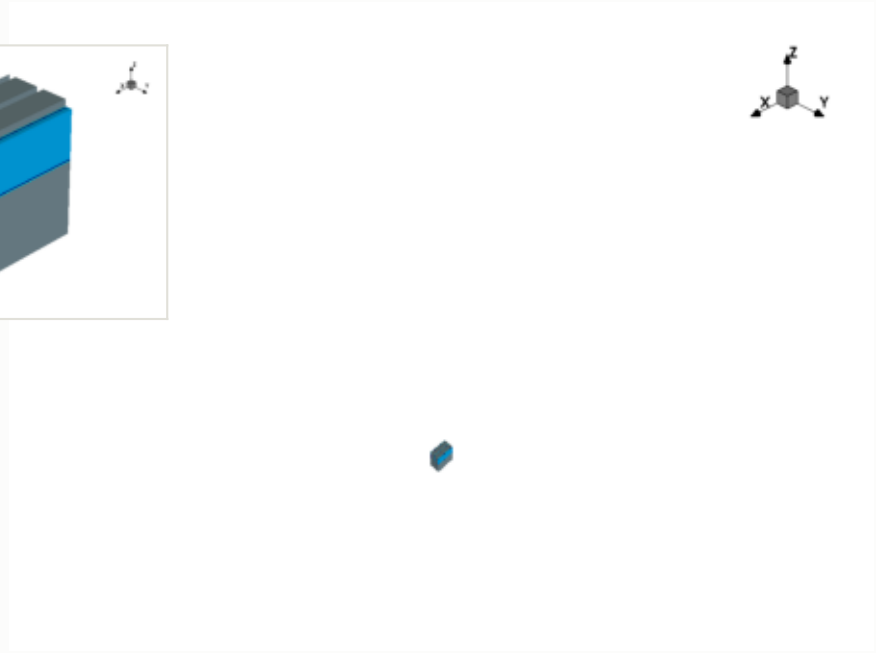
etch_anisotropic · RIE

1.04	58.3	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



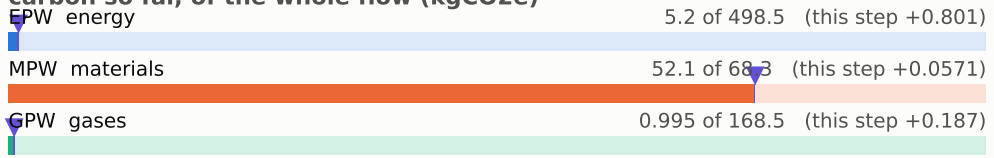


this step

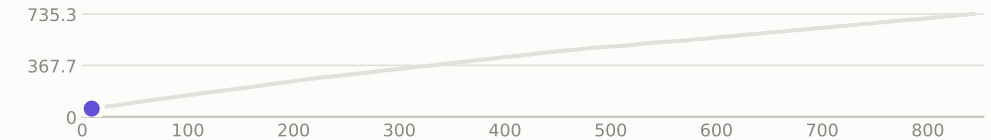
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

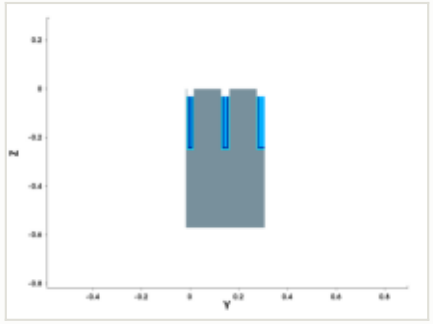
carbon so far, of the whole flow (kgCO2e)



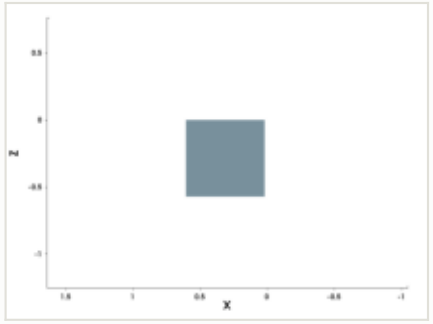
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



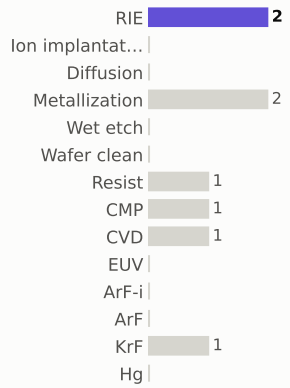
y cut at 0.184 µm



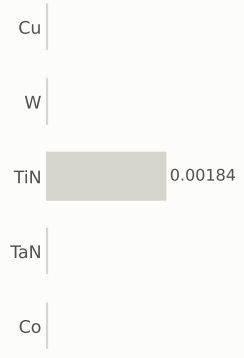
mask for this step

no mask for this step

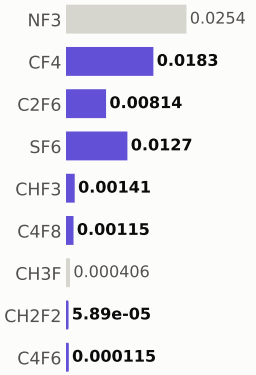
steps by process area



materials deposited (g)



gases emitted (g)



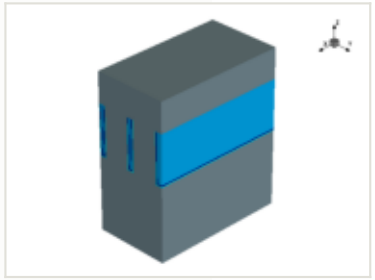
Step #120: CVD

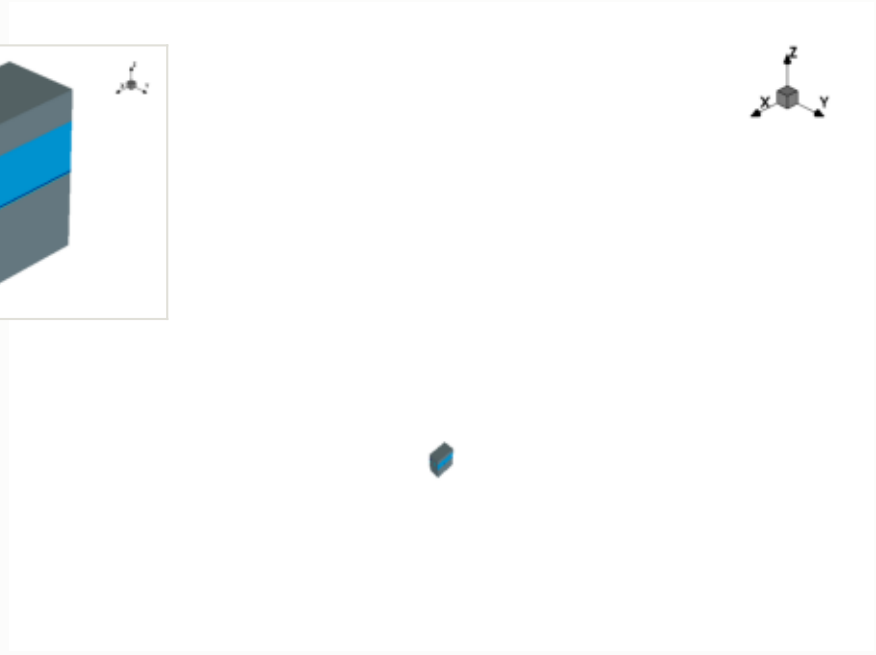
depo_fill · CVD

1.31	59.7	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



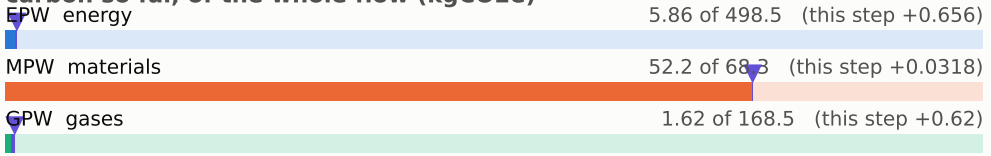


this step

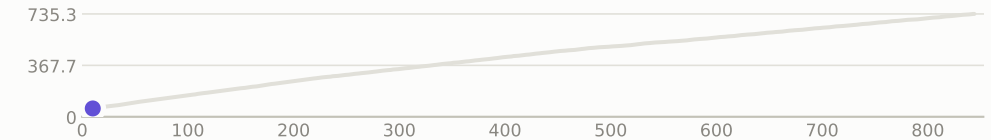
process area	CVD
material	—
thickness	0.1 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

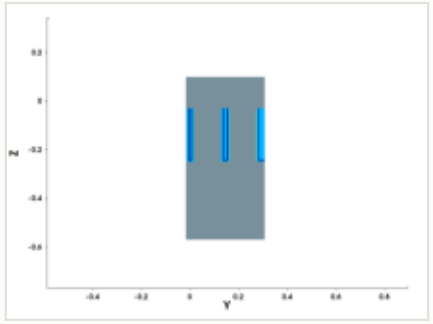
carbon so far, of the whole flow (kgCO2e)



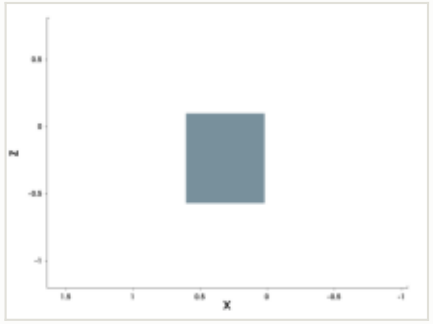
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



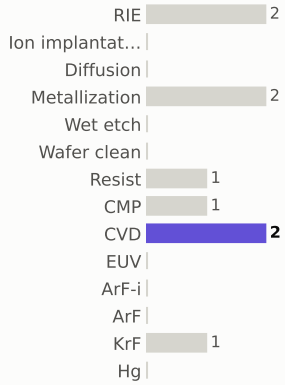
y cut at 0.184 µm



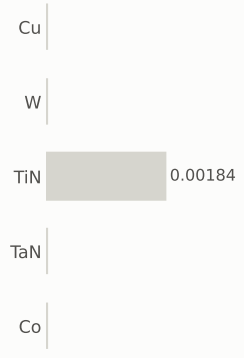
mask for this step

no mask for this step

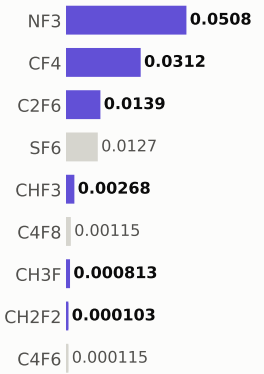
steps by process area



materials deposited (g)



gases emitted (g)



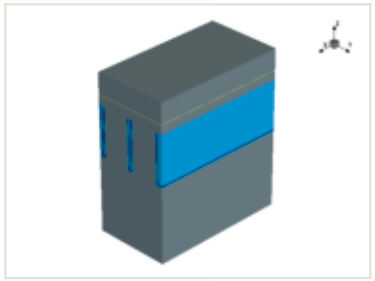
Step #140: Ion_implantation

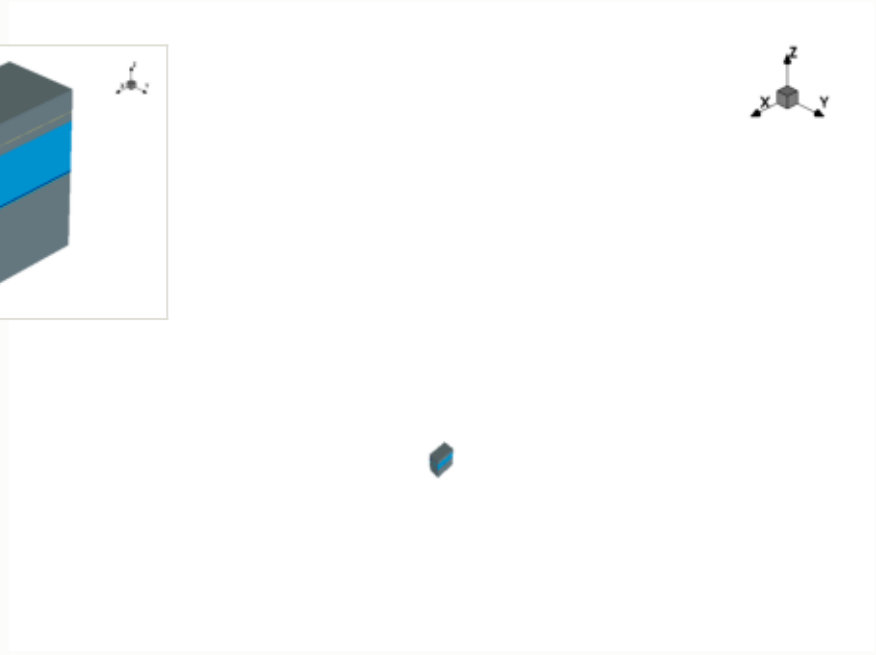
implant_ellipsoid · Ion_implantation

0.801	60.5	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

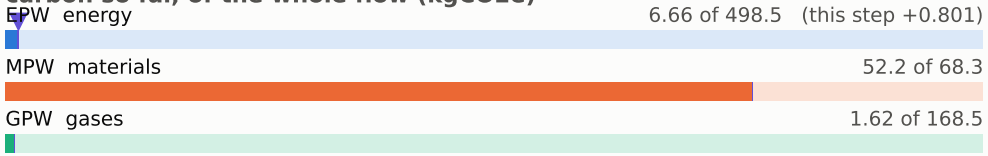




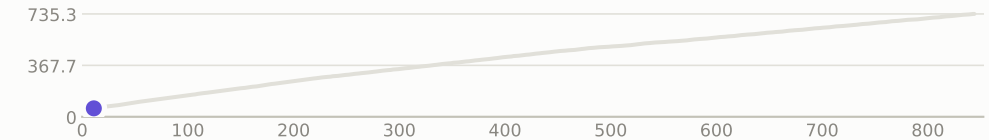
this step

process area	Ion_implantation	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

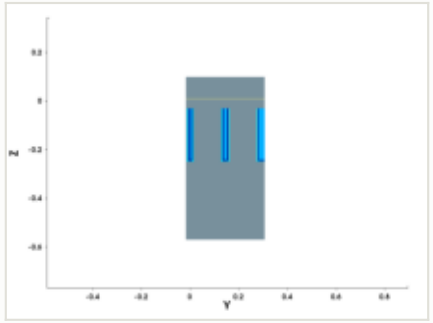
carbon so far, of the whole flow (kgCO2e)



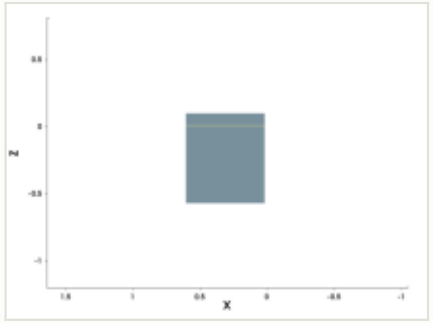
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



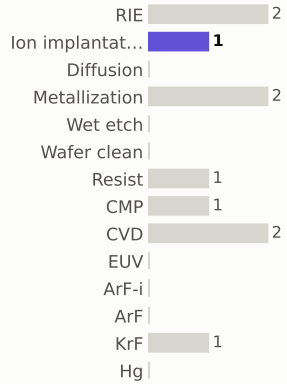
y cut at 0.184 µm



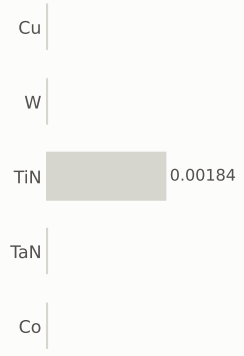
mask for this step

no mask for this step

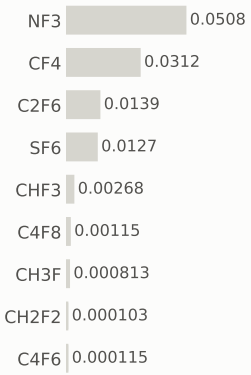
steps by process area



materials deposited (g)



gases emitted (g)



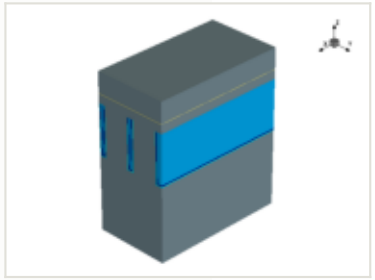
Step #150: Diffusion

treatment · Diffusion

0.801	61.3	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



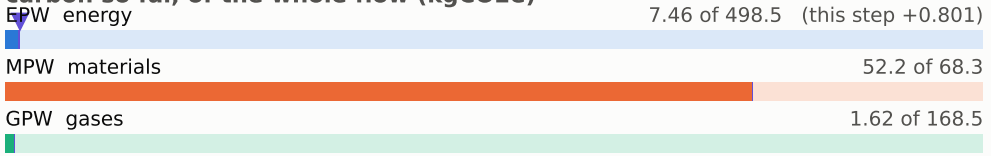
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

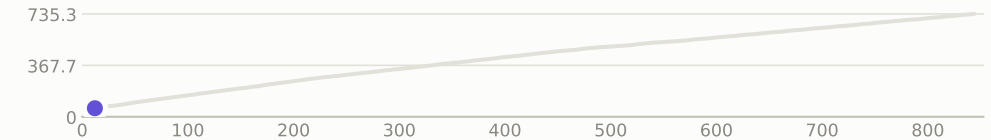
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

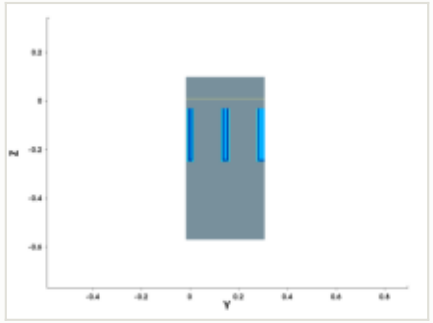
carbon so far, of the whole flow (kgCO2e)



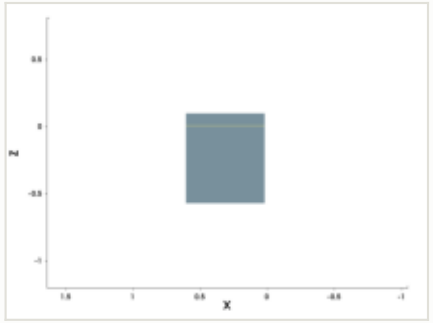
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



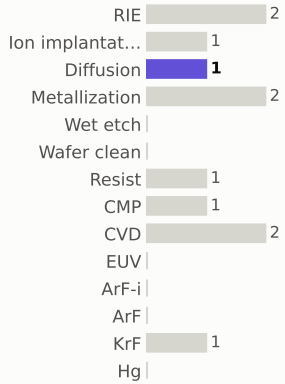
y cut at 0.184 µm



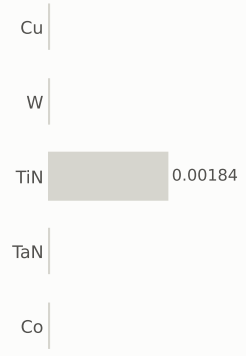
mask for this step

no mask for this step

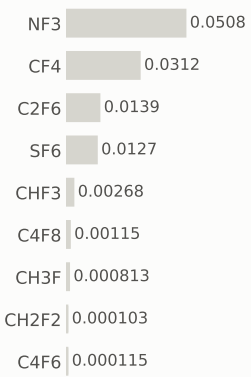
steps by process area



materials deposited (g)



gases emitted (g)



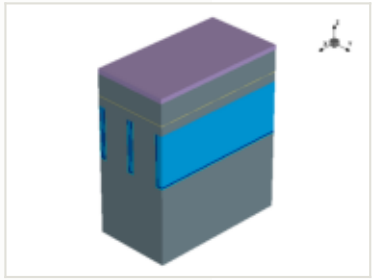
Step #170: CVD

depo_fill · CVD

1.31	62.6	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

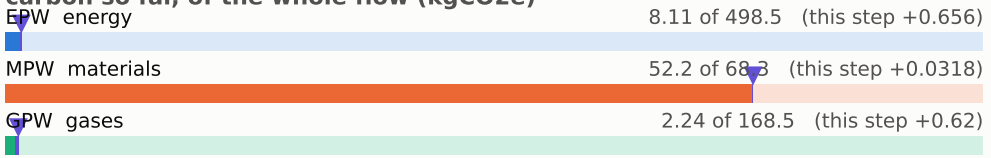


this step

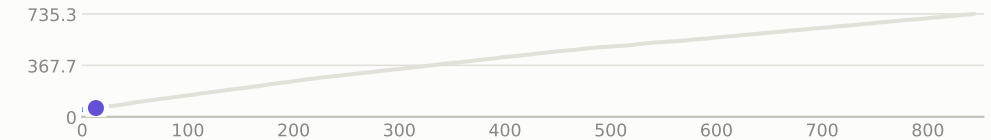
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

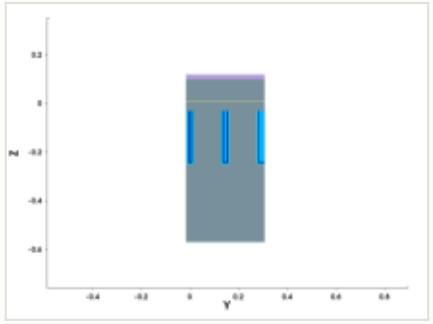
carbon so far, of the whole flow (kgCO2e)



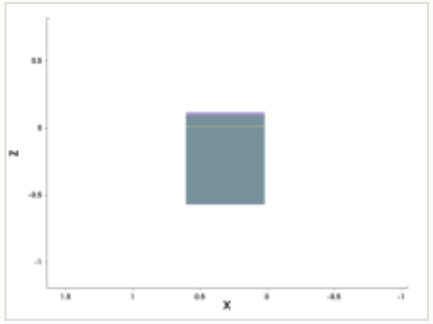
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



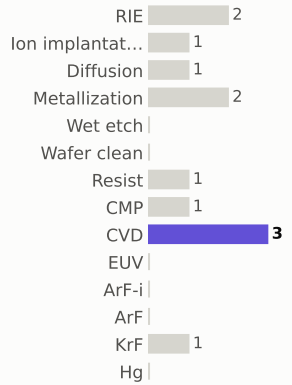
y cut at 0.184 µm



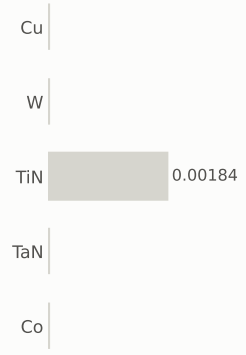
mask for this step

no mask for this step

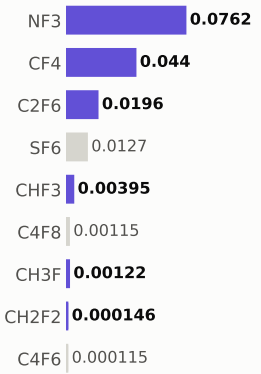
steps by process area



materials deposited (g)



gases emitted (g)



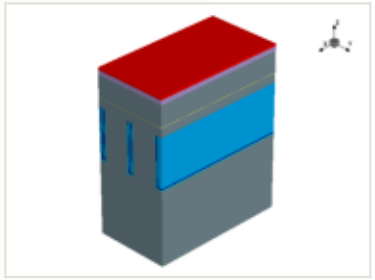
Step #180: CVD

depo_isotropic · CVD

1.31	63.9	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

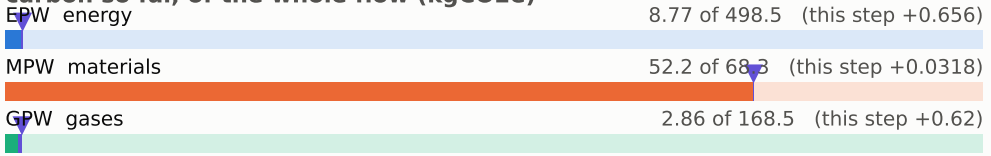
zoomed to fit



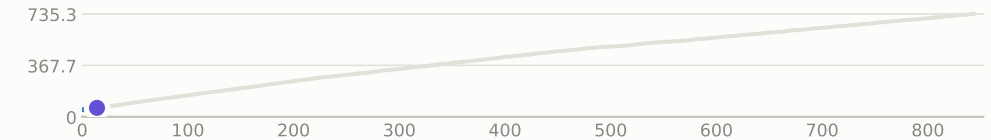
this step

process area	CVD	EPW	0.656 kgCO2e
material	—	EPW tool	0.469 kgCO2e
thickness	0.01 µm	EPW overhead	0.187 kgCO2e
material GWP	—	MPW	0.0318 kgCO2e (gas 0.0318)
input gases	NF3 2.54 g	GPW	0.62 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**1.31 kgCO2e**

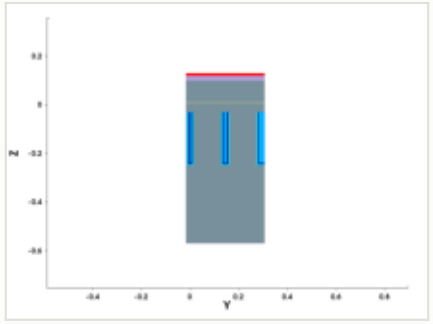
carbon so far, of the whole flow (kgCO2e)



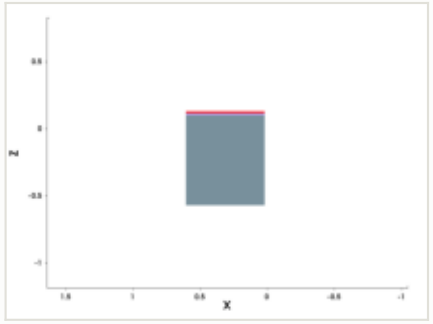
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



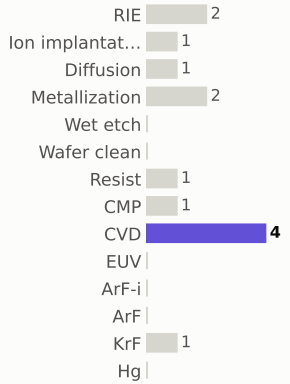
y cut at 0.184 µm



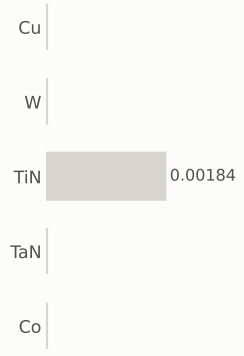
mask for this step

no mask for this step

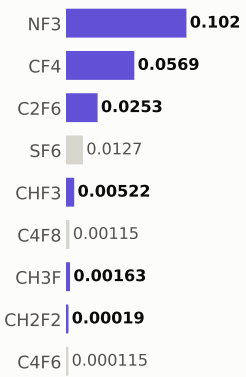
steps by process area



materials deposited (g)



gases emitted (g)



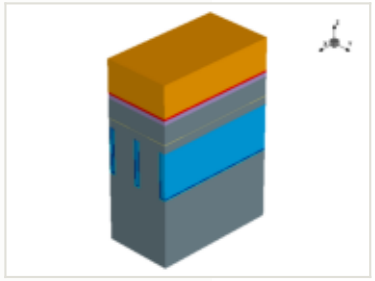
Step #190: CVD

depo_fill · CVD

1.31	65.2	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

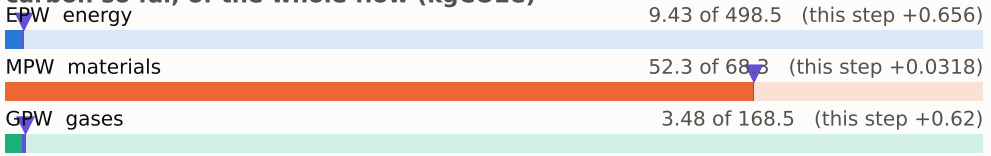


this step

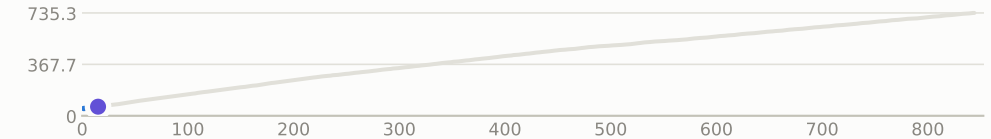
process area	CVD
material	—
thickness	0.15 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

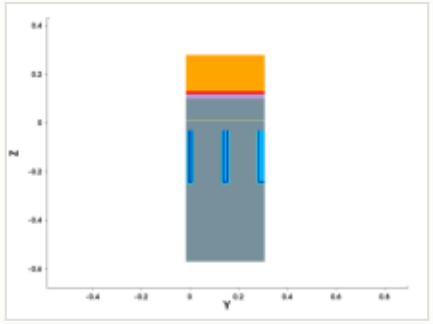
carbon so far, of the whole flow (kgCO2e)



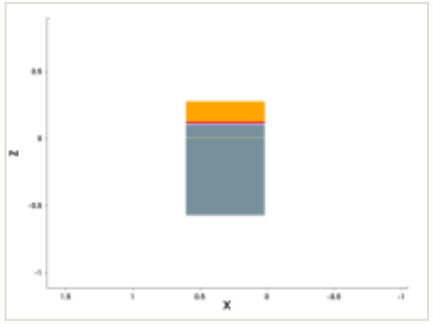
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



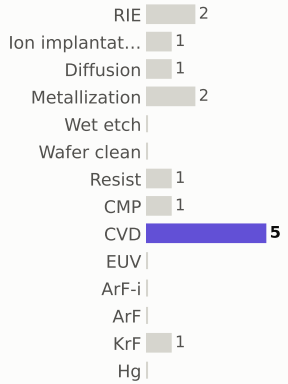
y cut at 0.184 µm



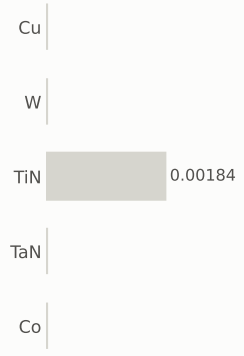
mask for this step

no mask for this step

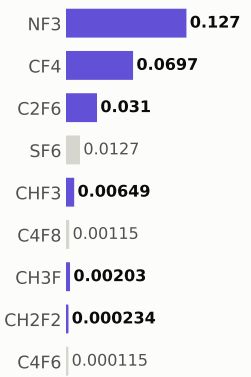
steps by process area



materials deposited (g)



gases emitted (g)



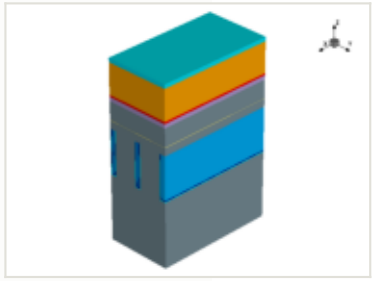
Step #200: CVD

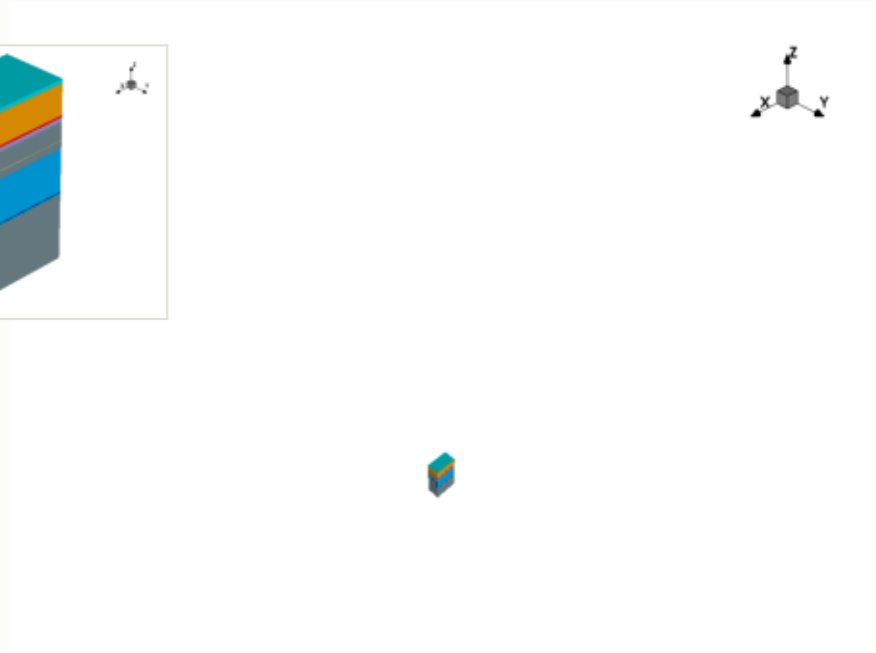
depo_isotropic · CVD

1.31	66.5	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

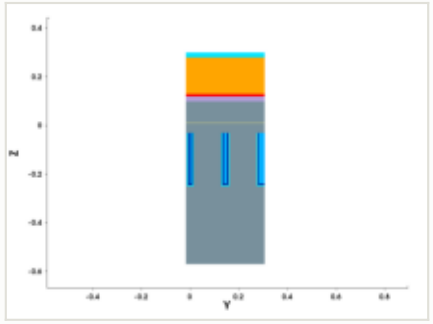
3D structure after this step

zoomed to fit

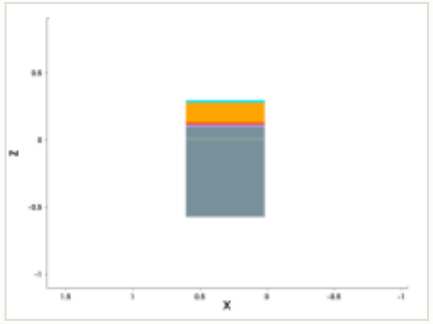




x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

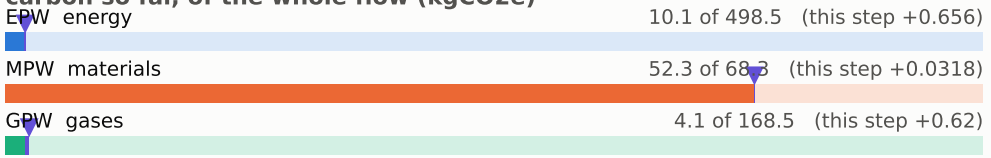
no mask for this step

this step

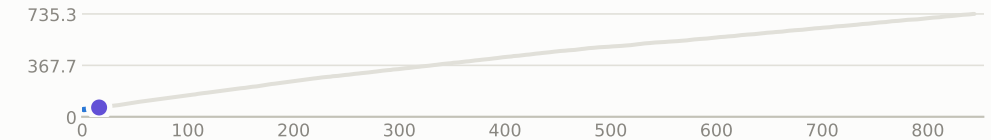
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

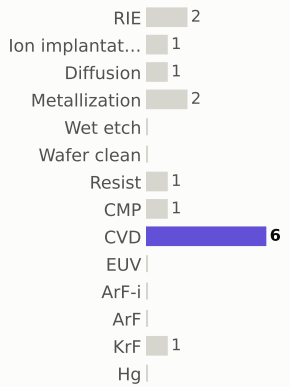
carbon so far, of the whole flow (kgCO2e)



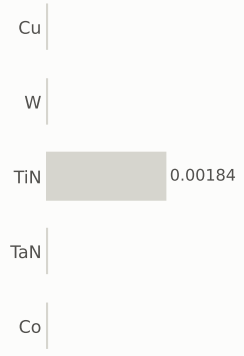
embodied carbon up to each step (kgCO2e per wafer)



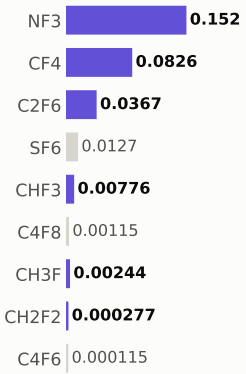
steps by process area



materials deposited (g)



gases emitted (g)



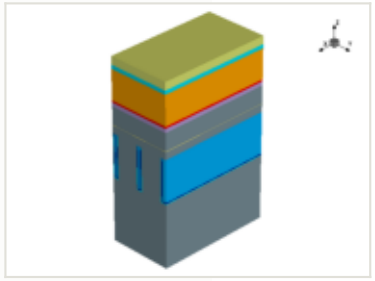
Step #210: CVD

depo_isotropic · CVD

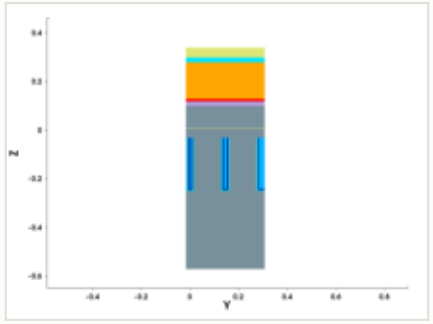
1.31	67.8	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

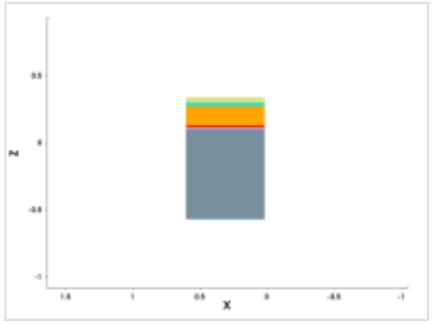
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

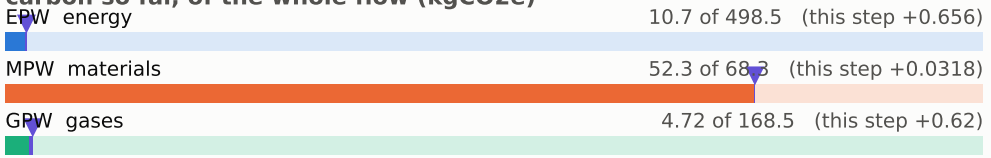
no mask for this step

this step

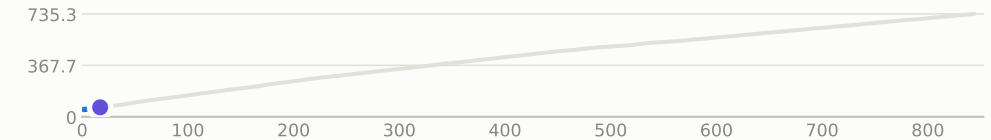
process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

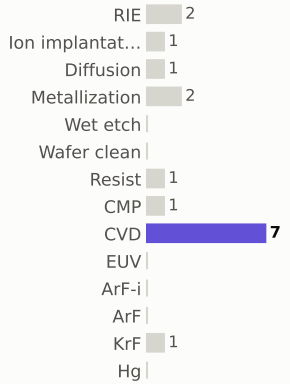
carbon so far, of the whole flow (kgCO2e)



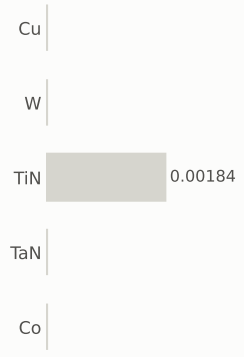
embodied carbon up to each step (kgCO2e per wafer)



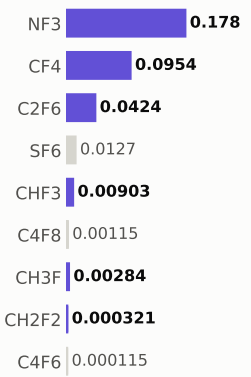
steps by process area



materials deposited (g)



gases emitted (g)



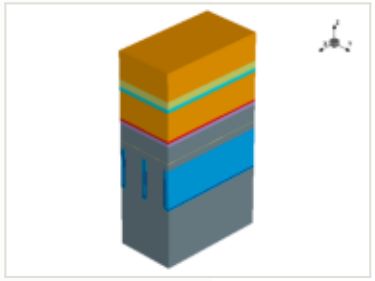
Step #220: CVD

depo_fill · CVD

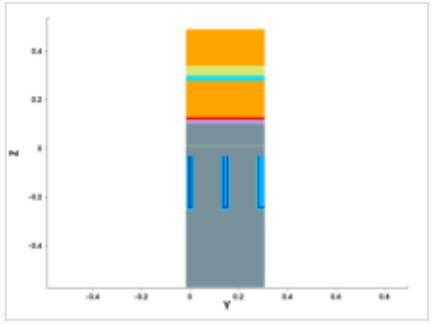
1.31	69.1	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

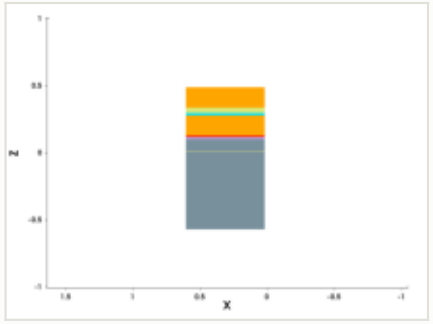
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

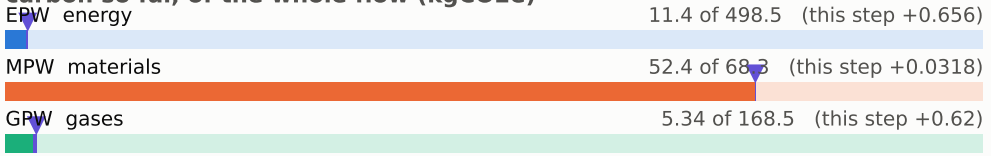
no mask for this step

this step

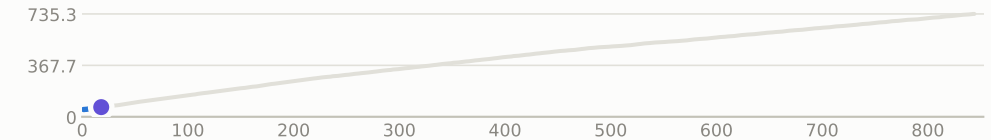
process area	CVD
material	—
thickness	0.15 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

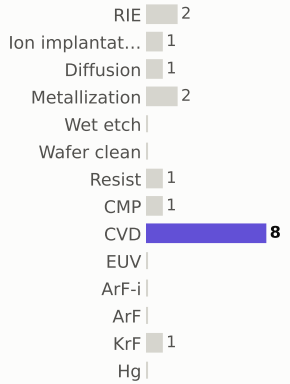
carbon so far, of the whole flow (kgCO2e)



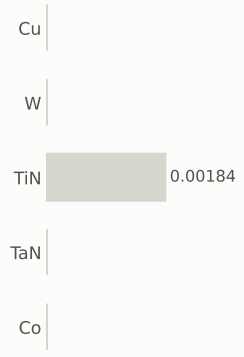
embodied carbon up to each step (kgCO2e per wafer)



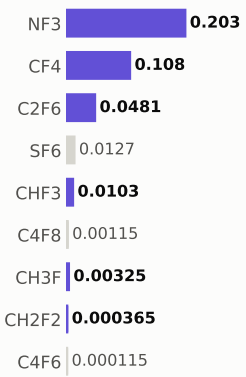
steps by process area



materials deposited (g)



gases emitted (g)



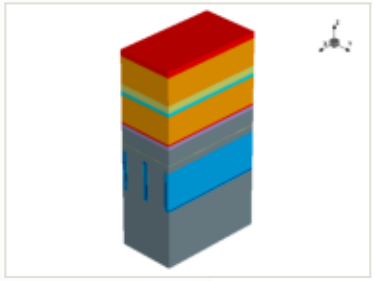
Step #230: CVD

depo_fill · CVD

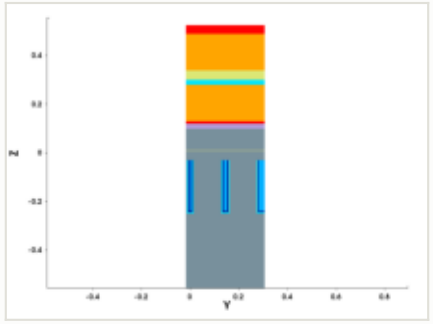
1.31	70.4	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

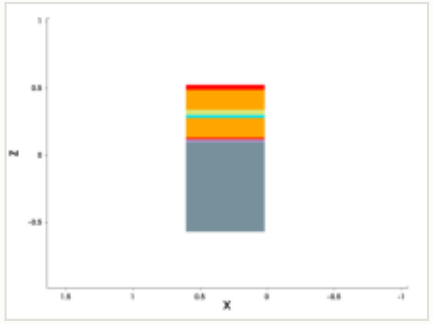
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



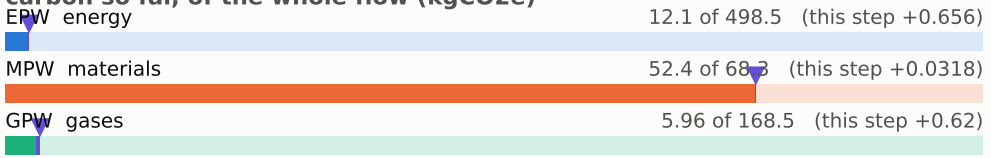
mask for this step

no mask for this step

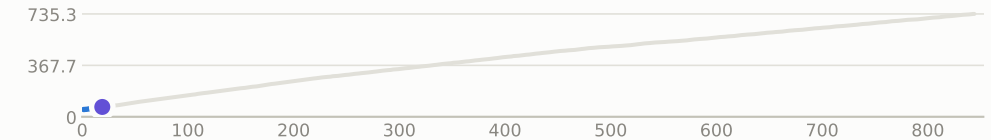
this step

process area	CVD	EPW	0.656 kgCO2e
material	—	EPW tool	0.469 kgCO2e
thickness	0.035 µm	EPW overhead	0.187 kgCO2e
material GWP	—	MPW	0.0318 kgCO2e (gas 0.0318)
input gases	NF3 2.54 g	GPW	0.62 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**1.31 kgCO2e**

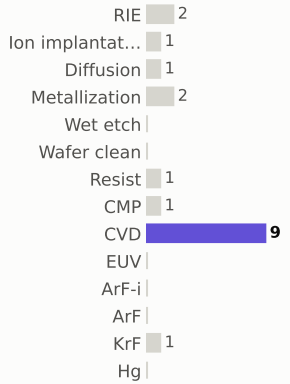
carbon so far, of the whole flow (kgCO2e)



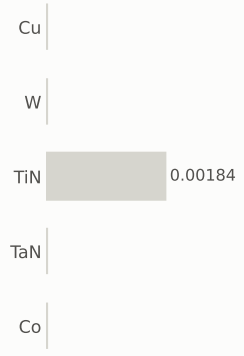
embodied carbon up to each step (kgCO2e per wafer)



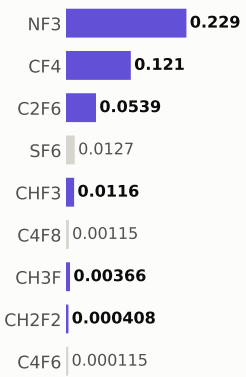
steps by process area



materials deposited (g)



gases emitted (g)



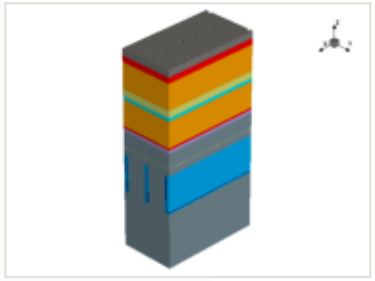
Step #240: KrF

litho_patterning · KrF

0.199	70.6	0.027%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

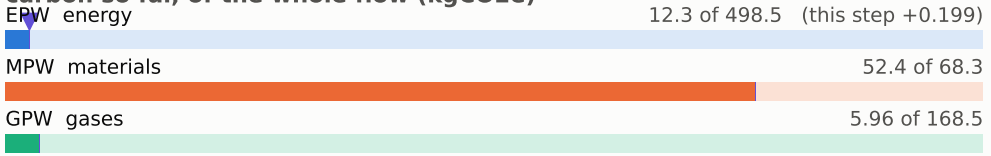


this step

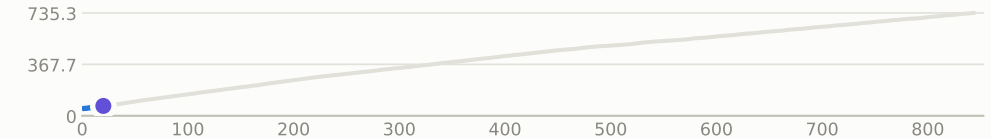
process area	KrF
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

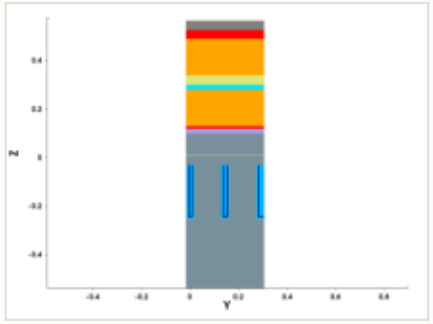
carbon so far, of the whole flow (kgCO2e)



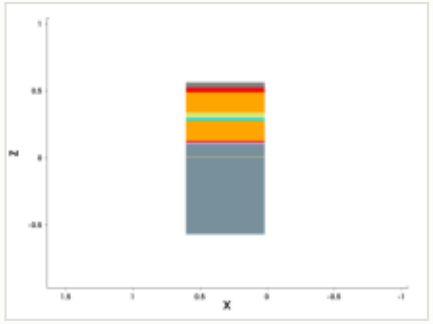
embodied carbon up to each step (kgCO2e per wafer)



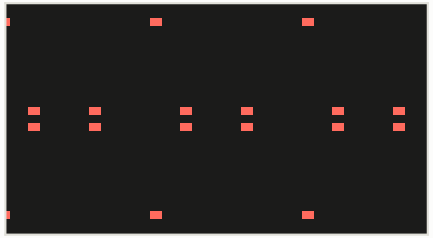
x cut at 0.294 µm



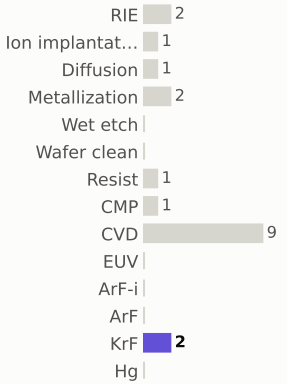
y cut at 0.184 µm



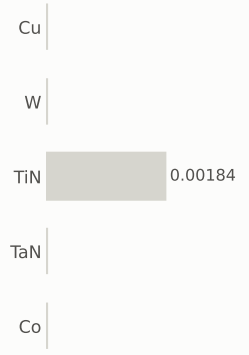
mask: layer 9:0 (GDSII)



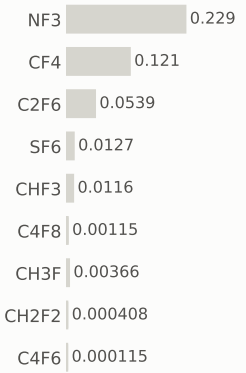
steps by process area



materials deposited (g)



gases emitted (g)



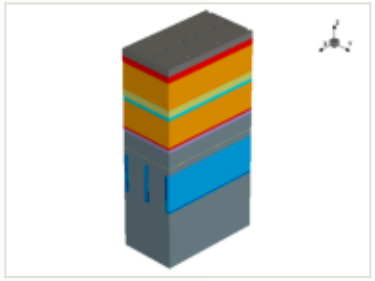
Step #250: RIE

etch_anisotropic · RIE

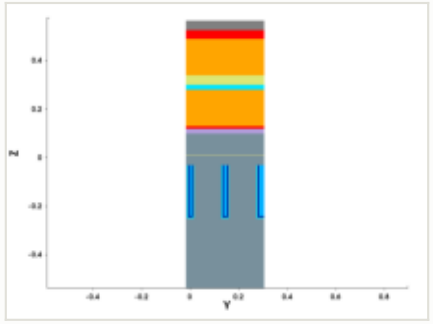
1.04	71.7	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

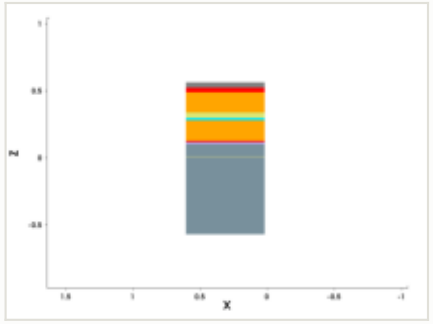
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



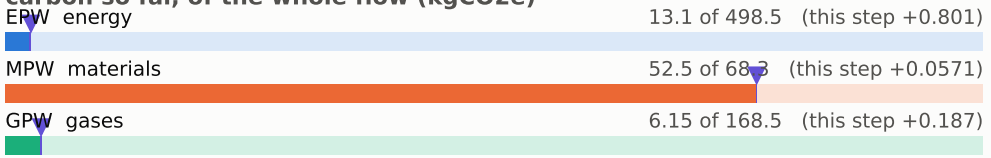
mask for this step

no mask for this step

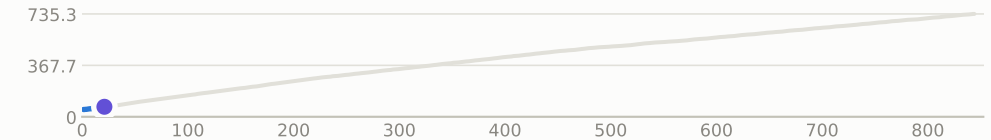
this step

process area	RIE	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

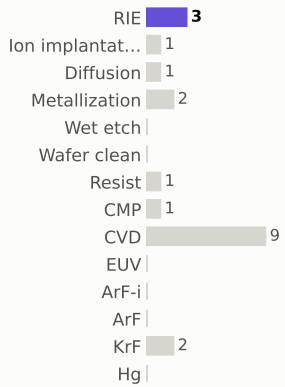
carbon so far, of the whole flow (kgCO2e)



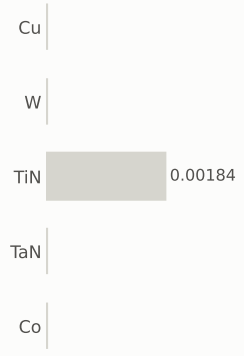
embodied carbon up to each step (kgCO2e per wafer)



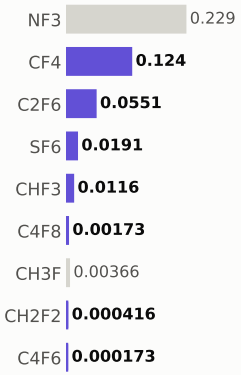
steps by process area



materials deposited (g)



gases emitted (g)



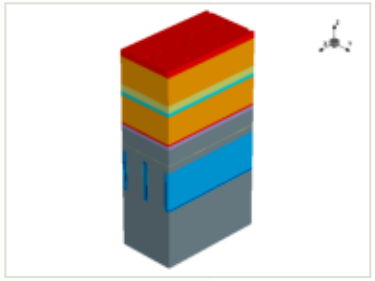
Step #260: RIE

etch_anisotropic · RIE

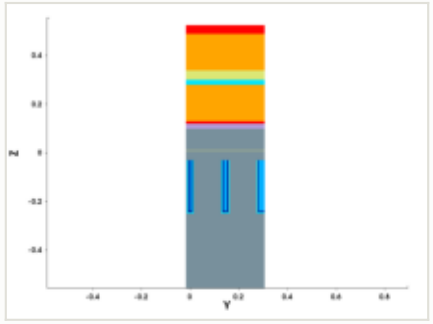
1.04	72.7	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

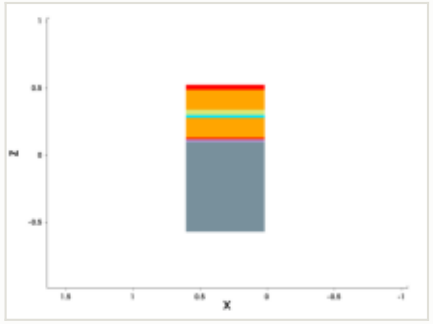
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



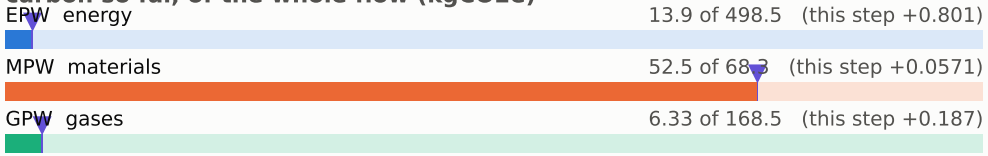
mask for this step

no mask for this step

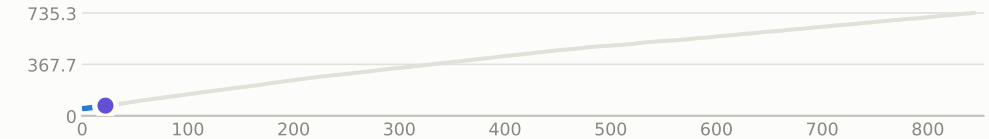
this step

process area	RIE	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

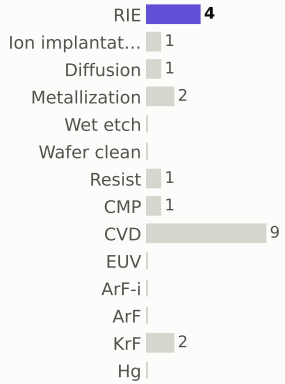
carbon so far, of the whole flow (kgCO2e)



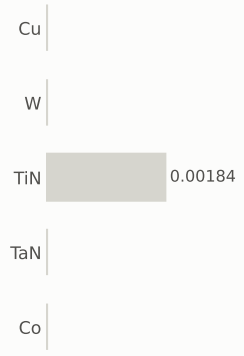
embodied carbon up to each step (kgCO2e per wafer)



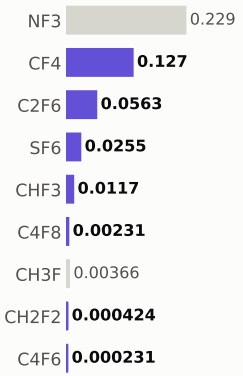
steps by process area



materials deposited (g)



gases emitted (g)



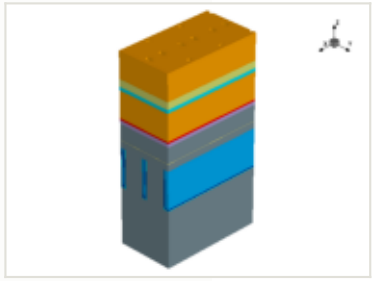
Step #270: RIE

etch_anisotropic · RIE

1.04	73.7	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

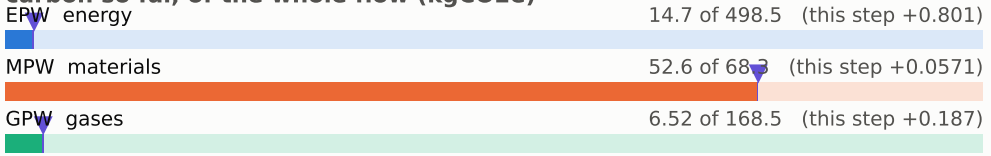


this step

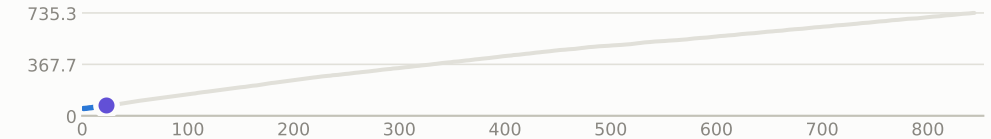
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

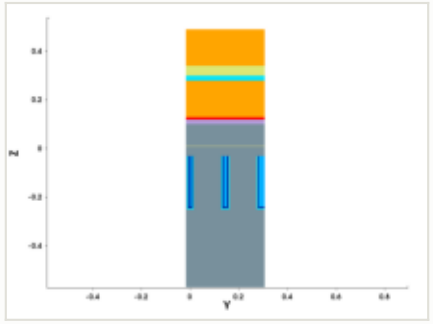
carbon so far, of the whole flow (kgCO2e)



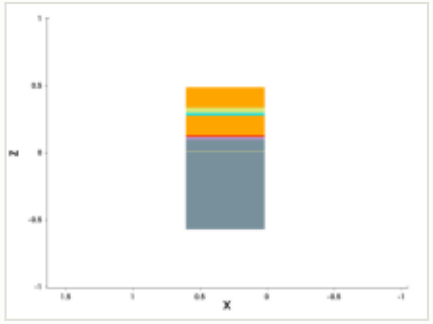
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



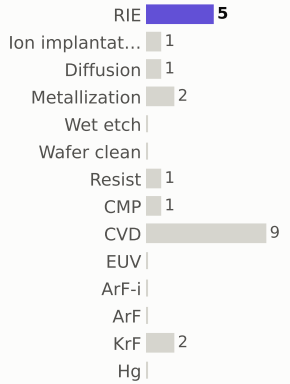
y cut at 0.184 µm



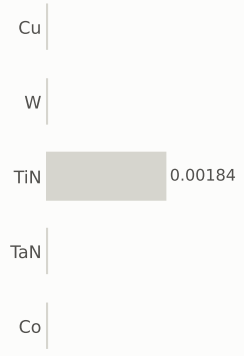
mask for this step

no mask for this step

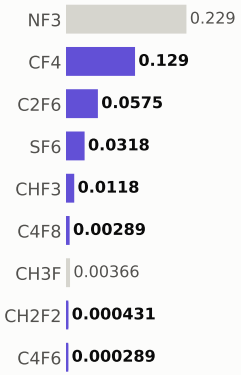
steps by process area



materials deposited (g)



gases emitted (g)



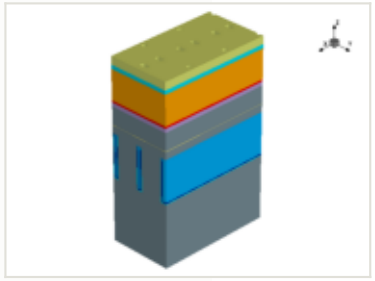
Step #280: Resist

etch_strip · Resist

0.169	73.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

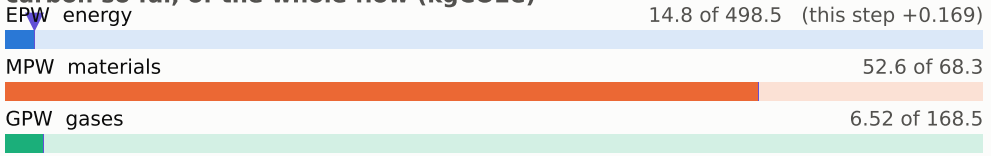


this step

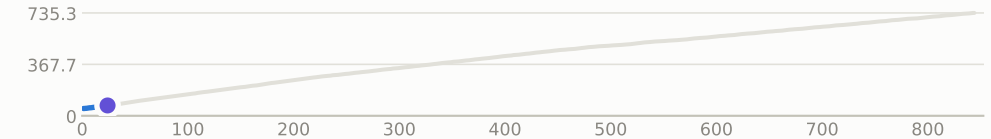
process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

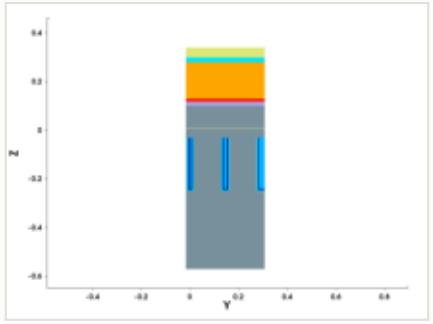
carbon so far, of the whole flow (kgCO2e)



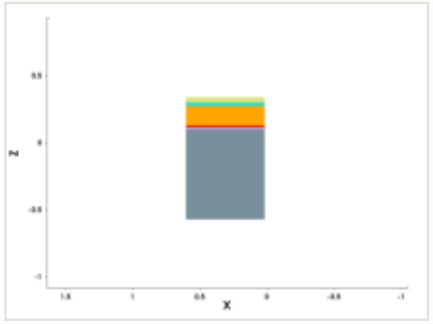
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



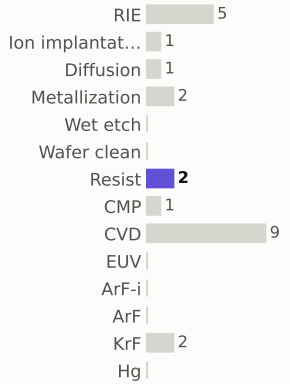
y cut at 0.184 µm



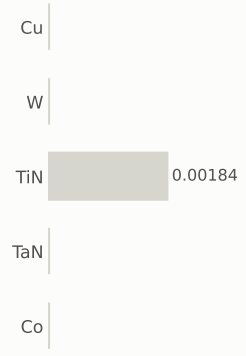
mask for this step

no mask for this step

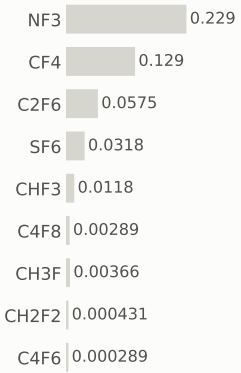
steps by process area



materials deposited (g)



gases emitted (g)



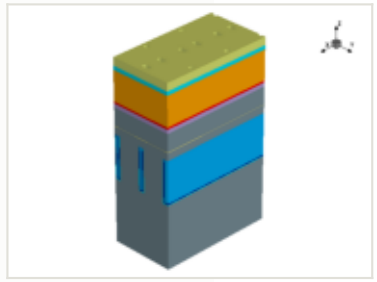
Step #290: RIE

etch_anisotropic · RIE

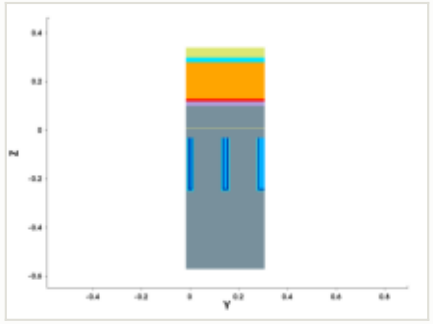
1.04	75	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

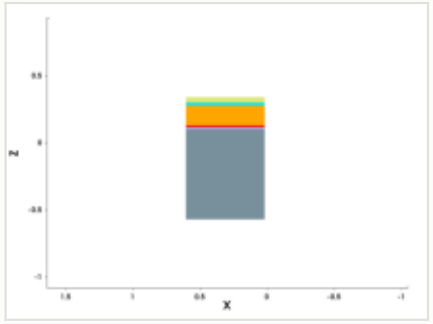
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

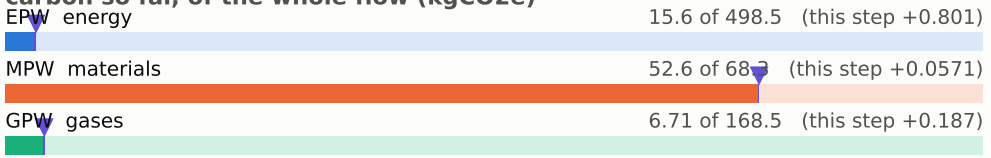
no mask for this step

this step

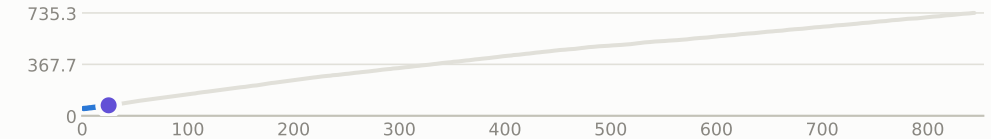
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

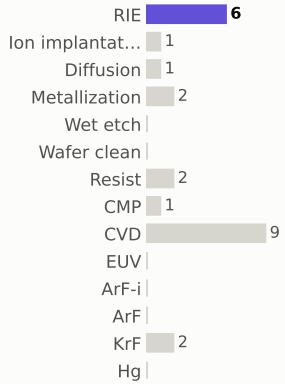
carbon so far, of the whole flow (kgCO2e)



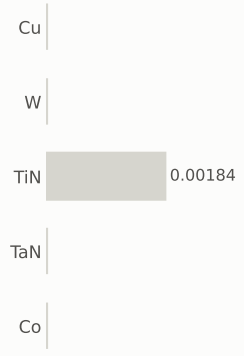
embodied carbon up to each step (kgCO2e per wafer)



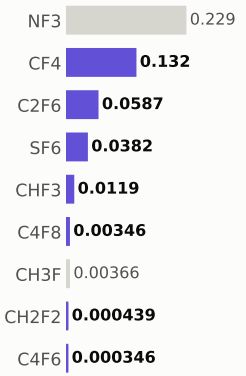
steps by process area



materials deposited (g)



gases emitted (g)



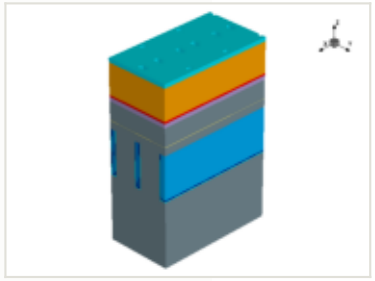
Step #300: RIE

etch_anisotropic · RIE

1.04	76	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

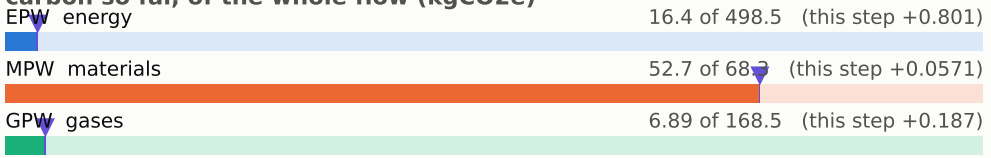


this step

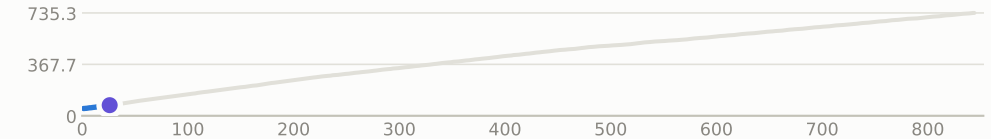
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

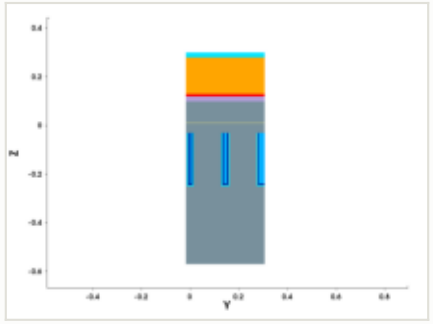
carbon so far, of the whole flow (kgCO2e)



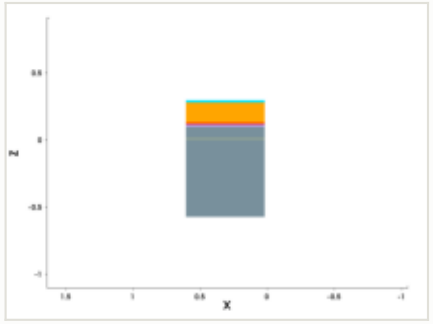
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



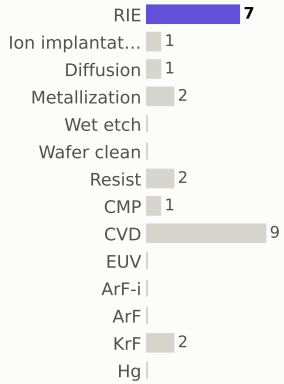
y cut at 0.184 µm



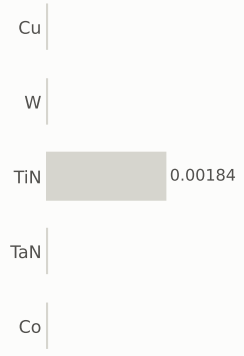
mask for this step

no mask for this step

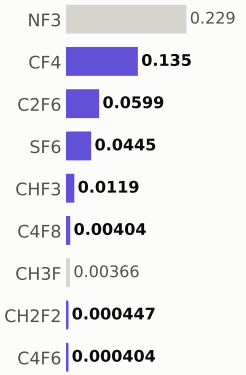
steps by process area



materials deposited (g)



gases emitted (g)



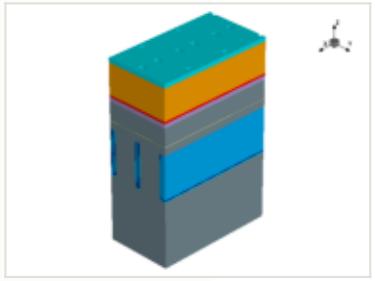
Step #310: RIE

etch_anisotropic · RIE

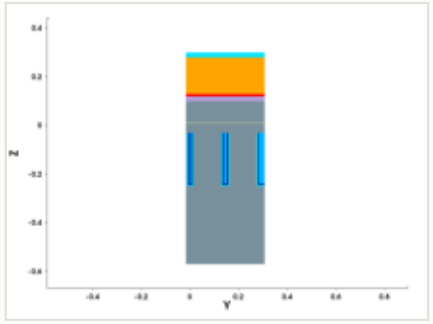
1.04	77	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

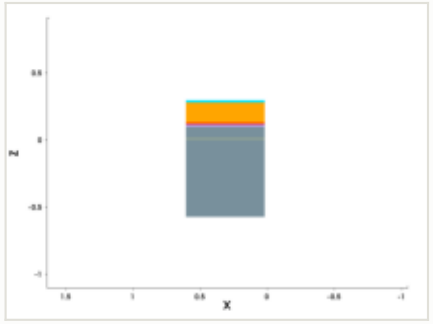
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

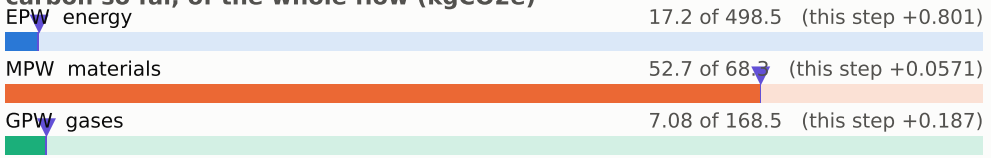
no mask for this step

this step

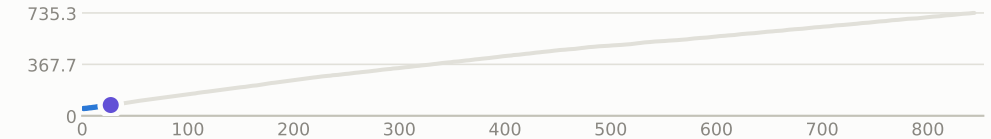
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

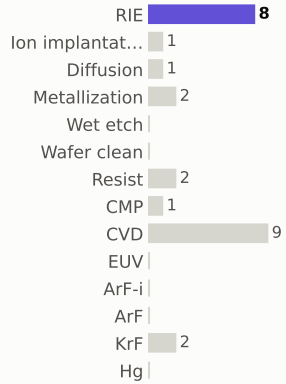
carbon so far, of the whole flow (kgCO2e)



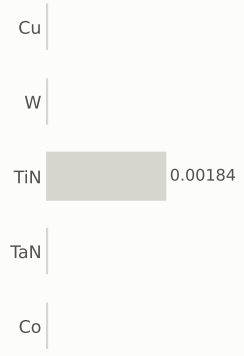
embodied carbon up to each step (kgCO2e per wafer)



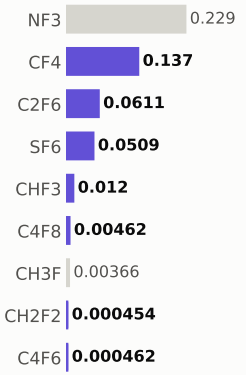
steps by process area



materials deposited (g)



gases emitted (g)



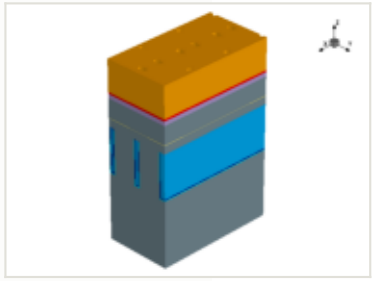
Step #320: RIE

etch_anisotropic · RIE

1.04	78.1	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

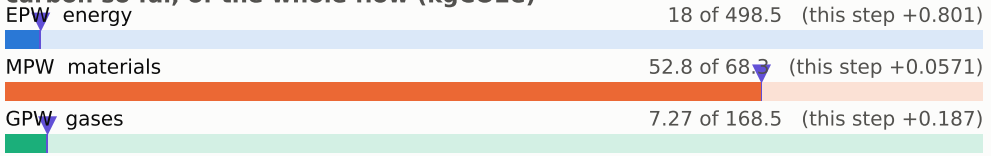


this step

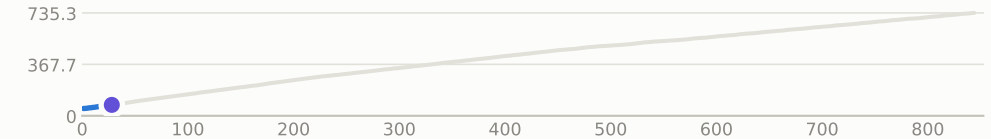
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

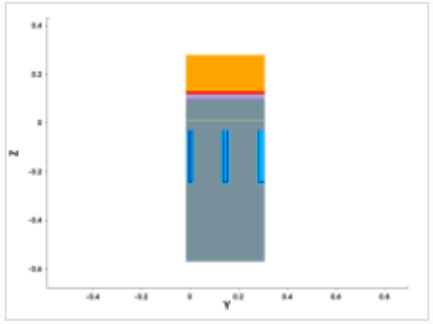
carbon so far, of the whole flow (kgCO2e)



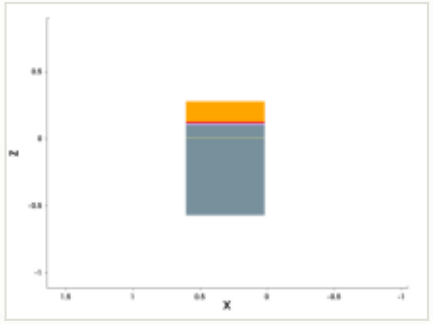
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



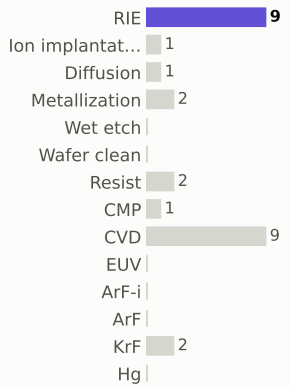
y cut at 0.184 µm



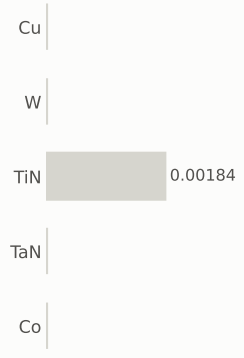
mask for this step

no mask for this step

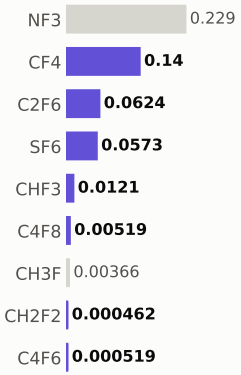
steps by process area



materials deposited (g)



gases emitted (g)



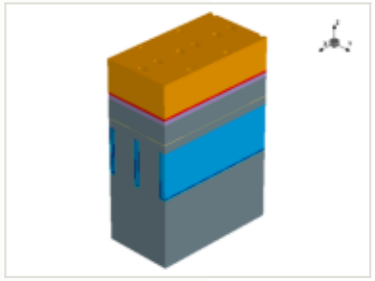
Step #330: RIE

etch_anisotropic · RIE

1.04	79.1	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

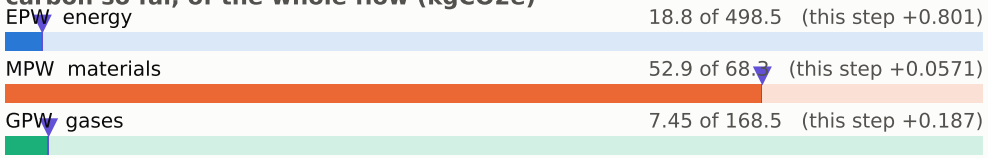
zoomed to fit



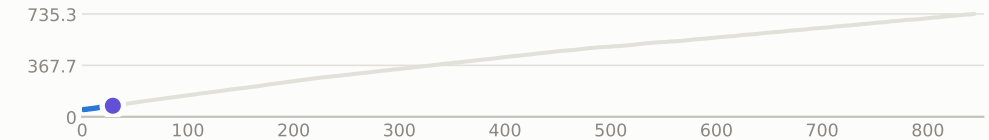
this step

process area	RIE	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

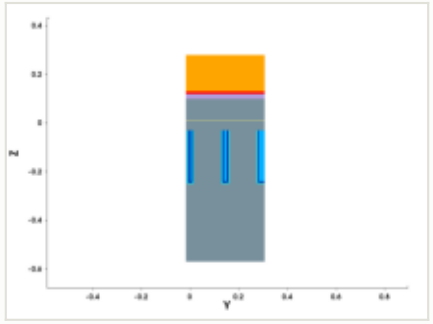
carbon so far, of the whole flow (kgCO2e)



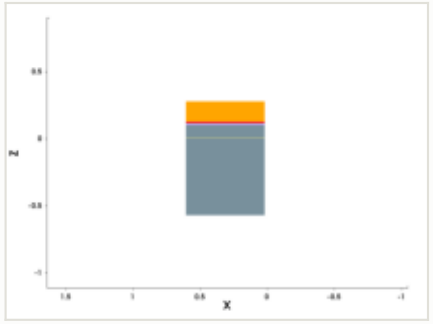
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



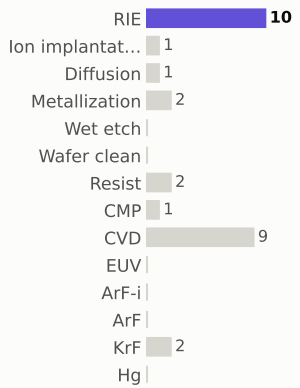
y cut at 0.184 µm



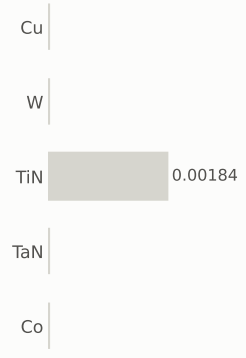
mask for this step

no mask for this step

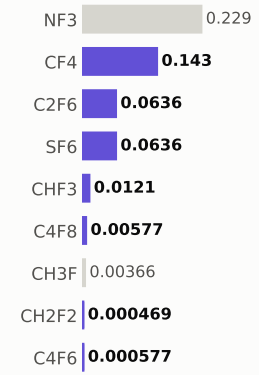
steps by process area



materials deposited (g)



gases emitted (g)



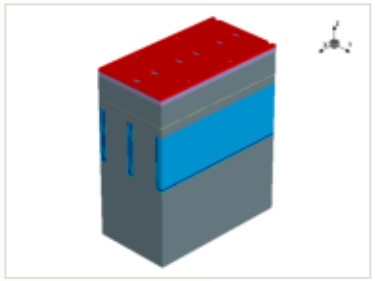
Step #340: Resist

etch_strip · Resist

0.169	79.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

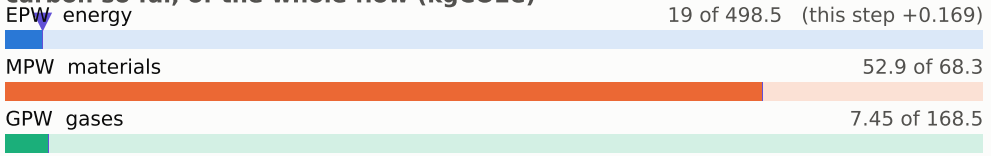


this step

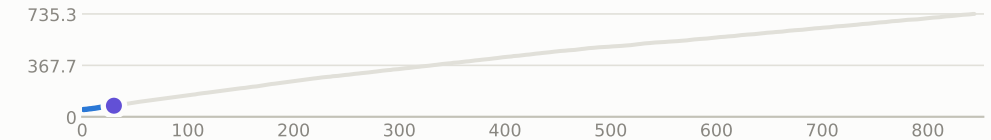
process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

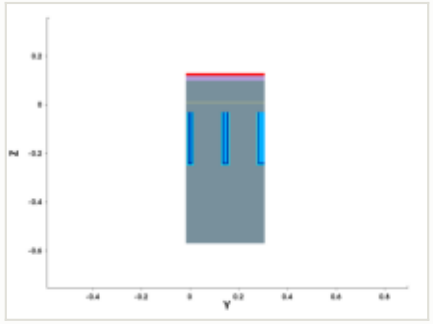
carbon so far, of the whole flow (kgCO2e)



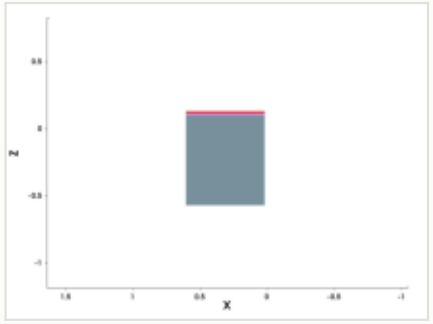
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



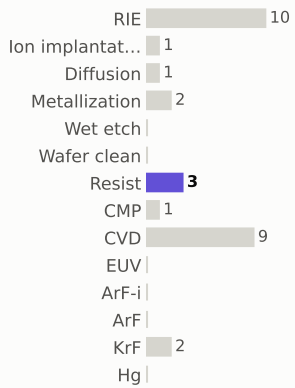
y cut at 0.184 µm



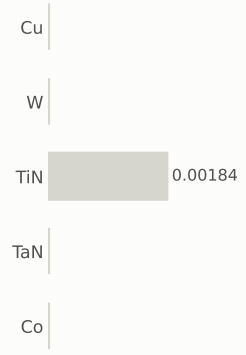
mask for this step

no mask for this step

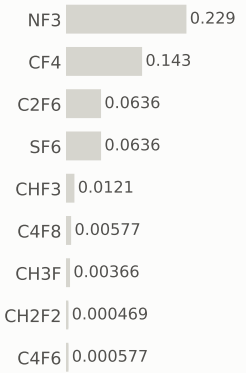
steps by process area



materials deposited (g)



gases emitted (g)



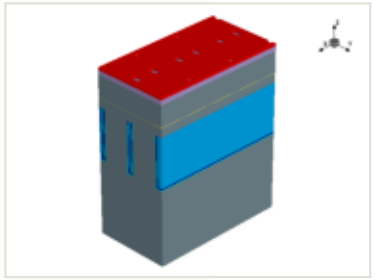
Step #350: RIE

etch_anisotropic · RIE

1.04	80.3	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

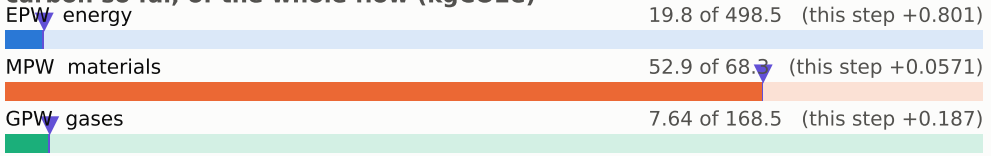


this step

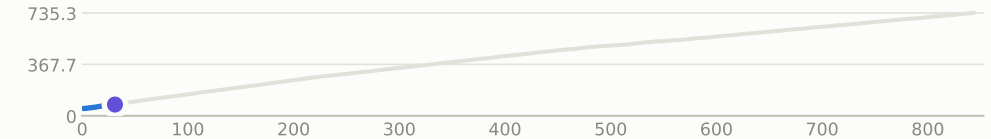
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

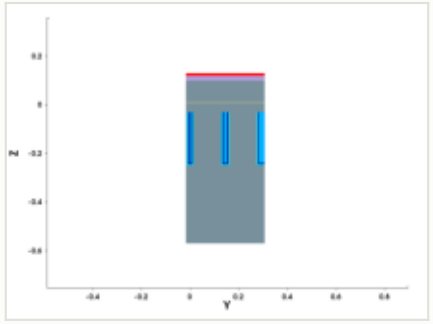
carbon so far, of the whole flow (kgCO2e)



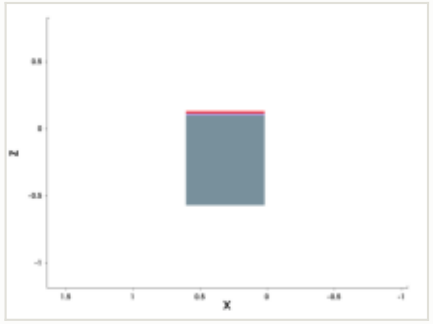
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



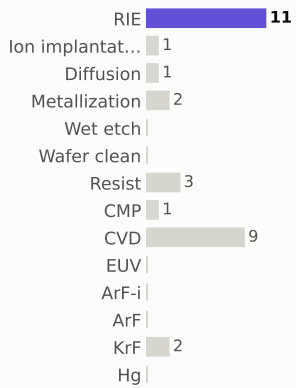
y cut at 0.184 µm



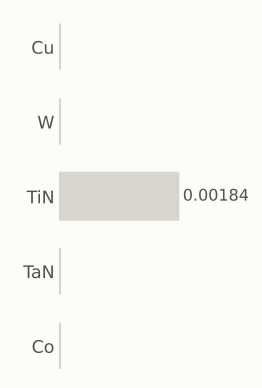
mask for this step

no mask for this step

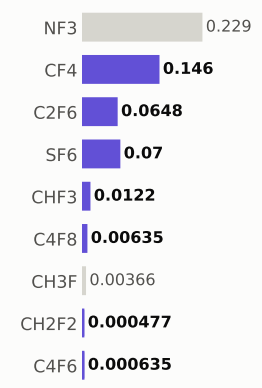
steps by process area



materials deposited (g)



gases emitted (g)



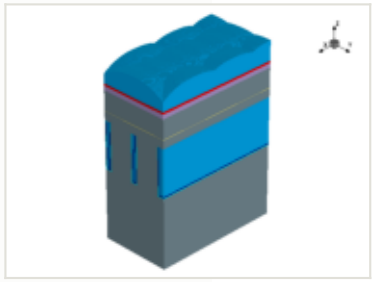
Step #360: Metallization

depo_isotropic · Metallization

1.23	81.6	0.17%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

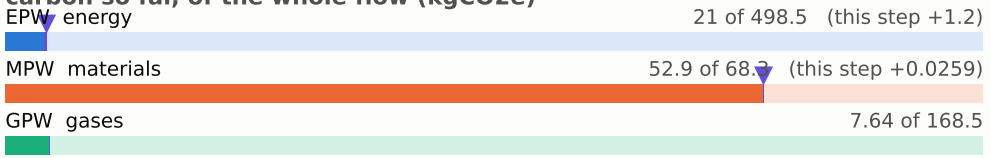


this step

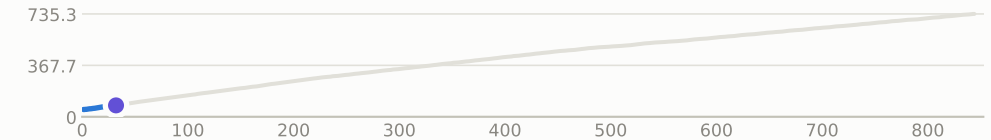
process area	Metallization
material	W
thickness	0.25 µm
material GWP	75.8149 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.0259 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.23 kgCO2e**

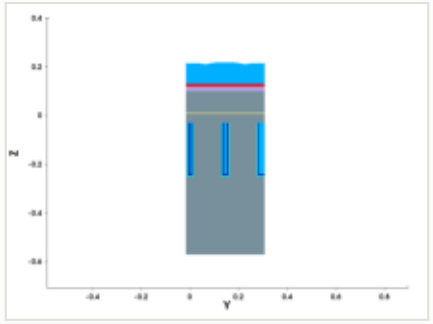
carbon so far, of the whole flow (kgCO2e)



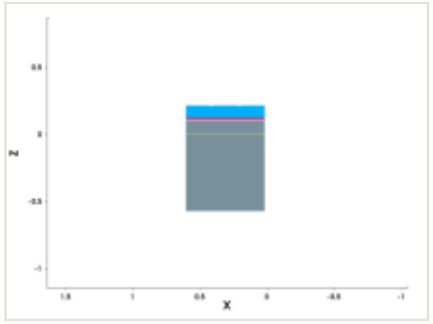
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



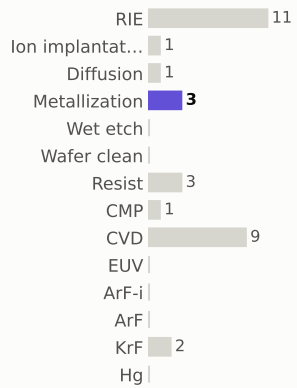
y cut at 0.184 µm



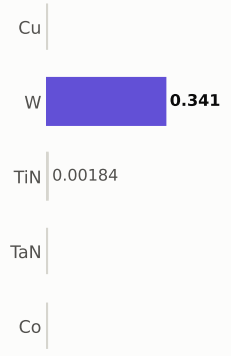
mask for this step

no mask for this step

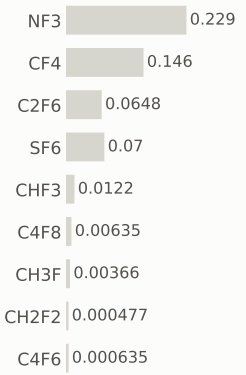
steps by process area



materials deposited (g)



gases emitted (g)



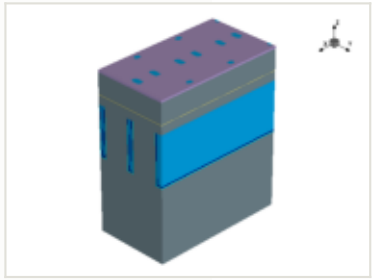
Step #370: CMP

etch_cmp · CMP

0.169	81.7	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

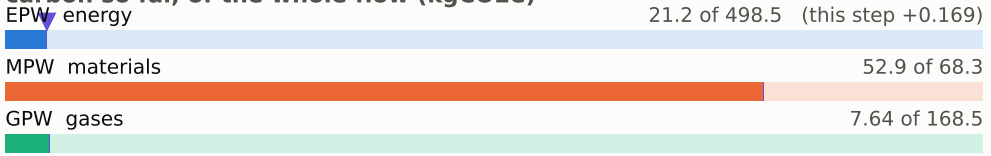


this step

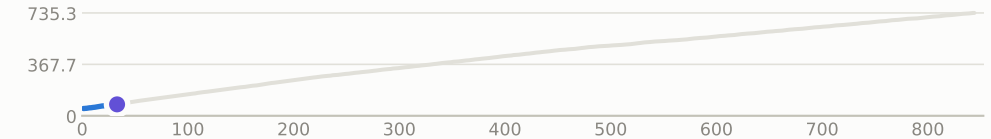
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

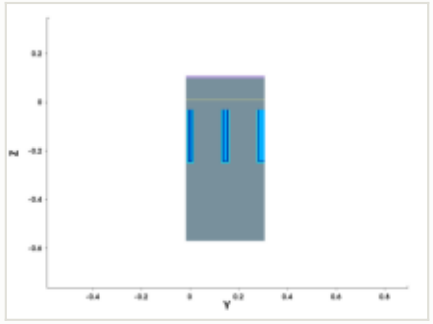
carbon so far, of the whole flow (kgCO2e)



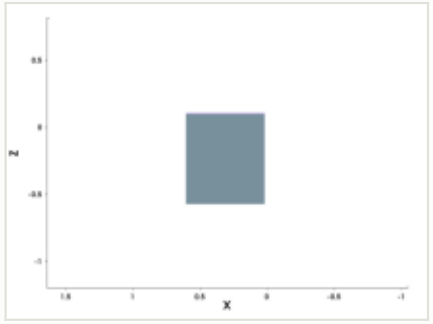
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



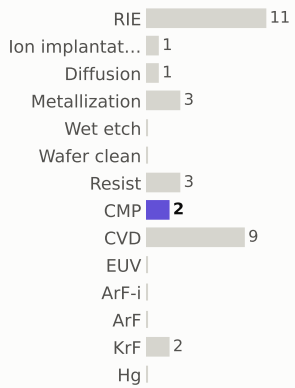
y cut at 0.184 µm



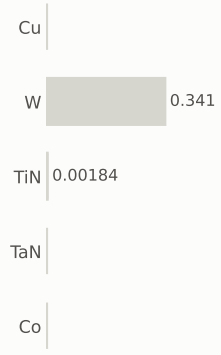
mask for this step

no mask for this step

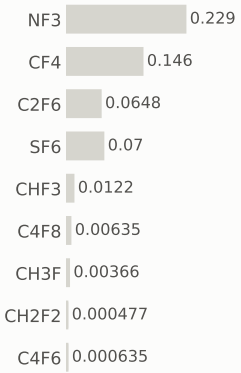
steps by process area



materials deposited (g)



gases emitted (g)



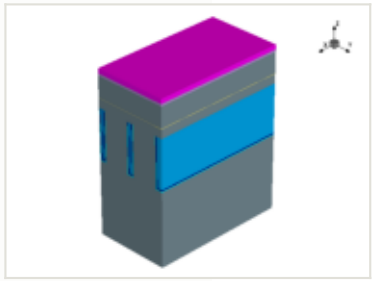
Step #390: CVD

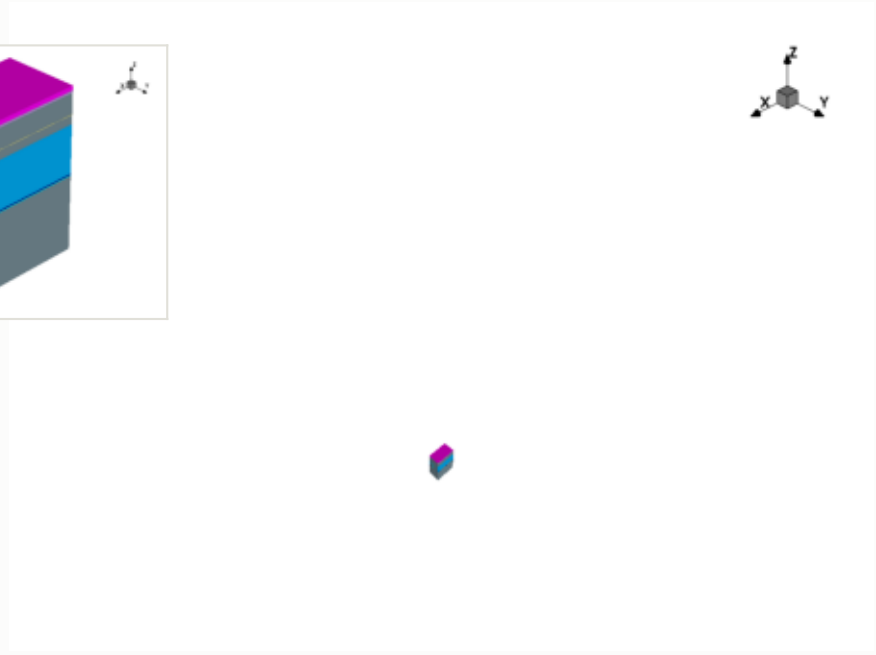
depo_epi · CVD

1.31	83.1	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



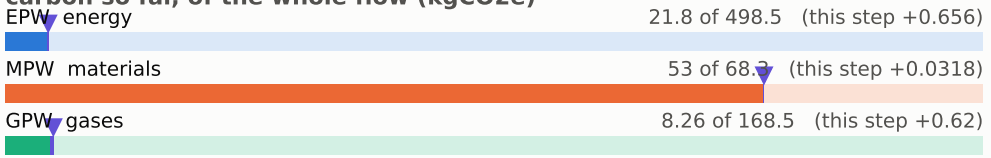


this step

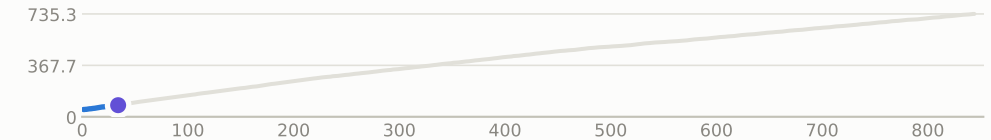
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

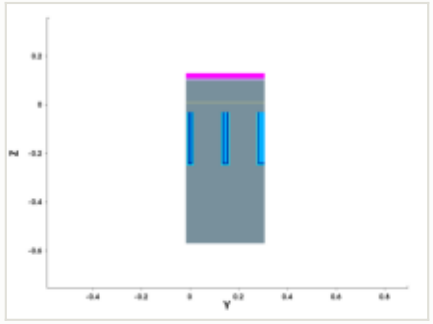
carbon so far, of the whole flow (kgCO2e)



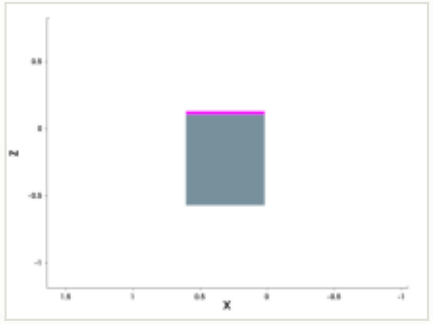
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



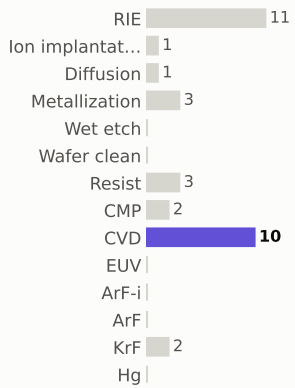
y cut at 0.184 µm



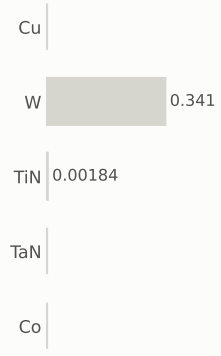
mask for this step

no mask for this step

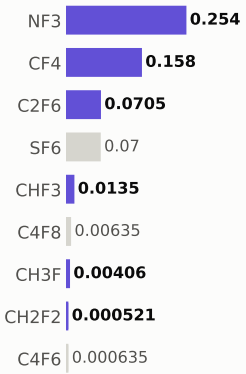
steps by process area



materials deposited (g)



gases emitted (g)



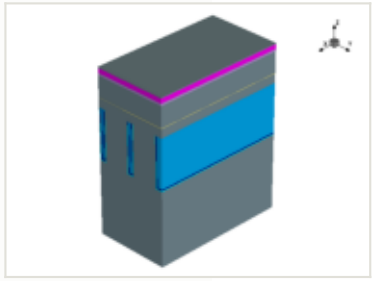
Step #400: CVD

depo_epi · CVD

1.31	84.4	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

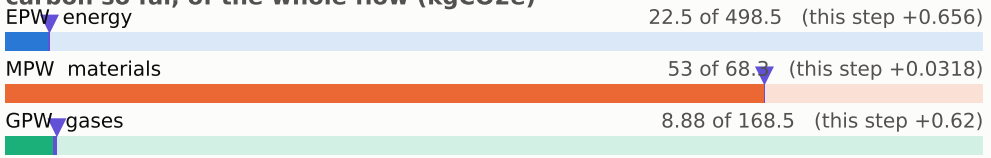


this step

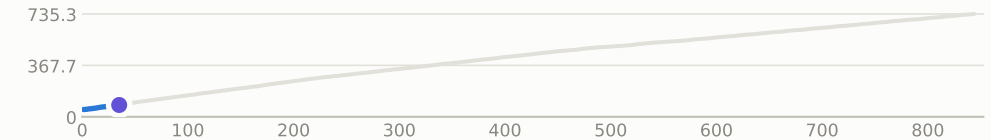
process area	CVD
material	—
thickness	0.005 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

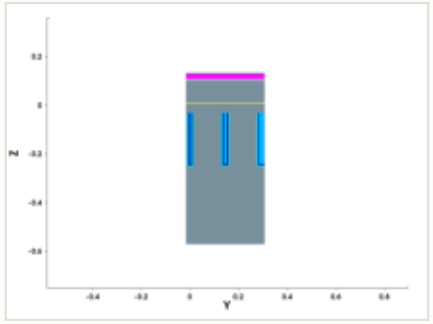
carbon so far, of the whole flow (kgCO2e)



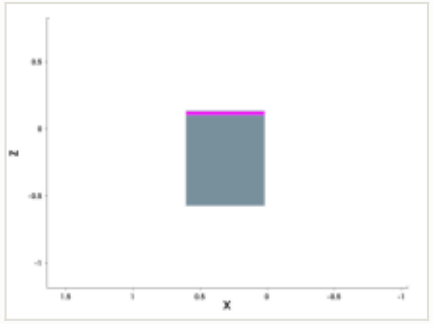
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



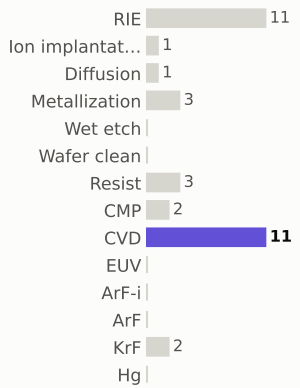
y cut at 0.184 µm



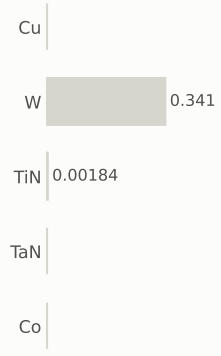
mask for this step

no mask for this step

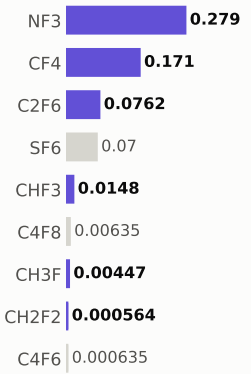
steps by process area



materials deposited (g)



gases emitted (g)



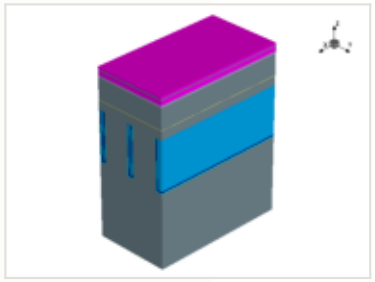
Step #410: CVD

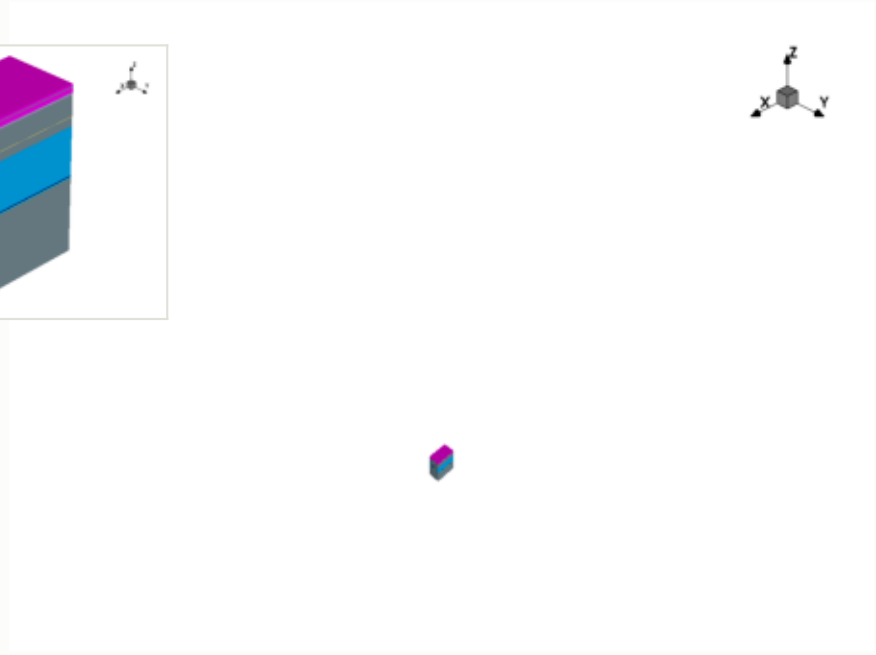
depo_epi · CVD

1.31	85.7	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



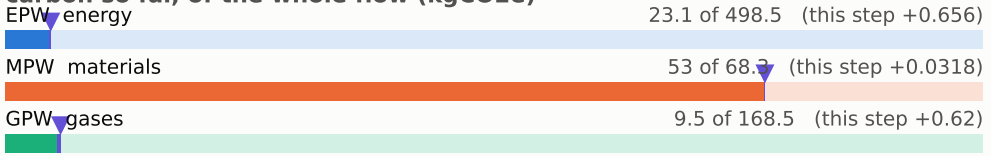


this step

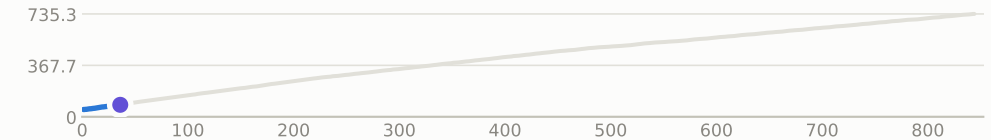
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

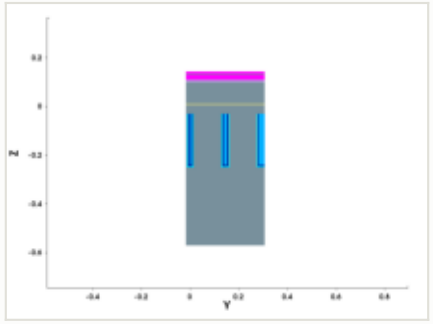
carbon so far, of the whole flow (kgCO2e)



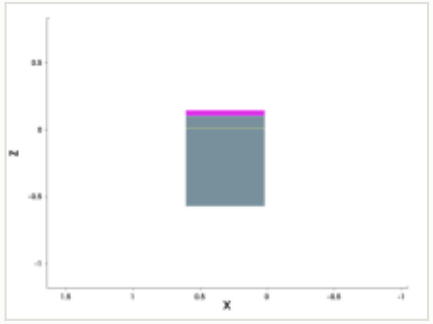
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



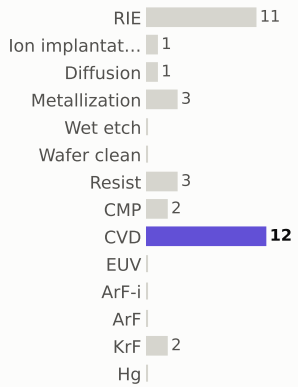
y cut at 0.184 µm



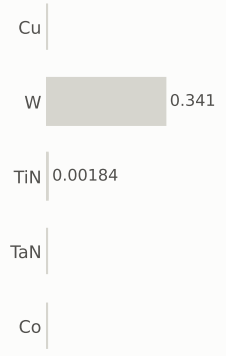
mask for this step

no mask for this step

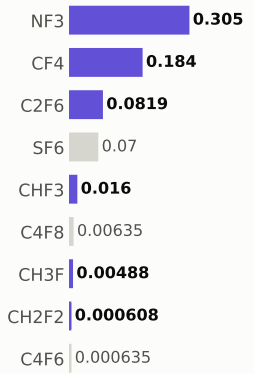
steps by process area



materials deposited (g)



gases emitted (g)



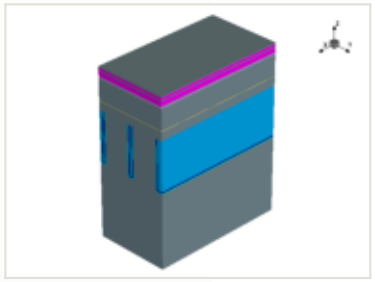
Step #420: CVD

depo_epi · CVD

1.31	87	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

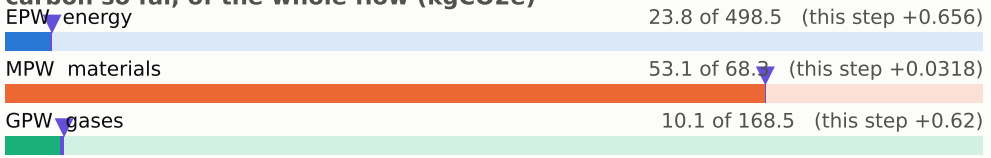


this step

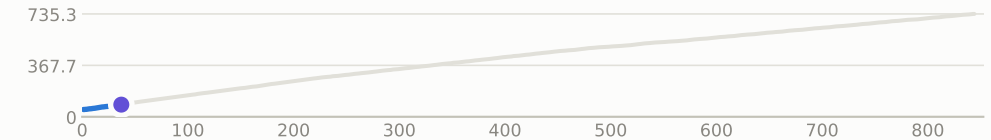
process area	CVD
material	—
thickness	0.005 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

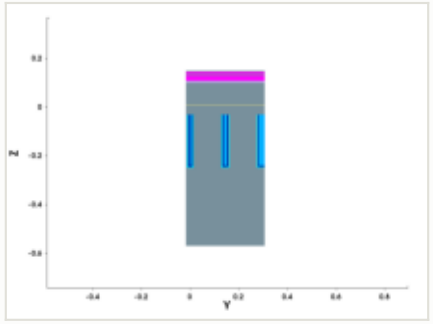
carbon so far, of the whole flow (kgCO2e)



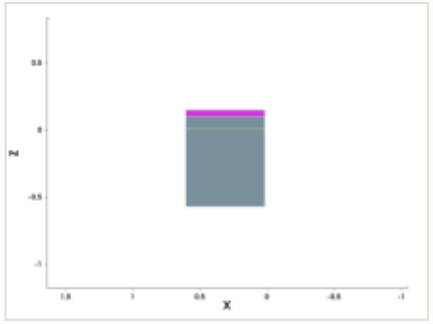
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



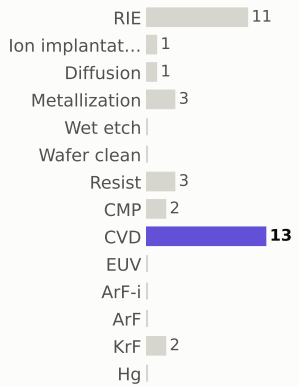
y cut at 0.184 µm



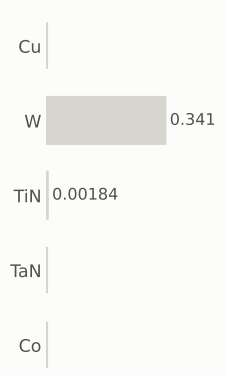
mask for this step

no mask for this step

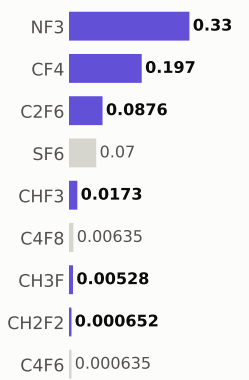
steps by process area



materials deposited (g)



gases emitted (g)



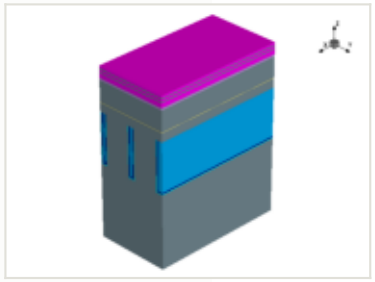
Step #430: CVD

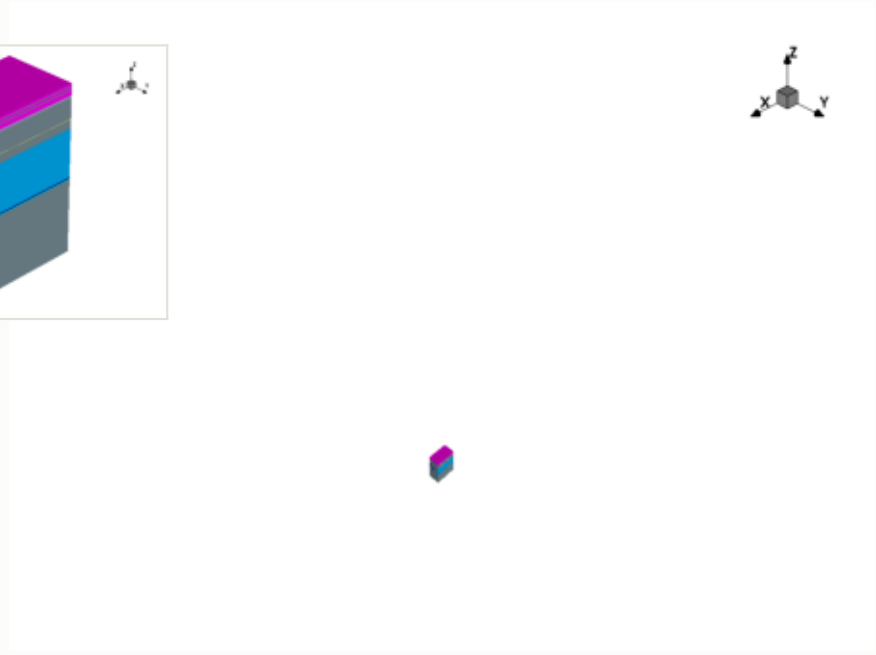
depo_epi · CVD

1.31	88.3	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



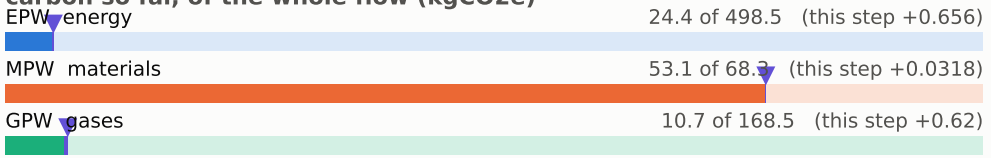


this step

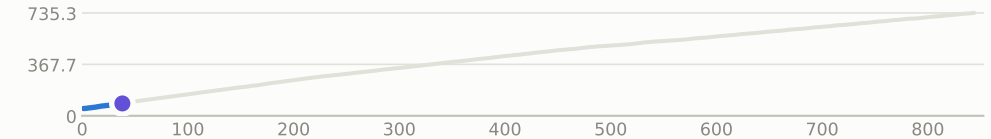
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

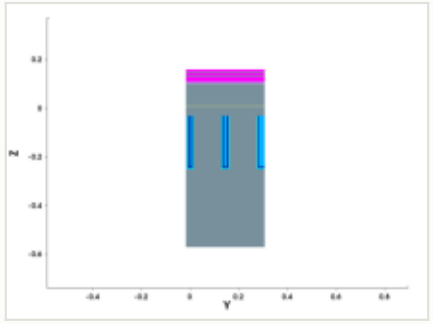
carbon so far, of the whole flow (kgCO2e)



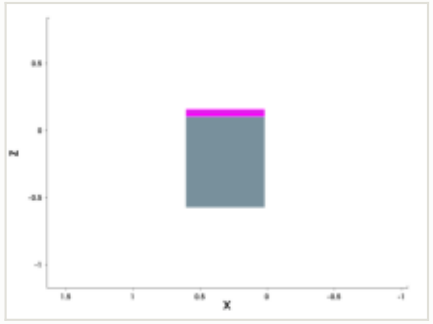
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



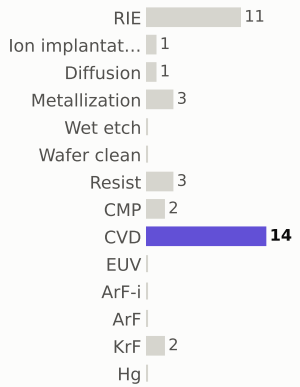
y cut at 0.184 µm



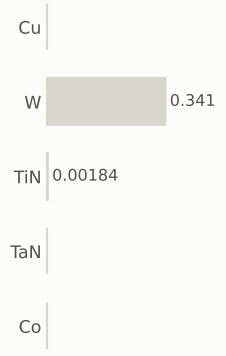
mask for this step

no mask for this step

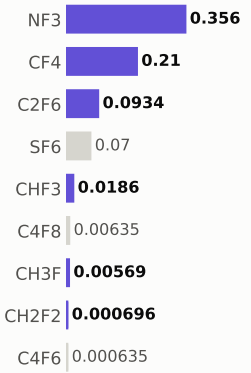
steps by process area



materials deposited (g)



gases emitted (g)



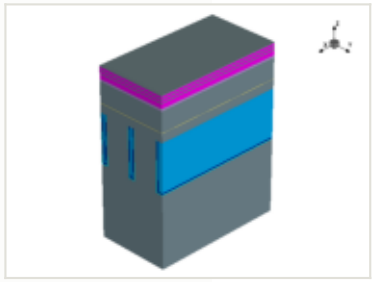
Step #440: CVD

depo_epi · CVD

1.31	89.6	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

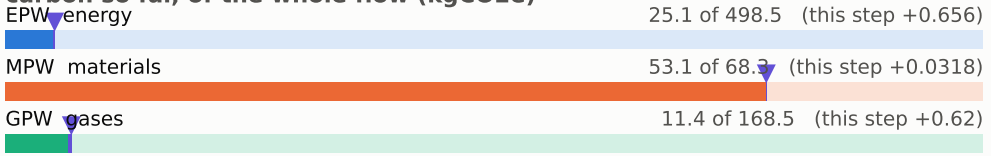


this step

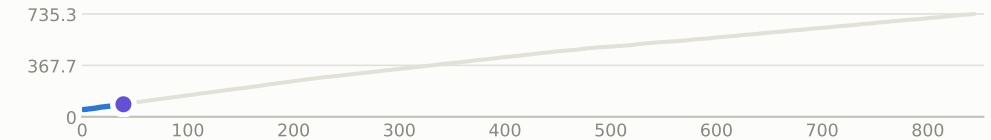
process area	CVD
material	—
thickness	0.005 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

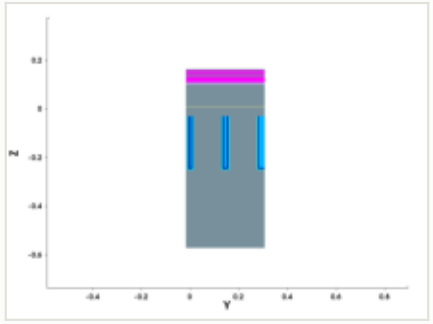
carbon so far, of the whole flow (kgCO2e)



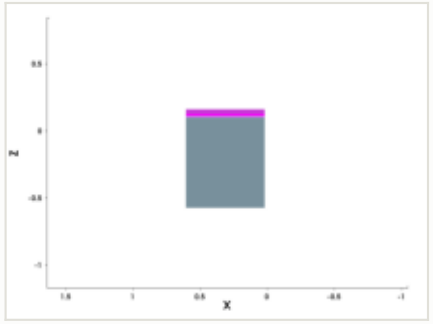
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



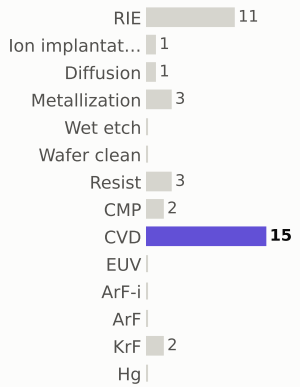
y cut at 0.184 µm



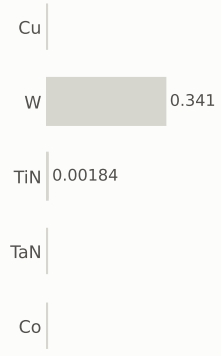
mask for this step

no mask for this step

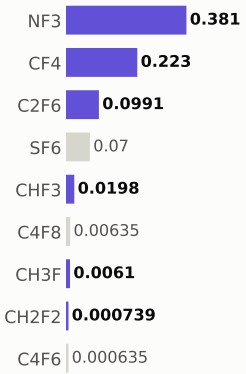
steps by process area



materials deposited (g)



gases emitted (g)



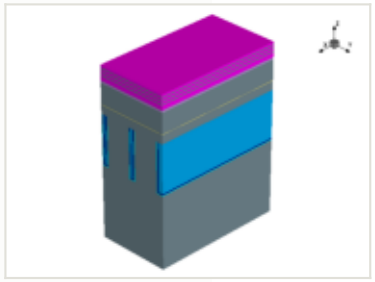
Step #450: CVD

depo_epi · CVD

1.31	90.9	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

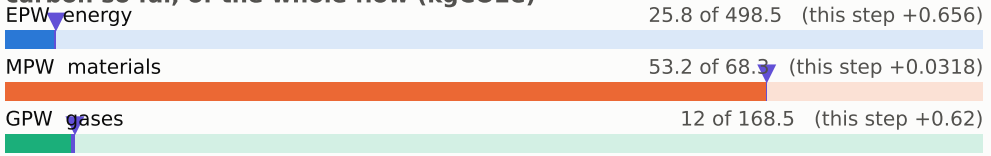


this step

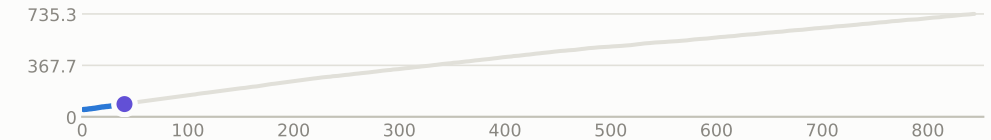
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

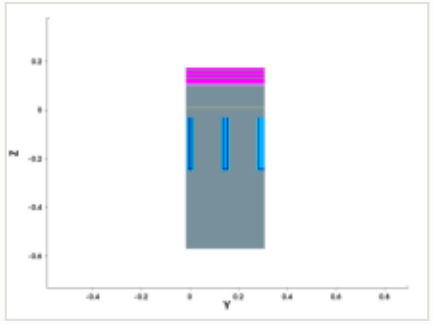
carbon so far, of the whole flow (kgCO2e)



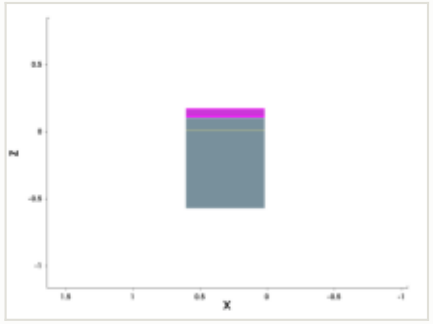
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



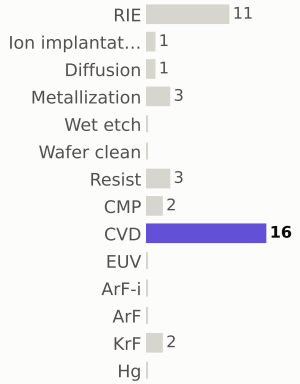
y cut at 0.184 µm



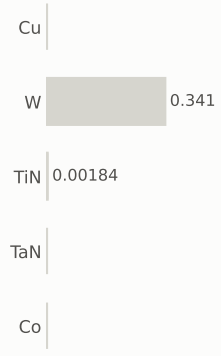
mask for this step

no mask for this step

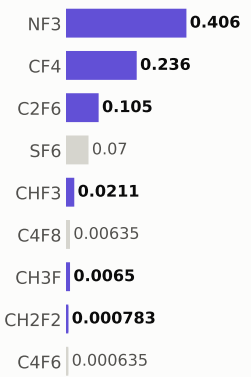
steps by process area



materials deposited (g)



gases emitted (g)



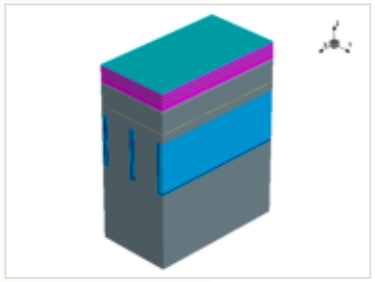
Step #470: CVD

depo_height_dependent · CVD

1.31	92.2	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

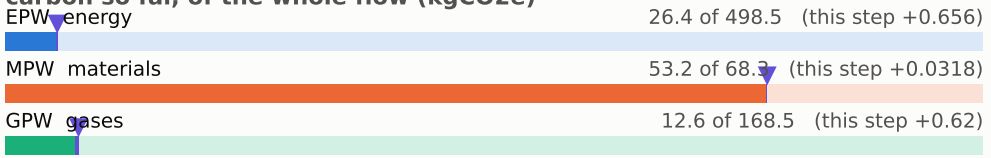


this step

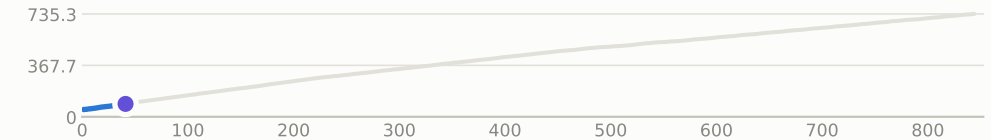
process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

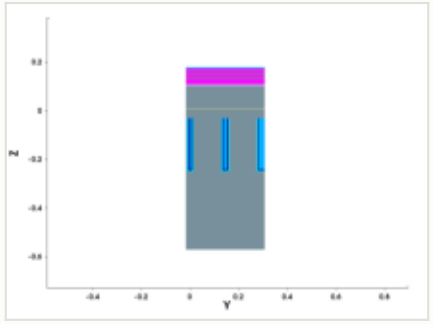
carbon so far, of the whole flow (kgCO2e)



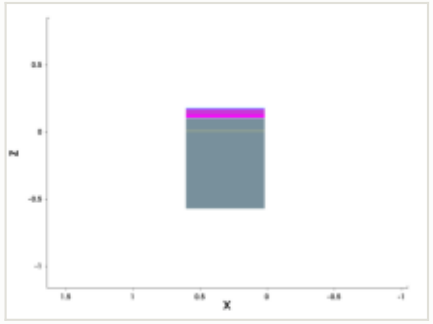
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



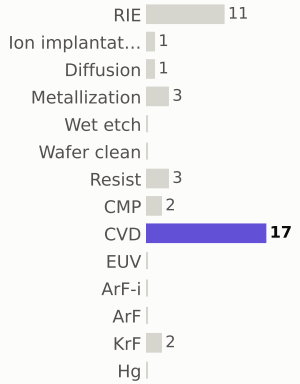
y cut at 0.184 µm



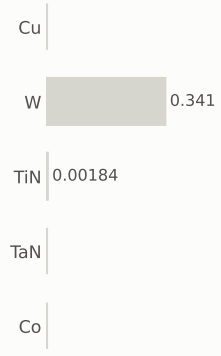
mask for this step

no mask for this step

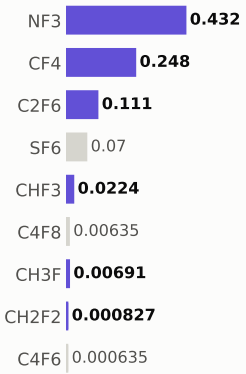
steps by process area



materials deposited (g)



gases emitted (g)



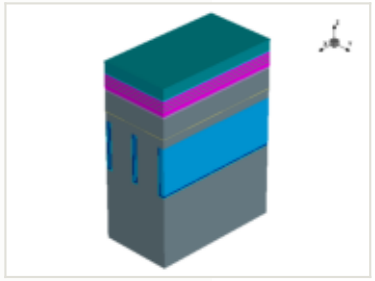
Step #480: CVD

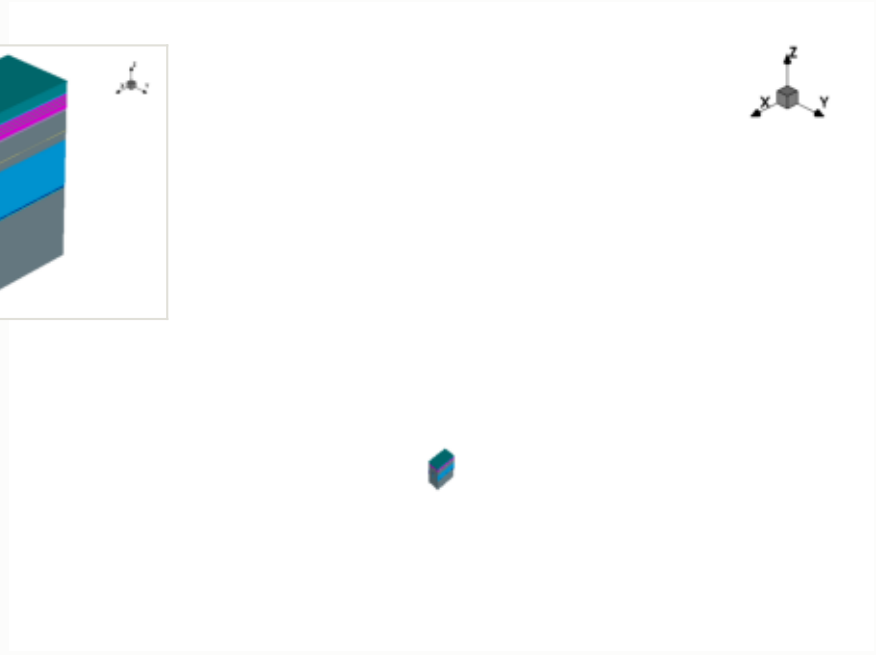
depo_height_dependent · CVD

1.31	93.5	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



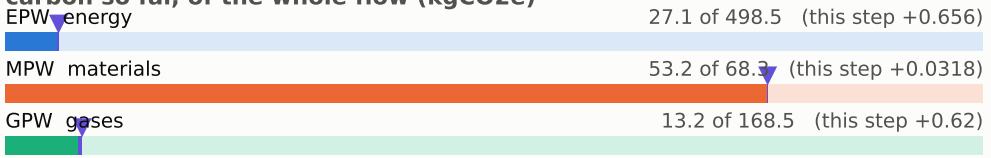


this step

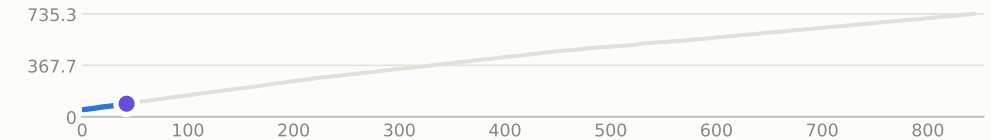
process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

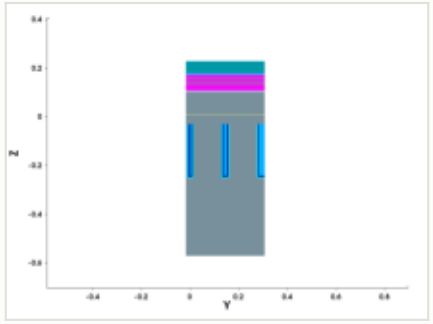
carbon so far, of the whole flow (kgCO2e)



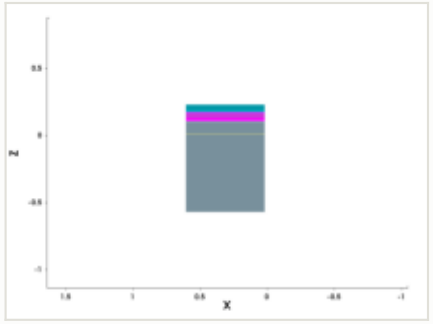
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



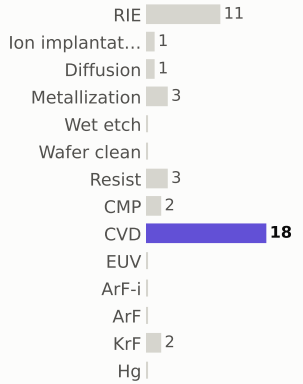
y cut at 0.184 µm



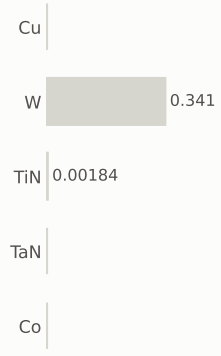
mask for this step

no mask for this step

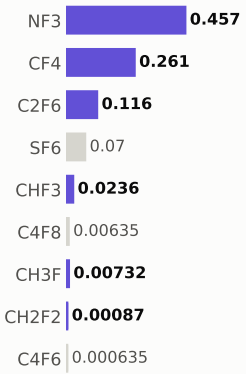
steps by process area



materials deposited (g)



gases emitted (g)



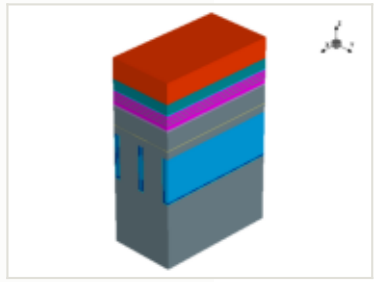
Step #490: CVD

depo_height_dependent · CVD

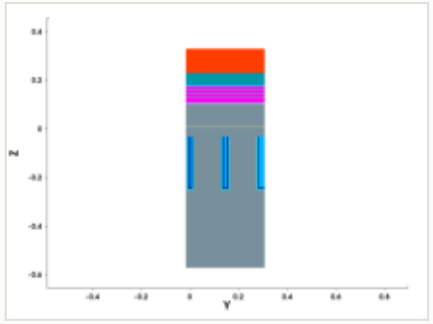
1.31	94.8	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

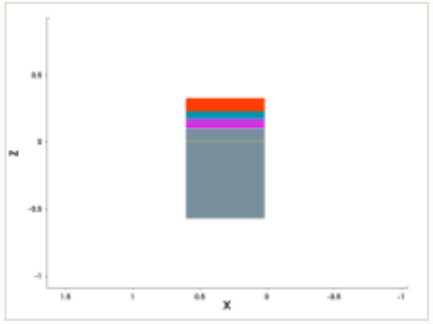
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

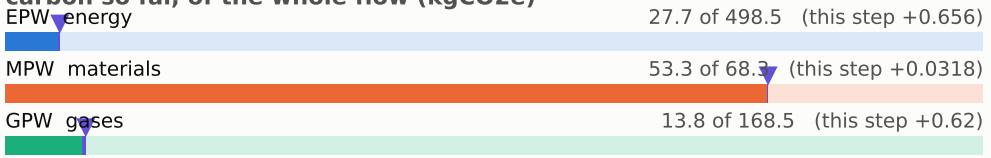
no mask for this step

this step

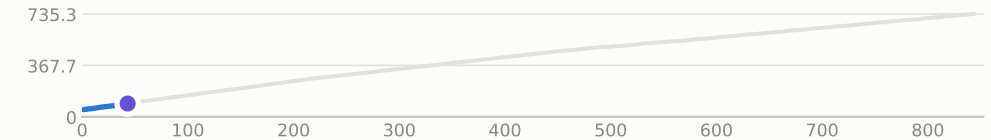
process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

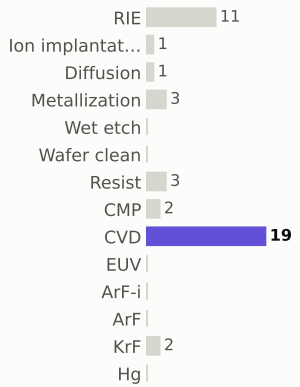
carbon so far, of the whole flow (kgCO2e)



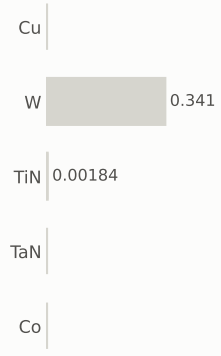
embodied carbon up to each step (kgCO2e per wafer)



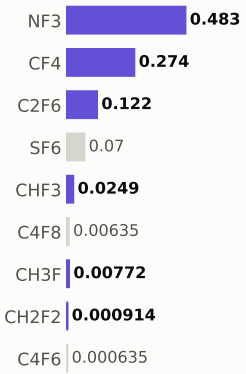
steps by process area



materials deposited (g)



gases emitted (g)



Step #500: CVD

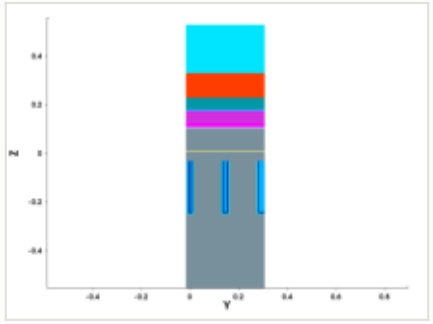
depo_height_dependent · CVD

1.31	96.1	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

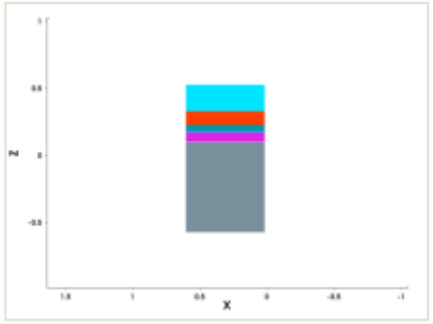
3D structure after this step

zoomed to fit

x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

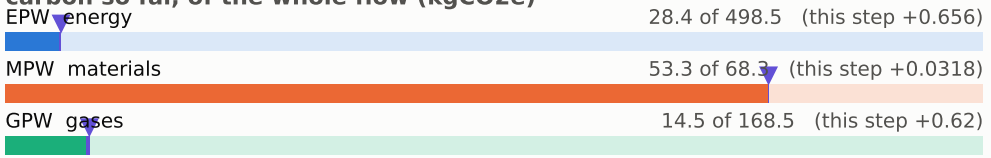
no mask for this step

this step

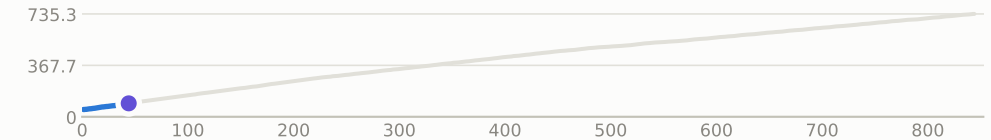
process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

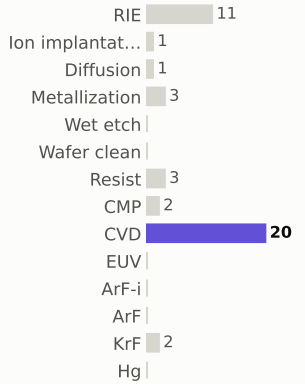
carbon so far, of the whole flow (kgCO2e)



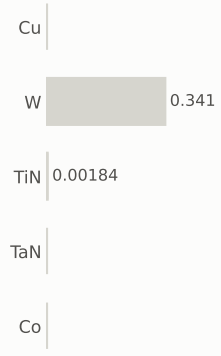
embodied carbon up to each step (kgCO2e per wafer)



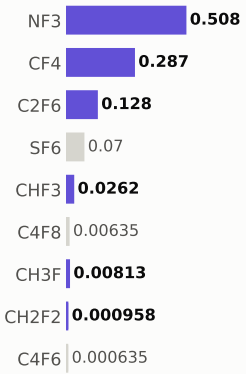
steps by process area



materials deposited (g)



gases emitted (g)



Step #510: CVD

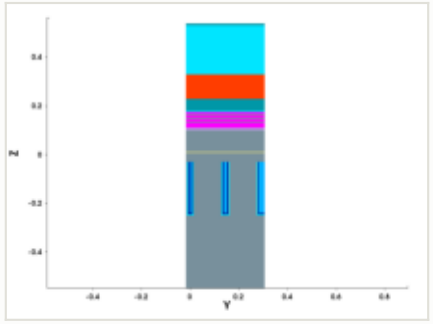
depo_height_dependent · CVD

1.31	97.4	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

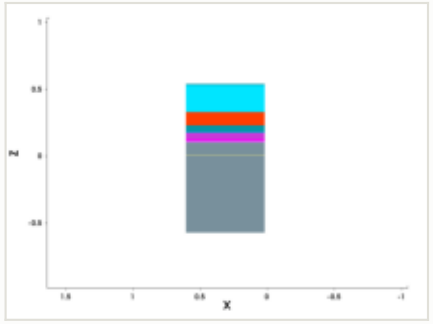
3D structure after this step

zoomed to fit

x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

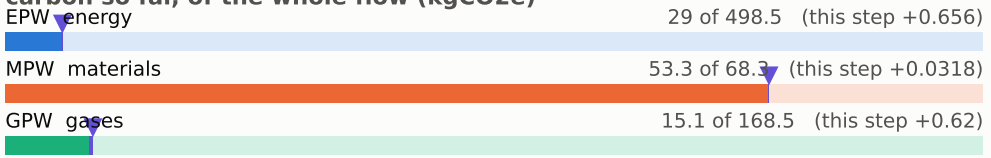
no mask for this step

this step

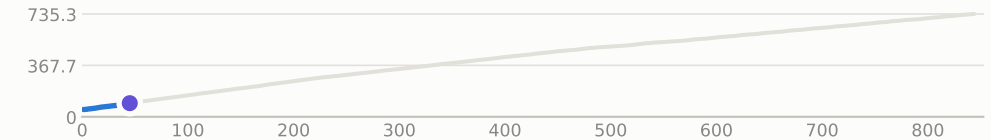
process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

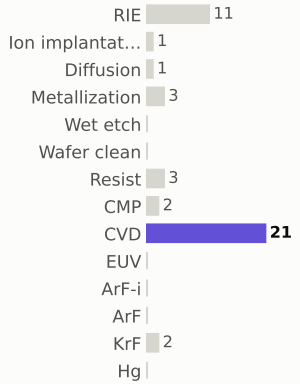
carbon so far, of the whole flow (kgCO2e)



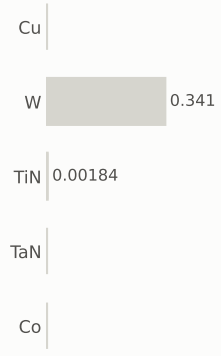
embodied carbon up to each step (kgCO2e per wafer)



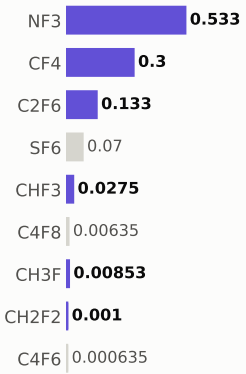
steps by process area



materials deposited (g)



gases emitted (g)



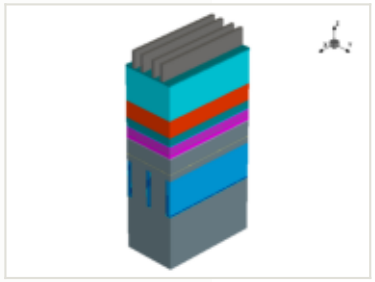
Step #520: EUV

litho_patterning · EUV

1.96	99.4	0.27%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

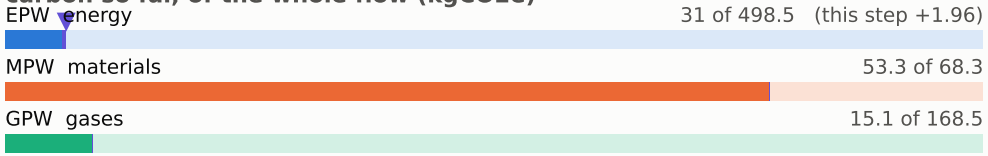


this step

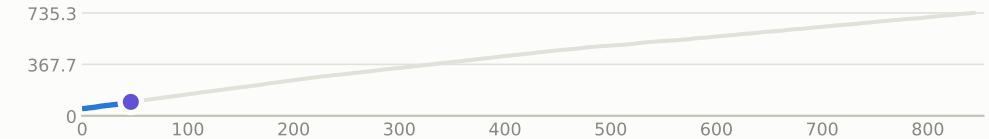
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

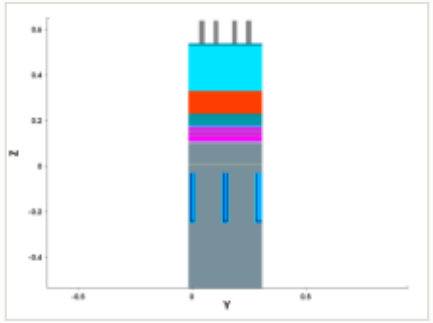
carbon so far, of the whole flow (kgCO2e)



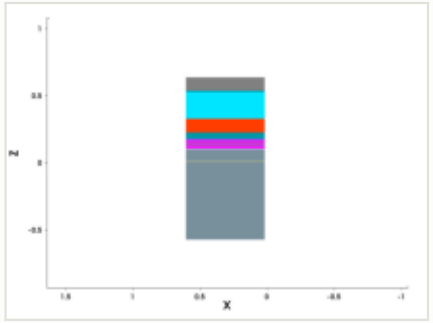
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm

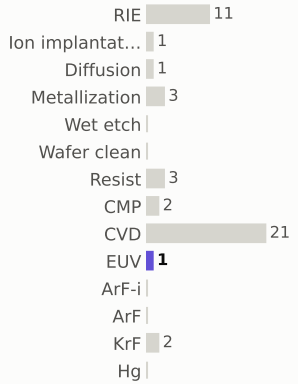


y cut at 0.184 µm

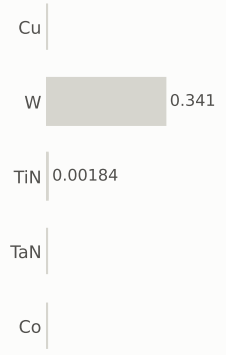


mask: layer 2:0, positive (GDSII)

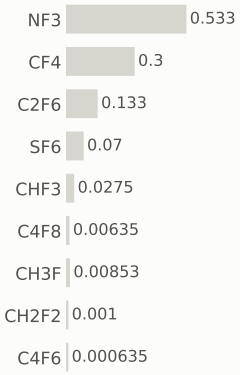
steps by process area



materials deposited (g)



gases emitted (g)



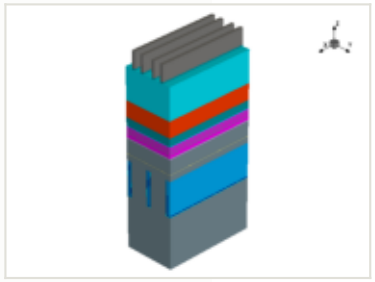
Step #530: RIE

etch_anisotropic · RIE

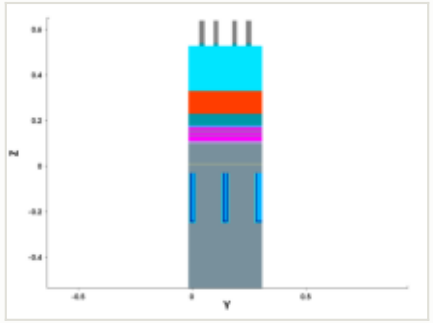
1.04	100.4	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

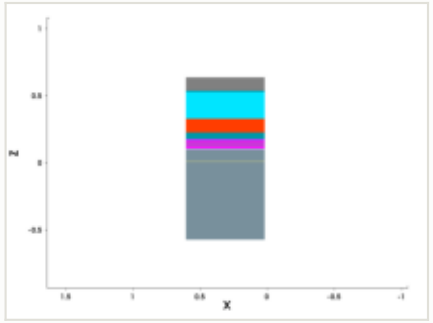
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

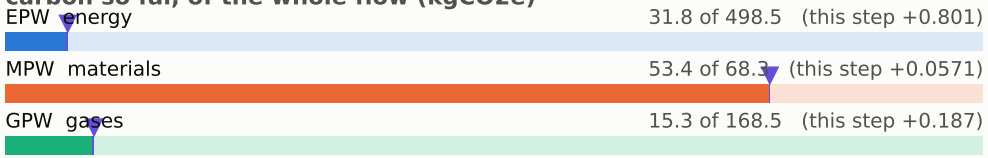
no mask for this step

this step

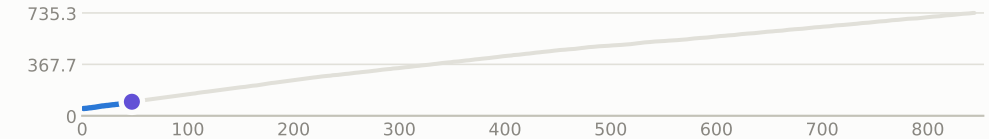
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

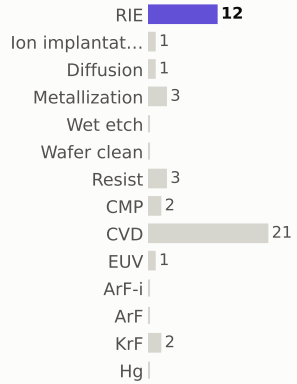
carbon so far, of the whole flow (kgCO2e)



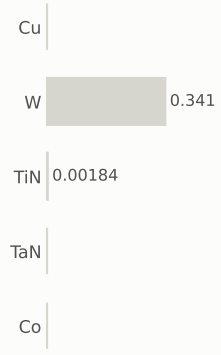
embodied carbon up to each step (kgCO2e per wafer)



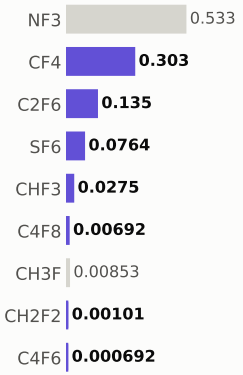
steps by process area



materials deposited (g)



gases emitted (g)



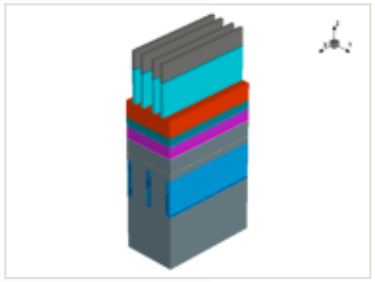
Step #540: RIE

etch_anisotropic · RIE

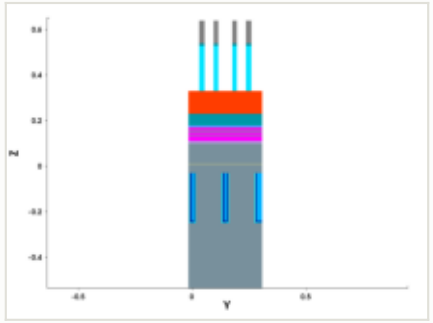
1.04	101.5	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

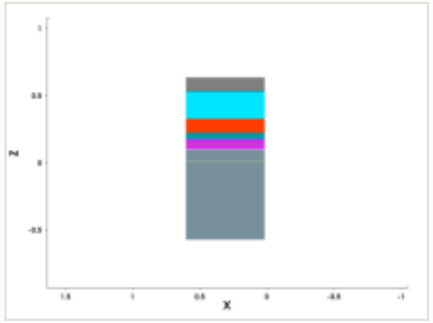
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



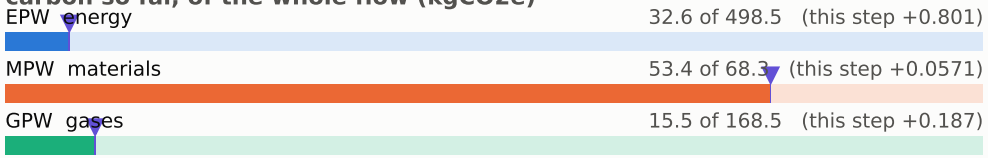
mask for this step

no mask for this step

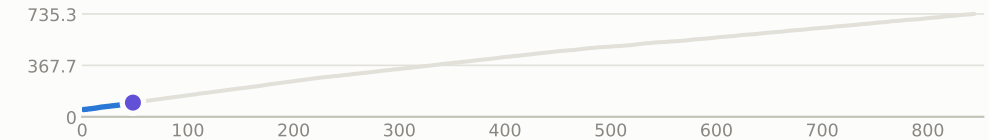
this step

process area	RIE	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

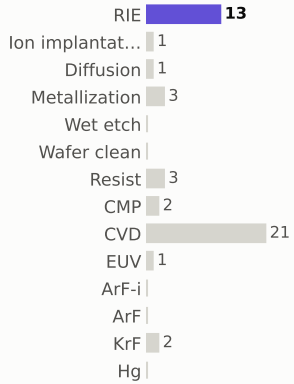
carbon so far, of the whole flow (kgCO2e)



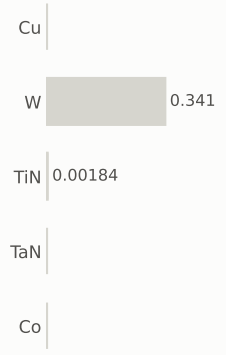
embodied carbon up to each step (kgCO2e per wafer)



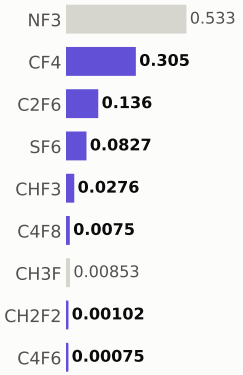
steps by process area



materials deposited (g)



gases emitted (g)



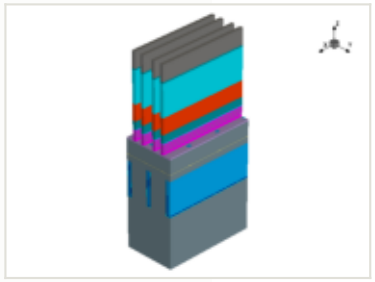
Step #550: RIE

etch_anisotropic · RIE

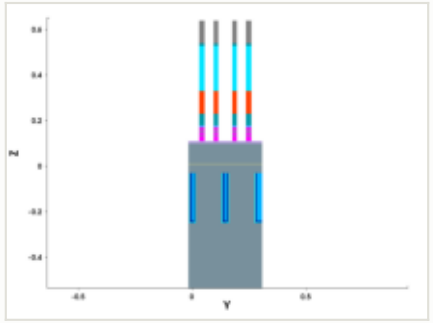
1.04	102.5	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

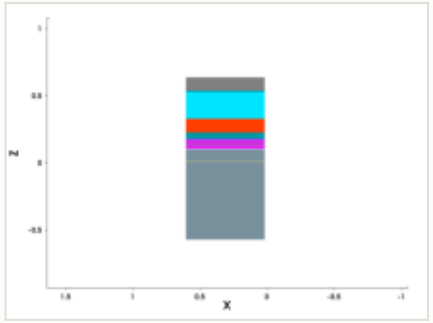
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

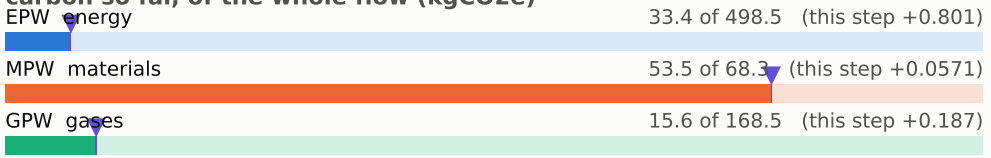
no mask for this step

this step

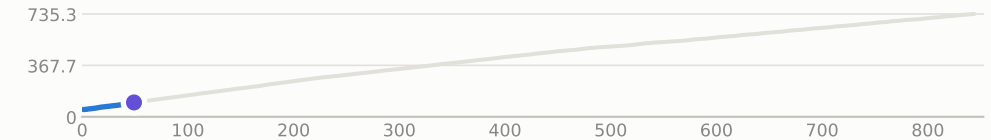
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

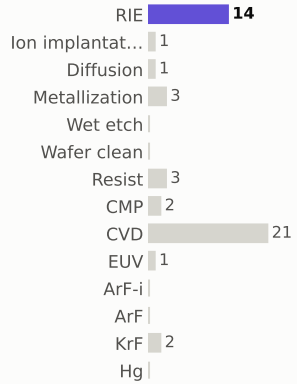
carbon so far, of the whole flow (kgCO2e)



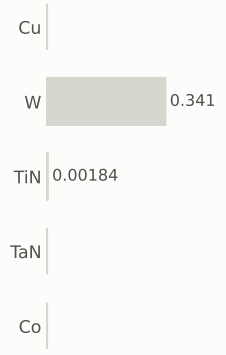
embodied carbon up to each step (kgCO2e per wafer)



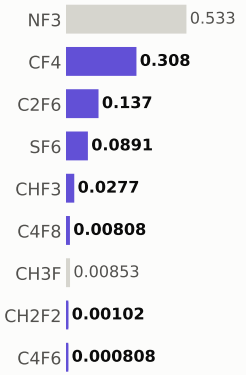
steps by process area



materials deposited (g)



gases emitted (g)



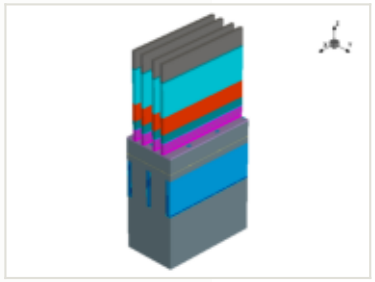
Step #560: RIE

etch_anisotropic · RIE

1.04	103.6	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

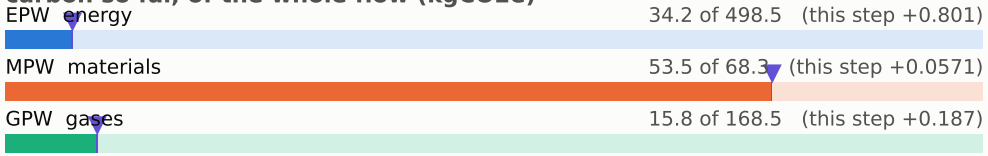
zoomed to fit



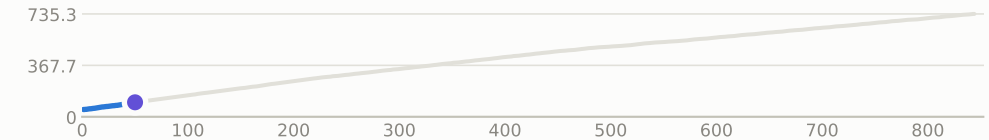
this step

process area	RIE	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

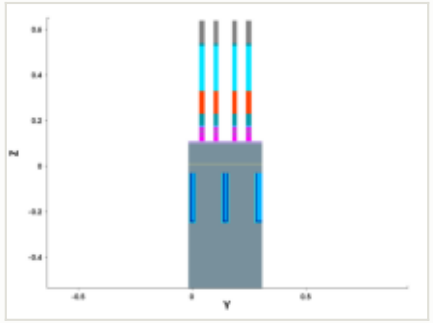
carbon so far, of the whole flow (kgCO2e)



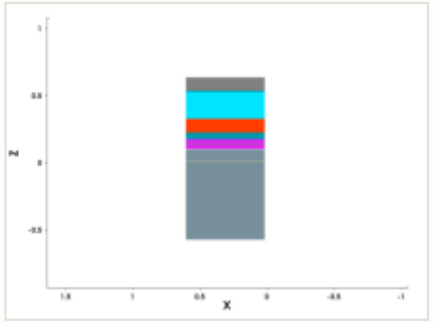
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



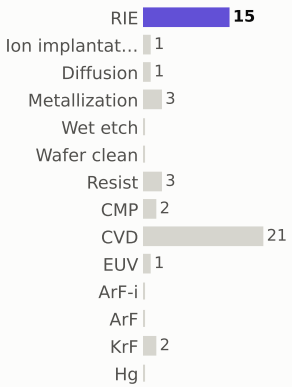
y cut at 0.184 µm



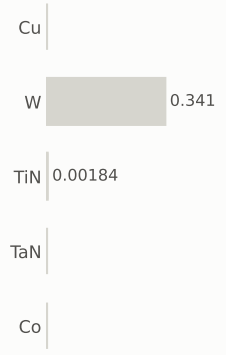
mask for this step

no mask for this step

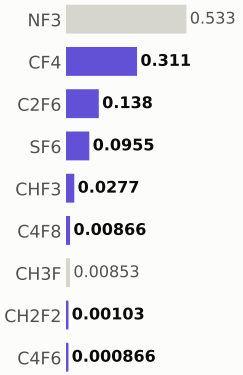
steps by process area



materials deposited (g)



gases emitted (g)



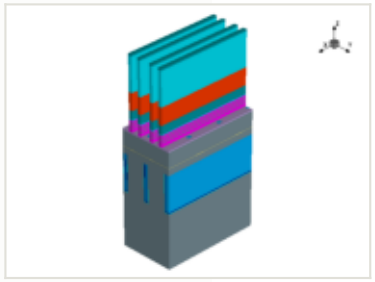
Step #570: Resist

etch_strip · Resist

0.169	103.8	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

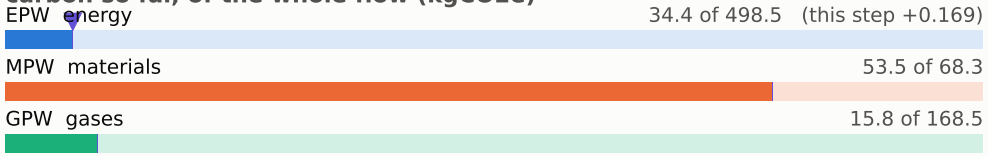


this step

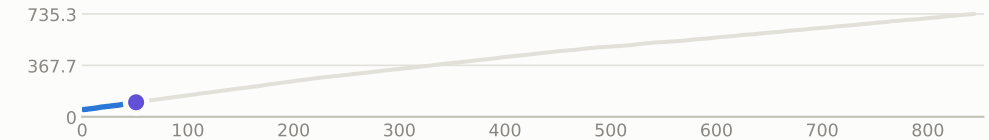
process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

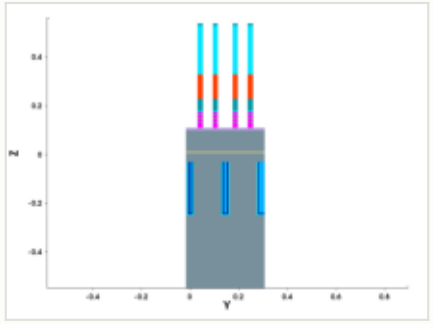
carbon so far, of the whole flow (kgCO2e)



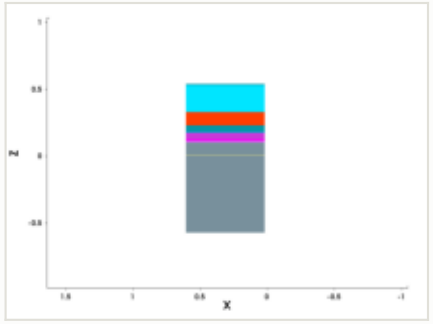
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



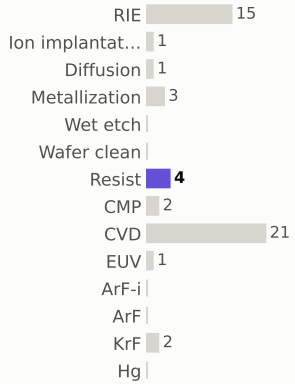
y cut at 0.184 µm



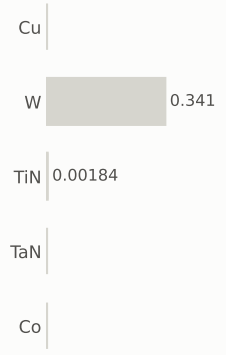
mask for this step

no mask for this step

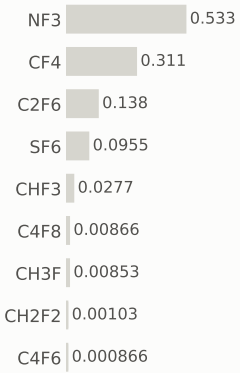
steps by process area



materials deposited (g)



gases emitted (g)



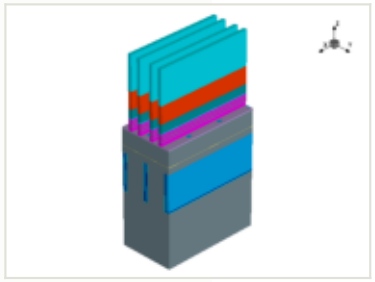
Step #580: RIE

etch_anisotropic · RIE

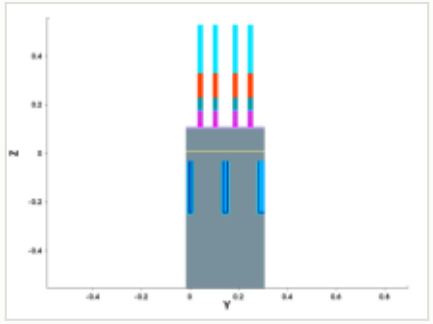
1.04	104.8	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

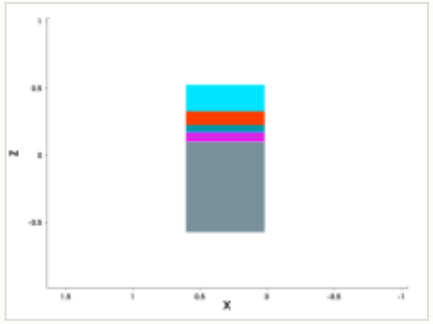
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

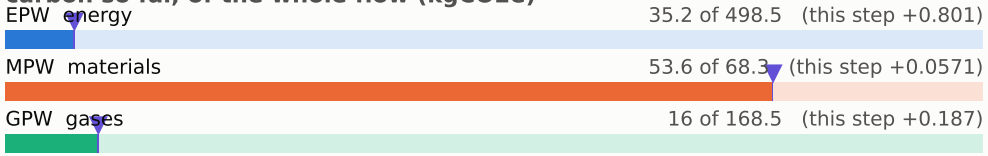
no mask for this step

this step

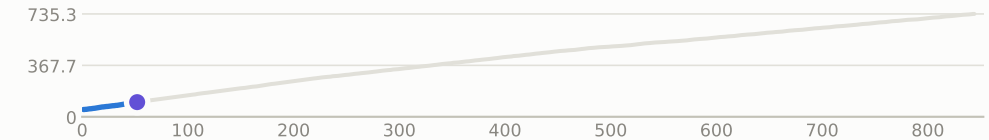
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

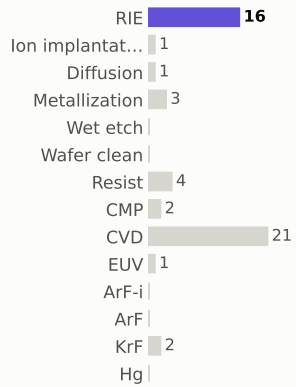
carbon so far, of the whole flow (kgCO2e)



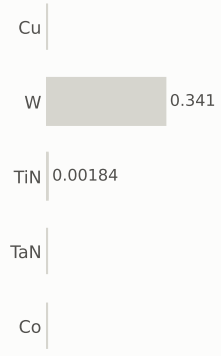
embodied carbon up to each step (kgCO2e per wafer)



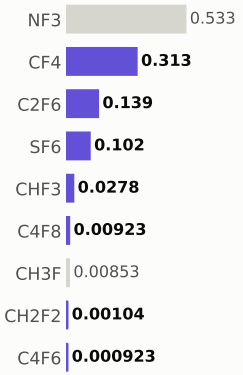
steps by process area



materials deposited (g)



gases emitted (g)



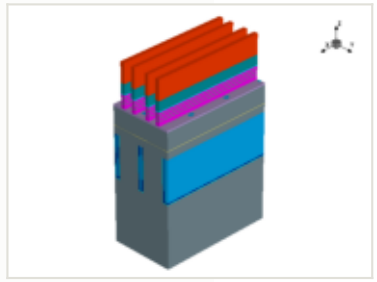
Step #590: RIE

etch_anisotropic · RIE

1.04	105.8	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

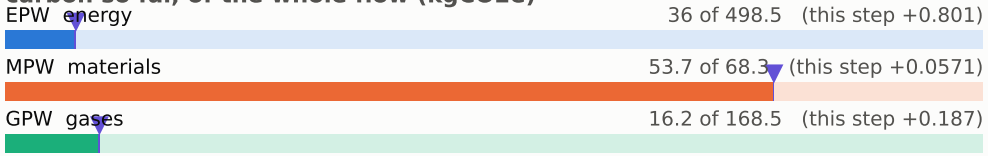


this step

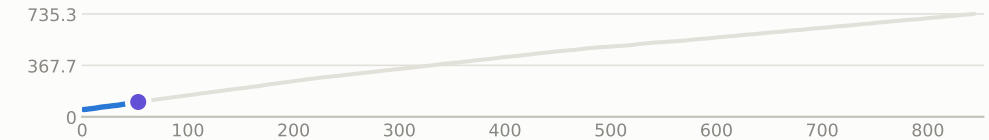
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

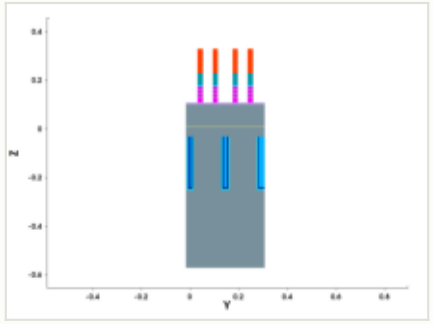
carbon so far, of the whole flow (kgCO2e)



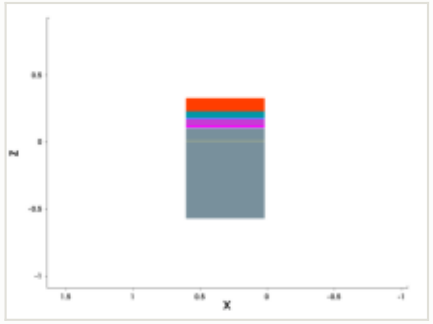
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



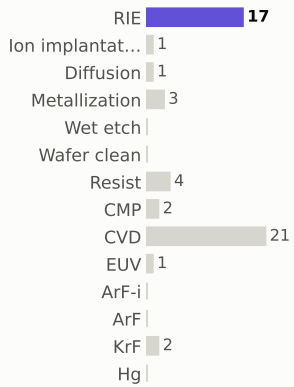
y cut at 0.184 µm



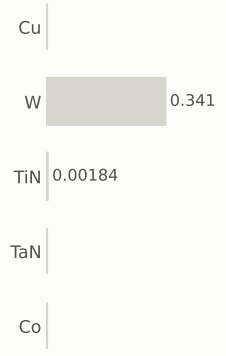
mask for this step

no mask for this step

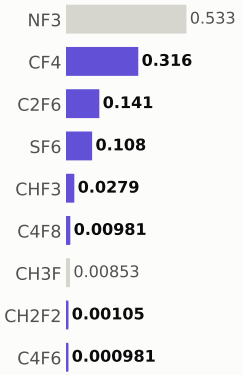
steps by process area



materials deposited (g)



gases emitted (g)



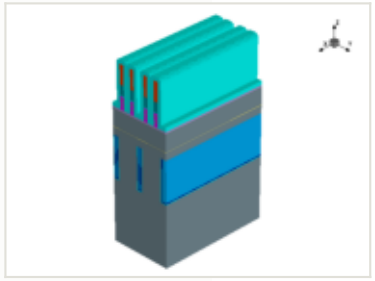
Step #600: CVD

depo_isotropic · CVD

1.31	107.2	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

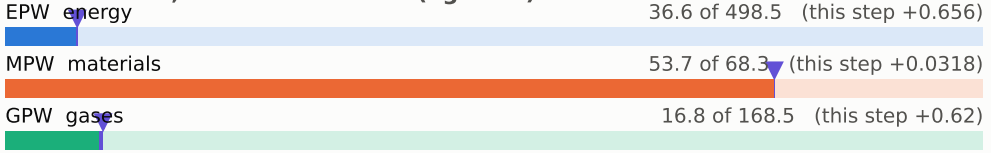


this step

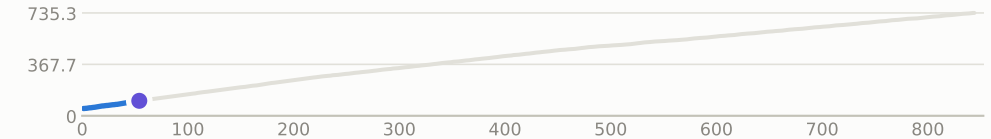
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

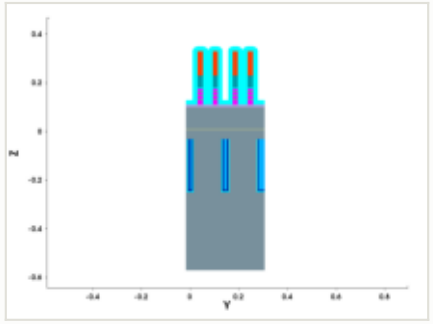
carbon so far, of the whole flow (kgCO2e)



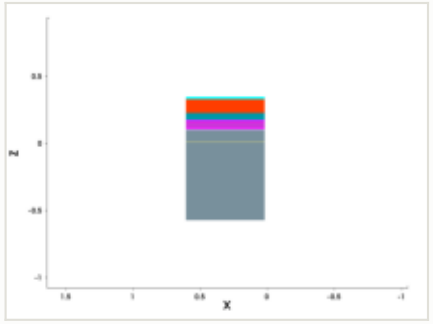
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



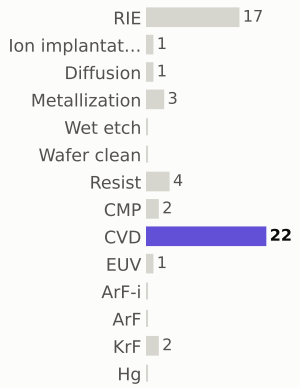
y cut at 0.184 µm



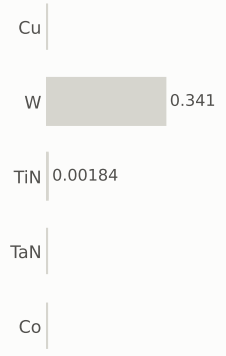
mask for this step

no mask for this step

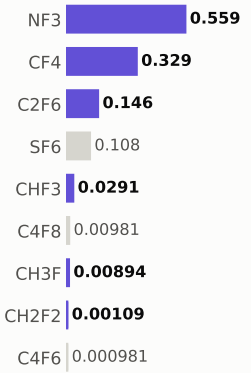
steps by process area



materials deposited (g)



gases emitted (g)



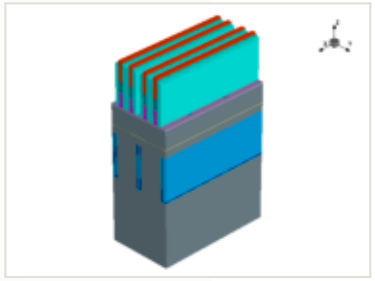
Step #610: RIE

etch_anisotropic · RIE

1.04	108.2	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

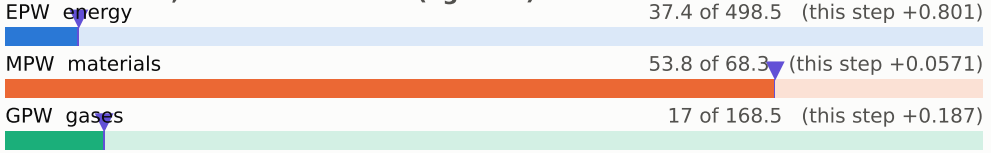


this step

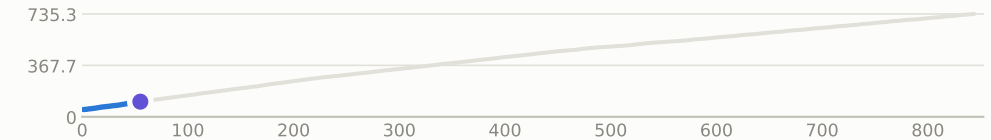
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

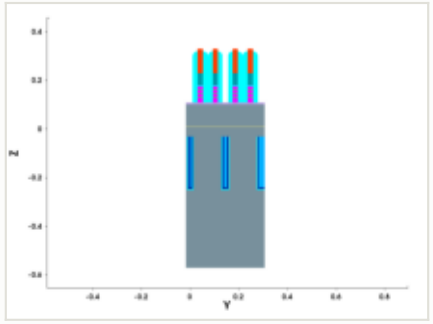
carbon so far, of the whole flow (kgCO2e)



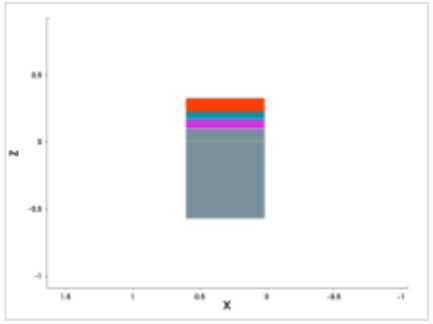
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



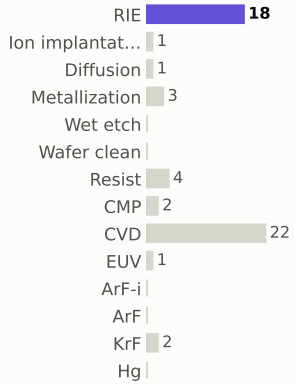
y cut at 0.184 µm



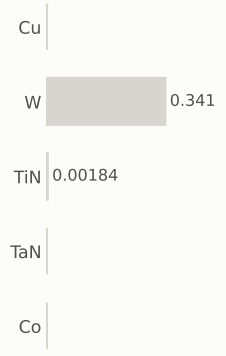
mask for this step

no mask for this step

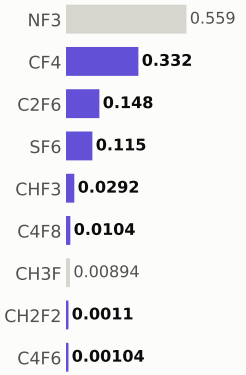
steps by process area



materials deposited (g)



gases emitted (g)



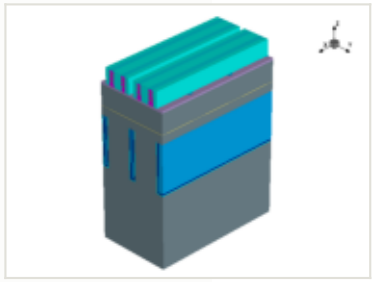
Step #620: CMP

etch_cmp · CMP

0.169	108.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

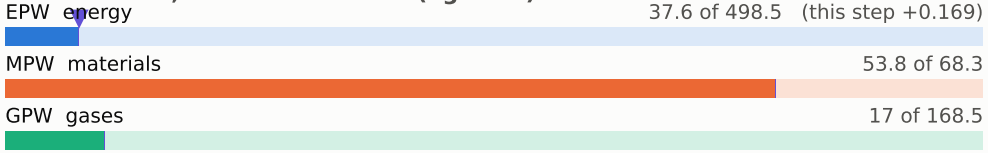


this step

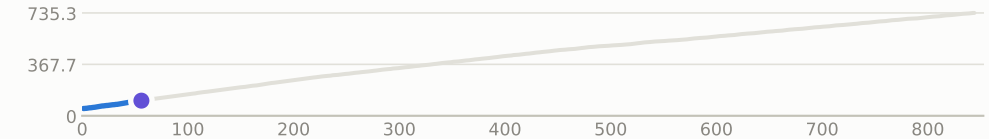
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

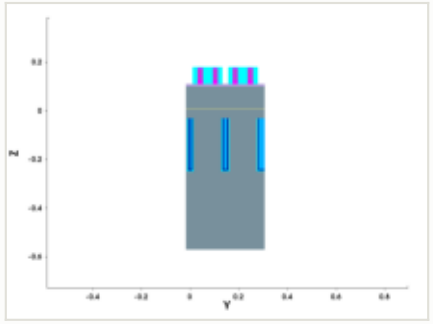
carbon so far, of the whole flow (kgCO2e)



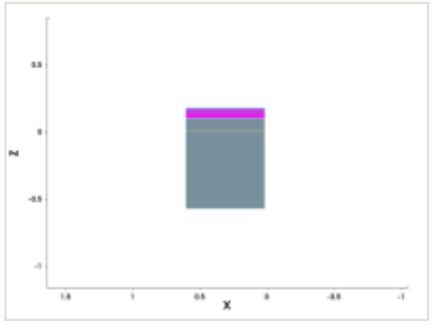
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



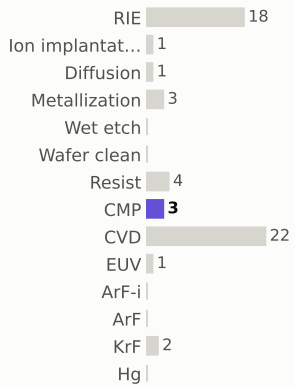
y cut at 0.184 µm



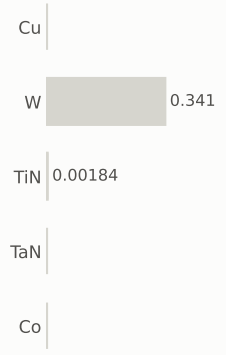
mask for this step

no mask for this step

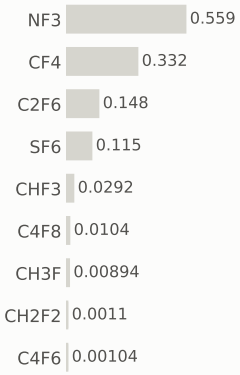
steps by process area



materials deposited (g)



gases emitted (g)



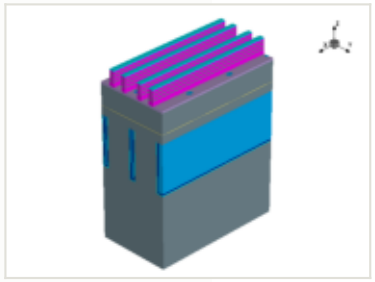
Step #630: RIE

etch_anisotropic · RIE

1.04	109.4	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

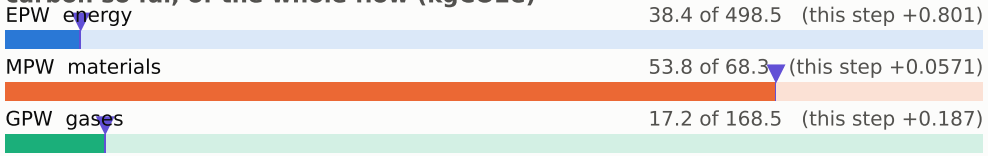


this step

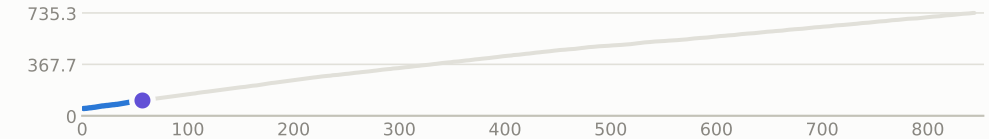
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

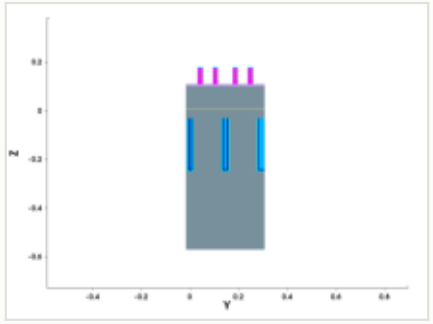
carbon so far, of the whole flow (kgCO2e)



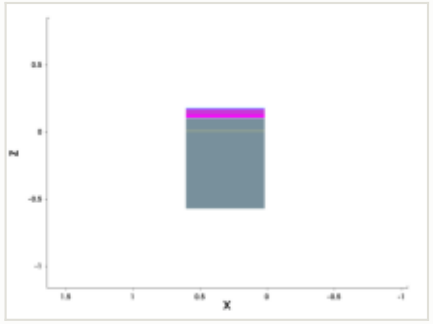
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



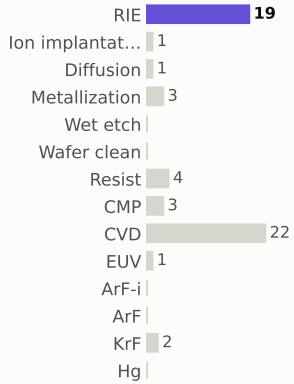
y cut at 0.184 µm



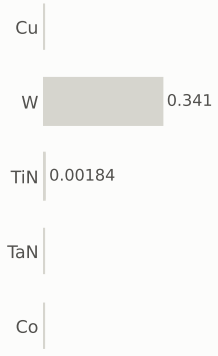
mask for this step

no mask for this step

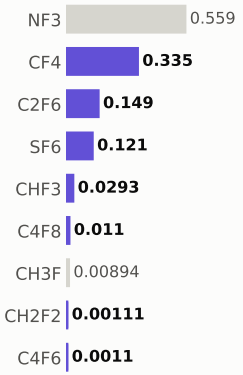
steps by process area



materials deposited (g)



gases emitted (g)



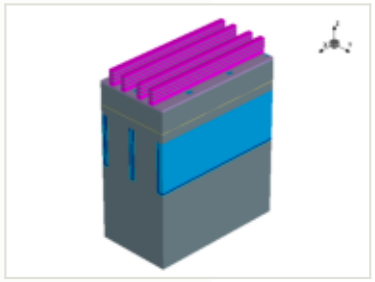
Step #640: RIE

etch_anisotropic · RIE

1.04	110.5	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

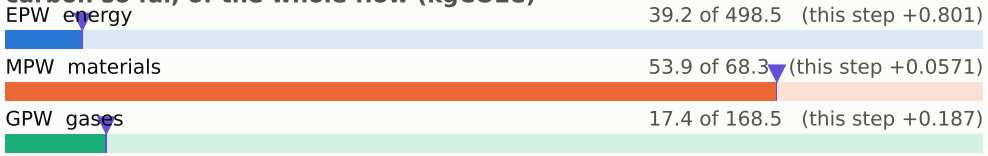


this step

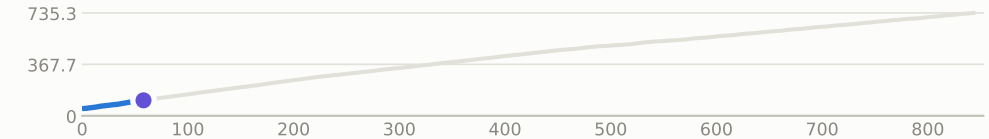
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

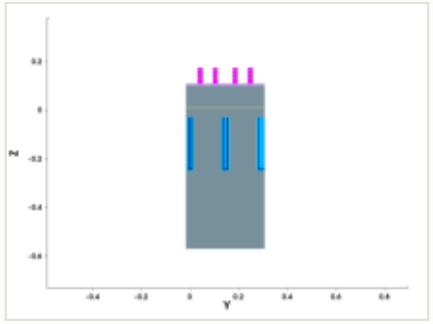
carbon so far, of the whole flow (kgCO2e)



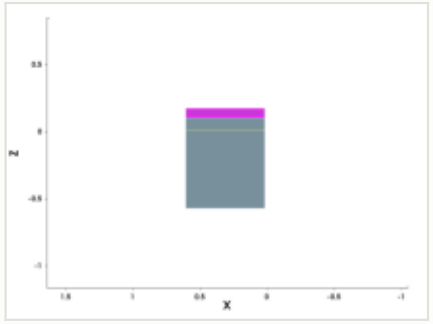
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



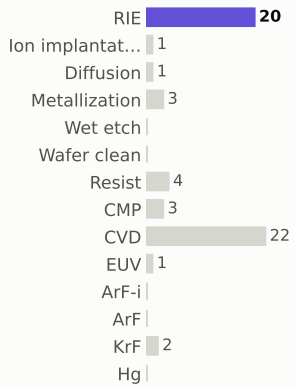
y cut at 0.184 µm



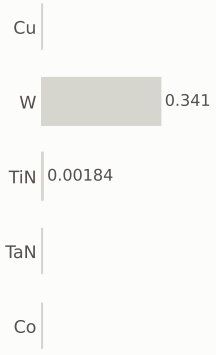
mask for this step

no mask for this step

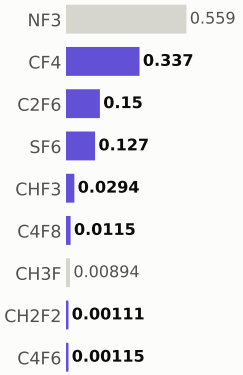
steps by process area



materials deposited (g)



gases emitted (g)



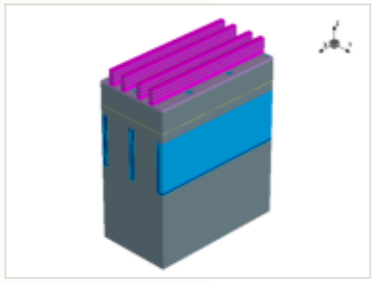
Step #660: CVD

depo_fill · CVD

1.31	111.8	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

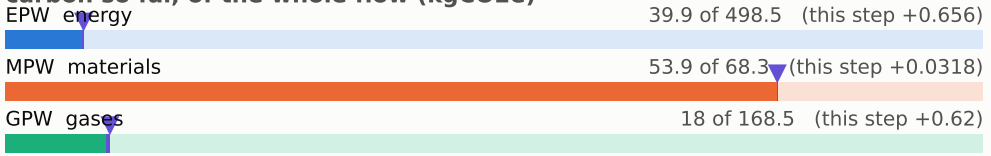


this step

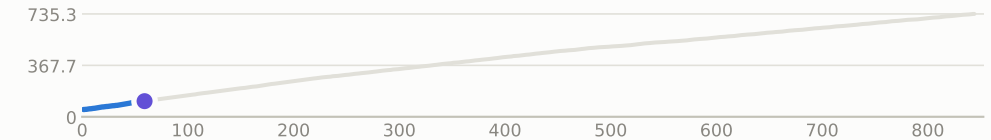
process area	CVD
material	—
thickness	-0.055 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

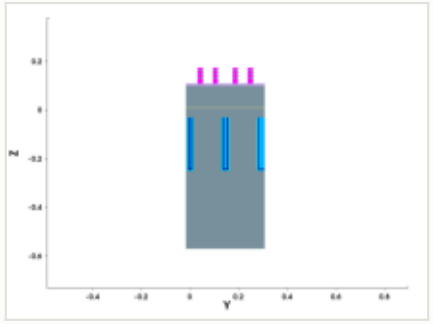
carbon so far, of the whole flow (kgCO2e)



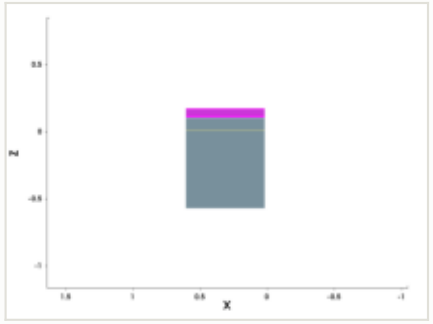
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



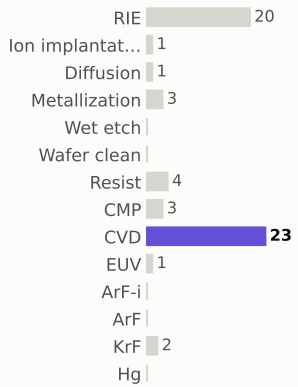
y cut at 0.184 µm



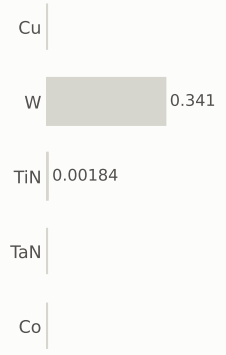
mask for this step

no mask for this step

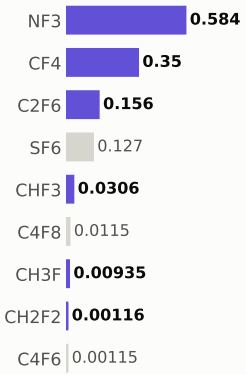
steps by process area



materials deposited (g)



gases emitted (g)



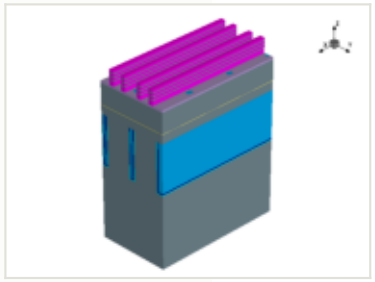
Step #670: CMP

etch_cmp · CMP

0.169	111.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

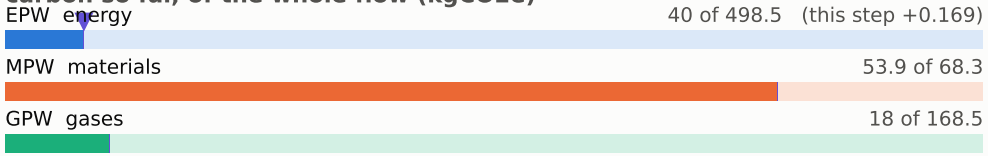


this step

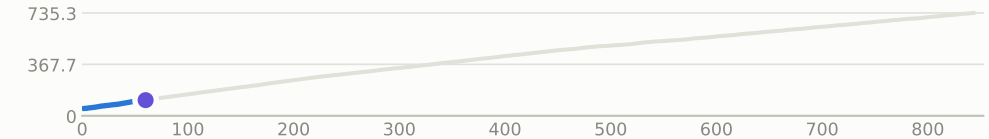
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

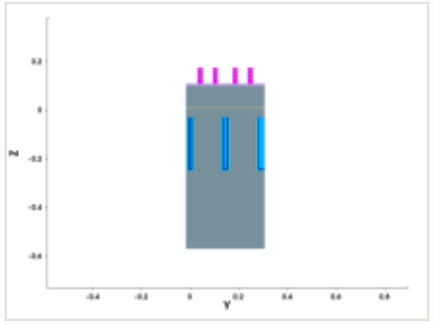
carbon so far, of the whole flow (kgCO2e)



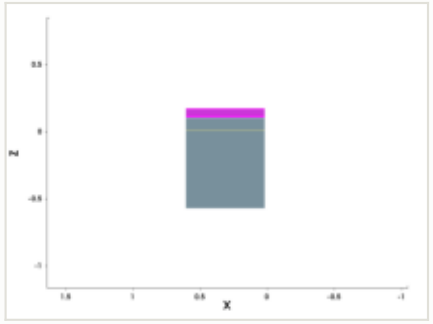
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



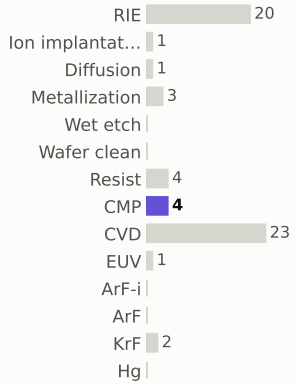
y cut at 0.184 µm



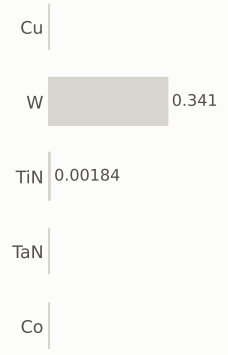
mask for this step

no mask for this step

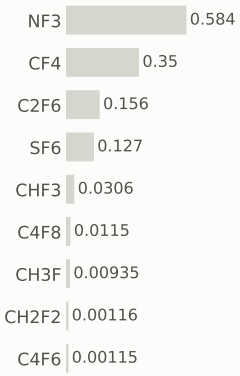
steps by process area



materials deposited (g)



gases emitted (g)



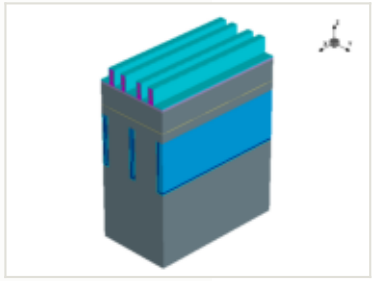
Step #690: CVD

depo_isotropic · CVD

1.31	113.2	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

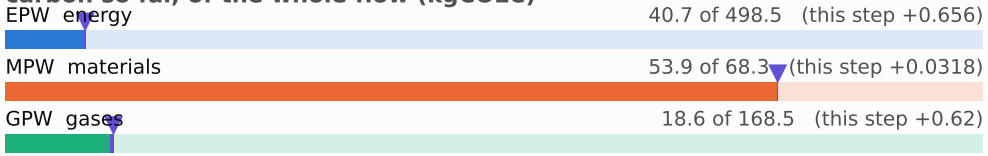
zoomed to fit



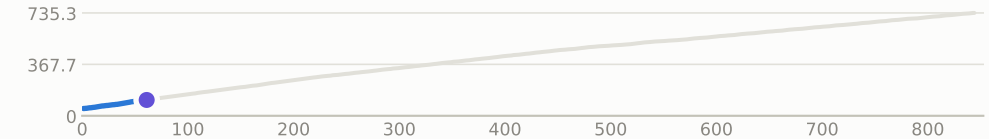
this step

process area	CVD	EPW	0.656 kgCO2e
material	—	EPW tool	0.469 kgCO2e
thickness	0.005 µm	EPW overhead	0.187 kgCO2e
material GWP	—	MPW	0.0318 kgCO2e (gas 0.0318)
input gases	NF3 2.54 g	GPW	0.62 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**1.31 kgCO2e**

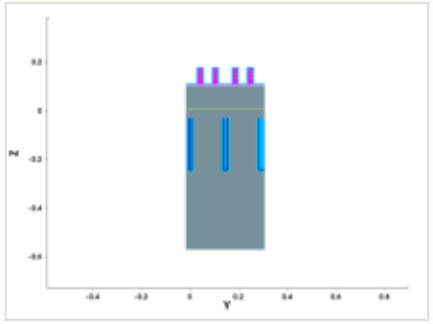
carbon so far, of the whole flow (kgCO2e)



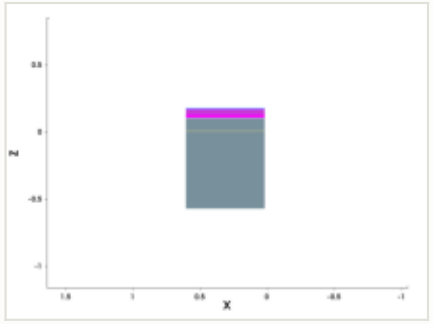
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



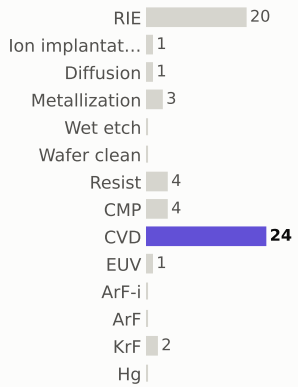
y cut at 0.184 µm



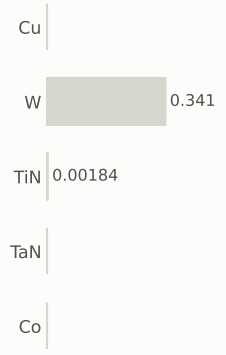
mask for this step

no mask for this step

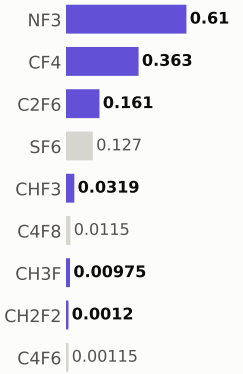
steps by process area



materials deposited (g)



gases emitted (g)



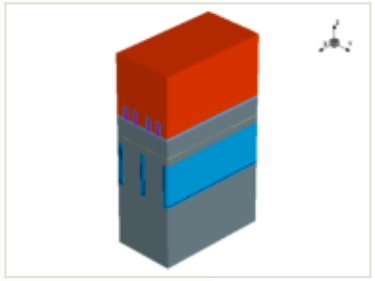
Step #700: CVD

depo_isotropic · CVD

1.31	114.5	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

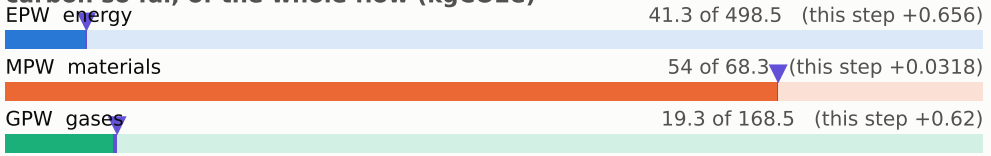


this step

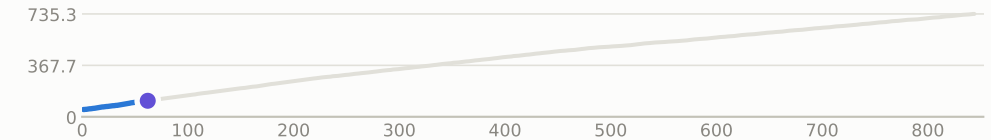
process area	CVD
material	—
thickness	0.25 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

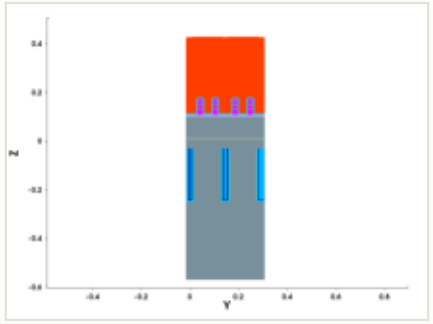
carbon so far, of the whole flow (kgCO2e)



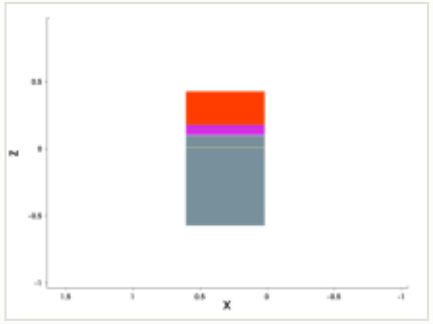
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



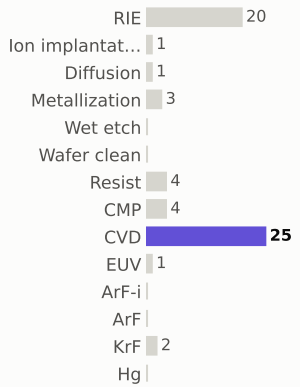
y cut at 0.184 µm



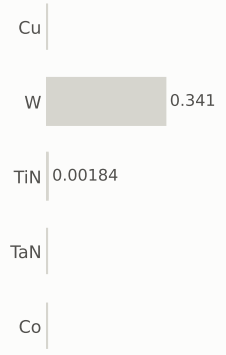
mask for this step

no mask for this step

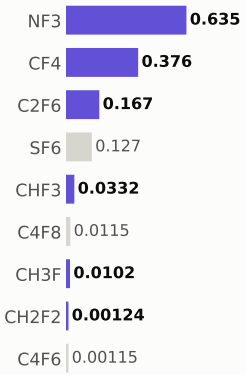
steps by process area



materials deposited (g)



gases emitted (g)



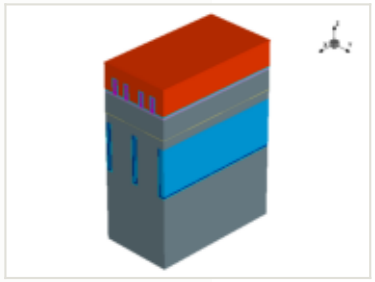
Step #710: CMP

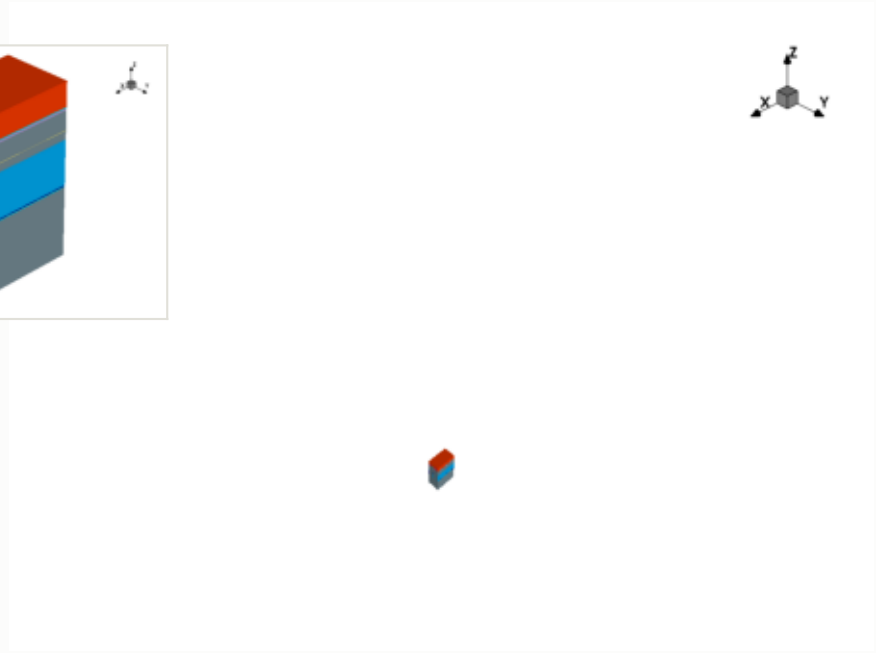
etch_cmp · CMP

0.169	114.7	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



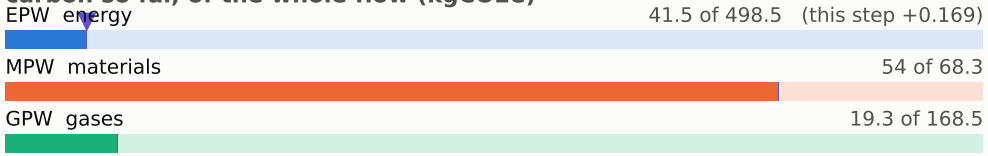


this step

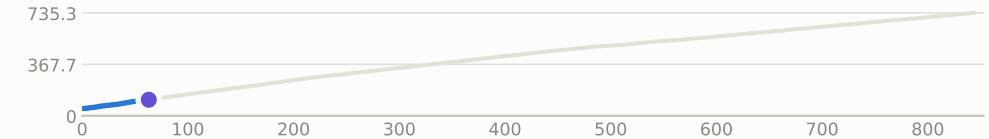
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

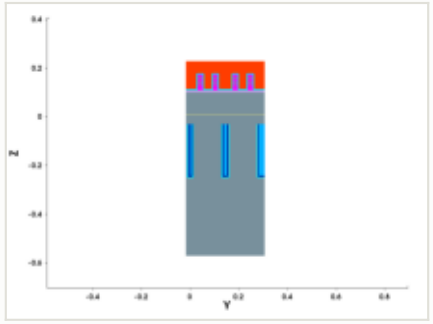
carbon so far, of the whole flow (kgCO2e)



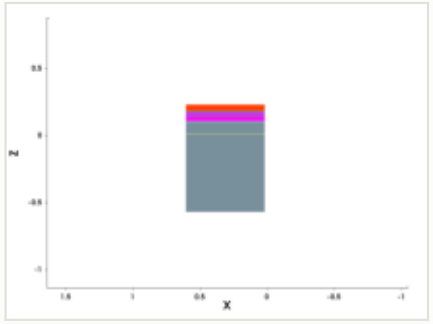
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



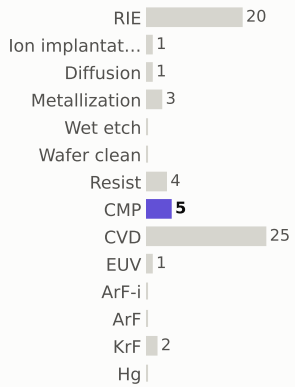
y cut at 0.184 µm



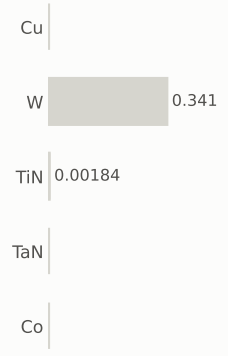
mask for this step

no mask for this step

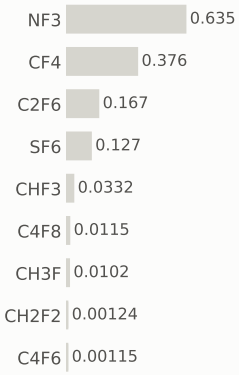
steps by process area



materials deposited (g)



gases emitted (g)



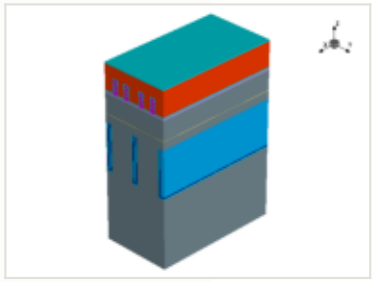
Step #730: CVD

depo_height_dependent · CVD

1.31	116.0	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

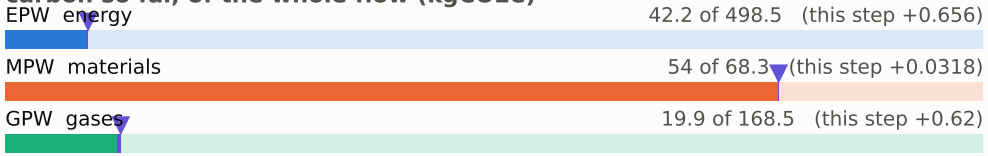


this step

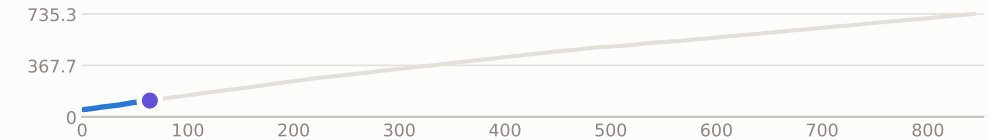
process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

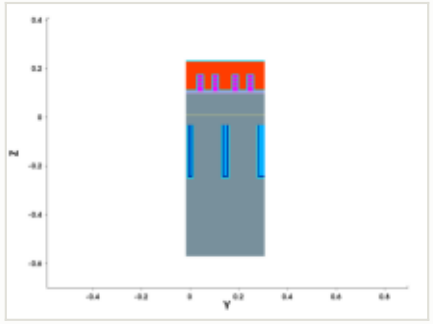
carbon so far, of the whole flow (kgCO2e)



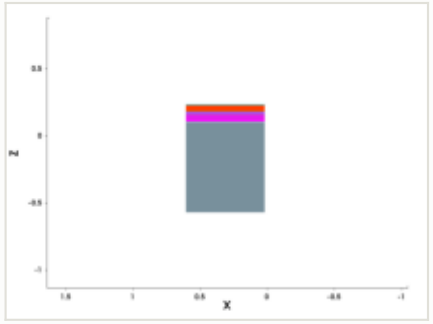
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



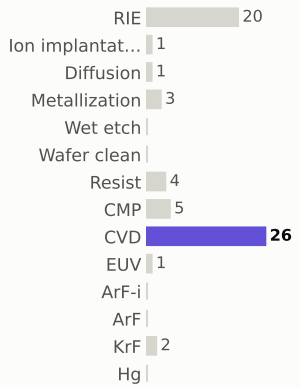
y cut at 0.184 µm



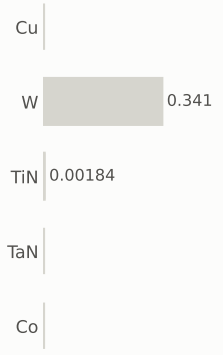
mask for this step

no mask for this step

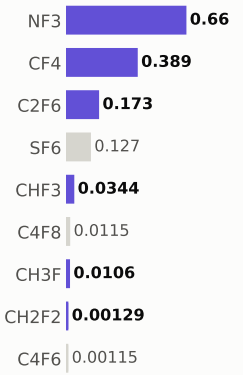
steps by process area



materials deposited (g)



gases emitted (g)



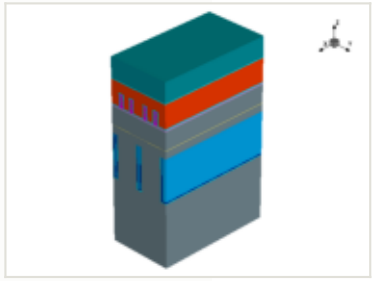
Step #740: CVD

depo_height_dependent · CVD

1.31	117.3	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

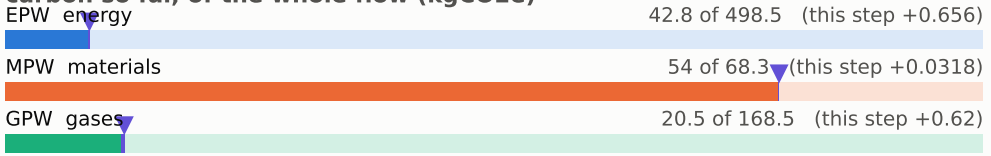


this step

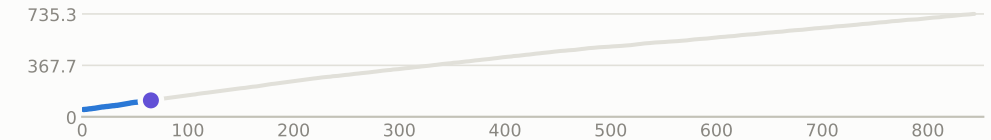
process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

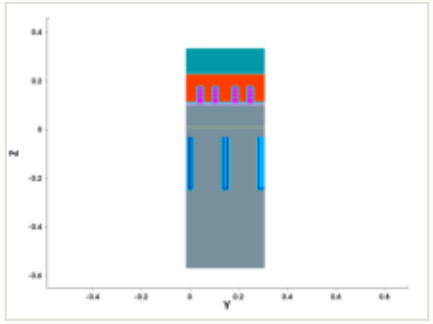
carbon so far, of the whole flow (kgCO2e)



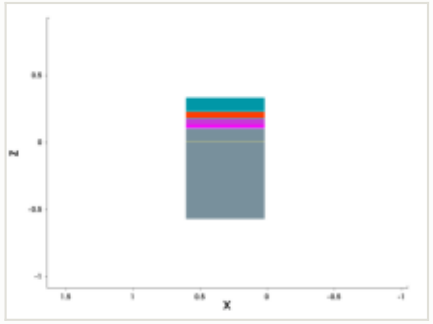
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



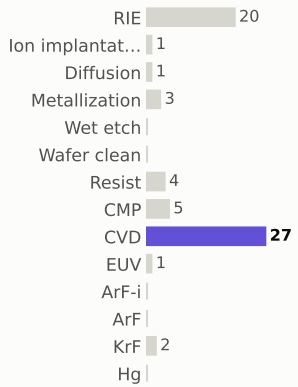
y cut at 0.184 µm



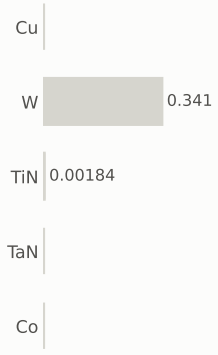
mask for this step

no mask for this step

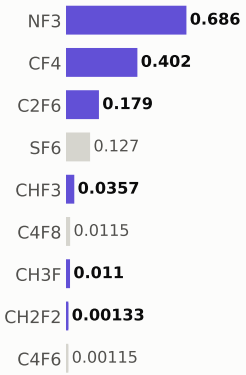
steps by process area



materials deposited (g)



gases emitted (g)



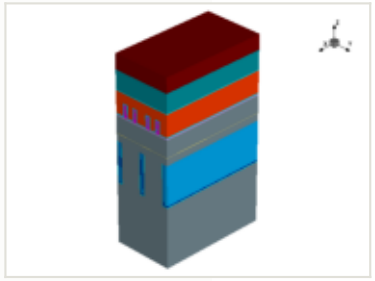
Step #750: CVD

depo_height_dependent · CVD

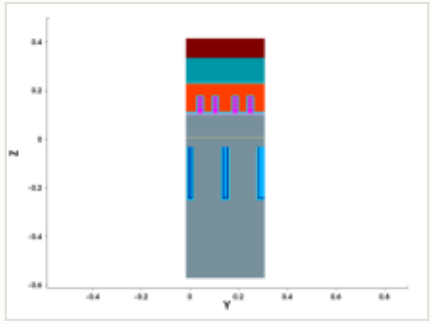
1.31	118.6	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

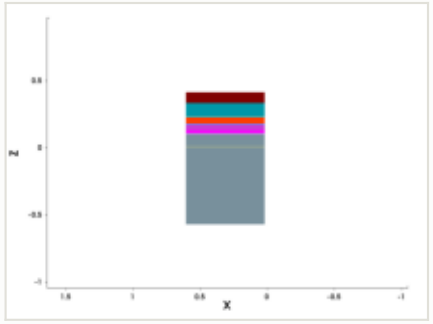
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

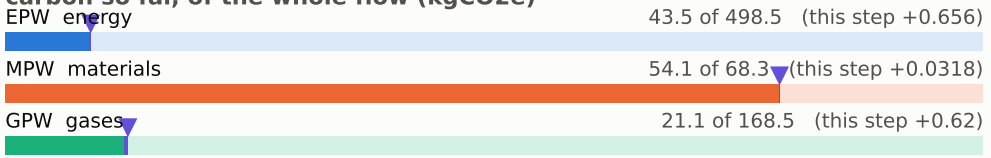
no mask for this step

this step

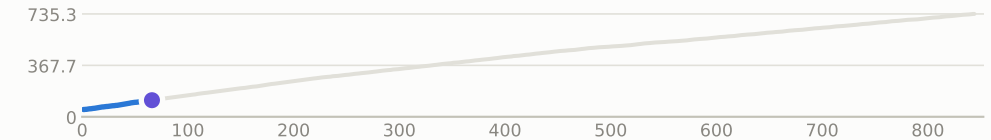
process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

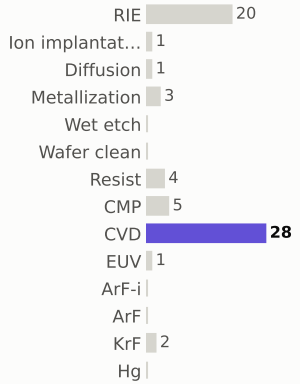
carbon so far, of the whole flow (kgCO2e)



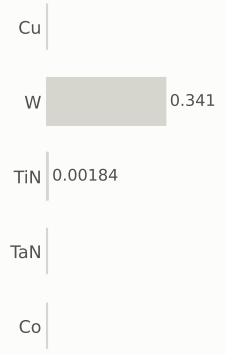
embodied carbon up to each step (kgCO2e per wafer)



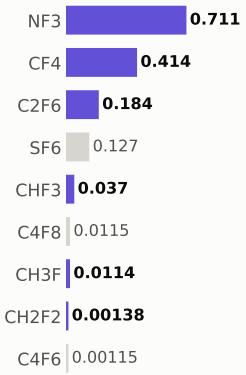
steps by process area



materials deposited (g)



gases emitted (g)



Step #760: CVD

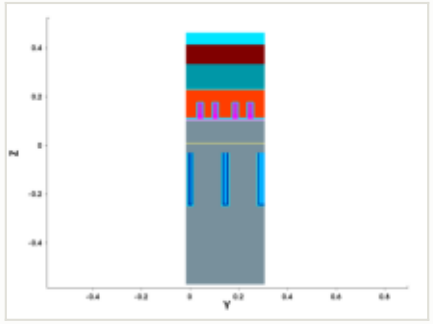
depo_height_dependent · CVD

1.31	119.9	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

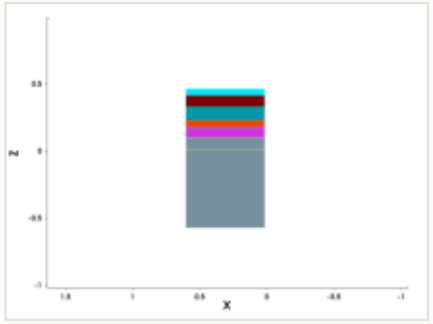
3D structure after this step

zoomed to fit

x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

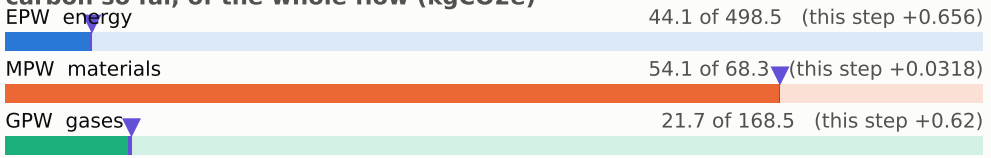
no mask for this step

this step

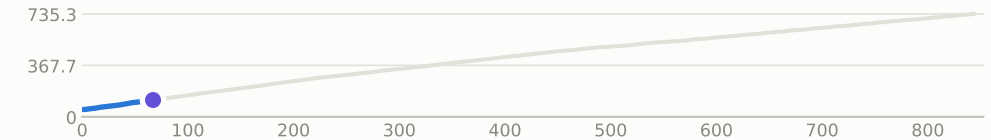
process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

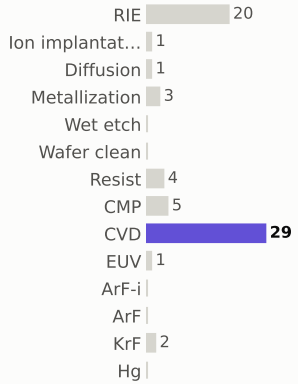
carbon so far, of the whole flow (kgCO2e)



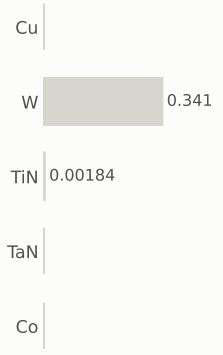
embodied carbon up to each step (kgCO2e per wafer)



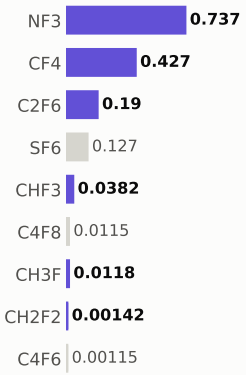
steps by process area



materials deposited (g)



gases emitted (g)



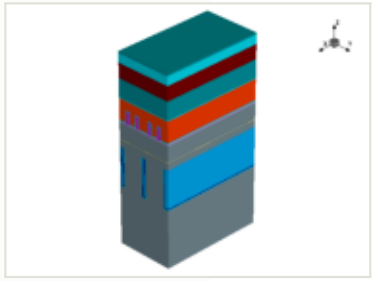
Step #770: CVD

depo_height_dependent · CVD

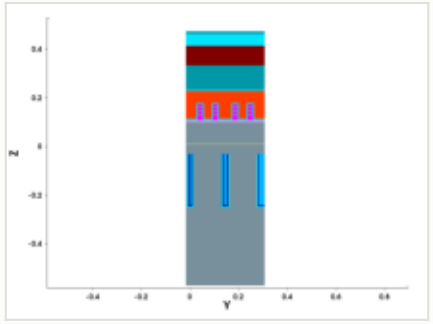
1.31	121.3	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

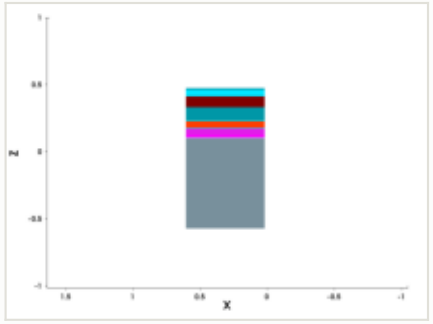
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



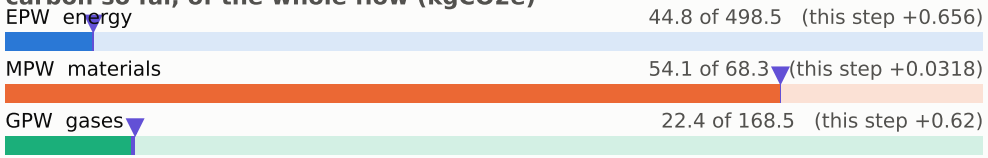
mask for this step

no mask for this step

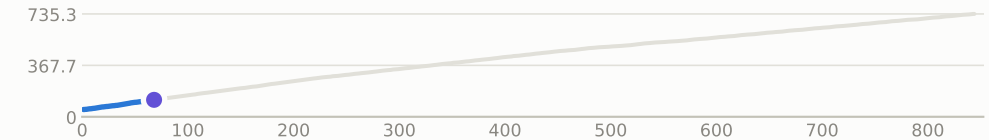
this step

process area	CVD	EPW	0.656 kgCO2e
material	—	EPW tool	0.469 kgCO2e
thickness	0 µm	EPW overhead	0.187 kgCO2e
material GWP	—	MPW	0.0318 kgCO2e (gas 0.0318)
input gases	NF3 2.54 g	GPW	0.62 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**1.31 kgCO2e**

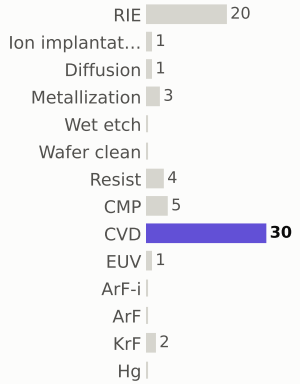
carbon so far, of the whole flow (kgCO2e)



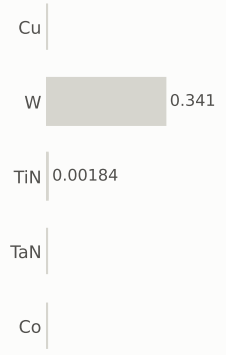
embodied carbon up to each step (kgCO2e per wafer)



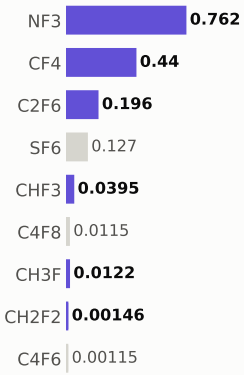
steps by process area



materials deposited (g)



gases emitted (g)



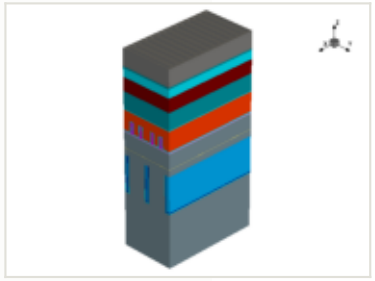
Step #780: EUV

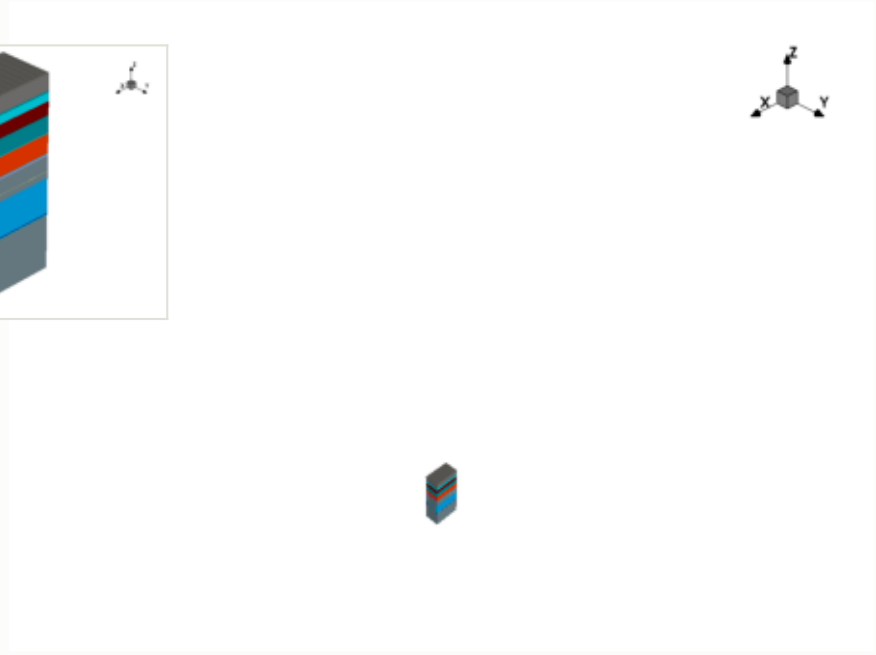
litho_patterning · EUV

1.96	123.2	0.27%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



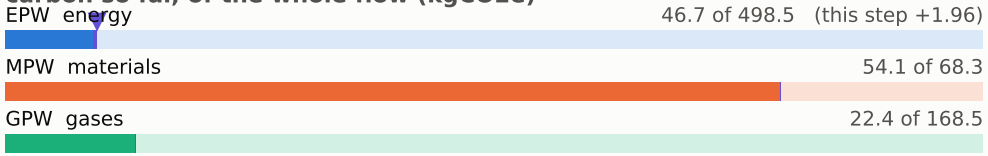


this step

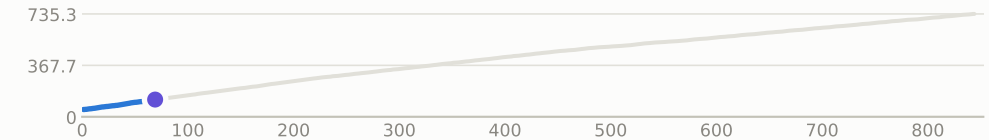
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

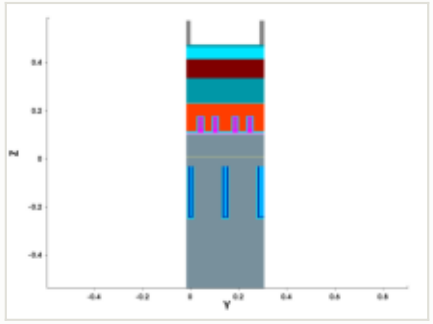
carbon so far, of the whole flow (kgCO2e)



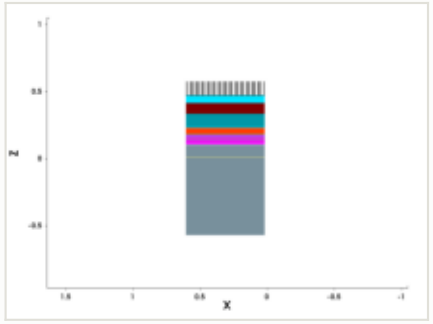
embodied carbon up to each step (kgCO2e per wafer)



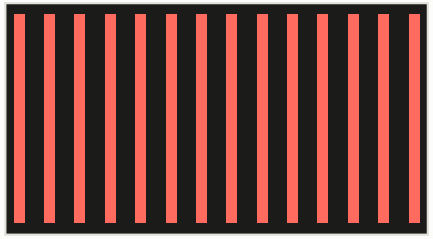
x cut at 0.294 µm



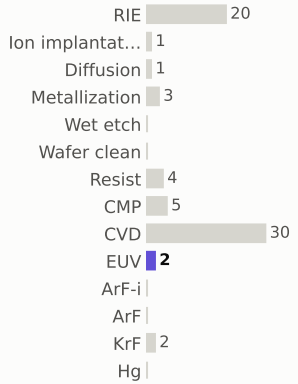
y cut at 0.184 µm



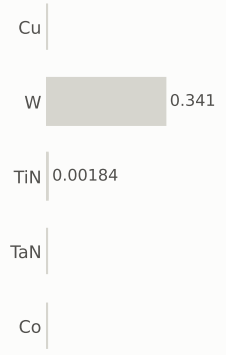
mask: layer 3:0 (GDSII)



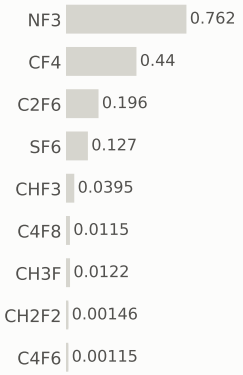
steps by process area



materials deposited (g)



gases emitted (g)



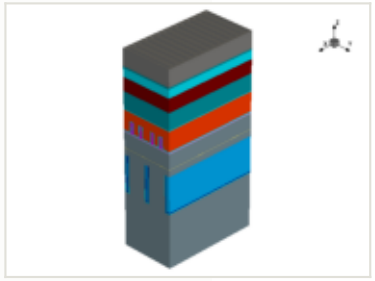
Step #790: RIE

etch_anisotropic · RIE

1.04	124.3	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

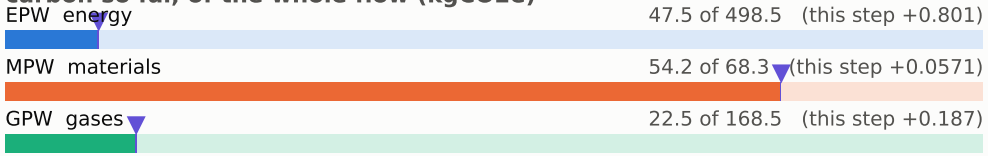


this step

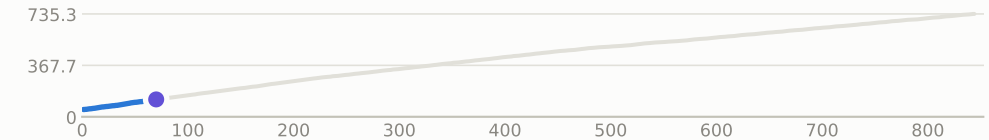
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

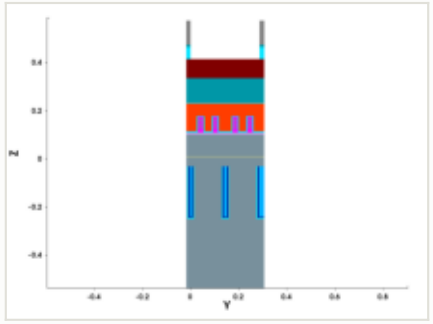
carbon so far, of the whole flow (kgCO2e)



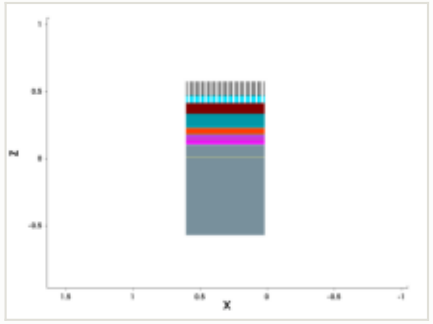
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



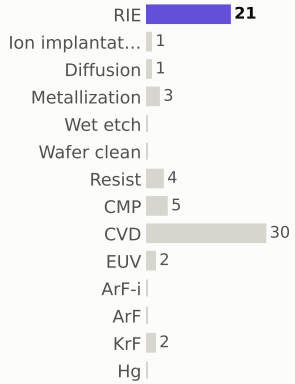
y cut at 0.184 µm



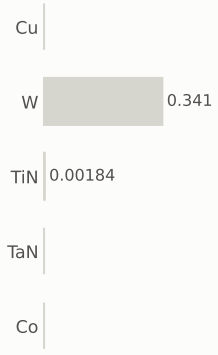
mask for this step

no mask for this step

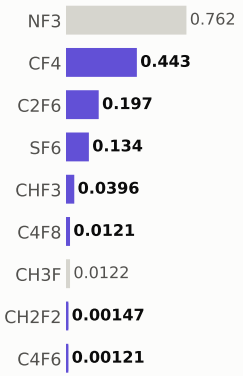
steps by process area



materials deposited (g)



gases emitted (g)



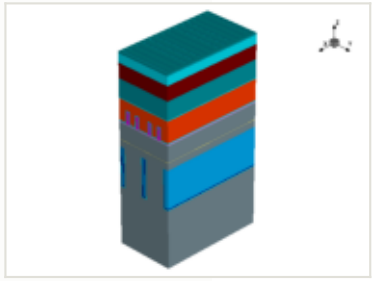
Step #800: Resist

etch_strip · Resist

0.169	124.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

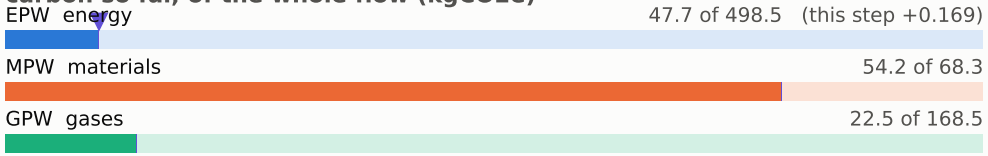


this step

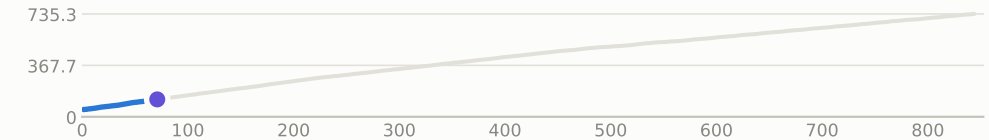
process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

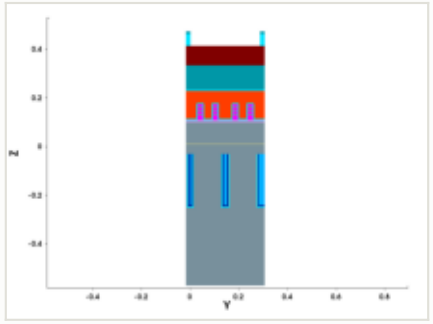
carbon so far, of the whole flow (kgCO2e)



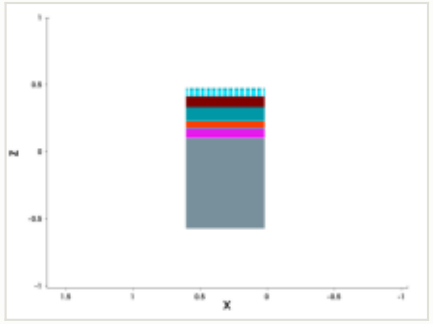
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



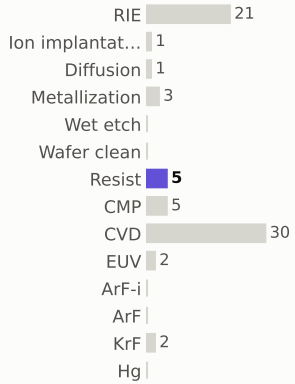
y cut at 0.184 µm



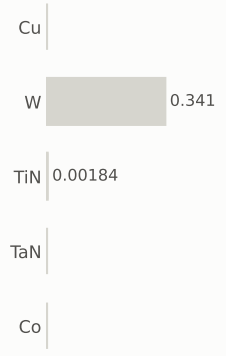
mask for this step

no mask for this step

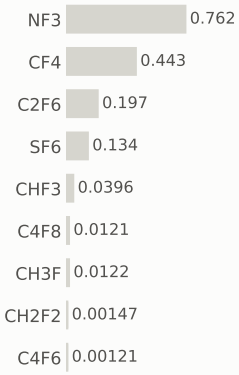
steps by process area



materials deposited (g)



gases emitted (g)



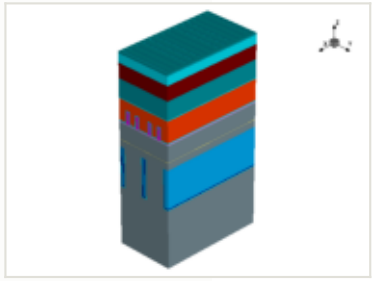
Step #810: RIE

etch_anisotropic · RIE

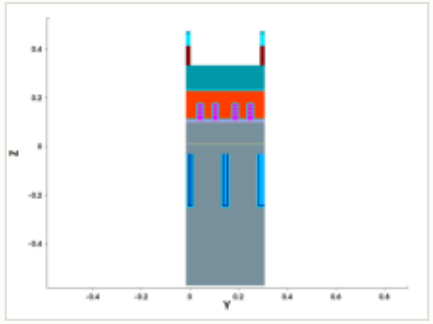
1.04	125.5	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

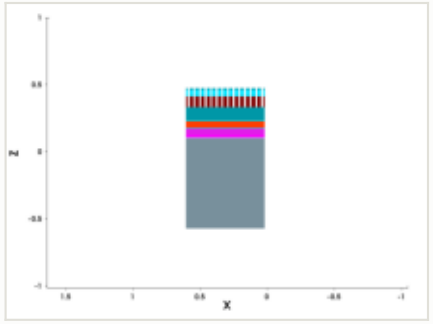
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

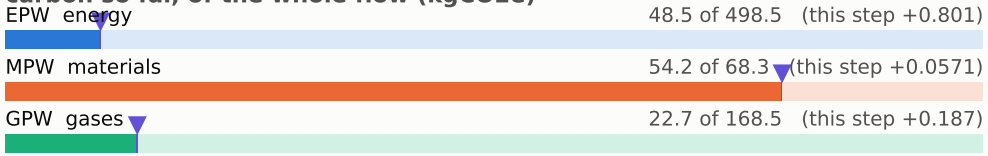
no mask for this step

this step

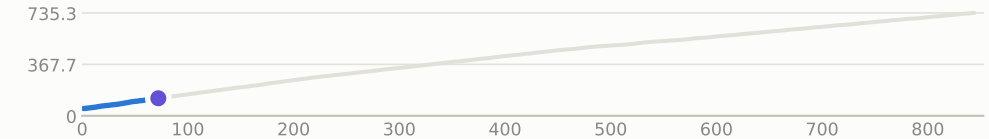
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

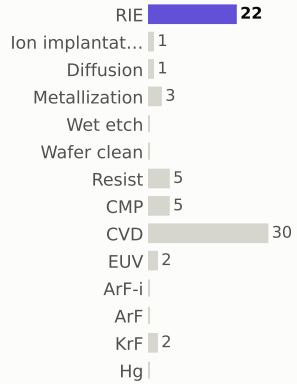
carbon so far, of the whole flow (kgCO2e)



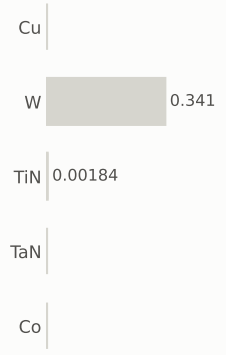
embodied carbon up to each step (kgCO2e per wafer)



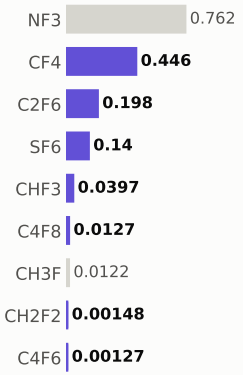
steps by process area



materials deposited (g)



gases emitted (g)



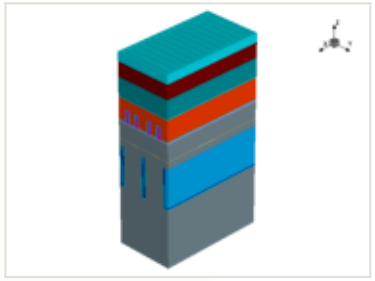
Step #820: CMP

etch_cmp · CMP

0.169	125.6	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

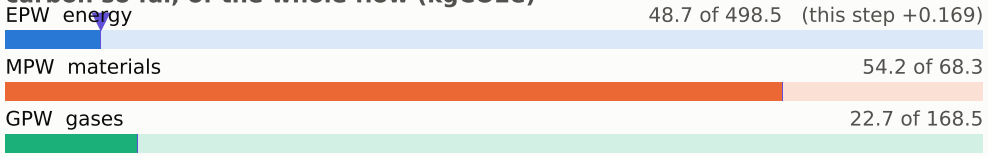


this step

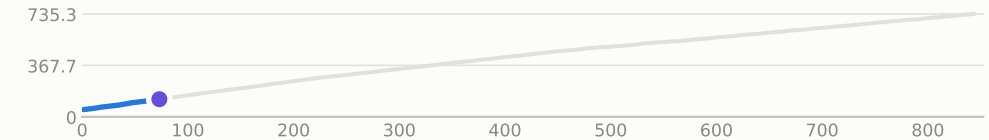
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

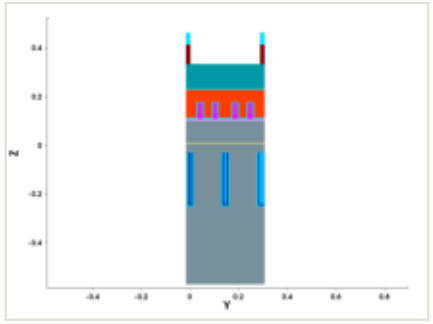
carbon so far, of the whole flow (kgCO2e)



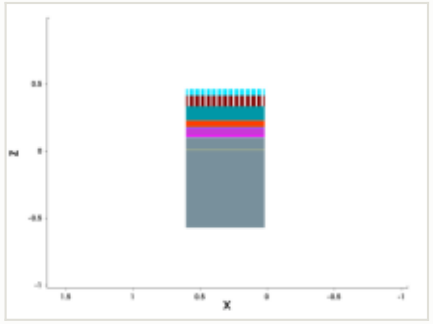
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



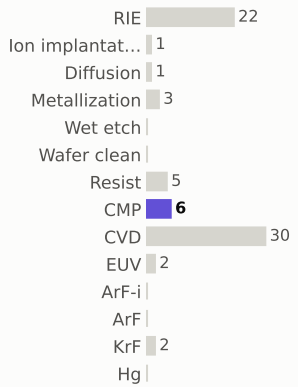
y cut at 0.184 µm



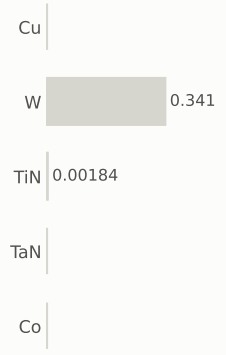
mask for this step

no mask for this step

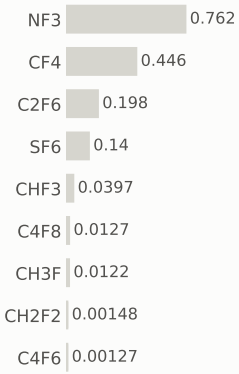
steps by process area



materials deposited (g)



gases emitted (g)



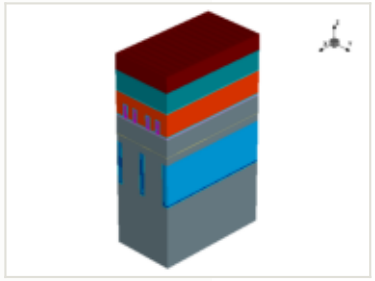
Step #830: RIE

etch_anisotropic · RIE

1.04	126.7	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

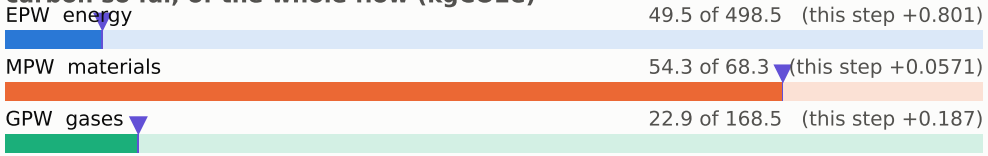


this step

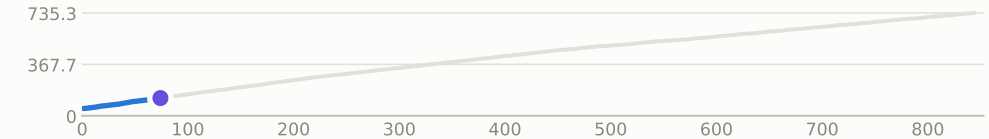
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

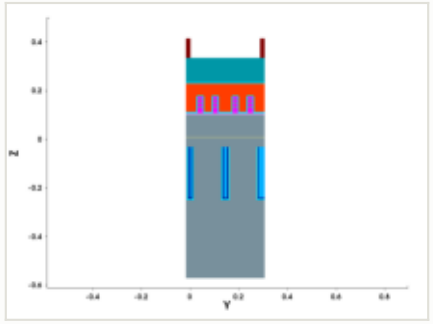
carbon so far, of the whole flow (kgCO2e)



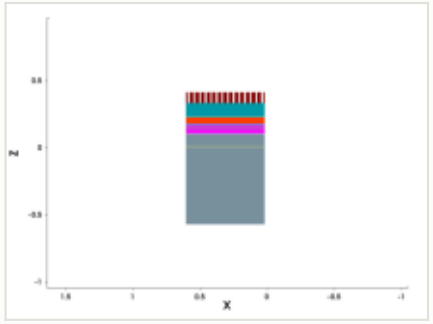
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



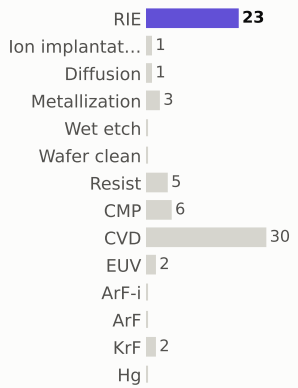
y cut at 0.184 µm



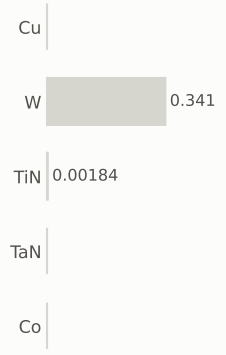
mask for this step

no mask for this step

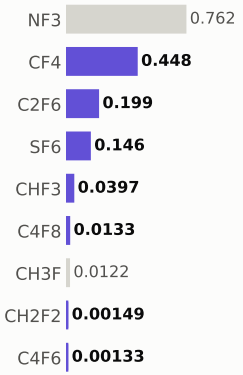
steps by process area



materials deposited (g)



gases emitted (g)



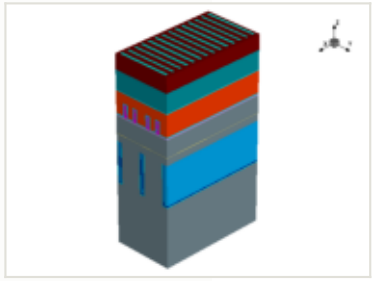
Step #840: CVD

depo_fill · CVD

1.31	128.0	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

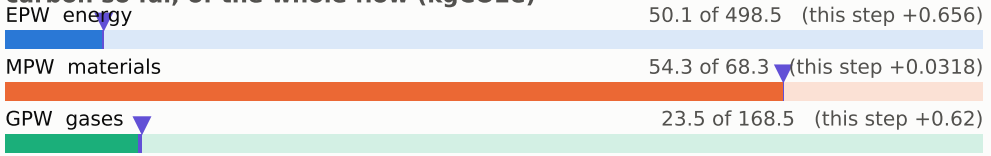


this step

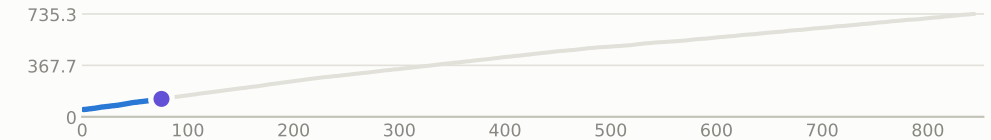
process area	CVD
material	—
thickness	0.08 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

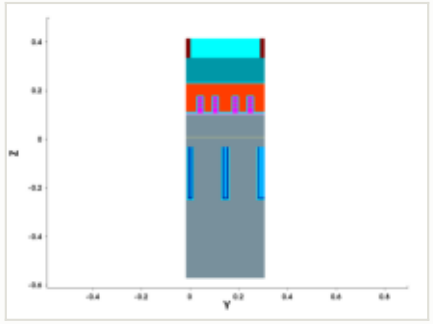
carbon so far, of the whole flow (kgCO2e)



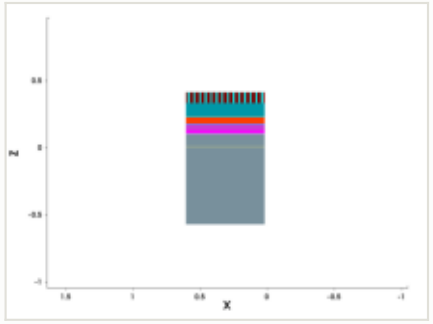
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



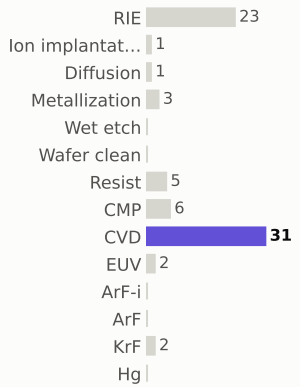
y cut at 0.184 µm



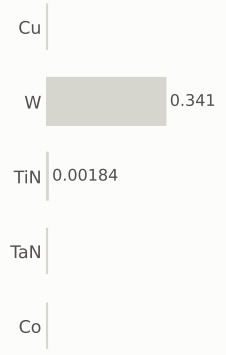
mask for this step

no mask for this step

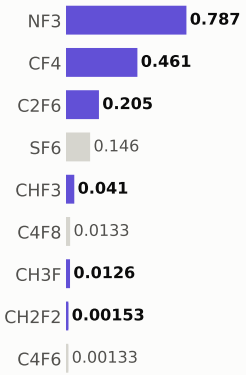
steps by process area



materials deposited (g)



gases emitted (g)



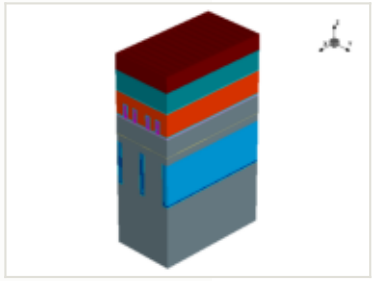
Step #850: RIE

etch_anisotropic · RIE

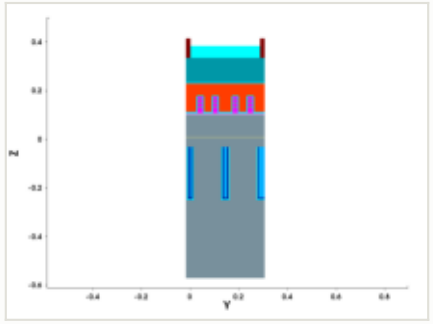
1.04	129.0	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

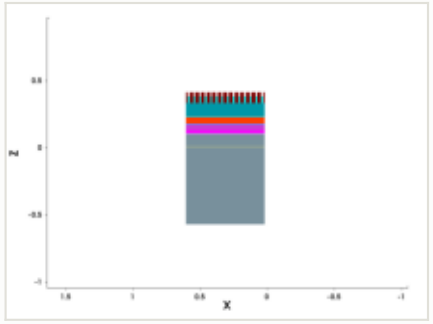
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

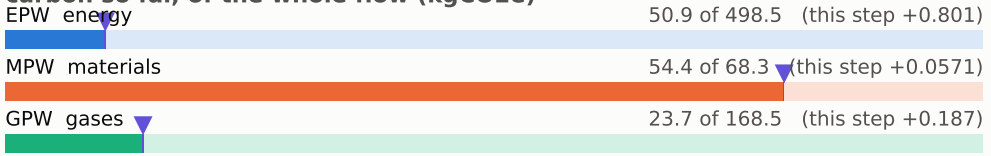
no mask for this step

this step

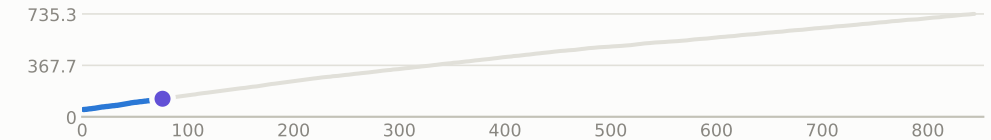
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

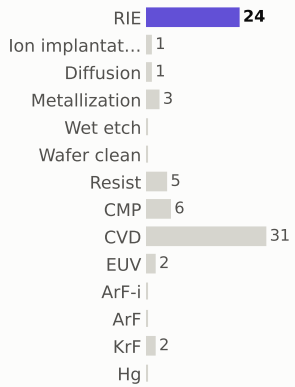
carbon so far, of the whole flow (kgCO2e)



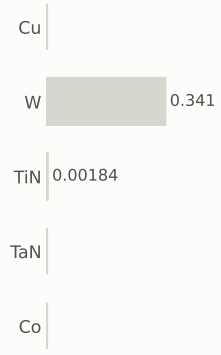
embodied carbon up to each step (kgCO2e per wafer)



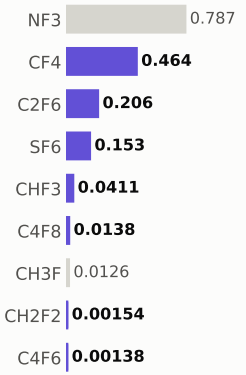
steps by process area



materials deposited (g)



gases emitted (g)



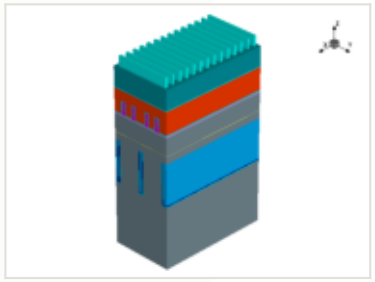
Step #860: RIE

etch_anisotropic · RIE

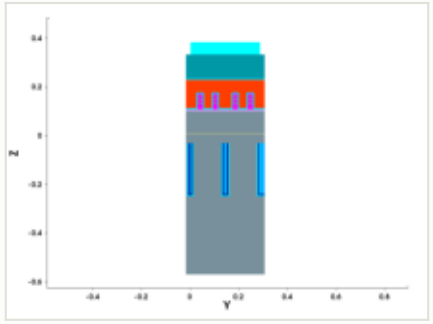
1.04	130.1	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

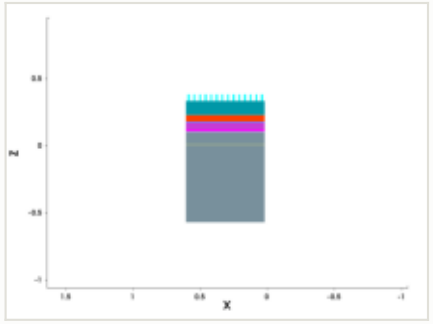
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



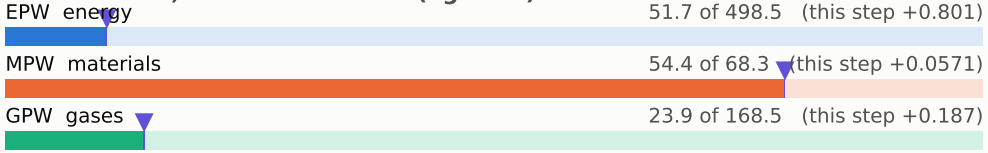
mask for this step

no mask for this step

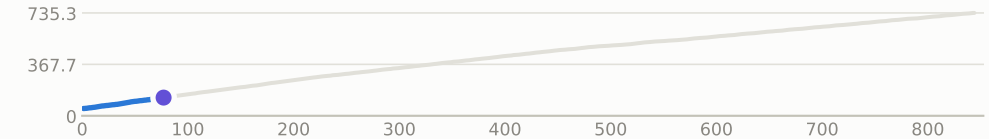
this step

process area	RIE	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

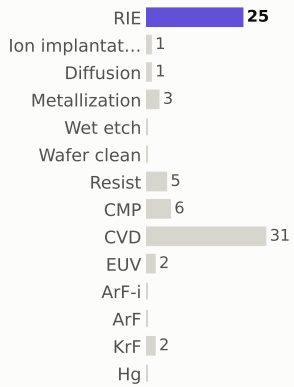
carbon so far, of the whole flow (kgCO2e)



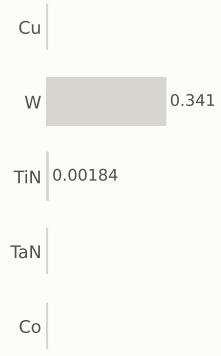
embodied carbon up to each step (kgCO2e per wafer)



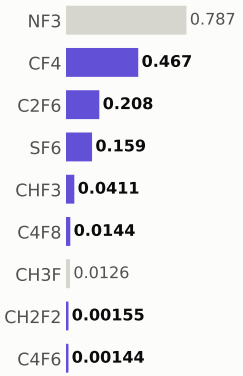
steps by process area



materials deposited (g)



gases emitted (g)



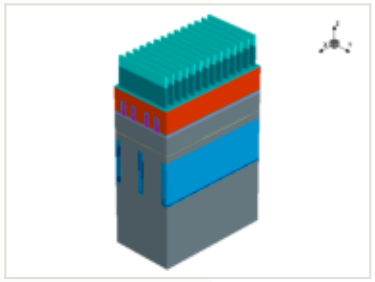
Step #870: RIE

etch_anisotropic · RIE

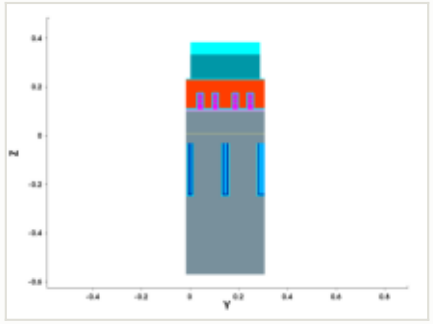
1.04	131.1	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

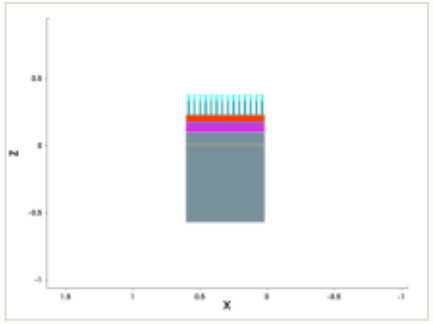
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

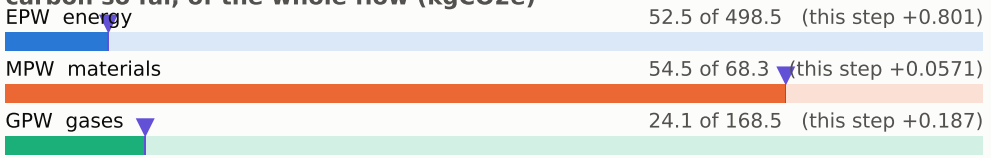
no mask for this step

this step

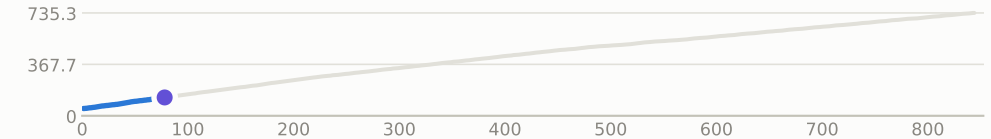
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

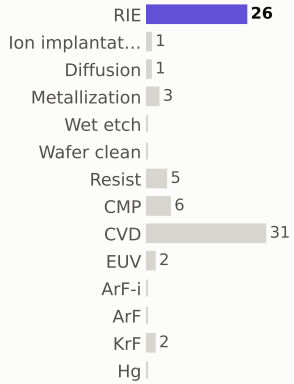
carbon so far, of the whole flow (kgCO2e)



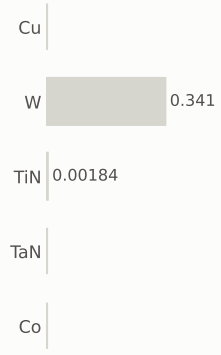
embodied carbon up to each step (kgCO2e per wafer)



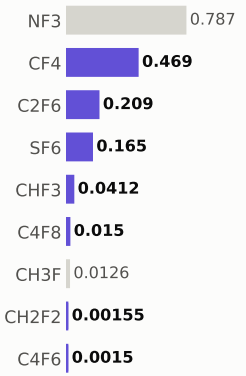
steps by process area



materials deposited (g)



gases emitted (g)



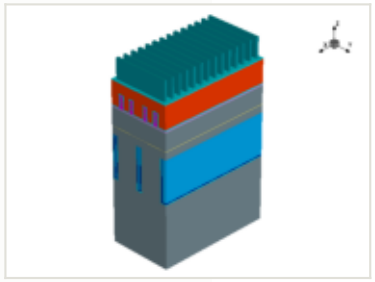
Step #880: RIE

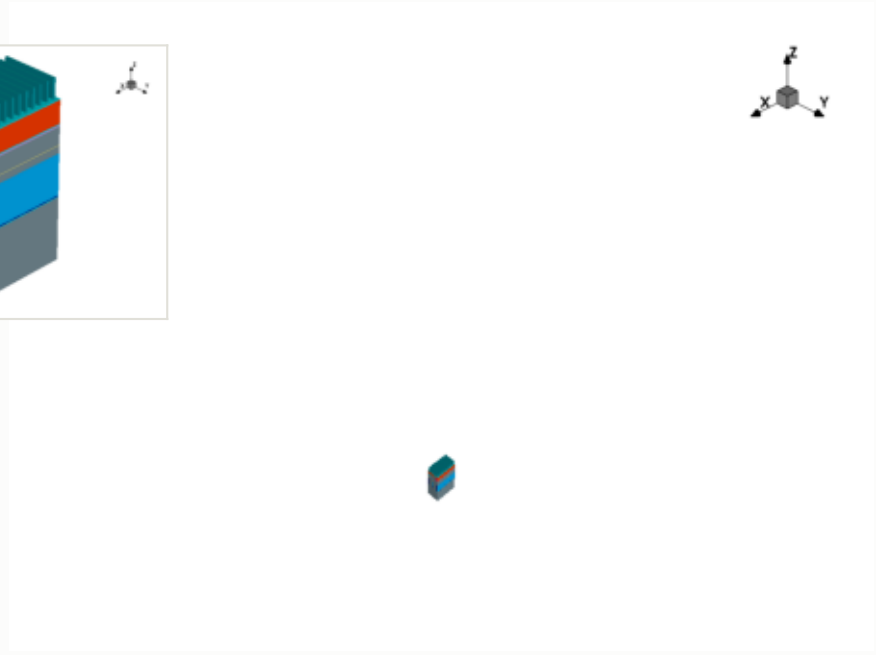
etch_anisotropic · RIE

1.04	132.2	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

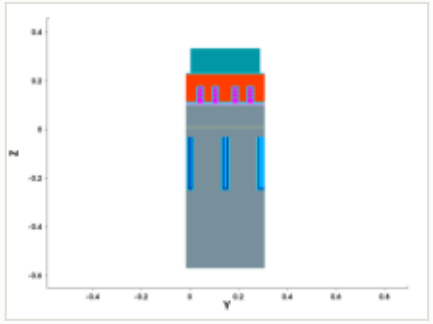
3D structure after this step

zoomed to fit

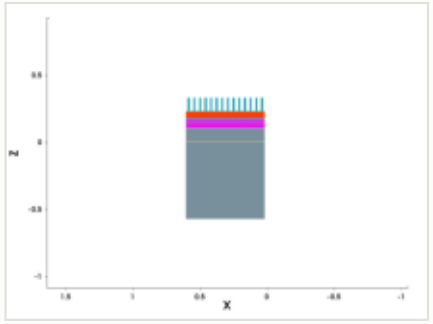




x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

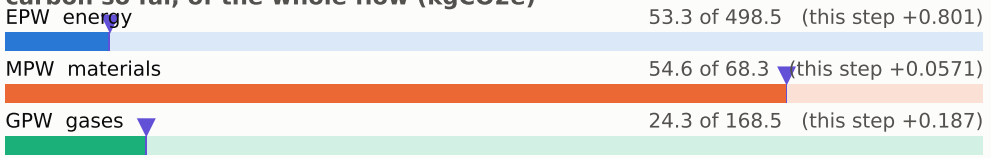
no mask for this step

this step

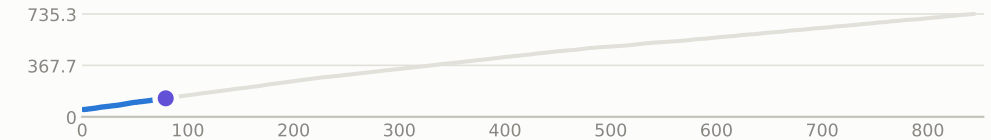
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

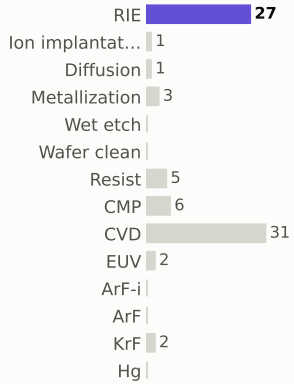
carbon so far, of the whole flow (kgCO2e)



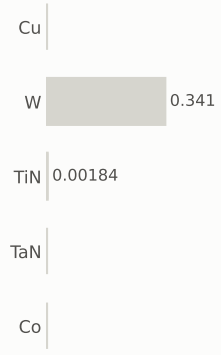
embodied carbon up to each step (kgCO2e per wafer)



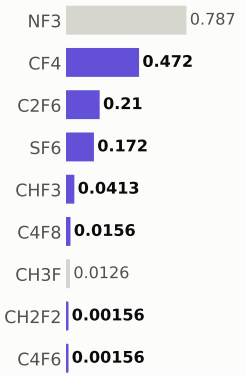
steps by process area



materials deposited (g)



gases emitted (g)



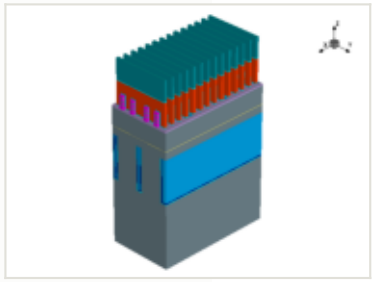
Step #890: RIE

etch_anisotropic · RIE

1.04	133.2	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

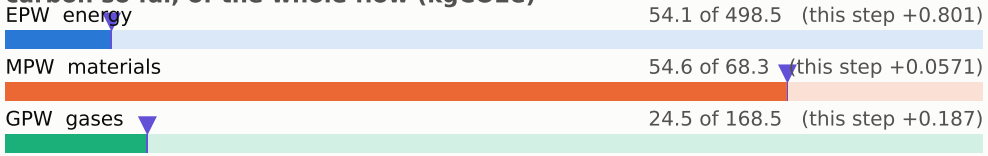


this step

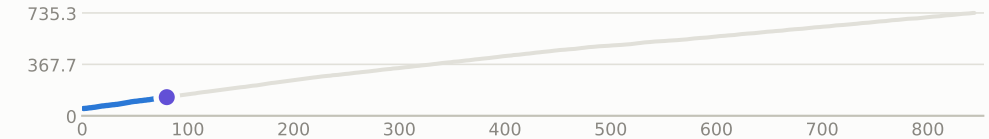
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

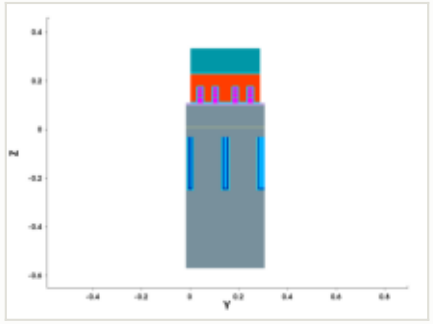
carbon so far, of the whole flow (kgCO2e)



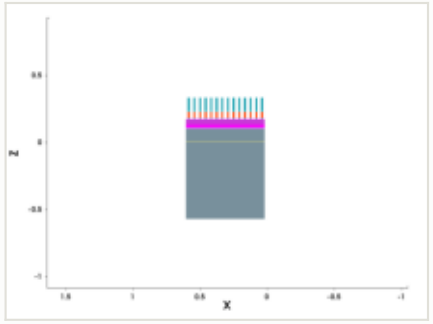
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



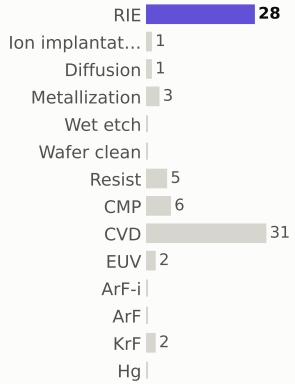
y cut at 0.184 µm



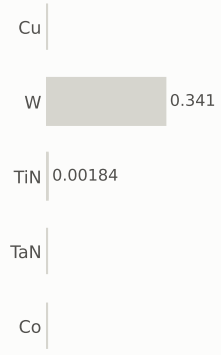
mask for this step

no mask for this step

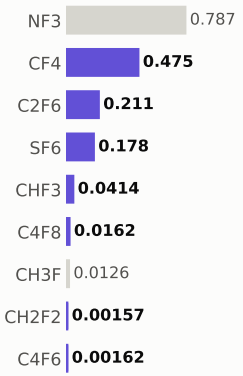
steps by process area



materials deposited (g)



gases emitted (g)



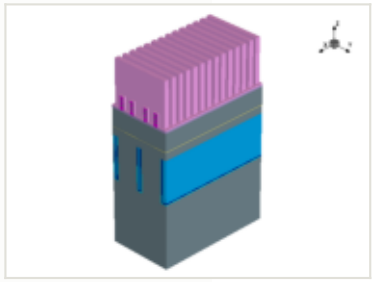
Step #900: CVD

depo_isotropic · CVD

1.31	134.5	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

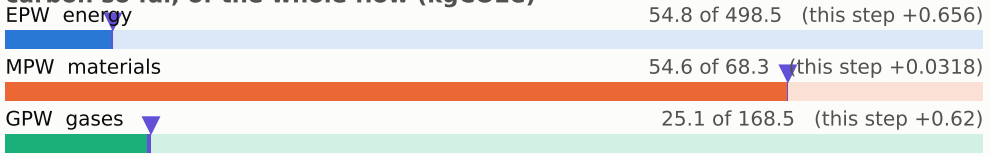


this step

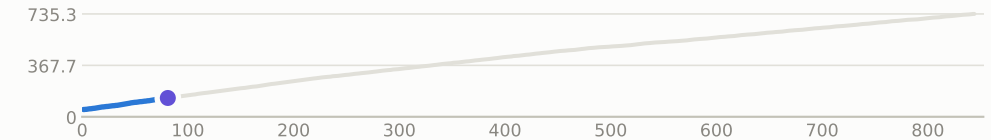
process area	CVD
material	—
thickness	0.006 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

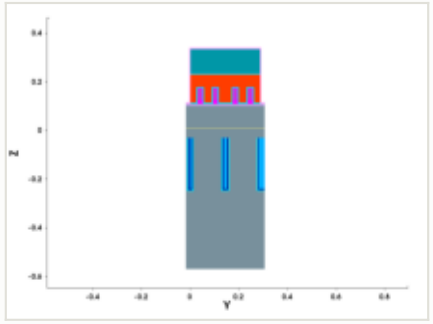
carbon so far, of the whole flow (kgCO2e)



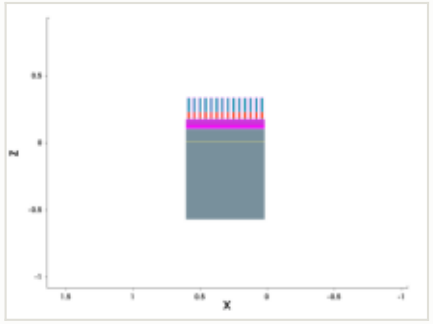
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



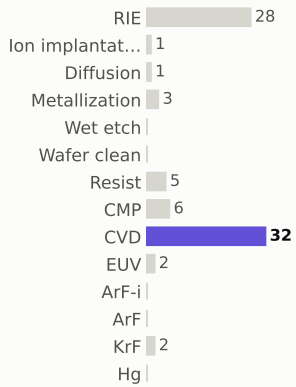
y cut at 0.184 µm



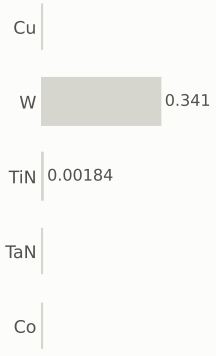
mask for this step

no mask for this step

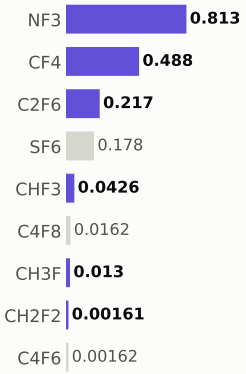
steps by process area



materials deposited (g)



gases emitted (g)



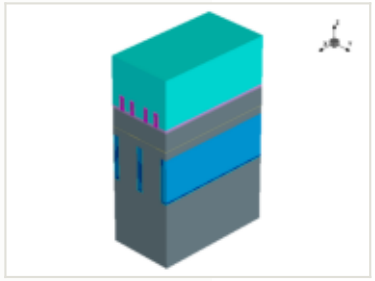
Step #920: CVD

depo_fill · CVD

1.31	135.8	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

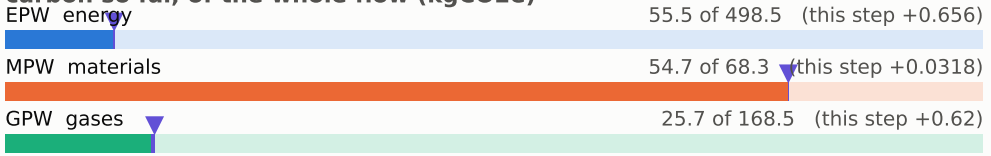


this step

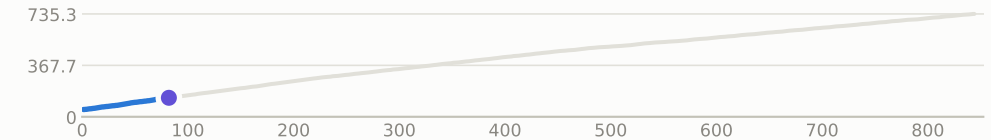
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

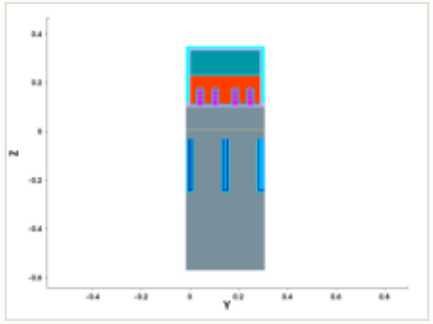
carbon so far, of the whole flow (kgCO2e)



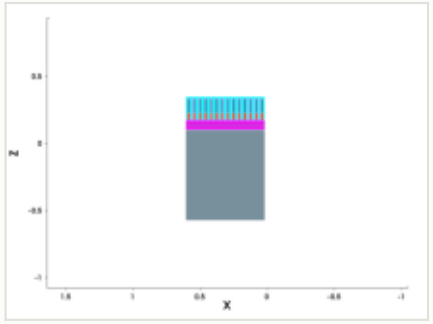
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



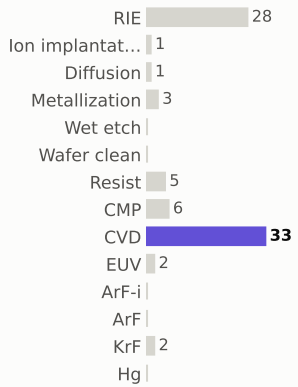
y cut at 0.184 µm



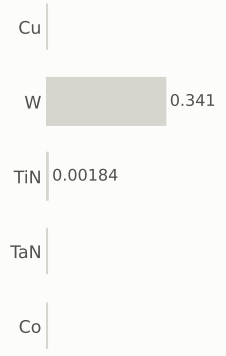
mask for this step

no mask for this step

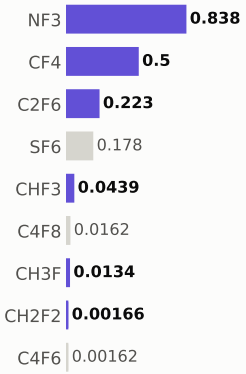
steps by process area



materials deposited (g)



gases emitted (g)



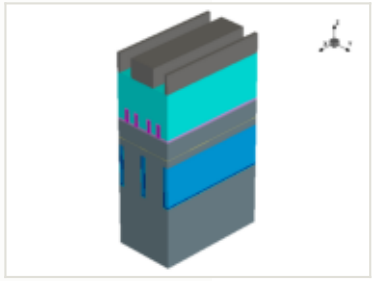
Step #930: EUV

litho_patterning · EUV

1.96	137.8	0.27%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

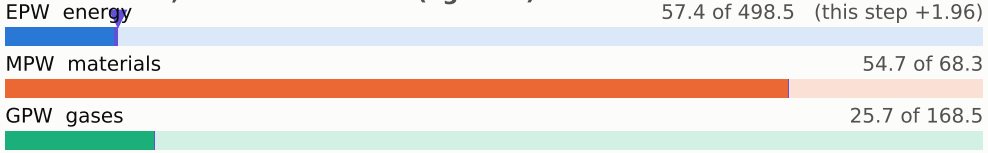


this step

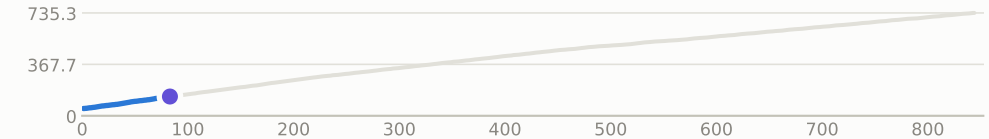
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

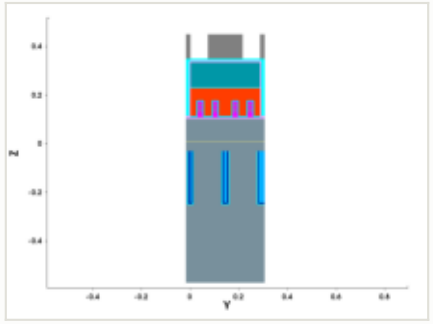
carbon so far, of the whole flow (kgCO2e)



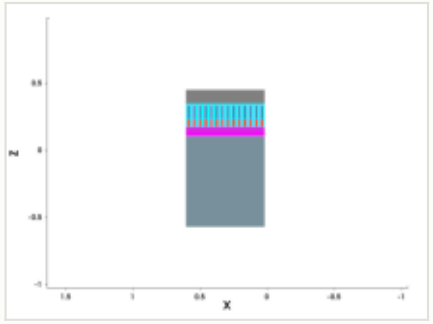
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm

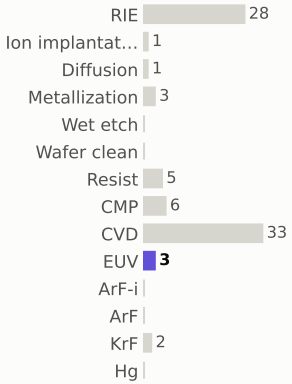


y cut at 0.184 µm

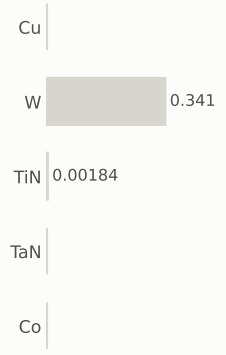


mask: layer 6:0 (GDSII)

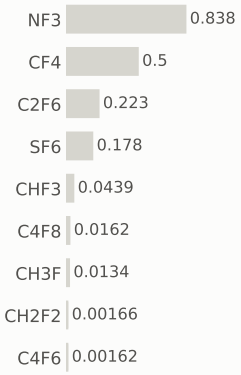
steps by process area



materials deposited (g)



gases emitted (g)



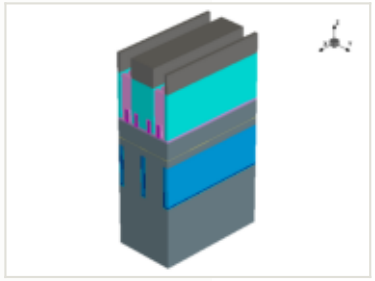
Step #940: RIE

etch_anisotropic · RIE

1.04	138.8	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

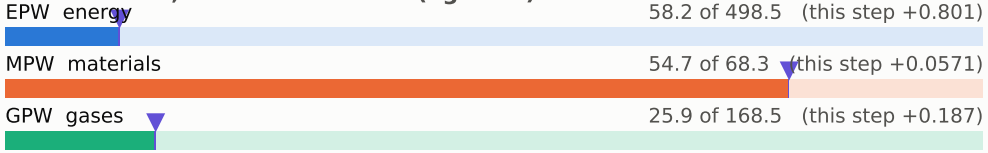


this step

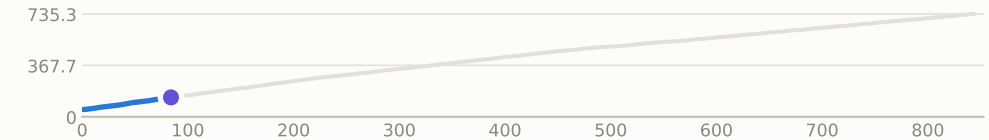
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

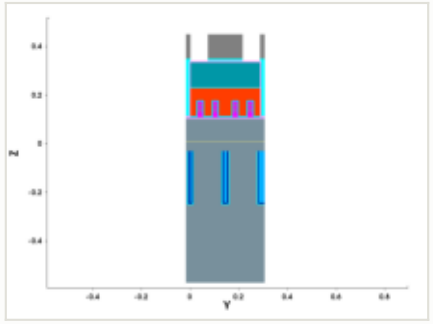
carbon so far, of the whole flow (kgCO2e)



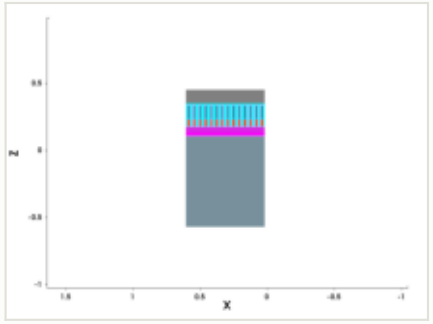
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



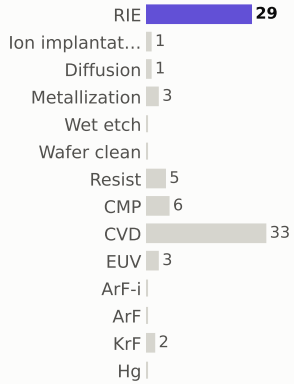
y cut at 0.184 µm



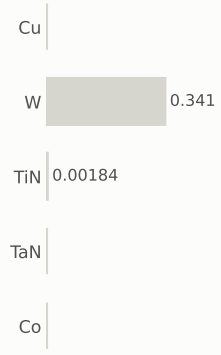
mask for this step

no mask for this step

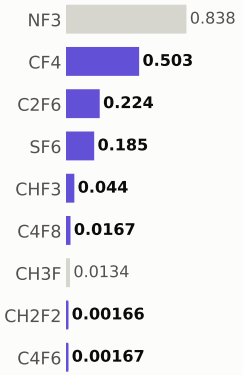
steps by process area



materials deposited (g)



gases emitted (g)



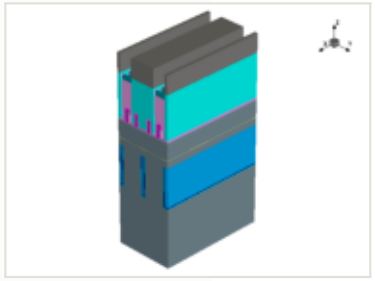
Step #950: RIE

etch_anisotropic · RIE

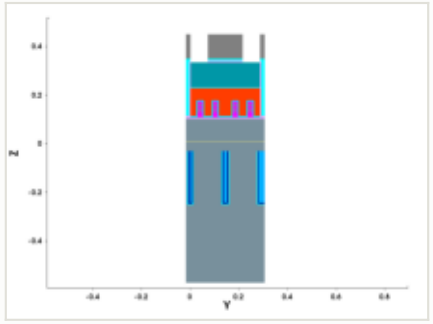
1.04	139.9	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

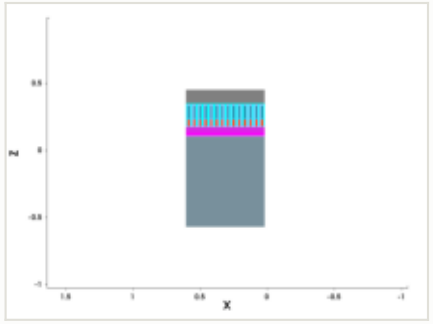
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

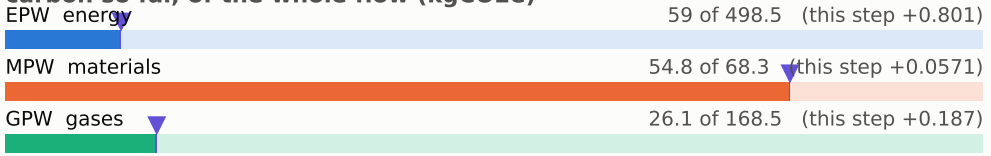
no mask for this step

this step

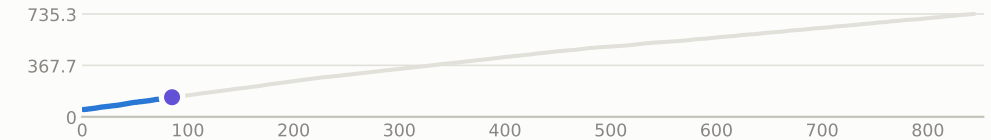
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

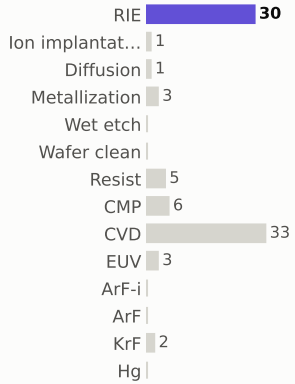
carbon so far, of the whole flow (kgCO2e)



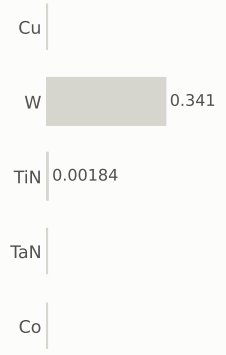
embodied carbon up to each step (kgCO2e per wafer)



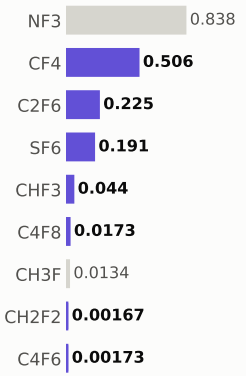
steps by process area



materials deposited (g)



gases emitted (g)



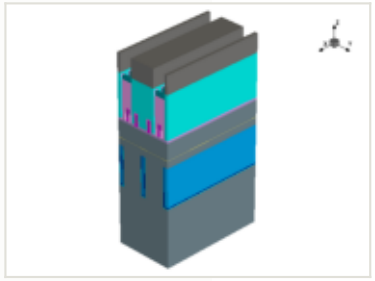
Step #960: RIE

etch_anisotropic · RIE

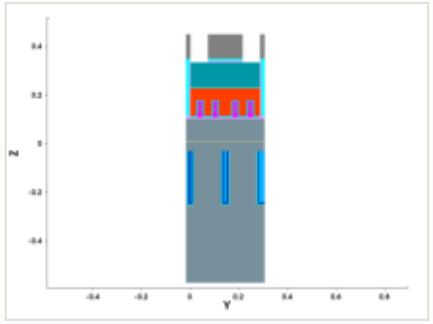
1.04	140.9	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

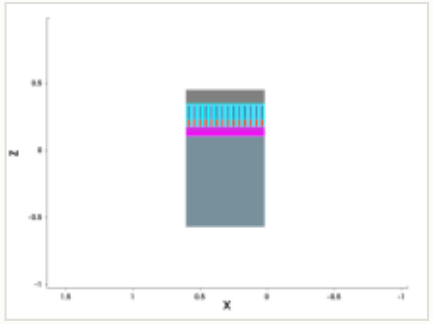
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

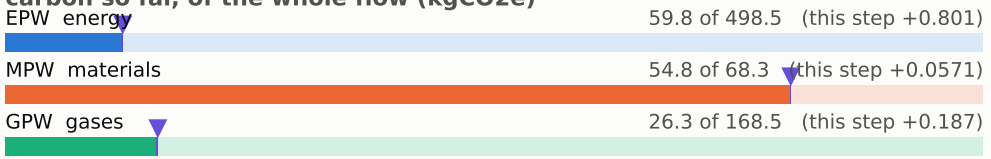
no mask for this step

this step

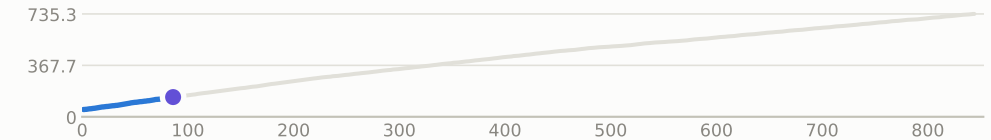
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

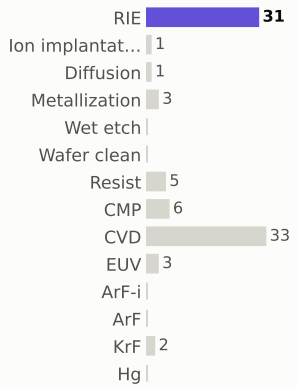
carbon so far, of the whole flow (kgCO2e)



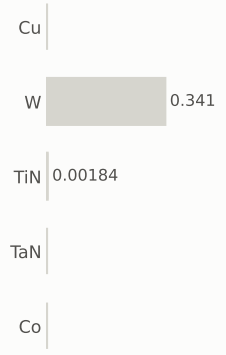
embodied carbon up to each step (kgCO2e per wafer)



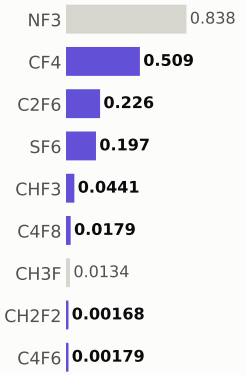
steps by process area



materials deposited (g)



gases emitted (g)



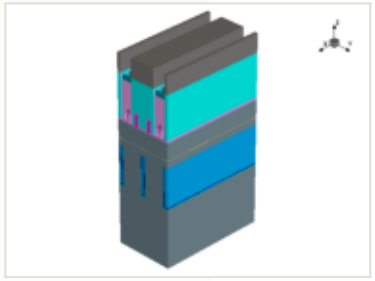
Step #970: Wet_etch

etch_isotropic · Wet_etch

0.169	141.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

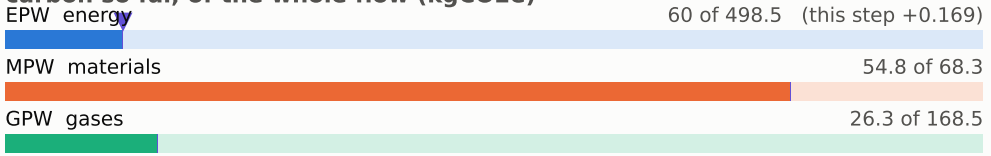


this step

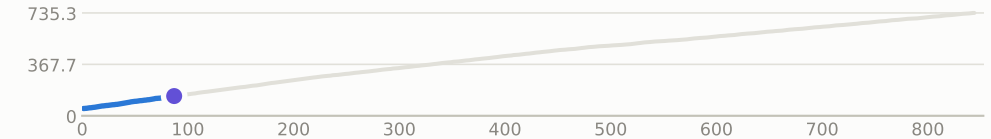
process area	Wet_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

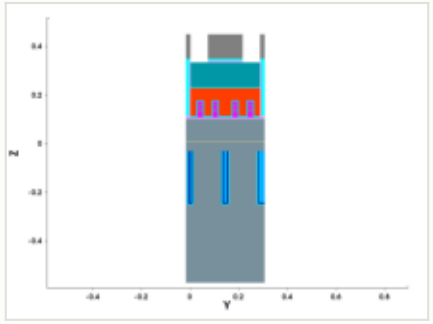
carbon so far, of the whole flow (kgCO2e)



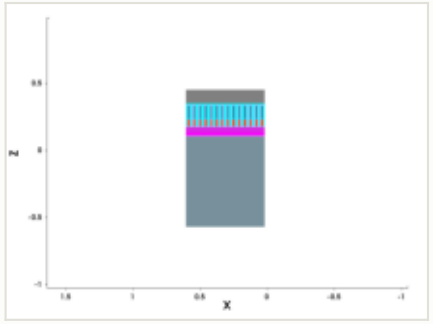
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



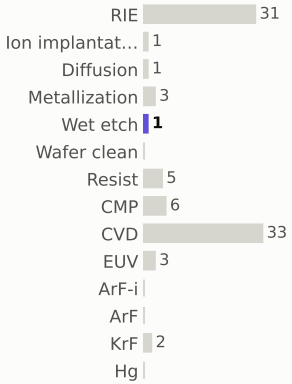
y cut at 0.184 µm



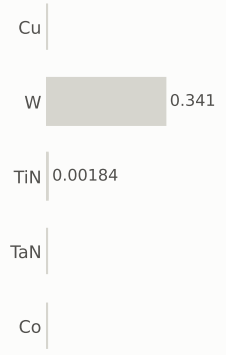
mask for this step

no mask for this step

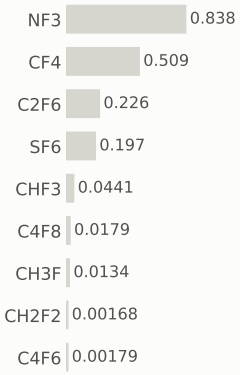
steps by process area



materials deposited (g)



gases emitted (g)



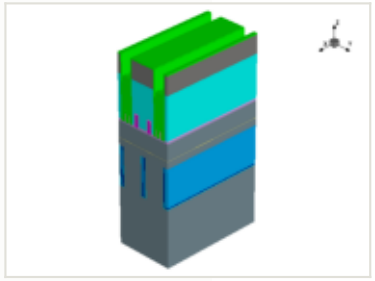
Step #980: CVD

depo_isotropic · CVD

1.31	142.4	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

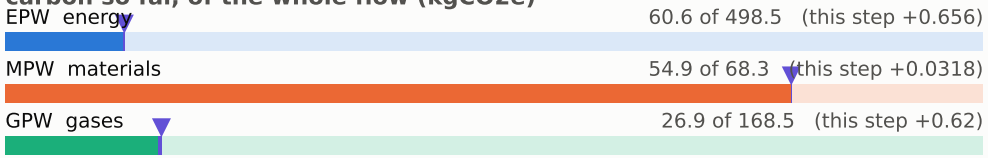


this step

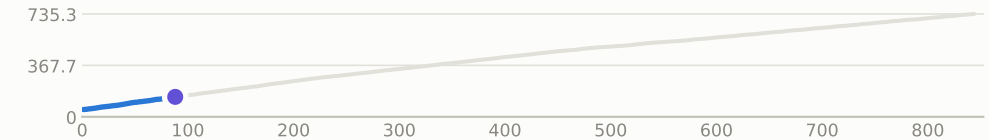
process area	CVD
material	—
thickness	0.007 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

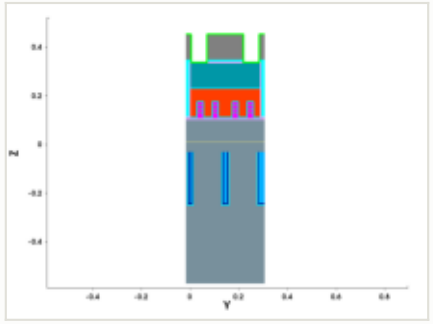
carbon so far, of the whole flow (kgCO2e)



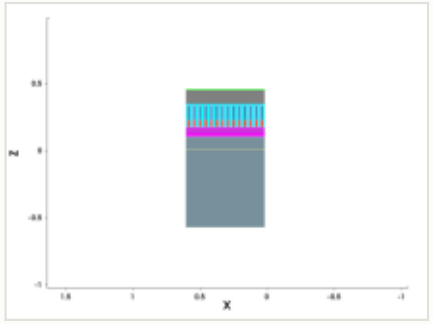
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



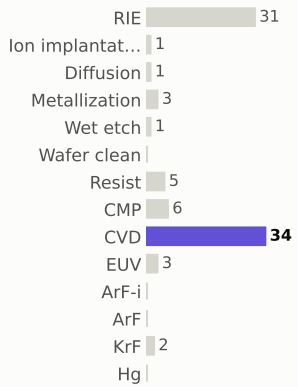
y cut at 0.184 µm



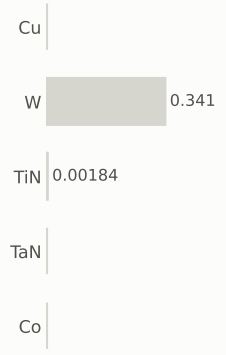
mask for this step

no mask for this step

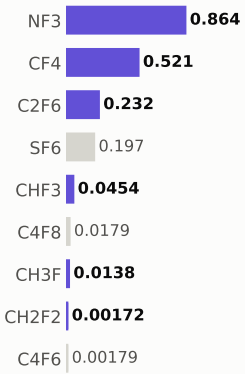
steps by process area



materials deposited (g)



gases emitted (g)



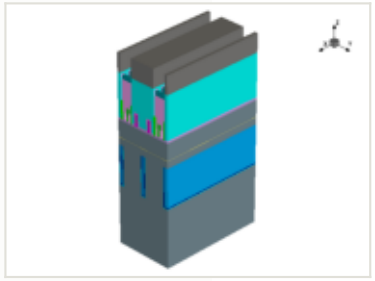
Step #990: RIE

etch_anisotropic · RIE

1.04	143.4	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

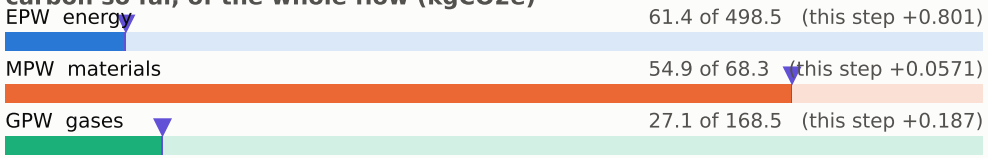


this step

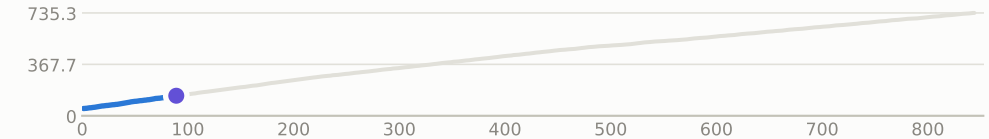
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

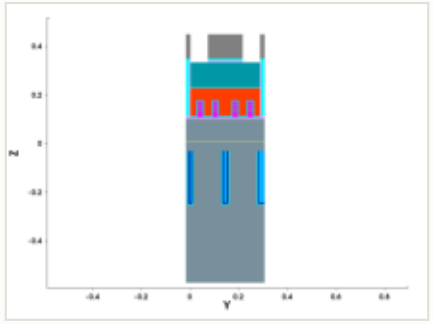
carbon so far, of the whole flow (kgCO2e)



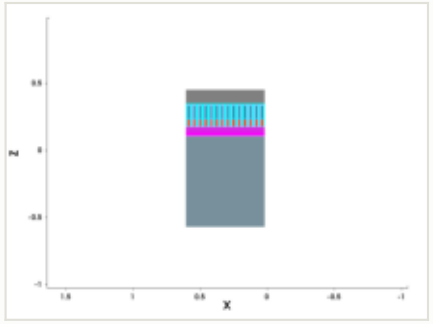
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



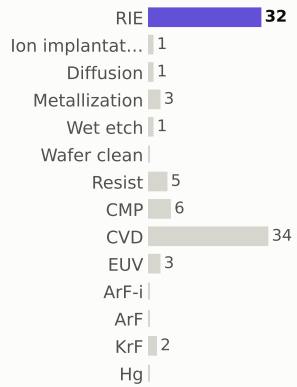
y cut at 0.184 µm



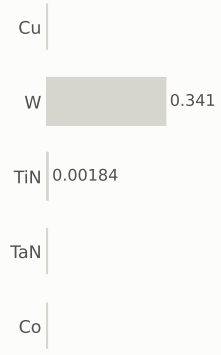
mask for this step

no mask for this step

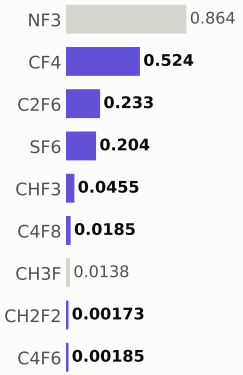
steps by process area



materials deposited (g)



gases emitted (g)



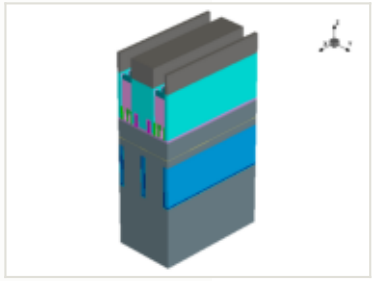
Step #1000: CMP

etch_cmp · CMP

0.169	143.6	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

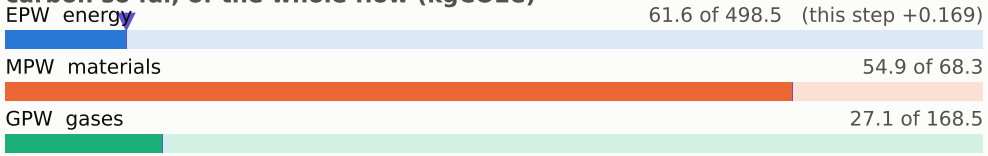


this step

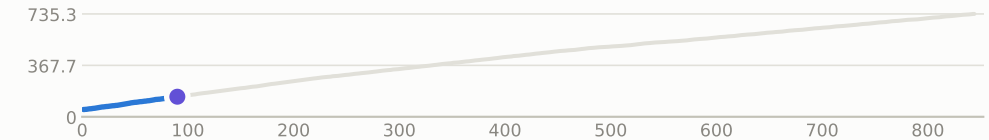
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

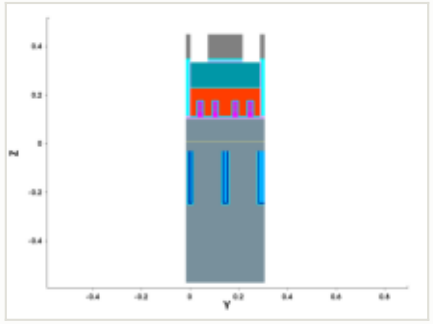
carbon so far, of the whole flow (kgCO2e)



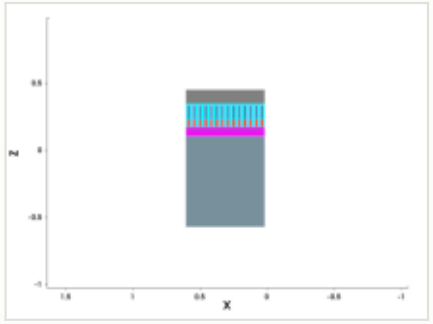
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



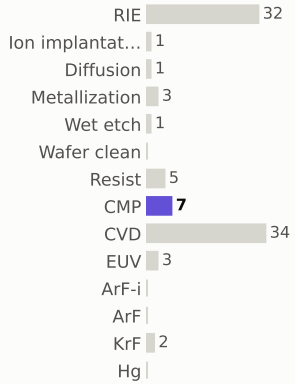
y cut at 0.184 µm



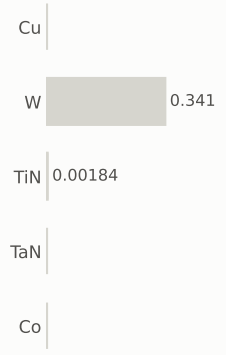
mask for this step

no mask for this step

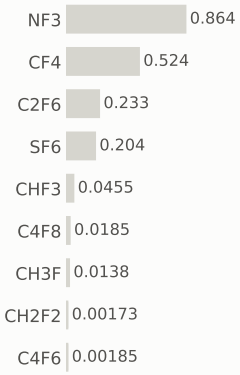
steps by process area



materials deposited (g)



gases emitted (g)



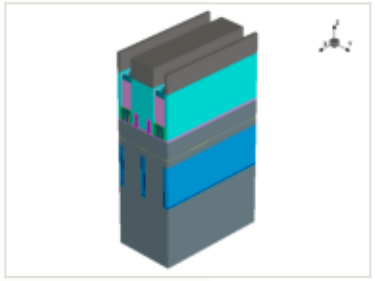
Step #1010: CVD

depo_fill · CVD

1.31	144.9	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

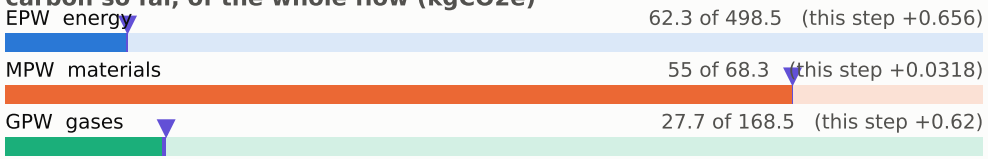


this step

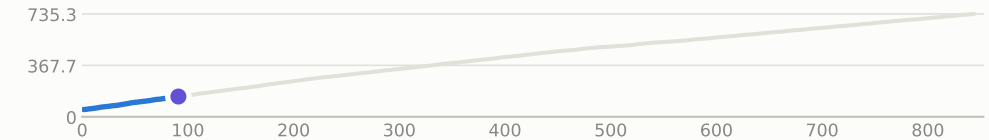
process area	CVD
material	—
thickness	0.005 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

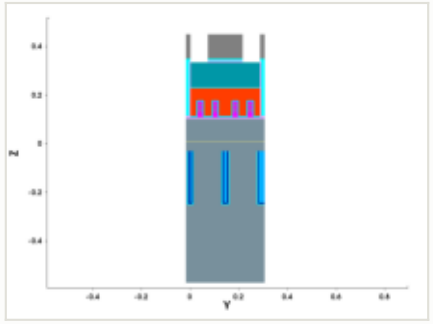
carbon so far, of the whole flow (kgCO2e)



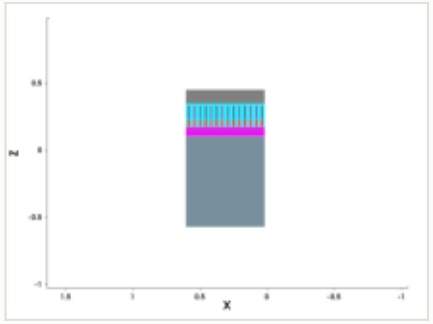
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



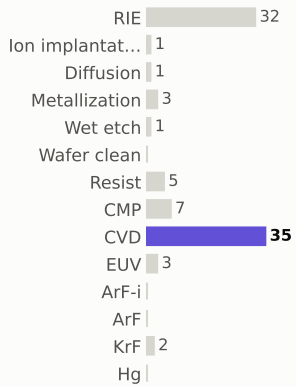
y cut at 0.184 µm



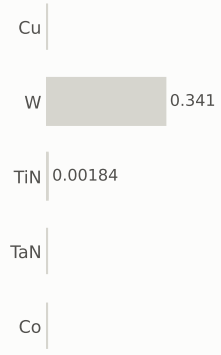
mask for this step

no mask for this step

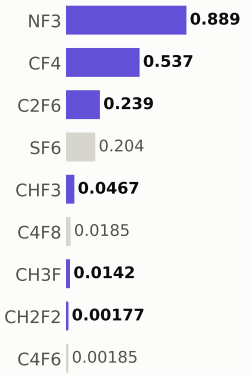
steps by process area



materials deposited (g)



gases emitted (g)



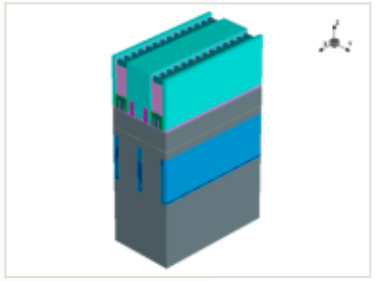
Step #1020: Resist

etch_strip · Resist

0.169	145.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

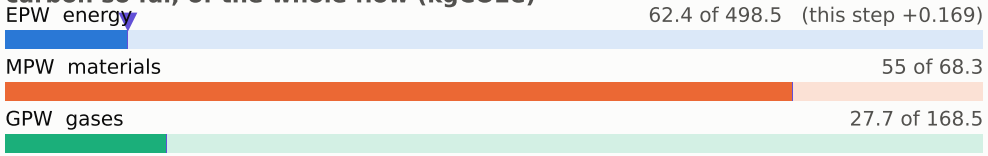


this step

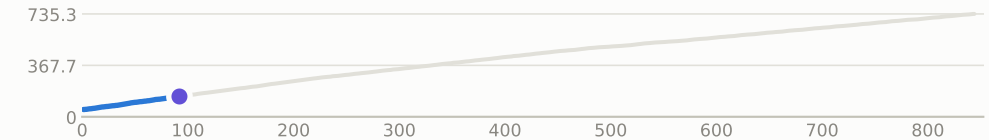
process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

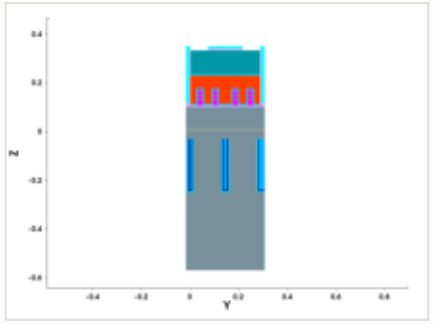
carbon so far, of the whole flow (kgCO2e)



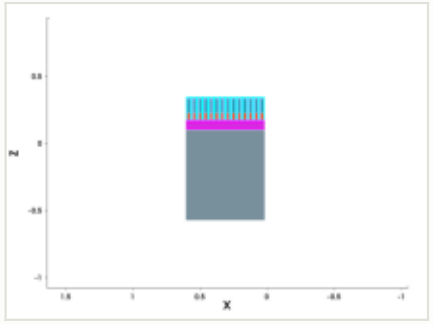
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



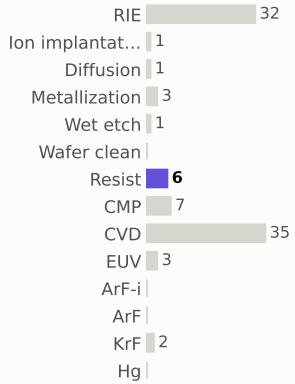
y cut at 0.184 µm



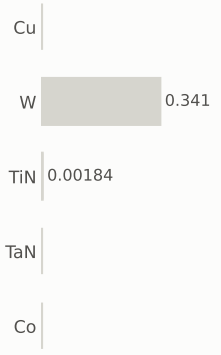
mask for this step

no mask for this step

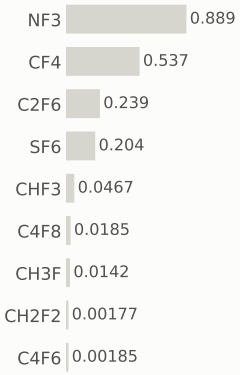
steps by process area



materials deposited (g)



gases emitted (g)



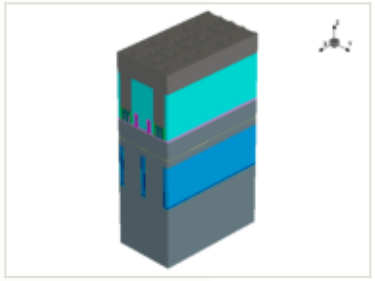
Step #1030: EUV

litho_patterning · EUV

1.96	147.0	0.27%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

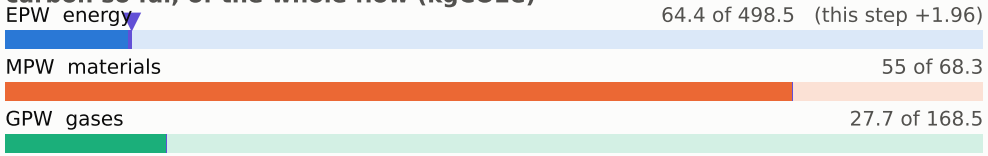


this step

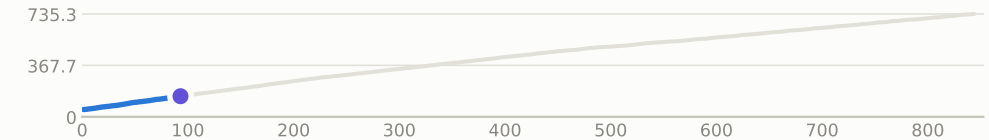
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

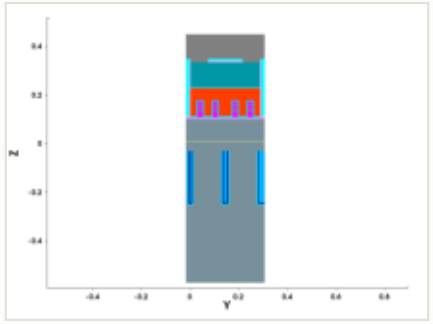
carbon so far, of the whole flow (kgCO2e)



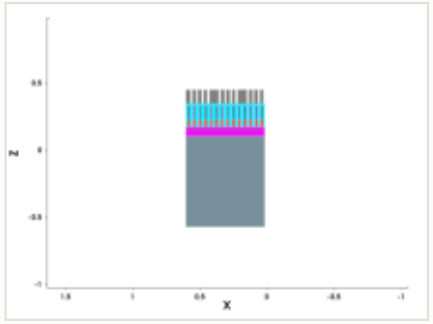
embodied carbon up to each step (kgCO2e per wafer)



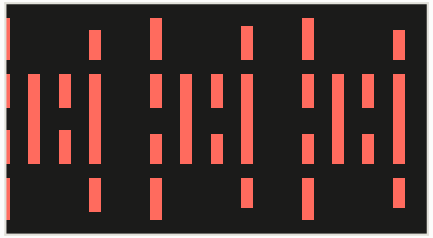
x cut at 0.294 µm



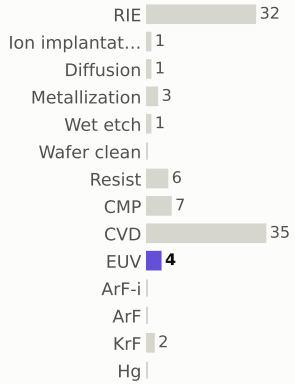
y cut at 0.184 µm



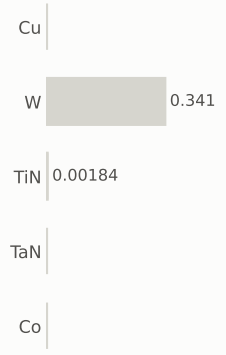
mask: layer 10:0 (GDSII)



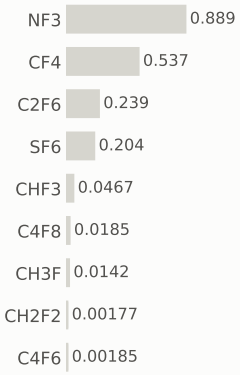
steps by process area



materials deposited (g)



gases emitted (g)



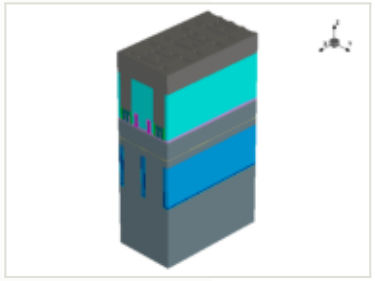
Step #1040: CVD

depo_anisotropic · CVD

1.31	148.4	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

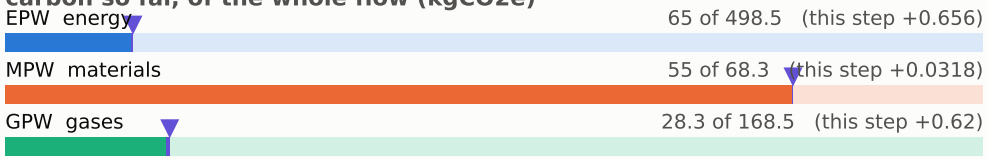


this step

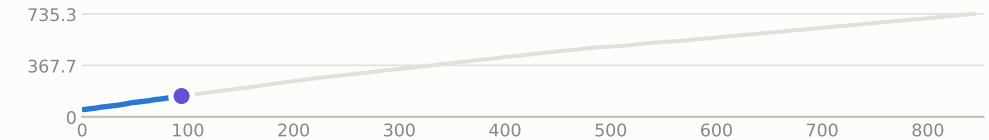
process area	CVD
material	TiN
thickness	0.005 µm
material GWP	43.9778 gCO2e/g
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

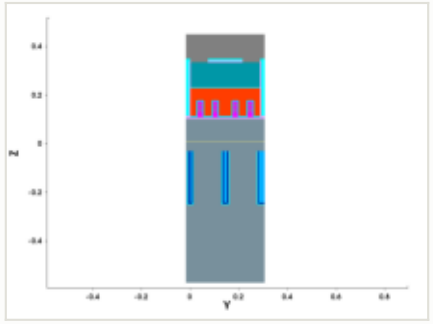
carbon so far, of the whole flow (kgCO2e)



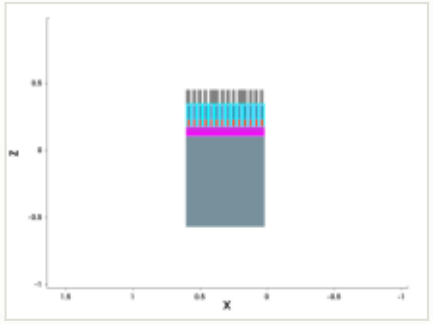
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



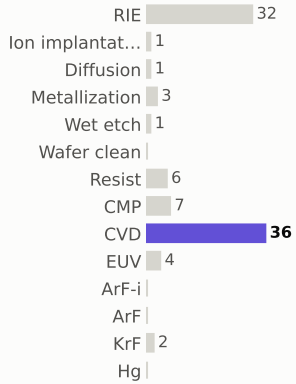
y cut at 0.184 µm



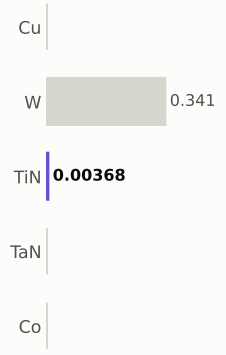
mask for this step

no mask for this step

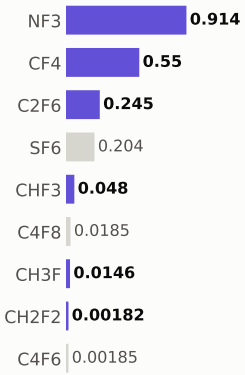
steps by process area



materials deposited (g)



gases emitted (g)



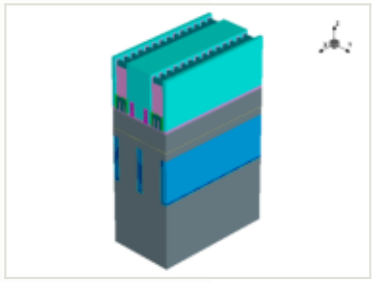
Step #1050: Resist

etch_strip · Resist

0.169	148.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

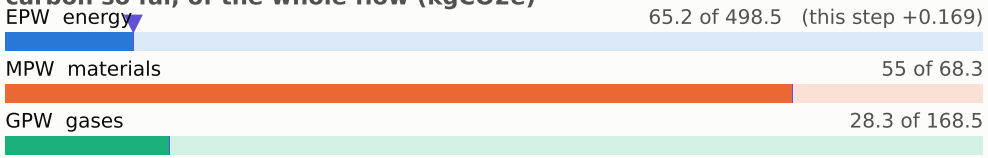


this step

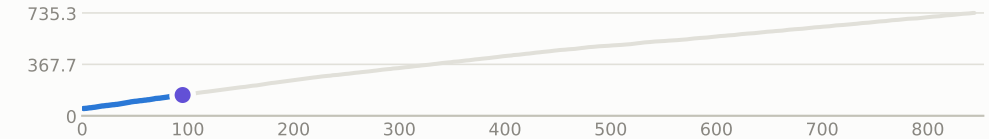
process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

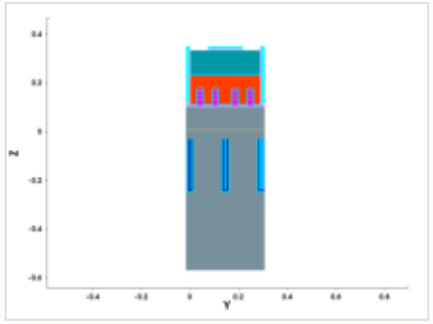
carbon so far, of the whole flow (kgCO2e)



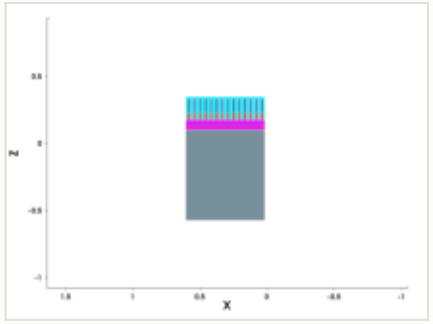
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



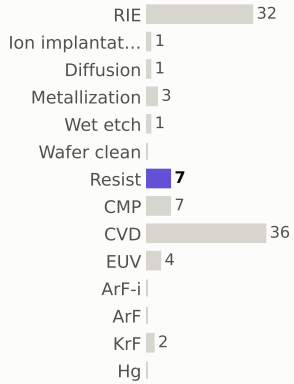
y cut at 0.184 µm



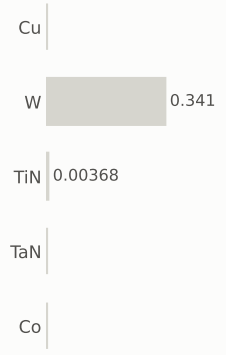
mask for this step

no mask for this step

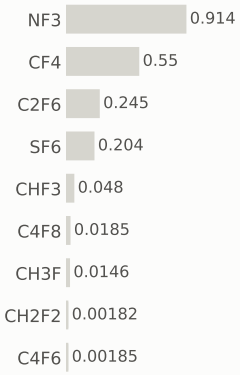
steps by process area



materials deposited (g)



gases emitted (g)



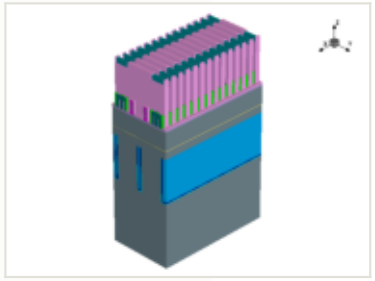
Step #1060: RIE

etch_anisotropic · RIE

1.04	149.6	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

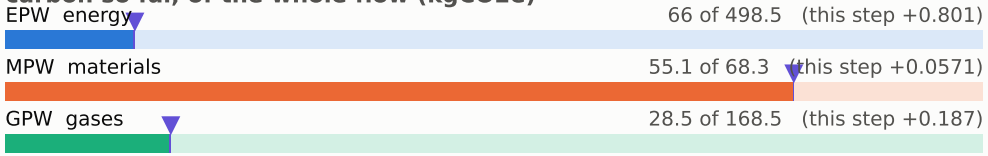


this step

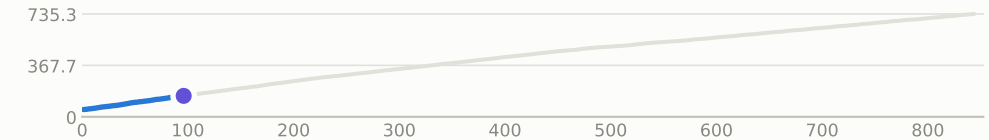
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

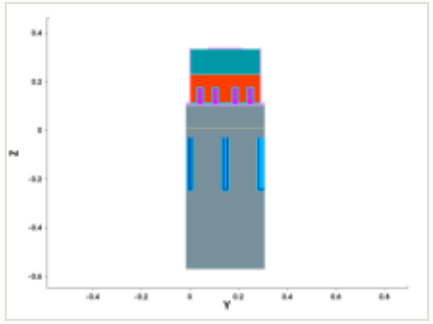
carbon so far, of the whole flow (kgCO2e)



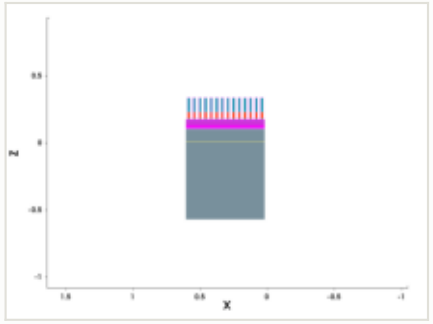
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



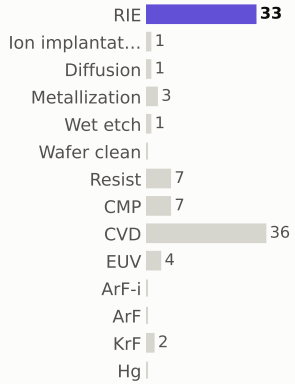
y cut at 0.184 µm



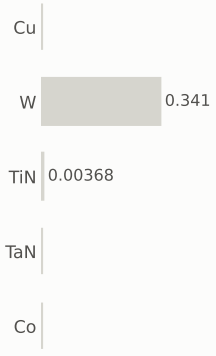
mask for this step

no mask for this step

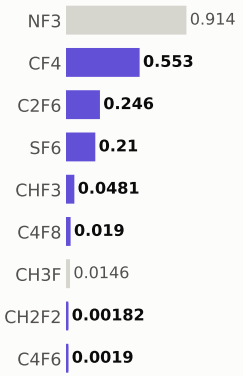
steps by process area



materials deposited (g)



gases emitted (g)



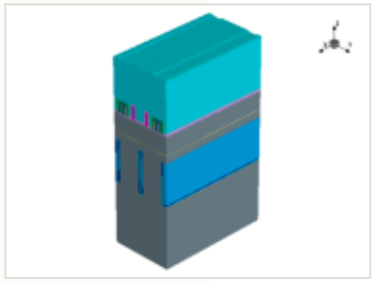
Step #1070: CVD

depo_isotropic · CVD

1.31	150.9	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

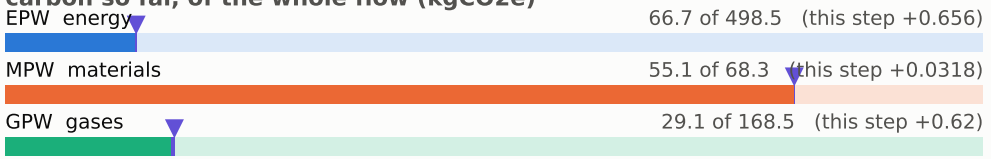


this step

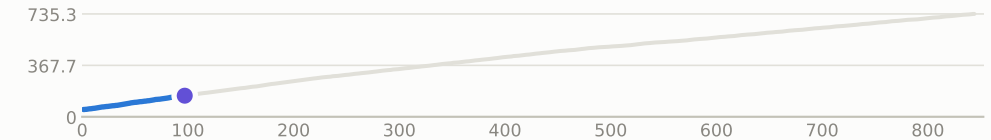
process area	CVD
material	—
thickness	0.035 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

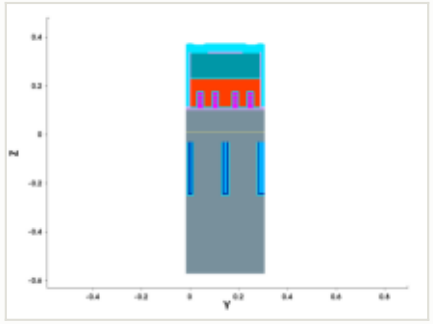
carbon so far, of the whole flow (kgCO2e)



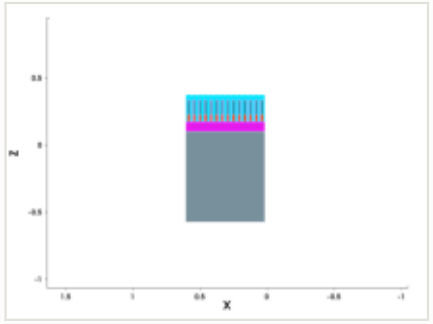
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



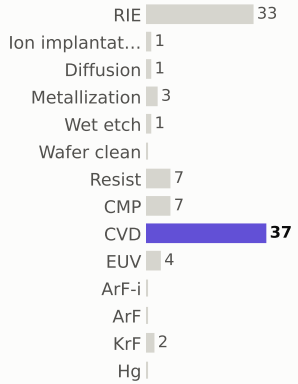
y cut at 0.184 µm



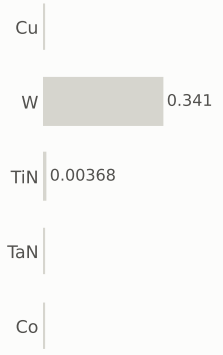
mask for this step

no mask for this step

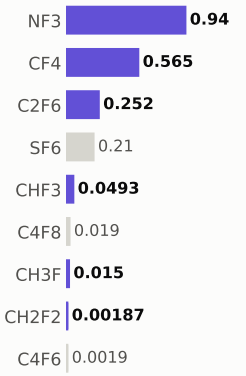
steps by process area



materials deposited (g)



gases emitted (g)



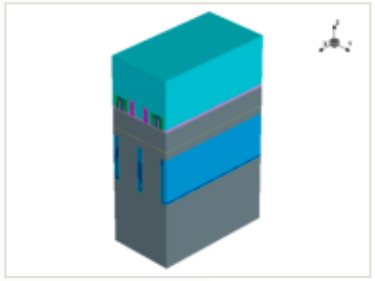
Step #1080: CMP

etch_cmp · CMP

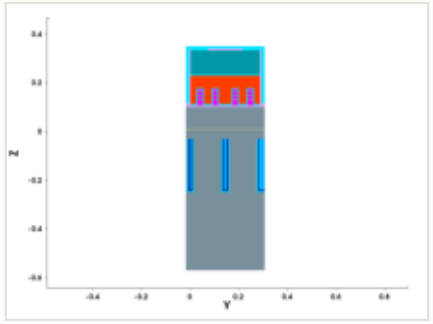
0.169	151.0	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

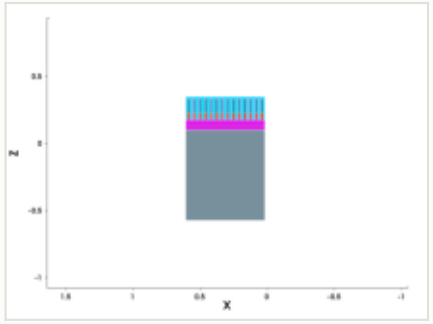
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



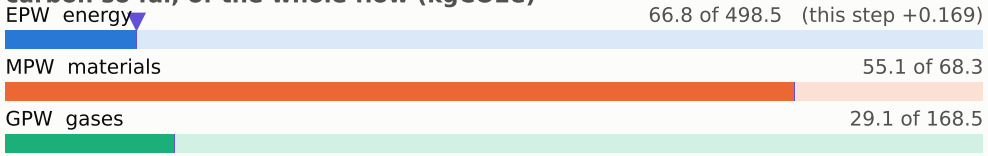
mask for this step

no mask for this step

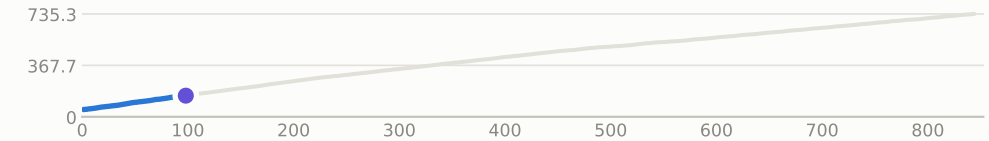
this step

process area	CMP	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

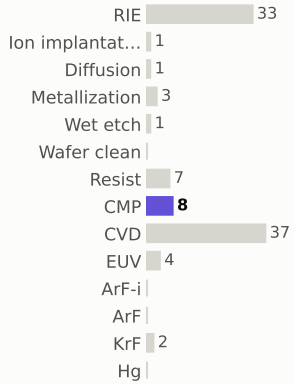
carbon so far, of the whole flow (kgCO2e)



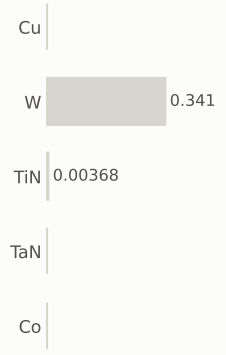
embodied carbon up to each step (kgCO2e per wafer)



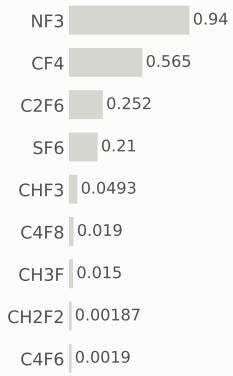
steps by process area



materials deposited (g)



gases emitted (g)



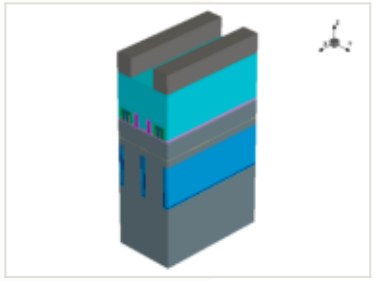
Step #1100: EUV

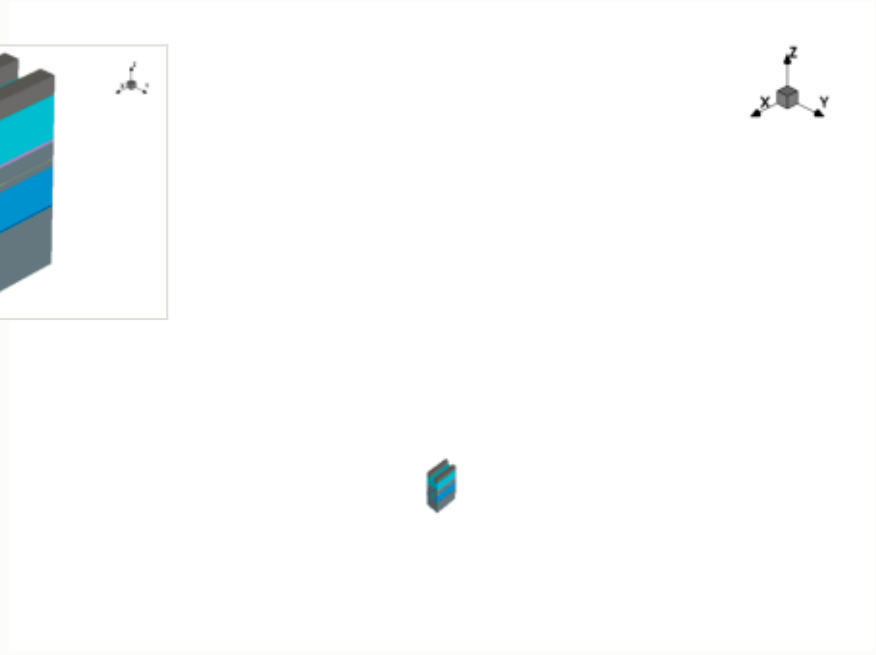
litho_patterning · EUV

1.96	153.0	0.27%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



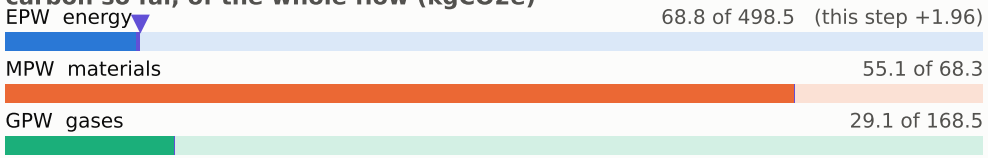


this step

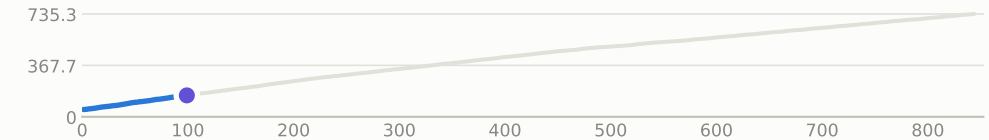
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

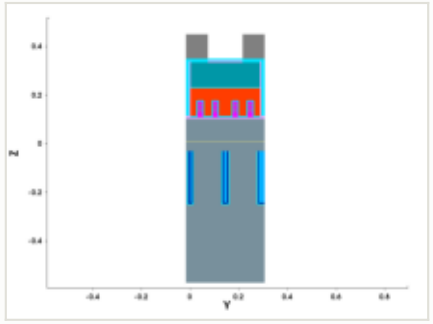
carbon so far, of the whole flow (kgCO2e)



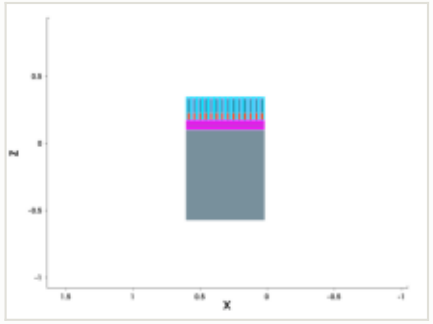
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm

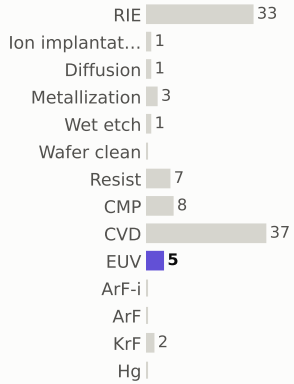


y cut at 0.184 µm

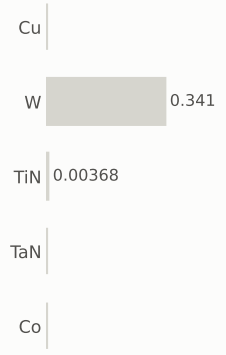


mask: layer 7:0 (GDSII)

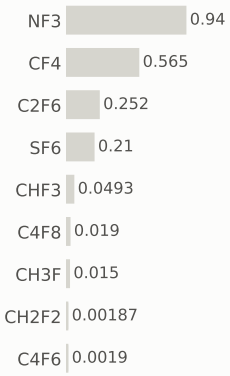
steps by process area



materials deposited (g)



gases emitted (g)



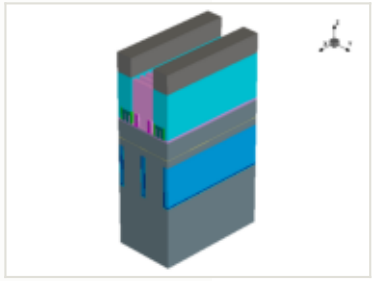
Step #1110: RIE

etch_anisotropic · RIE

1.04	154.0	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

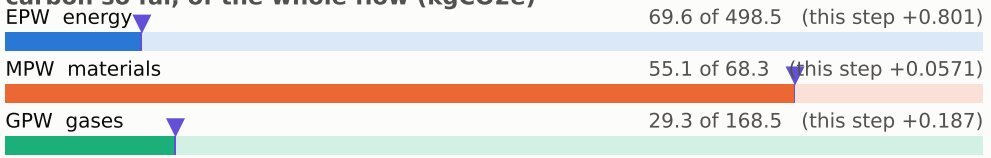


this step

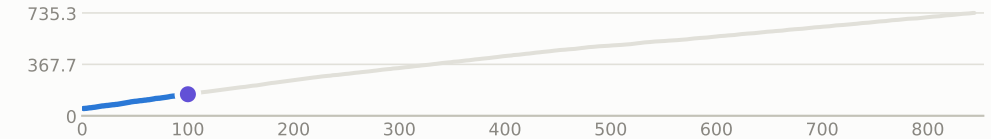
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

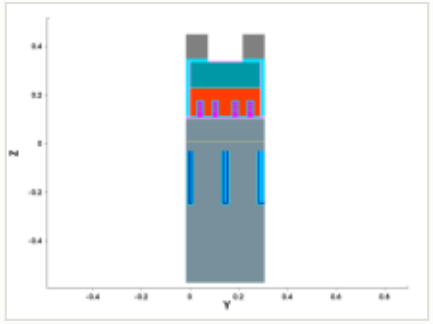
carbon so far, of the whole flow (kgCO2e)



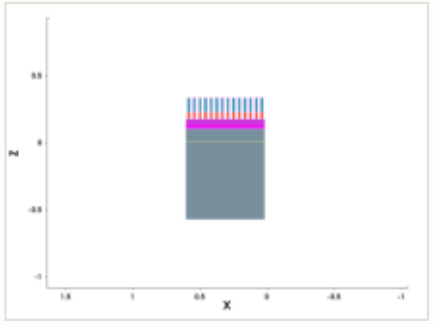
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



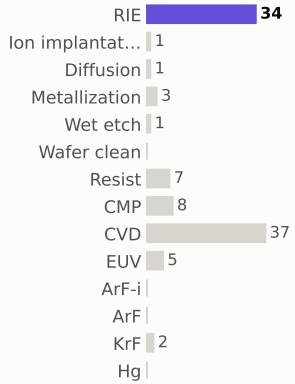
y cut at 0.184 µm



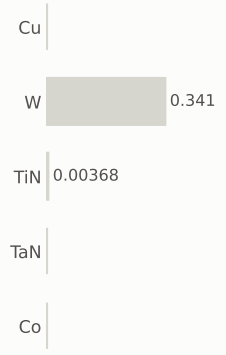
mask for this step

no mask for this step

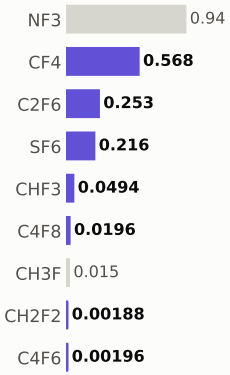
steps by process area



materials deposited (g)



gases emitted (g)



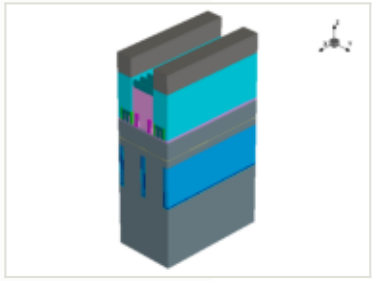
Step #1120: RIE

etch_anisotropic · RIE

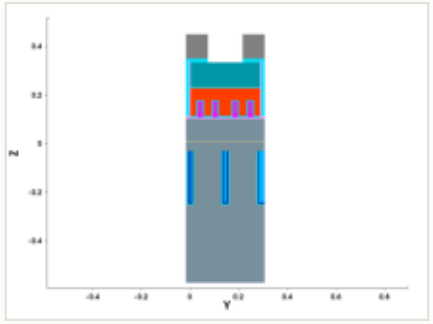
1.04	155.1	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

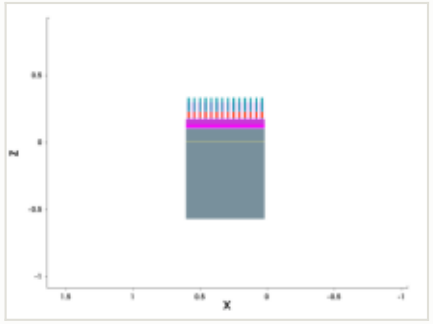
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

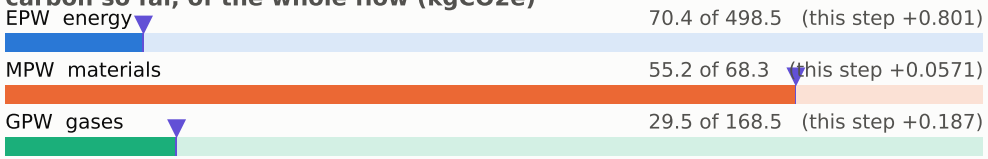
no mask for this step

this step

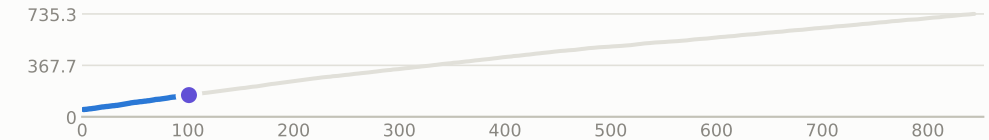
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

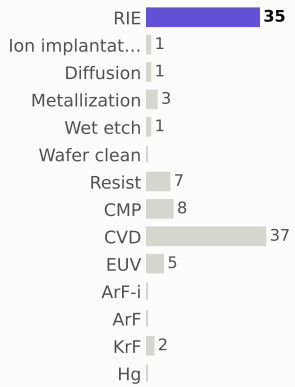
carbon so far, of the whole flow (kgCO2e)



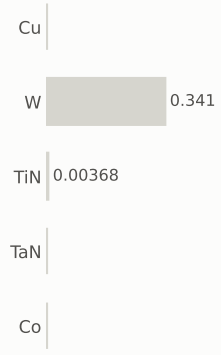
embodied carbon up to each step (kgCO2e per wafer)



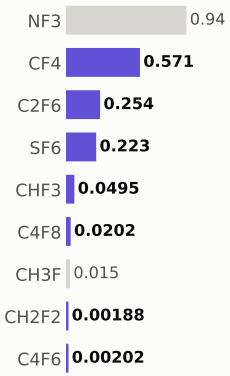
steps by process area



materials deposited (g)



gases emitted (g)



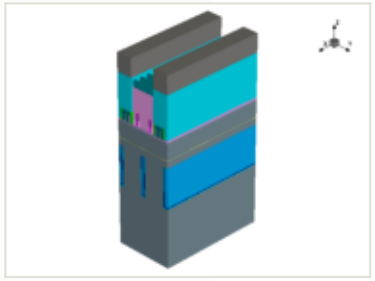
Step #1130: RIE

etch_anisotropic · RIE

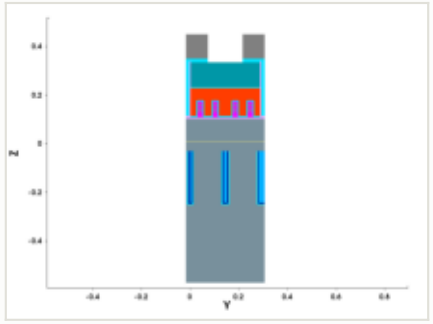
1.04	156.1	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

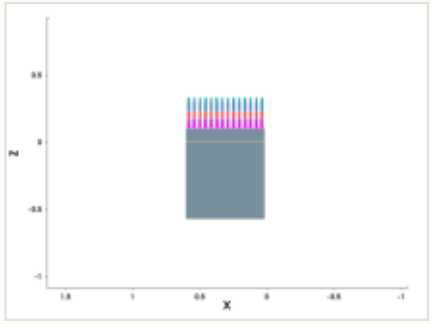
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

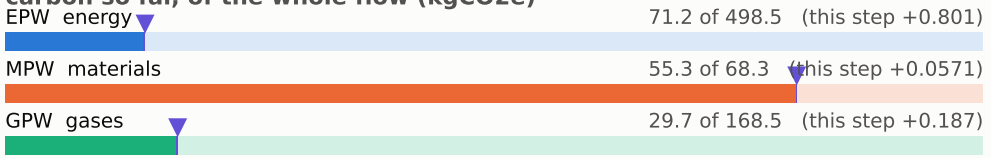
no mask for this step

this step

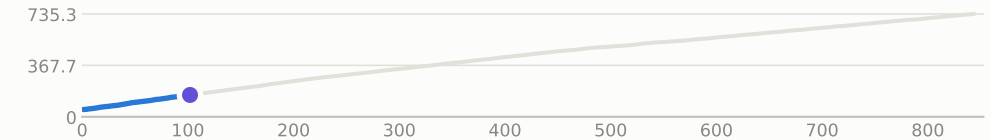
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

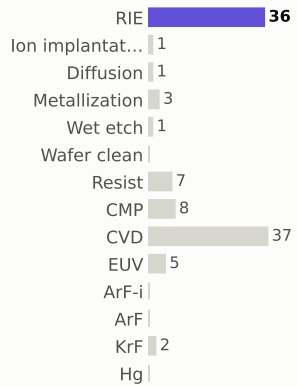
carbon so far, of the whole flow (kgCO2e)



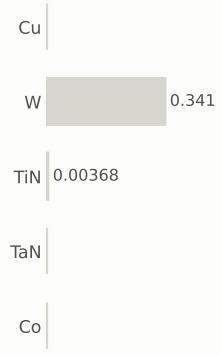
embodied carbon up to each step (kgCO2e per wafer)



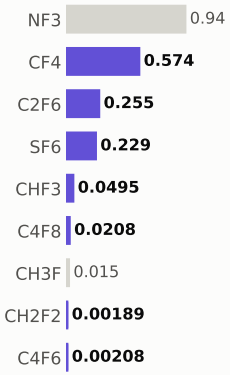
steps by process area



materials deposited (g)



gases emitted (g)



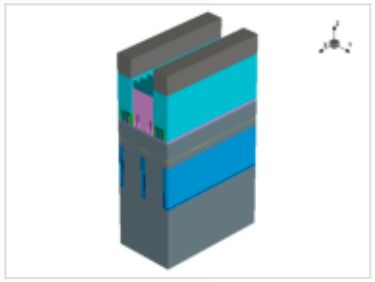
Step #1140: Wet_etch

etch_isotropic · Wet_etch

0.169	156.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

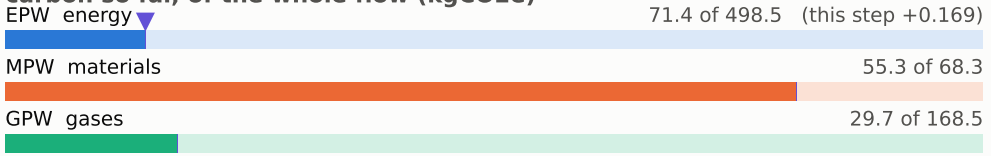


this step

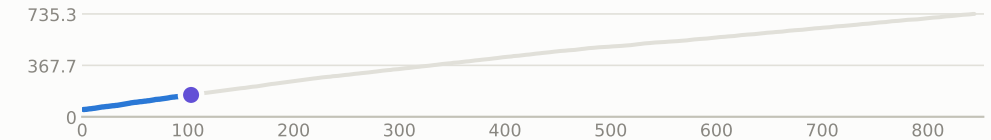
process area	Wet_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

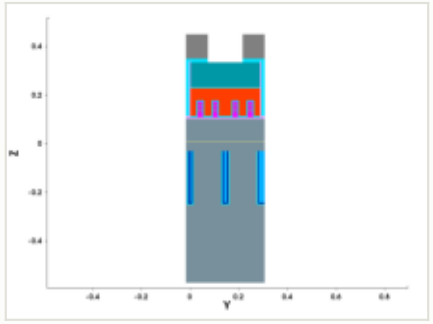
carbon so far, of the whole flow (kgCO2e)



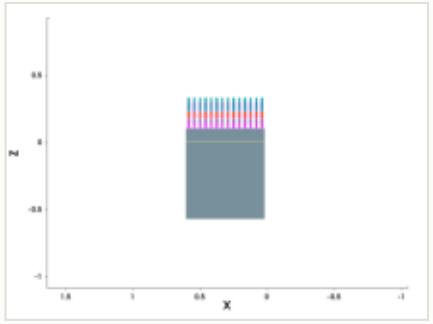
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



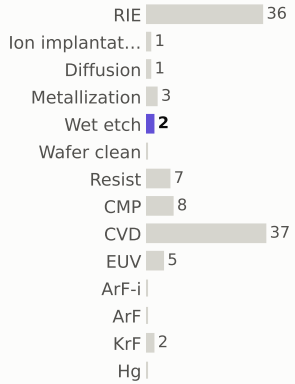
y cut at 0.184 µm



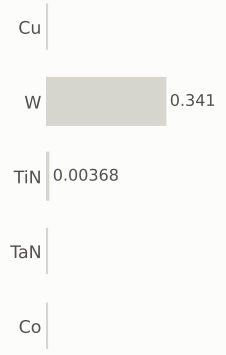
mask for this step

no mask for this step

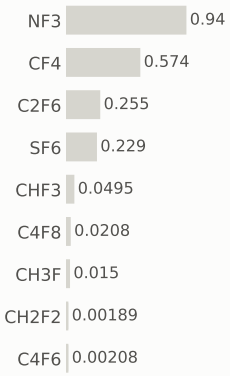
steps by process area



materials deposited (g)



gases emitted (g)



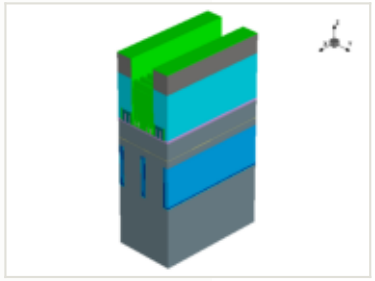
Step #1150: CVD

depo_isotropic · CVD

1.31	157.6	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

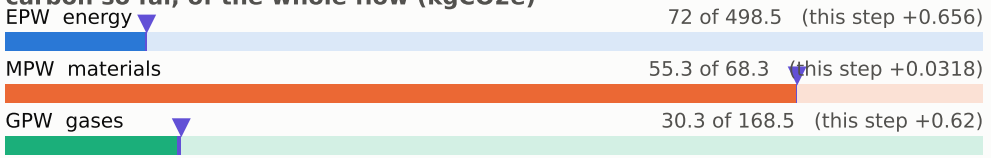


this step

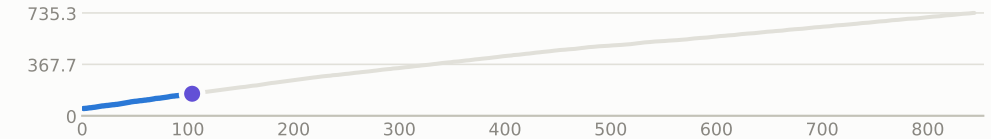
process area	CVD
material	—
thickness	0.0035 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

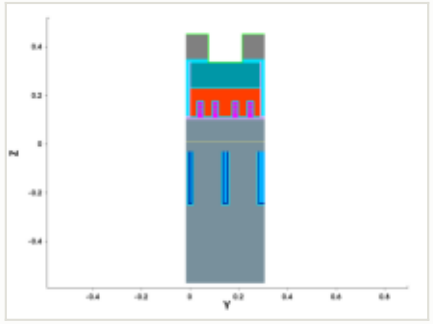
carbon so far, of the whole flow (kgCO2e)



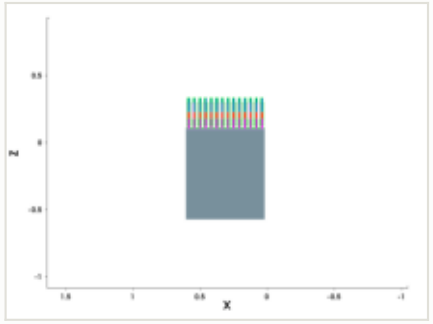
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



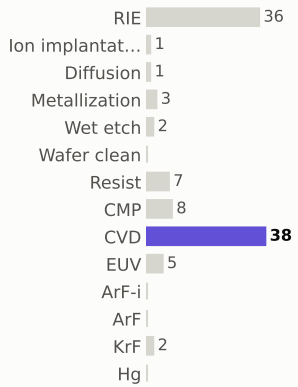
y cut at 0.184 µm



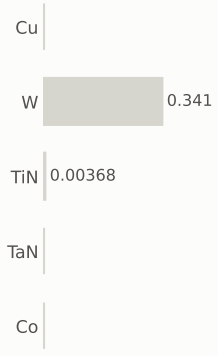
mask for this step

no mask for this step

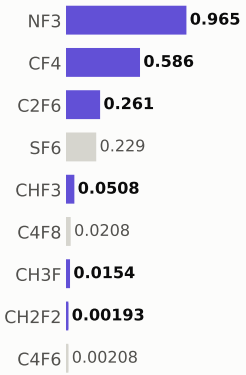
steps by process area



materials deposited (g)



gases emitted (g)



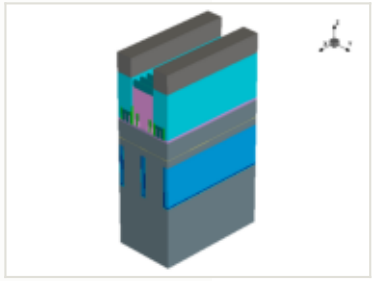
Step #1160: RIE

etch_anisotropic · RIE

1.04	158.7	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

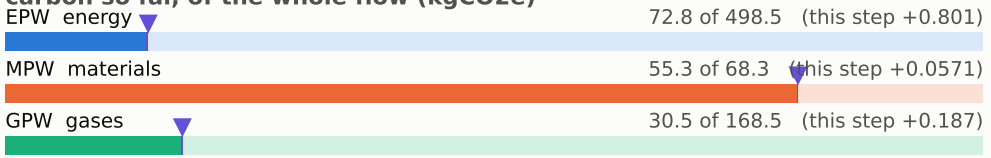


this step

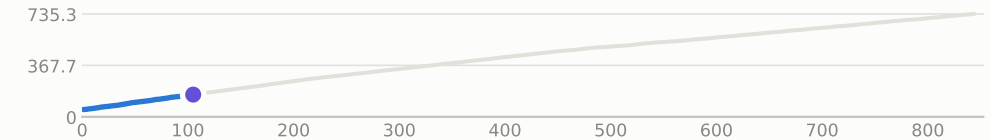
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

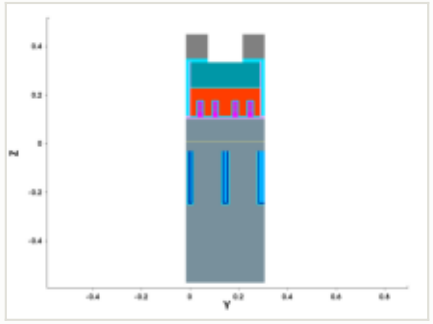
carbon so far, of the whole flow (kgCO2e)



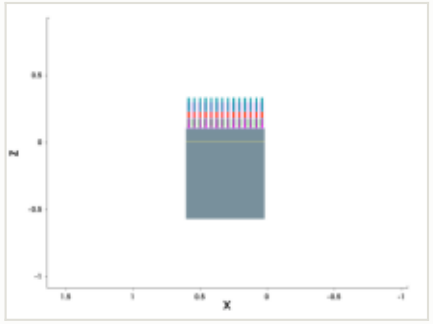
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



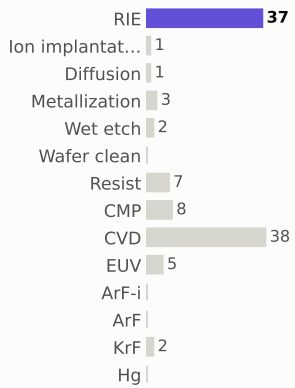
y cut at 0.184 µm



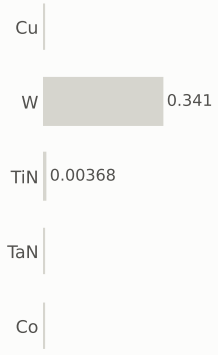
mask for this step

no mask for this step

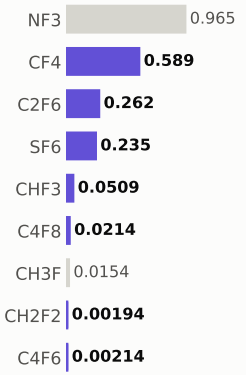
steps by process area



materials deposited (g)



gases emitted (g)



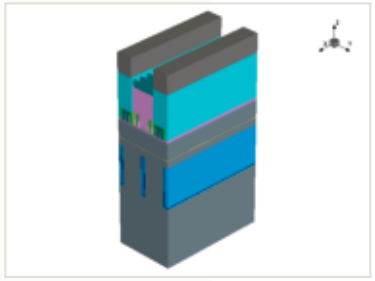
Step #1170: CMP

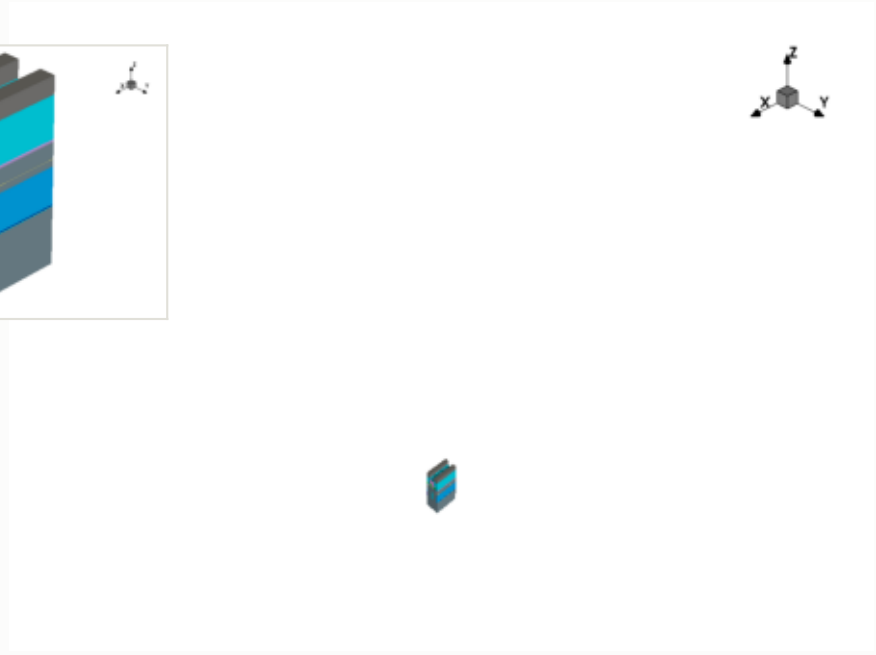
etch_cmp · CMP

0.169	158.8	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



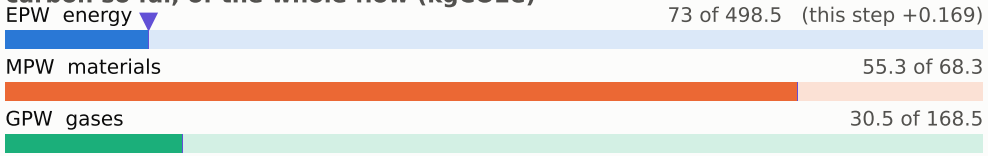


this step

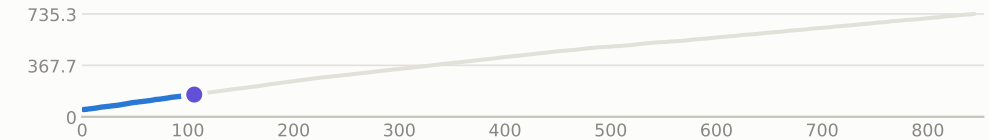
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

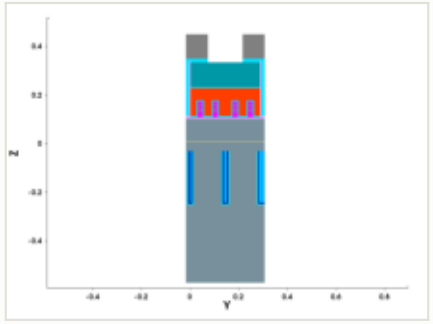
carbon so far, of the whole flow (kgCO2e)



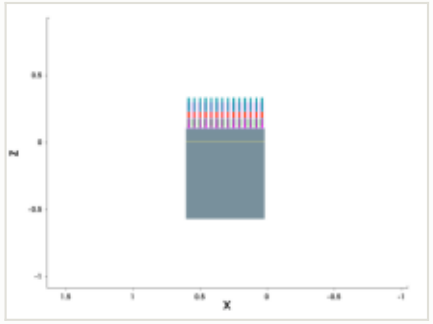
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



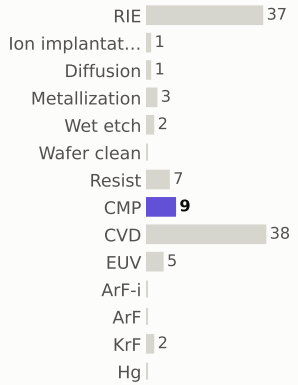
y cut at 0.184 µm



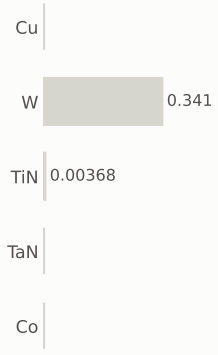
mask for this step

no mask for this step

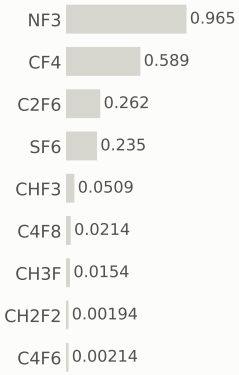
steps by process area



materials deposited (g)



gases emitted (g)



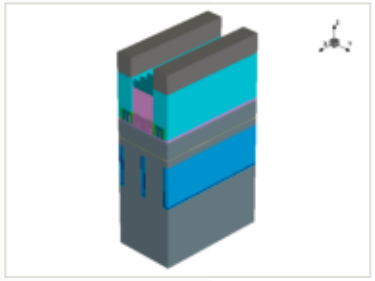
Step #1180: CVD

depo_fill · CVD

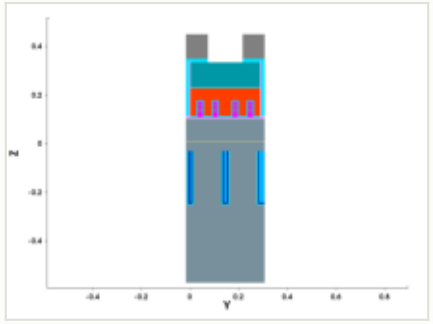
1.31	160.1	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

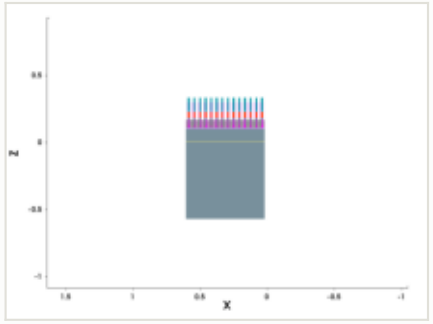
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

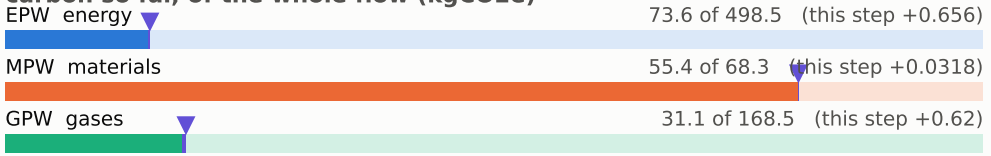
no mask for this step

this step

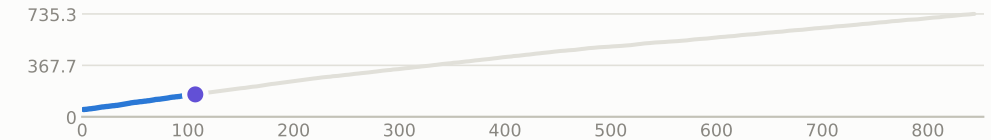
process area	CVD
material	—
thickness	0.005 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

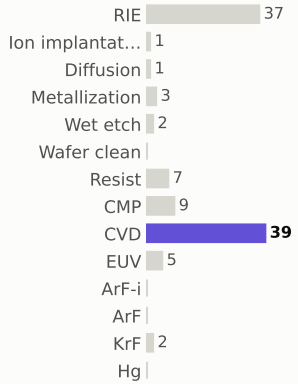
carbon so far, of the whole flow (kgCO2e)



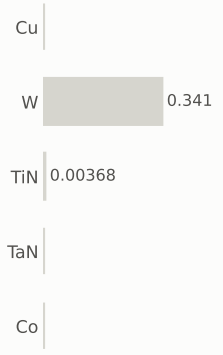
embodied carbon up to each step (kgCO2e per wafer)



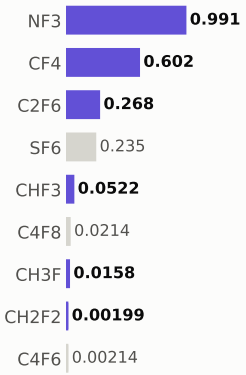
steps by process area



materials deposited (g)



gases emitted (g)



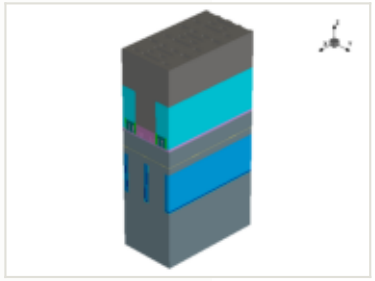
Step #1190: EUV

litho_patterning · EUV

1.96	162.1	0.27%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

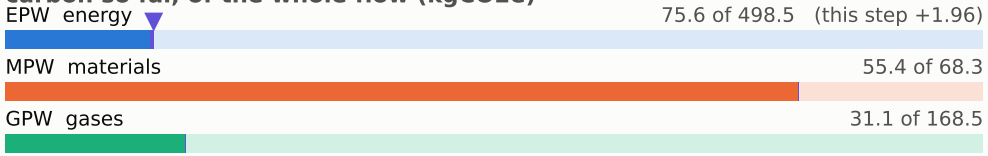


this step

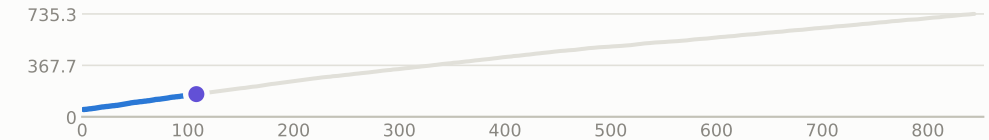
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

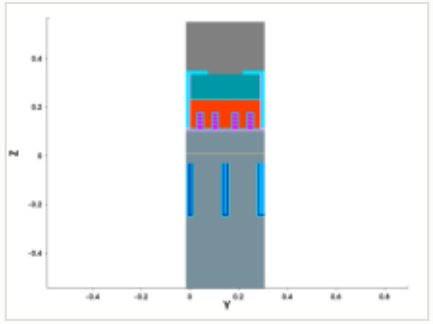
carbon so far, of the whole flow (kgCO2e)



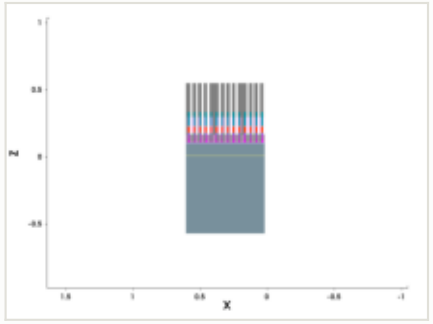
embodied carbon up to each step (kgCO2e per wafer)



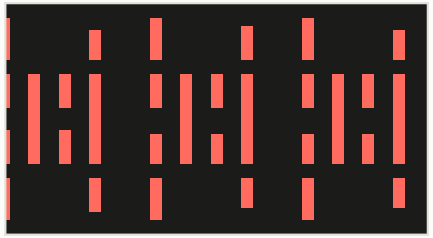
x cut at 0.294 µm



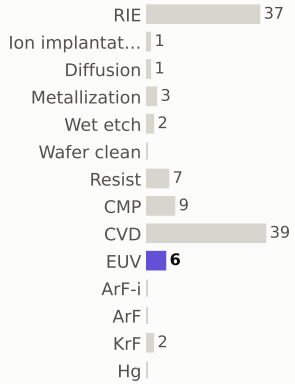
y cut at 0.184 µm



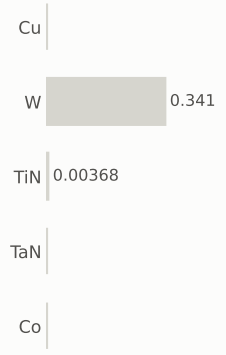
mask: layer 10:0 (GDSII)



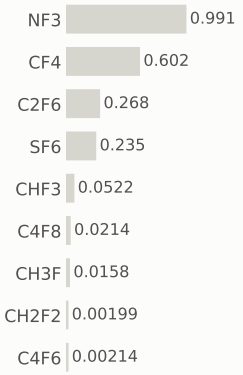
steps by process area



materials deposited (g)



gases emitted (g)



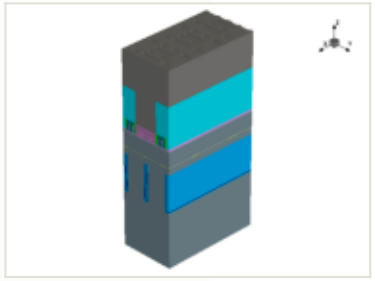
Step #1200: CVD

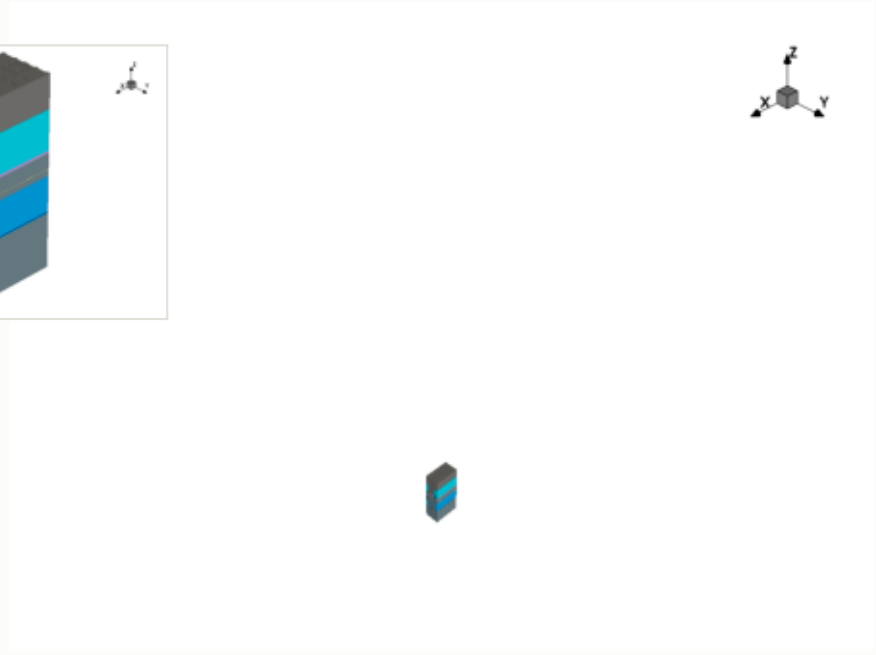
depo_anisotropic · CVD

1.31	163.4	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



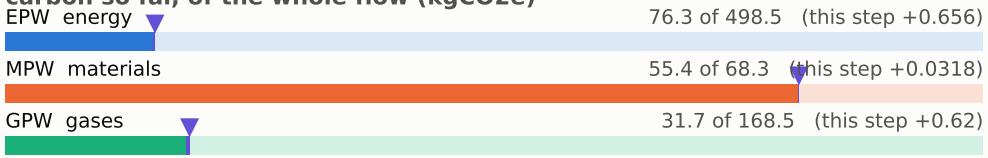


this step

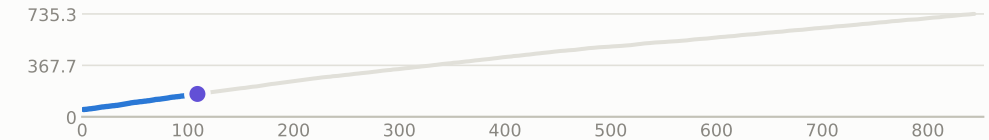
process area	CVD
material	TiN
thickness	0.005 µm
material GWP	43.9778 gCO2e/g
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

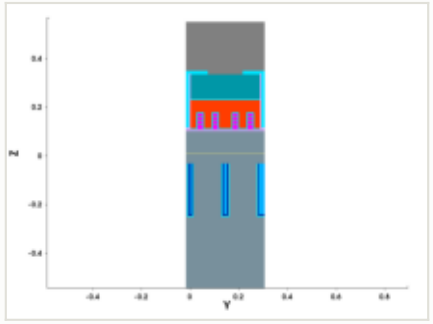
carbon so far, of the whole flow (kgCO2e)



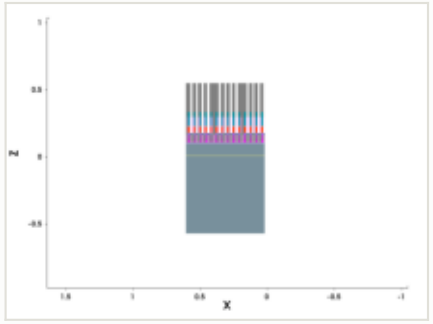
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



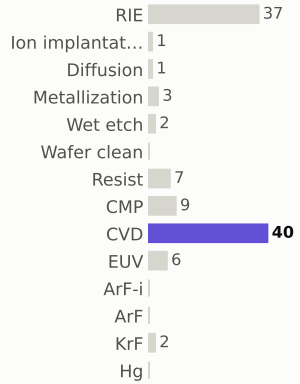
y cut at 0.184 µm



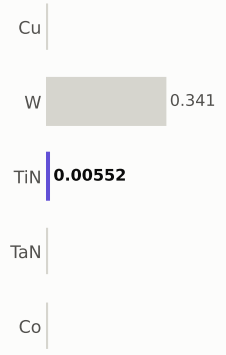
mask for this step

no mask for this step

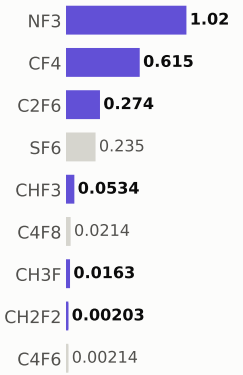
steps by process area



materials deposited (g)



gases emitted (g)



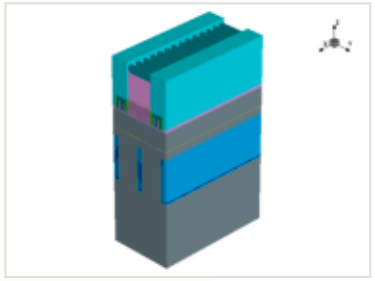
Step #1210: Resist

etch_strip · Resist

0.169	163.6	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

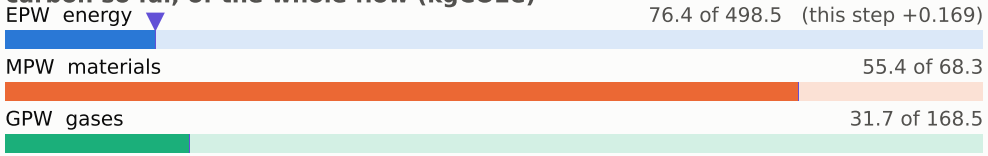


this step

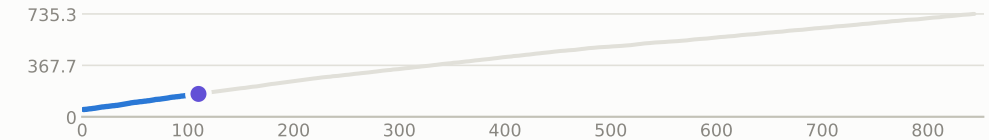
process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

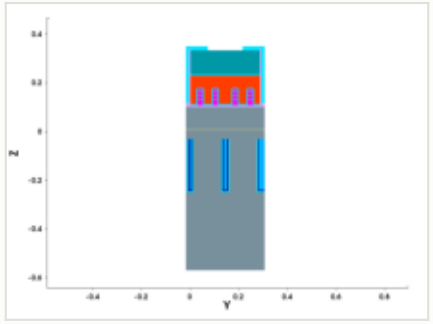
carbon so far, of the whole flow (kgCO2e)



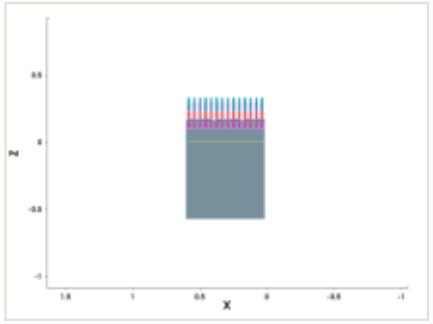
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



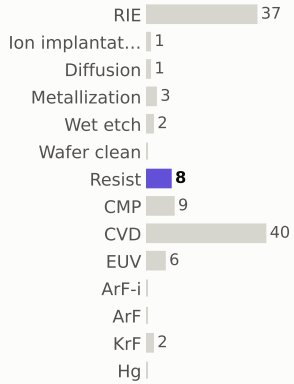
y cut at 0.184 µm



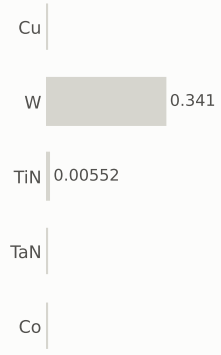
mask for this step

no mask for this step

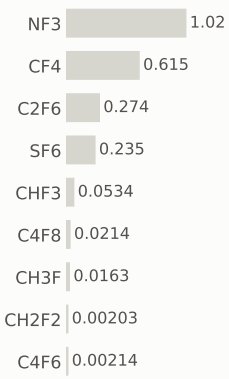
steps by process area



materials deposited (g)



gases emitted (g)



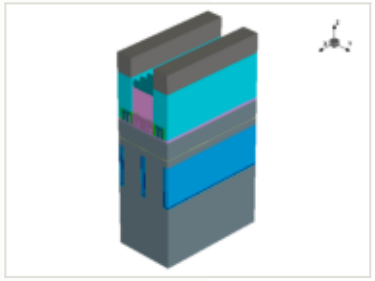
Step #1220: EUV

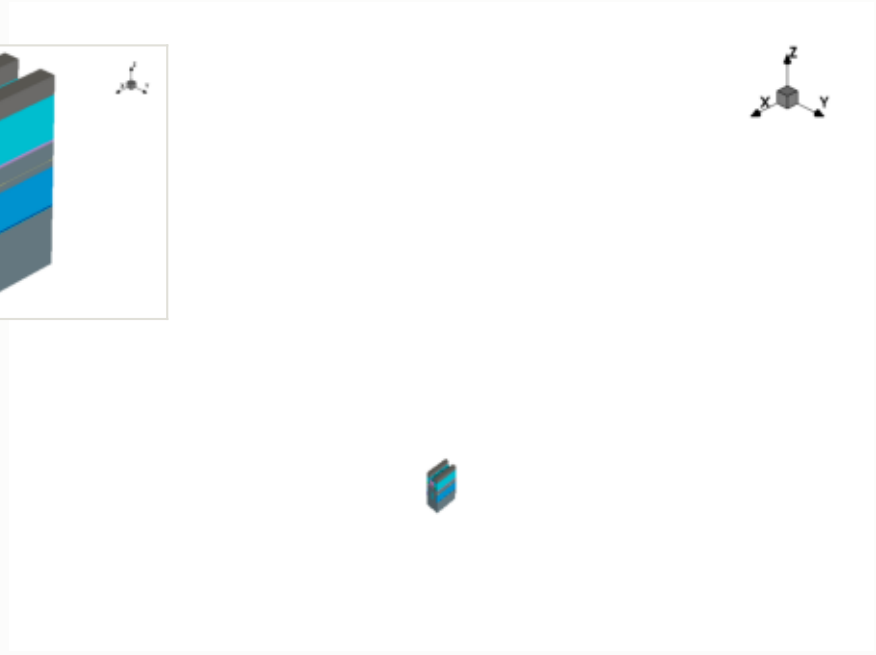
litho_patterning · EUV

1.96	165.5	0.27%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



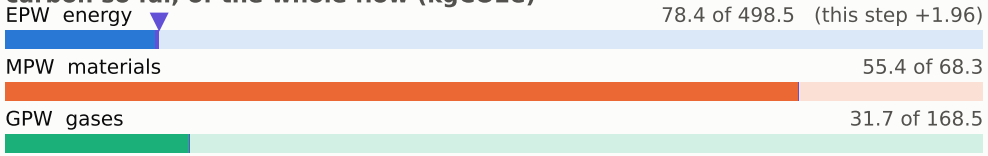


this step

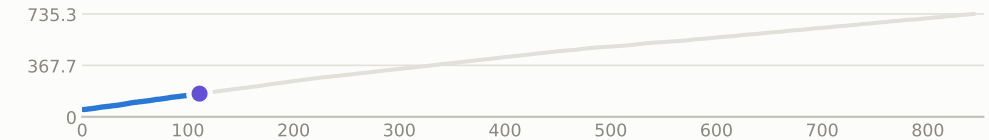
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

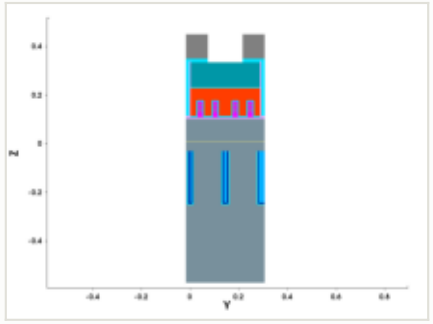
carbon so far, of the whole flow (kgCO2e)



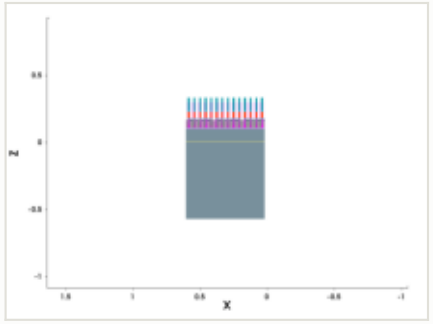
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm

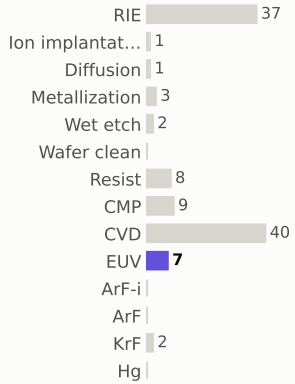


y cut at 0.184 µm

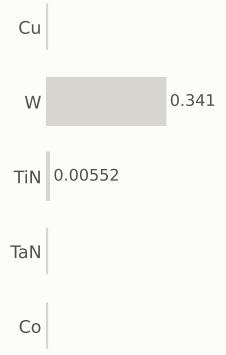


mask: layer 7:0 (GDSII)

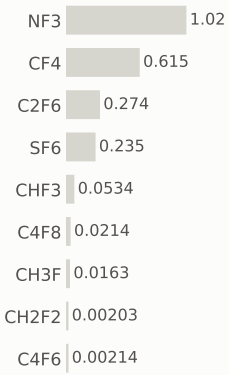
steps by process area



materials deposited (g)



gases emitted (g)



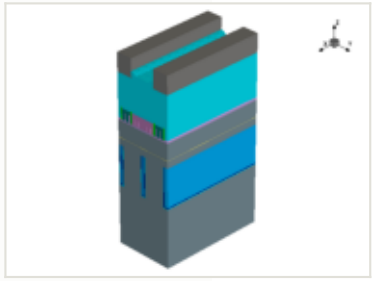
Step #1230: CVD

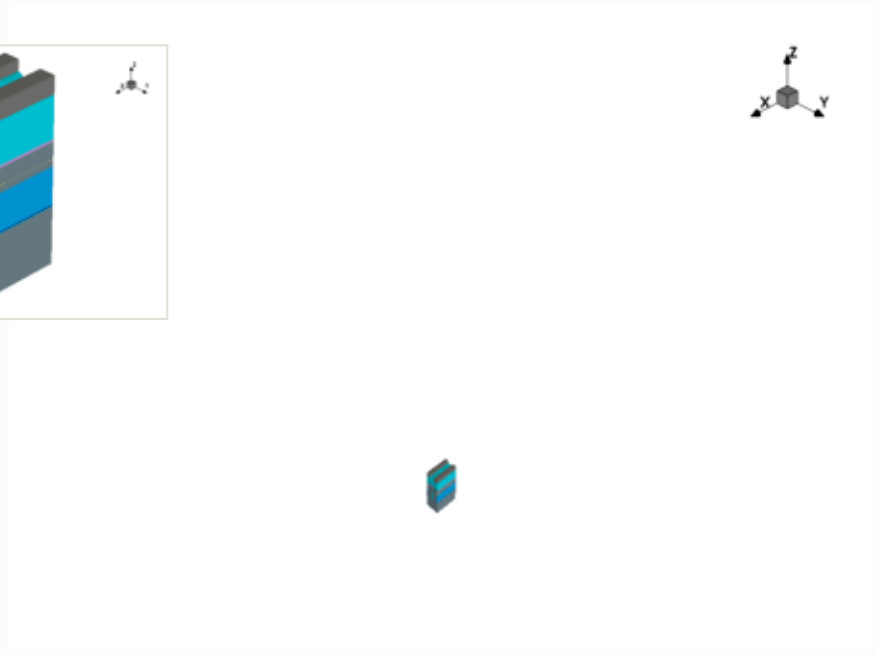
depo_isotropic · CVD

1.31	166.8	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



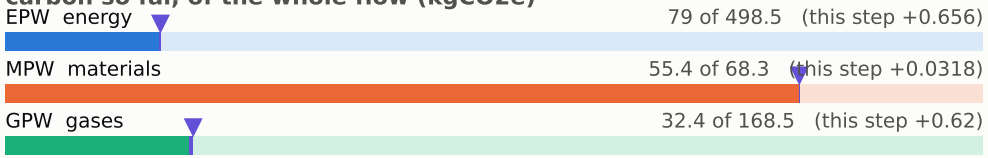


this step

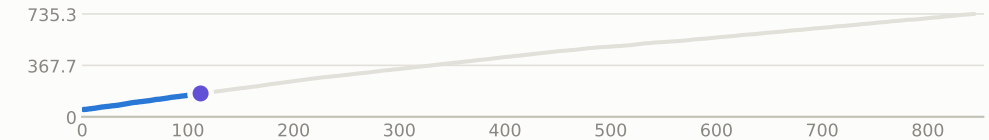
process area	CVD
material	—
thickness	0.035 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

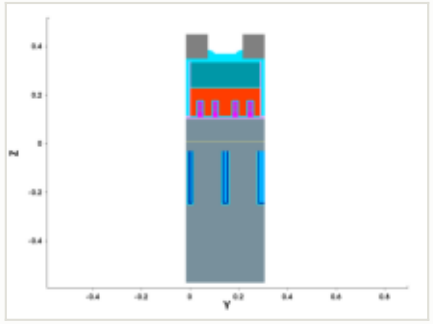
carbon so far, of the whole flow (kgCO2e)



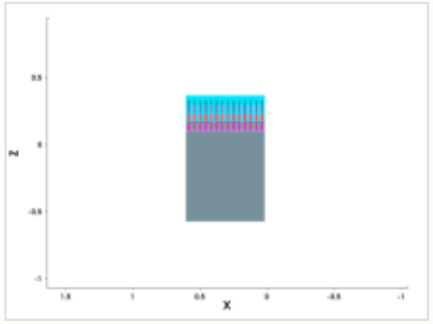
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



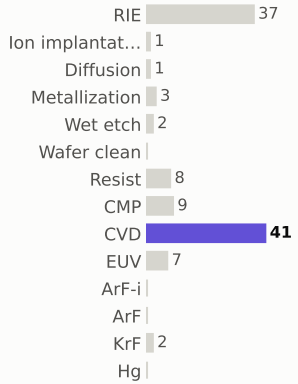
y cut at 0.184 µm



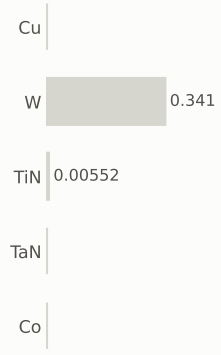
mask for this step

no mask for this step

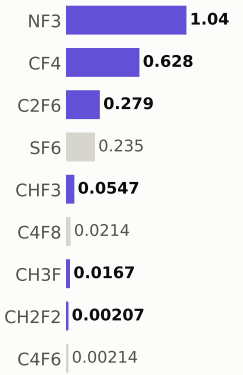
steps by process area



materials deposited (g)



gases emitted (g)



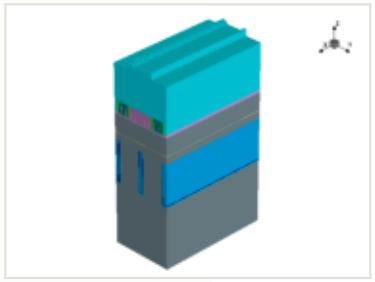
Step #1240: Resist

etch_strip · Resist

0.169	167.0	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

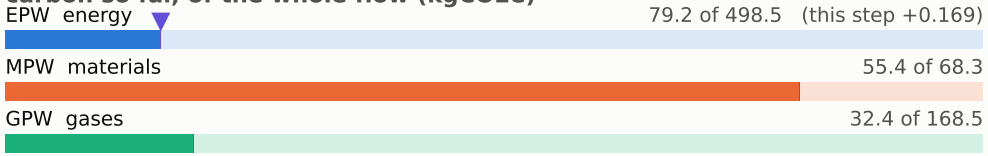


this step

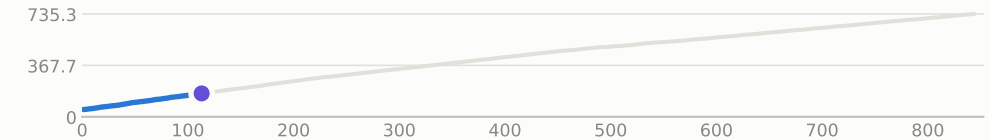
process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

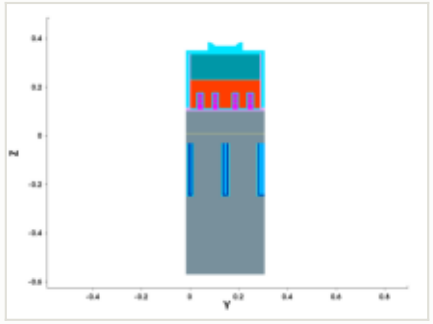
carbon so far, of the whole flow (kgCO2e)



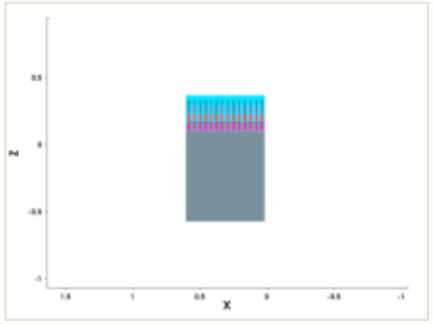
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



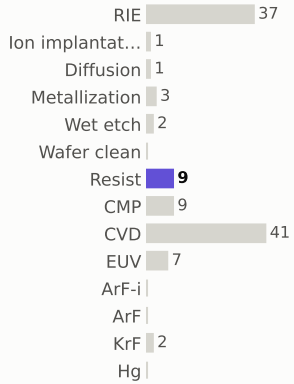
y cut at 0.184 µm



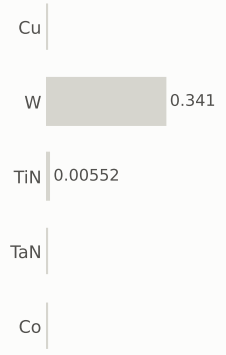
mask for this step

no mask for this step

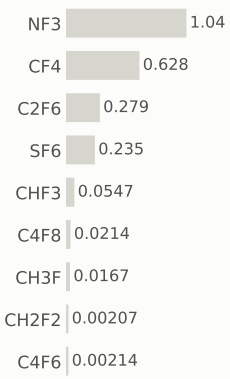
steps by process area



materials deposited (g)



gases emitted (g)



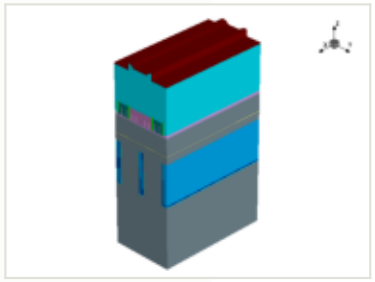
Step #1260: CVD

depo_isotropic · CVD

1.31	168.3	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

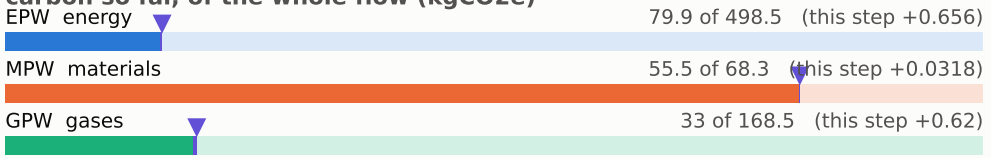


this step

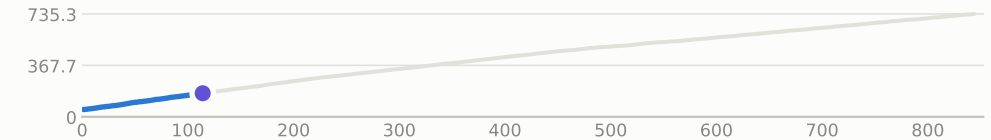
process area	CVD
material	—
thickness	0.006 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

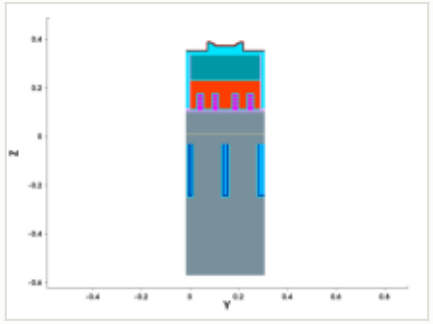
carbon so far, of the whole flow (kgCO2e)



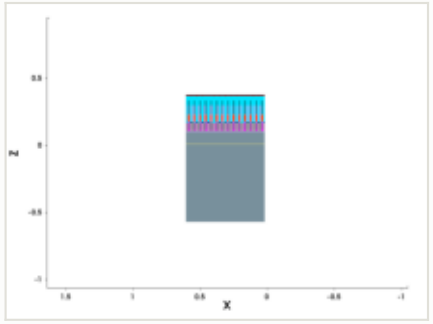
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



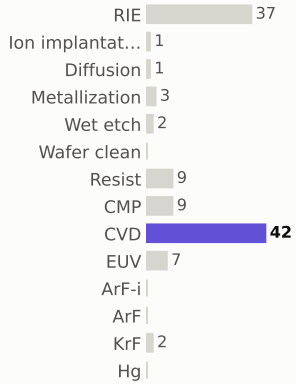
y cut at 0.184 µm



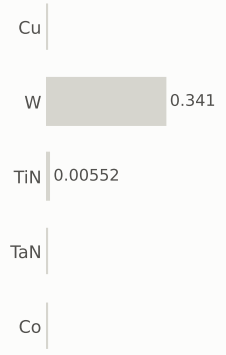
mask for this step

no mask for this step

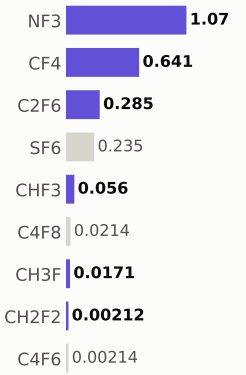
steps by process area



materials deposited (g)



gases emitted (g)



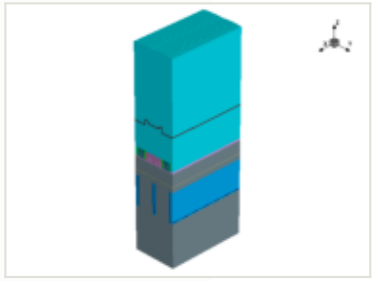
Step #1270: CVD

depo_isotropic · CVD

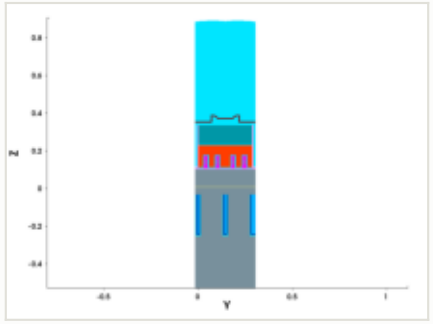
1.31	169.6	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

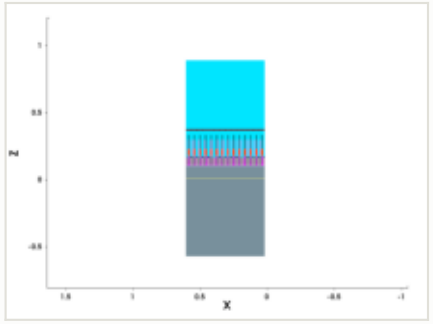
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

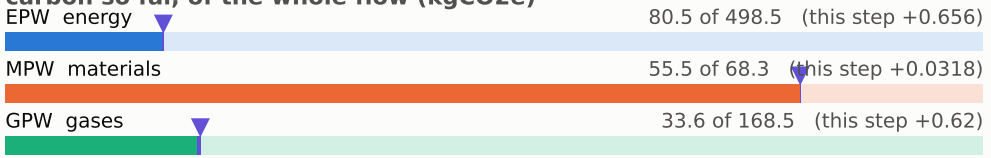
no mask for this step

this step

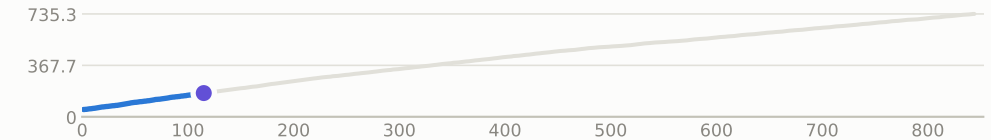
process area	CVD
material	—
thickness	0.5 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

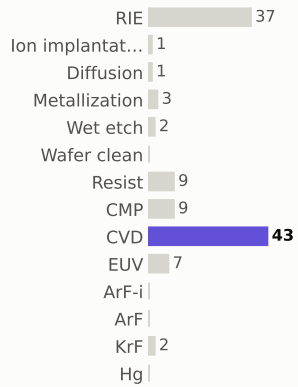
carbon so far, of the whole flow (kgCO2e)



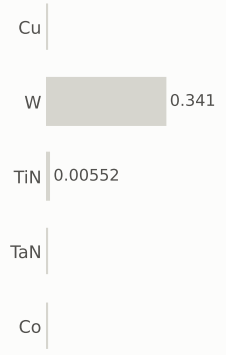
embodied carbon up to each step (kgCO2e per wafer)



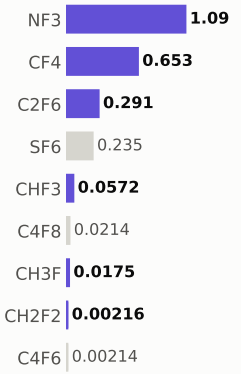
steps by process area



materials deposited (g)



gases emitted (g)



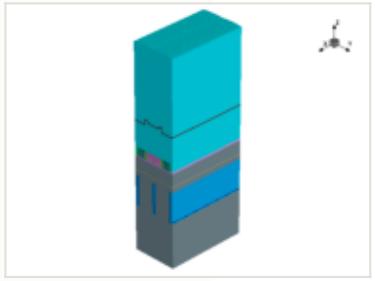
Step #1280: CMP

etch_cmp · CMP

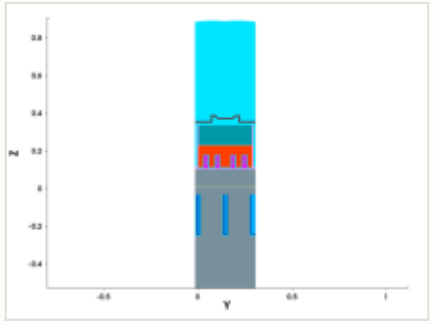
0.169	169.8	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

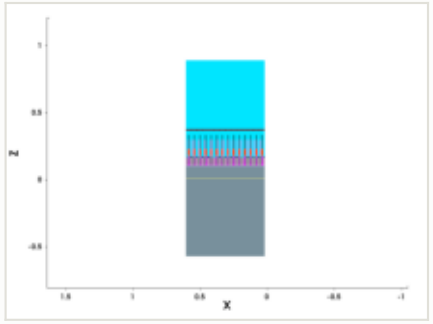
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

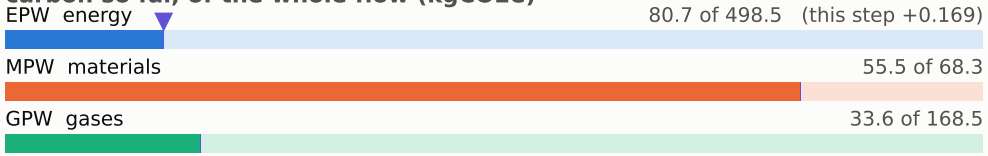
no mask for this step

this step

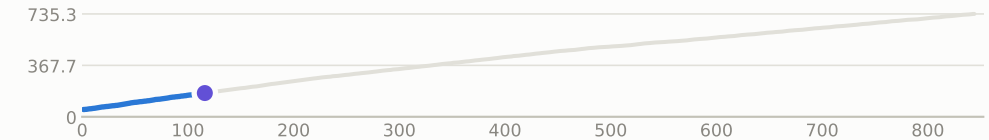
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

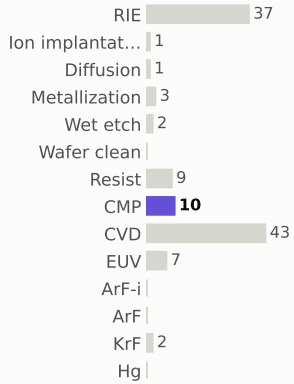
carbon so far, of the whole flow (kgCO2e)



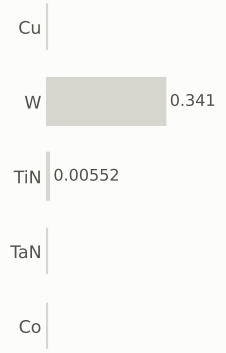
embodied carbon up to each step (kgCO2e per wafer)



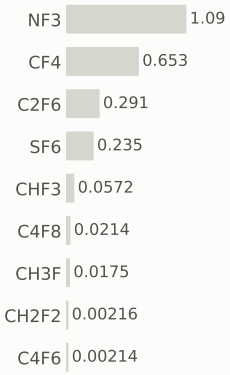
steps by process area



materials deposited (g)



gases emitted (g)



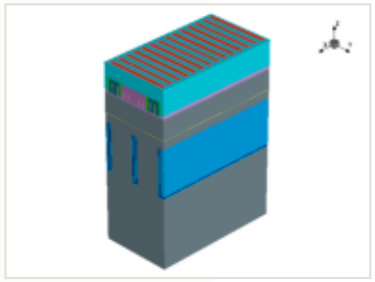
Step #1290: CMP

etch_cmp · CMP

0.169	170.0	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

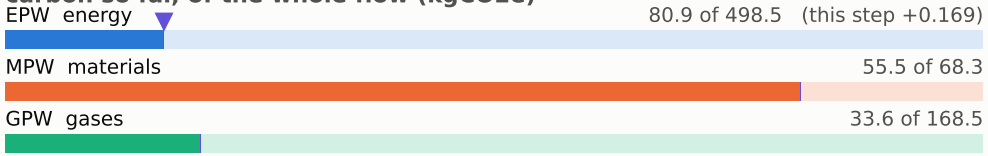


this step

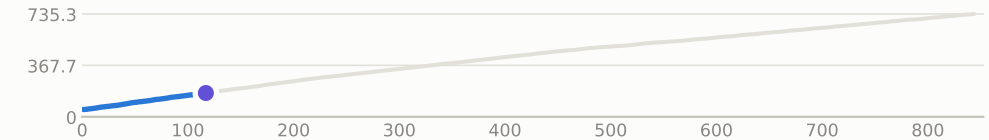
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

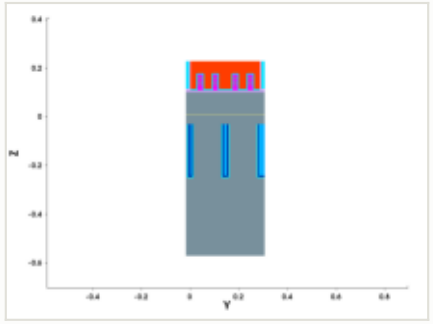
carbon so far, of the whole flow (kgCO2e)



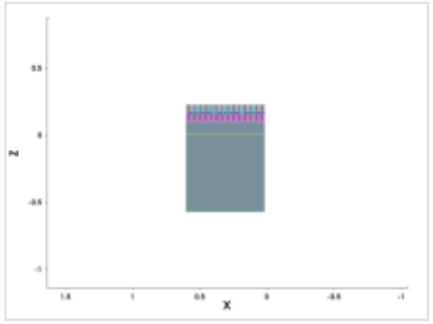
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



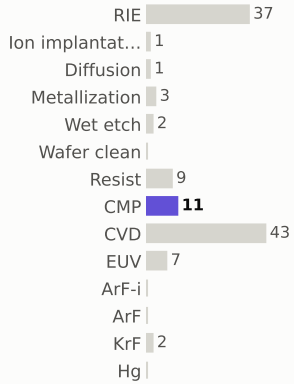
y cut at 0.184 µm



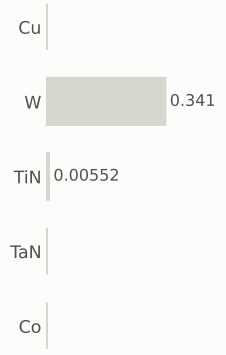
mask for this step

no mask for this step

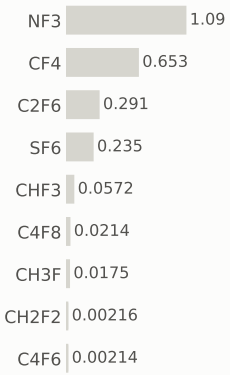
steps by process area



materials deposited (g)



gases emitted (g)



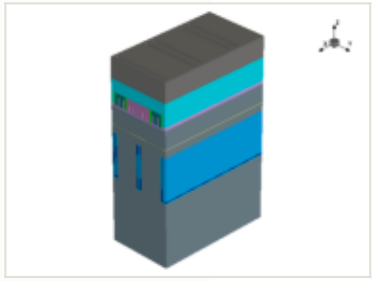
Step #1300: EUV

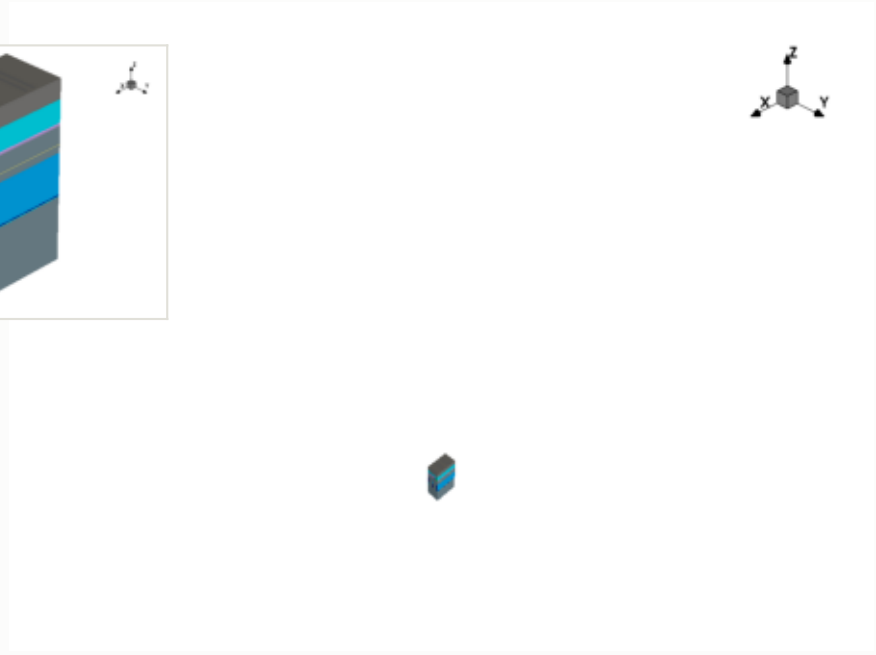
litho_patterning · EUV

1.96	171.9	0.27%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



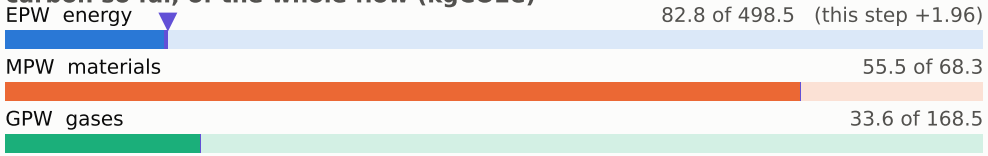


this step

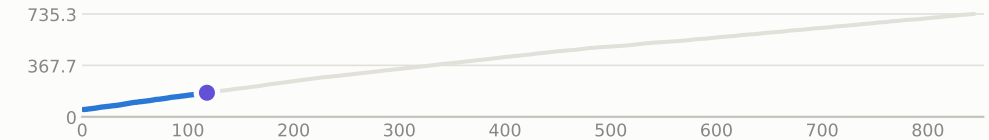
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

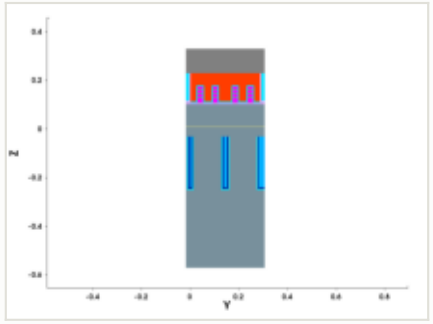
carbon so far, of the whole flow (kgCO2e)



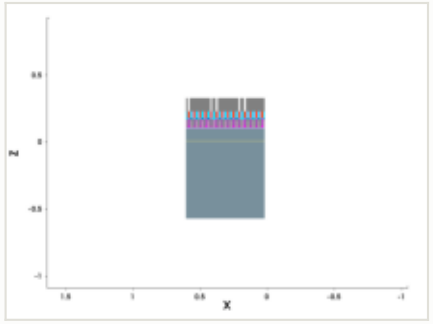
embodied carbon up to each step (kgCO2e per wafer)



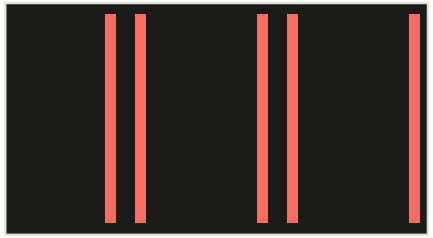
x cut at 0.294 µm



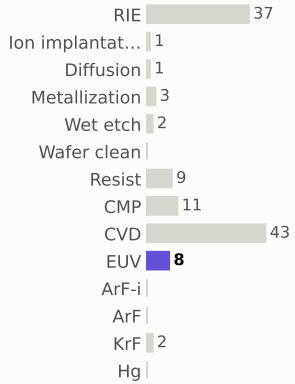
y cut at 0.184 µm



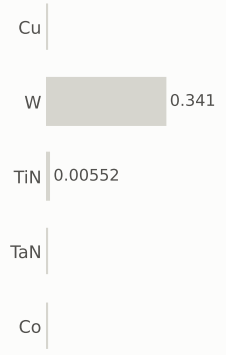
mask: layer 4:0 (GDSII)



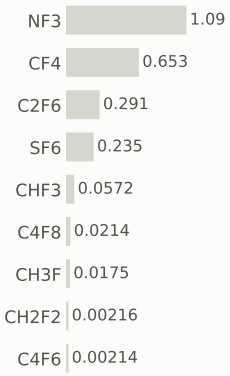
steps by process area



materials deposited (g)



gases emitted (g)



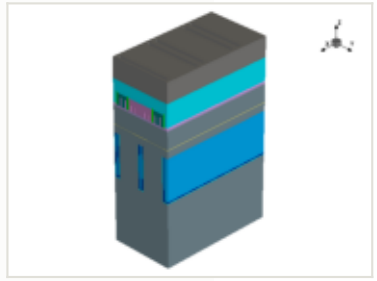
Step #1310: RIE

etch_anisotropic · RIE

1.04	173.0	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

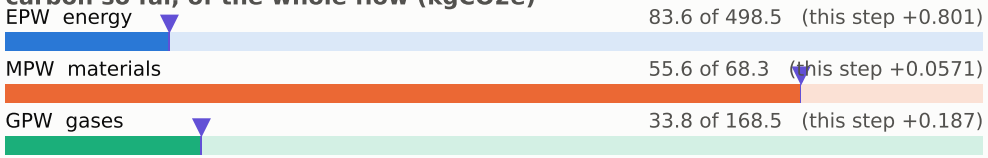


this step

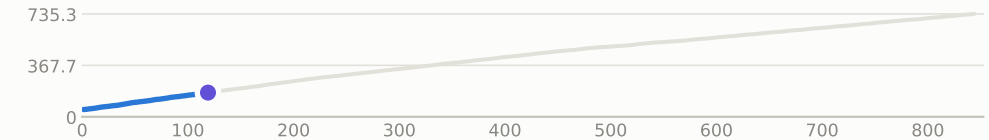
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

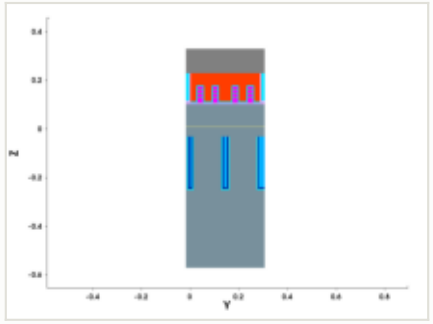
carbon so far, of the whole flow (kgCO2e)



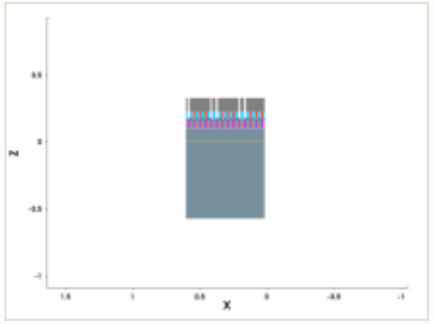
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



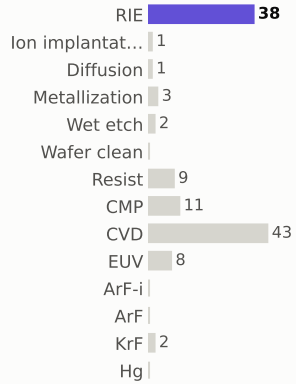
y cut at 0.184 µm



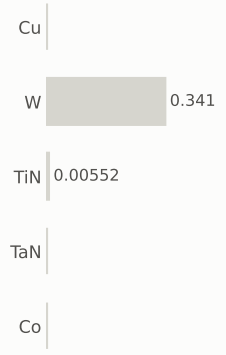
mask for this step

no mask for this step

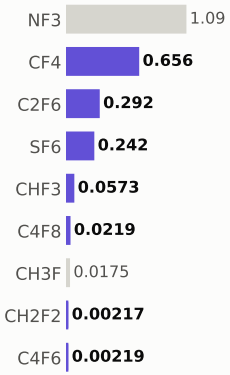
steps by process area



materials deposited (g)



gases emitted (g)



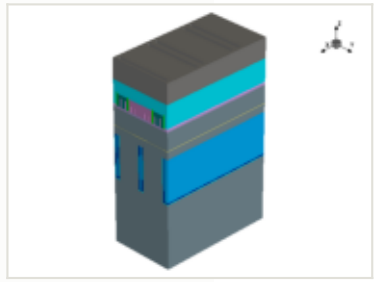
Step #1320: RIE

etch_anisotropic · RIE

1.04	174.0	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

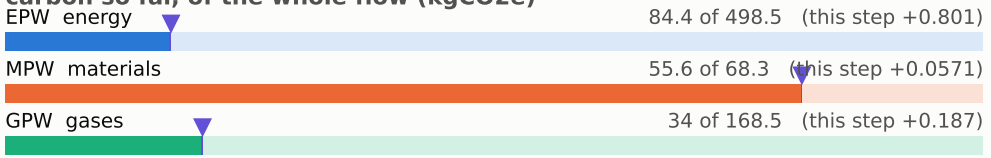


this step

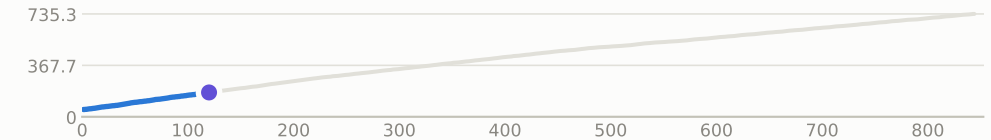
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

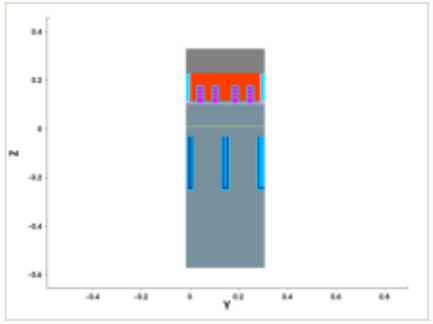
carbon so far, of the whole flow (kgCO2e)



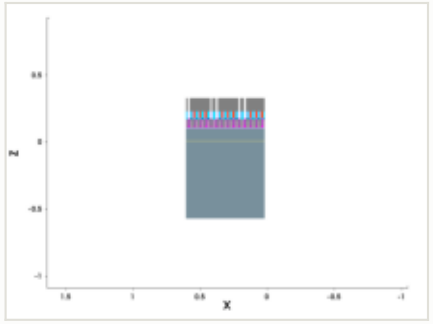
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



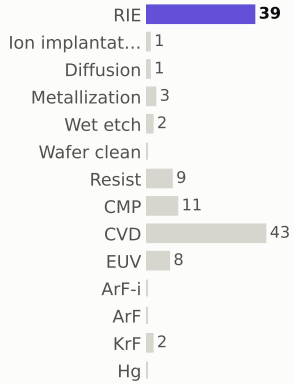
y cut at 0.184 µm



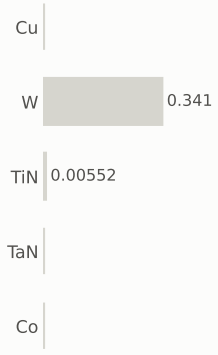
mask for this step

no mask for this step

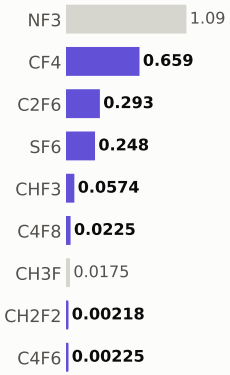
steps by process area



materials deposited (g)



gases emitted (g)



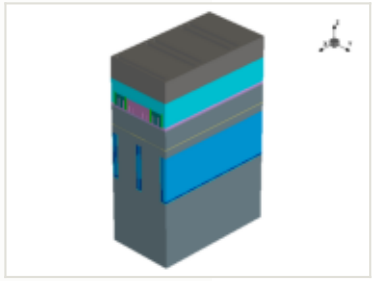
Step #1330: RIE

etch_anisotropic · RIE

1.04	175.0	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

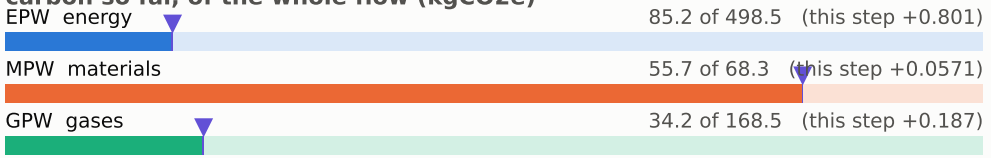


this step

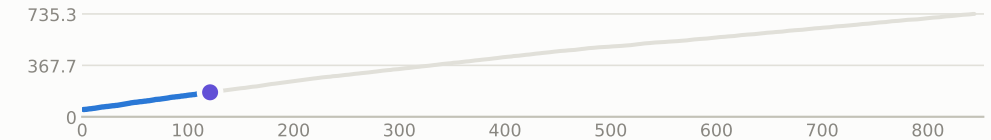
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

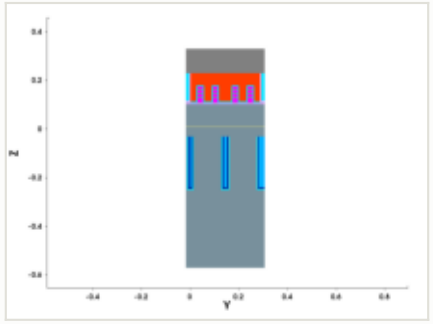
carbon so far, of the whole flow (kgCO2e)



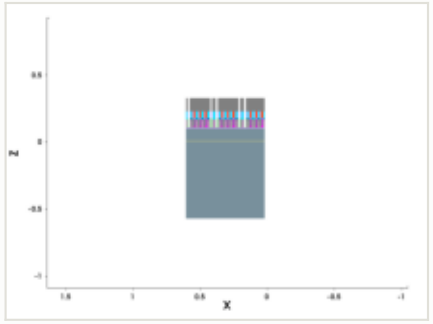
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



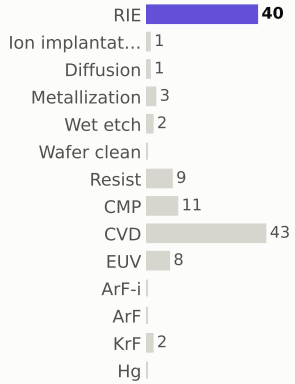
y cut at 0.184 µm



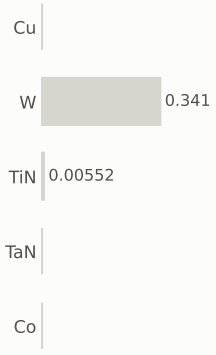
mask for this step

no mask for this step

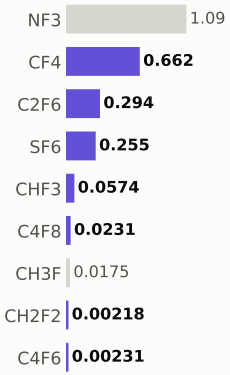
steps by process area



materials deposited (g)



gases emitted (g)



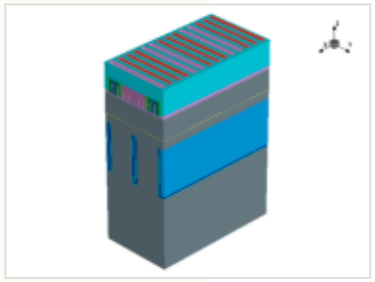
Step #1340: Resist

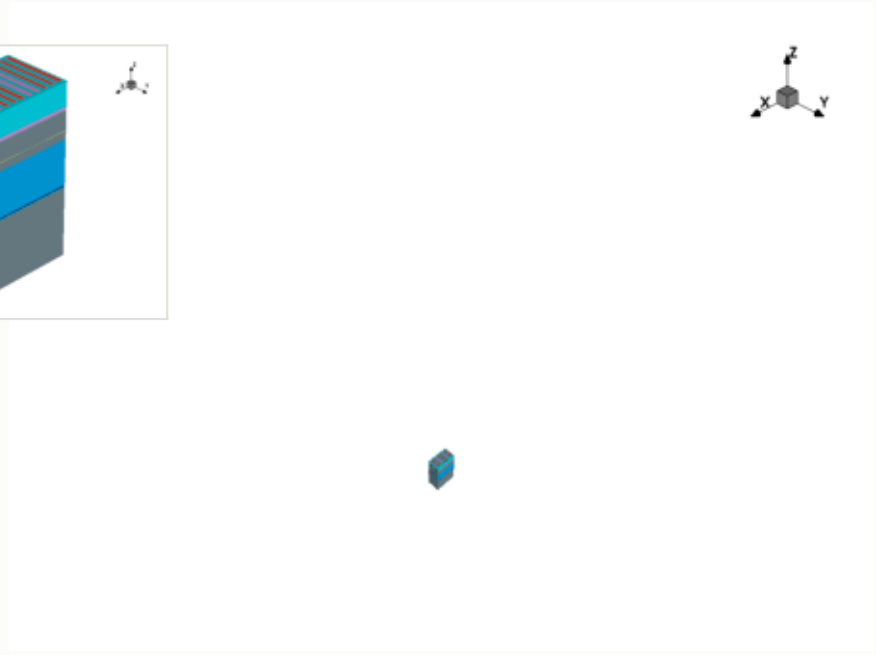
etch_strip · Resist

0.169	175.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



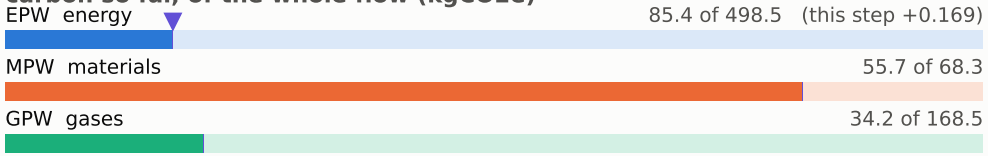


this step

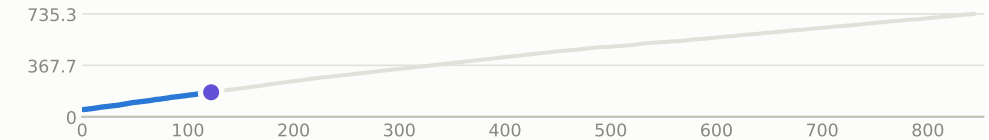
process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

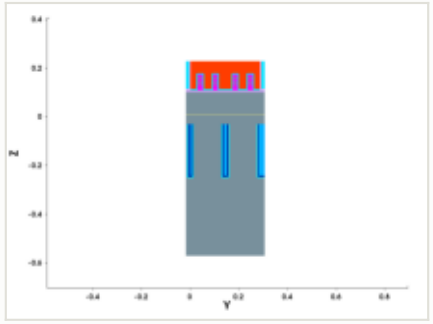
carbon so far, of the whole flow (kgCO2e)



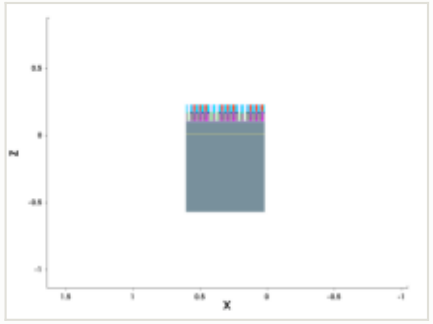
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



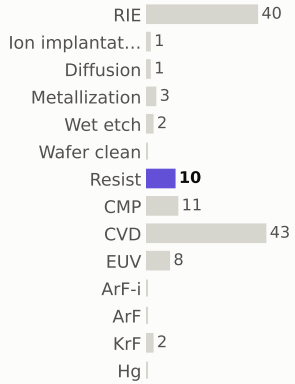
y cut at 0.184 µm



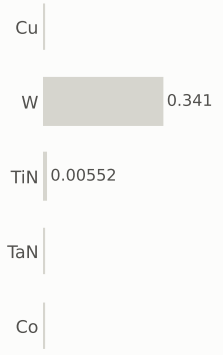
mask for this step

no mask for this step

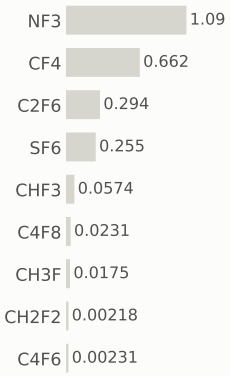
steps by process area



materials deposited (g)



gases emitted (g)



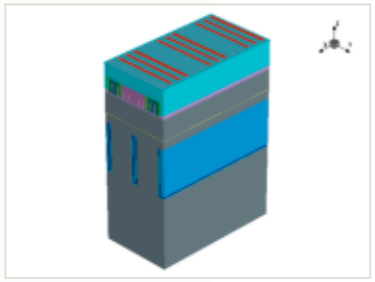
Step #1350: CVD

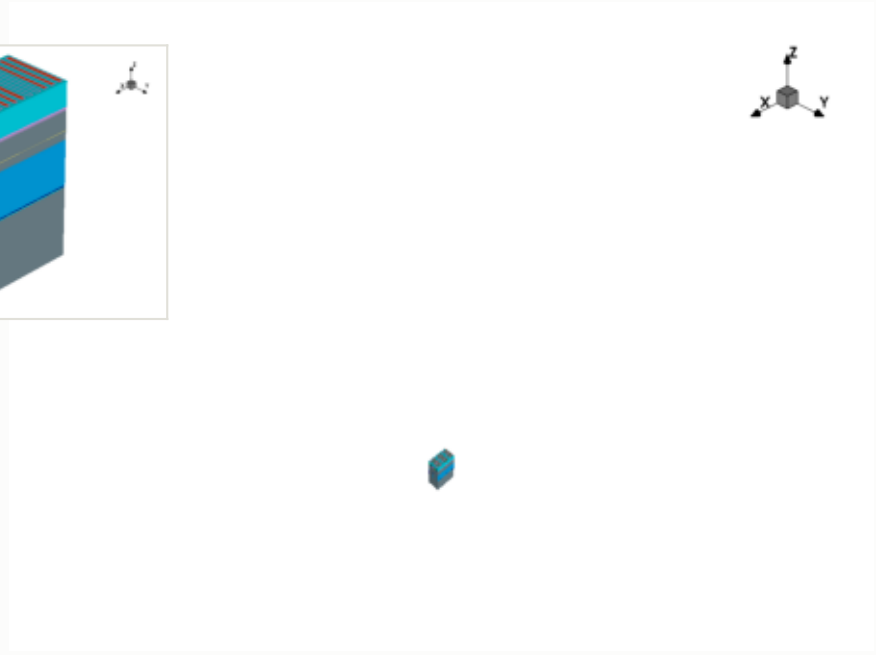
depo_fill · CVD

1.31	176.5	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



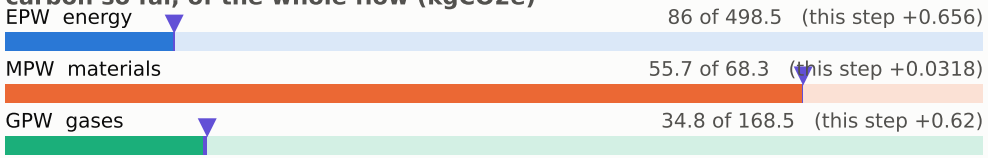


this step

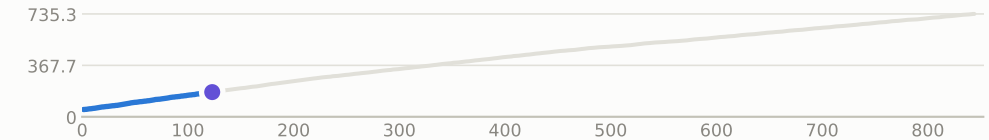
process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

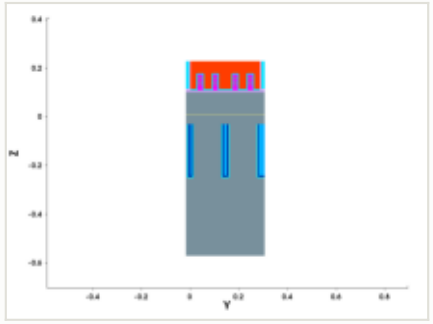
carbon so far, of the whole flow (kgCO2e)



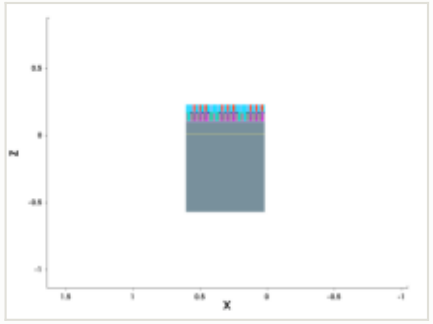
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



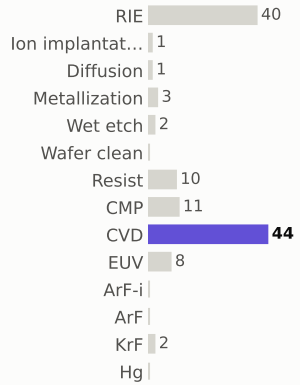
y cut at 0.184 µm



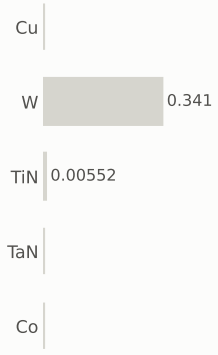
mask for this step

no mask for this step

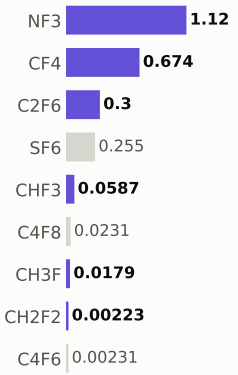
steps by process area



materials deposited (g)



gases emitted (g)



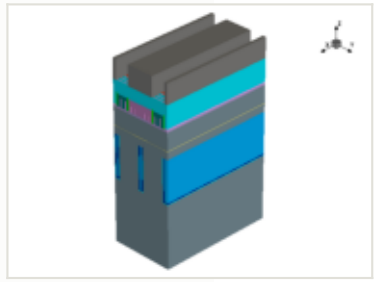
Step #1370: EUV

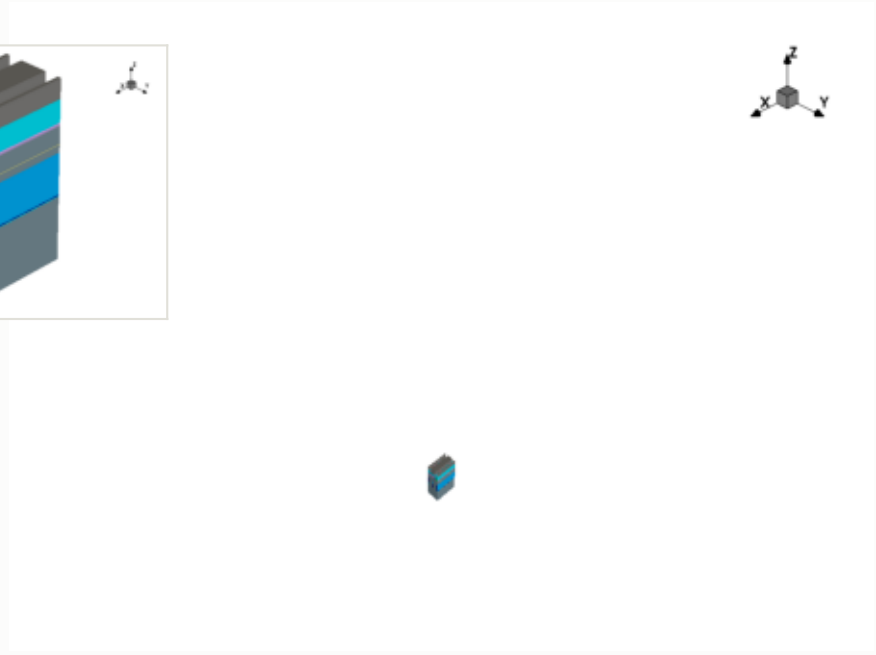
litho_patterning · EUV

1.96	178.5	0.27%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



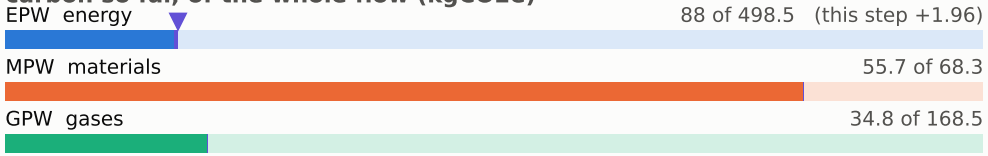


this step

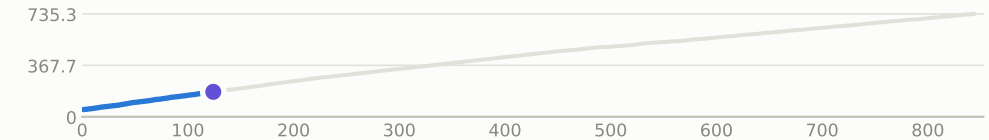
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

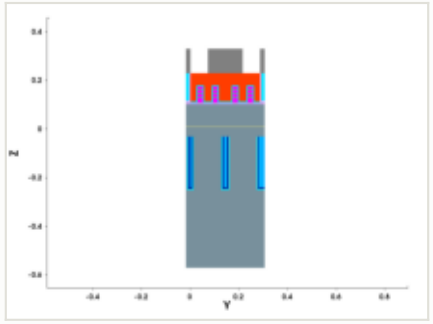
carbon so far, of the whole flow (kgCO2e)



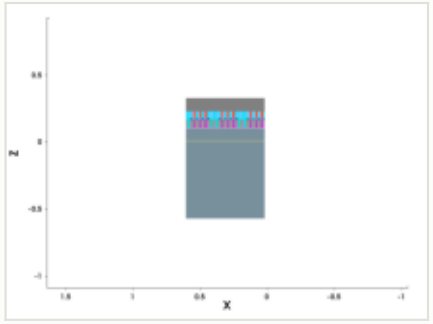
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm

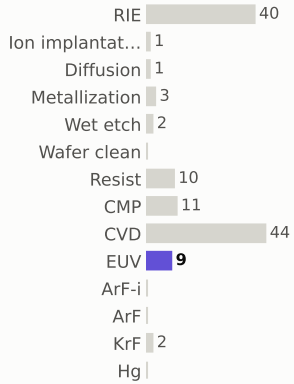


y cut at 0.184 µm

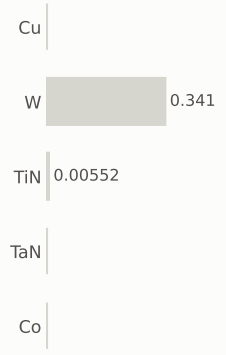


mask: layer 6:0 (GDSII)

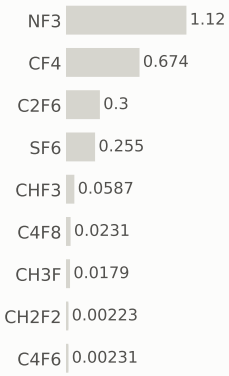
steps by process area



materials deposited (g)



gases emitted (g)



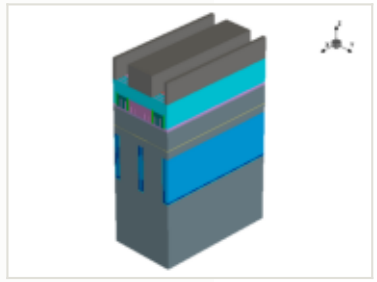
Step #1380: RIE

etch_anisotropic · RIE

1.04	179.5	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

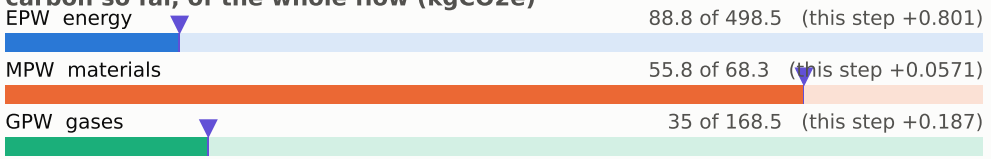


this step

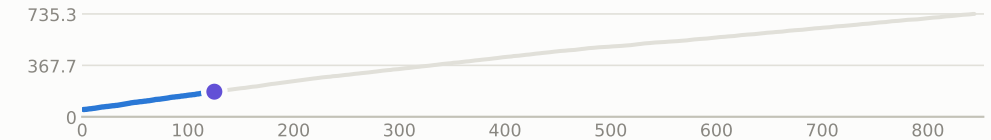
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

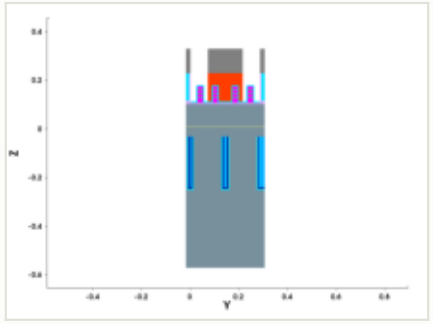
carbon so far, of the whole flow (kgCO2e)



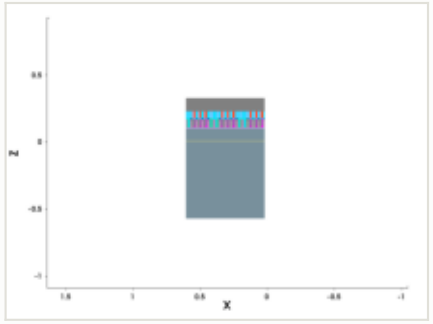
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



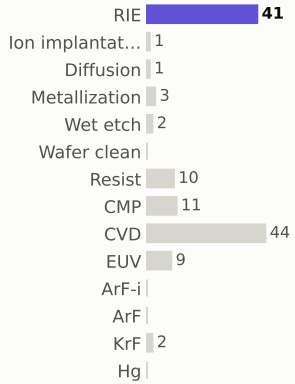
y cut at 0.184 µm



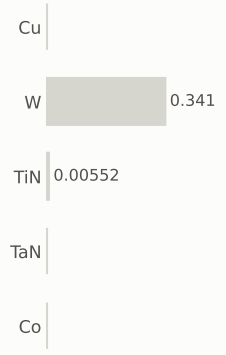
mask for this step

no mask for this step

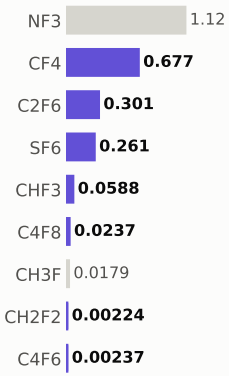
steps by process area



materials deposited (g)



gases emitted (g)



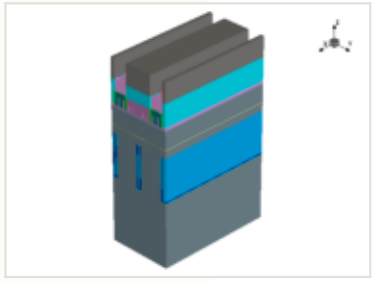
Step #1390: RIE

etch_anisotropic · RIE

1.04	180.6	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

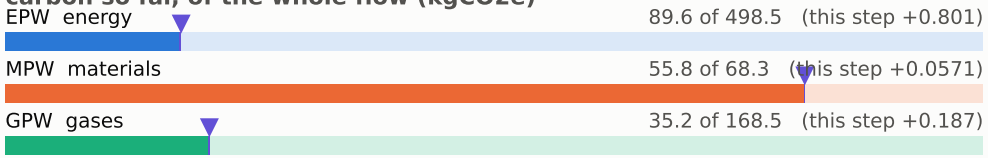


this step

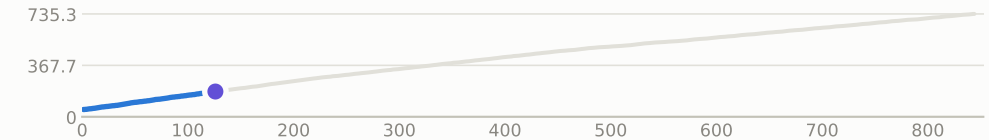
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

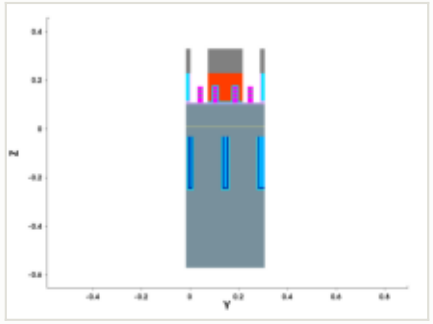
carbon so far, of the whole flow (kgCO2e)



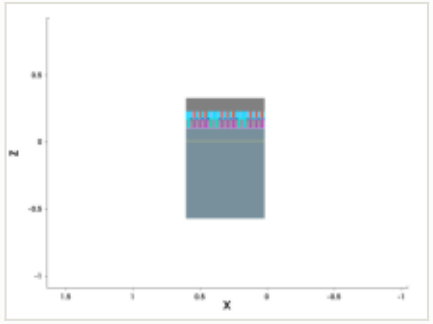
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



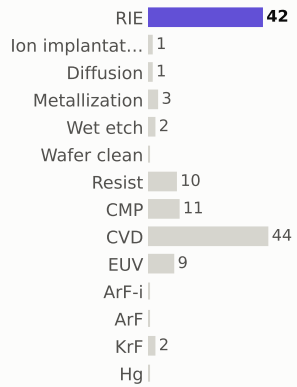
y cut at 0.184 µm



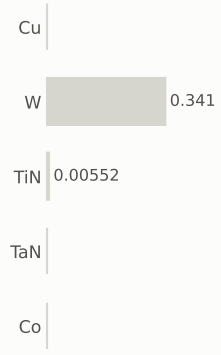
mask for this step

no mask for this step

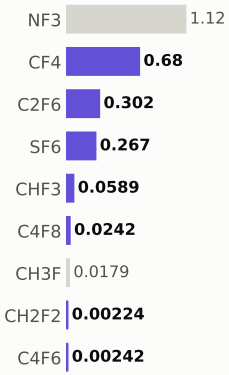
steps by process area



materials deposited (g)



gases emitted (g)



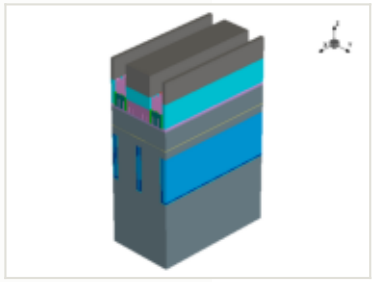
Step #1400: Resist

etch_strip · Resist

0.169	180.7	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

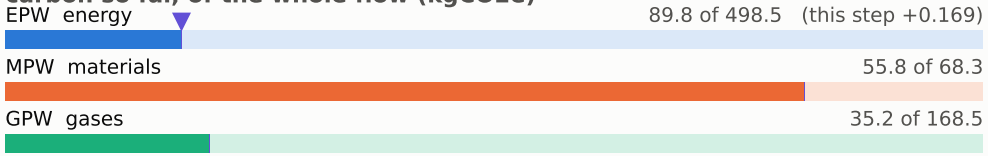


this step

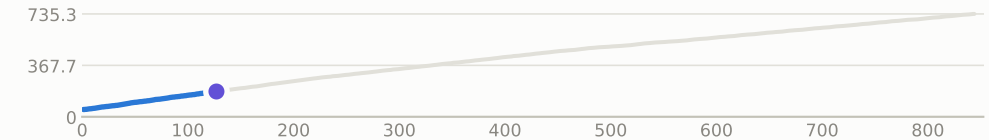
process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

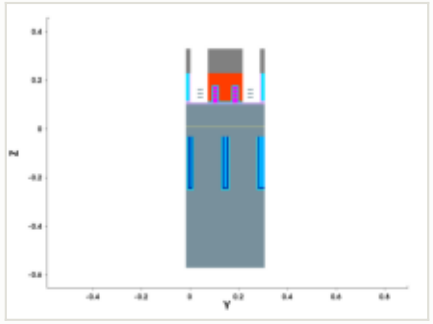
carbon so far, of the whole flow (kgCO2e)



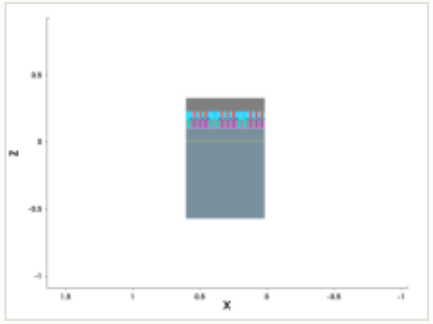
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



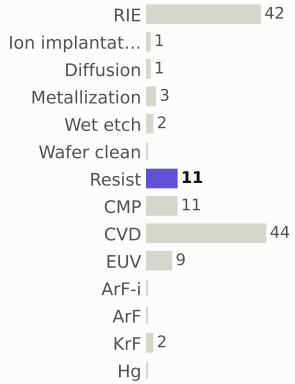
y cut at 0.184 µm



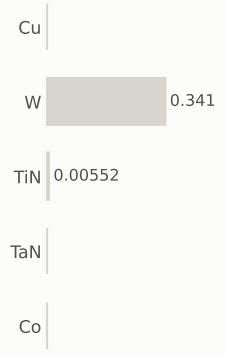
mask for this step

no mask for this step

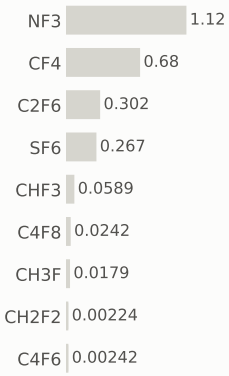
steps by process area



materials deposited (g)



gases emitted (g)



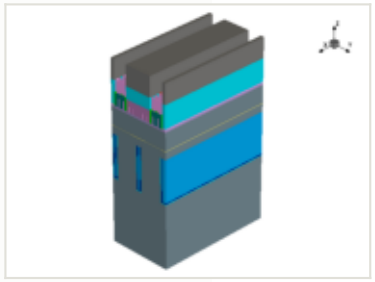
Step #1410: CVD

depo_fill · CVD

1.31	182.0	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

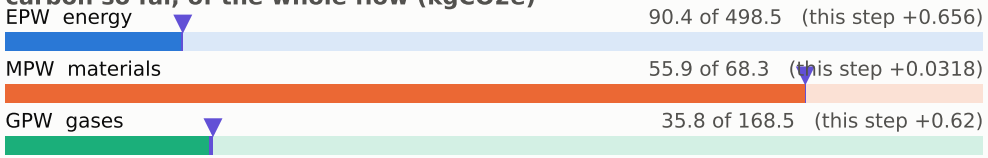


this step

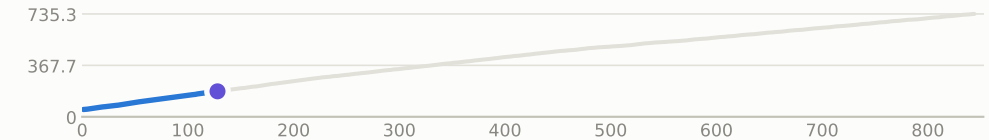
process area	CVD
material	—
thickness	-0.053 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

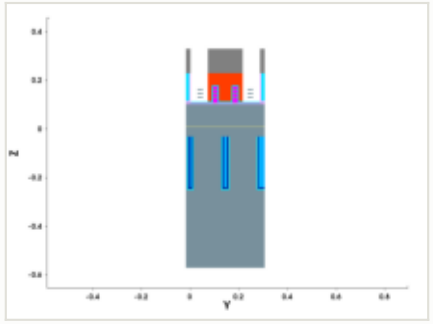
carbon so far, of the whole flow (kgCO2e)



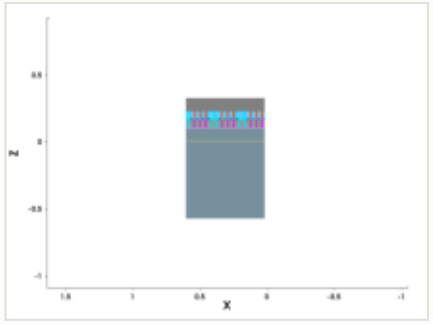
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



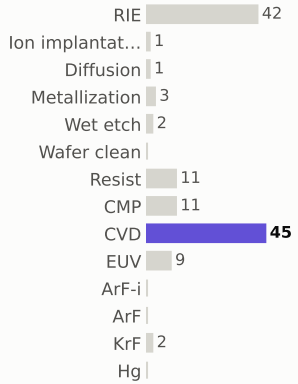
y cut at 0.184 µm



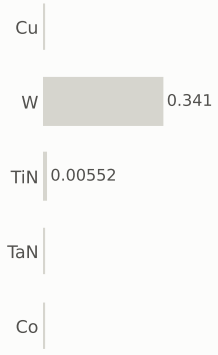
mask for this step

no mask for this step

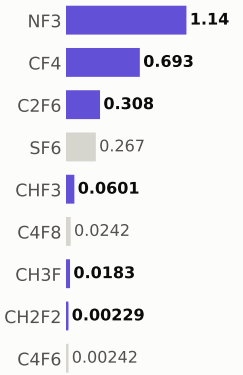
steps by process area



materials deposited (g)



gases emitted (g)



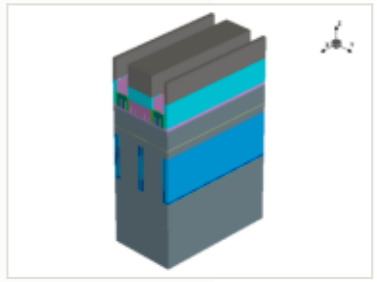
Step #1420: CVD

depo_isotropic · CVD

1.31	183.4	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

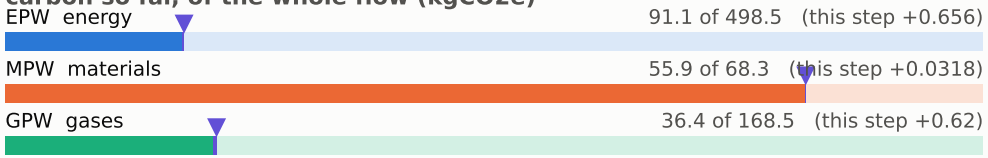
zoomed to fit



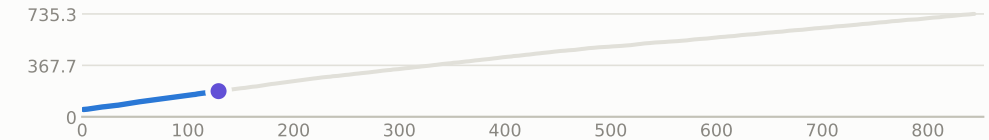
this step

process area	CVD	EPW	0.656 kgCO2e
material	—	EPW tool	0.469 kgCO2e
thickness	0.0028 µm	EPW overhead	0.187 kgCO2e
material GWP	—	MPW	0.0318 kgCO2e (gas 0.0318)
input gases	NF3 2.54 g	GPW	0.62 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**1.31 kgCO2e**

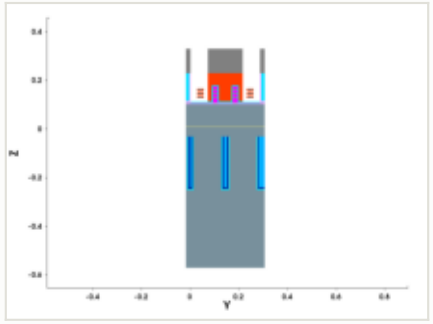
carbon so far, of the whole flow (kgCO2e)



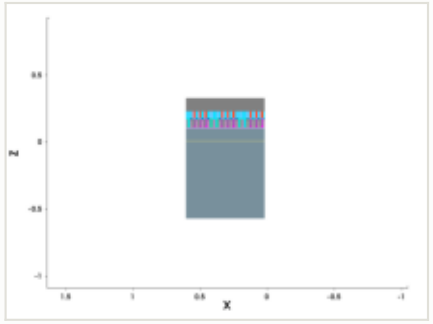
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



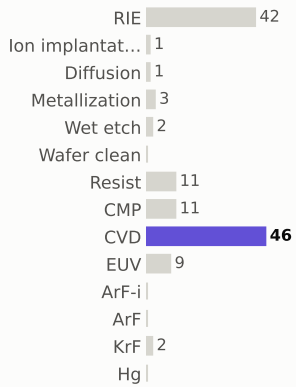
y cut at 0.184 µm



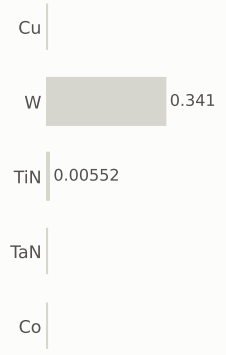
mask for this step

no mask for this step

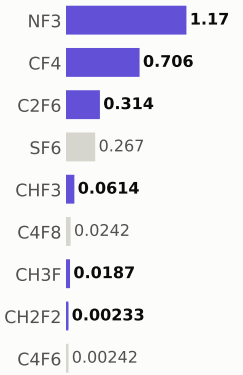
steps by process area



materials deposited (g)



gases emitted (g)



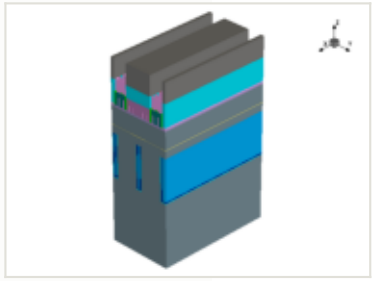
Step #1430: Wet_etch

etch_isotropic · Wet_etch

0.169	183.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

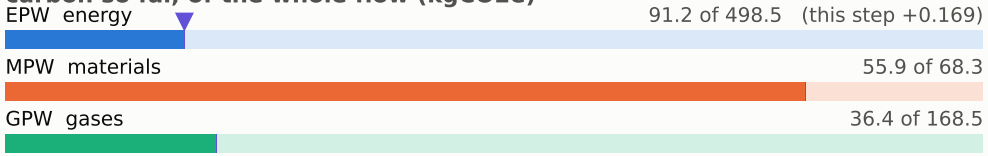


this step

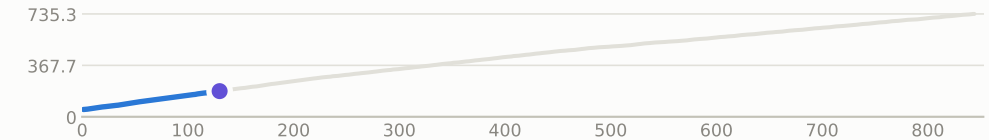
process area	Wet_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

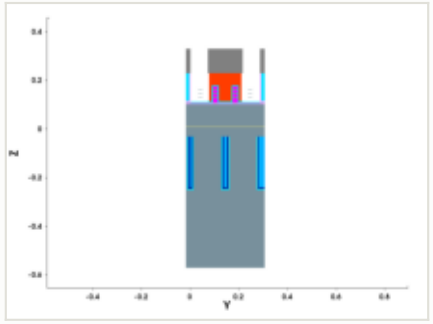
carbon so far, of the whole flow (kgCO2e)



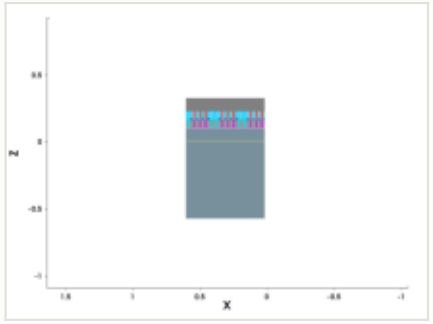
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



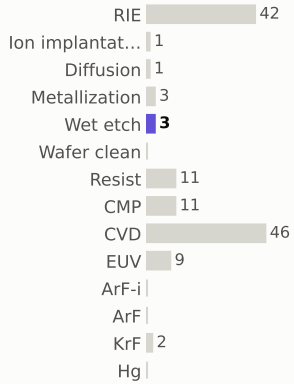
y cut at 0.184 µm



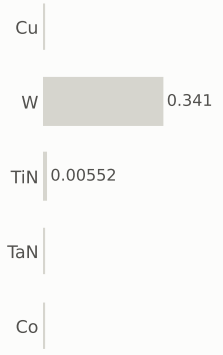
mask for this step

no mask for this step

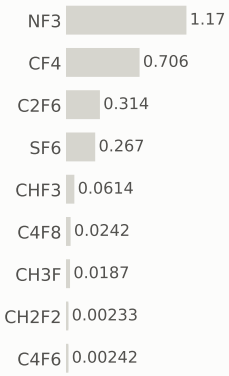
steps by process area



materials deposited (g)



gases emitted (g)



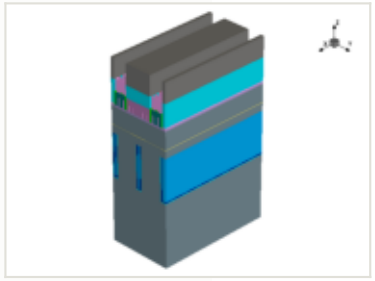
Step #1440: CVD

depo_isotropic · CVD

1.31	184.8	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

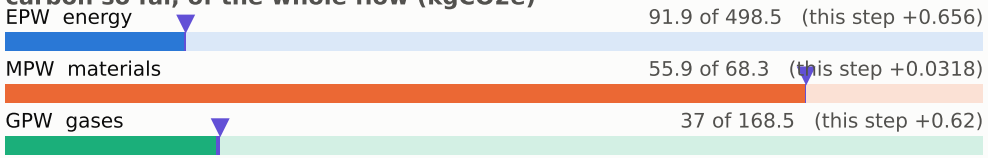


this step

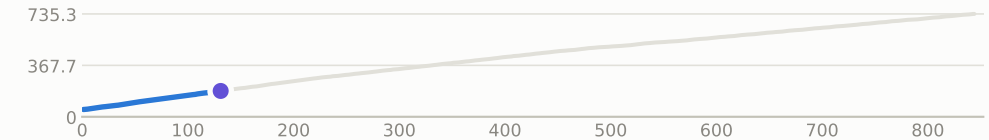
process area	CVD
material	—
thickness	0.0015 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

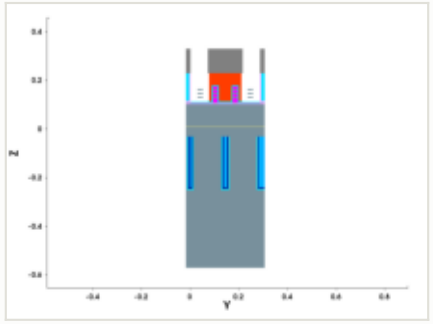
carbon so far, of the whole flow (kgCO2e)



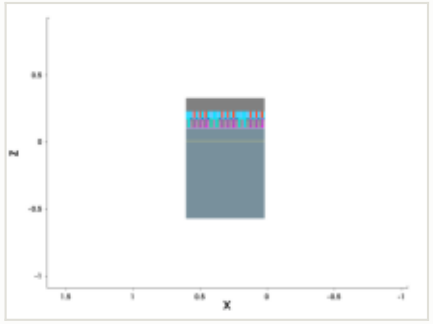
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



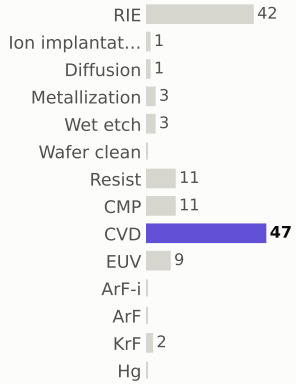
y cut at 0.184 µm



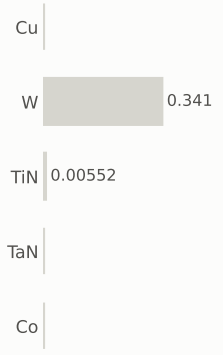
mask for this step

no mask for this step

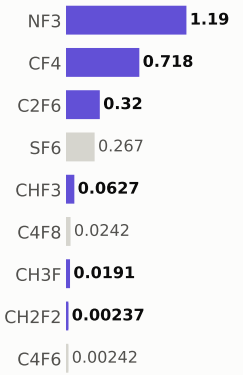
steps by process area



materials deposited (g)



gases emitted (g)



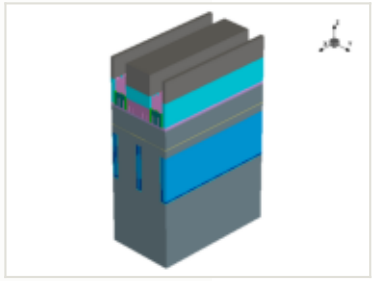
Step #1450: CVD

depo_isotropic · CVD

1.31	186.1	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

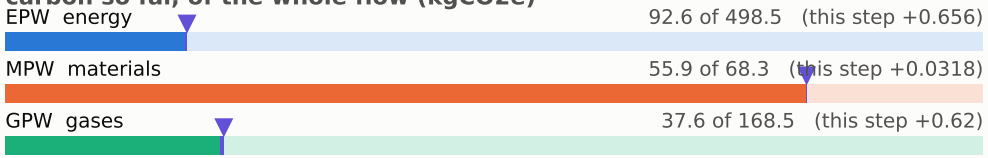


this step

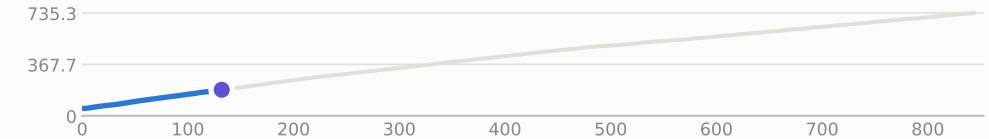
process area	CVD
material	—
thickness	0.0005 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

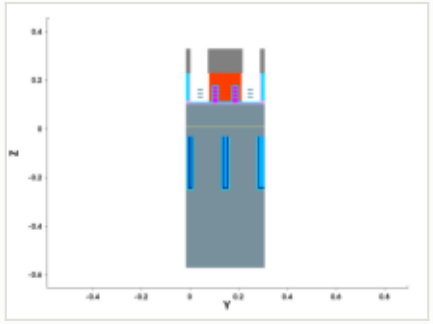
carbon so far, of the whole flow (kgCO2e)



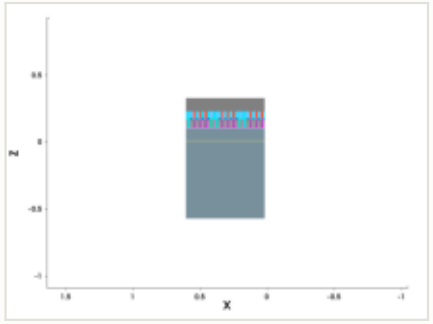
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



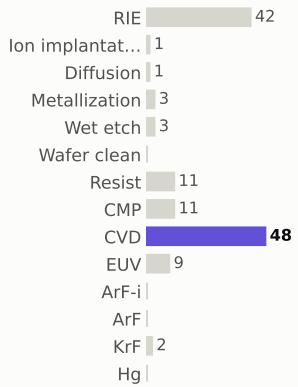
y cut at 0.184 µm



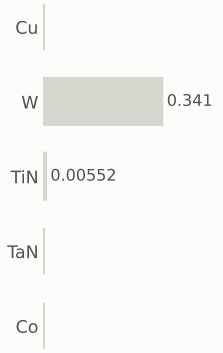
mask for this step

no mask for this step

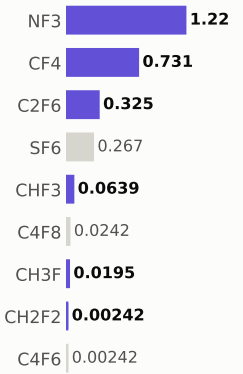
steps by process area



materials deposited (g)



gases emitted (g)



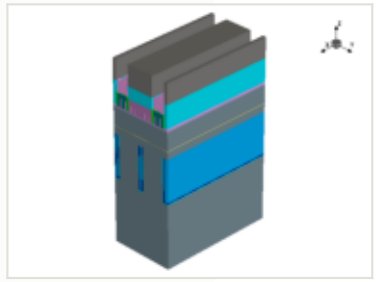
Step #1460: CVD

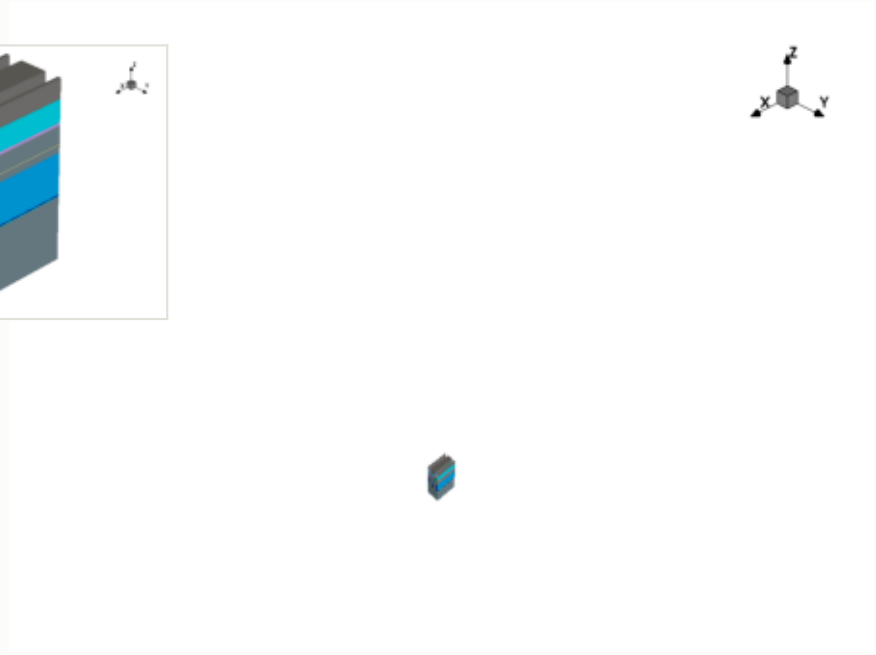
depo_isotropic · CVD

1.31	187.5	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



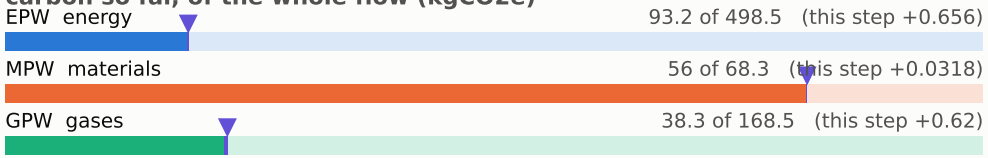


this step

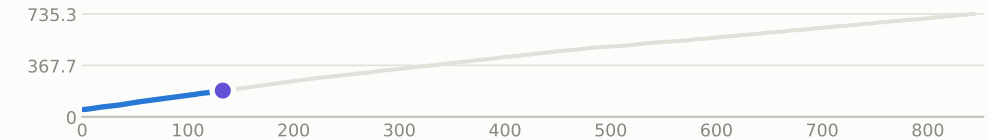
process area	CVD
material	—
thickness	0.002 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

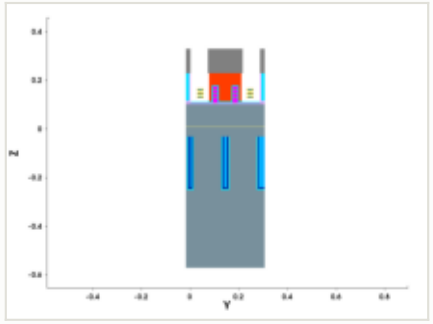
carbon so far, of the whole flow (kgCO2e)



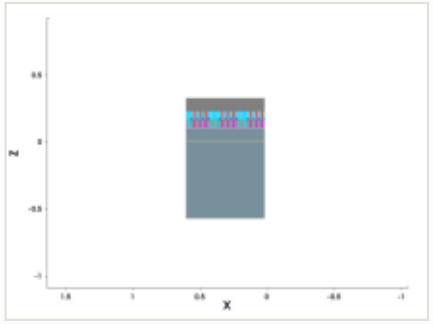
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



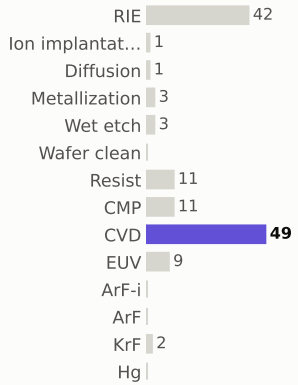
y cut at 0.184 µm



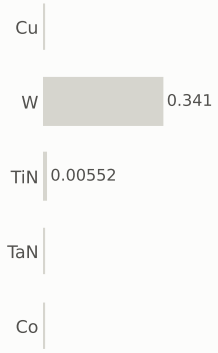
mask for this step

no mask for this step

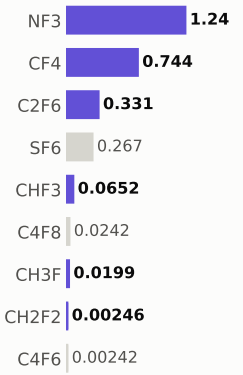
steps by process area



materials deposited (g)



gases emitted (g)



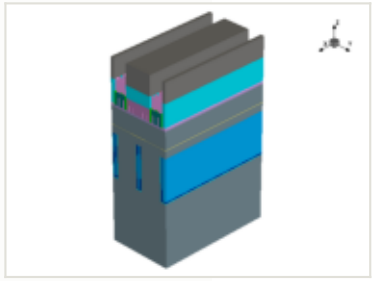
Step #1470: CVD

depo_isotropic · CVD

1.31	188.8	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

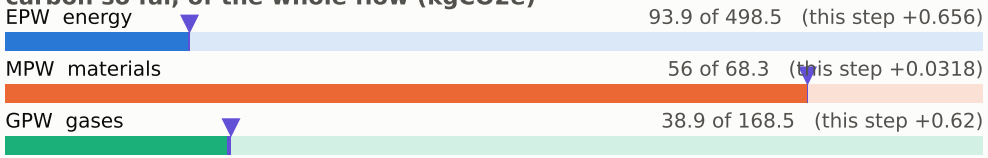
zoomed to fit



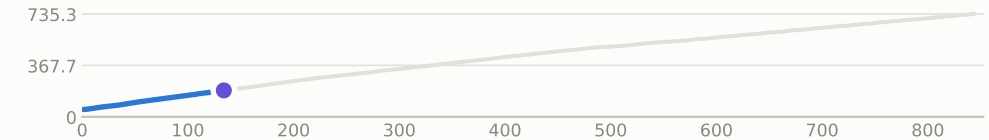
this step

process area	CVD	EPW	0.656 kgCO2e
material	—	EPW tool	0.469 kgCO2e
thickness	0.0005 µm	EPW overhead	0.187 kgCO2e
material GWP	—	MPW	0.0318 kgCO2e (gas 0.0318)
input gases	NF3 2.54 g	GPW	0.62 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**1.31 kgCO2e**

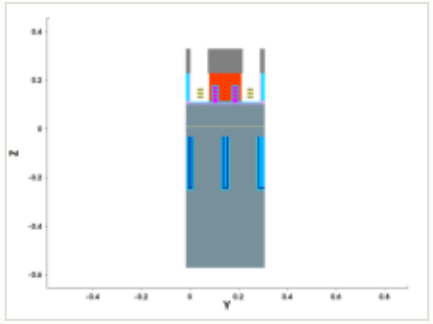
carbon so far, of the whole flow (kgCO2e)



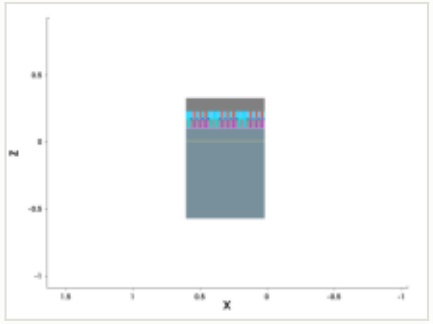
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



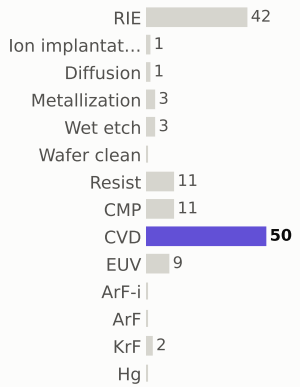
y cut at 0.184 µm



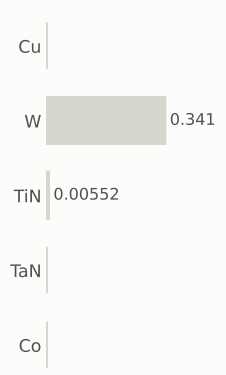
mask for this step

no mask for this step

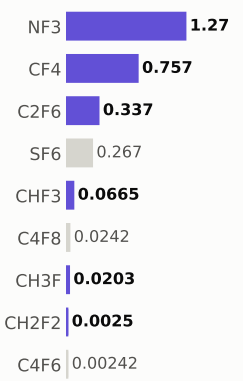
steps by process area



materials deposited (g)



gases emitted (g)



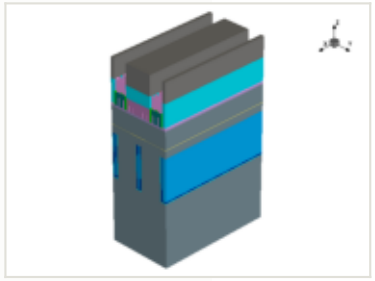
Step #1480: CVD

depo_isotropic · CVD

1.31	190.1	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

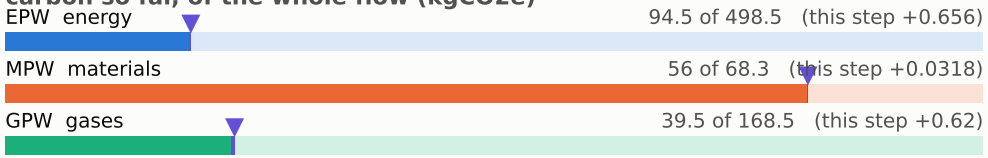


this step

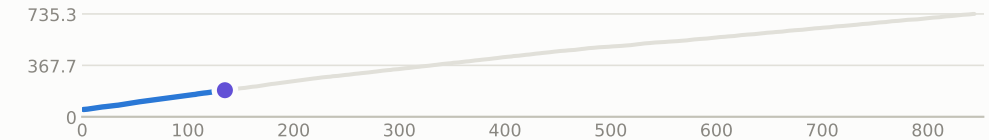
process area	CVD
material	—
thickness	0.0005 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

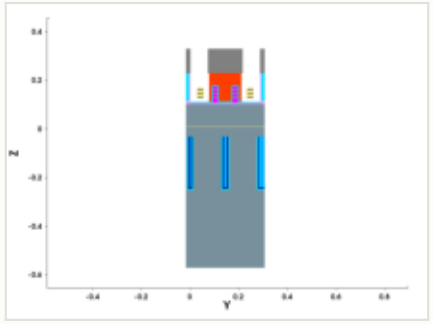
carbon so far, of the whole flow (kgCO2e)



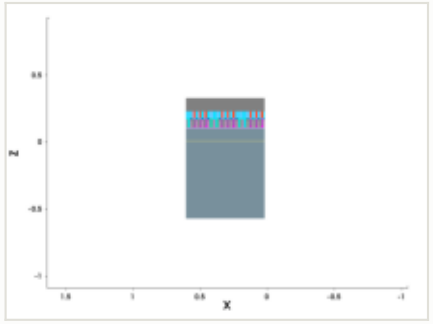
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



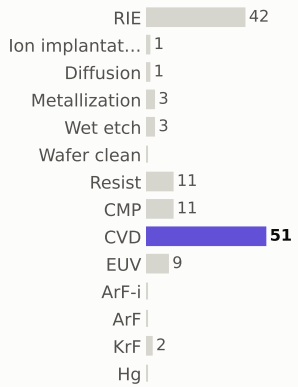
y cut at 0.184 µm



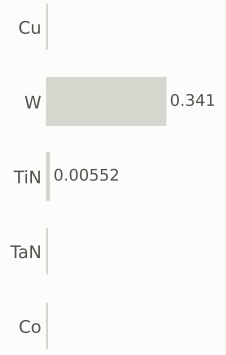
mask for this step

no mask for this step

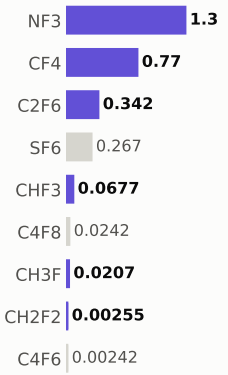
steps by process area



materials deposited (g)



gases emitted (g)



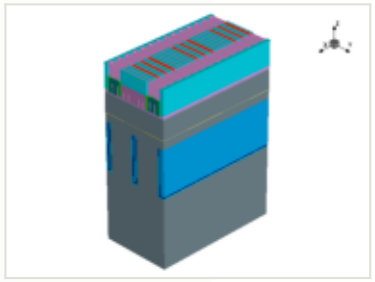
Step #1490: CMP

etch_cmp · CMP

0.169	190.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

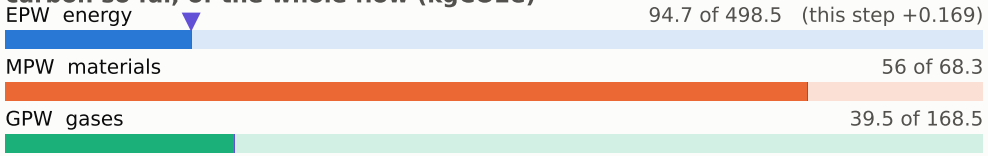


this step

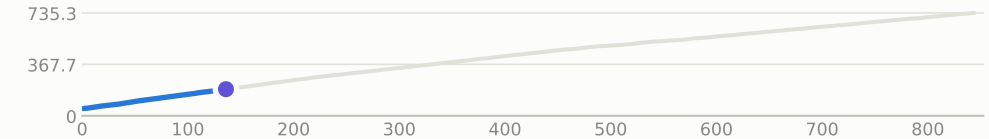
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

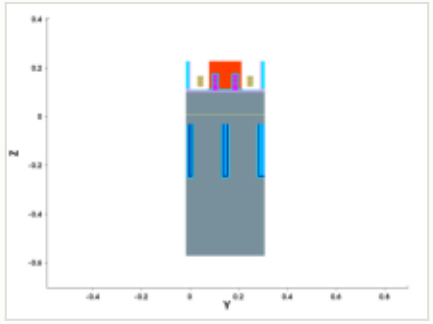
carbon so far, of the whole flow (kgCO2e)



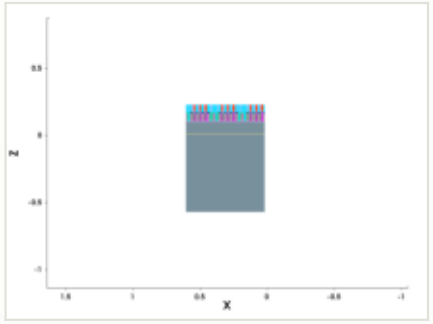
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



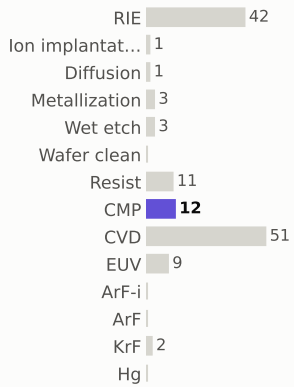
y cut at 0.184 µm



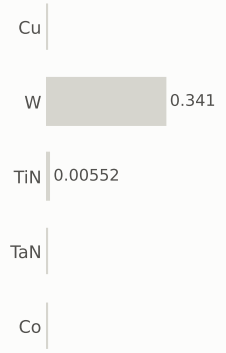
mask for this step

no mask for this step

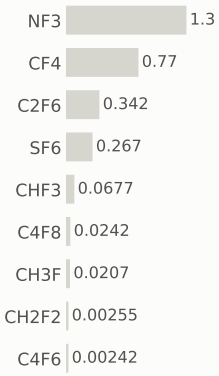
steps by process area



materials deposited (g)



gases emitted (g)



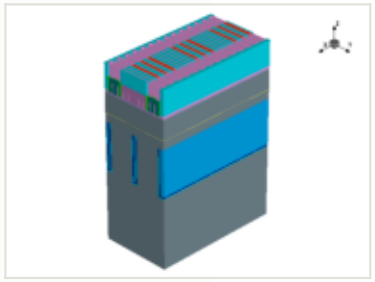
Step #1500: Resist

etch_strip · Resist

0.169	190.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

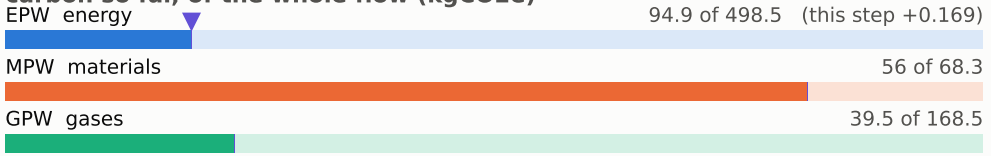


this step

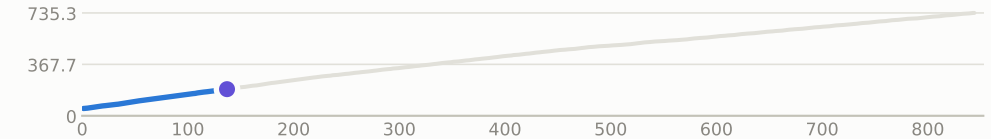
process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

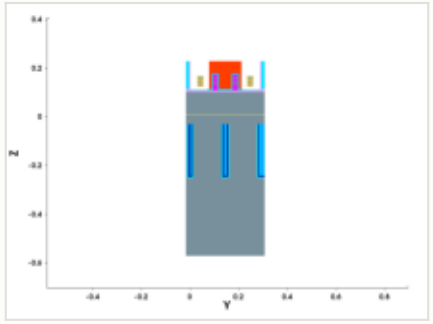
carbon so far, of the whole flow (kgCO2e)



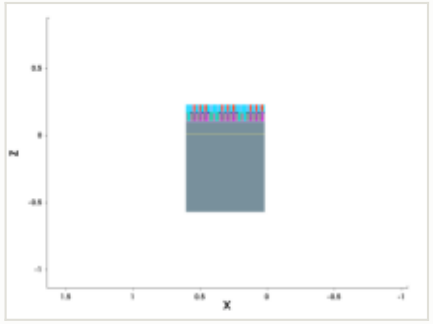
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



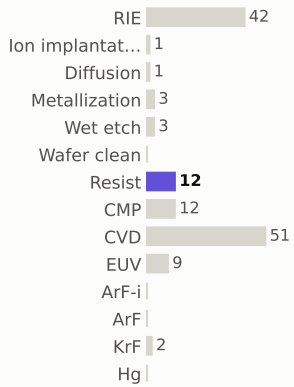
y cut at 0.184 µm



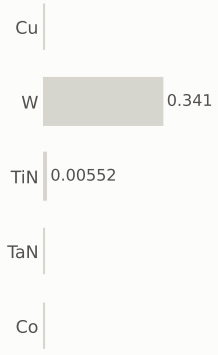
mask for this step

no mask for this step

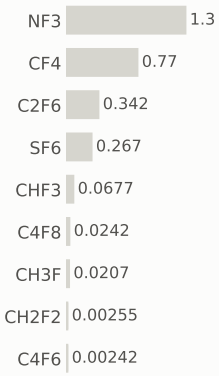
steps by process area



materials deposited (g)



gases emitted (g)



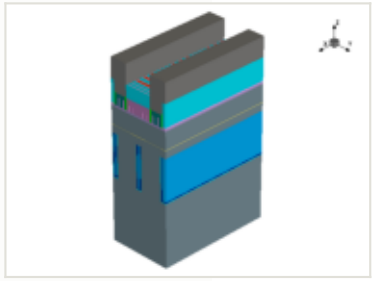
Step #1520: EUV

litho_patterning · EUV

1.96	192.4	0.27%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

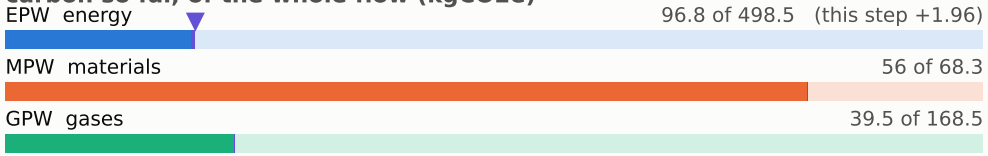


this step

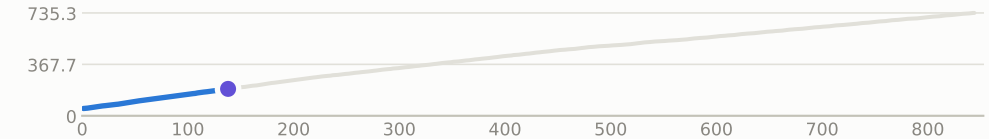
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

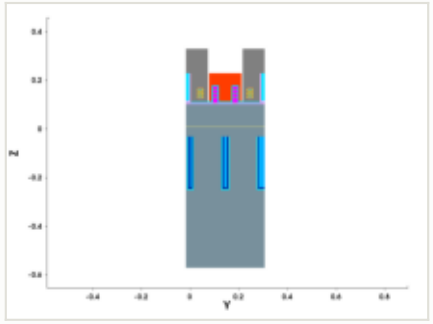
carbon so far, of the whole flow (kgCO2e)



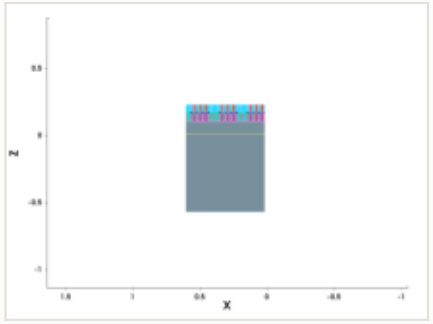
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm

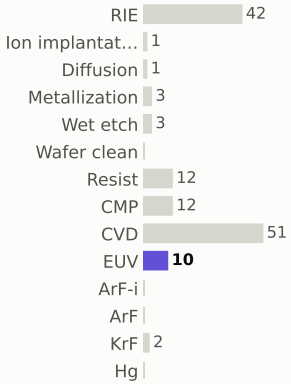


y cut at 0.184 µm

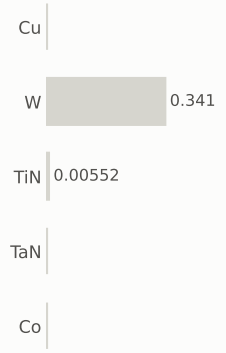


mask: layer 7:0 (GDSII)

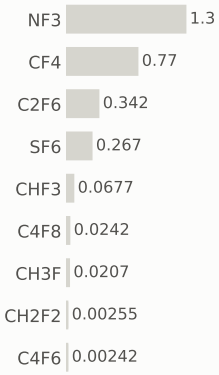
steps by process area



materials deposited (g)



gases emitted (g)



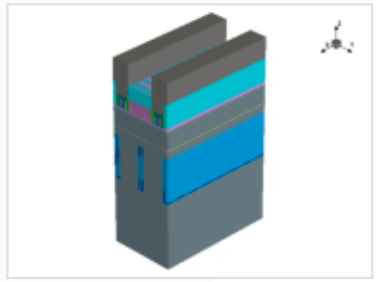
Step #1530: RIE

etch_anisotropic · RIE

1.04	193.4	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

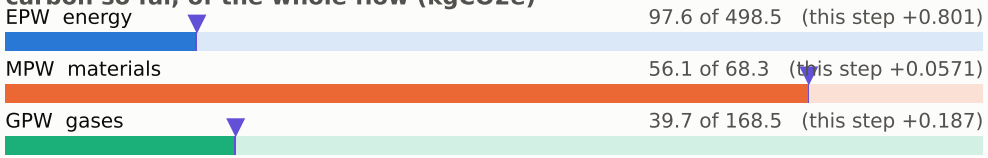


this step

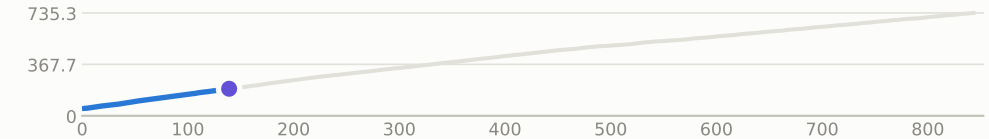
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

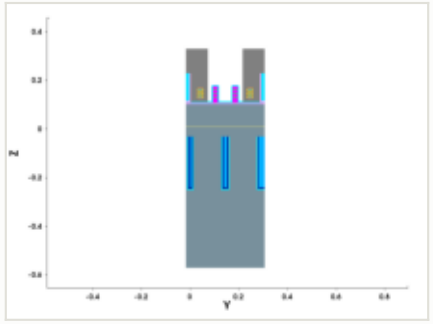
carbon so far, of the whole flow (kgCO2e)



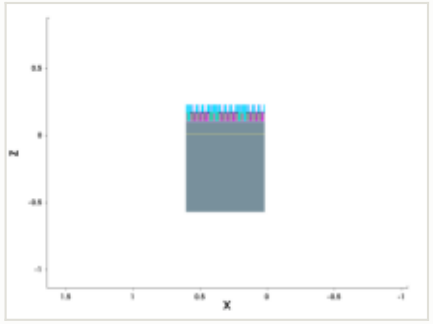
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



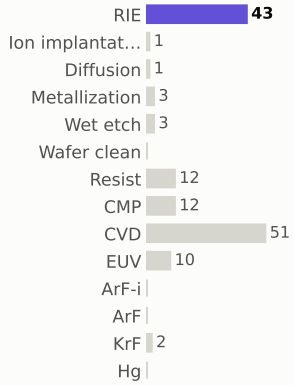
y cut at 0.184 µm



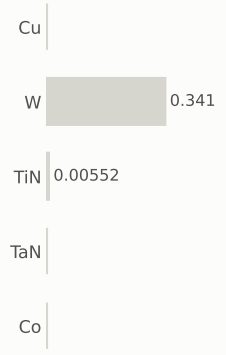
mask for this step

no mask for this step

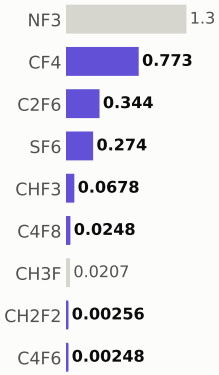
steps by process area



materials deposited (g)



gases emitted (g)



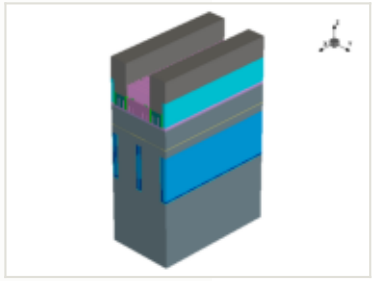
Step #1540: RIE

etch_anisotropic · RIE

1.04	194.5	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

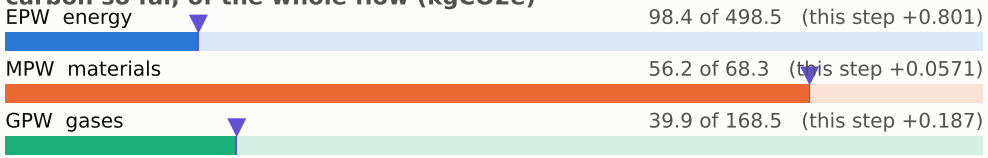


this step

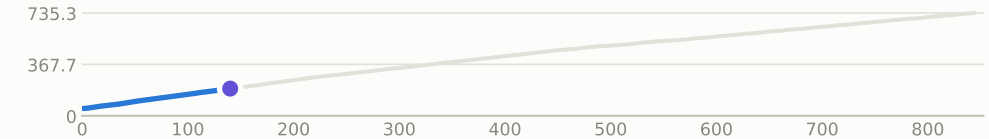
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

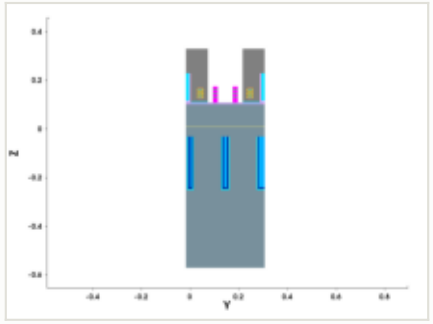
carbon so far, of the whole flow (kgCO2e)



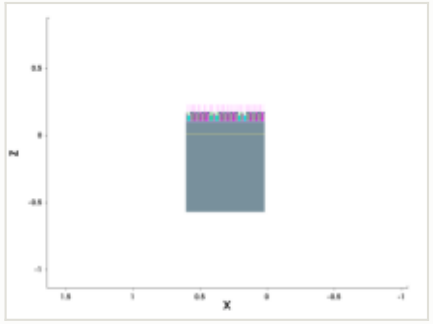
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



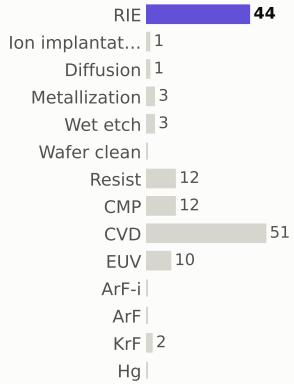
y cut at 0.184 µm



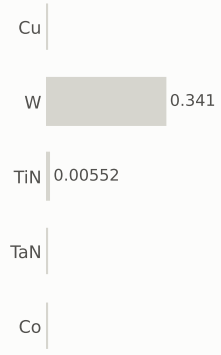
mask for this step

no mask for this step

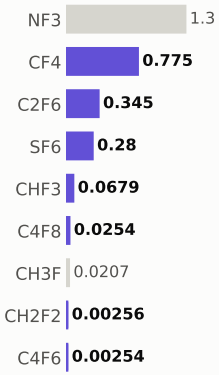
steps by process area



materials deposited (g)



gases emitted (g)



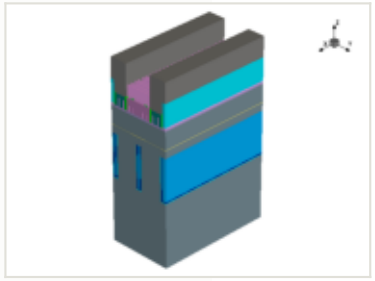
Step #1550: Resist

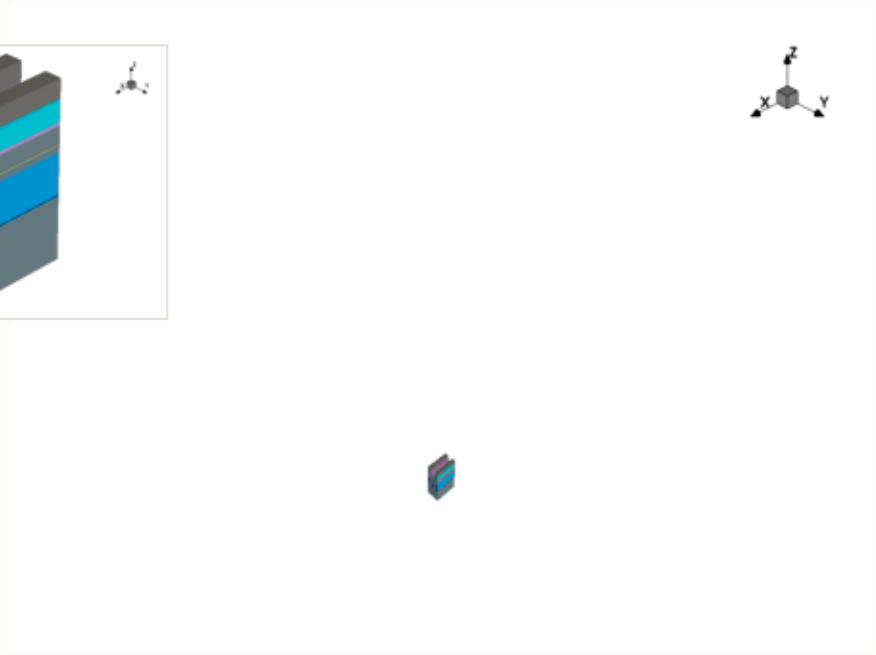
etch_strip · Resist

0.169	194.6	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



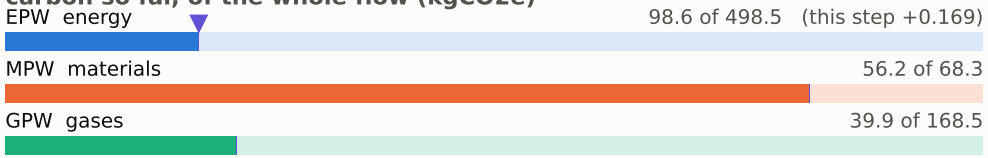


this step

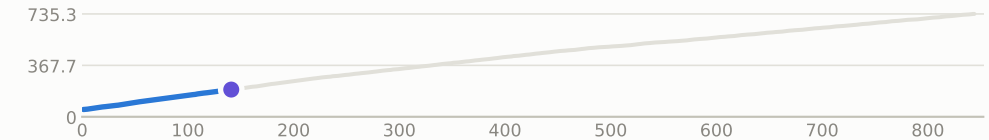
process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

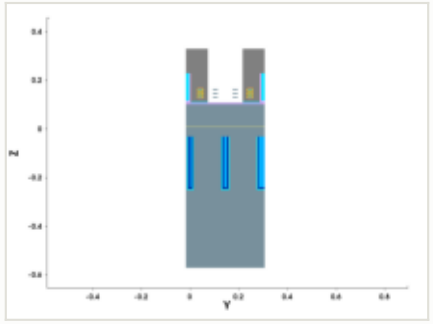
carbon so far, of the whole flow (kgCO2e)



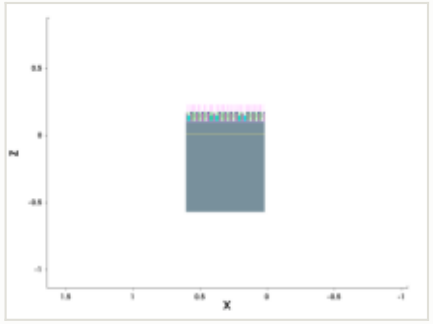
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



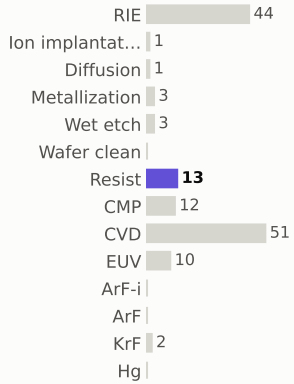
y cut at 0.184 µm



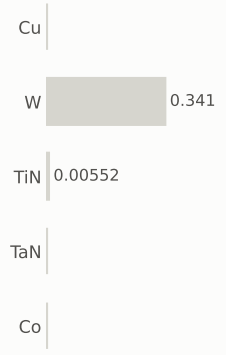
mask for this step

no mask for this step

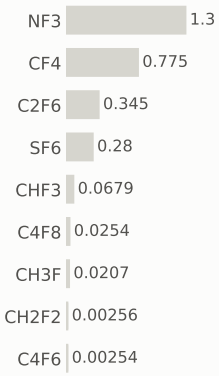
steps by process area



materials deposited (g)



gases emitted (g)



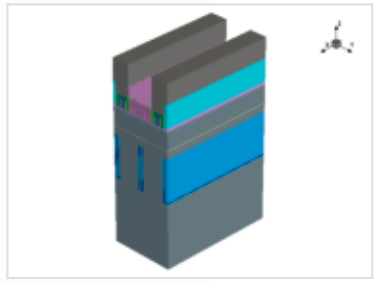
Step #1560: CVD

depo_fill · CVD

1.31	195.9	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

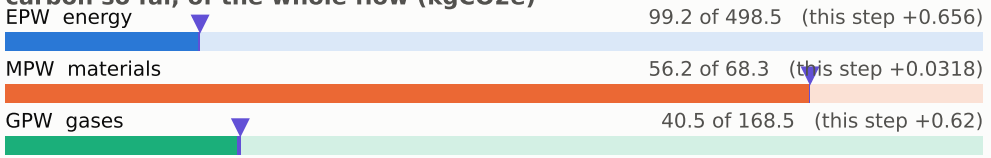


this step

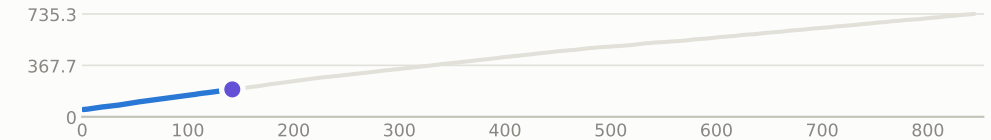
process area	CVD
material	—
thickness	-0.053 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

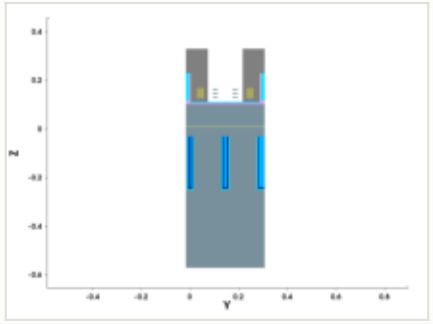
carbon so far, of the whole flow (kgCO2e)



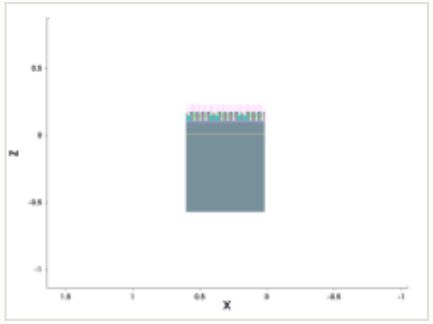
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



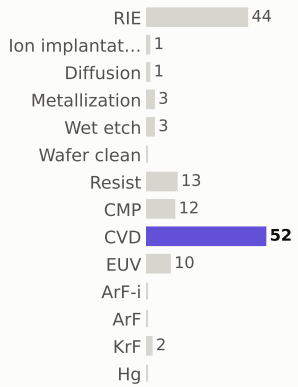
y cut at 0.184 µm



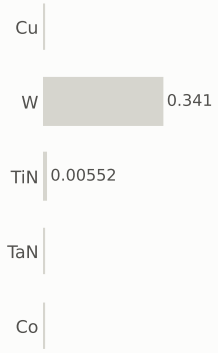
mask for this step

no mask for this step

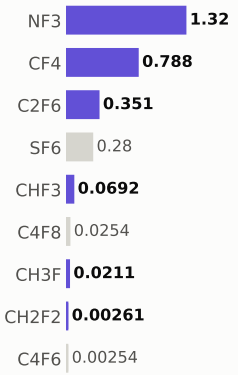
steps by process area



materials deposited (g)



gases emitted (g)



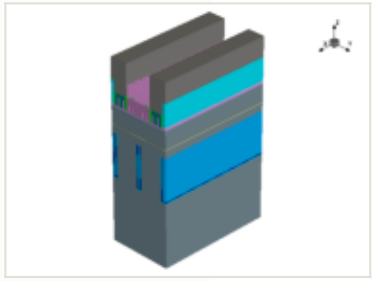
Step #1570: CVD

depo_isotropic · CVD

1.31	197.2	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

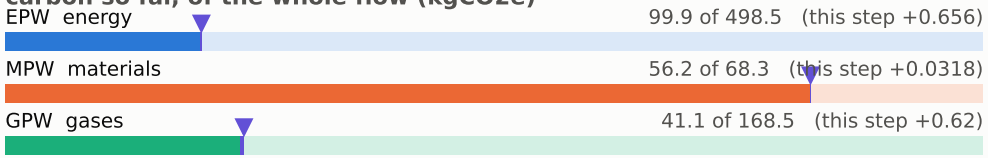


this step

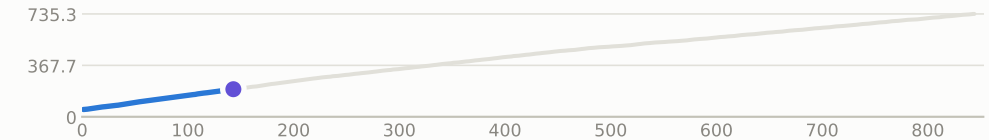
process area	CVD
material	—
thickness	0.0028 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

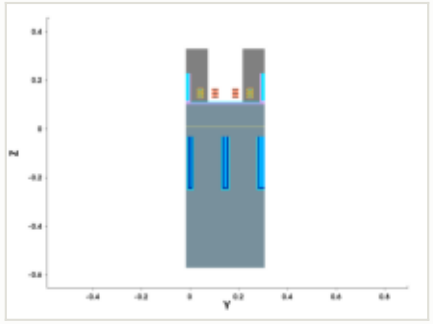
carbon so far, of the whole flow (kgCO2e)



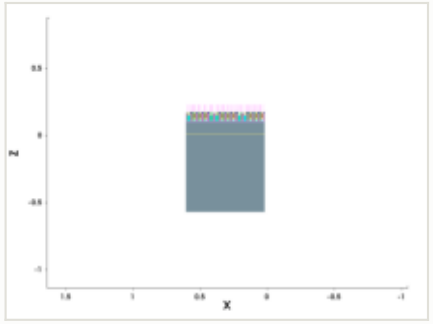
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



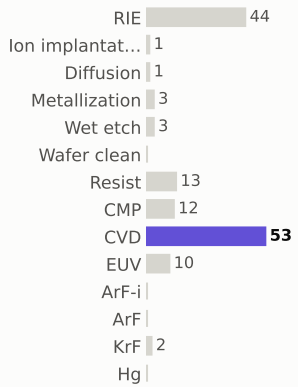
y cut at 0.184 µm



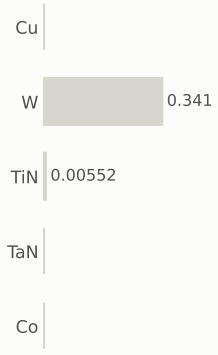
mask for this step

no mask for this step

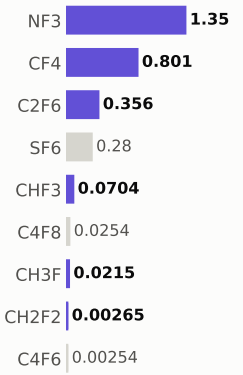
steps by process area



materials deposited (g)



gases emitted (g)



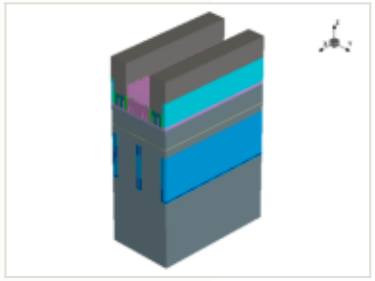
Step #1580: Wet_etch

etch_isotropic · Wet_etch

0.169	197.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

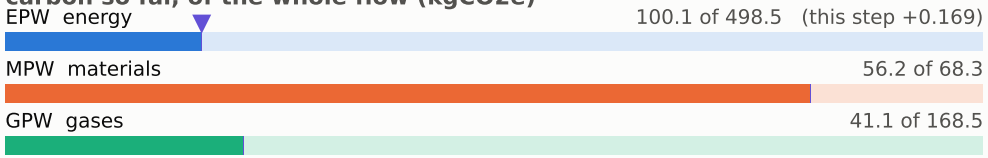


this step

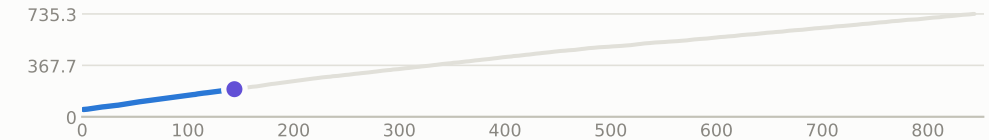
process area	Wet_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

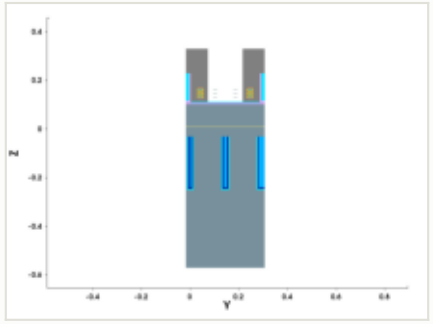
carbon so far, of the whole flow (kgCO2e)



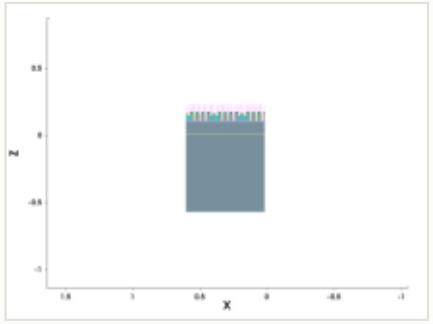
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



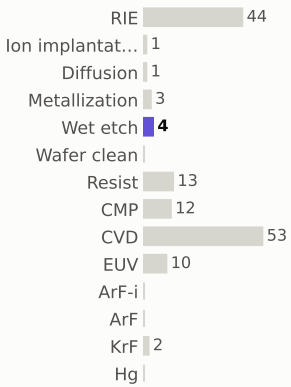
y cut at 0.184 µm



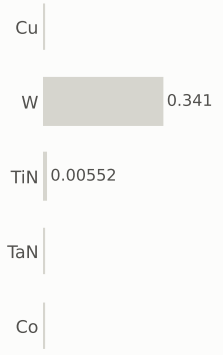
mask for this step

no mask for this step

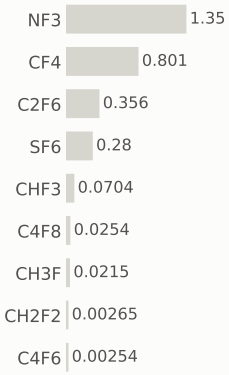
steps by process area



materials deposited (g)



gases emitted (g)



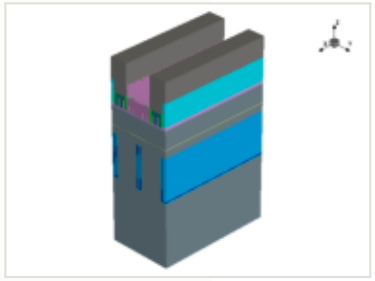
Step #1590: CVD

depo_isotropic · CVD

1.31	198.7	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

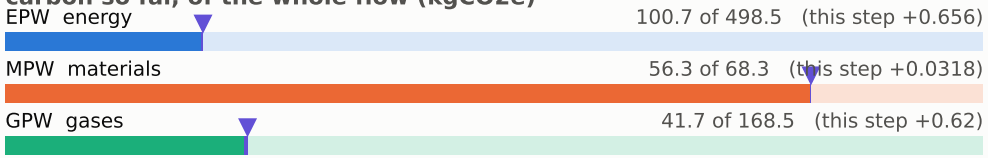
zoomed to fit



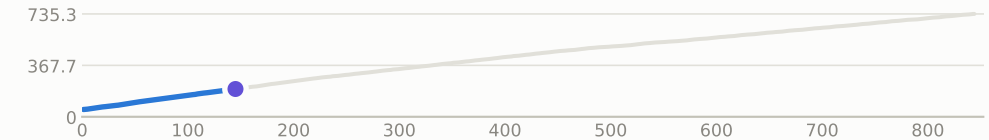
this step

process area	CVD	EPW	0.656 kgCO2e
material	—	EPW tool	0.469 kgCO2e
thickness	0.0015 µm	EPW overhead	0.187 kgCO2e
material GWP	—	MPW	0.0318 kgCO2e (gas 0.0318)
input gases	NF3 2.54 g	GPW	0.62 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**1.31 kgCO2e**

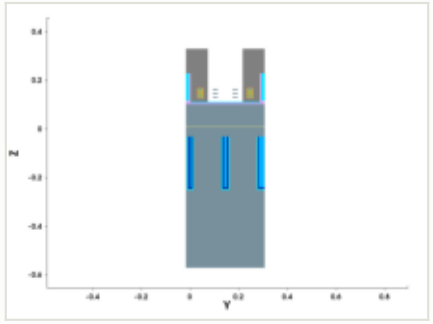
carbon so far, of the whole flow (kgCO2e)



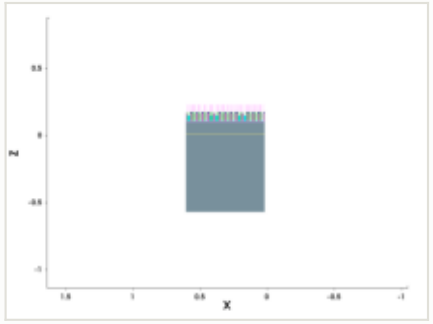
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



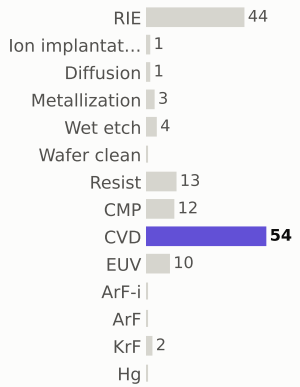
y cut at 0.184 µm



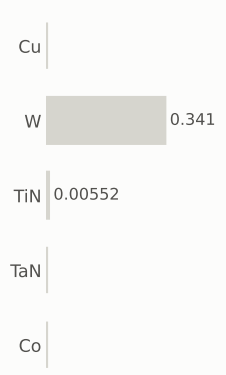
mask for this step

no mask for this step

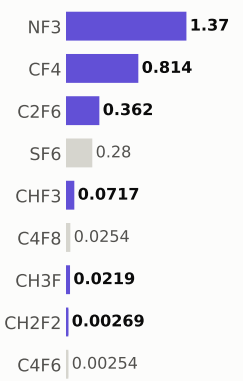
steps by process area



materials deposited (g)



gases emitted (g)



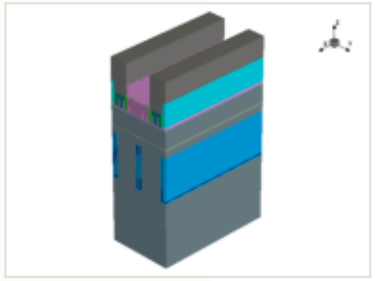
Step #1600: CVD

depo_isotropic · CVD

1.31	200.0	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

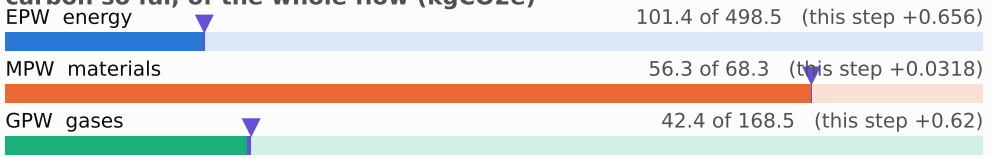


this step

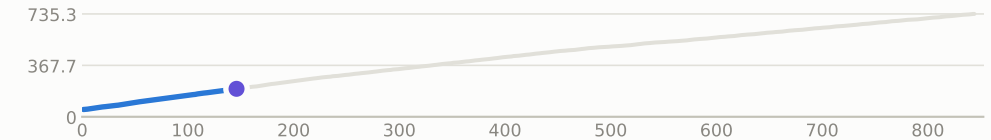
process area	CVD
material	—
thickness	0.0005 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

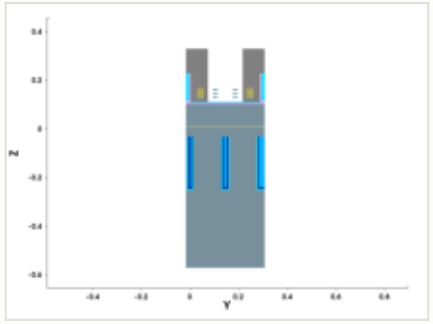
carbon so far, of the whole flow (kgCO2e)



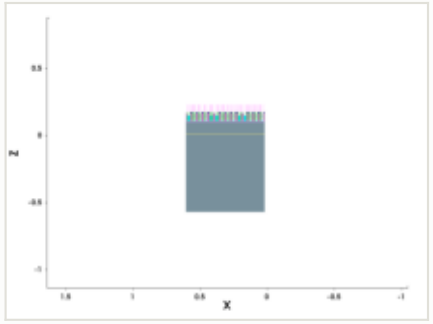
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



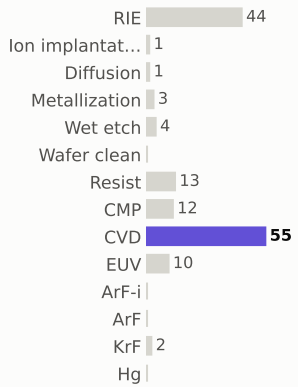
y cut at 0.184 µm



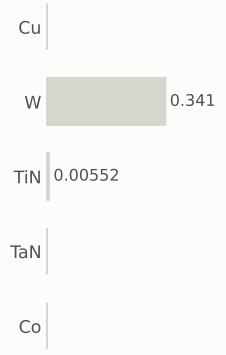
mask for this step

no mask for this step

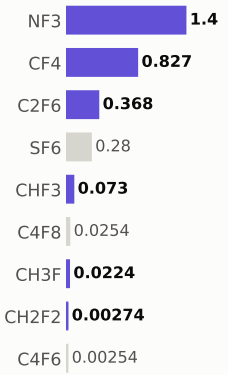
steps by process area



materials deposited (g)



gases emitted (g)



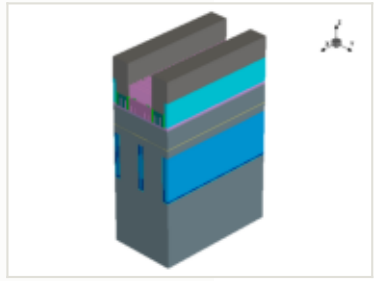
Step #1610: CVD

depo_isotropic · CVD

1.31	201.3	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

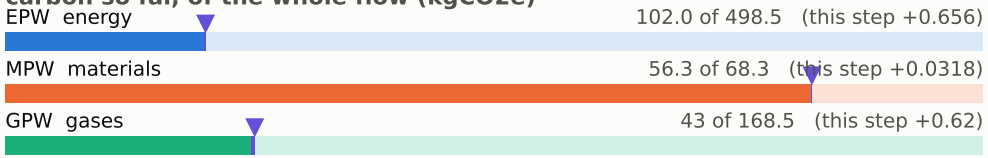
zoomed to fit



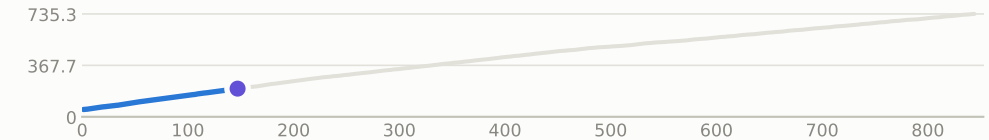
this step

process area	CVD	EPW	0.656 kgCO2e
material	—	EPW tool	0.469 kgCO2e
thickness	0.002 µm	EPW overhead	0.187 kgCO2e
material GWP	—	MPW	0.0318 kgCO2e (gas 0.0318)
input gases	NF3 2.54 g	GPW	0.62 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**1.31 kgCO2e**

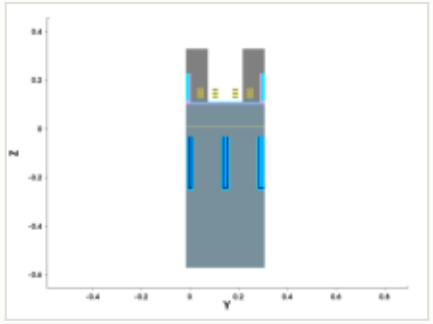
carbon so far, of the whole flow (kgCO2e)



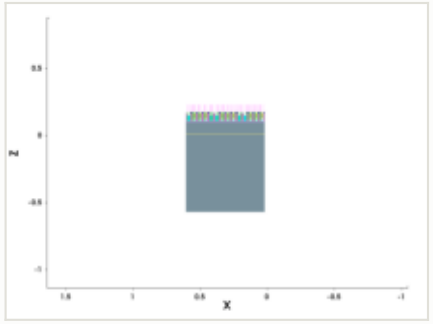
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



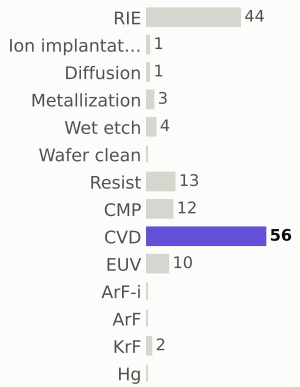
y cut at 0.184 µm



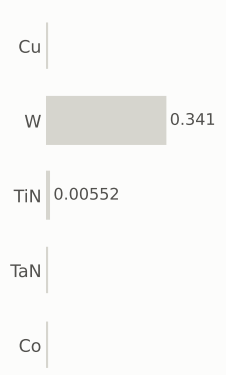
mask for this step

no mask for this step

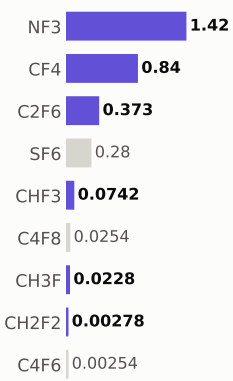
steps by process area



materials deposited (g)



gases emitted (g)



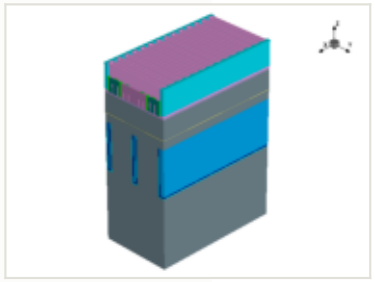
Step #1620: Resist

etch_strip · Resist

0.169	201.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

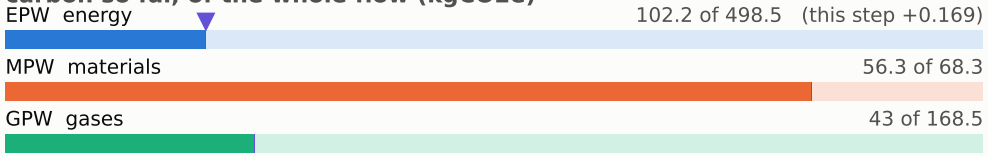


this step

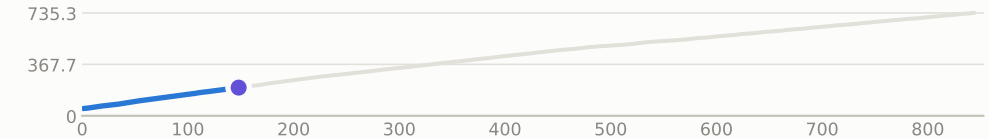
process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

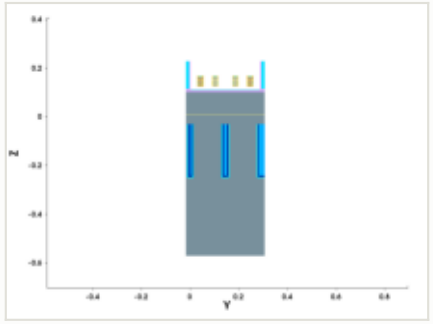
carbon so far, of the whole flow (kgCO2e)



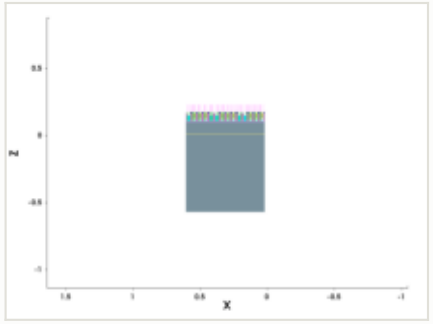
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



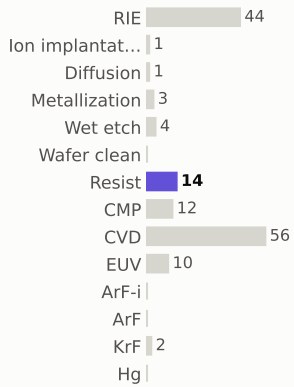
y cut at 0.184 µm



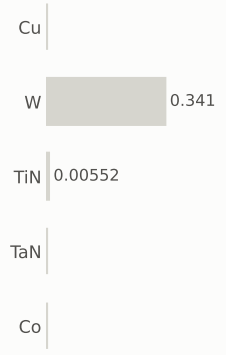
mask for this step

no mask for this step

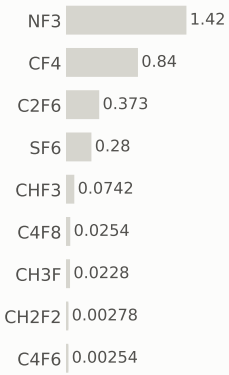
steps by process area



materials deposited (g)



gases emitted (g)



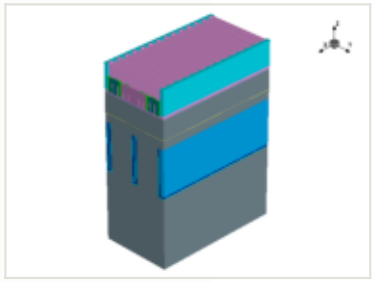
Step #1630: CVD

depo_isotropic · CVD

1.31	202.8	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

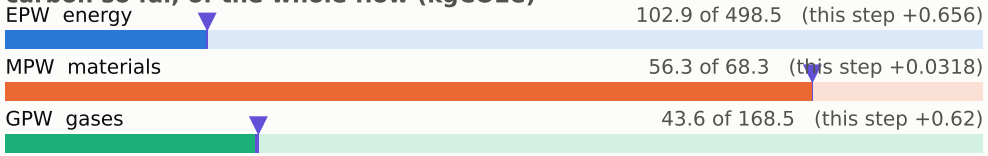
zoomed to fit



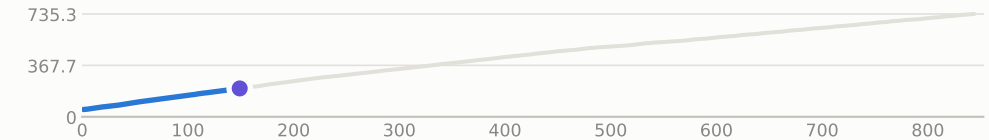
this step

process area	CVD	EPW	0.656 kgCO2e
material	TiN	EPW tool	0.469 kgCO2e
thickness	0.0005 µm	EPW overhead	0.187 kgCO2e
material GWP	43.9778 gCO2e/g	MPW	0.0318 kgCO2e (gas 0.0318)
input gases	NF3 2.54 g	GPW	0.62 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**1.31 kgCO2e**

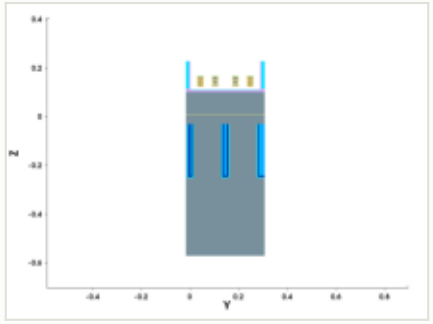
carbon so far, of the whole flow (kgCO2e)



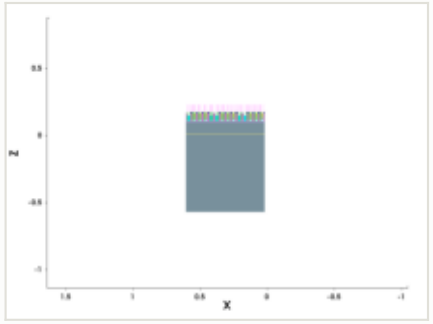
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



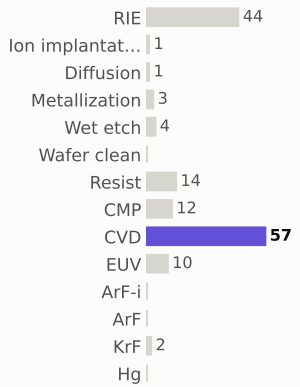
y cut at 0.184 µm



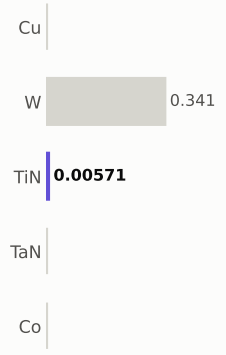
mask for this step

no mask for this step

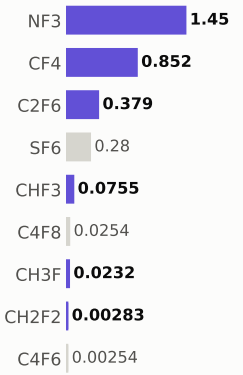
steps by process area



materials deposited (g)



gases emitted (g)



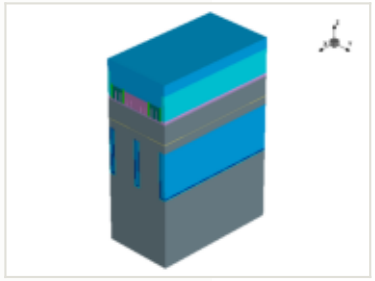
Step #1640: Metallization

depo_fill · Metallization

1.21	204.0	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

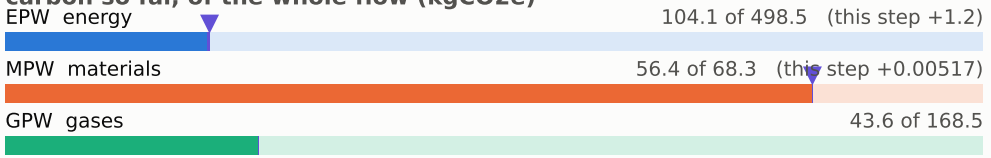


this step

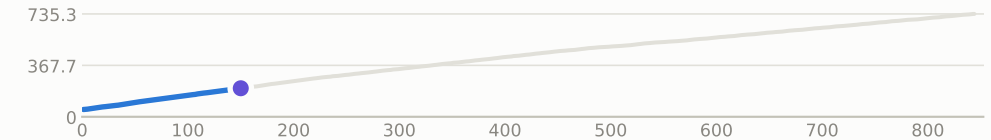
process area	Metallization
material	W
thickness	0.05 µm
material GWP	75.8149 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00517 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

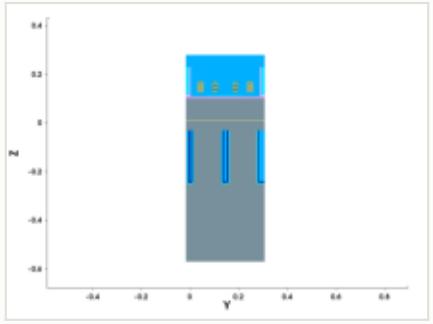
carbon so far, of the whole flow (kgCO2e)



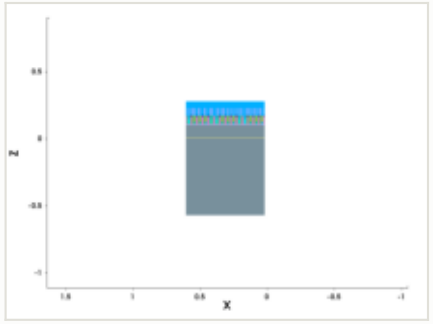
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



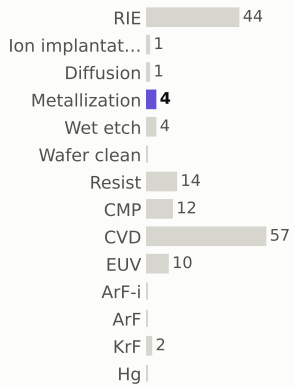
y cut at 0.184 µm



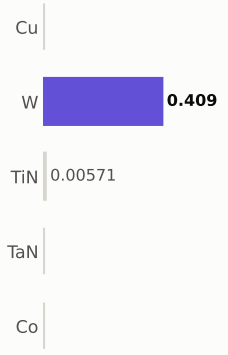
mask for this step

no mask for this step

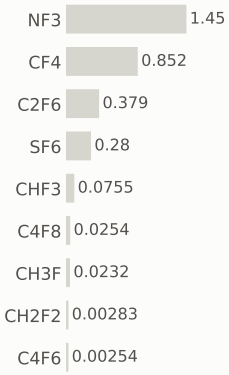
steps by process area



materials deposited (g)



gases emitted (g)



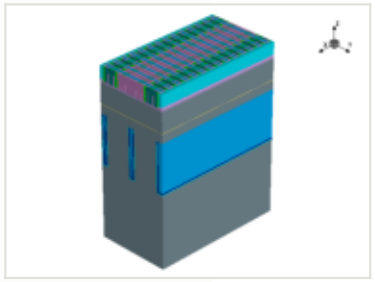
Step #1650: CMP

etch_cmp · CMP

0.169	204.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

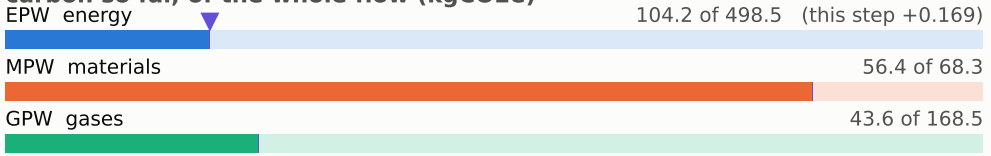


this step

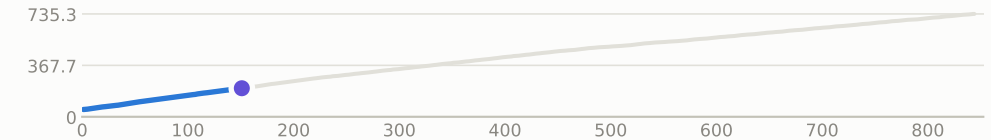
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

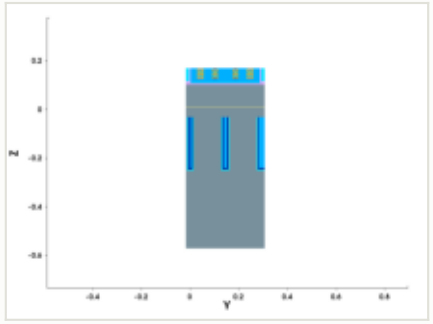
carbon so far, of the whole flow (kgCO2e)



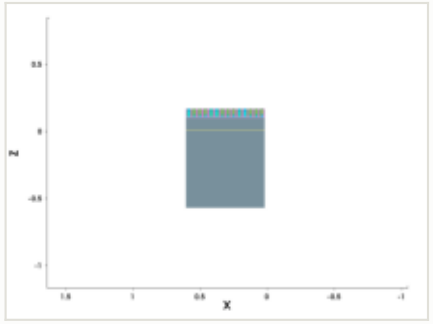
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



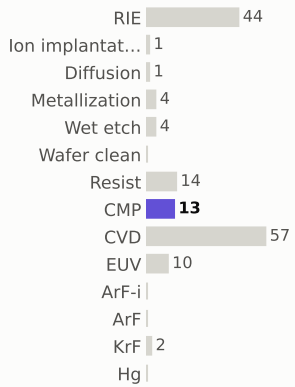
y cut at 0.184 µm



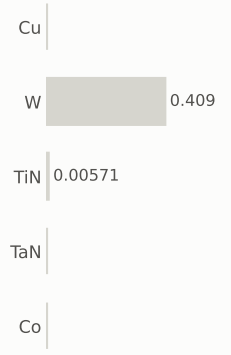
mask for this step

no mask for this step

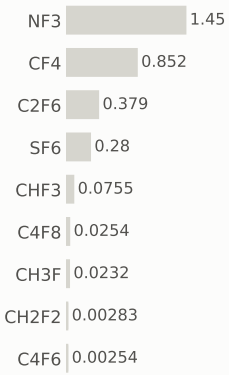
steps by process area



materials deposited (g)



gases emitted (g)



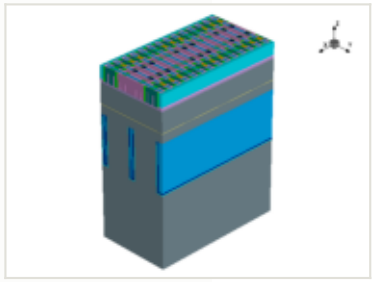
Step #1660: RIE

etch_height_dependent · RIE

1.04	205.2	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

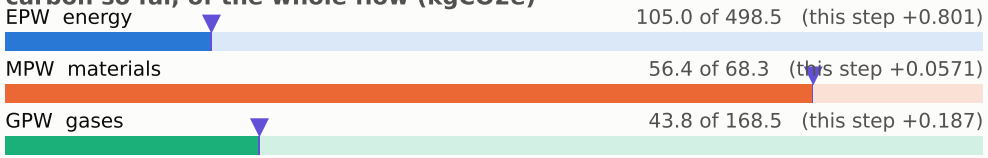
zoomed to fit



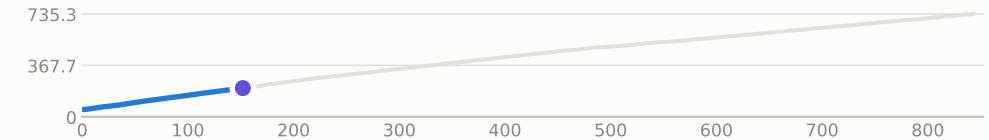
this step

process area	RIE	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

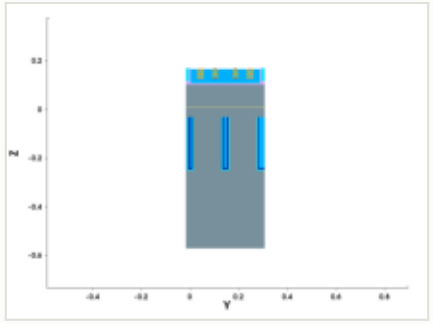
carbon so far, of the whole flow (kgCO2e)



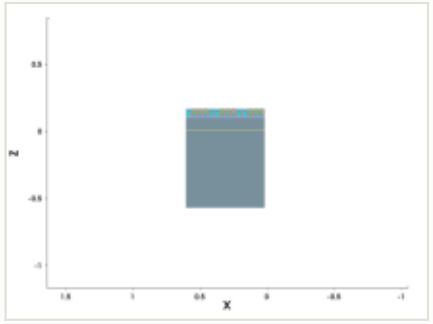
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



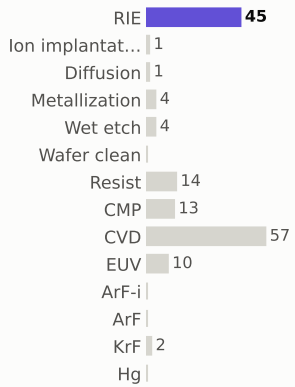
y cut at 0.184 µm



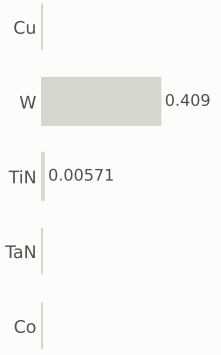
mask for this step

no mask for this step

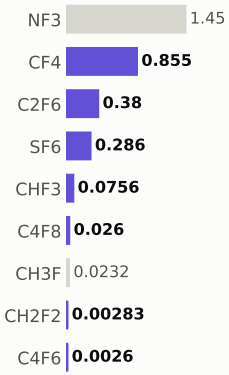
steps by process area



materials deposited (g)



gases emitted (g)



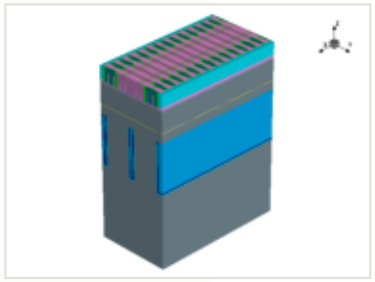
Step #1670: CVD

depo_fill · CVD

1.31	206.5	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

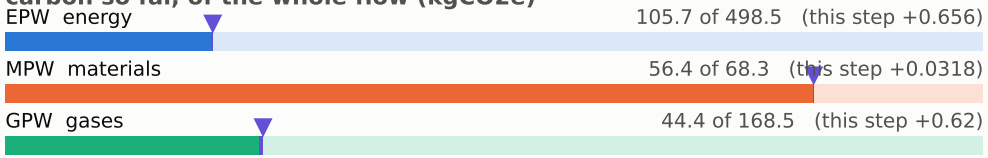


this step

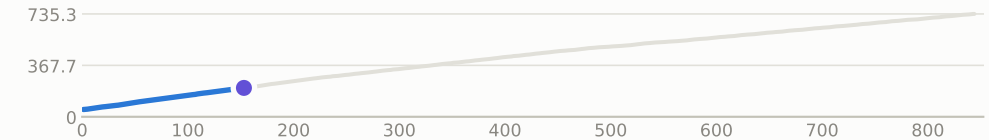
process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

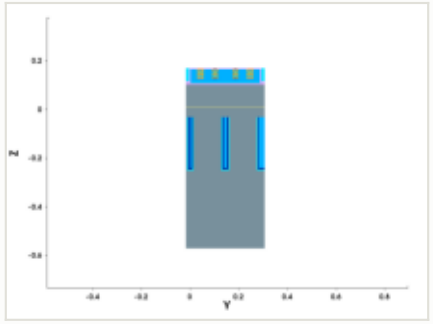
carbon so far, of the whole flow (kgCO2e)



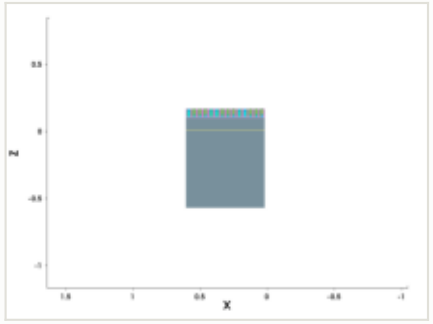
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



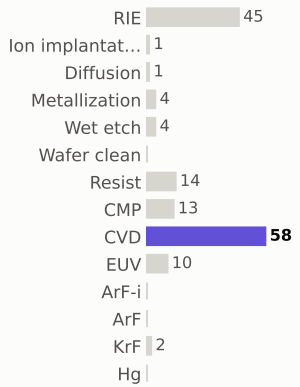
y cut at 0.184 µm



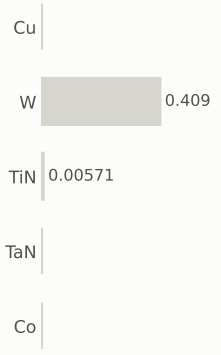
mask for this step

no mask for this step

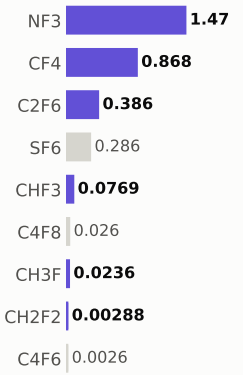
steps by process area



materials deposited (g)



gases emitted (g)



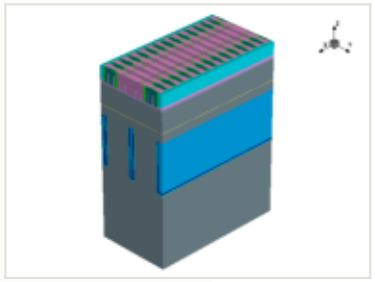
Step #1680: CMP

etch_cmp · CMP

0.169	206.7	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

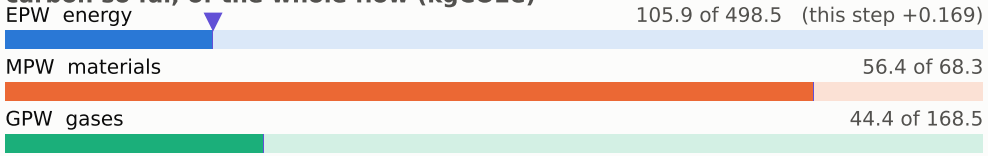


this step

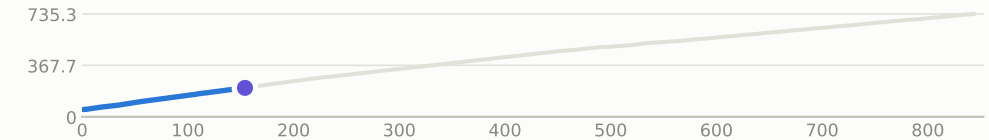
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

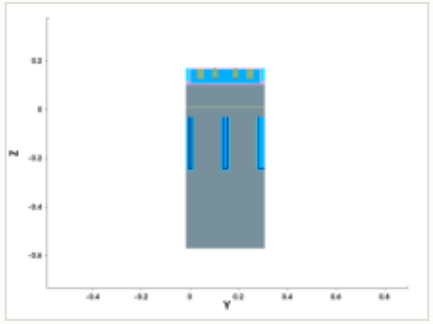
carbon so far, of the whole flow (kgCO2e)



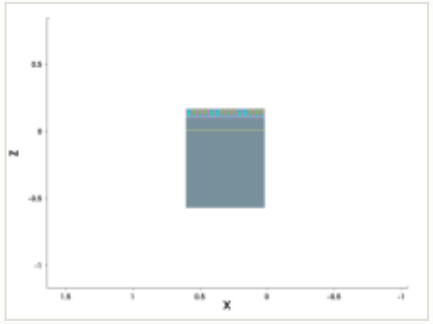
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



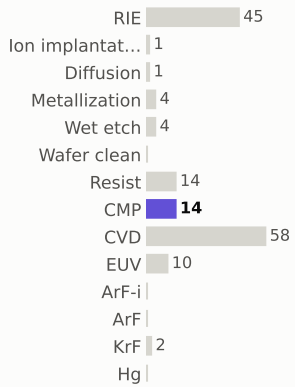
y cut at 0.184 µm



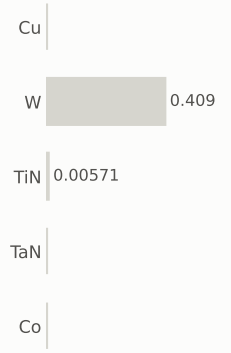
mask for this step

no mask for this step

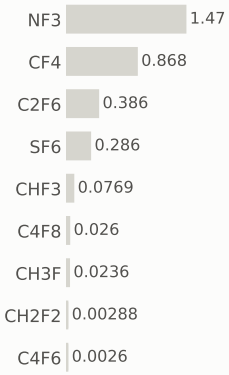
steps by process area



materials deposited (g)



gases emitted (g)



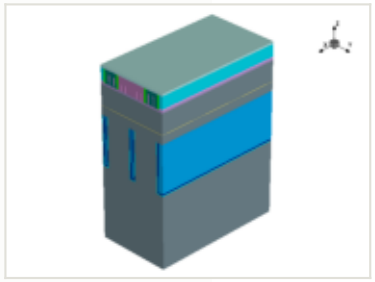
Step #1690: CVD

depo_anisotropic · CVD

1.31	208.0	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

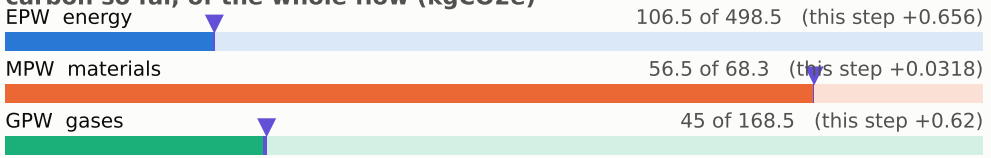


this step

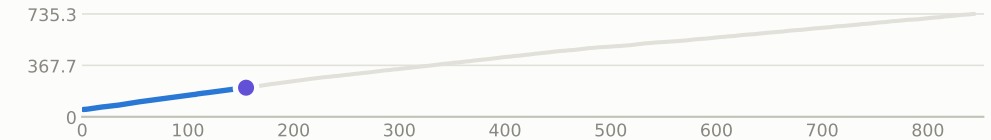
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

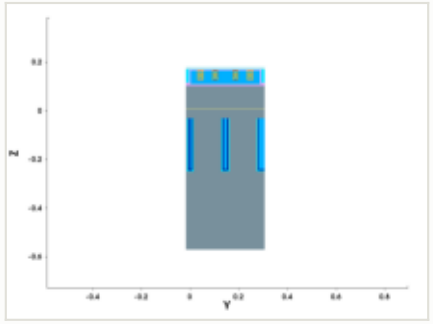
carbon so far, of the whole flow (kgCO2e)



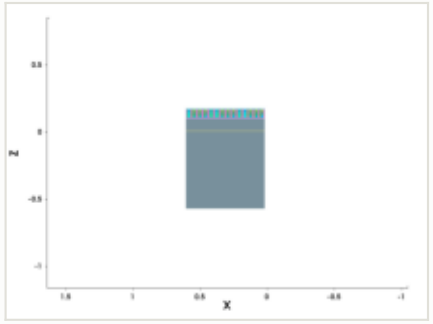
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



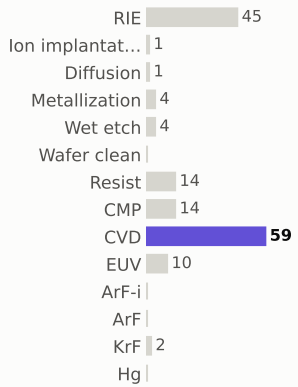
y cut at 0.184 µm



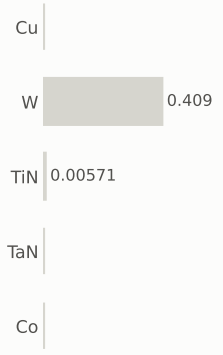
mask for this step

no mask for this step

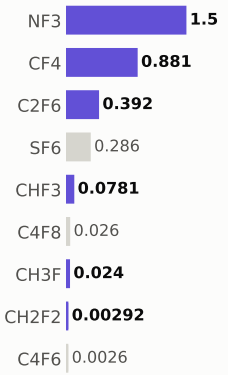
steps by process area



materials deposited (g)



gases emitted (g)



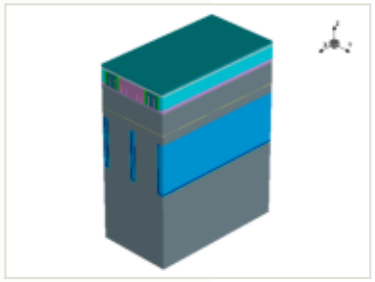
Step #1700: CVD

depo_anisotropic · CVD

1.31	209.3	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

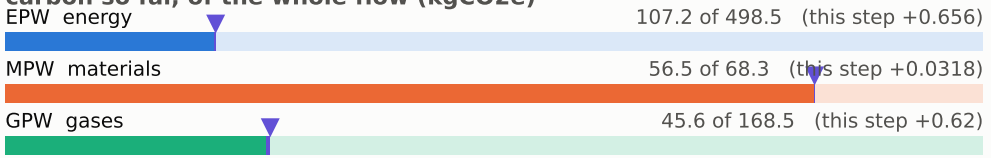


this step

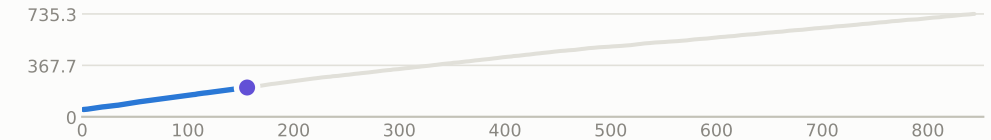
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

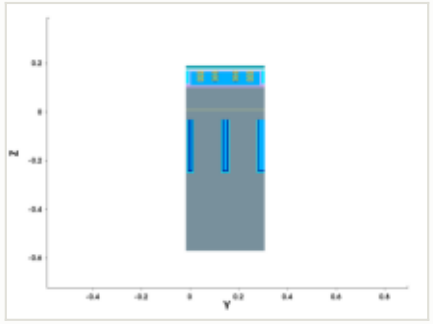
carbon so far, of the whole flow (kgCO2e)



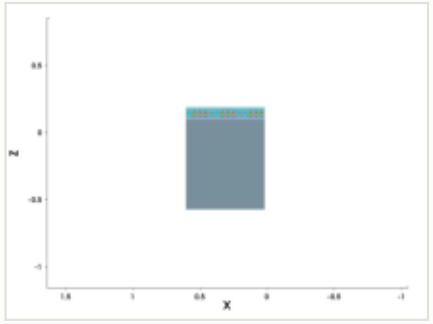
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



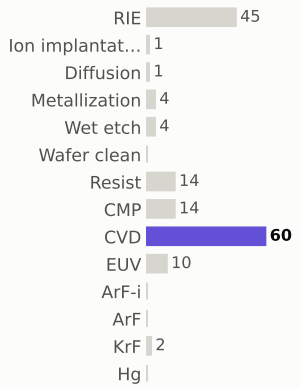
y cut at 0.184 µm



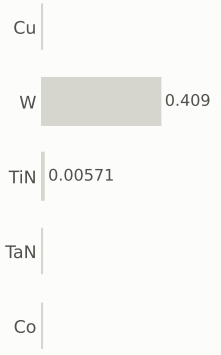
mask for this step

no mask for this step

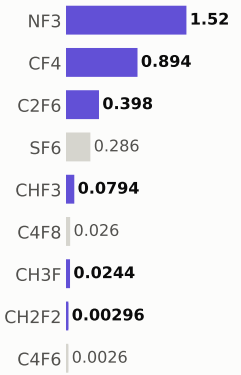
steps by process area



materials deposited (g)



gases emitted (g)



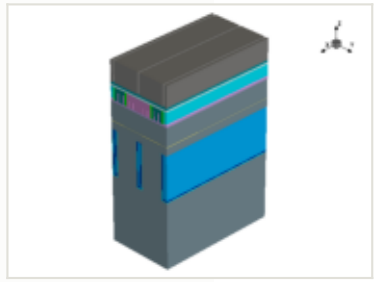
Step #1720: EUV

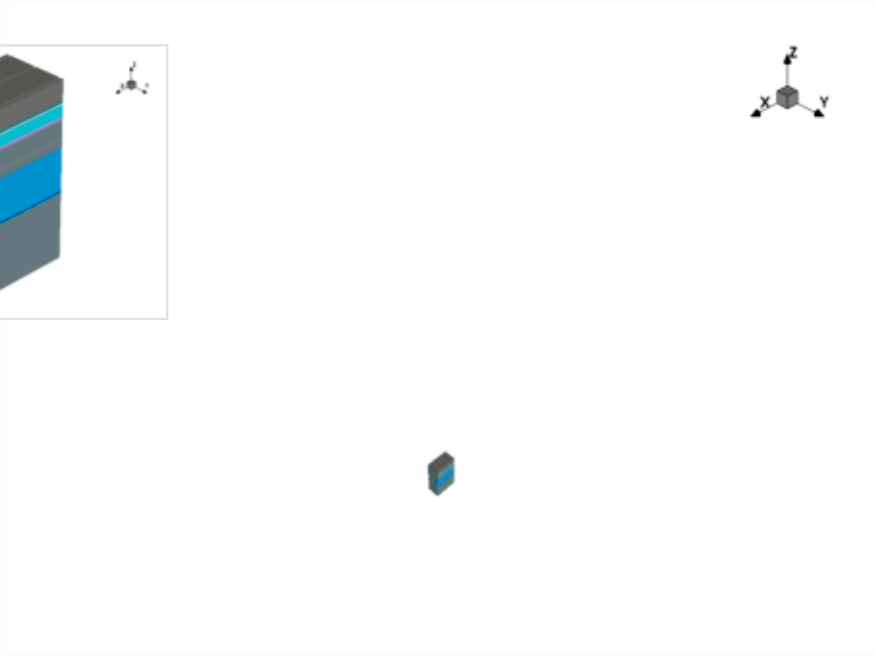
litho_patterning · EUV

1.96	211.3	0.27%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



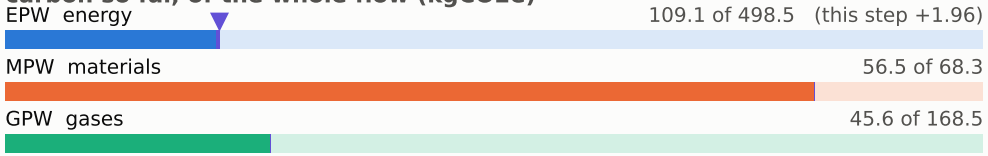


this step

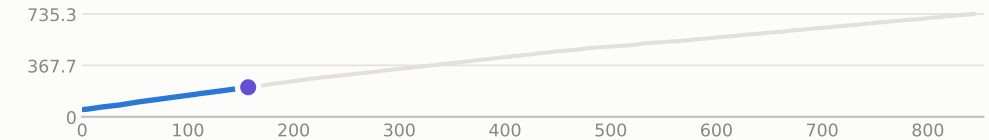
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

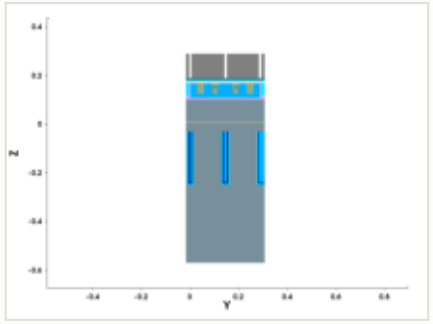
carbon so far, of the whole flow (kgCO2e)



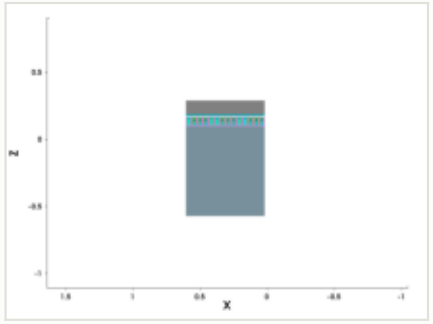
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm

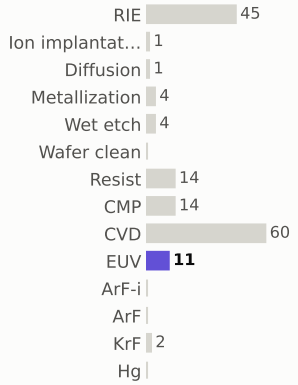


y cut at 0.184 µm

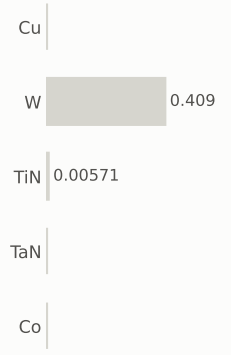


mask: layer 5:0 (GDSII)

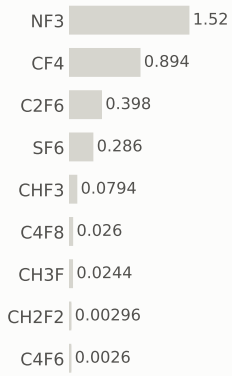
steps by process area



materials deposited (g)



gases emitted (g)



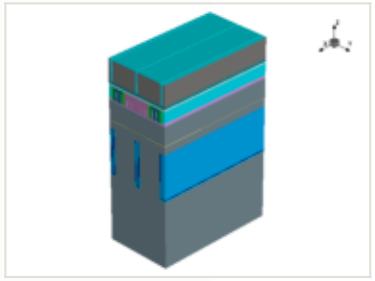
Step #1730: CVD

depo_isotropic · CVD

1.31	212.6	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

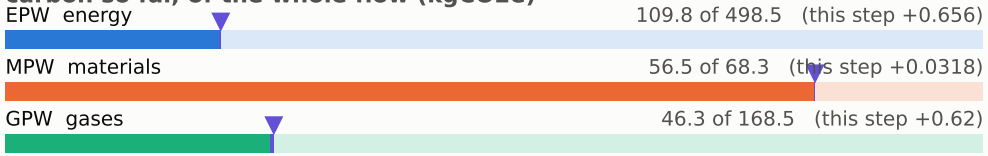
zoomed to fit



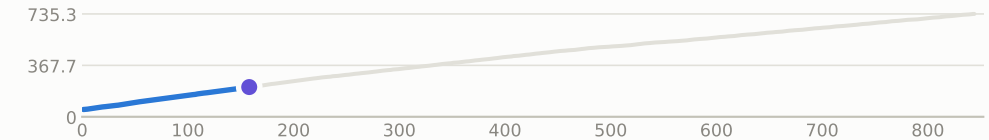
this step

process area	CVD	EPW	0.656 kgCO2e
material	—	EPW tool	0.469 kgCO2e
thickness	0.0025 µm	EPW overhead	0.187 kgCO2e
material GWP	—	MPW	0.0318 kgCO2e (gas 0.0318)
input gases	NF3 2.54 g	GPW	0.62 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**1.31 kgCO2e**

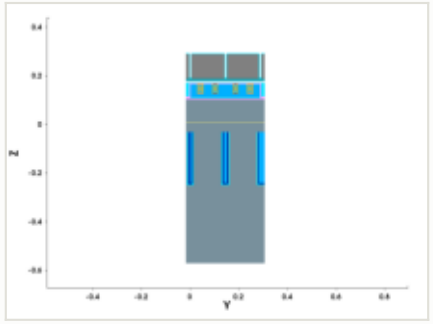
carbon so far, of the whole flow (kgCO2e)



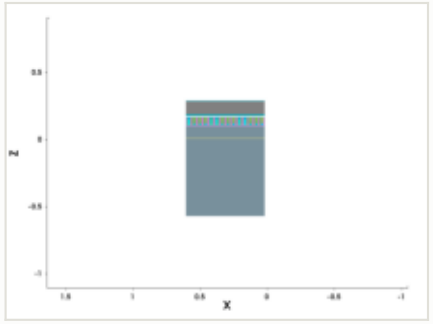
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



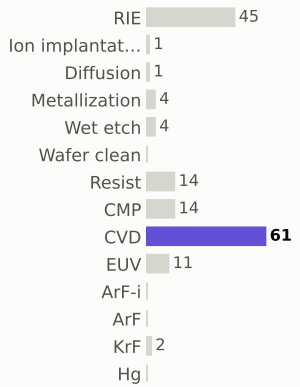
y cut at 0.184 µm



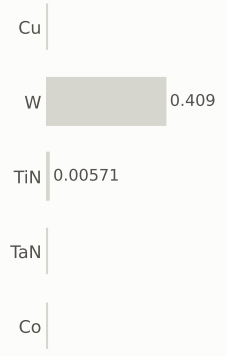
mask for this step

no mask for this step

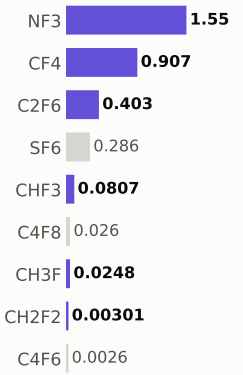
steps by process area



materials deposited (g)



gases emitted (g)



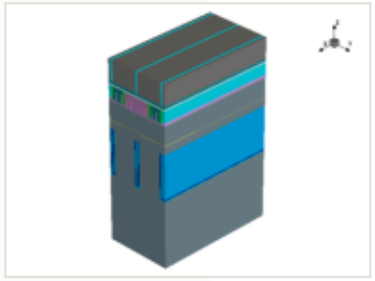
Step #1740: RIE

etch_anisotropic · RIE

1.04	213.6	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

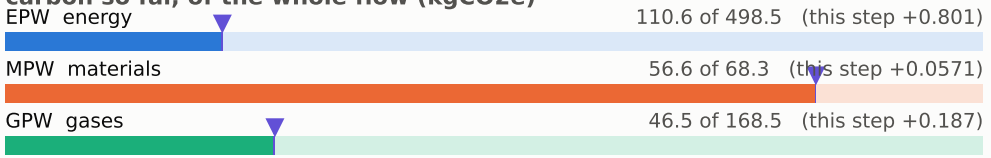


this step

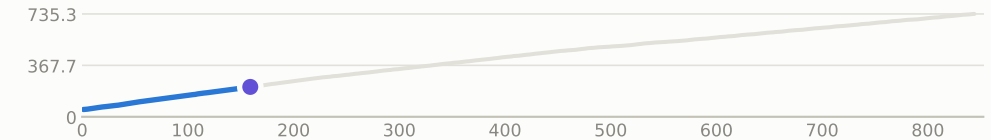
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

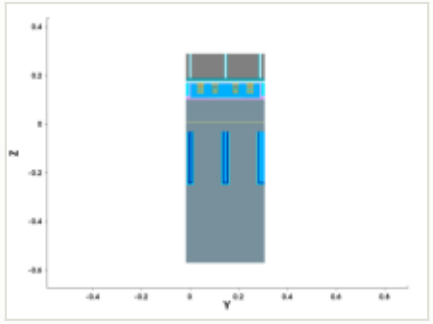
carbon so far, of the whole flow (kgCO2e)



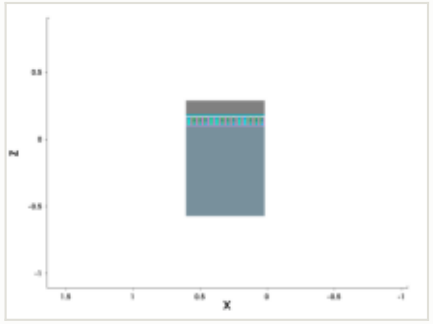
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



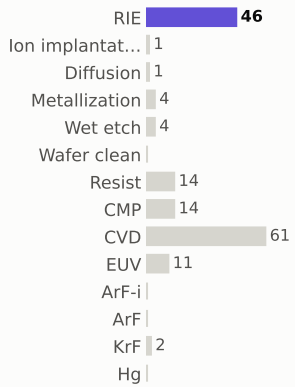
y cut at 0.184 µm



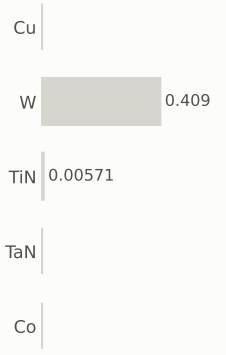
mask for this step

no mask for this step

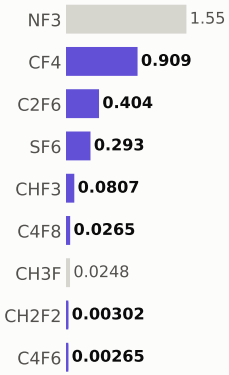
steps by process area



materials deposited (g)



gases emitted (g)



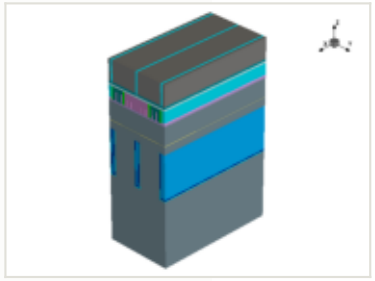
Step #1750: RIE

etch_anisotropic · RIE

1.04	214.7	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

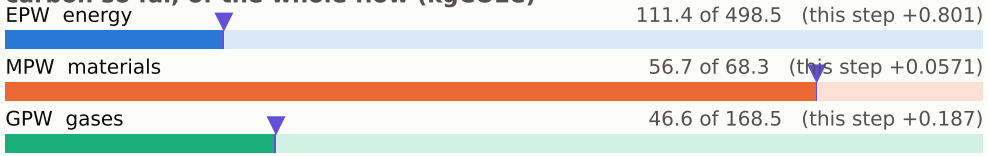


this step

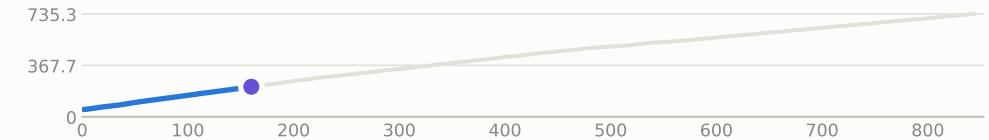
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

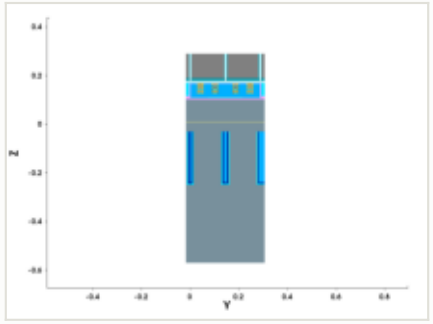
carbon so far, of the whole flow (kgCO2e)



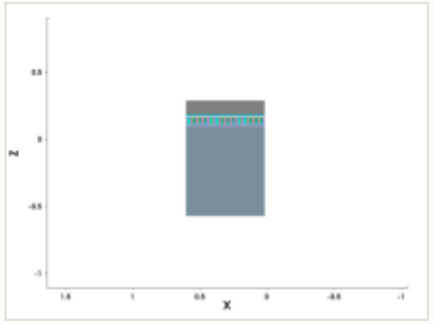
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



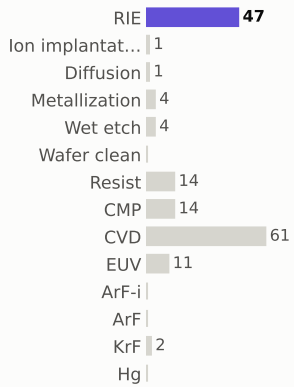
y cut at 0.184 µm



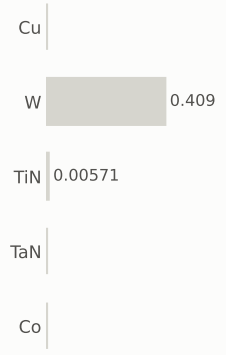
mask for this step

no mask for this step

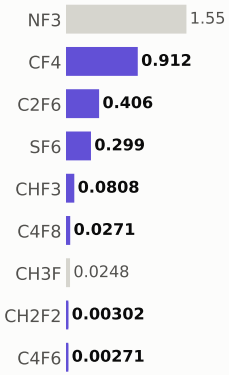
steps by process area



materials deposited (g)



gases emitted (g)



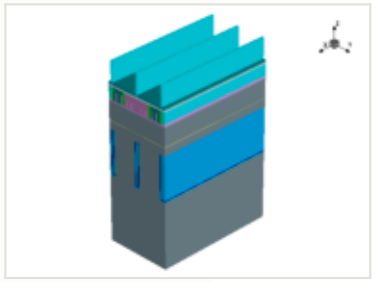
Step #1760: Resist

etch_strip · Resist

0.169	214.8	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

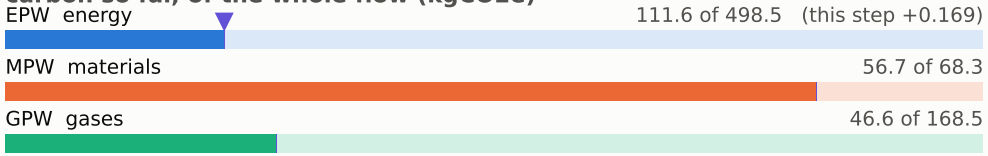


this step

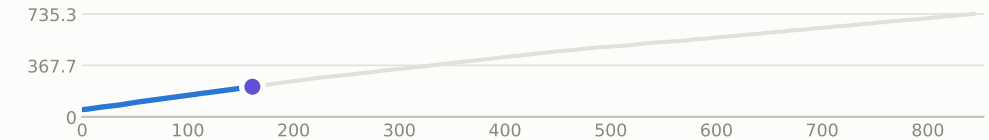
process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

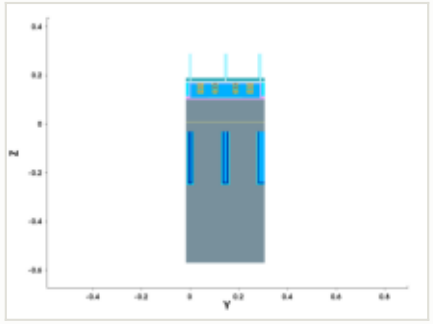
carbon so far, of the whole flow (kgCO2e)



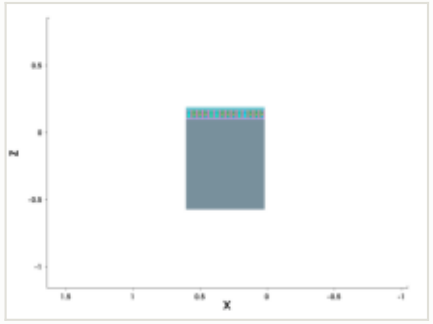
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



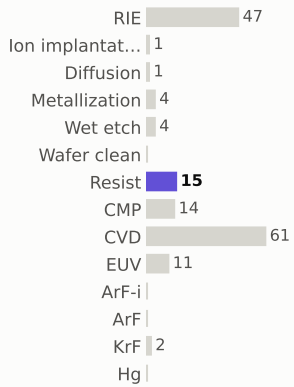
y cut at 0.184 µm



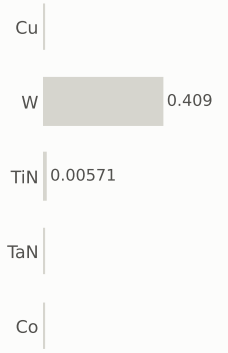
mask for this step

no mask for this step

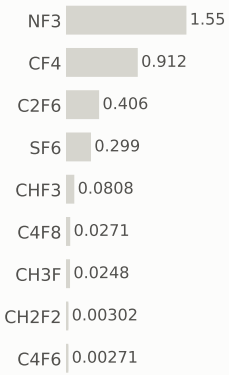
steps by process area



materials deposited (g)



gases emitted (g)



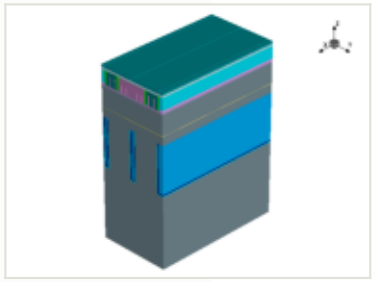
Step #1770: Resist

etch_strip · Resist

0.169	215.0	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

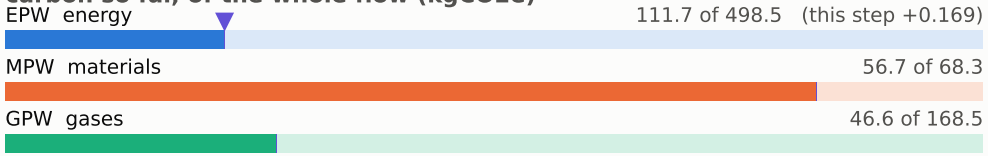


this step

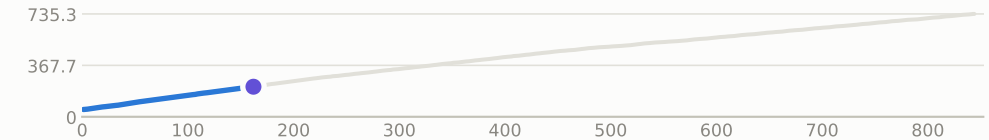
process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

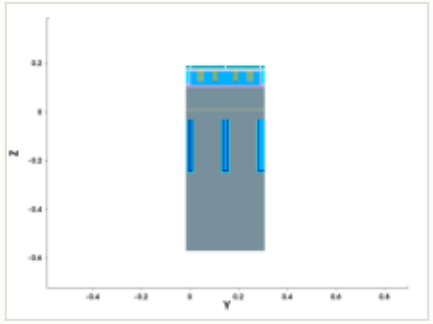
carbon so far, of the whole flow (kgCO2e)



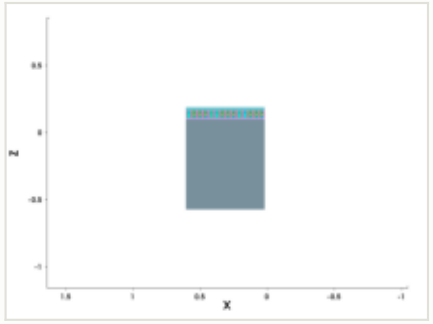
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



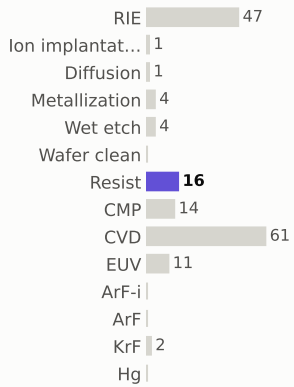
y cut at 0.184 µm



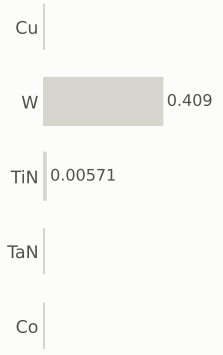
mask for this step

no mask for this step

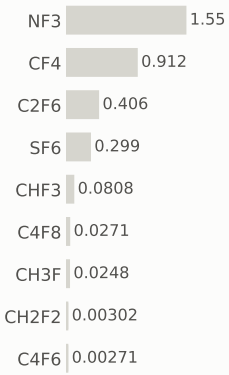
steps by process area



materials deposited (g)



gases emitted (g)



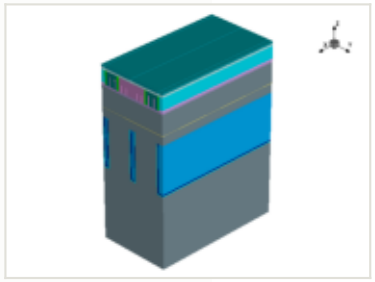
Step #1780: RIE

etch_anisotropic · RIE

1.04	216.1	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

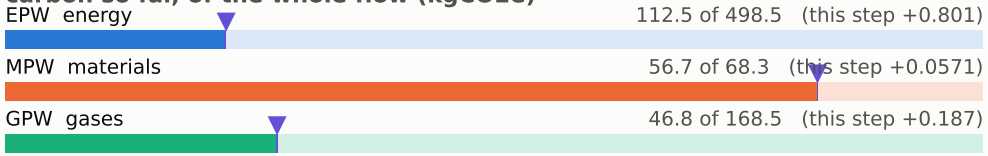


this step

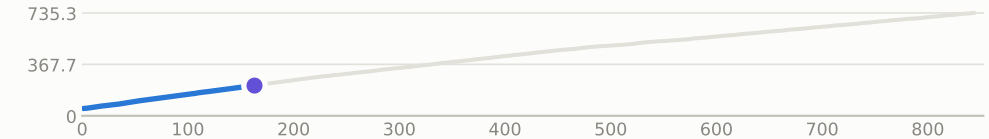
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

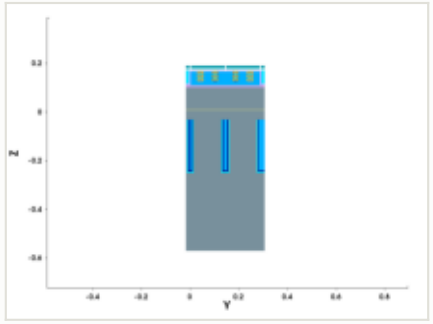
carbon so far, of the whole flow (kgCO2e)



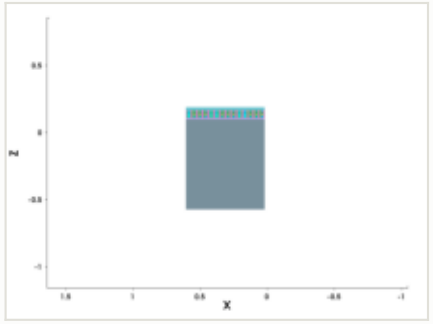
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



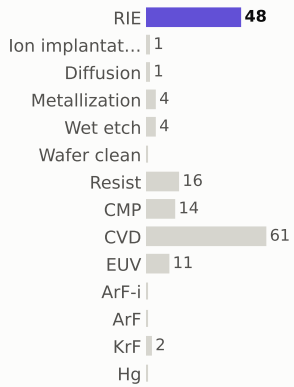
y cut at 0.184 µm



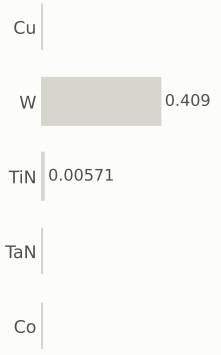
mask for this step

no mask for this step

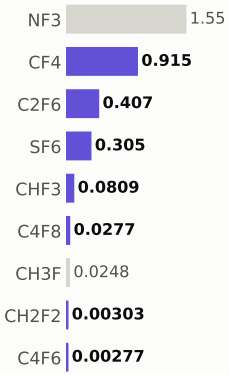
steps by process area



materials deposited (g)



gases emitted (g)



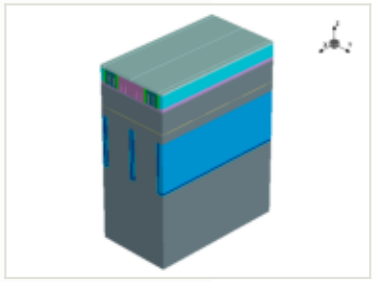
Step #1790: Resist

etch_strip · Resist

0.169	216.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

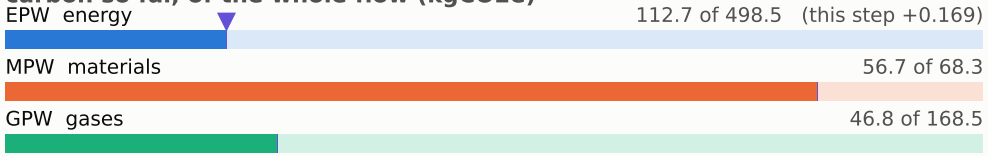


this step

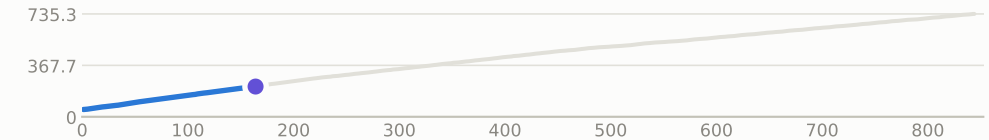
process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

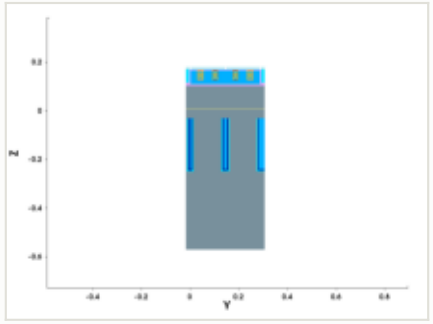
carbon so far, of the whole flow (kgCO2e)



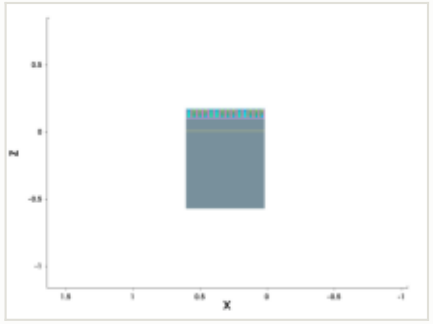
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



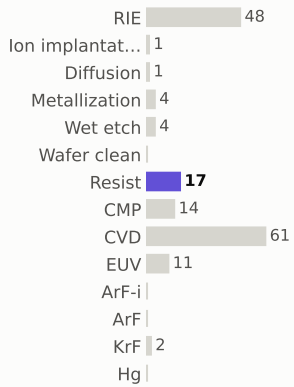
y cut at 0.184 µm



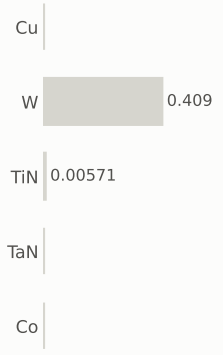
mask for this step

no mask for this step

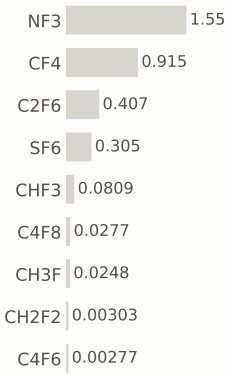
steps by process area



materials deposited (g)



gases emitted (g)



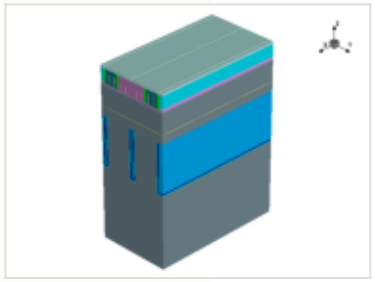
Step #1800: RIE

etch_anisotropic · RIE

1.04	217.3	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

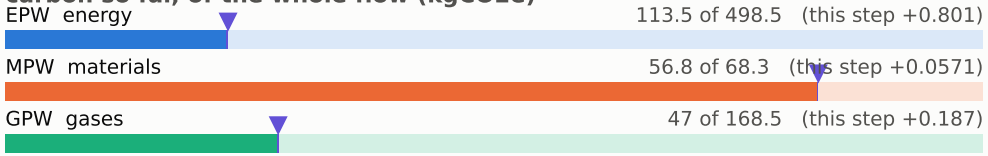
zoomed to fit



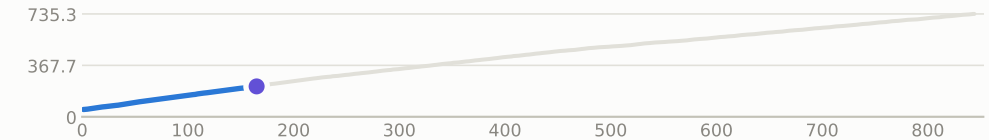
this step

process area	RIE	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

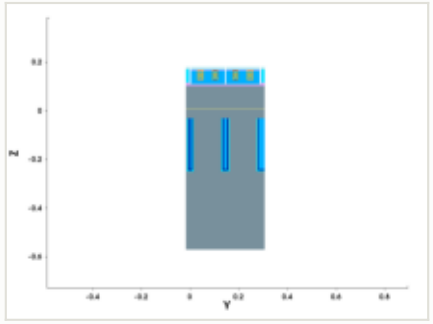
carbon so far, of the whole flow (kgCO2e)



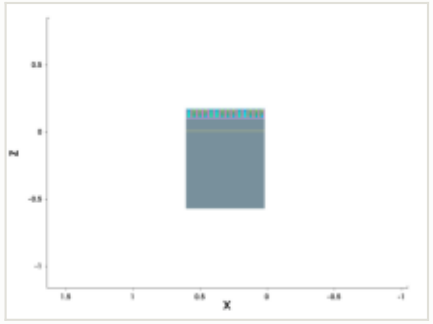
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



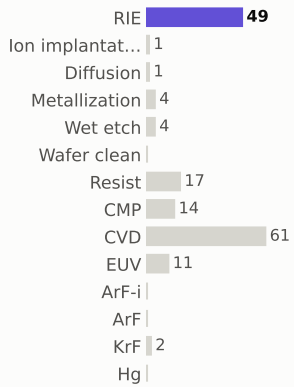
y cut at 0.184 µm



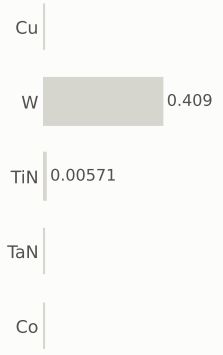
mask for this step

no mask for this step

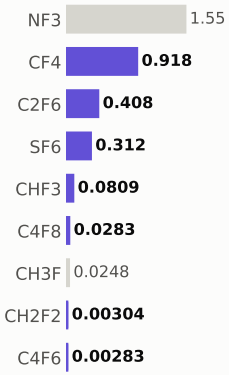
steps by process area



materials deposited (g)



gases emitted (g)



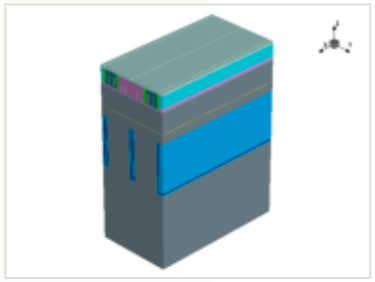
Step #1810: CVD

depo_fill · CVD

1.31	218.6	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

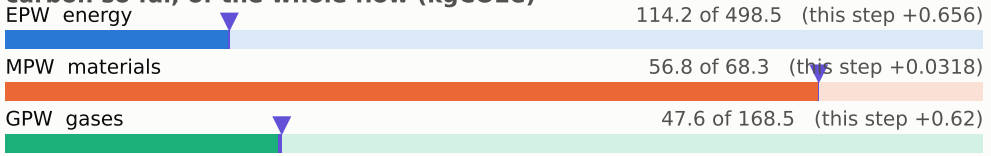


this step

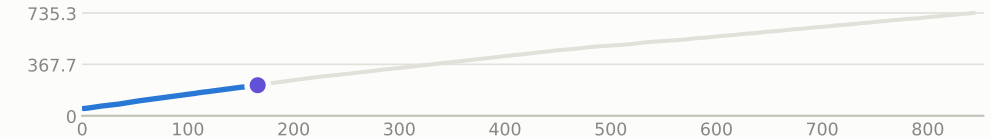
process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

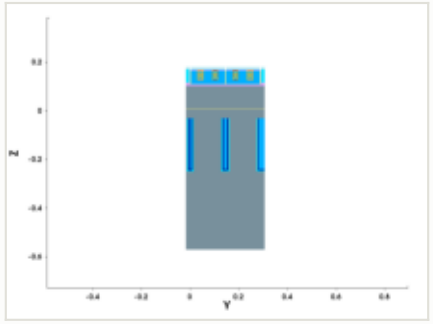
carbon so far, of the whole flow (kgCO2e)



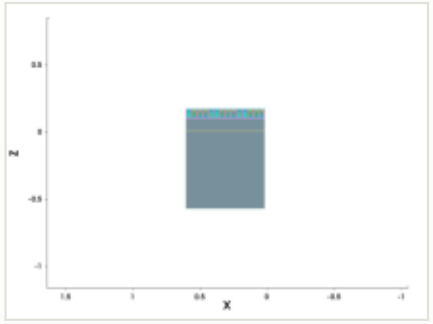
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



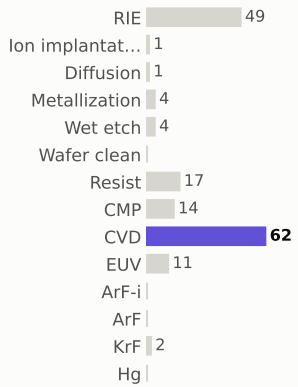
y cut at 0.184 µm



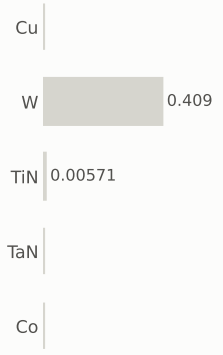
mask for this step

no mask for this step

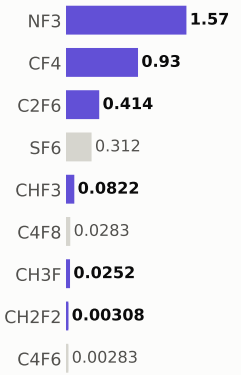
steps by process area



materials deposited (g)



gases emitted (g)



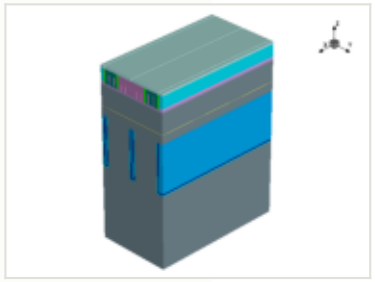
Step #1820: CVD

depo_fill · CVD

1.31	219.9	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

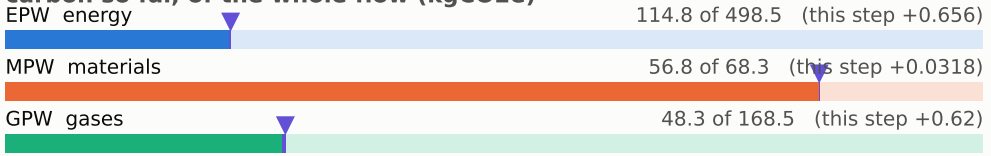


this step

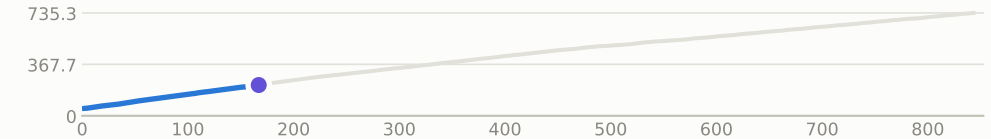
process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

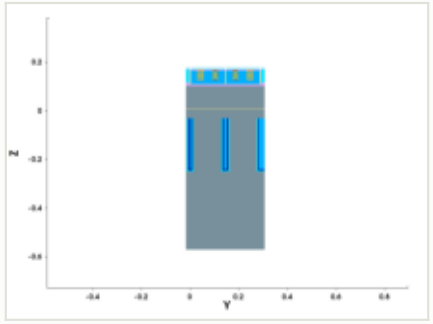
carbon so far, of the whole flow (kgCO2e)



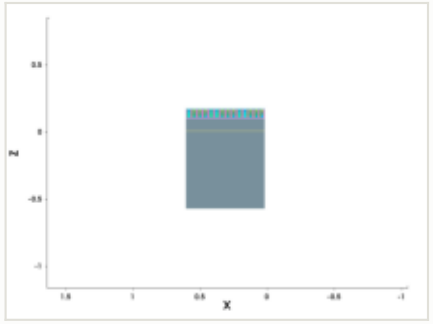
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



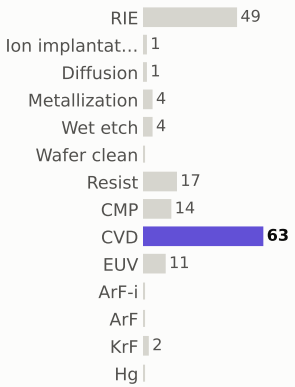
y cut at 0.184 µm



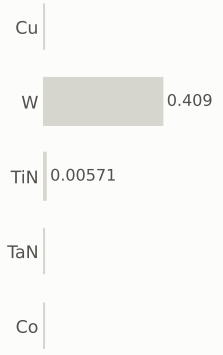
mask for this step

no mask for this step

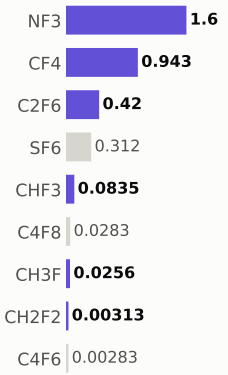
steps by process area



materials deposited (g)



gases emitted (g)



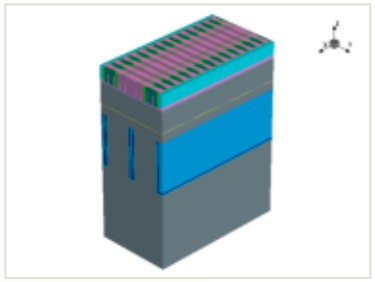
Step #1830: Resist

etch_strip · Resist

0.169	220.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

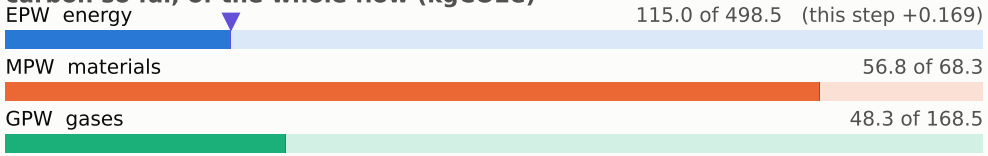


this step

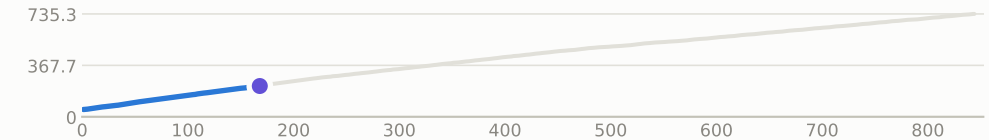
process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

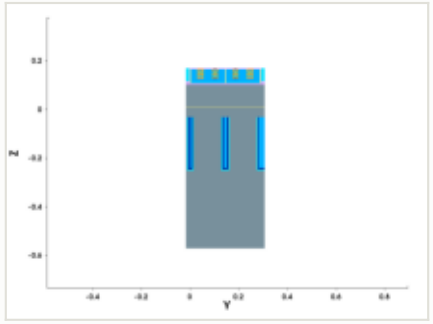
carbon so far, of the whole flow (kgCO2e)



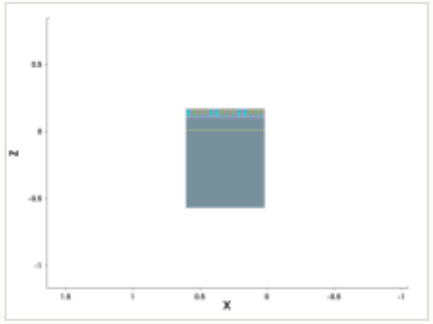
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



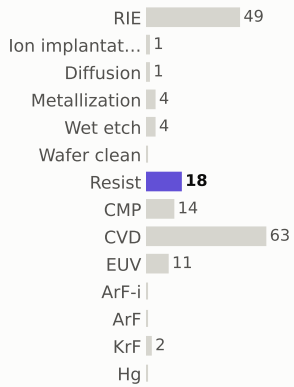
y cut at 0.184 µm



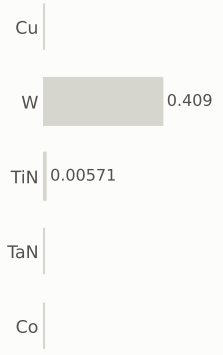
mask for this step

no mask for this step

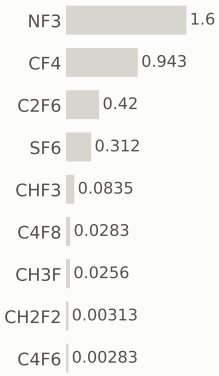
steps by process area



materials deposited (g)



gases emitted (g)



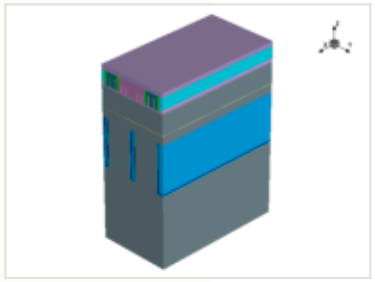
Step #1850: CVD

depo_fill · CVD

1.31	221.4	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

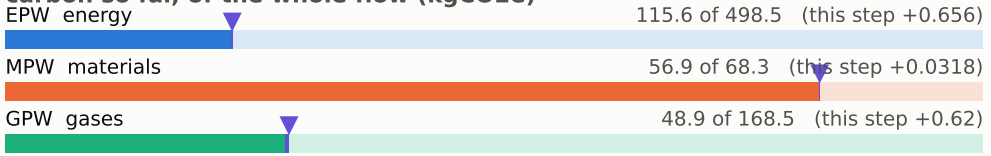


this step

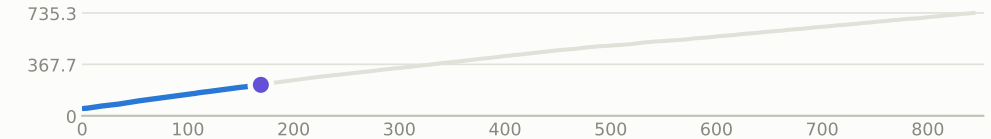
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

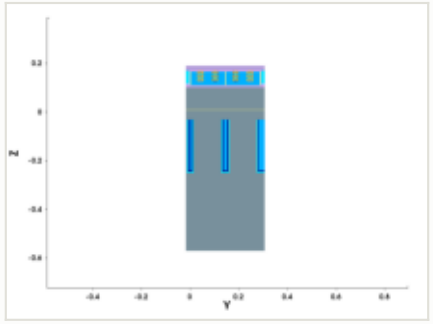
carbon so far, of the whole flow (kgCO2e)



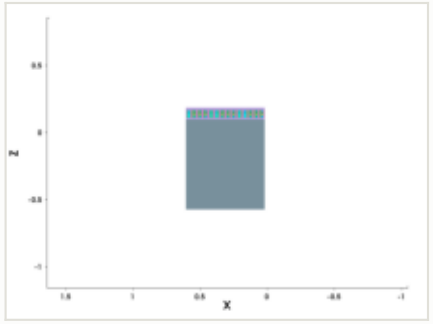
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



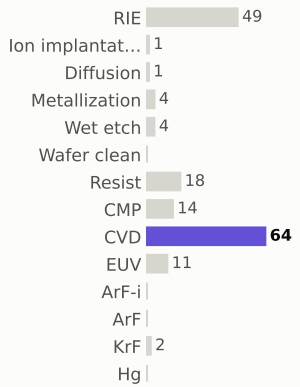
y cut at 0.184 µm



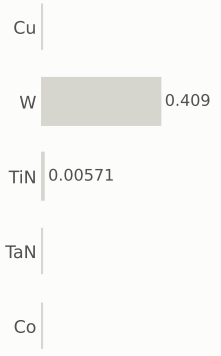
mask for this step

no mask for this step

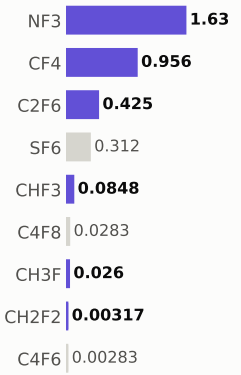
steps by process area



materials deposited (g)



gases emitted (g)



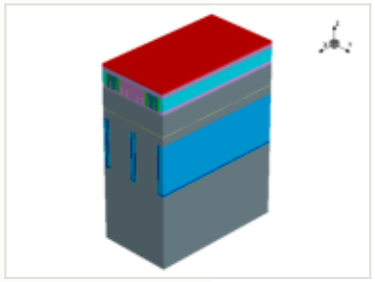
Step #1860: CVD

depo_isotropic · CVD

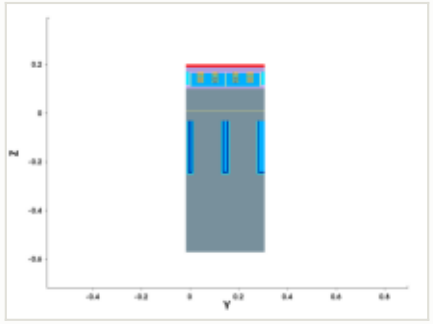
1.31	222.7	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

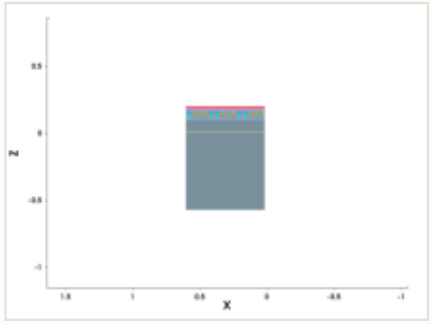
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



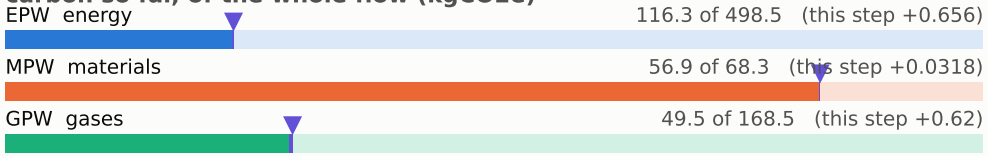
mask for this step

no mask for this step

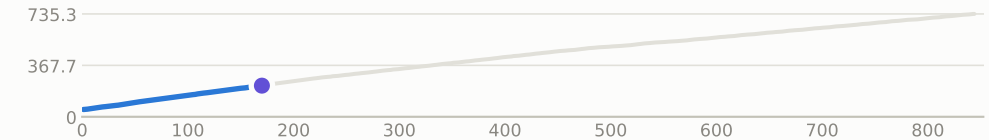
this step

process area	CVD	EPW	0.656 kgCO2e
material	—	EPW tool	0.469 kgCO2e
thickness	0.01 µm	EPW overhead	0.187 kgCO2e
material GWP	—	MPW	0.0318 kgCO2e (gas 0.0318)
input gases	NF3 2.54 g	GPW	0.62 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**1.31 kgCO2e**

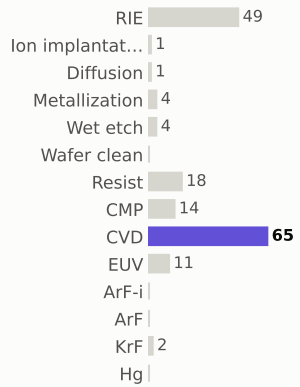
carbon so far, of the whole flow (kgCO2e)



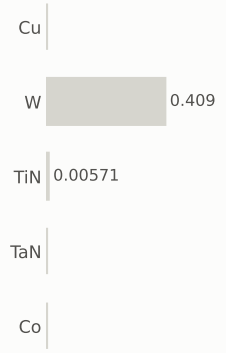
embodied carbon up to each step (kgCO2e per wafer)



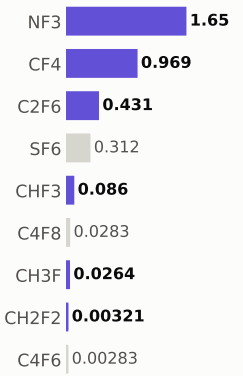
steps by process area



materials deposited (g)



gases emitted (g)



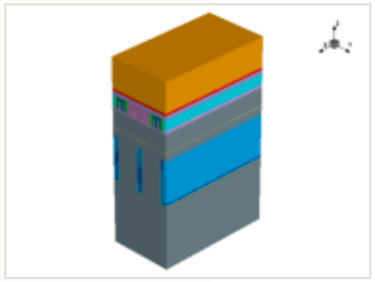
Step #1870: CVD

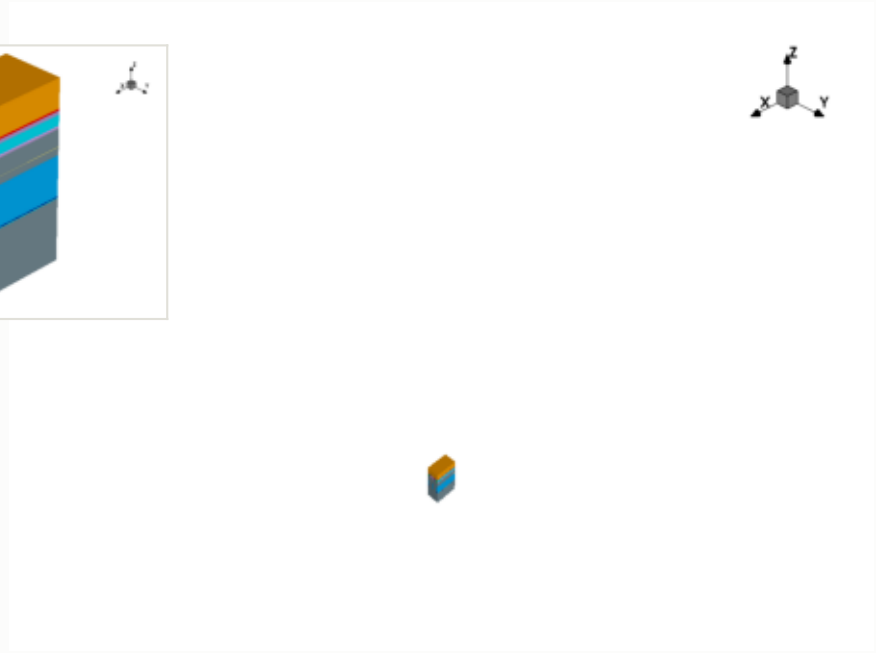
depo_fill · CVD

1.31	224.0	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



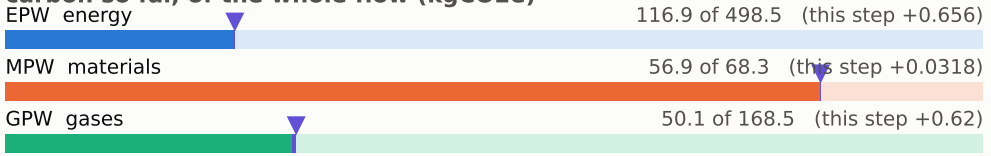


this step

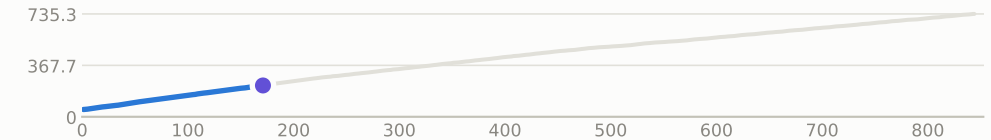
process area	CVD
material	—
thickness	0.15 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

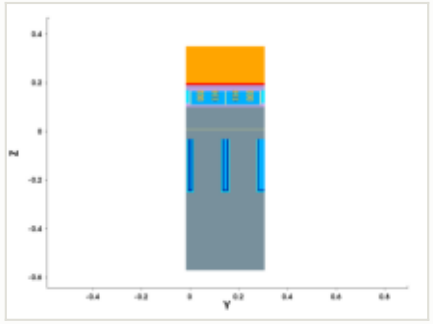
carbon so far, of the whole flow (kgCO2e)



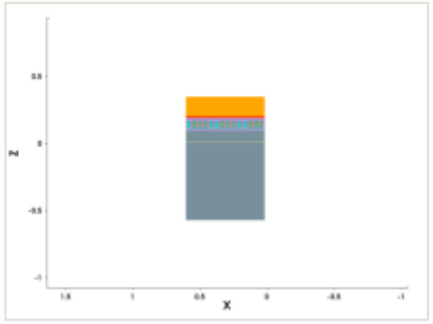
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



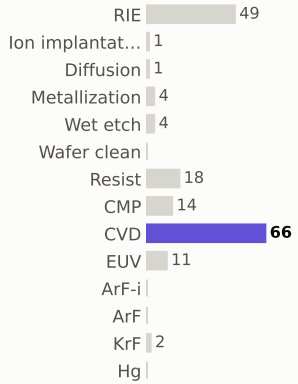
y cut at 0.184 µm



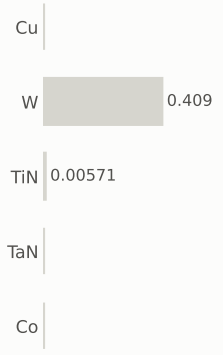
mask for this step

no mask for this step

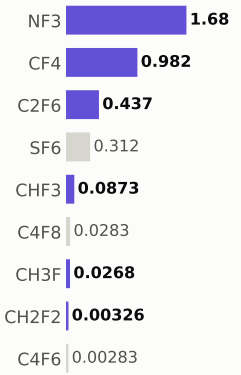
steps by process area



materials deposited (g)



gases emitted (g)



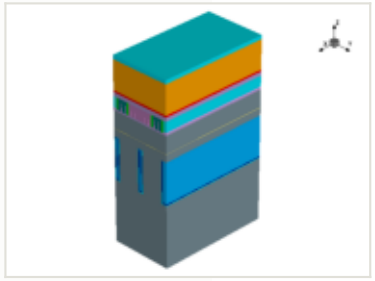
Step #1880: CVD

depo_isotropic · CVD

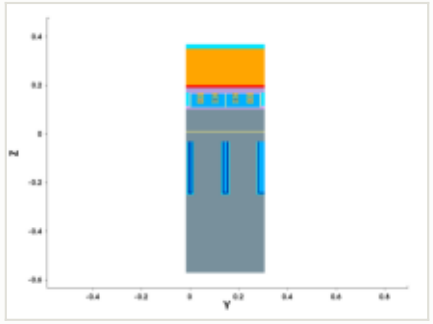
1.31	225.3	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

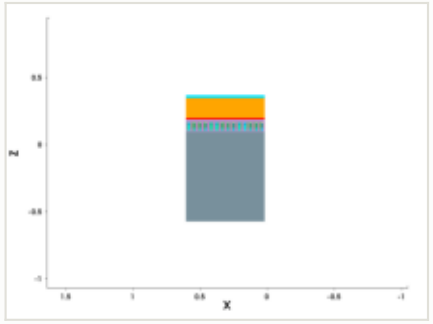
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

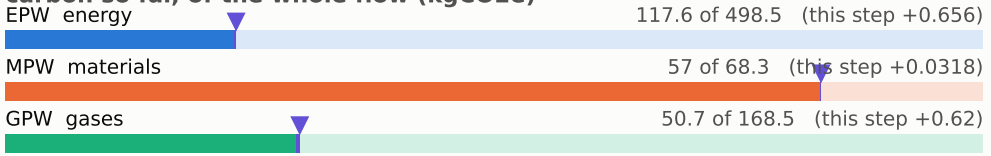
no mask for this step

this step

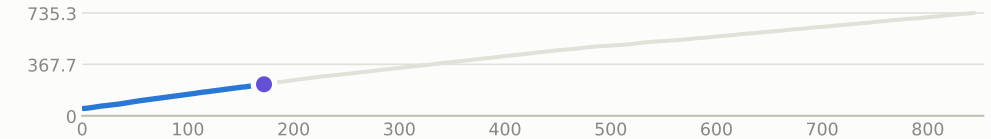
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

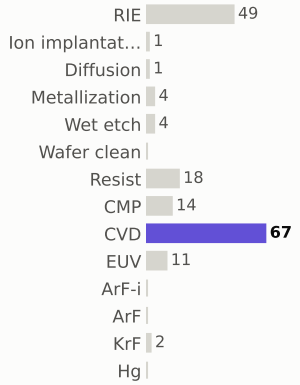
carbon so far, of the whole flow (kgCO2e)



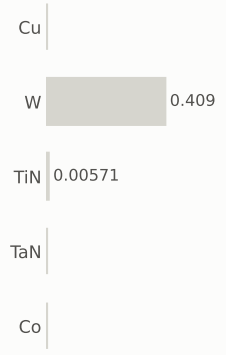
embodied carbon up to each step (kgCO2e per wafer)



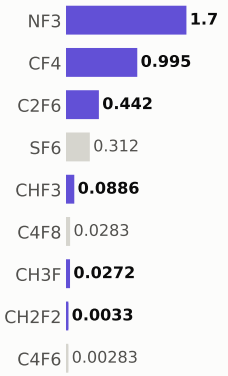
steps by process area



materials deposited (g)



gases emitted (g)



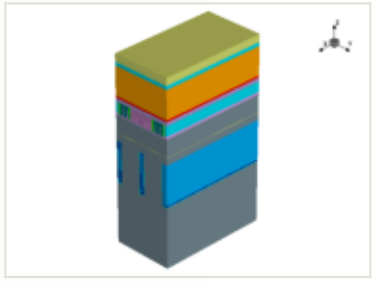
Step #1890: CVD

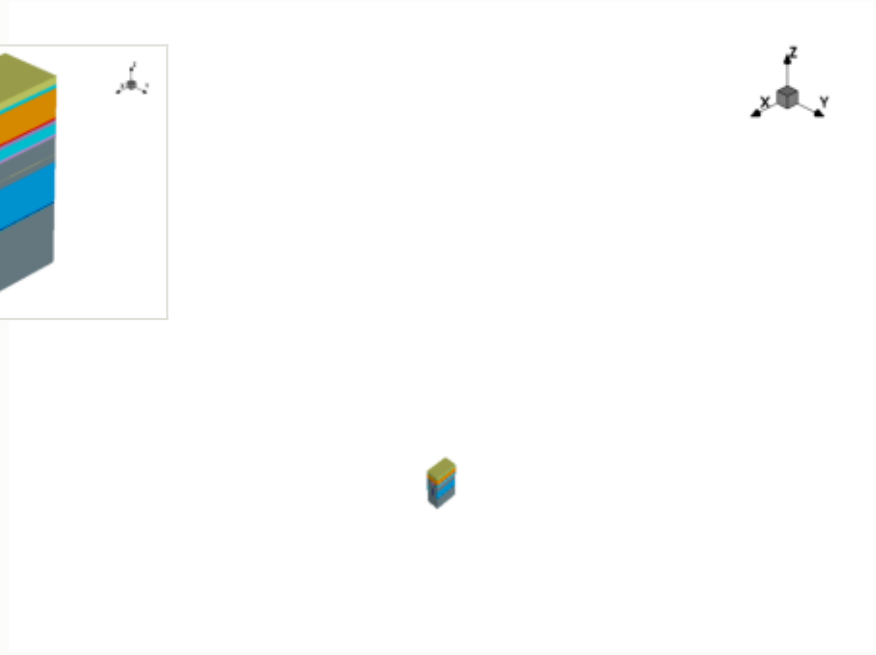
depo_isotropic · CVD

1.31	226.6	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



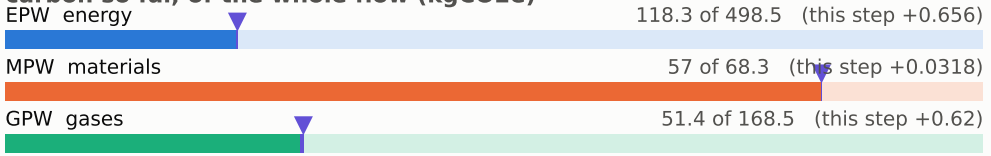


this step

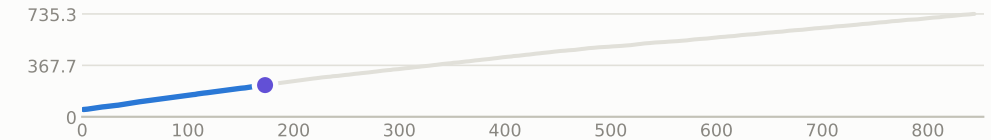
process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

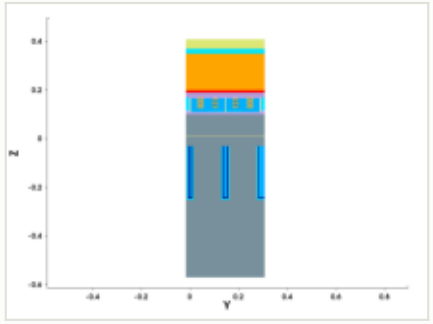
carbon so far, of the whole flow (kgCO2e)



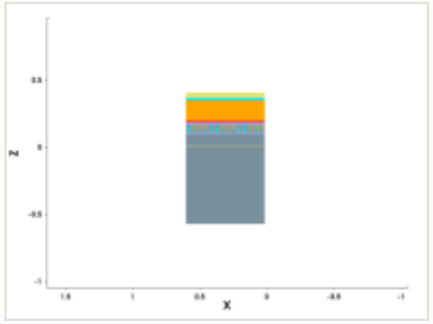
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



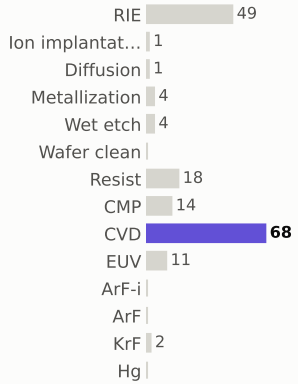
y cut at 0.184 µm



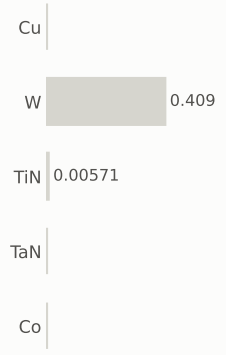
mask for this step

no mask for this step

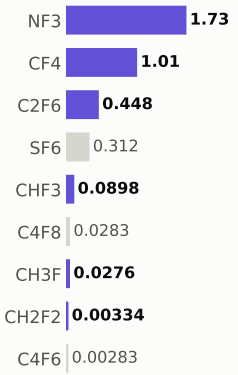
steps by process area



materials deposited (g)



gases emitted (g)



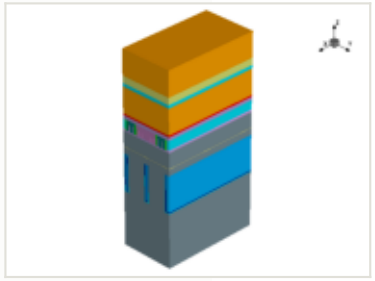
Step #1900: CVD

depo_fill · CVD

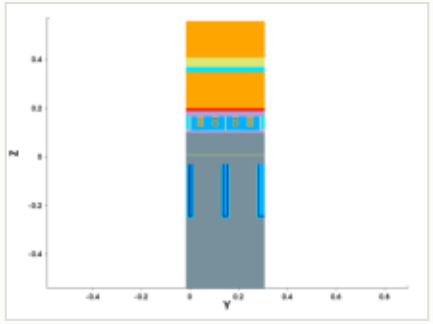
1.31	227.9	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

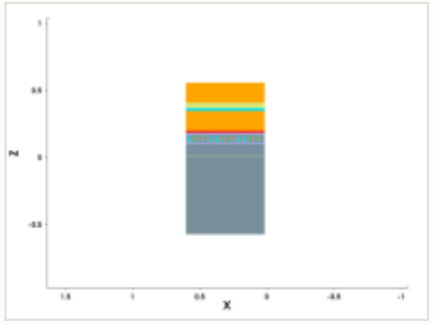
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

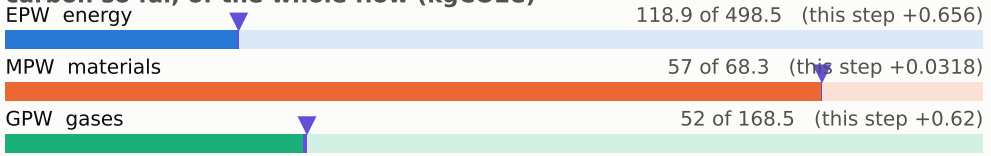
no mask for this step

this step

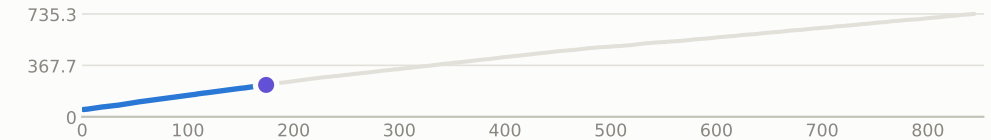
process area	CVD
material	—
thickness	0.15 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

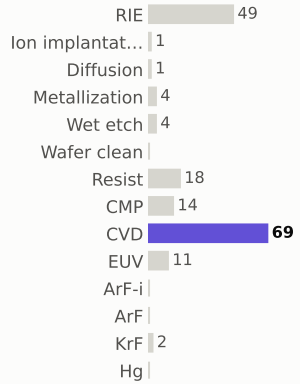
carbon so far, of the whole flow (kgCO2e)



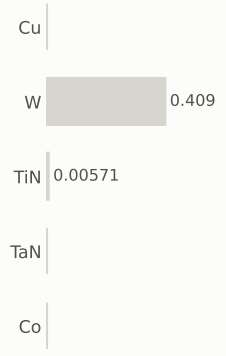
embodied carbon up to each step (kgCO2e per wafer)



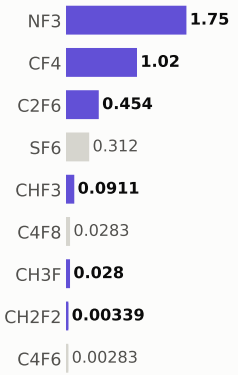
steps by process area



materials deposited (g)



gases emitted (g)



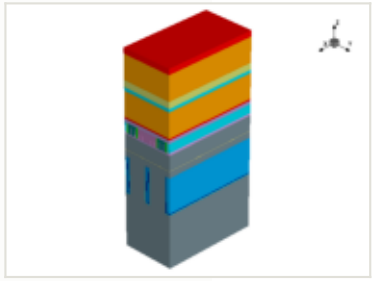
Step #1910: CVD

depo_fill · CVD

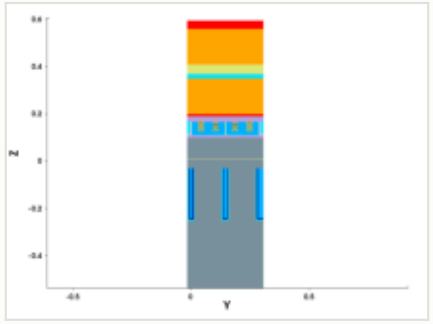
1.31	229.2	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

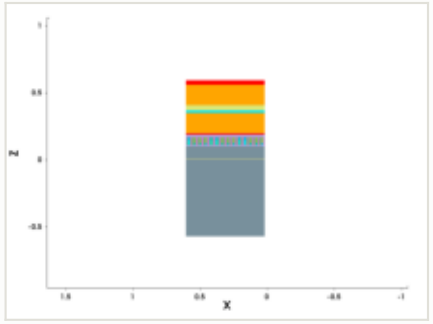
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

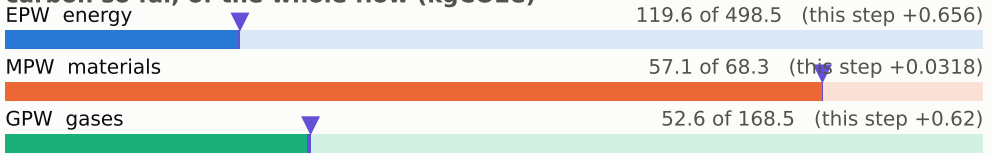
no mask for this step

this step

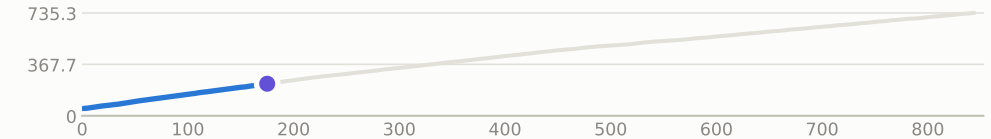
process area	CVD
material	—
thickness	0.035 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

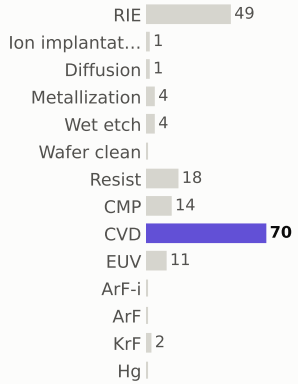
carbon so far, of the whole flow (kgCO2e)



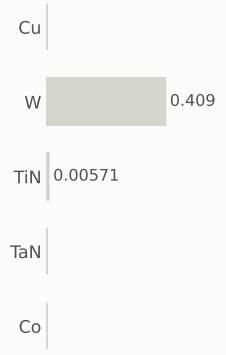
embodied carbon up to each step (kgCO2e per wafer)



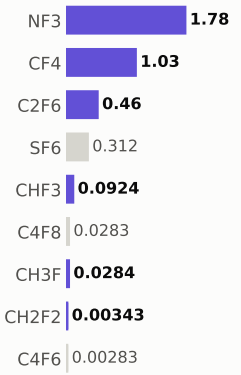
steps by process area



materials deposited (g)



gases emitted (g)



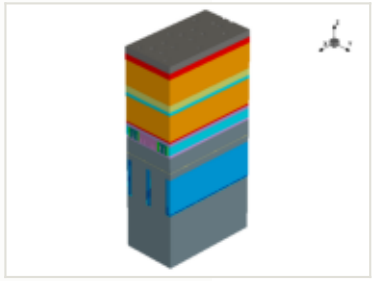
Step #1920: EUV

litho_patterning · EUV

1.96	231.2	0.27%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

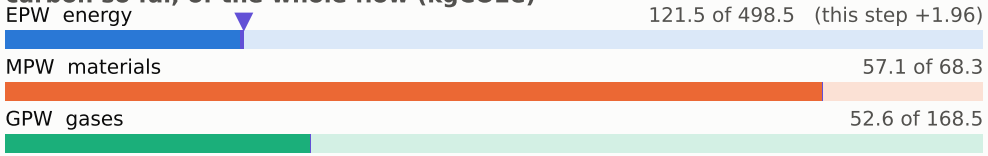


this step

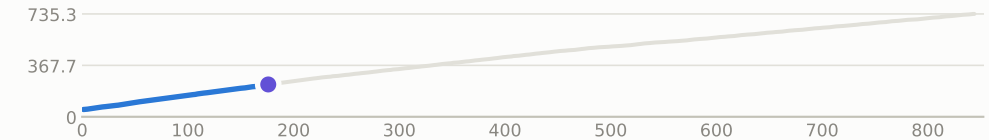
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

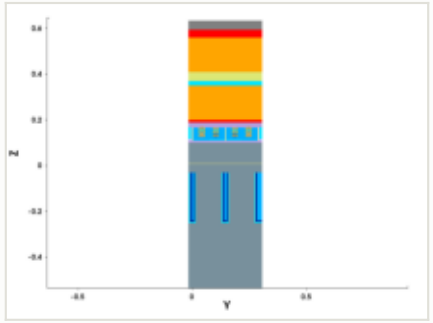
carbon so far, of the whole flow (kgCO2e)



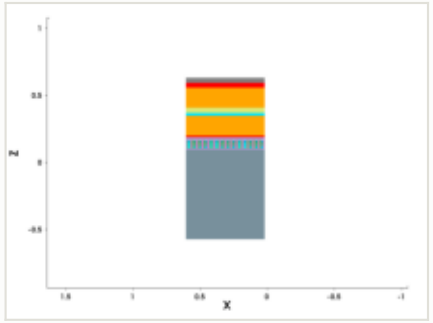
embodied carbon up to each step (kgCO2e per wafer)



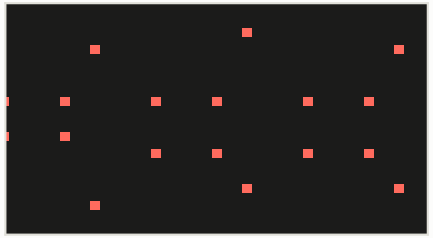
x cut at 0.294 µm



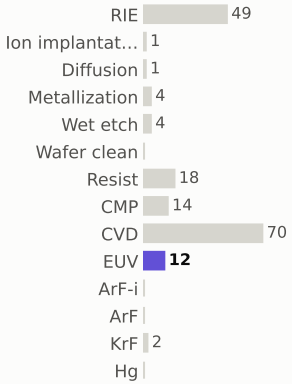
y cut at 0.184 µm



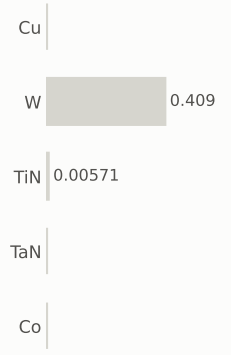
mask: layer 11:0 (GDSII)



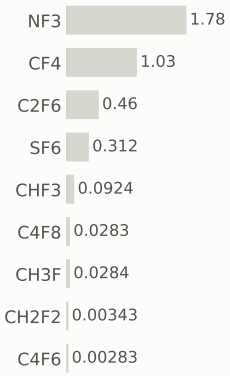
steps by process area



materials deposited (g)



gases emitted (g)



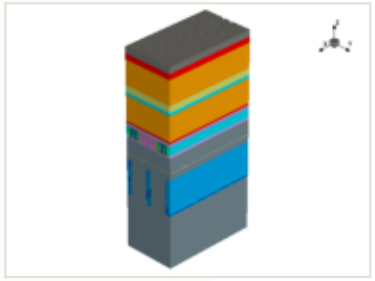
Step #1930: RIE

etch_anisotropic · RIE

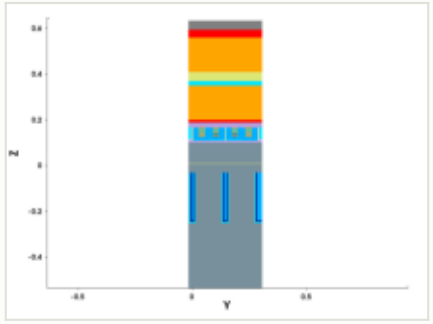
1.04	232.2	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

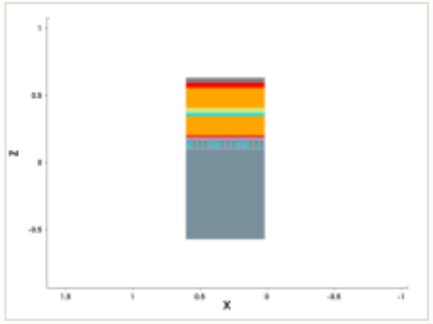
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

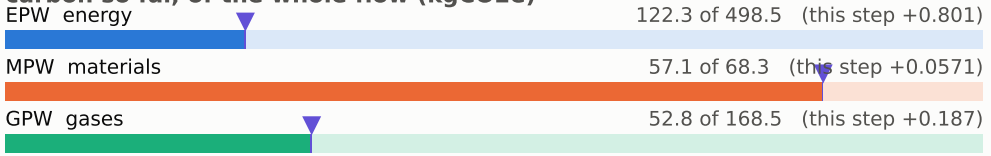
no mask for this step

this step

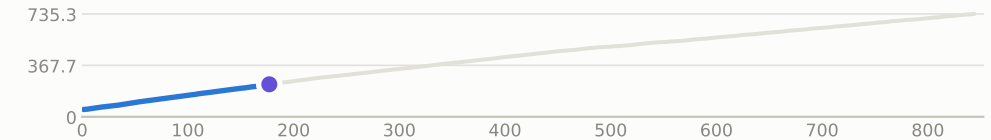
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

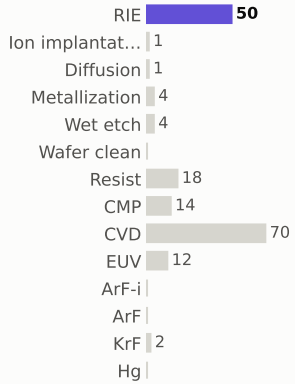
carbon so far, of the whole flow (kgCO2e)



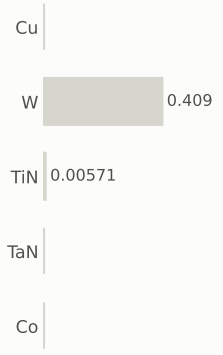
embodied carbon up to each step (kgCO2e per wafer)



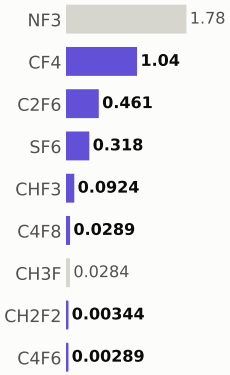
steps by process area



materials deposited (g)



gases emitted (g)



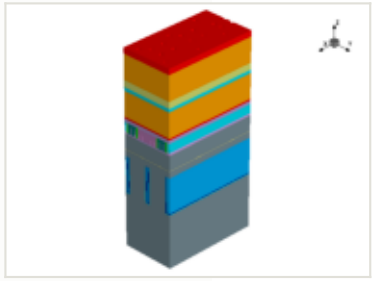
Step #1940: RIE

etch_anisotropic · RIE

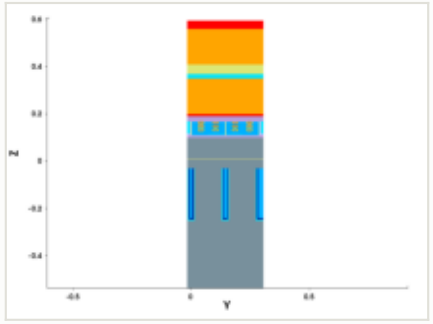
1.04	233.3	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

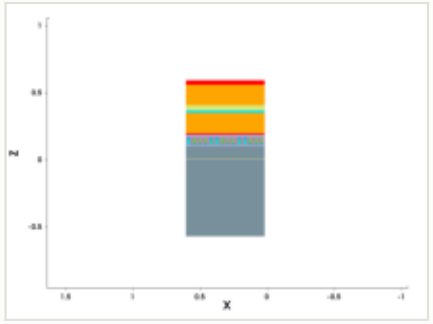
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

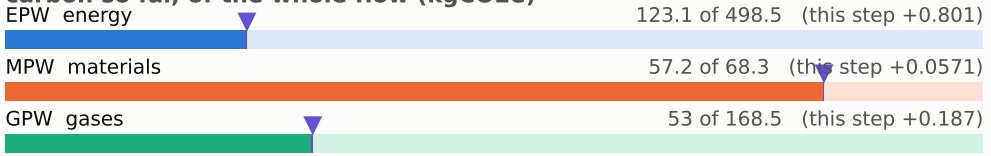
no mask for this step

this step

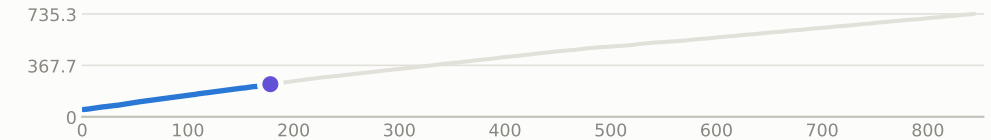
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

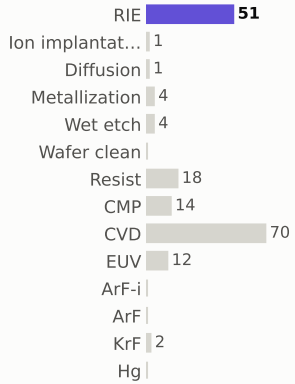
carbon so far, of the whole flow (kgCO2e)



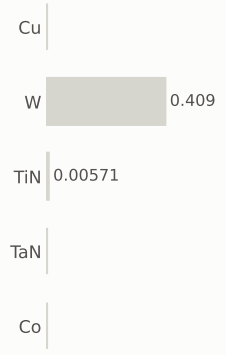
embodied carbon up to each step (kgCO2e per wafer)



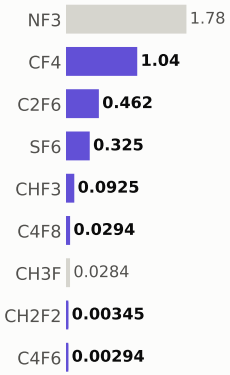
steps by process area



materials deposited (g)



gases emitted (g)



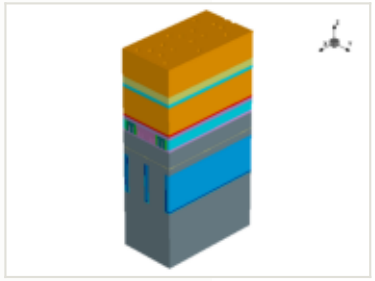
Step #1950: RIE

etch_anisotropic · RIE

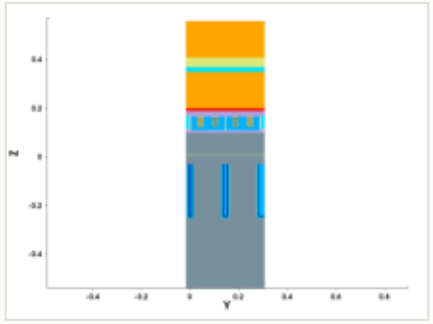
1.04	234.3	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

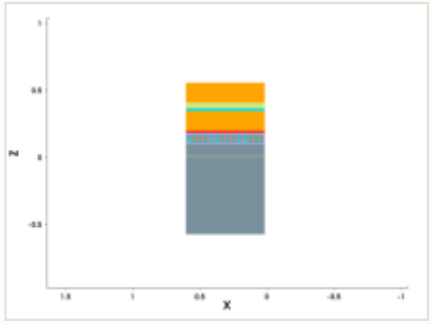
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

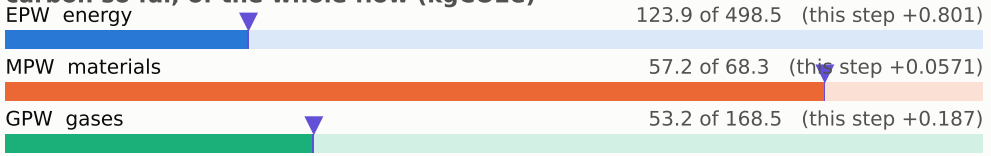
no mask for this step

this step

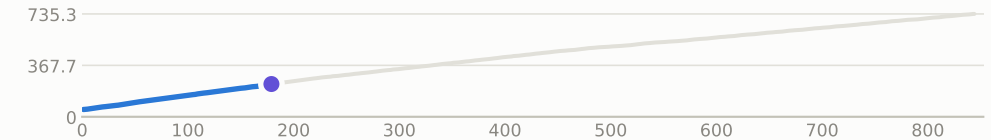
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

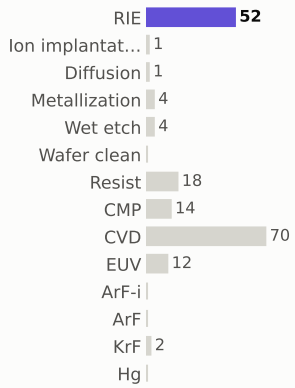
carbon so far, of the whole flow (kgCO2e)



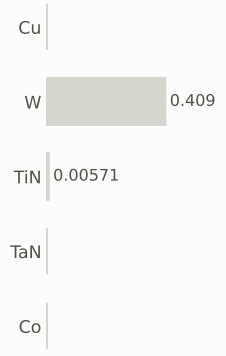
embodied carbon up to each step (kgCO2e per wafer)



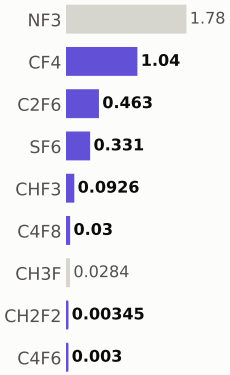
steps by process area



materials deposited (g)



gases emitted (g)



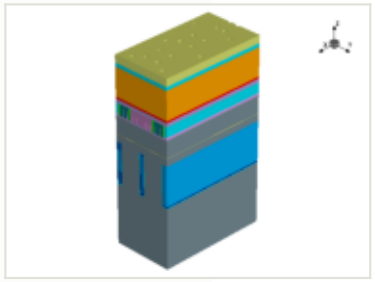
Step #1960: Resist

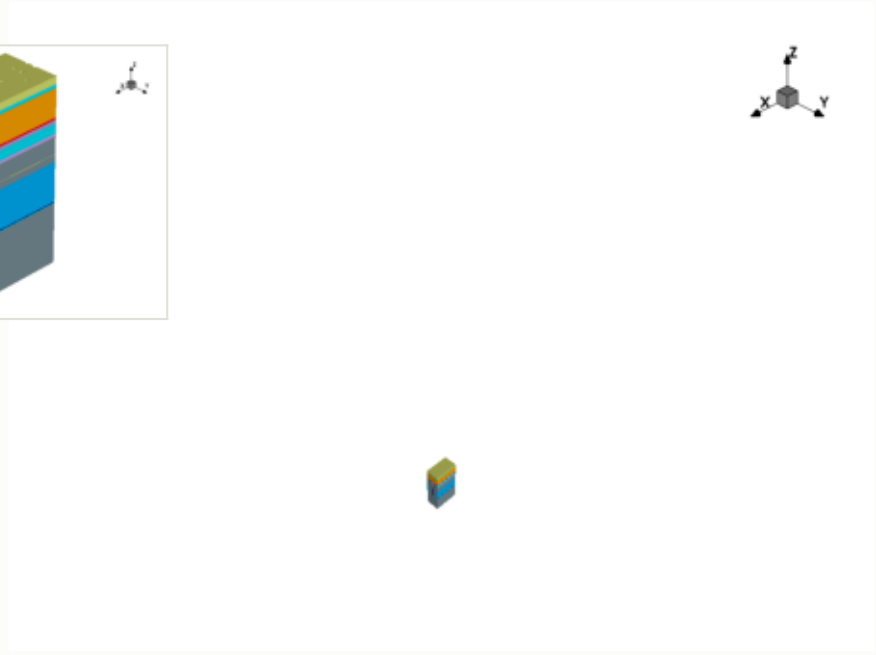
etch_strip · Resist

0.169	234.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



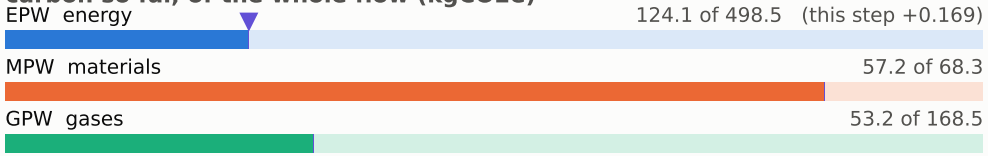


this step

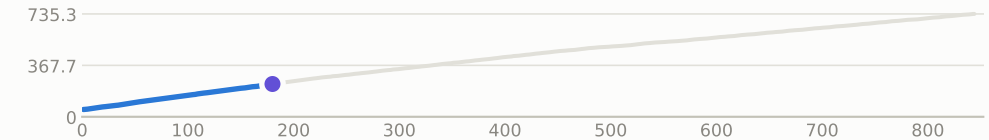
process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

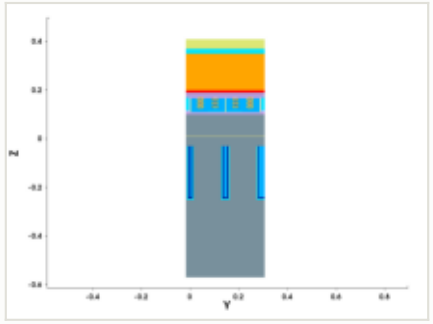
carbon so far, of the whole flow (kgCO2e)



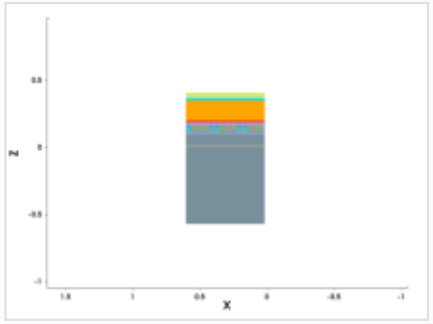
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



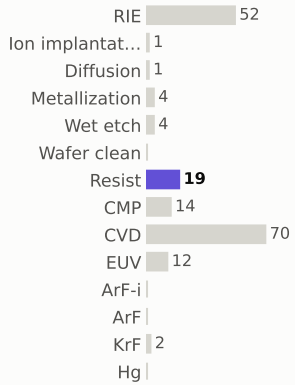
y cut at 0.184 µm



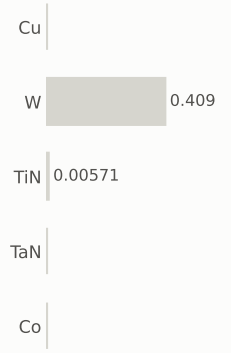
mask for this step

no mask for this step

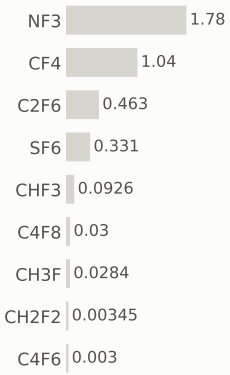
steps by process area



materials deposited (g)



gases emitted (g)



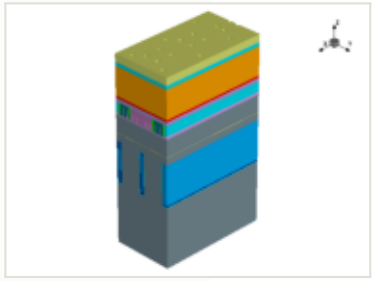
Step #1970: RIE

etch_anisotropic · RIE

1.04	235.5	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

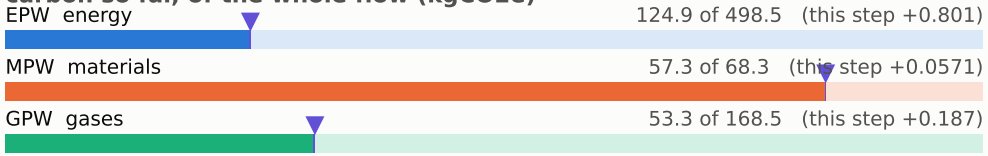


this step

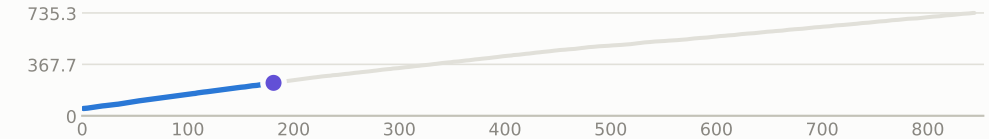
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

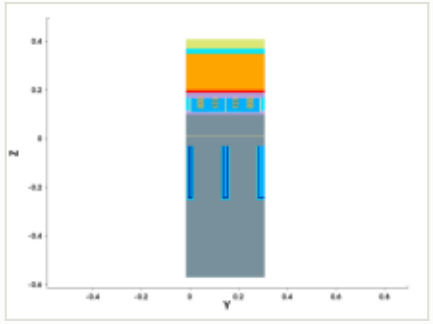
carbon so far, of the whole flow (kgCO2e)



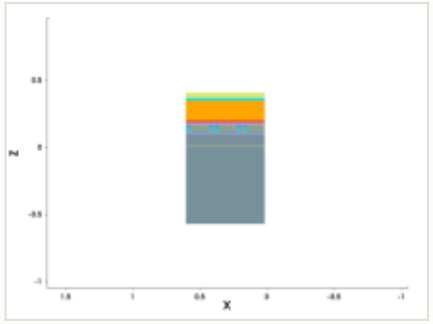
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



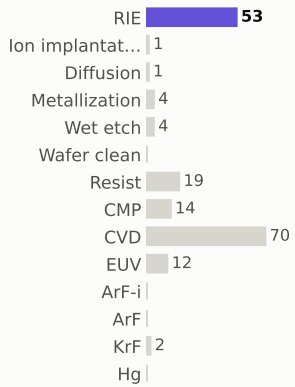
y cut at 0.184 µm



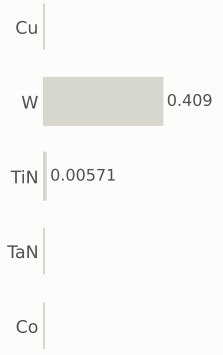
mask for this step

no mask for this step

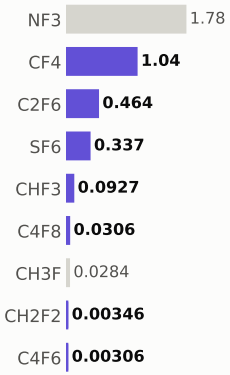
steps by process area



materials deposited (g)



gases emitted (g)



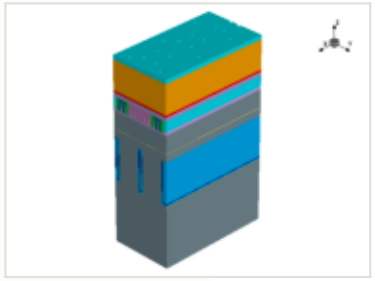
Step #1980: RIE

etch_anisotropic · RIE

1.04	236.6	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

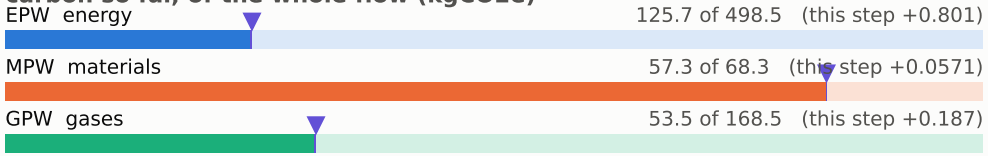


this step

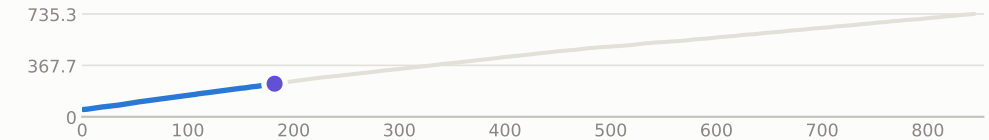
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

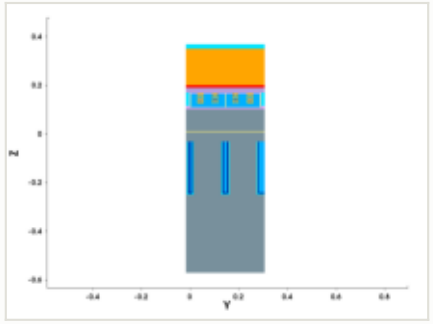
carbon so far, of the whole flow (kgCO2e)



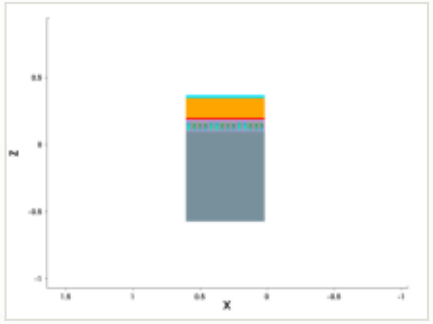
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



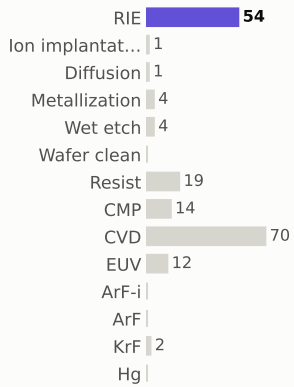
y cut at 0.184 µm



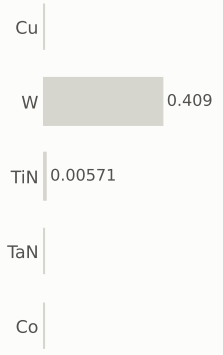
mask for this step

no mask for this step

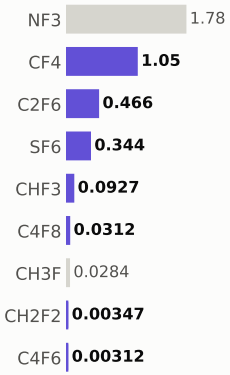
steps by process area



materials deposited (g)



gases emitted (g)



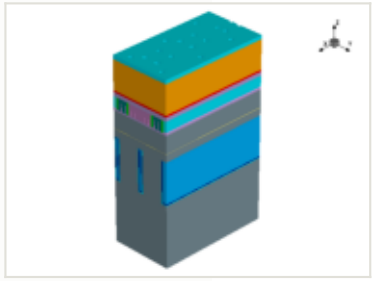
Step #1990: RIE

etch_anisotropic · RIE

1.04	237.6	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

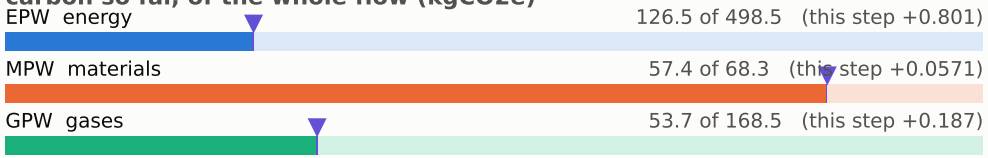


this step

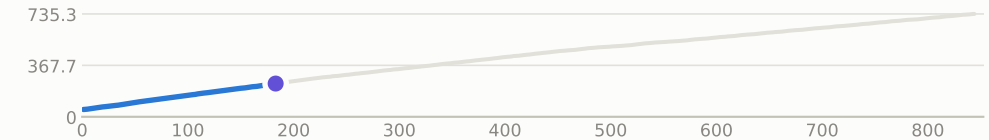
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

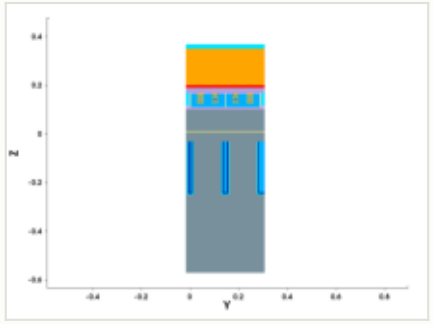
carbon so far, of the whole flow (kgCO2e)



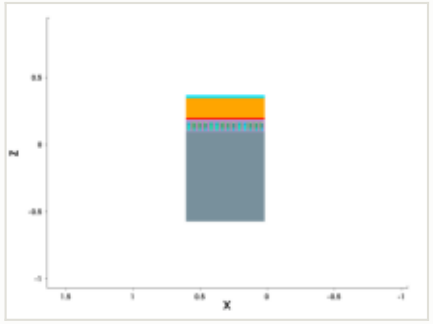
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



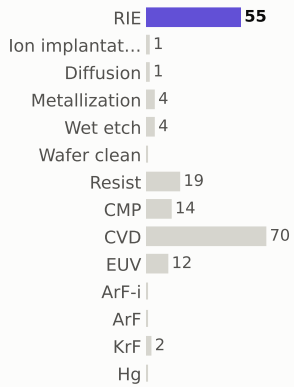
y cut at 0.184 µm



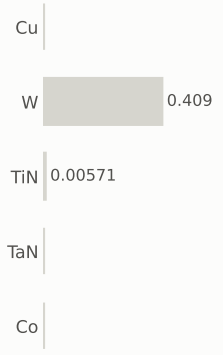
mask for this step

no mask for this step

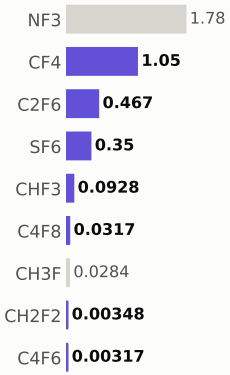
steps by process area



materials deposited (g)



gases emitted (g)



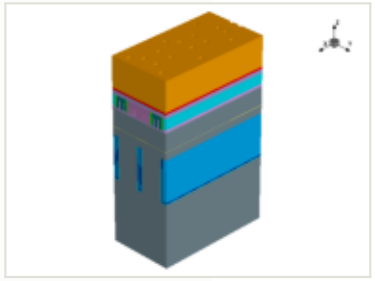
Step #2000: RIE

etch_anisotropic · RIE

1.04	238.7	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

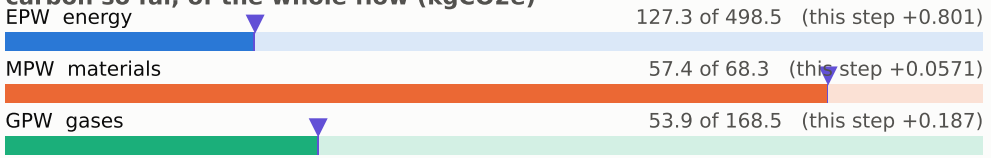


this step

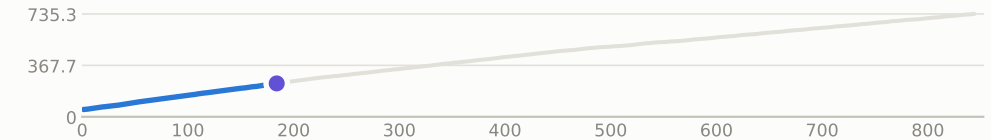
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

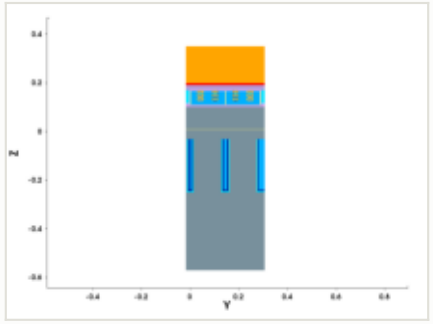
carbon so far, of the whole flow (kgCO2e)



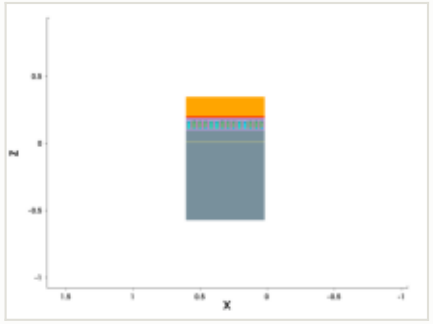
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



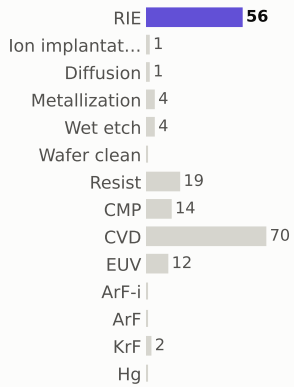
y cut at 0.184 µm



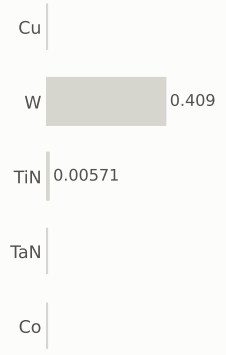
mask for this step

no mask for this step

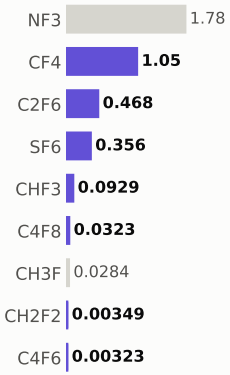
steps by process area



materials deposited (g)



gases emitted (g)



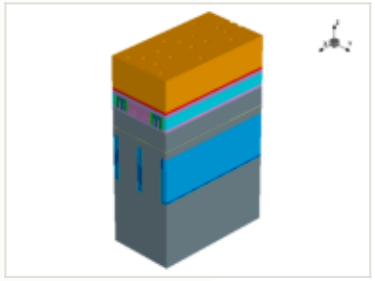
Step #2010: RIE

etch_anisotropic · RIE

1.04	239.7	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

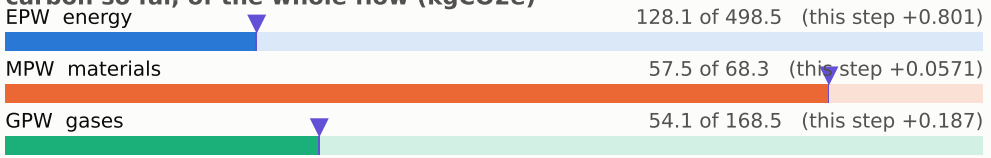


this step

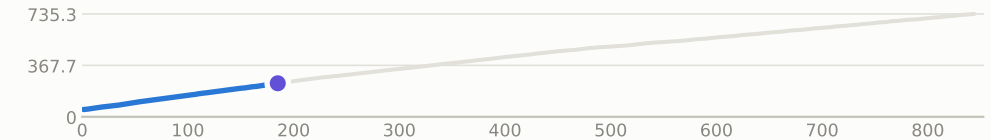
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

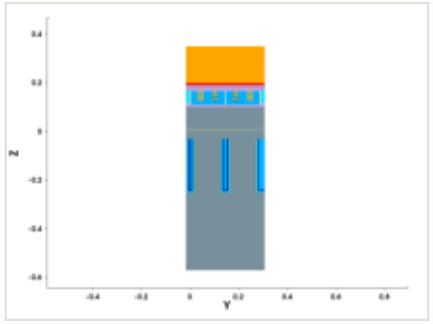
carbon so far, of the whole flow (kgCO2e)



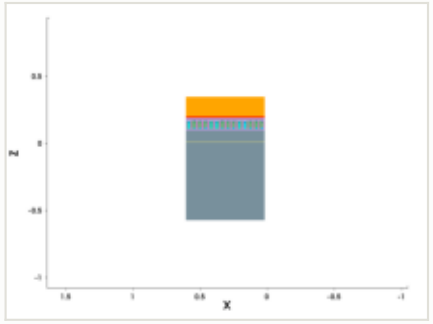
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



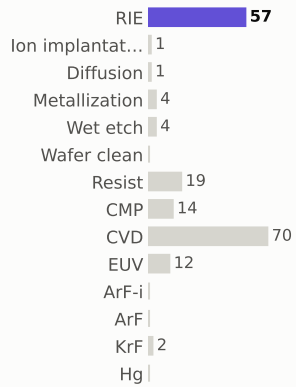
y cut at 0.184 µm



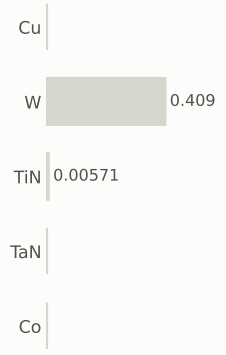
mask for this step

no mask for this step

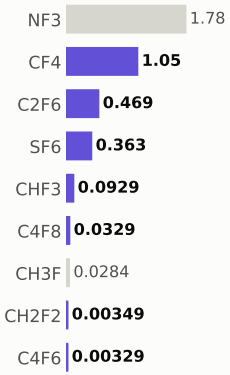
steps by process area



materials deposited (g)



gases emitted (g)



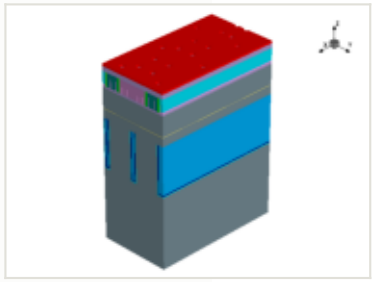
Step #2020: Resist

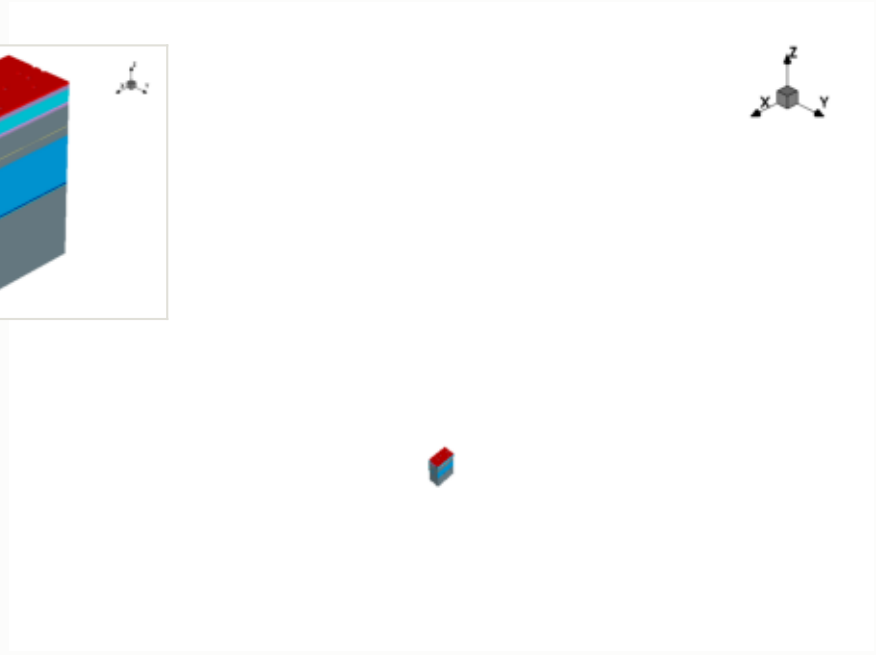
etch_strip · Resist

0.169	239.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



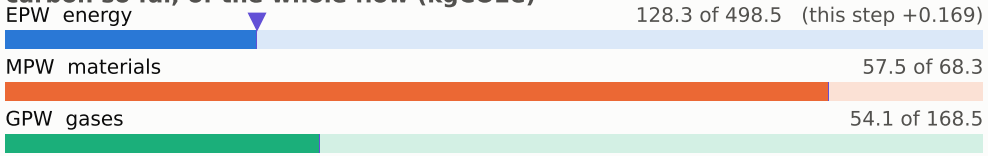


this step

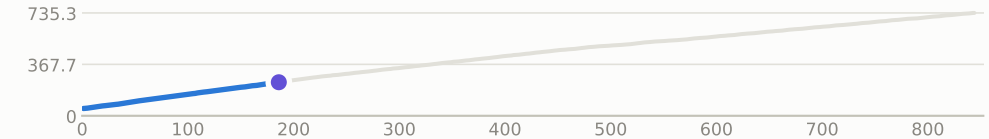
process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

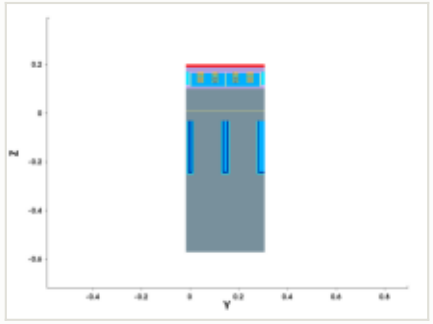
carbon so far, of the whole flow (kgCO2e)



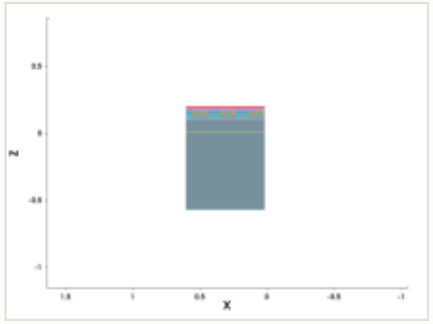
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



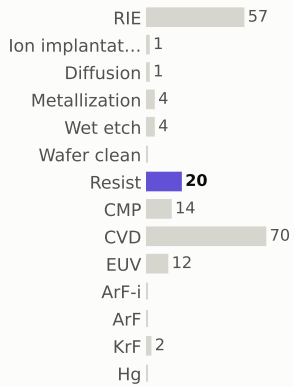
y cut at 0.184 µm



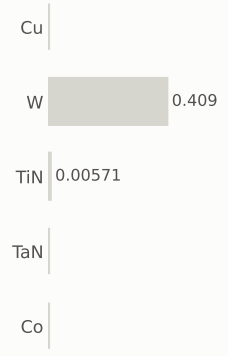
mask for this step

no mask for this step

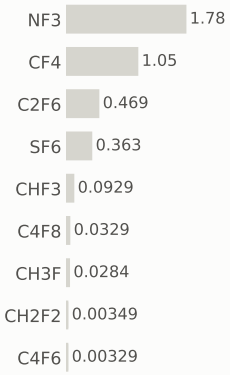
steps by process area



materials deposited (g)



gases emitted (g)



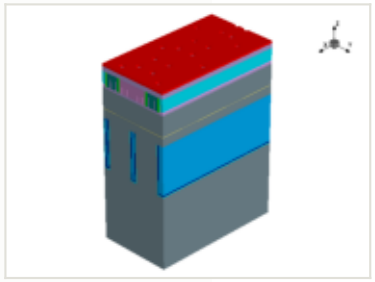
Step #2030: RIE

etch_anisotropic · RIE

1.04	240.9	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

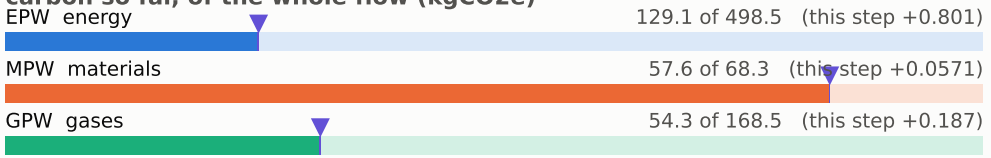


this step

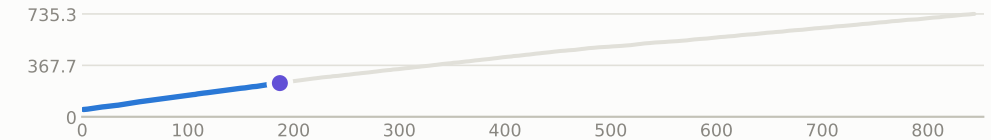
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

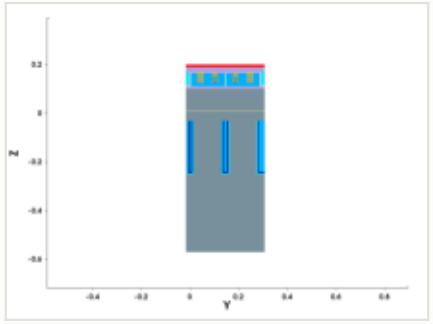
carbon so far, of the whole flow (kgCO2e)



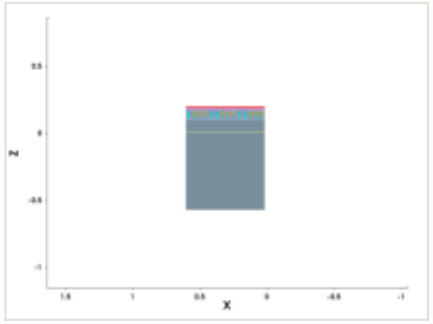
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



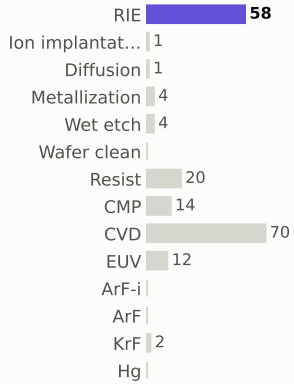
y cut at 0.184 µm



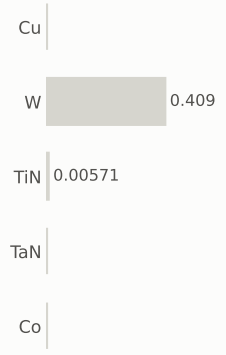
mask for this step

no mask for this step

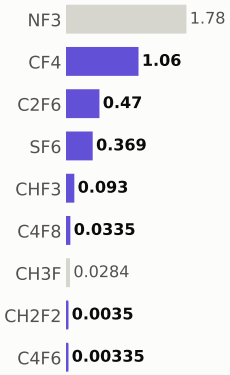
steps by process area



materials deposited (g)



gases emitted (g)



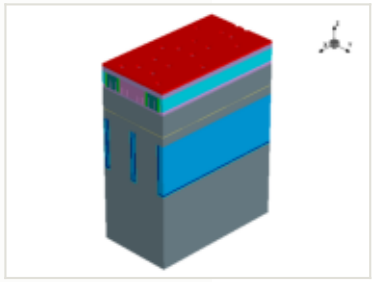
Step #2040: RIE

etch_anisotropic · RIE

1.04	242.0	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

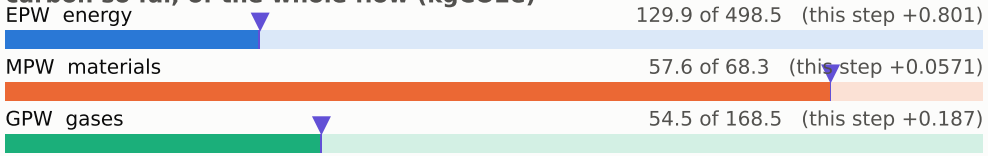


this step

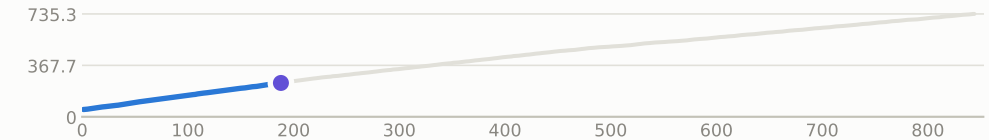
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

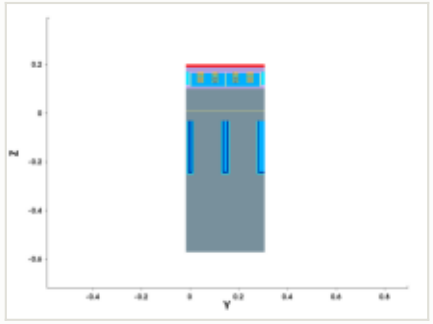
carbon so far, of the whole flow (kgCO2e)



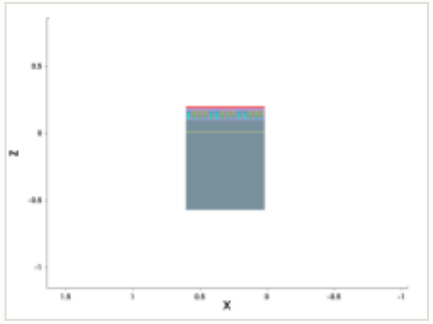
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



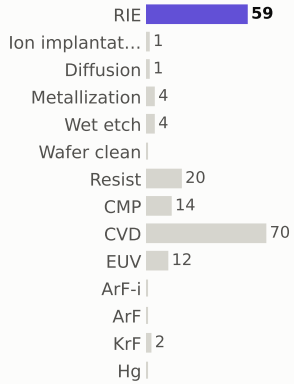
y cut at 0.184 µm



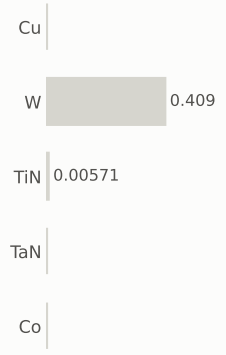
mask for this step

no mask for this step

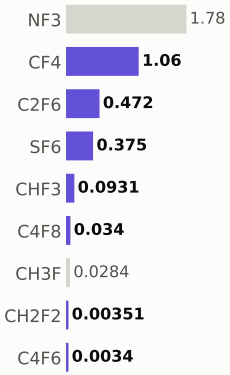
steps by process area



materials deposited (g)



gases emitted (g)



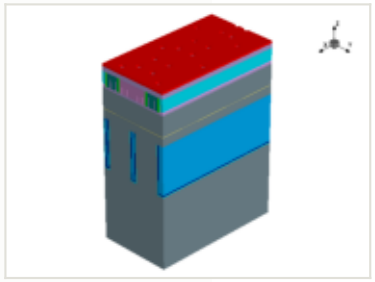
Step #2050: Metallization

depo_isotropic · Metallization

1.23	243.2	0.17%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

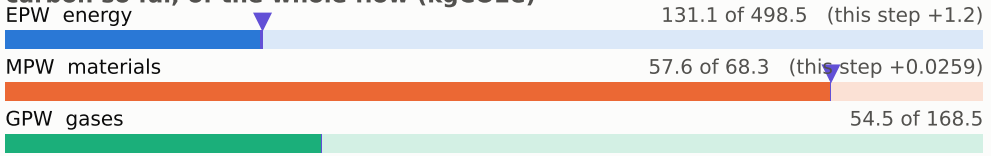


this step

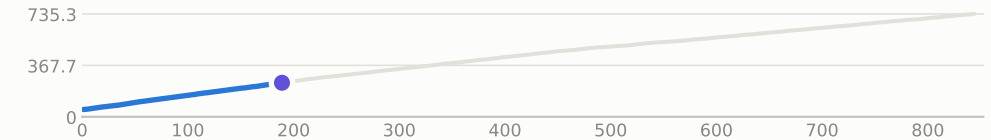
process area	Metallization
material	W
thickness	0.25 µm
material GWP	75.8149 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.0259 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.23 kgCO2e**

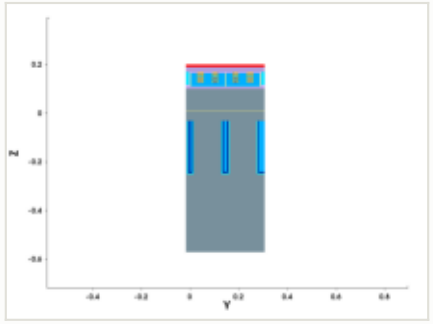
carbon so far, of the whole flow (kgCO2e)



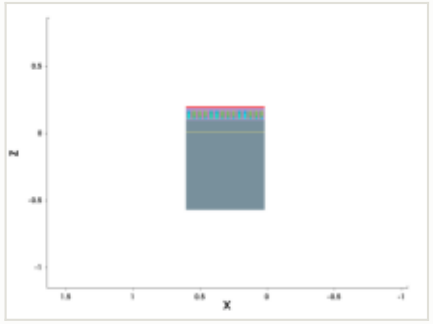
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



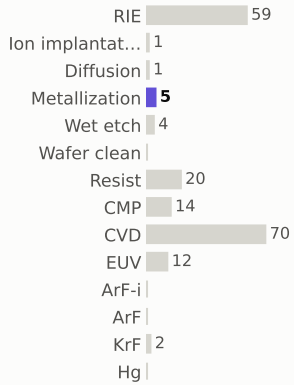
y cut at 0.184 µm



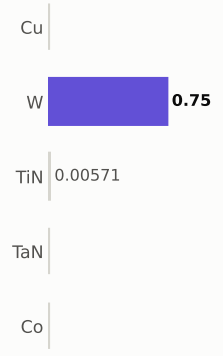
mask for this step

no mask for this step

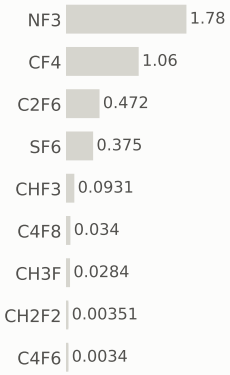
steps by process area



materials deposited (g)



gases emitted (g)



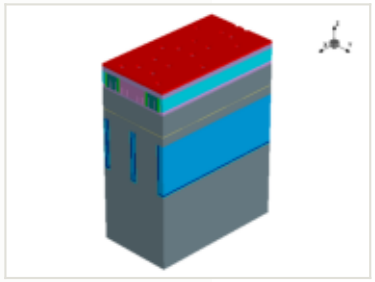
Step #2060: CMP

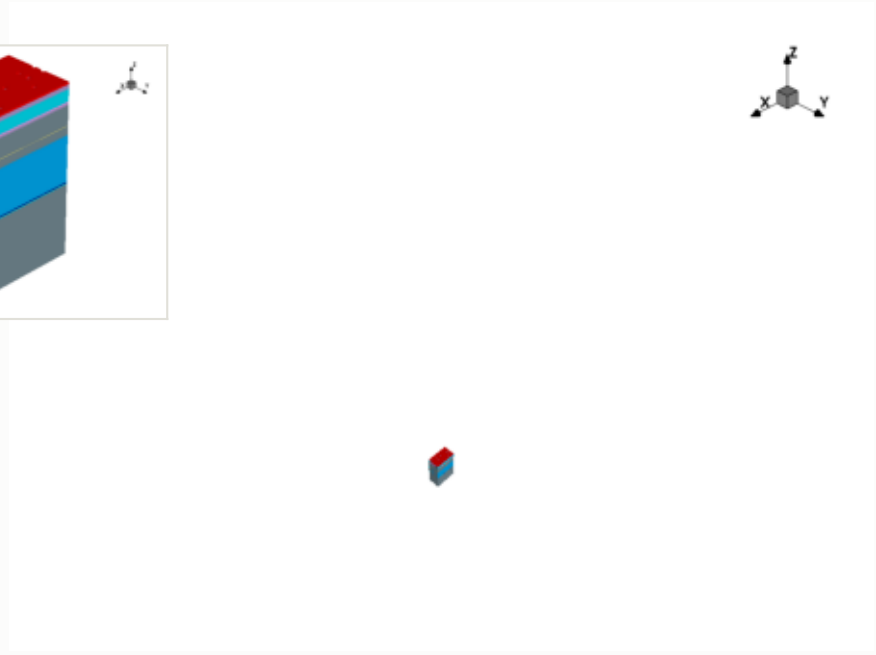
etch_cmp · CMP

0.169	243.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



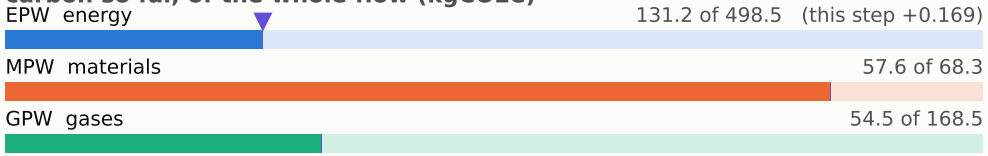


this step

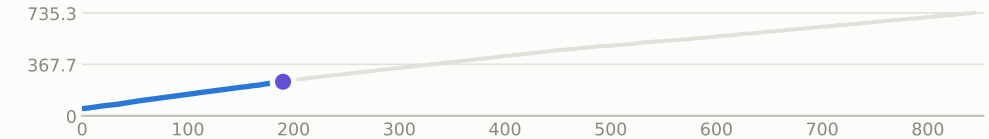
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

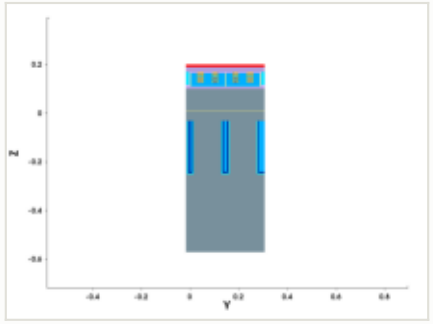
carbon so far, of the whole flow (kgCO2e)



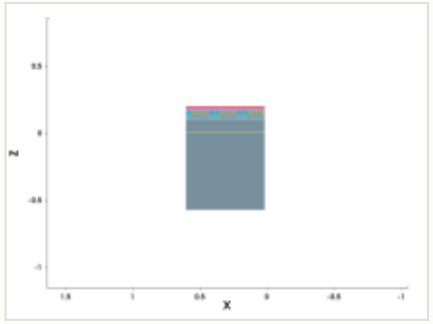
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



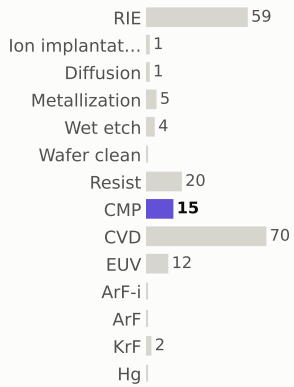
y cut at 0.184 µm



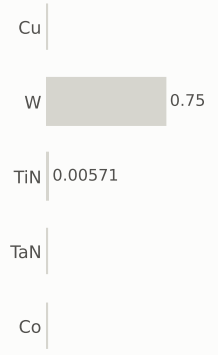
mask for this step

no mask for this step

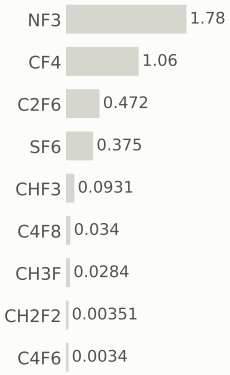
steps by process area



materials deposited (g)



gases emitted (g)



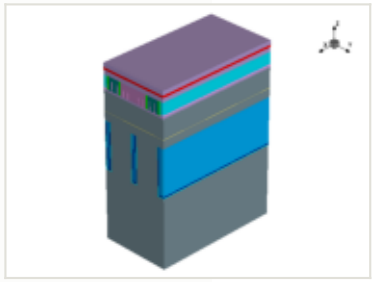
Step #2080: CVD

depo_fill · CVD

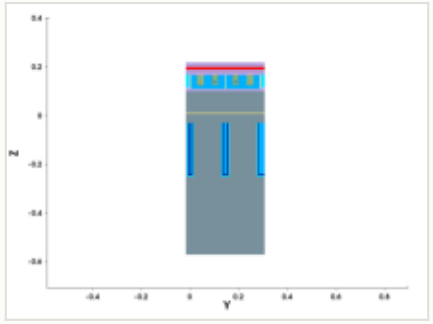
1.31	244.7	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

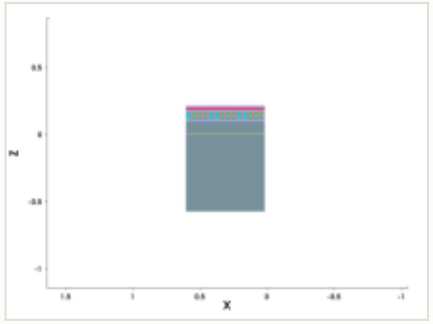
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

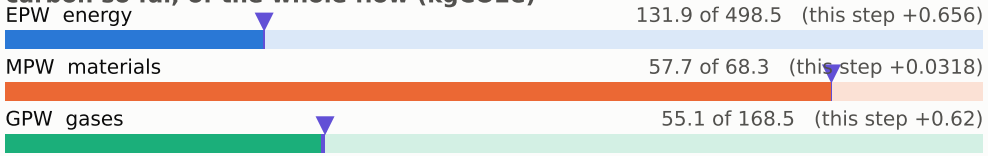
no mask for this step

this step

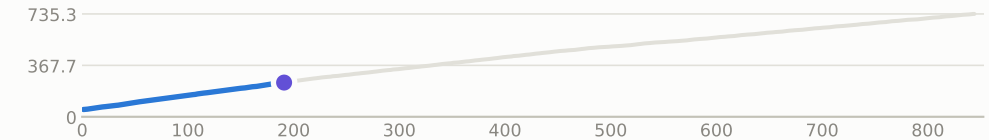
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

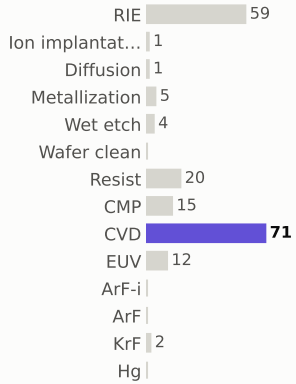
carbon so far, of the whole flow (kgCO2e)



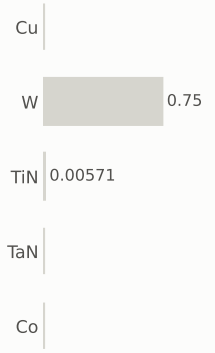
embodied carbon up to each step (kgCO2e per wafer)



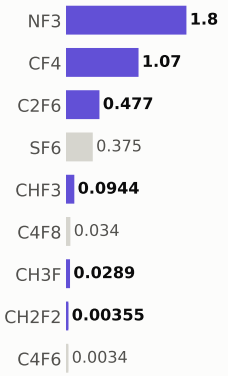
steps by process area



materials deposited (g)



gases emitted (g)



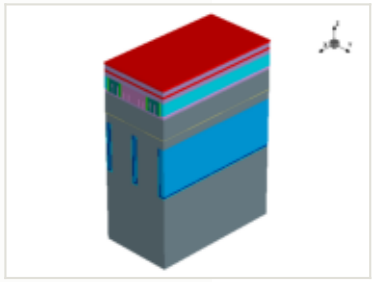
Step #2090: CVD

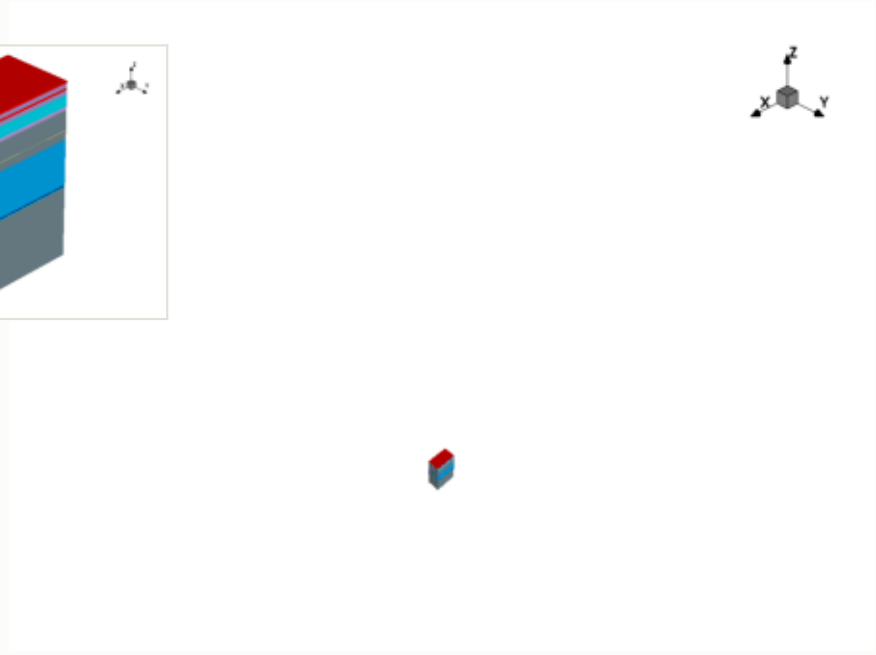
depo_isotropic · CVD

1.31	246.0	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



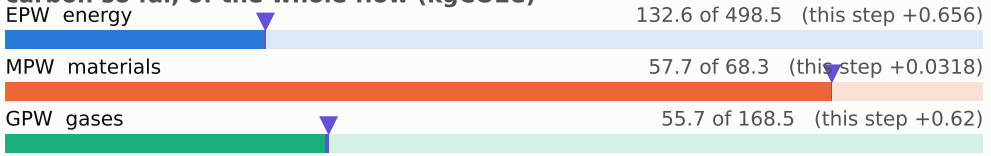


this step

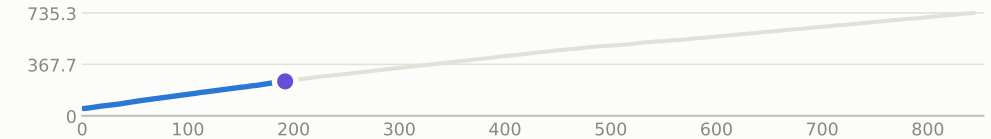
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

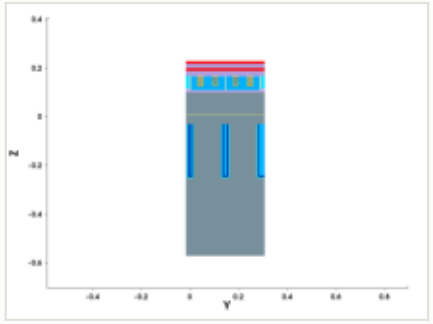
carbon so far, of the whole flow (kgCO2e)



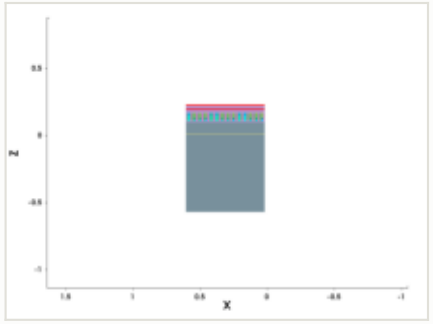
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



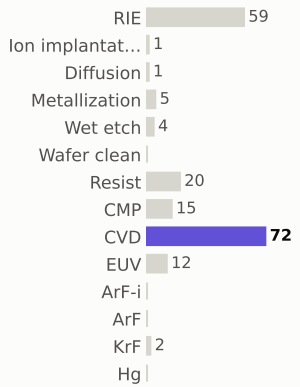
y cut at 0.184 µm



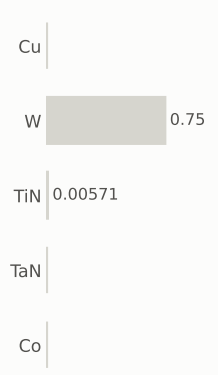
mask for this step

no mask for this step

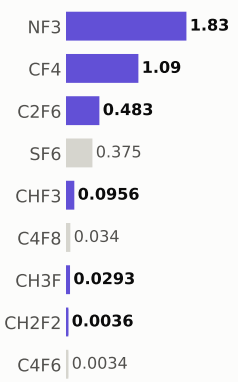
steps by process area



materials deposited (g)



gases emitted (g)



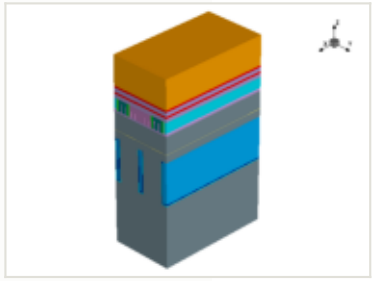
Step #2100: CVD

depo_fill · CVD

1.31	247.3	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

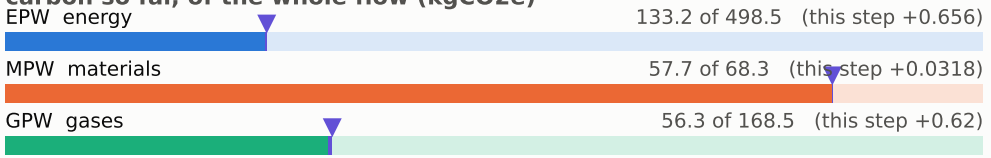


this step

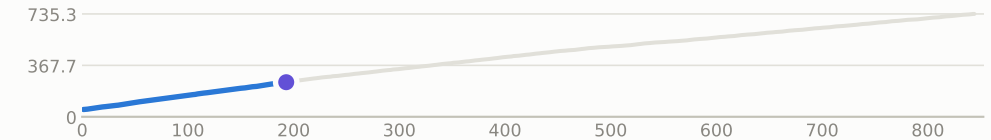
process area	CVD
material	—
thickness	0.15 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

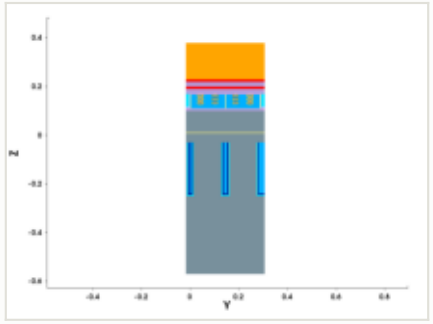
carbon so far, of the whole flow (kgCO2e)



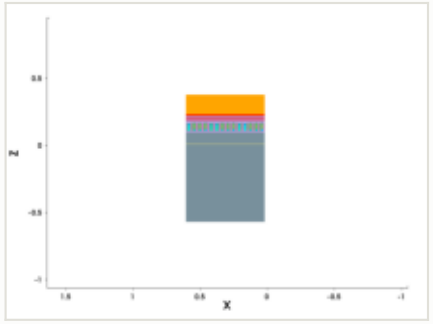
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



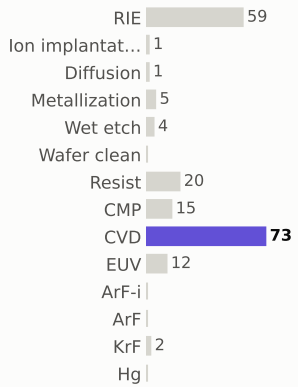
y cut at 0.184 µm



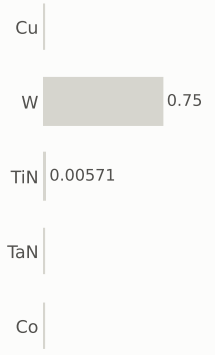
mask for this step

no mask for this step

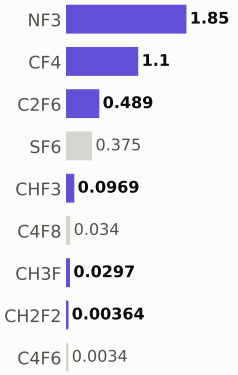
steps by process area



materials deposited (g)



gases emitted (g)



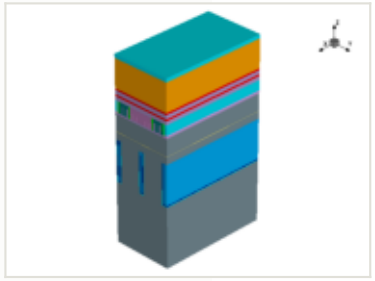
Step #2110: CVD

depo_isotropic · CVD

1.31	248.6	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

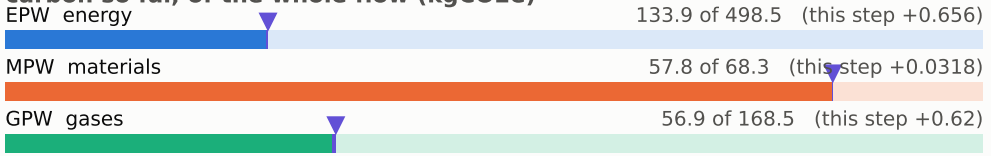


this step

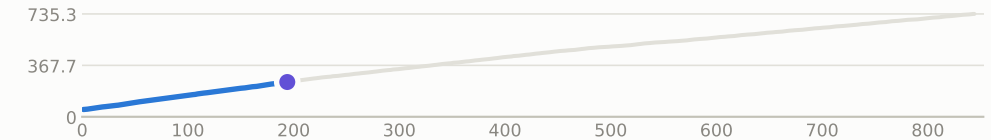
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

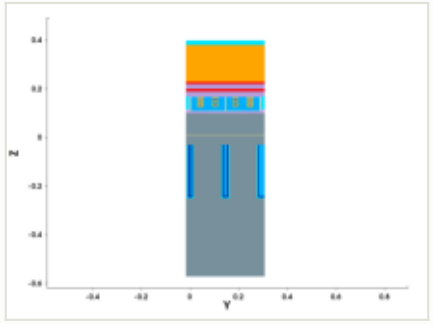
carbon so far, of the whole flow (kgCO2e)



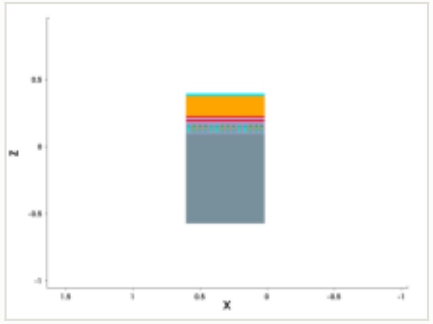
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



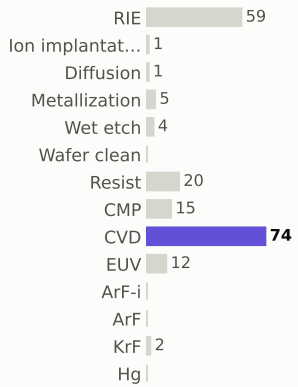
y cut at 0.184 µm



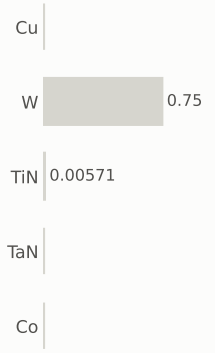
mask for this step

no mask for this step

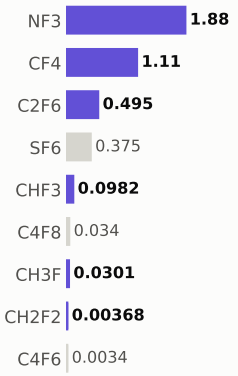
steps by process area



materials deposited (g)



gases emitted (g)



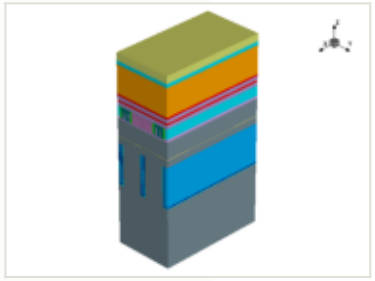
Step #2120: CVD

depo_isotropic · CVD

1.31	249.9	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

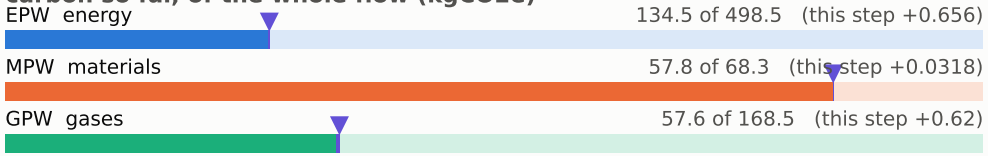


this step

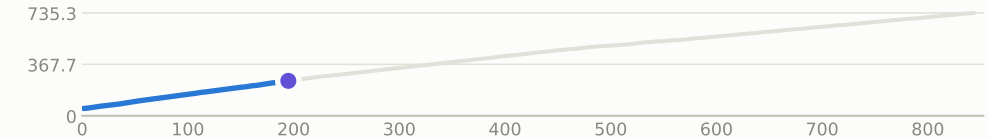
process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

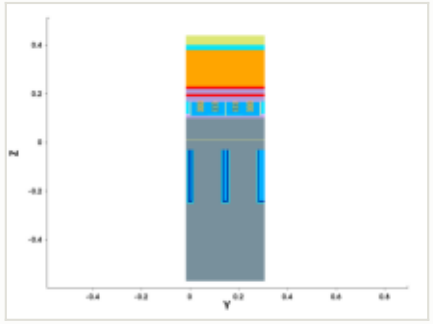
carbon so far, of the whole flow (kgCO2e)



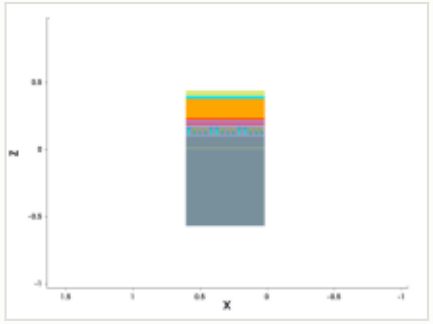
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



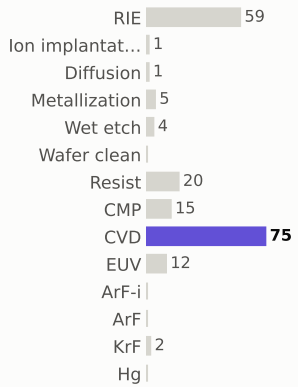
y cut at 0.184 µm



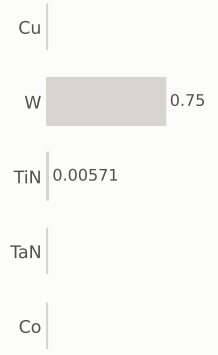
mask for this step

no mask for this step

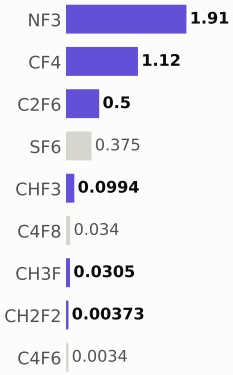
steps by process area



materials deposited (g)



gases emitted (g)



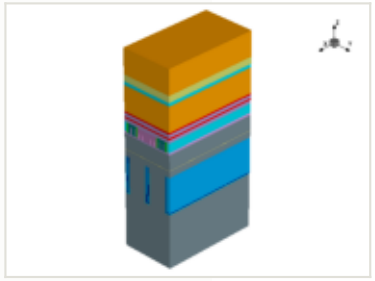
Step #2130: CVD

depo_fill · CVD

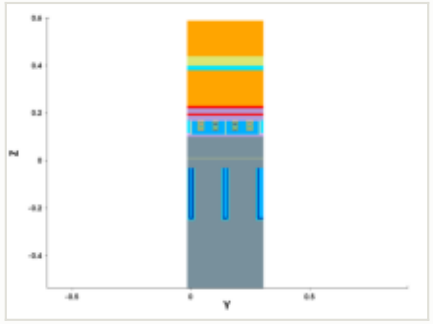
1.31	251.2	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

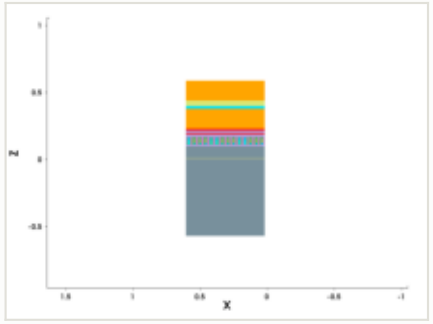
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

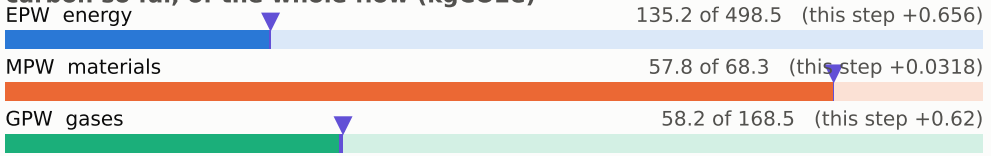
no mask for this step

this step

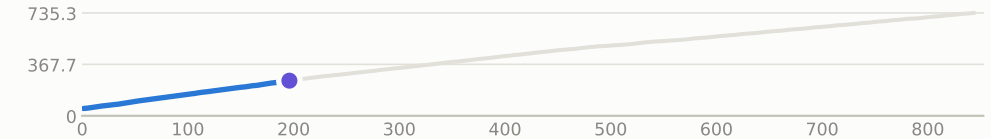
process area	CVD
material	—
thickness	0.15 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

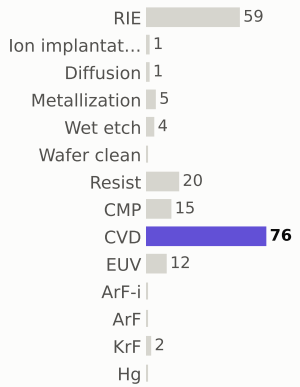
carbon so far, of the whole flow (kgCO2e)



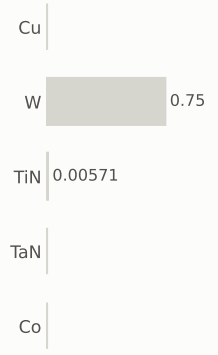
embodied carbon up to each step (kgCO2e per wafer)



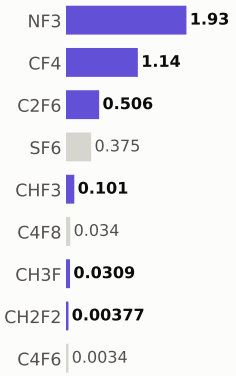
steps by process area



materials deposited (g)



gases emitted (g)



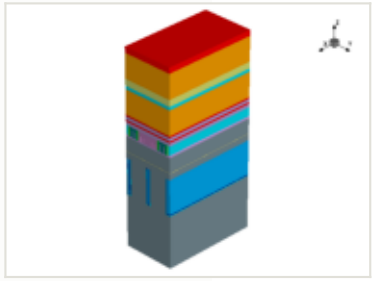
Step #2140: CVD

depo_fill · CVD

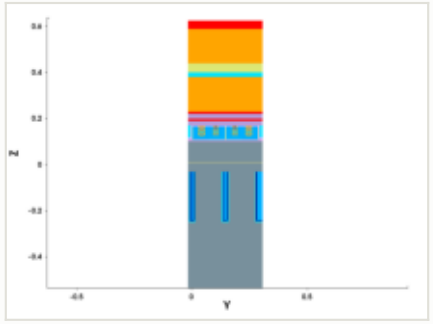
1.31	252.5	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

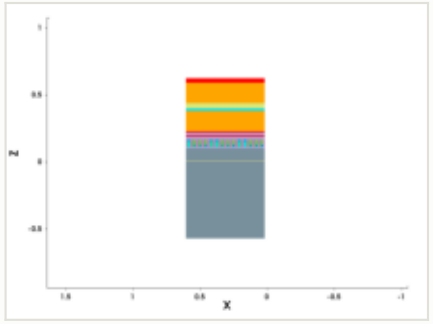
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

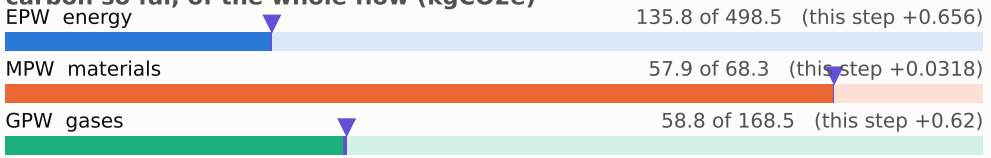
no mask for this step

this step

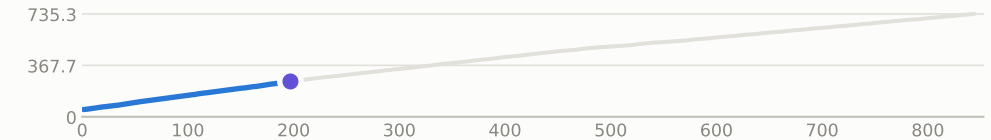
process area	CVD
material	—
thickness	0.035 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

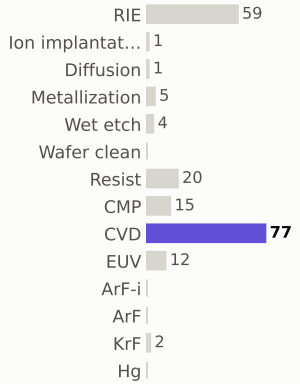
carbon so far, of the whole flow (kgCO2e)



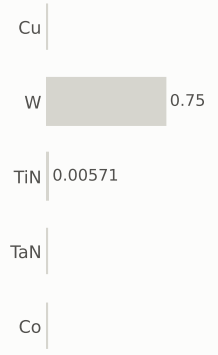
embodied carbon up to each step (kgCO2e per wafer)



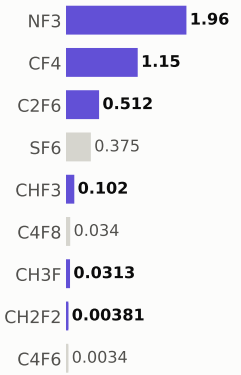
steps by process area



materials deposited (g)



gases emitted (g)



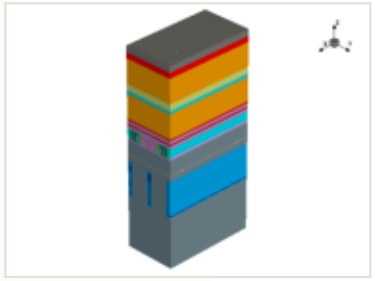
Step #2150: EUV

litho_patterning · EUV

1.96	254.5	0.27%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

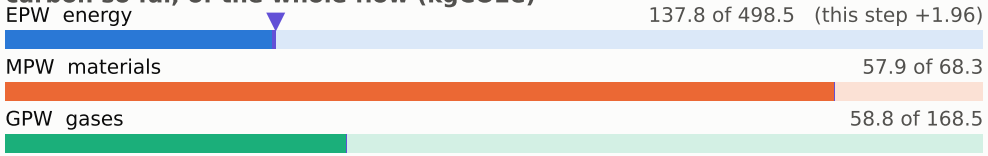


this step

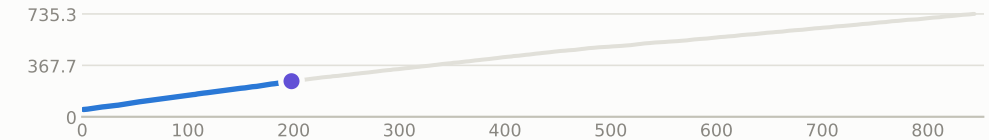
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

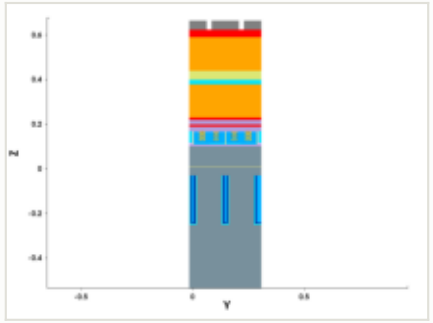
carbon so far, of the whole flow (kgCO2e)



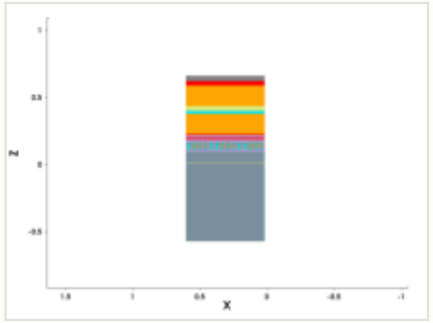
embodied carbon up to each step (kgCO2e per wafer)



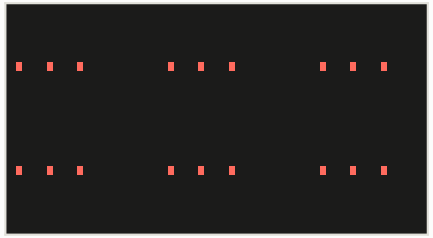
x cut at 0.294 µm



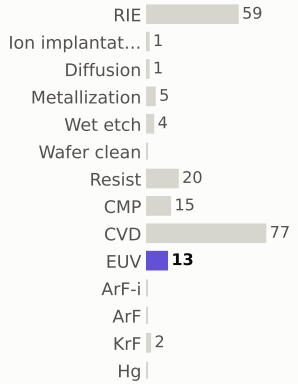
y cut at 0.184 µm



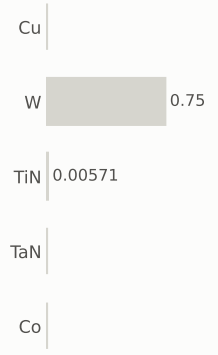
mask: layer 3012:0 (GDSII)



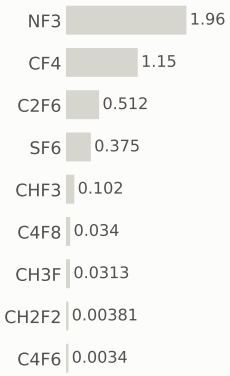
steps by process area



materials deposited (g)



gases emitted (g)



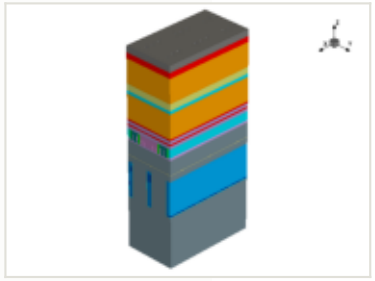
Step #2160: RIE

etch_anisotropic · RIE

1.04	255.5	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

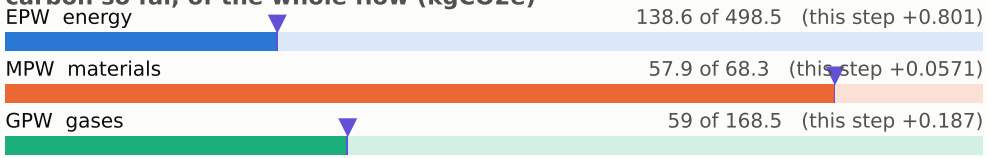


this step

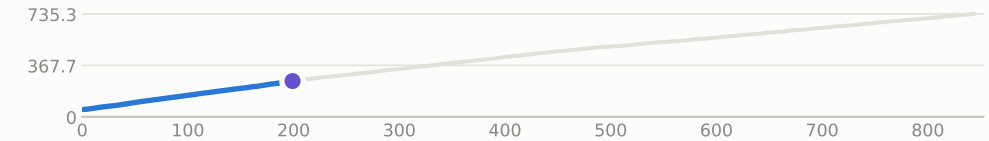
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

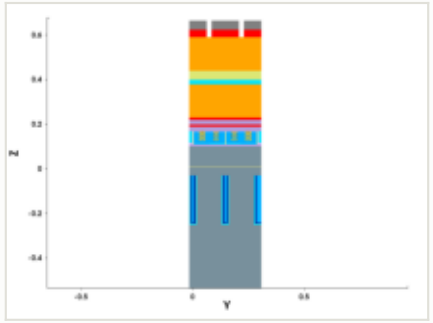
carbon so far, of the whole flow (kgCO2e)



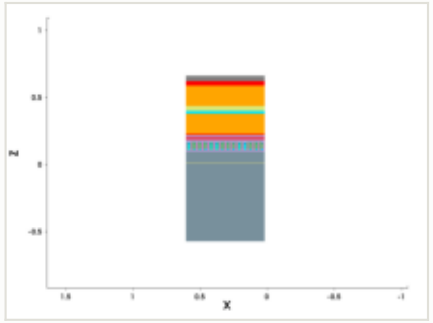
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



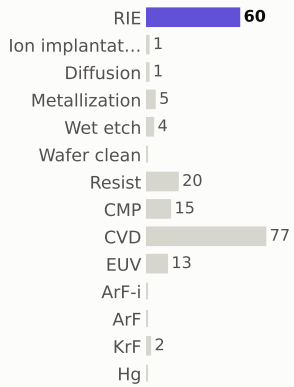
y cut at 0.184 µm



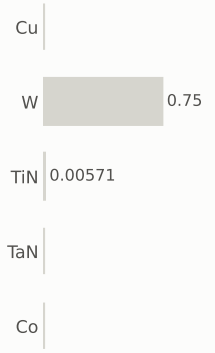
mask for this step

no mask for this step

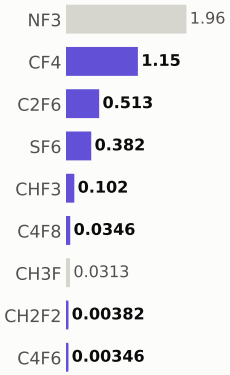
steps by process area



materials deposited (g)



gases emitted (g)



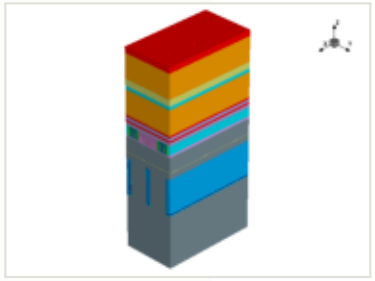
Step #2170: RIE

etch_anisotropic · RIE

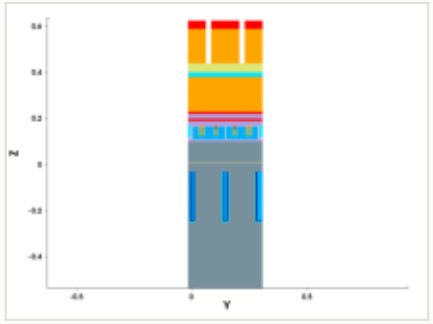
1.04	256.6	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

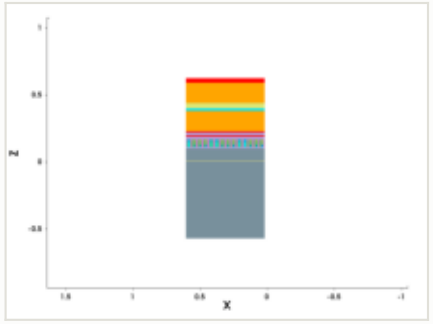
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

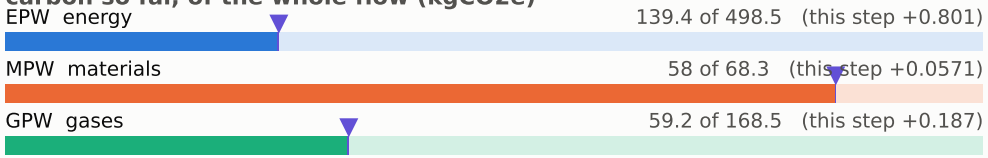
no mask for this step

this step

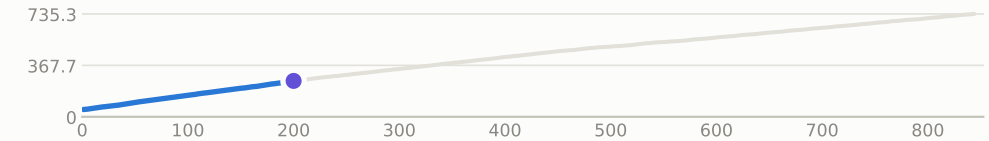
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

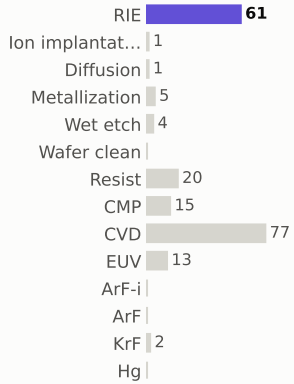
carbon so far, of the whole flow (kgCO2e)



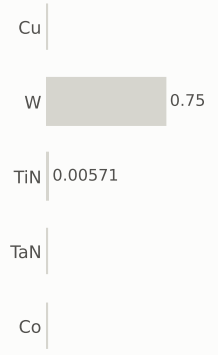
embodied carbon up to each step (kgCO2e per wafer)



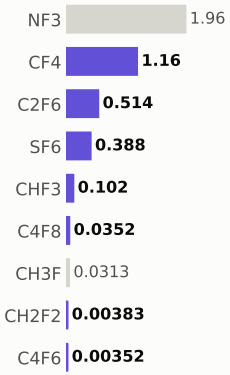
steps by process area



materials deposited (g)



gases emitted (g)



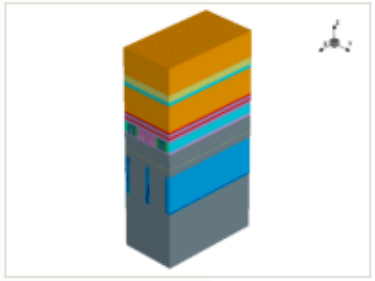
Step #2180: RIE

etch_anisotropic · RIE

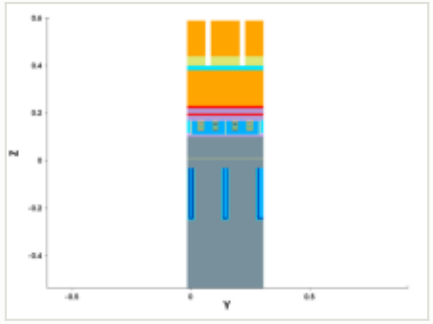
1.04	257.6	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

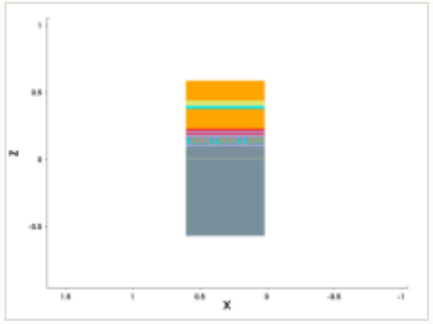
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

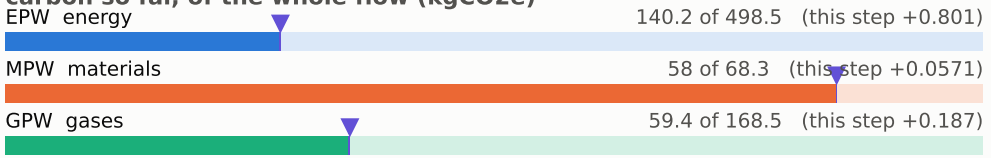
no mask for this step

this step

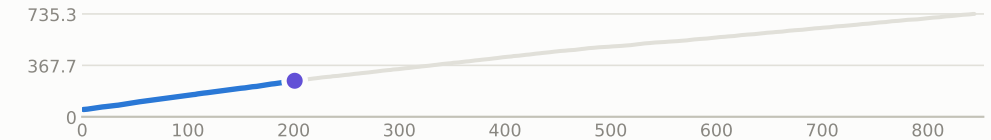
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

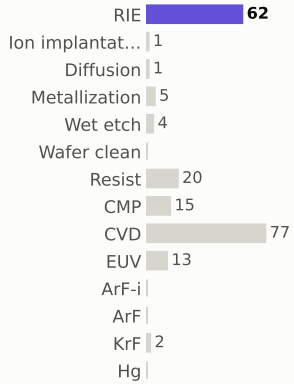
carbon so far, of the whole flow (kgCO2e)



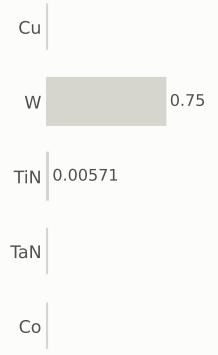
embodied carbon up to each step (kgCO2e per wafer)



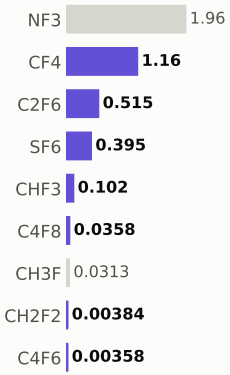
steps by process area



materials deposited (g)



gases emitted (g)



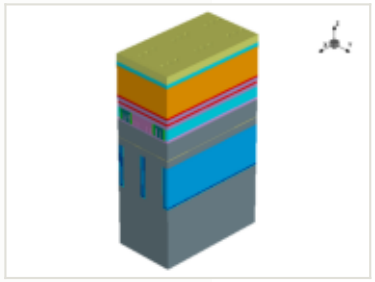
Step #2190: Resist

etch_strip · Resist

0.169	257.8	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

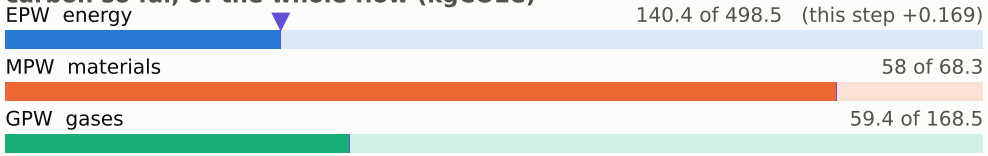


this step

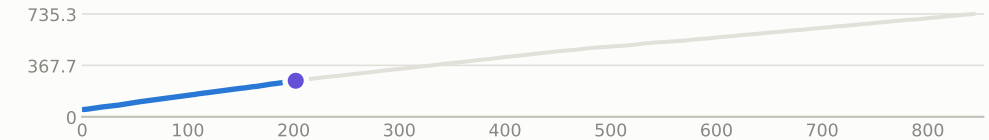
process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

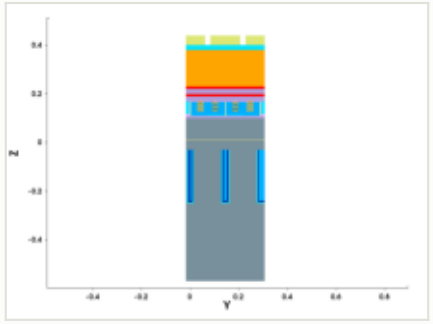
carbon so far, of the whole flow (kgCO2e)



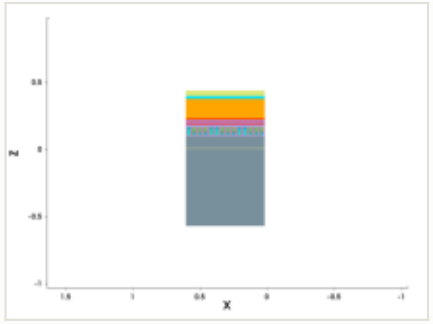
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



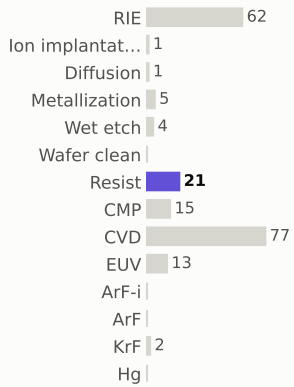
y cut at 0.184 µm



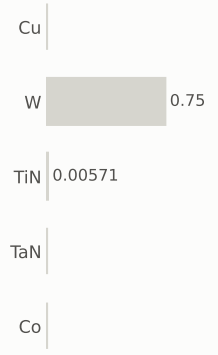
mask for this step

no mask for this step

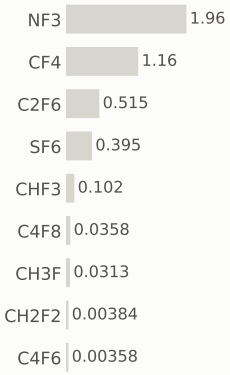
steps by process area



materials deposited (g)



gases emitted (g)



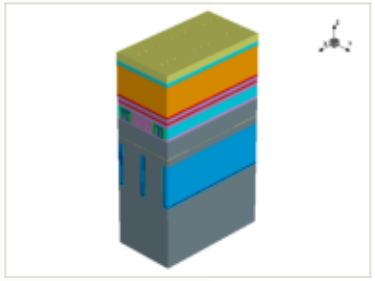
Step #2200: RIE

etch_anisotropic · RIE

1.04	258.8	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

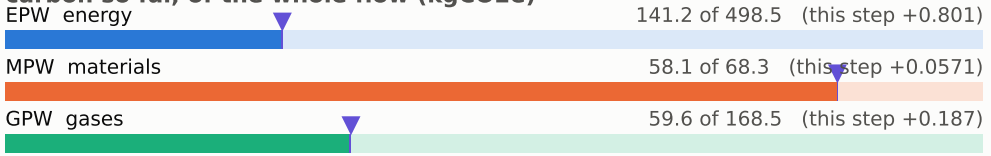


this step

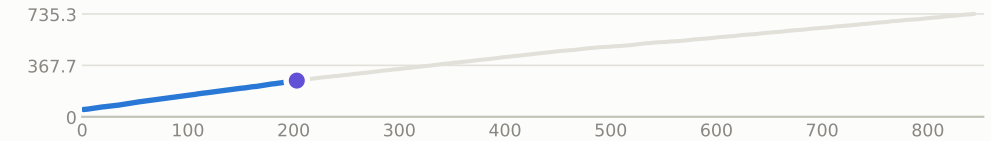
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

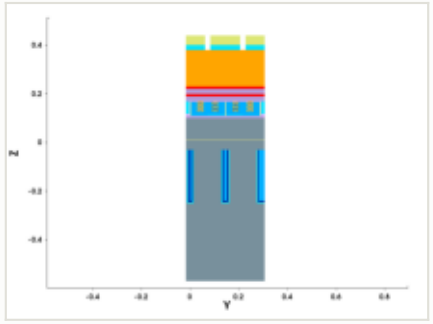
carbon so far, of the whole flow (kgCO2e)



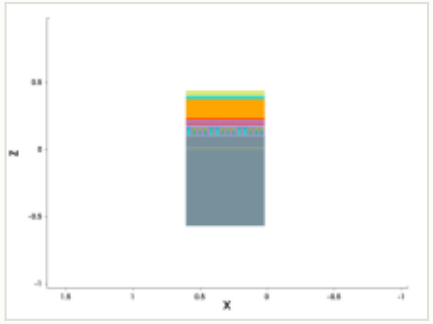
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



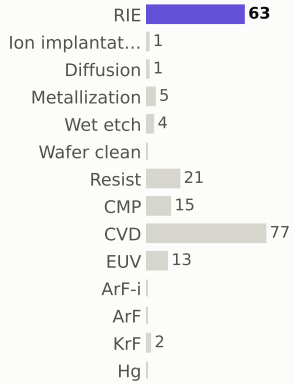
y cut at 0.184 µm



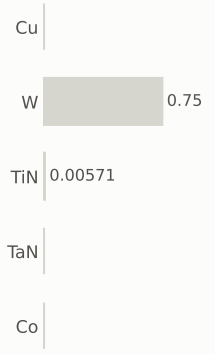
mask for this step

no mask for this step

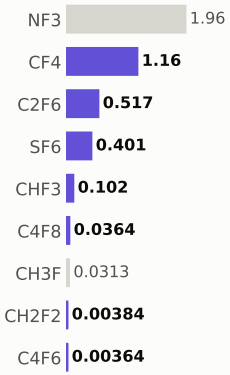
steps by process area



materials deposited (g)



gases emitted (g)



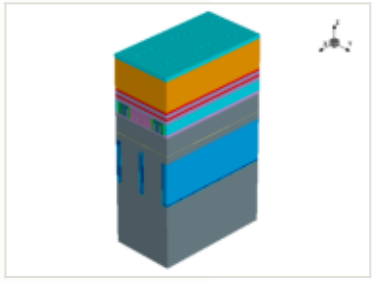
Step #2210: RIE

etch_anisotropic · RIE

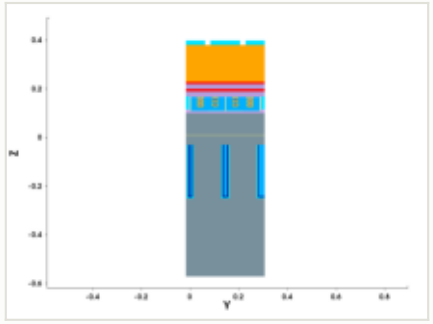
1.04	259.9	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

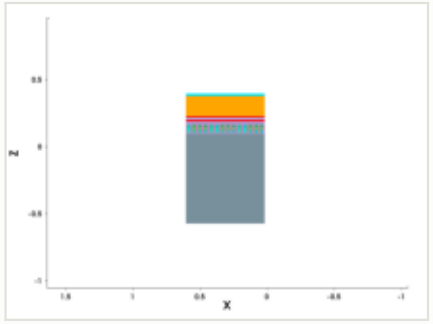
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

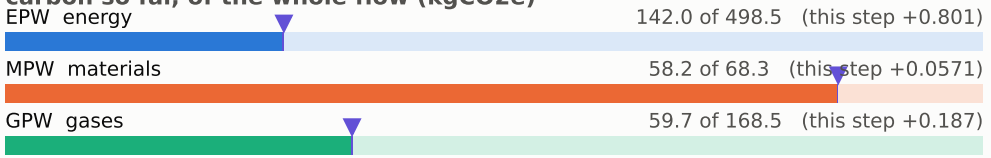
no mask for this step

this step

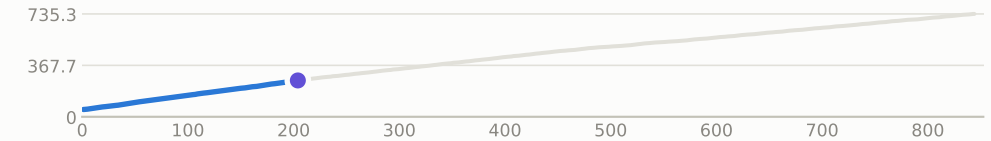
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

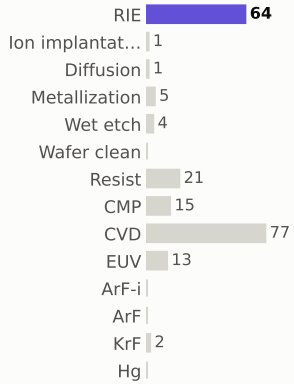
carbon so far, of the whole flow (kgCO2e)



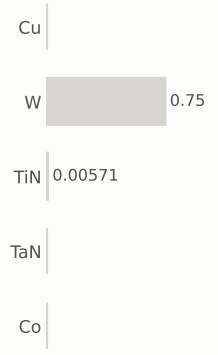
embodied carbon up to each step (kgCO2e per wafer)



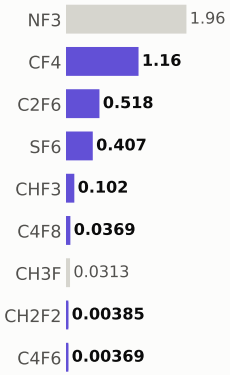
steps by process area



materials deposited (g)



gases emitted (g)



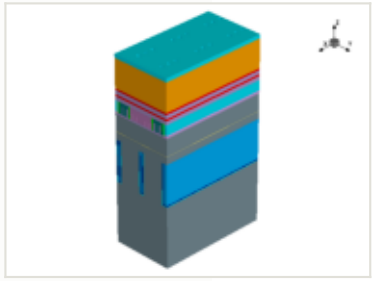
Step #2220: RIE

etch_anisotropic · RIE

1.04	260.9	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

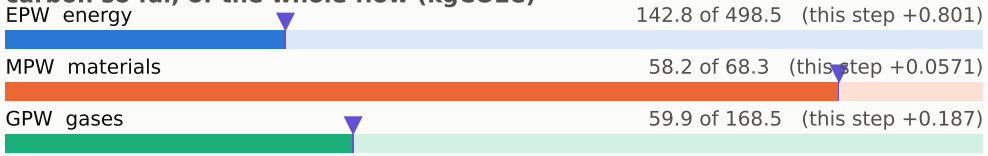


this step

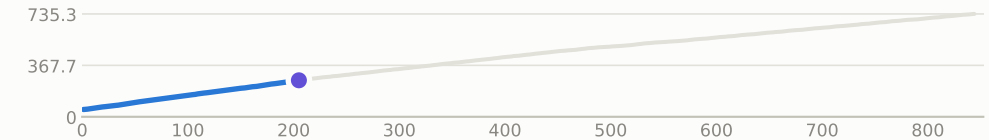
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

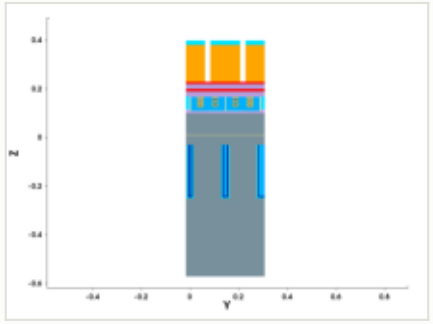
carbon so far, of the whole flow (kgCO2e)



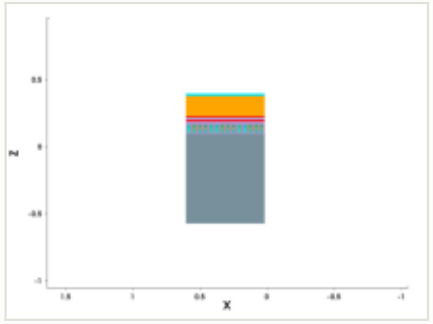
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



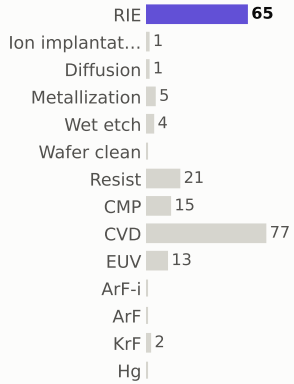
y cut at 0.184 µm



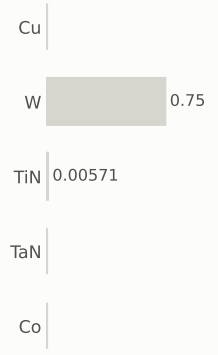
mask for this step

no mask for this step

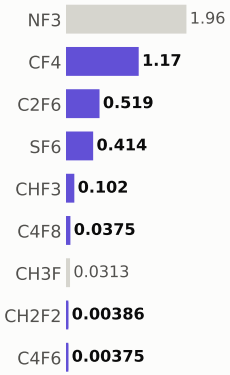
steps by process area



materials deposited (g)



gases emitted (g)



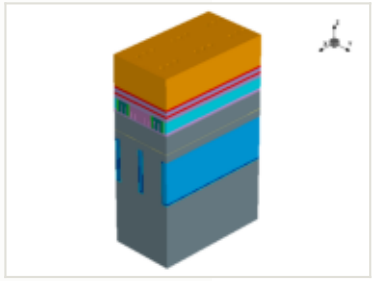
Step #2230: RIE

etch_anisotropic · RIE

1.04	262.0	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

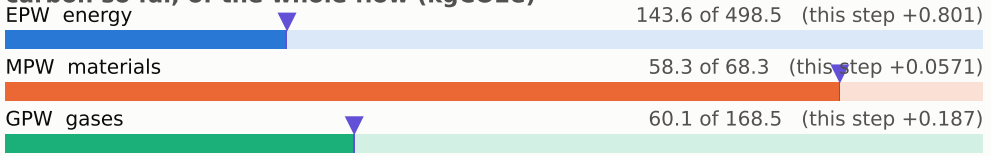


this step

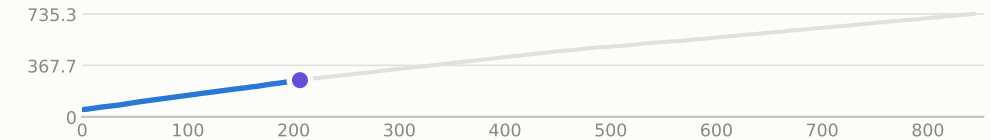
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

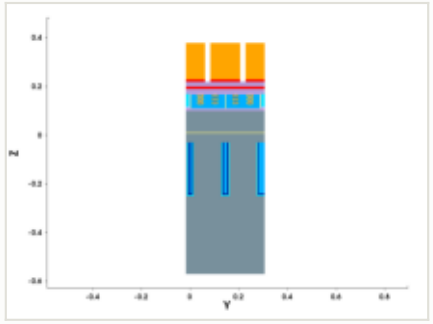
carbon so far, of the whole flow (kgCO2e)



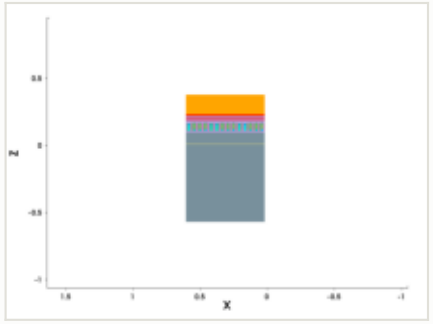
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



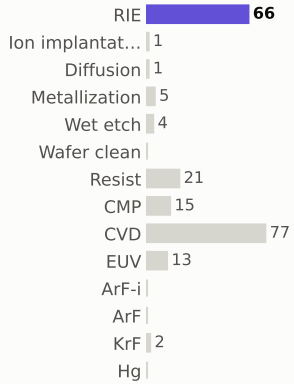
y cut at 0.184 µm



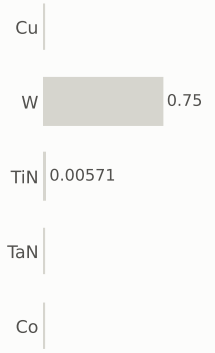
mask for this step

no mask for this step

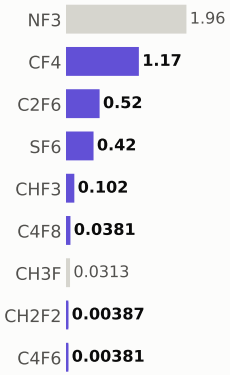
steps by process area



materials deposited (g)



gases emitted (g)



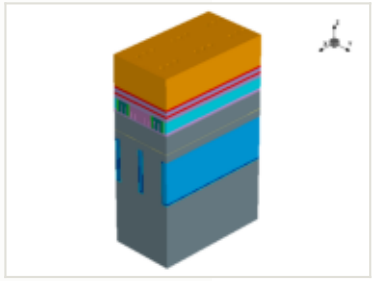
Step #2240: RIE

etch_anisotropic · RIE

1.04	263.0	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

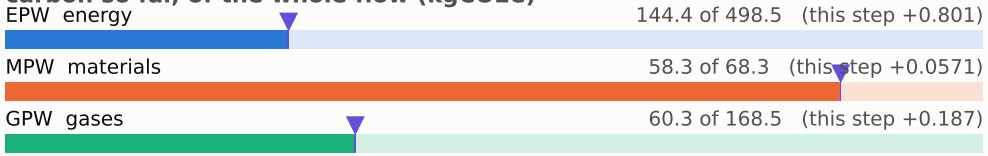


this step

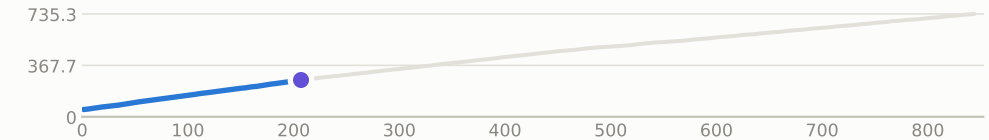
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

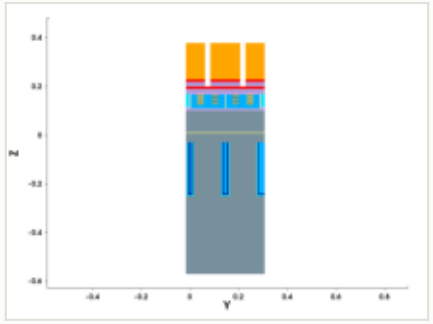
carbon so far, of the whole flow (kgCO2e)



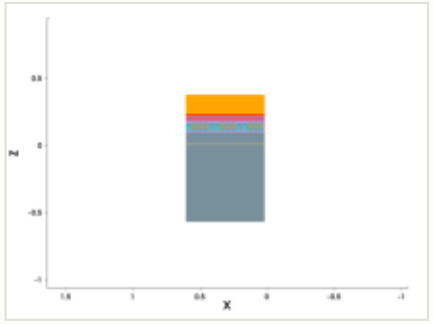
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



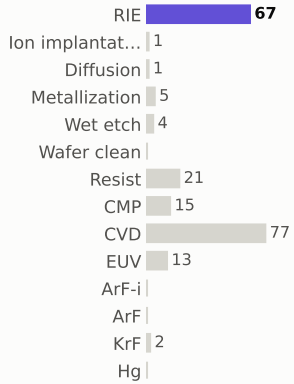
y cut at 0.184 µm



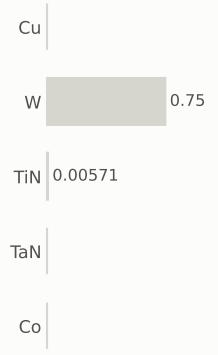
mask for this step

no mask for this step

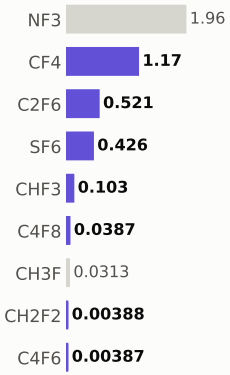
steps by process area



materials deposited (g)



gases emitted (g)



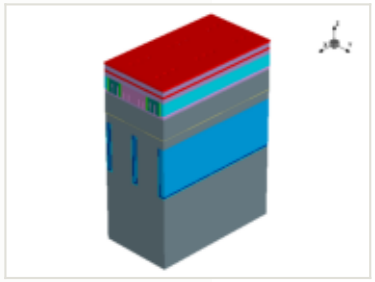
Step #2250: Resist

etch_strip · Resist

0.169	263.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

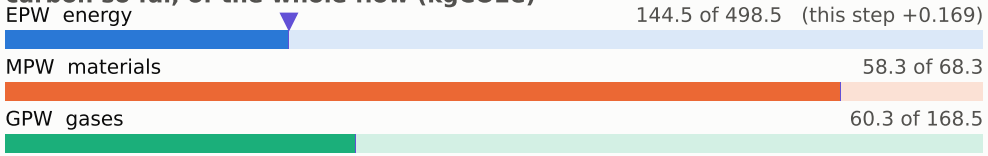


this step

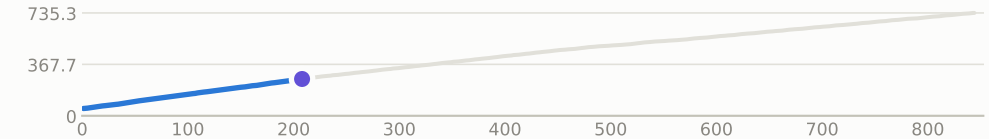
process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

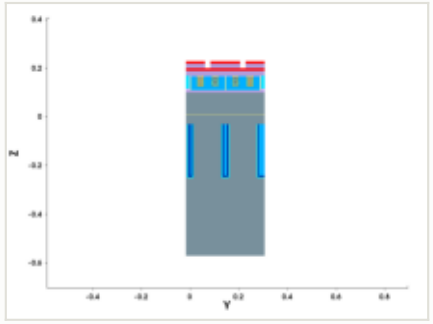
carbon so far, of the whole flow (kgCO2e)



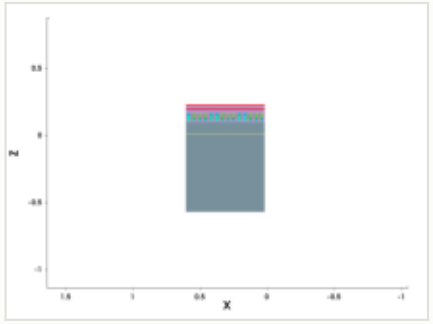
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



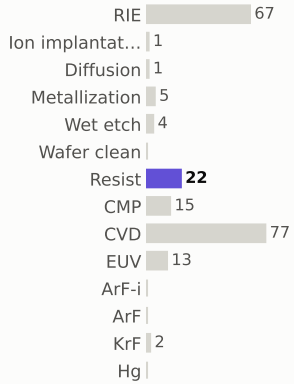
y cut at 0.184 µm



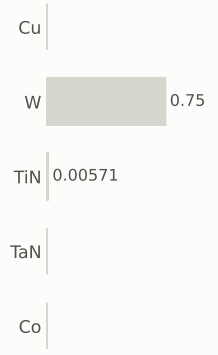
mask for this step

no mask for this step

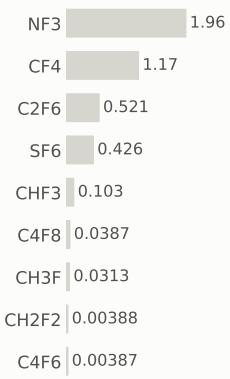
steps by process area



materials deposited (g)



gases emitted (g)



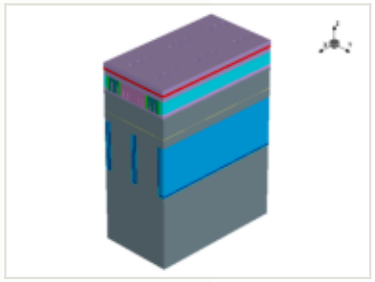
Step #2260: RIE

etch_anisotropic · RIE

1.04	264.2	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

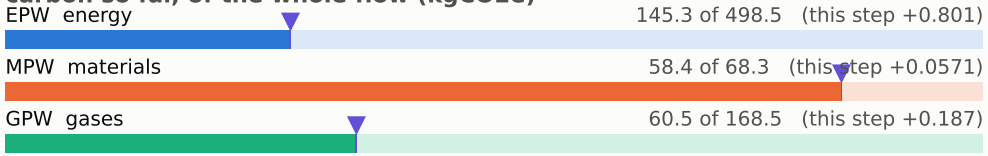


this step

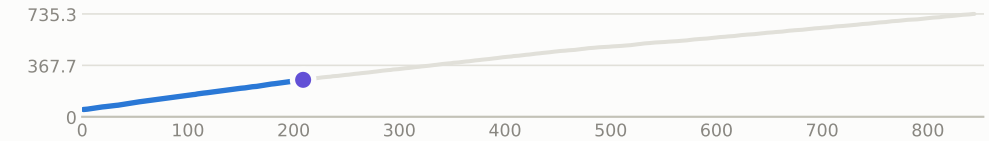
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

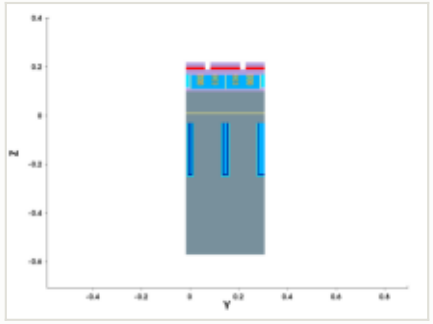
carbon so far, of the whole flow (kgCO2e)



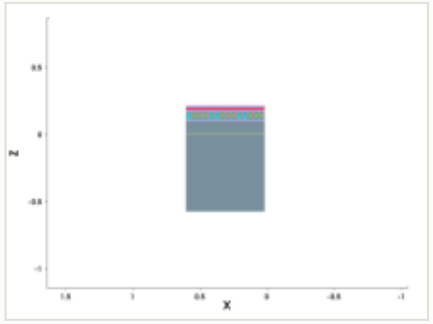
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



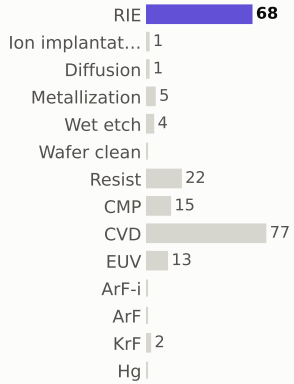
y cut at 0.184 µm



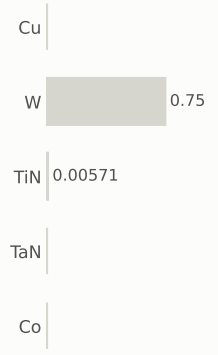
mask for this step

no mask for this step

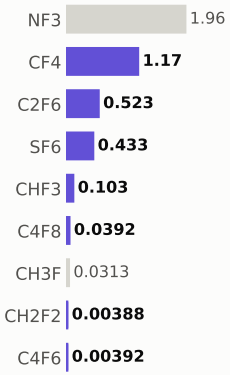
steps by process area



materials deposited (g)



gases emitted (g)



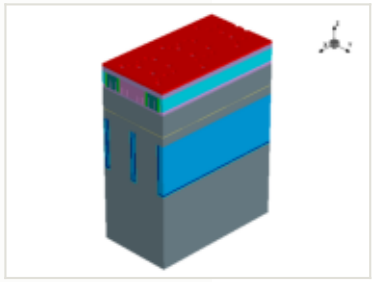
Step #2270: RIE

etch_anisotropic · RIE

1.04	265.3	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

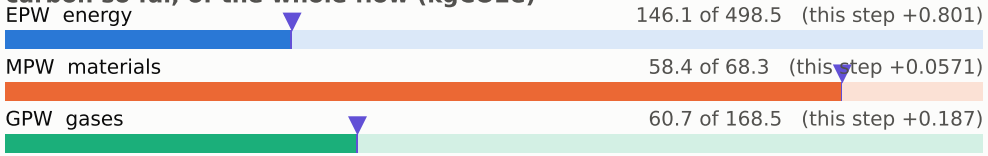


this step

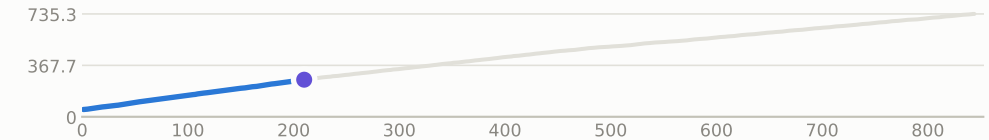
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

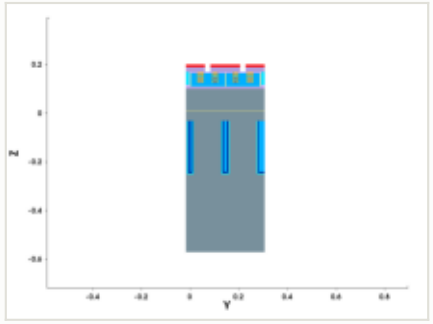
carbon so far, of the whole flow (kgCO2e)



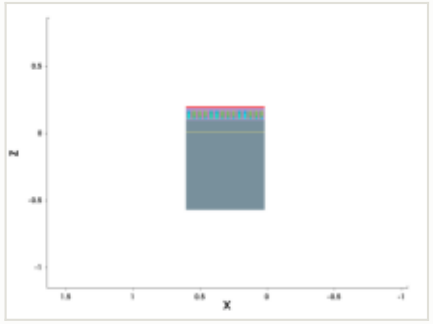
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



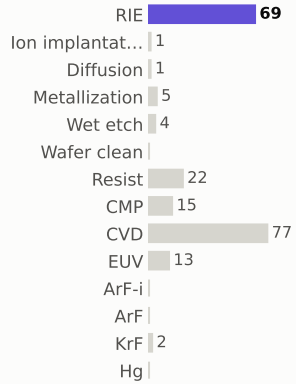
y cut at 0.184 µm



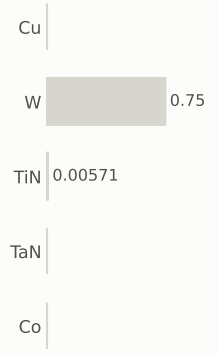
mask for this step

no mask for this step

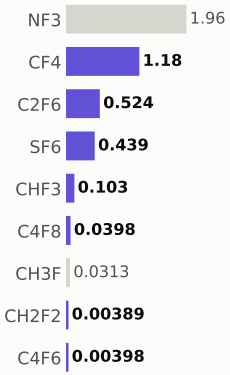
steps by process area



materials deposited (g)



gases emitted (g)



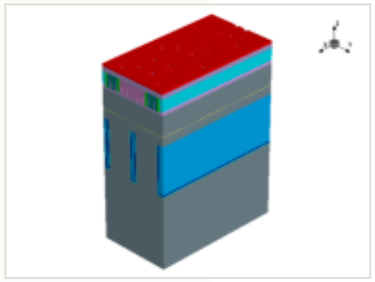
Step #2280: RIE

etch_anisotropic · RIE

1.04	266.3	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

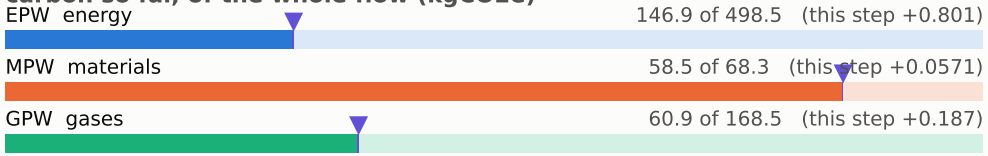
zoomed to fit



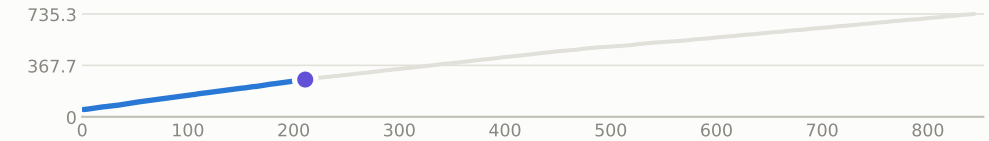
this step

process area	RIE	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

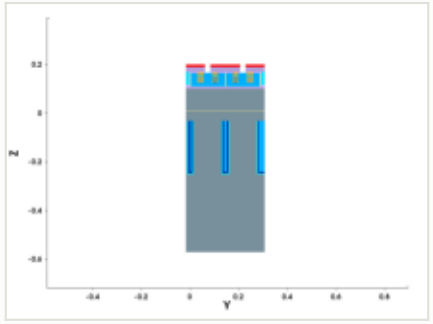
carbon so far, of the whole flow (kgCO2e)



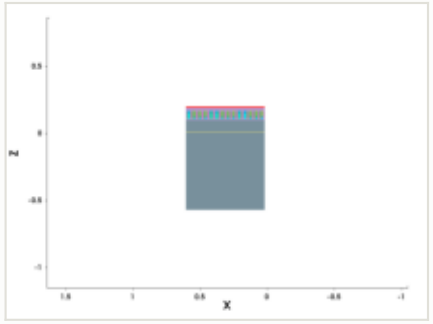
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



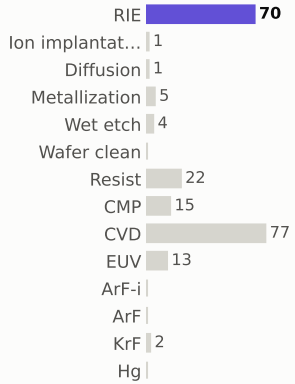
y cut at 0.184 µm



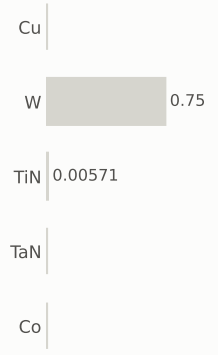
mask for this step

no mask for this step

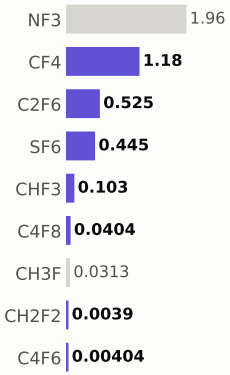
steps by process area



materials deposited (g)



gases emitted (g)



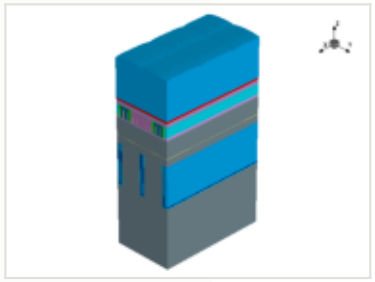
Step #2290: Metallization

depo_isotropic · Metallization

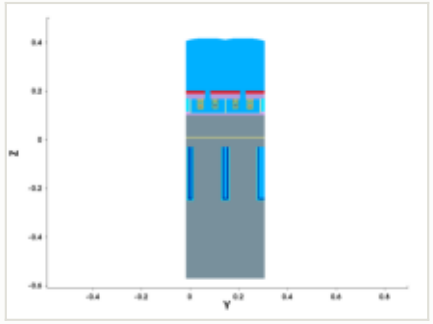
1.23	267.5	0.17%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

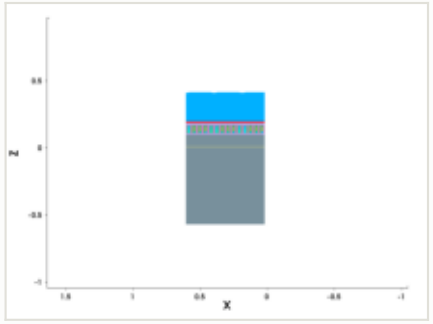
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

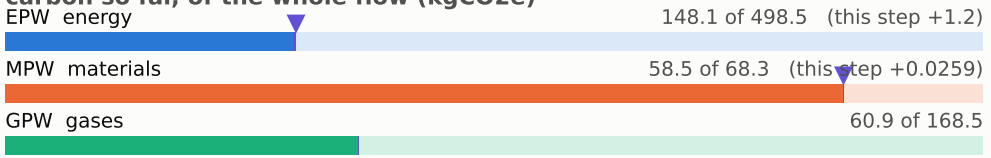
no mask for this step

this step

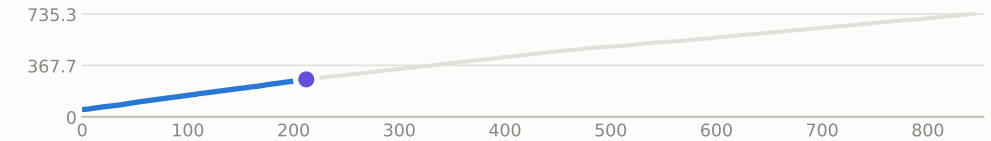
process area	Metallization
material	W
thickness	0.25 µm
material GWP	75.8149 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.0259 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.23 kgCO2e**

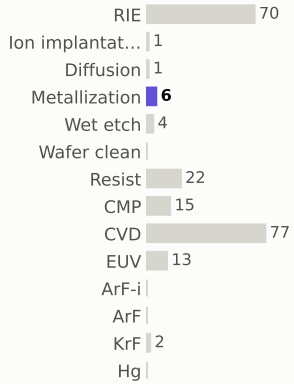
carbon so far, of the whole flow (kgCO2e)



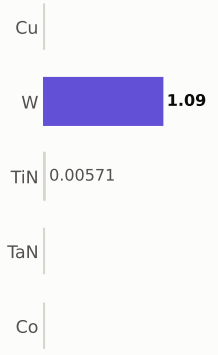
embodied carbon up to each step (kgCO2e per wafer)



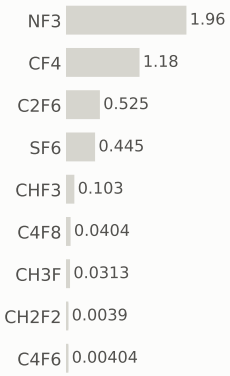
steps by process area



materials deposited (g)



gases emitted (g)



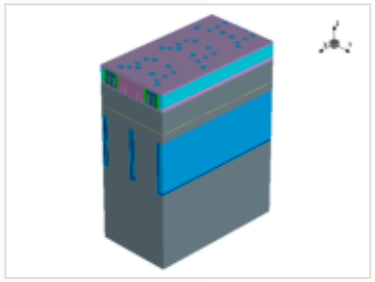
Step #2300: CMP

etch_cmp · CMP

0.169	267.7	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

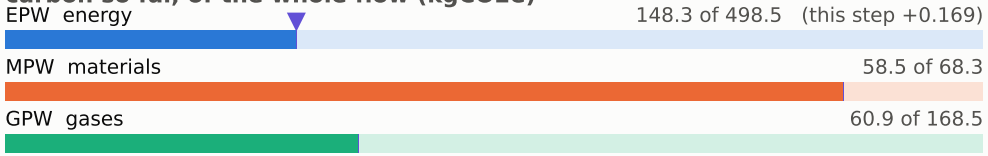


this step

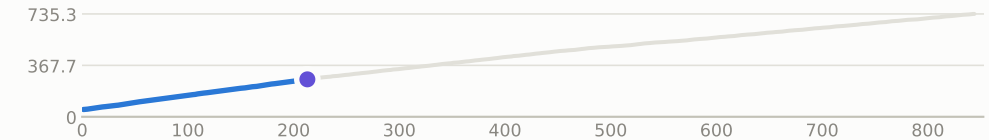
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

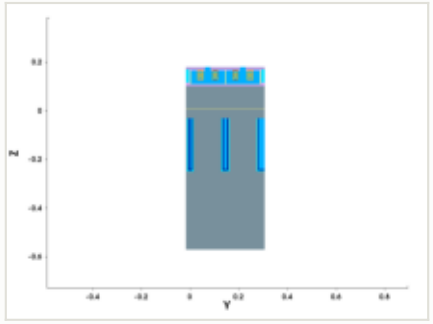
carbon so far, of the whole flow (kgCO2e)



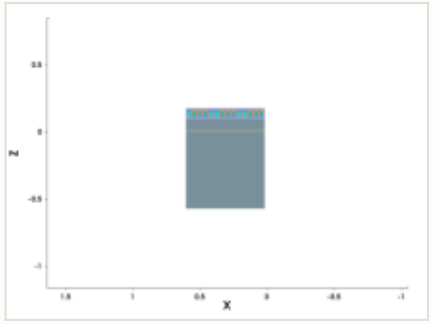
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



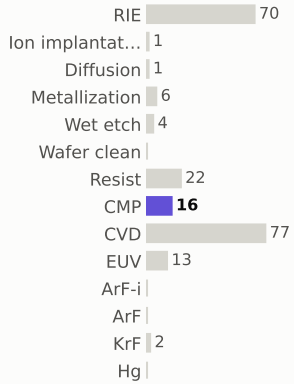
y cut at 0.184 µm



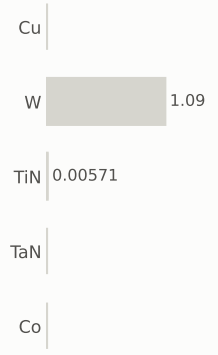
mask for this step

no mask for this step

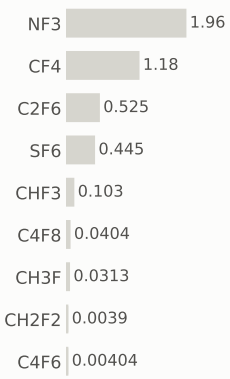
steps by process area



materials deposited (g)



gases emitted (g)



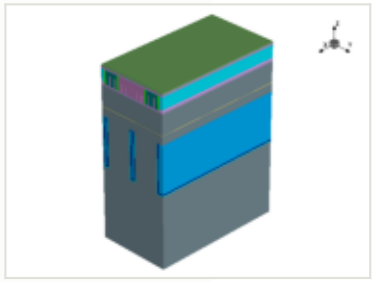
Step #2320: CVD

depo_fill · CVD

1.31	269.0	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

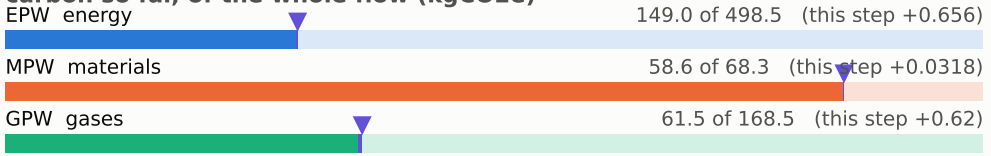


this step

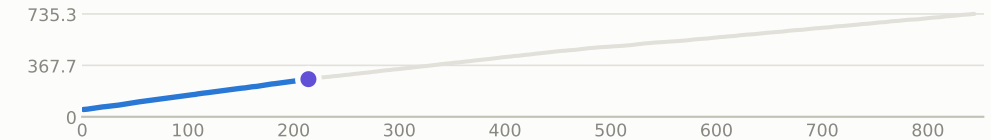
process area	CVD
material	—
thickness	0.006 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

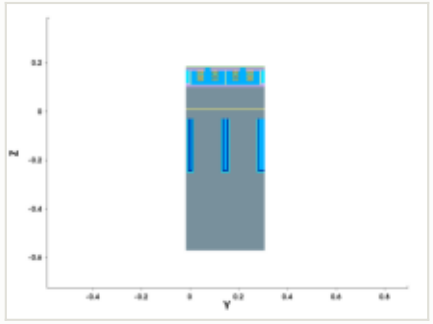
carbon so far, of the whole flow (kgCO2e)



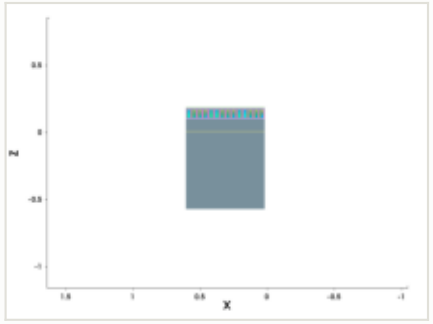
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



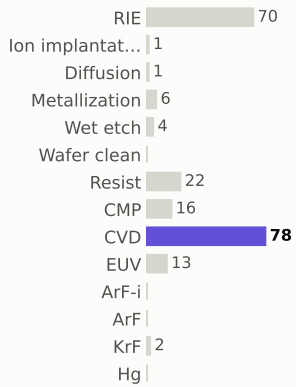
y cut at 0.184 µm



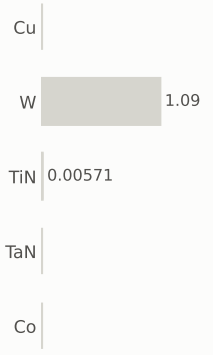
mask for this step

no mask for this step

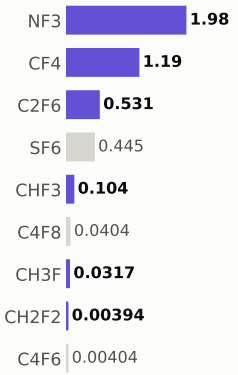
steps by process area



materials deposited (g)



gases emitted (g)



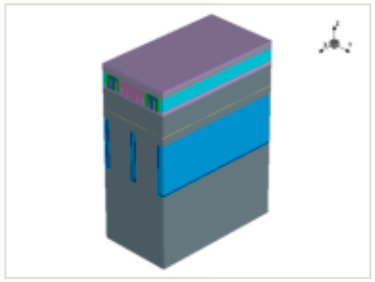
Step #2330: CVD

depo_fill · CVD

1.31	270.3	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

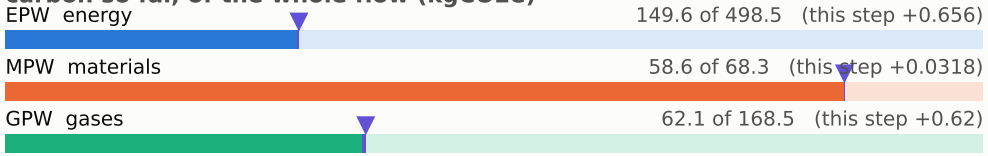


this step

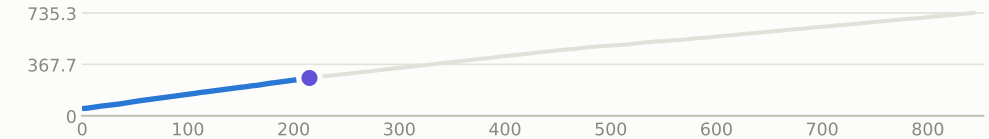
process area	CVD
material	—
thickness	0.018 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

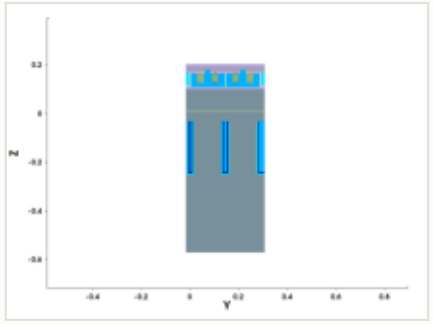
carbon so far, of the whole flow (kgCO2e)



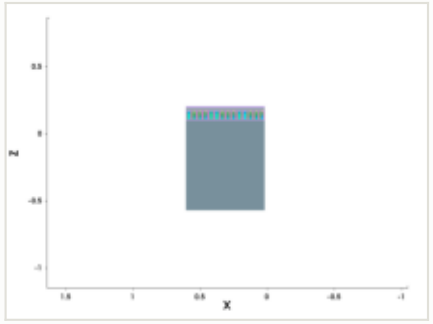
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



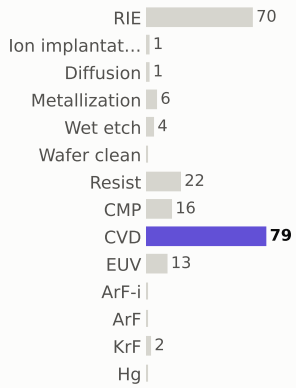
y cut at 0.184 µm



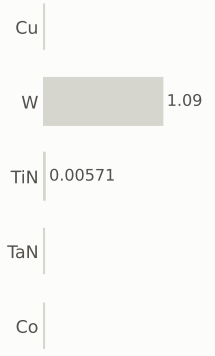
mask for this step

no mask for this step

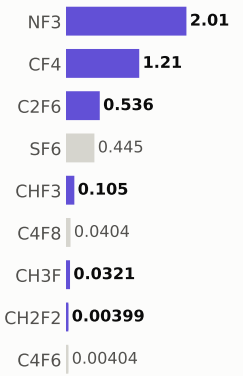
steps by process area



materials deposited (g)



gases emitted (g)



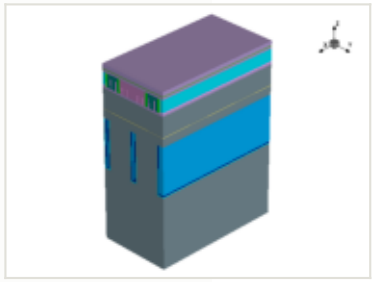
Step #2340: Diffusion

treatment · Diffusion

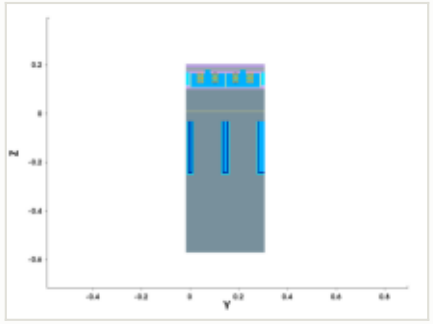
0.801	271.1	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

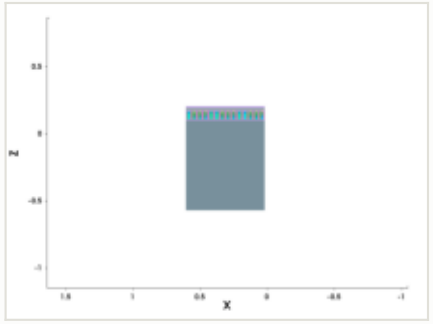
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

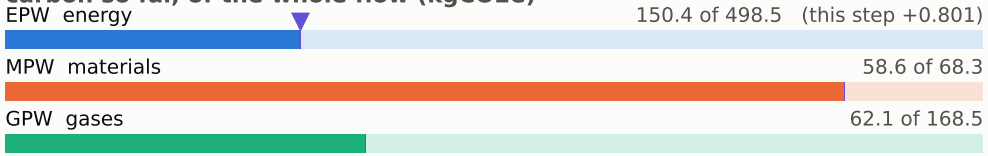
no mask for this step

this step

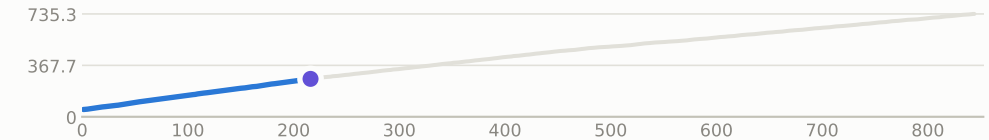
process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

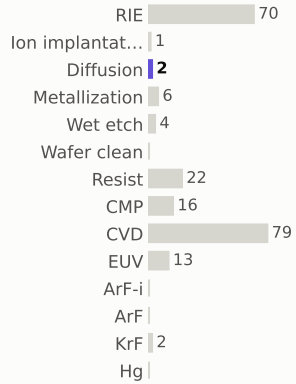
carbon so far, of the whole flow (kgCO2e)



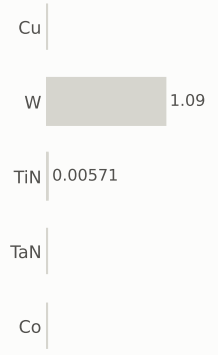
embodied carbon up to each step (kgCO2e per wafer)



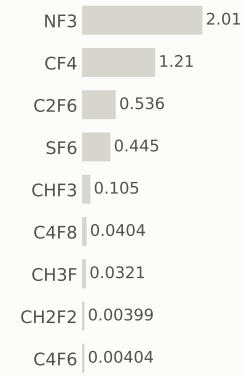
steps by process area



materials deposited (g)



gases emitted (g)



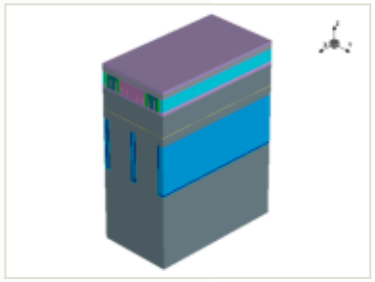
Step #2350: Wafer_clean

clean · Wafer_clean

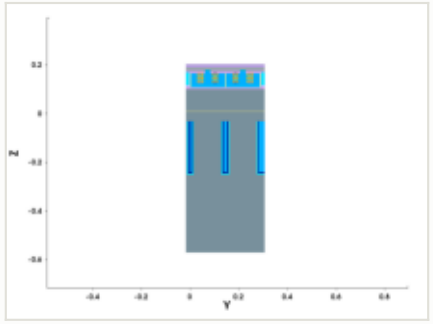
0.169	271.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

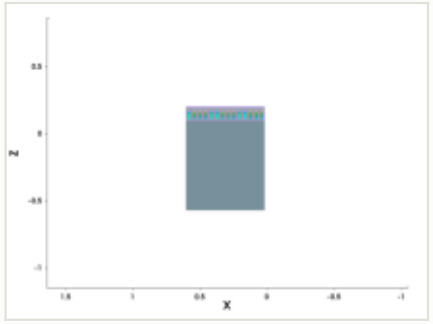
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

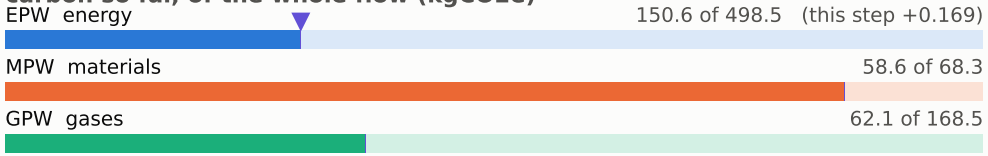
no mask for this step

this step

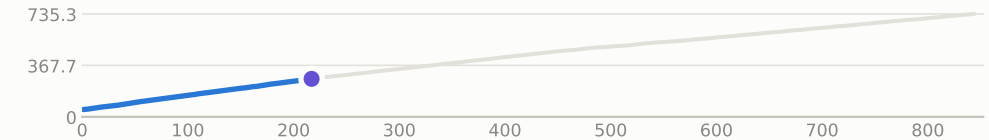
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

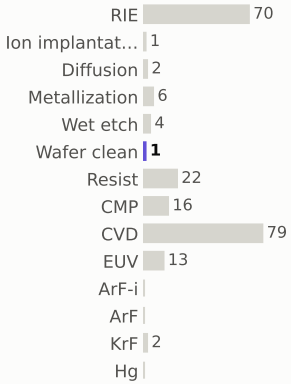
carbon so far, of the whole flow (kgCO2e)



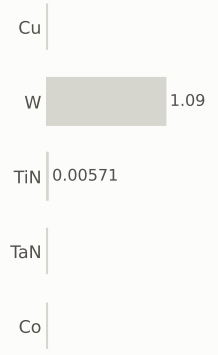
embodied carbon up to each step (kgCO2e per wafer)



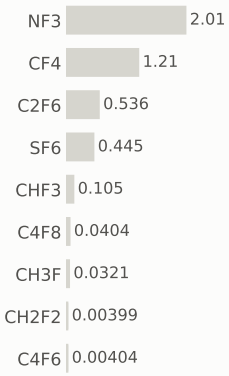
steps by process area



materials deposited (g)



gases emitted (g)



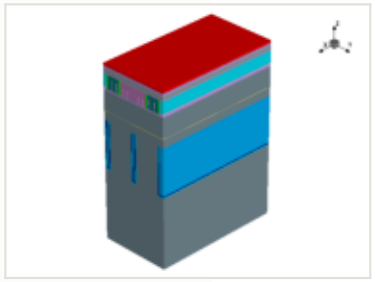
Step #2360: CVD

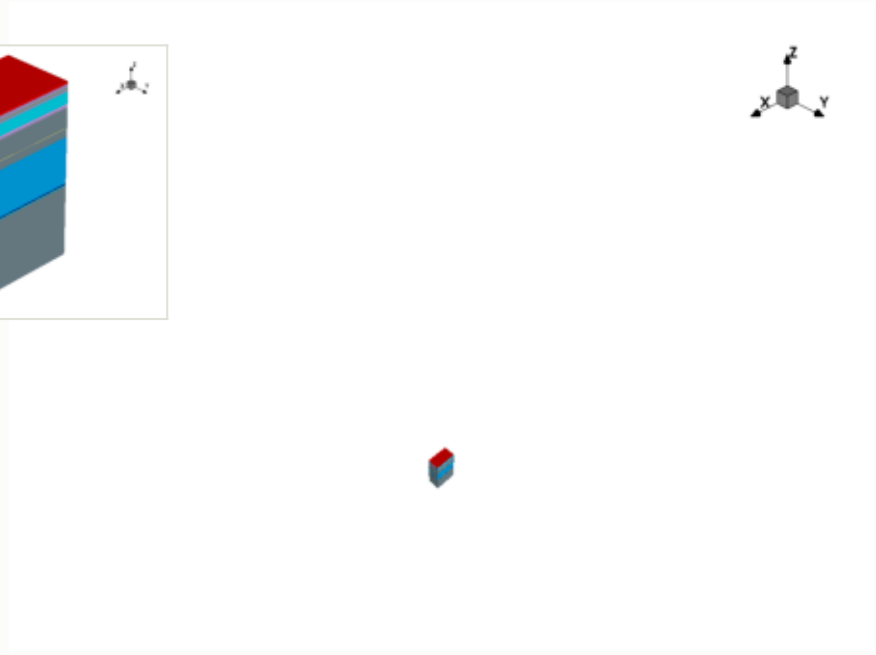
depo_fill · CVD

1.31	272.6	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



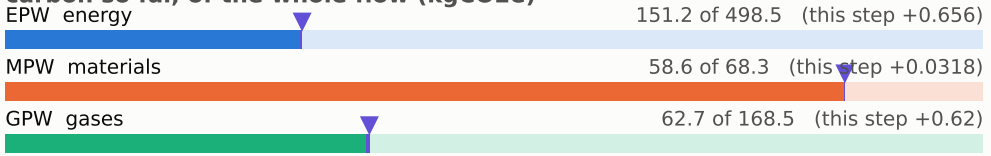


this step

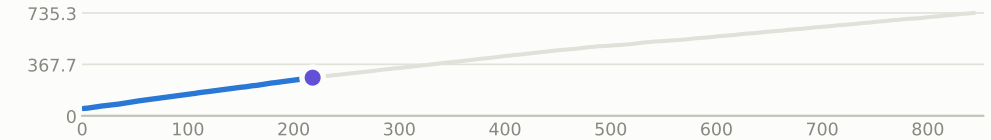
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

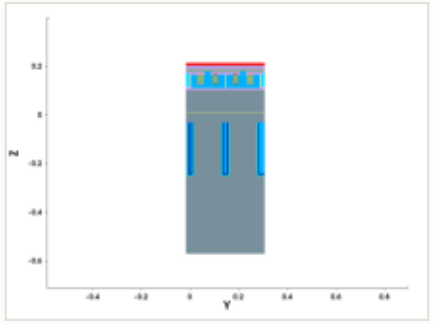
carbon so far, of the whole flow (kgCO2e)



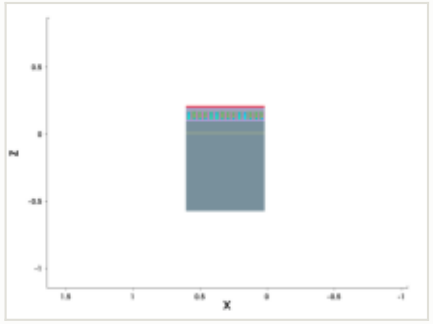
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



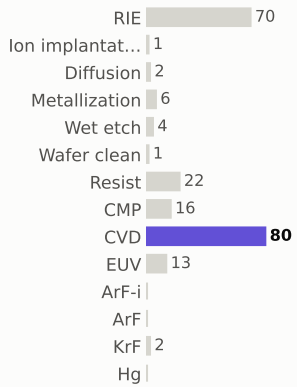
y cut at 0.184 µm



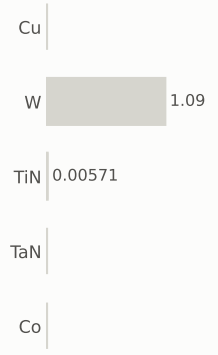
mask for this step

no mask for this step

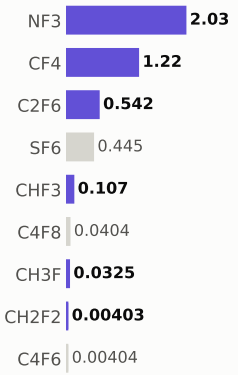
steps by process area



materials deposited (g)



gases emitted (g)



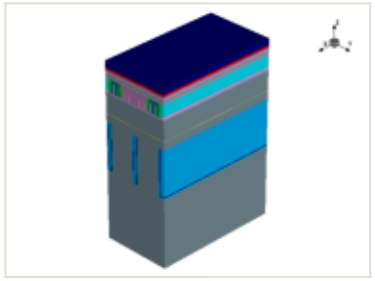
Step #2370: CVD

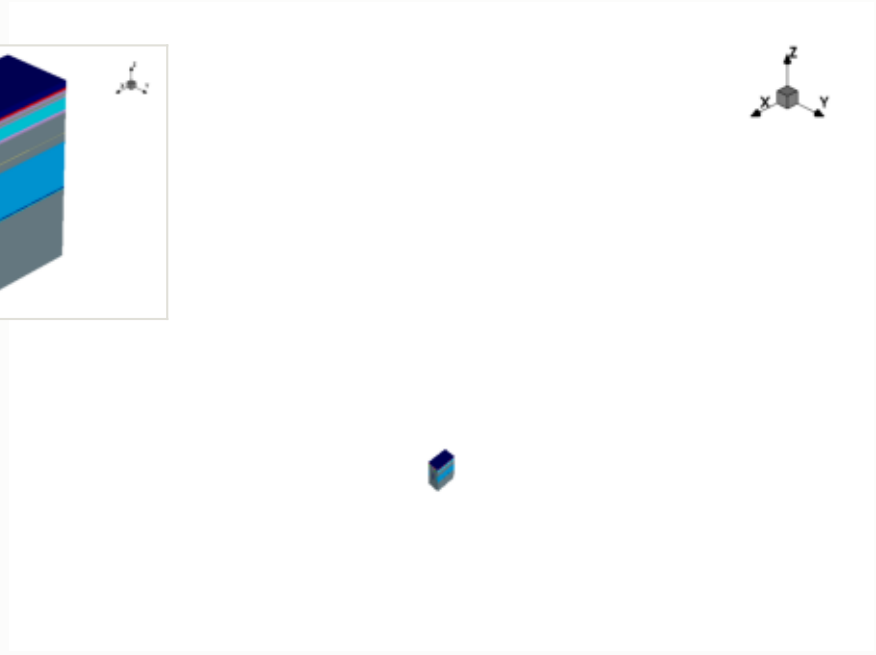
depo_fill · CVD

1.31	273.9	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

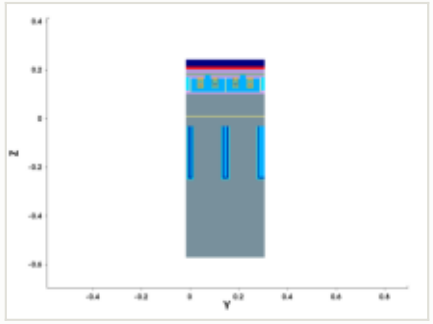
3D structure after this step

zoomed to fit

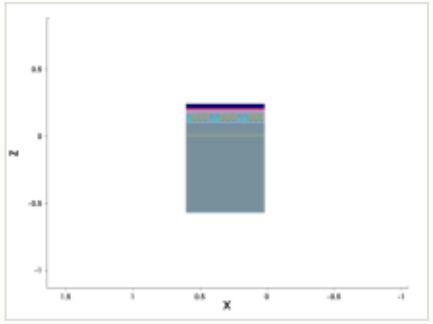




x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

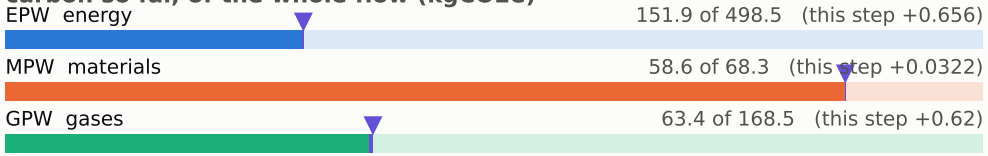
no mask for this step

this step

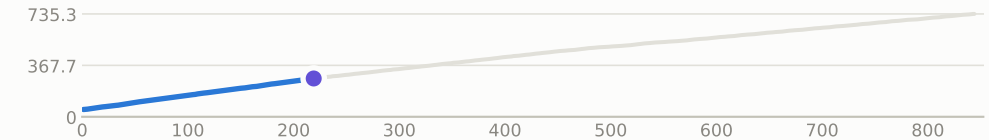
process area	CVD
material	TiN
thickness	0.03 µm
material GWP	43.9778 gCO2e/g
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0322 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

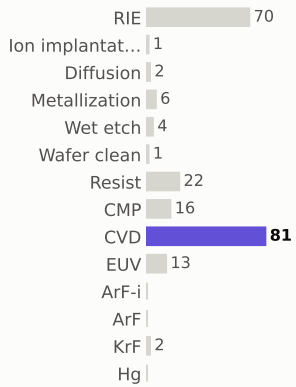
carbon so far, of the whole flow (kgCO2e)



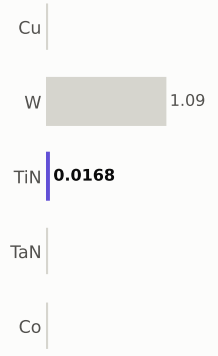
embodied carbon up to each step (kgCO2e per wafer)



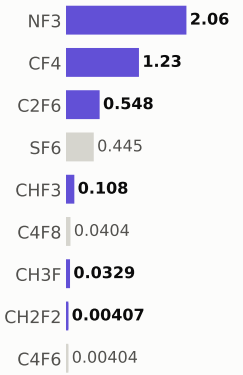
steps by process area



materials deposited (g)



gases emitted (g)



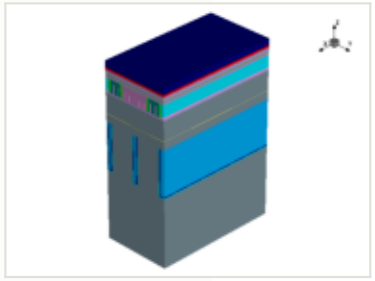
Step #2380: Wafer_clean

clean · Wafer_clean

0.169	274.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



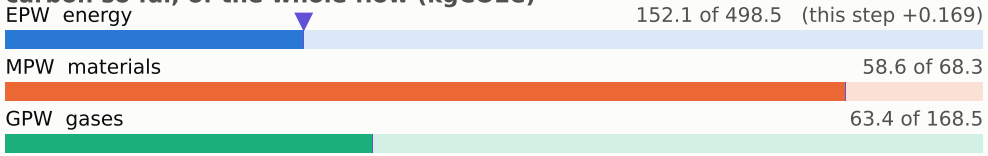
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

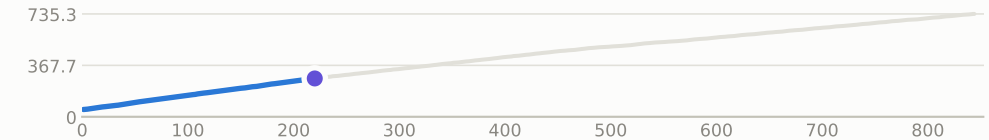
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

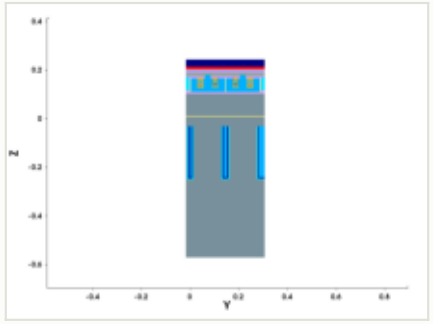
carbon so far, of the whole flow (kgCO2e)



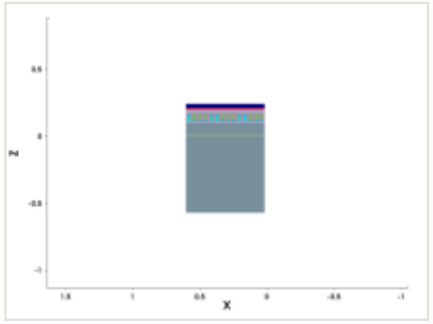
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



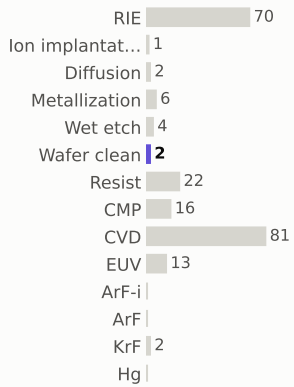
y cut at 0.184 µm



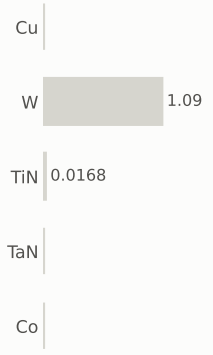
mask for this step

no mask for this step

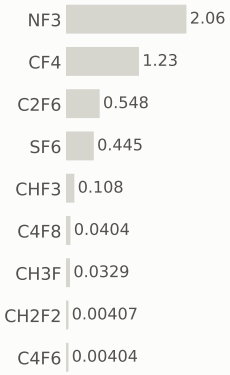
steps by process area



materials deposited (g)



gases emitted (g)



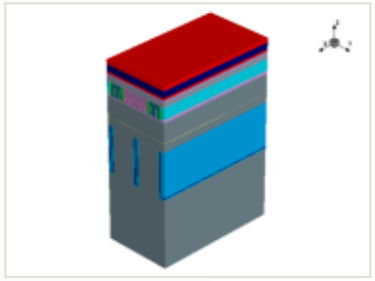
Step #2390: CVD

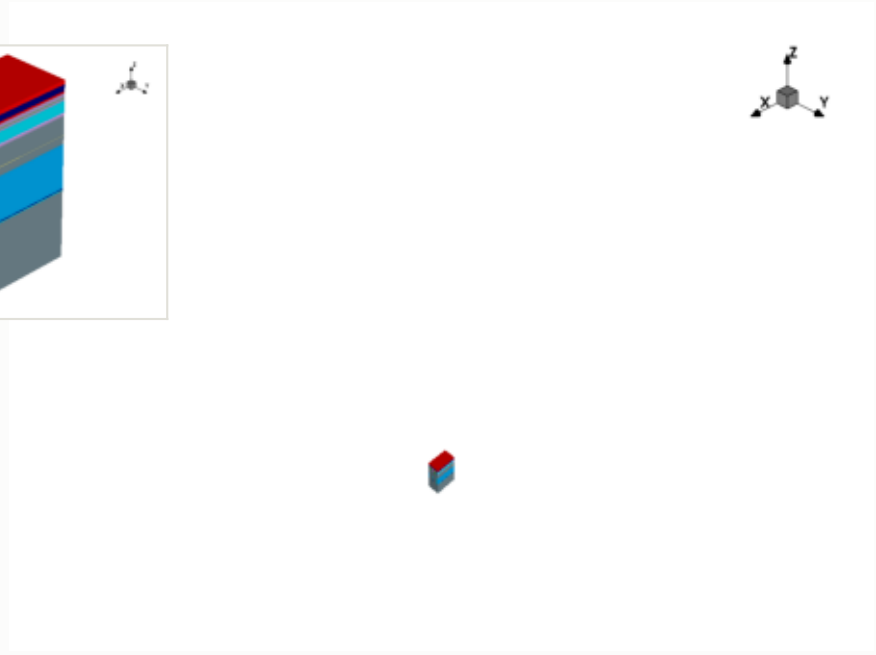
depo_fill · CVD

1.31	275.4	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



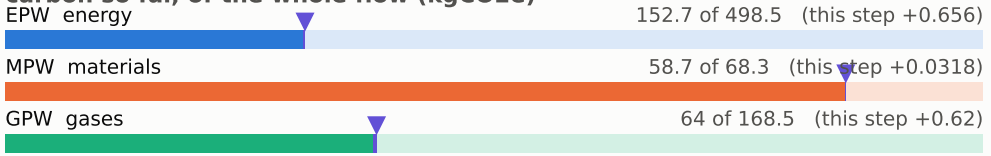


this step

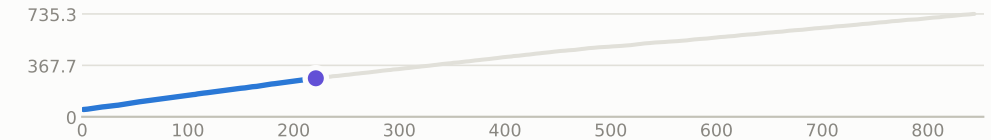
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

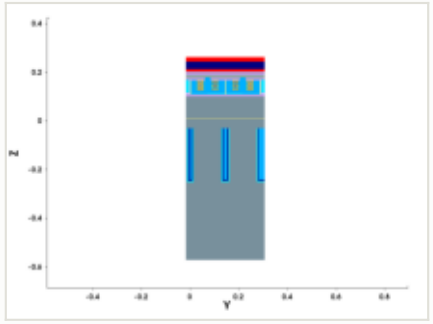
carbon so far, of the whole flow (kgCO2e)



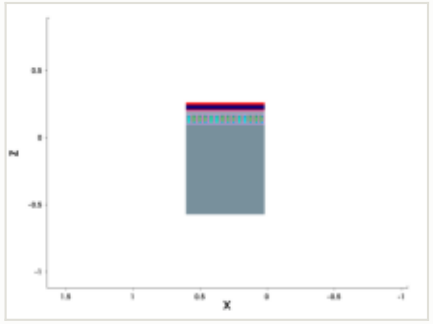
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



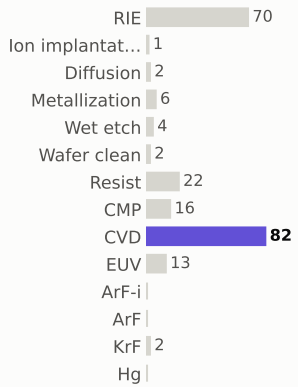
y cut at 0.184 µm



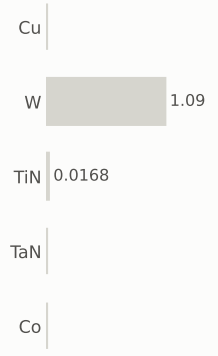
mask for this step

no mask for this step

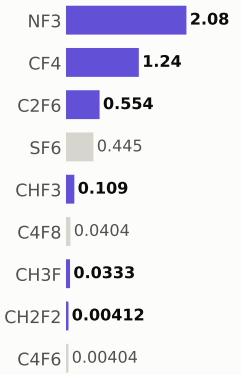
steps by process area



materials deposited (g)



gases emitted (g)



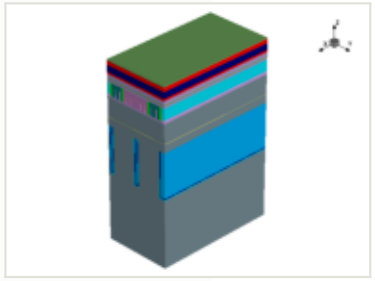
Step #2400: CVD

depo_fill · CVD

1.31	276.7	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

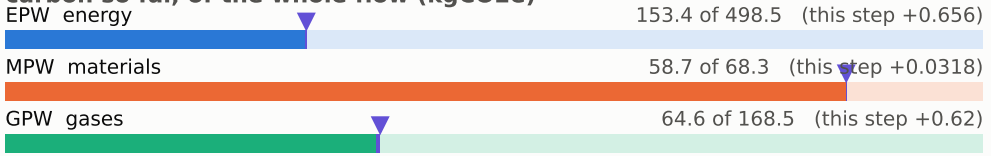


this step

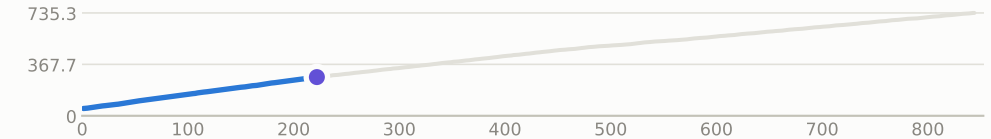
process area	CVD
material	—
thickness	0.005 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

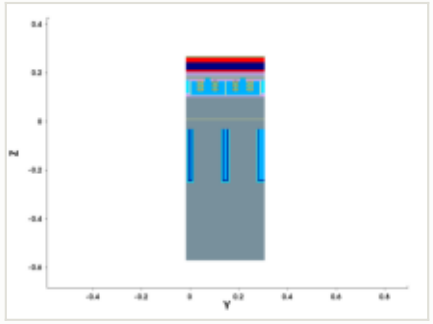
carbon so far, of the whole flow (kgCO2e)



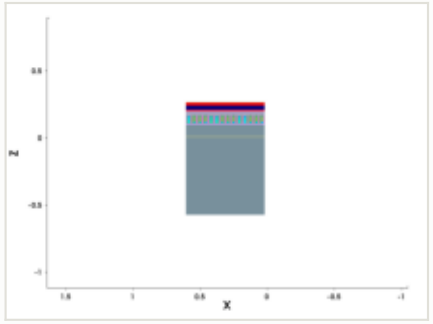
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



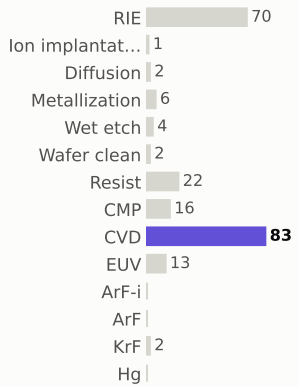
y cut at 0.184 µm



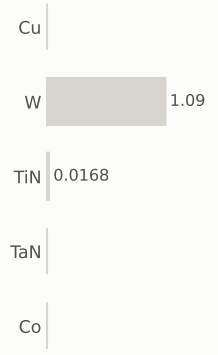
mask for this step

no mask for this step

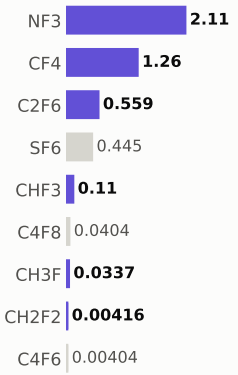
steps by process area



materials deposited (g)



gases emitted (g)



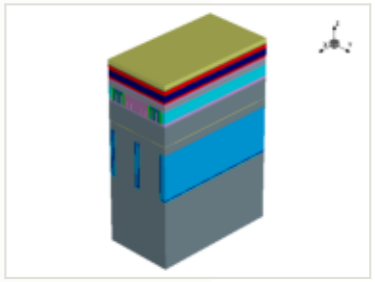
Step #2410: CVD

depo_fill · CVD

1.31	278.0	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

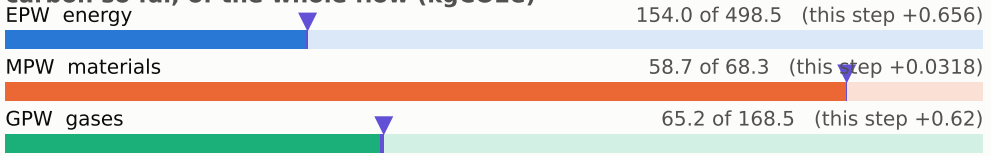


this step

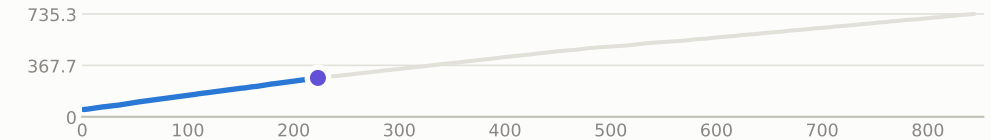
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

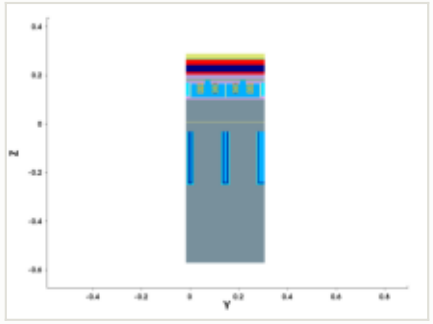
carbon so far, of the whole flow (kgCO2e)



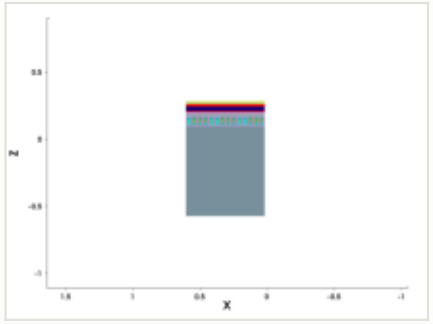
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



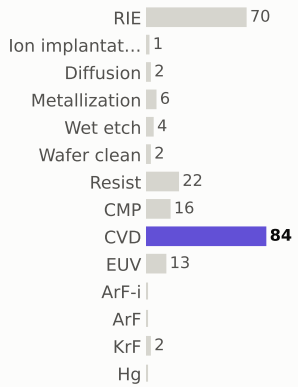
y cut at 0.184 µm



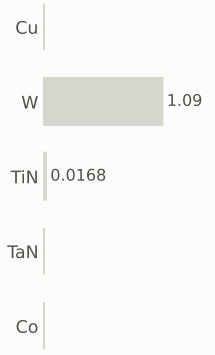
mask for this step

no mask for this step

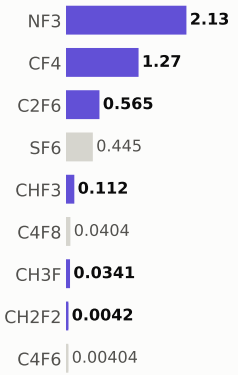
steps by process area



materials deposited (g)



gases emitted (g)



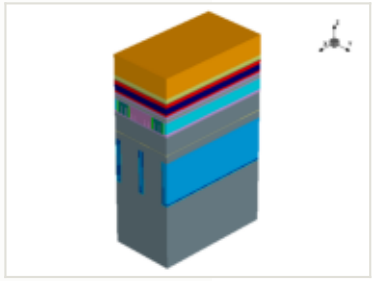
Step #2420: Resist

depo_fill · Resist

0.169	278.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

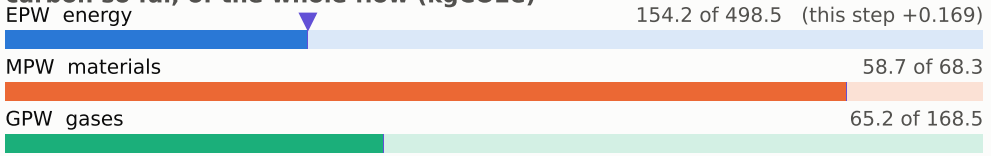


this step

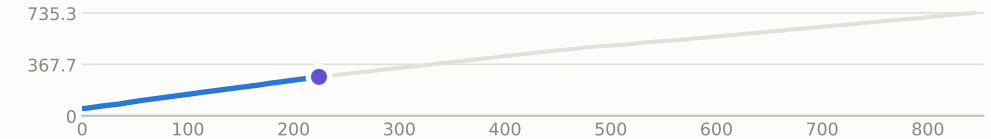
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

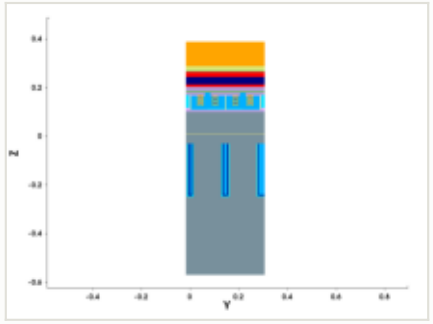
carbon so far, of the whole flow (kgCO2e)



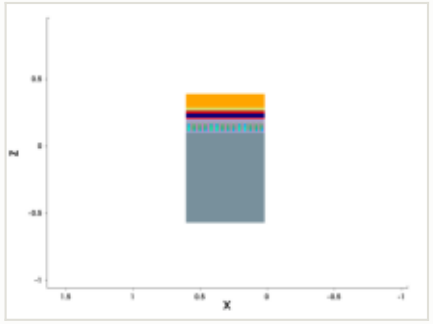
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



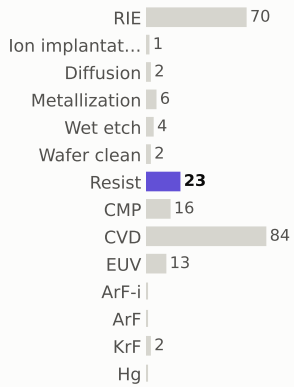
y cut at 0.184 µm



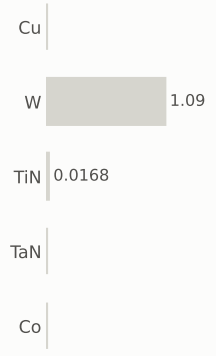
mask for this step

no mask for this step

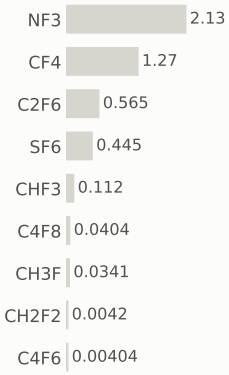
steps by process area



materials deposited (g)



gases emitted (g)



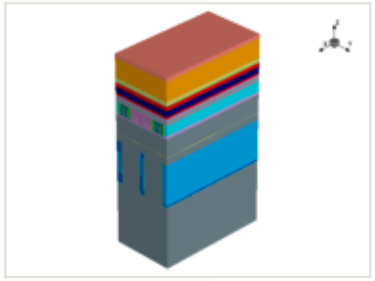
Step #2430: CVD

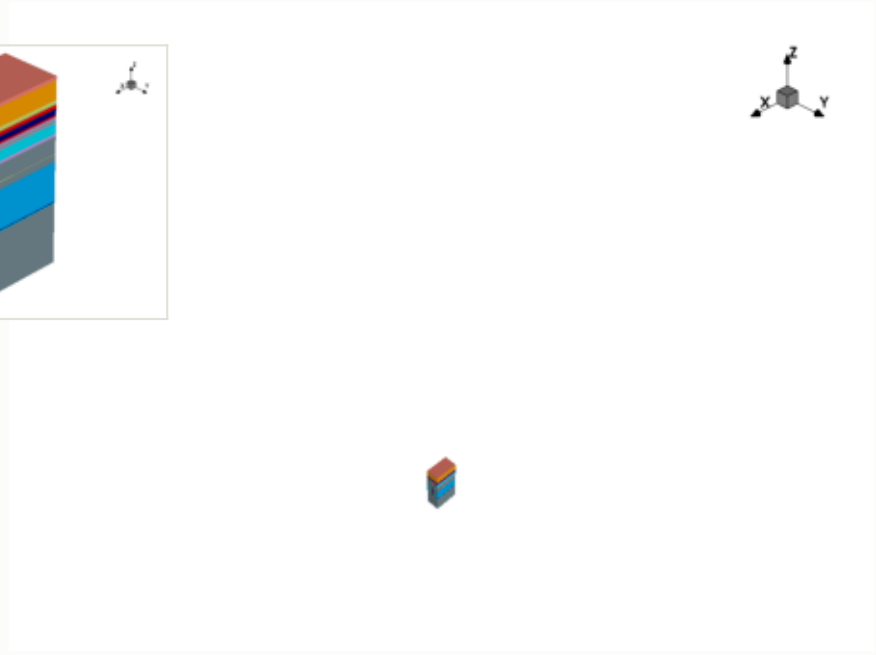
depo_fill · CVD

1.31	279.5	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



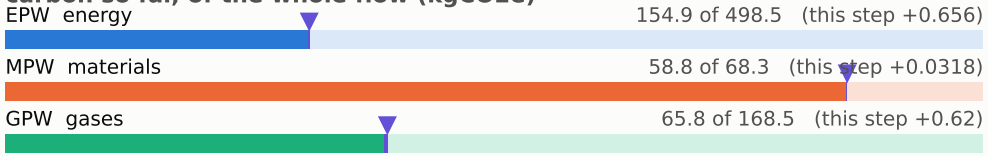


this step

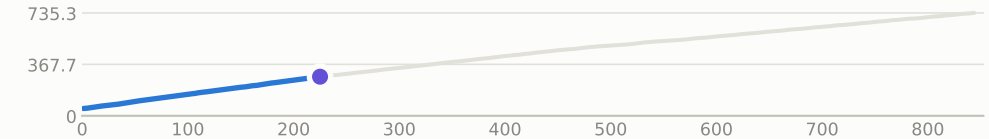
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

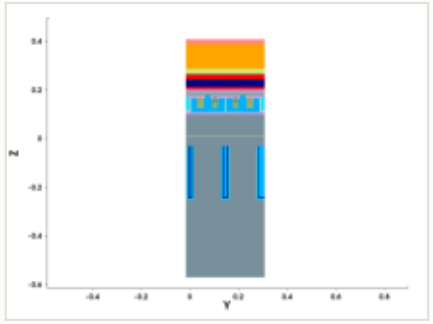
carbon so far, of the whole flow (kgCO2e)



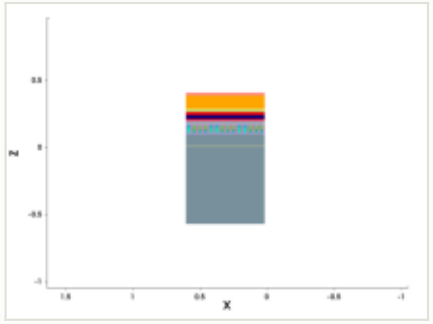
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



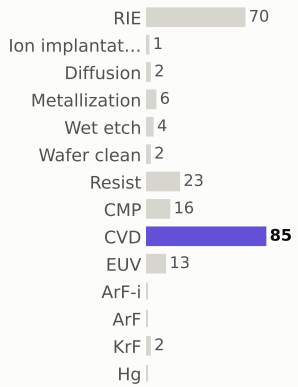
y cut at 0.184 µm



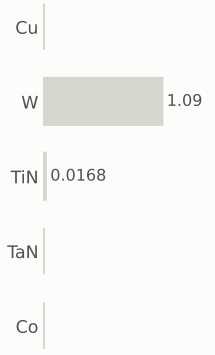
mask for this step

no mask for this step

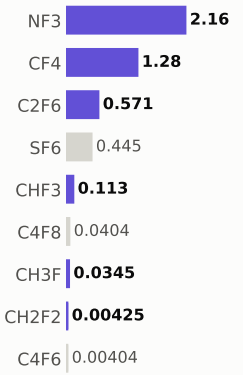
steps by process area



materials deposited (g)



gases emitted (g)



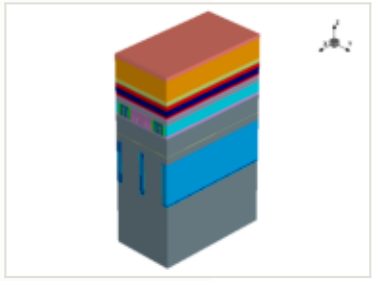
Step #2440: Wafer_clean

clean · Wafer_clean

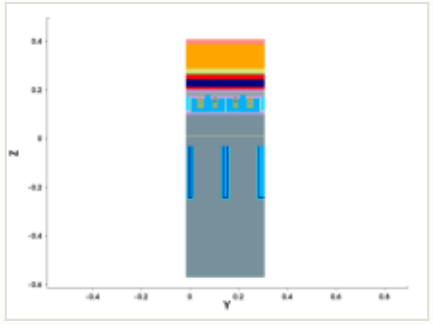
0.169	279.6	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

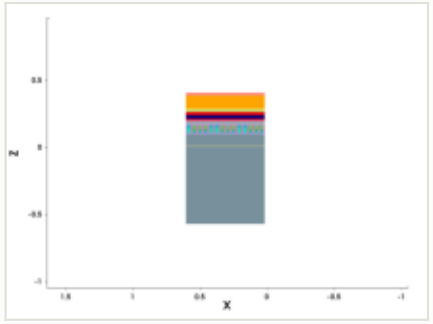
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

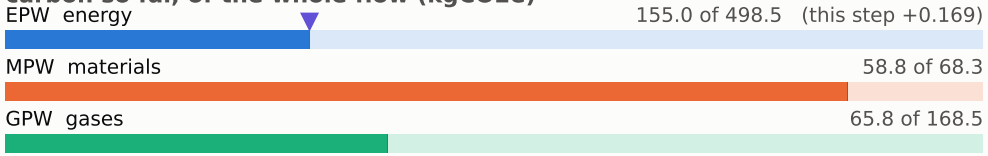
no mask for this step

this step

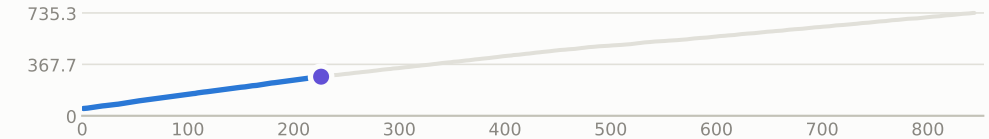
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

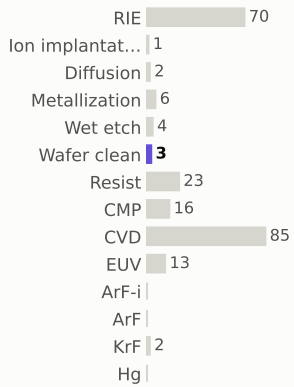
carbon so far, of the whole flow (kgCO2e)



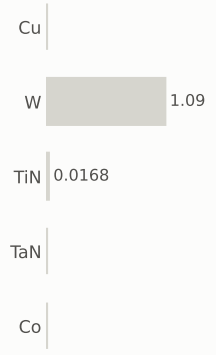
embodied carbon up to each step (kgCO2e per wafer)



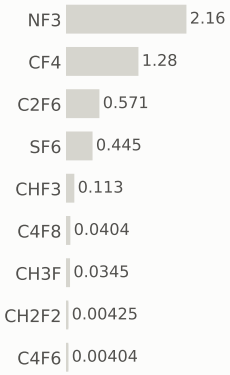
steps by process area



materials deposited (g)



gases emitted (g)



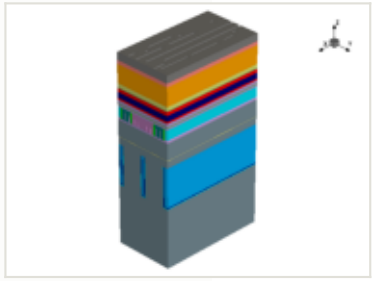
Step #2450: EUV

litho_patterning · EUV

1.96	281.6	0.27%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

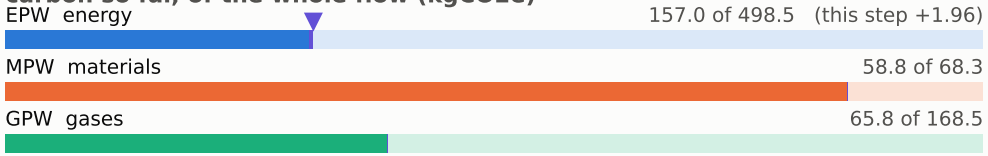


this step

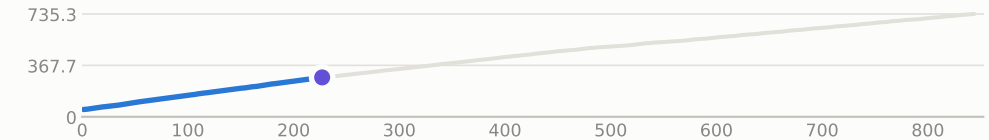
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

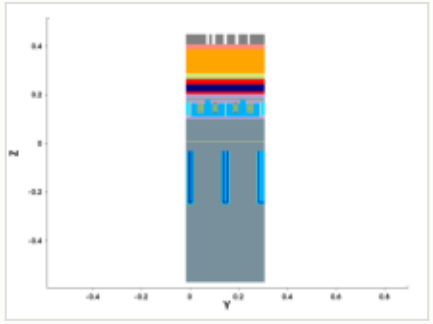
carbon so far, of the whole flow (kgCO2e)



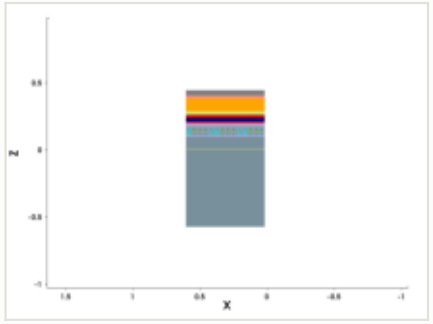
embodied carbon up to each step (kgCO2e per wafer)



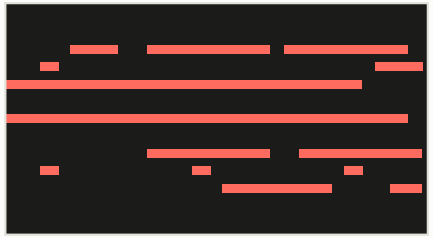
x cut at 0.294 µm



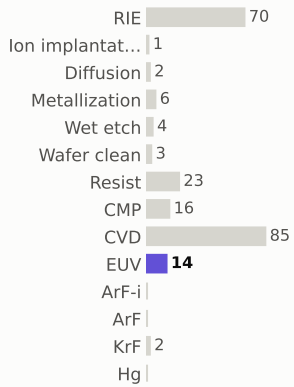
y cut at 0.184 µm



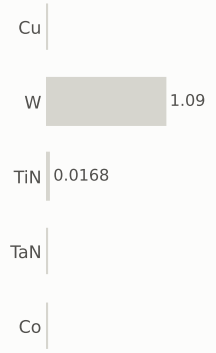
mask: layer 10201:0 (GDSII)



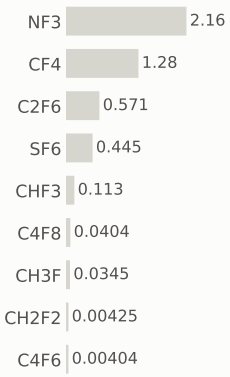
steps by process area



materials deposited (g)



gases emitted (g)



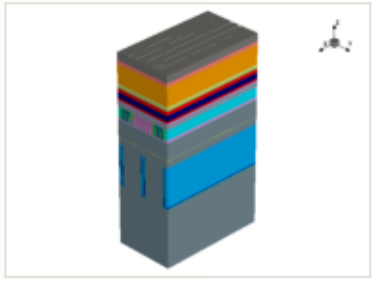
Step #2460: RIE

etch_anisotropic · RIE

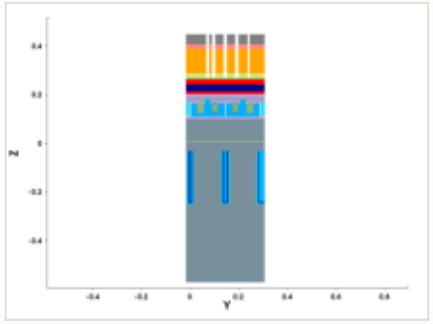
1.04	282.6	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

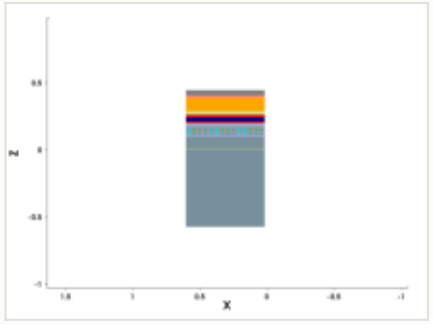
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

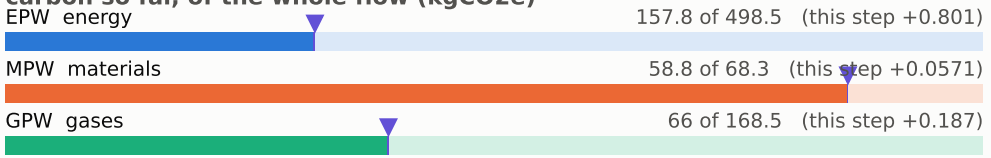
no mask for this step

this step

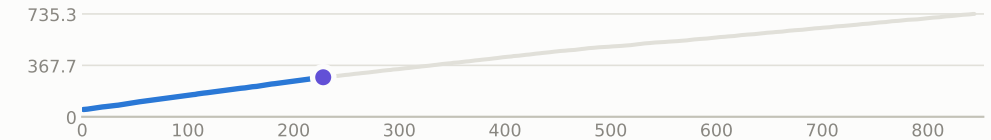
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

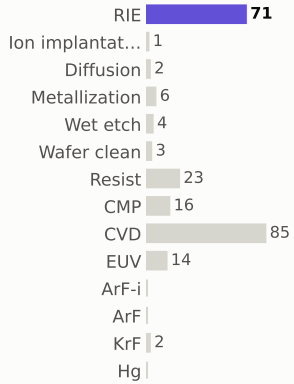
carbon so far, of the whole flow (kgCO2e)



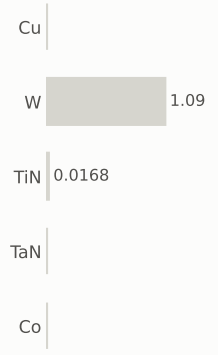
embodied carbon up to each step (kgCO2e per wafer)



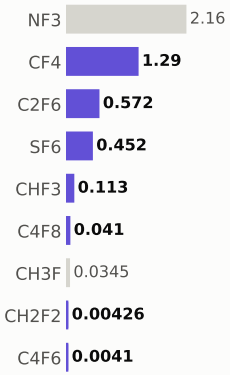
steps by process area



materials deposited (g)



gases emitted (g)



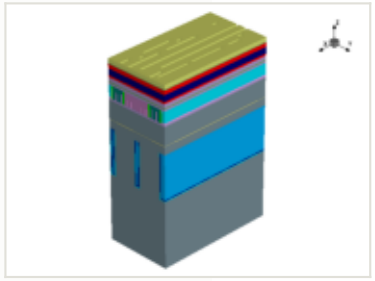
Step #2470: Wafer_clean

etch_strip · Wafer_clean

0.169	282.8	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

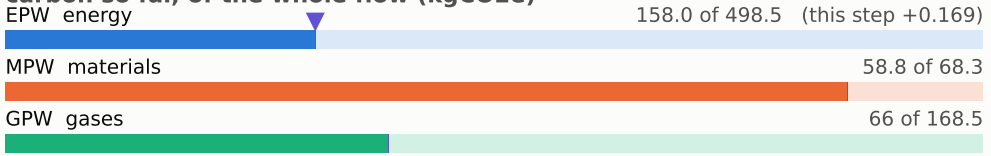


this step

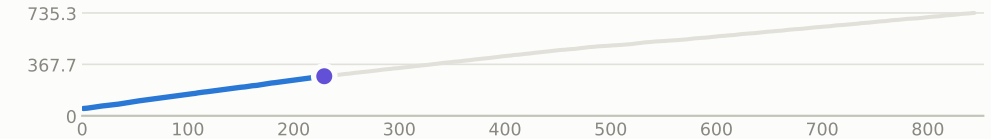
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

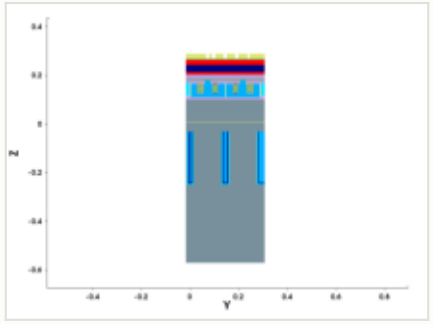
carbon so far, of the whole flow (kgCO2e)



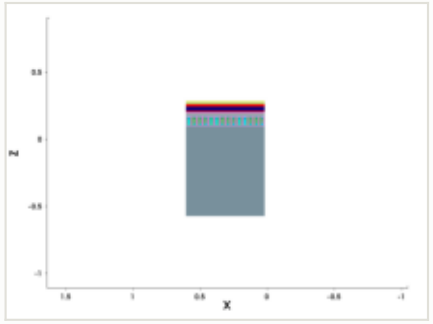
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



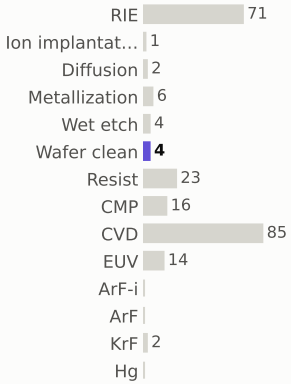
y cut at 0.184 µm



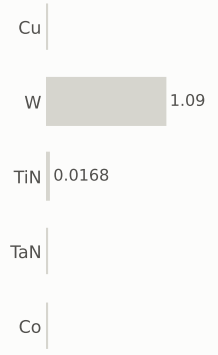
mask for this step

no mask for this step

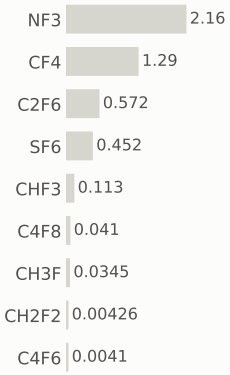
steps by process area



materials deposited (g)



gases emitted (g)



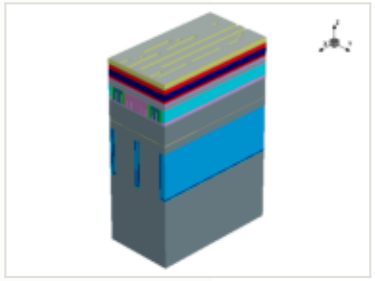
Step #2480: CVD

depo_isotropic · CVD

1.31	284.1	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

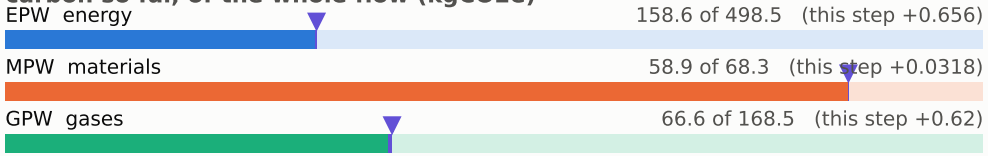


this step

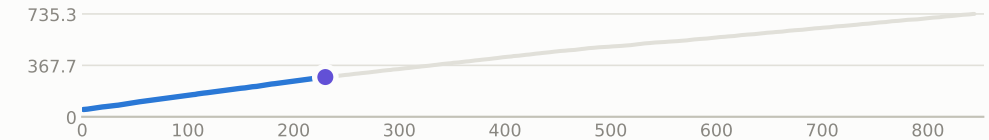
process area	CVD
material	—
thickness	0.001 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

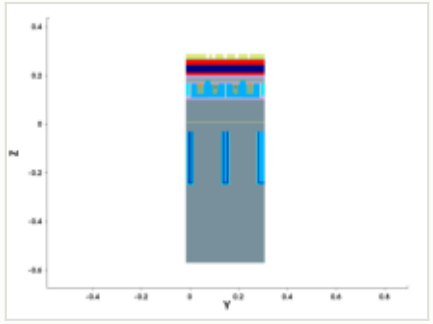
carbon so far, of the whole flow (kgCO2e)



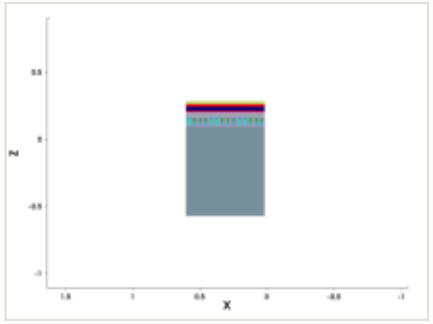
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



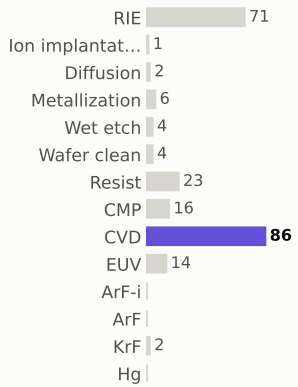
y cut at 0.184 µm



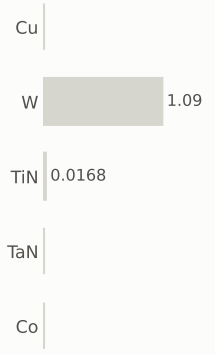
mask for this step

no mask for this step

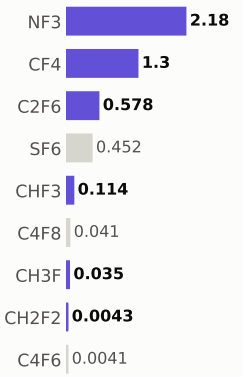
steps by process area



materials deposited (g)



gases emitted (g)



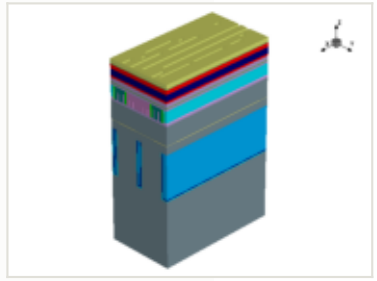
Step #2490: RIE

etch_anisotropic · RIE

1.04	285.2	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

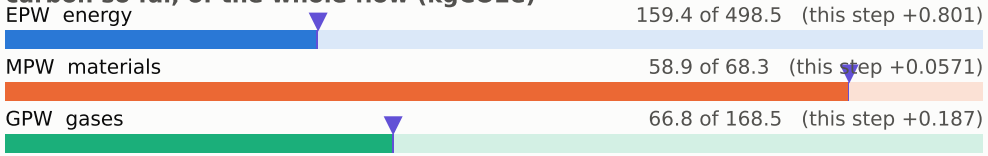


this step

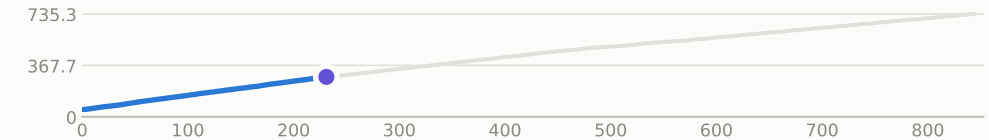
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

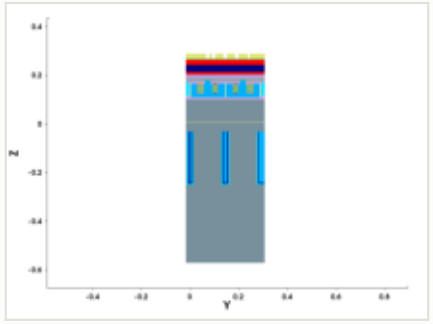
carbon so far, of the whole flow (kgCO2e)



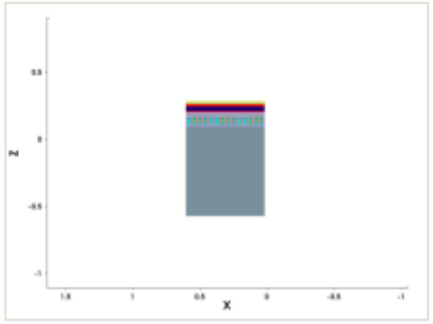
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



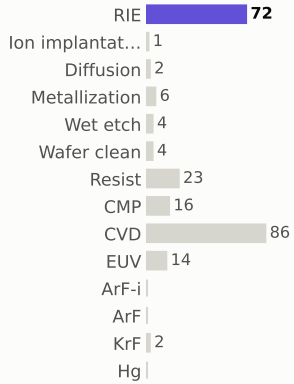
y cut at 0.184 µm



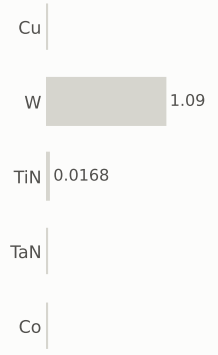
mask for this step

no mask for this step

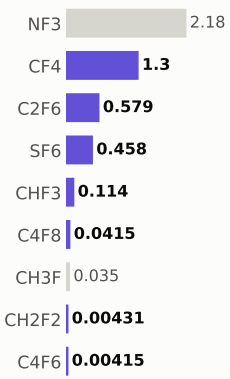
steps by process area



materials deposited (g)



gases emitted (g)



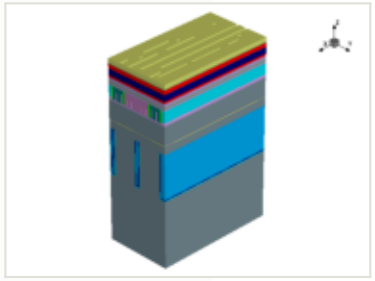
Step #2500: Wafer_clean

clean · Wafer_clean

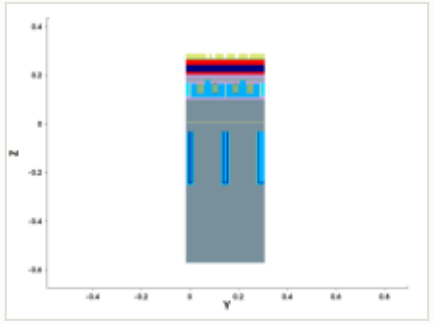
0.169	285.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

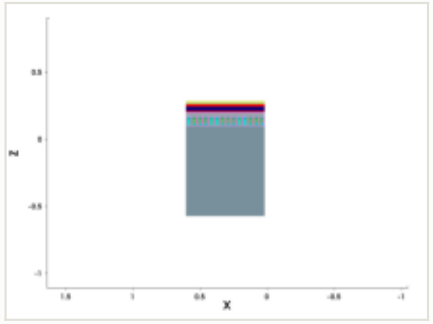
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

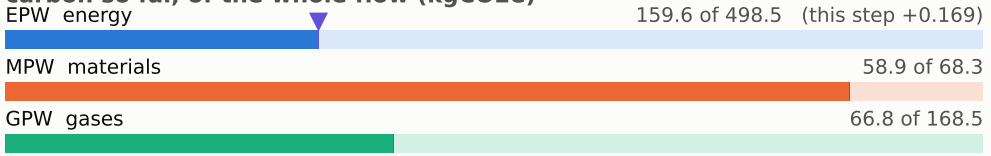
no mask for this step

this step

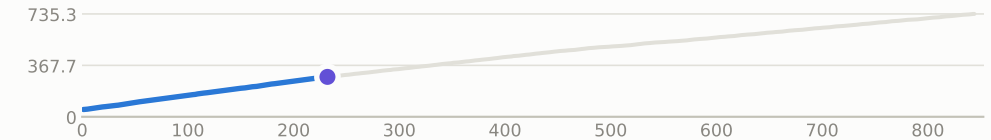
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

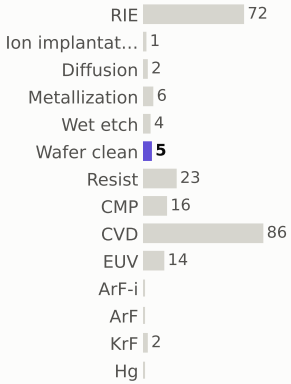
carbon so far, of the whole flow (kgCO2e)



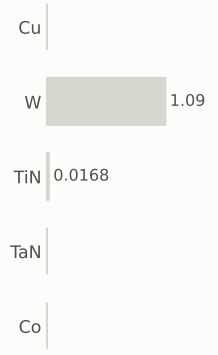
embodied carbon up to each step (kgCO2e per wafer)



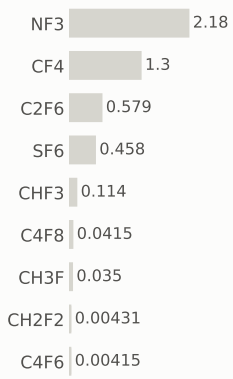
steps by process area



materials deposited (g)



gases emitted (g)



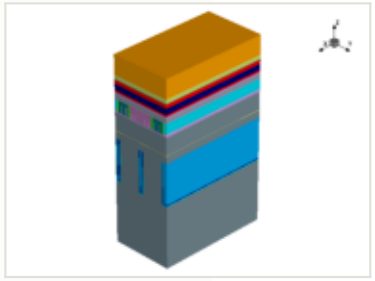
Step #2510: Resist

depo_fill · Resist

0.169	285.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

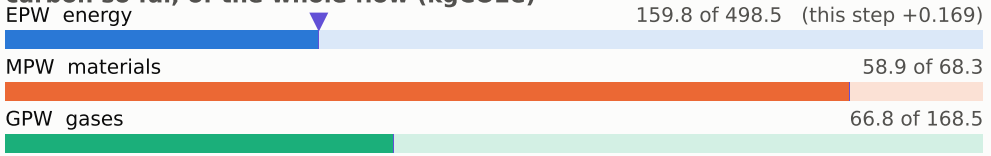


this step

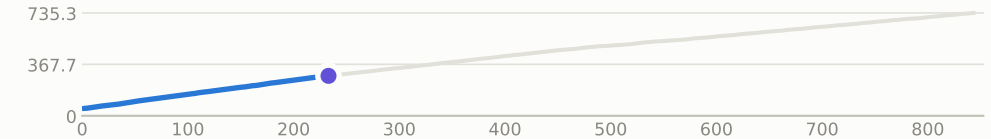
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

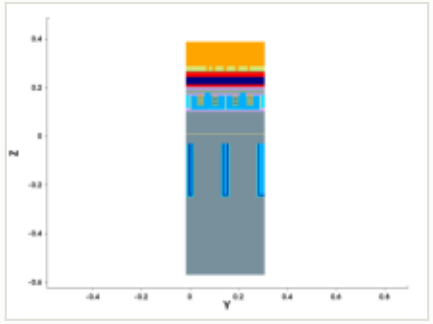
carbon so far, of the whole flow (kgCO2e)



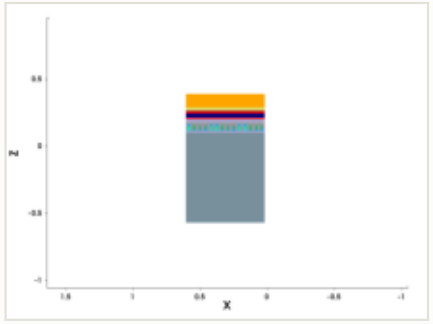
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



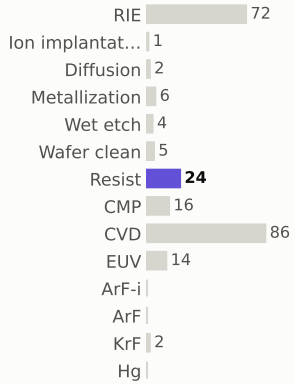
y cut at 0.184 µm



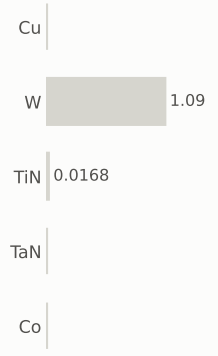
mask for this step

no mask for this step

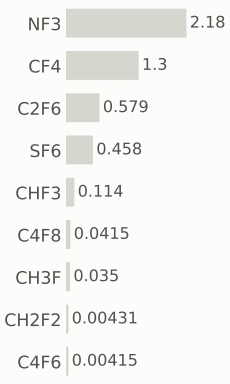
steps by process area



materials deposited (g)



gases emitted (g)



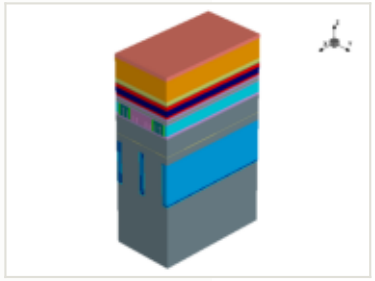
Step #2520: CVD

depo_fill · CVD

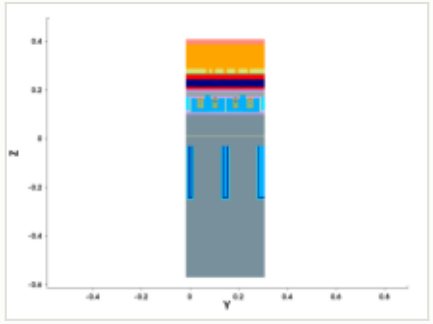
1.31	286.8	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

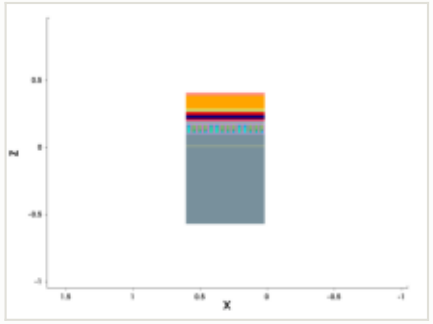
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

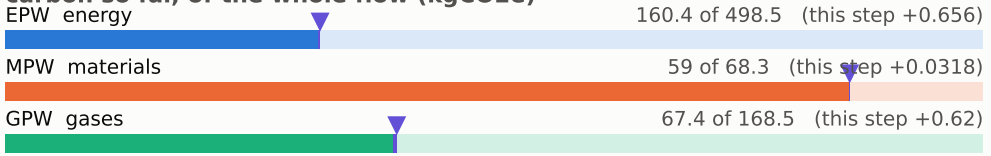
no mask for this step

this step

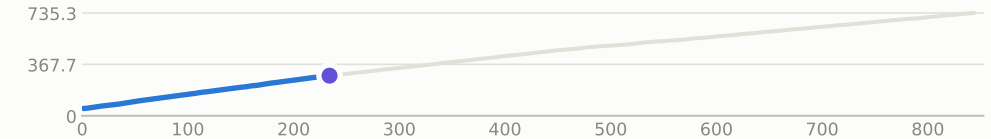
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

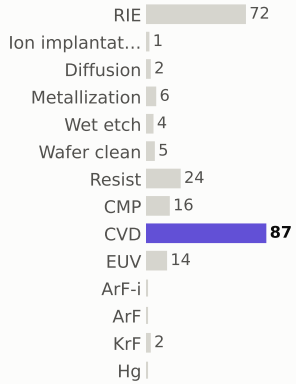
carbon so far, of the whole flow (kgCO2e)



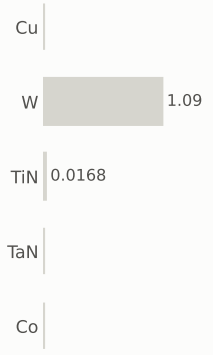
embodied carbon up to each step (kgCO2e per wafer)



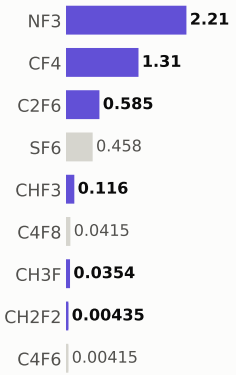
steps by process area



materials deposited (g)



gases emitted (g)



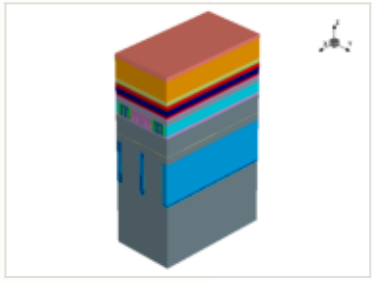
Step #2530: Wafer_clean

clean · Wafer_clean

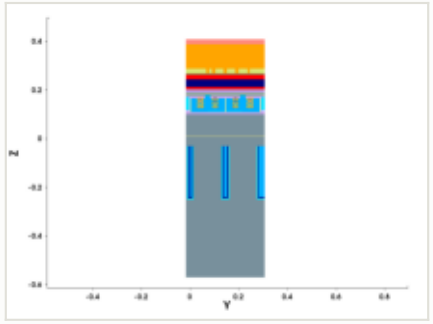
0.169	287.0	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

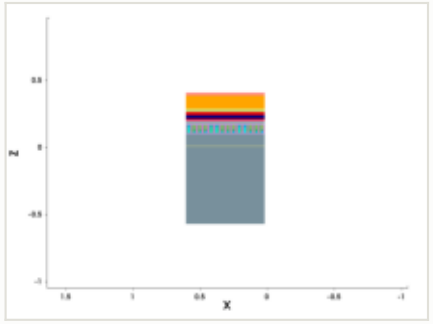
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

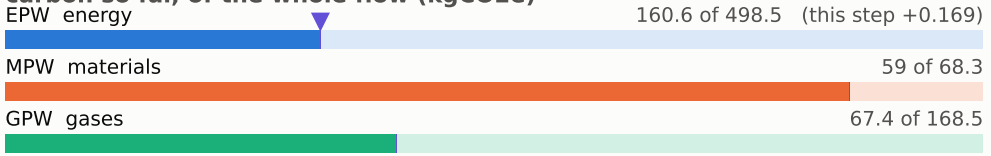
no mask for this step

this step

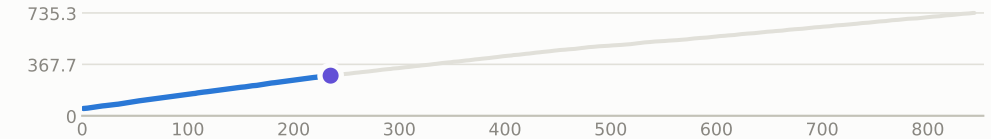
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

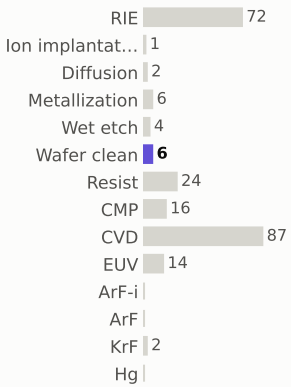
carbon so far, of the whole flow (kgCO2e)



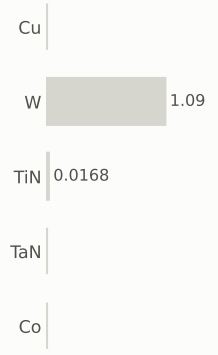
embodied carbon up to each step (kgCO2e per wafer)



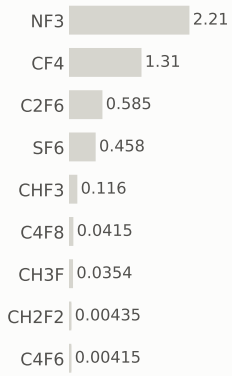
steps by process area



materials deposited (g)



gases emitted (g)



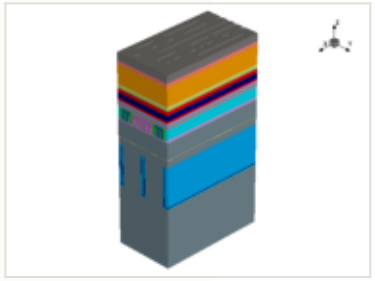
Step #2540: EUV

litho_patterning · EUV

1.96	288.9	0.27%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

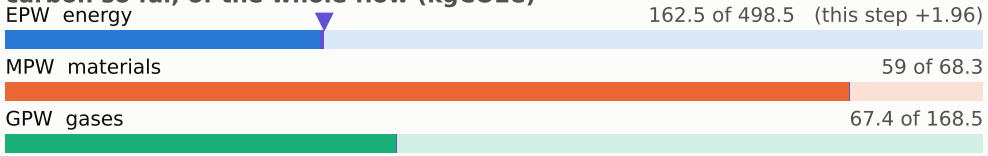


this step

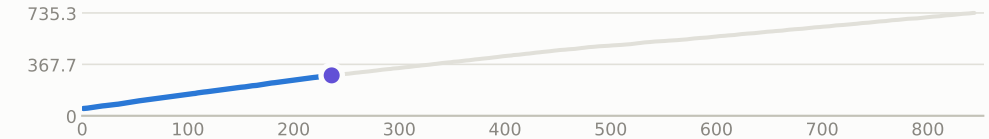
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

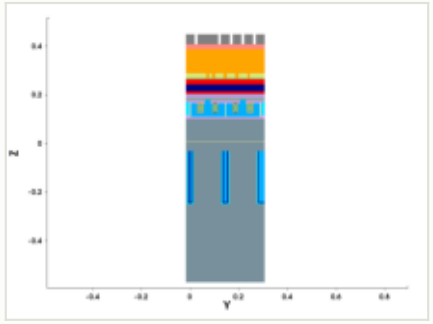
carbon so far, of the whole flow (kgCO2e)



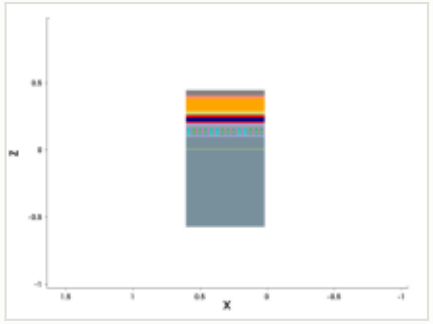
embodied carbon up to each step (kgCO2e per wafer)



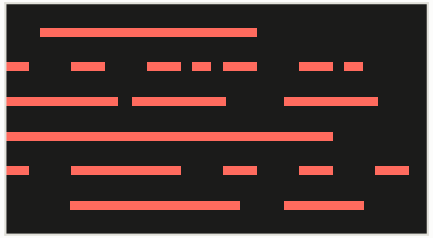
x cut at 0.294 µm



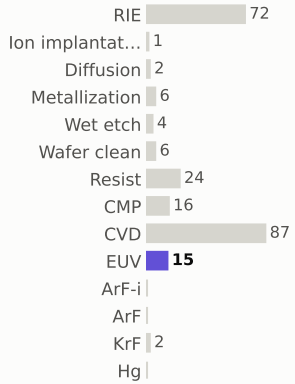
y cut at 0.184 µm



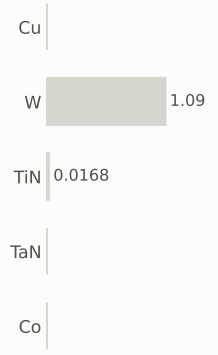
mask: layer 10202:0 (GDSII)



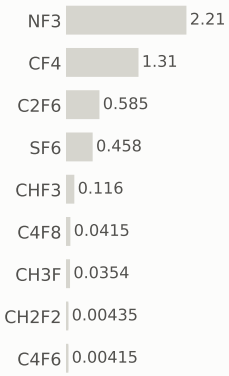
steps by process area



materials deposited (g)



gases emitted (g)



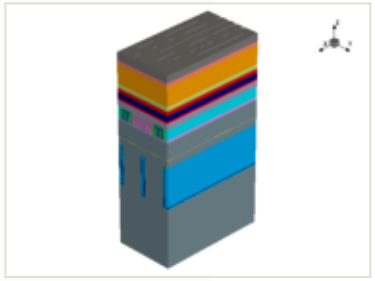
Step #2550: RIE

etch_anisotropic · RIE

1.04	290.0	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

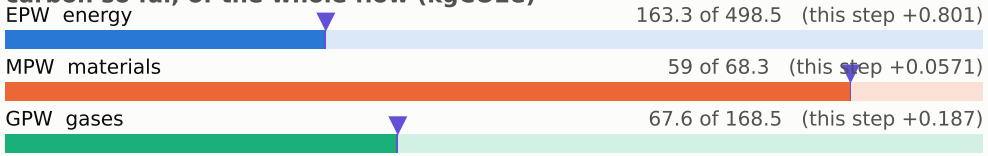


this step

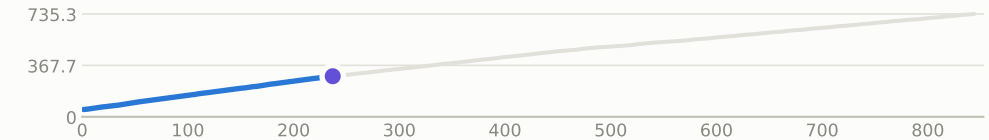
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

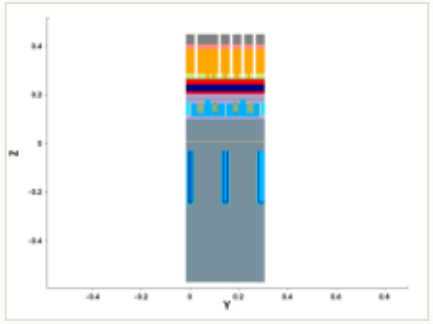
carbon so far, of the whole flow (kgCO2e)



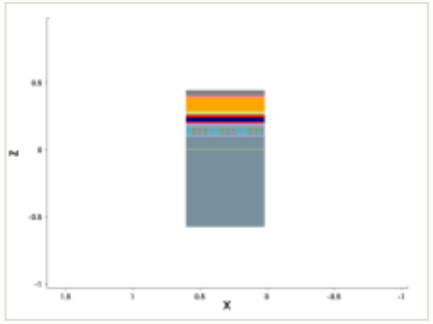
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



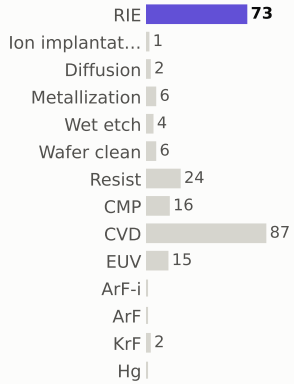
y cut at 0.184 µm



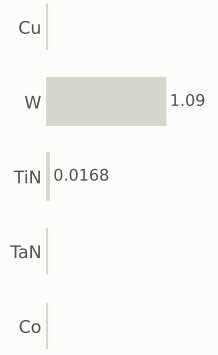
mask for this step

no mask for this step

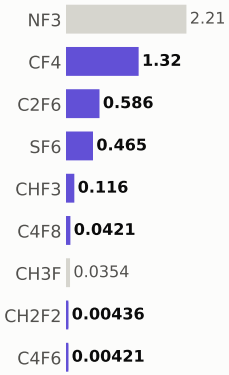
steps by process area



materials deposited (g)



gases emitted (g)



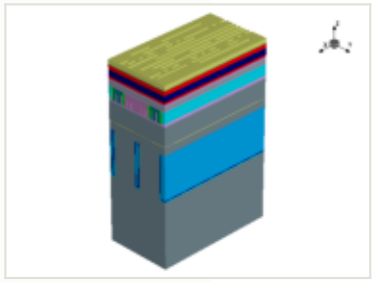
Step #2560: Wafer_clean

etch_strip · Wafer_clean

0.169	290.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

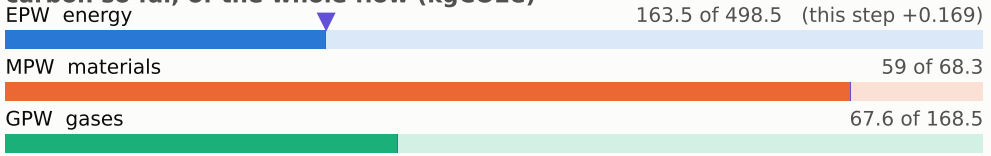


this step

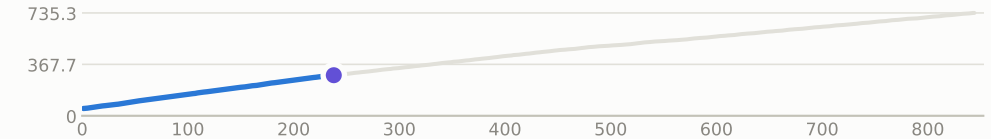
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

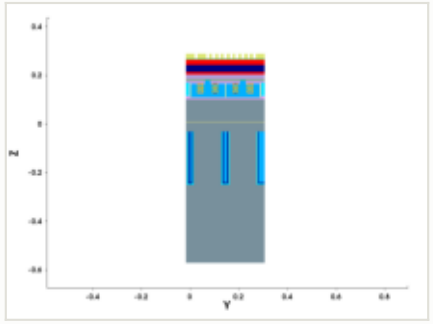
carbon so far, of the whole flow (kgCO2e)



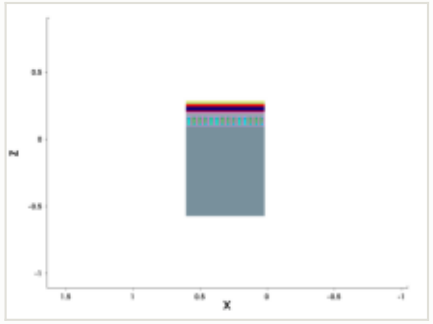
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



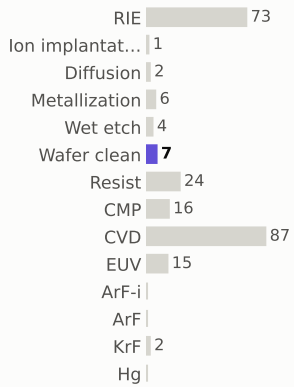
y cut at 0.184 µm



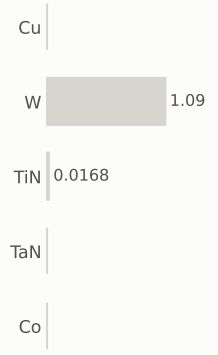
mask for this step

no mask for this step

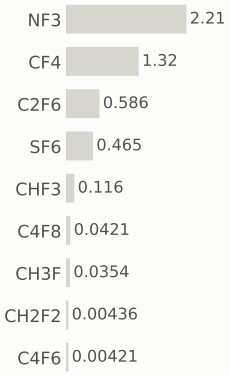
steps by process area



materials deposited (g)



gases emitted (g)



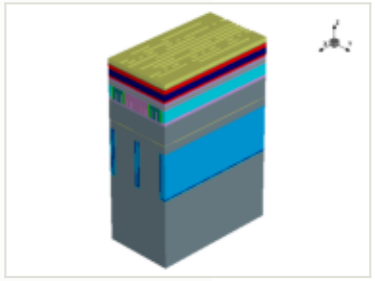
Step #2570: RIE

etch_anisotropic · RIE

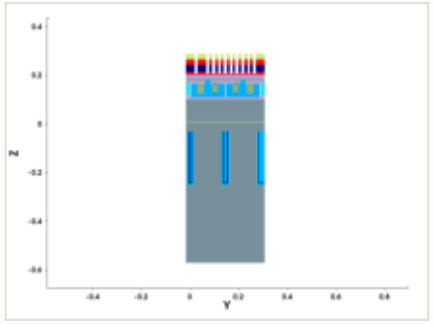
1.04	291.2	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

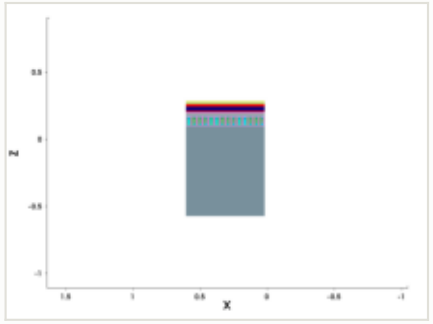
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

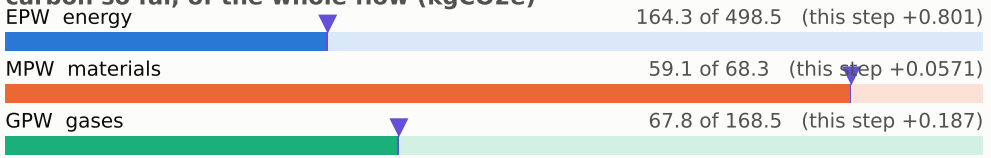
no mask for this step

this step

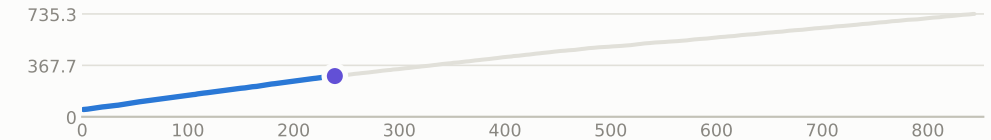
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

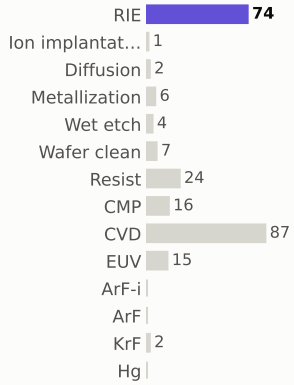
carbon so far, of the whole flow (kgCO2e)



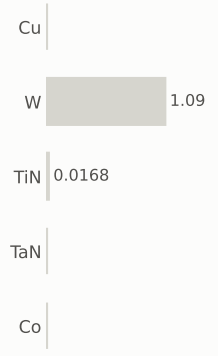
embodied carbon up to each step (kgCO2e per wafer)



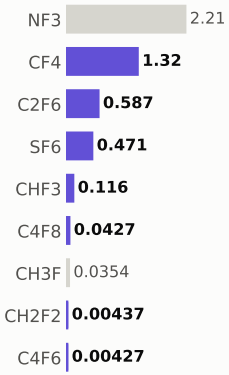
steps by process area



materials deposited (g)



gases emitted (g)



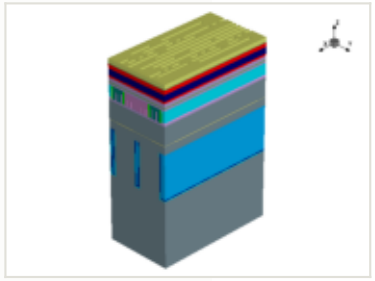
Step #2580: Wafer_clean

clean · Wafer_clean

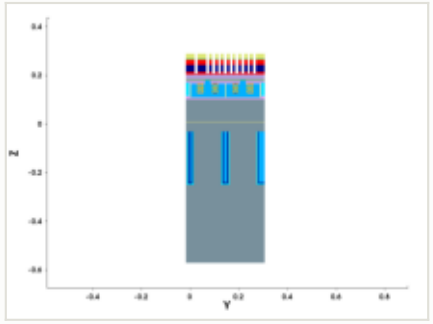
0.169	291.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

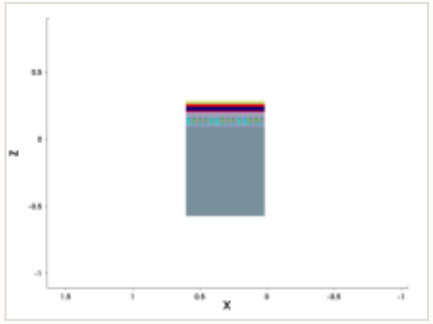
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

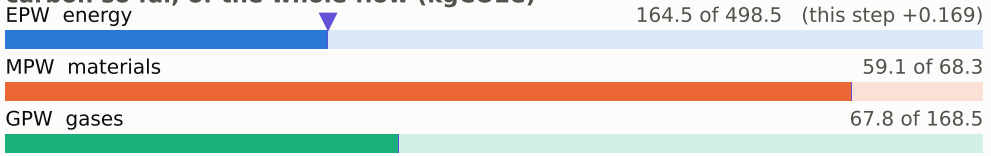
no mask for this step

this step

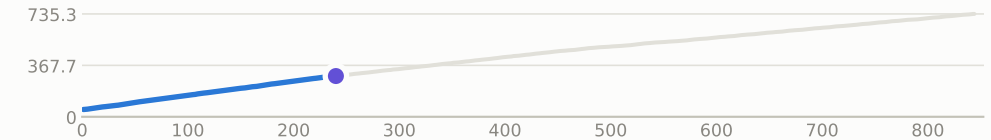
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

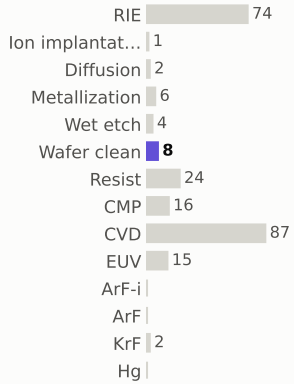
carbon so far, of the whole flow (kgCO2e)



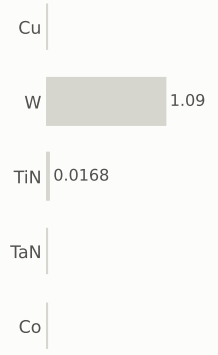
embodied carbon up to each step (kgCO2e per wafer)



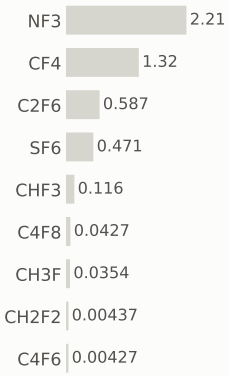
steps by process area



materials deposited (g)



gases emitted (g)



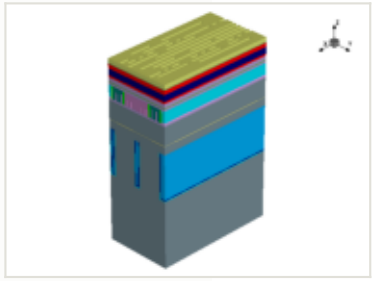
Step #2590: RIE

etch_anisotropic · RIE

1.04	292.4	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

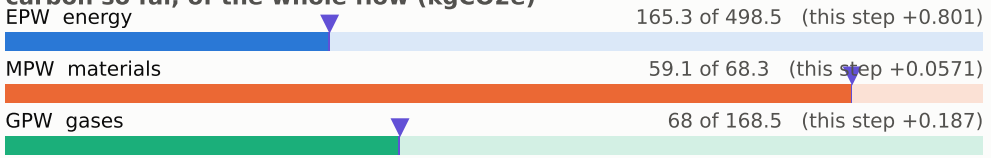


this step

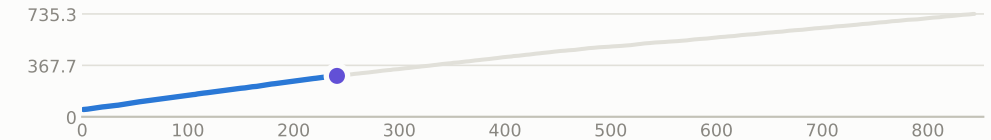
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

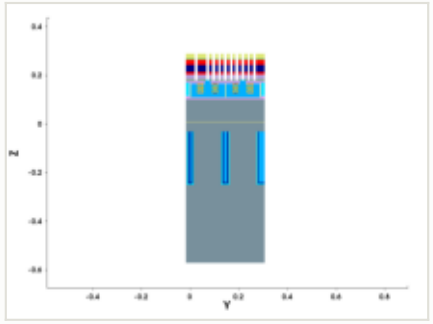
carbon so far, of the whole flow (kgCO2e)



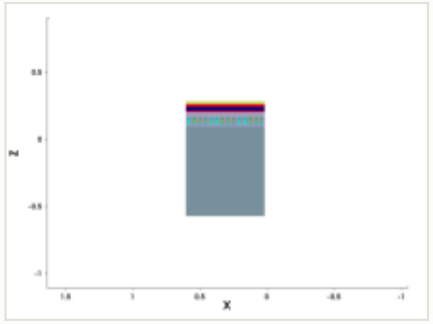
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



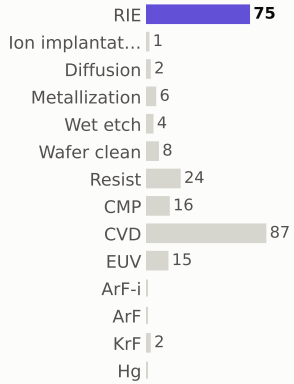
y cut at 0.184 µm



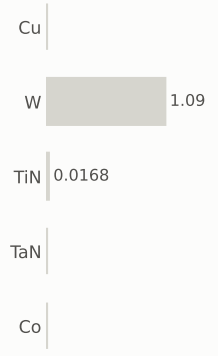
mask for this step

no mask for this step

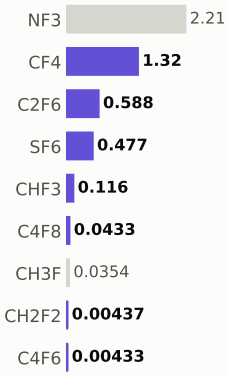
steps by process area



materials deposited (g)



gases emitted (g)



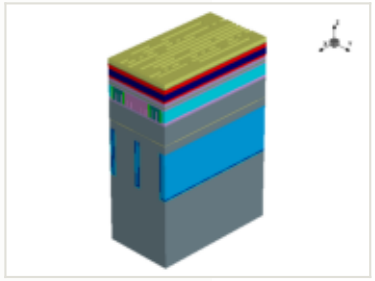
Step #2600: Wafer_clean

clean · Wafer_clean

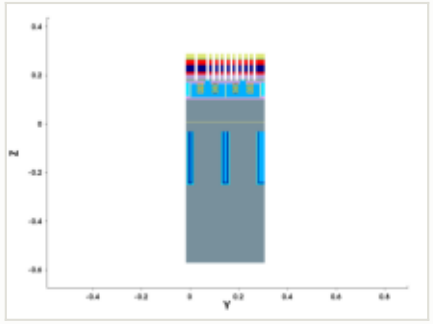
0.169	292.6	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

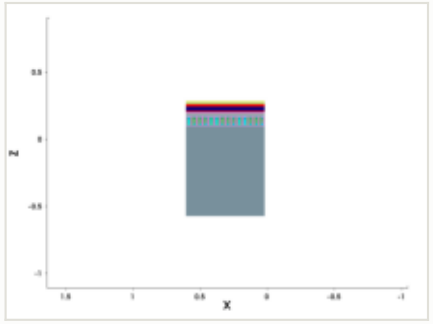
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

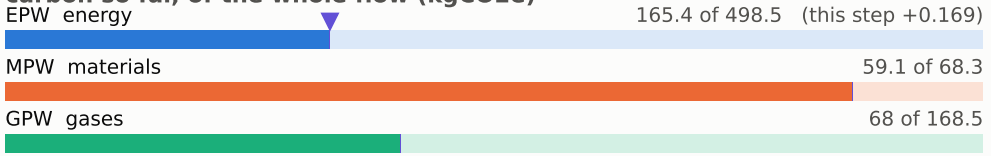
no mask for this step

this step

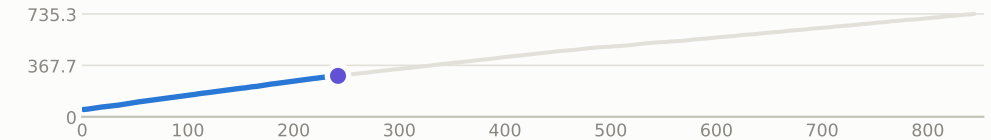
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

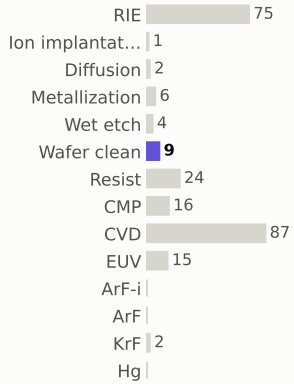
carbon so far, of the whole flow (kgCO2e)



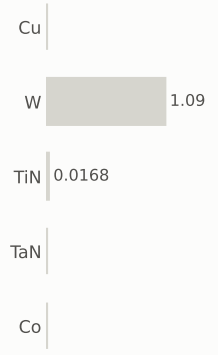
embodied carbon up to each step (kgCO2e per wafer)



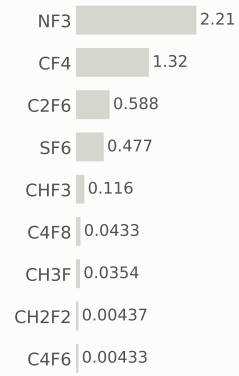
steps by process area



materials deposited (g)



gases emitted (g)



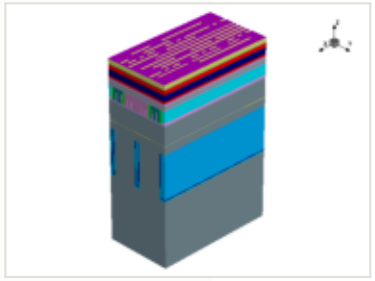
Step #2610: Metallization

depo_ellipsoidal · Metallization

1.2	293.8	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

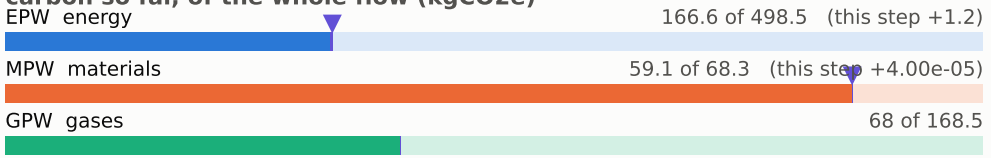


this step

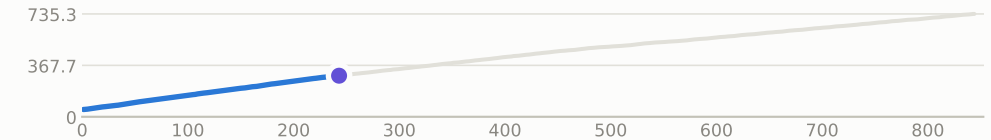
process area	Metallization
material	TaN
thickness	0.0016 µm
material GWP	25.8159 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.00e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

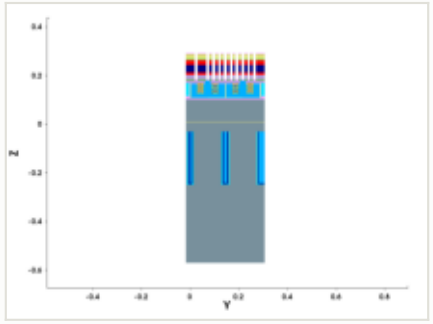
carbon so far, of the whole flow (kgCO2e)



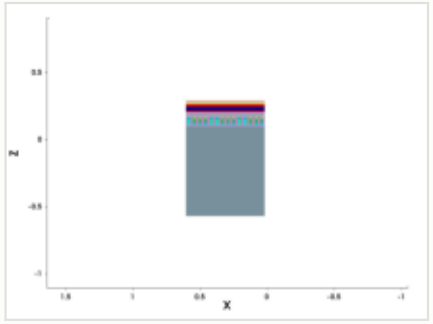
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



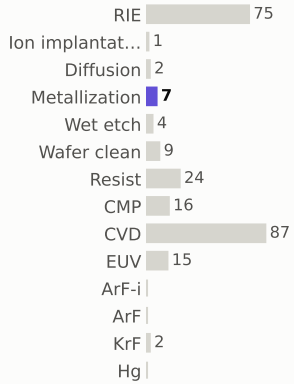
y cut at 0.184 µm



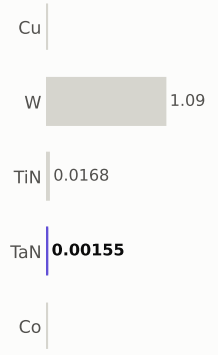
mask for this step

no mask for this step

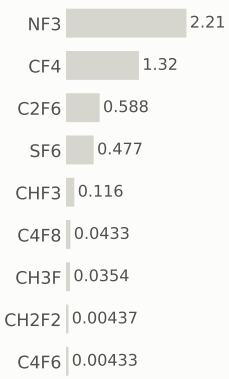
steps by process area



materials deposited (g)



gases emitted (g)



Step #2620: Metallization

depo_fill · Metallization

1.21	295.0	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

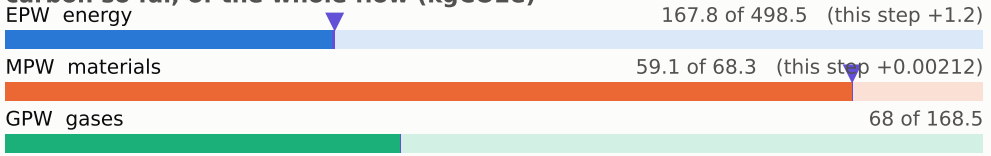
zoomed to fit

this step

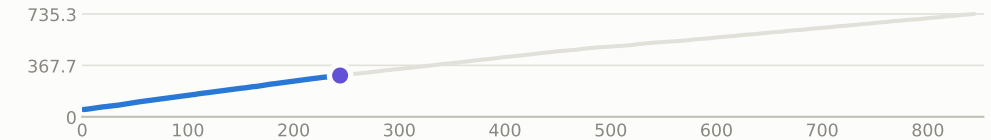
process area	Metallization
material	Cu
thickness	0.55 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00212 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

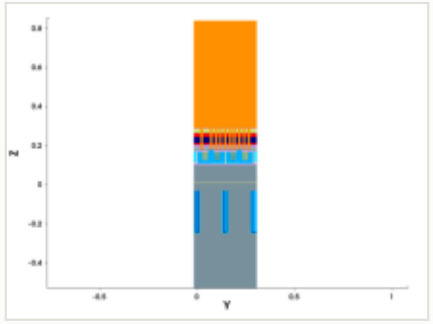
carbon so far, of the whole flow (kgCO2e)



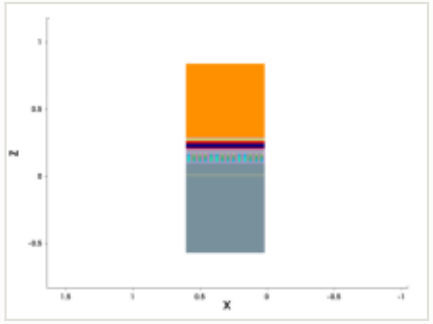
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



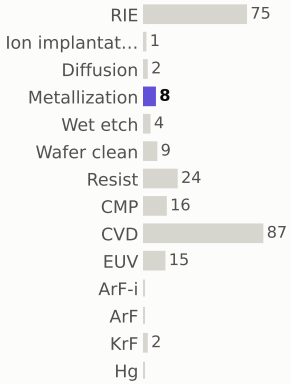
y cut at 0.184 µm



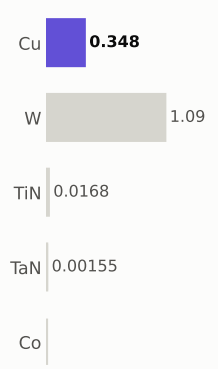
mask for this step

no mask for this step

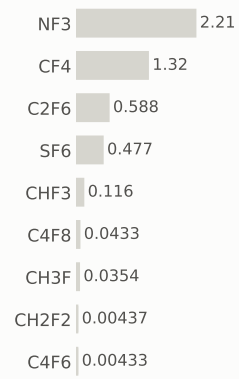
steps by process area



materials deposited (g)



gases emitted (g)



Step #2630: Diffusion

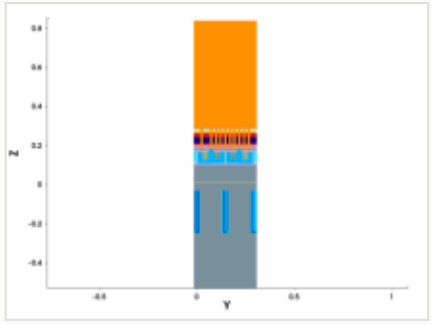
treatment · Diffusion

0.801	295.8	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

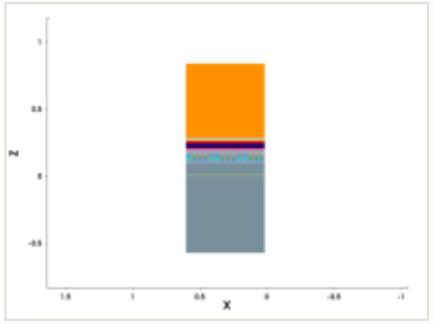
3D structure after this step

zoomed to fit

x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

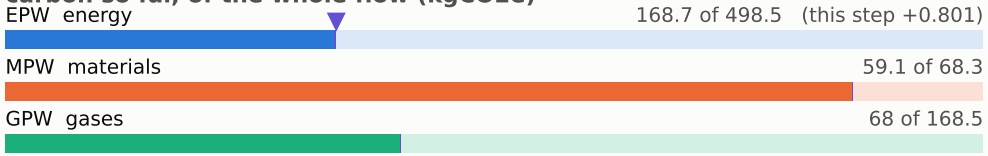
no mask for this step

this step

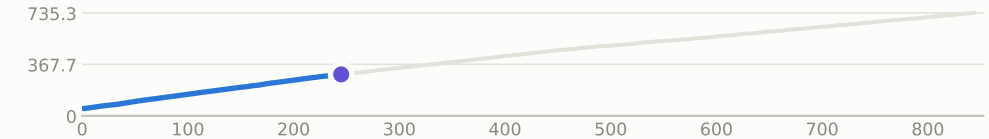
process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

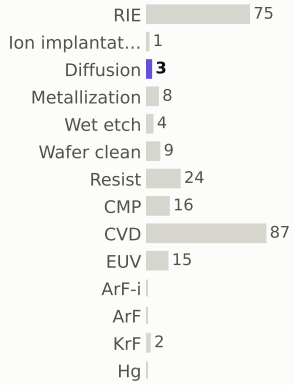
carbon so far, of the whole flow (kgCO2e)



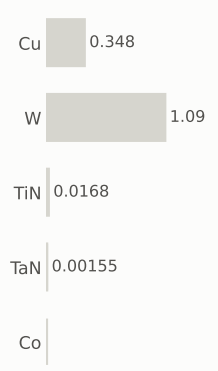
embodied carbon up to each step (kgCO2e per wafer)



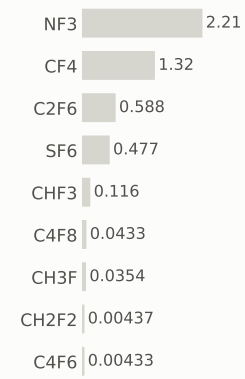
steps by process area



materials deposited (g)



gases emitted (g)



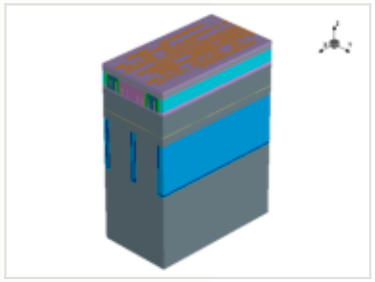
Step #2640: CMP

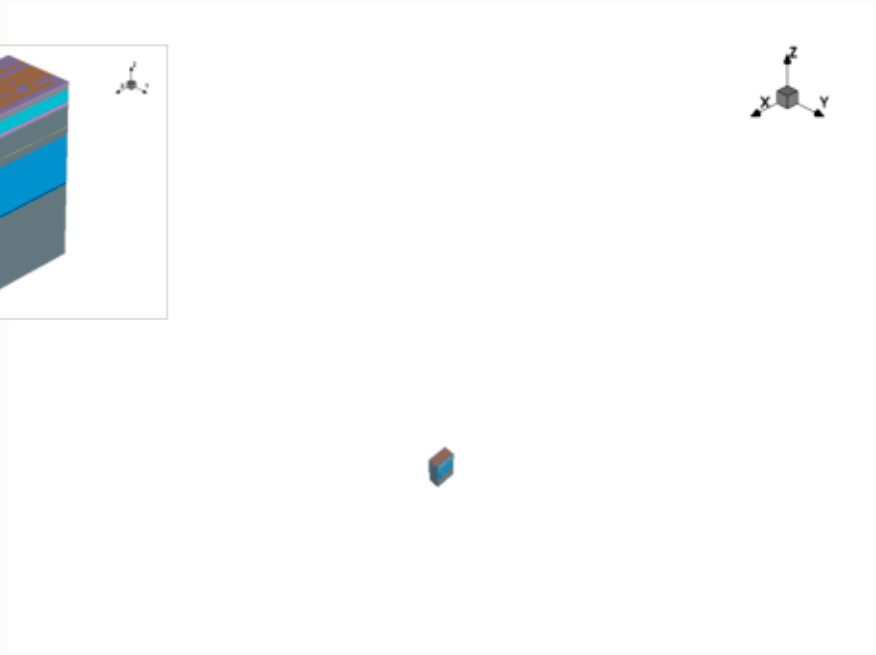
etch_cmp · CMP

0.169	296.0	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



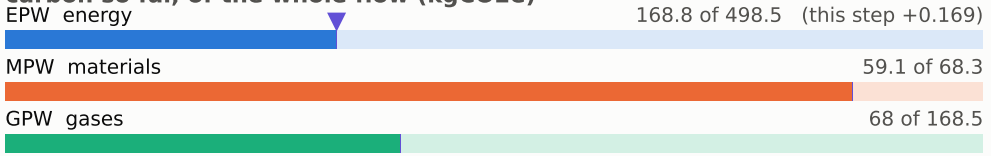


this step

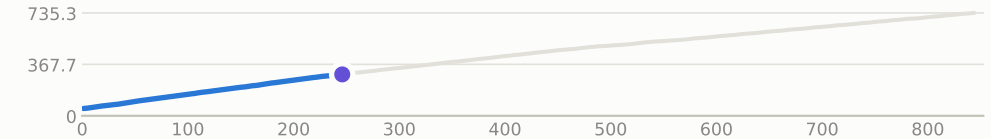
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

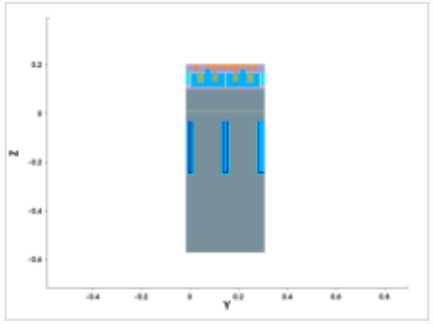
carbon so far, of the whole flow (kgCO2e)



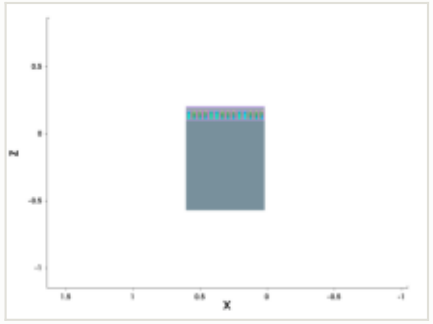
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



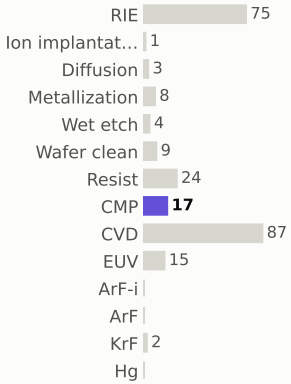
y cut at 0.184 µm



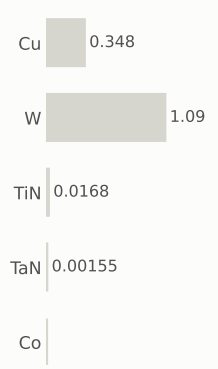
mask for this step

no mask for this step

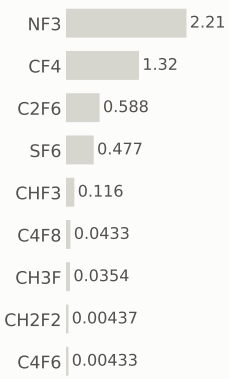
steps by process area



materials deposited (g)



gases emitted (g)



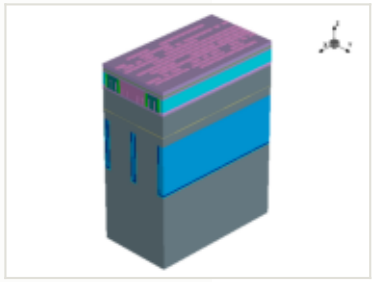
Step #2650: Metallization

depo_isotropic · Metallization

1.2	297.2	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

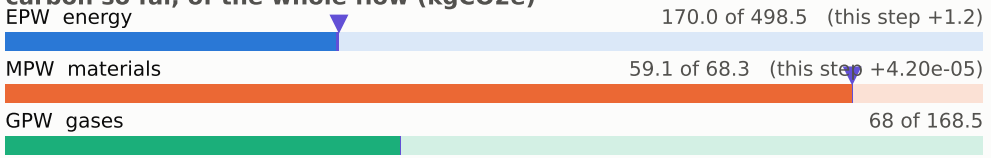


this step

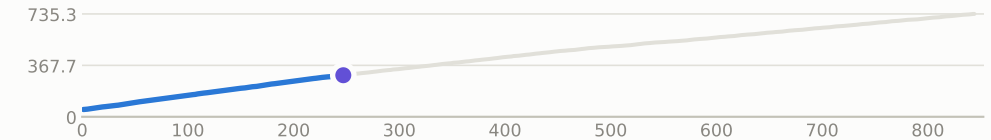
process area	Metallization
material	Co
thickness	0.0015 µm
material GWP	44.5 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.20e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

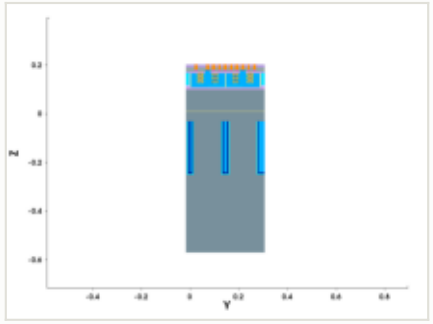
carbon so far, of the whole flow (kgCO2e)



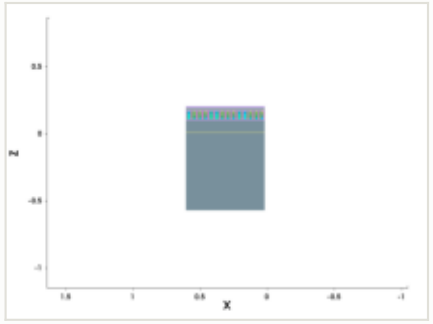
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



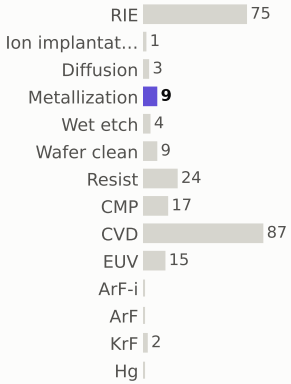
y cut at 0.184 µm



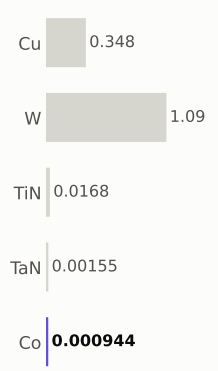
mask for this step

no mask for this step

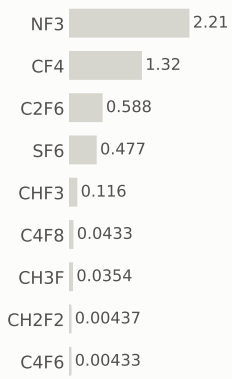
steps by process area



materials deposited (g)



gases emitted (g)



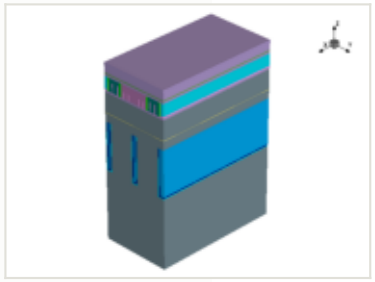
Step #2670: CVD

depo_fill · CVD

1.31	298.5	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

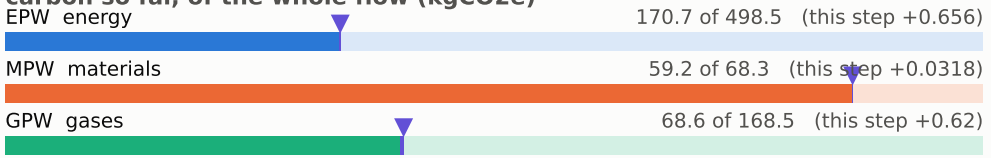


this step

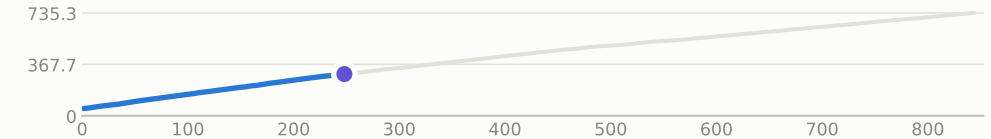
process area	CVD
material	—
thickness	0.024 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

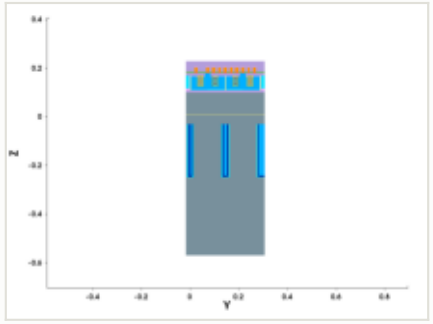
carbon so far, of the whole flow (kgCO2e)



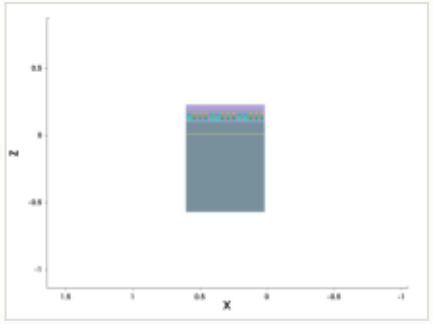
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



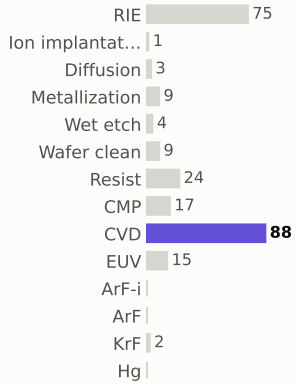
y cut at 0.184 µm



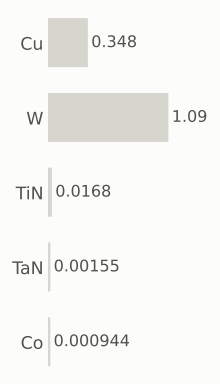
mask for this step

no mask for this step

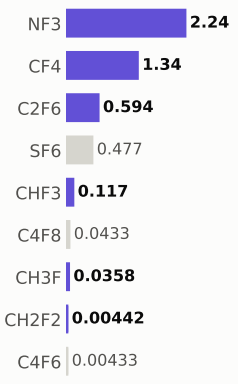
steps by process area



materials deposited (g)



gases emitted (g)



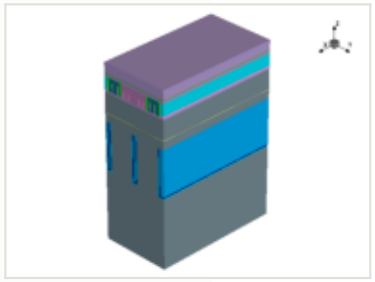
Step #2680: Diffusion

treatment · Diffusion

0.801	299.3	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

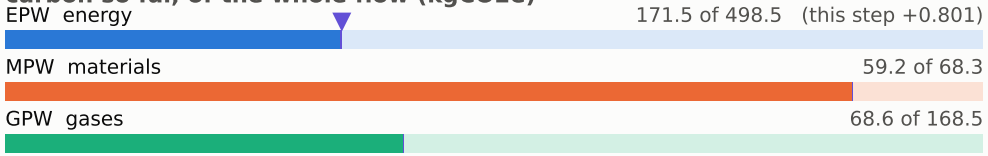


this step

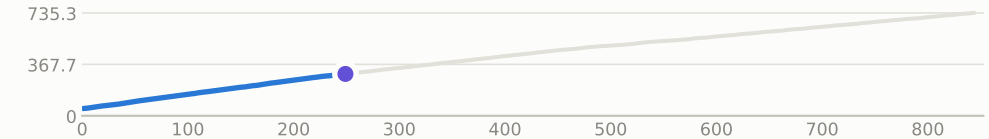
process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

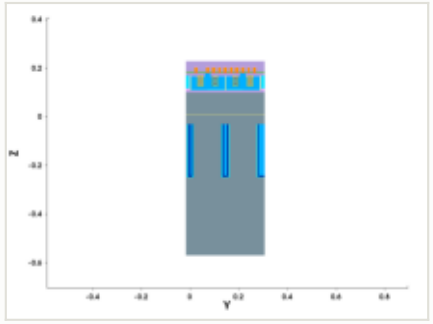
carbon so far, of the whole flow (kgCO2e)



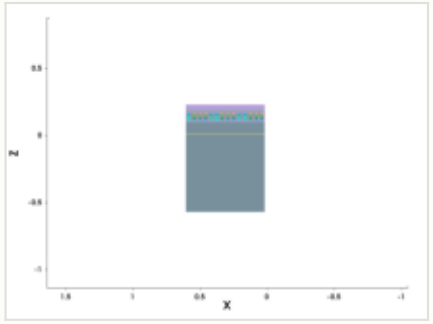
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



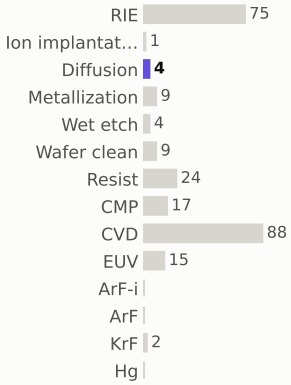
y cut at 0.184 µm



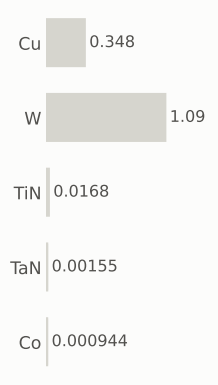
mask for this step

no mask for this step

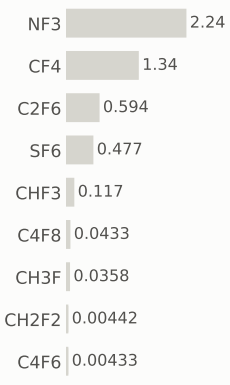
steps by process area



materials deposited (g)



gases emitted (g)



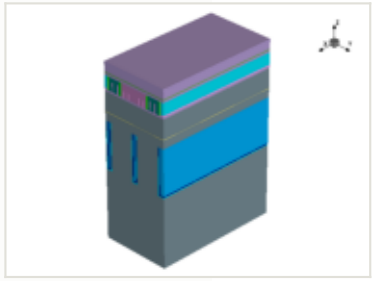
Step #2690: Wafer_clean

clean · Wafer_clean

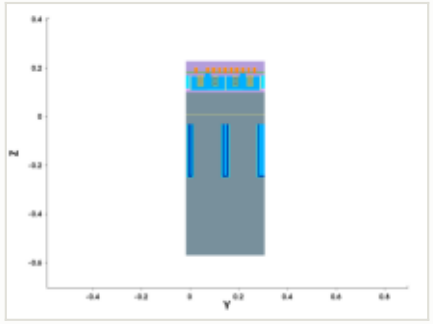
0.169	299.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

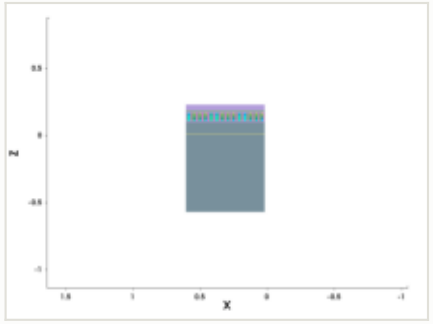
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

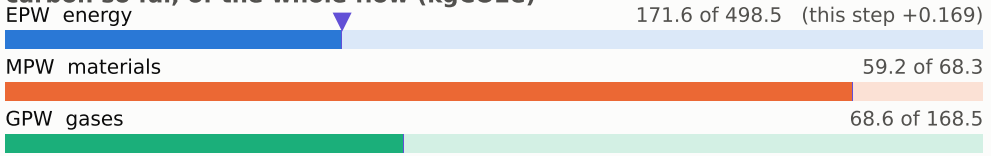
no mask for this step

this step

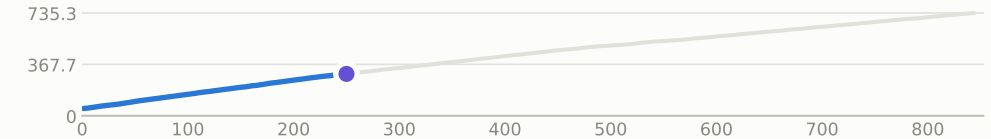
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

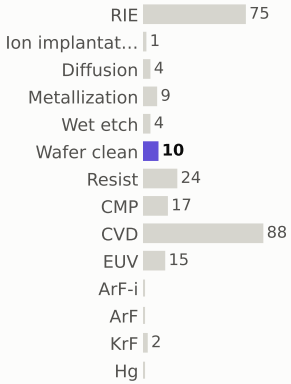
carbon so far, of the whole flow (kgCO2e)



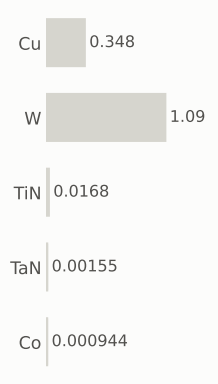
embodied carbon up to each step (kgCO2e per wafer)



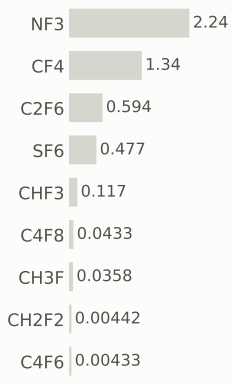
steps by process area



materials deposited (g)



gases emitted (g)



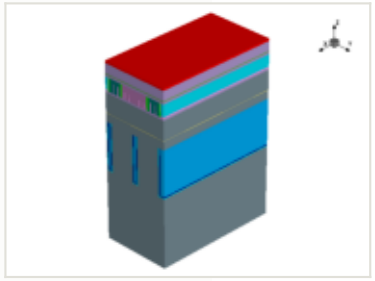
Step #2700: CVD

depo_fill · CVD

1.31	300.7	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

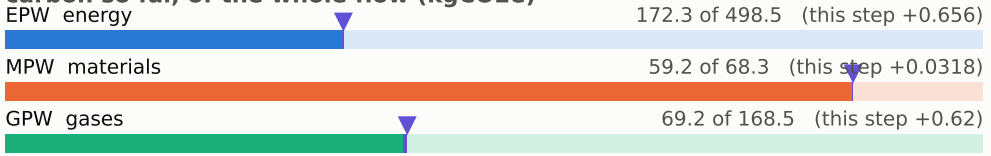


this step

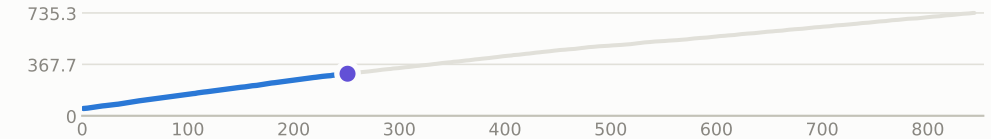
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

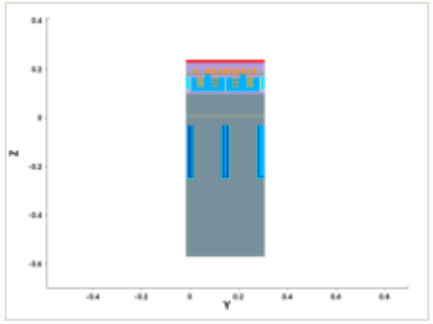
carbon so far, of the whole flow (kgCO2e)



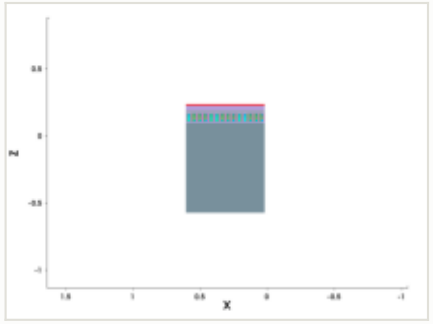
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



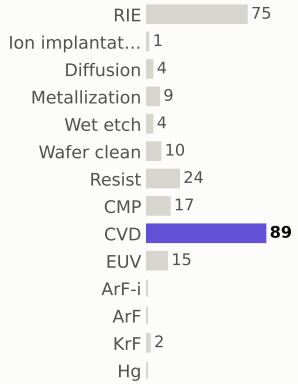
y cut at 0.184 µm



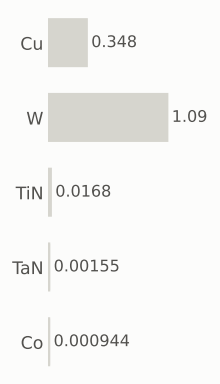
mask for this step

no mask for this step

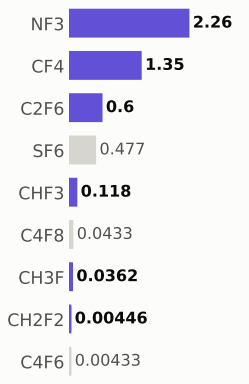
steps by process area



materials deposited (g)



gases emitted (g)



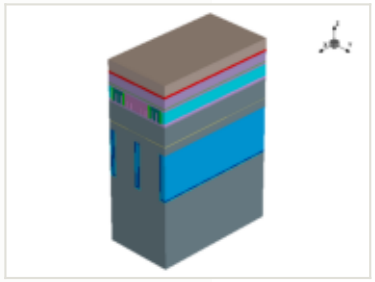
Step #2710: Resist

depo_fill · Resist

0.169	300.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

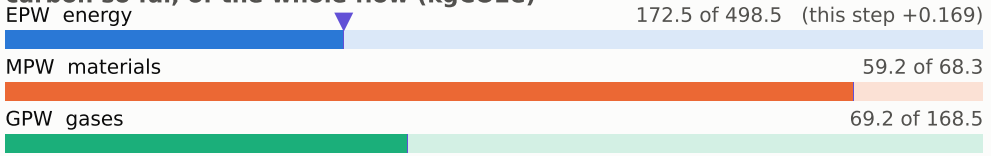


this step

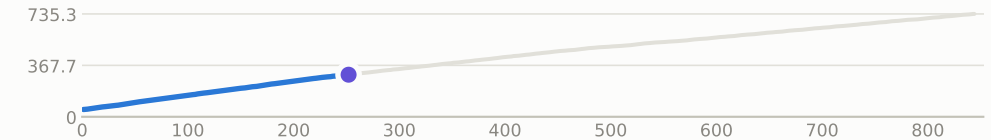
process area	Resist
material	—
thickness	0.05 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

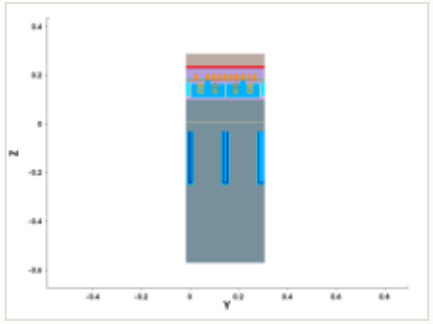
carbon so far, of the whole flow (kgCO2e)



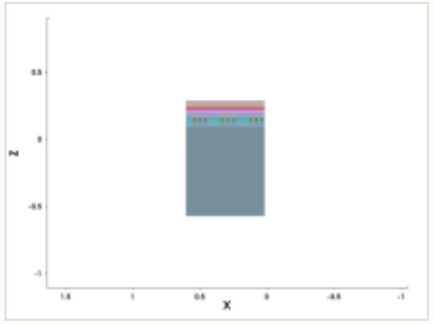
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



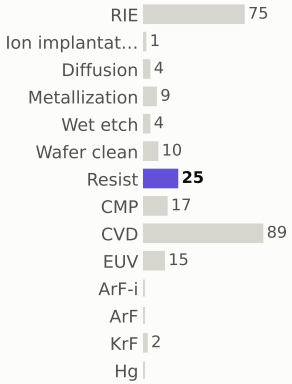
y cut at 0.184 µm



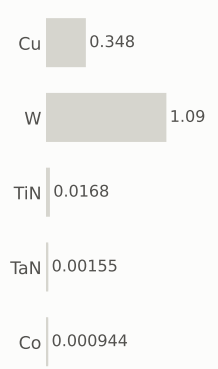
mask for this step

no mask for this step

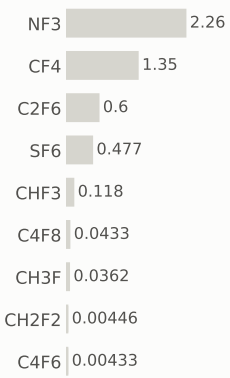
steps by process area



materials deposited (g)



gases emitted (g)



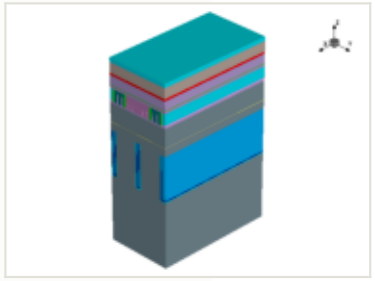
Step #2720: CVD

depo_fill · CVD

1.31	302.2	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

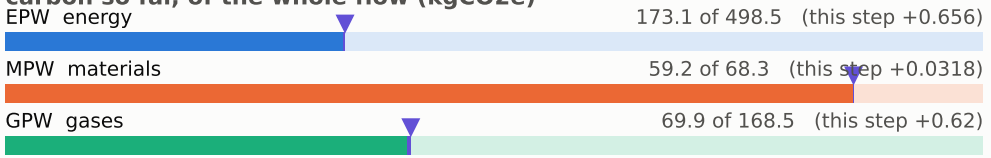


this step

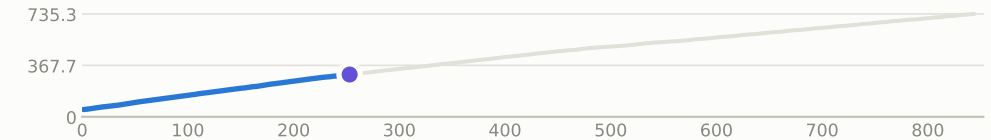
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

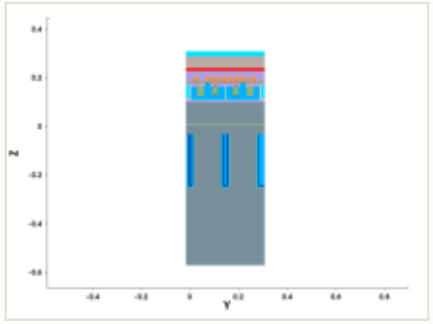
carbon so far, of the whole flow (kgCO2e)



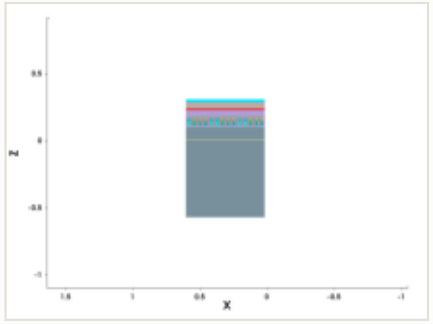
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



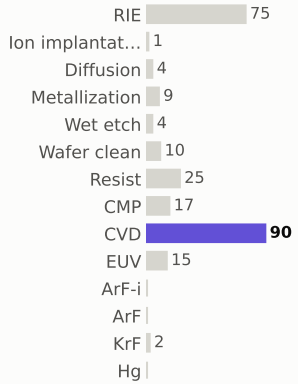
y cut at 0.184 µm



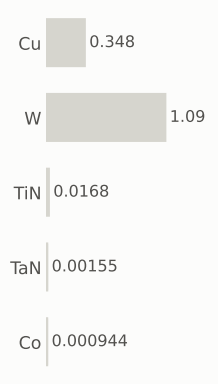
mask for this step

no mask for this step

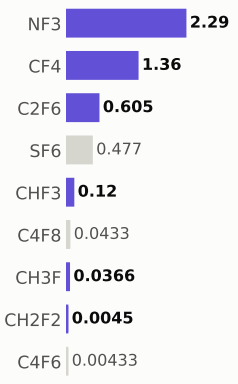
steps by process area



materials deposited (g)



gases emitted (g)



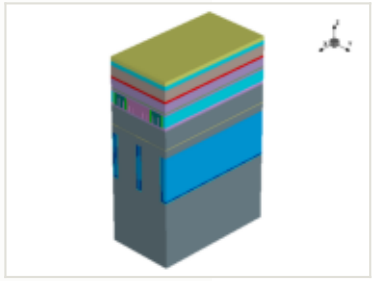
Step #2730: CVD

depo_fill · CVD

1.31	303.5	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

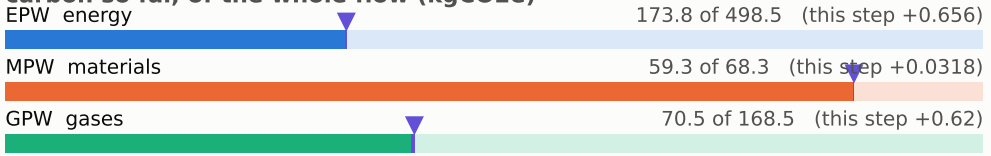


this step

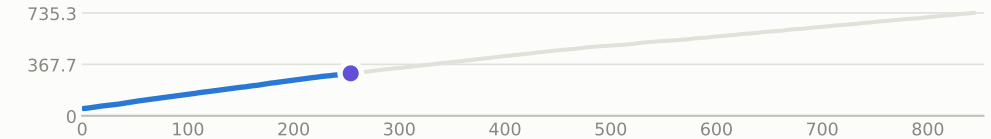
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

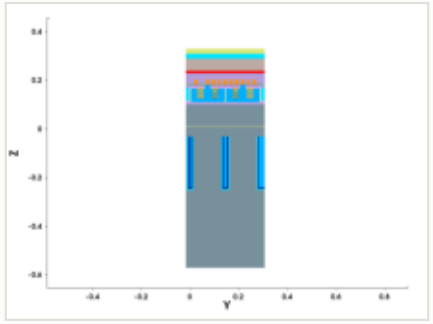
carbon so far, of the whole flow (kgCO2e)



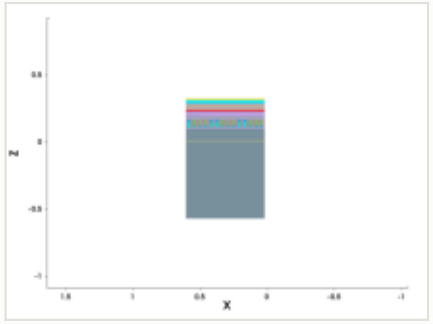
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



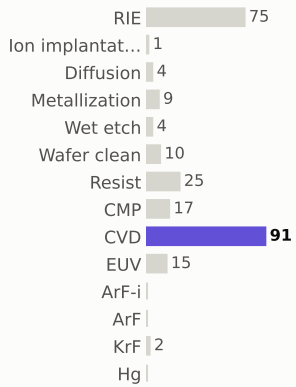
y cut at 0.184 µm



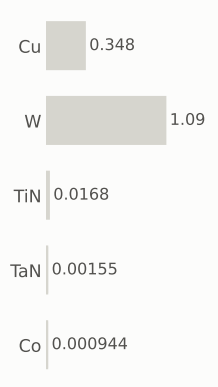
mask for this step

no mask for this step

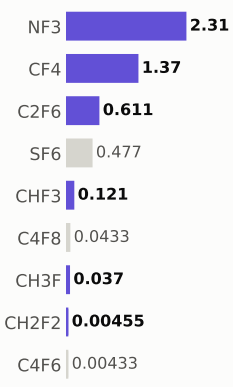
steps by process area



materials deposited (g)



gases emitted (g)



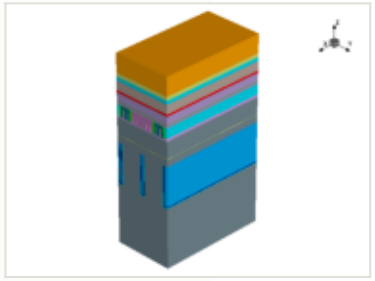
Step #2740: Resist

depo_fill · Resist

0.169	303.7	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

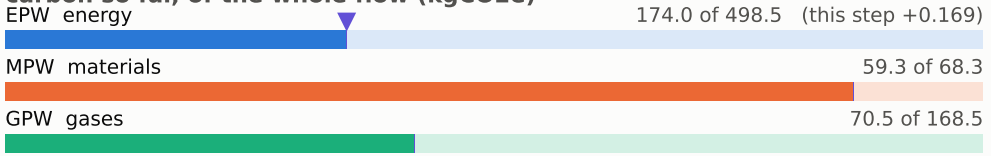


this step

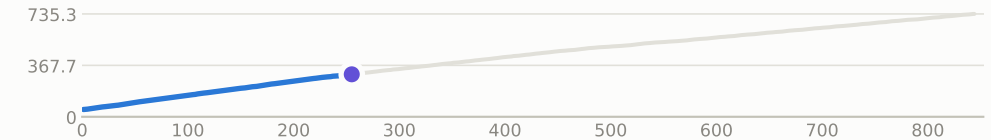
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

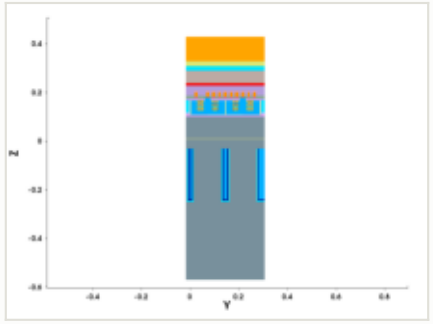
carbon so far, of the whole flow (kgCO2e)



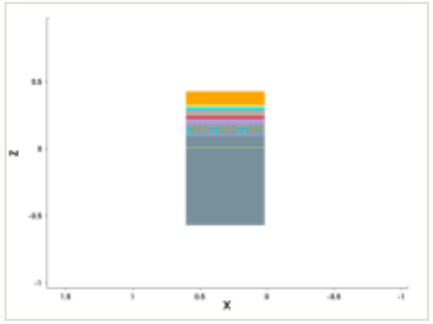
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



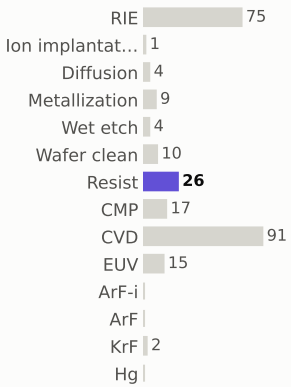
y cut at 0.184 µm



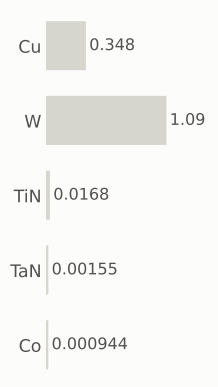
mask for this step

no mask for this step

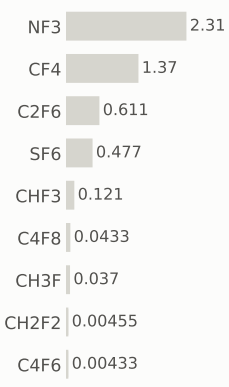
steps by process area



materials deposited (g)



gases emitted (g)



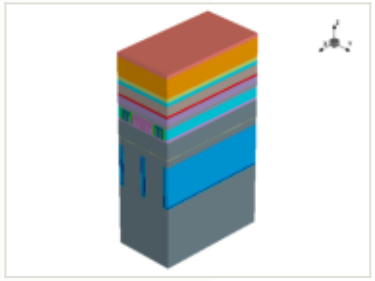
Step #2750: CVD

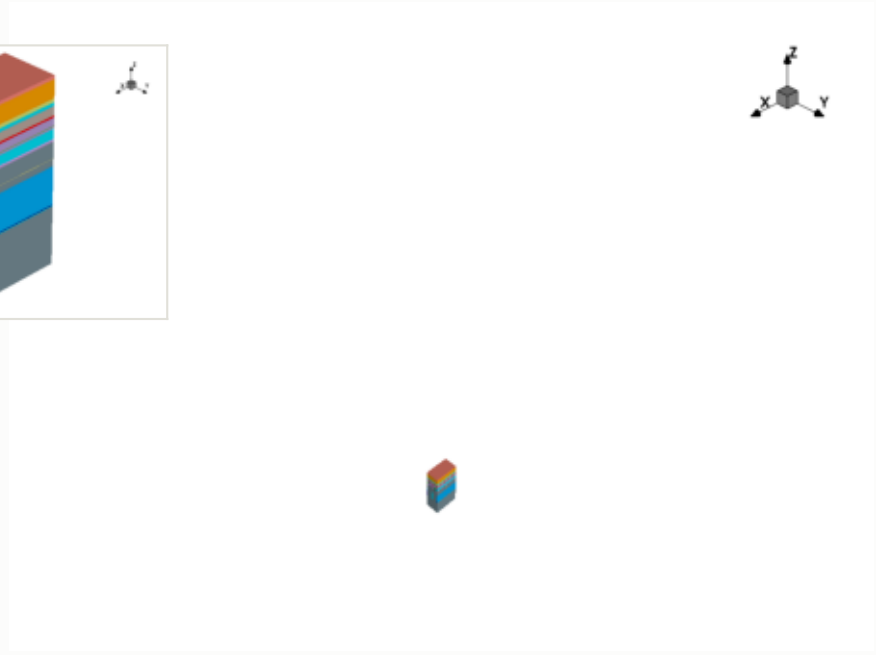
depo_fill · CVD

1.31	305.0	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



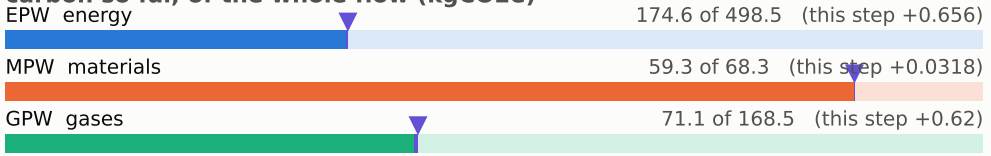


this step

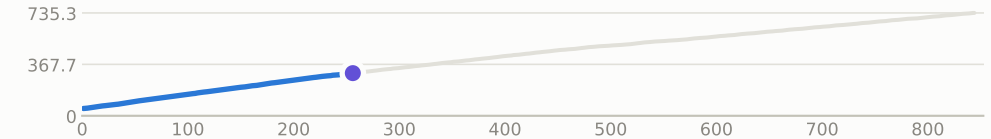
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

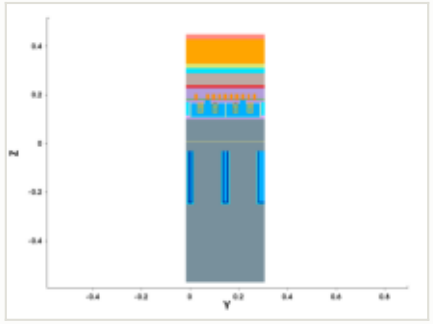
carbon so far, of the whole flow (kgCO2e)



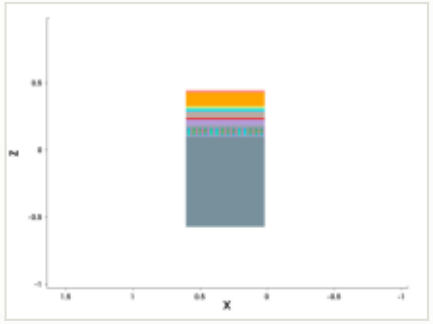
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



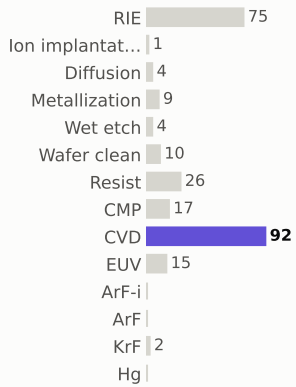
y cut at 0.184 µm



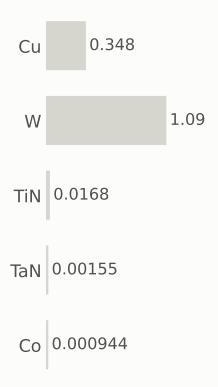
mask for this step

no mask for this step

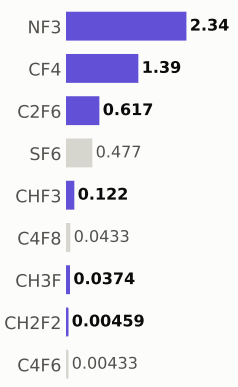
steps by process area



materials deposited (g)



gases emitted (g)



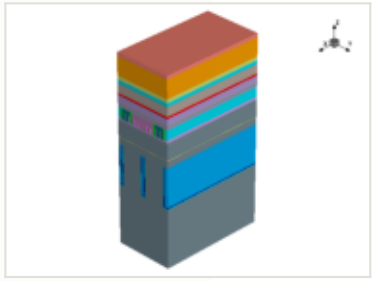
Step #2760: Wafer_clean

clean · Wafer_clean

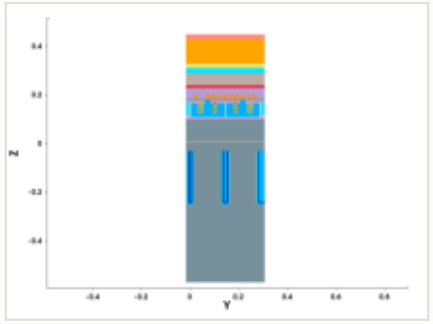
0.169	305.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

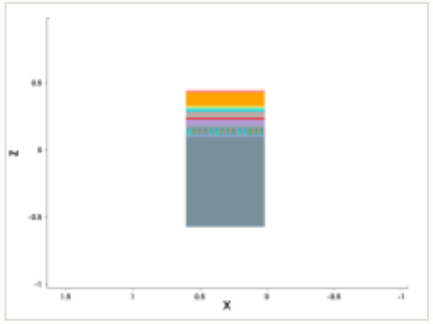
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

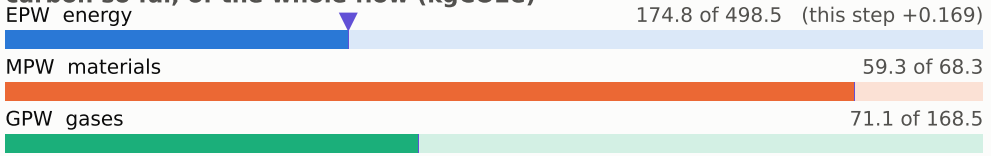
no mask for this step

this step

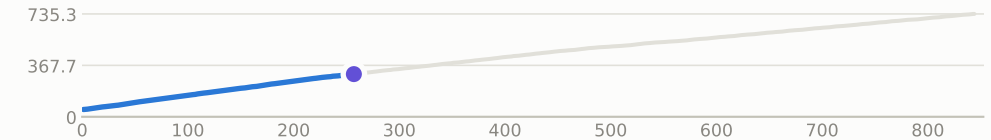
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

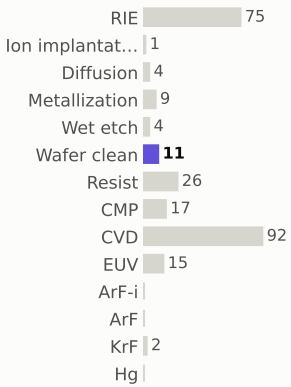
carbon so far, of the whole flow (kgCO2e)



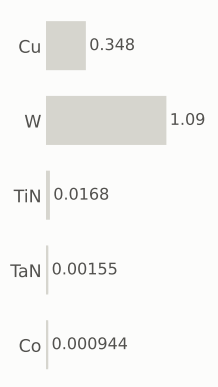
embodied carbon up to each step (kgCO2e per wafer)



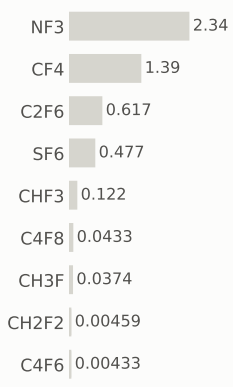
steps by process area



materials deposited (g)



gases emitted (g)



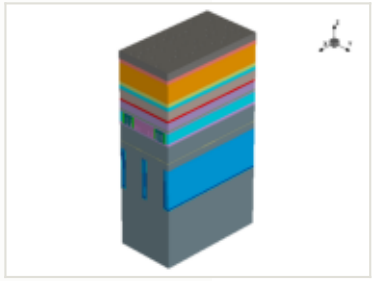
Step #2770: EUV

litho_patterning · EUV

1.96	307.1	0.27%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

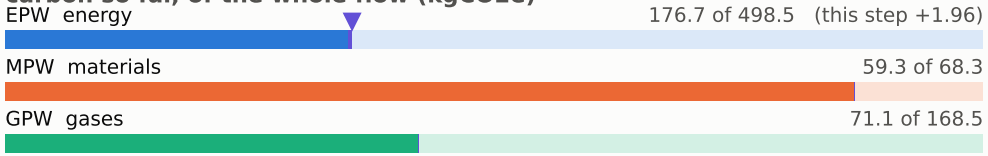


this step

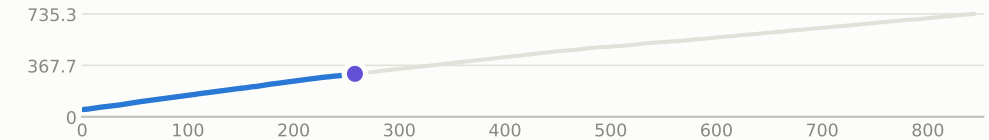
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

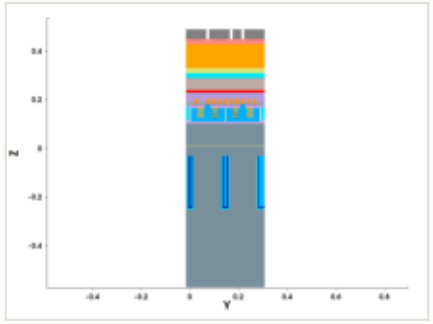
carbon so far, of the whole flow (kgCO2e)



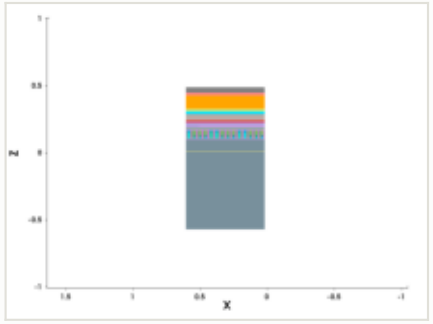
embodied carbon up to each step (kgCO2e per wafer)



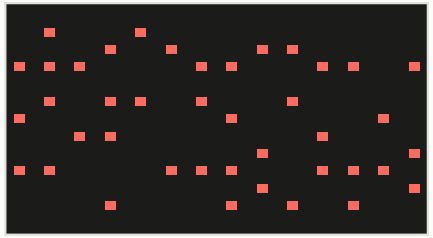
x cut at 0.294 µm



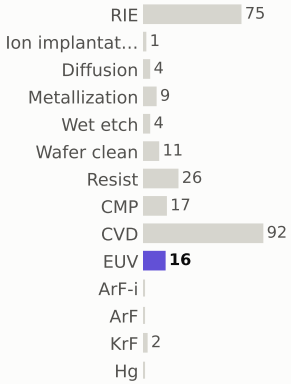
y cut at 0.184 µm



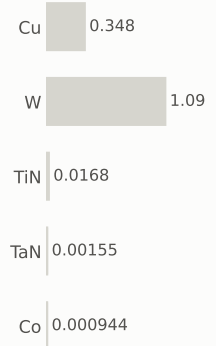
mask: layer 10221:0 (GDSII)



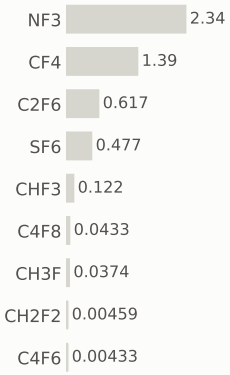
steps by process area



materials deposited (g)



gases emitted (g)



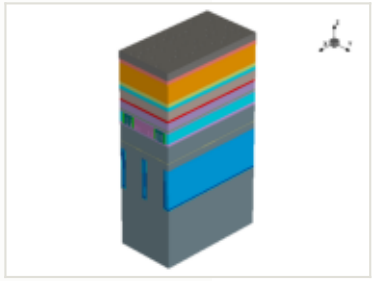
Step #2780: RIE

etch_anisotropic · RIE

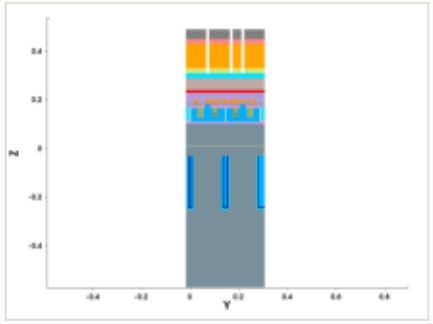
1.04	308.2	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

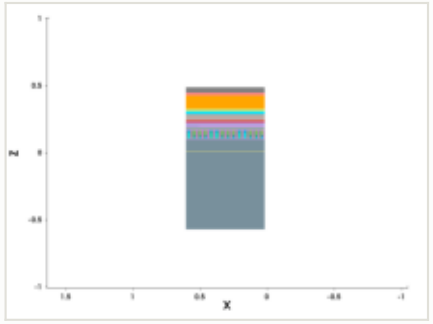
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

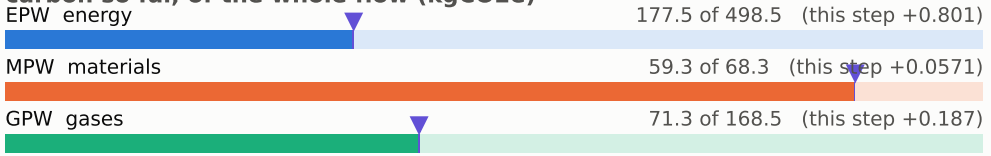
no mask for this step

this step

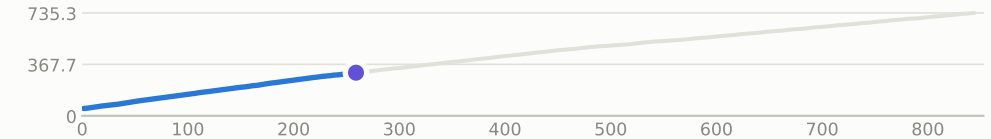
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

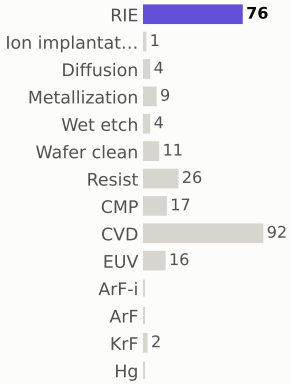
carbon so far, of the whole flow (kgCO2e)



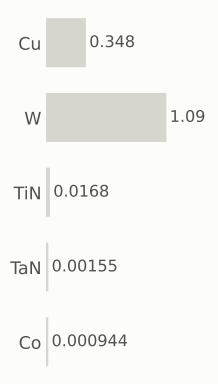
embodied carbon up to each step (kgCO2e per wafer)



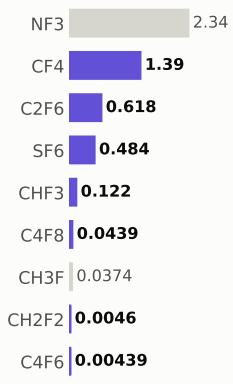
steps by process area



materials deposited (g)



gases emitted (g)



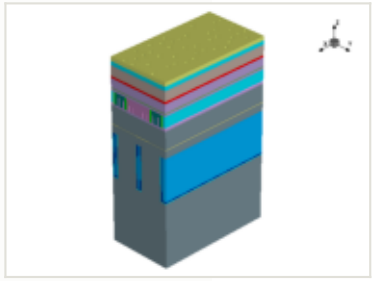
Step #2790: Wafer_clean

etch_strip · Wafer_clean

0.169	308.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

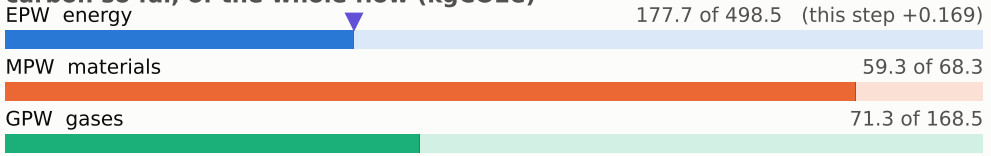


this step

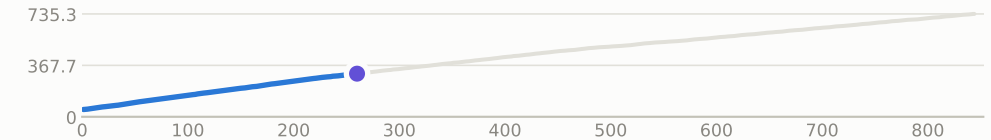
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

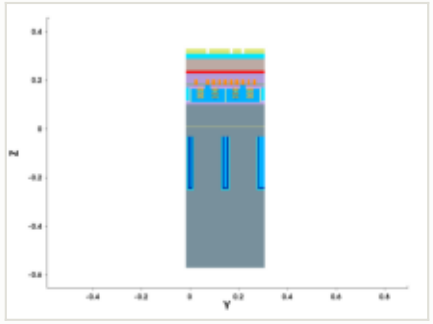
carbon so far, of the whole flow (kgCO2e)



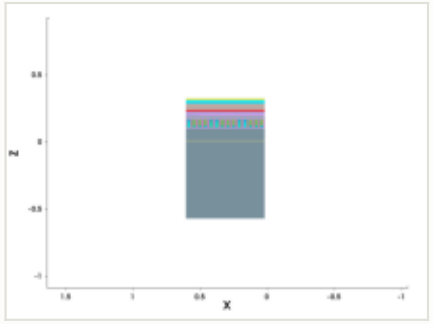
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



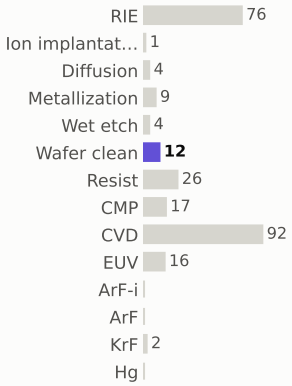
y cut at 0.184 µm



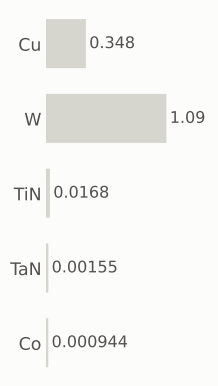
mask for this step

no mask for this step

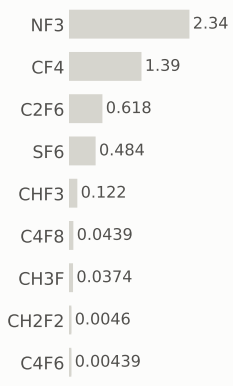
steps by process area



materials deposited (g)



gases emitted (g)



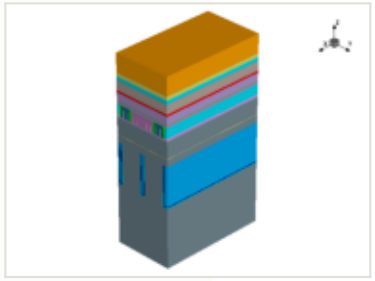
Step #2800: Resist

depo_fill · Resist

0.169	308.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

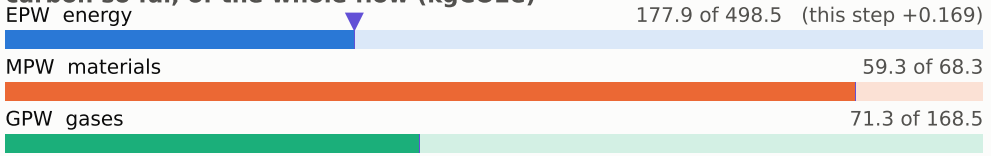


this step

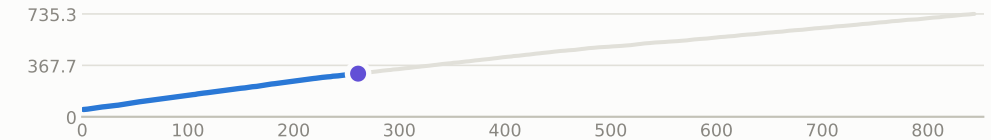
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

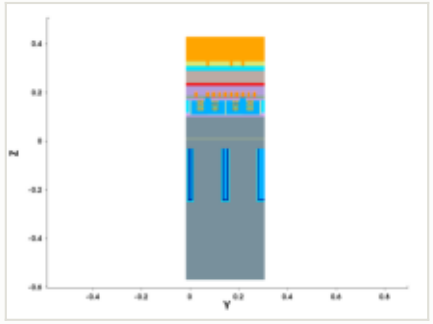
carbon so far, of the whole flow (kgCO2e)



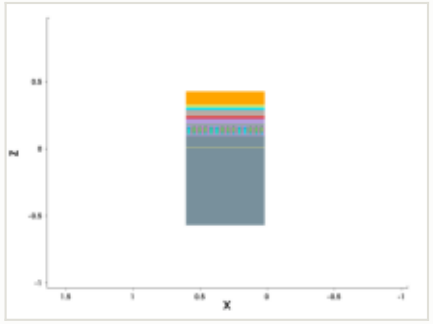
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



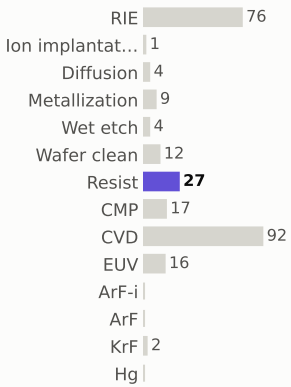
y cut at 0.184 µm



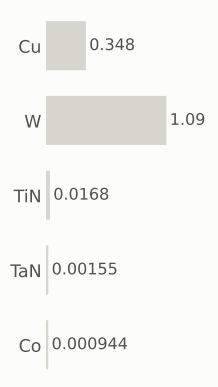
mask for this step

no mask for this step

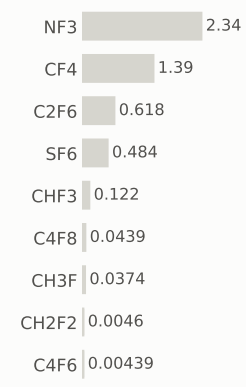
steps by process area



materials deposited (g)



gases emitted (g)



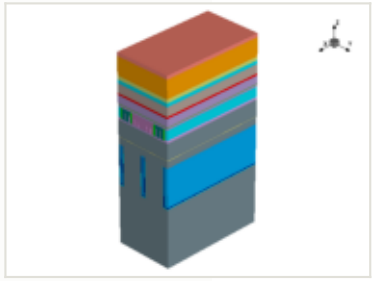
Step #2810: CVD

depo_fill · CVD

1.31	309.8	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

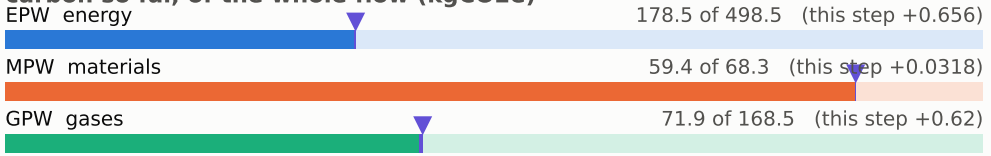


this step

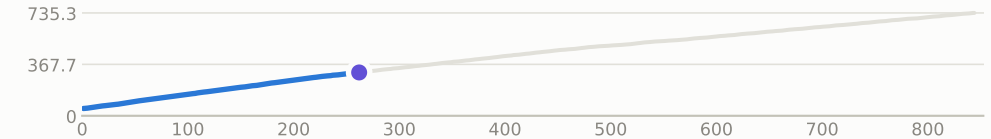
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

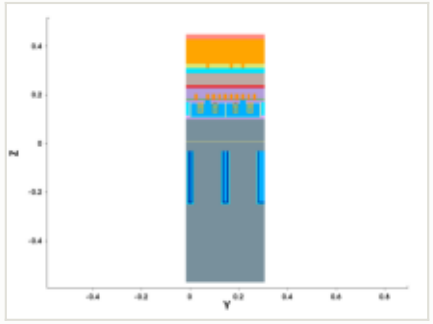
carbon so far, of the whole flow (kgCO2e)



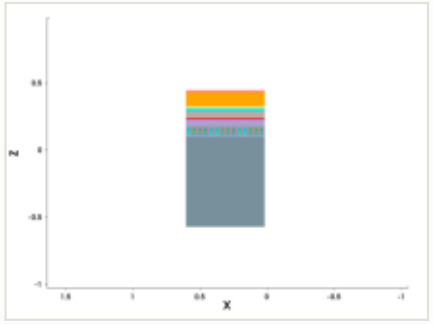
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



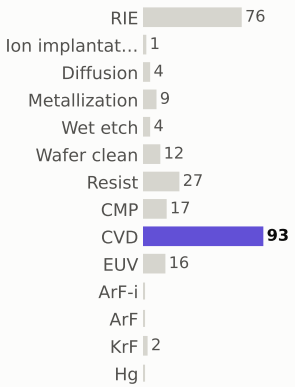
y cut at 0.184 µm



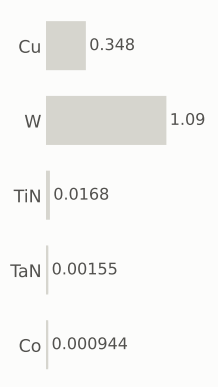
mask for this step

no mask for this step

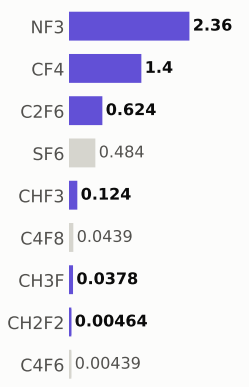
steps by process area



materials deposited (g)



gases emitted (g)



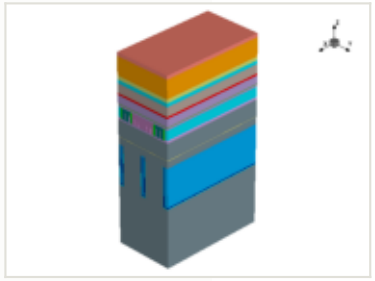
Step #2820: Wafer_clean

clean · Wafer_clean

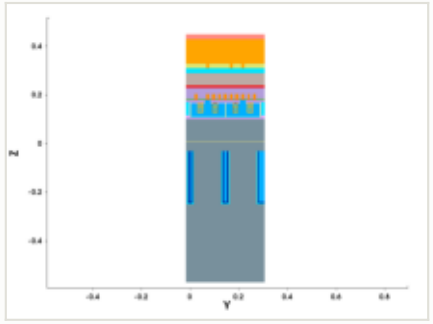
0.169	310.0	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

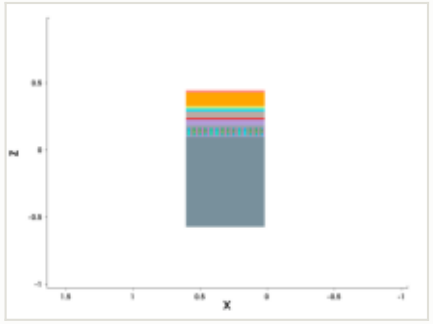
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

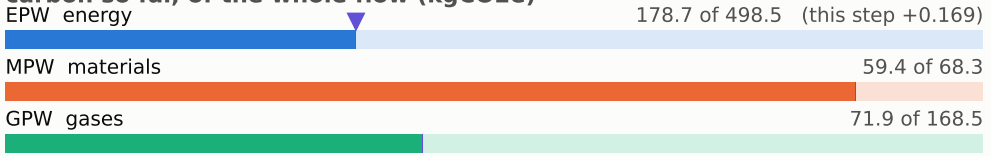
no mask for this step

this step

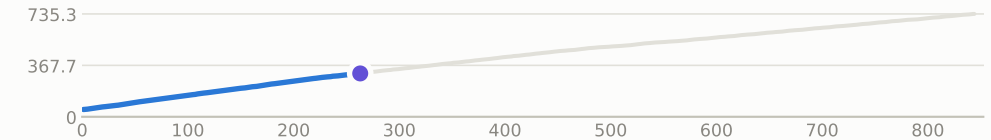
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

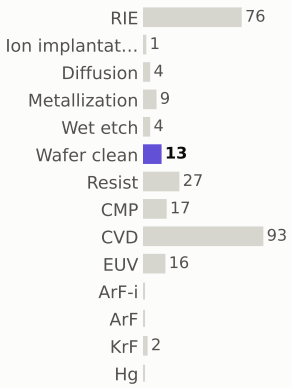
carbon so far, of the whole flow (kgCO2e)



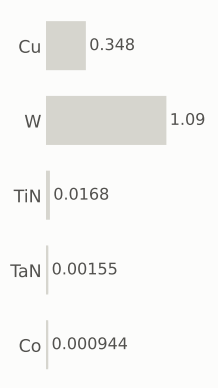
embodied carbon up to each step (kgCO2e per wafer)



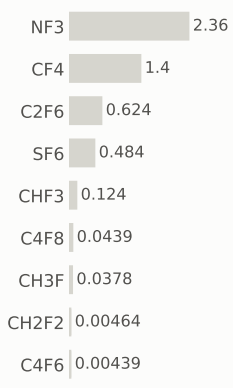
steps by process area



materials deposited (g)



gases emitted (g)



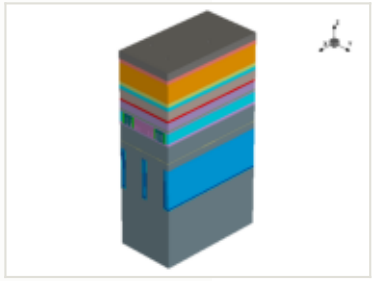
Step #2830: EUV

litho_patterning · EUV

1.96	311.9	0.27%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

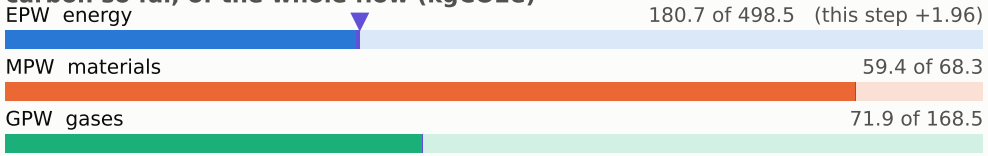


this step

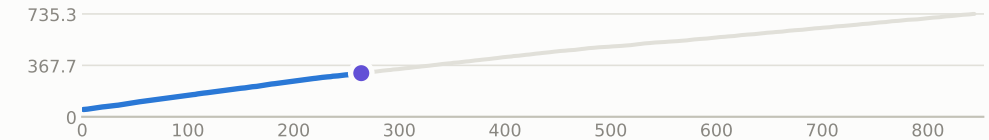
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

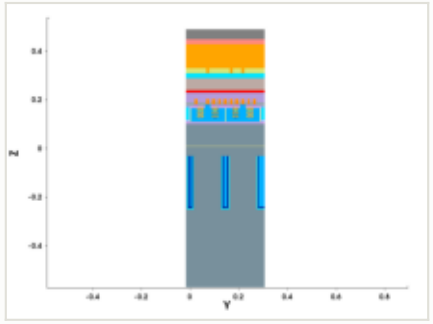
carbon so far, of the whole flow (kgCO2e)



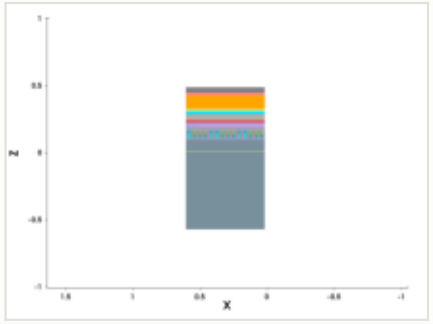
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm

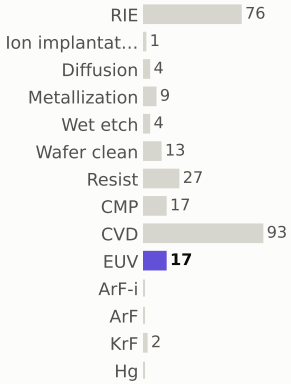


y cut at 0.184 µm

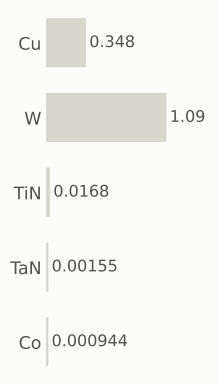


mask: layer 10222:0 (GDSII)

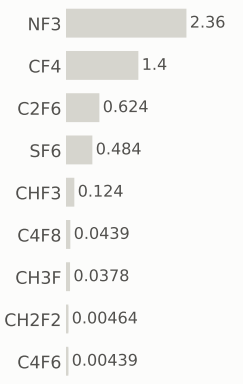
steps by process area



materials deposited (g)



gases emitted (g)



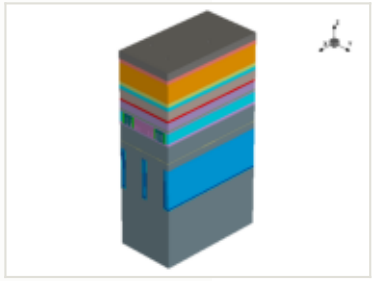
Step #2840: RIE

etch_anisotropic · RIE

1.04	313.0	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

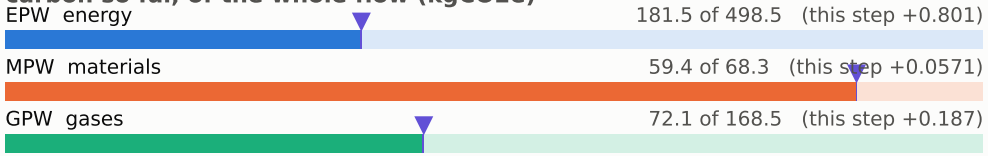


this step

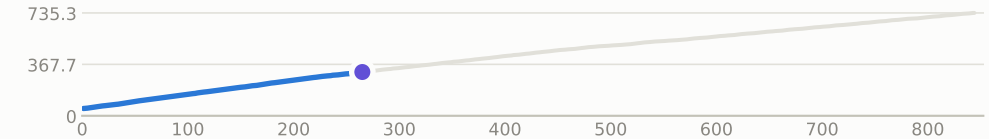
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

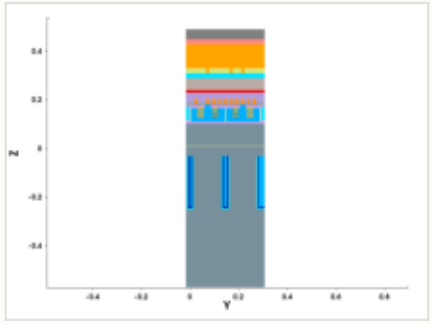
carbon so far, of the whole flow (kgCO2e)



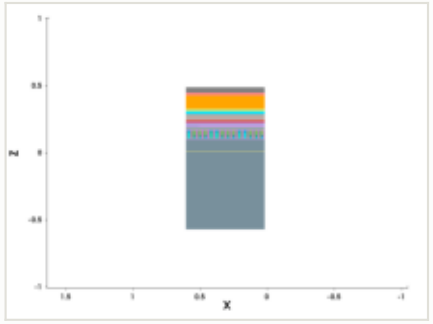
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



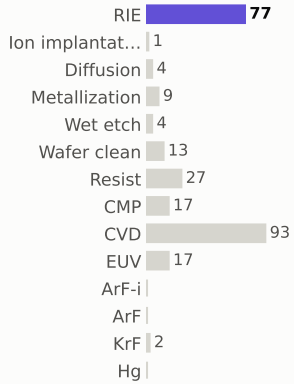
y cut at 0.184 µm



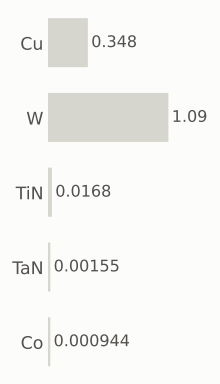
mask for this step

no mask for this step

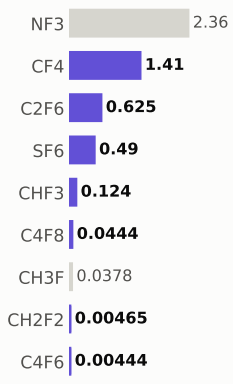
steps by process area



materials deposited (g)



gases emitted (g)



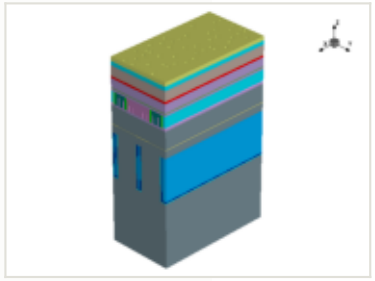
Step #2850: Wafer_clean

etch_strip · Wafer_clean

0.169	313.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

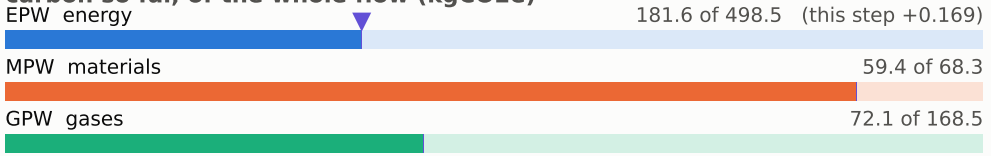


this step

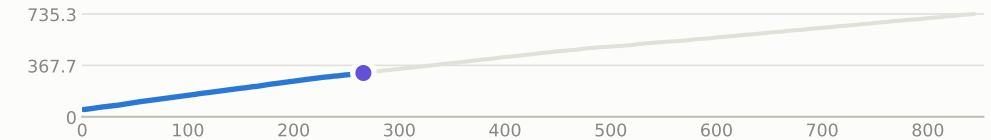
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

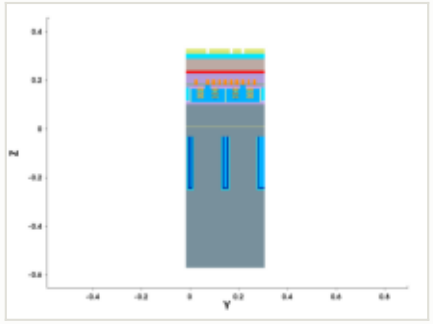
carbon so far, of the whole flow (kgCO2e)



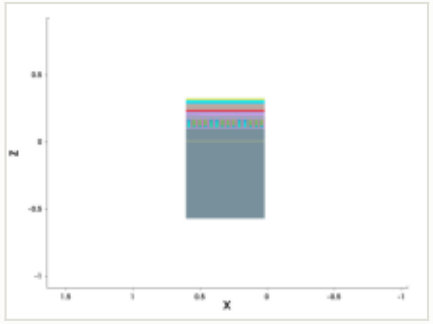
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



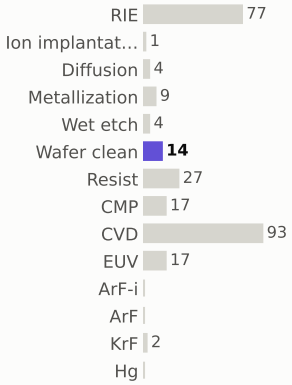
y cut at 0.184 µm



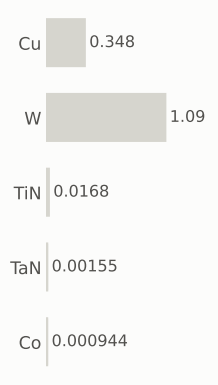
mask for this step

no mask for this step

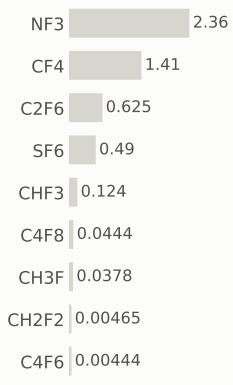
steps by process area



materials deposited (g)



gases emitted (g)



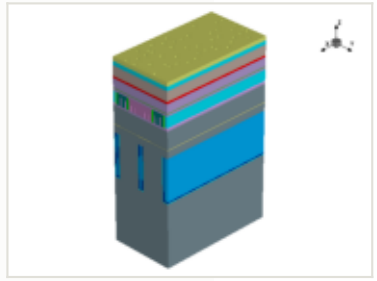
Step #2860: RIE

etch_anisotropic · RIE

1.04	314.2	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

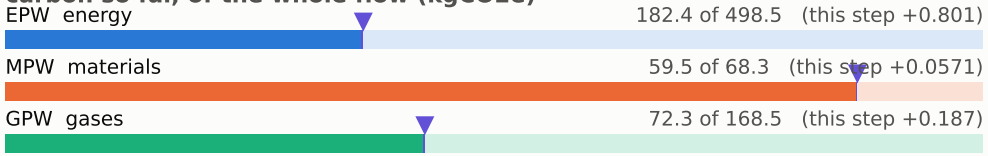


this step

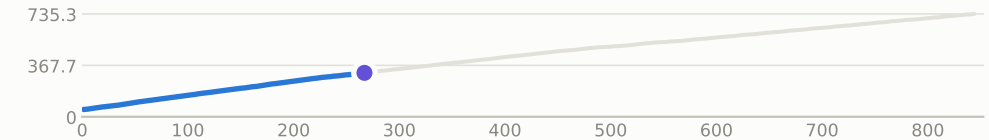
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

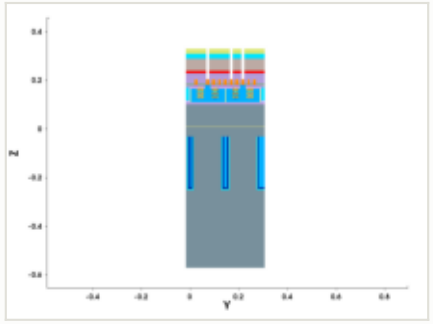
carbon so far, of the whole flow (kgCO2e)



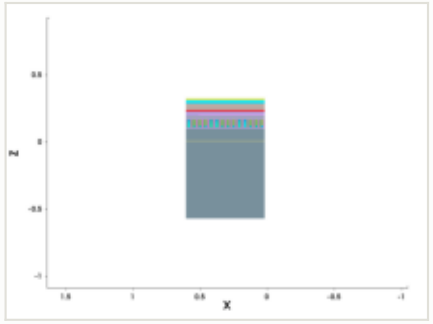
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



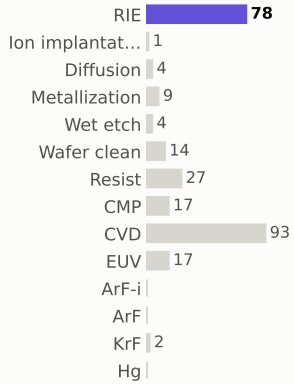
y cut at 0.184 µm



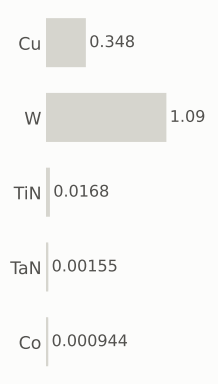
mask for this step

no mask for this step

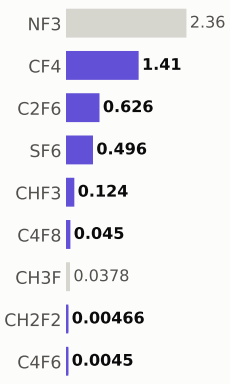
steps by process area



materials deposited (g)



gases emitted (g)



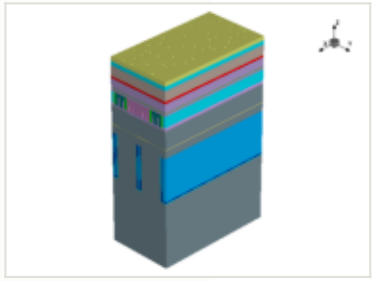
Step #2870: Wafer_clean

clean · Wafer_clean

0.169	314.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



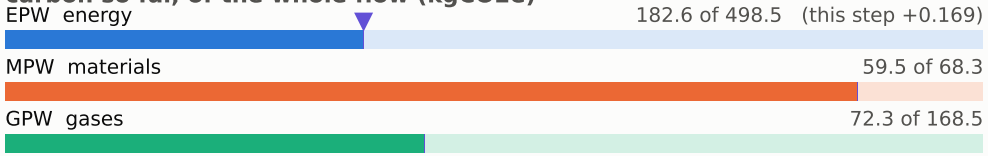
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

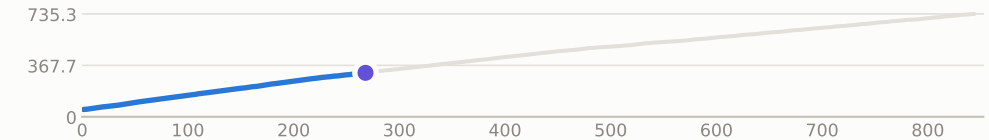
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

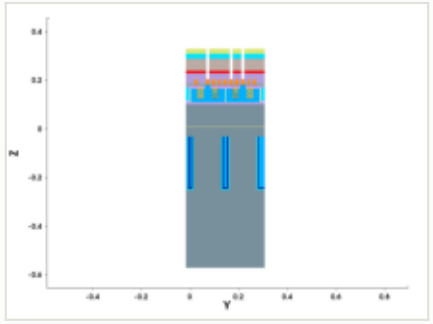
carbon so far, of the whole flow (kgCO2e)



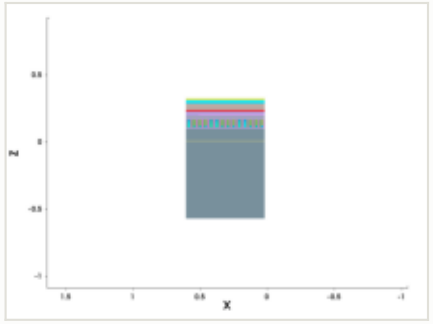
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



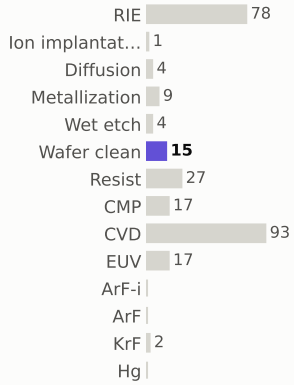
y cut at 0.184 µm



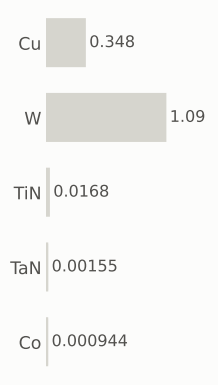
mask for this step

no mask for this step

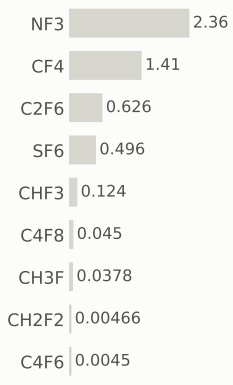
steps by process area



materials deposited (g)



gases emitted (g)



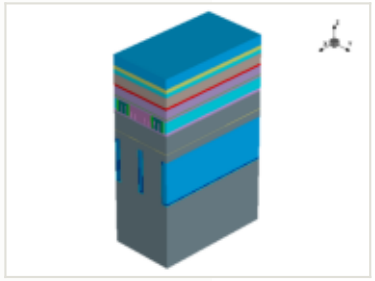
Step #2880: Metallization

depo_fill · Metallization

1.21	315.6	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

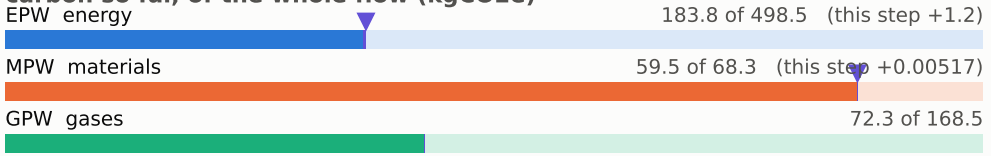


this step

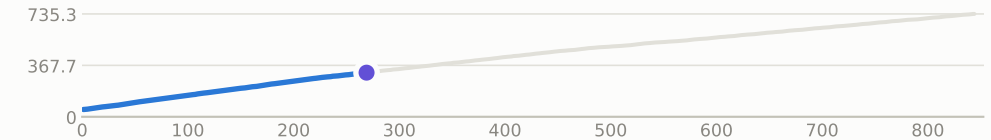
process area	Metallization
material	W
thickness	0.05 µm
material GWP	75.8149 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00517 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

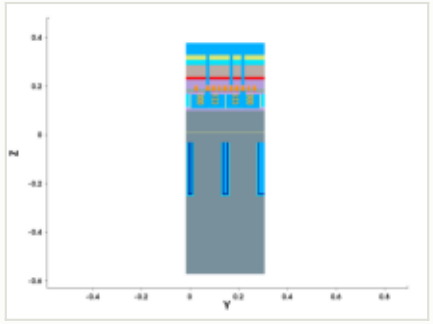
carbon so far, of the whole flow (kgCO2e)



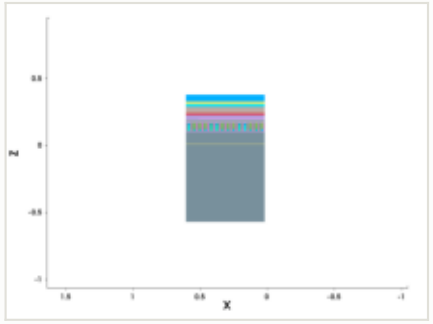
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



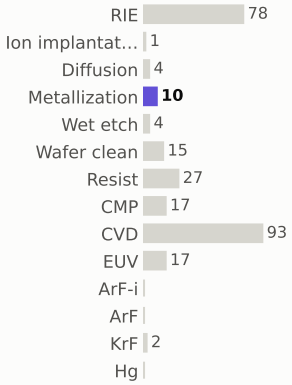
y cut at 0.184 µm



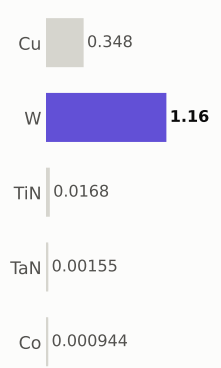
mask for this step

no mask for this step

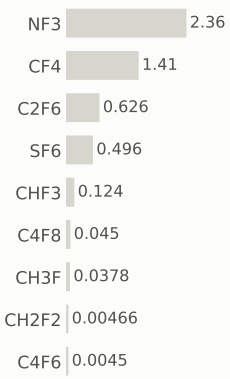
steps by process area



materials deposited (g)



gases emitted (g)



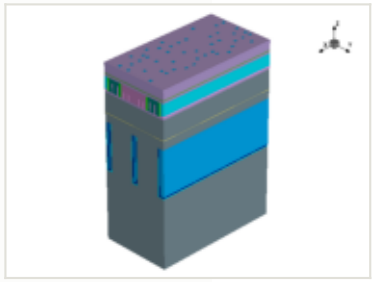
Step #2890: CMP

etch_cmp · CMP

0.169	315.8	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

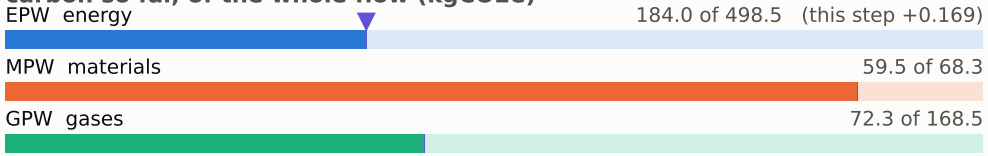


this step

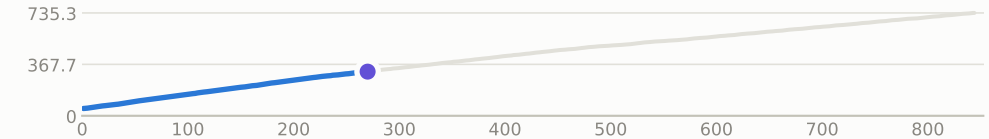
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

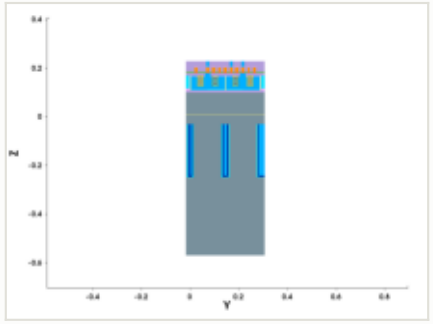
carbon so far, of the whole flow (kgCO2e)



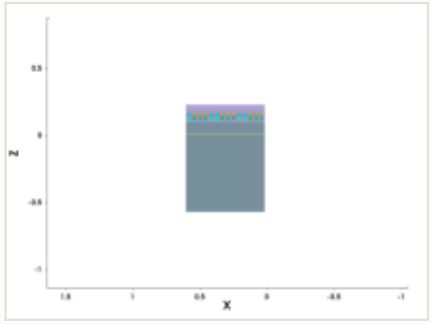
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



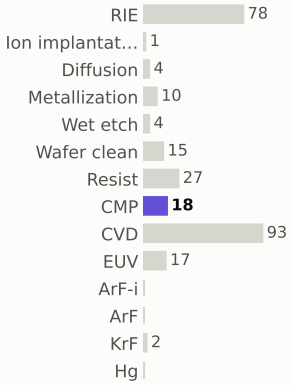
y cut at 0.184 µm



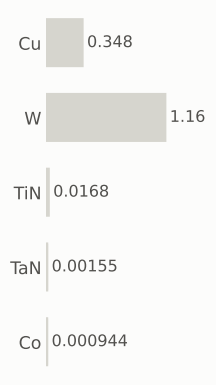
mask for this step

no mask for this step

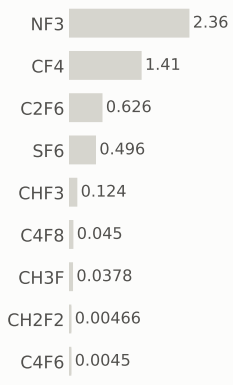
steps by process area



materials deposited (g)



gases emitted (g)



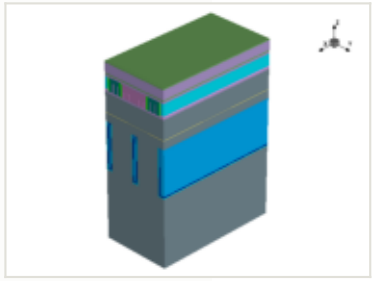
Step #2900: CVD

depo_fill · CVD

1.31	317.1	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

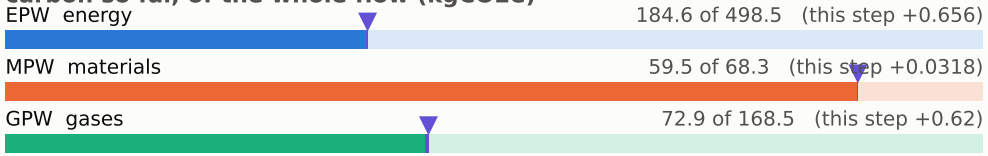


this step

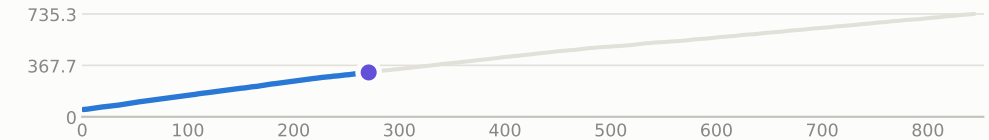
process area	CVD
material	—
thickness	0.006 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

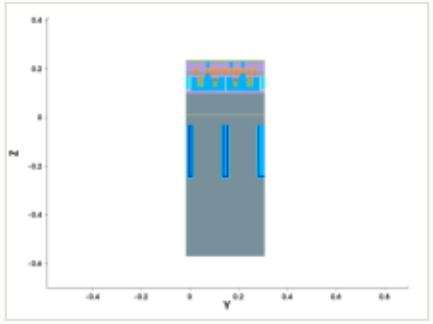
carbon so far, of the whole flow (kgCO2e)



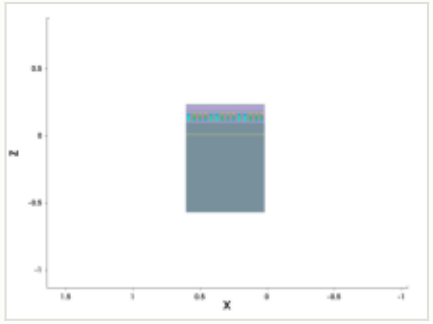
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



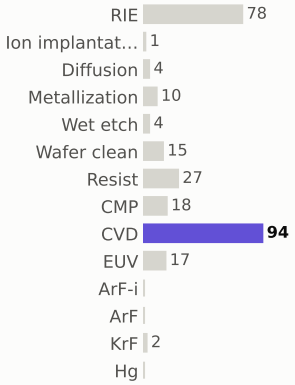
y cut at 0.184 µm



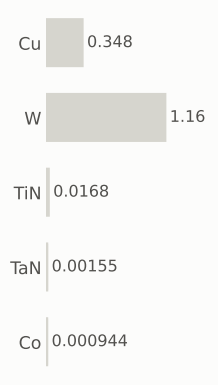
mask for this step

no mask for this step

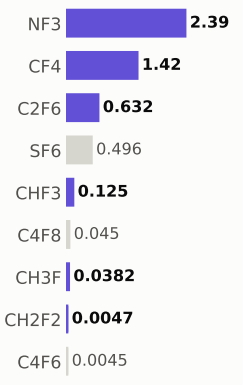
steps by process area



materials deposited (g)



gases emitted (g)



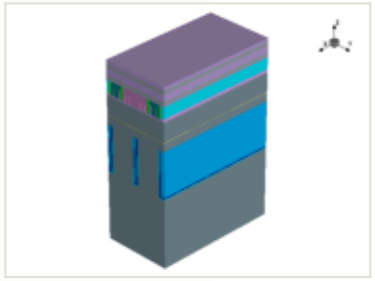
Step #2910: CVD

depo_fill · CVD

1.31	318.4	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

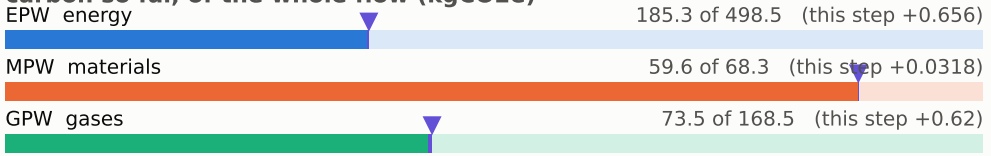


this step

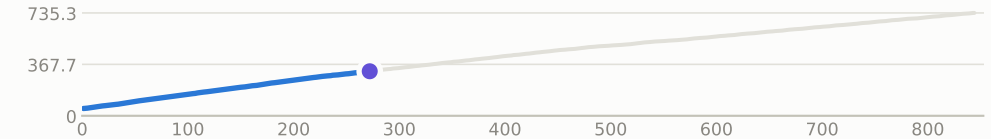
process area	CVD
material	—
thickness	0.022 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

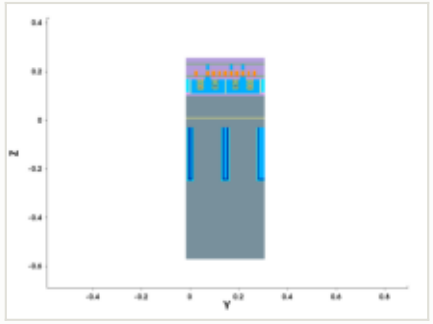
carbon so far, of the whole flow (kgCO2e)



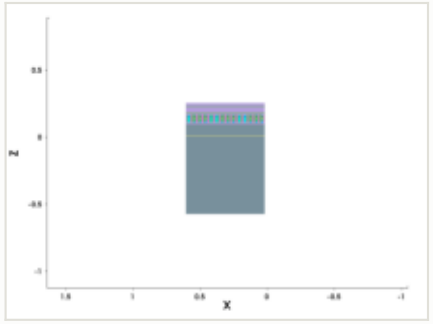
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



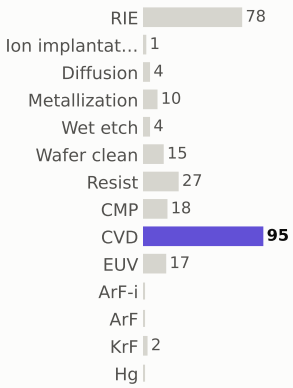
y cut at 0.184 µm



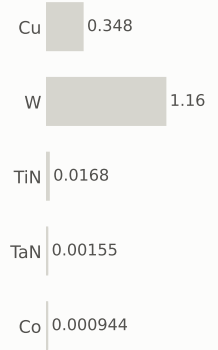
mask for this step

no mask for this step

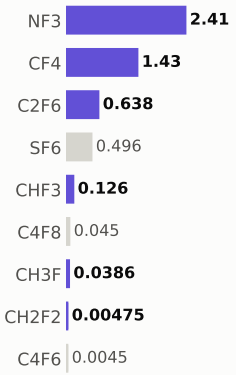
steps by process area



materials deposited (g)



gases emitted (g)



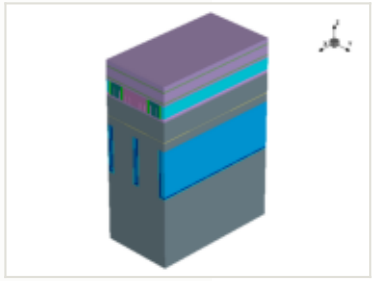
Step #2920: Diffusion

treatment · Diffusion

0.801	319.2	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



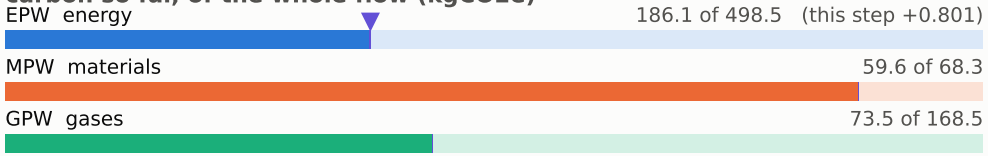
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

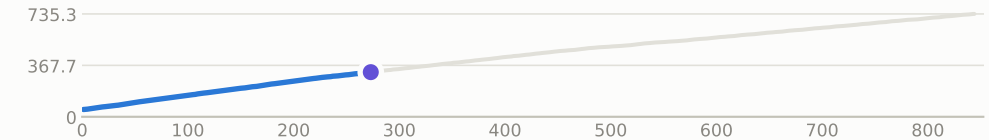
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

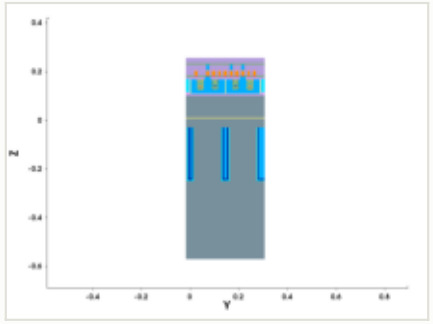
carbon so far, of the whole flow (kgCO2e)



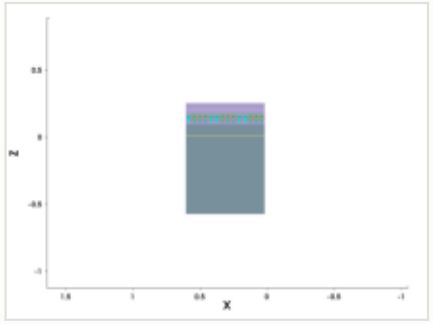
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



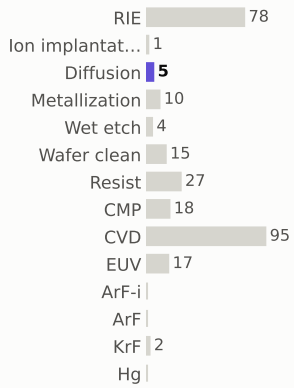
y cut at 0.184 µm



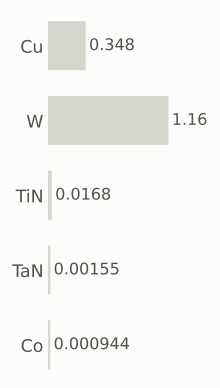
mask for this step

no mask for this step

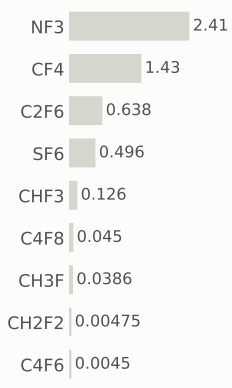
steps by process area



materials deposited (g)



gases emitted (g)



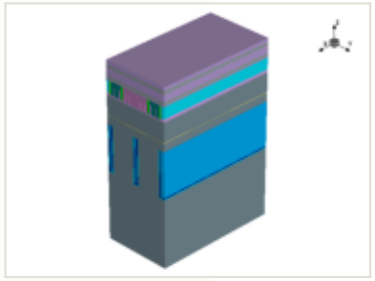
Step #2930: Wafer_clean

clean · Wafer_clean

0.169	319.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



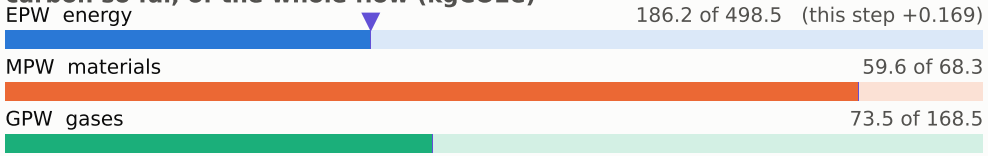
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

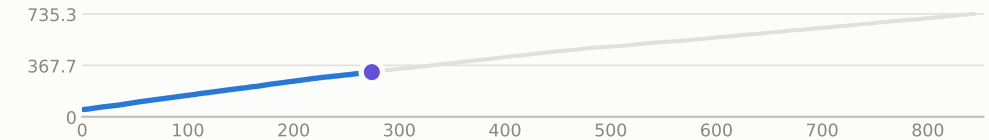
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

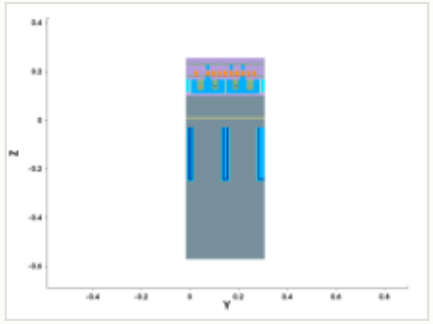
carbon so far, of the whole flow (kgCO2e)



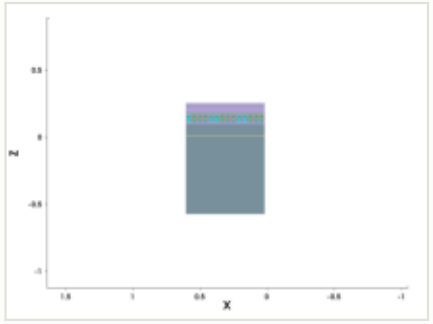
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



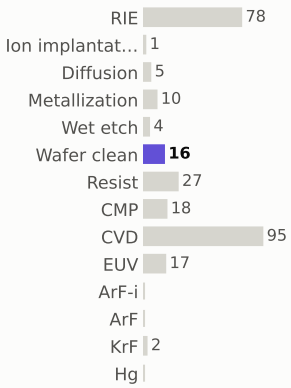
y cut at 0.184 µm



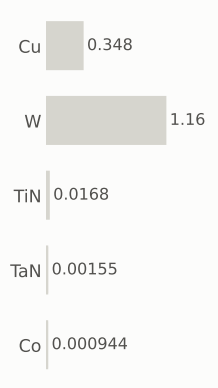
mask for this step

no mask for this step

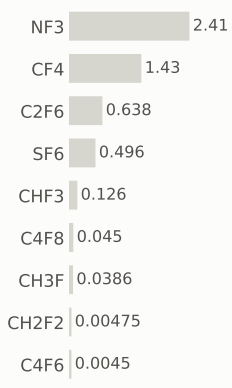
steps by process area



materials deposited (g)



gases emitted (g)



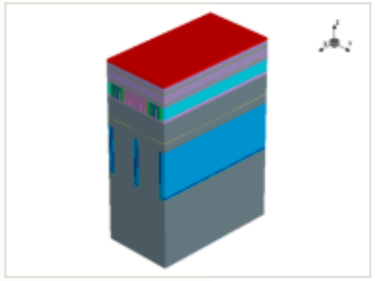
Step #2940: CVD

depo_fill · CVD

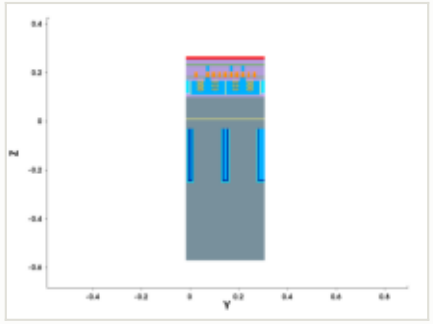
1.31	320.6	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

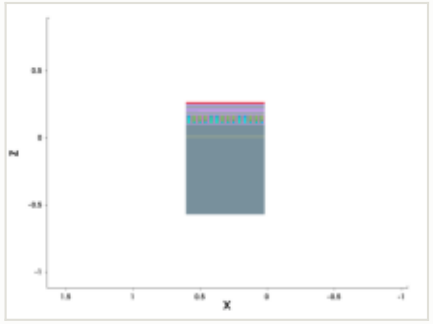
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

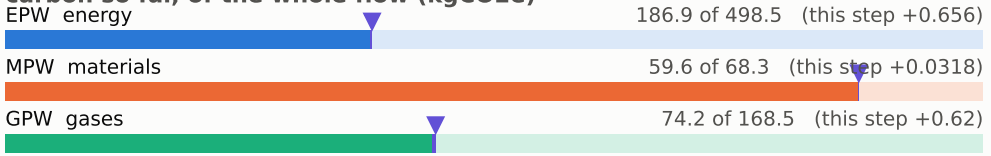
no mask for this step

this step

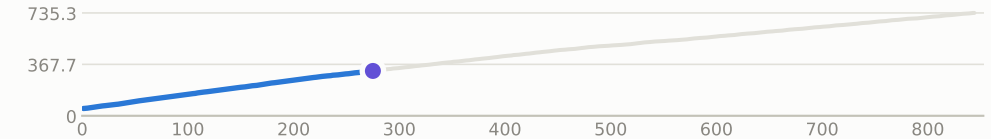
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

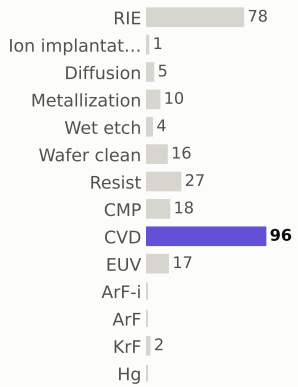
carbon so far, of the whole flow (kgCO2e)



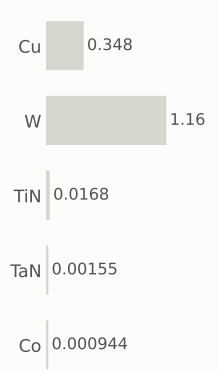
embodied carbon up to each step (kgCO2e per wafer)



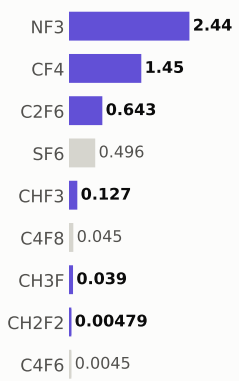
steps by process area



materials deposited (g)



gases emitted (g)



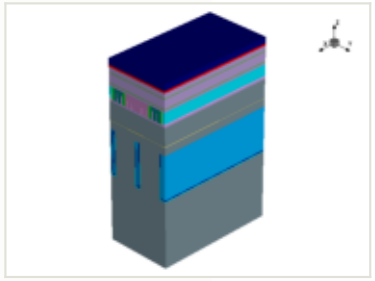
Step #2950: CVD

depo_fill · CVD

1.31	322.0	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

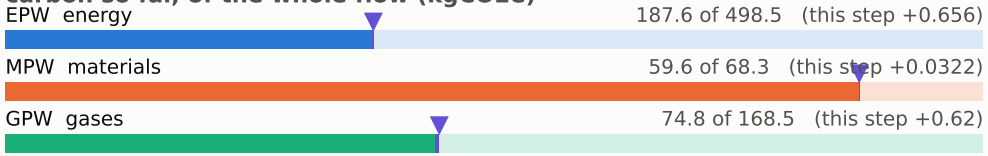
zoomed to fit



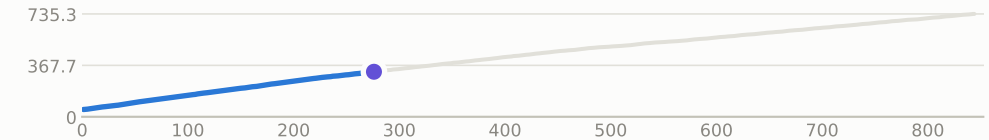
this step

process area	CVD	EPW	0.656 kgCO2e
material	TiN	EPW tool	0.469 kgCO2e
thickness	0.03 µm	EPW overhead	0.187 kgCO2e
material GWP	43.9778 gCO2e/g	MPW	0.0322 kgCO2e (gas 0.0318)
input gases	NF3 2.54 g	GPW	0.62 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**1.31 kgCO2e**

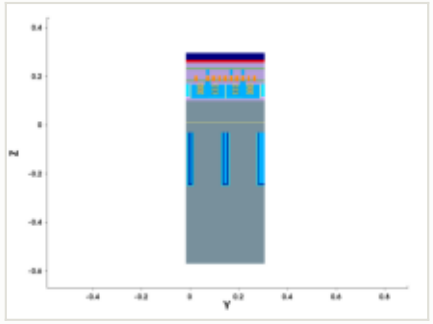
carbon so far, of the whole flow (kgCO2e)



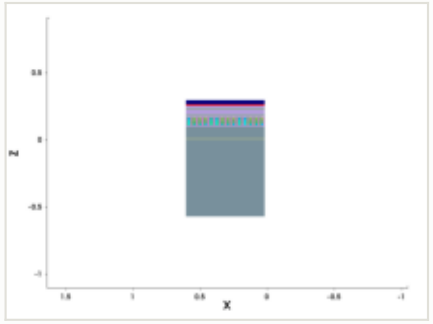
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



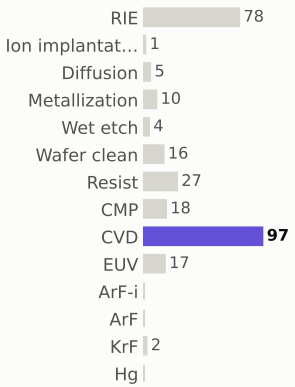
y cut at 0.184 µm



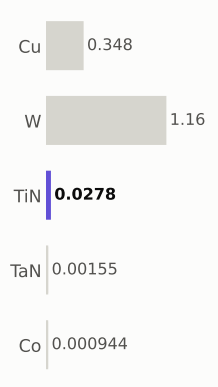
mask for this step

no mask for this step

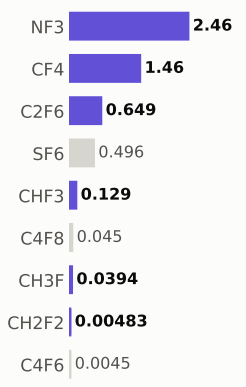
steps by process area



materials deposited (g)



gases emitted (g)



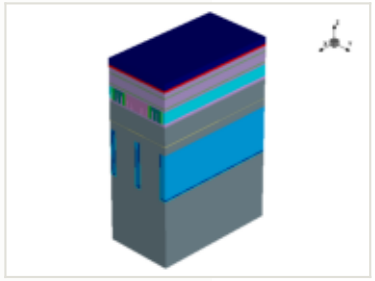
Step #2960: Wafer_clean

clean · Wafer_clean

0.169	322.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



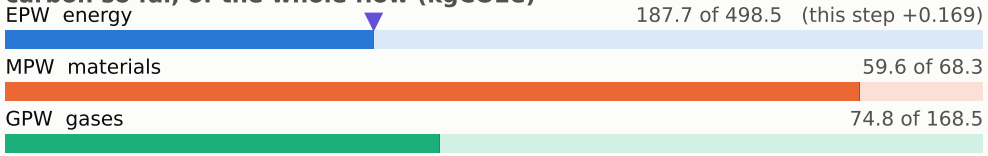
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

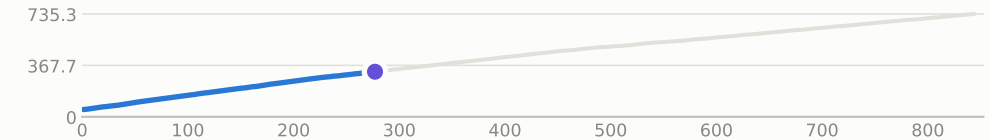
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

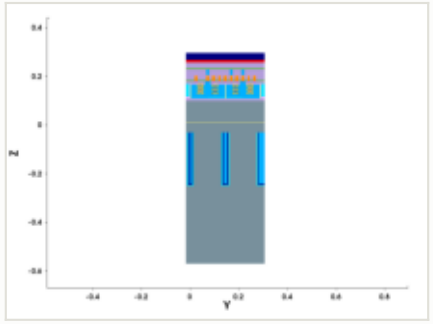
carbon so far, of the whole flow (kgCO2e)



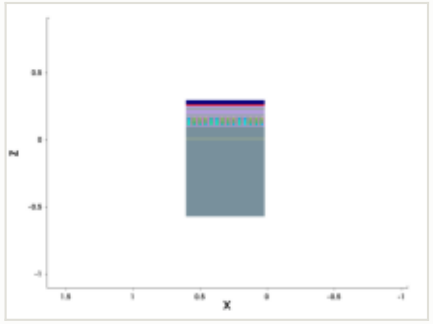
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



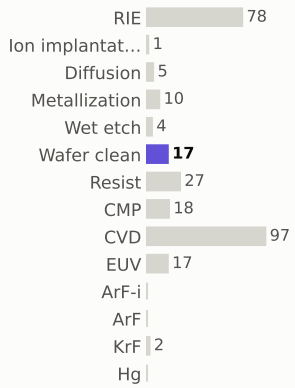
y cut at 0.184 µm



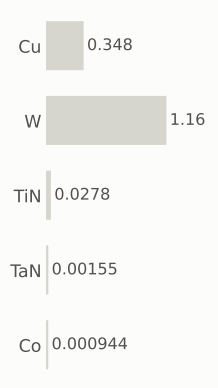
mask for this step

no mask for this step

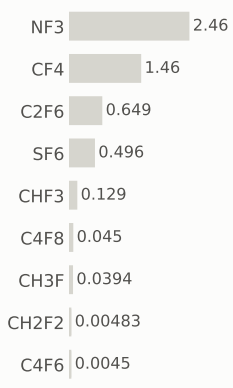
steps by process area



materials deposited (g)



gases emitted (g)



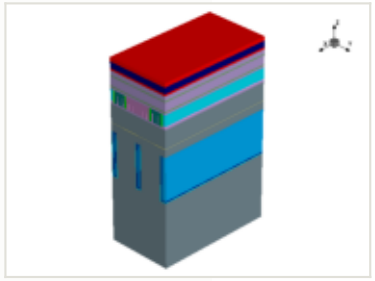
Step #2970: CVD

depo_fill · CVD

1.31	323.4	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

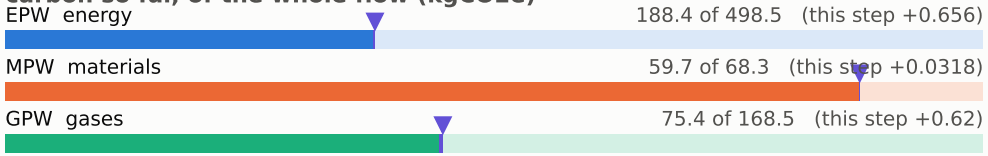


this step

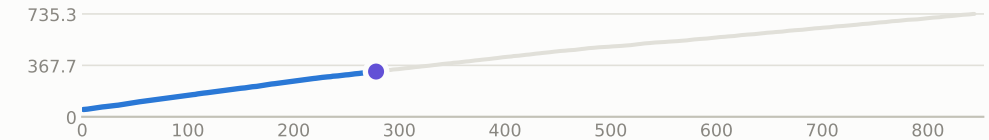
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

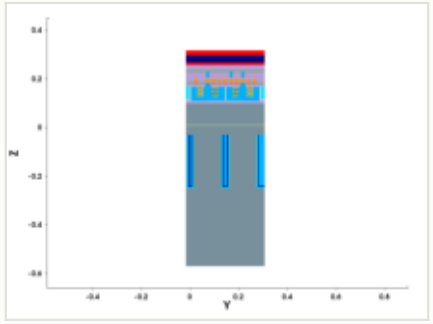
carbon so far, of the whole flow (kgCO2e)



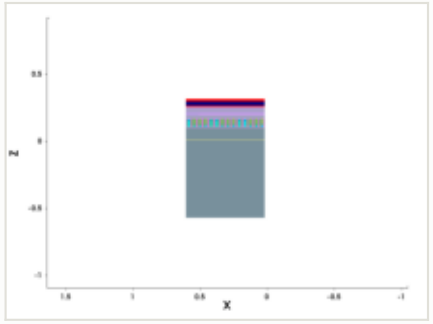
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



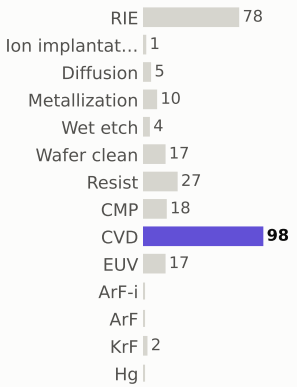
y cut at 0.184 µm



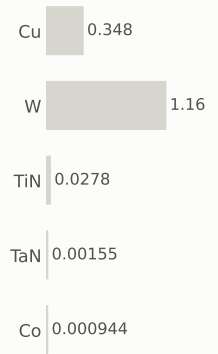
mask for this step

no mask for this step

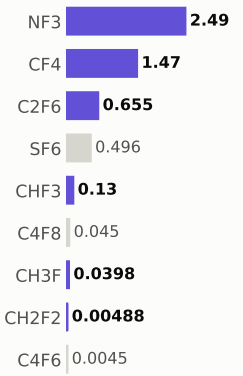
steps by process area



materials deposited (g)



gases emitted (g)



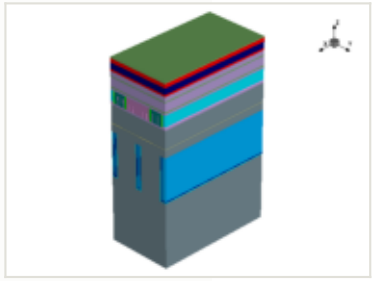
Step #2980: CVD

depo_fill · CVD

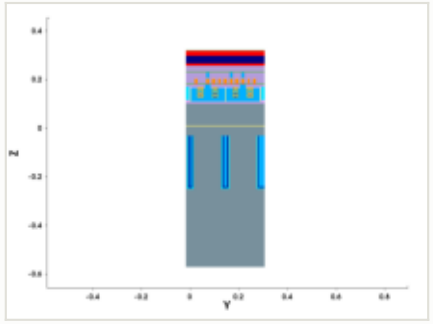
1.31	324.7	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

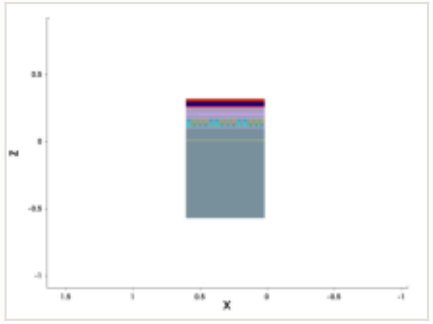
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

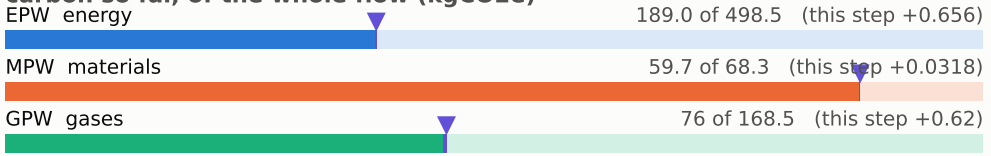
no mask for this step

this step

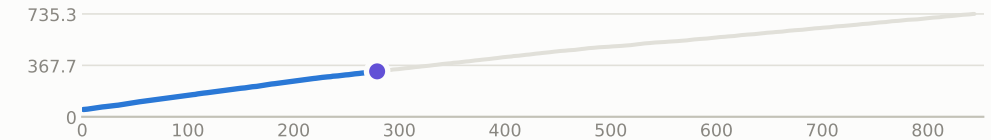
process area	CVD
material	—
thickness	0.005 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

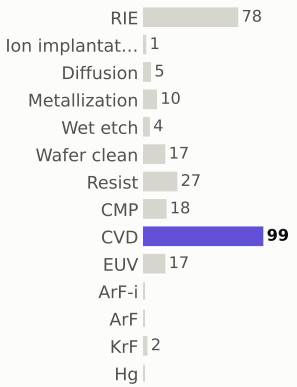
carbon so far, of the whole flow (kgCO2e)



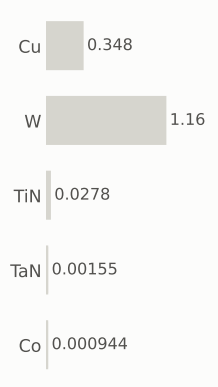
embodied carbon up to each step (kgCO2e per wafer)



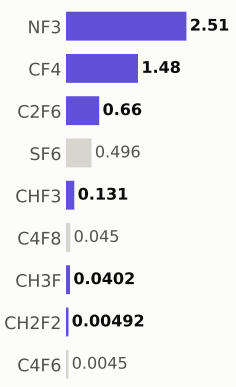
steps by process area



materials deposited (g)



gases emitted (g)



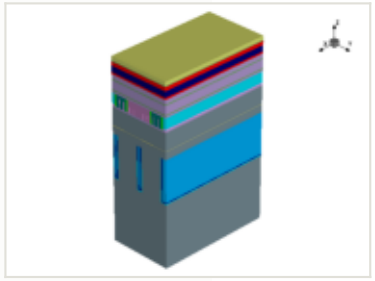
Step #2990: CVD

depo_fill · CVD

1.31	326.0	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

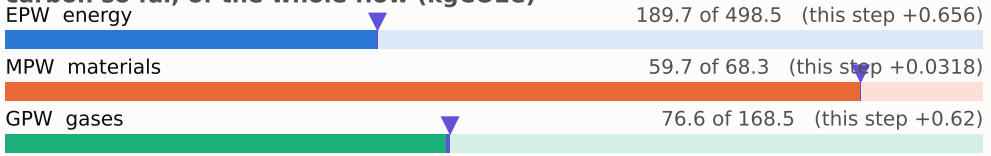


this step

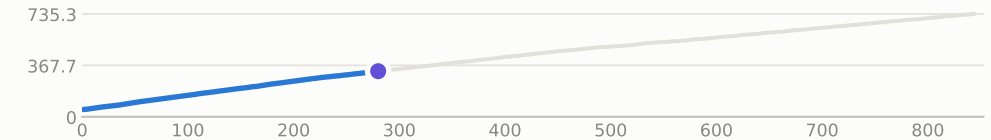
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

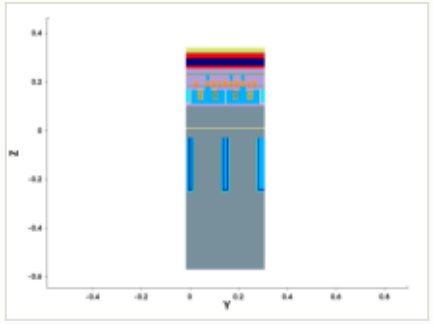
carbon so far, of the whole flow (kgCO2e)



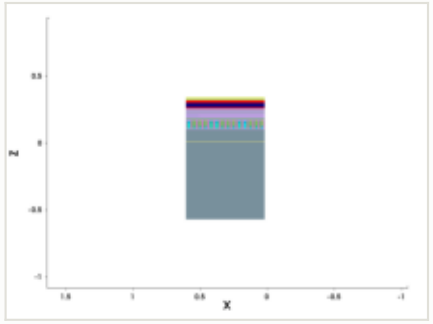
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



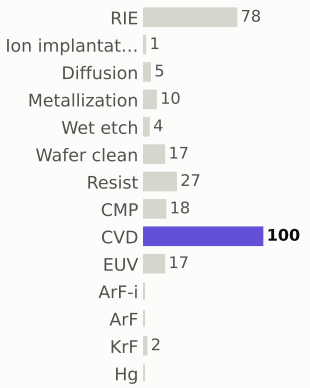
y cut at 0.184 µm



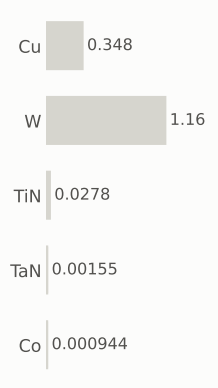
mask for this step

no mask for this step

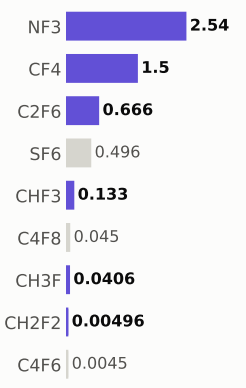
steps by process area



materials deposited (g)



gases emitted (g)



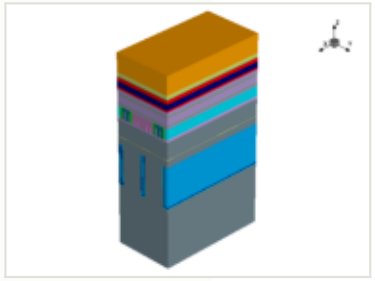
Step #3000: Resist

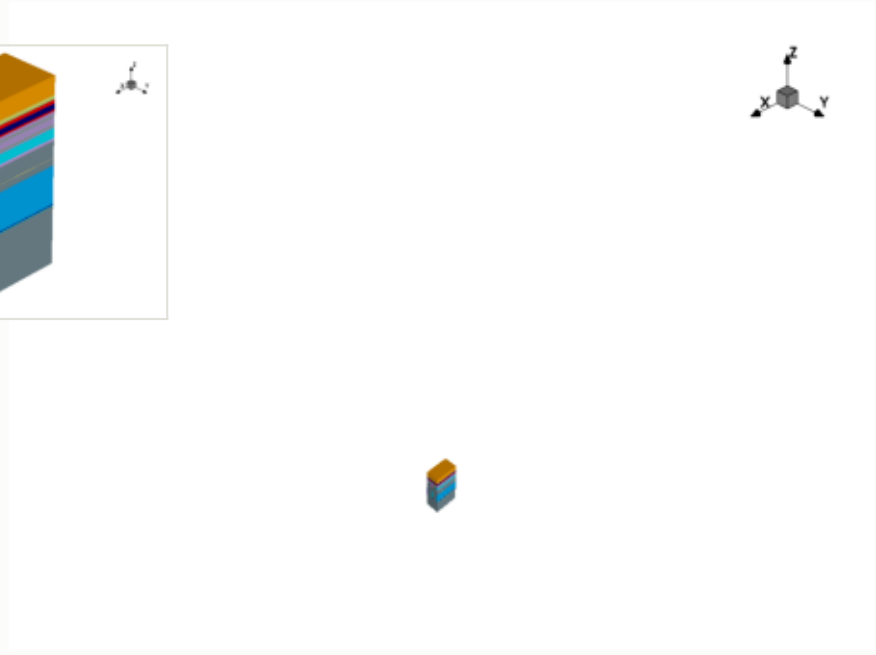
depo_fill · Resist

0.169	326.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



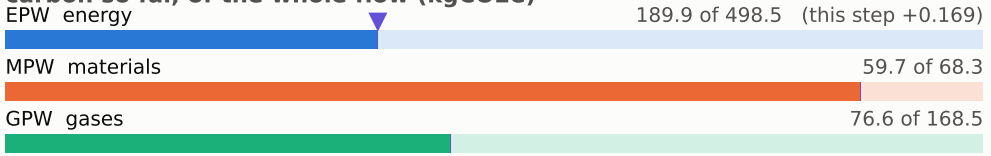


this step

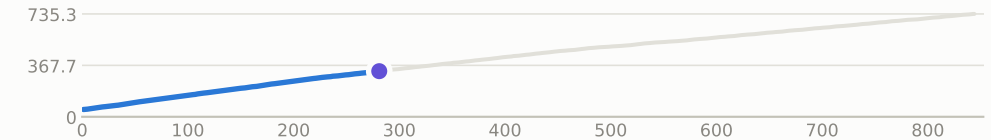
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

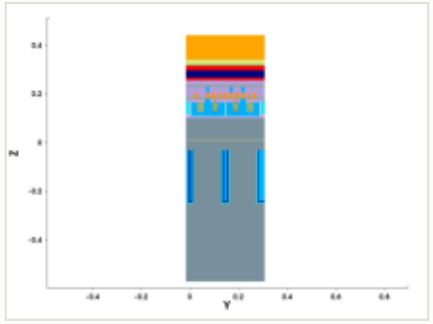
carbon so far, of the whole flow (kgCO2e)



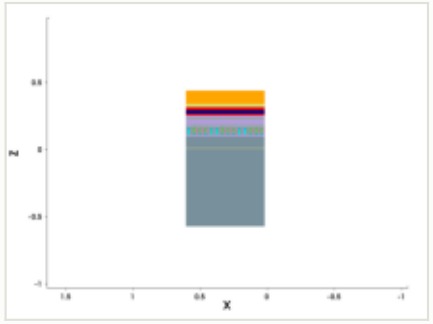
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



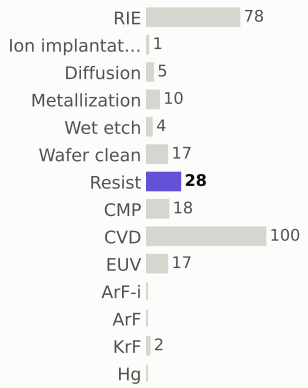
y cut at 0.184 µm



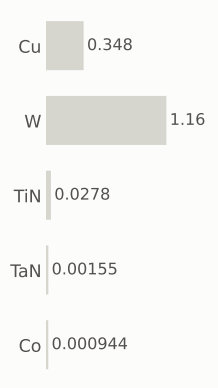
mask for this step

no mask for this step

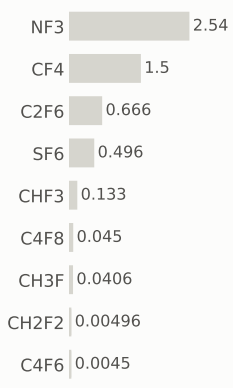
steps by process area



materials deposited (g)



gases emitted (g)



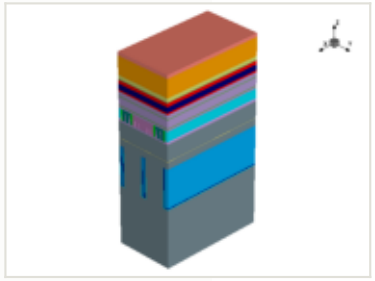
Step #3010: CVD

depo_fill · CVD

1.31	327.5	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

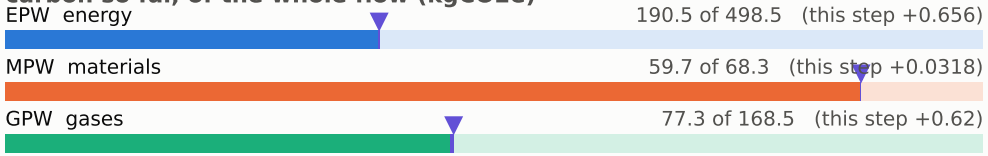


this step

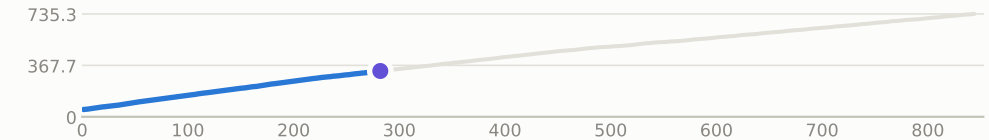
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

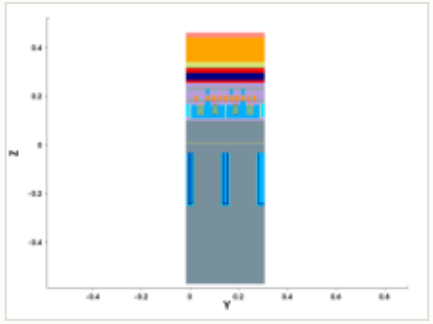
carbon so far, of the whole flow (kgCO2e)



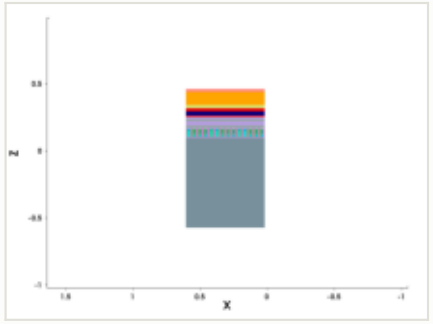
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



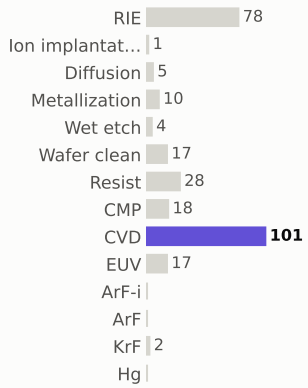
y cut at 0.184 µm



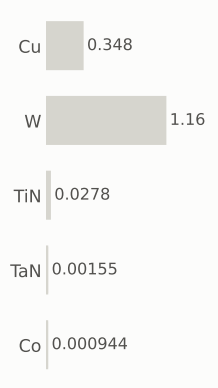
mask for this step

no mask for this step

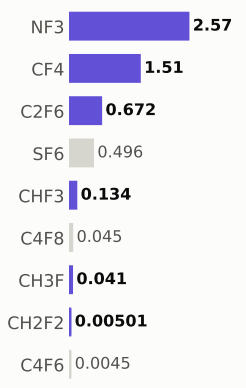
steps by process area



materials deposited (g)



gases emitted (g)



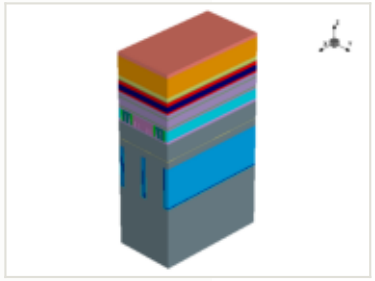
Step #3020: Wafer_clean

clean · Wafer_clean

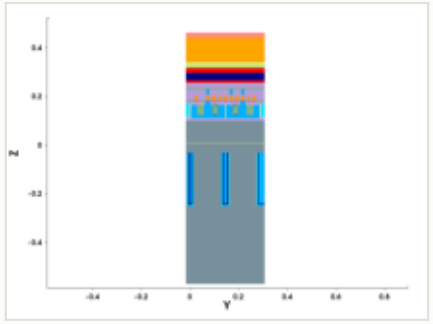
0.169	327.7	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

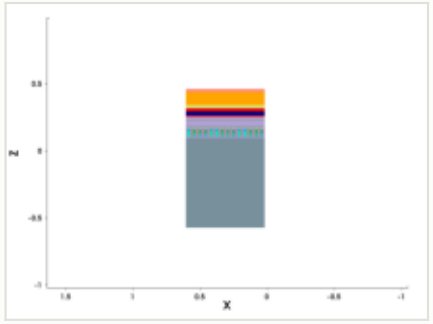
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

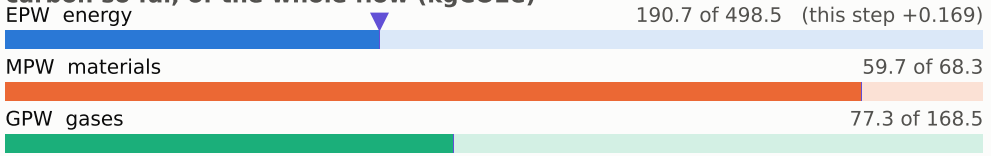
no mask for this step

this step

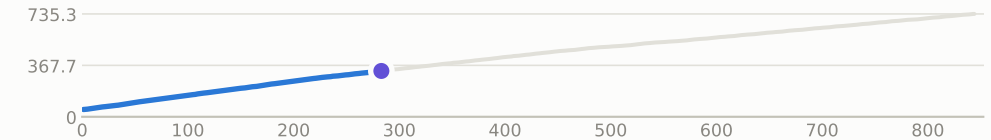
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

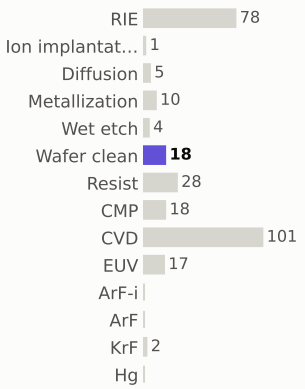
carbon so far, of the whole flow (kgCO2e)



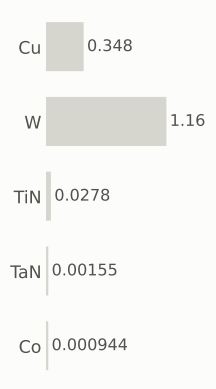
embodied carbon up to each step (kgCO2e per wafer)



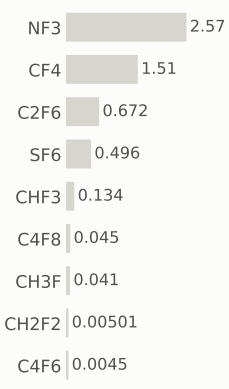
steps by process area



materials deposited (g)



gases emitted (g)



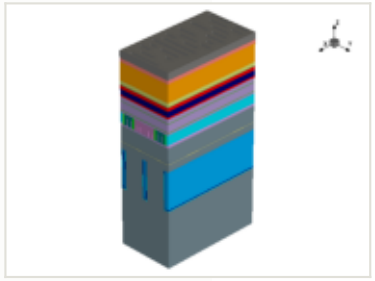
Step #3030: EUV

litho_patterning · EUV

1.96	329.7	0.27%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

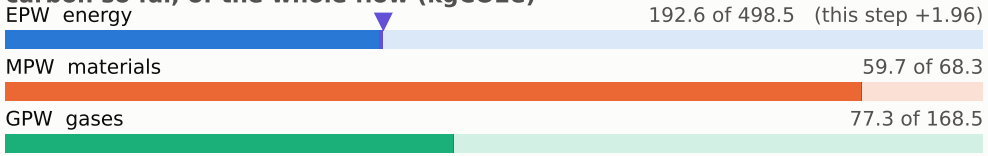


this step

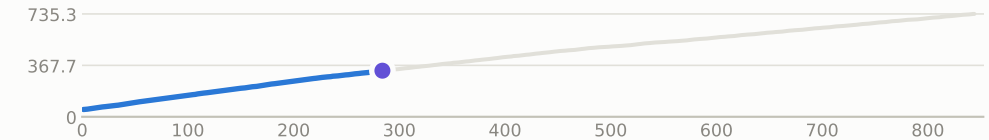
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

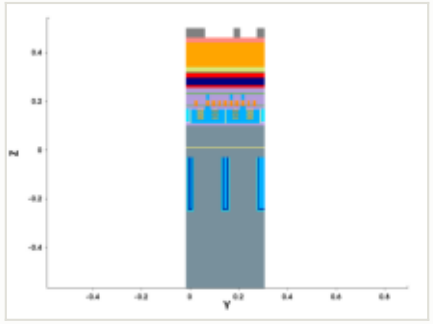
carbon so far, of the whole flow (kgCO2e)



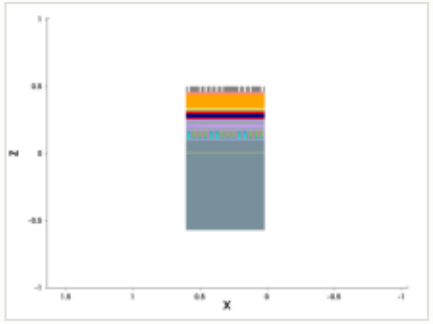
embodied carbon up to each step (kgCO2e per wafer)



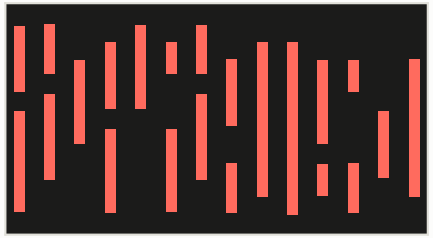
x cut at 0.294 µm



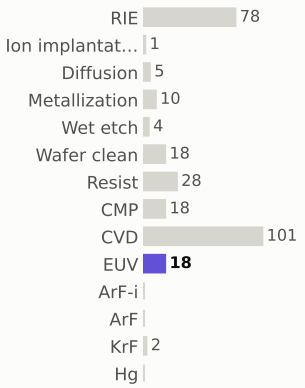
y cut at 0.184 µm



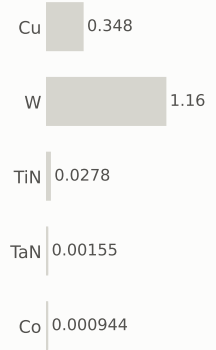
mask: layer 10251:0 (GDSII)



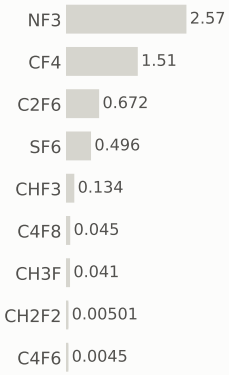
steps by process area



materials deposited (g)



gases emitted (g)



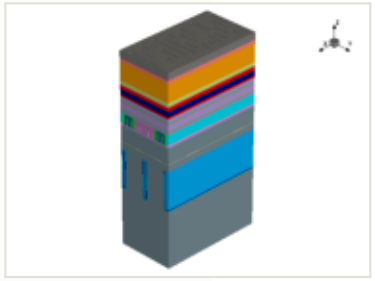
Step #3040: RIE

etch_anisotropic · RIE

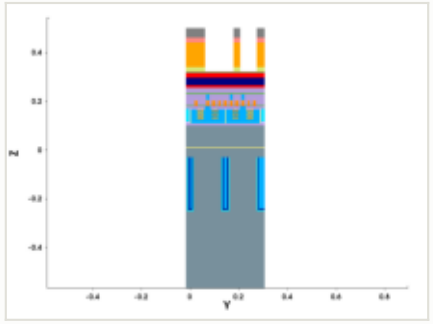
1.04	330.7	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

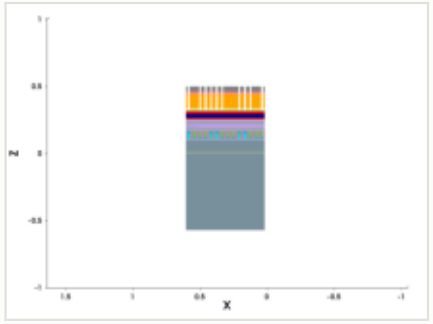
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

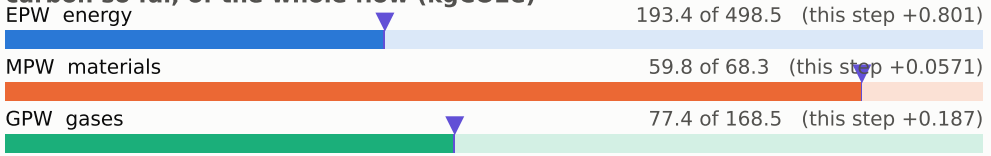
no mask for this step

this step

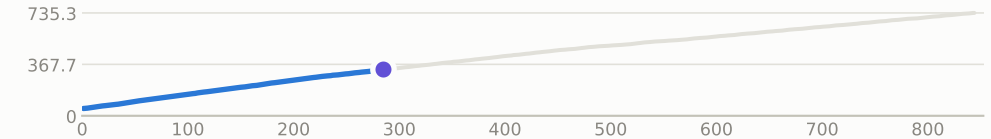
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

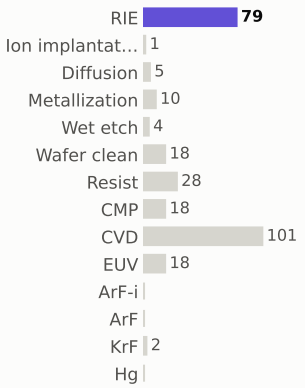
carbon so far, of the whole flow (kgCO2e)



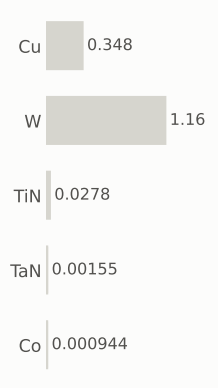
embodied carbon up to each step (kgCO2e per wafer)



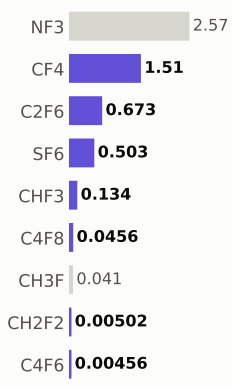
steps by process area



materials deposited (g)



gases emitted (g)



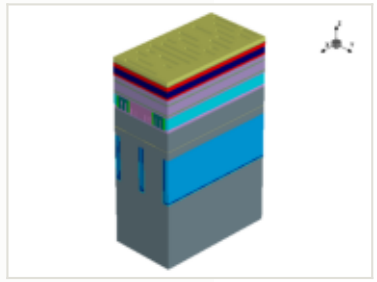
Step #3050: Wafer_clean

etch_strip · Wafer_clean

0.169	330.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

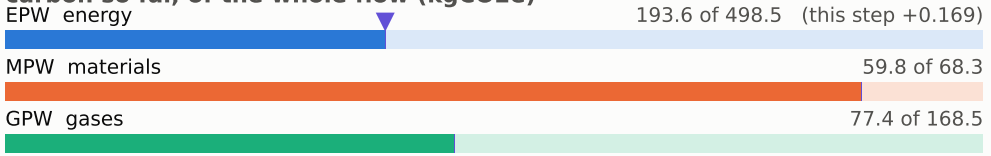


this step

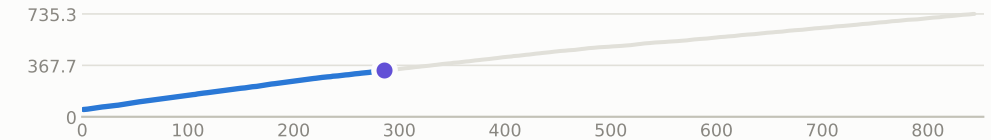
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

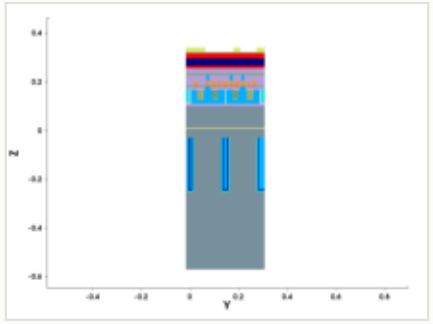
carbon so far, of the whole flow (kgCO2e)



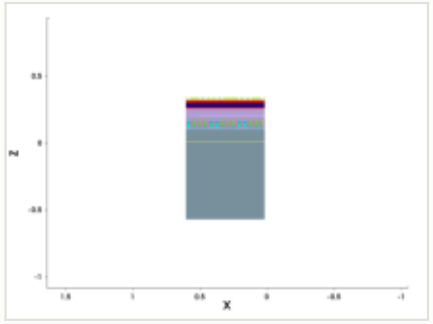
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



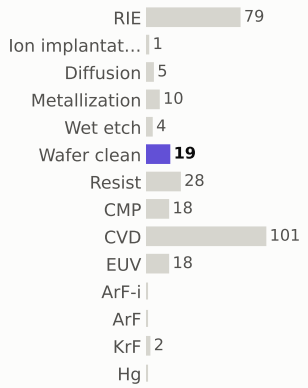
y cut at 0.184 µm



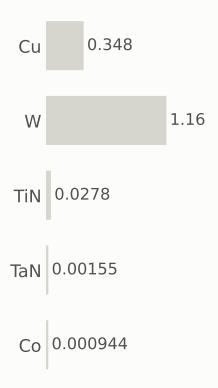
mask for this step

no mask for this step

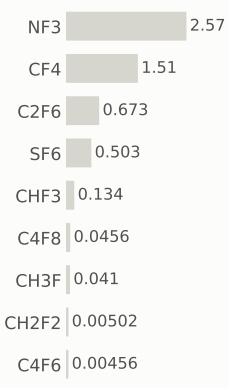
steps by process area



materials deposited (g)



gases emitted (g)



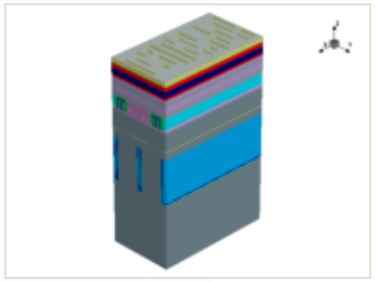
Step #3060: CVD

depo_isotropic · CVD

1.31	332.2	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

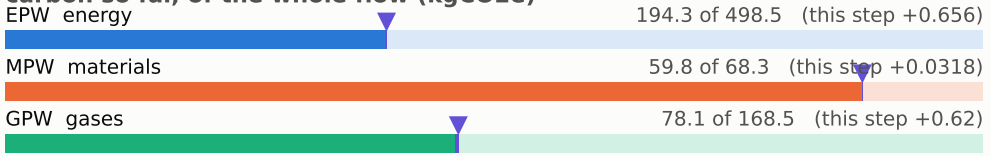


this step

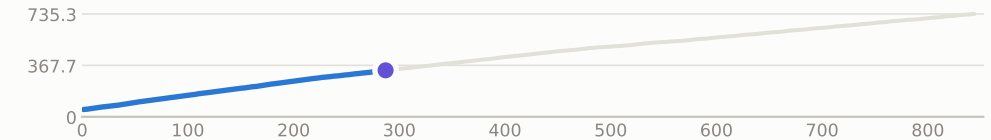
process area	CVD
material	—
thickness	0.001 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

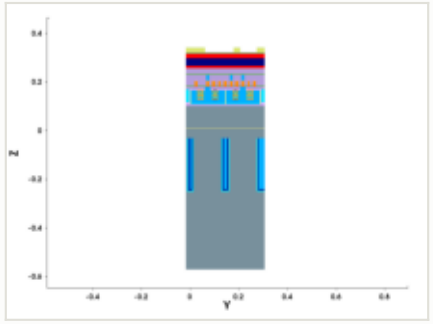
carbon so far, of the whole flow (kgCO2e)



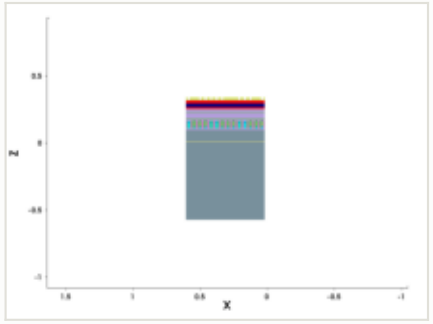
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



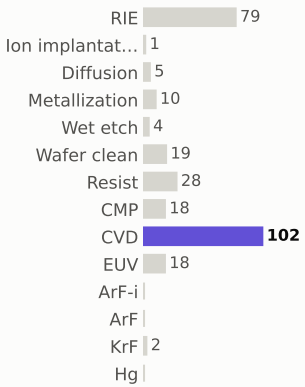
y cut at 0.184 µm



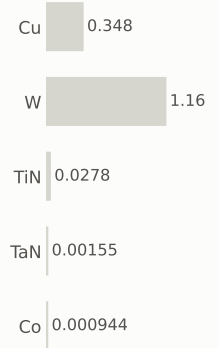
mask for this step

no mask for this step

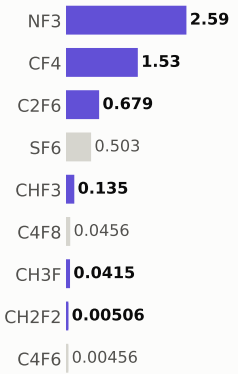
steps by process area



materials deposited (g)



gases emitted (g)



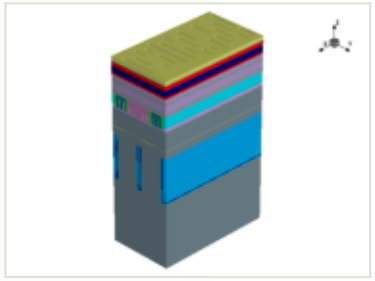
Step #3070: RIE

etch_anisotropic · RIE

1.04	333.2	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

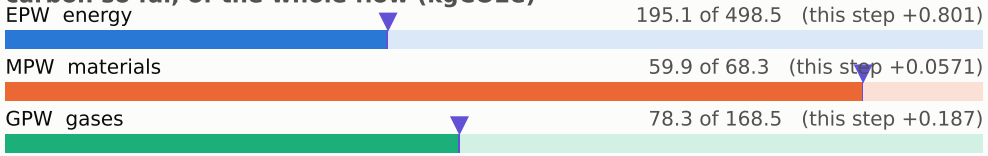


this step

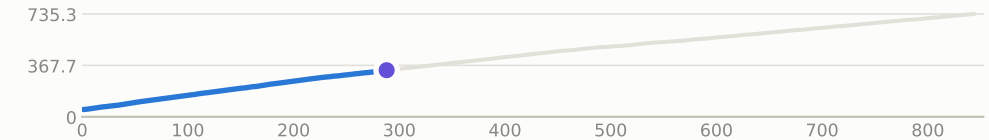
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

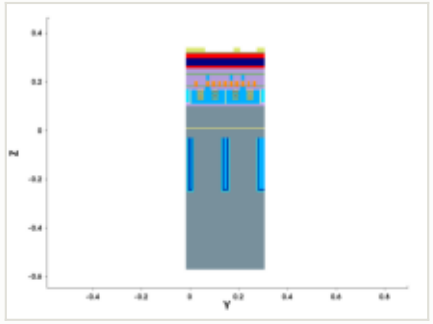
carbon so far, of the whole flow (kgCO2e)



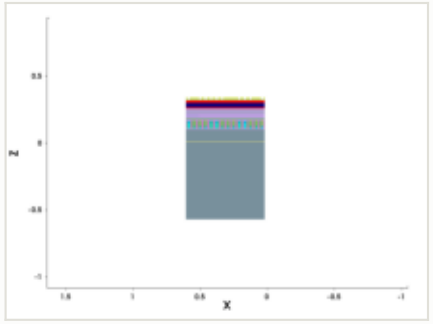
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



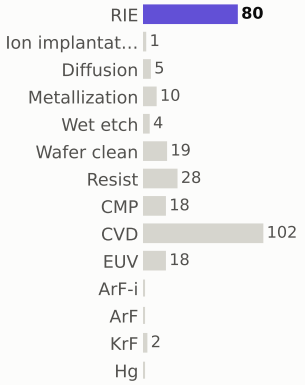
y cut at 0.184 µm



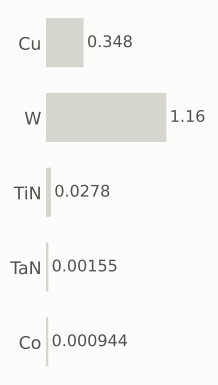
mask for this step

no mask for this step

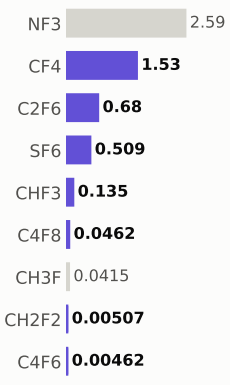
steps by process area



materials deposited (g)



gases emitted (g)



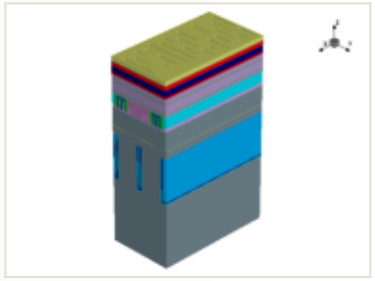
Step #3080: Wafer_clean

clean · Wafer_clean

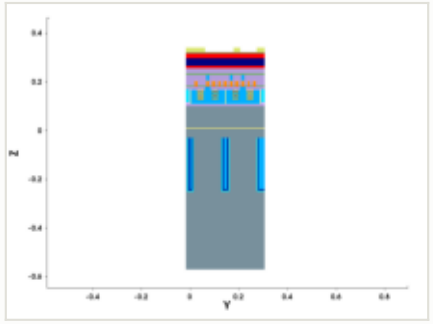
0.169	333.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

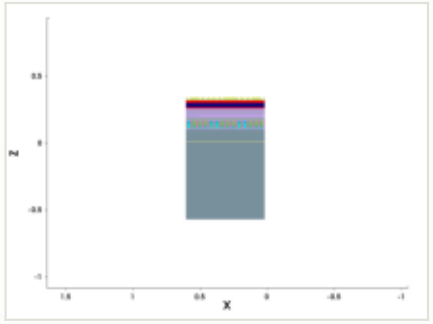
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

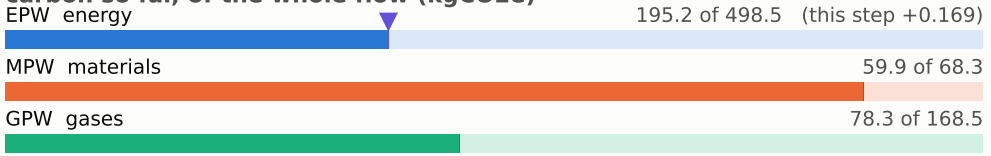
no mask for this step

this step

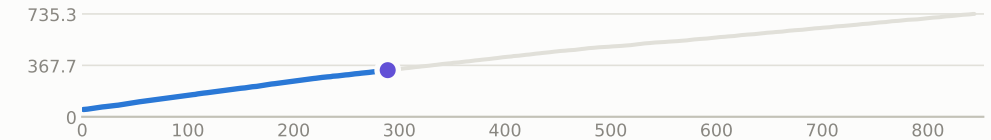
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

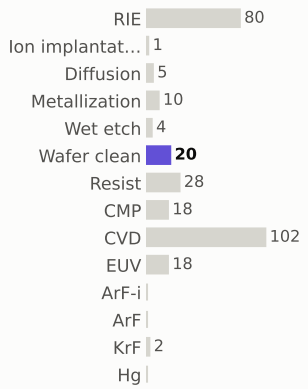
carbon so far, of the whole flow (kgCO2e)



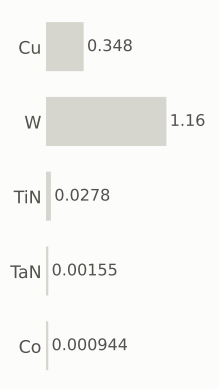
embodied carbon up to each step (kgCO2e per wafer)



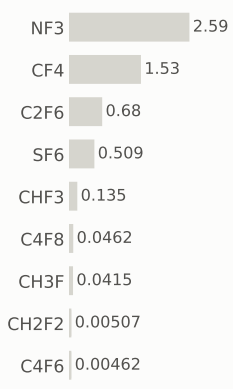
steps by process area



materials deposited (g)



gases emitted (g)



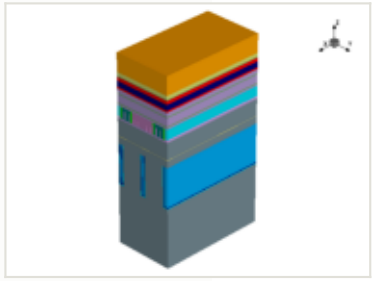
Step #3090: Resist

depo_fill · Resist

0.169	333.6	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

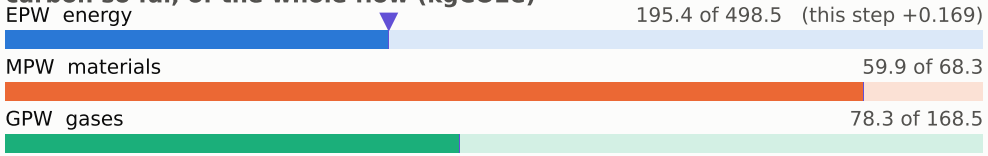


this step

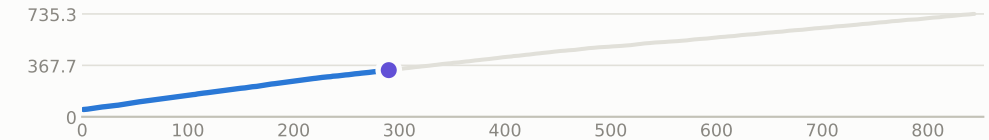
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

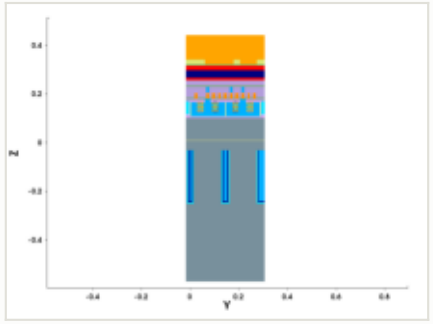
carbon so far, of the whole flow (kgCO2e)



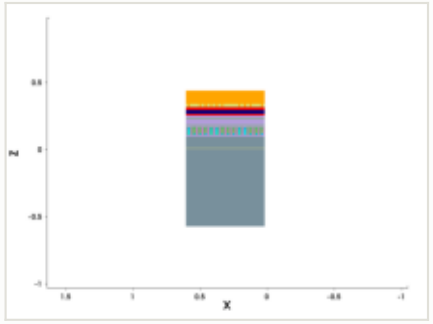
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



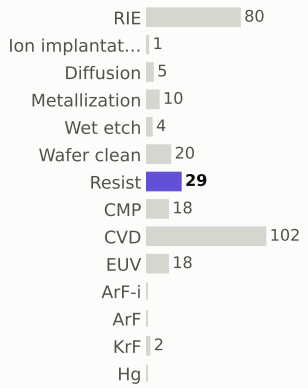
y cut at 0.184 µm



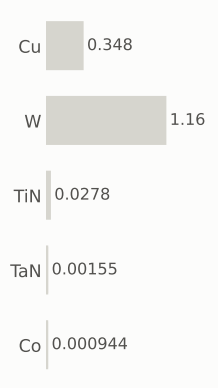
mask for this step

no mask for this step

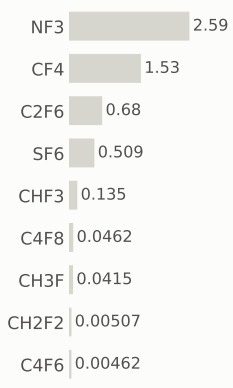
steps by process area



materials deposited (g)



gases emitted (g)



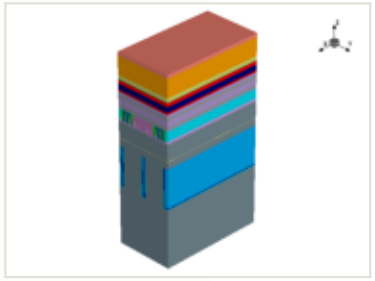
Step #3100: CVD

depo_fill · CVD

1.31	334.9	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

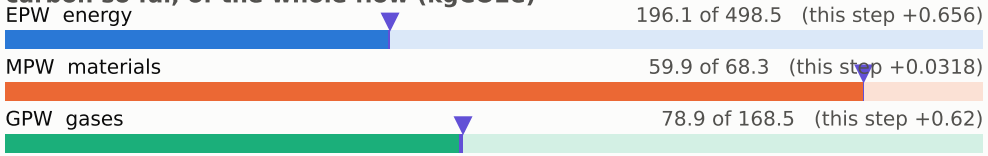


this step

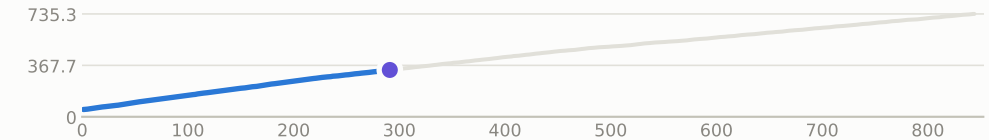
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

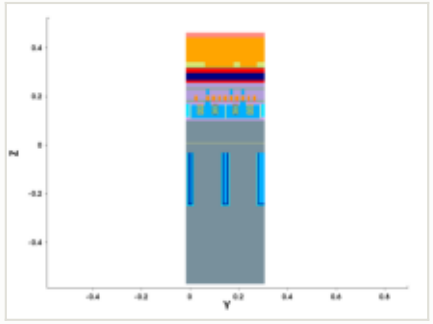
carbon so far, of the whole flow (kgCO2e)



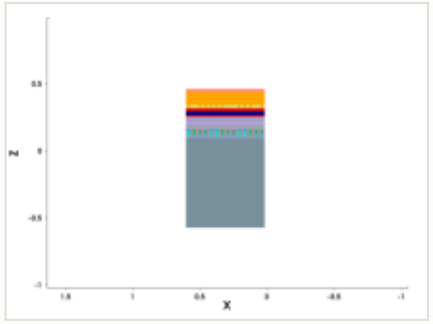
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



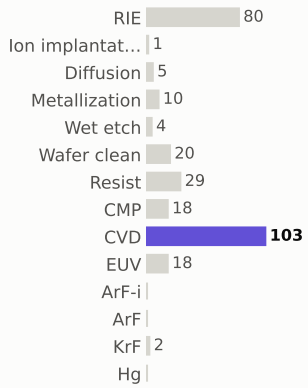
y cut at 0.184 µm



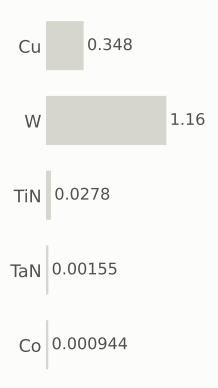
mask for this step

no mask for this step

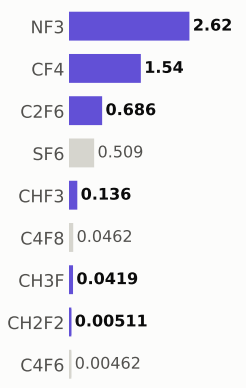
steps by process area



materials deposited (g)



gases emitted (g)



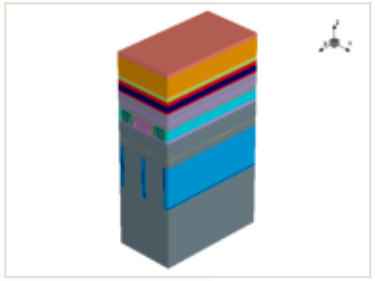
Step #3110: Wafer_clean

clean · Wafer_clean

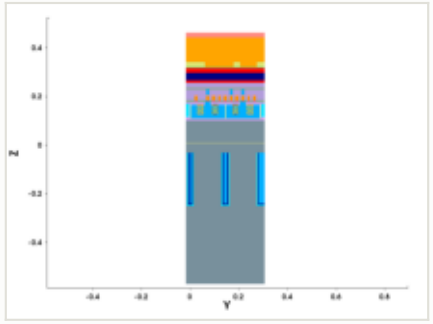
0.169	335.0	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

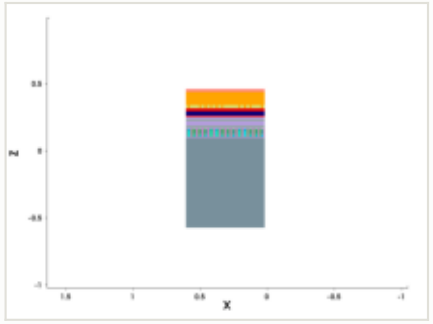
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

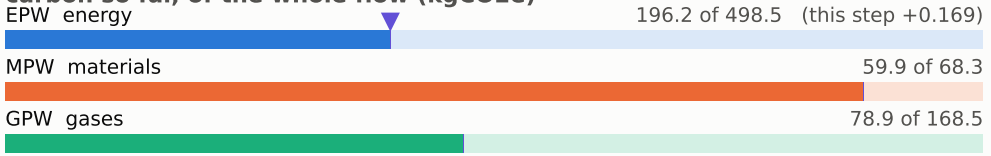
no mask for this step

this step

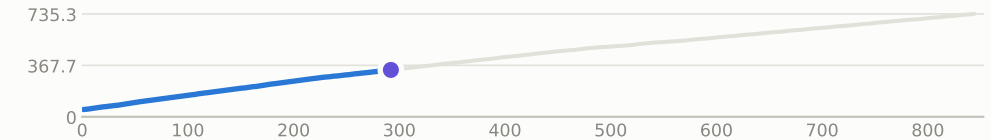
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

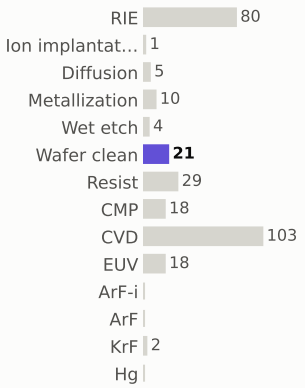
carbon so far, of the whole flow (kgCO2e)



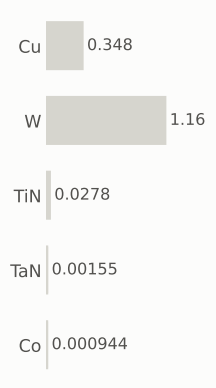
embodied carbon up to each step (kgCO2e per wafer)



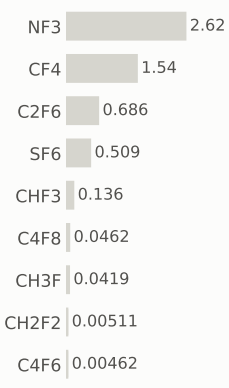
steps by process area



materials deposited (g)



gases emitted (g)



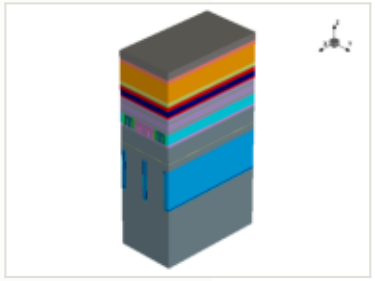
Step #3120: EUV

litho_patterning · EUV

1.96	337.0	0.27%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

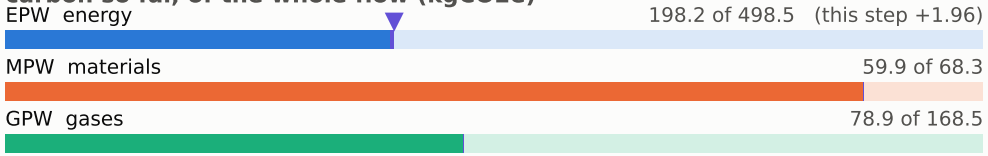


this step

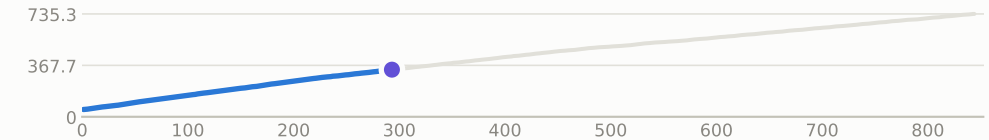
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

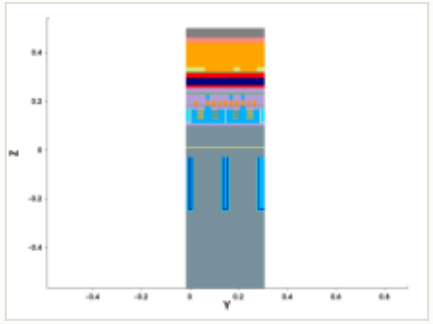
carbon so far, of the whole flow (kgCO2e)



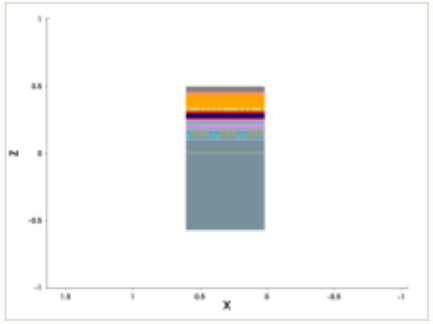
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



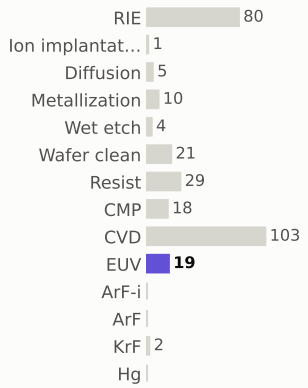
y cut at 0.184 µm



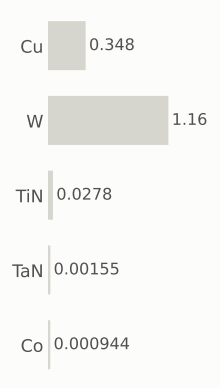
mask: layer 10252:0 (GDSII)

nothing in the clip window

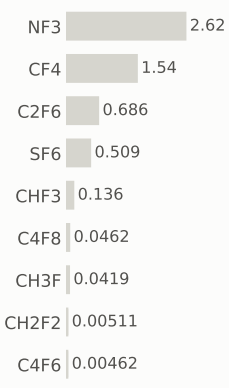
steps by process area



materials deposited (g)



gases emitted (g)



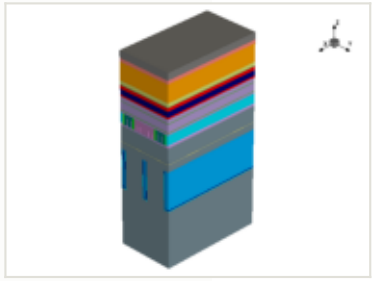
Step #3130: RIE

etch_anisotropic · RIE

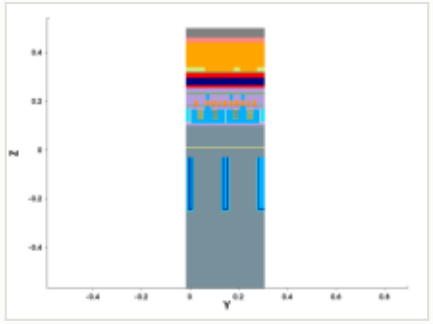
1.04	338.0	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

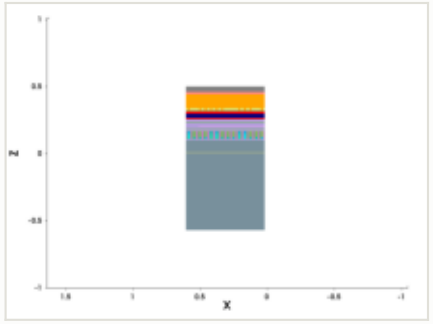
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

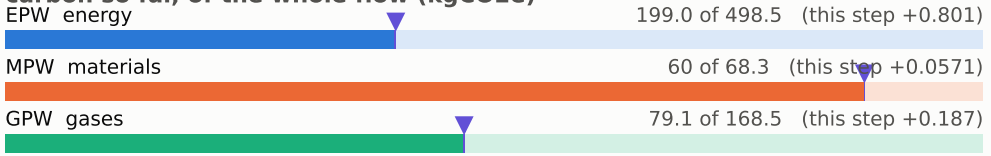
no mask for this step

this step

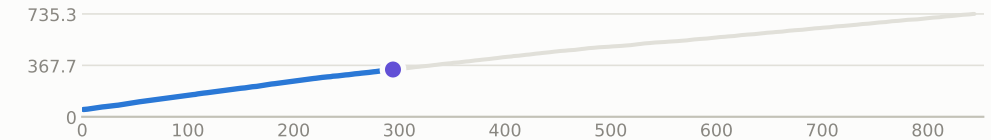
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

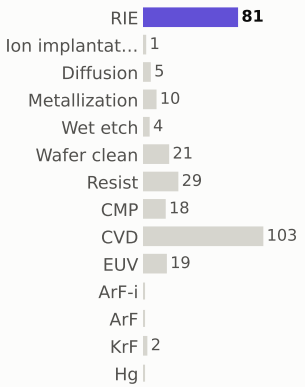
carbon so far, of the whole flow (kgCO2e)



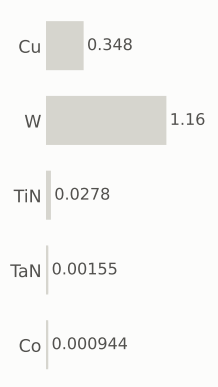
embodied carbon up to each step (kgCO2e per wafer)



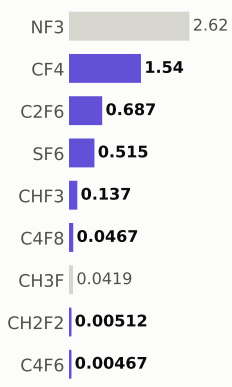
steps by process area



materials deposited (g)



gases emitted (g)



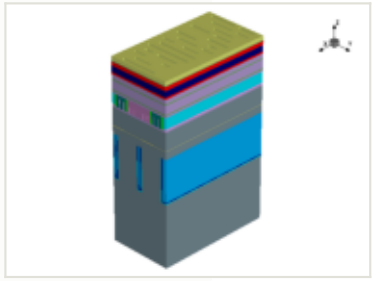
Step #3140: Wafer_clean

etch_strip · Wafer_clean

0.169	338.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

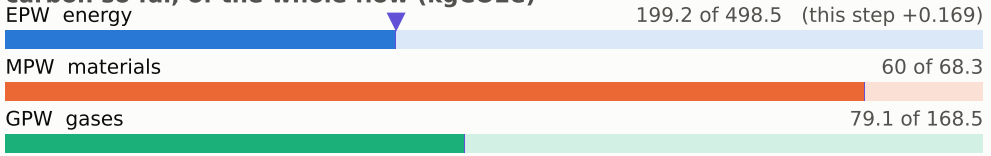


this step

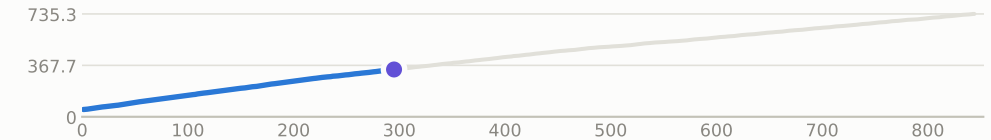
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

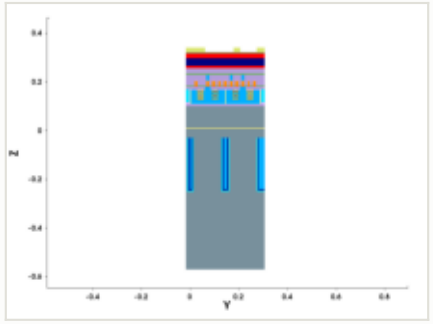
carbon so far, of the whole flow (kgCO2e)



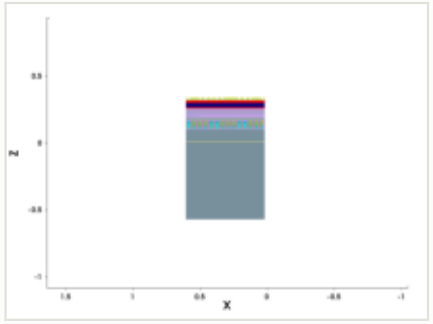
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



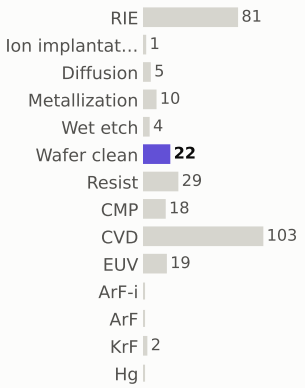
y cut at 0.184 µm



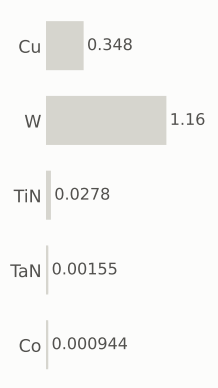
mask for this step

no mask for this step

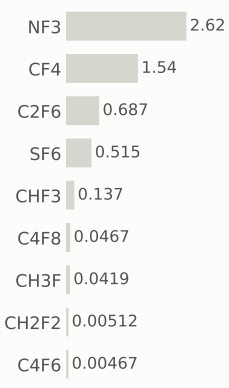
steps by process area



materials deposited (g)



gases emitted (g)



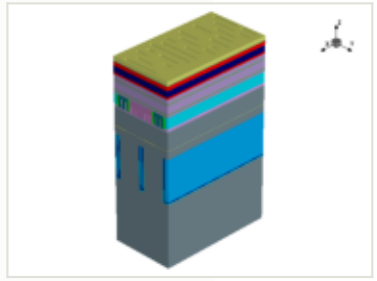
Step #3150: RIE

etch_anisotropic · RIE

1.04	339.2	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

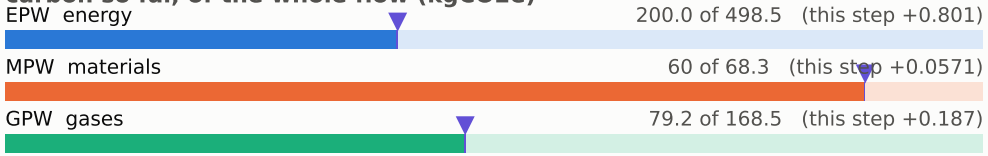


this step

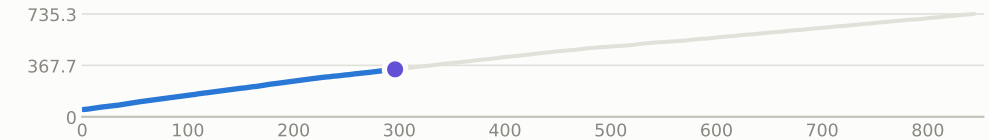
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

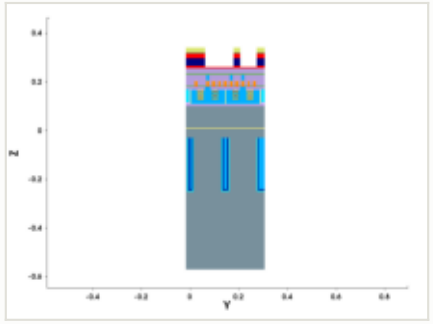
carbon so far, of the whole flow (kgCO2e)



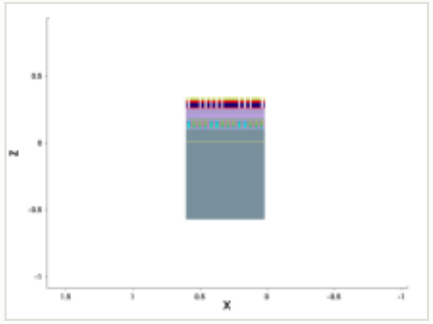
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



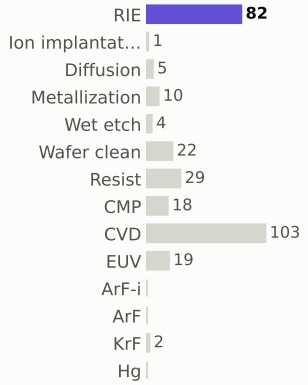
y cut at 0.184 µm



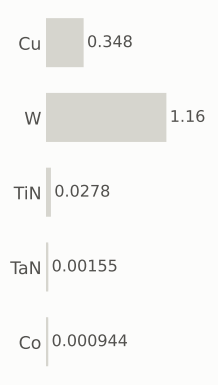
mask for this step

no mask for this step

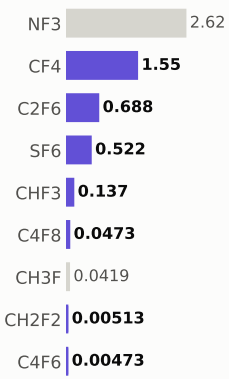
steps by process area



materials deposited (g)



gases emitted (g)



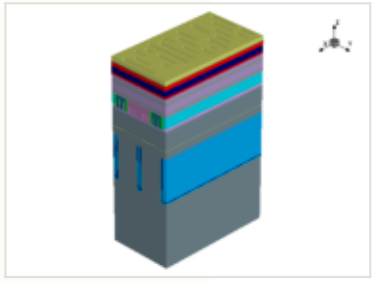
Step #3160: Wafer_clean

clean · Wafer_clean

0.169	339.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



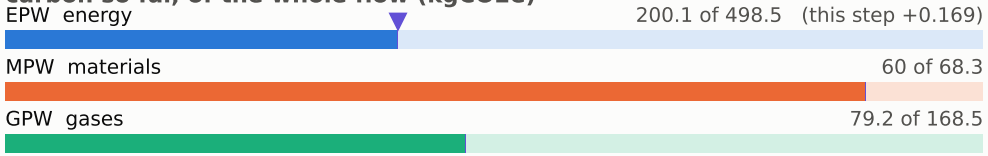
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

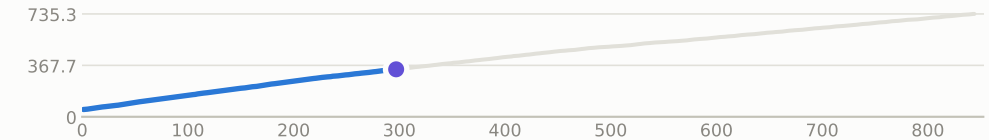
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

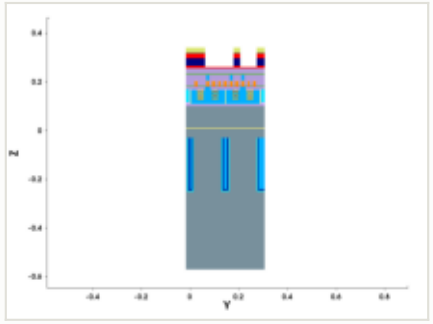
carbon so far, of the whole flow (kgCO2e)



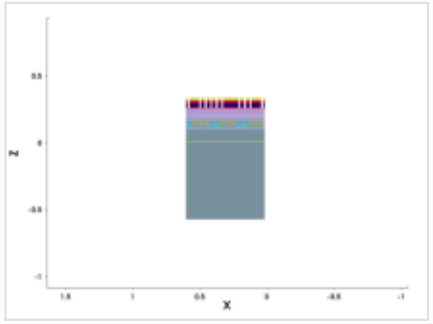
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



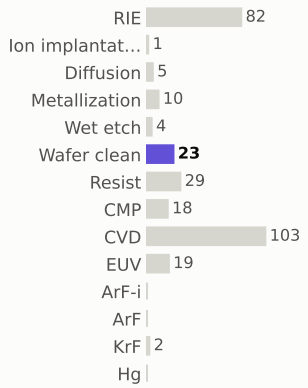
y cut at 0.184 µm



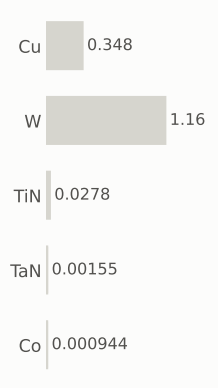
mask for this step

no mask for this step

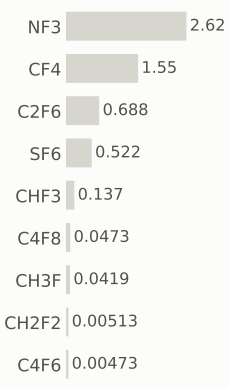
steps by process area



materials deposited (g)



gases emitted (g)



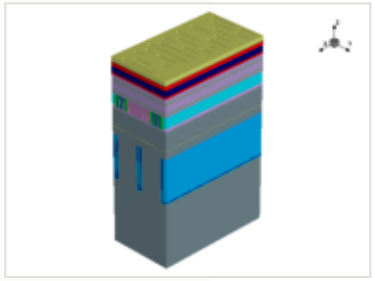
Step #3170: RIE

etch_anisotropic · RIE

1.04	340.5	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

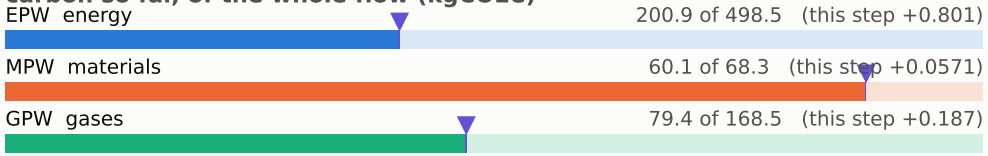


this step

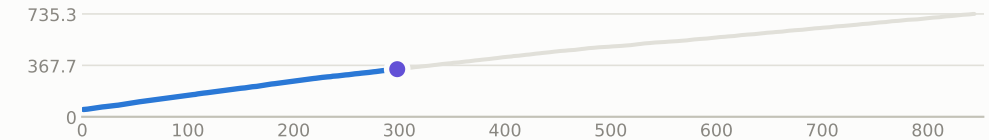
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

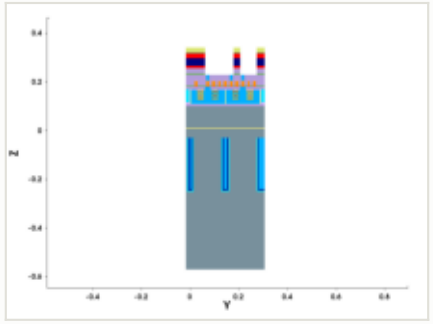
carbon so far, of the whole flow (kgCO2e)



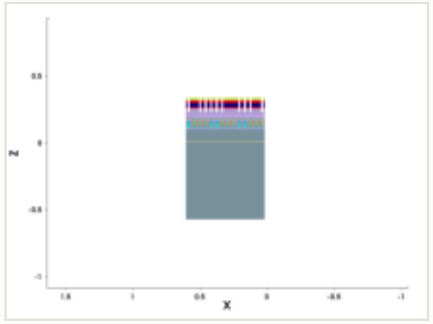
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



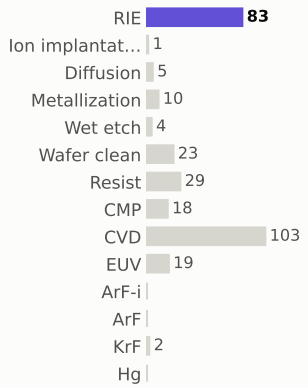
y cut at 0.184 µm



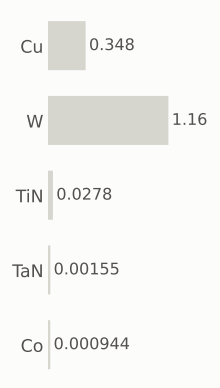
mask for this step

no mask for this step

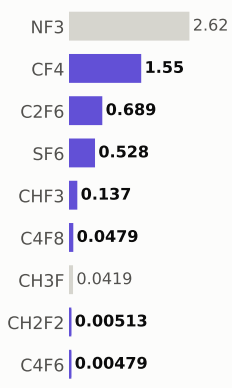
steps by process area



materials deposited (g)



gases emitted (g)



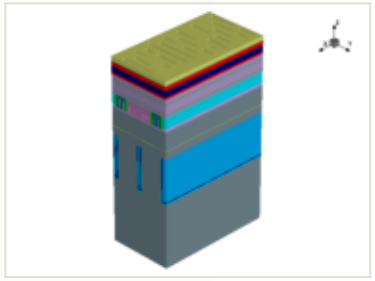
Step #3180: Wafer_clean

clean · Wafer_clean

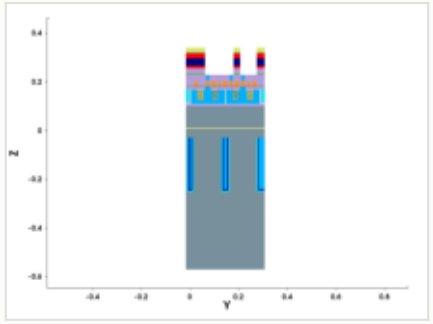
0.169	340.6	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

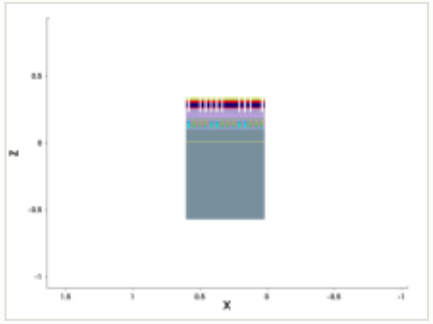
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

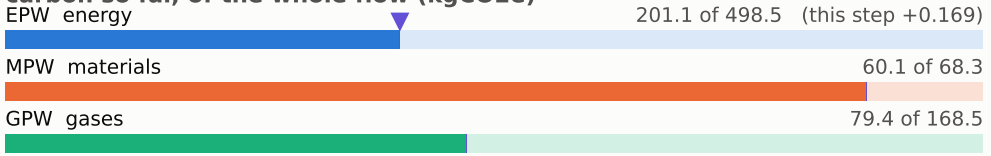
no mask for this step

this step

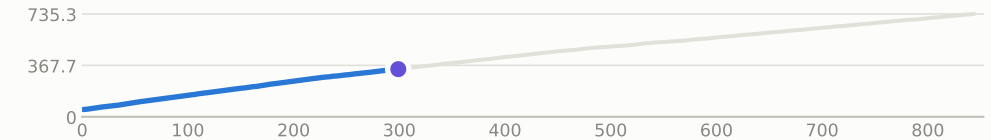
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

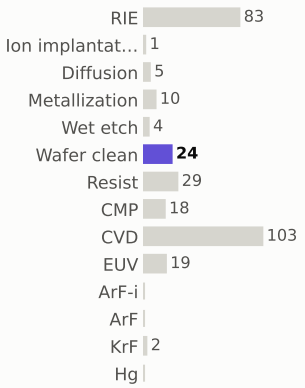
carbon so far, of the whole flow (kgCO2e)



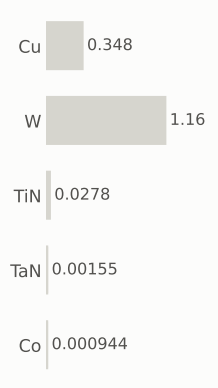
embodied carbon up to each step (kgCO2e per wafer)



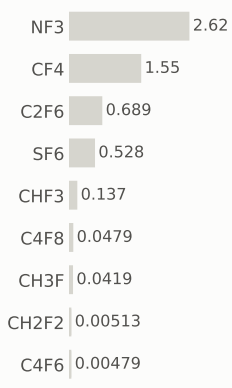
steps by process area



materials deposited (g)



gases emitted (g)



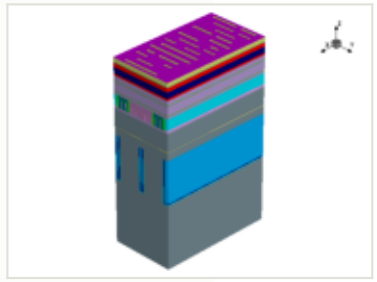
Step #3190: Metallization

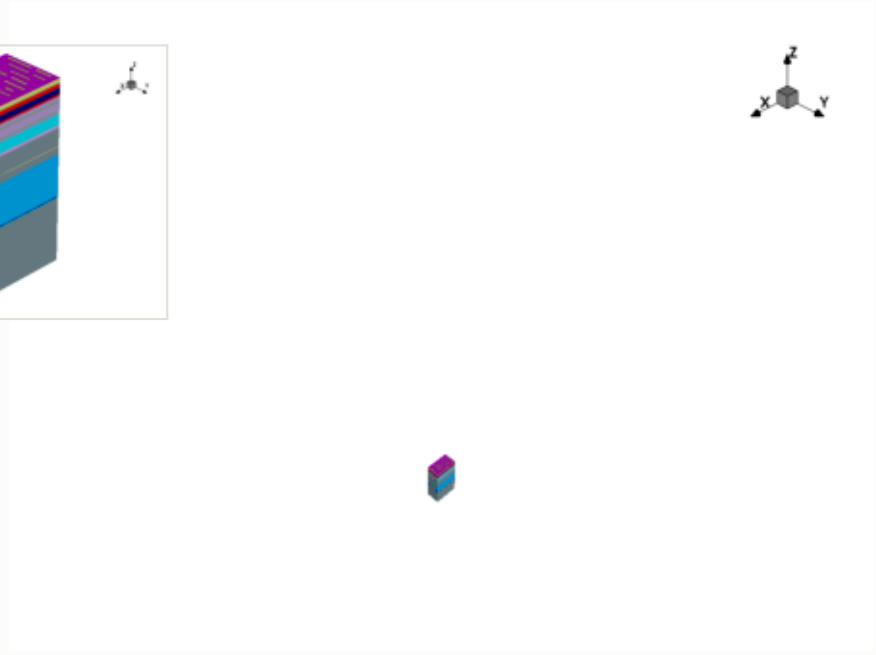
depo_ellipsoidal · Metallization

1.2	341.8	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



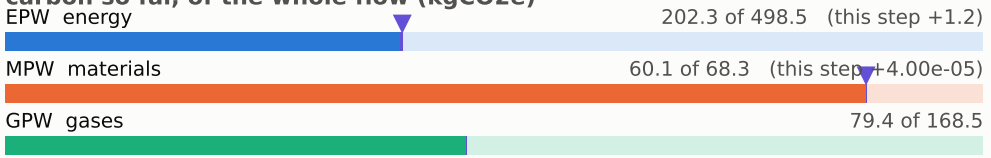


this step

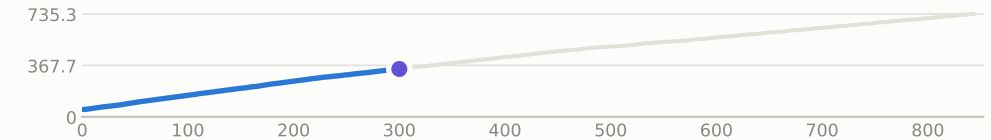
process area	Metallization
material	TaN
thickness	0.0016 µm
material GWP	25.8159 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.00e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

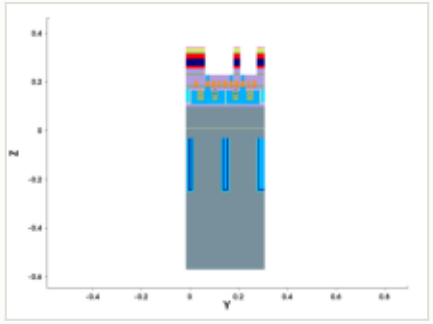
carbon so far, of the whole flow (kgCO2e)



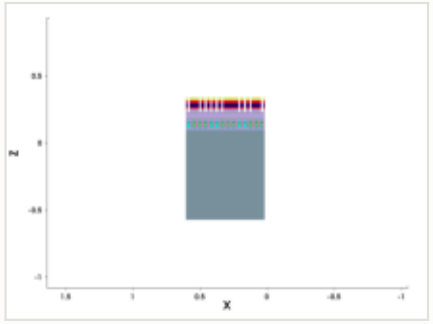
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



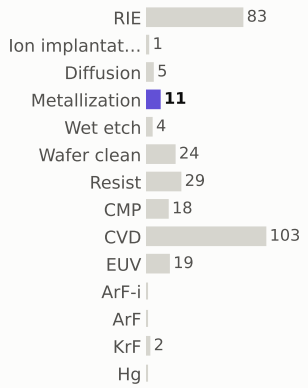
y cut at 0.184 µm



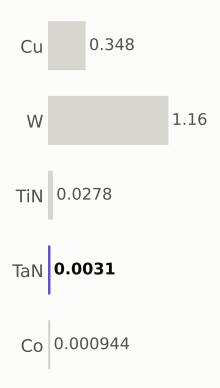
mask for this step

no mask for this step

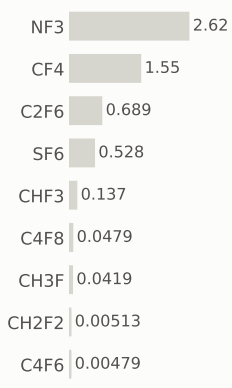
steps by process area



materials deposited (g)



gases emitted (g)



Step #3200: Metallization

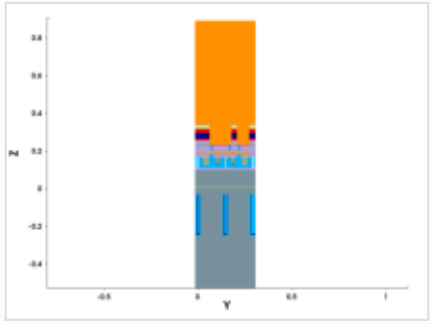
depo_fill · Metallization

1.21	343.0	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

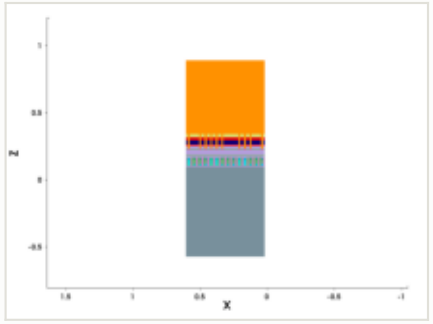
3D structure after this step

zoomed to fit

x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

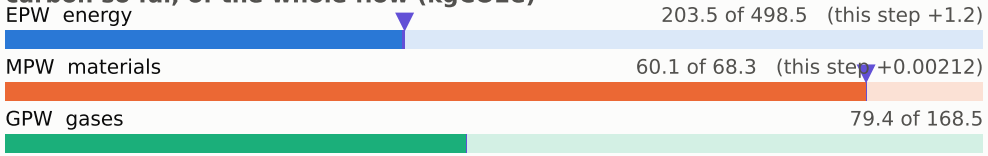
no mask for this step

this step

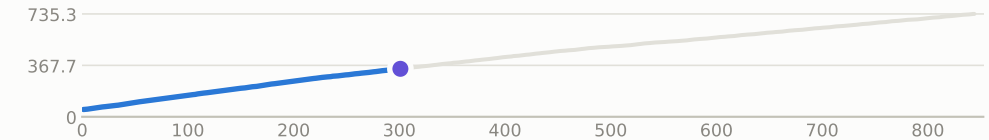
process area	Metallization
material	Cu
thickness	0.55 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00212 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

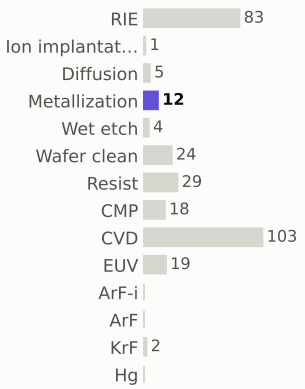
carbon so far, of the whole flow (kgCO2e)



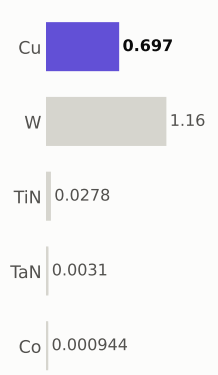
embodied carbon up to each step (kgCO2e per wafer)



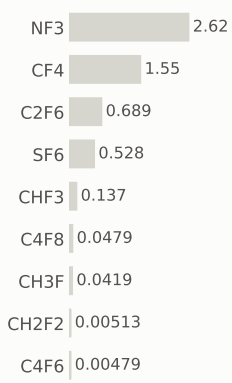
steps by process area



materials deposited (g)



gases emitted (g)



Step #3210: Diffusion

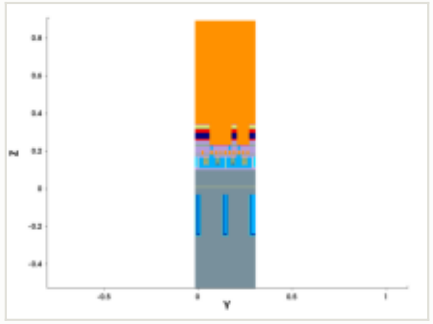
treatment · Diffusion

0.801	343.8	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

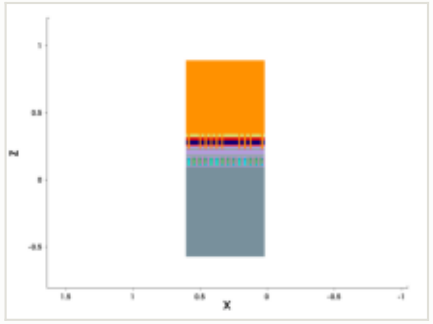
3D structure after this step

zoomed to fit

x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

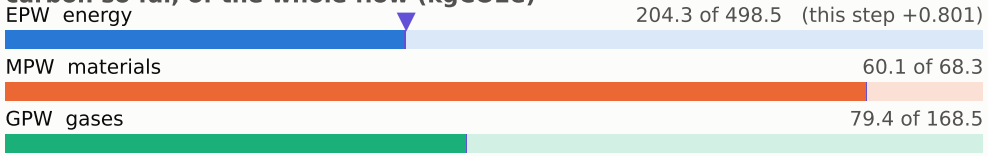
no mask for this step

this step

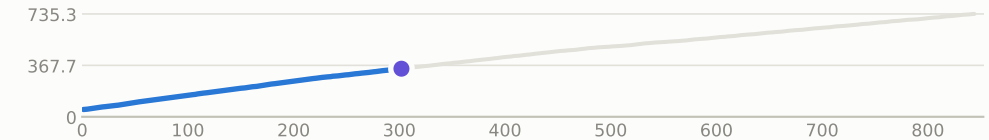
process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

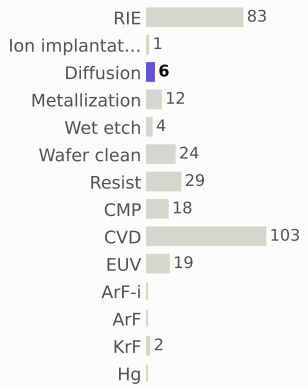
carbon so far, of the whole flow (kgCO2e)



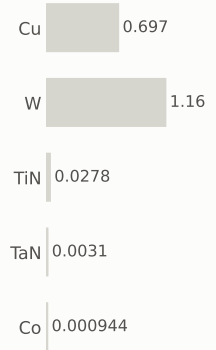
embodied carbon up to each step (kgCO2e per wafer)



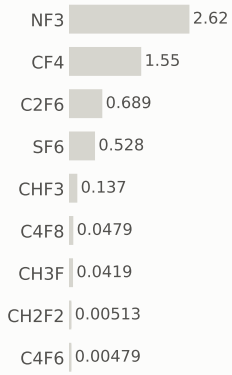
steps by process area



materials deposited (g)



gases emitted (g)



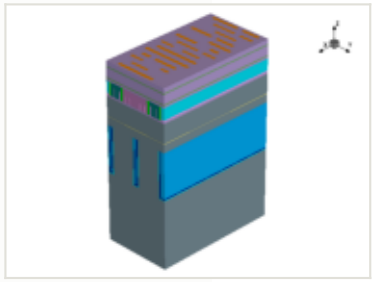
Step #3220: CMP

etch_cmp · CMP

0.169	344.0	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

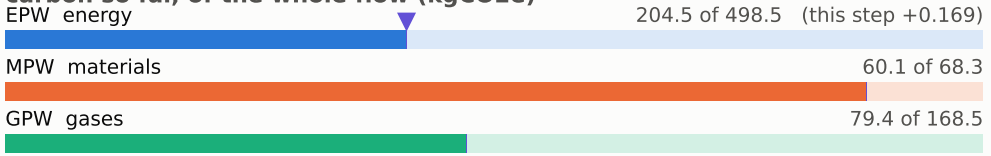


this step

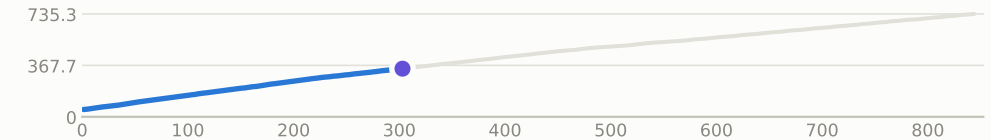
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

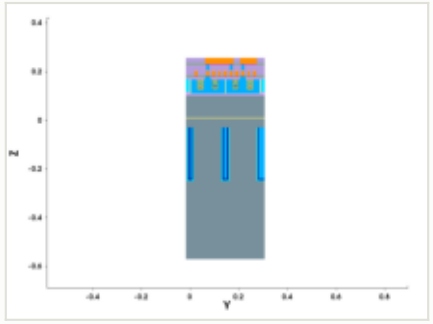
carbon so far, of the whole flow (kgCO2e)



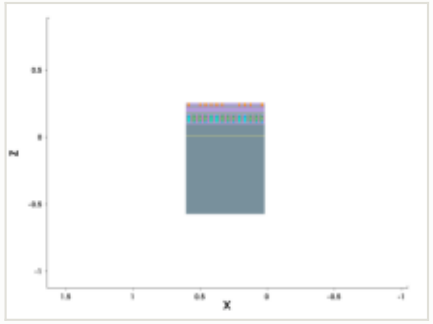
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



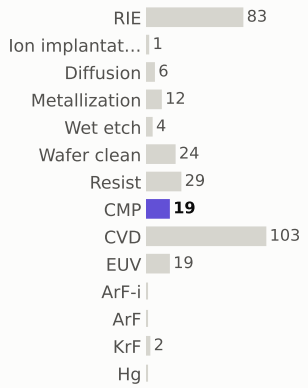
y cut at 0.184 µm



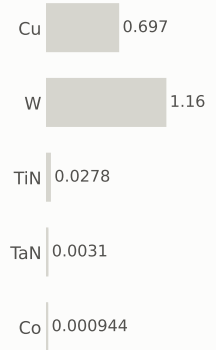
mask for this step

no mask for this step

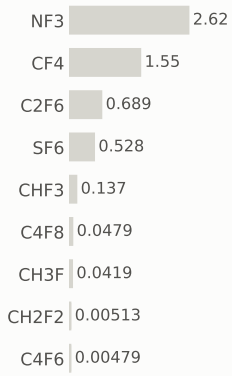
steps by process area



materials deposited (g)



gases emitted (g)



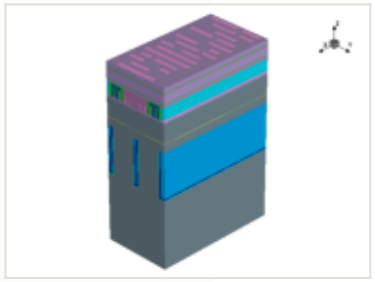
Step #3230: Metallization

depo_isotropic · Metallization

1.2	345.2	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

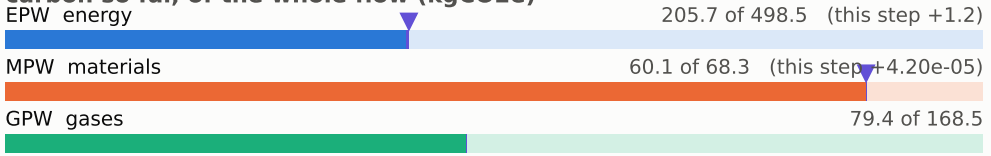


this step

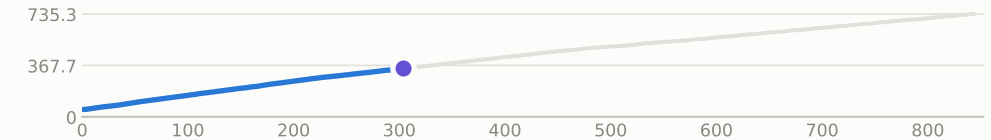
process area	Metallization
material	Co
thickness	0.0015 µm
material GWP	44.5 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.20e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

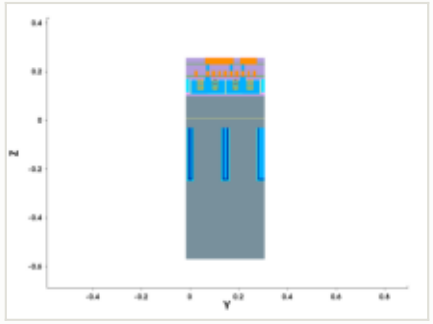
carbon so far, of the whole flow (kgCO2e)



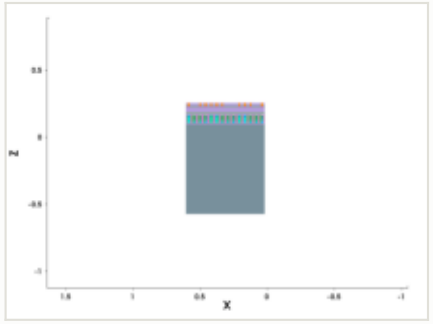
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



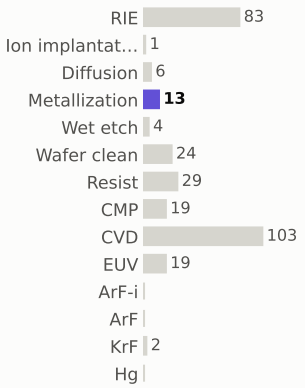
y cut at 0.184 µm



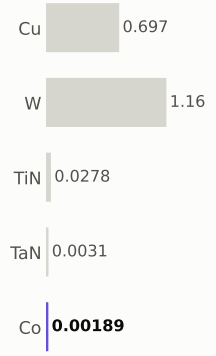
mask for this step

no mask for this step

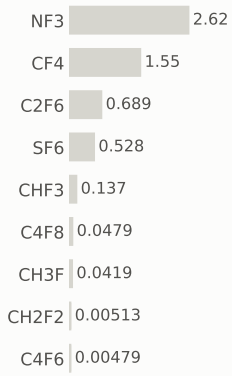
steps by process area



materials deposited (g)



gases emitted (g)



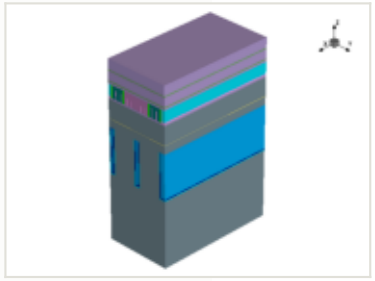
Step #3250: CVD

depo_fill · CVD

1.31	346.5	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

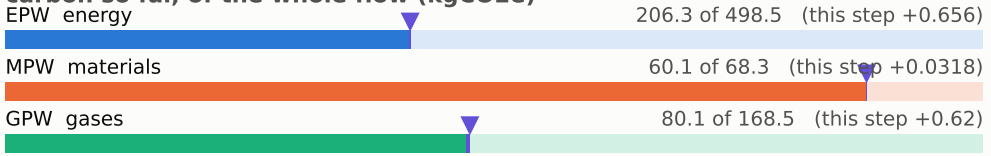


this step

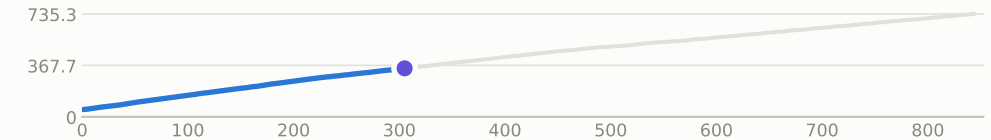
process area	CVD
material	—
thickness	0.024 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

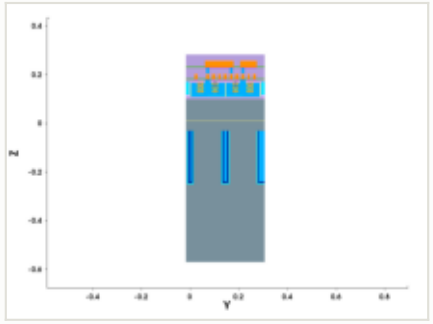
carbon so far, of the whole flow (kgCO2e)



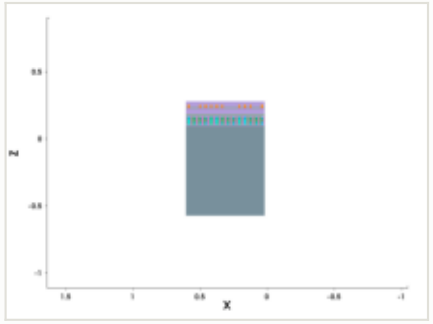
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



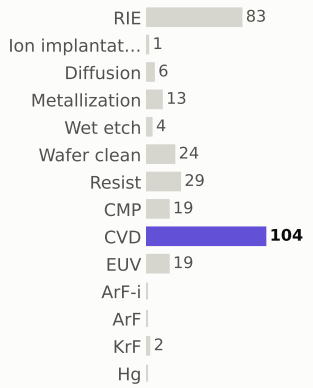
y cut at 0.184 µm



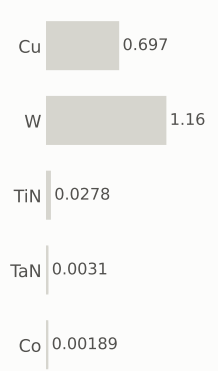
mask for this step

no mask for this step

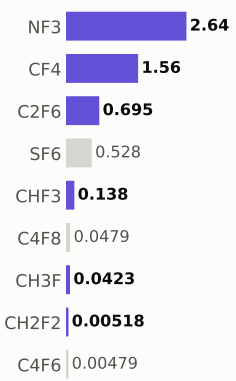
steps by process area



materials deposited (g)



gases emitted (g)



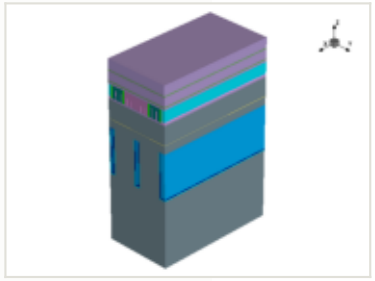
Step #3260: Diffusion

treatment · Diffusion

0.801	347.3	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

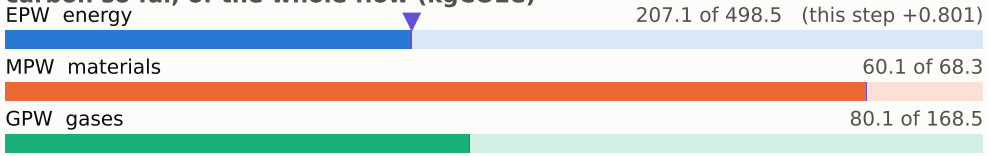


this step

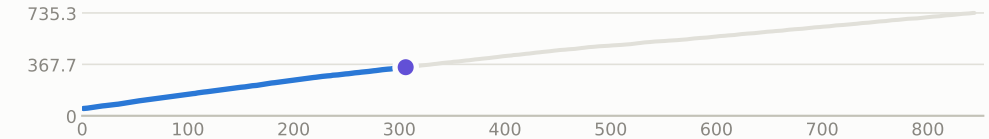
process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

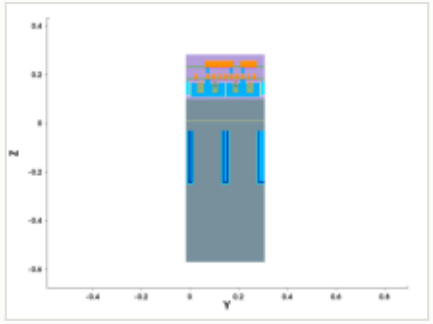
carbon so far, of the whole flow (kgCO2e)



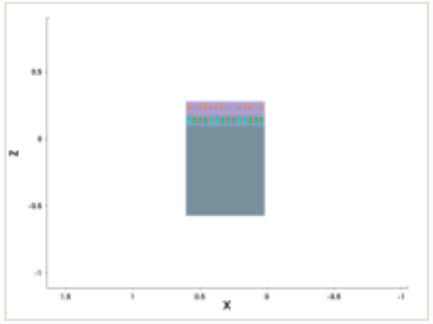
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



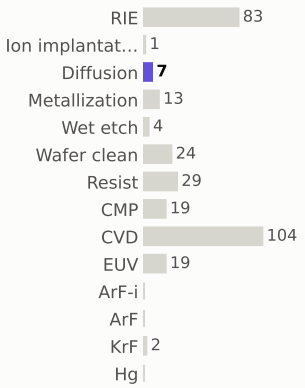
y cut at 0.184 µm



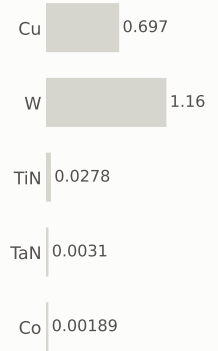
mask for this step

no mask for this step

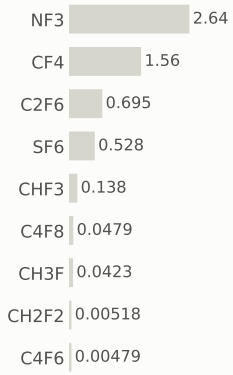
steps by process area



materials deposited (g)



gases emitted (g)



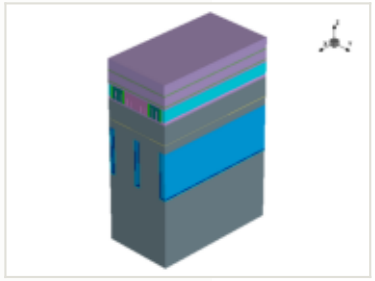
Step #3270: Wafer_clean

clean · Wafer_clean

0.169	347.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



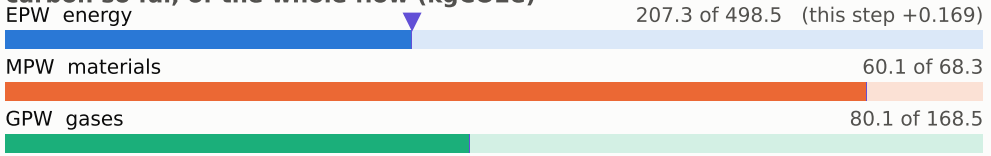
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

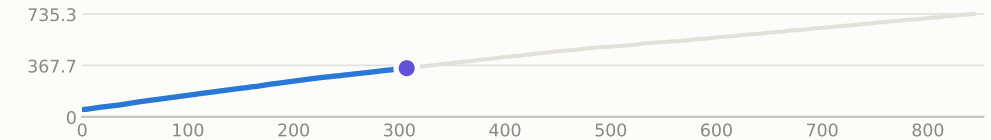
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

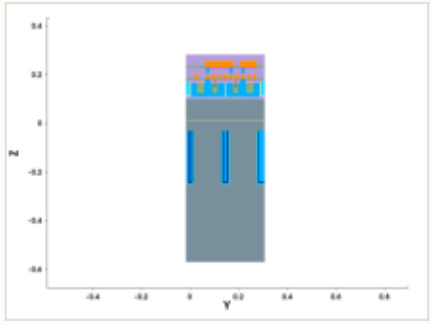
carbon so far, of the whole flow (kgCO2e)



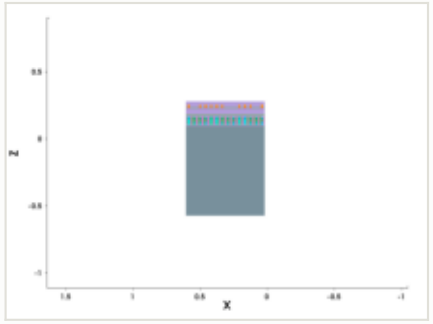
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



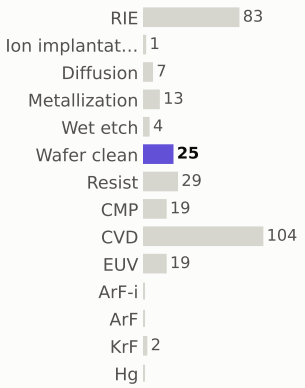
y cut at 0.184 µm



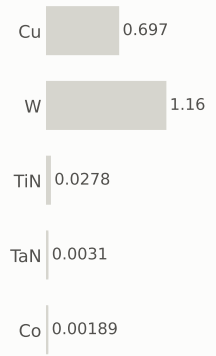
mask for this step

no mask for this step

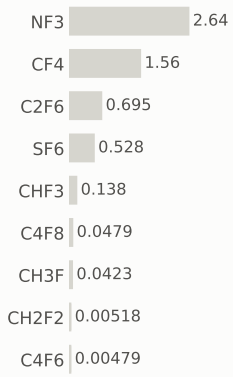
steps by process area



materials deposited (g)



gases emitted (g)



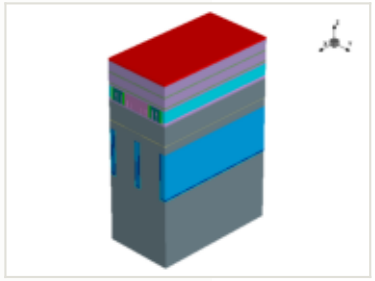
Step #3280: CVD

depo_fill · CVD

1.31	348.8	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

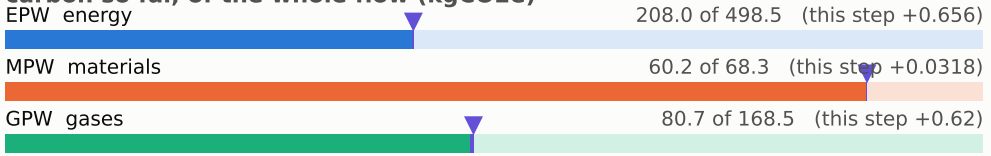


this step

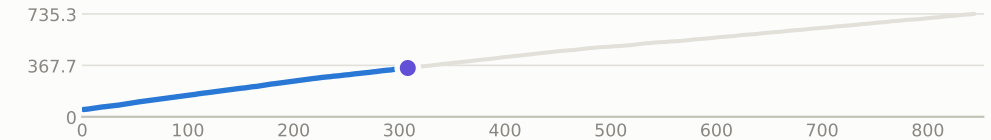
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

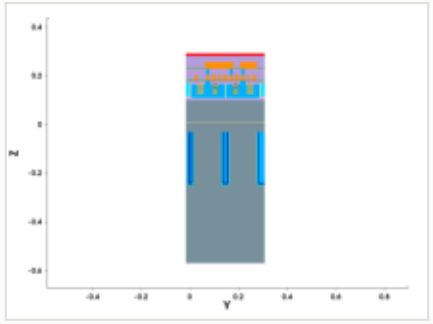
carbon so far, of the whole flow (kgCO2e)



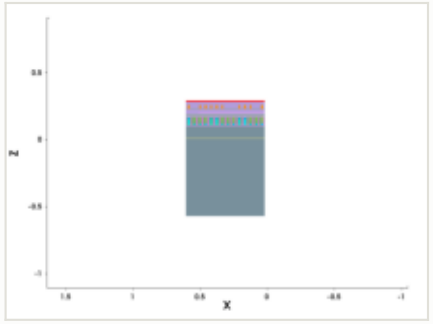
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



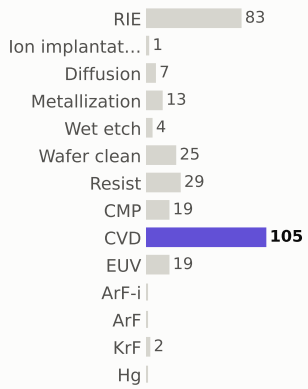
y cut at 0.184 µm



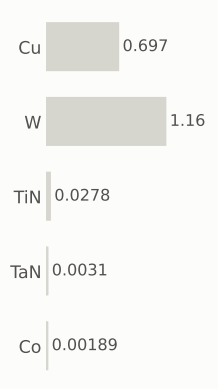
mask for this step

no mask for this step

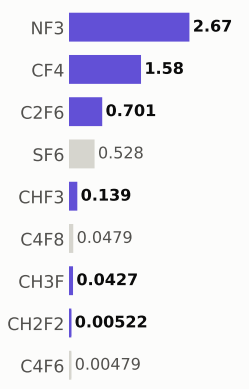
steps by process area



materials deposited (g)



gases emitted (g)



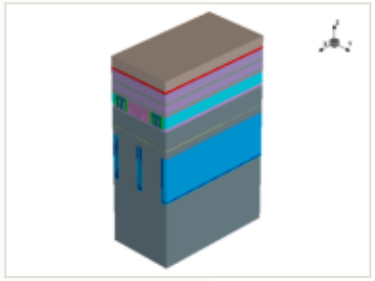
Step #3290: Resist

depo_fill · Resist

0.169	349.0	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

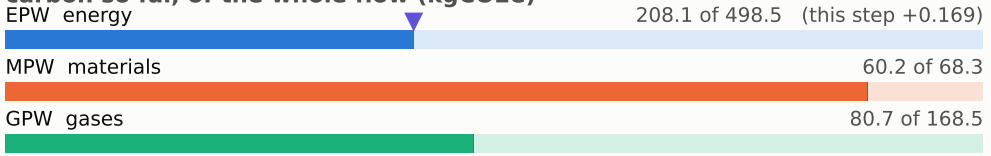
zoomed to fit



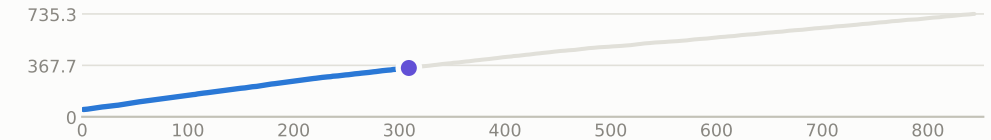
this step

process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	0.05 µm	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

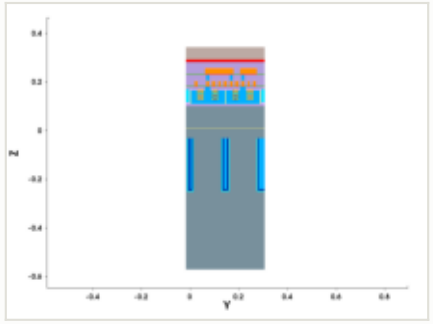
carbon so far, of the whole flow (kgCO2e)



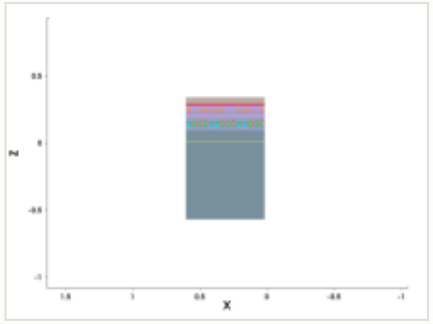
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



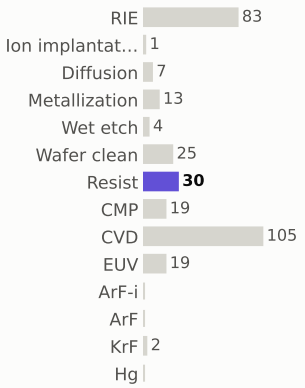
y cut at 0.184 µm



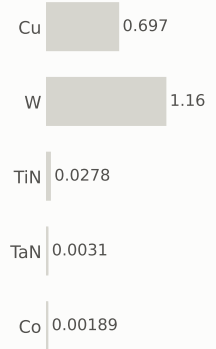
mask for this step

no mask for this step

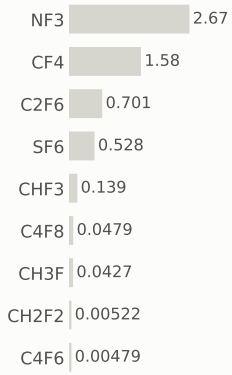
steps by process area



materials deposited (g)



gases emitted (g)



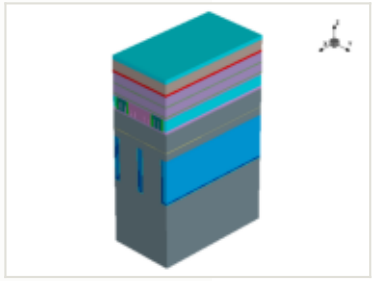
Step #3300: CVD

depo_fill · CVD

1.31	350.3	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

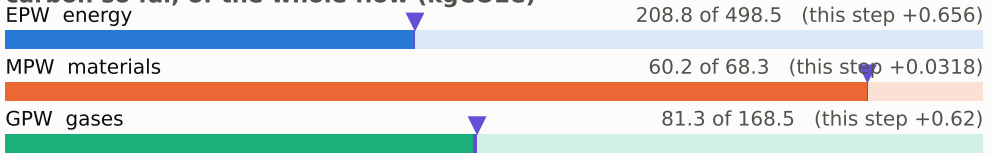


this step

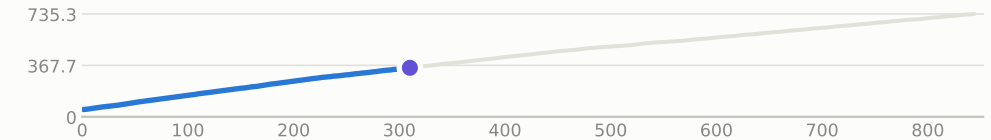
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

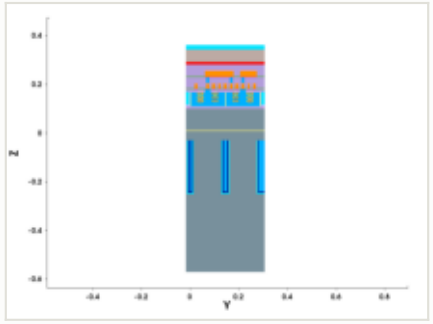
carbon so far, of the whole flow (kgCO2e)



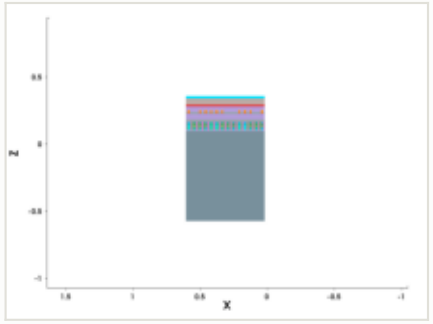
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



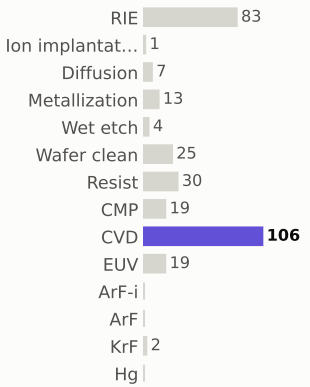
y cut at 0.184 µm



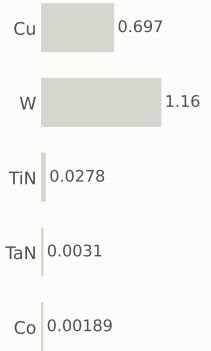
mask for this step

no mask for this step

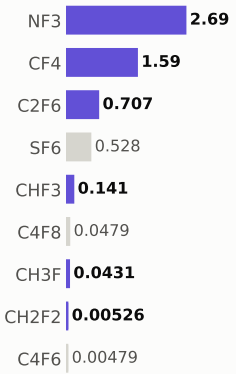
steps by process area



materials deposited (g)



gases emitted (g)



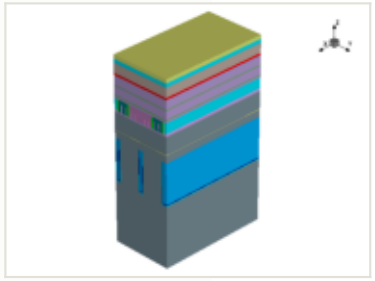
Step #3310: CVD

depo_fill · CVD

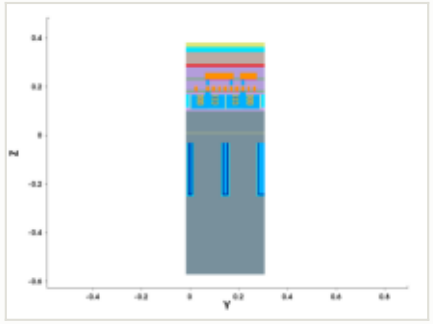
1.31	351.6	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

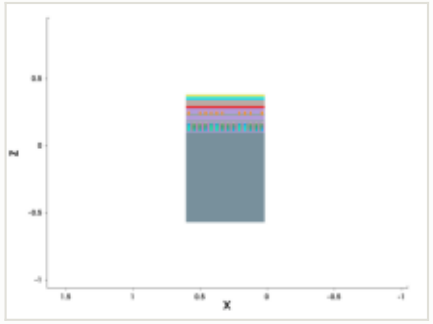
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

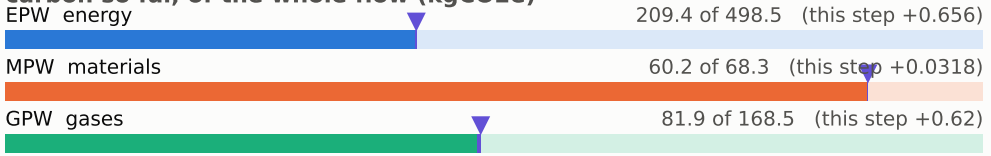
no mask for this step

this step

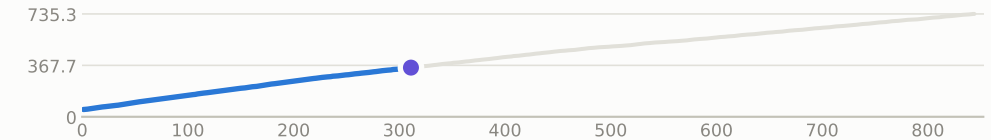
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

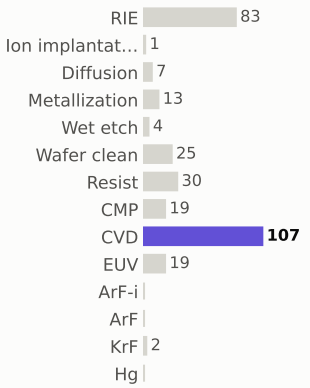
carbon so far, of the whole flow (kgCO2e)



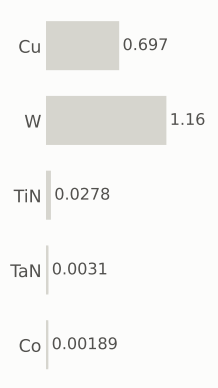
embodied carbon up to each step (kgCO2e per wafer)



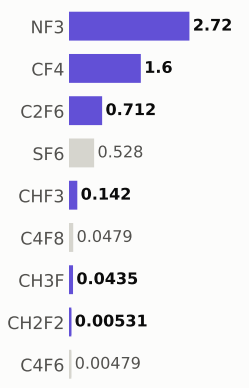
steps by process area



materials deposited (g)



gases emitted (g)



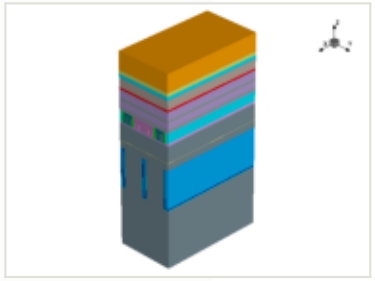
Step #3320: Resist

depo_fill · Resist

0.169	351.8	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

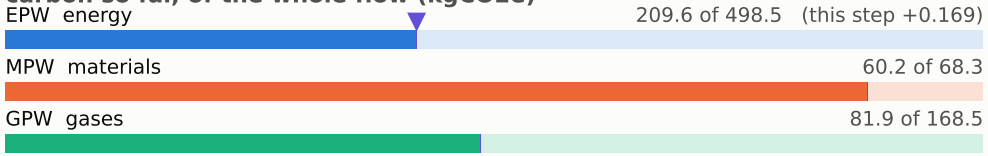


this step

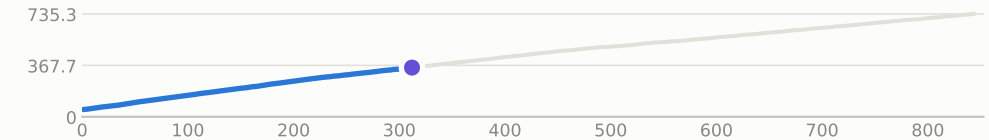
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

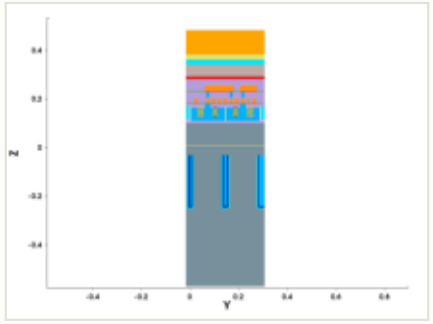
carbon so far, of the whole flow (kgCO2e)



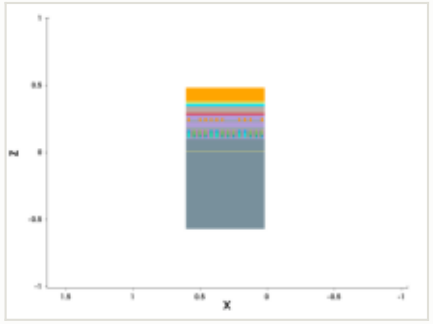
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



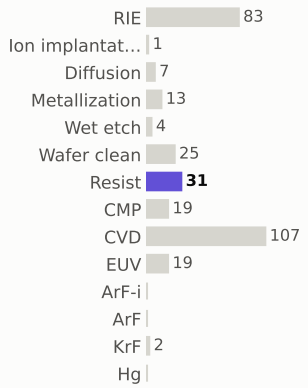
y cut at 0.184 µm



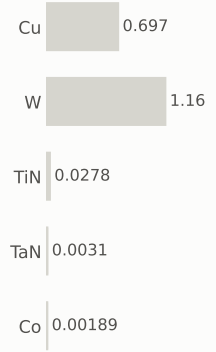
mask for this step

no mask for this step

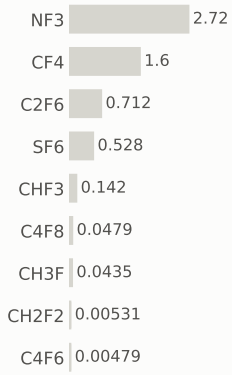
steps by process area



materials deposited (g)



gases emitted (g)



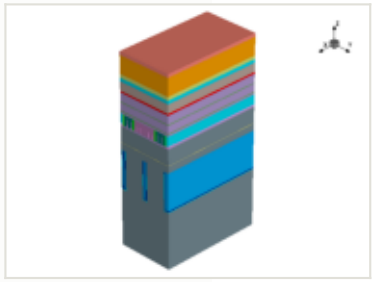
Step #3330: CVD

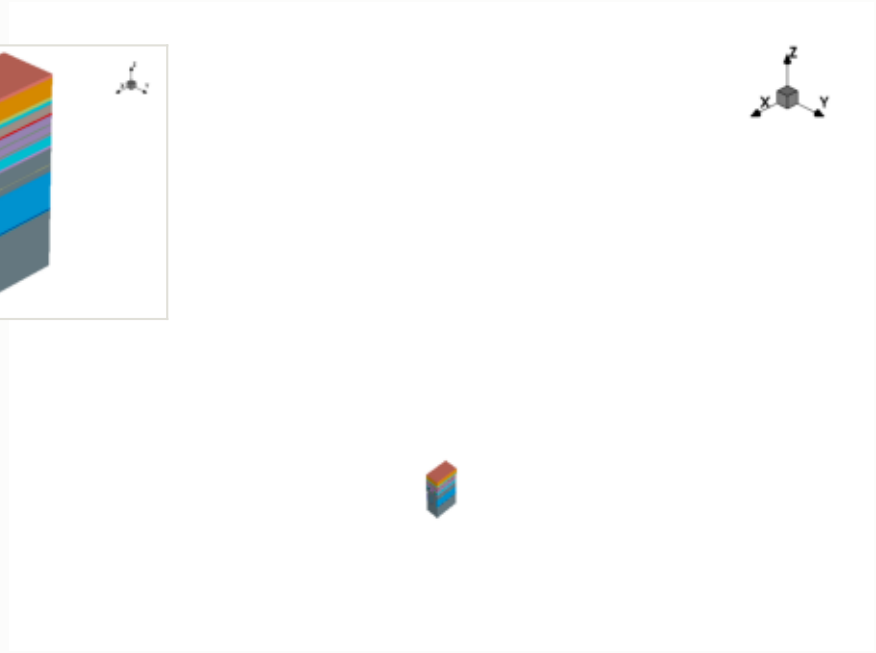
depo_fill · CVD

1.31	353.1	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



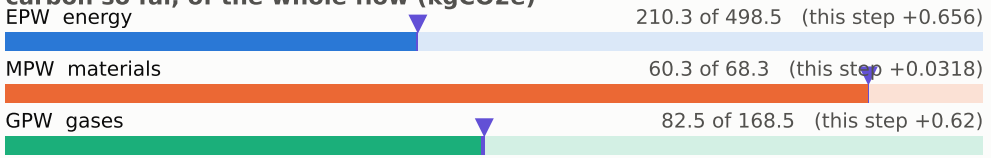


this step

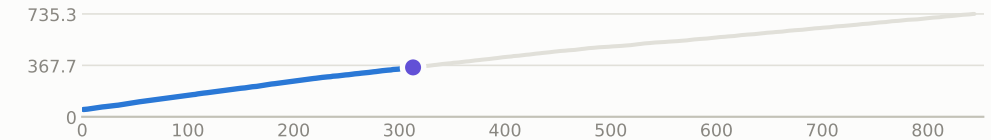
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

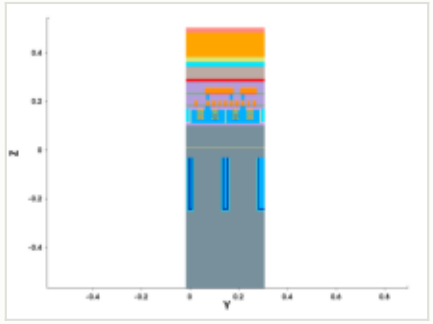
carbon so far, of the whole flow (kgCO2e)



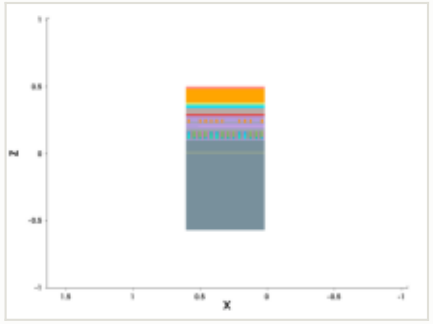
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



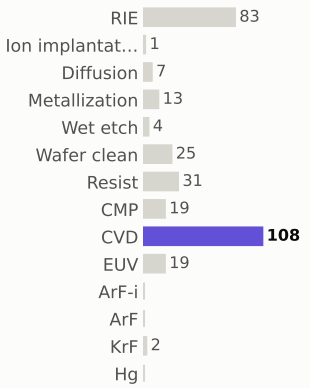
y cut at 0.184 µm



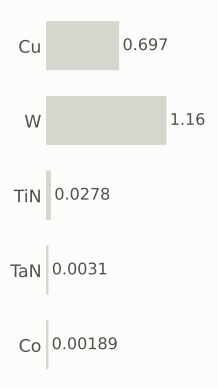
mask for this step

no mask for this step

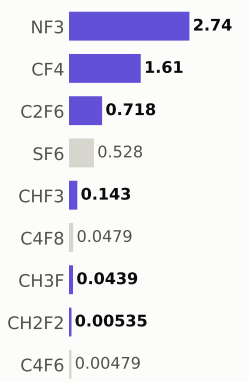
steps by process area



materials deposited (g)



gases emitted (g)



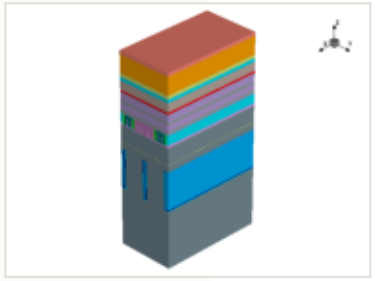
Step #3340: Wafer_clean

clean · Wafer_clean

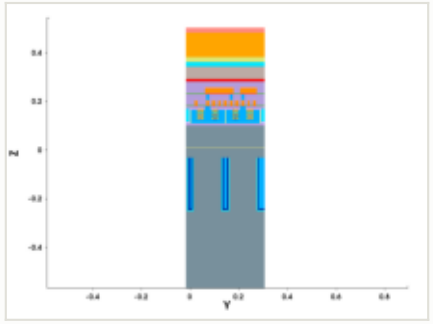
0.169	353.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

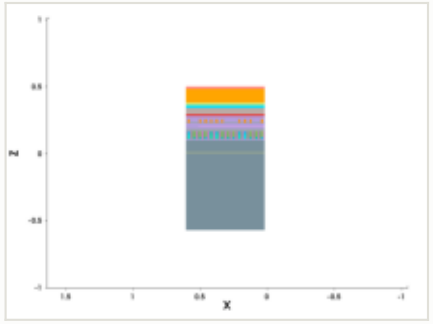
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

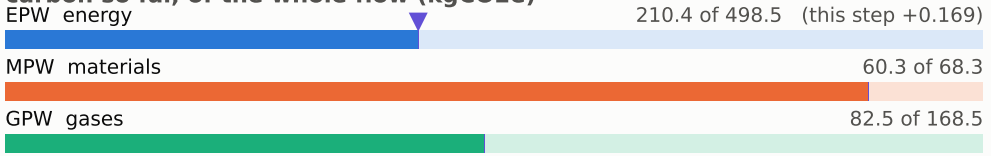
no mask for this step

this step

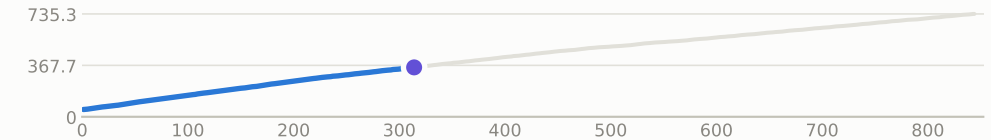
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

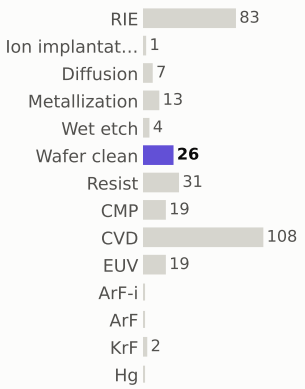
carbon so far, of the whole flow (kgCO2e)



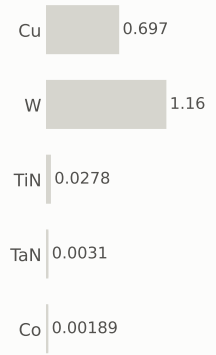
embodied carbon up to each step (kgCO2e per wafer)



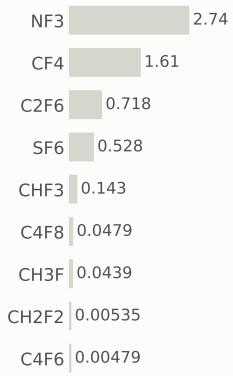
steps by process area



materials deposited (g)



gases emitted (g)



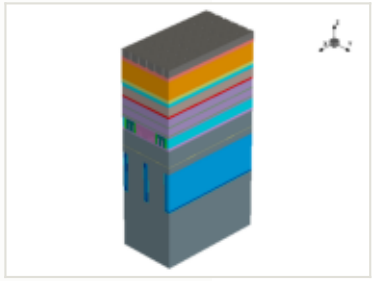
Step #3350: EUV

litho_patterning · EUV

1.96	355.2	0.27%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

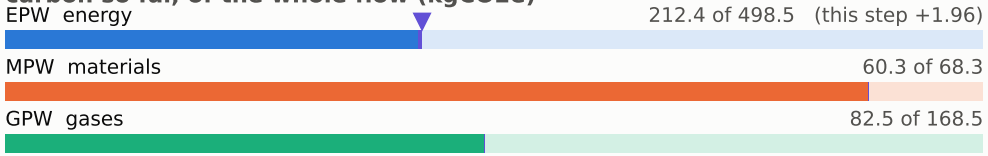


this step

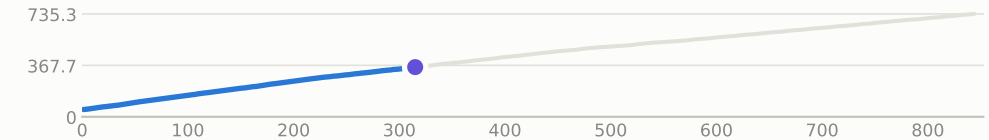
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

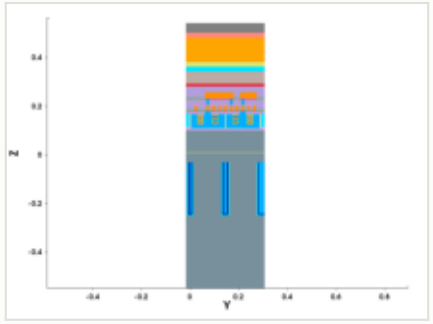
carbon so far, of the whole flow (kgCO2e)



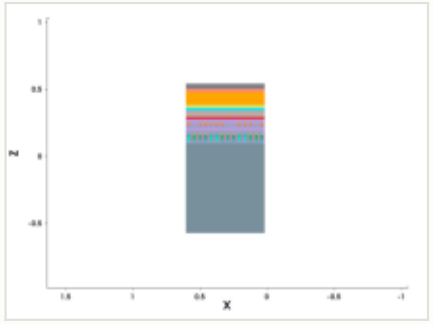
embodied carbon up to each step (kgCO2e per wafer)



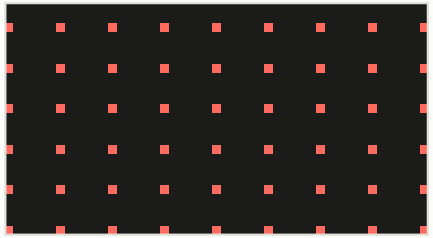
x cut at 0.294 µm



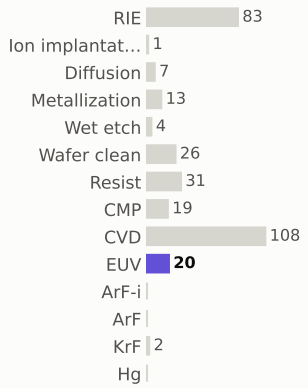
y cut at 0.184 µm



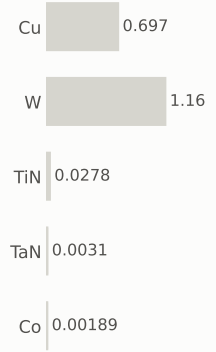
mask: layer 10271:0 (GDSII)



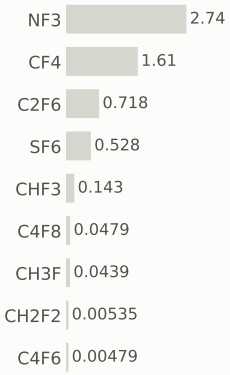
steps by process area



materials deposited (g)



gases emitted (g)



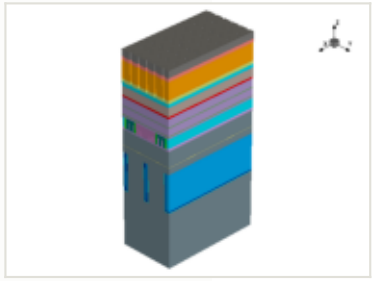
Step #3360: RIE

etch_anisotropic · RIE

1.04	356.2	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

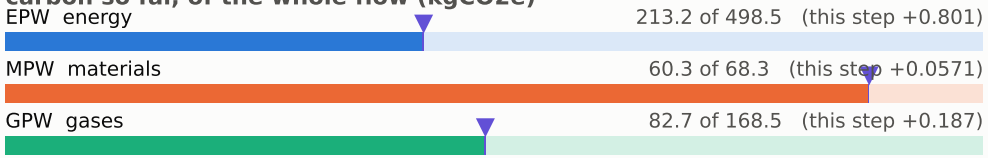


this step

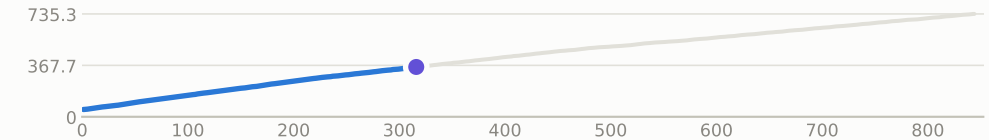
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

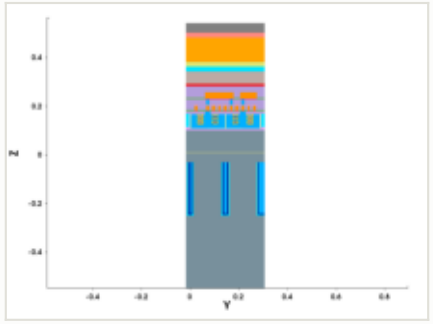
carbon so far, of the whole flow (kgCO2e)



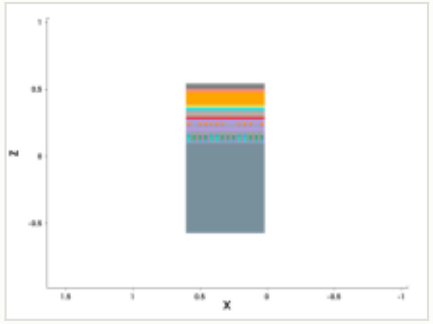
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



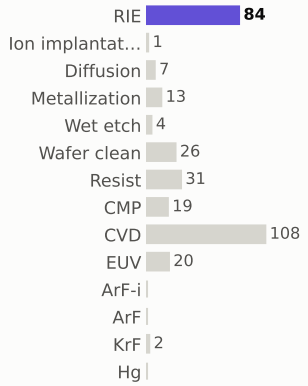
y cut at 0.184 µm



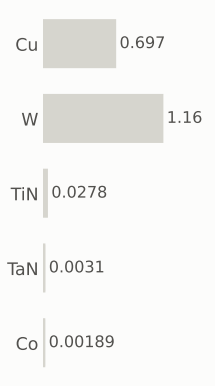
mask for this step

no mask for this step

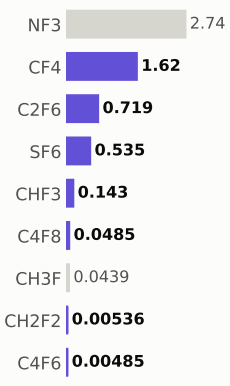
steps by process area



materials deposited (g)



gases emitted (g)



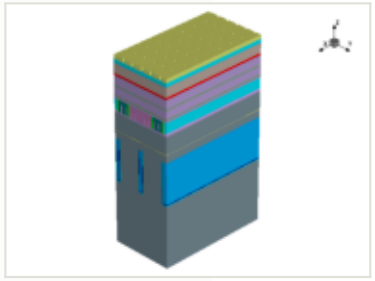
Step #3370: Wafer_clean

etch_strip · Wafer_clean

0.169	356.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

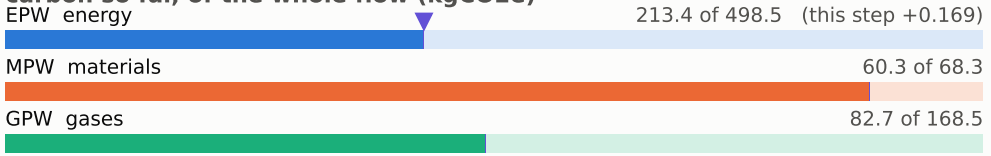


this step

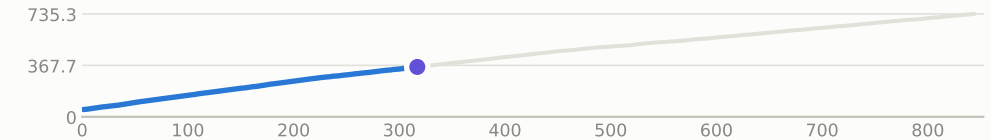
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

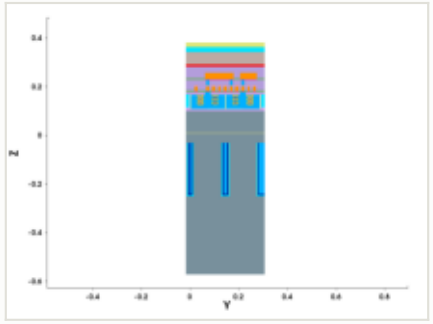
carbon so far, of the whole flow (kgCO2e)



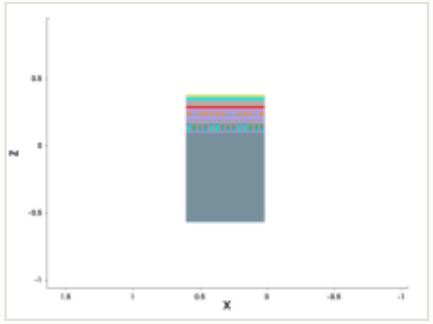
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



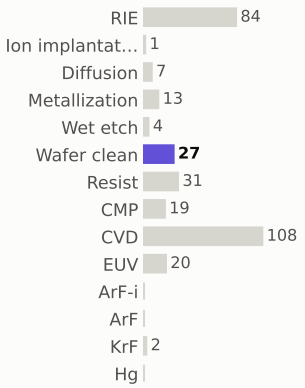
y cut at 0.184 µm



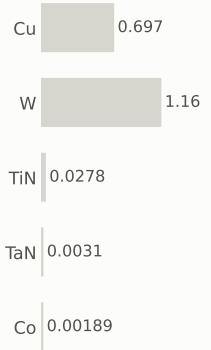
mask for this step

no mask for this step

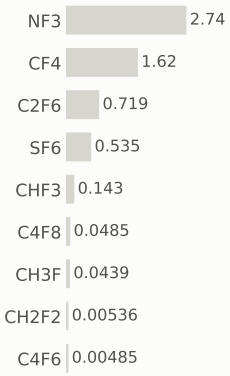
steps by process area



materials deposited (g)



gases emitted (g)



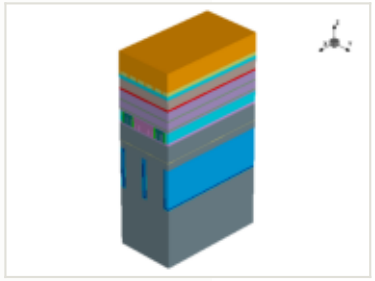
Step #3380: Resist

depo_fill · Resist

0.169	356.6	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

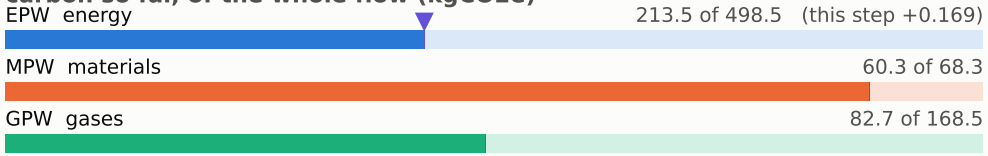


this step

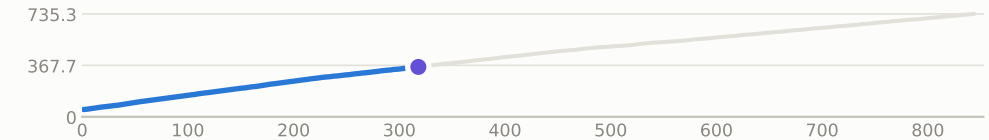
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

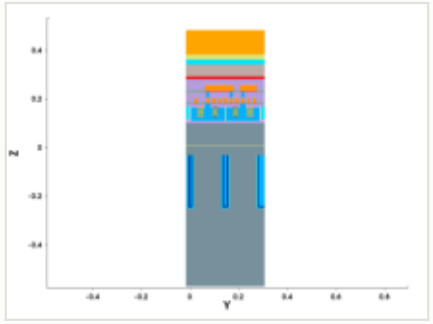
carbon so far, of the whole flow (kgCO2e)



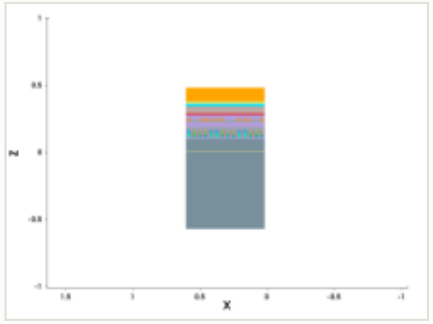
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



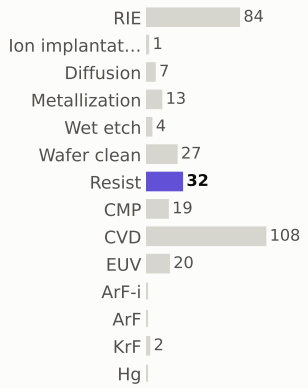
y cut at 0.184 µm



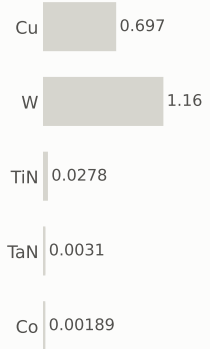
mask for this step

no mask for this step

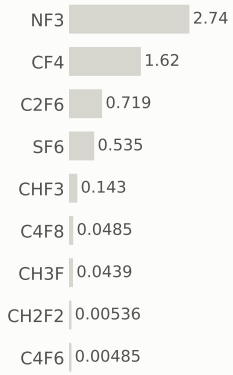
steps by process area



materials deposited (g)



gases emitted (g)



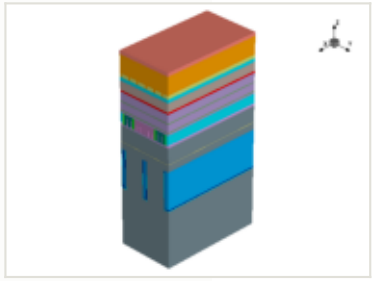
Step #3390: CVD

depo_fill · CVD

1.31	357.9	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

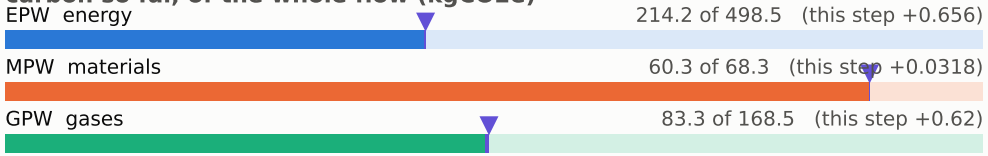


this step

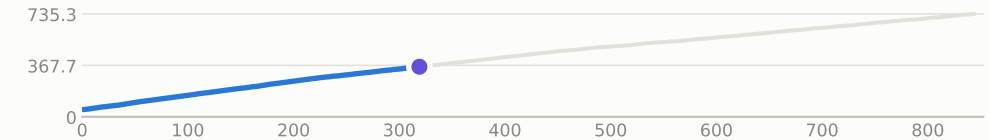
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

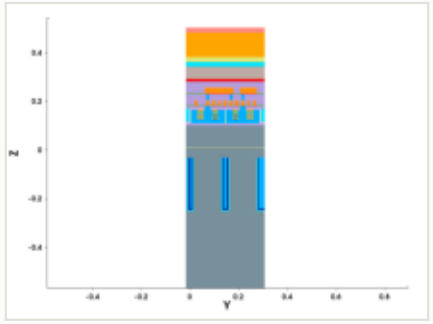
carbon so far, of the whole flow (kgCO2e)



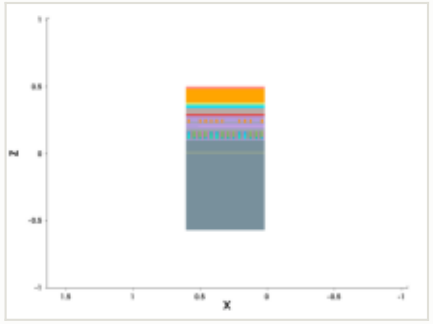
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



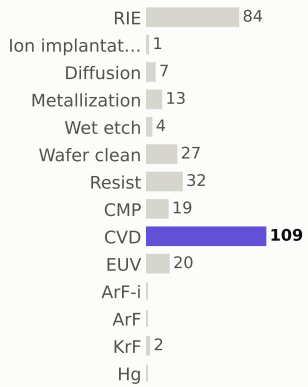
y cut at 0.184 µm



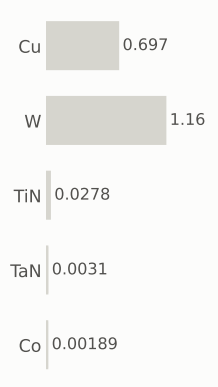
mask for this step

no mask for this step

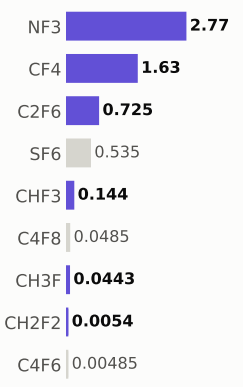
steps by process area



materials deposited (g)



gases emitted (g)



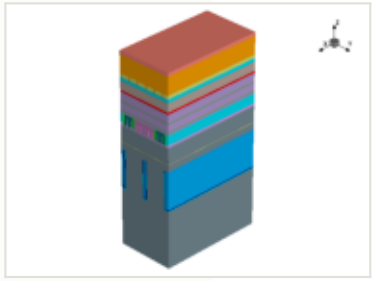
Step #3400: Wafer_clean

clean · Wafer_clean

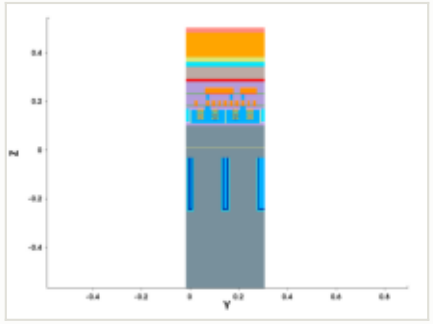
0.169	358.0	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

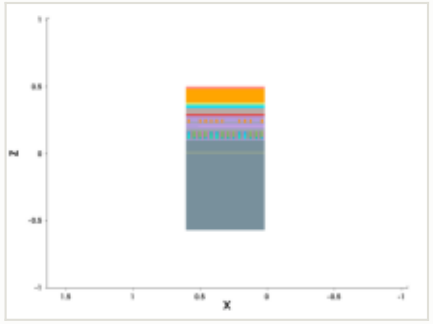
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

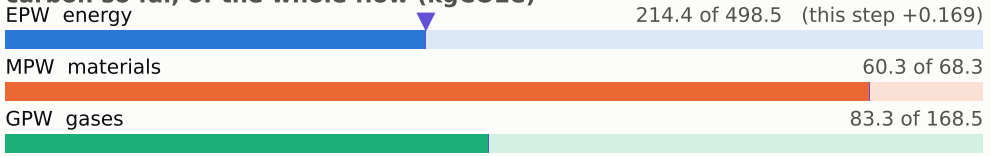
no mask for this step

this step

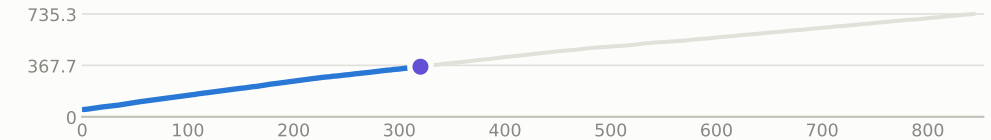
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

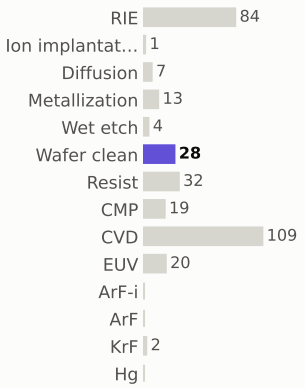
carbon so far, of the whole flow (kgCO2e)



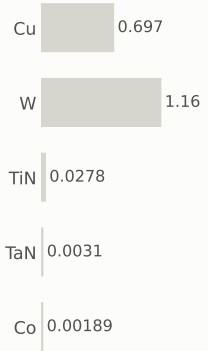
embodied carbon up to each step (kgCO2e per wafer)



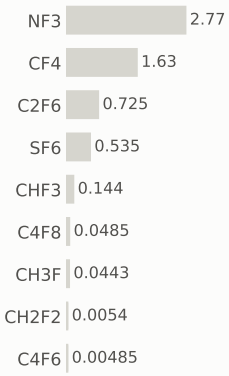
steps by process area



materials deposited (g)



gases emitted (g)



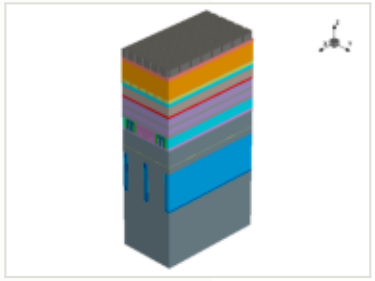
Step #3410: EUV

litho_patterning · EUV

1.96	360.0	0.27%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

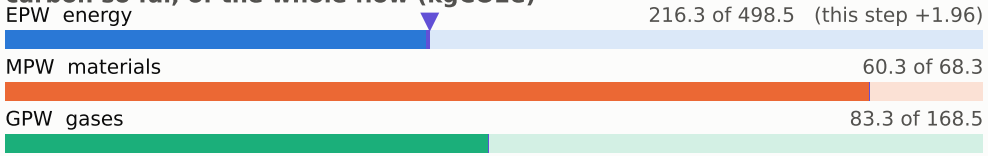


this step

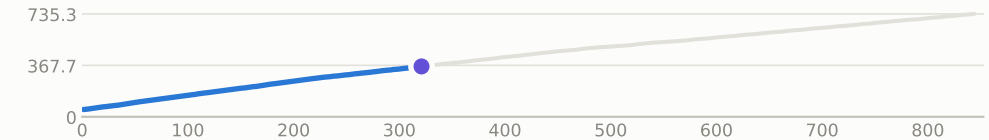
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

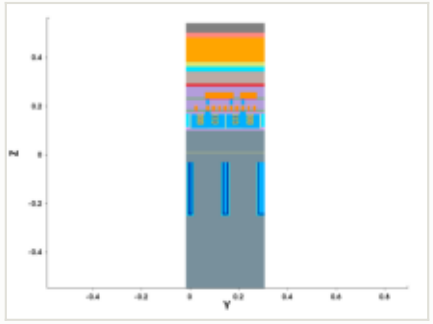
carbon so far, of the whole flow (kgCO2e)



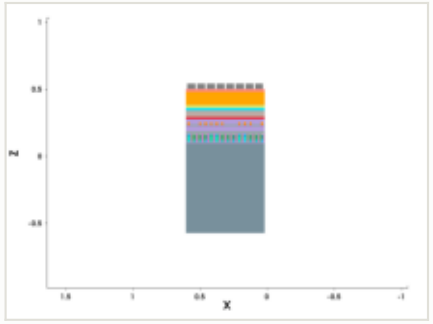
embodied carbon up to each step (kgCO2e per wafer)



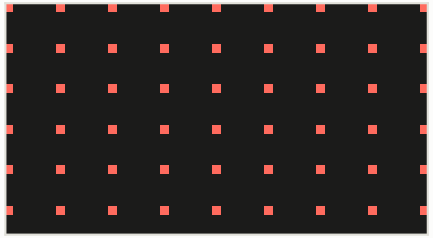
x cut at 0.294 µm



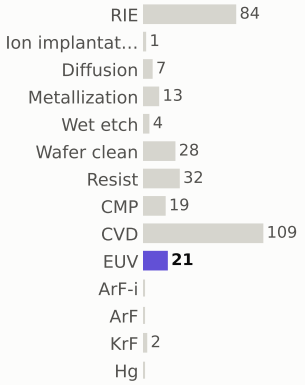
y cut at 0.184 µm



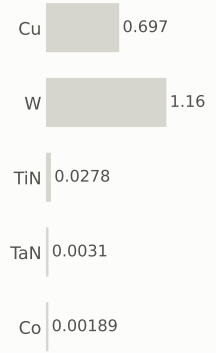
mask: layer 10272:0 (GDSII)



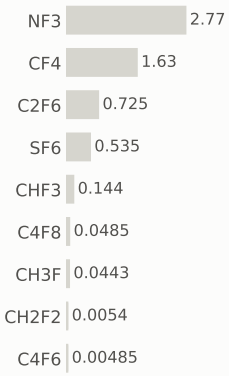
steps by process area



materials deposited (g)



gases emitted (g)



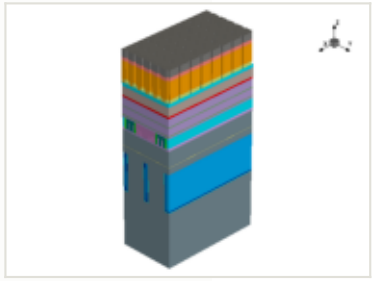
Step #3420: RIE

etch_anisotropic · RIE

1.04	361.0	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

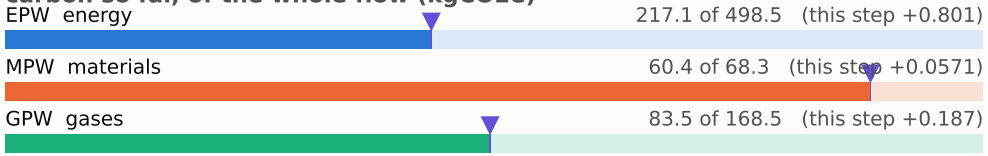


this step

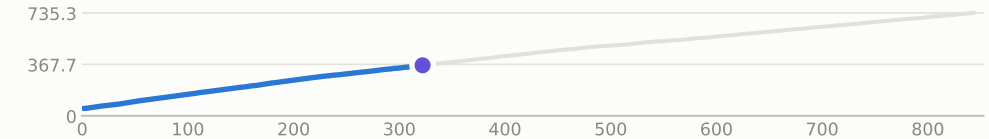
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

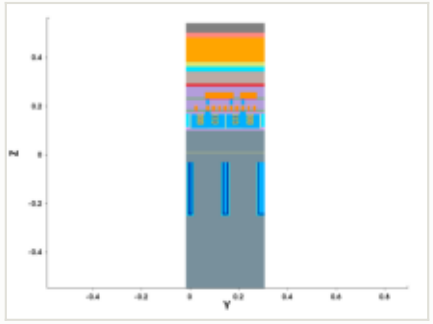
carbon so far, of the whole flow (kgCO2e)



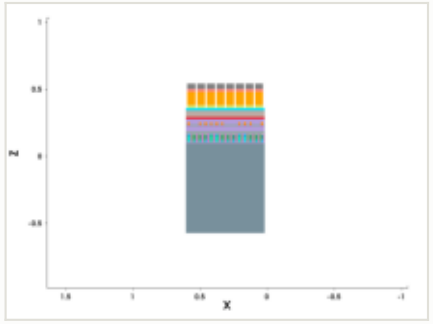
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



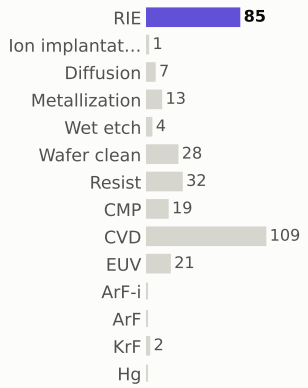
y cut at 0.184 µm



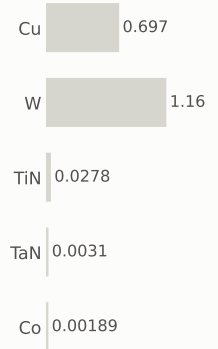
mask for this step

no mask for this step

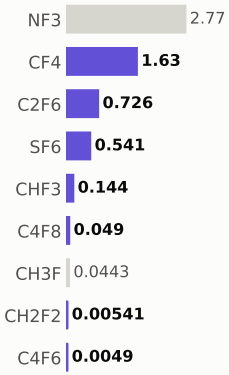
steps by process area



materials deposited (g)



gases emitted (g)



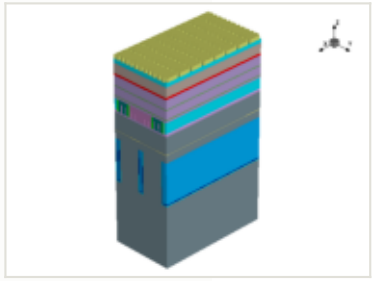
Step #3430: Wafer_clean

etch_strip · Wafer_clean

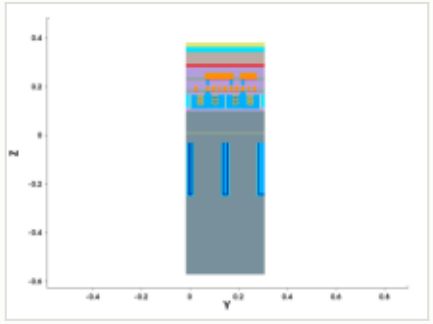
0.169	361.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

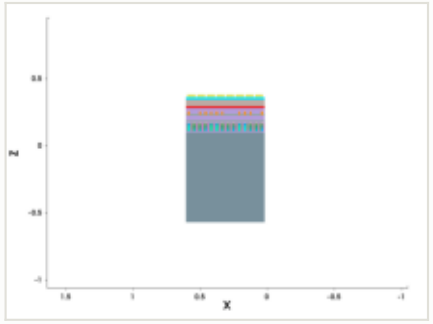
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



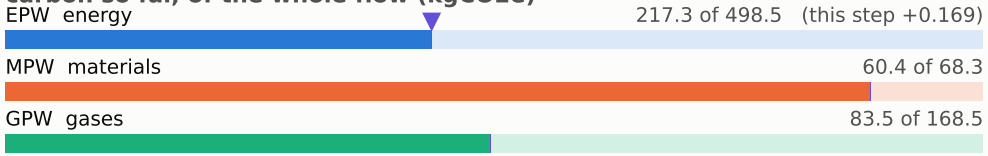
mask for this step

no mask for this step

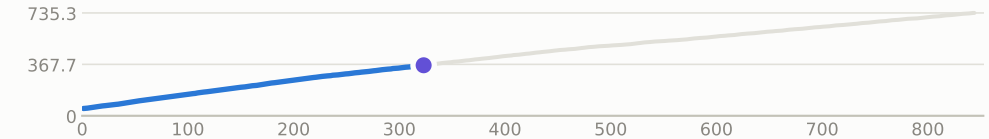
this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

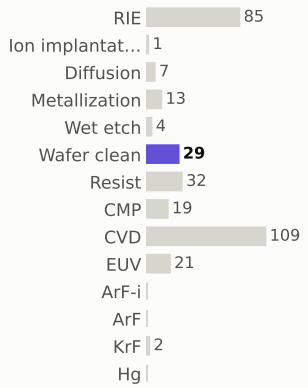
carbon so far, of the whole flow (kgCO2e)



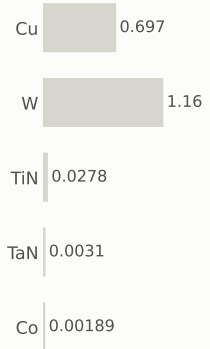
embodied carbon up to each step (kgCO2e per wafer)



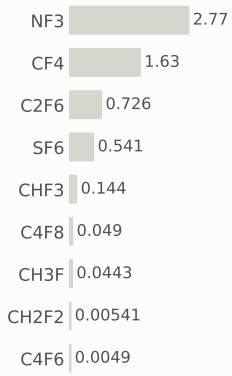
steps by process area



materials deposited (g)



gases emitted (g)



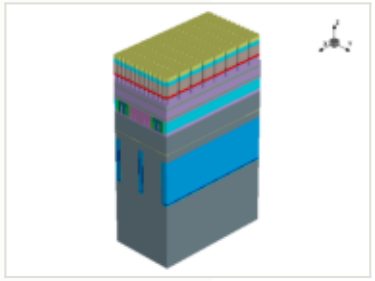
Step #3440: RIE

etch_anisotropic · RIE

1.04	362.3	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

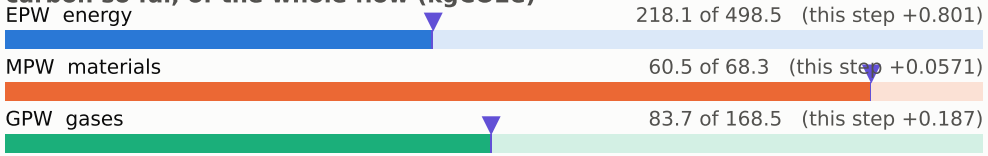


this step

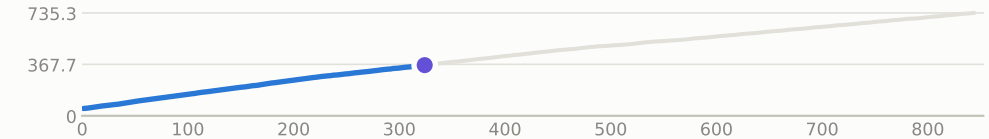
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

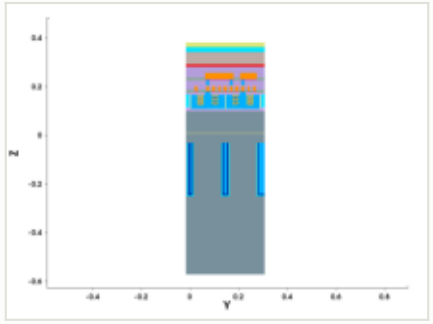
carbon so far, of the whole flow (kgCO2e)



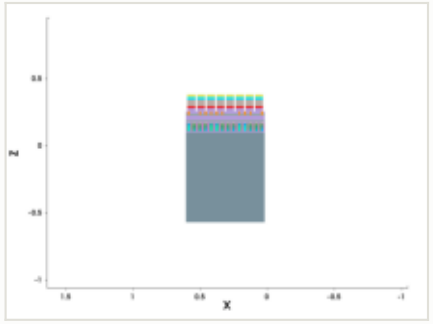
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



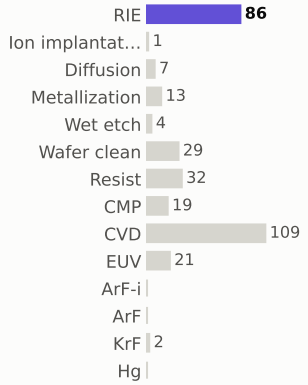
y cut at 0.184 µm



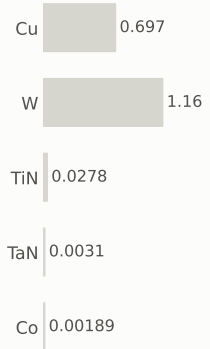
mask for this step

no mask for this step

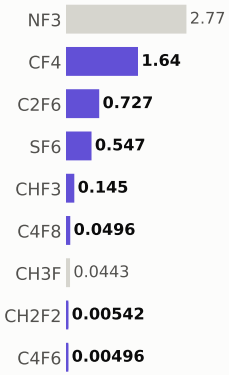
steps by process area



materials deposited (g)



gases emitted (g)



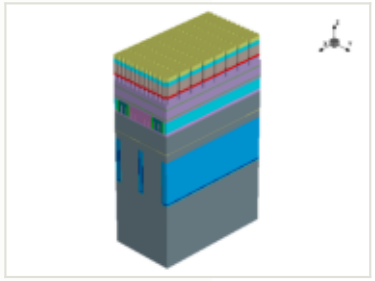
Step #3450: Wafer_clean

clean · Wafer_clean

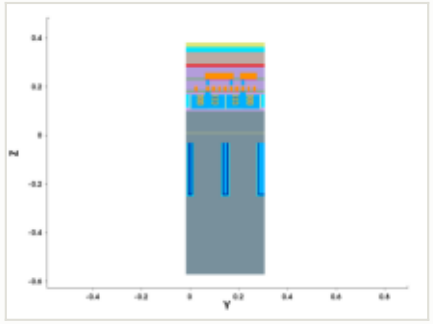
0.169	362.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

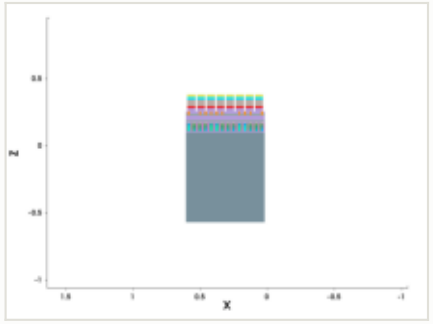
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

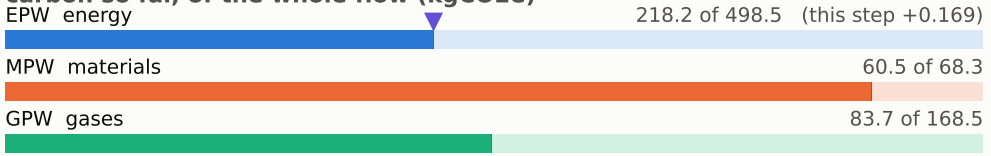
no mask for this step

this step

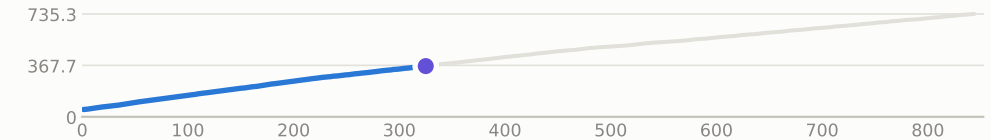
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

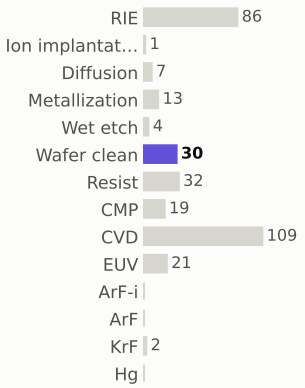
carbon so far, of the whole flow (kgCO2e)



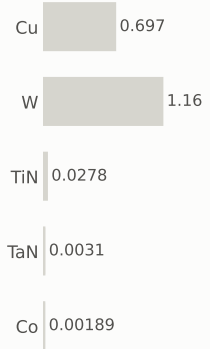
embodied carbon up to each step (kgCO2e per wafer)



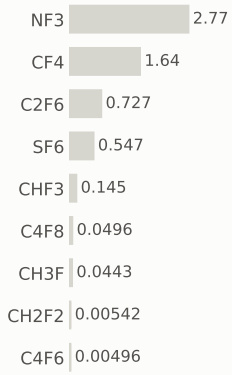
steps by process area



materials deposited (g)



gases emitted (g)



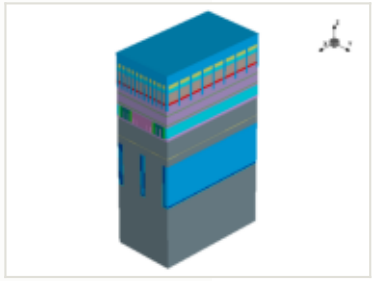
Step #3460: Metallization

depo_fill · Metallization

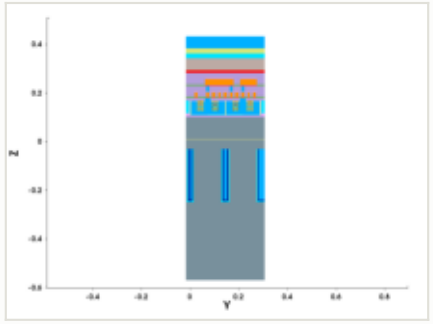
1.21	363.6	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

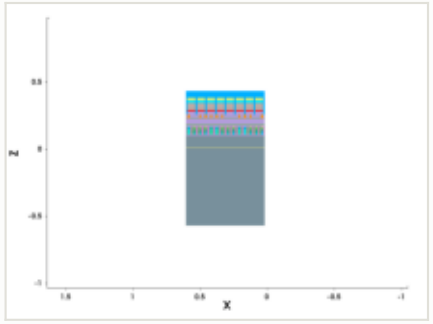
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

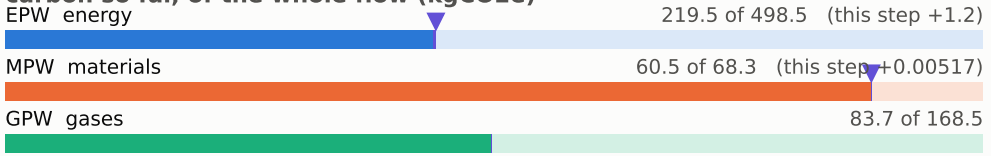
no mask for this step

this step

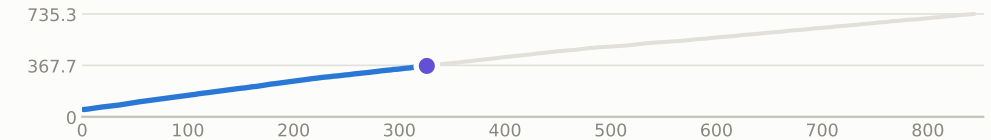
process area	Metallization
material	W
thickness	0.05 µm
material GWP	75.8149 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00517 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

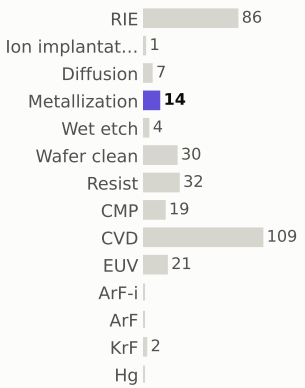
carbon so far, of the whole flow (kgCO2e)



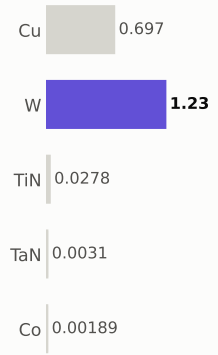
embodied carbon up to each step (kgCO2e per wafer)



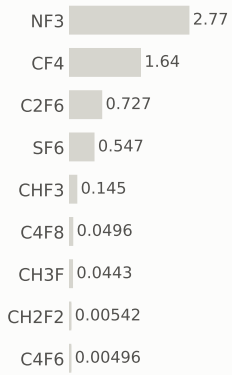
steps by process area



materials deposited (g)



gases emitted (g)



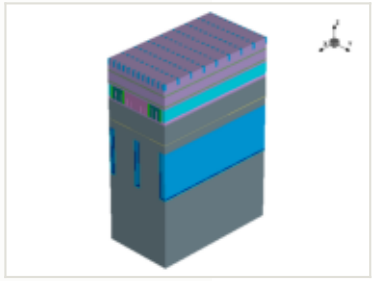
Step #3470: CMP

etch_cmp · CMP

0.169	363.8	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

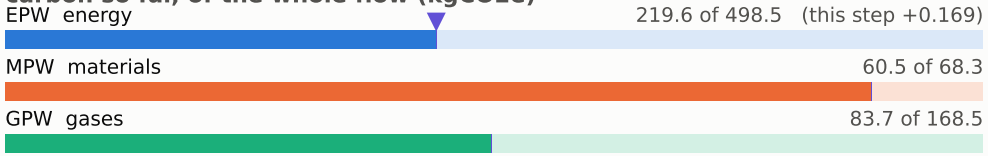


this step

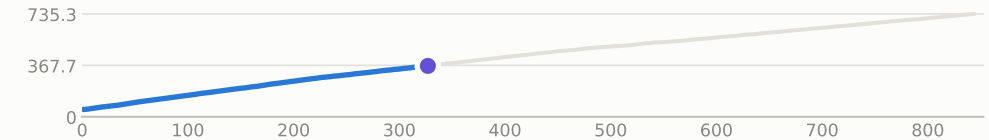
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

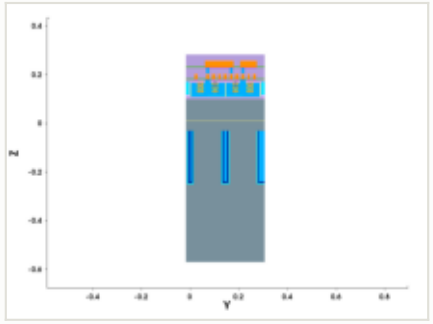
carbon so far, of the whole flow (kgCO2e)



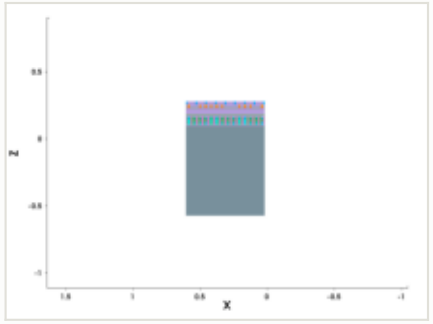
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



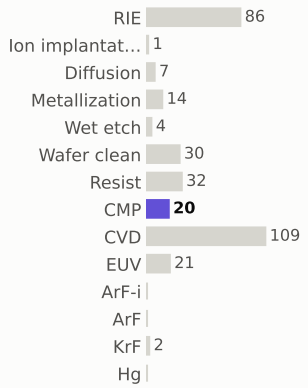
y cut at 0.184 µm



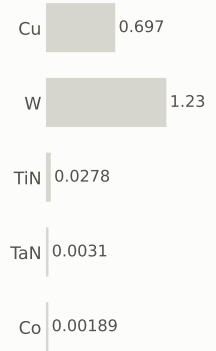
mask for this step

no mask for this step

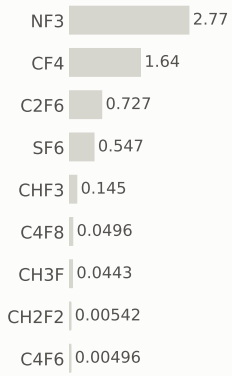
steps by process area



materials deposited (g)



gases emitted (g)



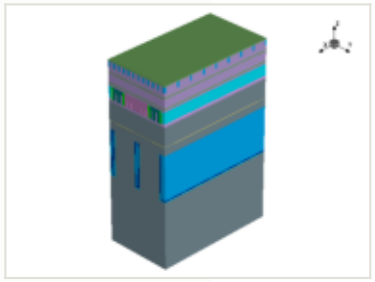
Step #3480: CVD

depo_fill · CVD

1.31	365.1	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

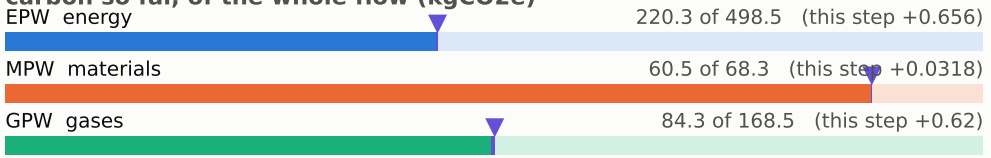


this step

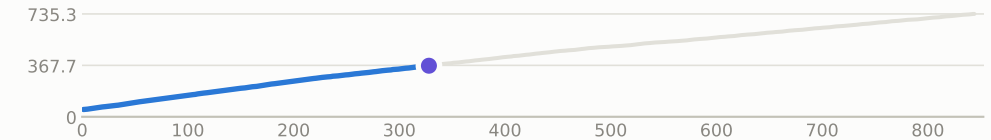
process area	CVD
material	—
thickness	0.006 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

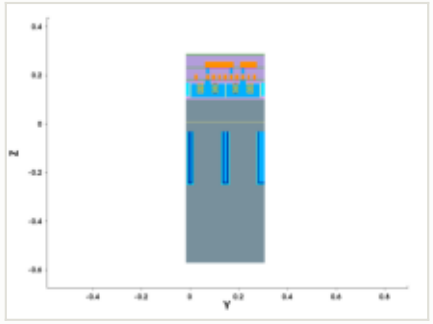
carbon so far, of the whole flow (kgCO2e)



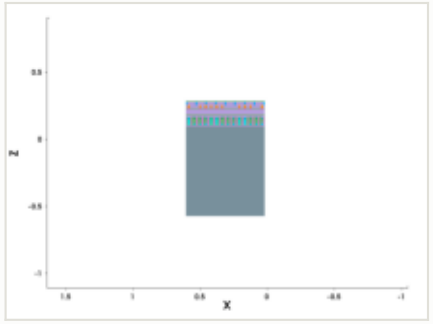
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



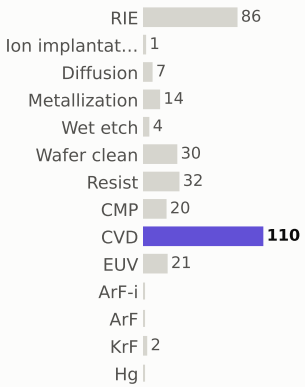
y cut at 0.184 µm



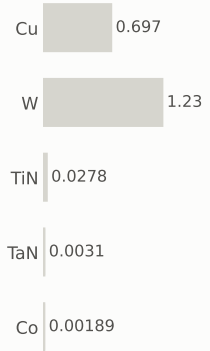
mask for this step

no mask for this step

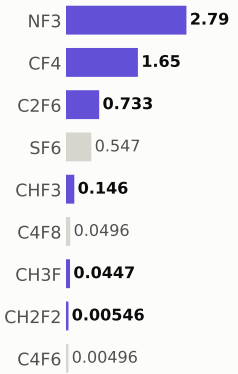
steps by process area



materials deposited (g)



gases emitted (g)



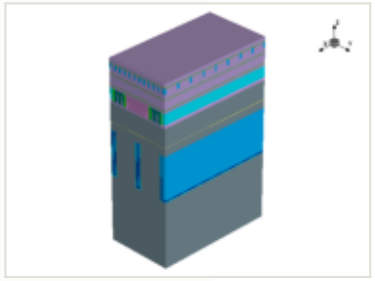
Step #3490: CVD

depo_fill · CVD

1.31	366.4	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

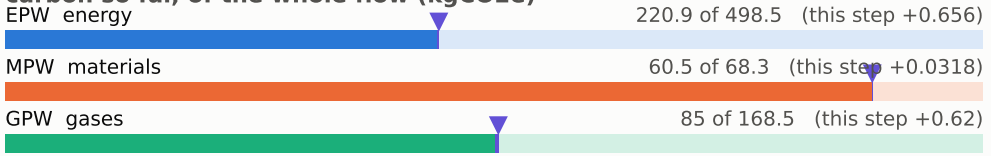


this step

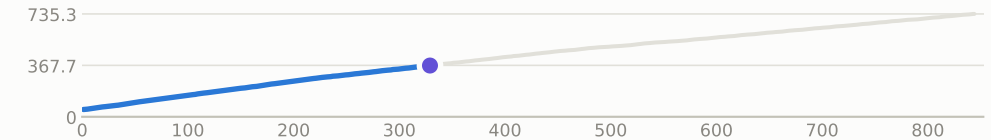
process area	CVD
material	—
thickness	0.018 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

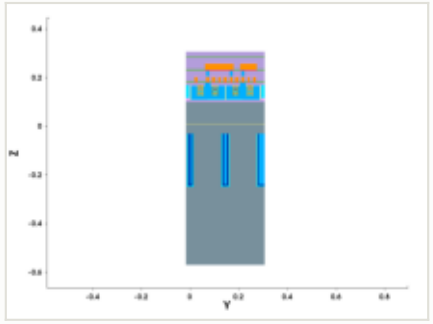
carbon so far, of the whole flow (kgCO2e)



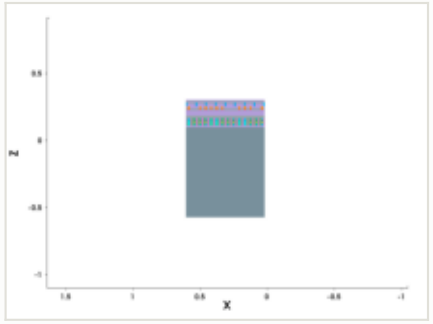
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



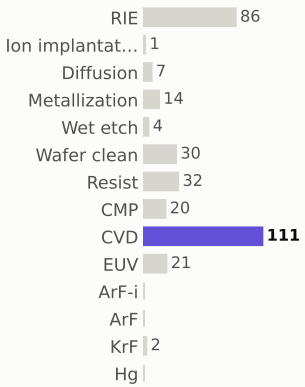
y cut at 0.184 µm



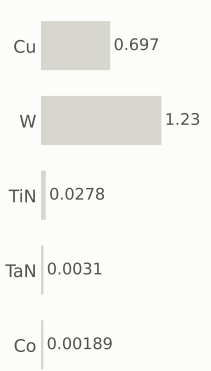
mask for this step

no mask for this step

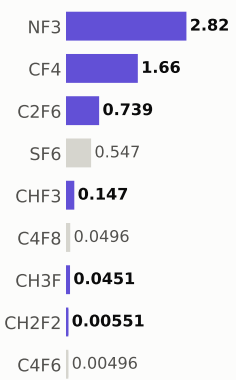
steps by process area



materials deposited (g)



gases emitted (g)



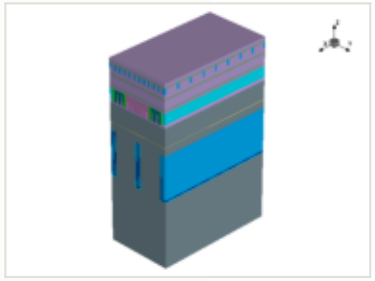
Step #3500: Diffusion

treatment · Diffusion

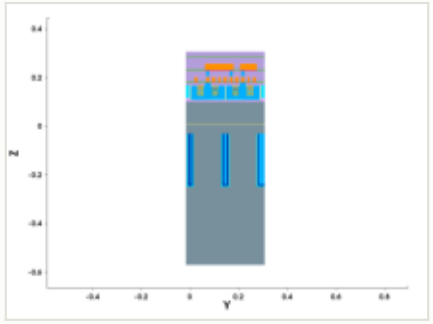
0.801	367.2	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

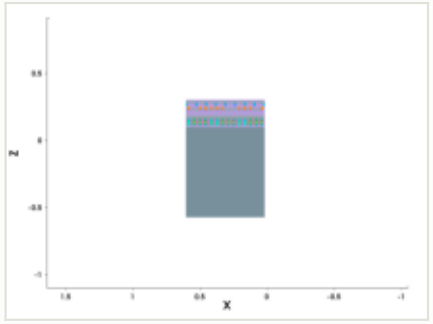
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

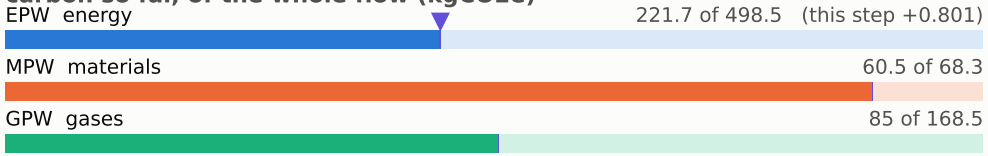
no mask for this step

this step

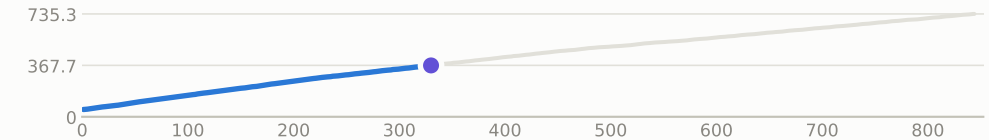
process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

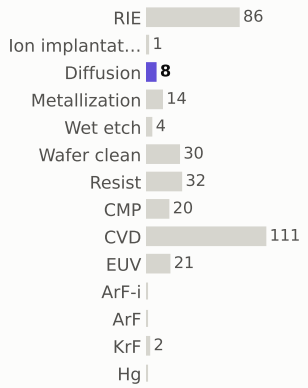
carbon so far, of the whole flow (kgCO2e)



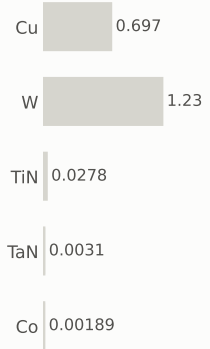
embodied carbon up to each step (kgCO2e per wafer)



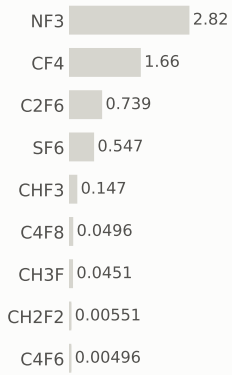
steps by process area



materials deposited (g)



gases emitted (g)



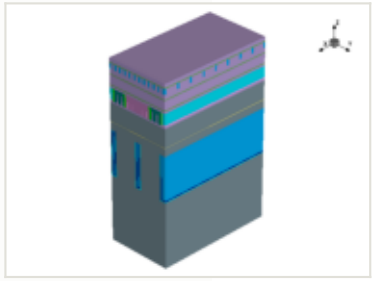
Step #3510: Wafer_clean

clean · Wafer_clean

0.169	367.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

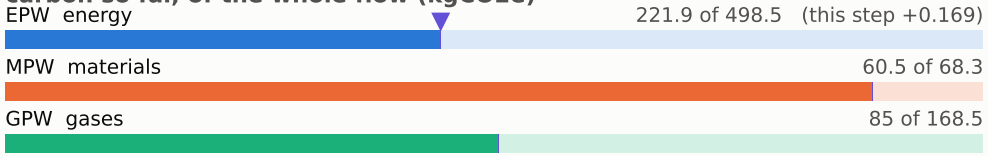


this step

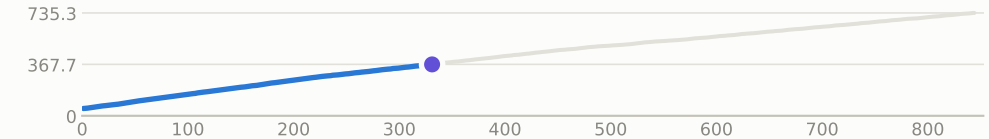
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

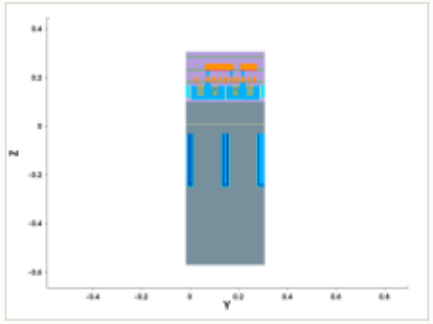
carbon so far, of the whole flow (kgCO2e)



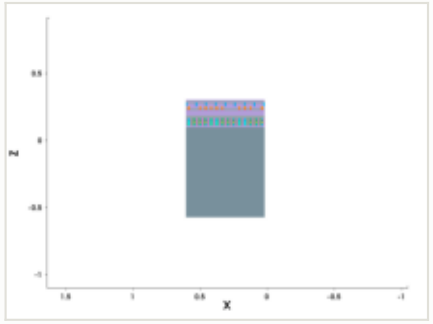
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



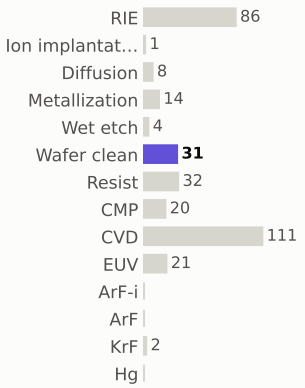
y cut at 0.184 µm



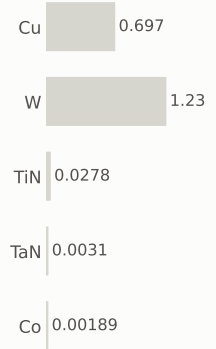
mask for this step

no mask for this step

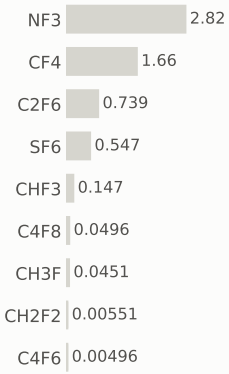
steps by process area



materials deposited (g)



gases emitted (g)



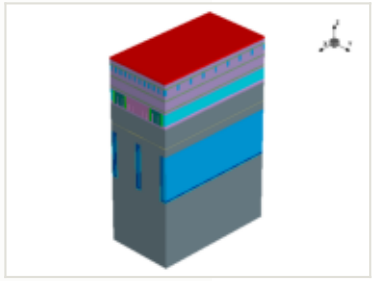
Step #3520: CVD

depo_fill · CVD

1.31	368.7	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

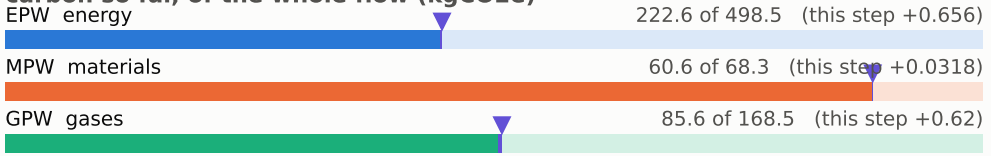


this step

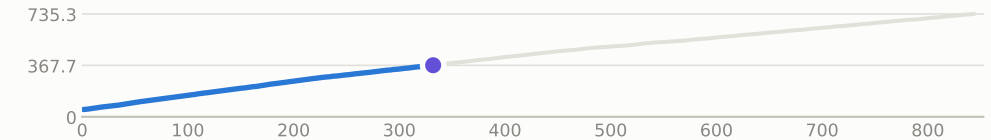
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

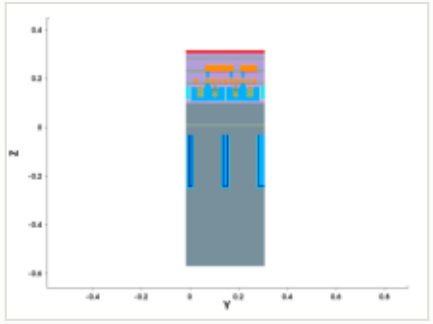
carbon so far, of the whole flow (kgCO2e)



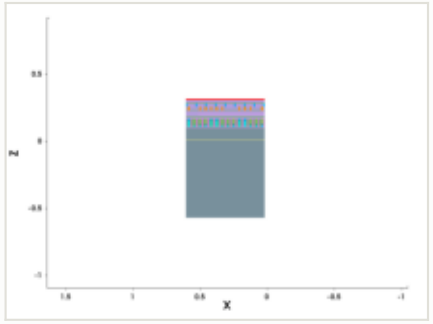
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



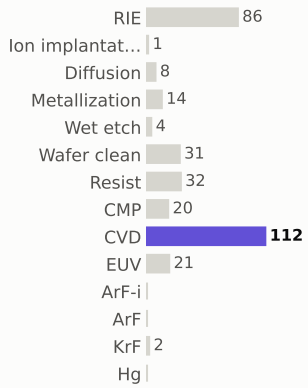
y cut at 0.184 µm



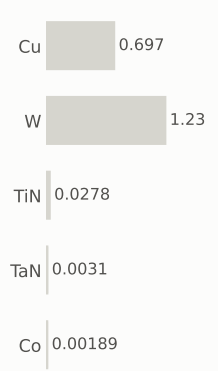
mask for this step

no mask for this step

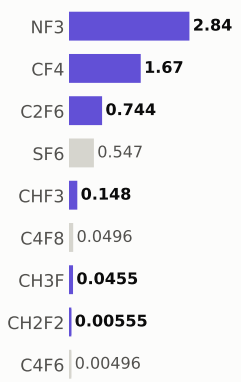
steps by process area



materials deposited (g)



gases emitted (g)



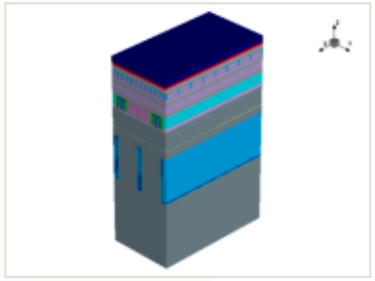
Step #3530: CVD

depo_fill · CVD

1.31	370.0	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

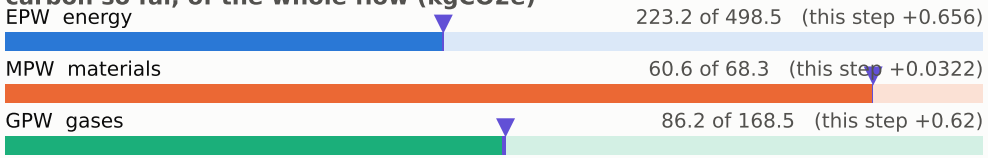


this step

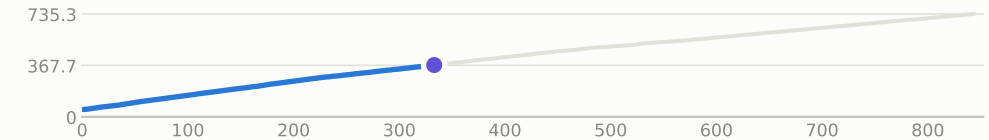
process area	CVD
material	TiN
thickness	0.03 µm
material GWP	43.9778 gCO2e/g
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0322 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

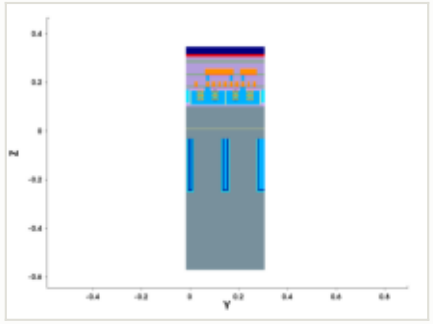
carbon so far, of the whole flow (kgCO2e)



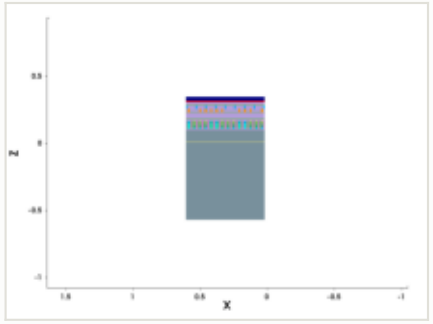
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



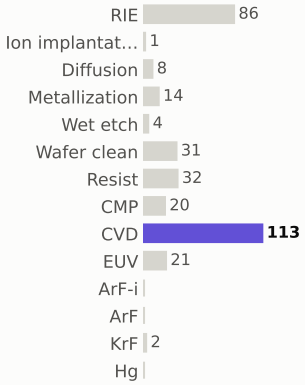
y cut at 0.184 µm



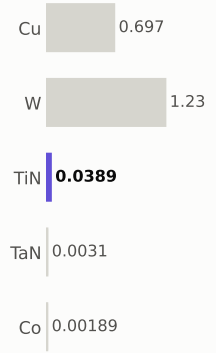
mask for this step

no mask for this step

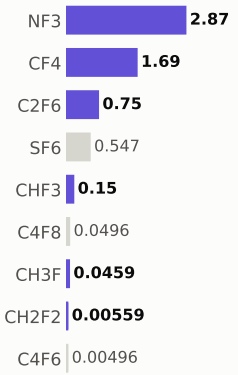
steps by process area



materials deposited (g)



gases emitted (g)



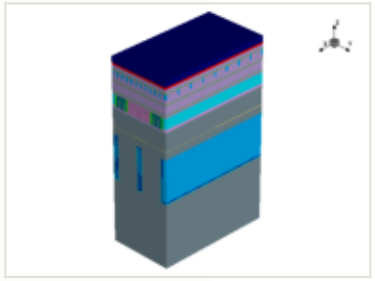
Step #3540: Wafer_clean

clean · Wafer_clean

0.169	370.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



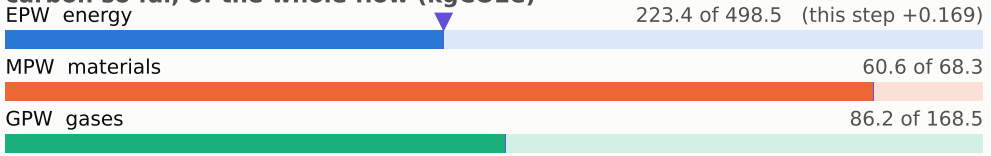
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

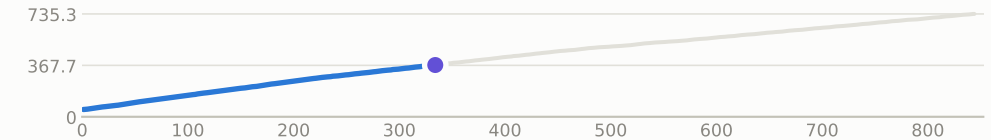
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

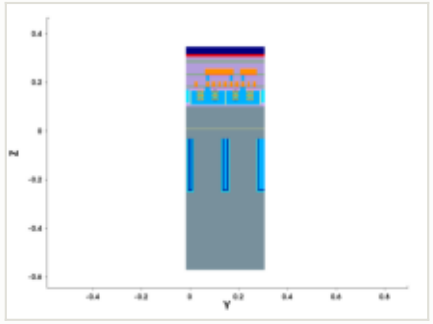
carbon so far, of the whole flow (kgCO2e)



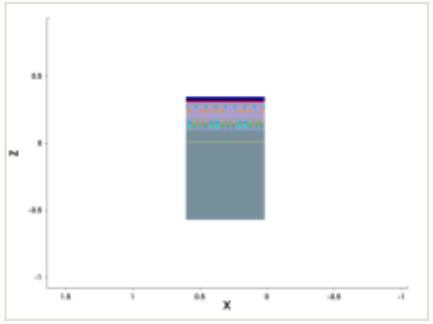
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



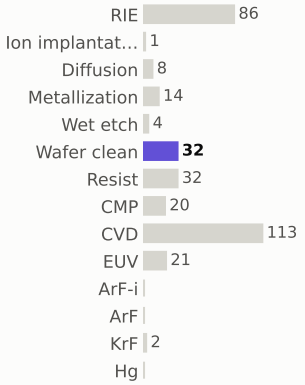
y cut at 0.184 µm



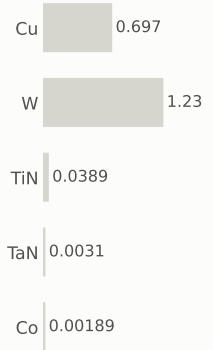
mask for this step

no mask for this step

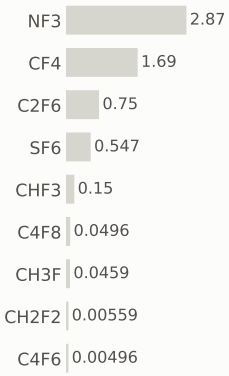
steps by process area



materials deposited (g)



gases emitted (g)



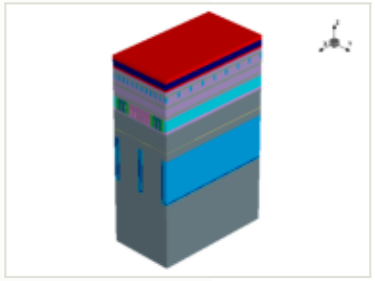
Step #3550: CVD

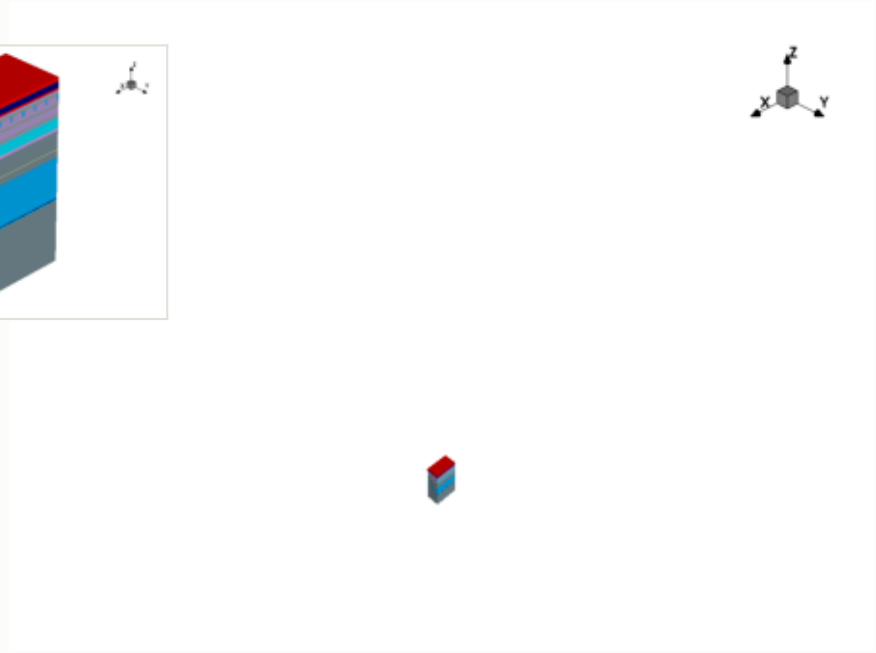
depo_fill · CVD

1.31	371.5	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

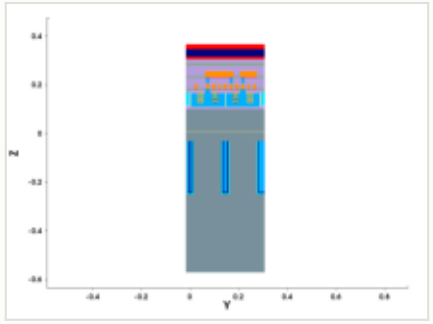
3D structure after this step

zoomed to fit

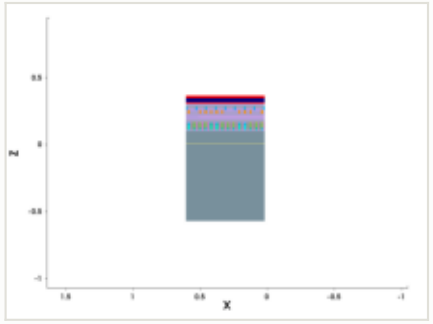




x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

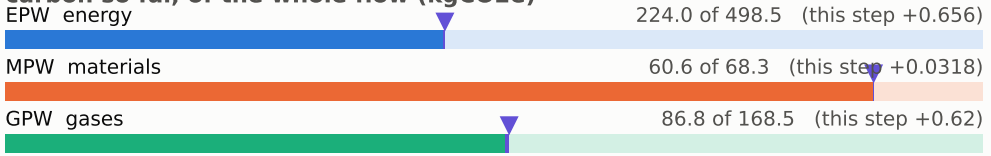
no mask for this step

this step

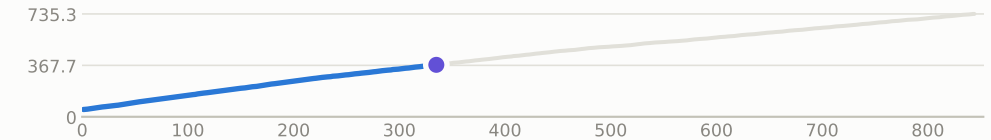
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

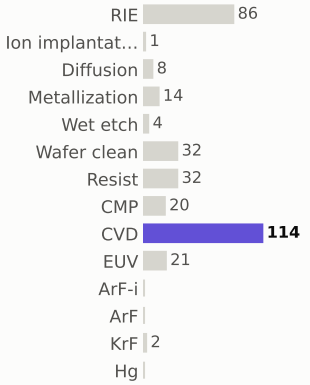
carbon so far, of the whole flow (kgCO2e)



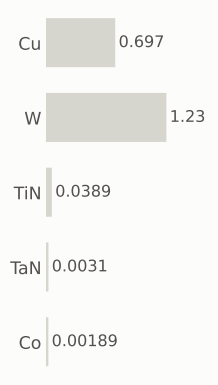
embodied carbon up to each step (kgCO2e per wafer)



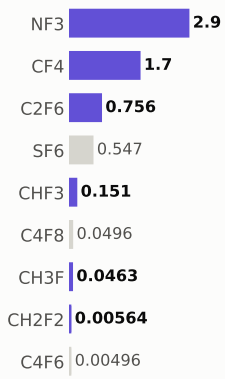
steps by process area



materials deposited (g)



gases emitted (g)



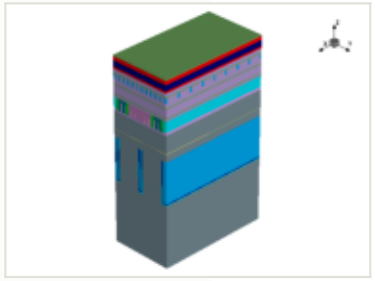
Step #3560: CVD

depo_fill · CVD

1.31	372.8	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

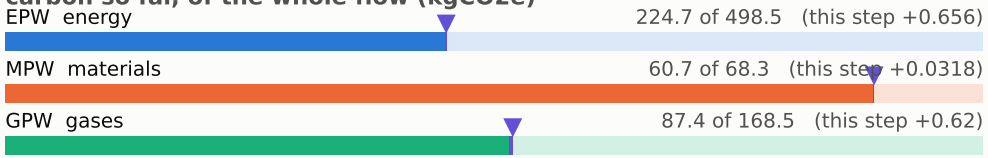
zoomed to fit



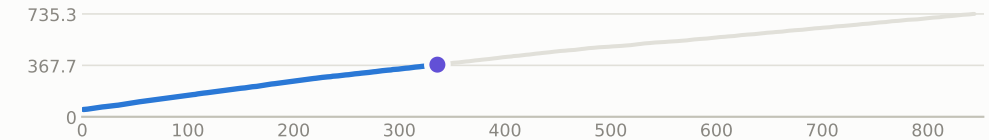
this step

process area	CVD	EPW	0.656 kgCO2e
material	—	EPW tool	0.469 kgCO2e
thickness	0.005 µm	EPW overhead	0.187 kgCO2e
material GWP	—	MPW	0.0318 kgCO2e (gas 0.0318)
input gases	NF3 2.54 g	GPW	0.62 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**1.31 kgCO2e**

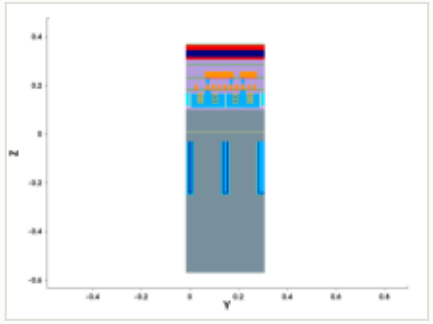
carbon so far, of the whole flow (kgCO2e)



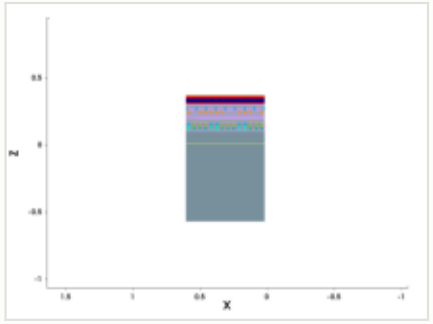
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



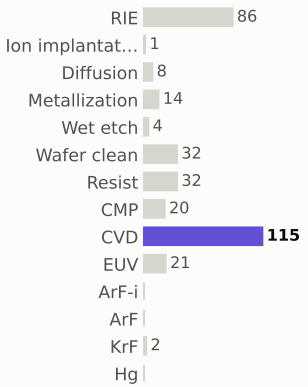
y cut at 0.184 µm



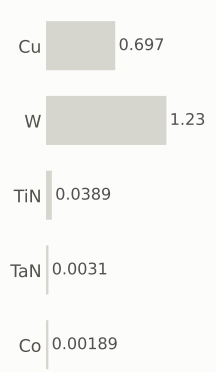
mask for this step

no mask for this step

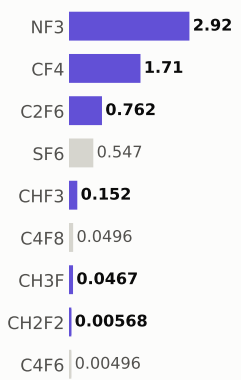
steps by process area



materials deposited (g)



gases emitted (g)



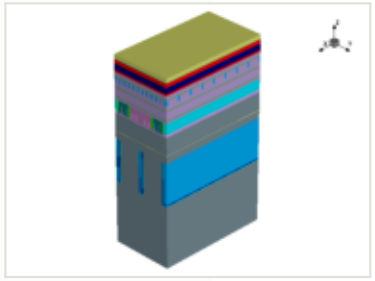
Step #3570: CVD

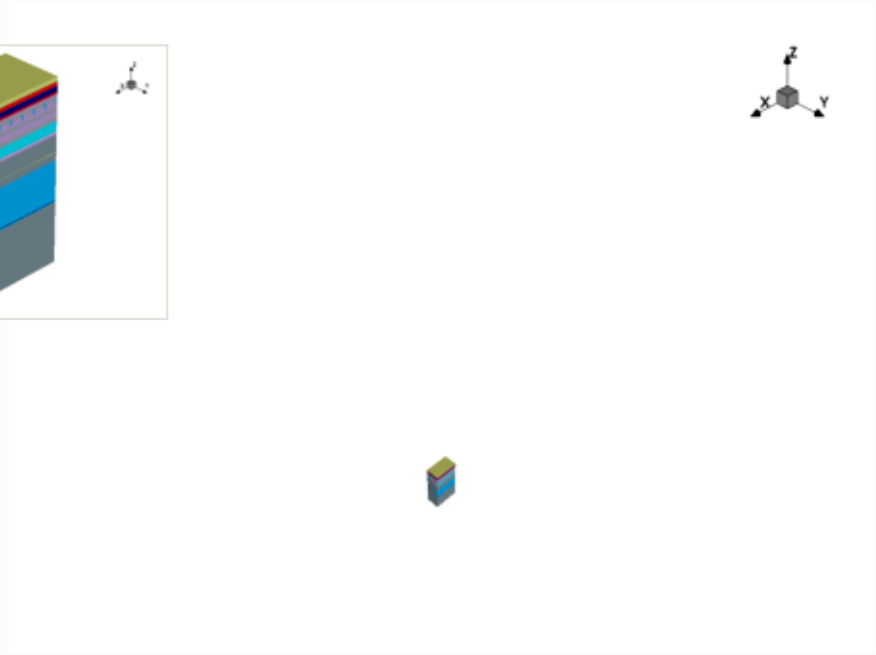
depo_fill · CVD

1.31	374.1	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



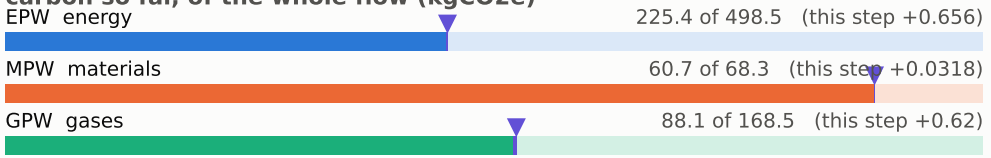


this step

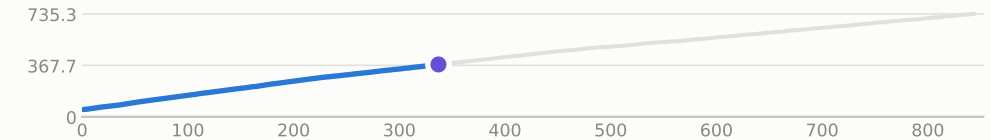
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

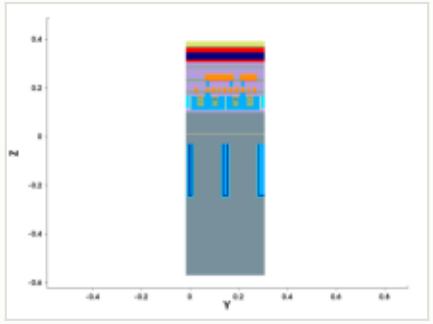
carbon so far, of the whole flow (kgCO2e)



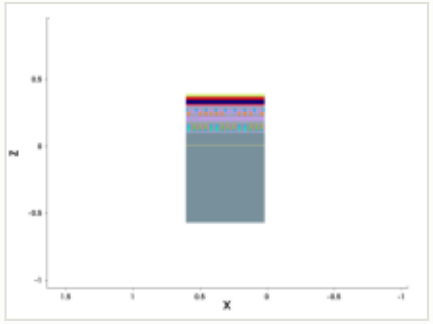
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



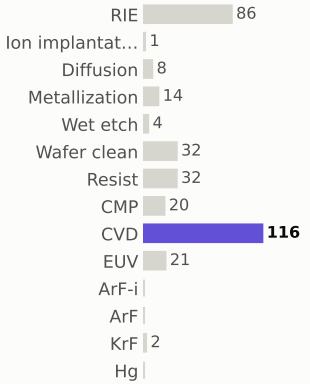
y cut at 0.184 µm



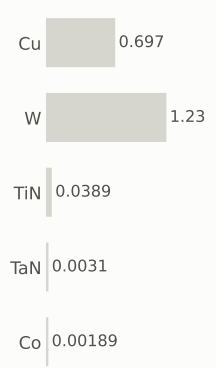
mask for this step

no mask for this step

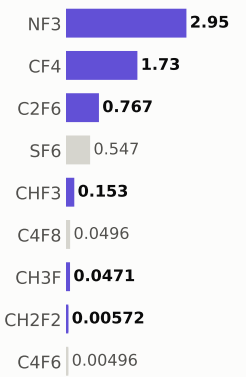
steps by process area



materials deposited (g)



gases emitted (g)



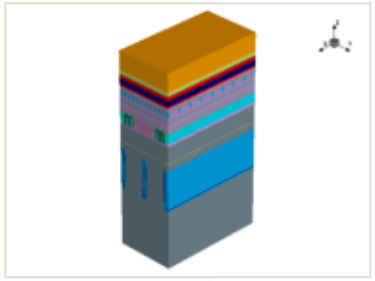
Step #3580: Resist

depo_fill · Resist

0.169	374.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

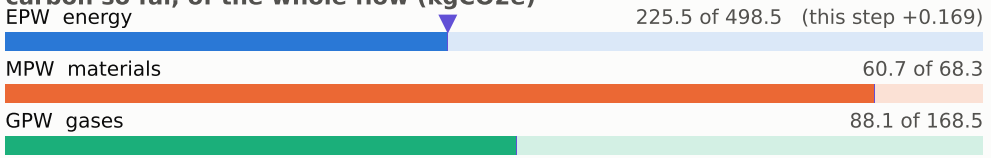


this step

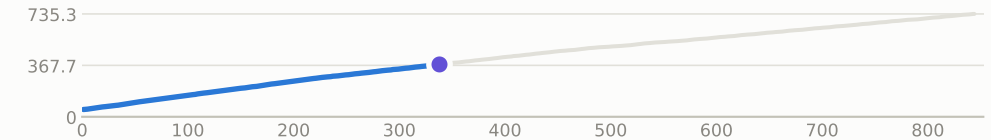
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

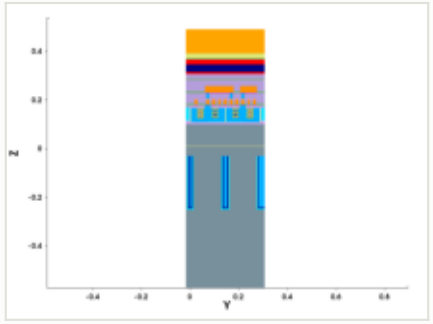
carbon so far, of the whole flow (kgCO2e)



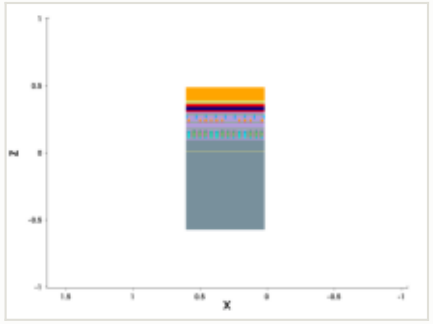
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



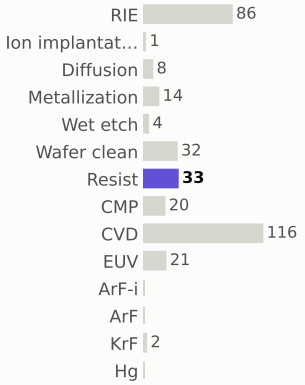
y cut at 0.184 µm



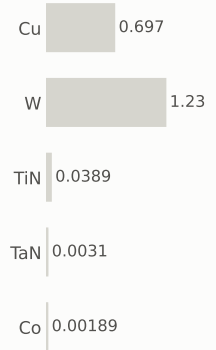
mask for this step

no mask for this step

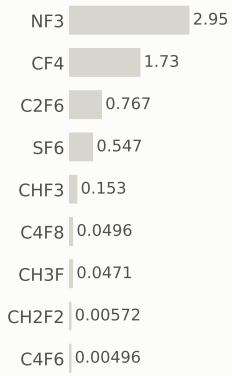
steps by process area



materials deposited (g)



gases emitted (g)



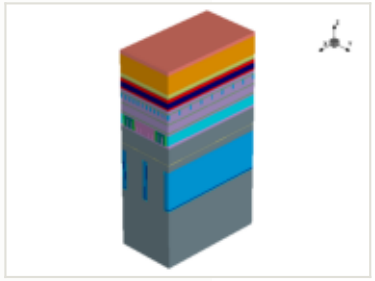
Step #3590: CVD

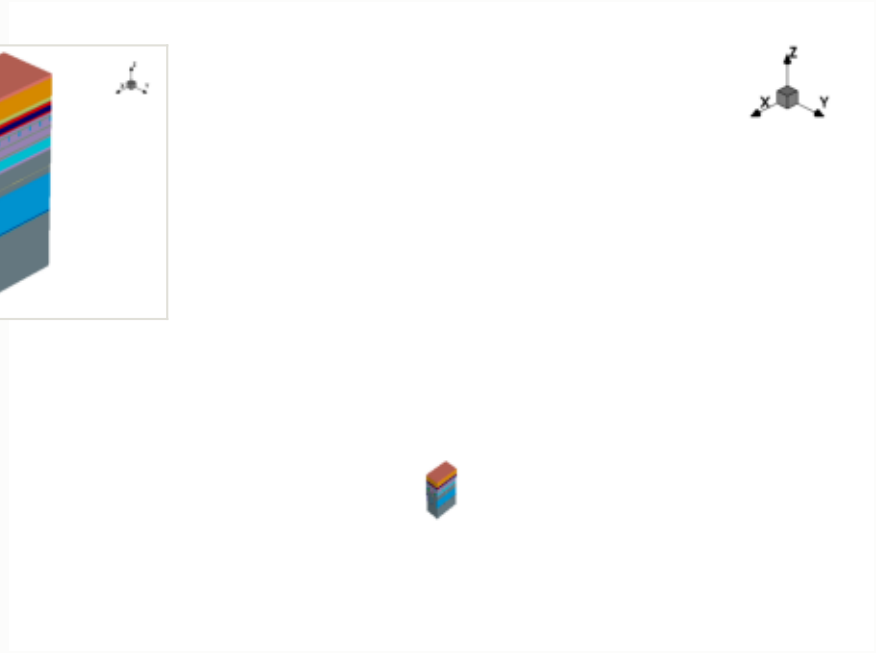
depo_fill · CVD

1.31	375.6	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



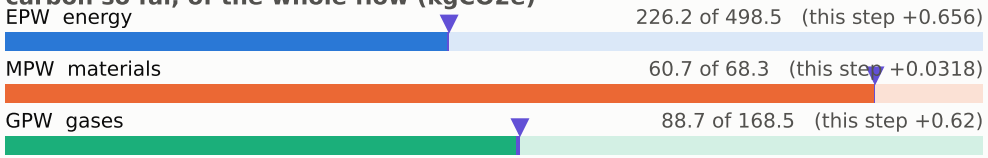


this step

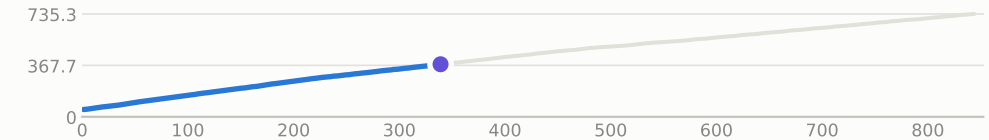
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

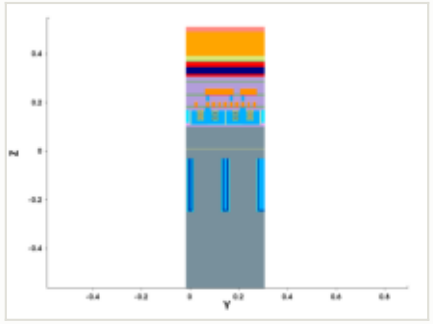
carbon so far, of the whole flow (kgCO2e)



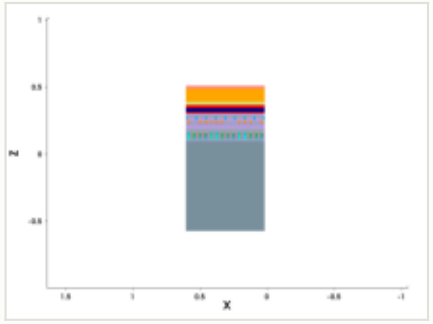
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



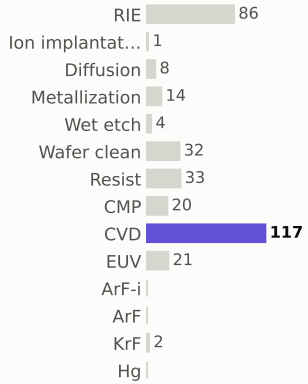
y cut at 0.184 µm



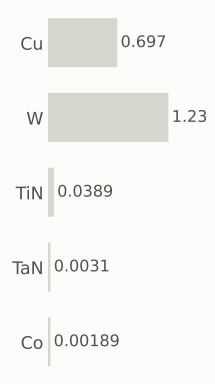
mask for this step

no mask for this step

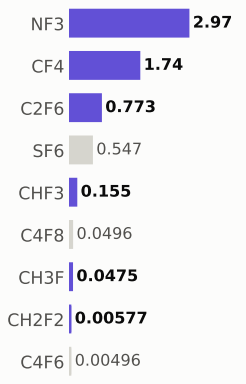
steps by process area



materials deposited (g)



gases emitted (g)



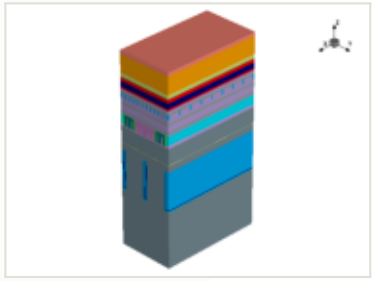
Step #3600: Wafer_clean

clean · Wafer_clean

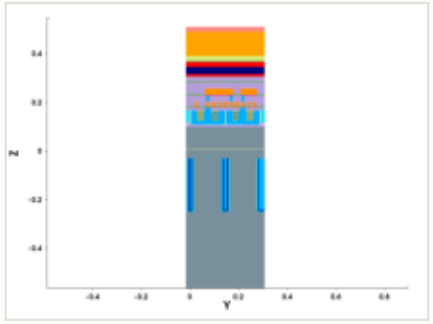
0.169	375.7	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

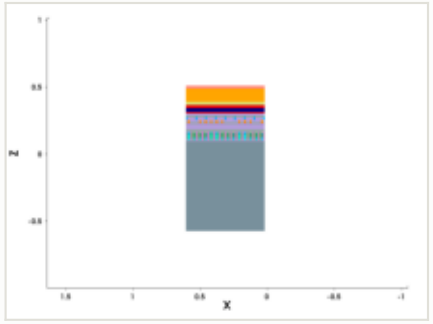
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

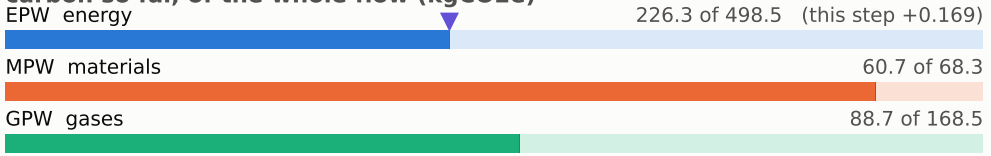
no mask for this step

this step

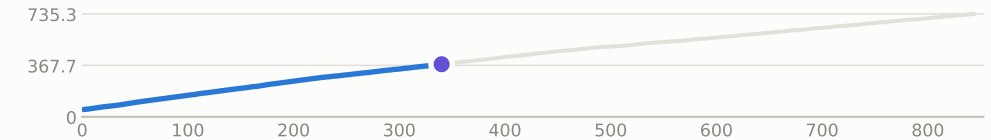
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

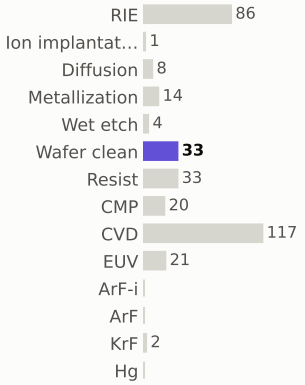
carbon so far, of the whole flow (kgCO2e)



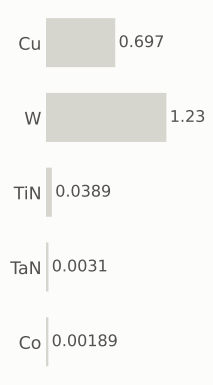
embodied carbon up to each step (kgCO2e per wafer)



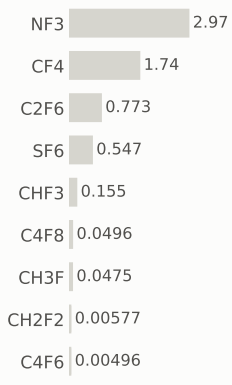
steps by process area



materials deposited (g)



gases emitted (g)



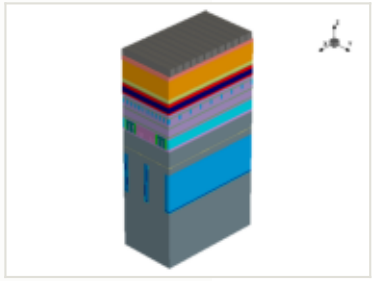
Step #3610: EUV

litho_patterning · EUV

1.96	377.7	0.27%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

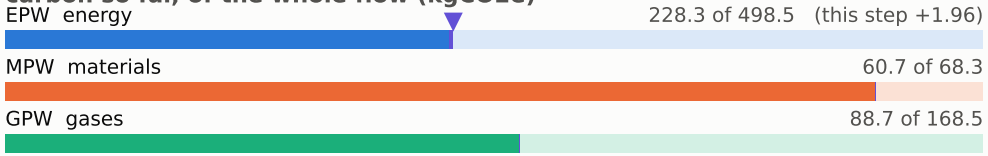


this step

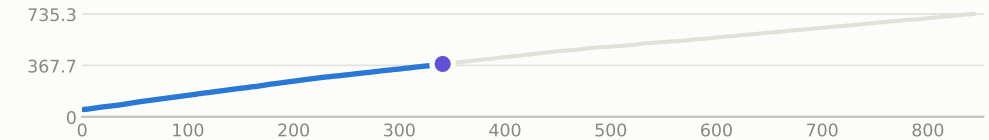
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

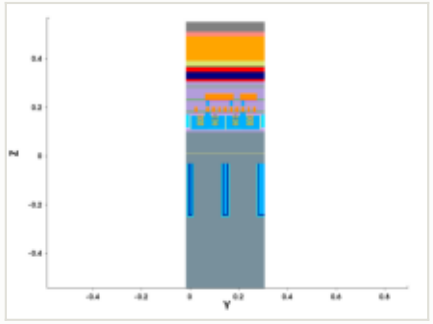
carbon so far, of the whole flow (kgCO2e)



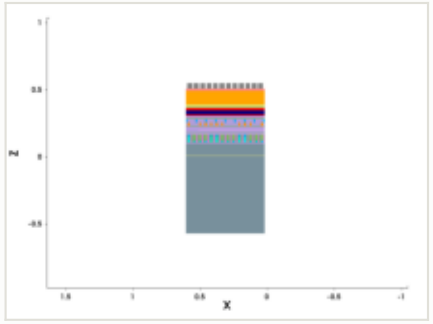
embodied carbon up to each step (kgCO2e per wafer)



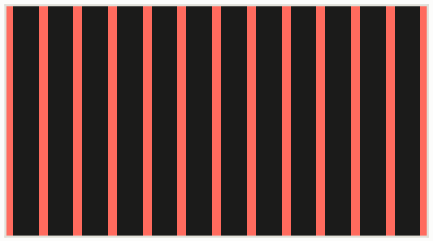
x cut at 0.294 µm



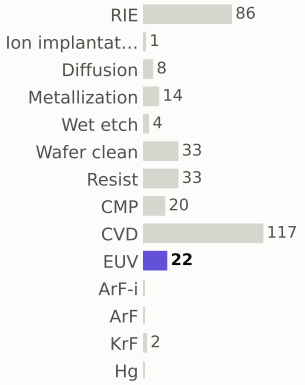
y cut at 0.184 µm



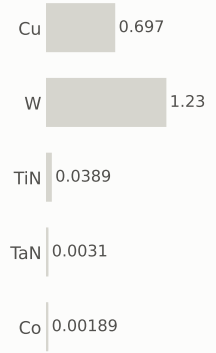
mask: layer 10301:0 (GDSII)



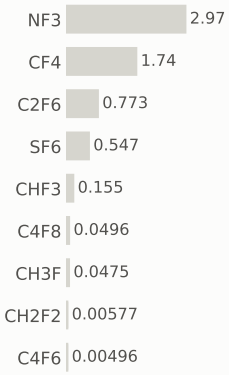
steps by process area



materials deposited (g)



gases emitted (g)



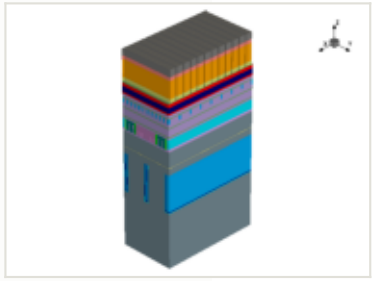
Step #3620: RIE

etch_anisotropic · RIE

1.04	378.7	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

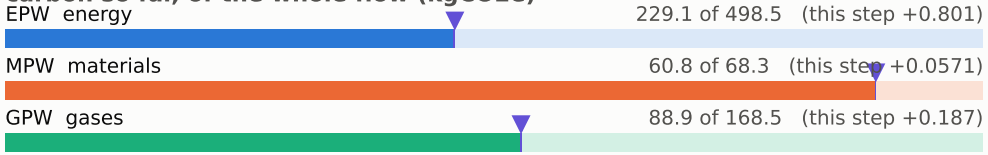


this step

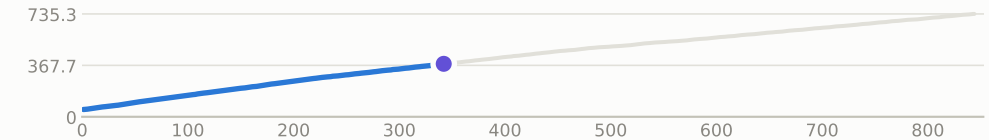
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

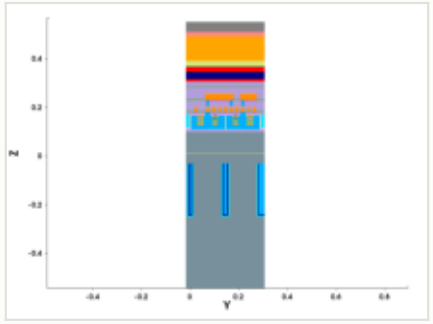
carbon so far, of the whole flow (kgCO2e)



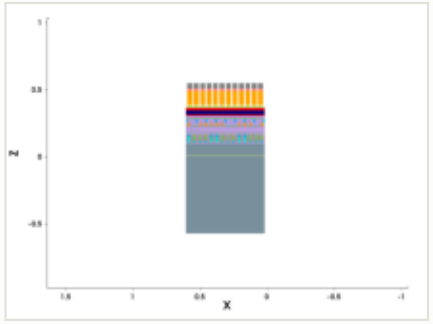
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



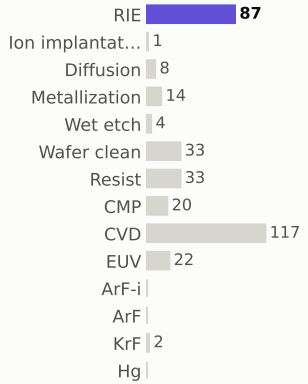
y cut at 0.184 µm



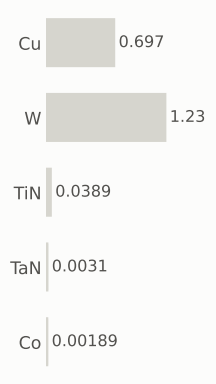
mask for this step

no mask for this step

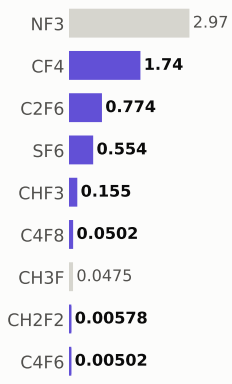
steps by process area



materials deposited (g)



gases emitted (g)



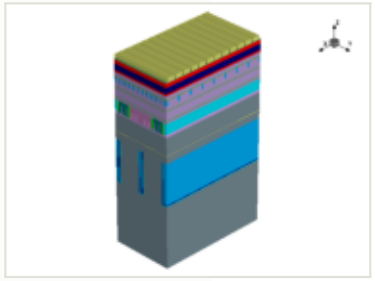
Step #3630: Wafer_clean

etch_strip · Wafer_clean

0.169	378.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

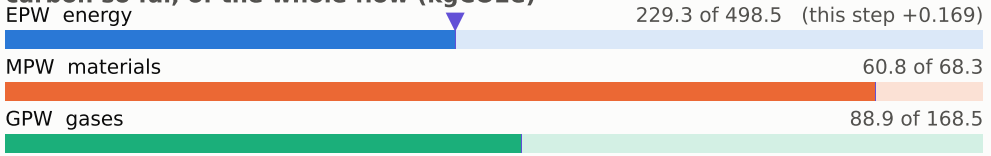


this step

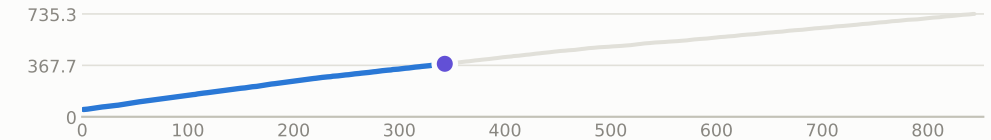
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

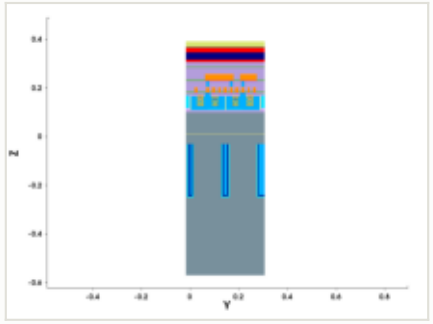
carbon so far, of the whole flow (kgCO2e)



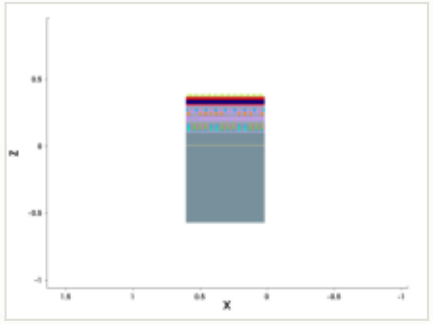
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



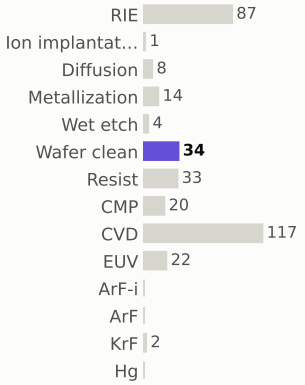
y cut at 0.184 µm



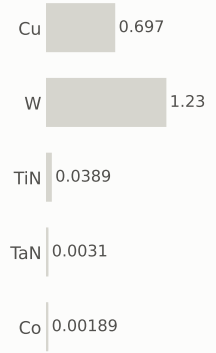
mask for this step

no mask for this step

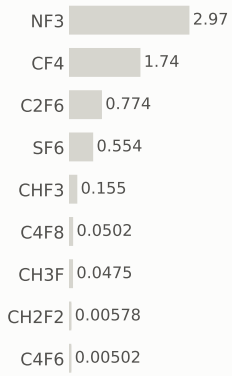
steps by process area



materials deposited (g)



gases emitted (g)



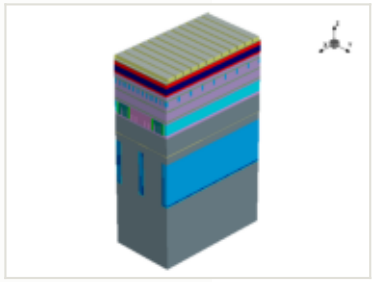
Step #3640: CVD

depo_isotropic · CVD

1.31	380.2	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

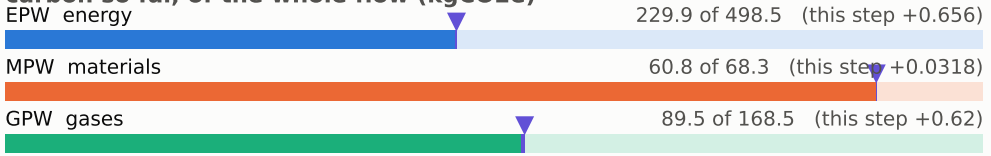


this step

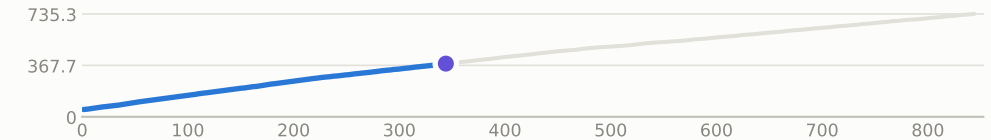
process area	CVD
material	—
thickness	0.001 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

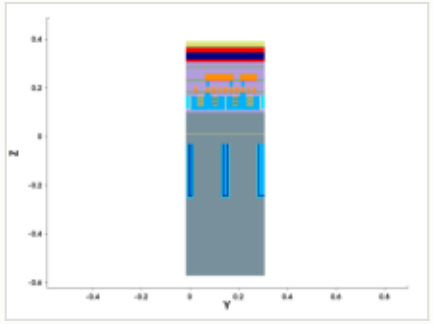
carbon so far, of the whole flow (kgCO2e)



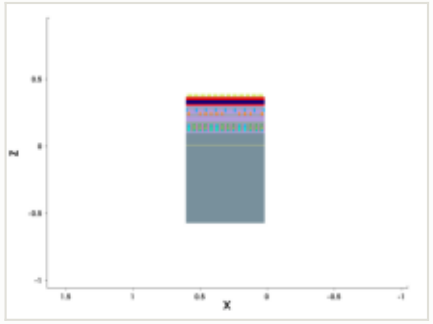
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



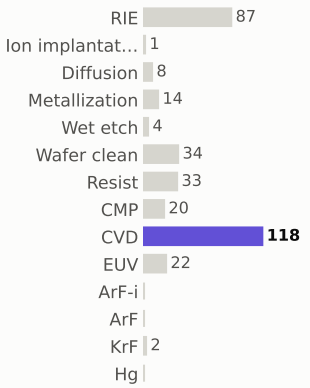
y cut at 0.184 µm



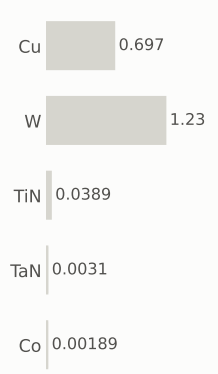
mask for this step

no mask for this step

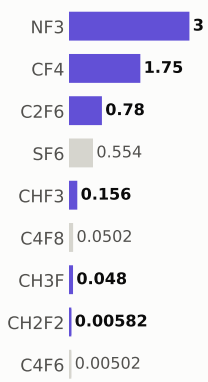
steps by process area



materials deposited (g)



gases emitted (g)



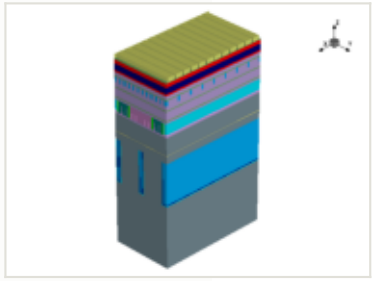
Step #3650: RIE

etch_anisotropic · RIE

1.04	381.3	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

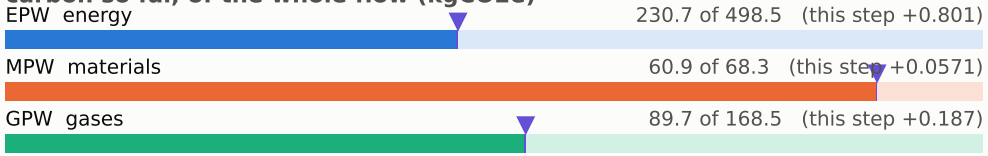


this step

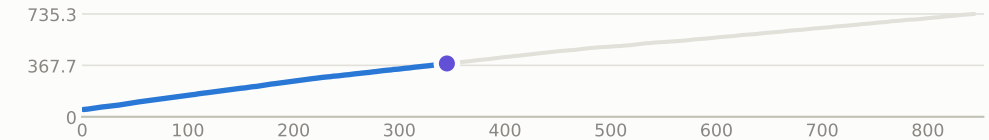
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

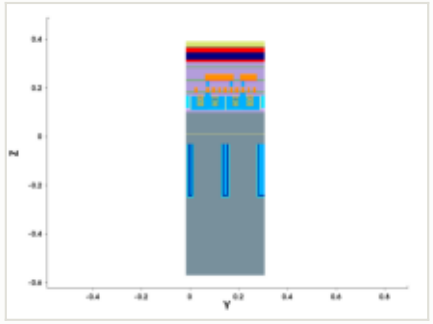
carbon so far, of the whole flow (kgCO2e)



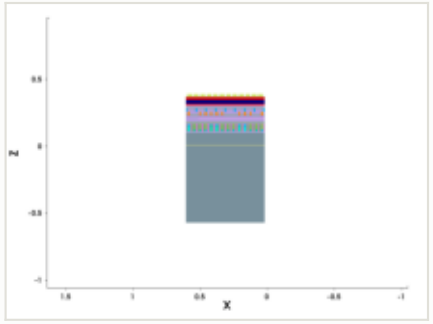
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



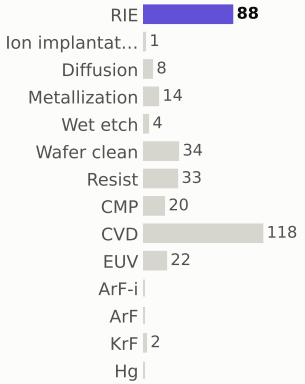
y cut at 0.184 µm



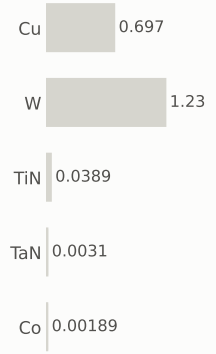
mask for this step

no mask for this step

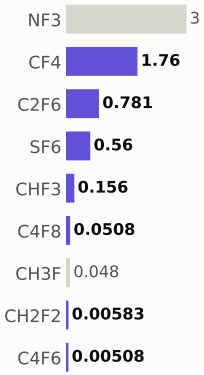
steps by process area



materials deposited (g)



gases emitted (g)



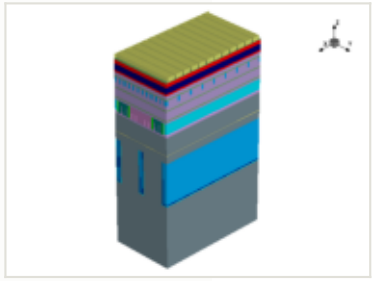
Step #3660: Wafer_clean

clean · Wafer_clean

0.169	381.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



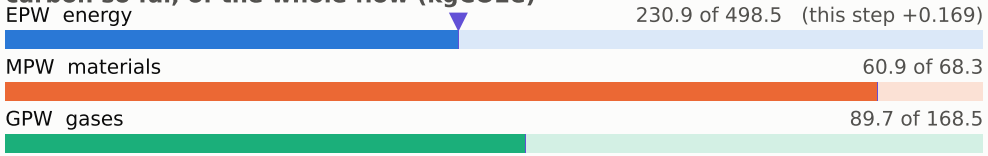
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

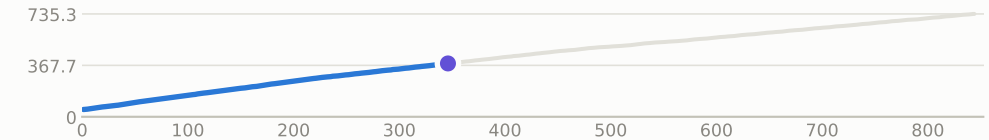
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

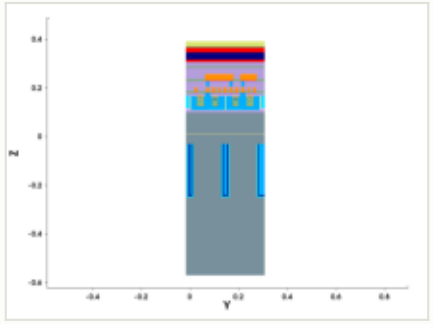
carbon so far, of the whole flow (kgCO2e)



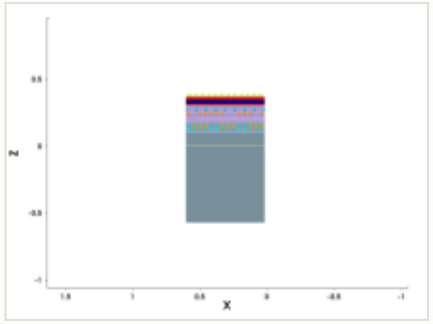
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



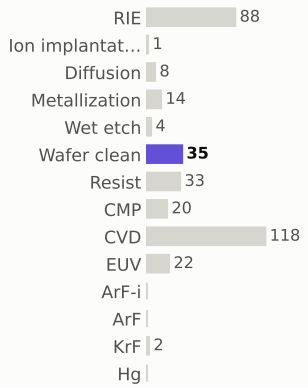
y cut at 0.184 µm



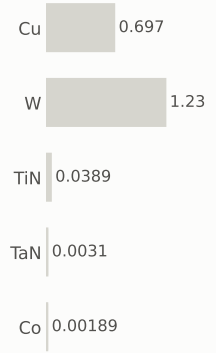
mask for this step

no mask for this step

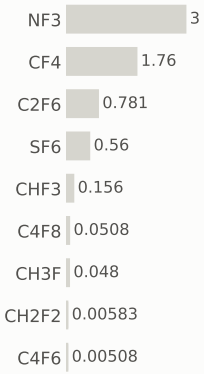
steps by process area



materials deposited (g)



gases emitted (g)



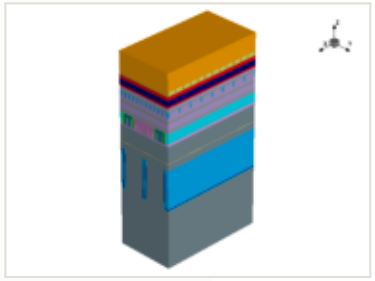
Step #3670: Resist

depo_fill · Resist

0.169	381.6	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

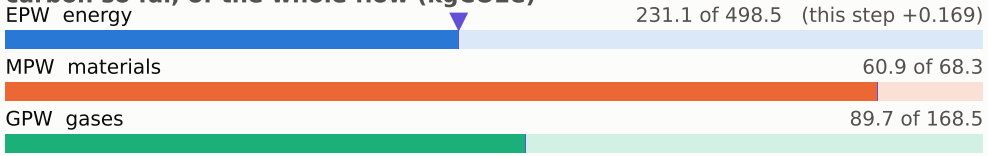


this step

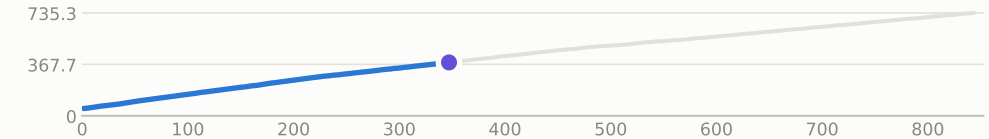
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

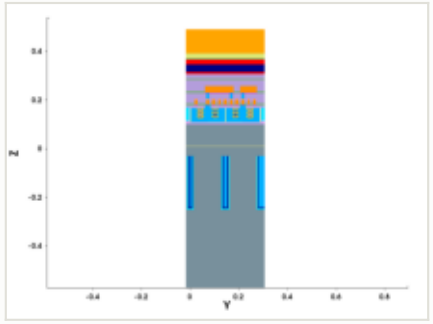
carbon so far, of the whole flow (kgCO2e)



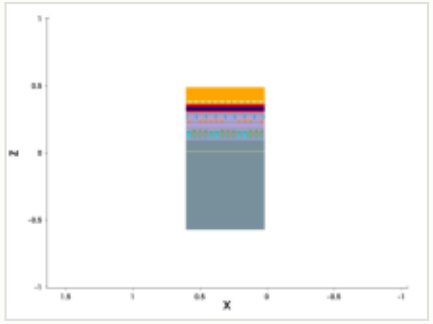
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



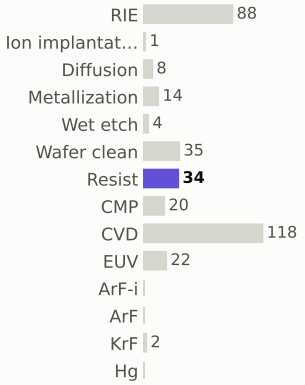
y cut at 0.184 µm



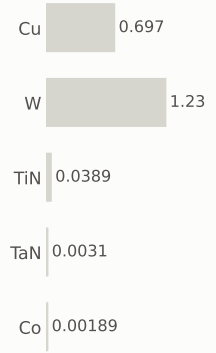
mask for this step

no mask for this step

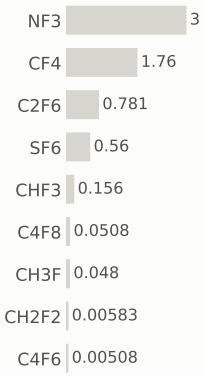
steps by process area



materials deposited (g)



gases emitted (g)



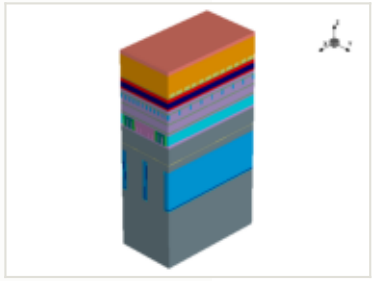
Step #3680: CVD

depo_fill · CVD

1.31	382.9	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

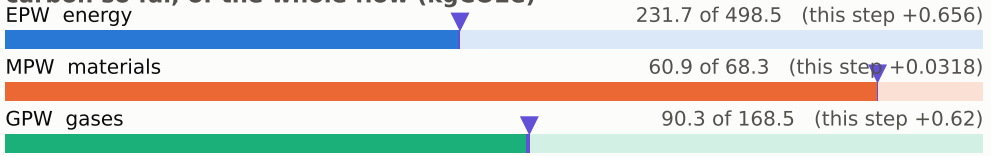


this step

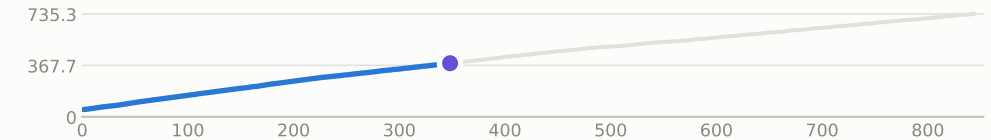
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

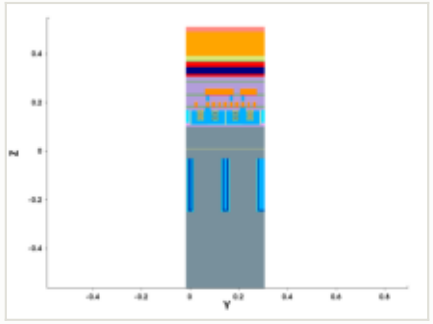
carbon so far, of the whole flow (kgCO2e)



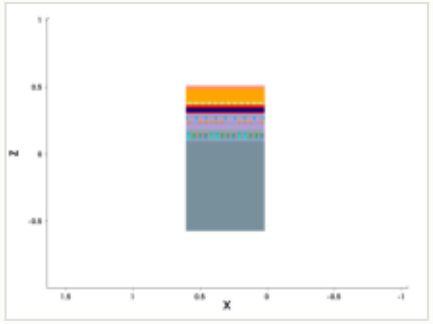
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



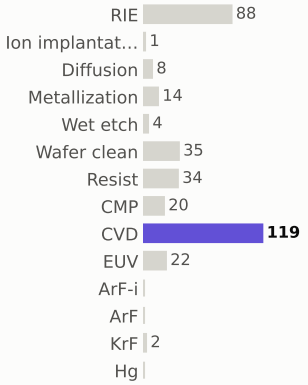
y cut at 0.184 µm



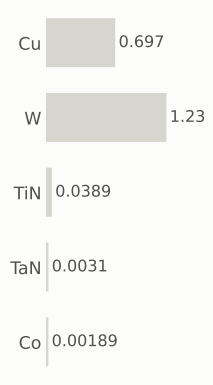
mask for this step

no mask for this step

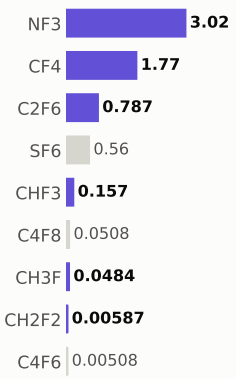
steps by process area



materials deposited (g)



gases emitted (g)



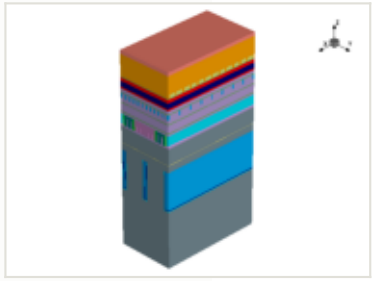
Step #3690: Wafer_clean

clean · Wafer_clean

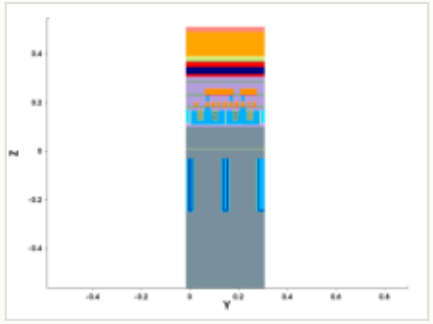
0.169	383.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

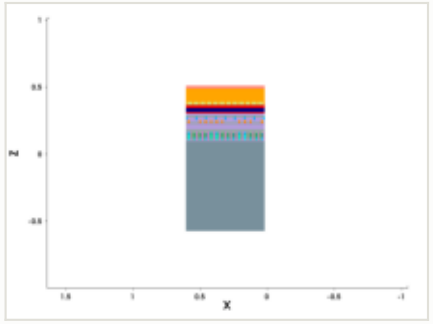
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

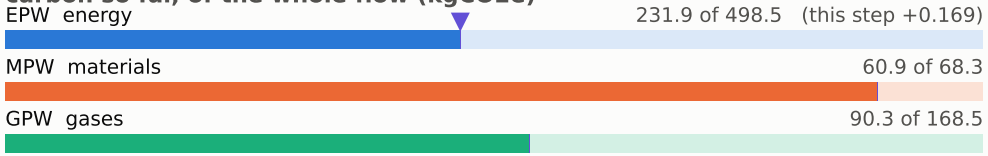
no mask for this step

this step

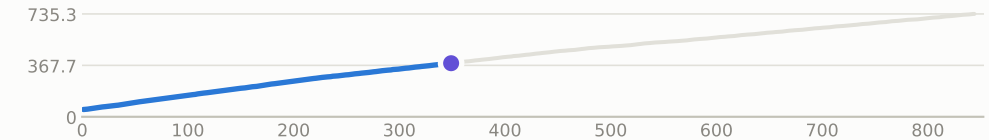
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

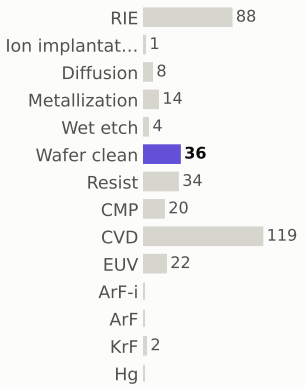
carbon so far, of the whole flow (kgCO2e)



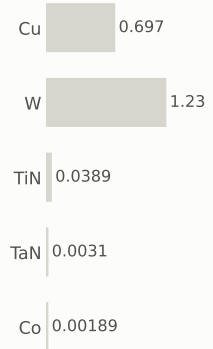
embodied carbon up to each step (kgCO2e per wafer)



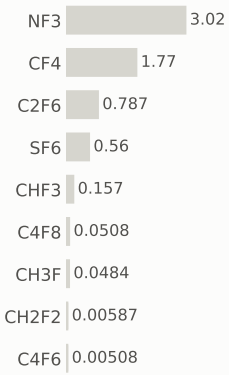
steps by process area



materials deposited (g)



gases emitted (g)



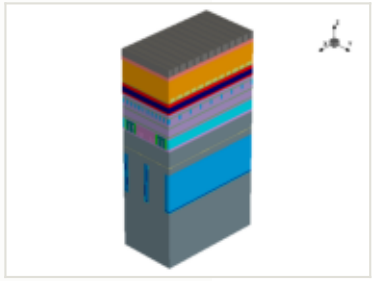
Step #3700: EUV

litho_patterning · EUV

1.96	385.0	0.27%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

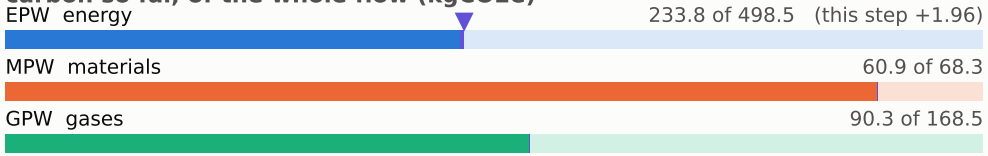


this step

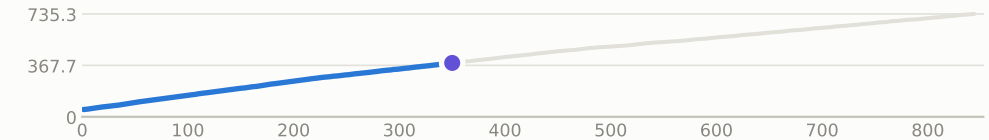
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

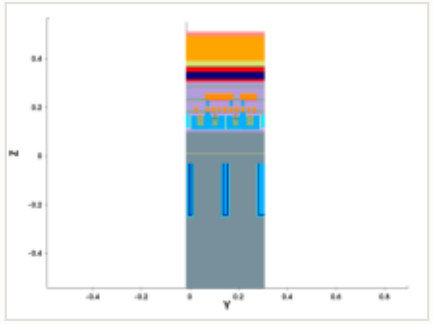
carbon so far, of the whole flow (kgCO2e)



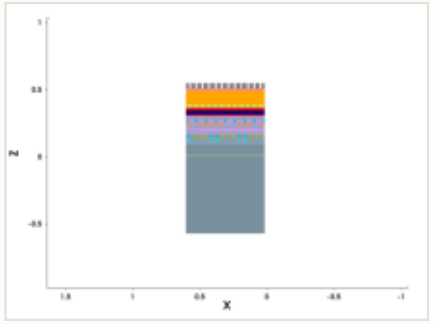
embodied carbon up to each step (kgCO2e per wafer)



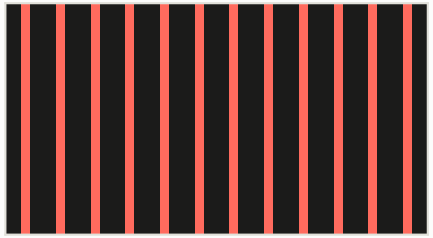
x cut at 0.294 µm



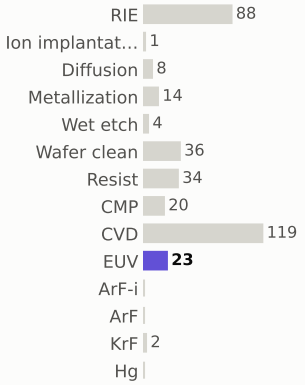
y cut at 0.184 µm



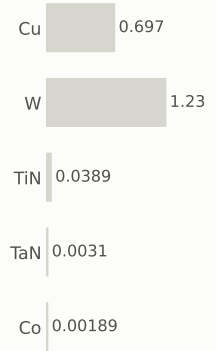
mask: layer 10302:0 (GDSII)



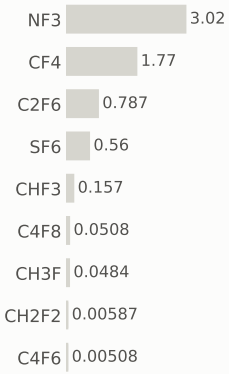
steps by process area



materials deposited (g)



gases emitted (g)



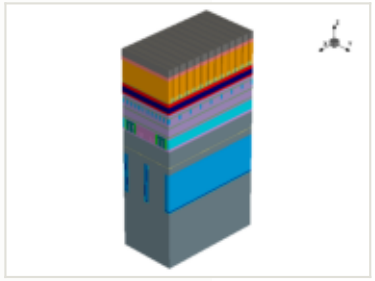
Step #3710: RIE

etch_anisotropic · RIE

1.04	386.1	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

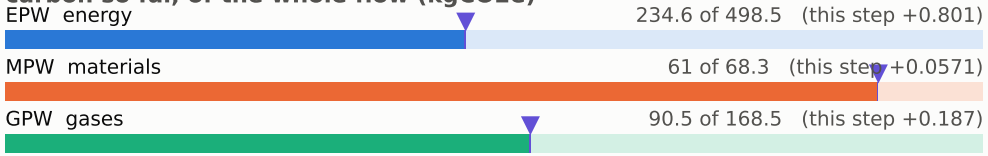


this step

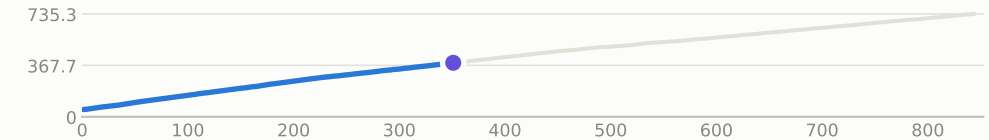
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

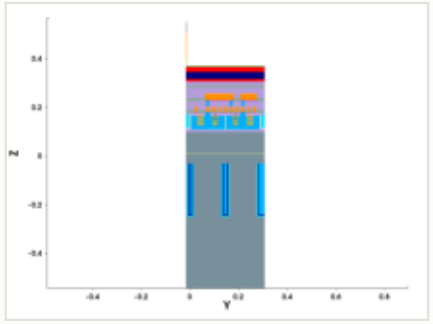
carbon so far, of the whole flow (kgCO2e)



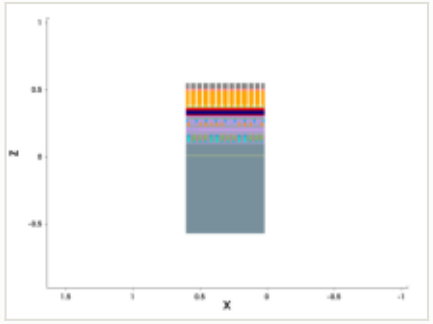
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



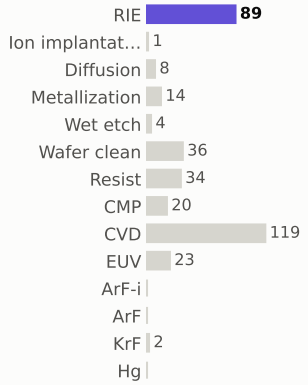
y cut at 0.184 µm



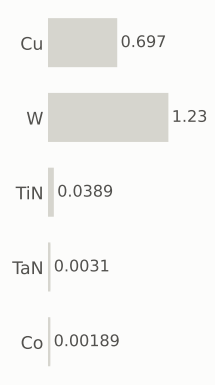
mask for this step

no mask for this step

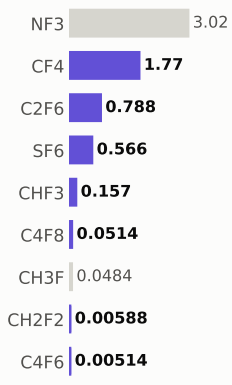
steps by process area



materials deposited (g)



gases emitted (g)



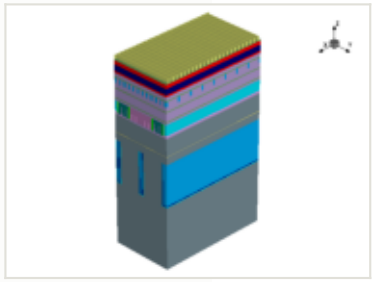
Step #3720: Wafer_clean

etch_strip · Wafer_clean

0.169	386.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

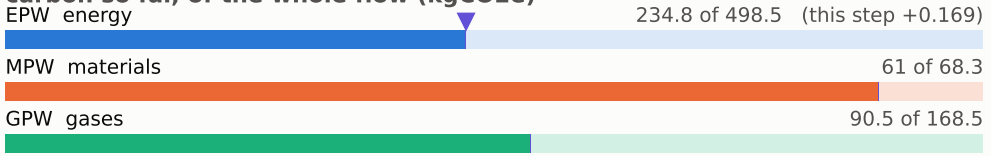


this step

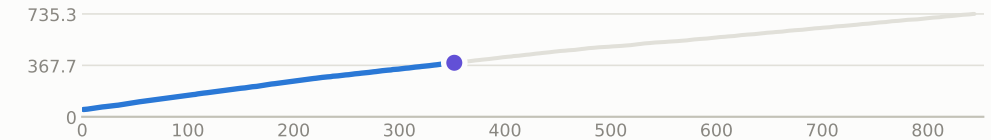
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

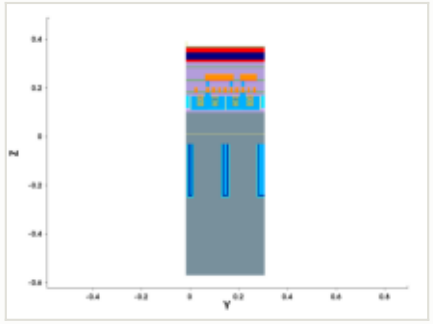
carbon so far, of the whole flow (kgCO2e)



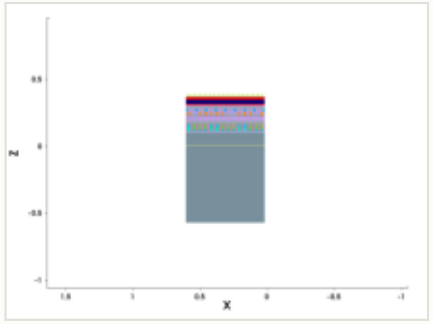
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



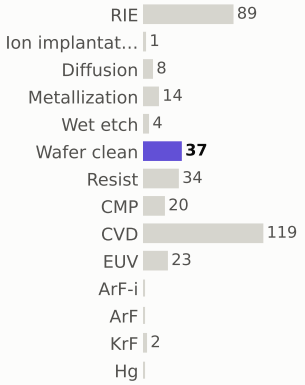
y cut at 0.184 µm



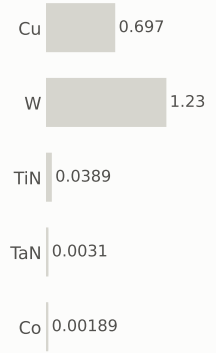
mask for this step

no mask for this step

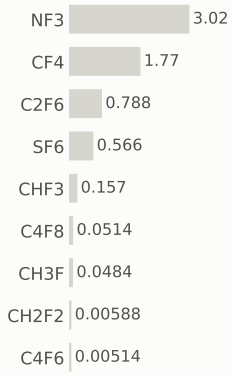
steps by process area



materials deposited (g)



gases emitted (g)



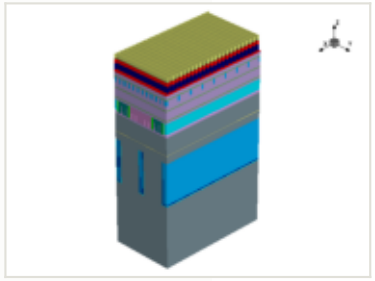
Step #3730: RIE

etch_anisotropic · RIE

1.04	387.3	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

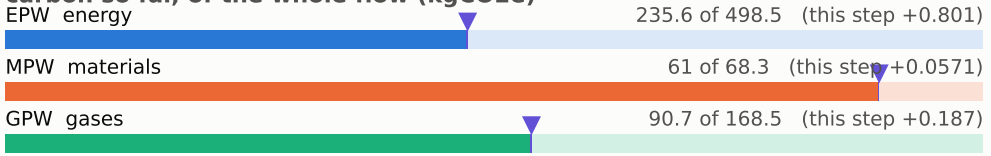


this step

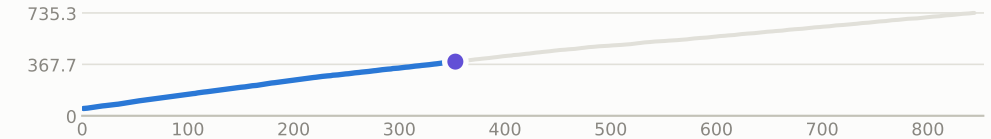
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

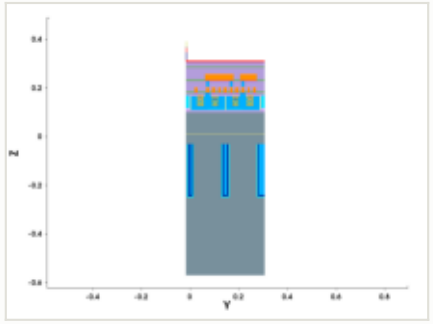
carbon so far, of the whole flow (kgCO2e)



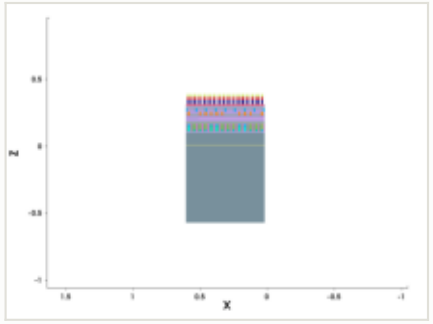
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



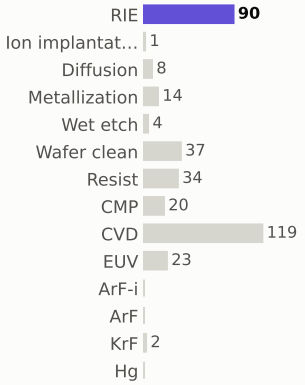
y cut at 0.184 µm



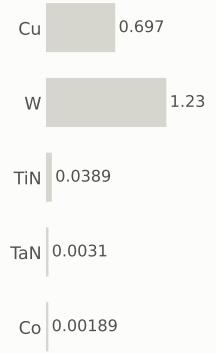
mask for this step

no mask for this step

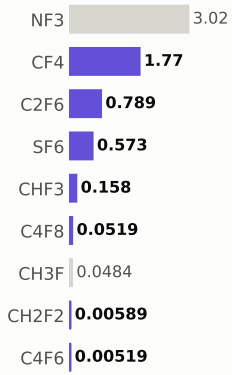
steps by process area



materials deposited (g)



gases emitted (g)



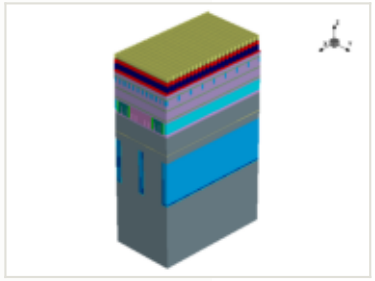
Step #3740: Wafer_clean

clean · Wafer_clean

0.169	387.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



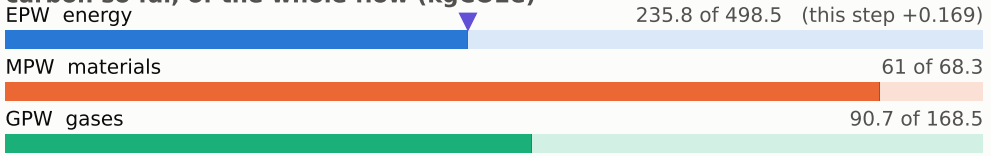
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

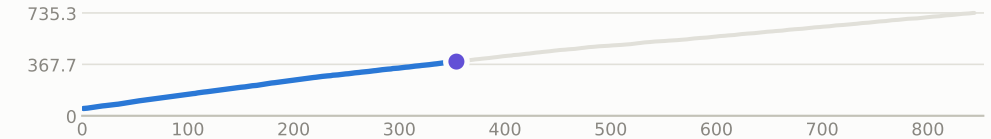
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

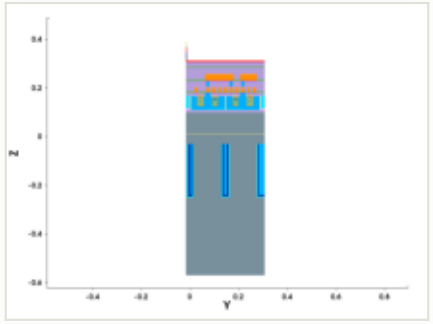
carbon so far, of the whole flow (kgCO2e)



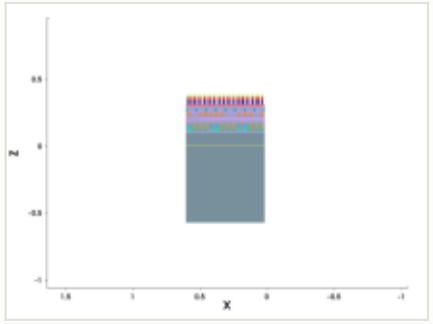
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



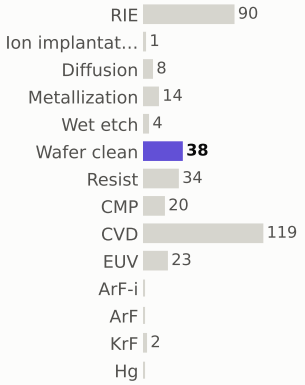
y cut at 0.184 µm



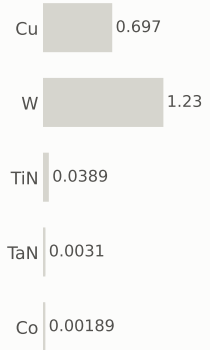
mask for this step

no mask for this step

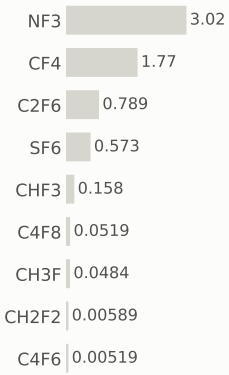
steps by process area



materials deposited (g)



gases emitted (g)



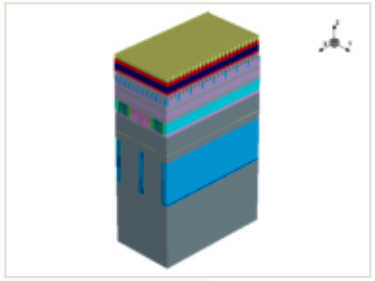
Step #3750: RIE

etch_anisotropic · RIE

1.04	388.5	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

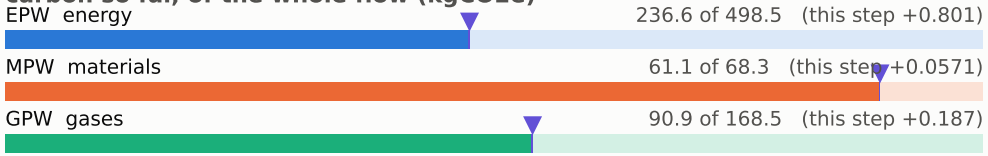


this step

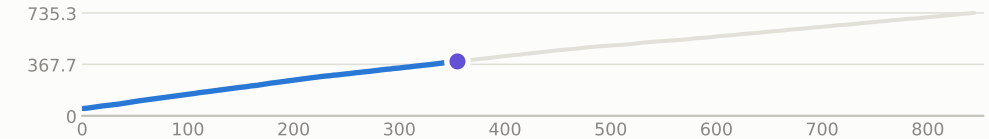
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

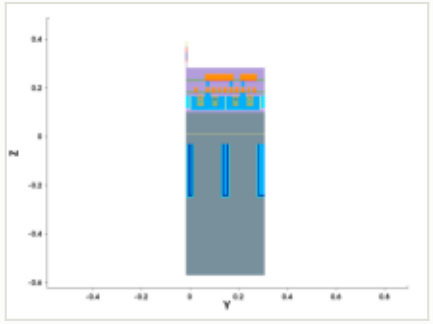
carbon so far, of the whole flow (kgCO2e)



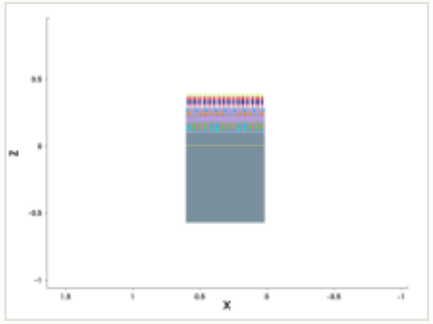
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



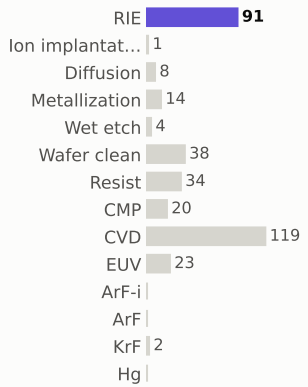
y cut at 0.184 µm



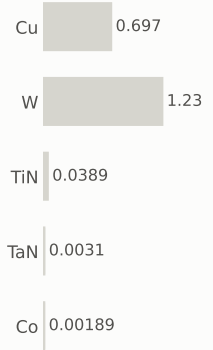
mask for this step

no mask for this step

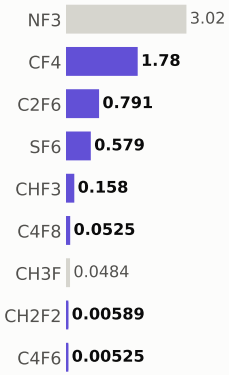
steps by process area



materials deposited (g)



gases emitted (g)



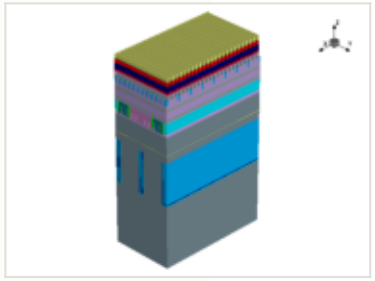
Step #3760: Wafer_clean

clean · Wafer_clean

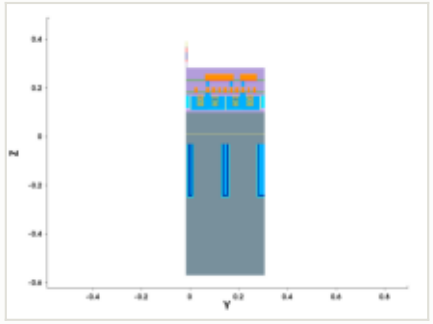
0.169	388.7	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

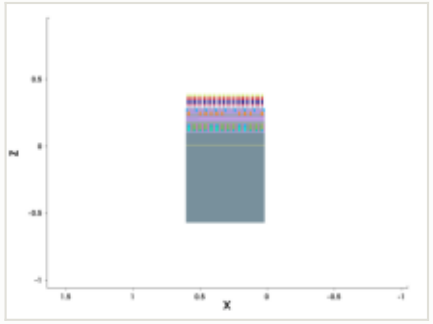
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

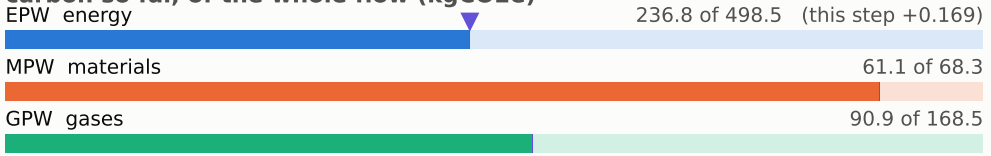
no mask for this step

this step

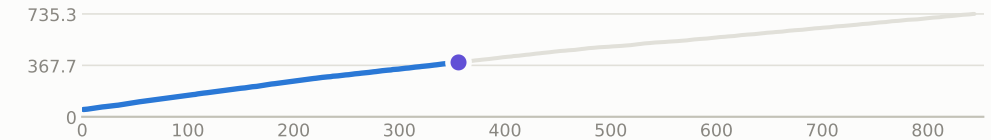
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

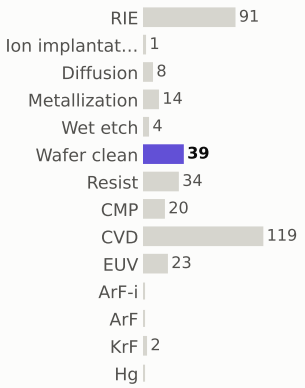
carbon so far, of the whole flow (kgCO2e)



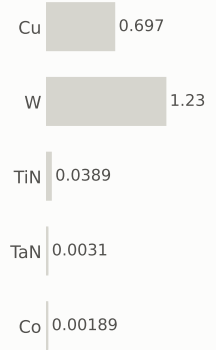
embodied carbon up to each step (kgCO2e per wafer)



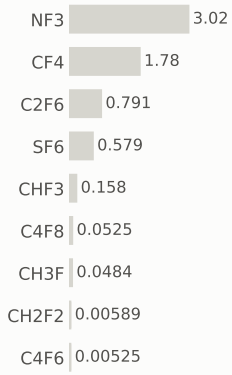
steps by process area



materials deposited (g)



gases emitted (g)



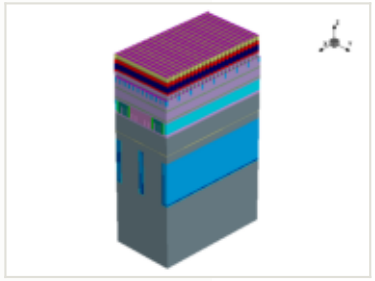
Step #3770: Metallization

depo_ellipsoidal · Metallization

1.2	389.9	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

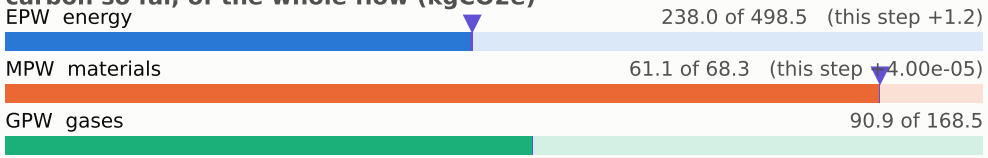


this step

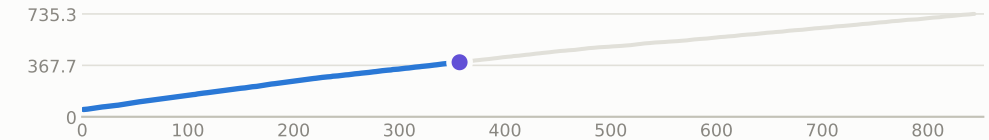
process area	Metallization
material	TaN
thickness	0.0016 µm
material GWP	25.8159 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.00e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

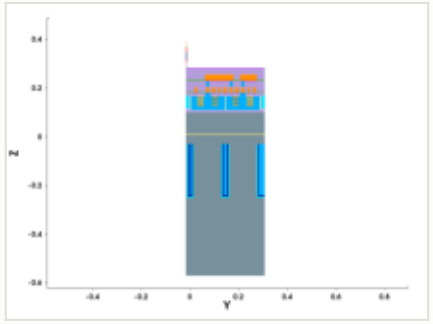
carbon so far, of the whole flow (kgCO2e)



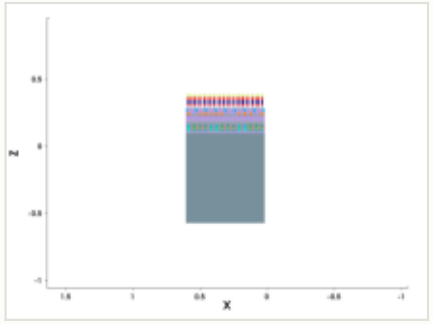
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



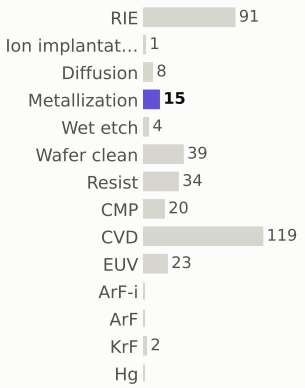
y cut at 0.184 µm



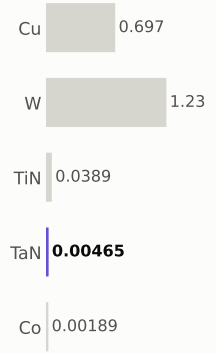
mask for this step

no mask for this step

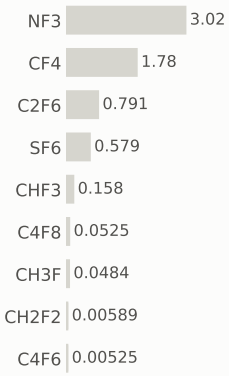
steps by process area



materials deposited (g)



gases emitted (g)



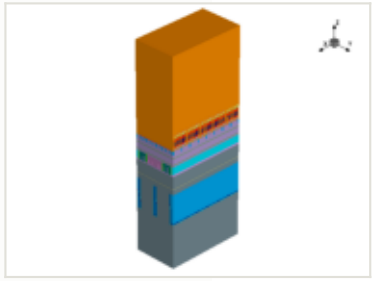
Step #3780: Metallization

depo_fill · Metallization

1.21	391.1	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

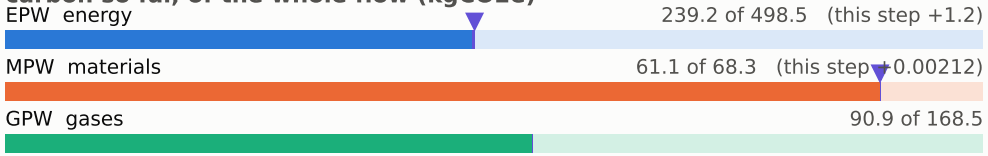


this step

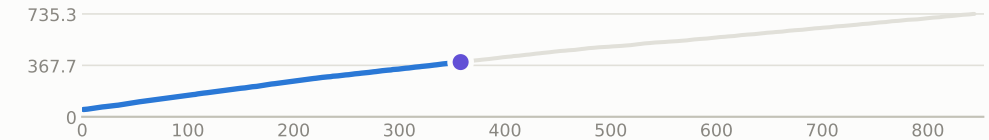
process area	Metallization
material	Cu
thickness	0.55 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00212 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

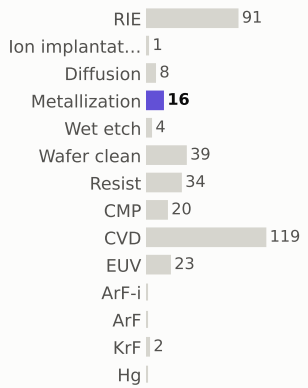
carbon so far, of the whole flow (kgCO2e)



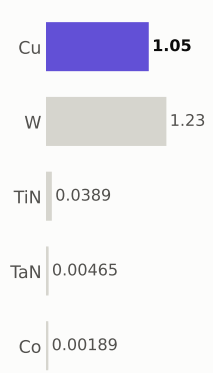
embodied carbon up to each step (kgCO2e per wafer)



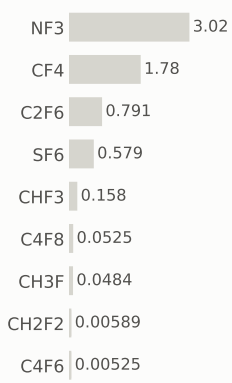
steps by process area



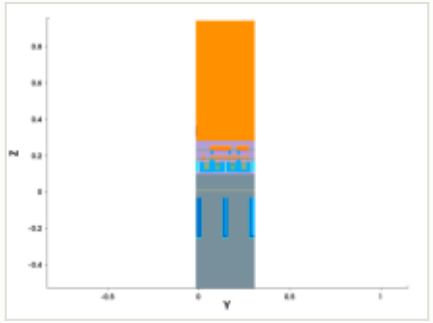
materials deposited (g)



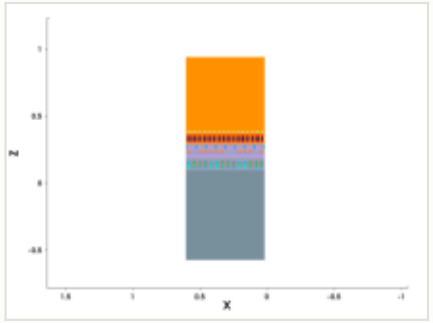
gases emitted (g)



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

no mask for this step

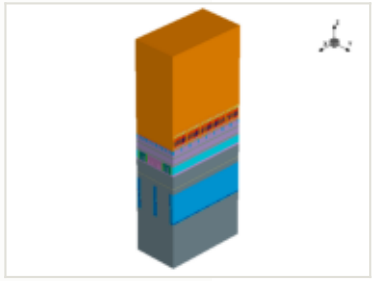
Step #3790: Diffusion

treatment · Diffusion

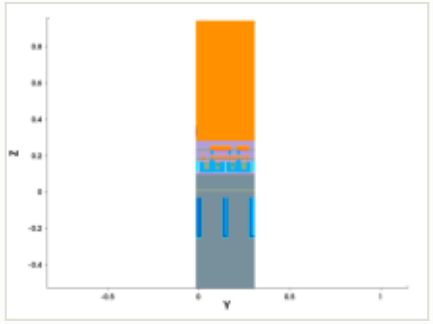
0.801	391.9	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

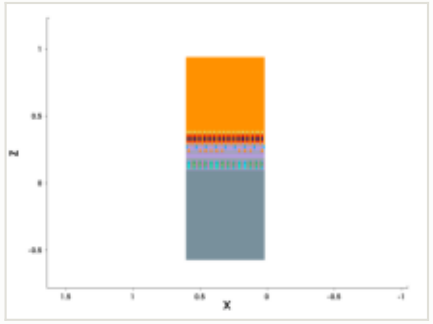
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

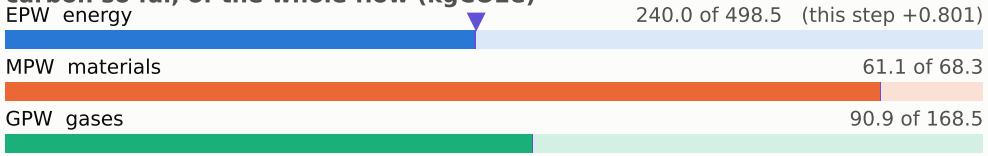
no mask for this step

this step

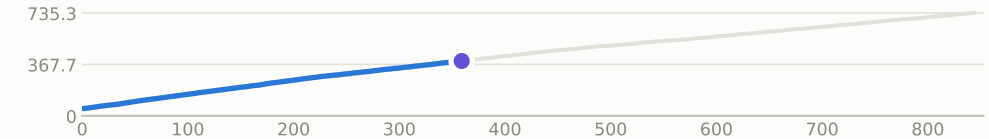
process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

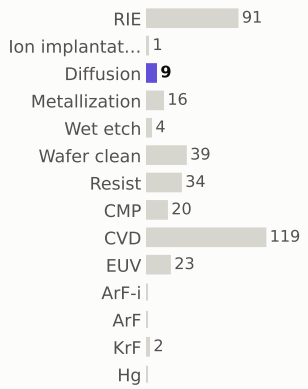
carbon so far, of the whole flow (kgCO2e)



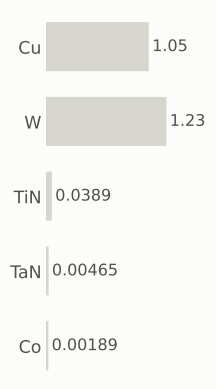
embodied carbon up to each step (kgCO2e per wafer)



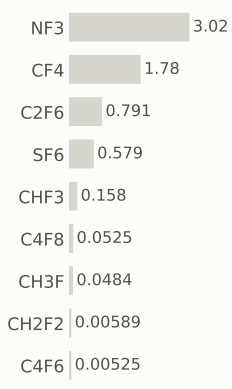
steps by process area



materials deposited (g)



gases emitted (g)



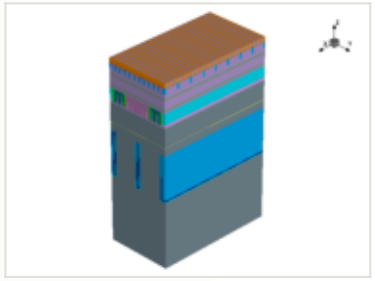
Step #3800: CMP

etch_cmp · CMP

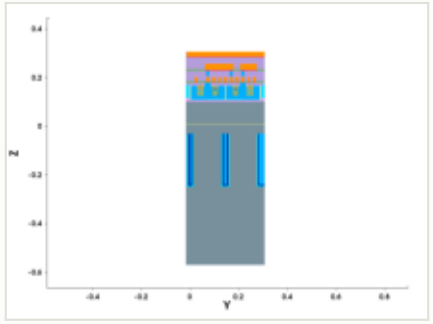
0.169	392.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

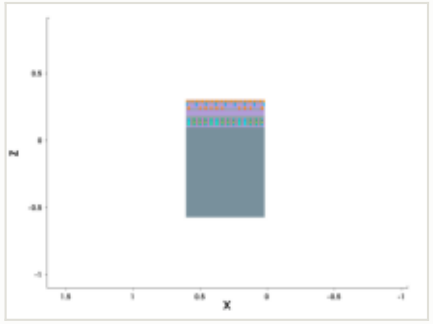
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

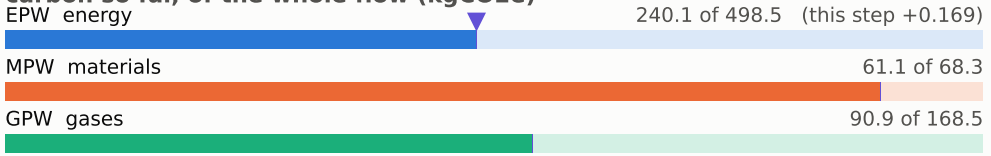
no mask for this step

this step

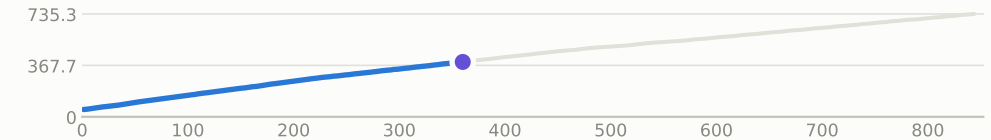
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

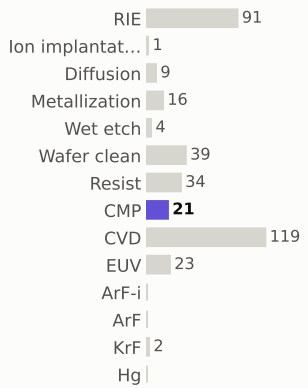
carbon so far, of the whole flow (kgCO2e)



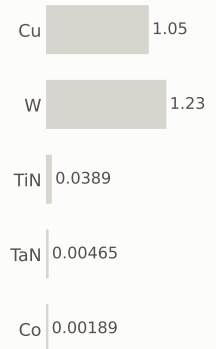
embodied carbon up to each step (kgCO2e per wafer)



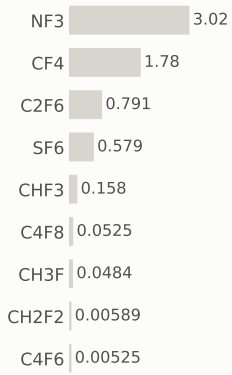
steps by process area



materials deposited (g)



gases emitted (g)



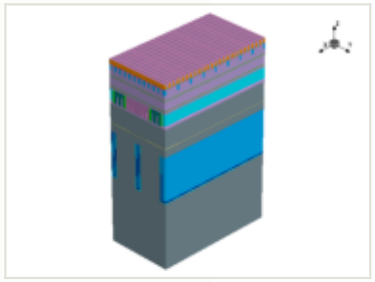
Step #3810: Metallization

depo_isotropic · Metallization

1.2	393.3	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

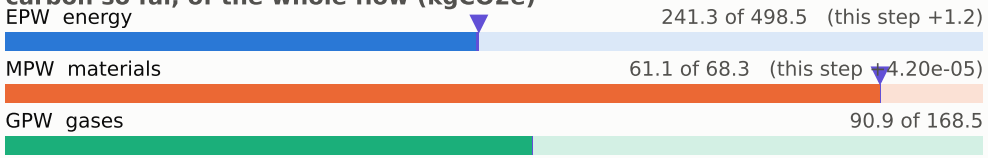


this step

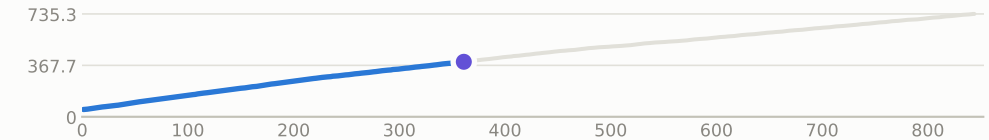
process area	Metallization
material	Co
thickness	0.0015 µm
material GWP	44.5 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.20e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

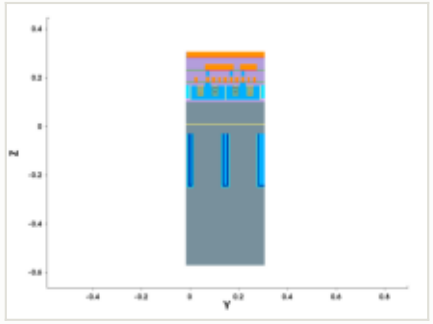
carbon so far, of the whole flow (kgCO2e)



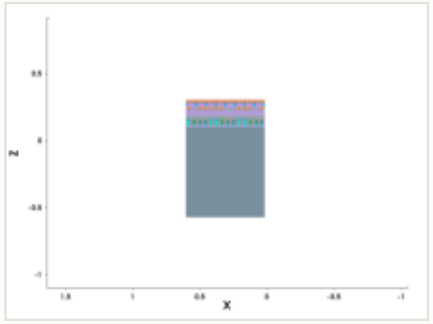
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



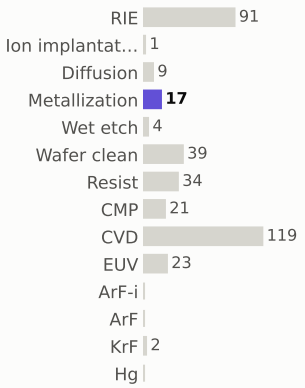
y cut at 0.184 µm



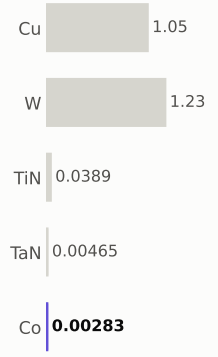
mask for this step

no mask for this step

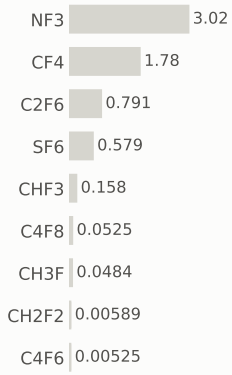
steps by process area



materials deposited (g)



gases emitted (g)



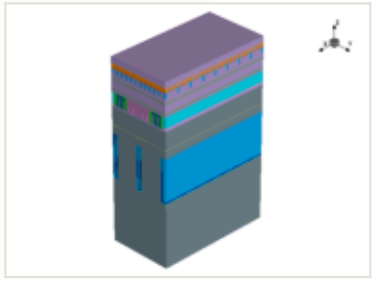
Step #3830: CVD

depo_fill · CVD

1.31	394.6	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

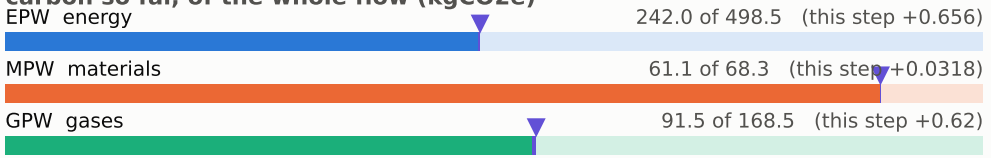


this step

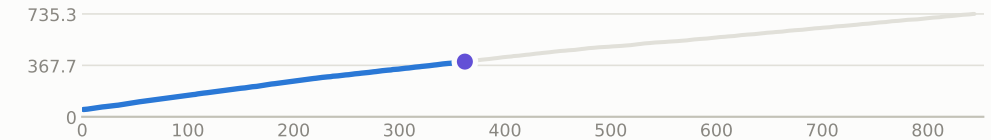
process area	CVD
material	—
thickness	0.028 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

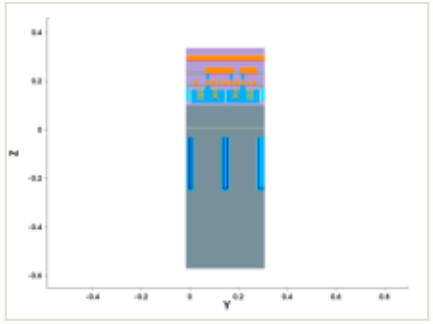
carbon so far, of the whole flow (kgCO2e)



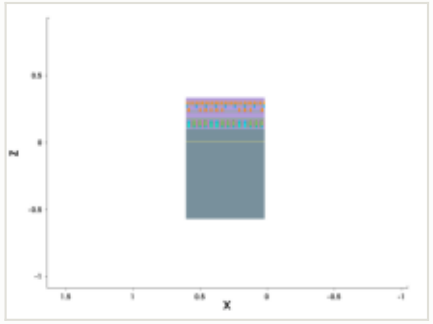
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



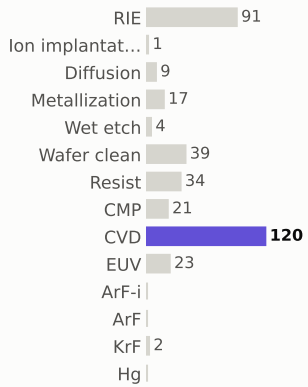
y cut at 0.184 µm



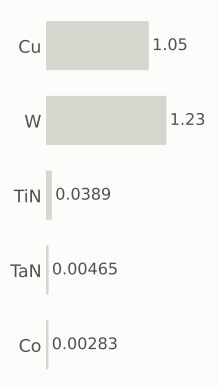
mask for this step

no mask for this step

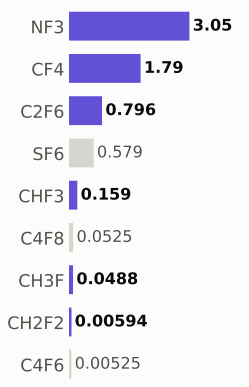
steps by process area



materials deposited (g)



gases emitted (g)



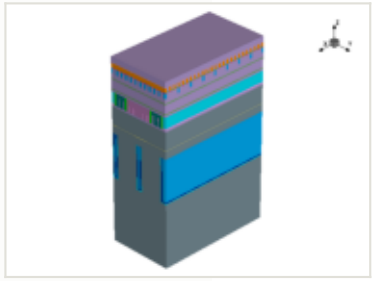
Step #3840: Diffusion

treatment · Diffusion

0.801	395.4	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



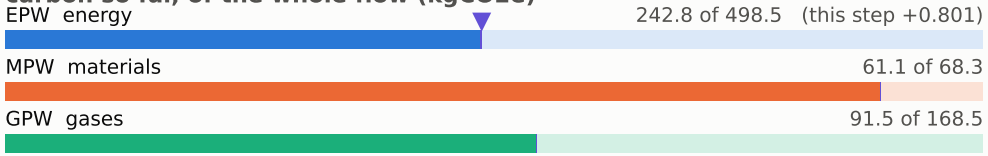
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

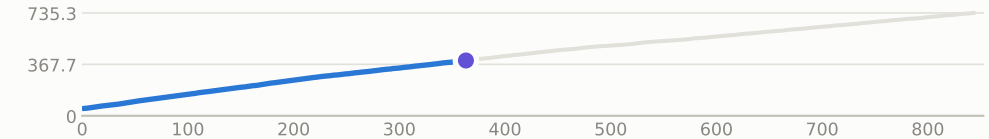
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

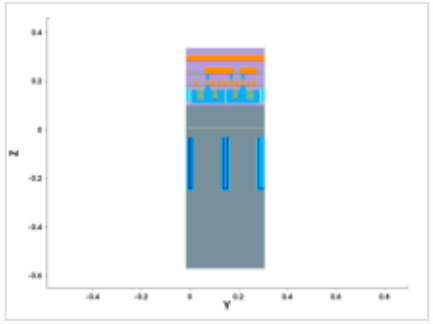
carbon so far, of the whole flow (kgCO2e)



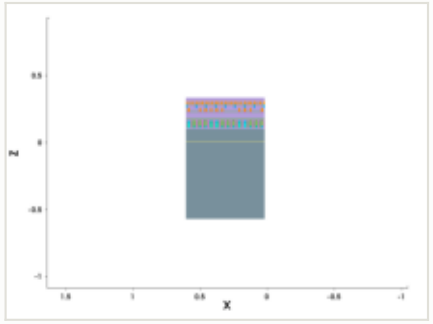
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



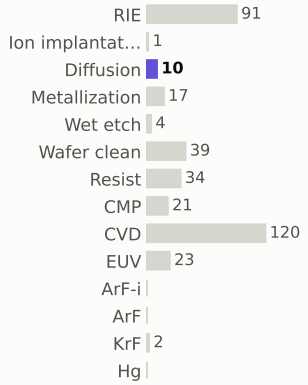
y cut at 0.184 µm



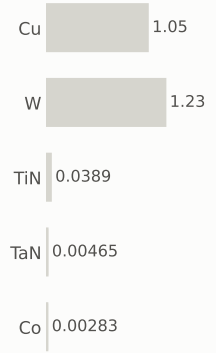
mask for this step

no mask for this step

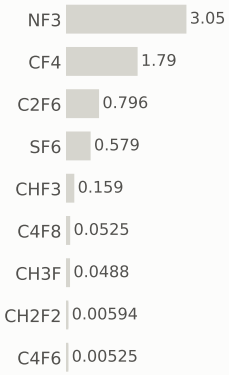
steps by process area



materials deposited (g)



gases emitted (g)



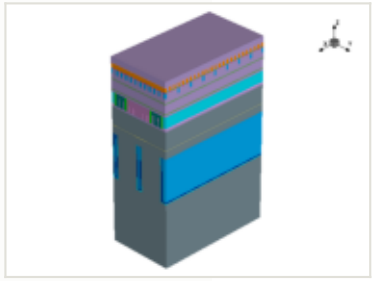
Step #3850: Wafer_clean

clean · Wafer_clean

0.169	395.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



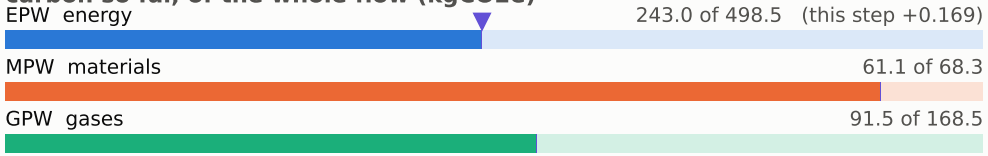
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

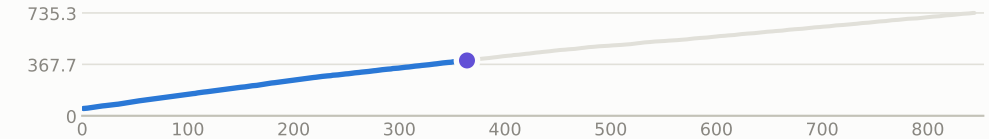
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

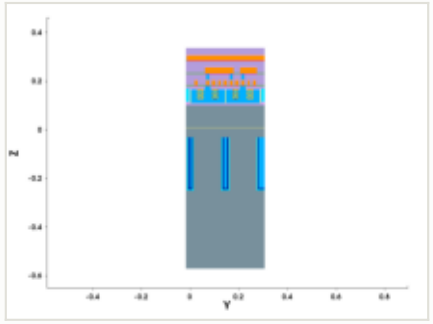
carbon so far, of the whole flow (kgCO2e)



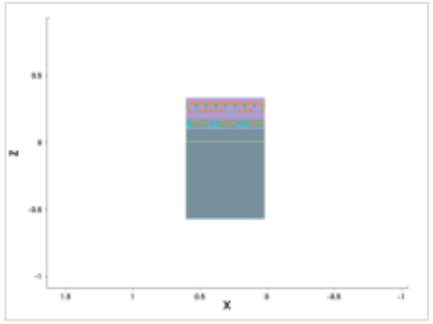
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



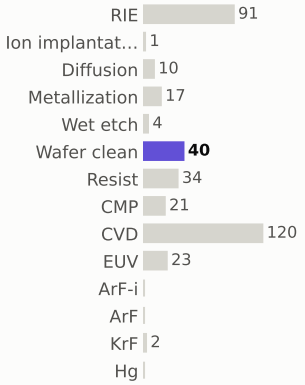
y cut at 0.184 µm



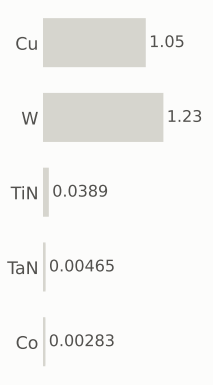
mask for this step

no mask for this step

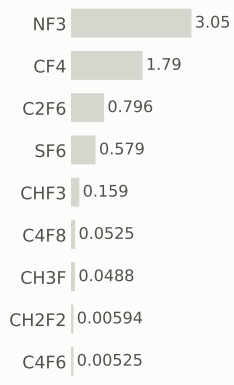
steps by process area



materials deposited (g)



gases emitted (g)



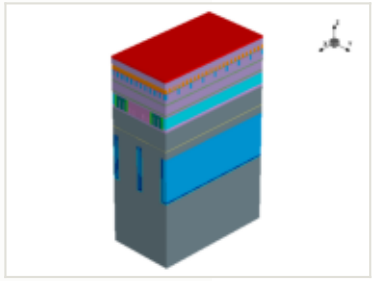
Step #3860: CVD

depo_fill · CVD

1.31	396.9	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

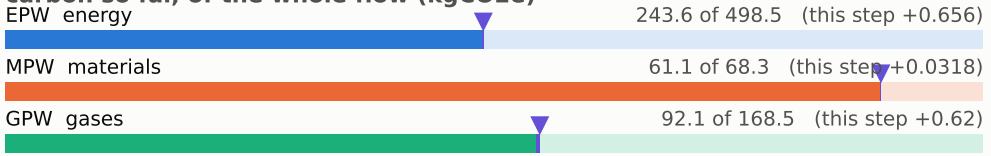


this step

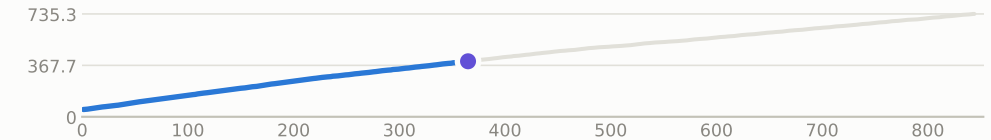
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

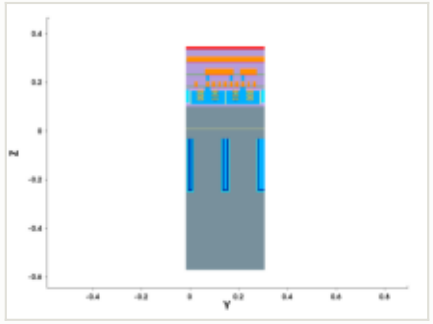
carbon so far, of the whole flow (kgCO2e)



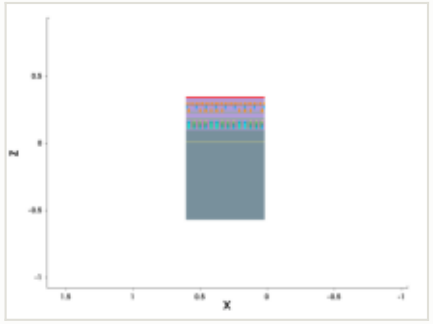
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



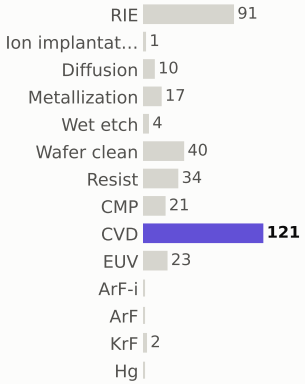
y cut at 0.184 µm



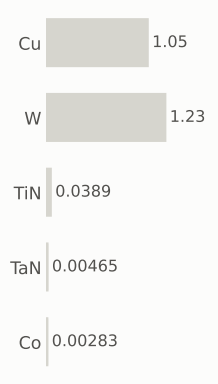
mask for this step

no mask for this step

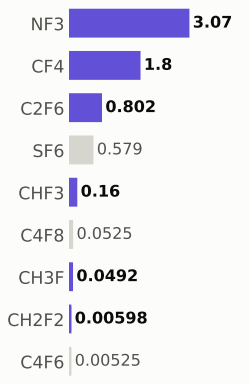
steps by process area



materials deposited (g)



gases emitted (g)



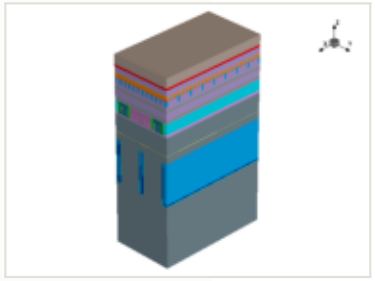
Step #3870: Resist

depo_fill · Resist

0.169	397.0	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

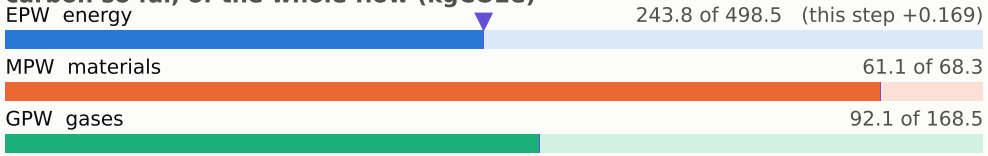


this step

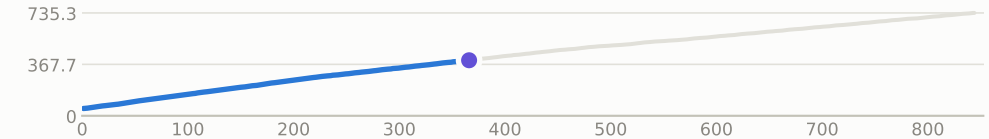
process area	Resist
material	—
thickness	0.05 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

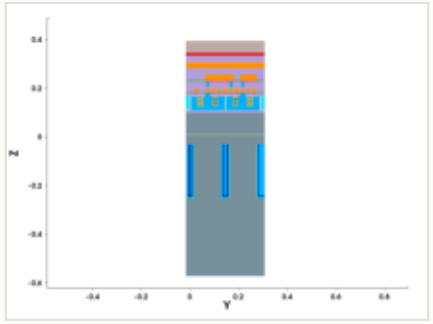
carbon so far, of the whole flow (kgCO2e)



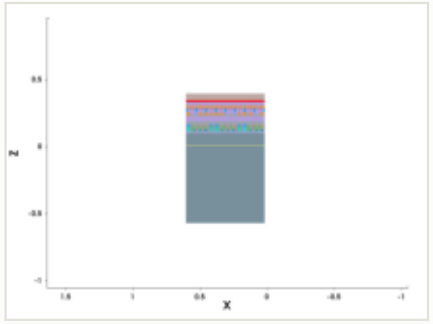
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



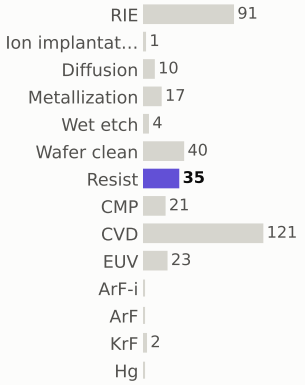
y cut at 0.184 µm



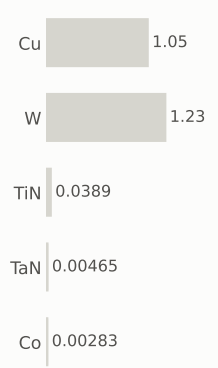
mask for this step

no mask for this step

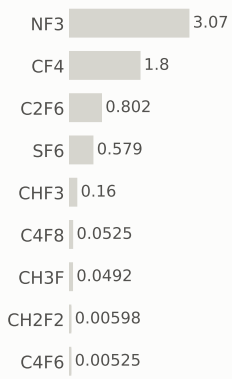
steps by process area



materials deposited (g)



gases emitted (g)



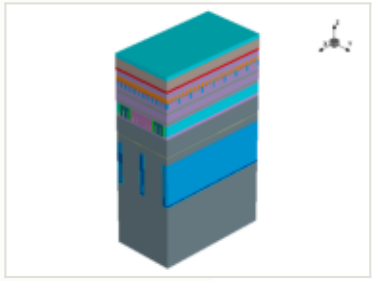
Step #3880: CVD

depo_fill · CVD

1.31	398.3	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



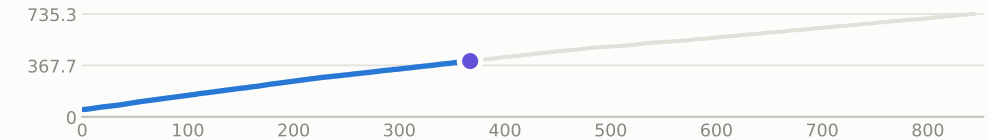
this step

process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

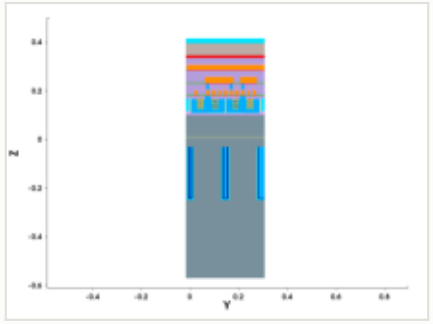
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

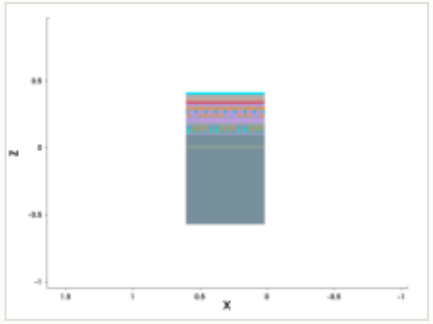
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



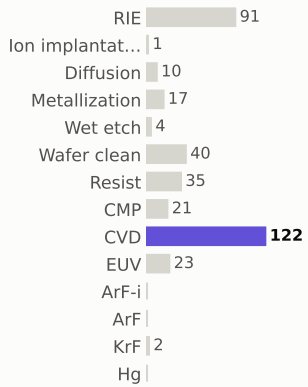
y cut at 0.184 µm



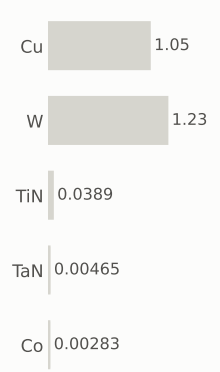
mask for this step

no mask for this step

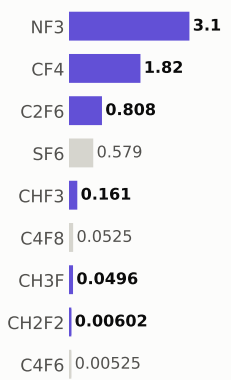
steps by process area



materials deposited (g)



gases emitted (g)



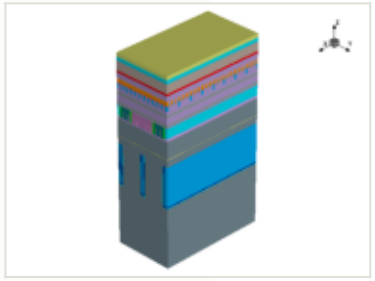
Step #3890: CVD

depo_fill · CVD

1.31	399.6	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



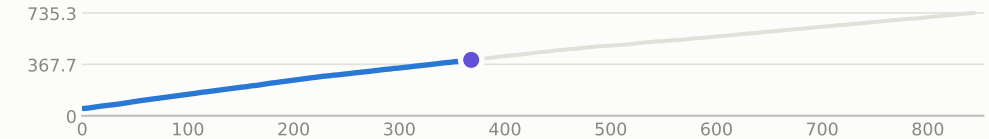
this step

process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

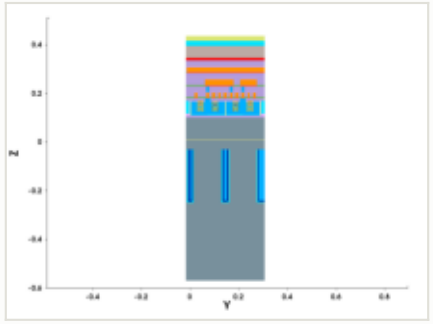
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

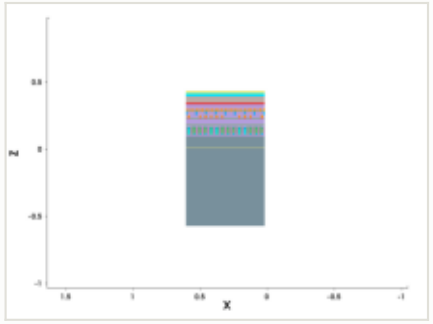
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



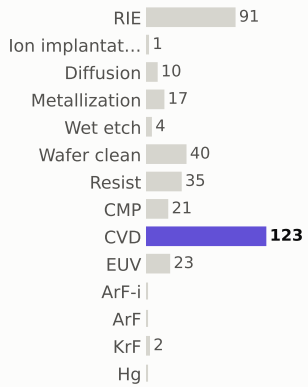
y cut at 0.184 µm



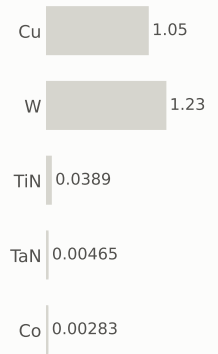
mask for this step

no mask for this step

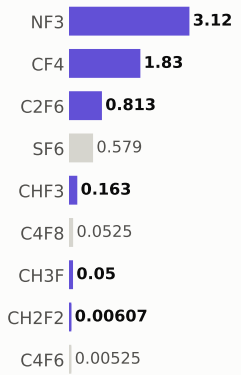
steps by process area



materials deposited (g)



gases emitted (g)



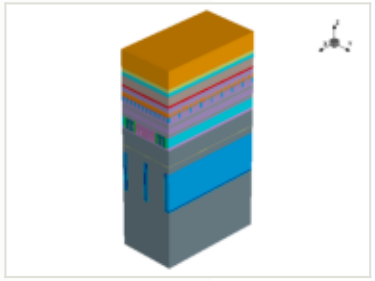
Step #3900: Resist

depo_fill · Resist

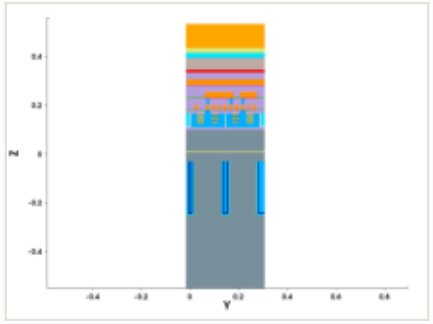
0.169	399.8	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

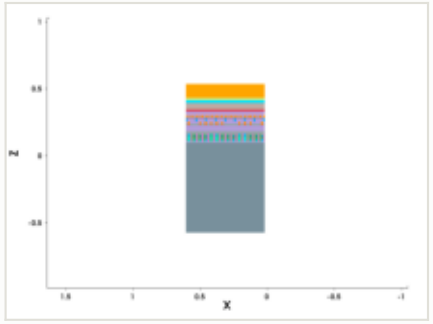
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



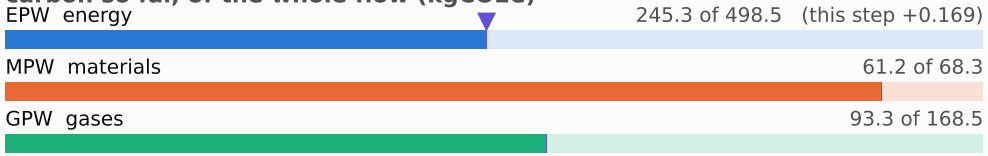
mask for this step

no mask for this step

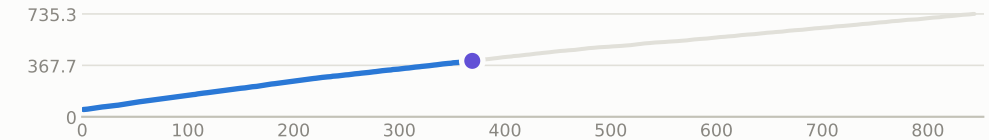
this step

process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	0.1 µm	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

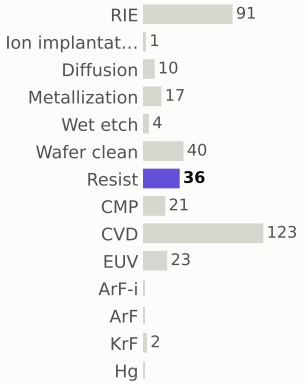
carbon so far, of the whole flow (kgCO2e)



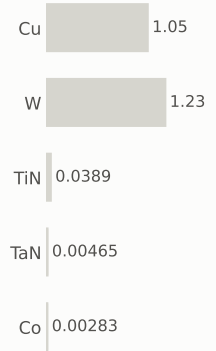
embodied carbon up to each step (kgCO2e per wafer)



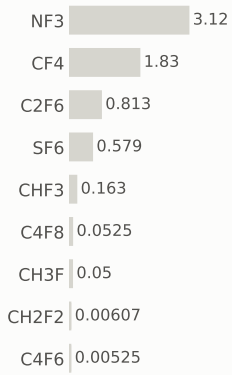
steps by process area



materials deposited (g)



gases emitted (g)



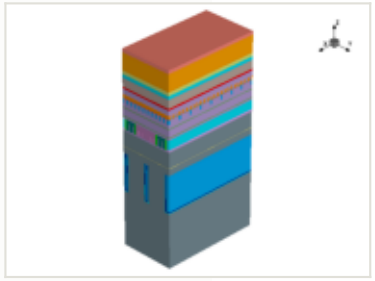
Step #3910: CVD

depo_fill · CVD

1.31	401.1	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

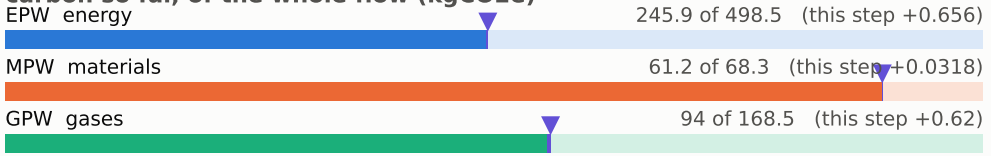


this step

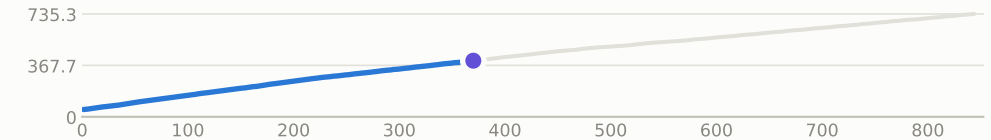
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

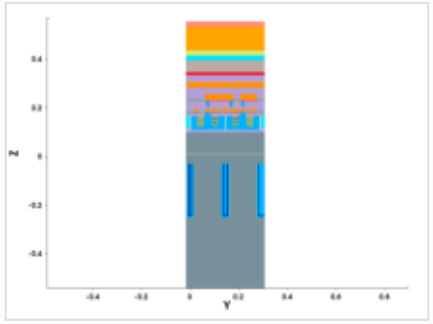
carbon so far, of the whole flow (kgCO2e)



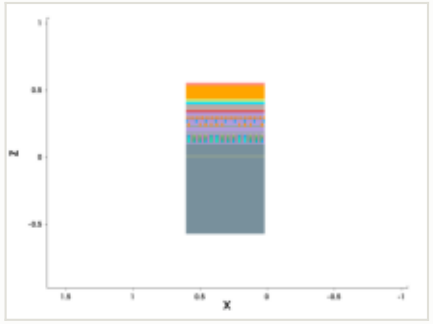
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



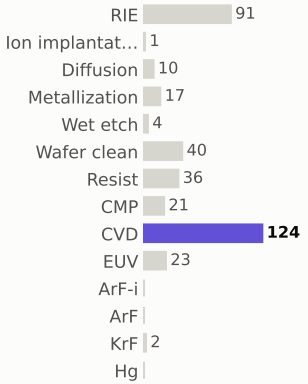
y cut at 0.184 µm



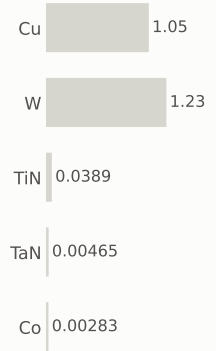
mask for this step

no mask for this step

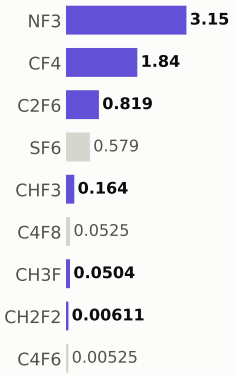
steps by process area



materials deposited (g)



gases emitted (g)



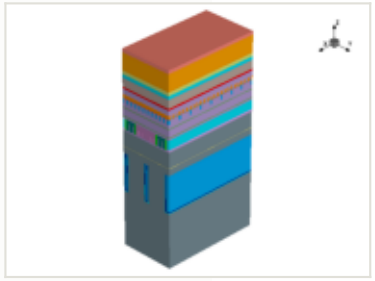
Step #3920: Wafer_clean

clean · Wafer_clean

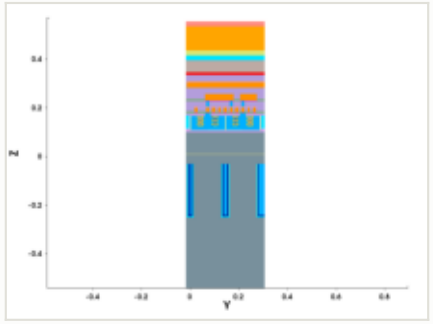
0.169	401.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

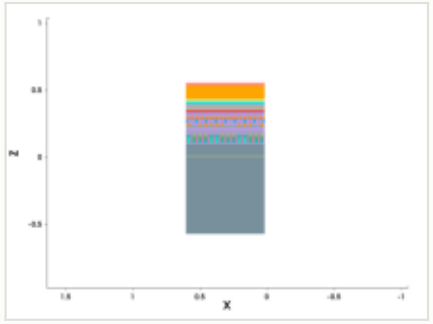
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

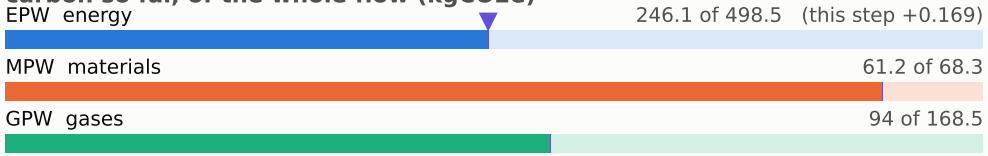
no mask for this step

this step

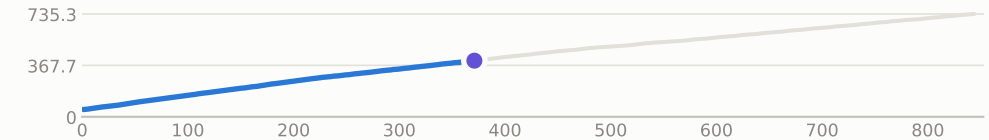
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

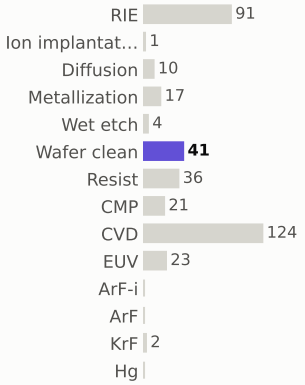
carbon so far, of the whole flow (kgCO2e)



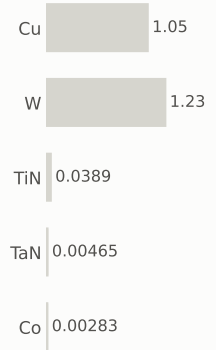
embodied carbon up to each step (kgCO2e per wafer)



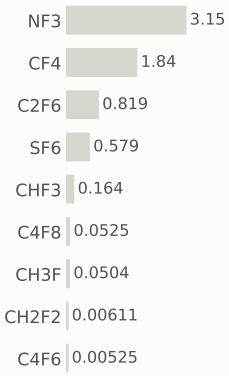
steps by process area



materials deposited (g)



gases emitted (g)



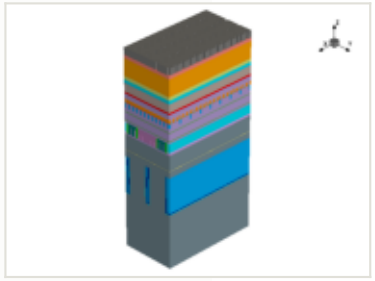
Step #3930: EUV

litho_patterning · EUV

1.96	403.2	0.27%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

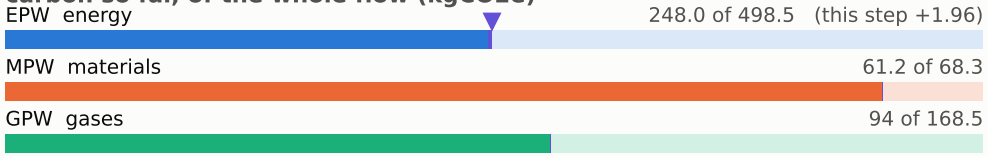


this step

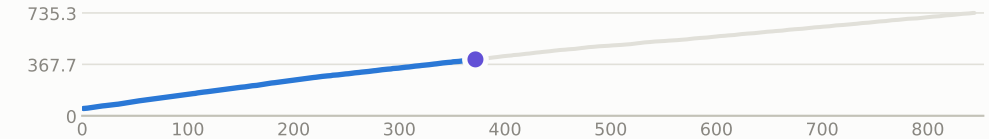
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

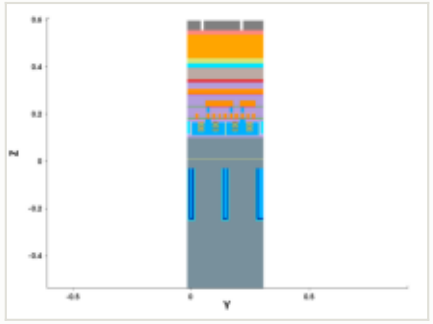
carbon so far, of the whole flow (kgCO2e)



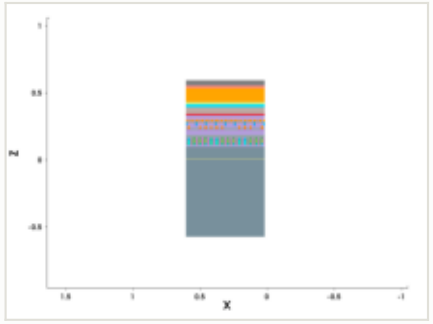
embodied carbon up to each step (kgCO2e per wafer)



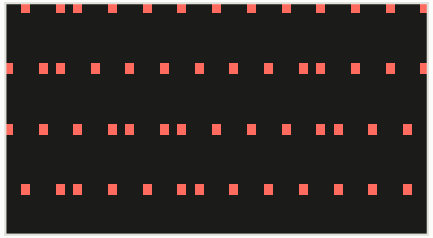
x cut at 0.294 µm



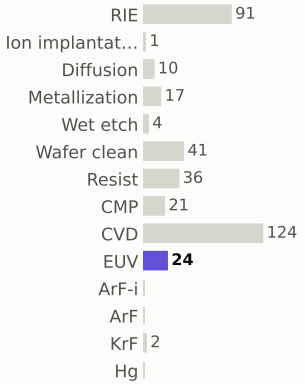
y cut at 0.184 µm



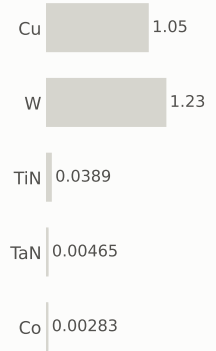
mask: layer 10321:0 (GDSII)



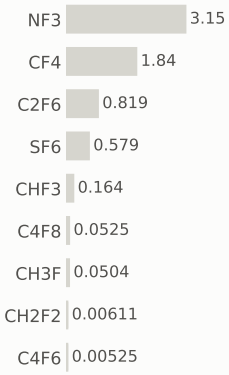
steps by process area



materials deposited (g)



gases emitted (g)



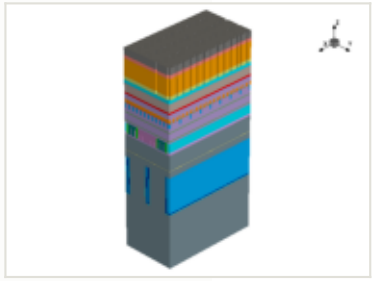
Step #3940: RIE

etch_anisotropic · RIE

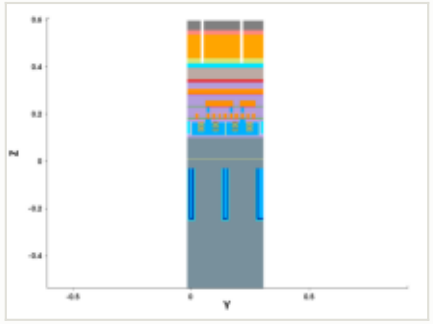
1.04	404.3	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

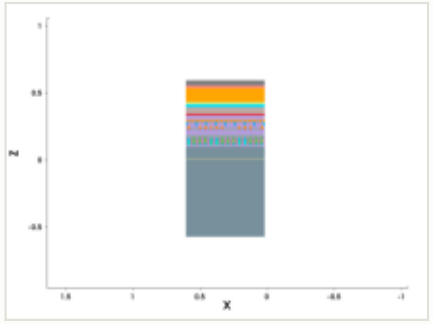
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

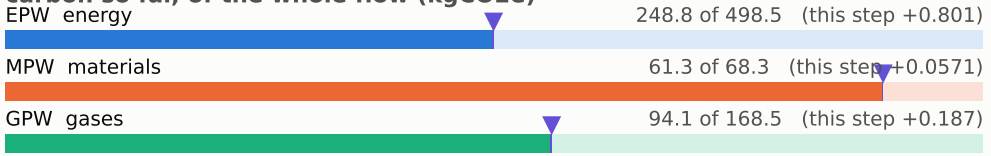
no mask for this step

this step

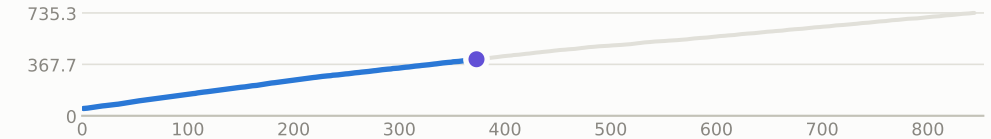
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

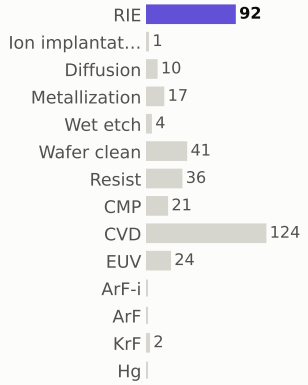
carbon so far, of the whole flow (kgCO2e)



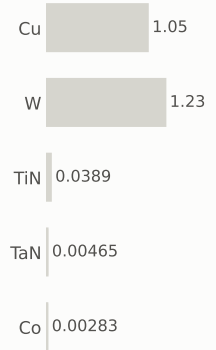
embodied carbon up to each step (kgCO2e per wafer)



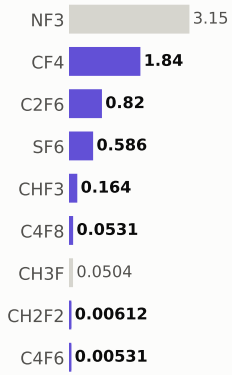
steps by process area



materials deposited (g)



gases emitted (g)



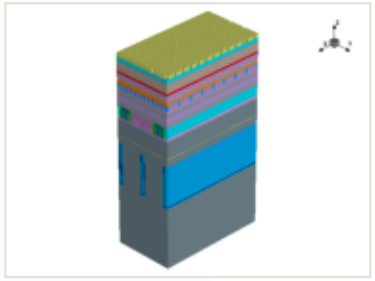
Step #3950: Wafer_clean

etch_strip · Wafer_clean

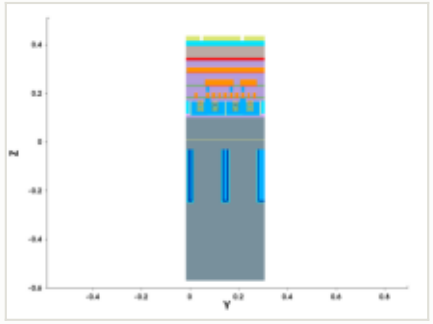
0.169	404.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

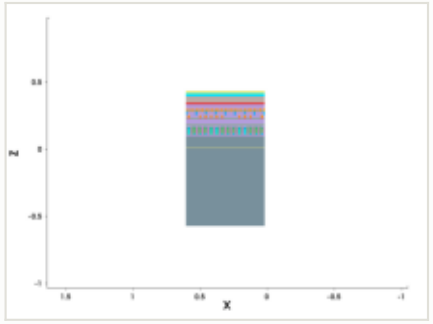
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



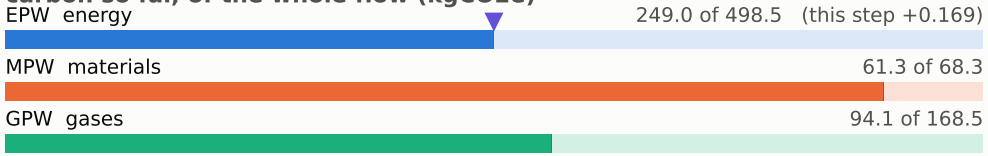
mask for this step

no mask for this step

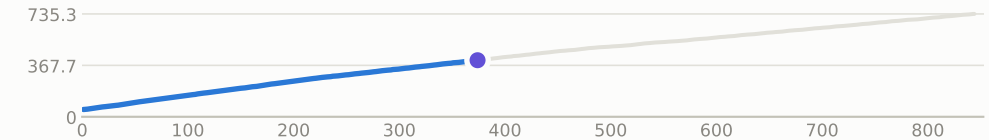
this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

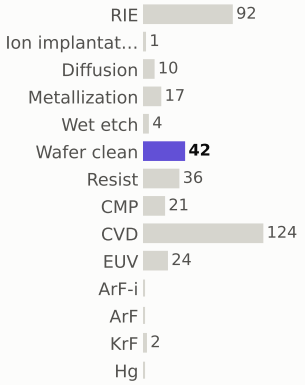
carbon so far, of the whole flow (kgCO2e)



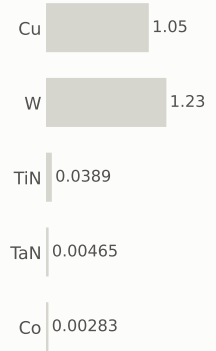
embodied carbon up to each step (kgCO2e per wafer)



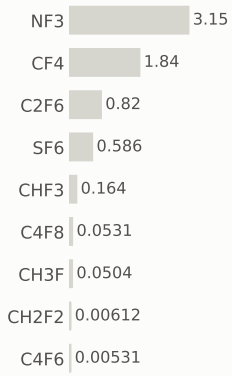
steps by process area



materials deposited (g)



gases emitted (g)



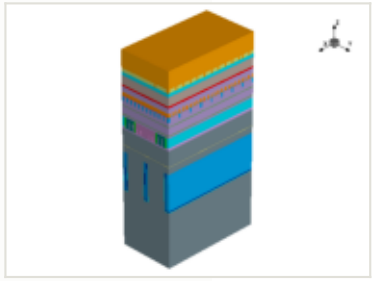
Step #3960: Resist

depo_fill · Resist

0.169	404.6	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

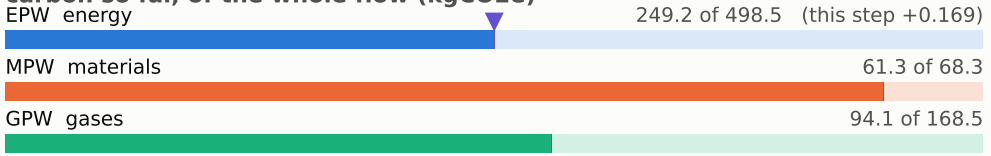


this step

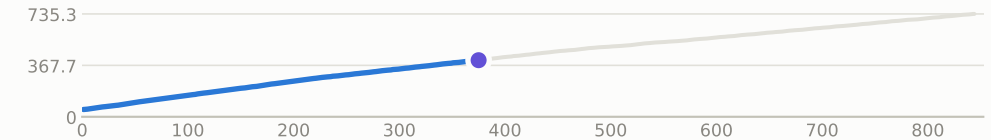
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

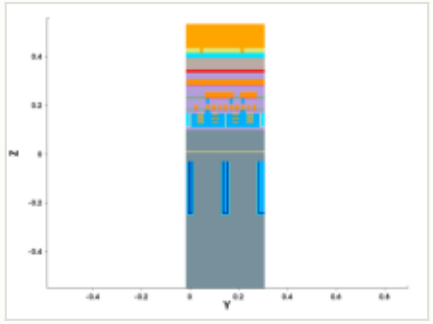
carbon so far, of the whole flow (kgCO2e)



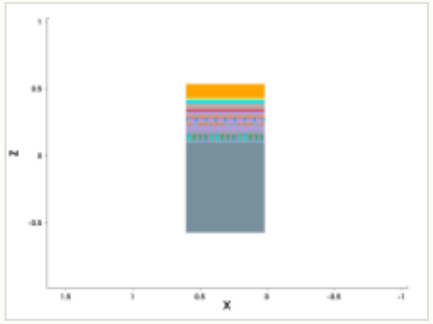
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



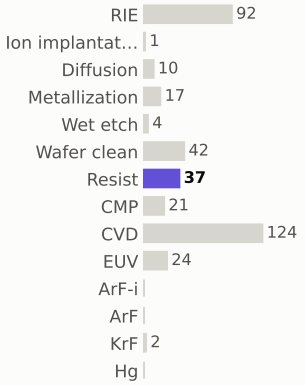
y cut at 0.184 µm



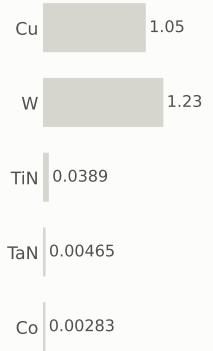
mask for this step

no mask for this step

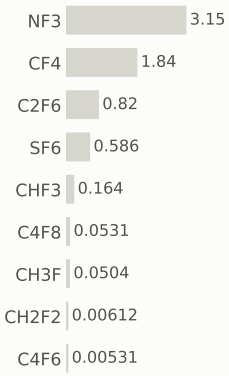
steps by process area



materials deposited (g)



gases emitted (g)



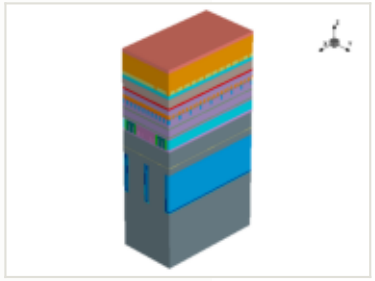
Step #3970: CVD

depo_fill · CVD

1.31	405.9	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

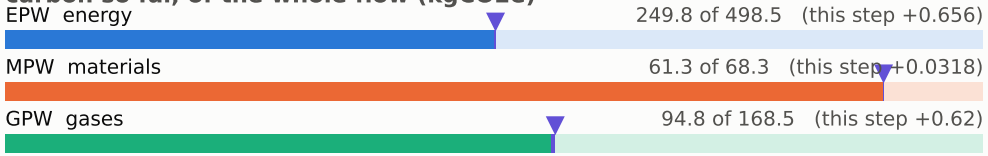


this step

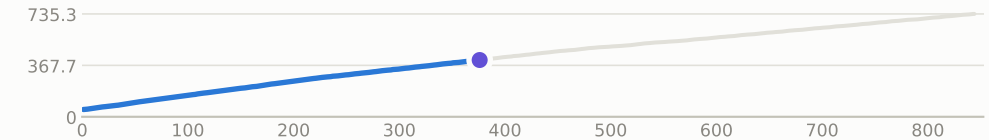
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

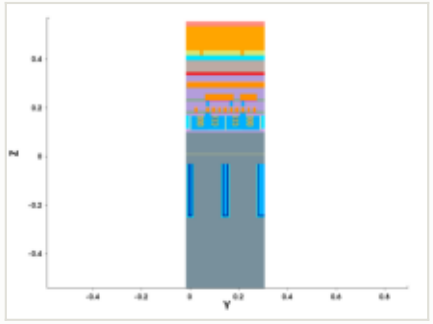
carbon so far, of the whole flow (kgCO2e)



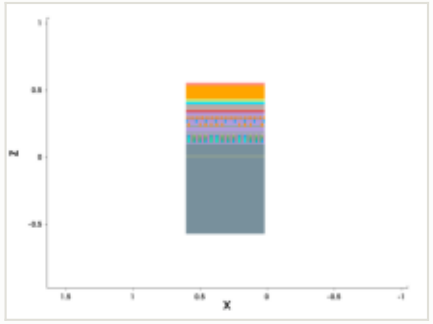
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



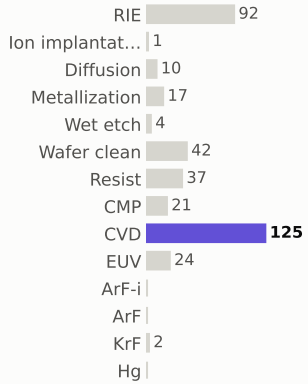
y cut at 0.184 µm



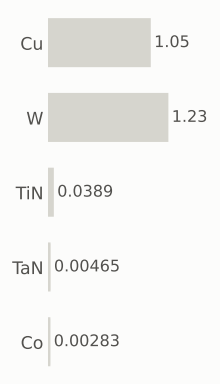
mask for this step

no mask for this step

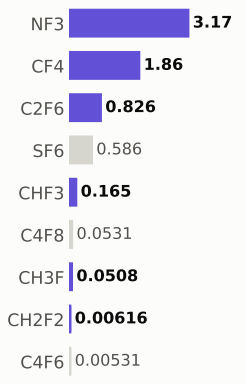
steps by process area



materials deposited (g)



gases emitted (g)



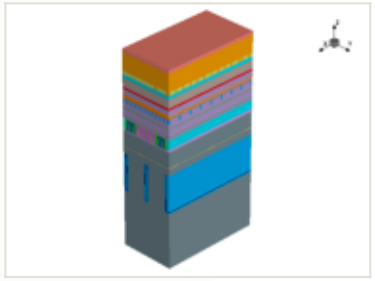
Step #3980: Wafer_clean

clean · Wafer_clean

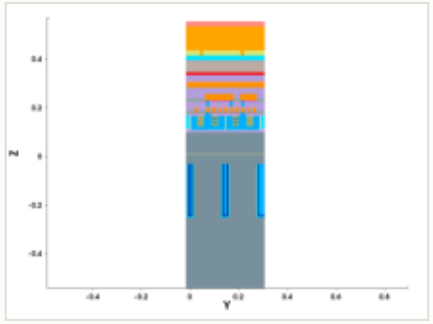
0.169	406.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

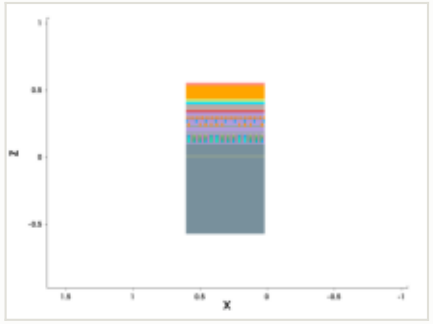
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

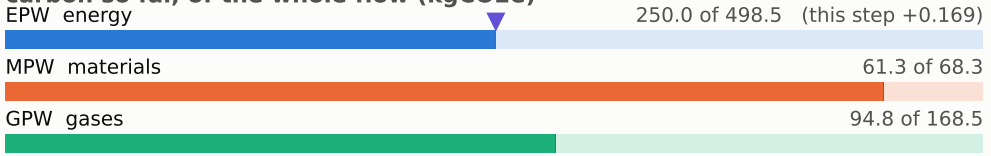
no mask for this step

this step

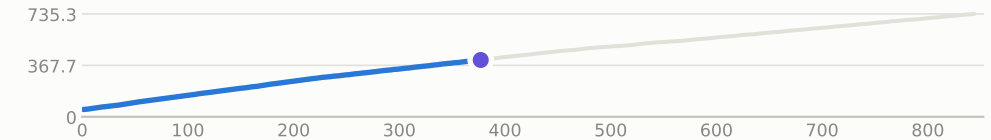
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

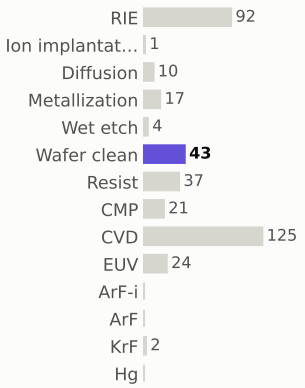
carbon so far, of the whole flow (kgCO2e)



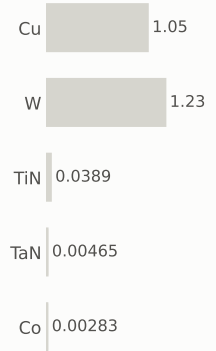
embodied carbon up to each step (kgCO2e per wafer)



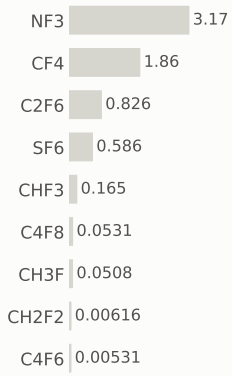
steps by process area



materials deposited (g)



gases emitted (g)



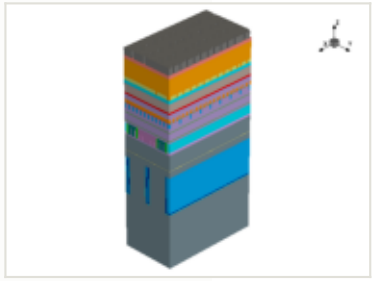
Step #3990: EUV

litho_patterning · EUV

1.96	408.1	0.27%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

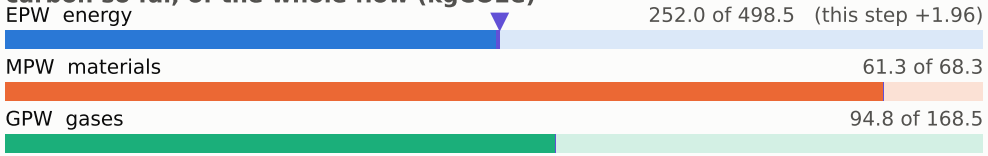


this step

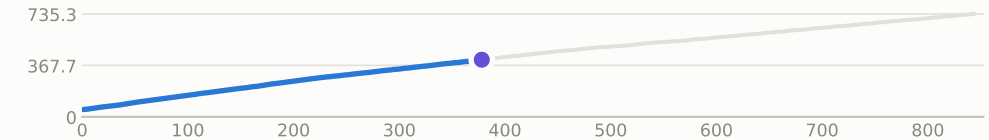
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

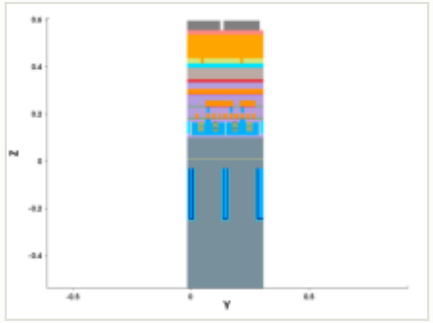
carbon so far, of the whole flow (kgCO2e)



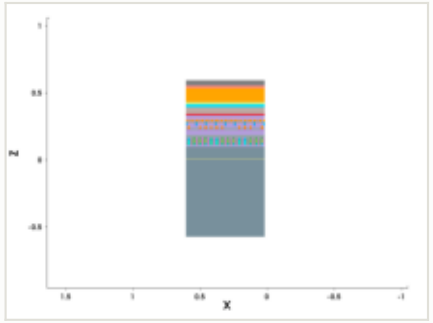
embodied carbon up to each step (kgCO2e per wafer)



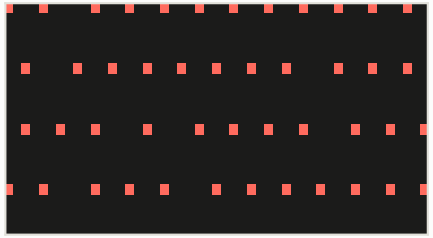
x cut at 0.294 µm



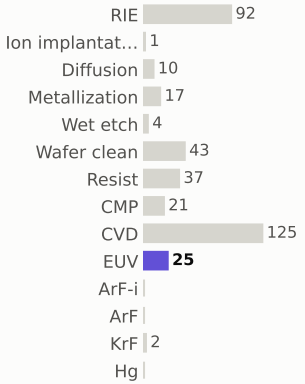
y cut at 0.184 µm



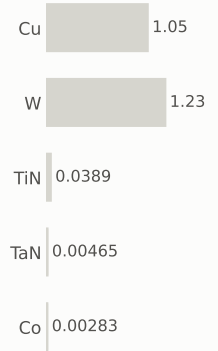
mask: layer 10322:0 (GDSII)



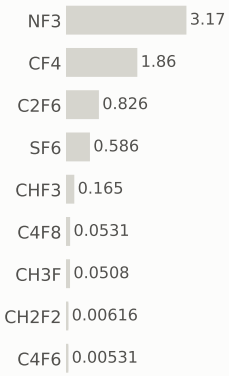
steps by process area



materials deposited (g)



gases emitted (g)



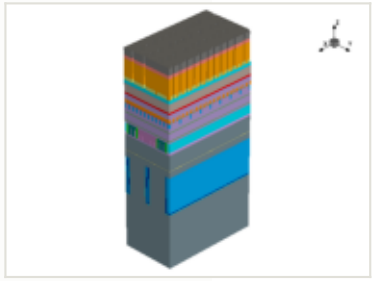
Step #4000: RIE

etch_anisotropic · RIE

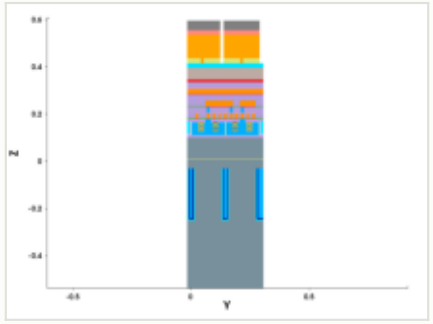
1.04	409.1	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

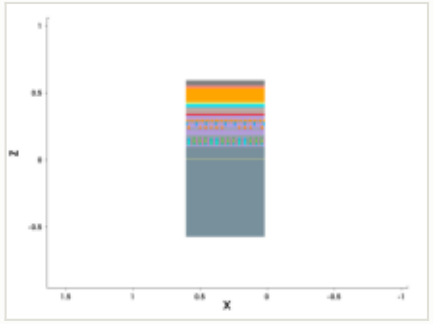
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

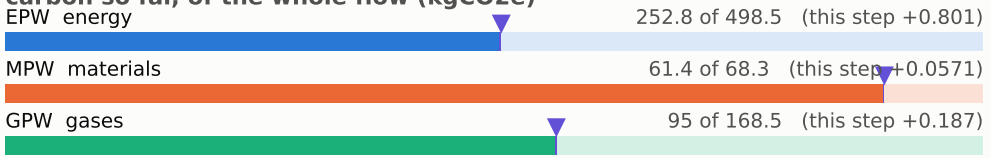
no mask for this step

this step

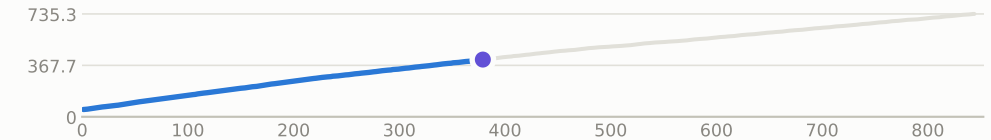
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

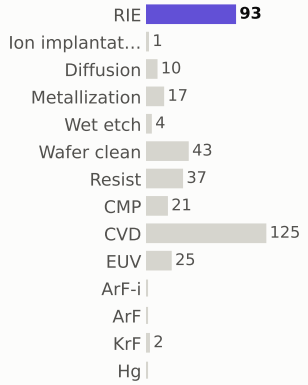
carbon so far, of the whole flow (kgCO2e)



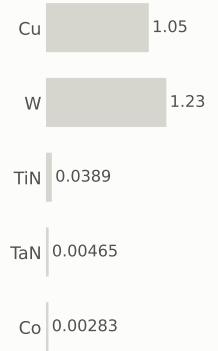
embodied carbon up to each step (kgCO2e per wafer)



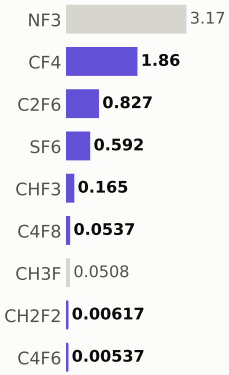
steps by process area



materials deposited (g)



gases emitted (g)



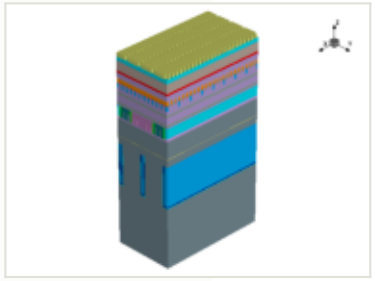
Step #4010: Wafer_clean

etch_strip · Wafer_clean

0.169	409.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

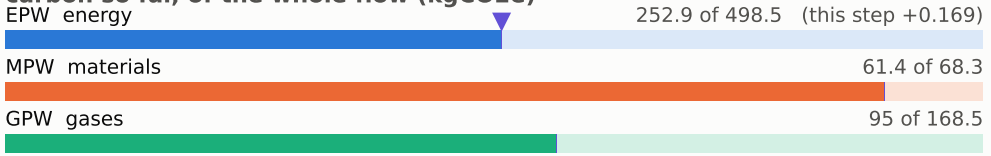


this step

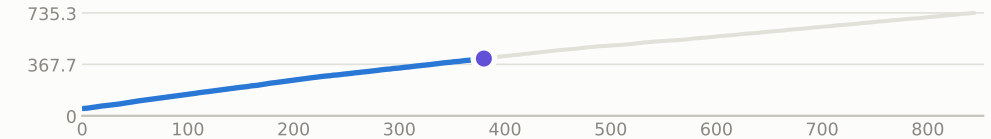
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

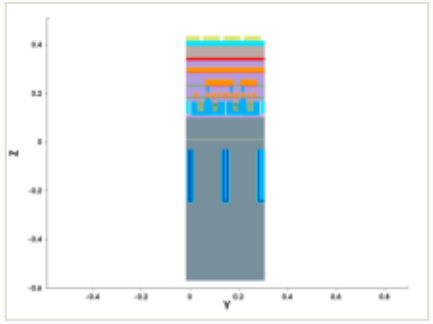
carbon so far, of the whole flow (kgCO2e)



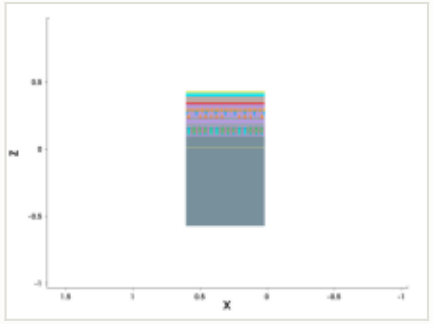
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



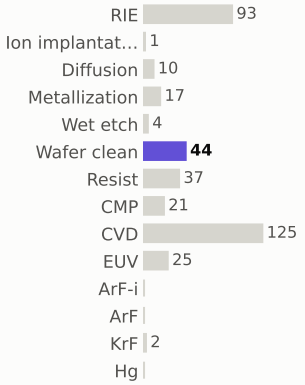
y cut at 0.184 µm



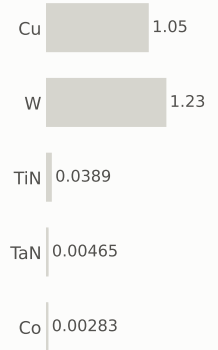
mask for this step

no mask for this step

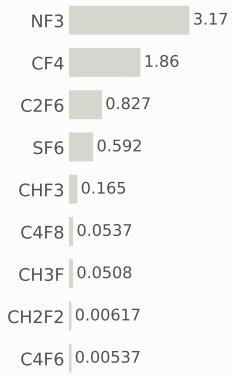
steps by process area



materials deposited (g)



gases emitted (g)



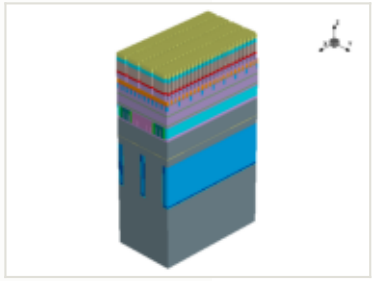
Step #4020: RIE

etch_anisotropic · RIE

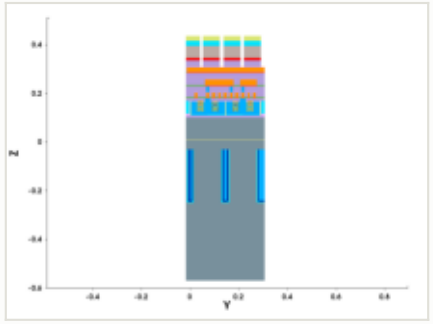
1.04	410.3	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

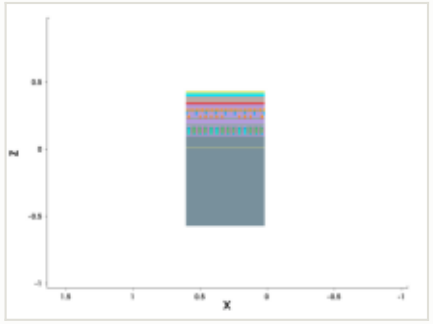
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

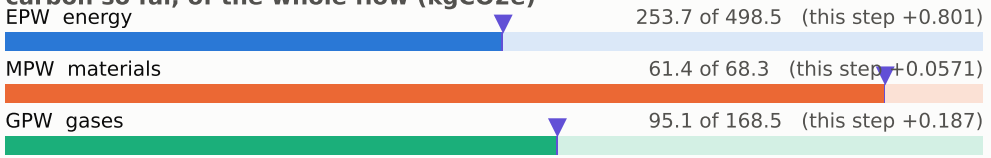
no mask for this step

this step

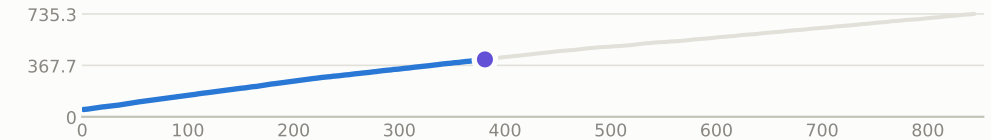
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

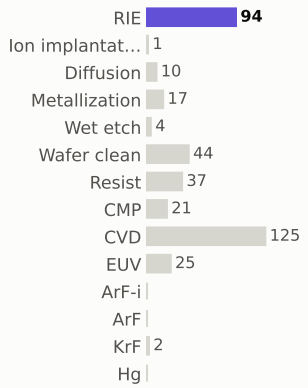
carbon so far, of the whole flow (kgCO2e)



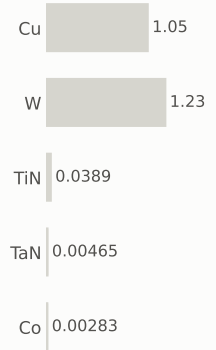
embodied carbon up to each step (kgCO2e per wafer)



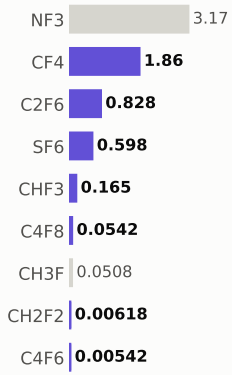
steps by process area



materials deposited (g)



gases emitted (g)



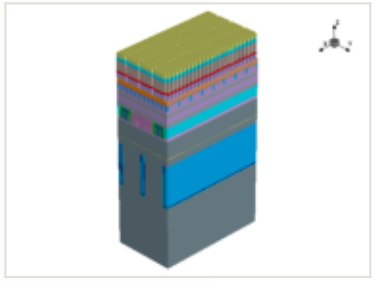
Step #4030: Wafer_clean

clean · Wafer_clean

0.169	410.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



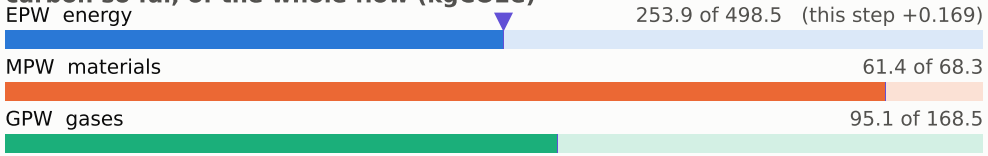
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

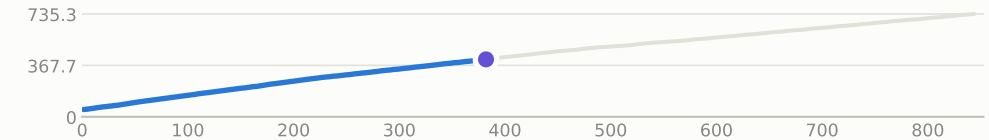
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

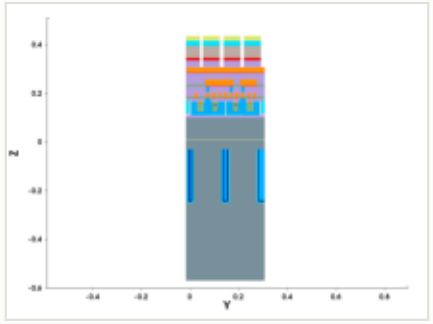
carbon so far, of the whole flow (kgCO2e)



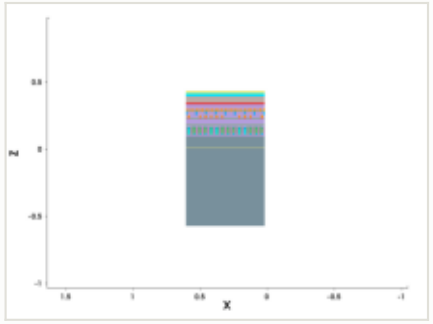
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



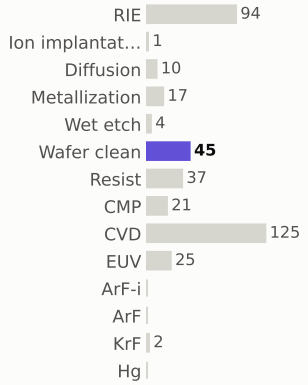
y cut at 0.184 µm



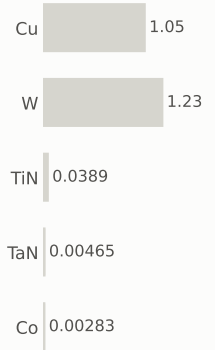
mask for this step

no mask for this step

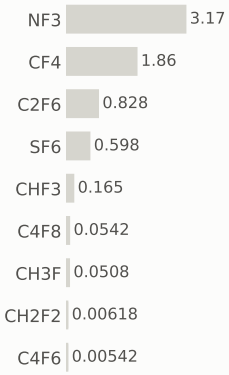
steps by process area



materials deposited (g)



gases emitted (g)



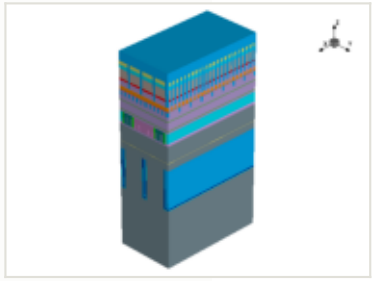
Step #4040: Metallization

depo_fill · Metallization

1.21	411.7	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

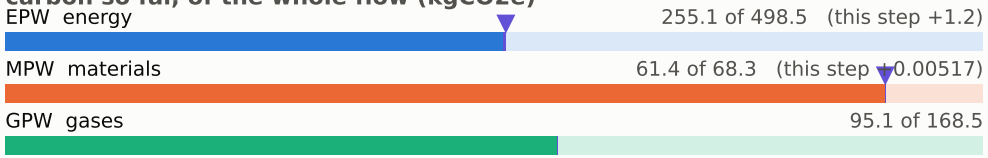


this step

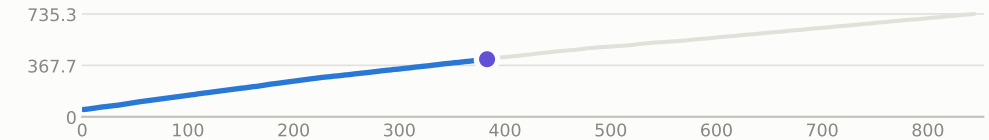
process area	Metallization
material	W
thickness	0.05 µm
material GWP	75.8149 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00517 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

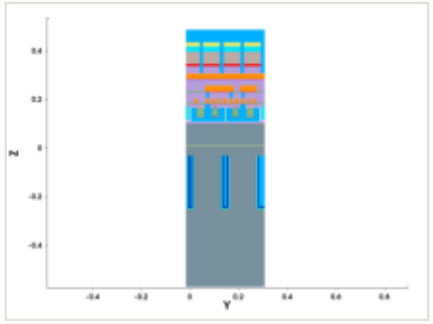
carbon so far, of the whole flow (kgCO2e)



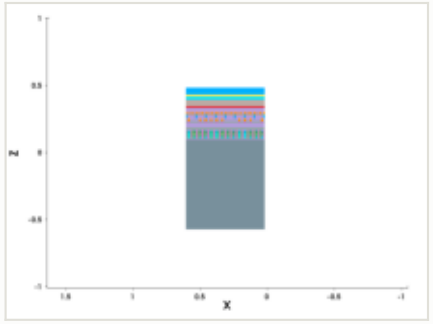
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



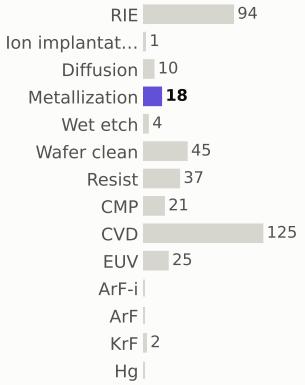
y cut at 0.184 µm



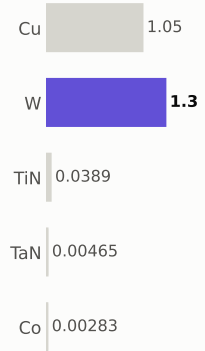
mask for this step

no mask for this step

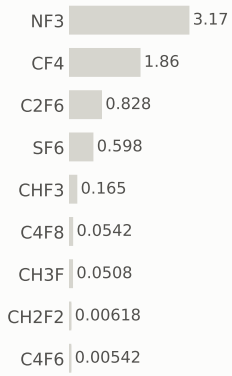
steps by process area



materials deposited (g)



gases emitted (g)



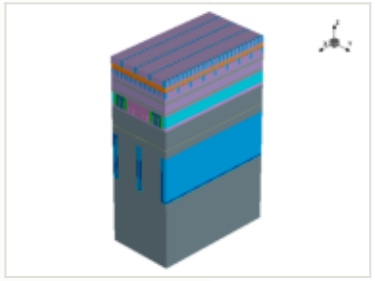
Step #4050: CMP

etch_cmp · CMP

0.169	411.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

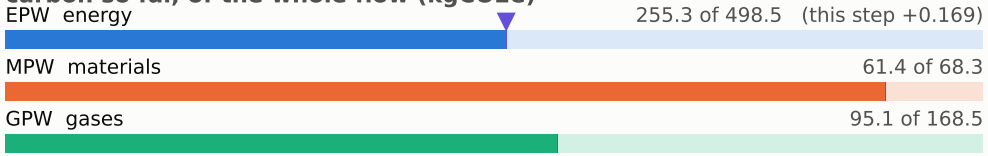


this step

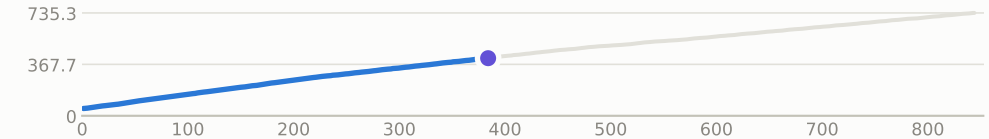
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

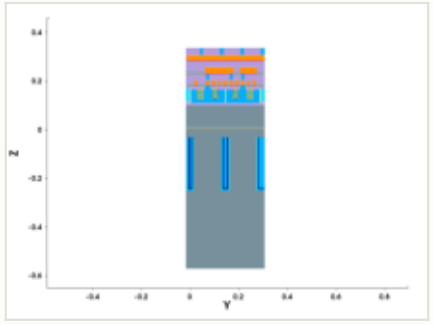
carbon so far, of the whole flow (kgCO2e)



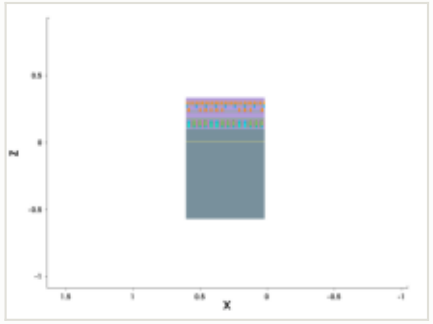
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



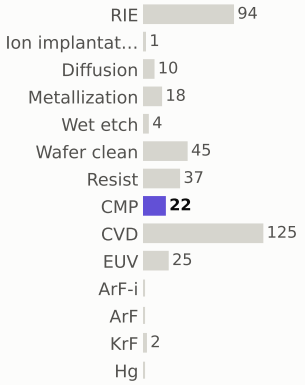
y cut at 0.184 µm



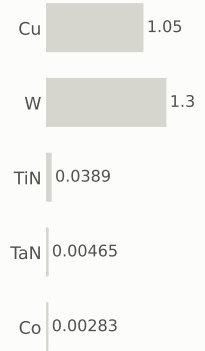
mask for this step

no mask for this step

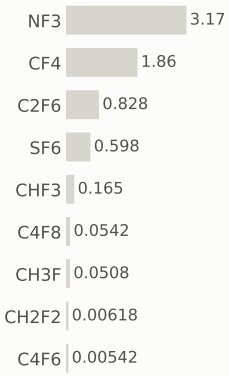
steps by process area



materials deposited (g)



gases emitted (g)



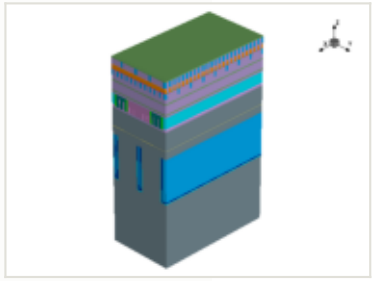
Step #4060: CVD

depo_fill · CVD

1.31	413.2	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

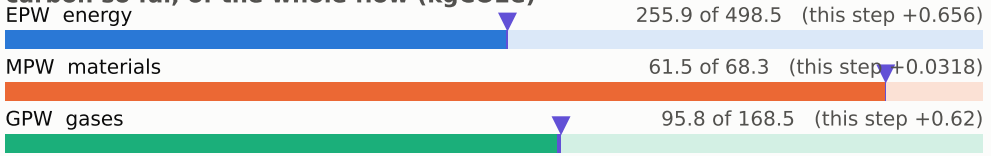


this step

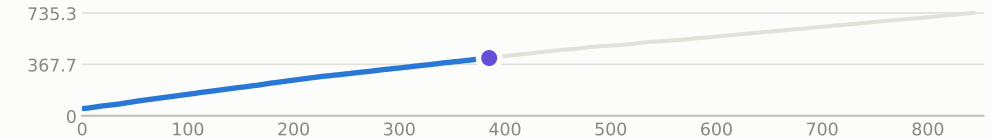
process area	CVD
material	—
thickness	0.006 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

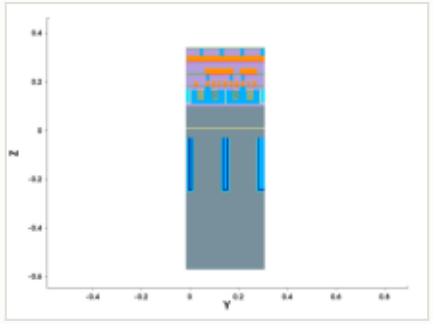
carbon so far, of the whole flow (kgCO2e)



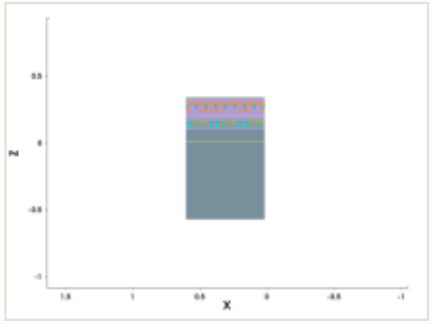
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



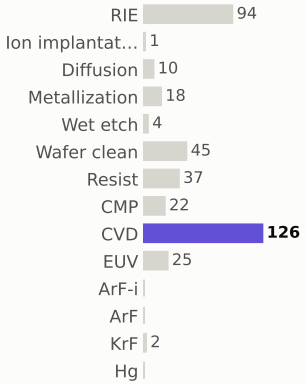
y cut at 0.184 µm



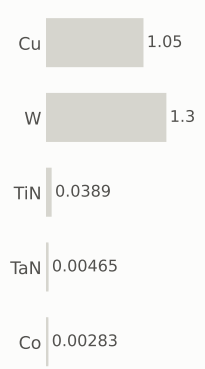
mask for this step

no mask for this step

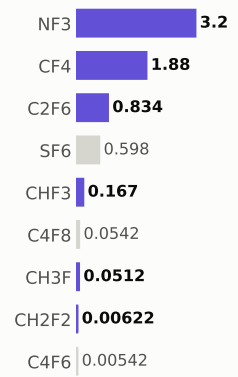
steps by process area



materials deposited (g)



gases emitted (g)



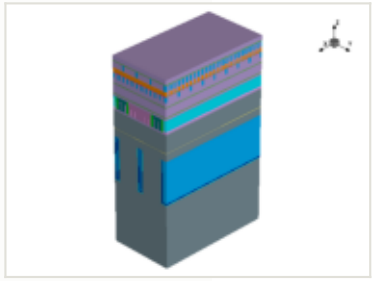
Step #4070: CVD

depo_fill · CVD

1.31	414.5	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

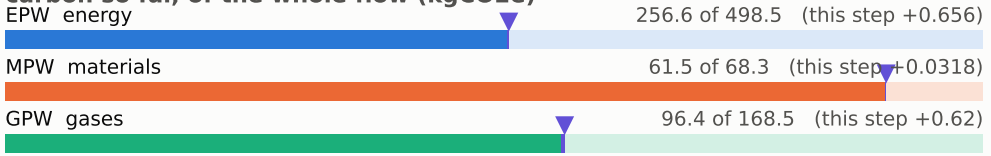


this step

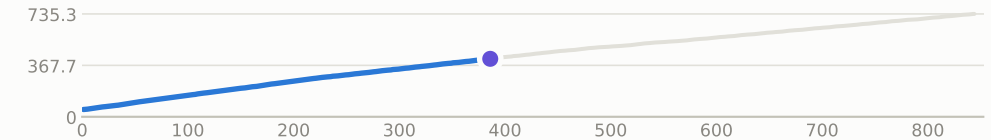
process area	CVD
material	—
thickness	0.022 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

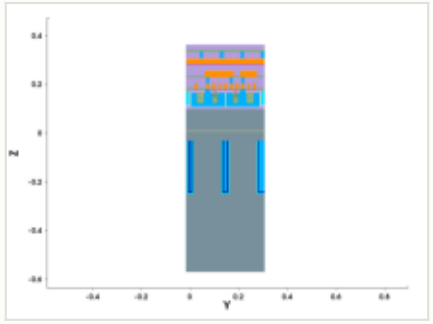
carbon so far, of the whole flow (kgCO2e)



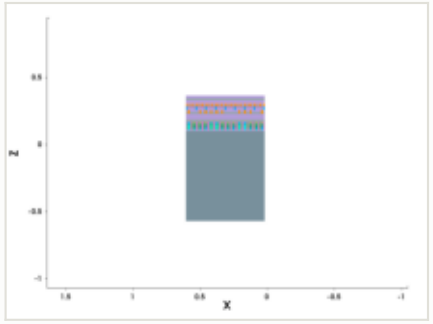
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



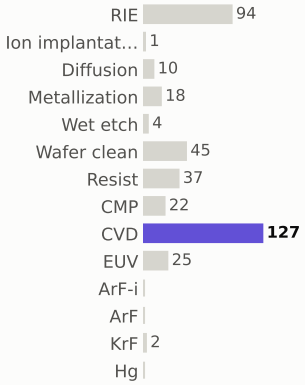
y cut at 0.184 µm



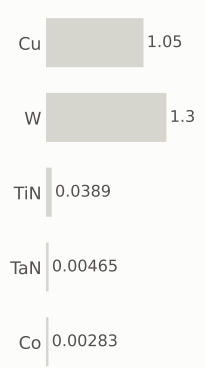
mask for this step

no mask for this step

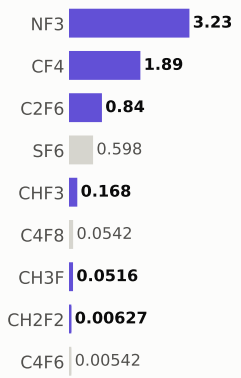
steps by process area



materials deposited (g)



gases emitted (g)



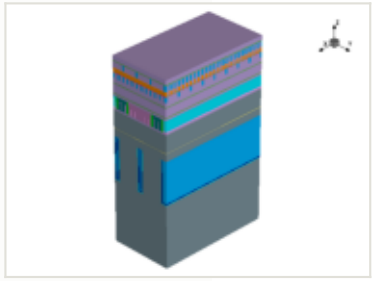
Step #4080: Diffusion

treatment · Diffusion

0.801	415.3	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



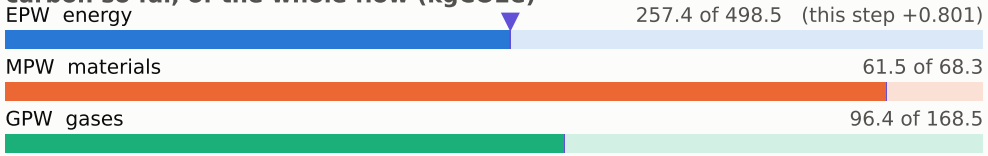
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

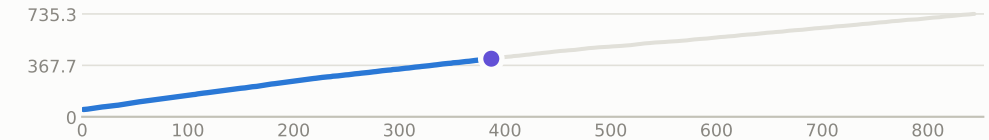
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

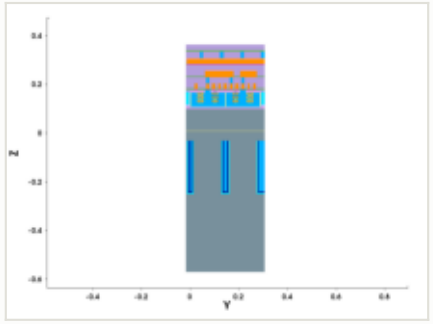
carbon so far, of the whole flow (kgCO2e)



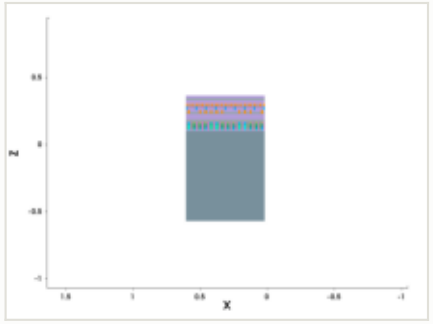
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



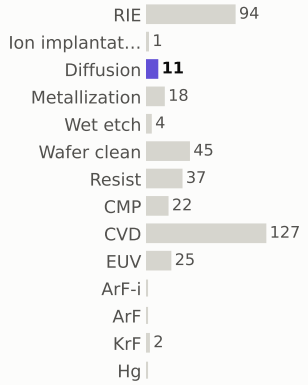
y cut at 0.184 µm



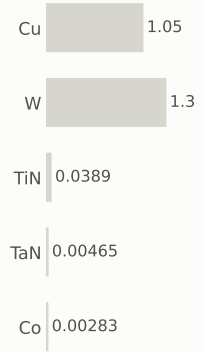
mask for this step

no mask for this step

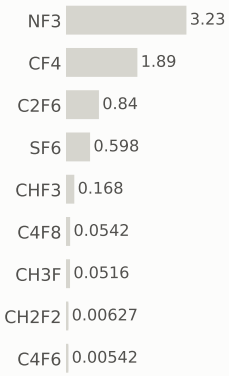
steps by process area



materials deposited (g)



gases emitted (g)



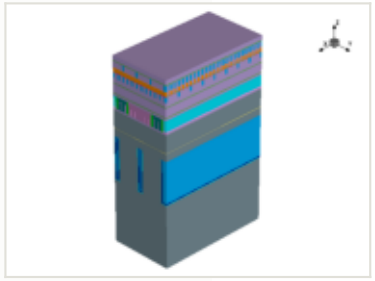
Step #4090: Wafer_clean

clean · Wafer_clean

0.169	415.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



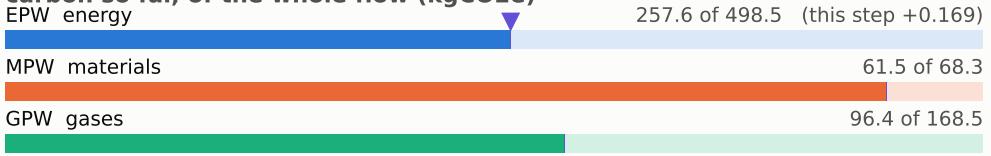
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

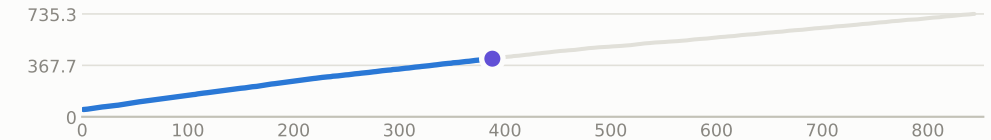
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

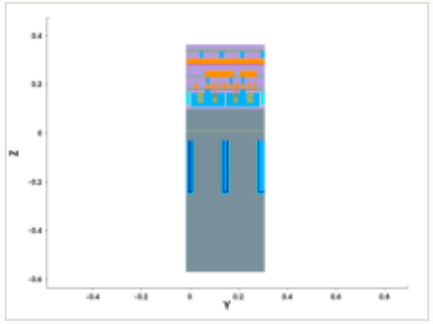
carbon so far, of the whole flow (kgCO2e)



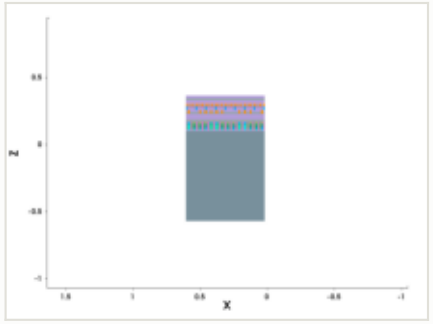
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



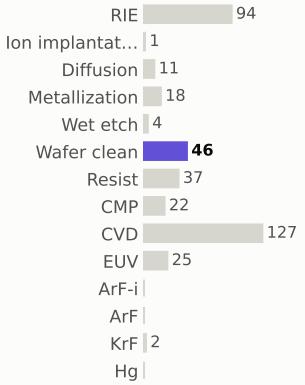
y cut at 0.184 µm



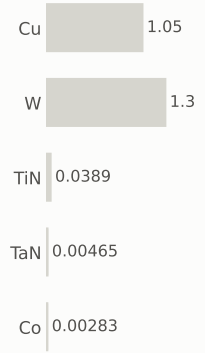
mask for this step

no mask for this step

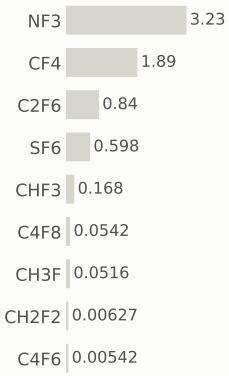
steps by process area



materials deposited (g)



gases emitted (g)



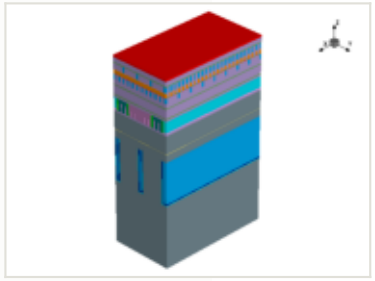
Step #4100: CVD

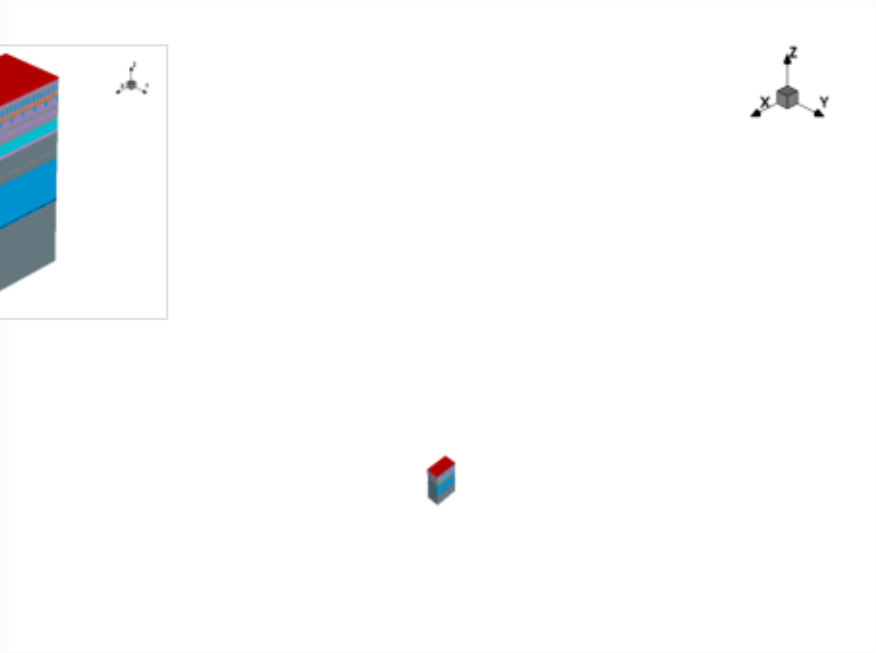
depo_fill · CVD

1.31	416.8	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



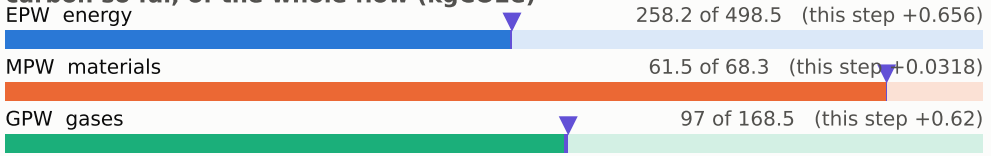


this step

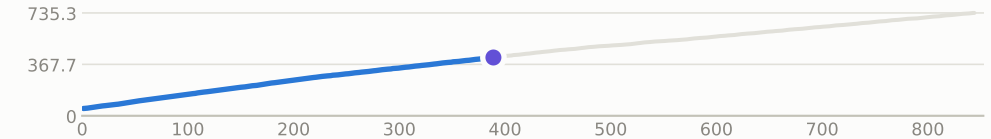
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

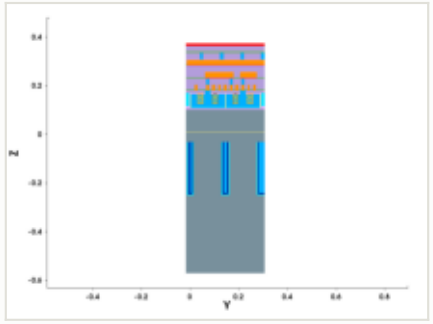
carbon so far, of the whole flow (kgCO2e)



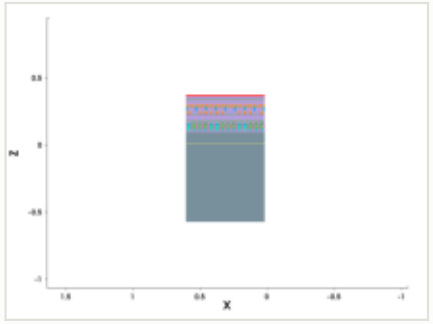
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



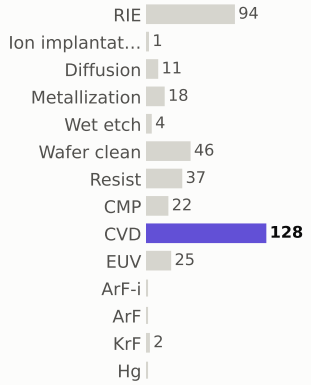
y cut at 0.184 µm



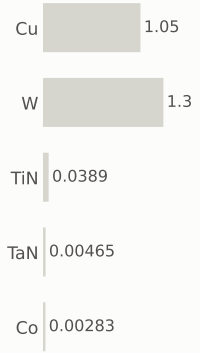
mask for this step

no mask for this step

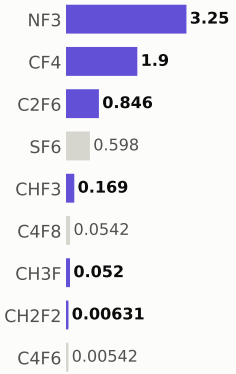
steps by process area



materials deposited (g)



gases emitted (g)



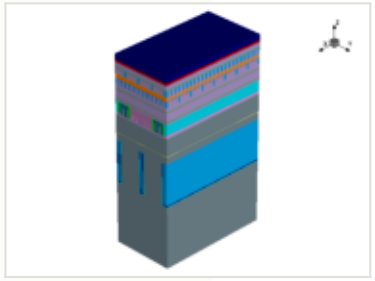
Step #4110: CVD

depo_fill · CVD

1.31	418.1	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



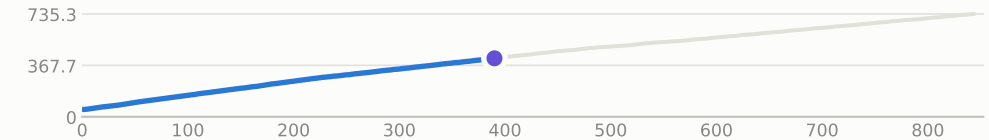
this step

process area	CVD
material	TiN
thickness	0.03 µm
material GWP	43.9778 gCO2e/g
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

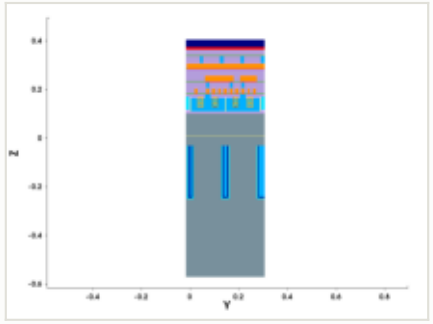
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0322 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

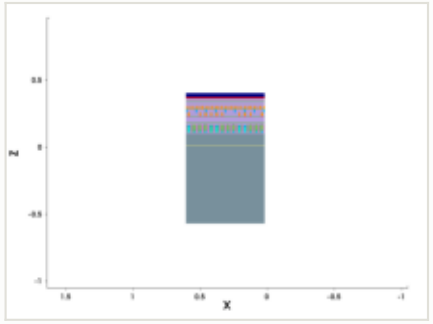
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



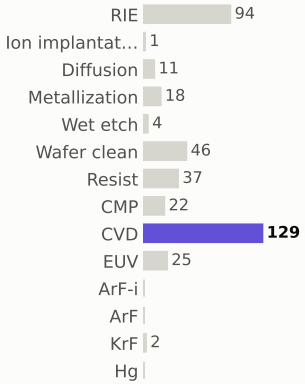
y cut at 0.184 µm



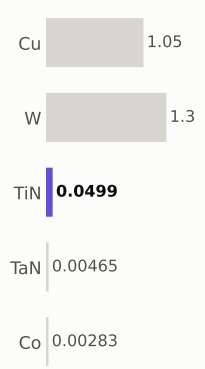
mask for this step

no mask for this step

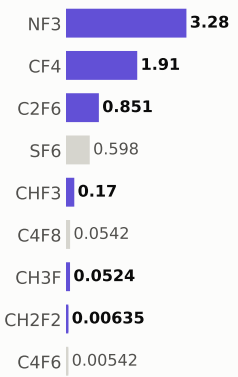
steps by process area



materials deposited (g)



gases emitted (g)



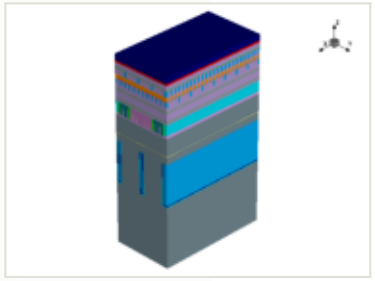
Step #4120: Wafer_clean

clean · Wafer_clean

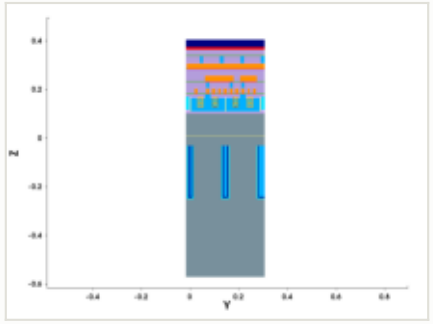
0.169	418.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

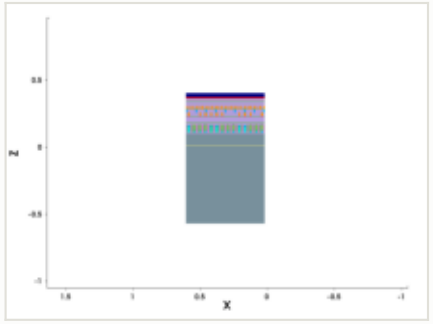
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

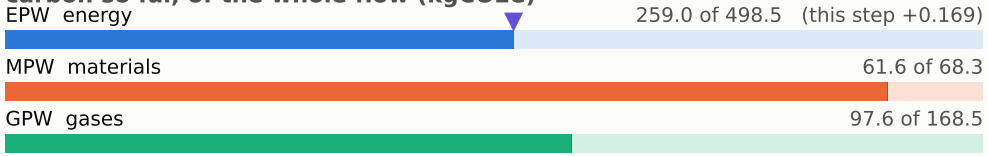
no mask for this step

this step

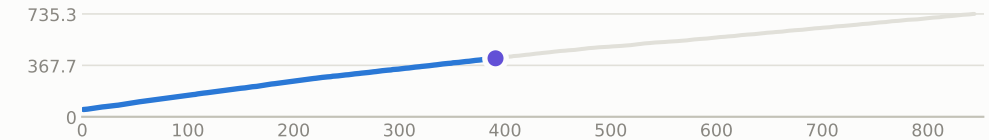
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

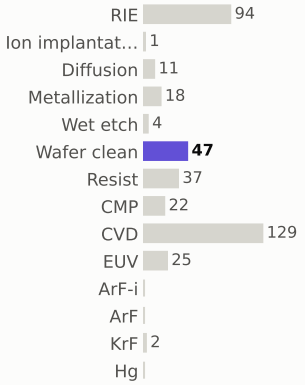
carbon so far, of the whole flow (kgCO2e)



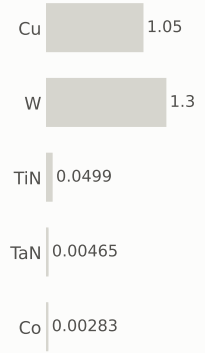
embodied carbon up to each step (kgCO2e per wafer)



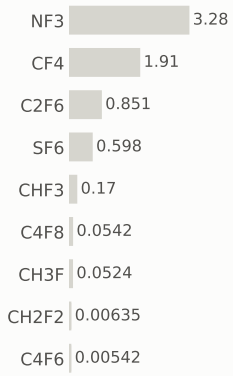
steps by process area



materials deposited (g)



gases emitted (g)



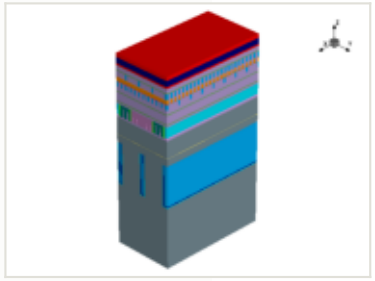
Step #4130: CVD

depo_fill · CVD

1.31	419.5	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

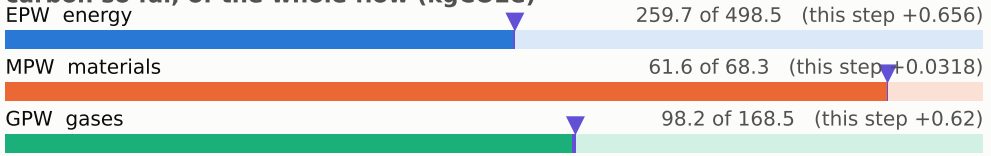


this step

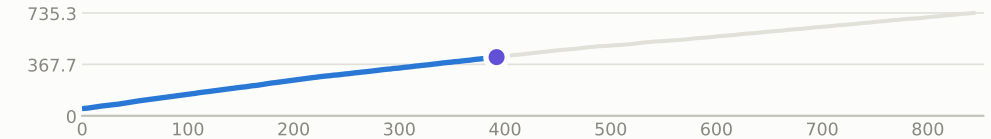
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

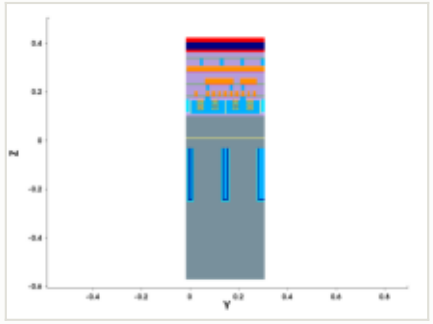
carbon so far, of the whole flow (kgCO2e)



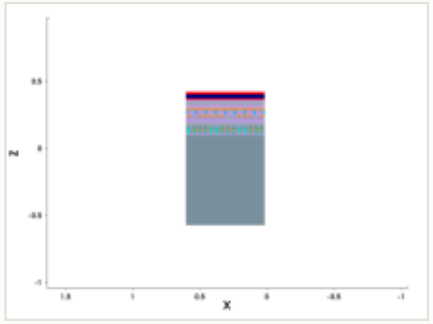
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



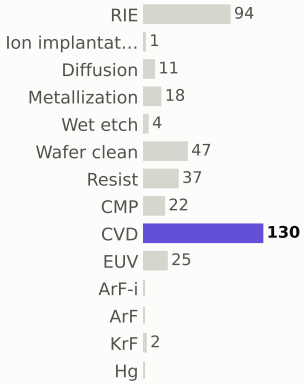
y cut at 0.184 µm



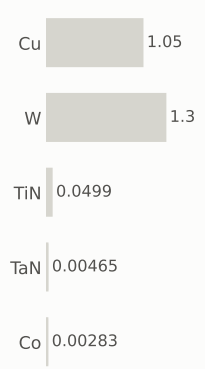
mask for this step

no mask for this step

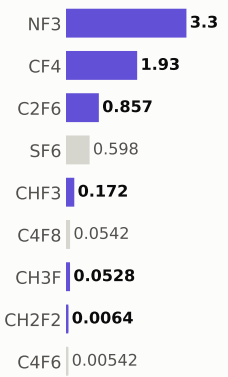
steps by process area



materials deposited (g)



gases emitted (g)



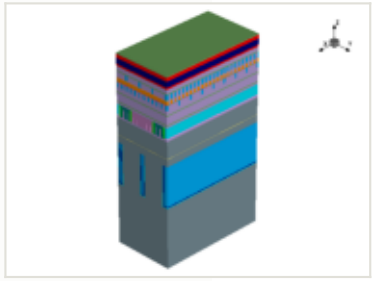
Step #4140: CVD

depo_fill · CVD

1.31	420.8	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

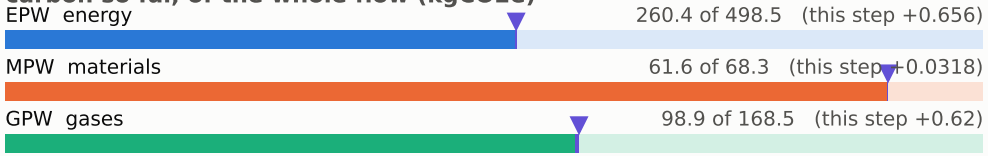


this step

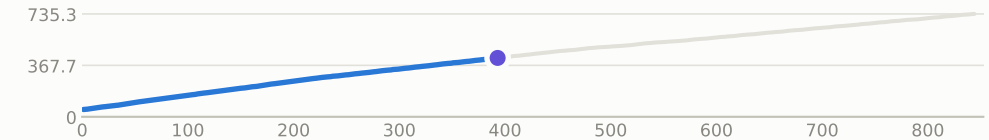
process area	CVD
material	—
thickness	0.005 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

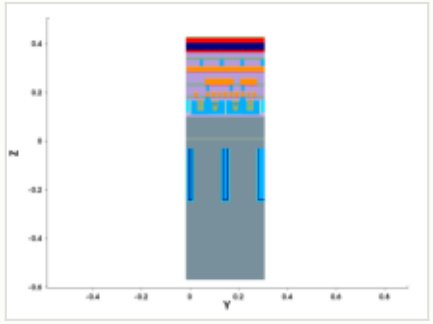
carbon so far, of the whole flow (kgCO2e)



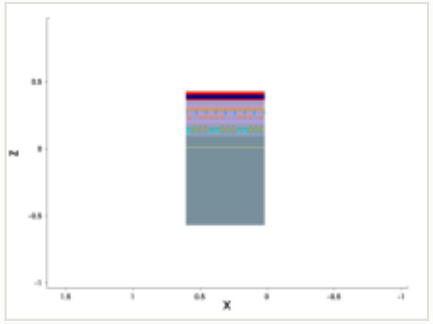
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



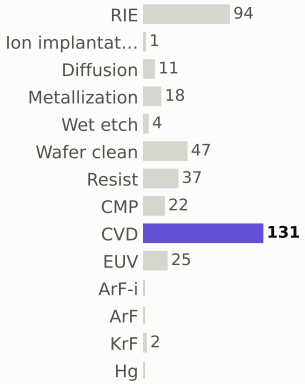
y cut at 0.184 µm



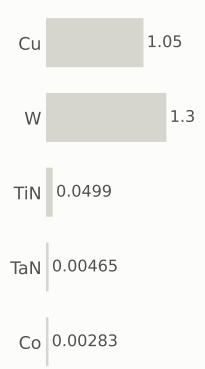
mask for this step

no mask for this step

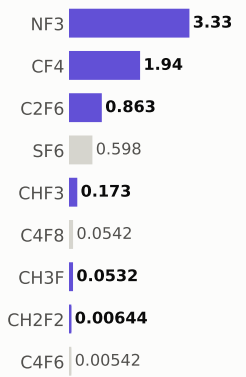
steps by process area



materials deposited (g)



gases emitted (g)



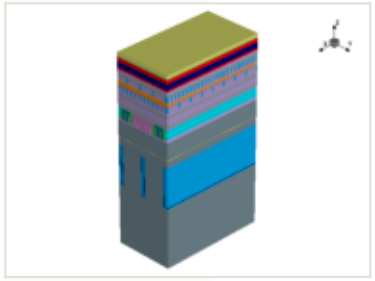
Step #4150: CVD

depo_fill · CVD

1.31	422.2	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

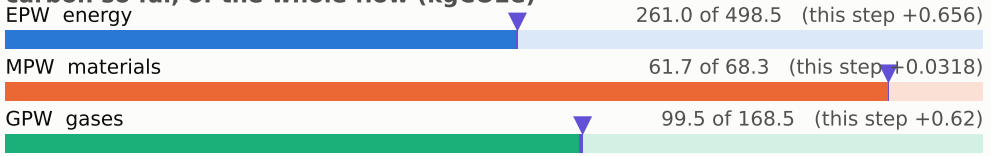


this step

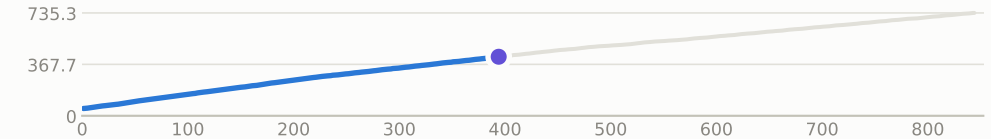
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

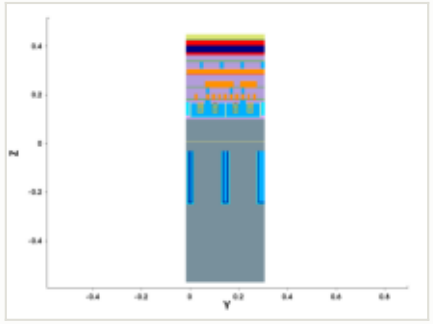
carbon so far, of the whole flow (kgCO2e)



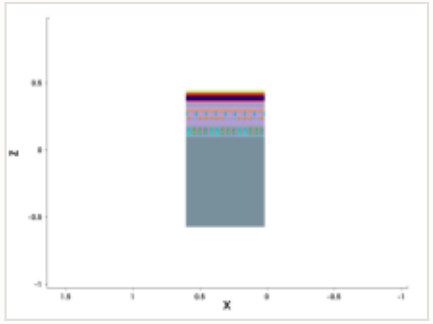
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



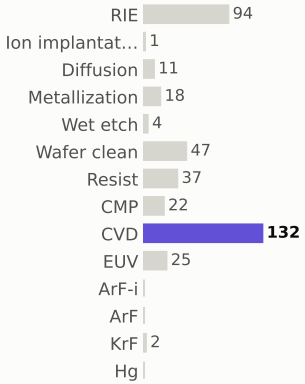
y cut at 0.184 µm



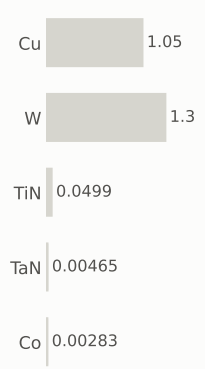
mask for this step

no mask for this step

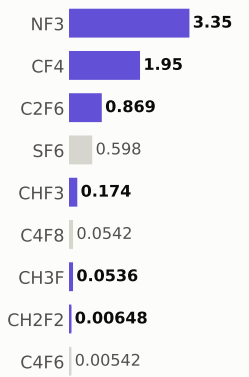
steps by process area



materials deposited (g)



gases emitted (g)



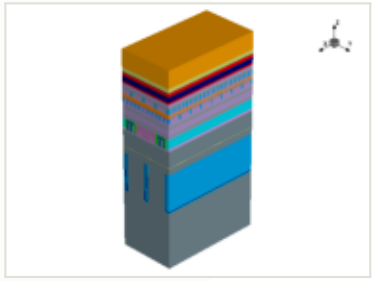
Step #4160: Resist

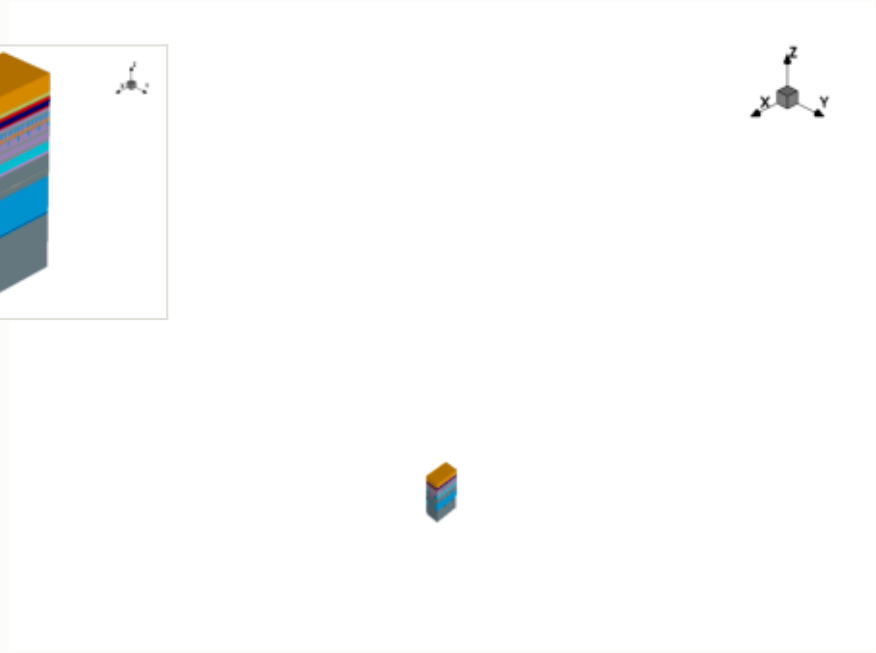
depo_fill · Resist

0.169	422.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



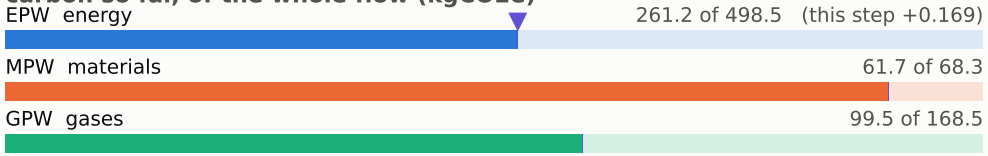


this step

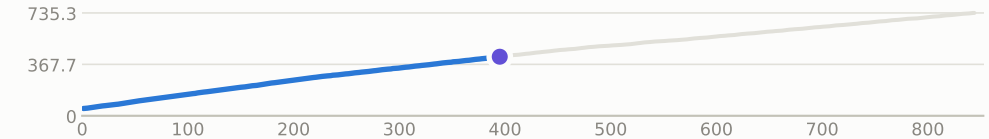
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

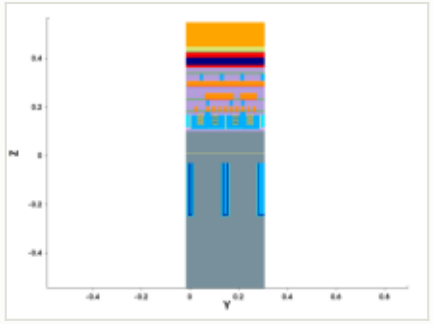
carbon so far, of the whole flow (kgCO2e)



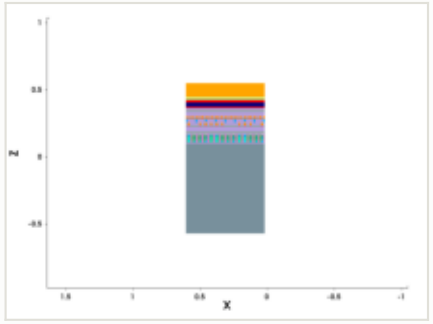
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



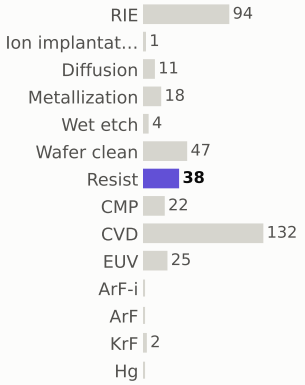
y cut at 0.184 µm



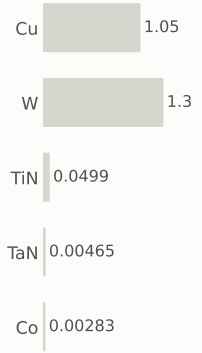
mask for this step

no mask for this step

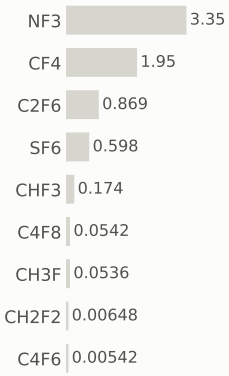
steps by process area



materials deposited (g)



gases emitted (g)



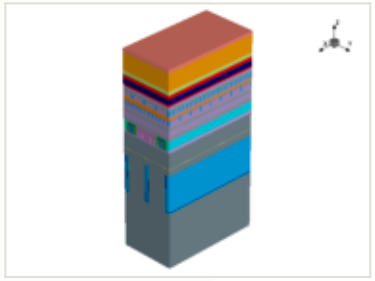
Step #4170: CVD

depo_fill · CVD

1.31	423.6	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

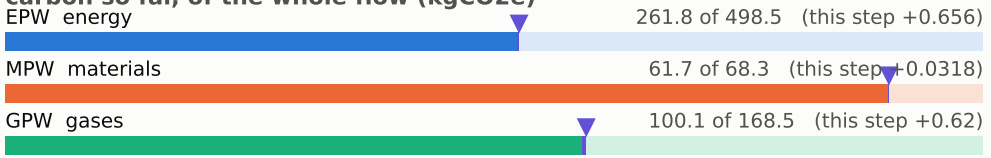


this step

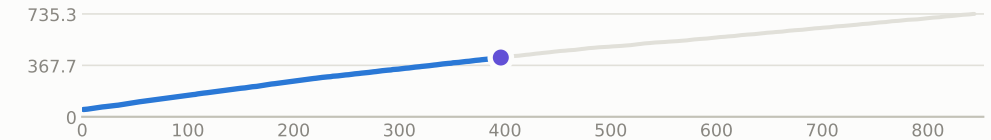
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

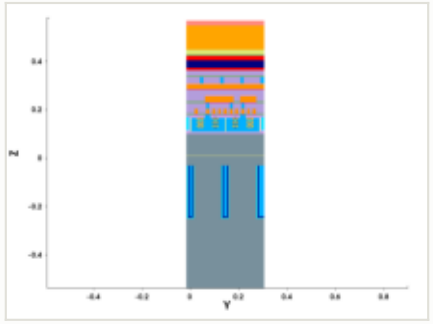
carbon so far, of the whole flow (kgCO2e)



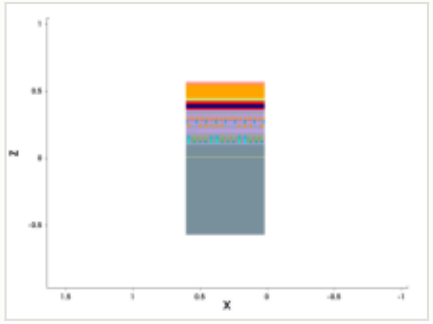
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



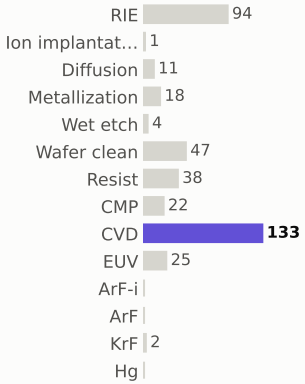
y cut at 0.184 µm



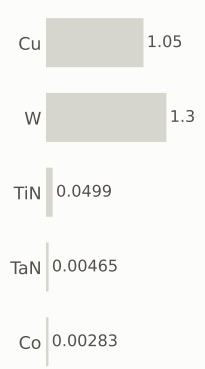
mask for this step

no mask for this step

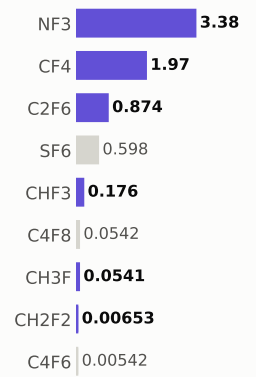
steps by process area



materials deposited (g)



gases emitted (g)



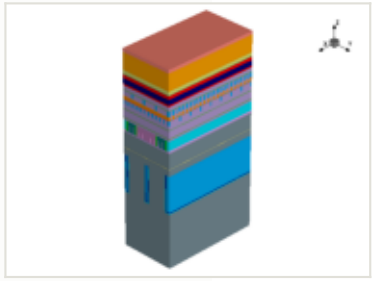
Step #4180: Wafer_clean

clean · Wafer_clean

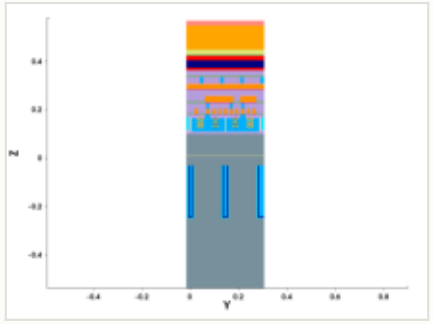
0.169	423.8	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

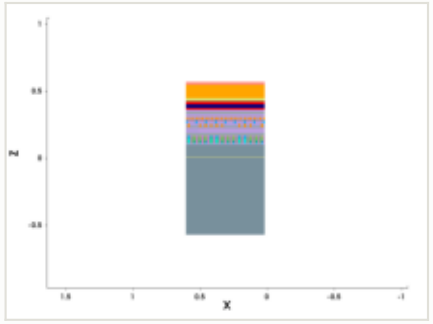
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

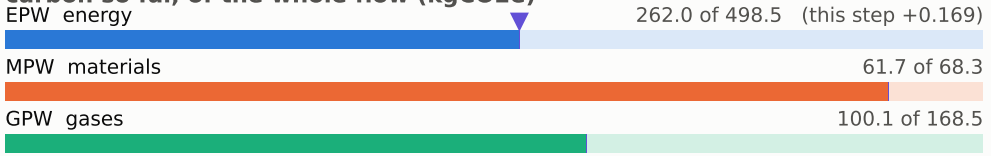
no mask for this step

this step

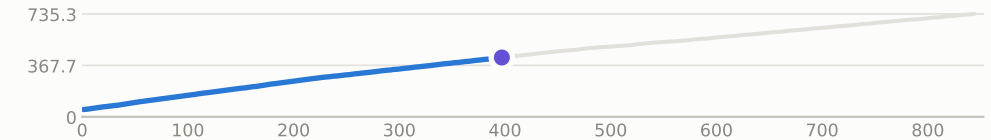
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

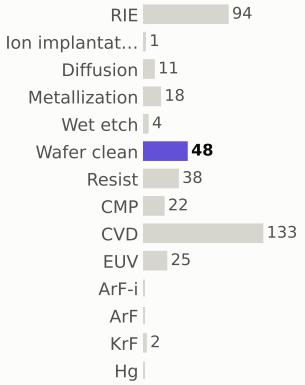
carbon so far, of the whole flow (kgCO2e)



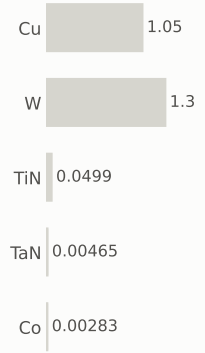
embodied carbon up to each step (kgCO2e per wafer)



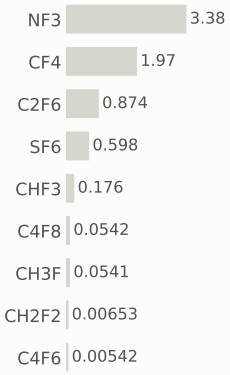
steps by process area



materials deposited (g)



gases emitted (g)



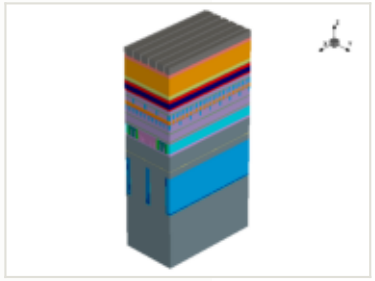
Step #4190: EUV

litho_patterning · EUV

1.96	425.8	0.27%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

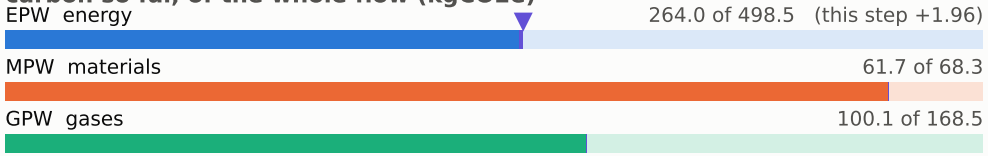


this step

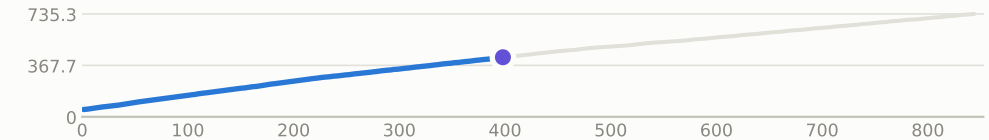
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

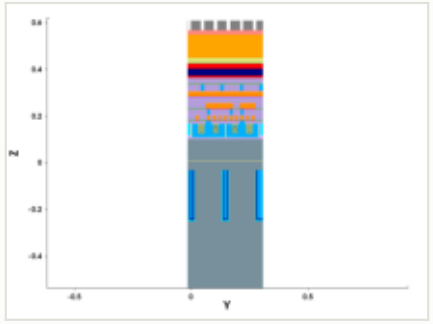
carbon so far, of the whole flow (kgCO2e)



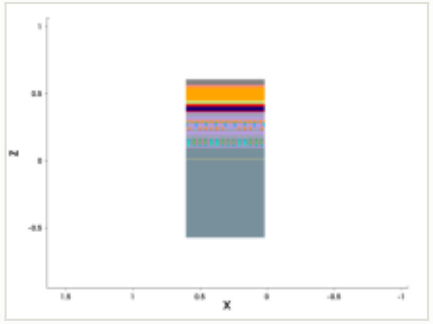
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm

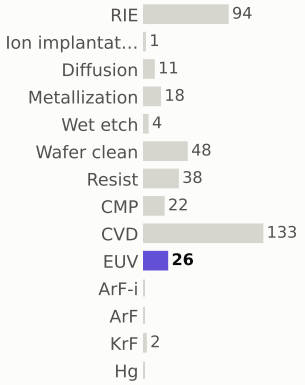


y cut at 0.184 µm

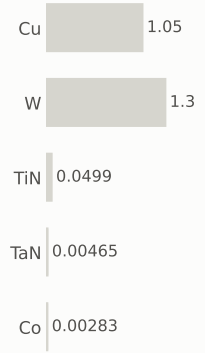


mask: layer 10351:0 (GDSII)

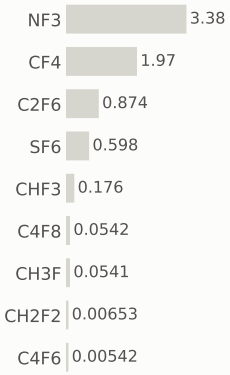
steps by process area



materials deposited (g)



gases emitted (g)



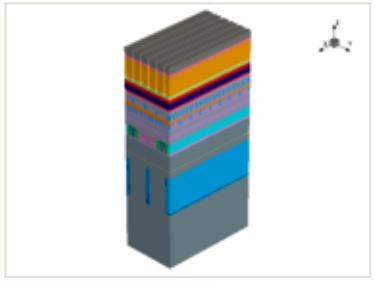
Step #4200: RIE

etch_anisotropic · RIE

1.04	426.8	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

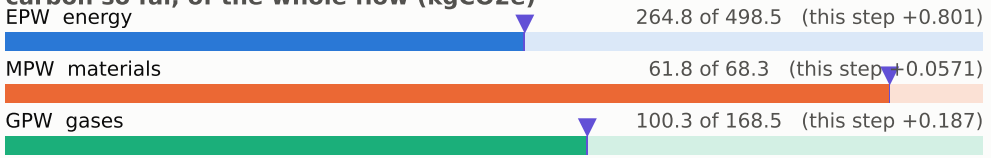


this step

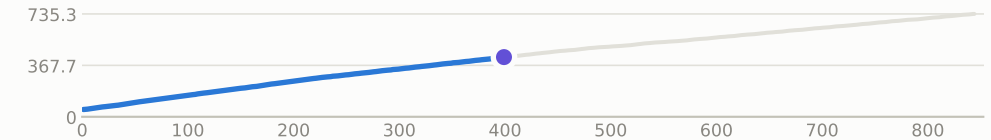
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

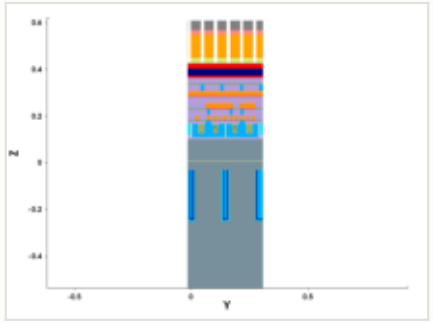
carbon so far, of the whole flow (kgCO2e)



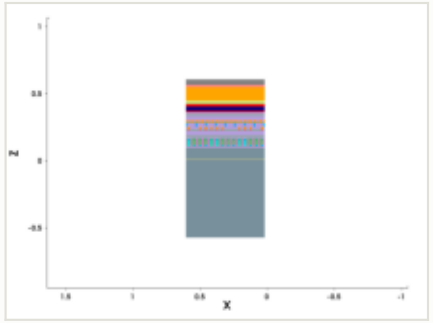
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



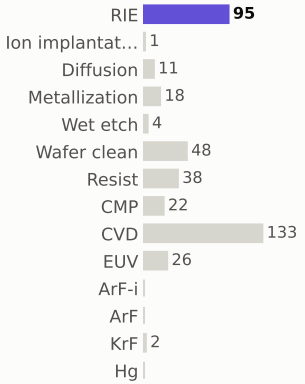
y cut at 0.184 µm



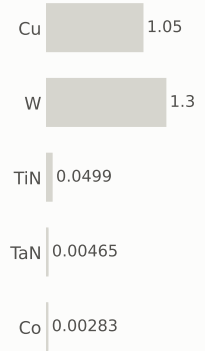
mask for this step

no mask for this step

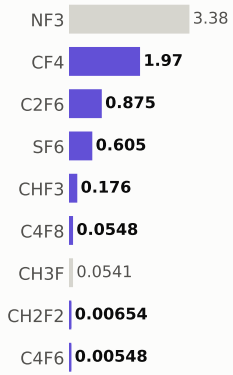
steps by process area



materials deposited (g)



gases emitted (g)



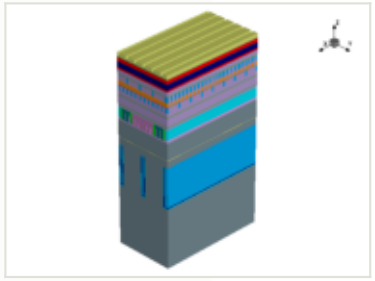
Step #4210: Wafer_clean

etch_strip · Wafer_clean

0.169	427.0	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

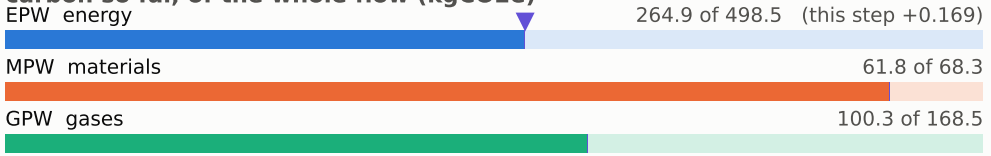


this step

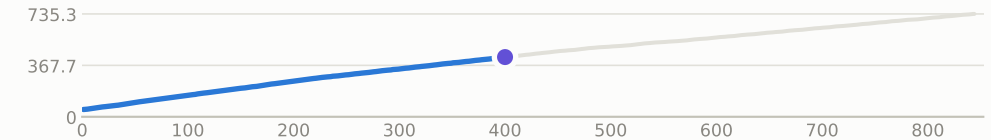
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

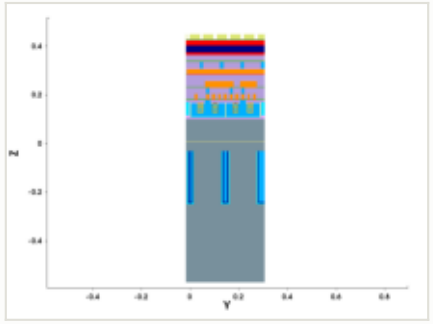
carbon so far, of the whole flow (kgCO2e)



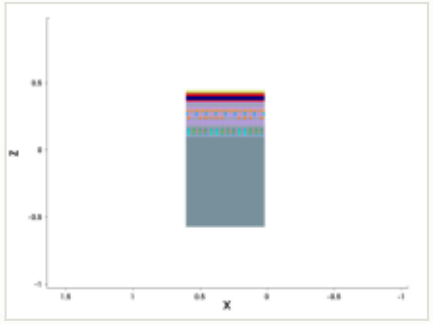
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



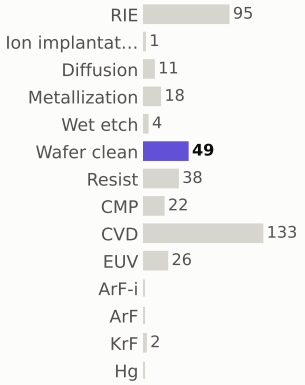
y cut at 0.184 µm



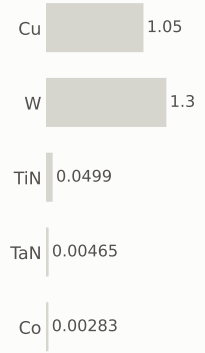
mask for this step

no mask for this step

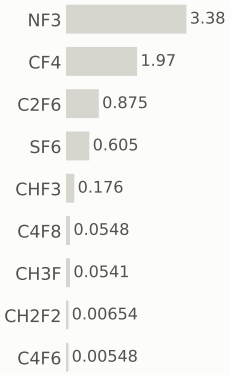
steps by process area



materials deposited (g)



gases emitted (g)



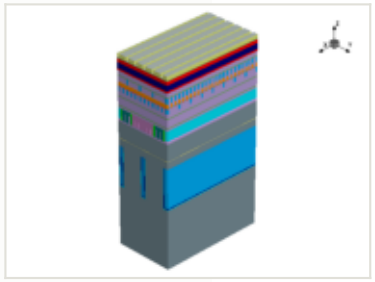
Step #4220: CVD

depo_isotropic · CVD

1.31	428.3	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

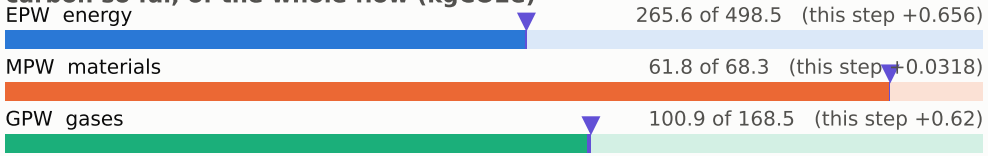
zoomed to fit



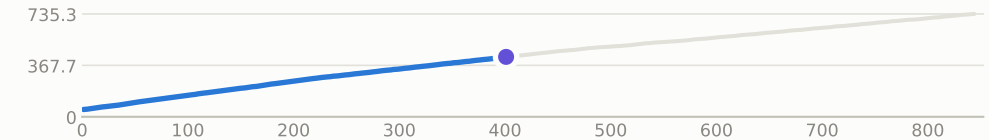
this step

process area	CVD	EPW	0.656 kgCO2e
material	—	EPW tool	0.469 kgCO2e
thickness	0.001 µm	EPW overhead	0.187 kgCO2e
material GWP	—	MPW	0.0318 kgCO2e (gas 0.0318)
input gases	NF3 2.54 g	GPW	0.62 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**1.31 kgCO2e**

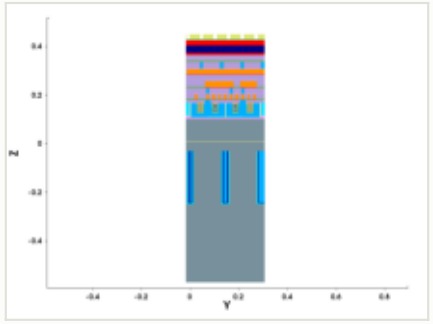
carbon so far, of the whole flow (kgCO2e)



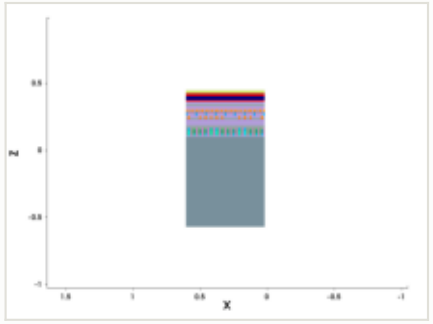
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



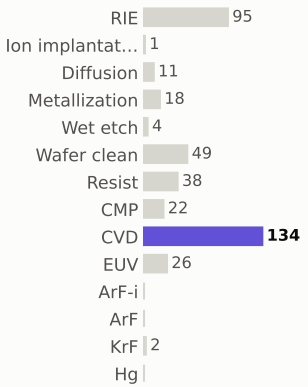
y cut at 0.184 µm



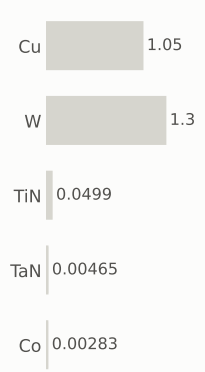
mask for this step

no mask for this step

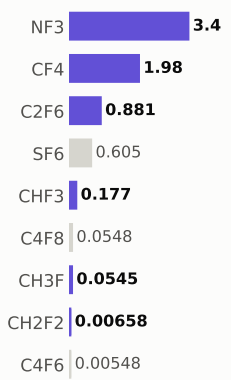
steps by process area



materials deposited (g)



gases emitted (g)



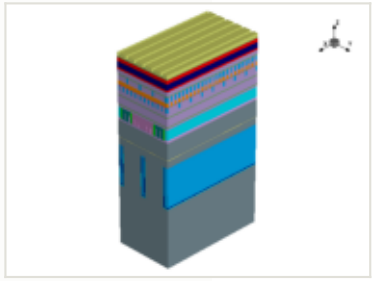
Step #4230: RIE

etch_anisotropic · RIE

1.04	429.3	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

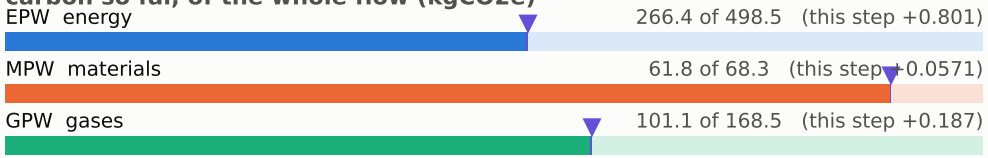


this step

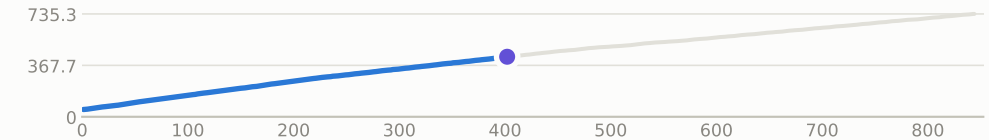
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

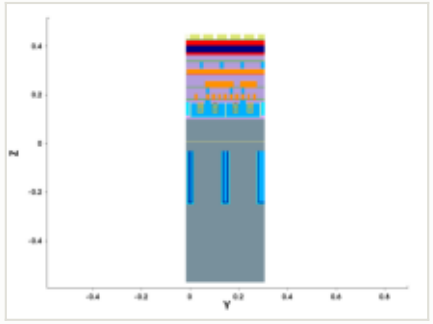
carbon so far, of the whole flow (kgCO2e)



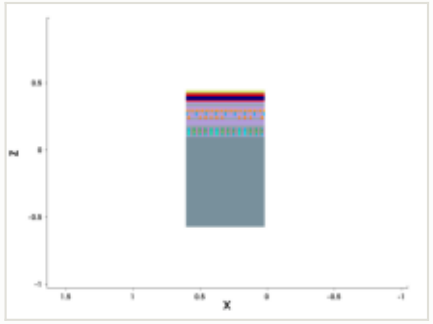
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



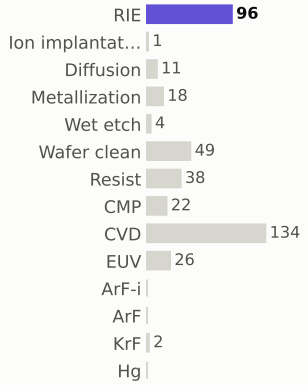
y cut at 0.184 µm



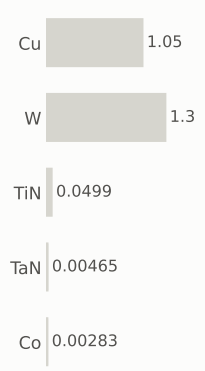
mask for this step

no mask for this step

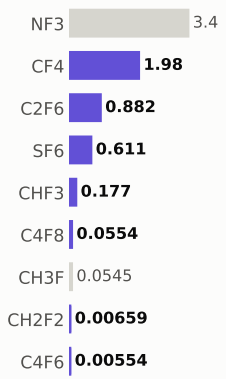
steps by process area



materials deposited (g)



gases emitted (g)



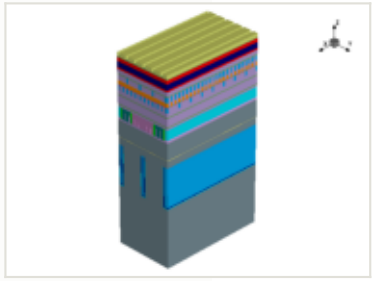
Step #4240: Wafer_clean

clean · Wafer_clean

0.169	429.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



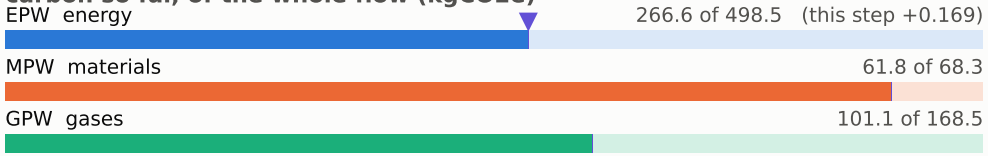
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

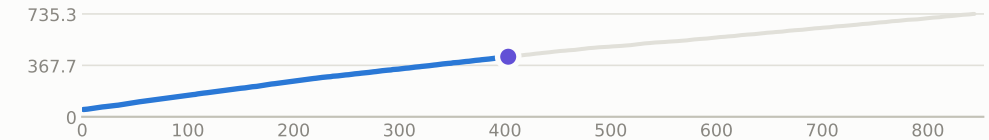
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

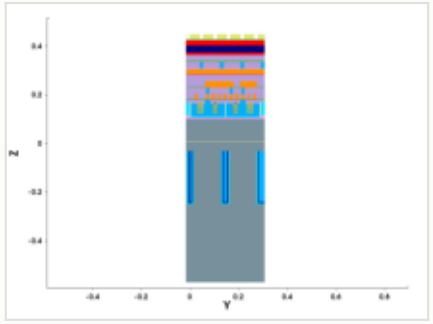
carbon so far, of the whole flow (kgCO2e)



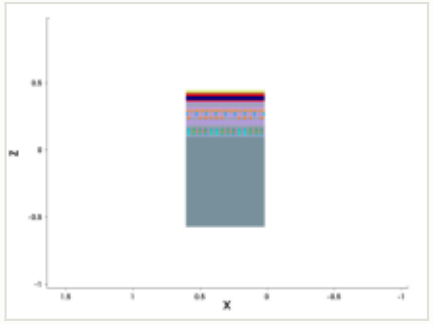
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



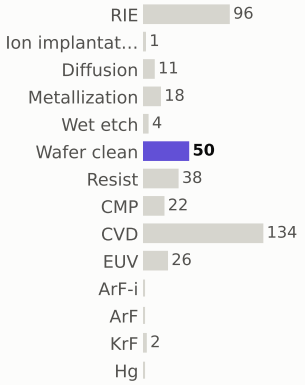
y cut at 0.184 µm



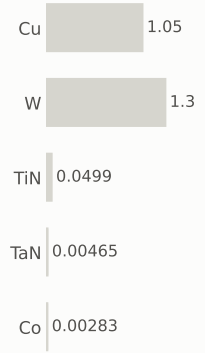
mask for this step

no mask for this step

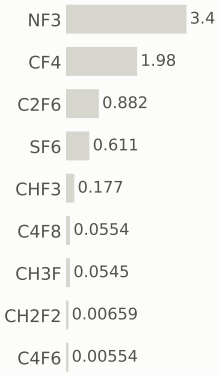
steps by process area



materials deposited (g)



gases emitted (g)



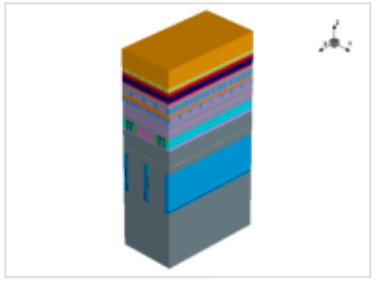
Step #4250: Resist

depo_fill · Resist

0.169	429.7	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

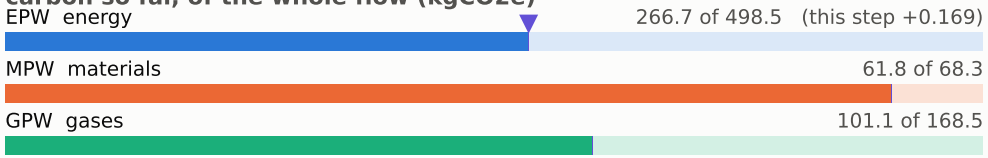


this step

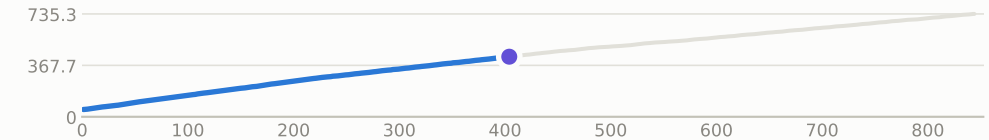
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

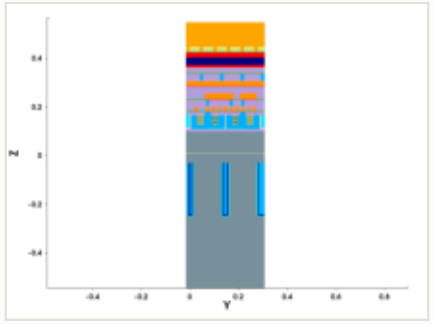
carbon so far, of the whole flow (kgCO2e)



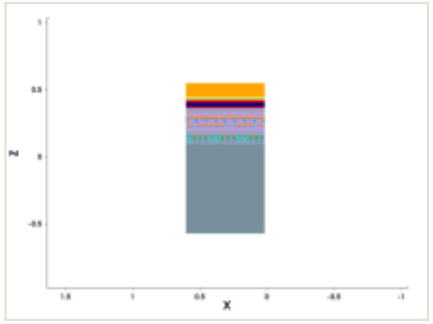
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



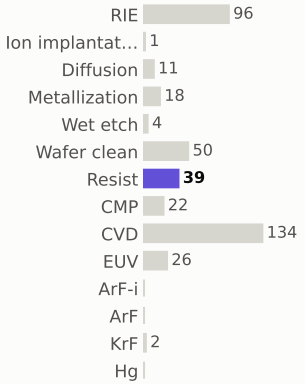
y cut at 0.184 µm



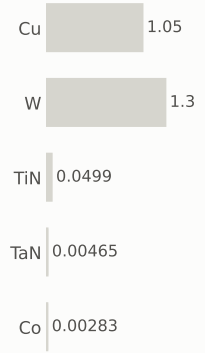
mask for this step

no mask for this step

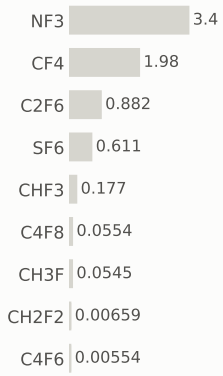
steps by process area



materials deposited (g)



gases emitted (g)



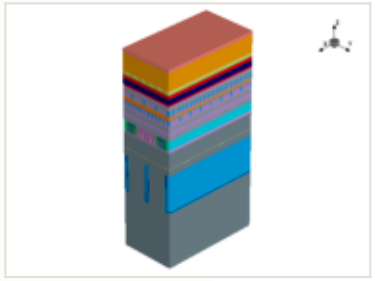
Step #4260: CVD

depo_fill · CVD

1.31	431.0	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

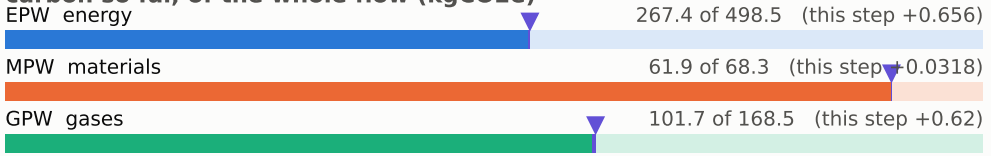


this step

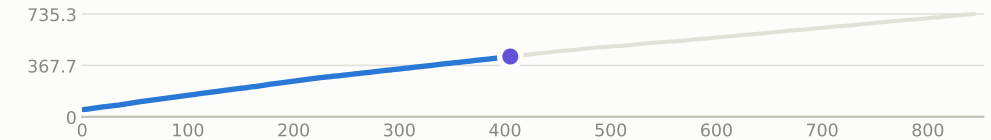
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

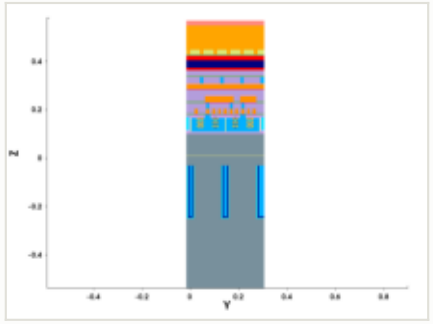
carbon so far, of the whole flow (kgCO2e)



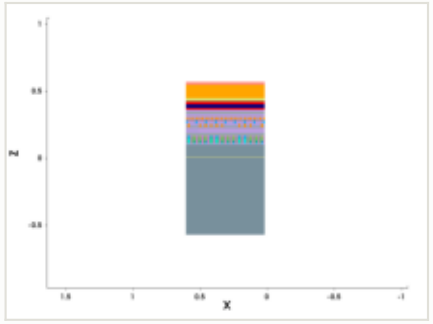
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



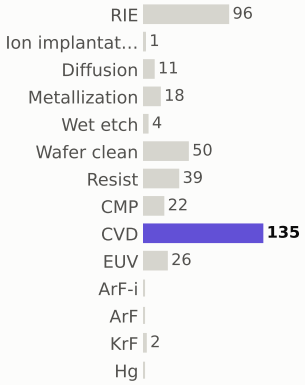
y cut at 0.184 µm



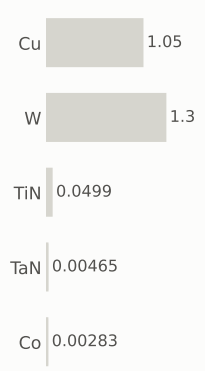
mask for this step

no mask for this step

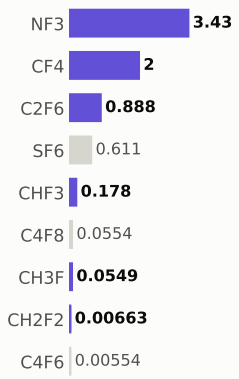
steps by process area



materials deposited (g)



gases emitted (g)



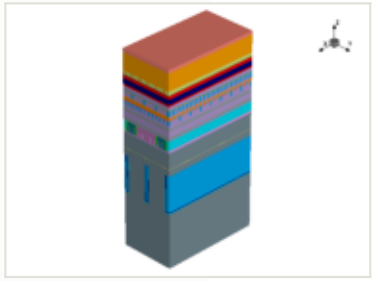
Step #4270: Wafer_clean

clean · Wafer_clean

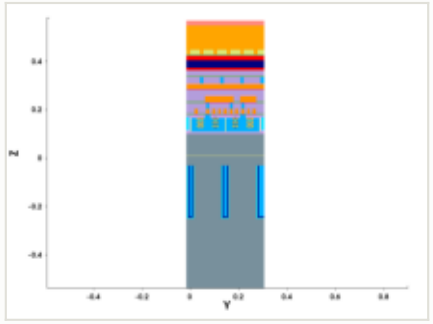
0.169	431.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

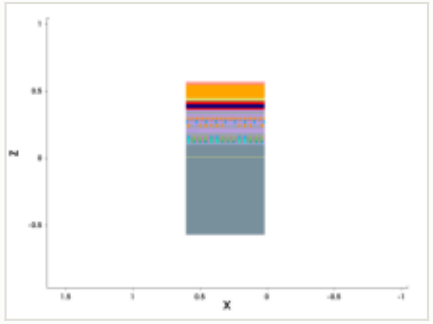
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

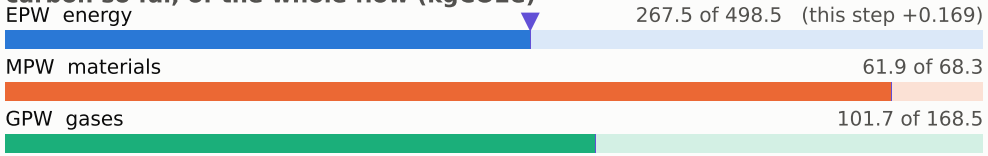
no mask for this step

this step

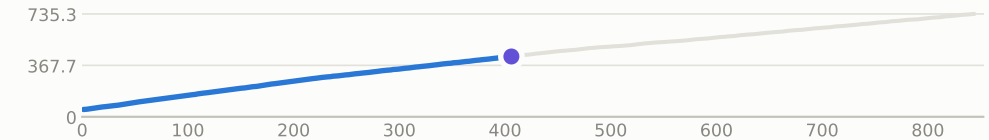
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

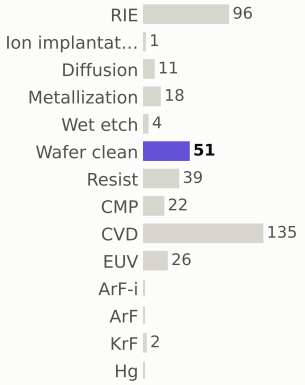
carbon so far, of the whole flow (kgCO2e)



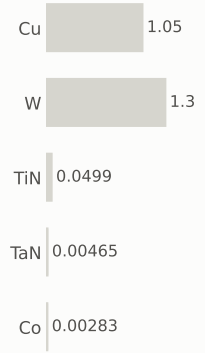
embodied carbon up to each step (kgCO2e per wafer)



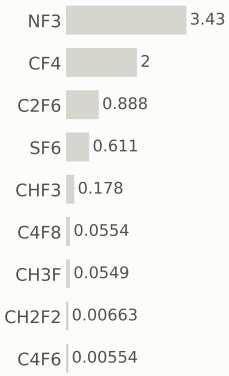
steps by process area



materials deposited (g)



gases emitted (g)



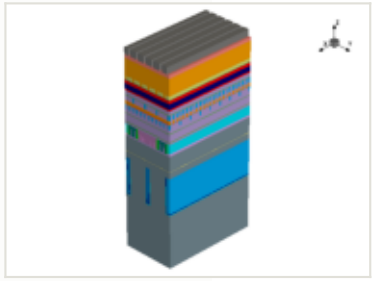
Step #4280: EUV

litho_patterning · EUV

1.96	433.1	0.27%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

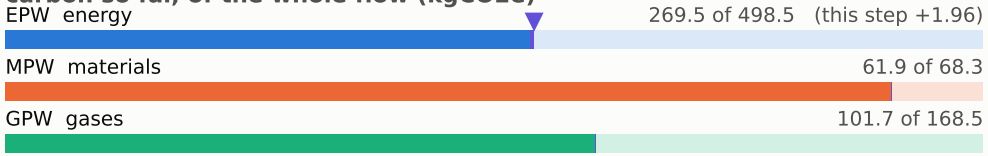


this step

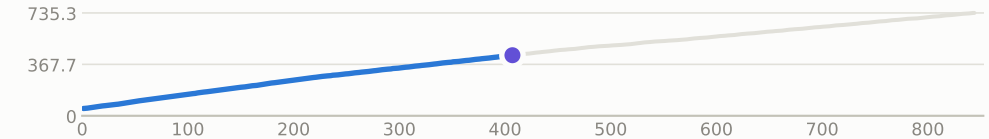
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

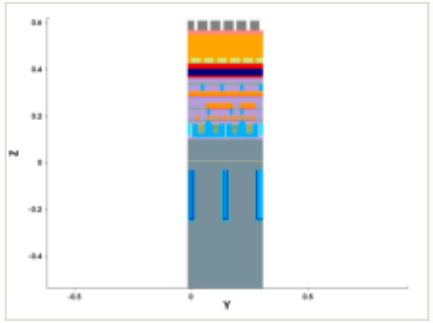
carbon so far, of the whole flow (kgCO2e)



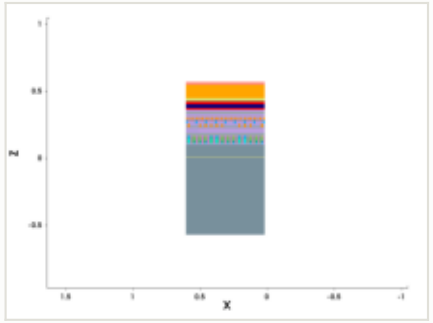
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm

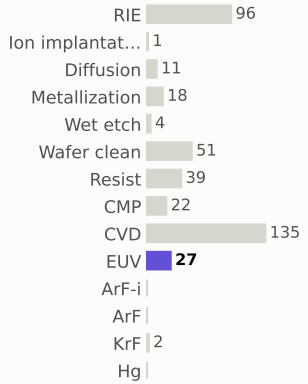


y cut at 0.184 µm

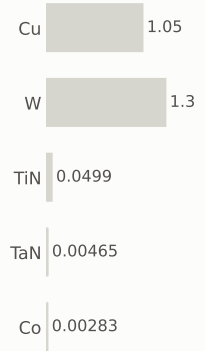


mask: layer 10352:0 (GDSII)

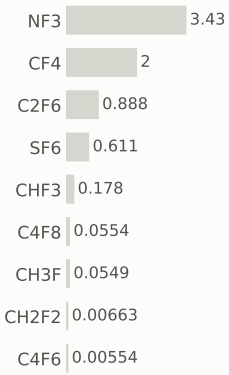
steps by process area



materials deposited (g)



gases emitted (g)



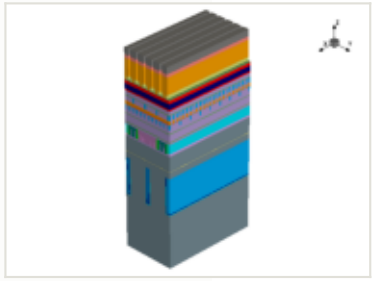
Step #4290: RIE

etch_anisotropic · RIE

1.04	434.1	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

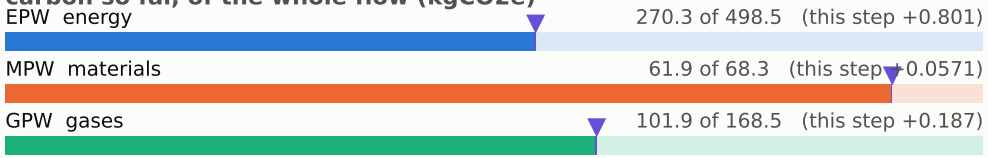


this step

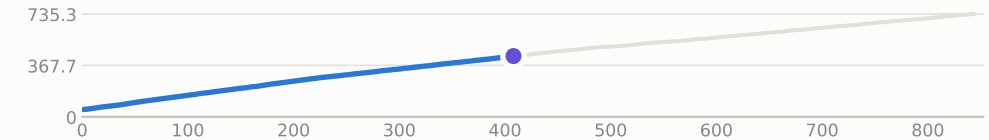
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

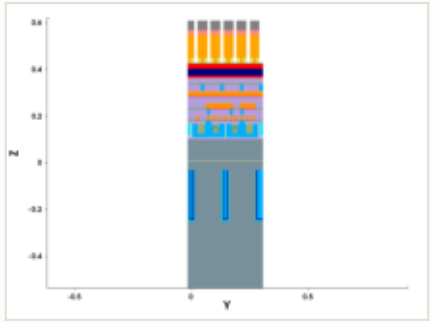
carbon so far, of the whole flow (kgCO2e)



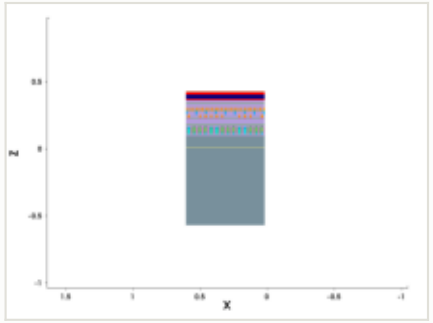
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



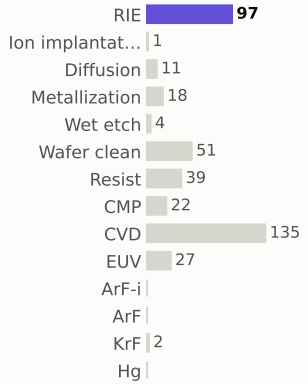
y cut at 0.184 µm



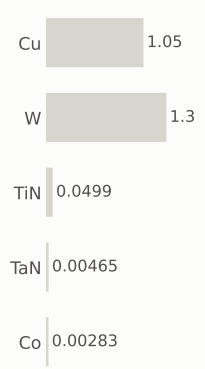
mask for this step

no mask for this step

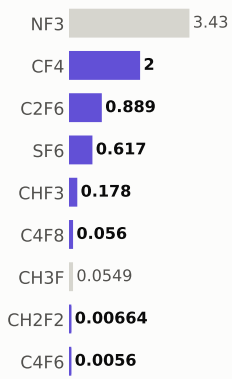
steps by process area



materials deposited (g)



gases emitted (g)



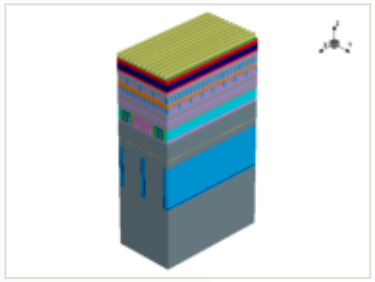
Step #4300: Wafer_clean

etch_strip · Wafer_clean

0.169	434.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

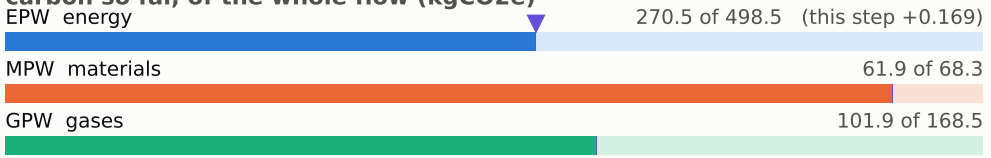


this step

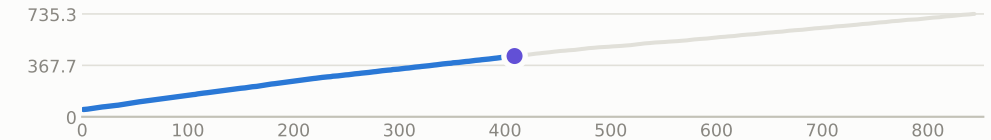
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

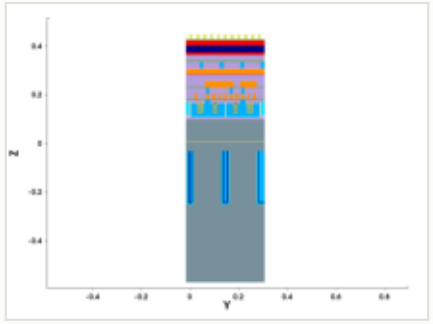
carbon so far, of the whole flow (kgCO2e)



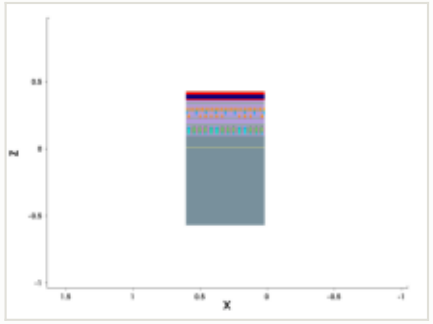
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



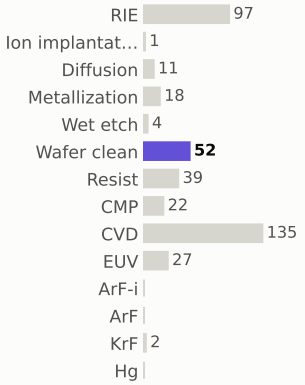
y cut at 0.184 µm



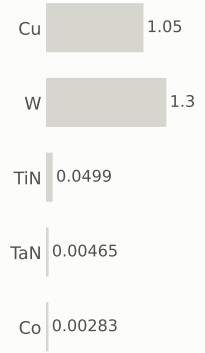
mask for this step

no mask for this step

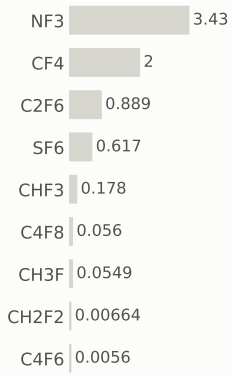
steps by process area



materials deposited (g)



gases emitted (g)



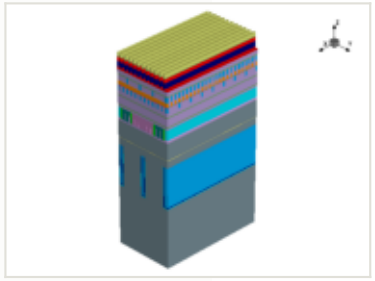
Step #4310: RIE

etch_anisotropic · RIE

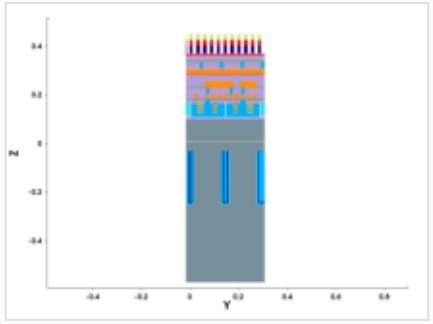
1.04	435.4	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

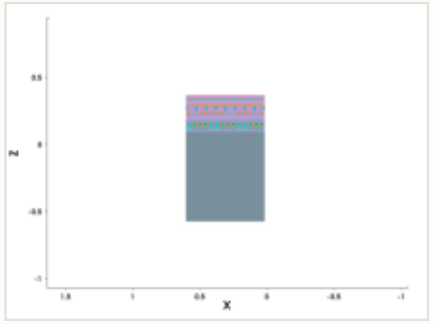
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

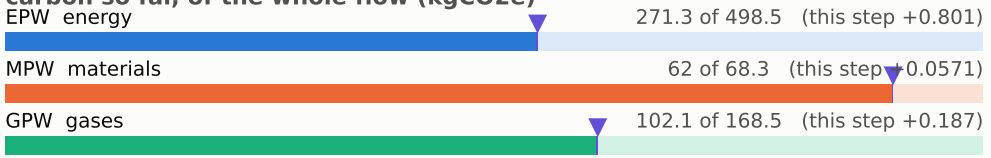
no mask for this step

this step

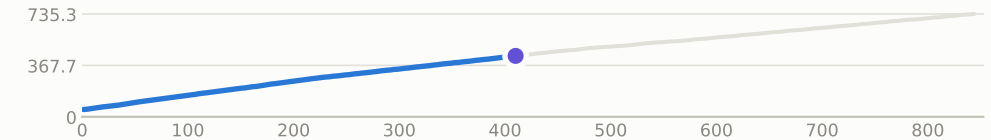
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

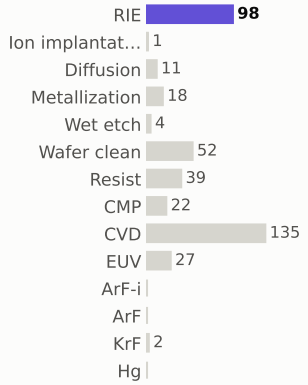
carbon so far, of the whole flow (kgCO2e)



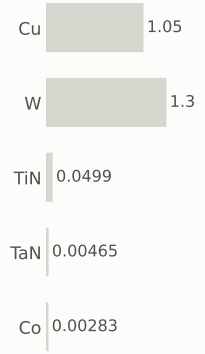
embodied carbon up to each step (kgCO2e per wafer)



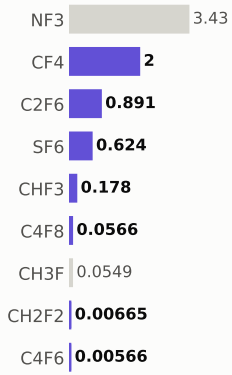
steps by process area



materials deposited (g)



gases emitted (g)



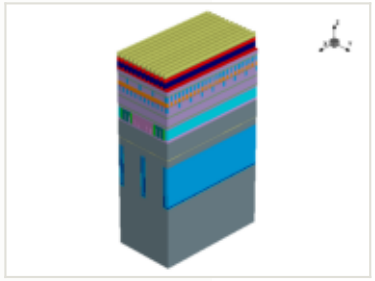
Step #4320: Wafer_clean

clean · Wafer_clean

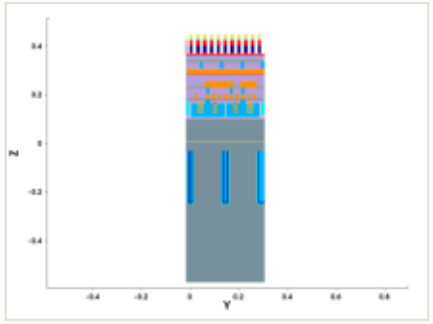
0.169	435.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

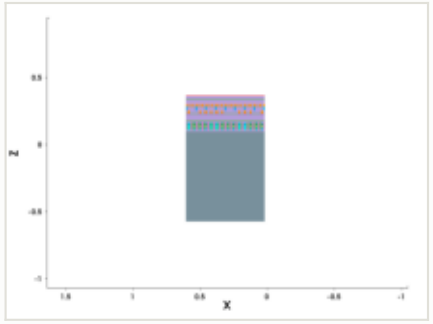
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

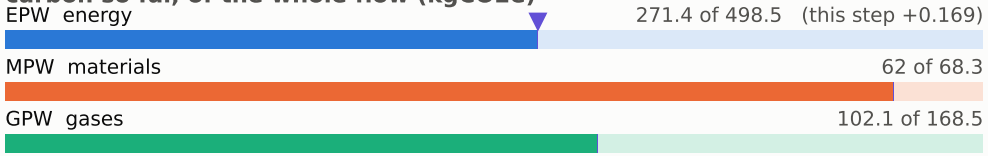
no mask for this step

this step

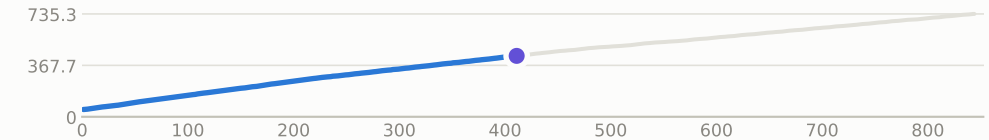
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

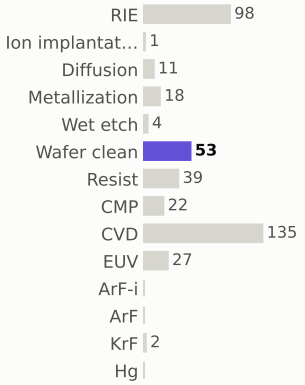
carbon so far, of the whole flow (kgCO2e)



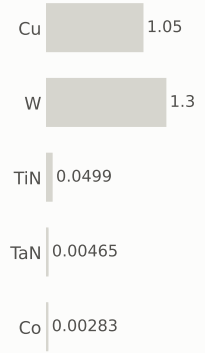
embodied carbon up to each step (kgCO2e per wafer)



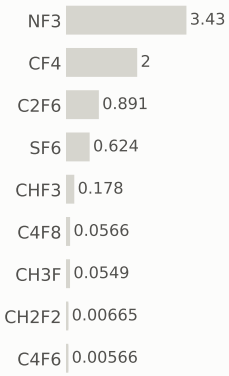
steps by process area



materials deposited (g)



gases emitted (g)



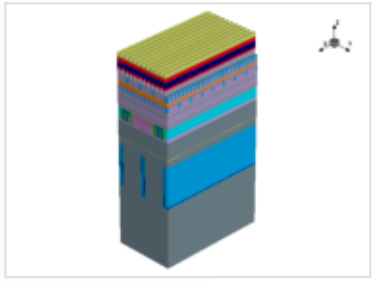
Step #4330: RIE

etch_anisotropic · RIE

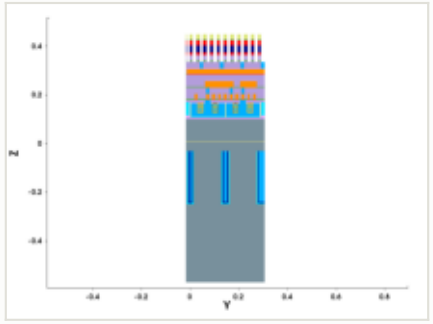
1.04	436.6	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

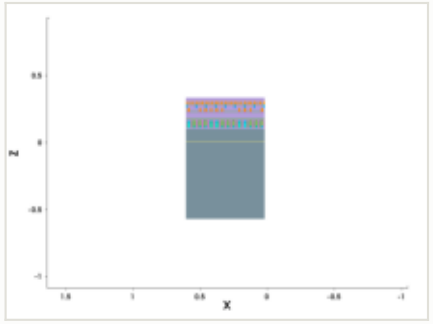
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

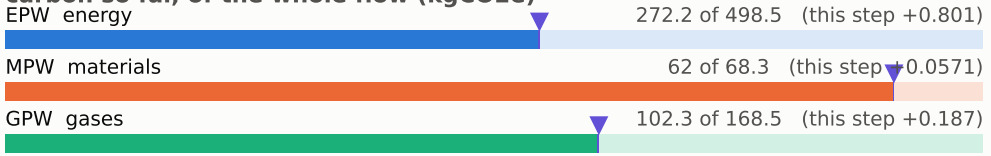
no mask for this step

this step

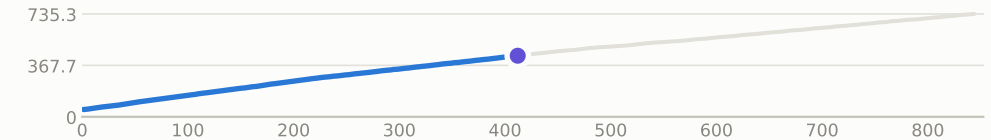
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

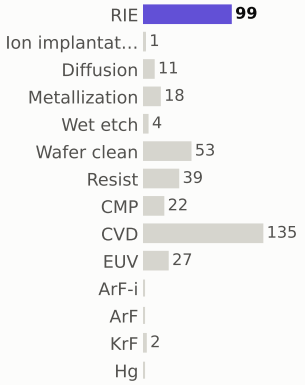
carbon so far, of the whole flow (kgCO2e)



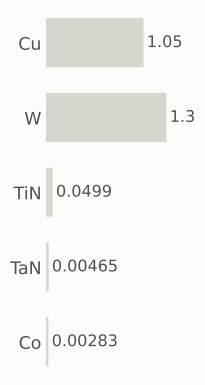
embodied carbon up to each step (kgCO2e per wafer)



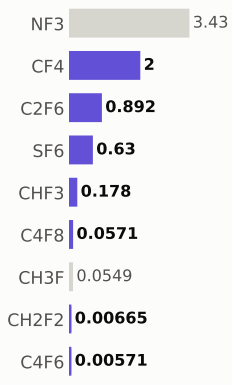
steps by process area



materials deposited (g)



gases emitted (g)



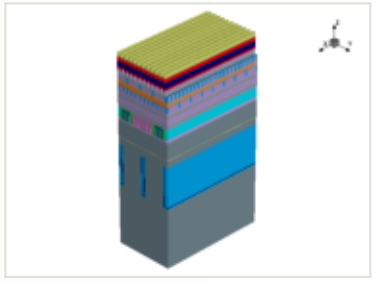
Step #4340: Wafer_clean

clean · Wafer_clean

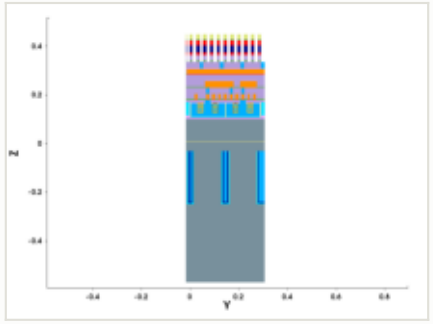
0.169	436.7	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

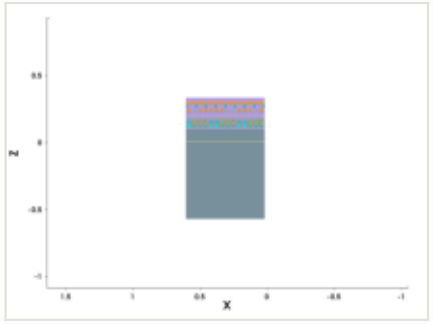
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

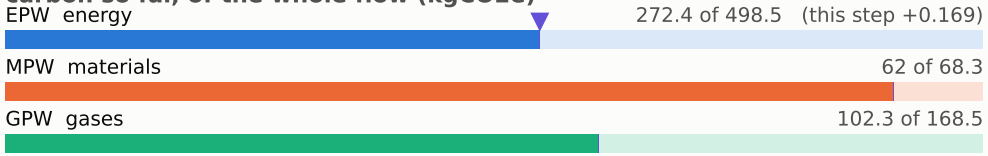
no mask for this step

this step

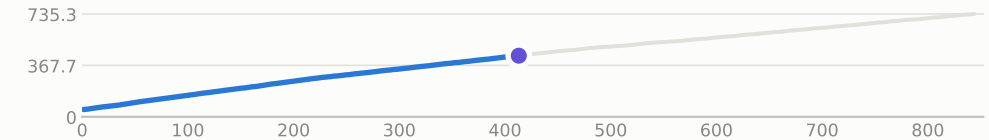
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

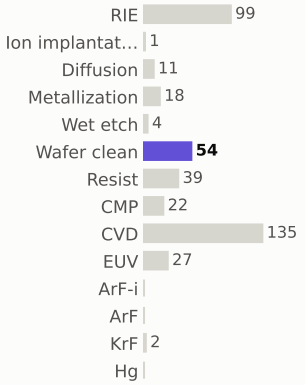
carbon so far, of the whole flow (kgCO2e)



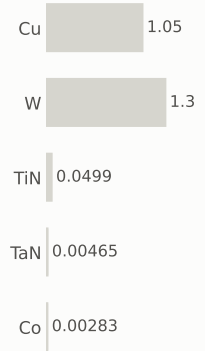
embodied carbon up to each step (kgCO2e per wafer)



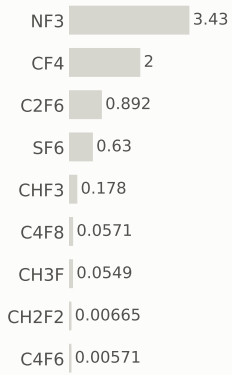
steps by process area



materials deposited (g)



gases emitted (g)



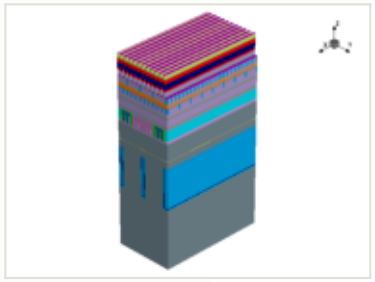
Step #4350: Metallization

depo_ellipsoidal · Metallization

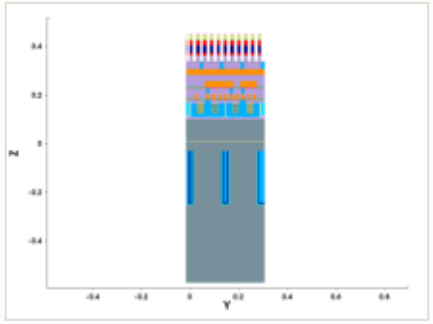
1.2	437.9	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

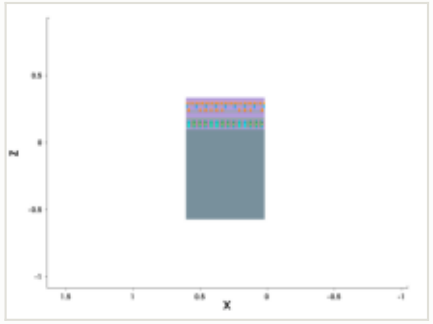
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

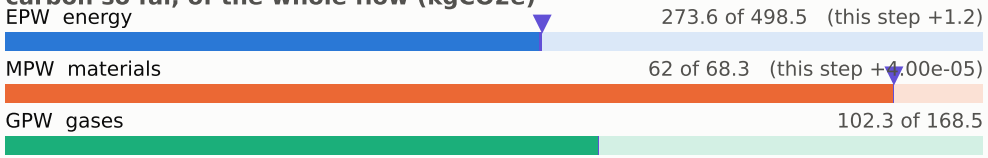
no mask for this step

this step

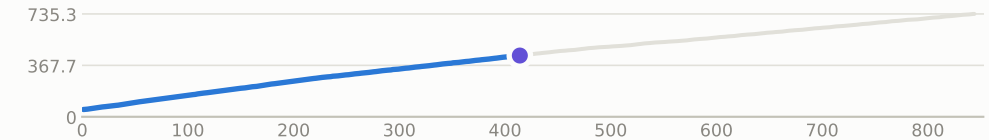
process area	Metallization
material	TaN
thickness	0.0016 µm
material GWP	25.8159 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.00e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

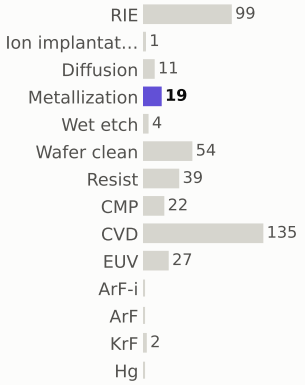
carbon so far, of the whole flow (kgCO2e)



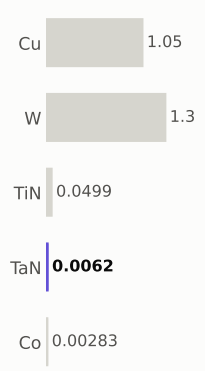
embodied carbon up to each step (kgCO2e per wafer)



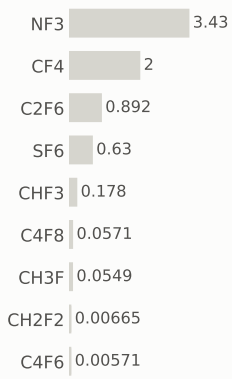
steps by process area



materials deposited (g)



gases emitted (g)



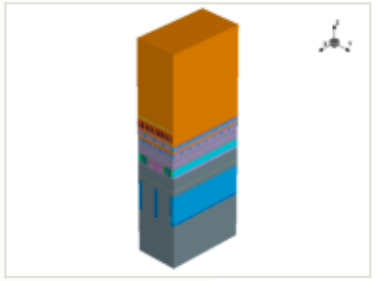
Step #4360: Metallization

depo_fill · Metallization

1.21	439.1	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

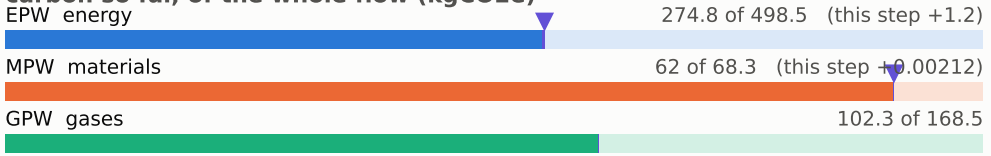


this step

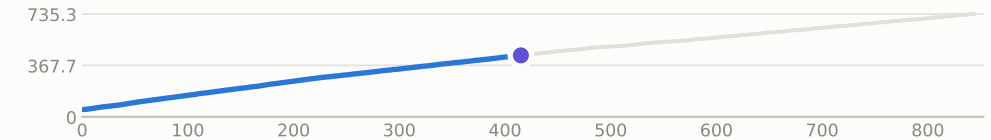
process area	Metallization
material	Cu
thickness	0.55 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00212 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

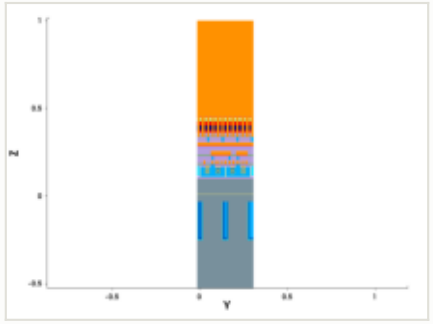
carbon so far, of the whole flow (kgCO2e)



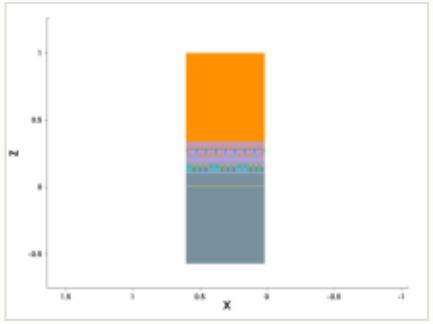
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



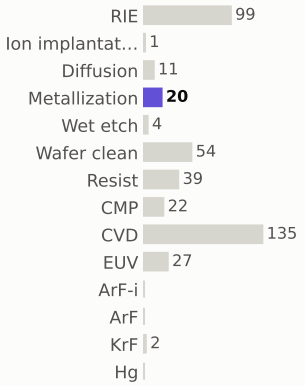
y cut at 0.184 µm



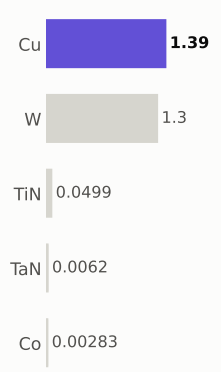
mask for this step

no mask for this step

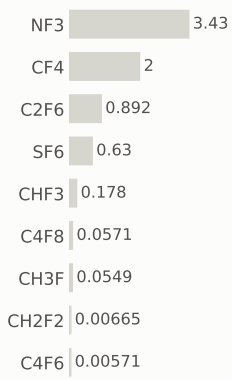
steps by process area



materials deposited (g)



gases emitted (g)



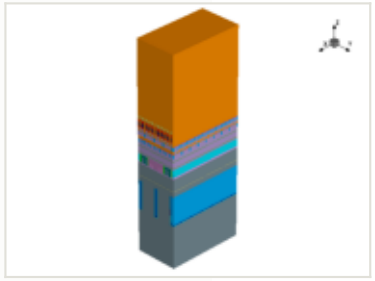
Step #4370: Diffusion

treatment · Diffusion

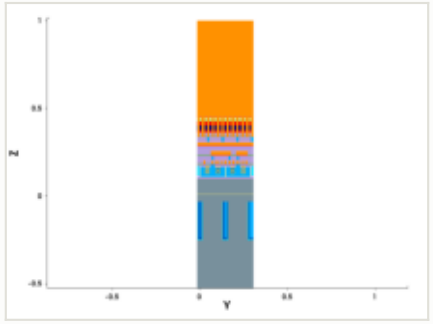
0.801	439.9	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

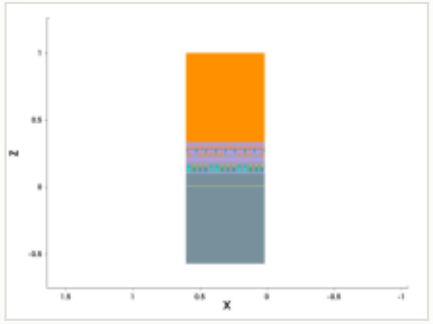
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

no mask for this step

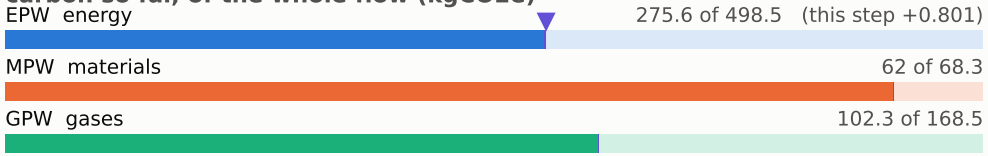
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

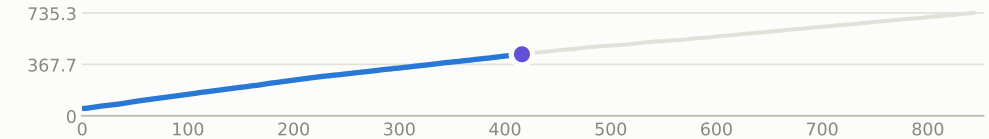
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

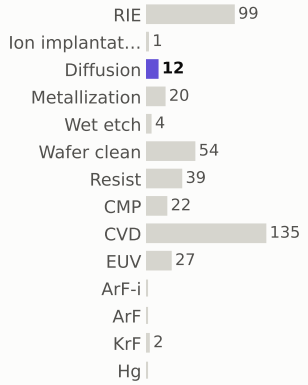
carbon so far, of the whole flow (kgCO2e)



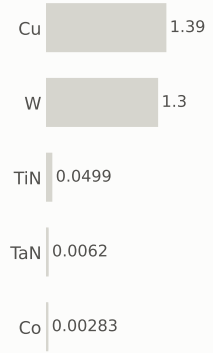
embodied carbon up to each step (kgCO2e per wafer)



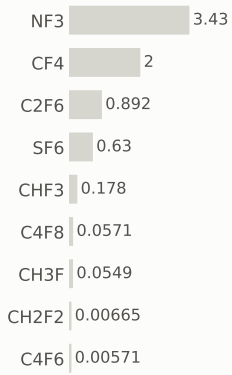
steps by process area



materials deposited (g)



gases emitted (g)



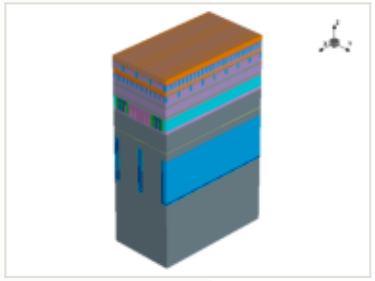
Step #4380: CMP

etch_cmp · CMP

0.169	440.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

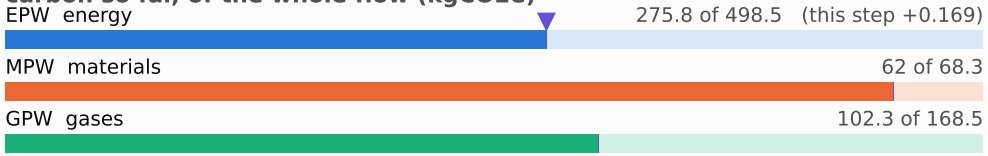
zoomed to fit



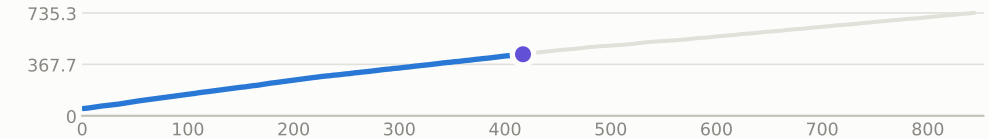
this step

process area	CMP	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

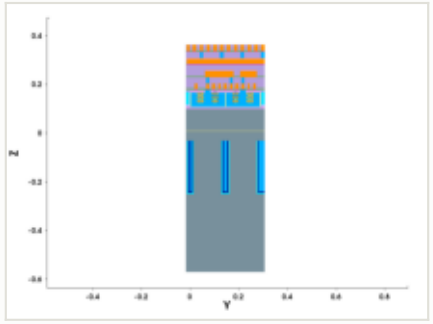
carbon so far, of the whole flow (kgCO2e)



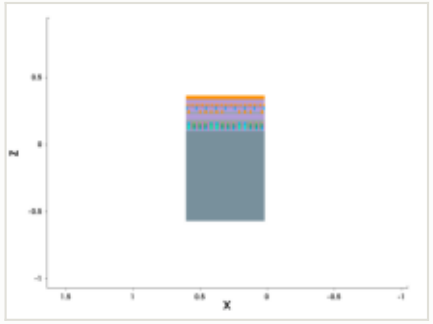
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



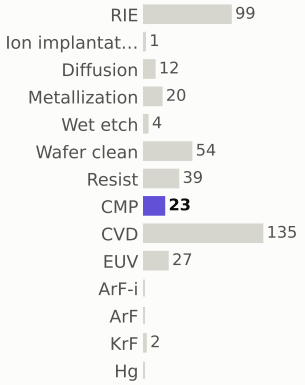
y cut at 0.184 µm



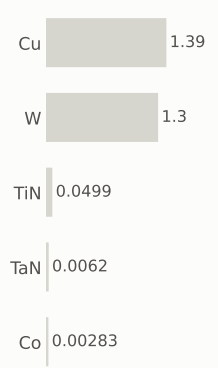
mask for this step

no mask for this step

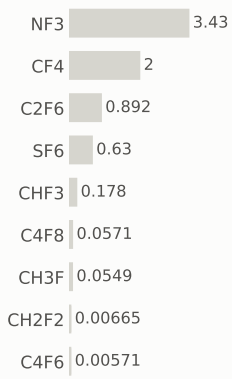
steps by process area



materials deposited (g)



gases emitted (g)



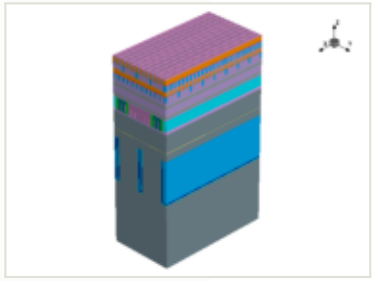
Step #4390: Metallization

depo_isotropic · Metallization

1.2	441.3	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

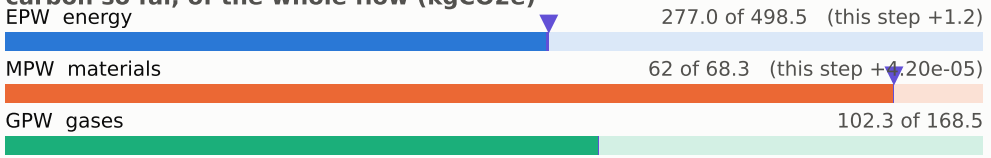


this step

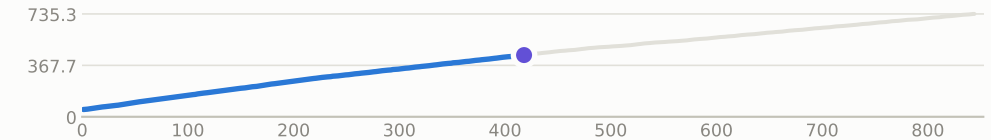
process area	Metallization
material	Co
thickness	0.0015 µm
material GWP	44.5 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.20e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

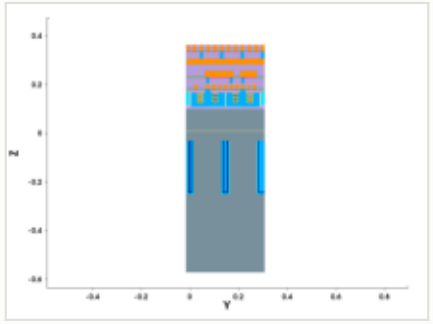
carbon so far, of the whole flow (kgCO2e)



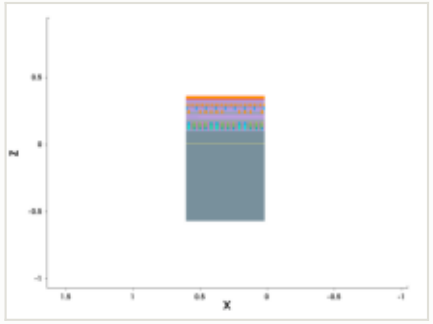
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



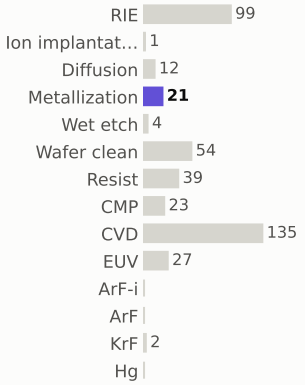
y cut at 0.184 µm



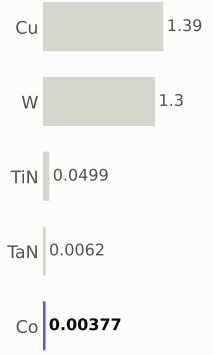
mask for this step

no mask for this step

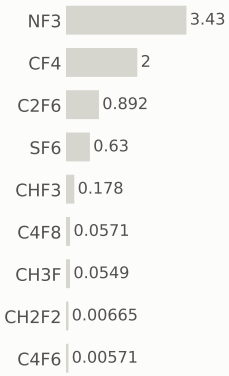
steps by process area



materials deposited (g)



gases emitted (g)



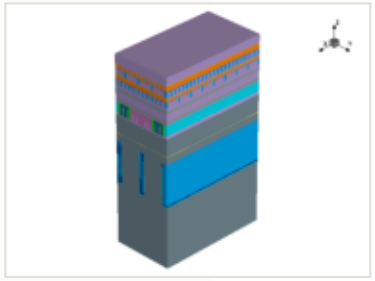
Step #4410: CVD

depo_fill · CVD

1.31	442.6	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

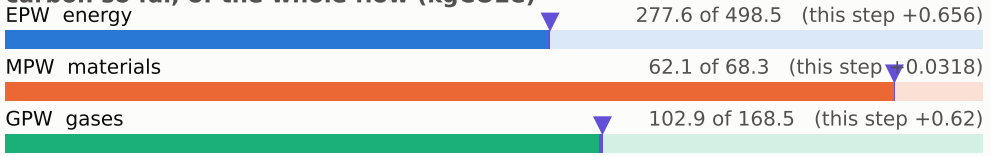


this step

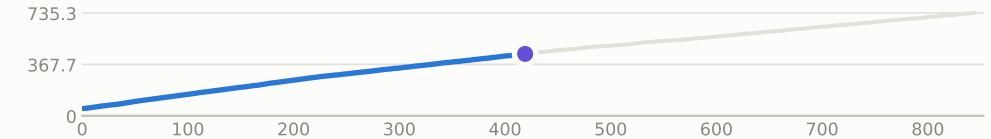
process area	CVD
material	—
thickness	0.042 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

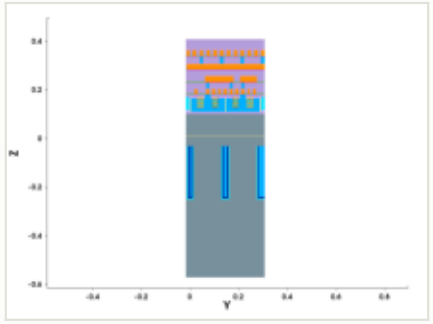
carbon so far, of the whole flow (kgCO2e)



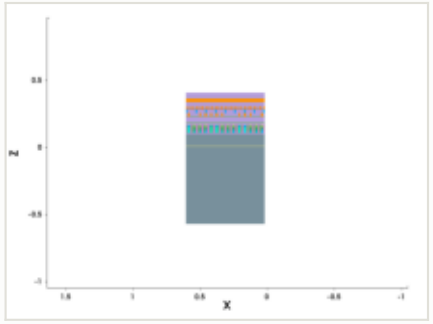
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



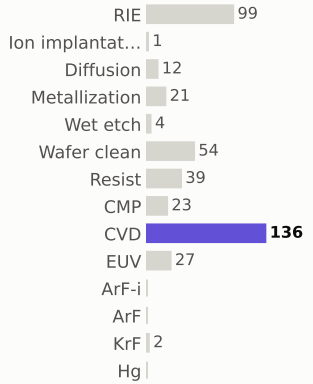
y cut at 0.184 µm



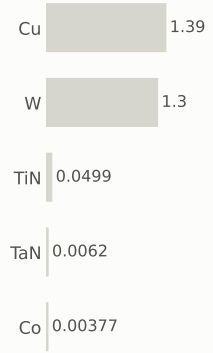
mask for this step

no mask for this step

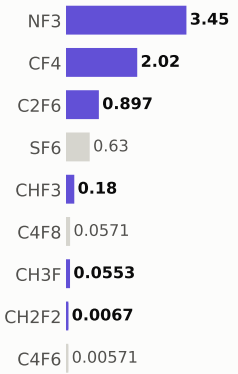
steps by process area



materials deposited (g)



gases emitted (g)



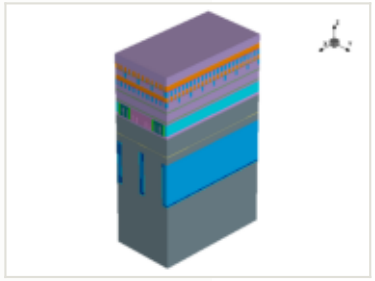
Step #4420: Diffusion

treatment · Diffusion

0.801	443.4	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



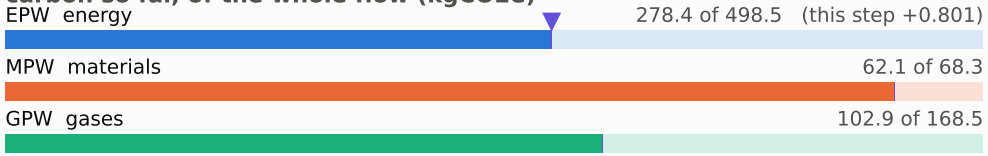
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

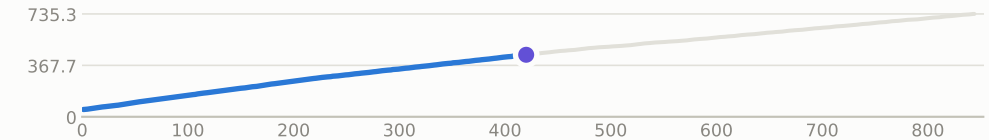
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

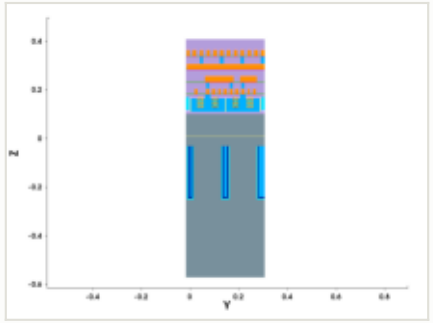
carbon so far, of the whole flow (kgCO2e)



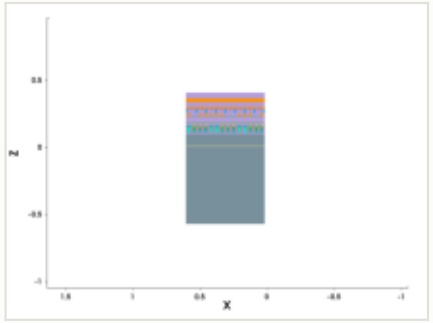
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



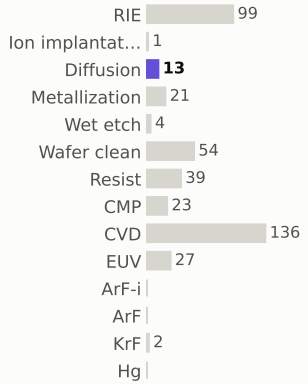
y cut at 0.184 µm



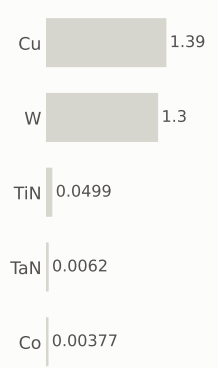
mask for this step

no mask for this step

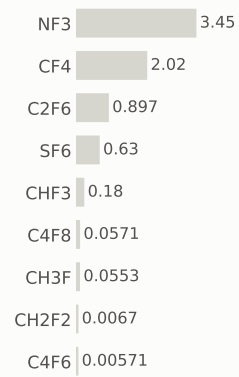
steps by process area



materials deposited (g)



gases emitted (g)



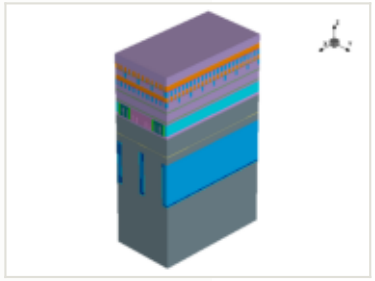
Step #4430: Wafer_clean

clean · Wafer_clean

0.169	443.6	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



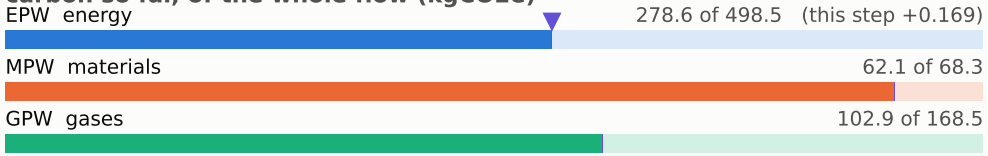
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

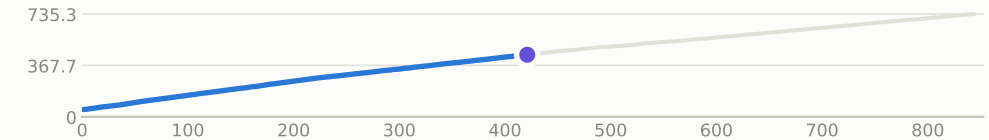
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

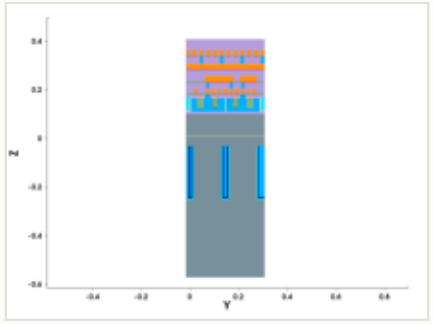
carbon so far, of the whole flow (kgCO2e)



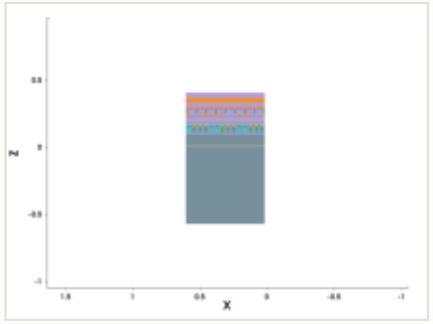
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



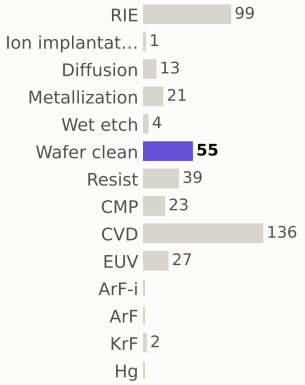
y cut at 0.184 µm



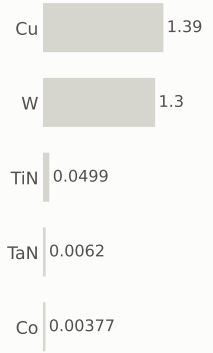
mask for this step

no mask for this step

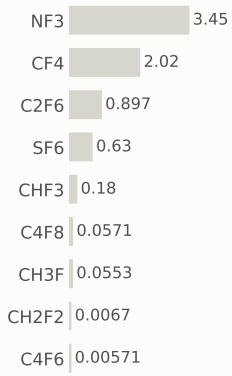
steps by process area



materials deposited (g)



gases emitted (g)



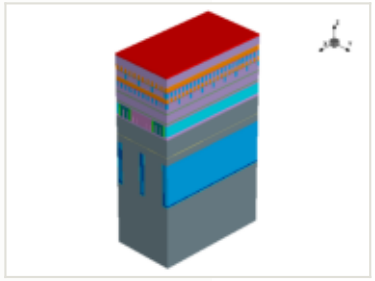
Step #4440: CVD

depo_fill · CVD

1.31	444.9	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

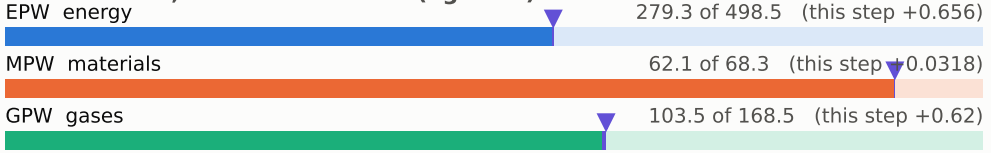


this step

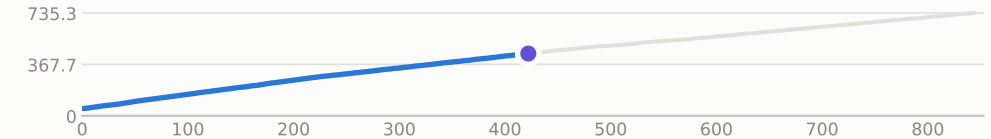
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

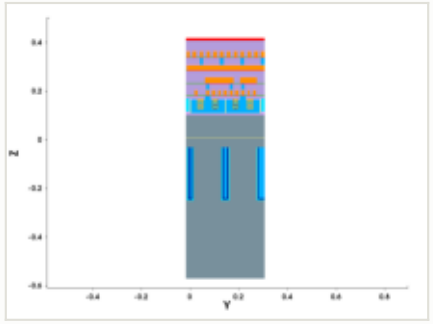
carbon so far, of the whole flow (kgCO2e)



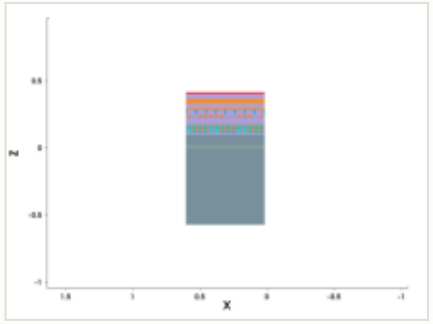
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



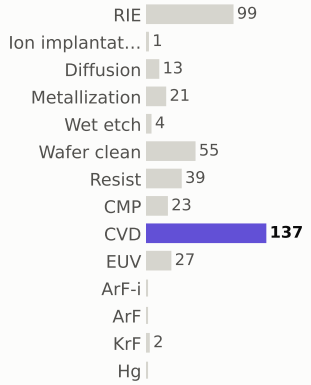
y cut at 0.184 µm



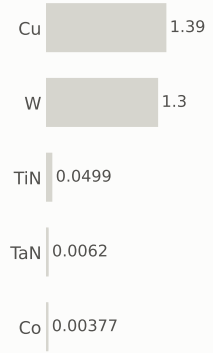
mask for this step

no mask for this step

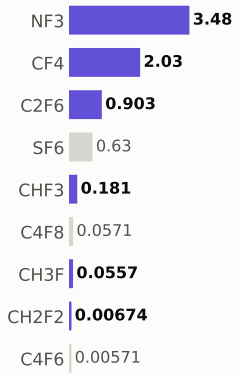
steps by process area



materials deposited (g)



gases emitted (g)



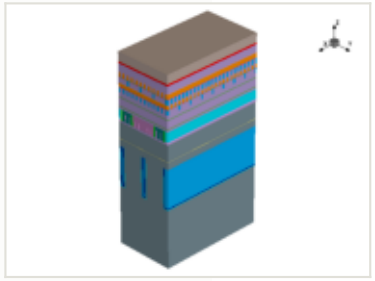
Step #4450: Resist

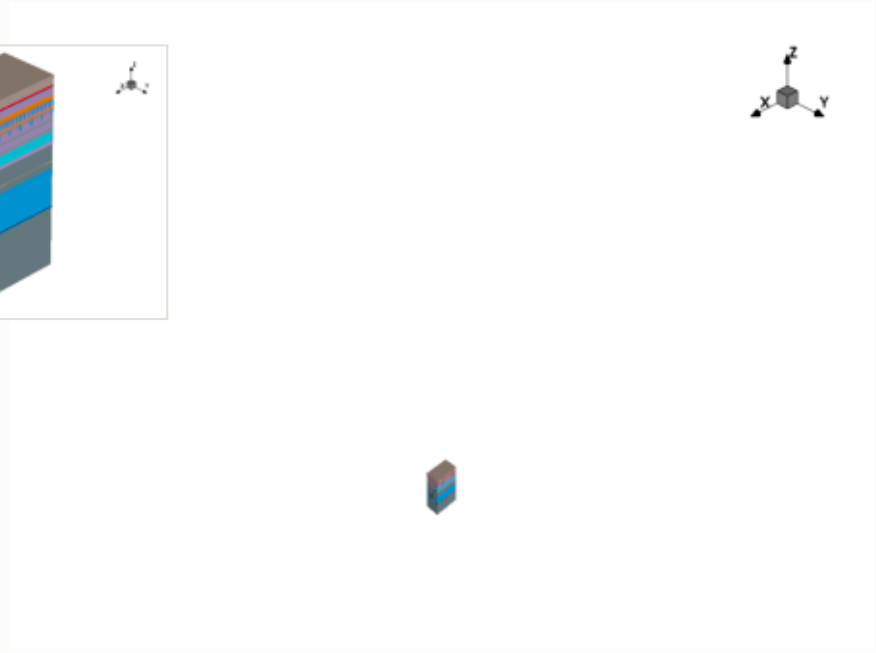
depo_fill · Resist

0.169	445.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



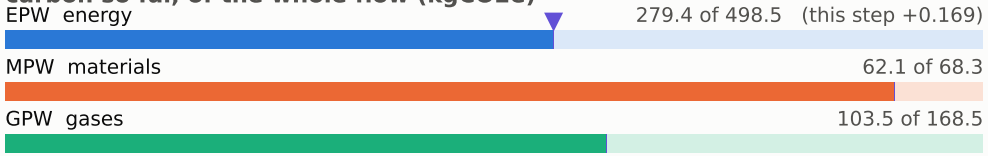


this step

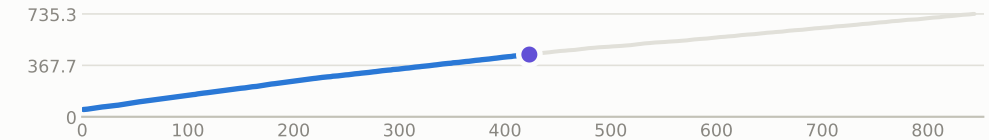
process area	Resist
material	—
thickness	0.05 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

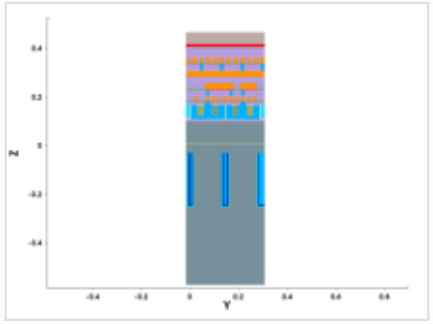
carbon so far, of the whole flow (kgCO2e)



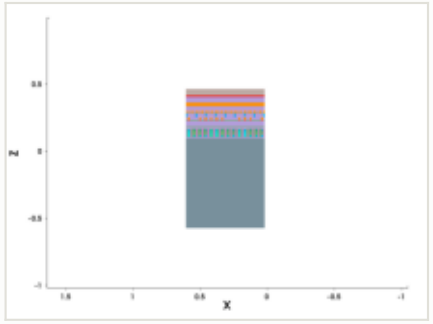
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



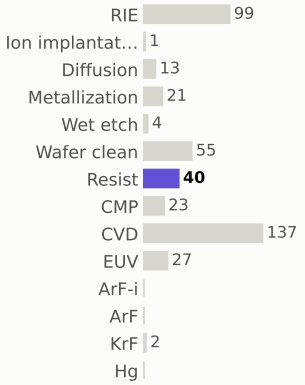
y cut at 0.184 µm



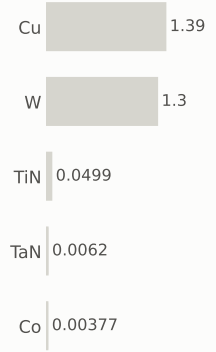
mask for this step

no mask for this step

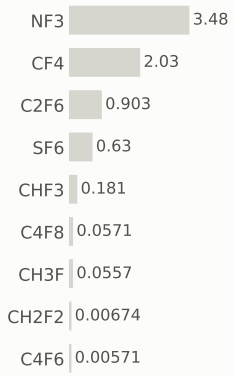
steps by process area



materials deposited (g)



gases emitted (g)



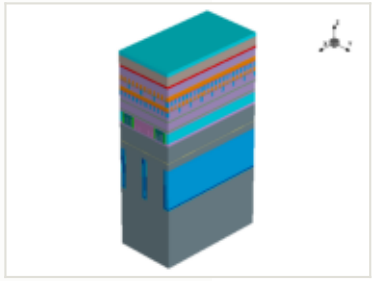
Step #4460: CVD

depo_fill · CVD

1.31	446.4	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

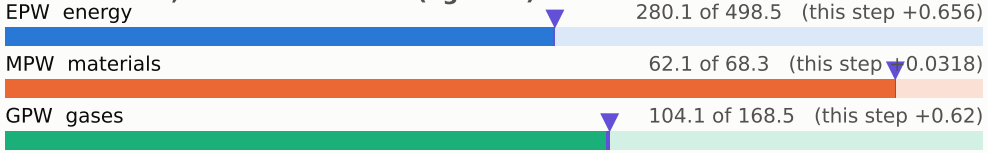


this step

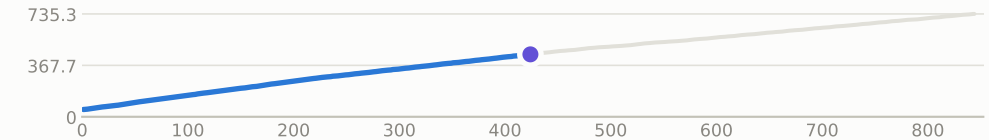
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

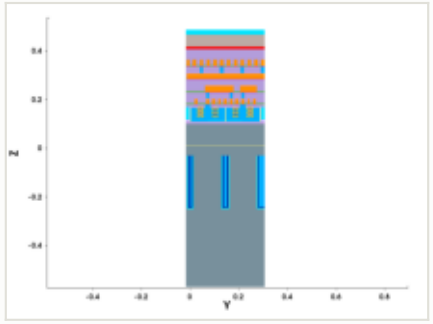
carbon so far, of the whole flow (kgCO2e)



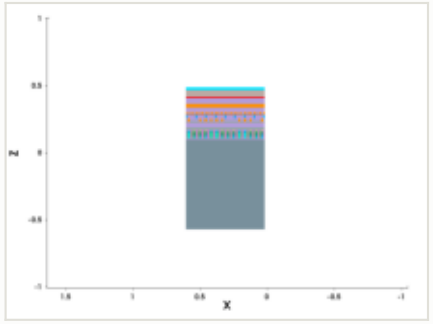
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



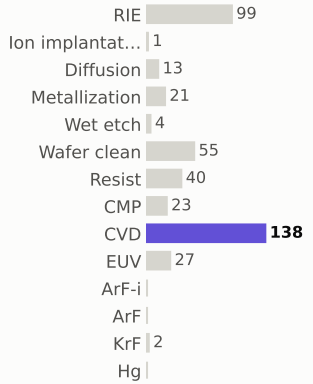
y cut at 0.184 µm



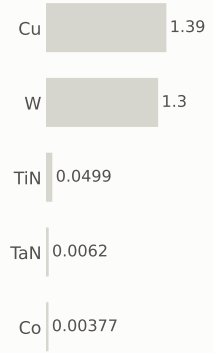
mask for this step

no mask for this step

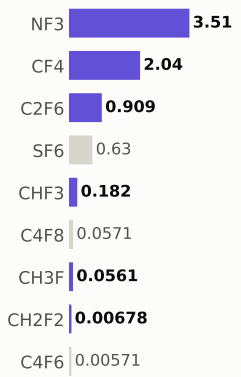
steps by process area



materials deposited (g)



gases emitted (g)



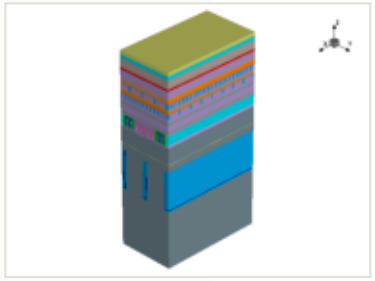
Step #4470: CVD

depo_fill · CVD

1.31	447.7	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

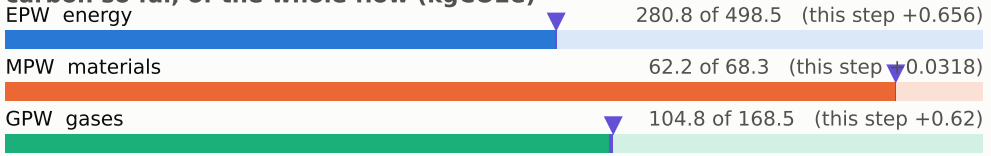


this step

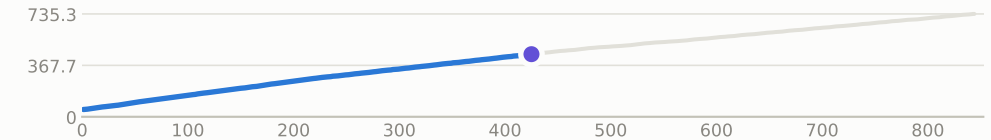
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

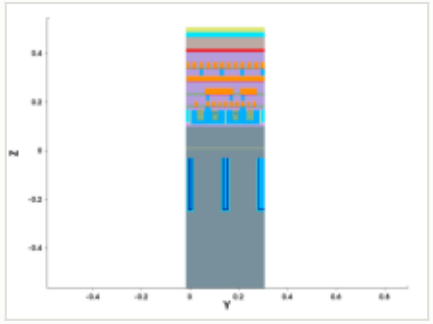
carbon so far, of the whole flow (kgCO2e)



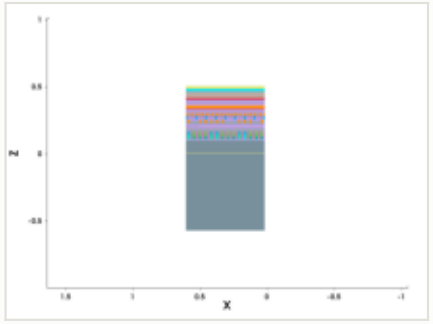
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



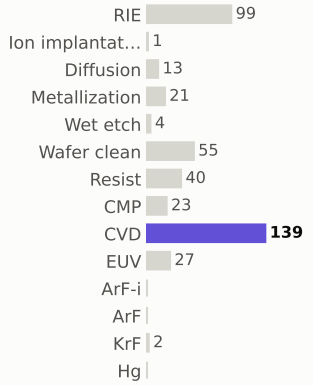
y cut at 0.184 µm



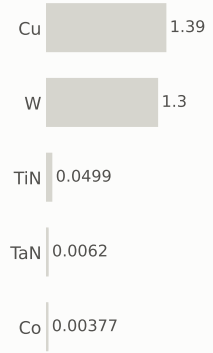
mask for this step

no mask for this step

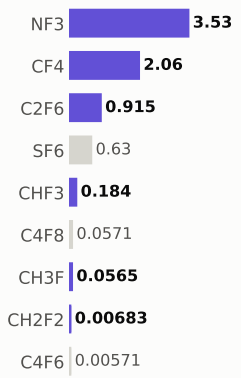
steps by process area



materials deposited (g)



gases emitted (g)



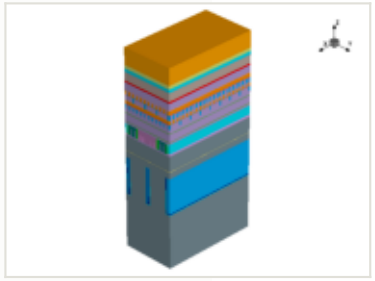
Step #4480: Resist

depo_fill · Resist

0.169	447.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

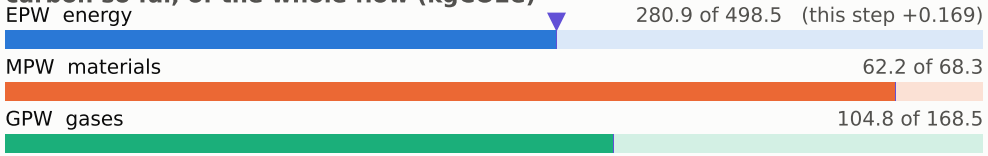


this step

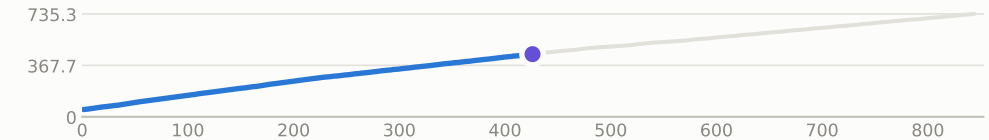
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

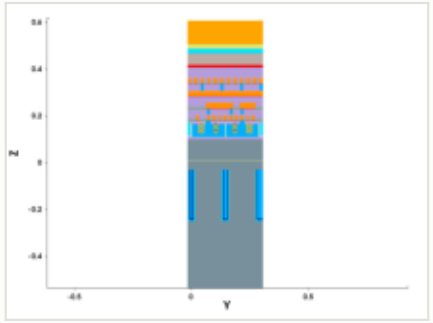
carbon so far, of the whole flow (kgCO2e)



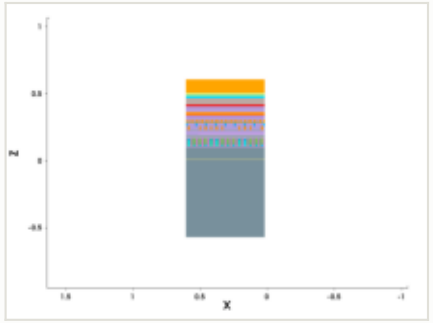
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



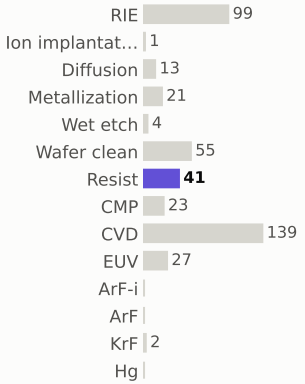
y cut at 0.184 µm



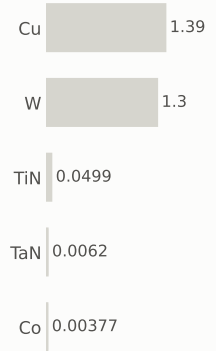
mask for this step

no mask for this step

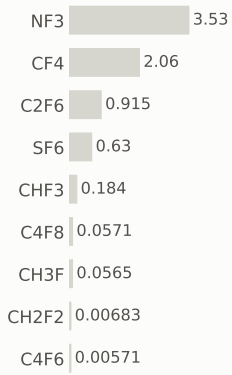
steps by process area



materials deposited (g)



gases emitted (g)



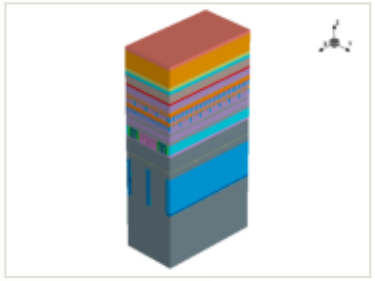
Step #4490: CVD

depo_fill · CVD

1.31	449.2	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

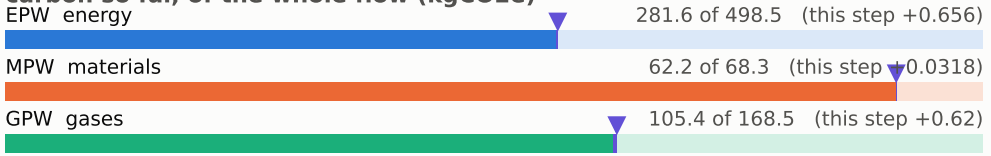


this step

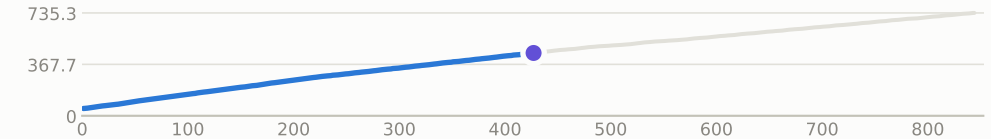
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

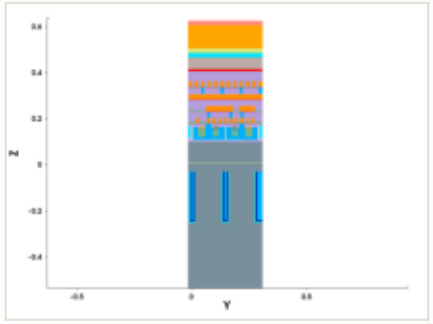
carbon so far, of the whole flow (kgCO2e)



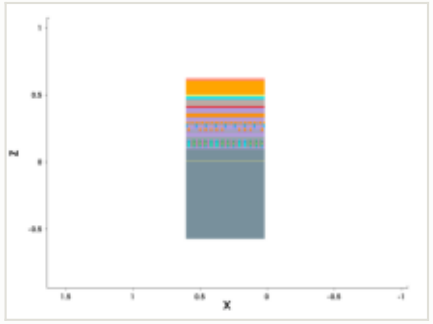
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



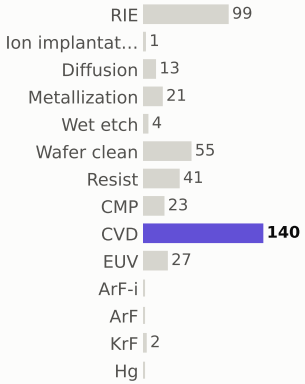
y cut at 0.184 µm



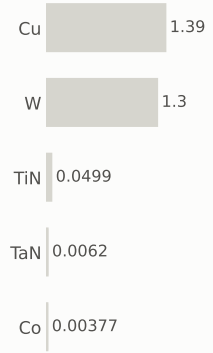
mask for this step

no mask for this step

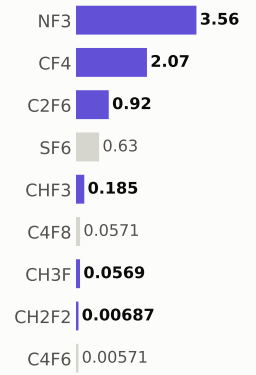
steps by process area



materials deposited (g)



gases emitted (g)



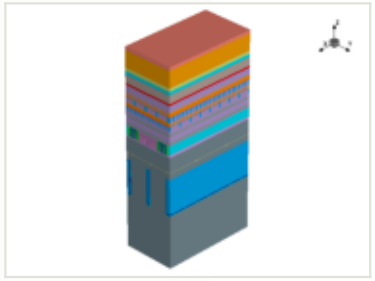
Step #4500: Wafer_clean

clean · Wafer_clean

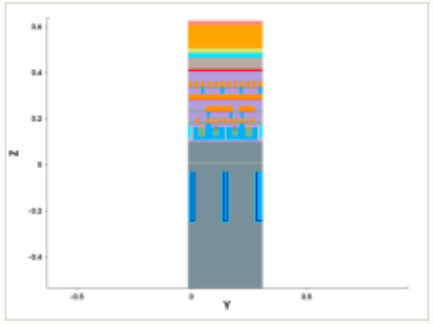
0.169	449.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

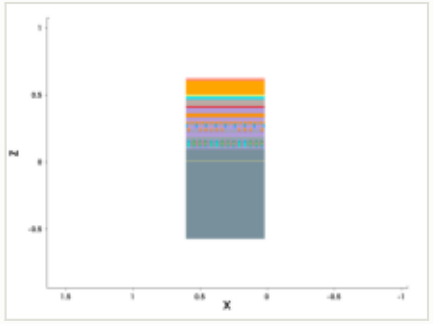
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

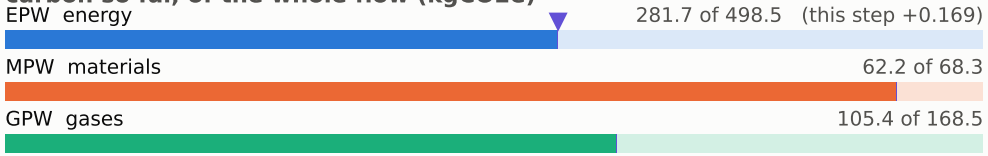
no mask for this step

this step

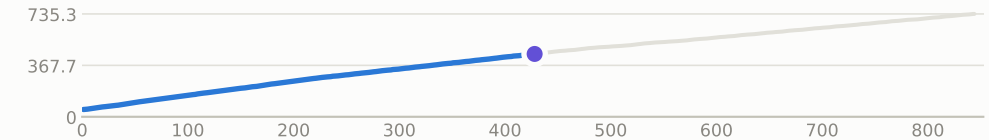
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

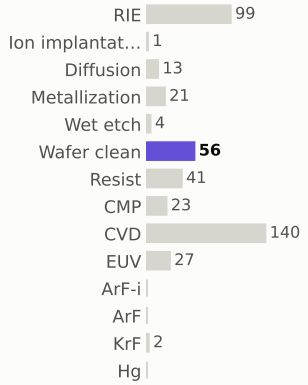
carbon so far, of the whole flow (kgCO2e)



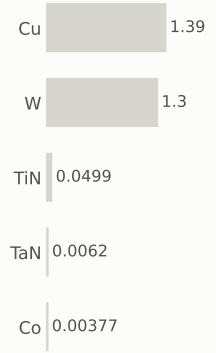
embodied carbon up to each step (kgCO2e per wafer)



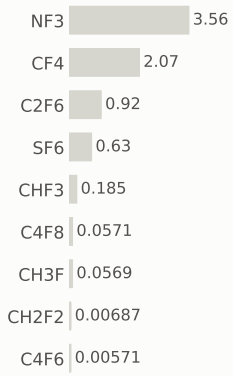
steps by process area



materials deposited (g)



gases emitted (g)



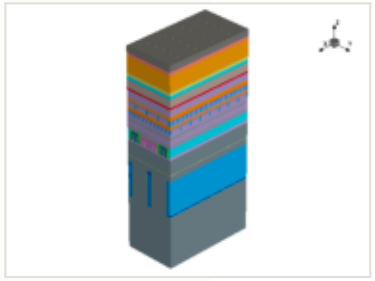
Step #4510: EUV

litho_patterning · EUV

1.96	451.3	0.27%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

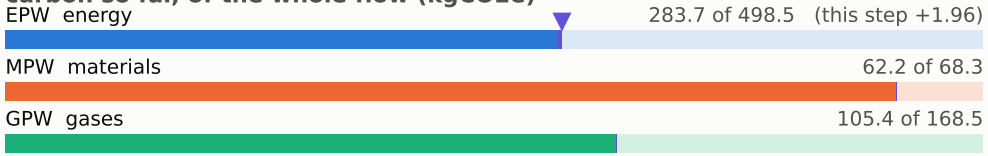


this step

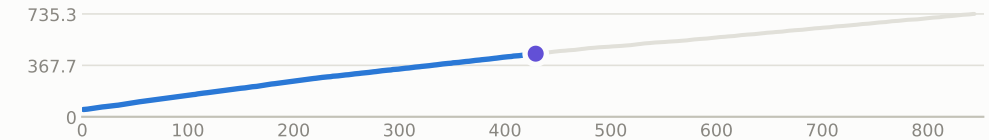
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

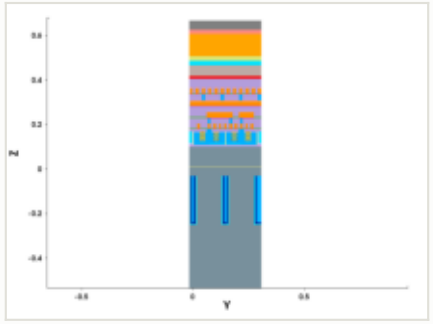
carbon so far, of the whole flow (kgCO2e)



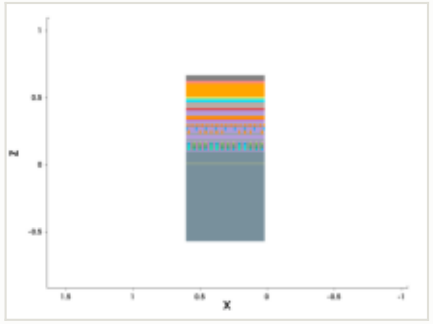
embodied carbon up to each step (kgCO2e per wafer)



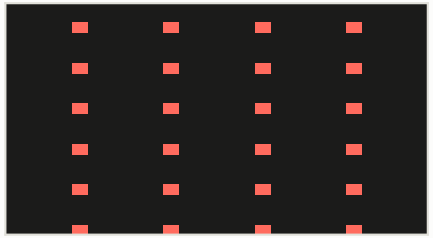
x cut at 0.294 µm



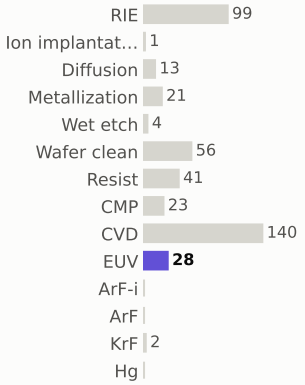
y cut at 0.184 µm



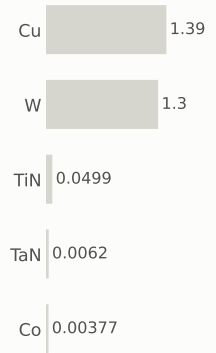
mask: layer 10371:0 (GDSII)



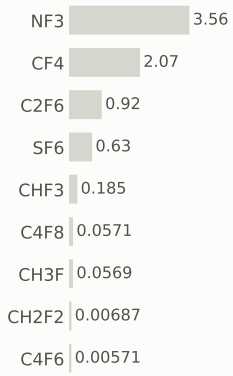
steps by process area



materials deposited (g)



gases emitted (g)



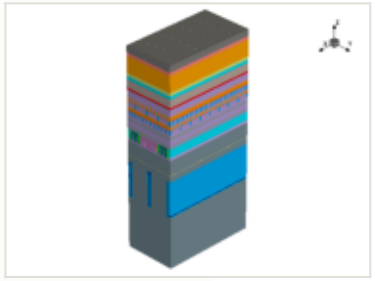
Step #4520: RIE

etch_anisotropic · RIE

1.04	452.3	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

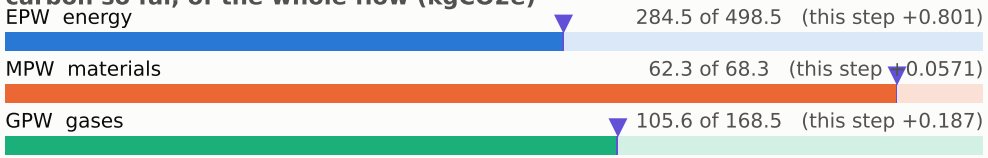


this step

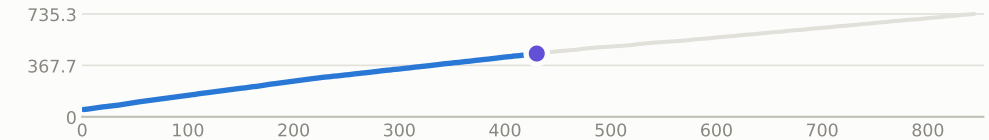
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

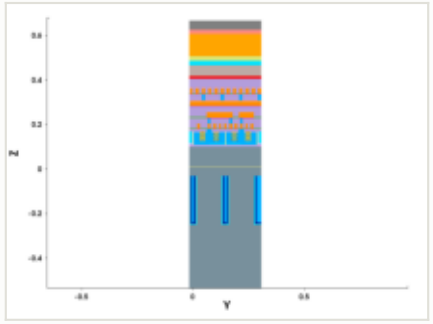
carbon so far, of the whole flow (kgCO2e)



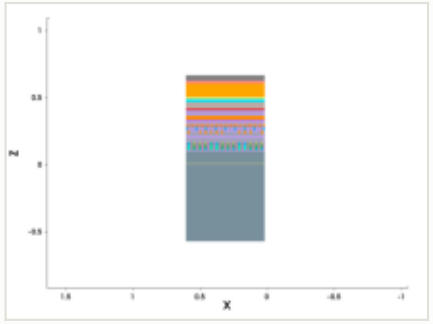
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



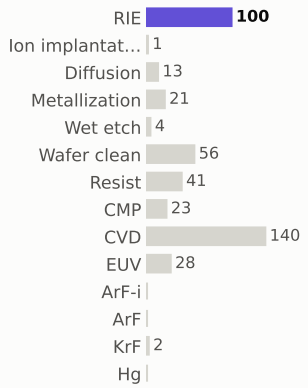
y cut at 0.184 µm



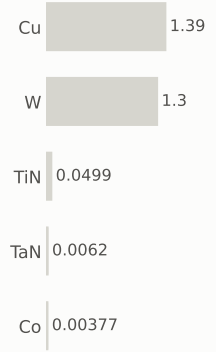
mask for this step

no mask for this step

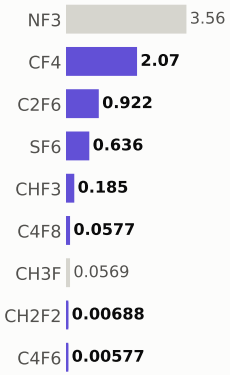
steps by process area



materials deposited (g)



gases emitted (g)



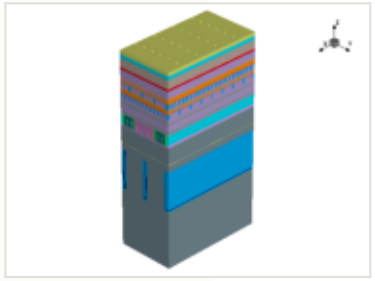
Step #4530: Wafer_clean

etch_strip · Wafer_clean

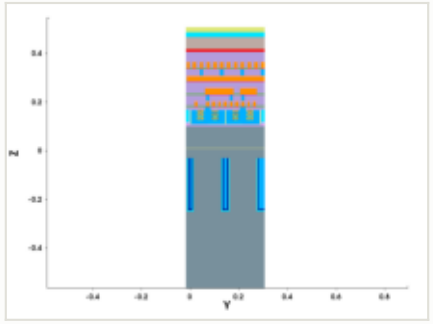
0.169	452.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

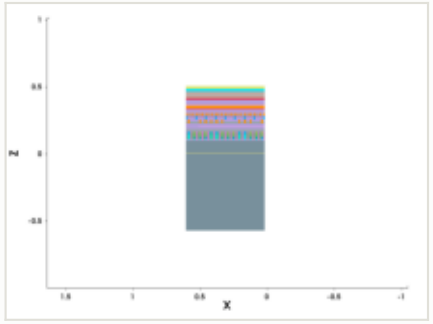
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

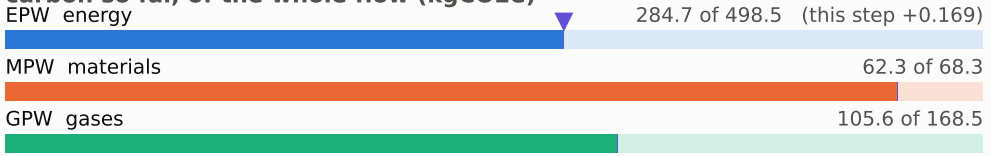
no mask for this step

this step

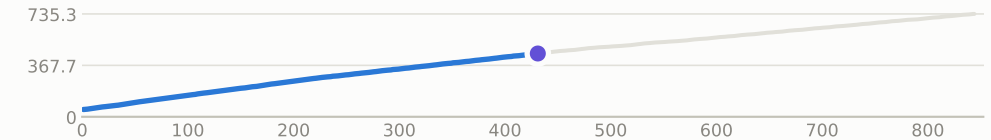
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

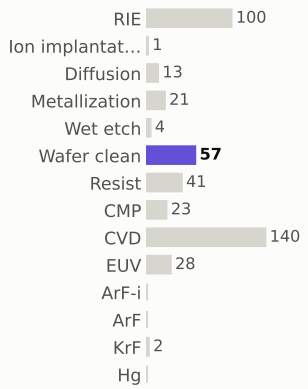
carbon so far, of the whole flow (kgCO2e)



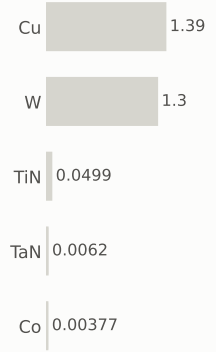
embodied carbon up to each step (kgCO2e per wafer)



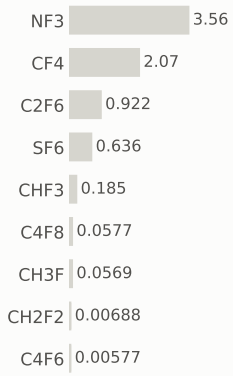
steps by process area



materials deposited (g)



gases emitted (g)



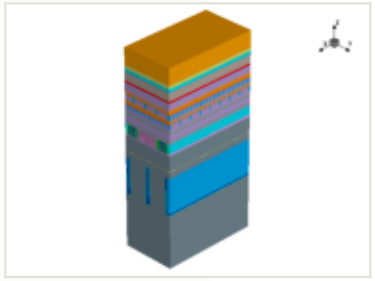
Step #4540: Resist

depo_fill · Resist

0.169	452.7	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

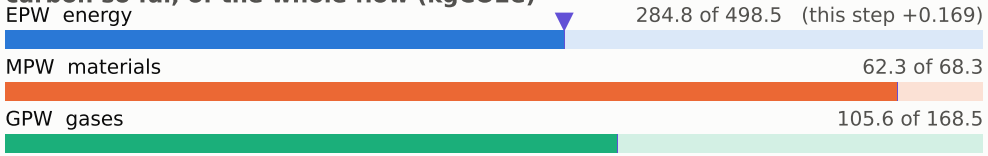


this step

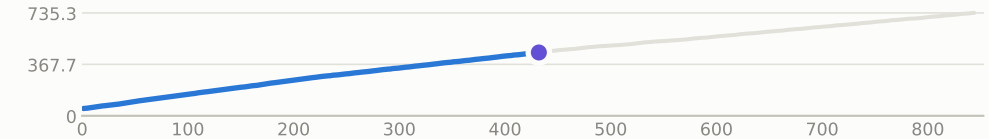
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

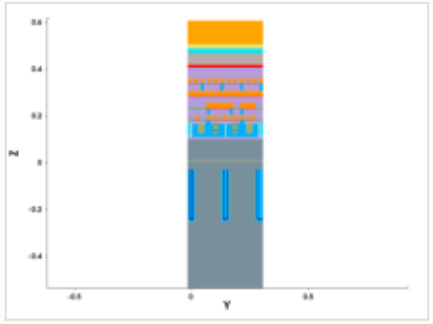
carbon so far, of the whole flow (kgCO2e)



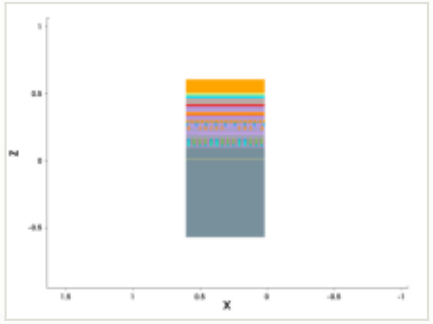
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



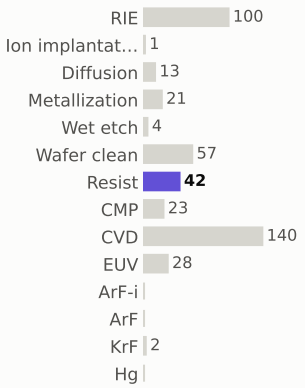
y cut at 0.184 µm



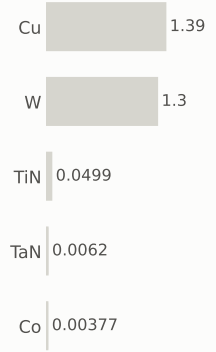
mask for this step

no mask for this step

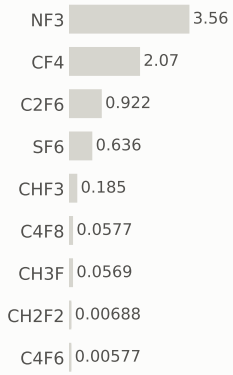
steps by process area



materials deposited (g)



gases emitted (g)



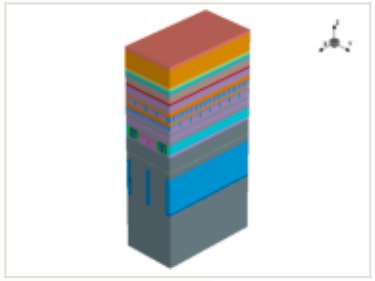
Step #4550: CVD

depo_fill · CVD

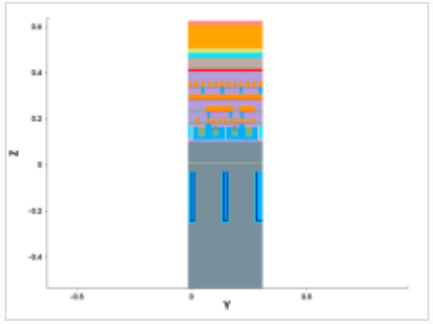
1.31	454.0	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

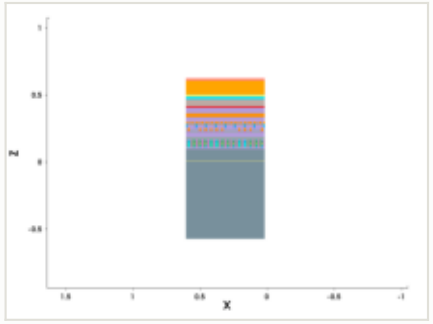
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

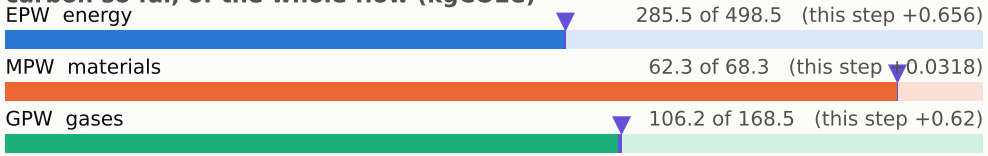
no mask for this step

this step

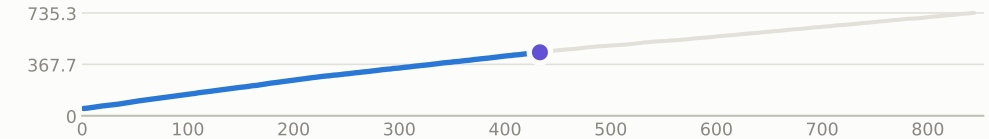
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

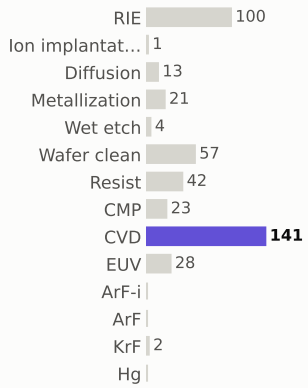
carbon so far, of the whole flow (kgCO2e)



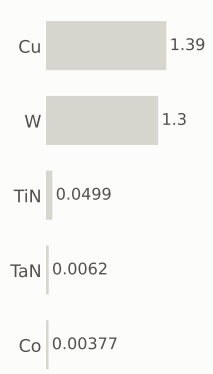
embodied carbon up to each step (kgCO2e per wafer)



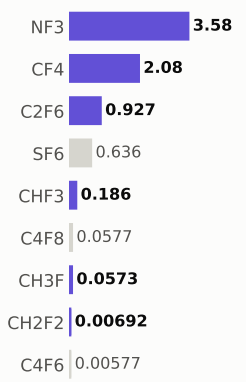
steps by process area



materials deposited (g)



gases emitted (g)



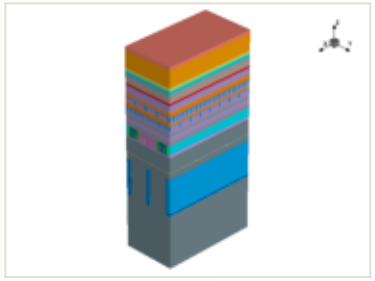
Step #4560: Wafer_clean

clean · Wafer_clean

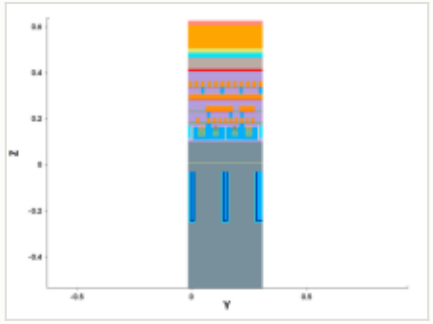
0.169	454.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

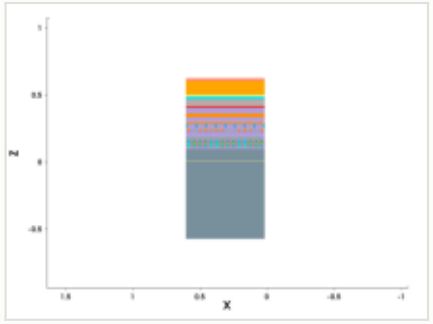
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

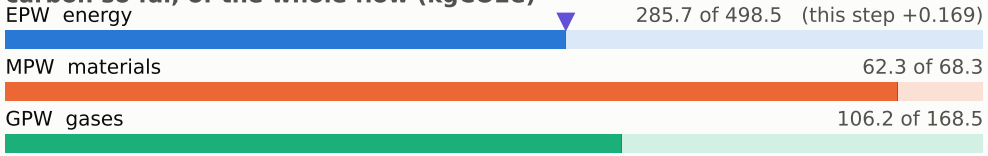
no mask for this step

this step

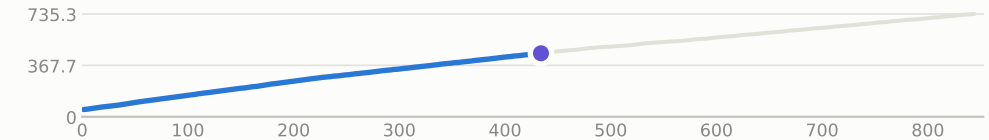
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

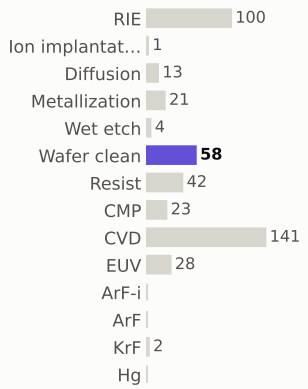
carbon so far, of the whole flow (kgCO2e)



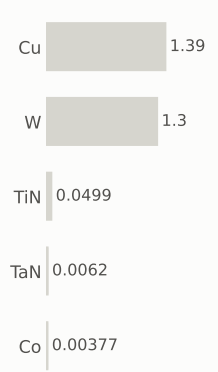
embodied carbon up to each step (kgCO2e per wafer)



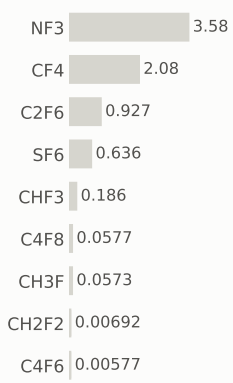
steps by process area



materials deposited (g)



gases emitted (g)



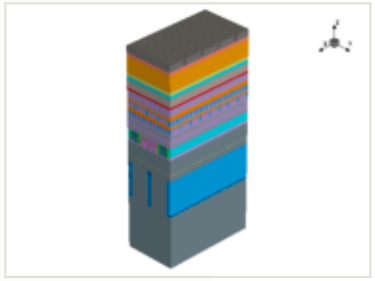
Step #4570: EUV

litho_patterning · EUV

1.96	456.1	0.27%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

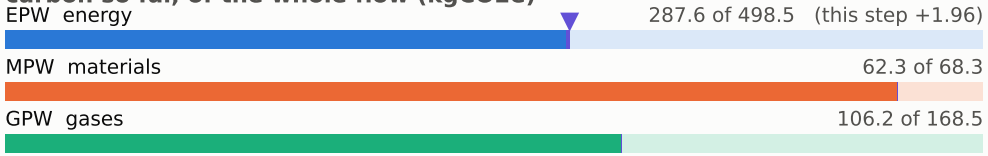


this step

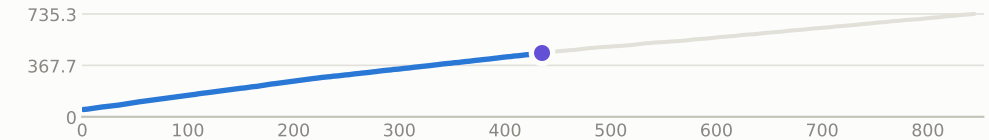
process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

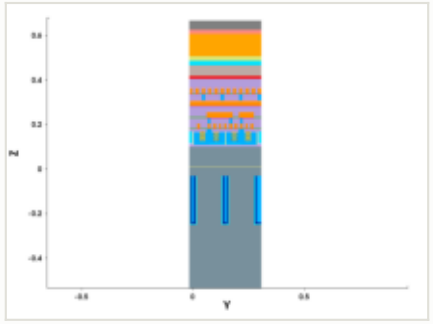
carbon so far, of the whole flow (kgCO2e)



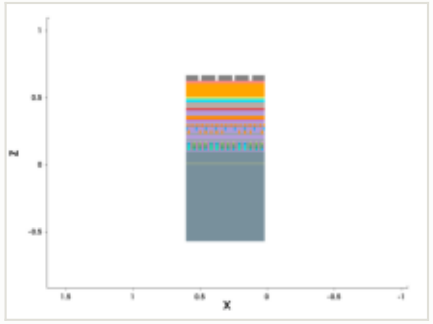
embodied carbon up to each step (kgCO2e per wafer)



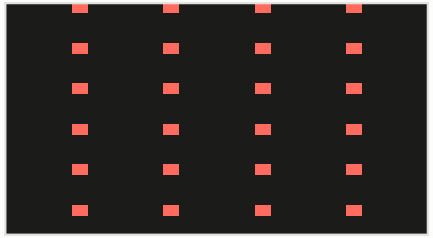
x cut at 0.294 µm



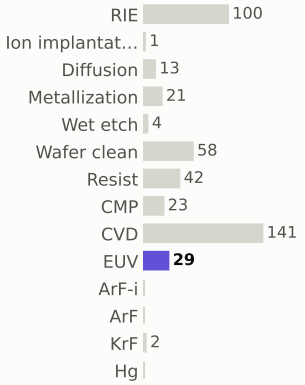
y cut at 0.184 µm



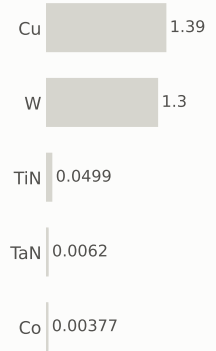
mask: layer 10372:0 (GDSII)



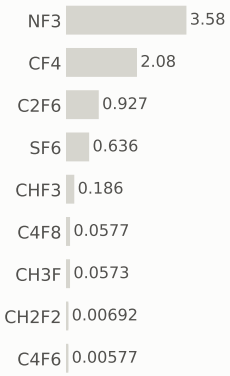
steps by process area



materials deposited (g)



gases emitted (g)



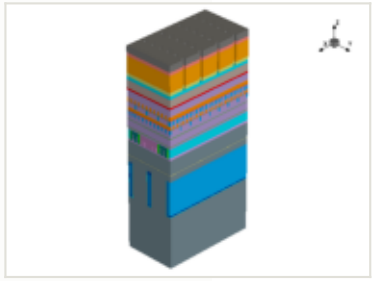
Step #4580: RIE

etch_anisotropic · RIE

1.04	457.1	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

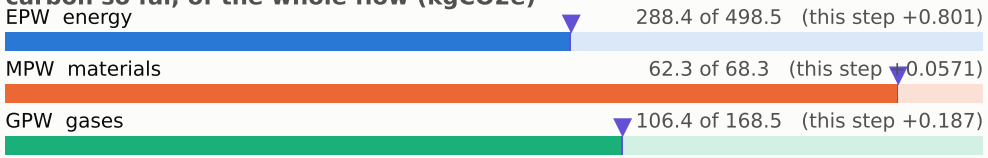


this step

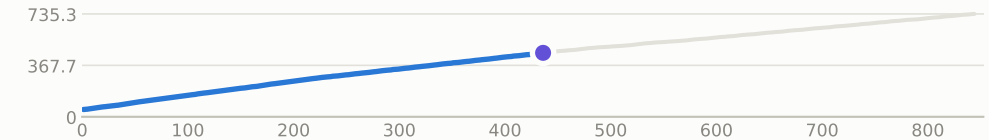
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

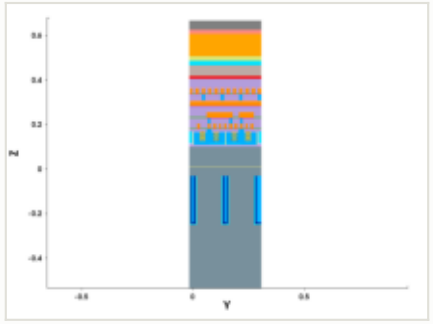
carbon so far, of the whole flow (kgCO2e)



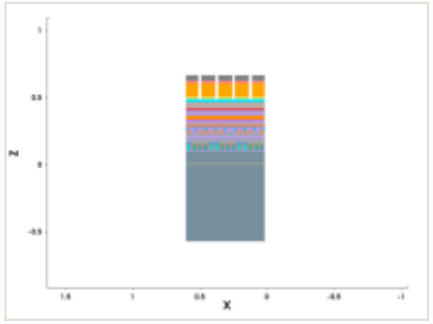
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



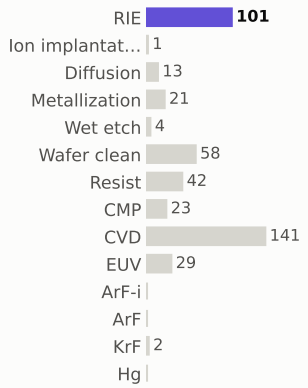
y cut at 0.184 µm



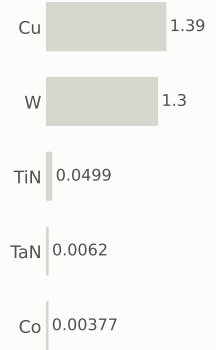
mask for this step

no mask for this step

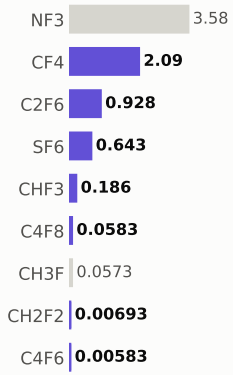
steps by process area



materials deposited (g)



gases emitted (g)



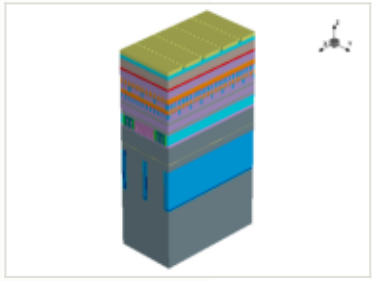
Step #4590: Wafer_clean

etch_strip · Wafer_clean

0.169	457.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

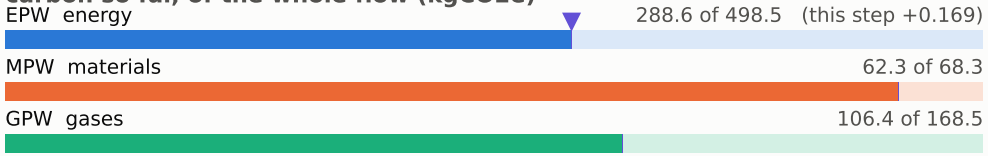
zoomed to fit



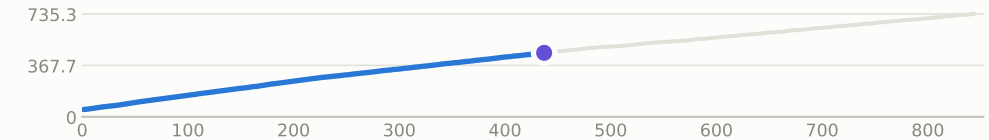
this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

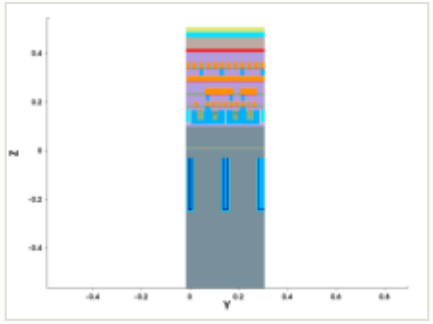
carbon so far, of the whole flow (kgCO2e)



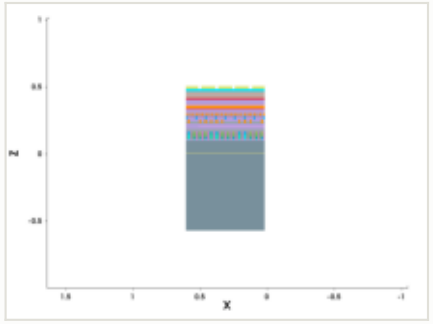
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



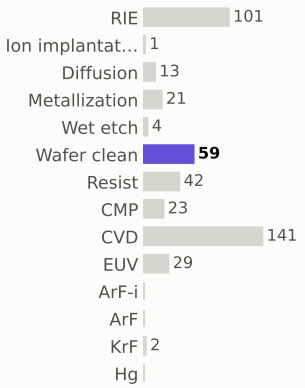
y cut at 0.184 µm



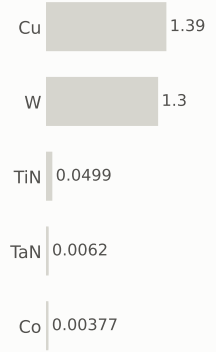
mask for this step

no mask for this step

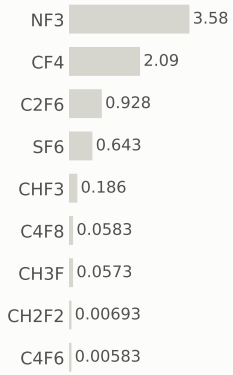
steps by process area



materials deposited (g)



gases emitted (g)



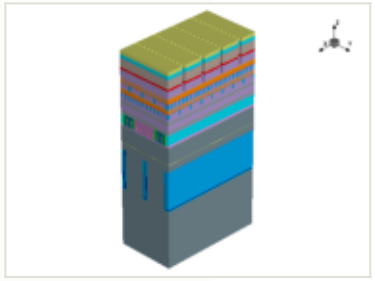
Step #4600: RIE

etch_anisotropic · RIE

1.04	458.4	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

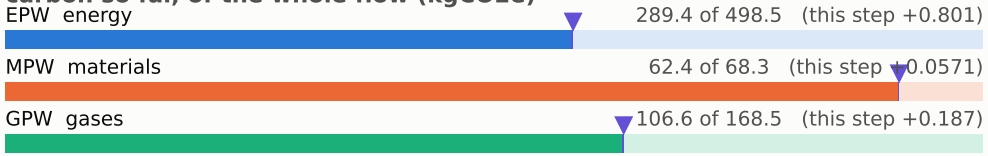


this step

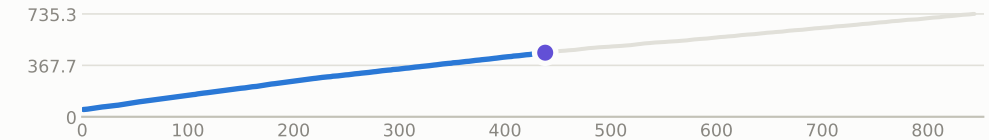
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

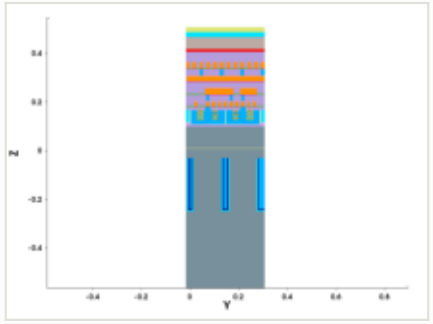
carbon so far, of the whole flow (kgCO2e)



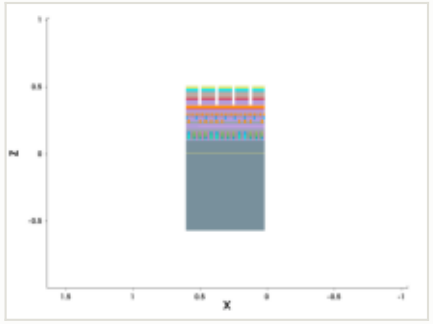
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



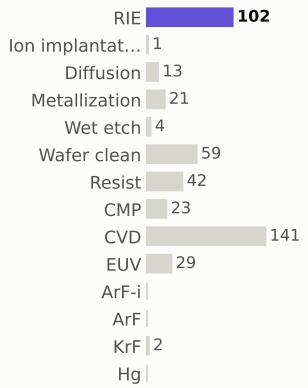
y cut at 0.184 µm



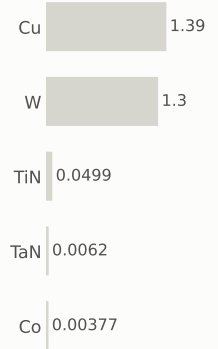
mask for this step

no mask for this step

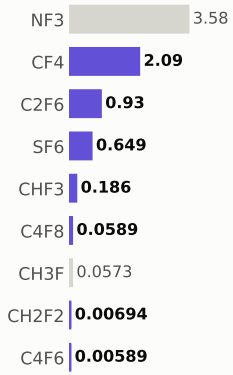
steps by process area



materials deposited (g)



gases emitted (g)



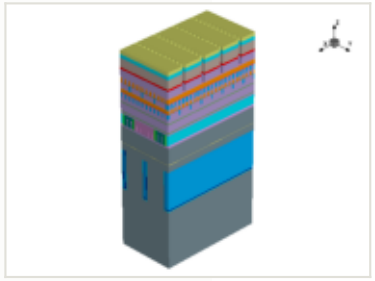
Step #4610: Wafer_clean

clean · Wafer_clean

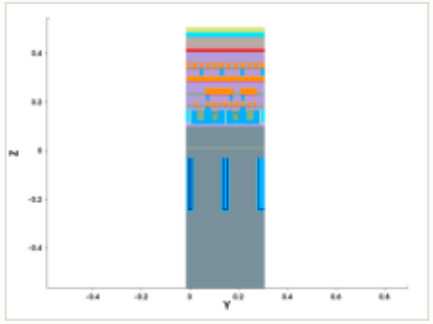
0.169	458.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

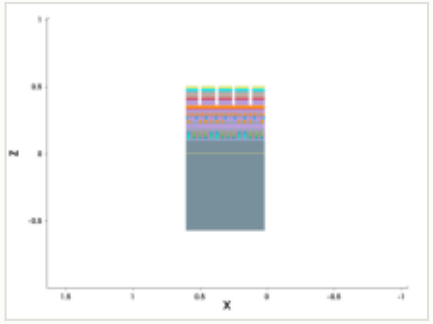
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

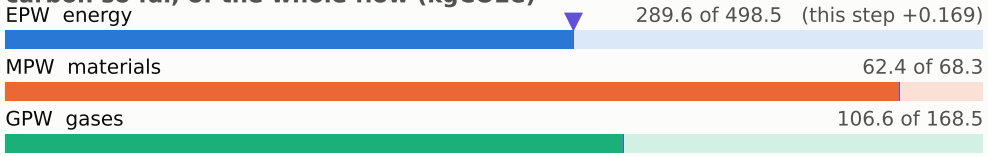
no mask for this step

this step

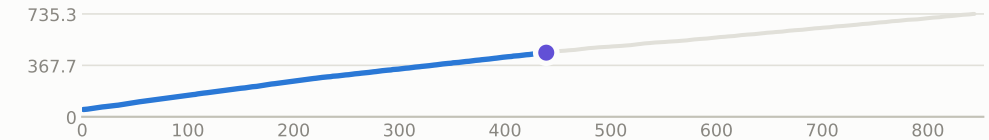
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

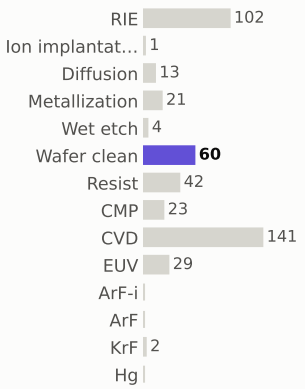
carbon so far, of the whole flow (kgCO2e)



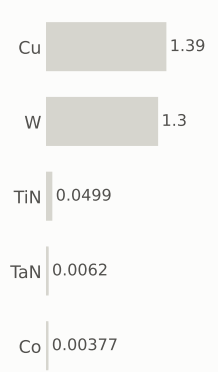
embodied carbon up to each step (kgCO2e per wafer)



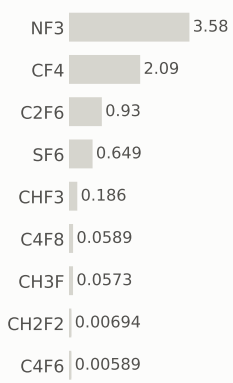
steps by process area



materials deposited (g)



gases emitted (g)



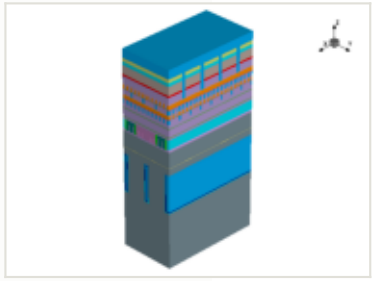
Step #4620: Metallization

depo_fill · Metallization

1.21	459.7	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

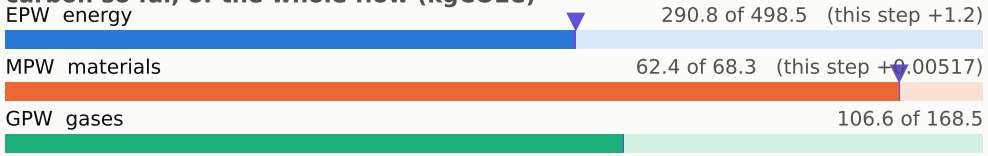


this step

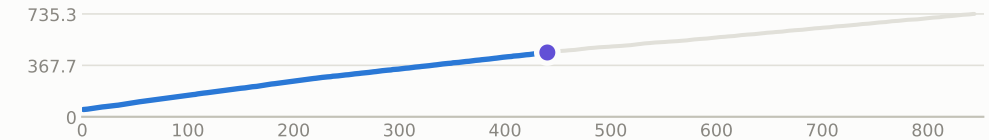
process area	Metallization
material	W
thickness	0.05 µm
material GWP	75.8149 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00517 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

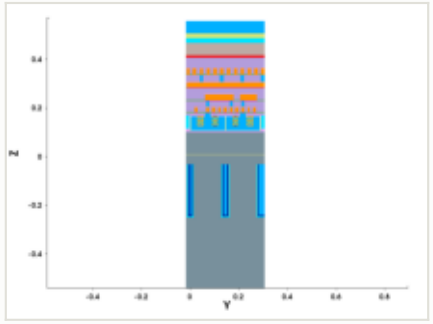
carbon so far, of the whole flow (kgCO2e)



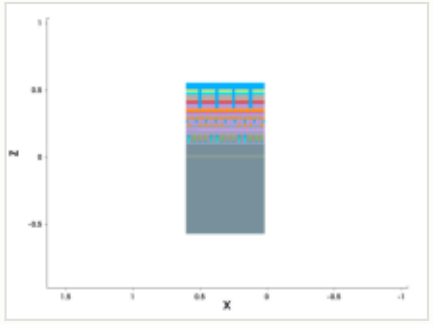
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



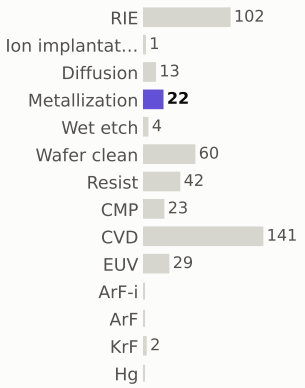
y cut at 0.184 µm



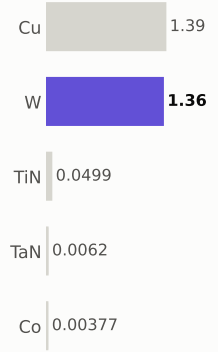
mask for this step

no mask for this step

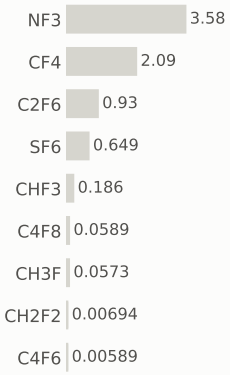
steps by process area



materials deposited (g)



gases emitted (g)



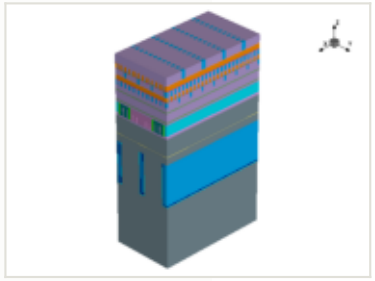
Step #4630: CMP

etch_cmp · CMP

0.169	459.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

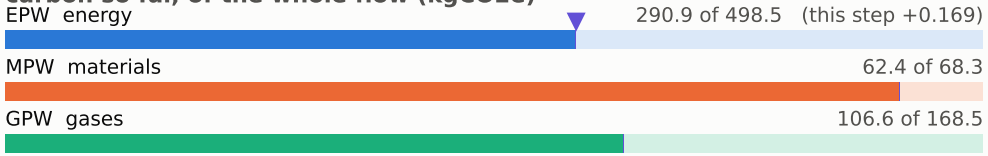


this step

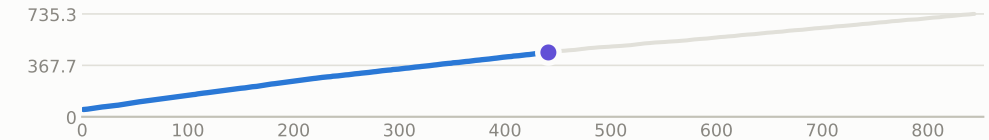
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

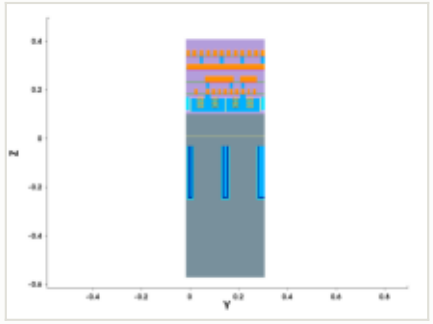
carbon so far, of the whole flow (kgCO2e)



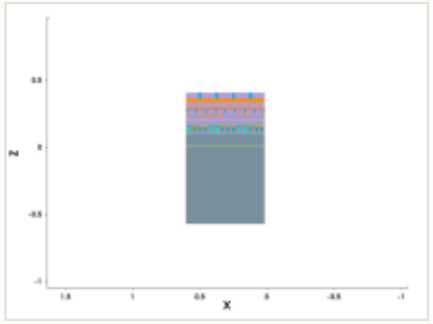
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



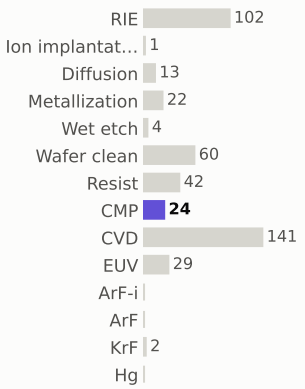
y cut at 0.184 µm



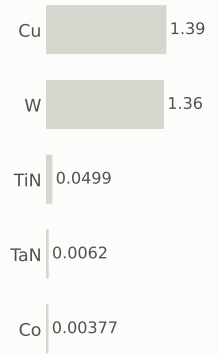
mask for this step

no mask for this step

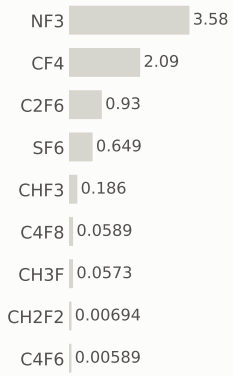
steps by process area



materials deposited (g)



gases emitted (g)



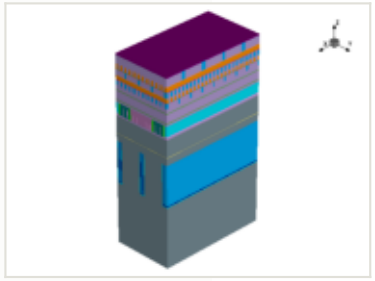
Step #4640: CVD

depo_fill · CVD

1.31	461.2	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

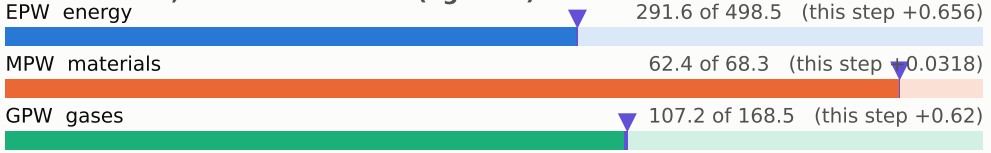


this step

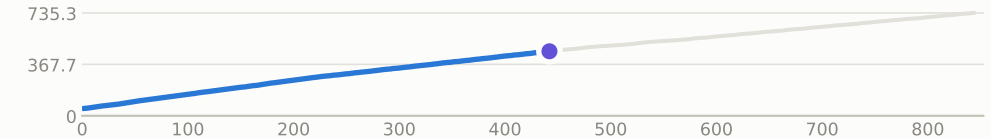
process area	CVD
material	—
thickness	0.006 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

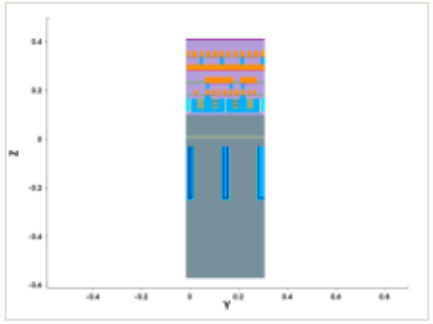
carbon so far, of the whole flow (kgCO2e)



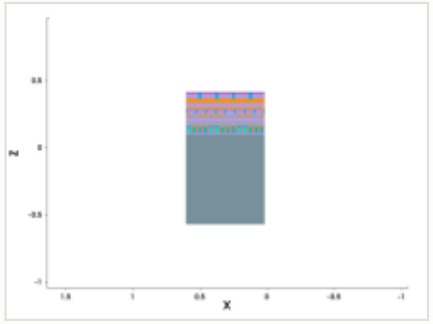
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



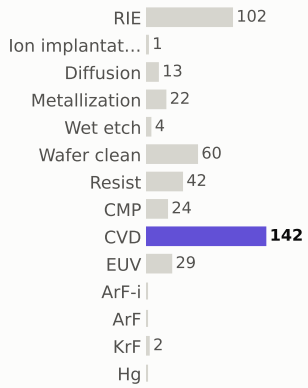
y cut at 0.184 µm



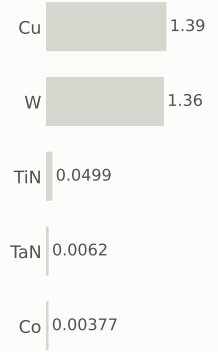
mask for this step

no mask for this step

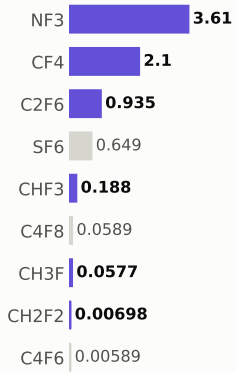
steps by process area



materials deposited (g)



gases emitted (g)



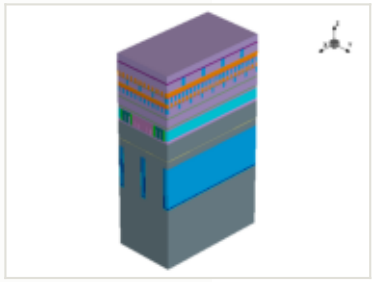
Step #4650: CVD

depo_fill · CVD

1.31	462.5	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

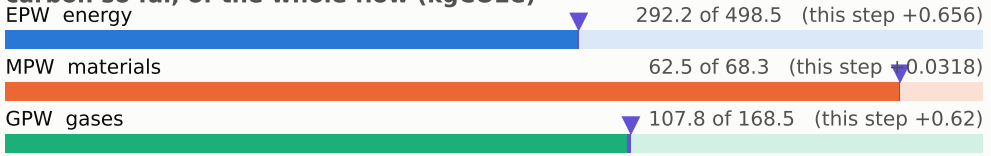


this step

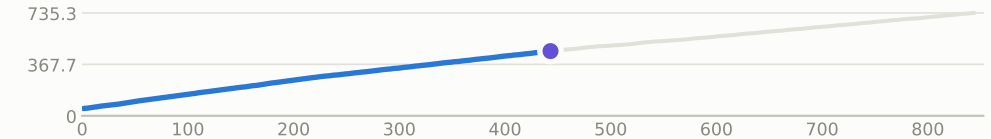
process area	CVD
material	—
thickness	0.036 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

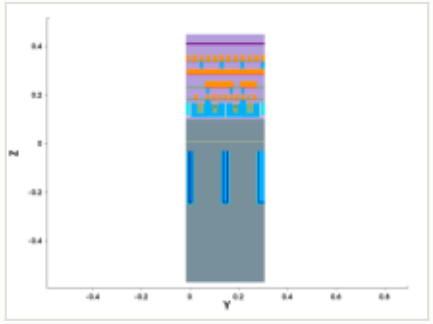
carbon so far, of the whole flow (kgCO2e)



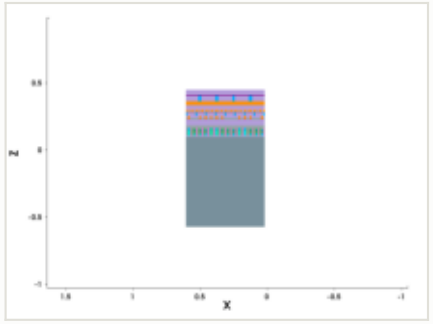
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



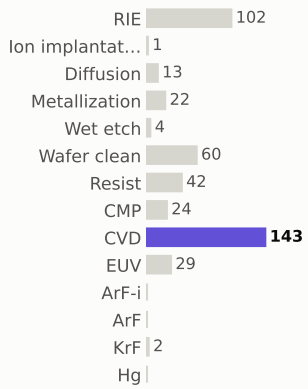
y cut at 0.184 µm



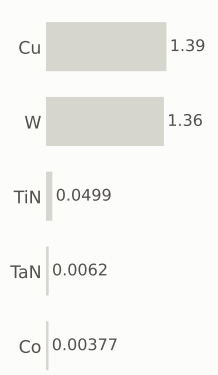
mask for this step

no mask for this step

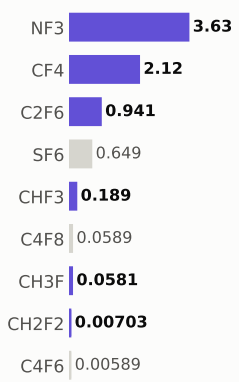
steps by process area



materials deposited (g)



gases emitted (g)



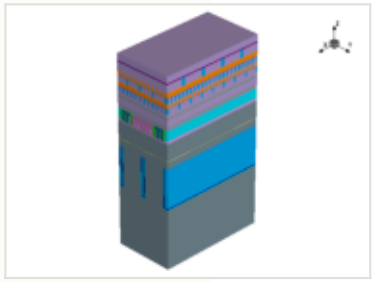
Step #4660: Diffusion

treatment · Diffusion

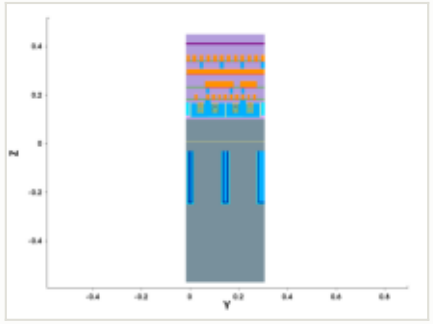
0.801	463.3	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

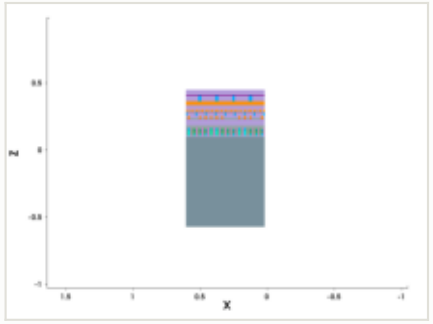
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

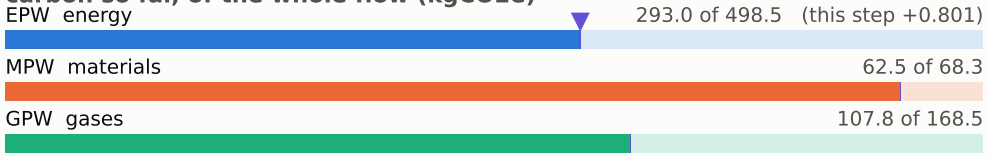
no mask for this step

this step

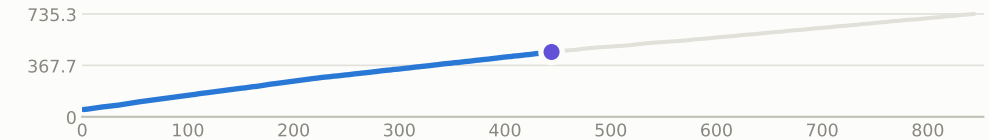
process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

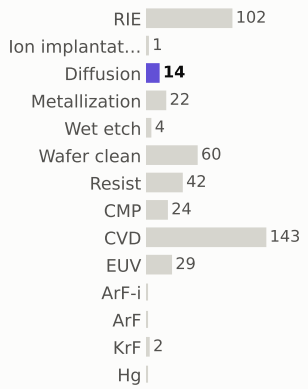
carbon so far, of the whole flow (kgCO2e)



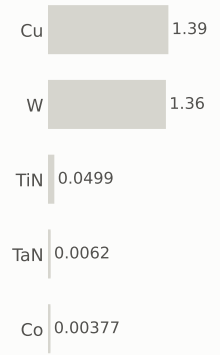
embodied carbon up to each step (kgCO2e per wafer)



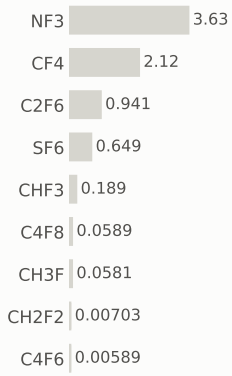
steps by process area



materials deposited (g)



gases emitted (g)



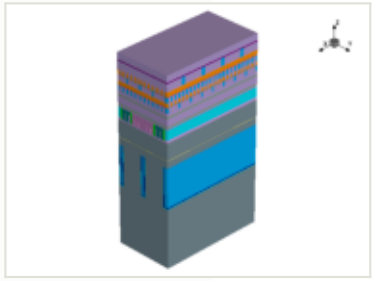
Step #4670: Wafer_clean

clean · Wafer_clean

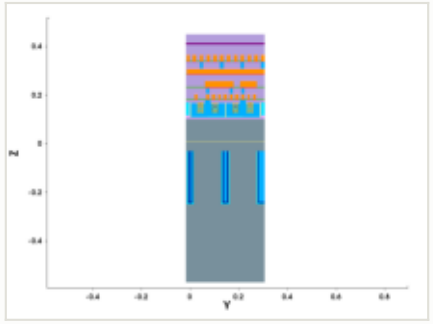
0.169	463.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

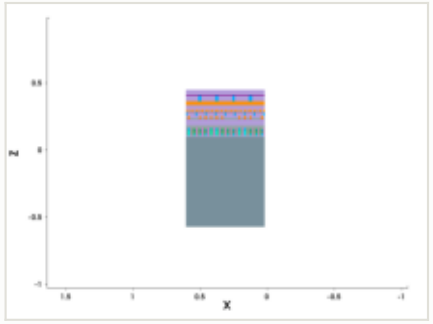
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

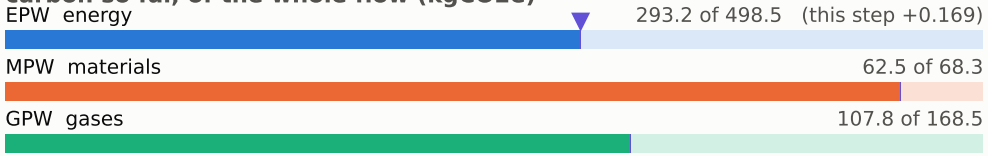
no mask for this step

this step

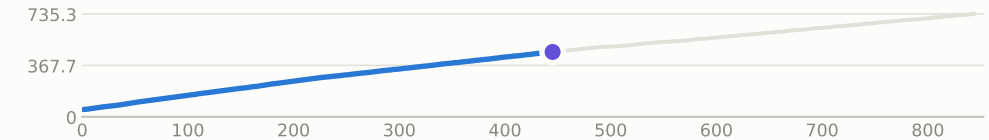
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

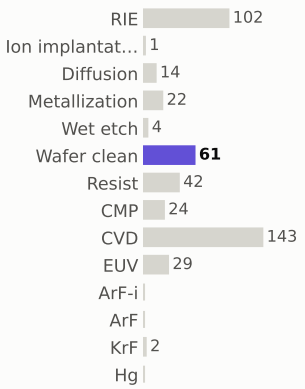
carbon so far, of the whole flow (kgCO2e)



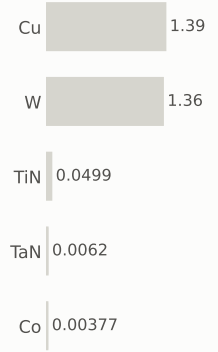
embodied carbon up to each step (kgCO2e per wafer)



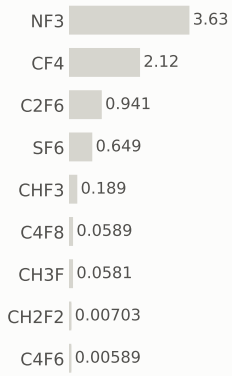
steps by process area



materials deposited (g)



gases emitted (g)



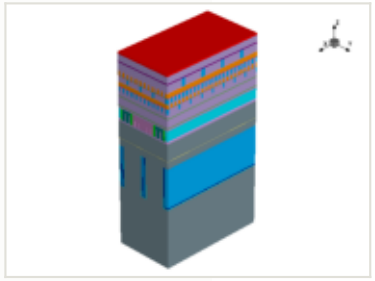
Step #4680: CVD

depo_fill · CVD

1.31	464.8	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

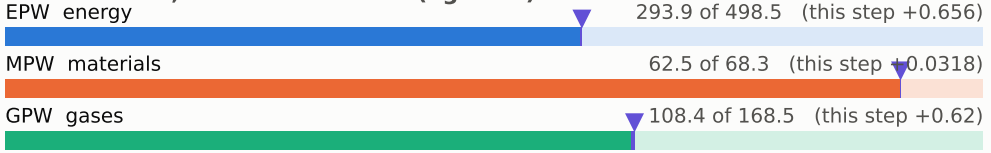


this step

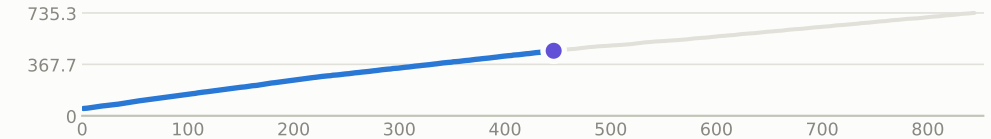
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

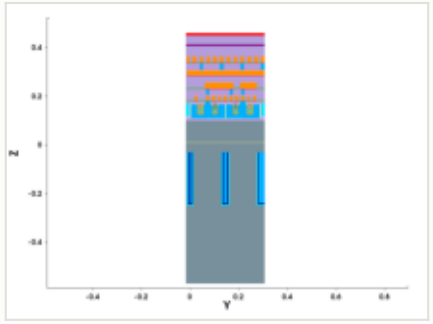
carbon so far, of the whole flow (kgCO2e)



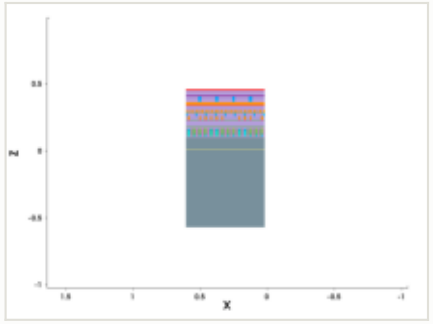
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



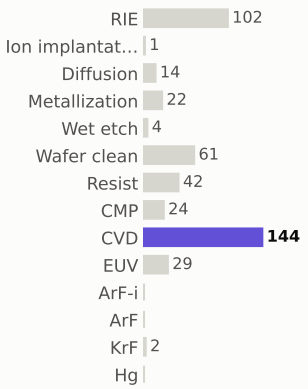
y cut at 0.184 µm



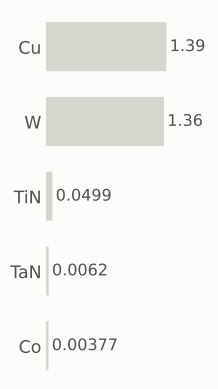
mask for this step

no mask for this step

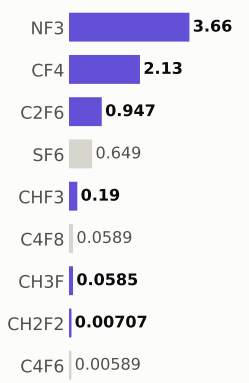
steps by process area



materials deposited (g)



gases emitted (g)



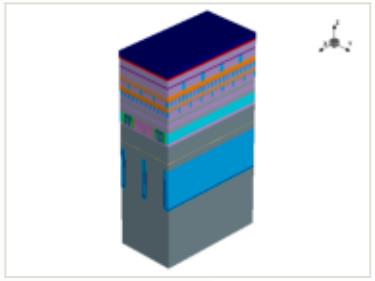
Step #4690: CVD

depo_fill · CVD

1.31	466.1	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

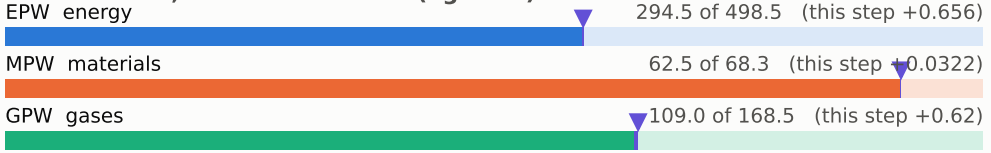


this step

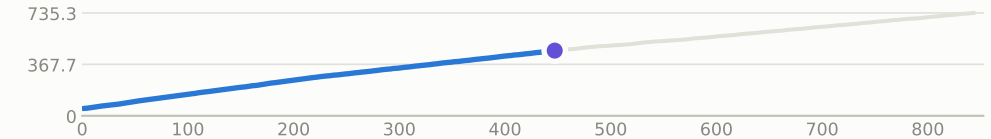
process area	CVD
material	TiN
thickness	0.03 µm
material GWP	43.9778 gCO2e/g
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0322 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

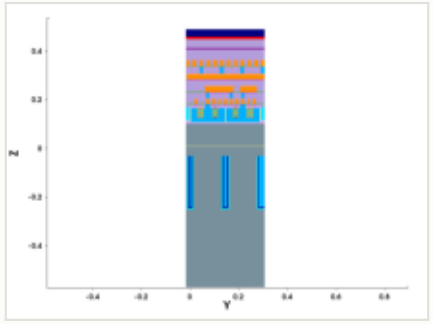
carbon so far, of the whole flow (kgCO2e)



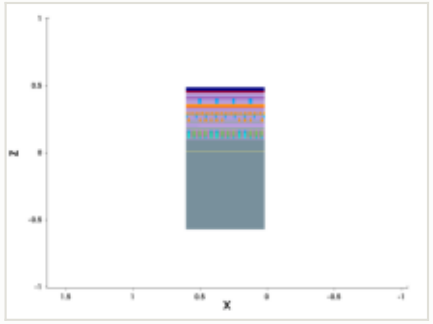
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



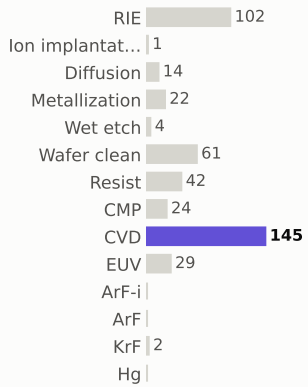
y cut at 0.184 µm



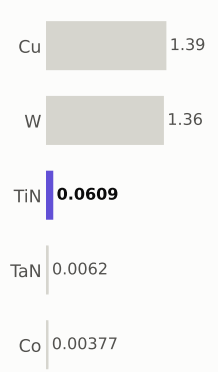
mask for this step

no mask for this step

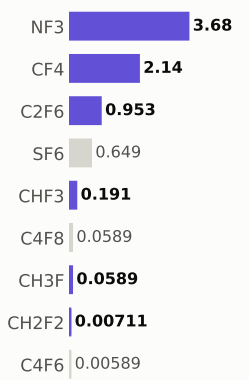
steps by process area



materials deposited (g)



gases emitted (g)



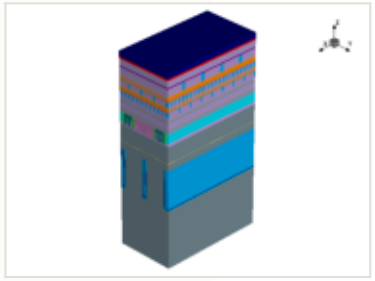
Step #4700: Wafer_clean

clean · Wafer_clean

0.169	466.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



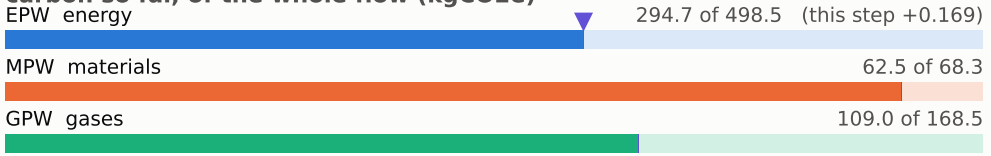
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

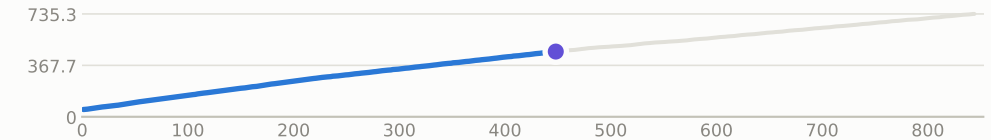
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

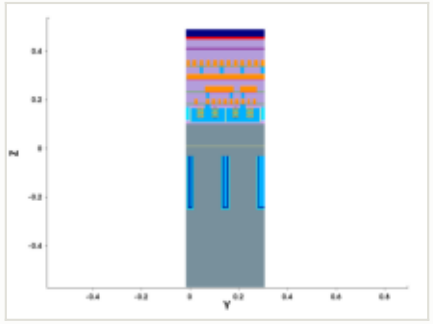
carbon so far, of the whole flow (kgCO2e)



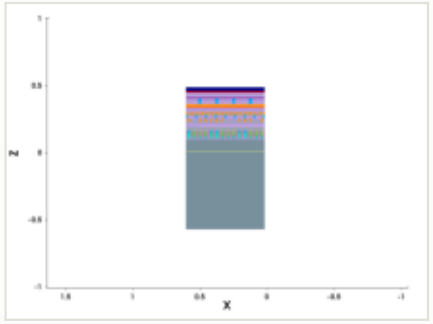
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



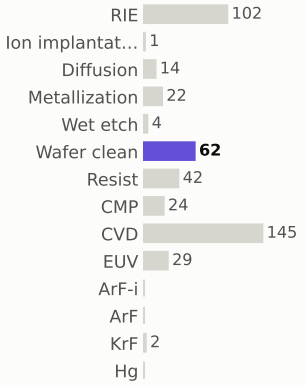
y cut at 0.184 µm



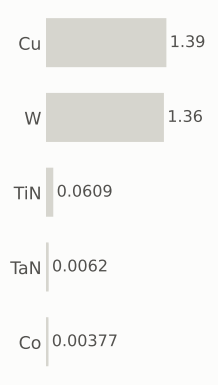
mask for this step

no mask for this step

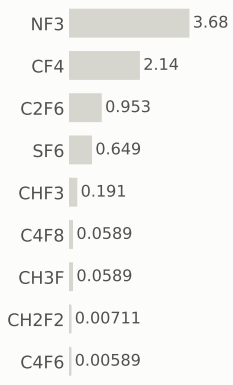
steps by process area



materials deposited (g)



gases emitted (g)



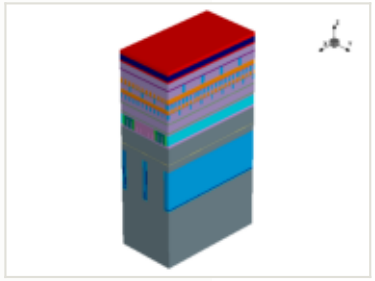
Step #4710: CVD

depo_fill · CVD

1.31	467.6	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

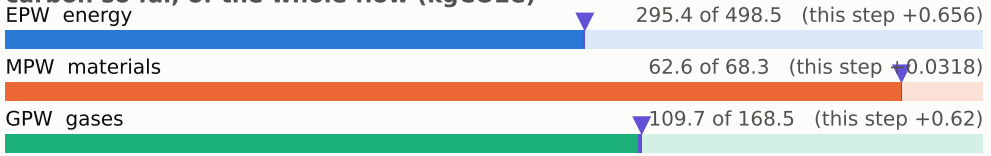


this step

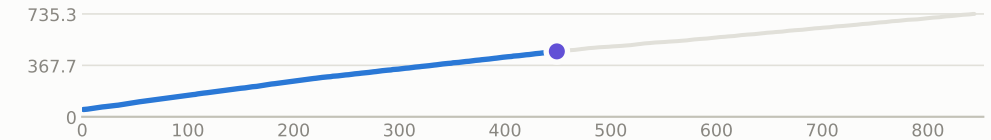
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

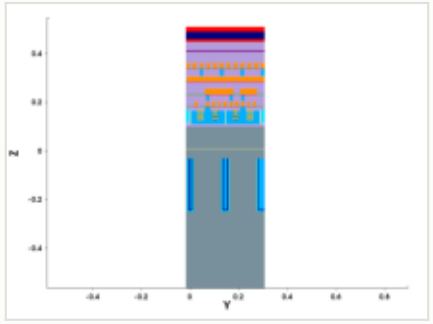
carbon so far, of the whole flow (kgCO2e)



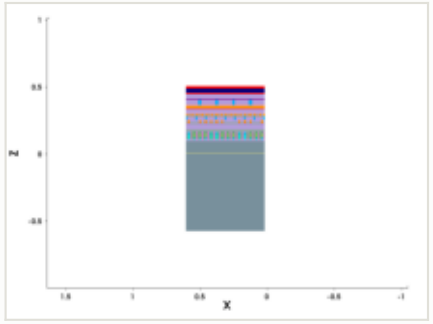
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



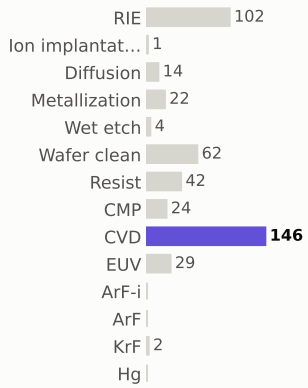
y cut at 0.184 µm



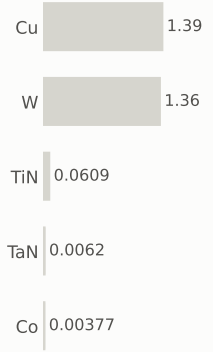
mask for this step

no mask for this step

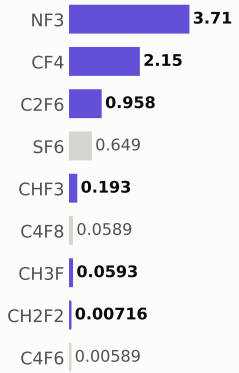
steps by process area



materials deposited (g)



gases emitted (g)



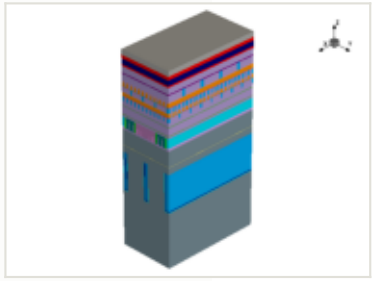
Step #4720: CVD

depo_fill · CVD

1.31	468.9	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

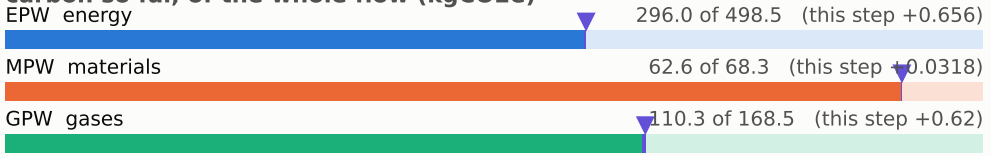


this step

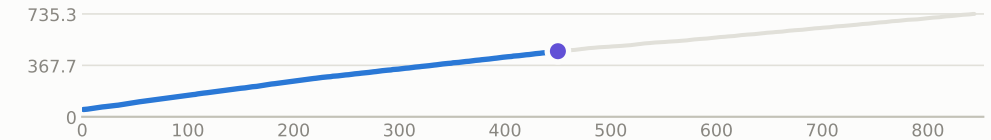
process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

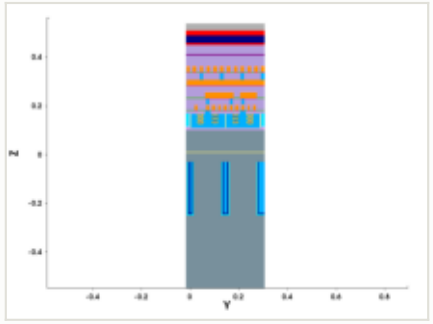
carbon so far, of the whole flow (kgCO2e)



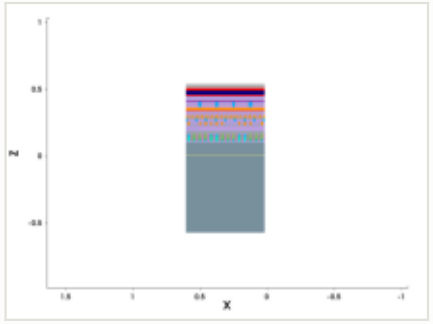
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



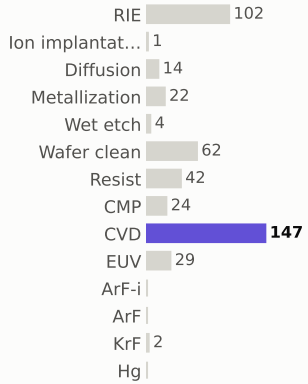
y cut at 0.184 µm



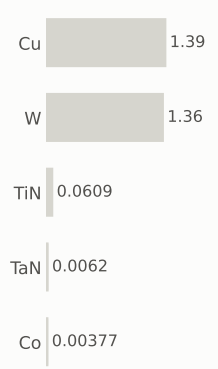
mask for this step

no mask for this step

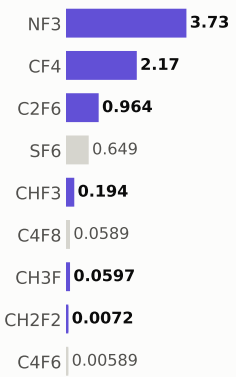
steps by process area



materials deposited (g)



gases emitted (g)



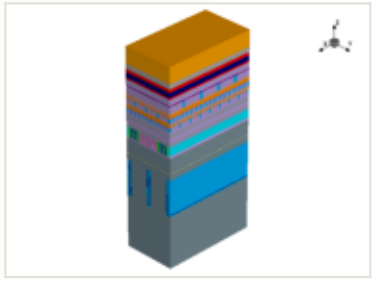
Step #4730: Resist

depo_fill · Resist

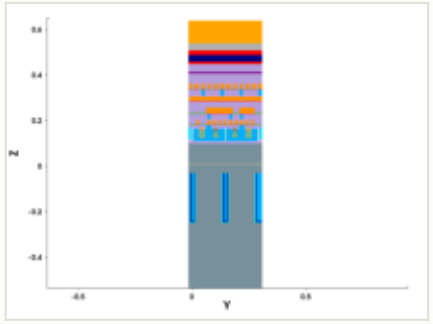
0.169	469.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

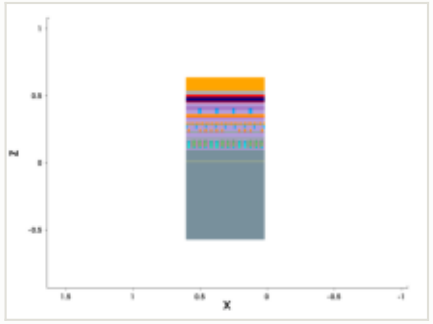
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



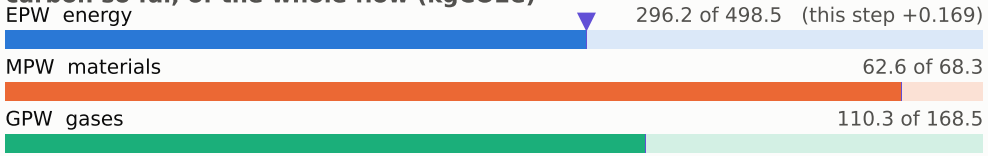
mask for this step

no mask for this step

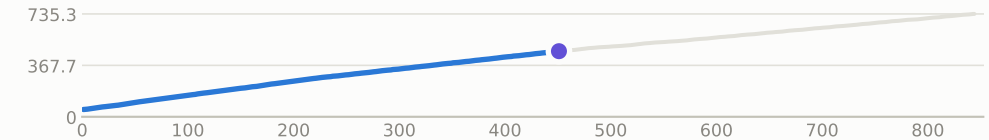
this step

process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	0.1 µm	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

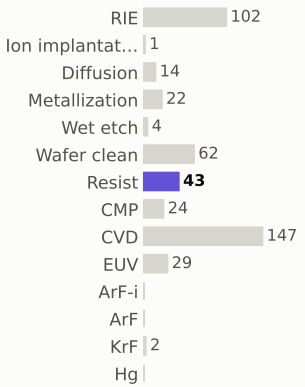
carbon so far, of the whole flow (kgCO2e)



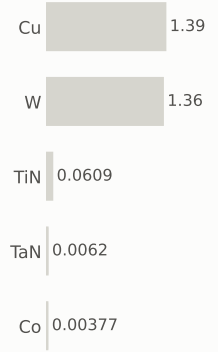
embodied carbon up to each step (kgCO2e per wafer)



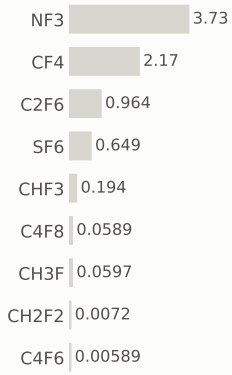
steps by process area



materials deposited (g)



gases emitted (g)



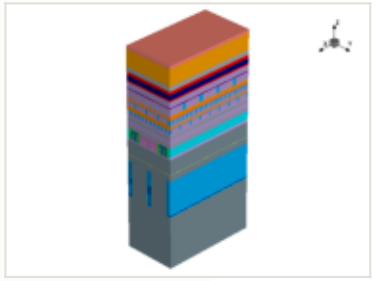
Step #4740: CVD

depo_fill · CVD

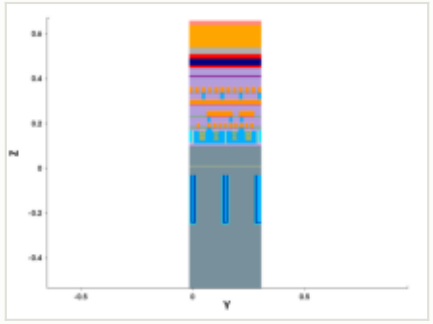
1.31	470.4	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

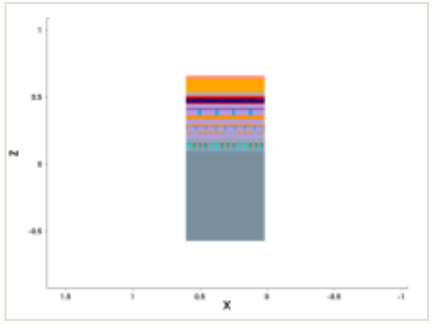
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

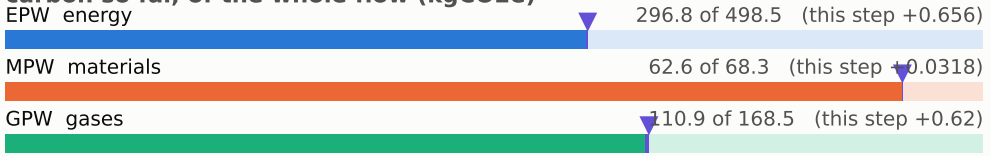
no mask for this step

this step

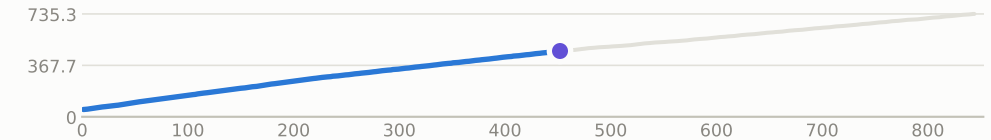
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

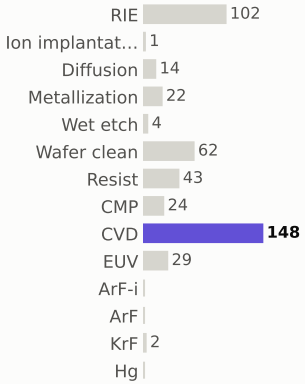
carbon so far, of the whole flow (kgCO2e)



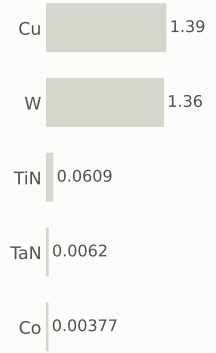
embodied carbon up to each step (kgCO2e per wafer)



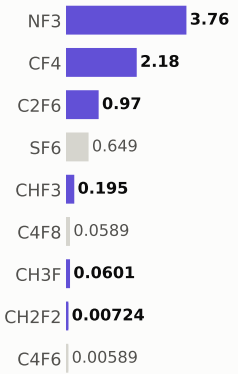
steps by process area



materials deposited (g)



gases emitted (g)



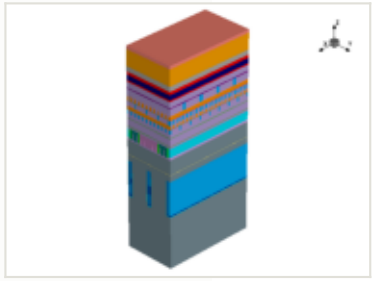
Step #4750: Wafer_clean

clean · Wafer_clean

0.169	470.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



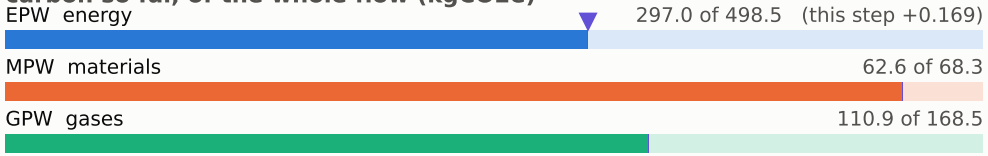
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

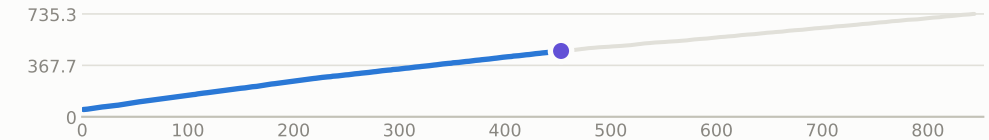
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

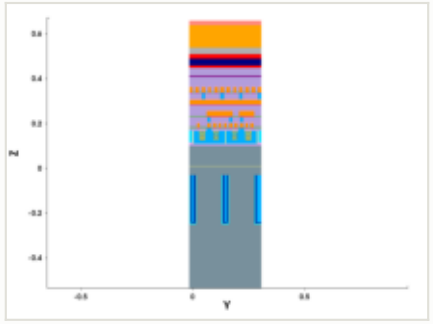
carbon so far, of the whole flow (kgCO2e)



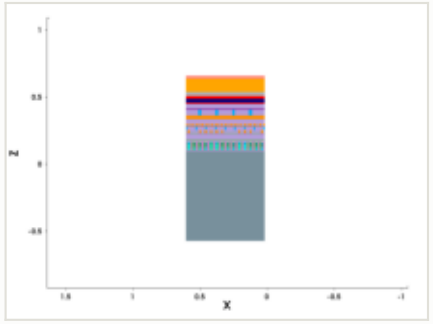
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



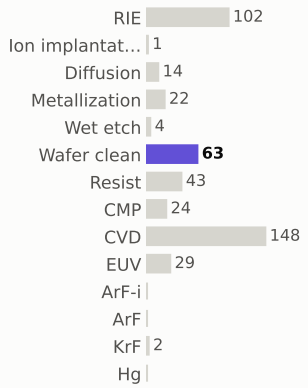
y cut at 0.184 µm



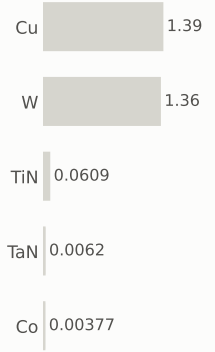
mask for this step

no mask for this step

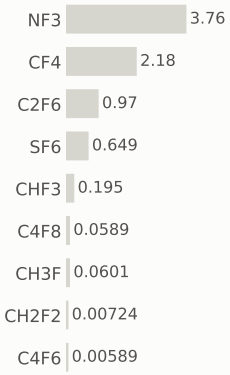
steps by process area



materials deposited (g)



gases emitted (g)



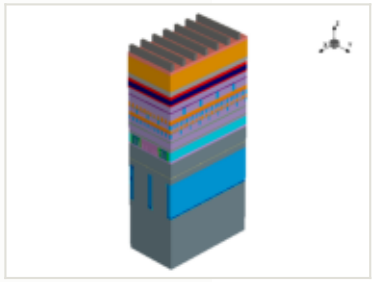
Step #4760: ArF-i

litho_patterning · ArF-i

0.217	470.8	0.029%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

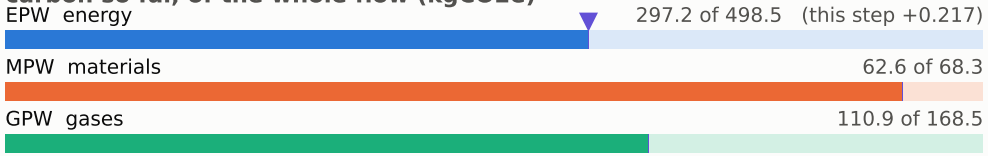


this step

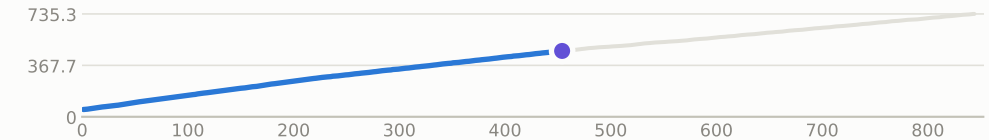
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

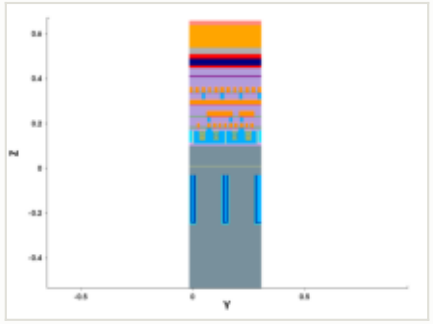
carbon so far, of the whole flow (kgCO2e)



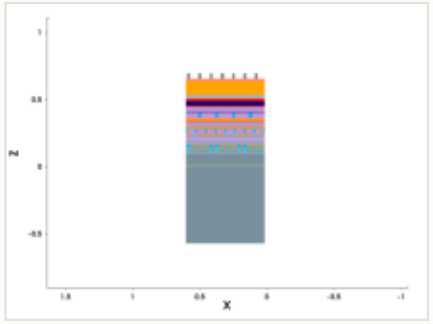
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm

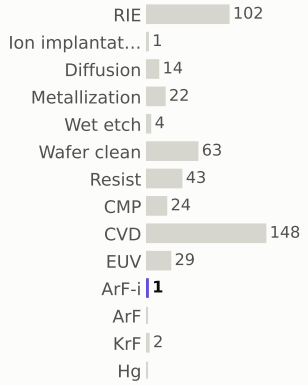


y cut at 0.184 µm

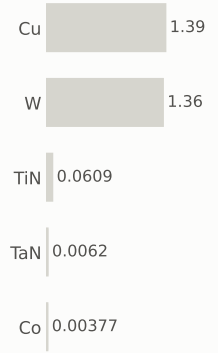


mask: layer 10401:0, positive (GDSII)

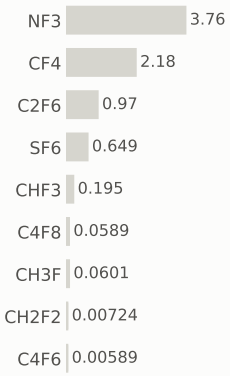
steps by process area



materials deposited (g)



gases emitted (g)



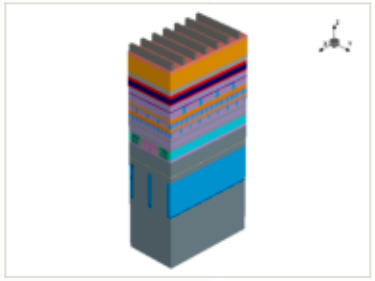
Step #4770: Ion_implantation

treatment · Ion_implantation

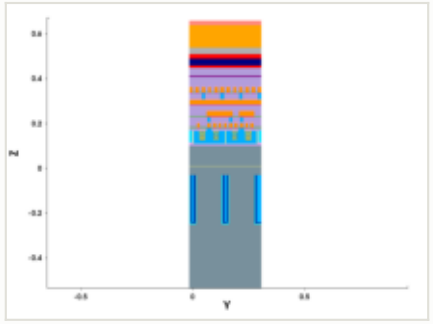
0.801	471.6	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

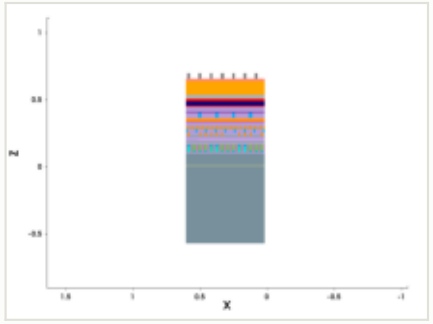
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

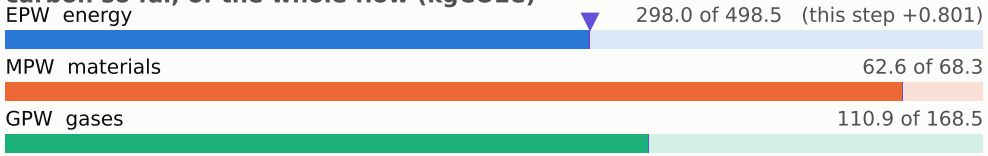
no mask for this step

this step

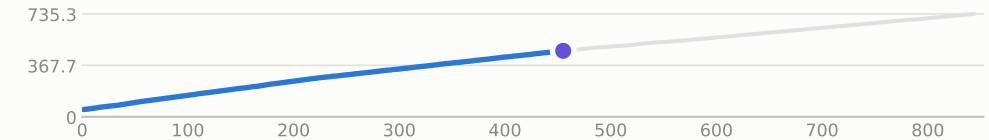
process area	Ion_implantation	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

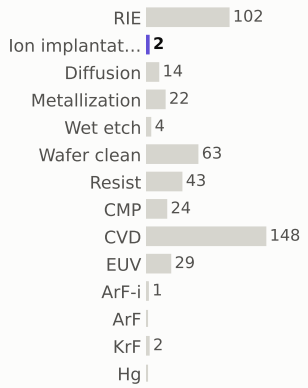
carbon so far, of the whole flow (kgCO2e)



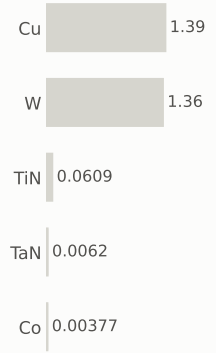
embodied carbon up to each step (kgCO2e per wafer)



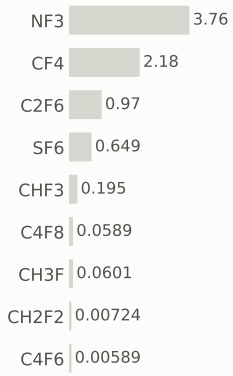
steps by process area



materials deposited (g)



gases emitted (g)



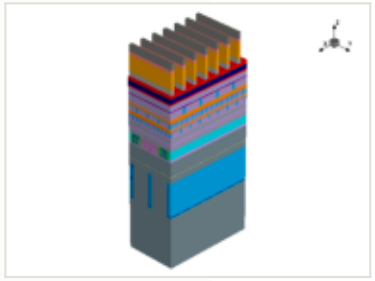
Step #4780: RIE

etch_anisotropic · RIE

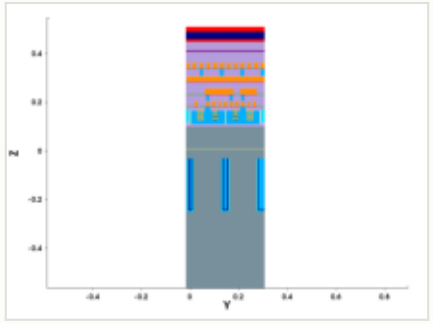
1.04	472.6	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

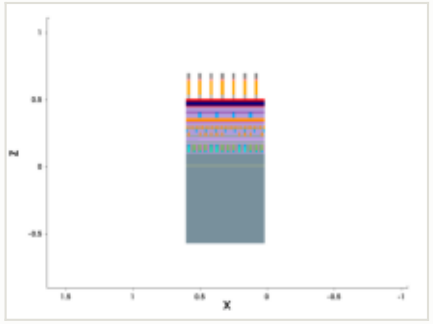
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

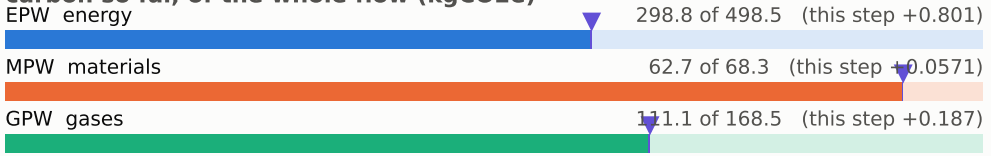
no mask for this step

this step

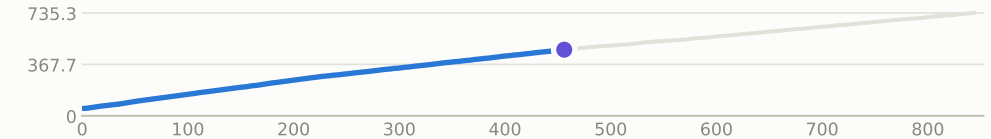
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

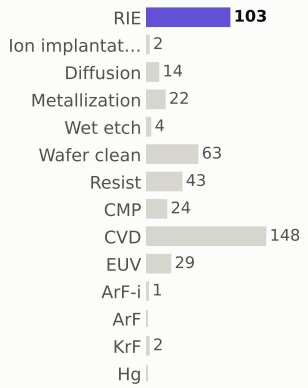
carbon so far, of the whole flow (kgCO2e)



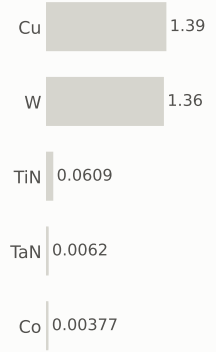
embodied carbon up to each step (kgCO2e per wafer)



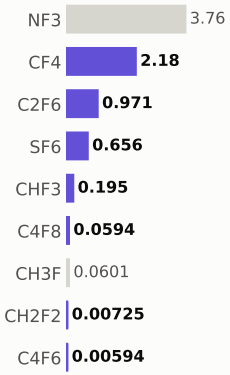
steps by process area



materials deposited (g)



gases emitted (g)



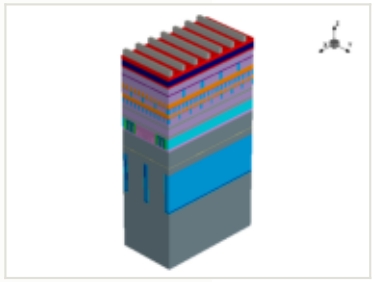
Step #4790: Wafer_clean

etch_strip · Wafer_clean

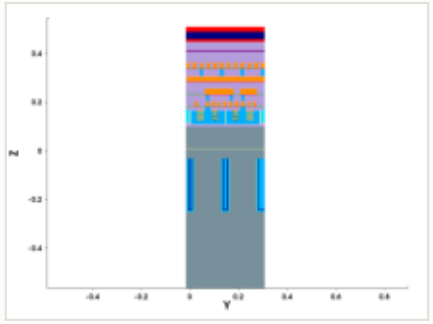
0.169	472.8	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

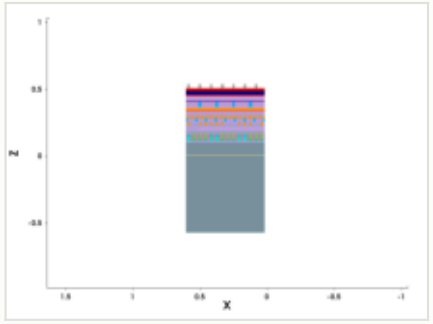
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



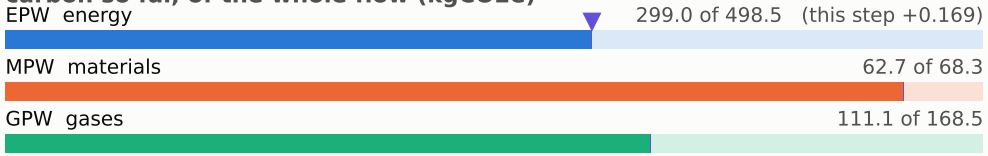
mask for this step

no mask for this step

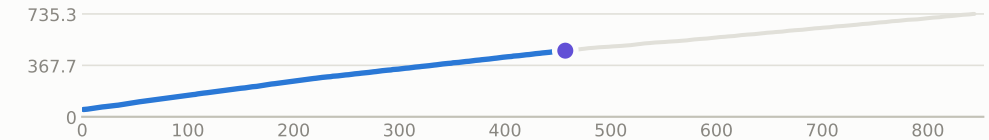
this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

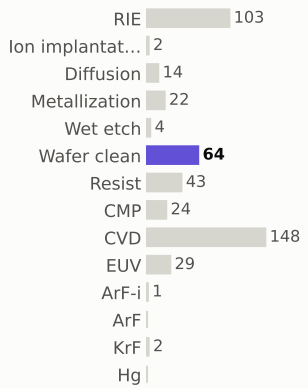
carbon so far, of the whole flow (kgCO2e)



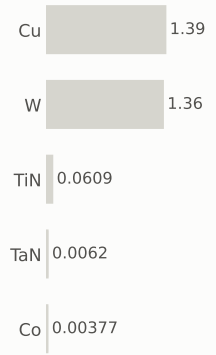
embodied carbon up to each step (kgCO2e per wafer)



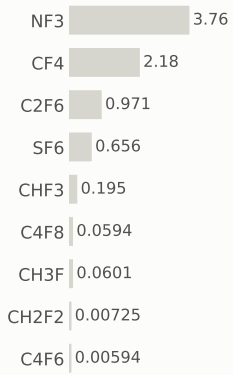
steps by process area



materials deposited (g)



gases emitted (g)



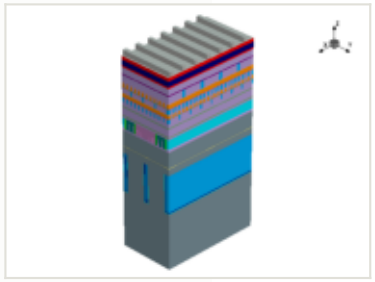
Step #4800: CVD

depo_isotropic · CVD

1.31	474.1	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

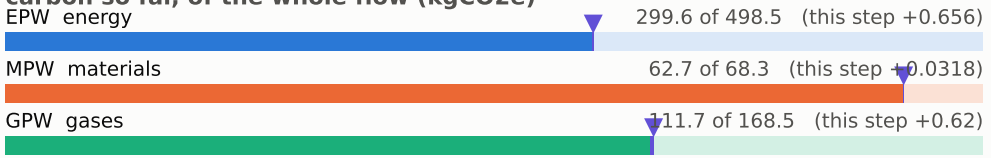


this step

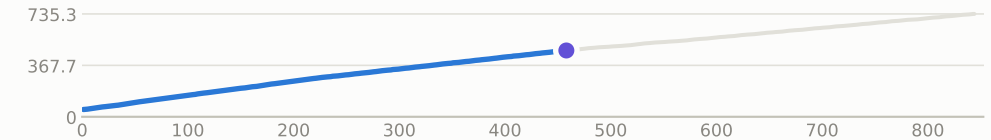
process area	CVD
material	—
thickness	0.001 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

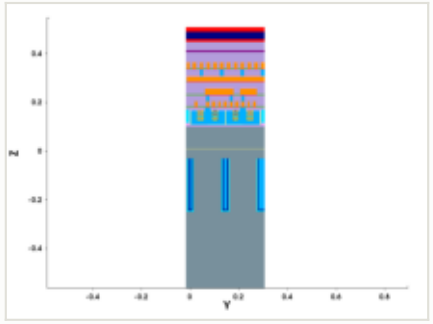
carbon so far, of the whole flow (kgCO2e)



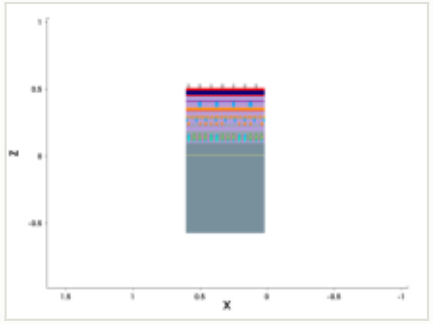
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



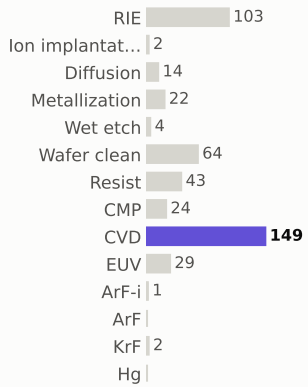
y cut at 0.184 µm



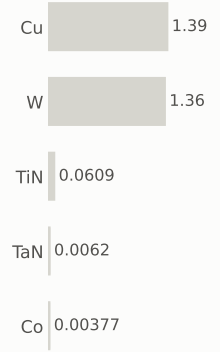
mask for this step

no mask for this step

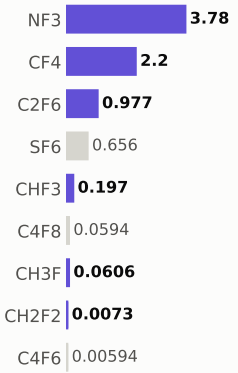
steps by process area



materials deposited (g)



gases emitted (g)



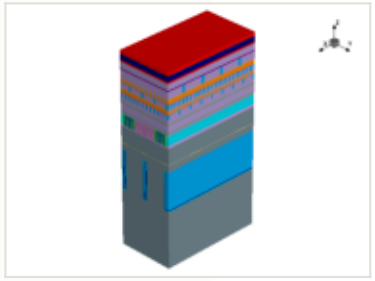
Step #4810: RIE

etch_anisotropic · RIE

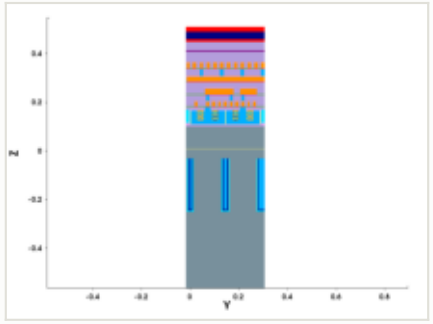
1.04	475.1	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

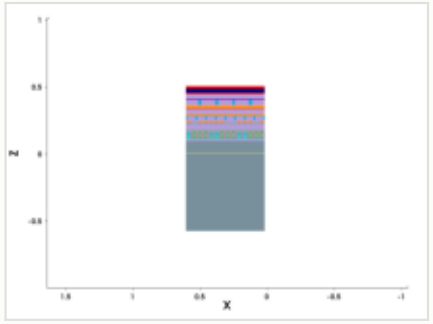
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

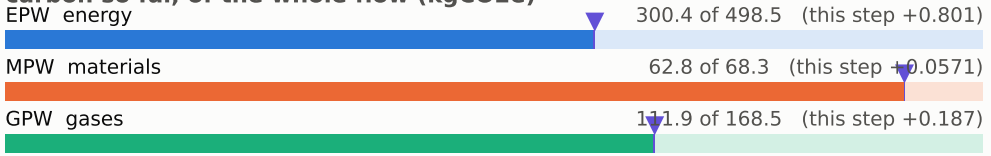
no mask for this step

this step

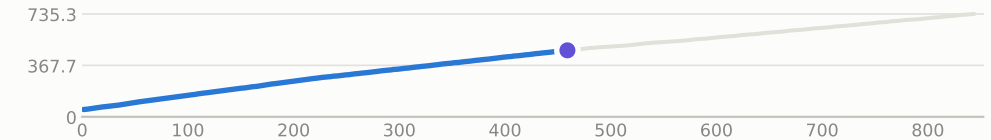
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

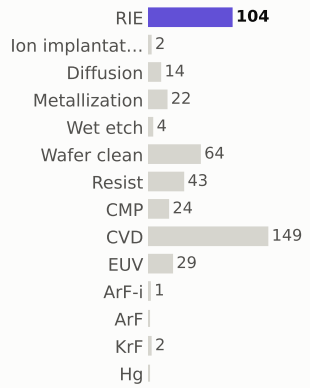
carbon so far, of the whole flow (kgCO2e)



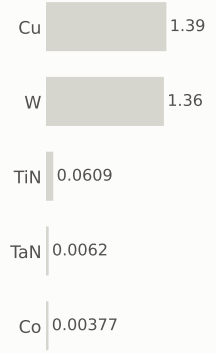
embodied carbon up to each step (kgCO2e per wafer)



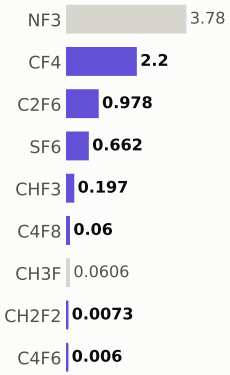
steps by process area



materials deposited (g)



gases emitted (g)



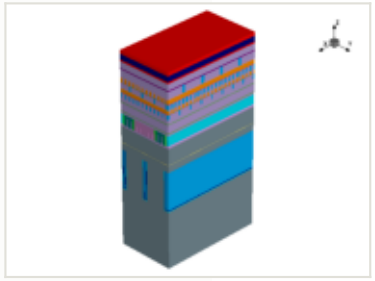
Step #4820: Wafer_clean

etch_strip · Wafer_clean

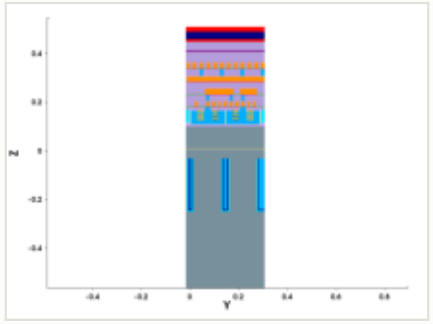
0.169	475.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

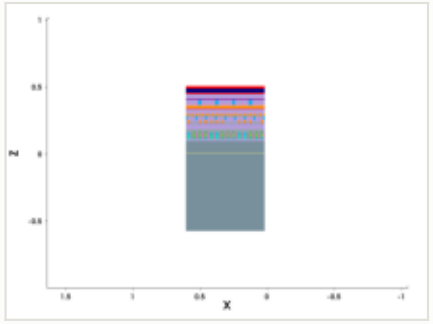
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



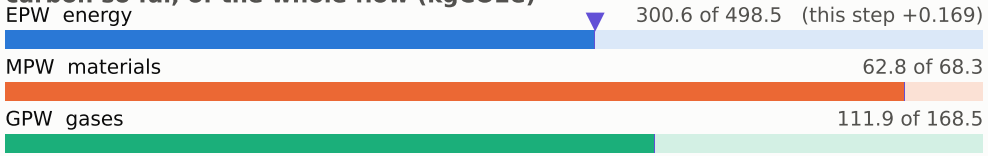
mask for this step

no mask for this step

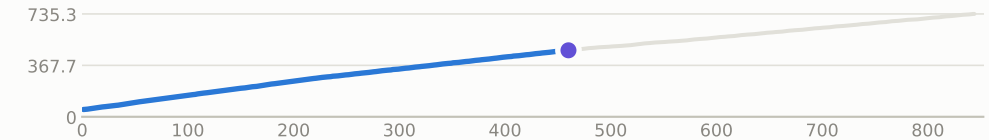
this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

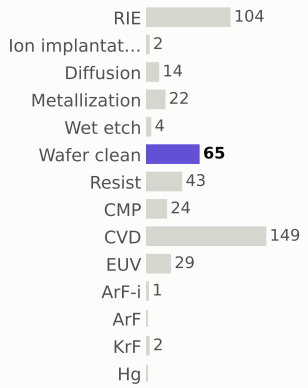
carbon so far, of the whole flow (kgCO2e)



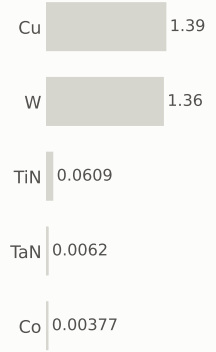
embodied carbon up to each step (kgCO2e per wafer)



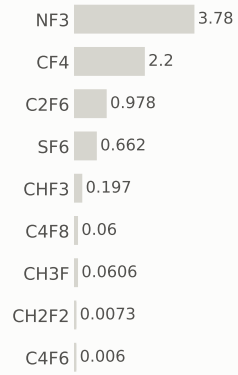
steps by process area



materials deposited (g)



gases emitted (g)



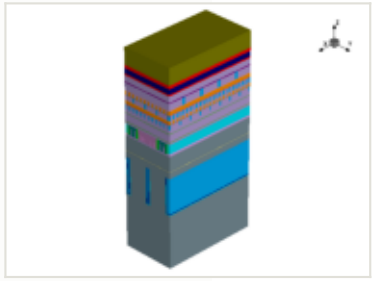
Step #4830: Resist

depo_fill · Resist

0.169	475.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

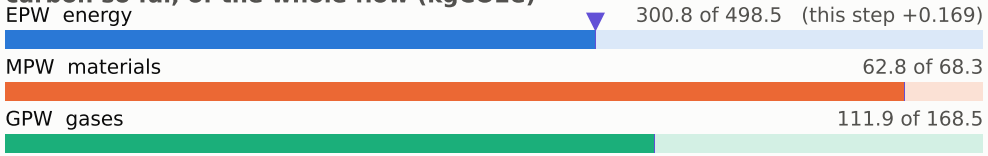


this step

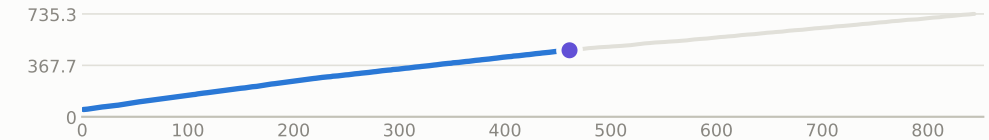
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

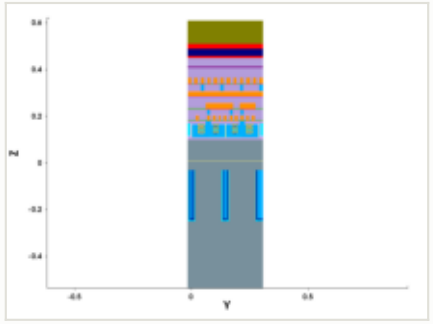
carbon so far, of the whole flow (kgCO2e)



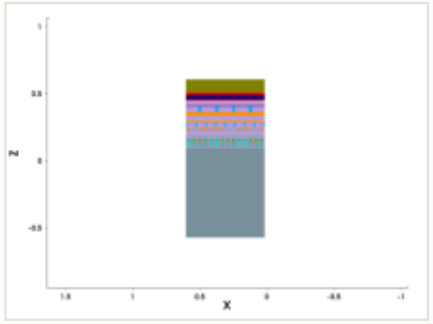
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



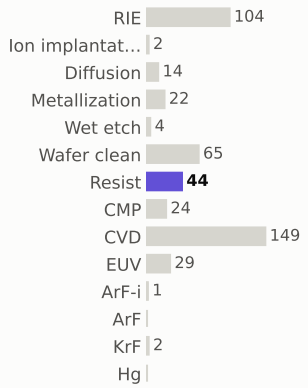
y cut at 0.184 µm



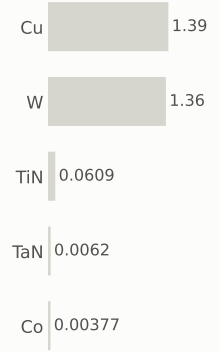
mask for this step

no mask for this step

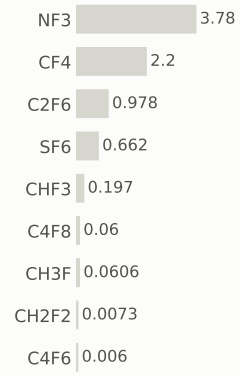
steps by process area



materials deposited (g)



gases emitted (g)



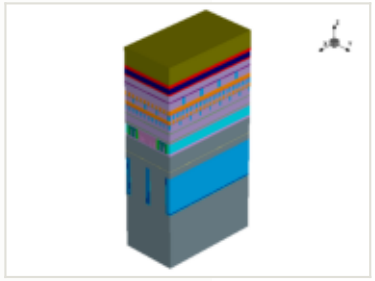
Step #4840: Wafer_clean

clean · Wafer_clean

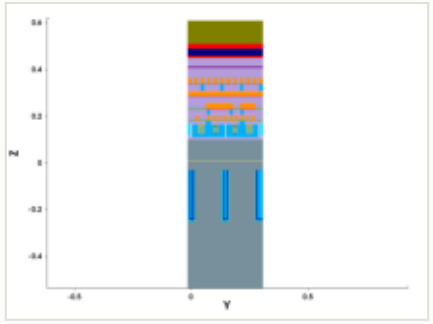
0.169	475.6	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

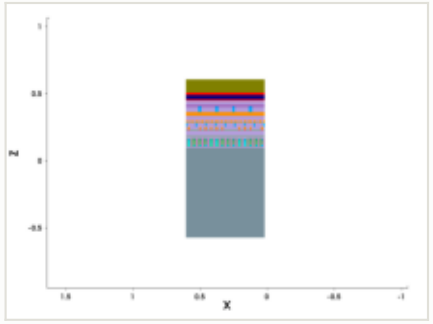
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

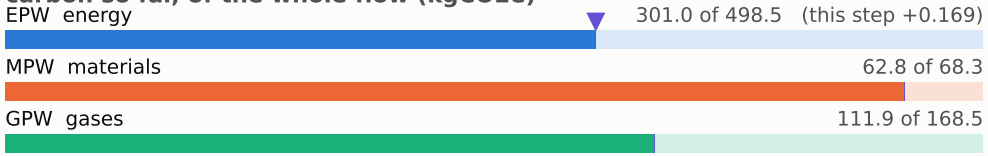
no mask for this step

this step

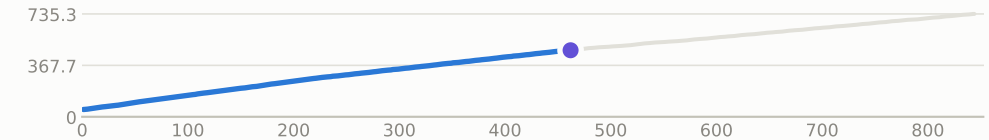
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

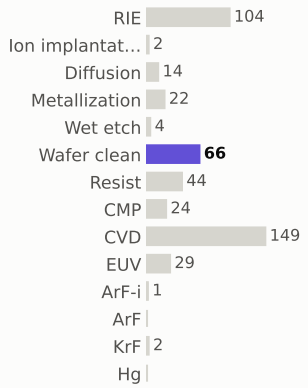
carbon so far, of the whole flow (kgCO2e)



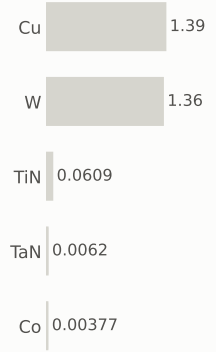
embodied carbon up to each step (kgCO2e per wafer)



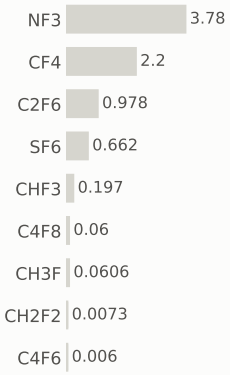
steps by process area



materials deposited (g)



gases emitted (g)



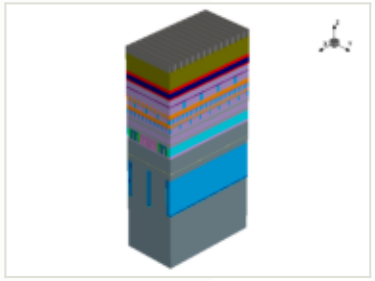
Step #4850: ArF-i

litho_patterning · ArF-i

0.217	475.8	0.029%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

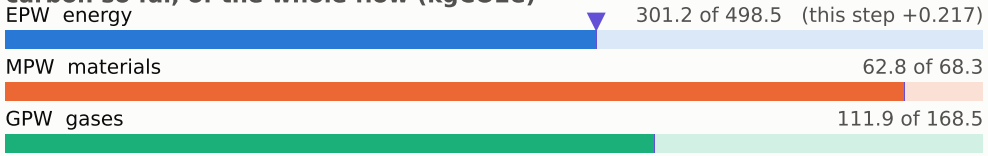


this step

process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

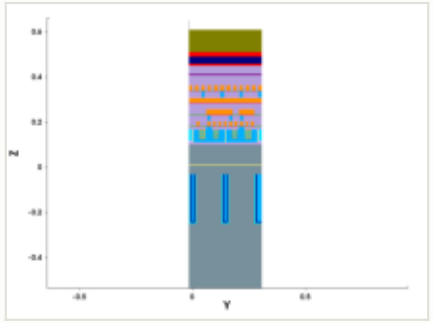
EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

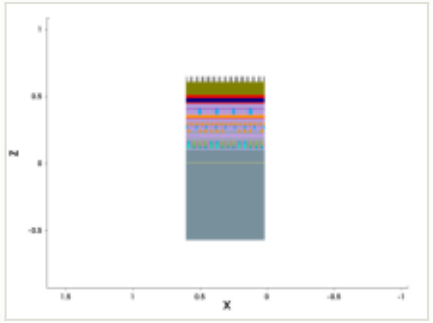


embodied carbon up to each step (kgCO2e per wafer)

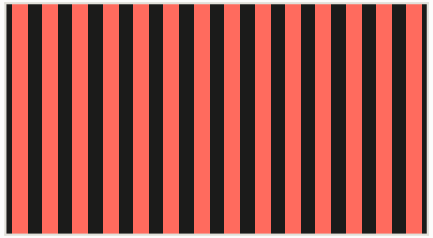
x cut at 0.294 µm



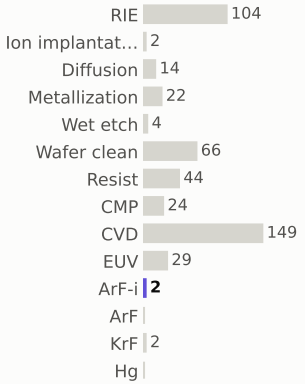
y cut at 0.184 µm



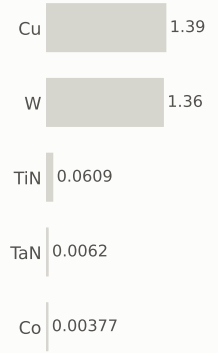
mask: layer 40:0 (GDSII)



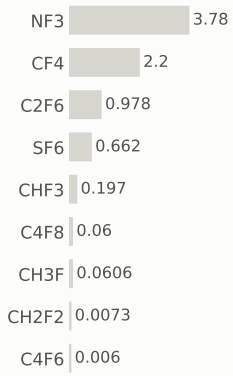
steps by process area



materials deposited (g)



gases emitted (g)



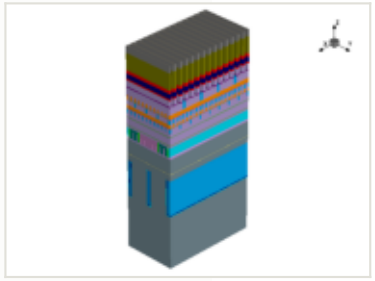
Step #4860: RIE

etch_anisotropic · RIE

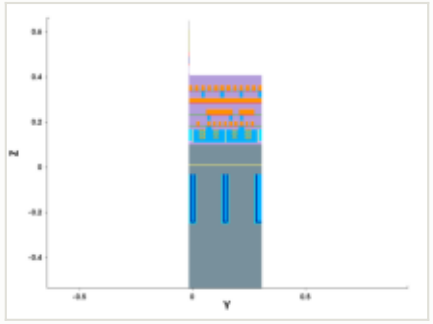
1.04	476.9	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

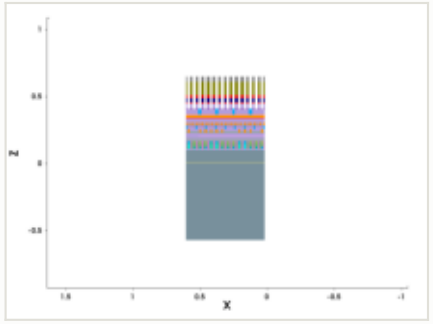
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

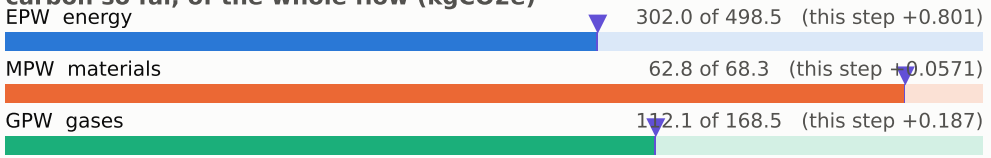
no mask for this step

this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

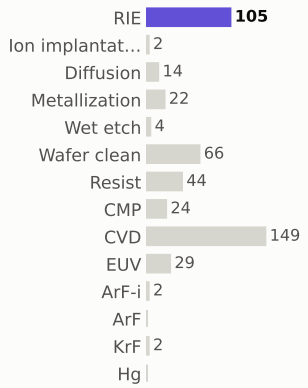
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

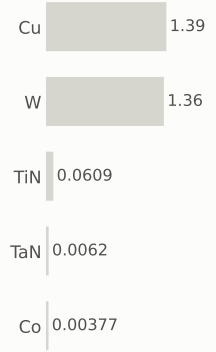


embodied carbon up to each step (kgCO2e per wafer)

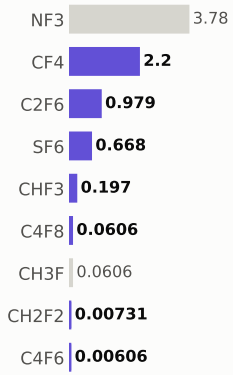
steps by process area



materials deposited (g)



gases emitted (g)



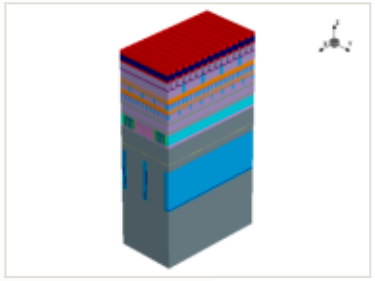
Step #4870: Wafer_clean

etch_strip · Wafer_clean

0.169	477.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

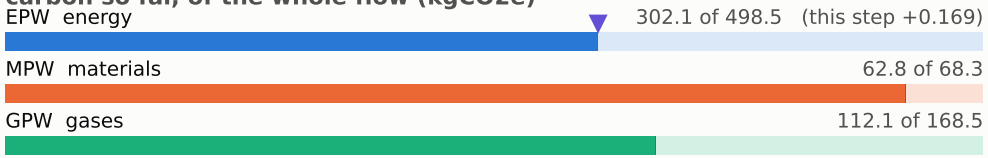


this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

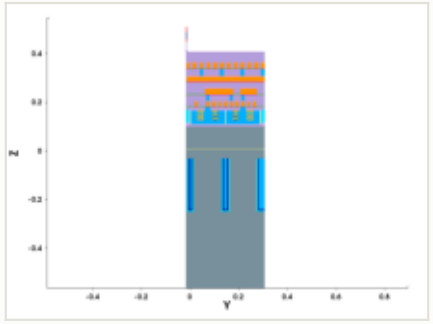
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

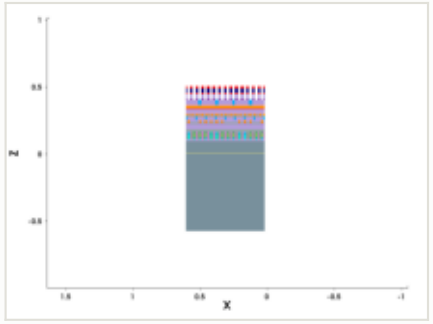


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



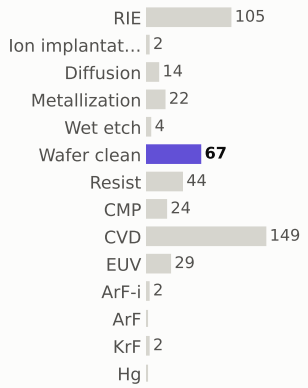
y cut at 0.184 µm



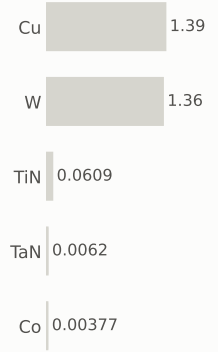
mask for this step

no mask for this step

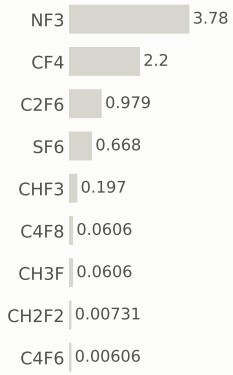
steps by process area



materials deposited (g)



gases emitted (g)



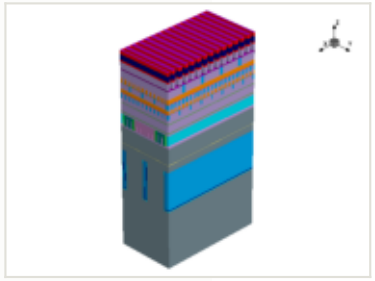
Step #4880: Metallization

depo_ellipsoidal · Metallization

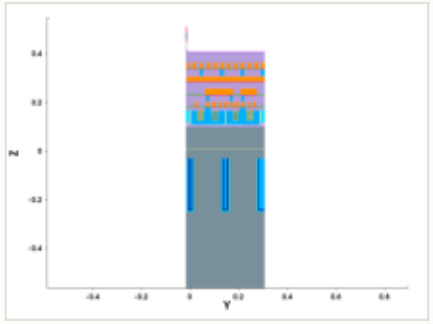
1.2	478.3	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

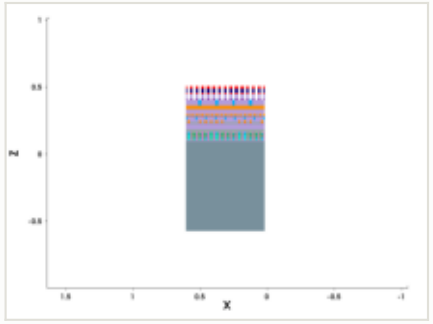
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

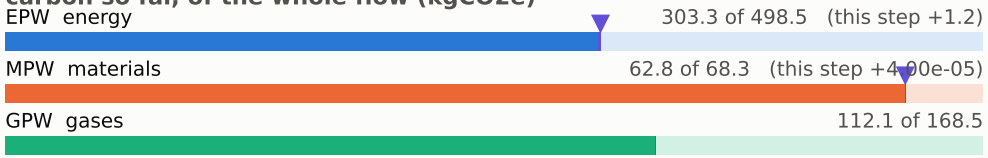
no mask for this step

this step

process area	Metallization
material	TaN
thickness	0.0016 µm
material GWP	25.8159 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

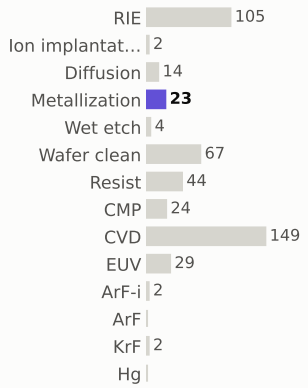
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.00e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

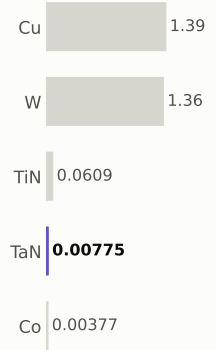


embodied carbon up to each step (kgCO2e per wafer)

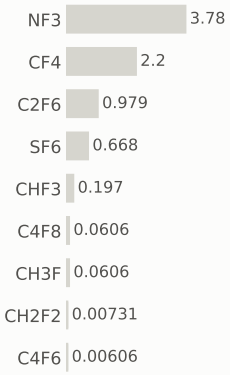
steps by process area



materials deposited (g)



gases emitted (g)



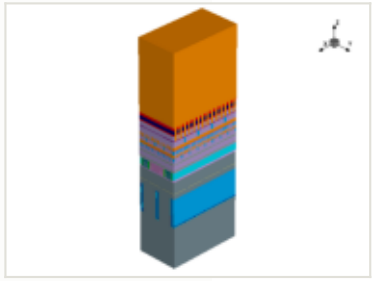
Step #4890: Metallization

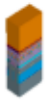
depo_fill · Metallization

1.21	479.5	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

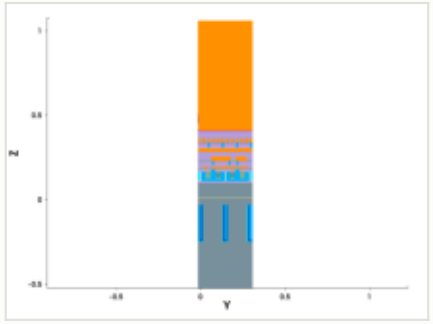
3D structure after this step

zoomed to fit

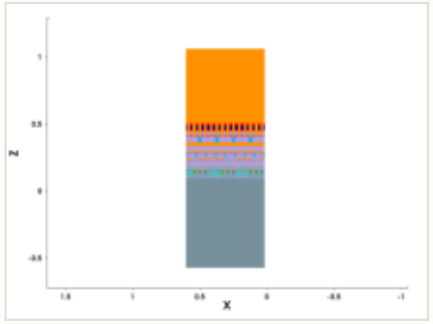




x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

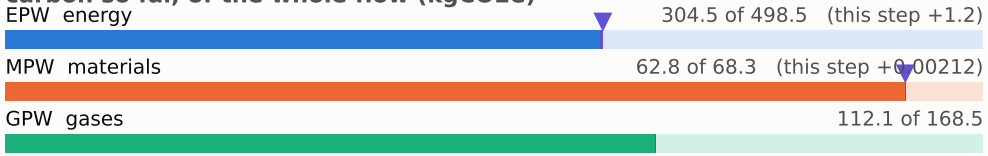
no mask for this step

this step

process area	Metallization
material	Cu
thickness	0.55 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

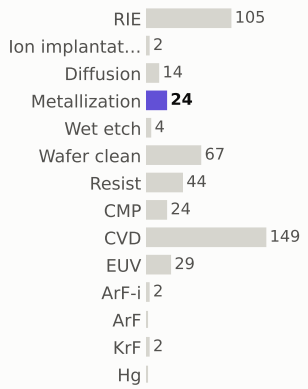
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00212 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

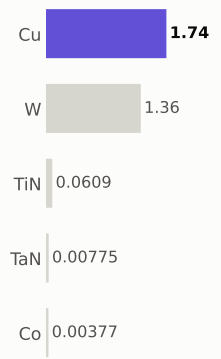


embodied carbon up to each step (kgCO2e per wafer)

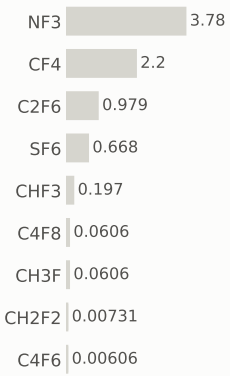
steps by process area



materials deposited (g)



gases emitted (g)



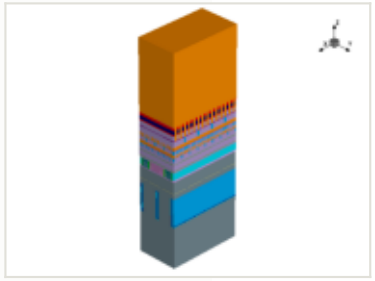
Step #4900: Diffusion

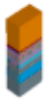
treatment · Diffusion

0.801	480.3	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

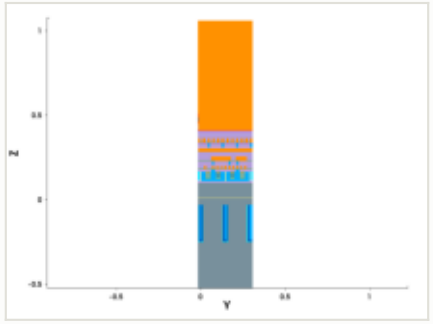
3D structure after this step

zoomed to fit

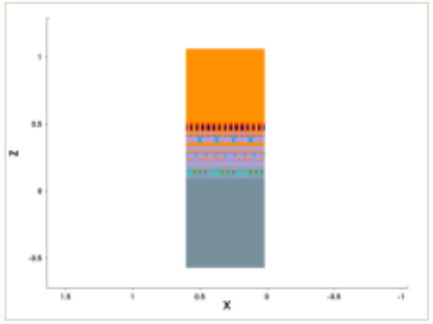




x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

no mask for this step

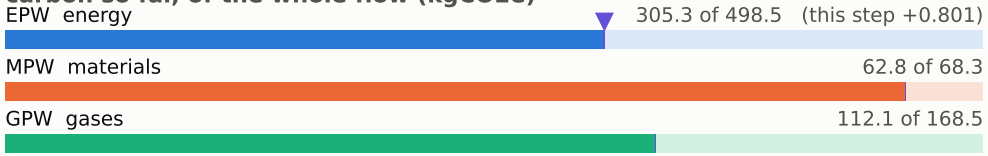
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

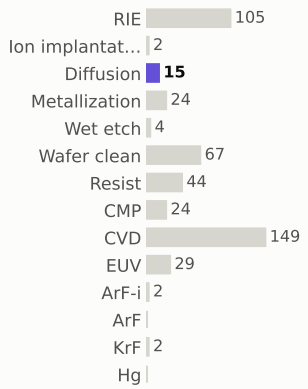
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

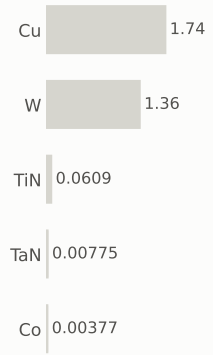


embodied carbon up to each step (kgCO2e per wafer)

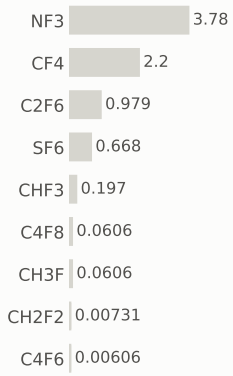
steps by process area



materials deposited (g)



gases emitted (g)



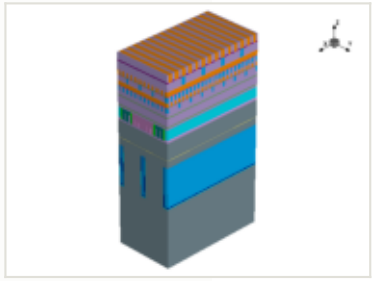
Step #4910: CMP

etch_cmp · CMP

0.169	480.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

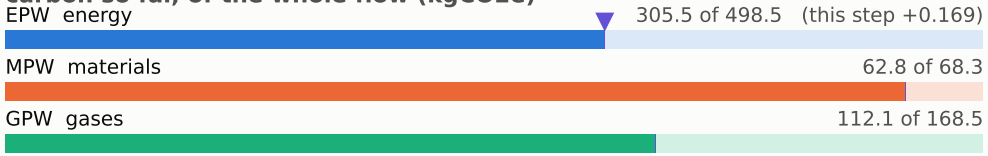


this step

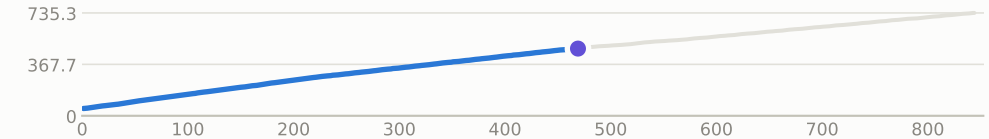
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

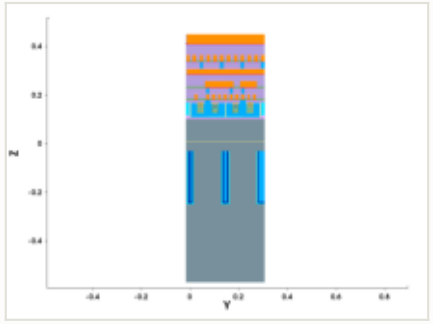
carbon so far, of the whole flow (kgCO2e)



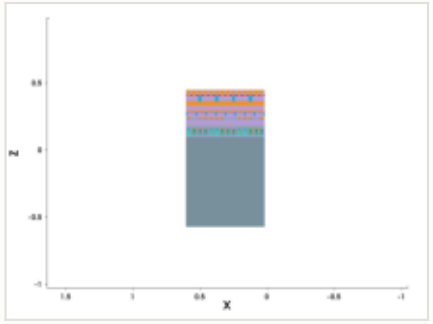
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



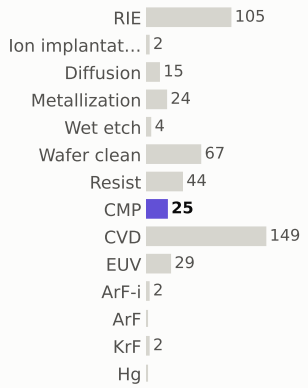
y cut at 0.184 µm



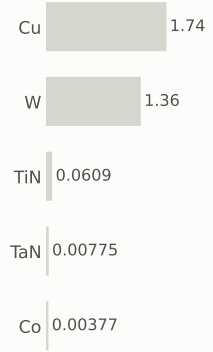
mask for this step

no mask for this step

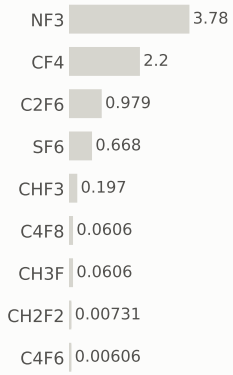
steps by process area



materials deposited (g)



gases emitted (g)



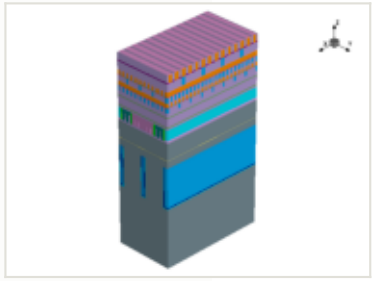
Step #4920: Metallization

depo_isotropic · Metallization

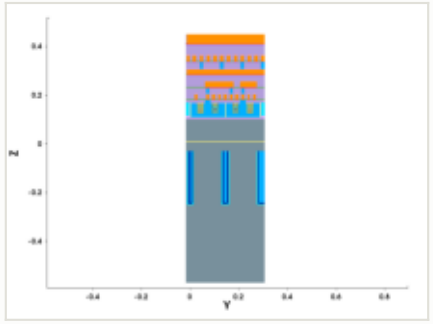
1.2	481.6	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

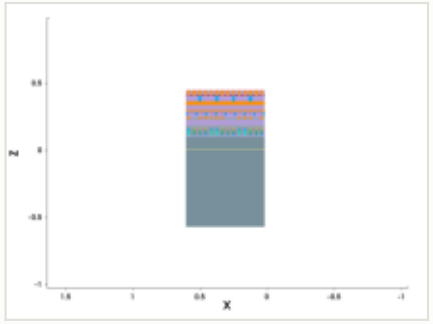
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

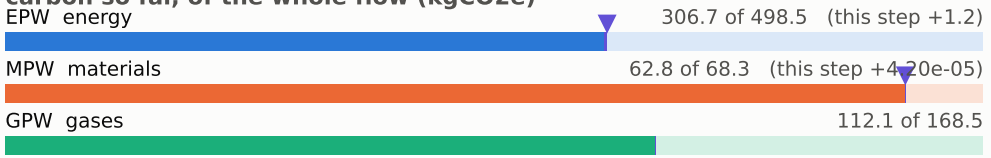
no mask for this step

this step

process area	Metallization
material	Co
thickness	0.0015 µm
material GWP	44.5 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

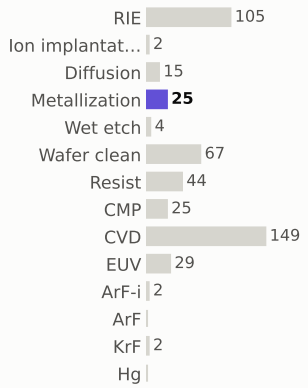
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.20e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

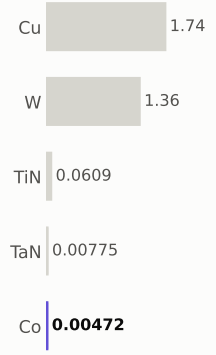


embodied carbon up to each step (kgCO2e per wafer)

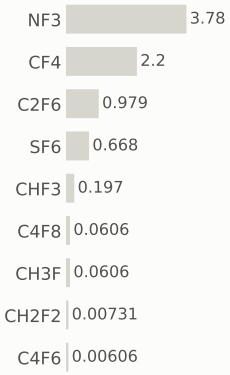
steps by process area



materials deposited (g)



gases emitted (g)



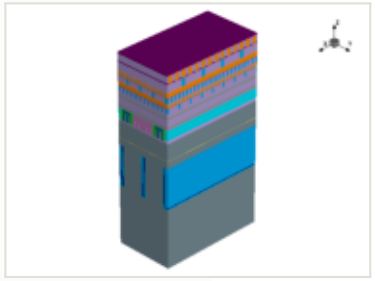
Step #4940: CVD

depo_fill · CVD

1.31	483.0	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

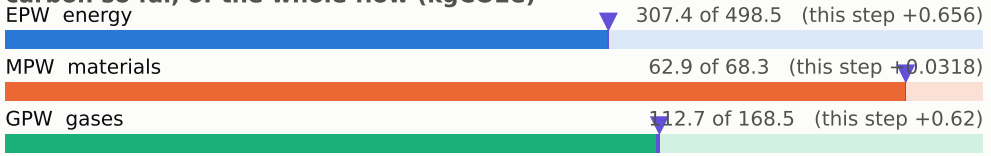


this step

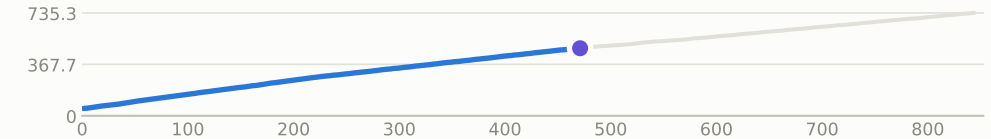
process area	CVD
material	—
thickness	0.006 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

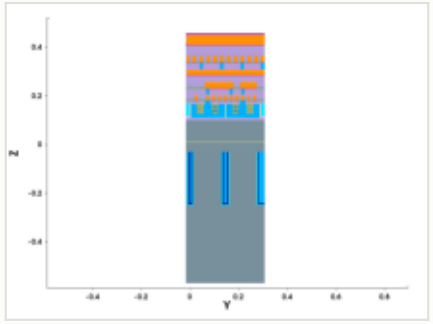
carbon so far, of the whole flow (kgCO2e)



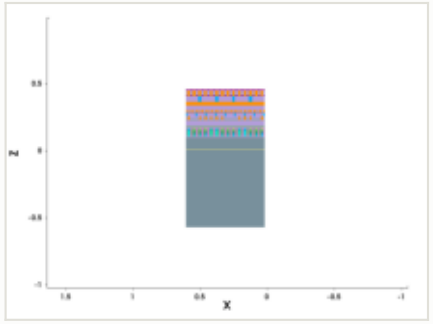
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



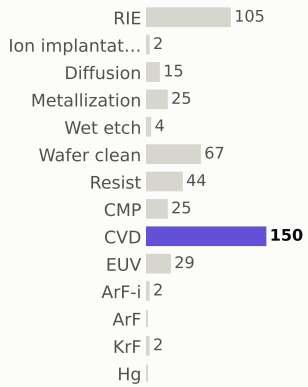
y cut at 0.184 µm



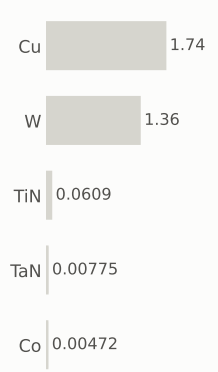
mask for this step

no mask for this step

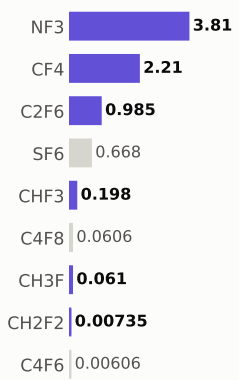
steps by process area



materials deposited (g)



gases emitted (g)



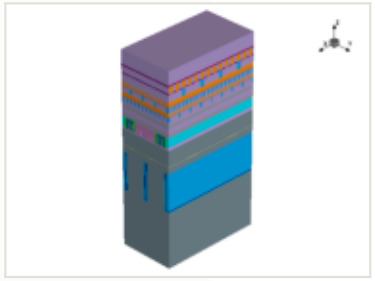
Step #4950: CVD

depo_fill · CVD

1.31	484.3	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

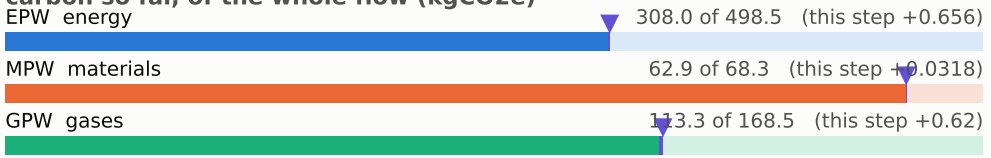


this step

process area	CVD
material	—
thickness	0.078 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

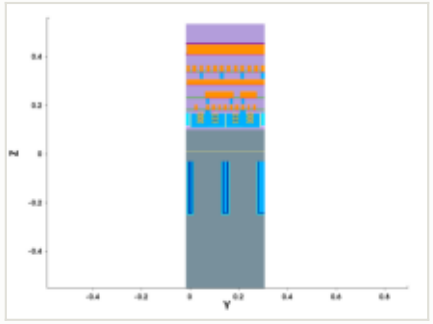
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

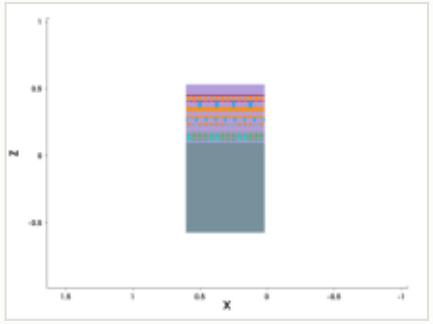


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



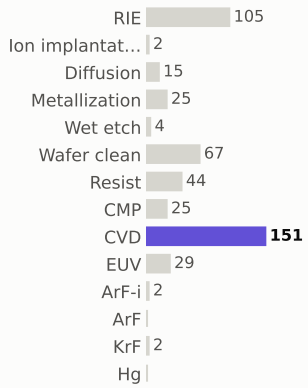
y cut at 0.184 µm



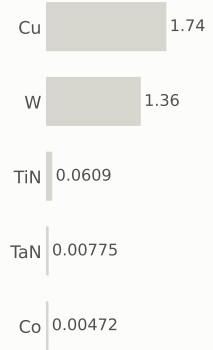
mask for this step

no mask for this step

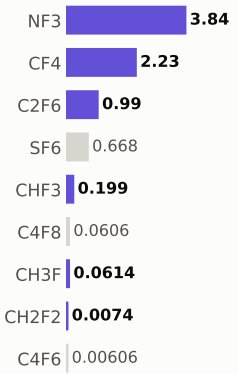
steps by process area



materials deposited (g)



gases emitted (g)



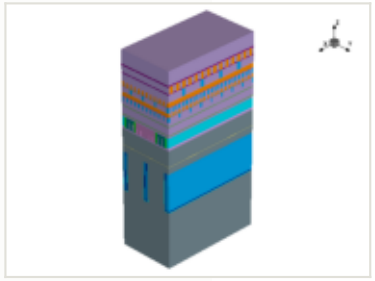
Step #4960: Diffusion

treatment · Diffusion

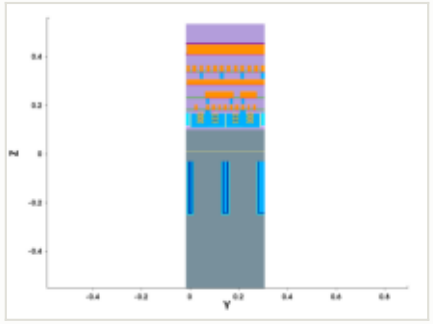
0.801	485.1	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

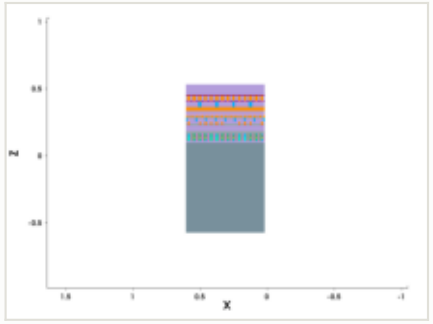
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

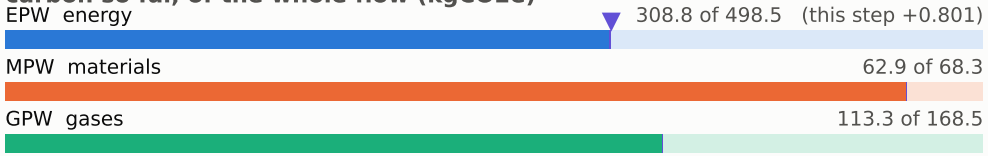
no mask for this step

this step

process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

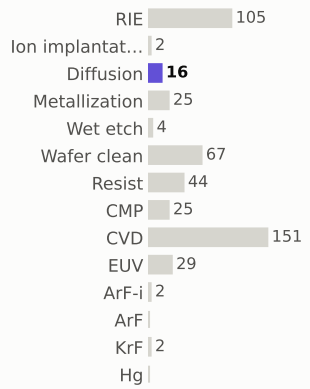
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

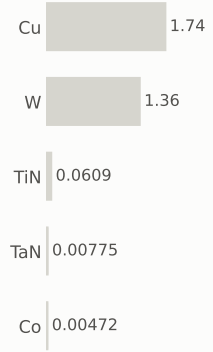


embodied carbon up to each step (kgCO2e per wafer)

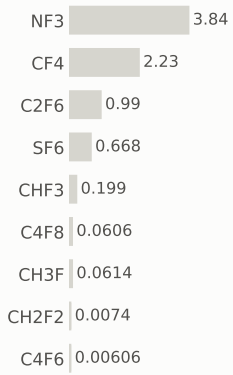
steps by process area



materials deposited (g)



gases emitted (g)



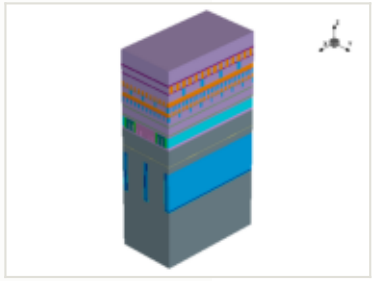
Step #4970: Wafer_clean

clean · Wafer_clean

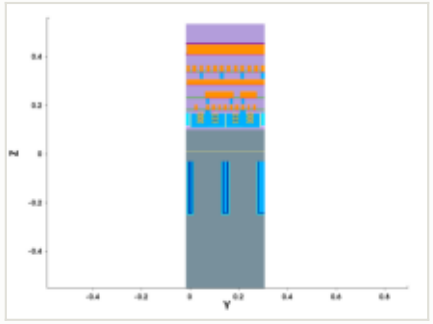
0.169	485.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

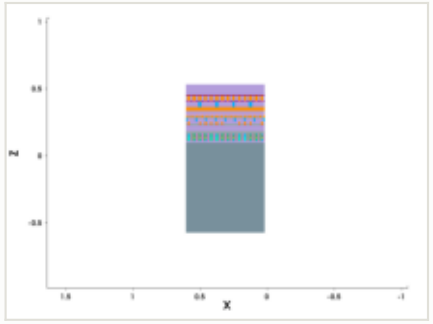
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

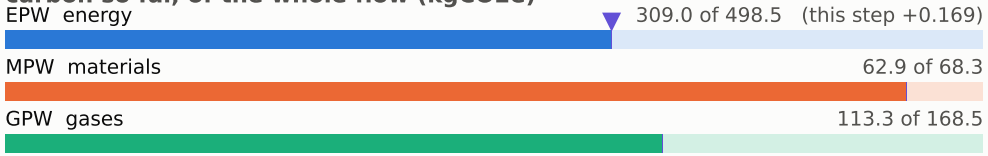
no mask for this step

this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

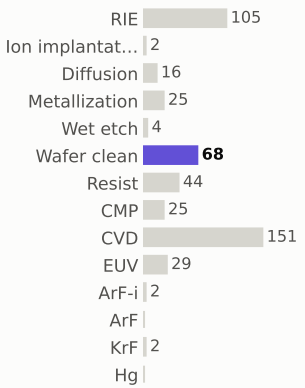
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

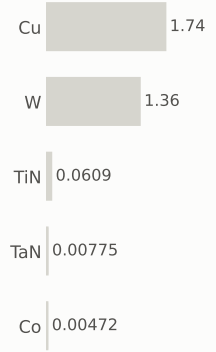


embodied carbon up to each step (kgCO2e per wafer)

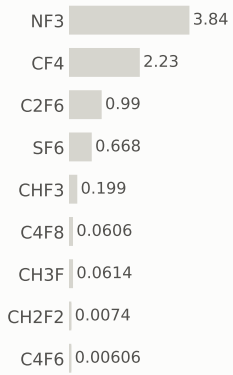
steps by process area



materials deposited (g)



gases emitted (g)



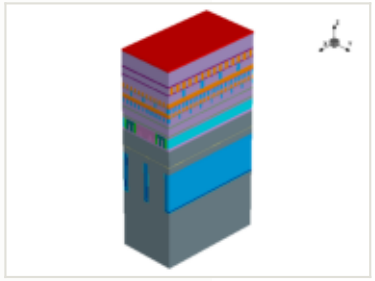
Step #4980: CVD

depo_fill · CVD

1.31	486.5	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

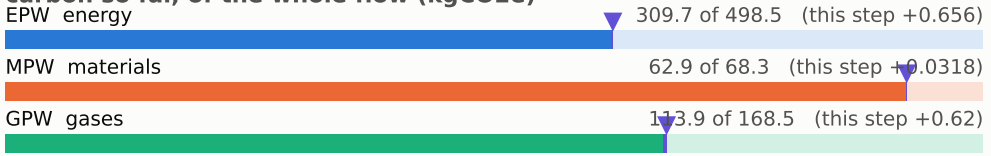


this step

process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

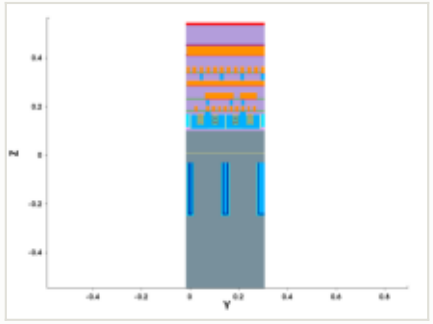
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

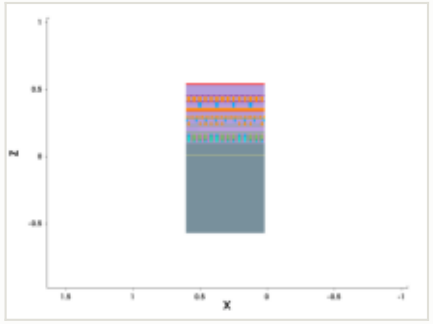


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



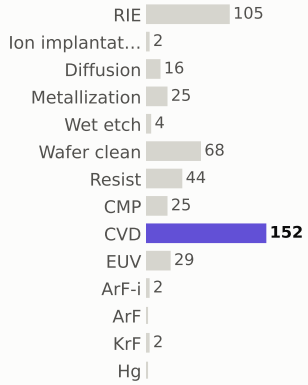
y cut at 0.184 µm



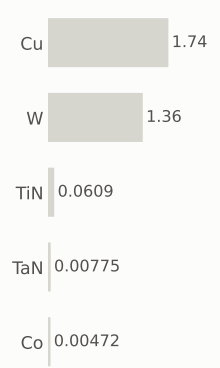
mask for this step

no mask for this step

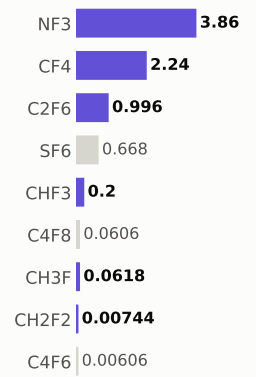
steps by process area



materials deposited (g)



gases emitted (g)



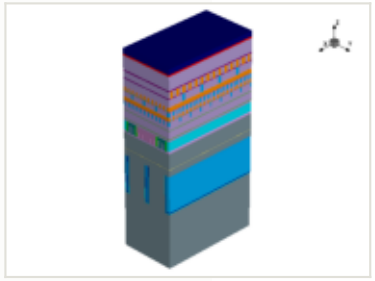
Step #4990: CVD

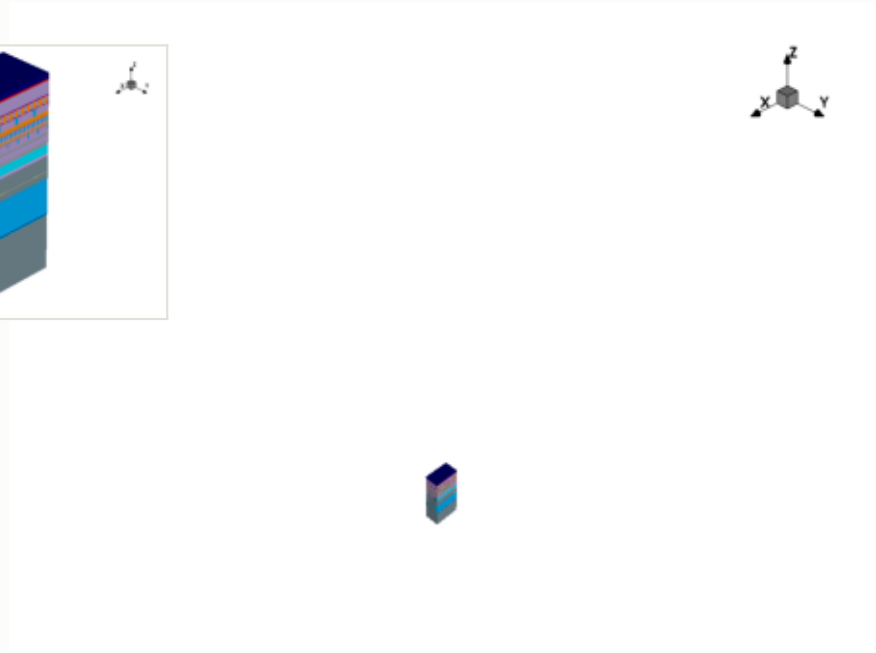
depo_fill · CVD

1.31	487.8	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



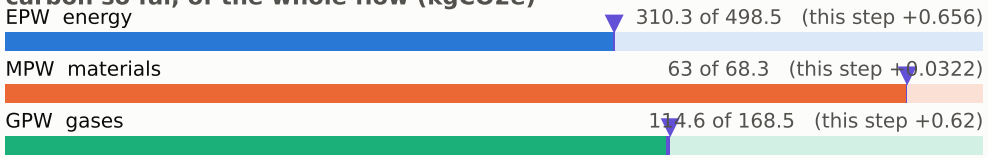


this step

process area	CVD
material	TiN
thickness	0.03 µm
material GWP	43.9778 gCO2e/g
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

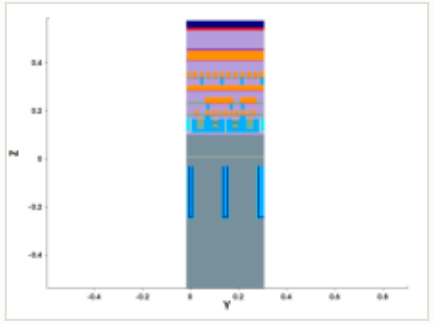
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0322 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

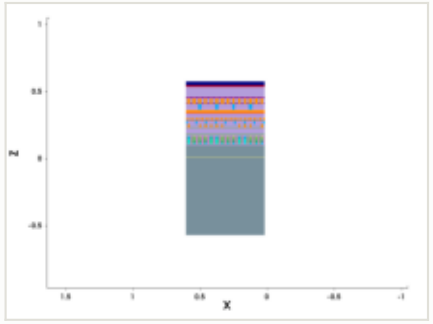


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



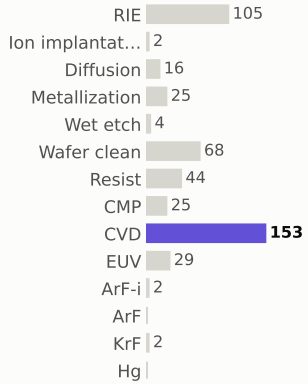
y cut at 0.184 µm



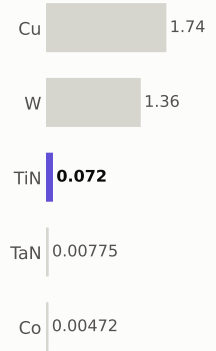
mask for this step

no mask for this step

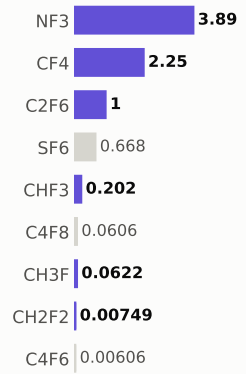
steps by process area



materials deposited (g)



gases emitted (g)



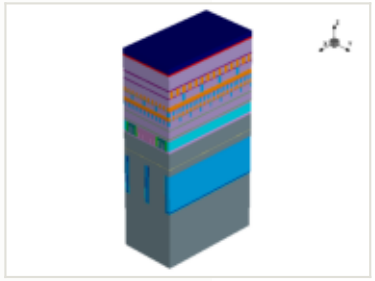
Step #5000: Wafer_clean

clean · Wafer_clean

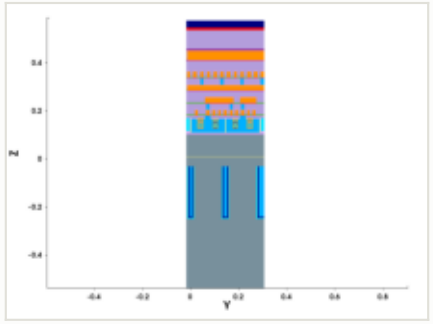
0.169	488.0	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

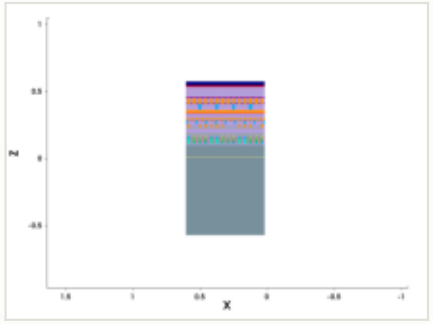
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

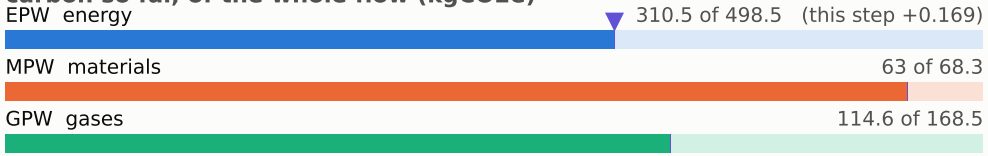
no mask for this step

this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

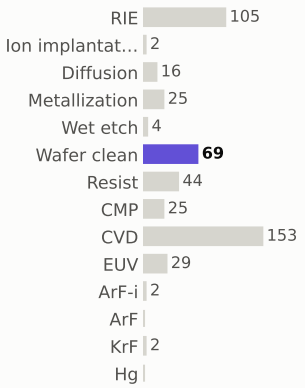
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

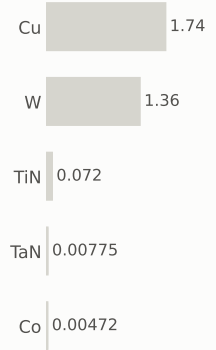


embodied carbon up to each step (kgCO2e per wafer)

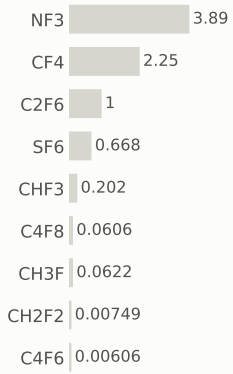
steps by process area



materials deposited (g)



gases emitted (g)



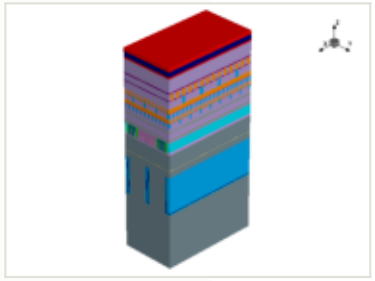
Step #5010: CVD

depo_fill · CVD

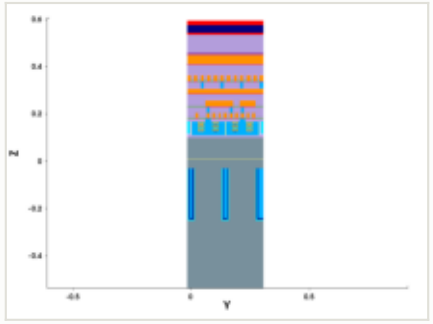
1.31	489.3	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

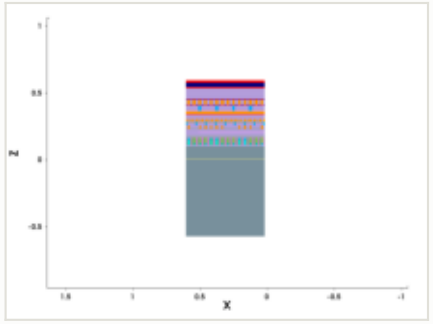
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

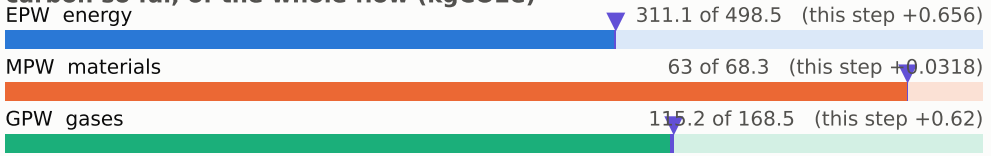
no mask for this step

this step

process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

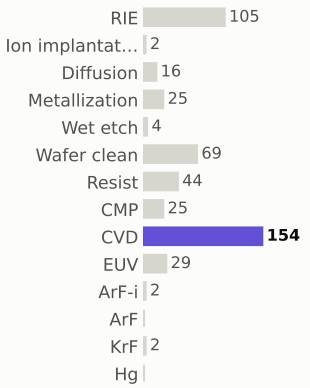
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

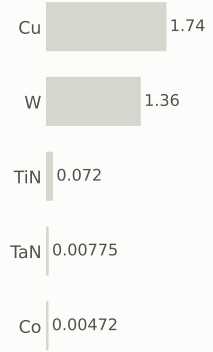


embodied carbon up to each step (kgCO2e per wafer)

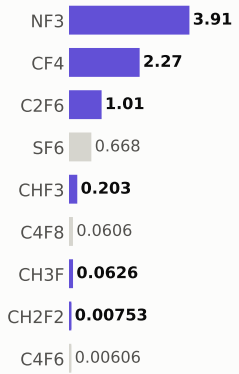
steps by process area



materials deposited (g)



gases emitted (g)



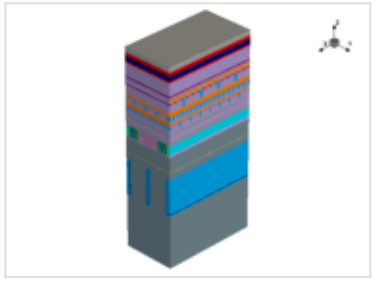
Step #5020: CVD

depo_fill · CVD

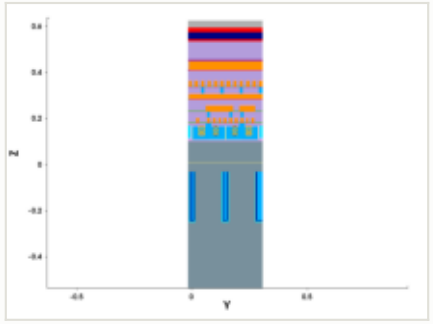
1.31	490.6	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

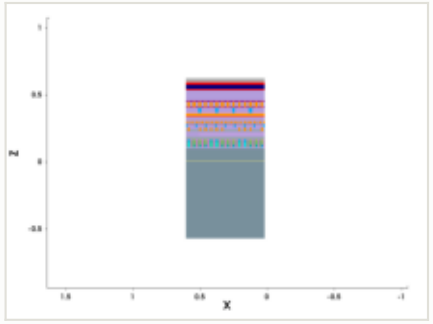
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

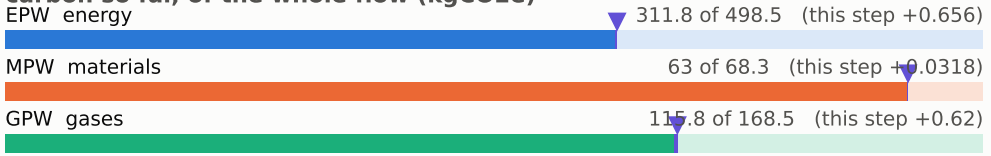
no mask for this step

this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

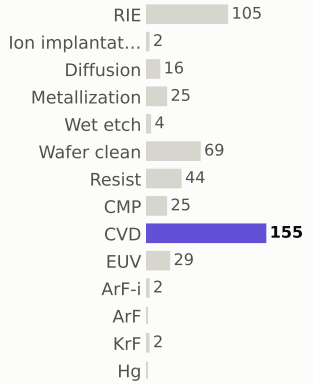
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

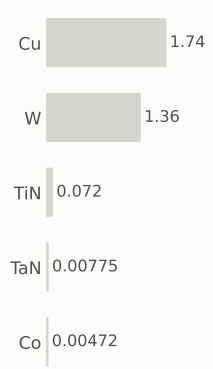


embodied carbon up to each step (kgCO2e per wafer)

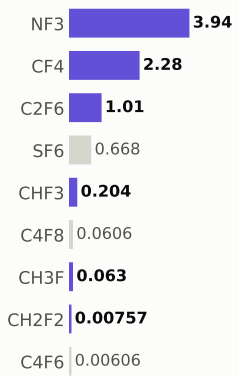
steps by process area



materials deposited (g)



gases emitted (g)



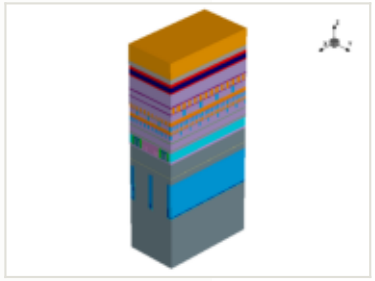
Step #5030: Resist

depo_fill · Resist

0.169	490.8	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

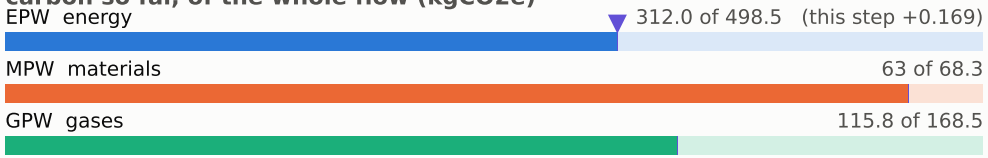


this step

process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

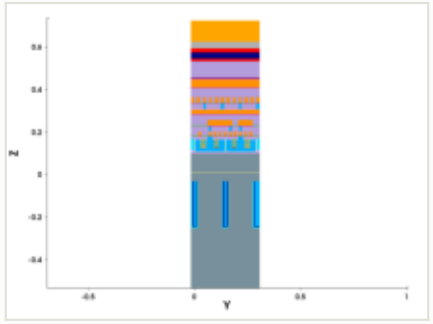
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

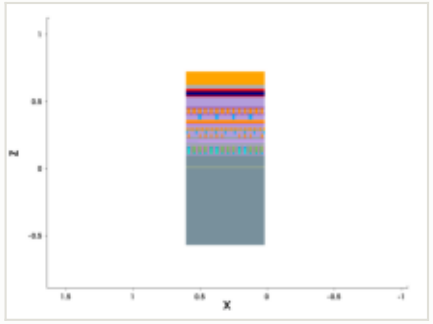


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



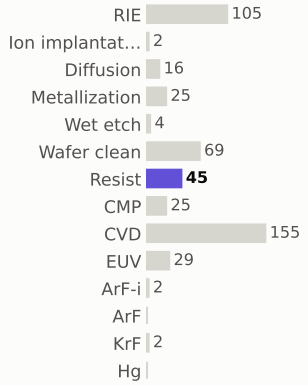
y cut at 0.184 µm



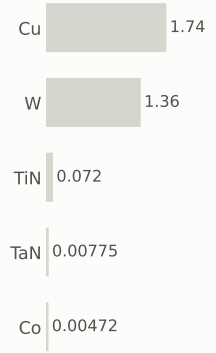
mask for this step

no mask for this step

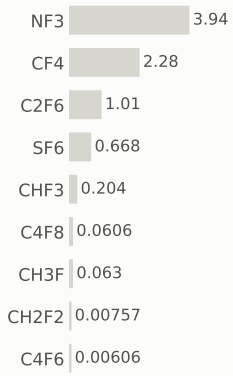
steps by process area



materials deposited (g)



gases emitted (g)



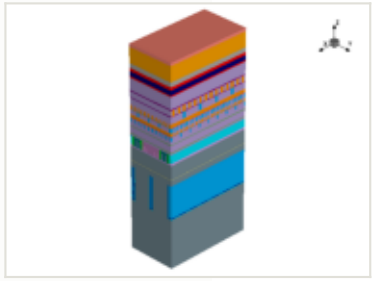
Step #5040: CVD

depo_fill · CVD

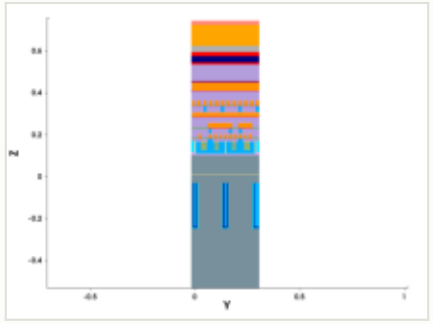
1.31	492.1	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

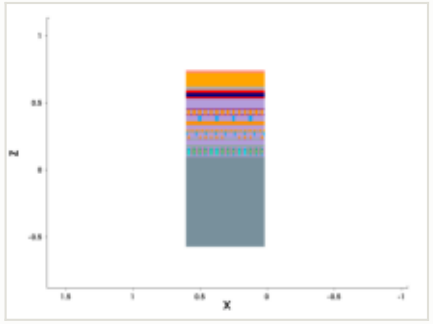
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

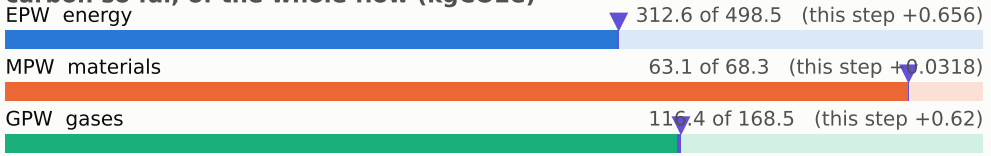
no mask for this step

this step

process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

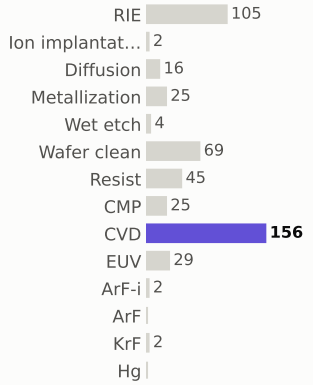
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

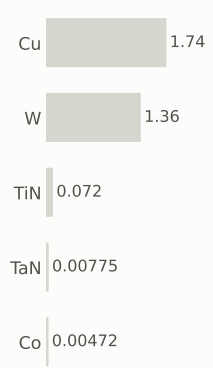


embodied carbon up to each step (kgCO2e per wafer)

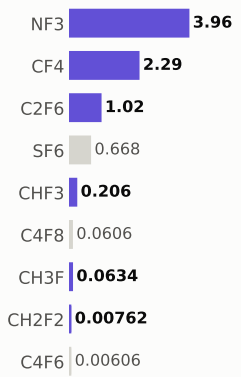
steps by process area



materials deposited (g)



gases emitted (g)



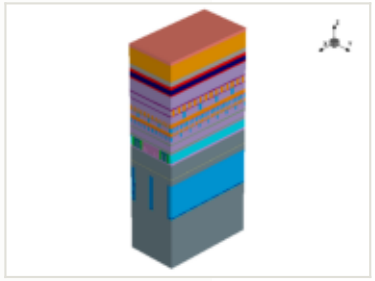
Step #5050: Wafer_clean

clean · Wafer_clean

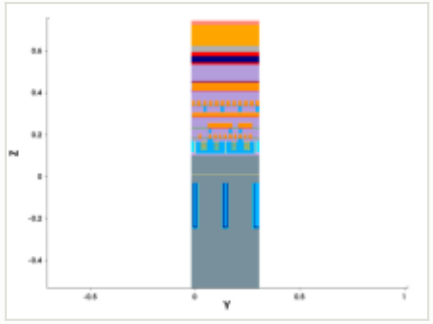
0.169	492.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

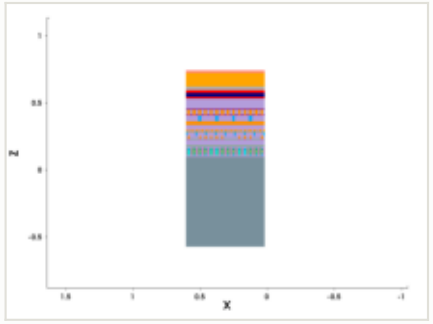
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

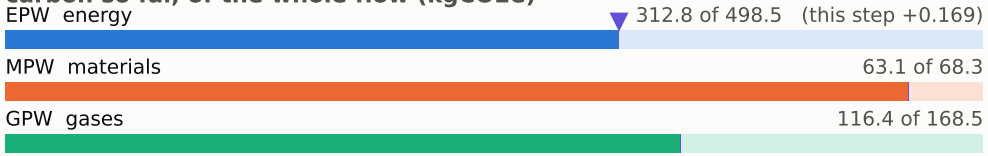
no mask for this step

this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

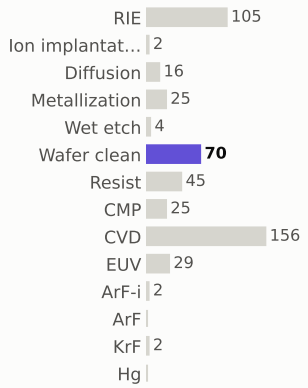
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

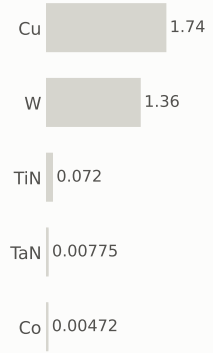


embodied carbon up to each step (kgCO2e per wafer)

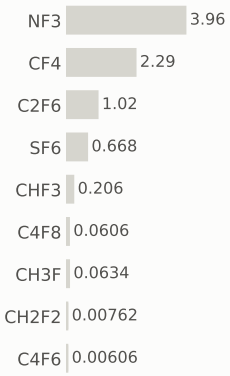
steps by process area



materials deposited (g)



gases emitted (g)



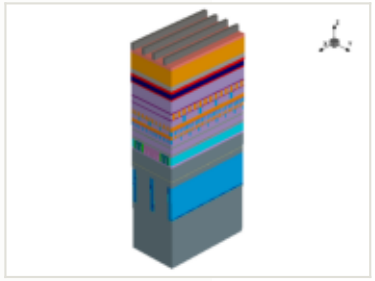
Step #5060: ArF-i

litho_patterning · ArF-i

0.217	492.5	0.029%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

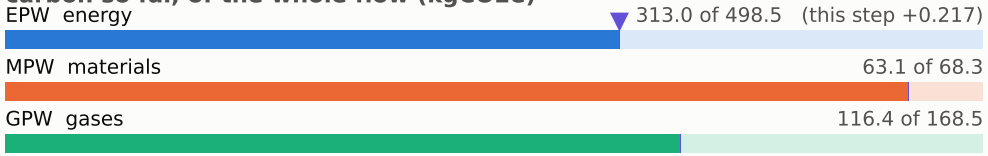


this step

process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

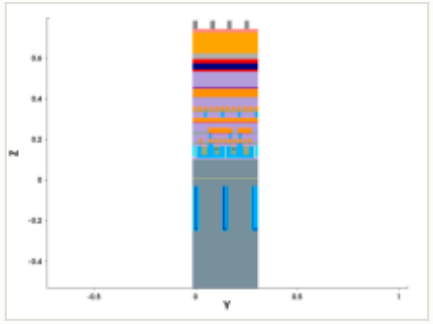
EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

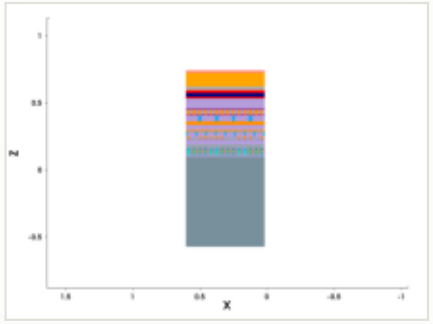


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm

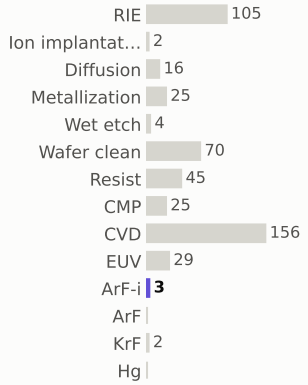


y cut at 0.184 µm

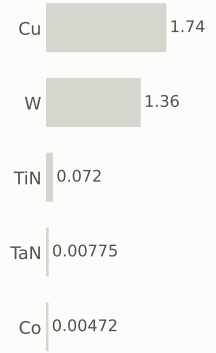


mask: layer 10451:0, positive (GDSII)

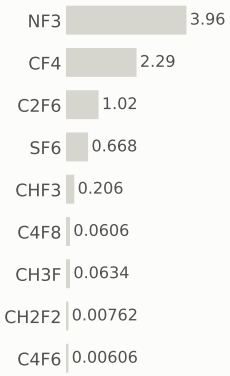
steps by process area



materials deposited (g)



gases emitted (g)



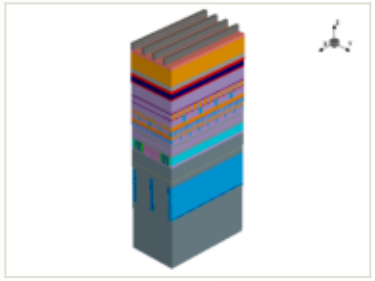
Step #5070: Ion_implantation

treatment · Ion_implantation

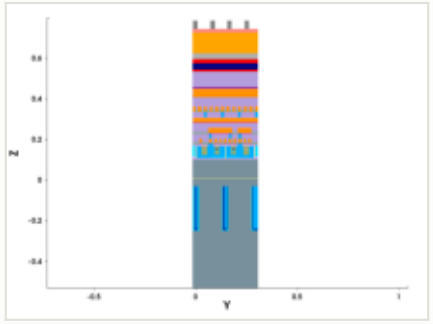
0.801	493.3	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

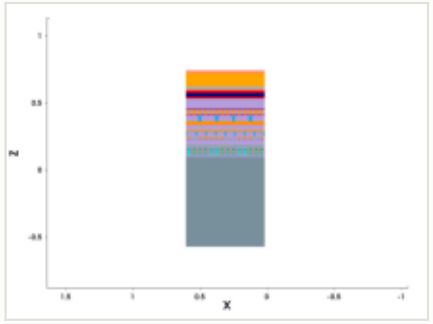
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

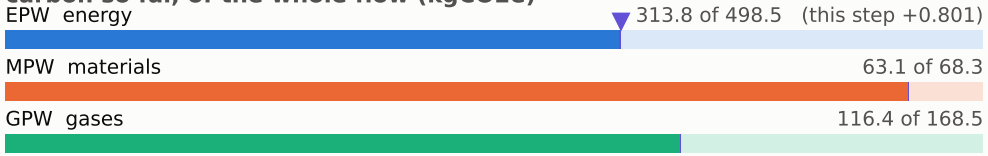
no mask for this step

this step

process area	Ion_implantation	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

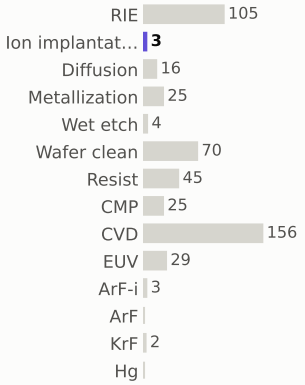
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

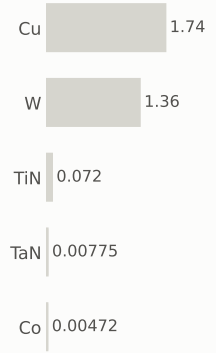


embodied carbon up to each step (kgCO2e per wafer)

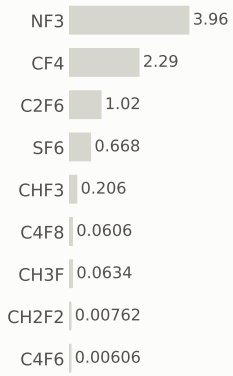
steps by process area



materials deposited (g)



gases emitted (g)



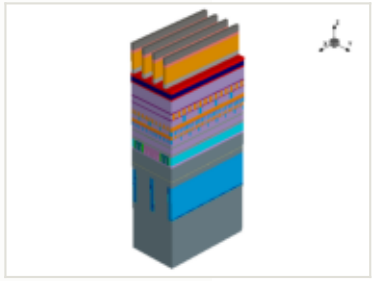
Step #5080: RIE

etch_anisotropic · RIE

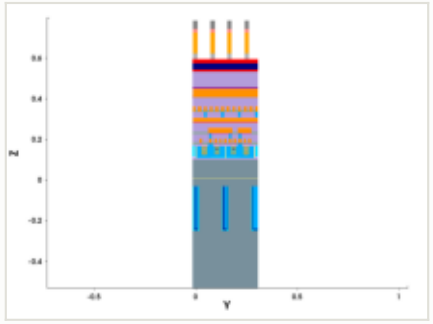
1.04	494.3	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

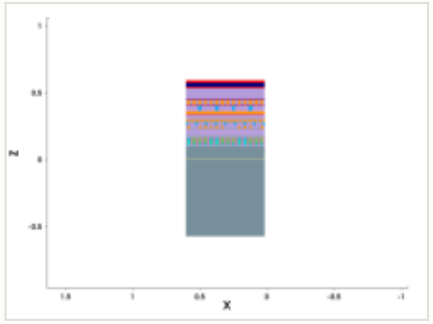
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

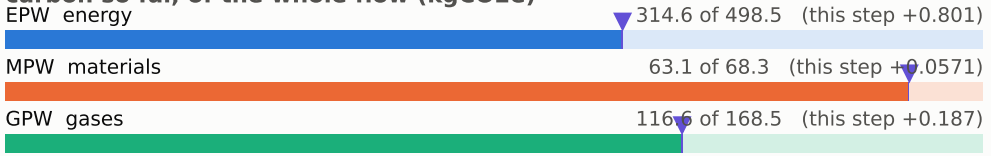
no mask for this step

this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

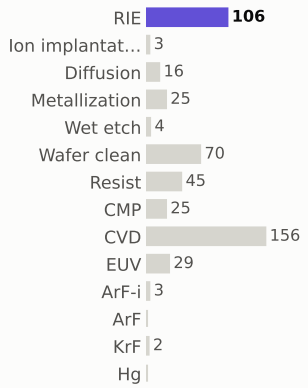
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

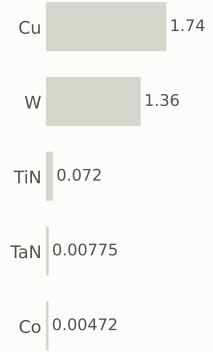


embodied carbon up to each step (kgCO2e per wafer)

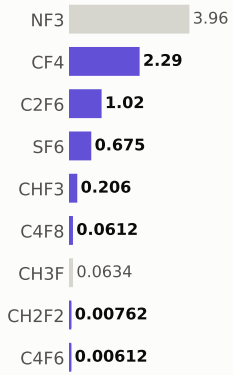
steps by process area



materials deposited (g)



gases emitted (g)



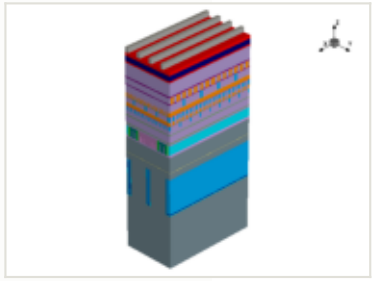
Step #5090: Wafer_clean

etch_strip · Wafer_clean

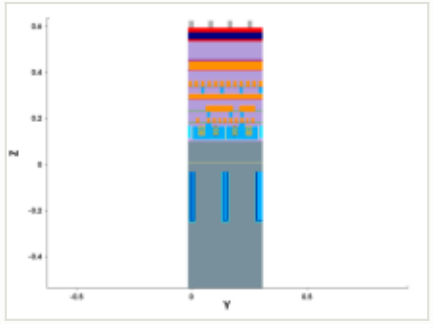
0.169	494.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

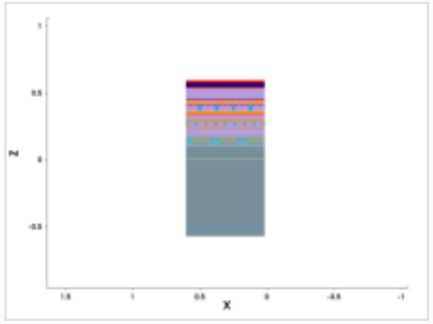
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

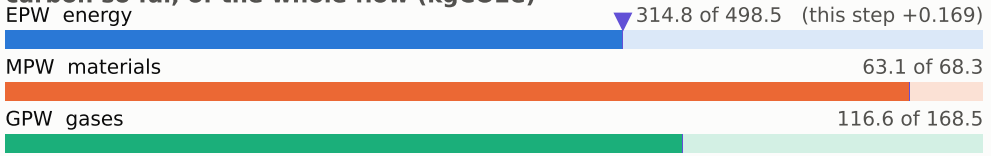
no mask for this step

this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

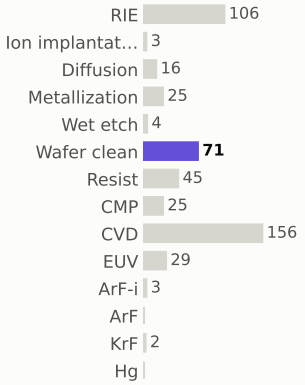
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

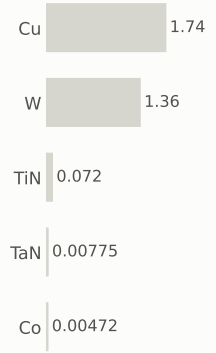


embodied carbon up to each step (kgCO2e per wafer)

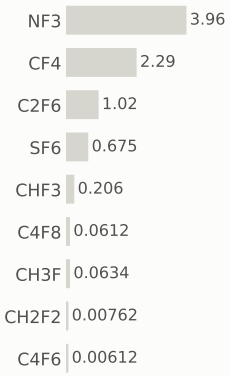
steps by process area



materials deposited (g)



gases emitted (g)



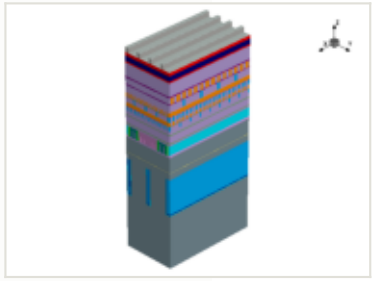
Step #5100: CVD

depo_isotropic · CVD

1.31	495.8	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

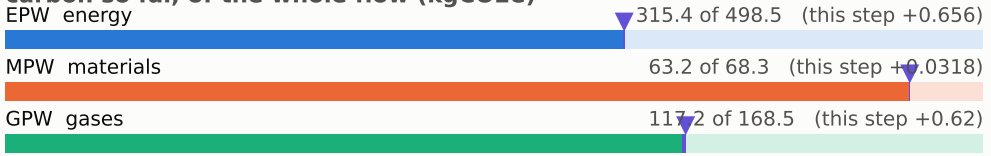


this step

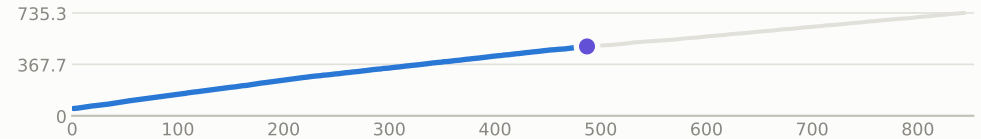
process area	CVD
material	—
thickness	0.001 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

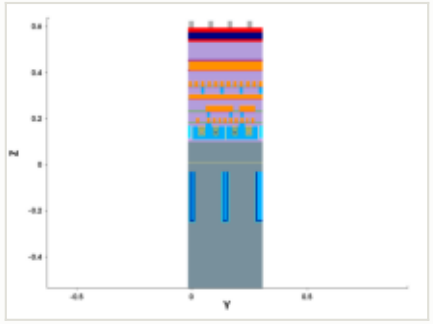
carbon so far, of the whole flow (kgCO2e)



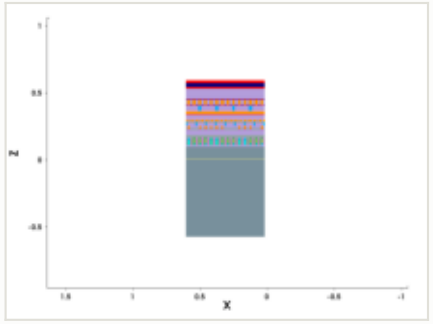
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



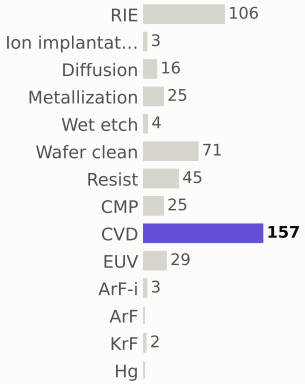
y cut at 0.184 µm



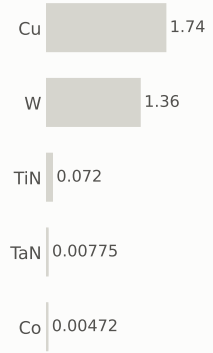
mask for this step

no mask for this step

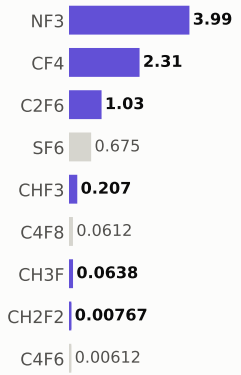
steps by process area



materials deposited (g)



gases emitted (g)



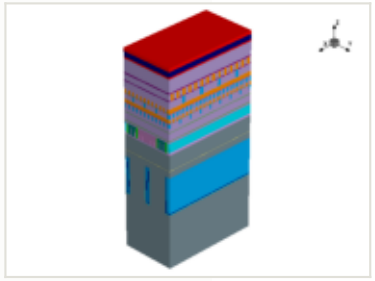
Step #5110: RIE

etch_anisotropic · RIE

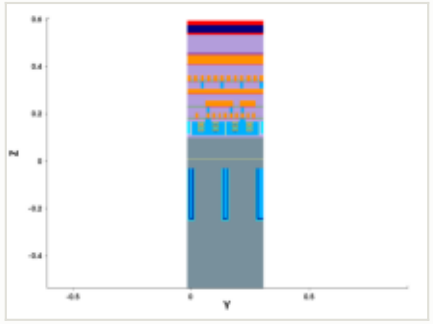
1.04	496.9	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

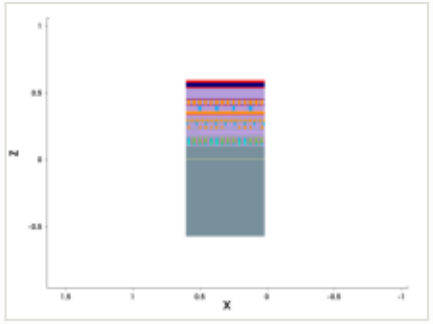
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

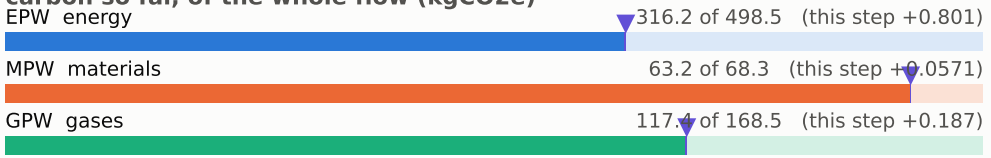
no mask for this step

this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

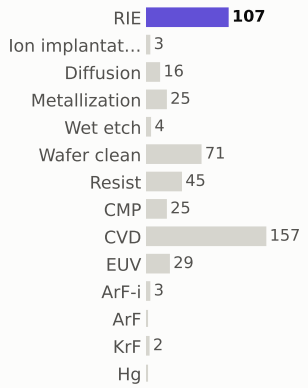
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

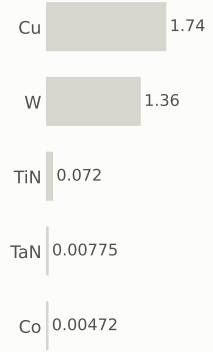


embodied carbon up to each step (kgCO2e per wafer)

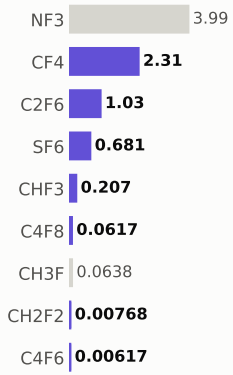
steps by process area



materials deposited (g)



gases emitted (g)



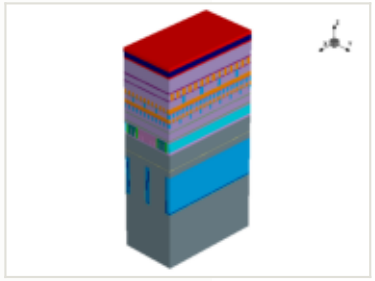
Step #5120: Wafer_clean

etch_strip · Wafer_clean

0.169	497.0	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

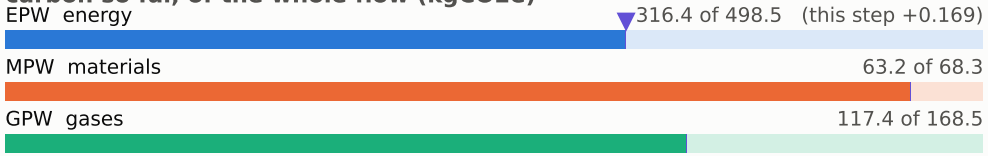


this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

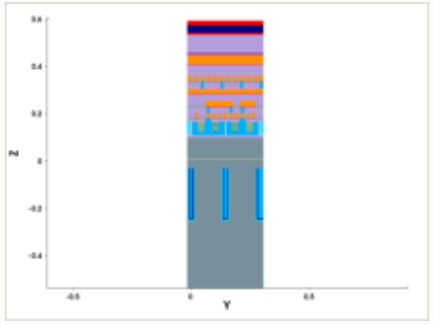
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

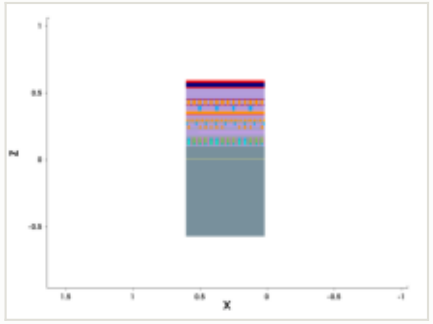


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



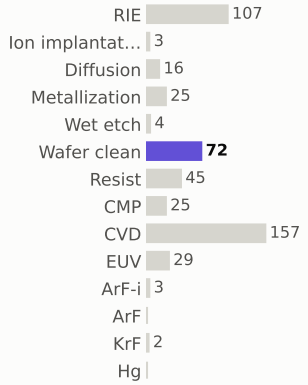
y cut at 0.184 µm



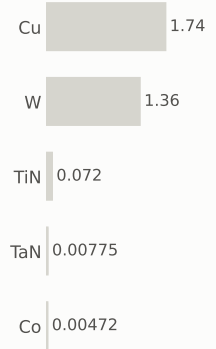
mask for this step

no mask for this step

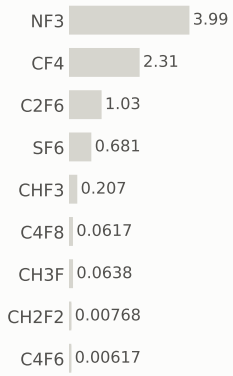
steps by process area



materials deposited (g)



gases emitted (g)



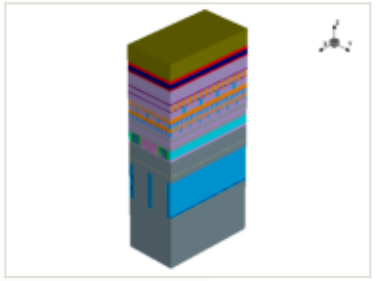
Step #5130: Resist

depo_fill · Resist

0.169	497.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

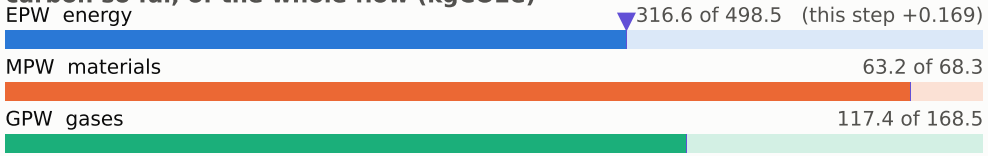


this step

process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

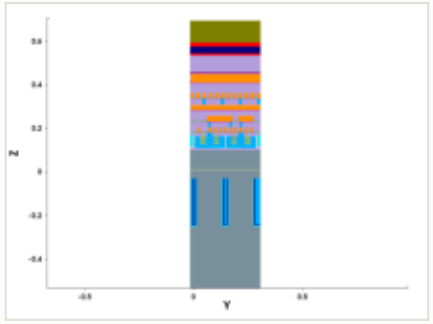
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

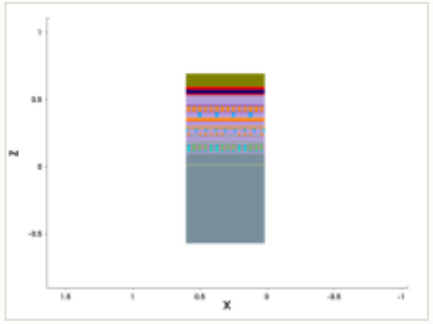


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



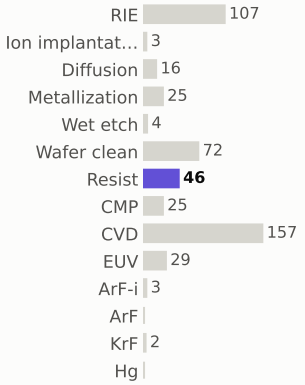
y cut at 0.184 µm



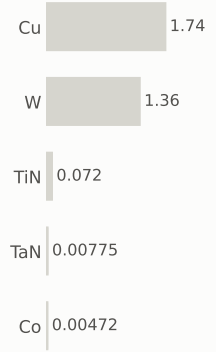
mask for this step

no mask for this step

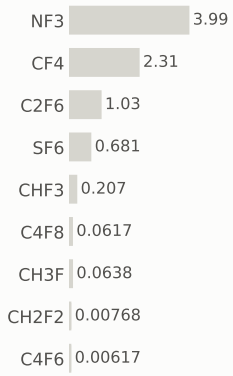
steps by process area



materials deposited (g)



gases emitted (g)



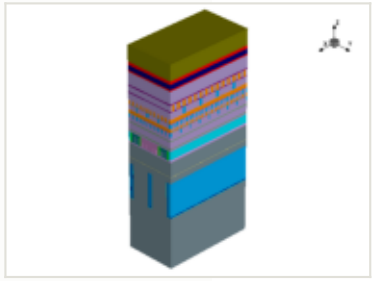
Step #5140: Wafer_clean

clean · Wafer_clean

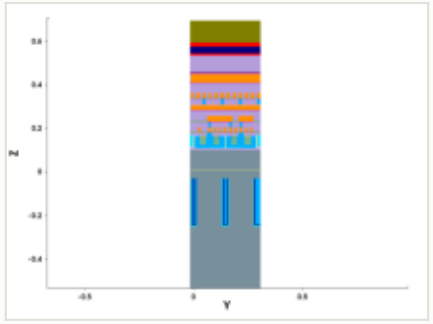
0.169	497.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

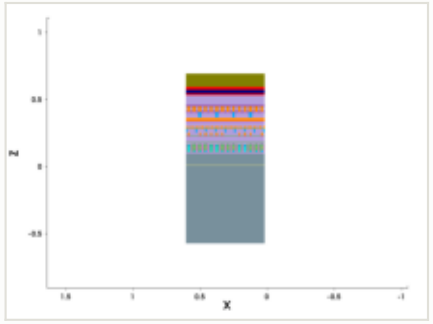
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

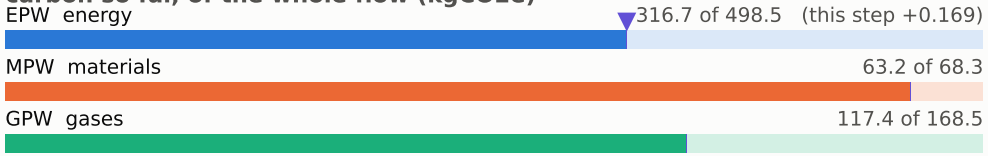
no mask for this step

this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

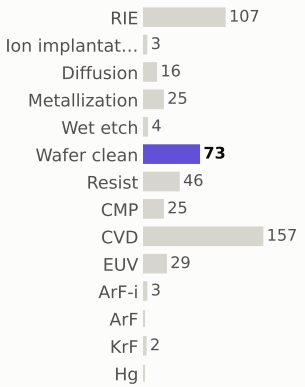
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

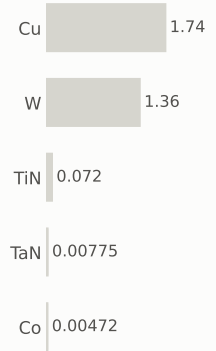


embodied carbon up to each step (kgCO2e per wafer)

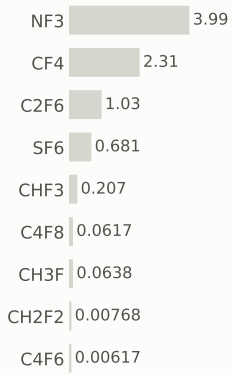
steps by process area



materials deposited (g)



gases emitted (g)



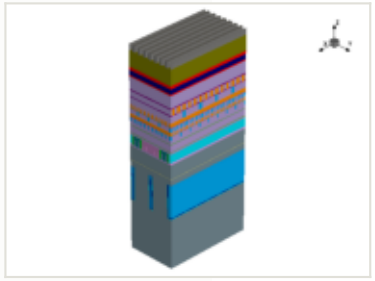
Step #5150: ArF-i

litho_patterning · ArF-i

0.217	497.6	0.029%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

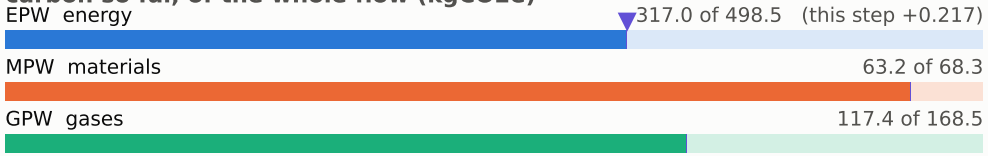


this step

process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

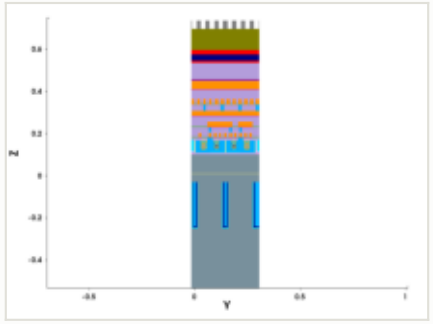
EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

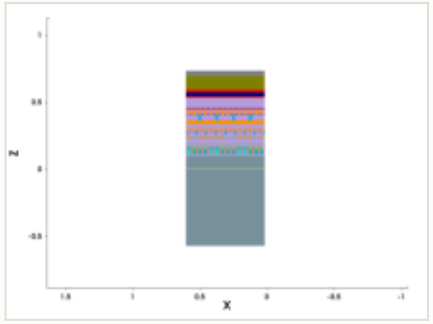


embodied carbon up to each step (kgCO2e per wafer)

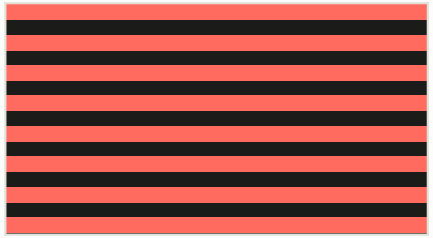
x cut at 0.294 µm



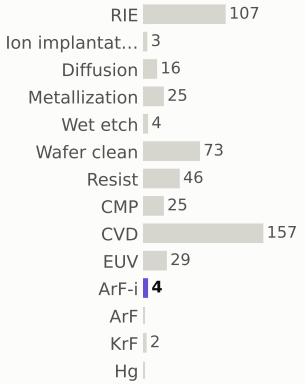
y cut at 0.184 µm



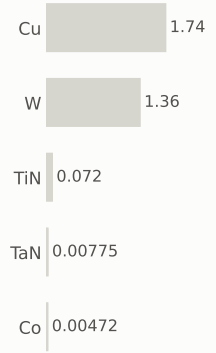
mask: layer 45:0 (GDSII)



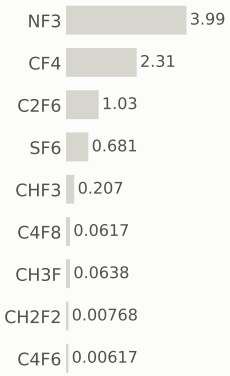
steps by process area



materials deposited (g)



gases emitted (g)



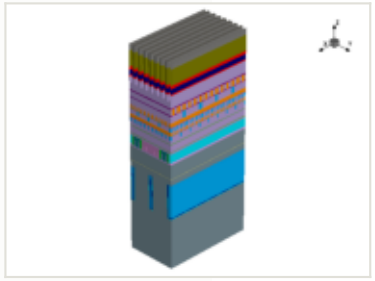
Step #5160: RIE

etch_anisotropic · RIE

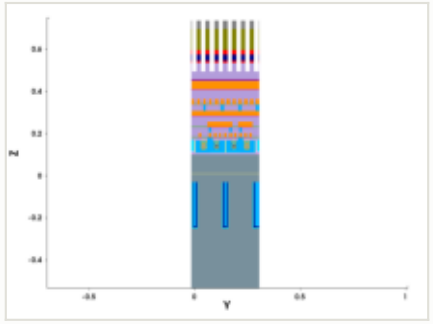
1.04	498.6	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

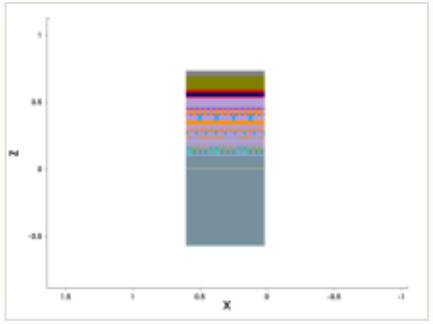
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

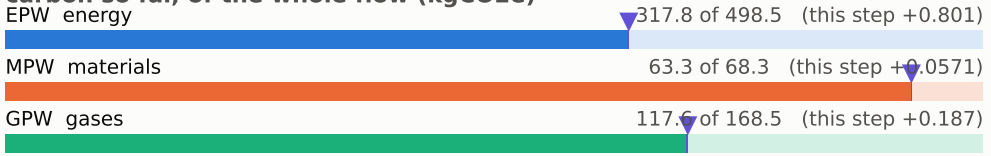
no mask for this step

this step

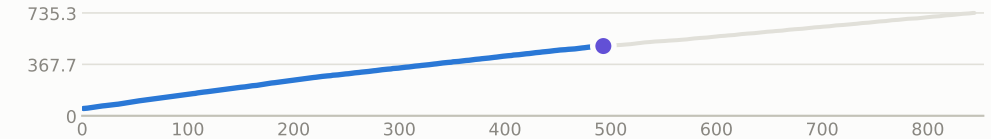
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

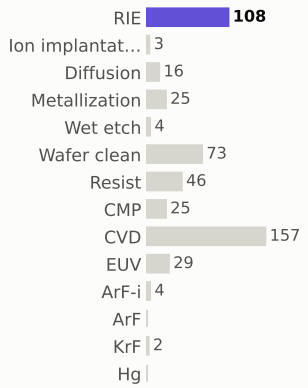
carbon so far, of the whole flow (kgCO2e)



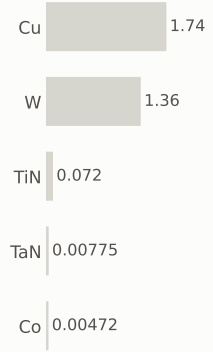
embodied carbon up to each step (kgCO2e per wafer)



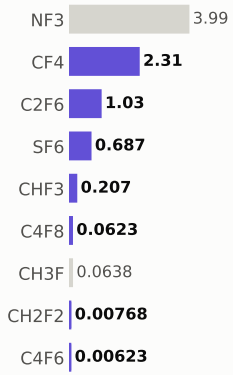
steps by process area



materials deposited (g)



gases emitted (g)



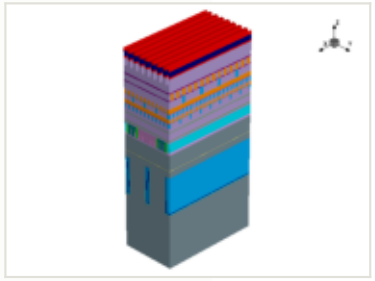
Step #5170: Wafer_clean

etch_strip · Wafer_clean

0.169	498.8	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

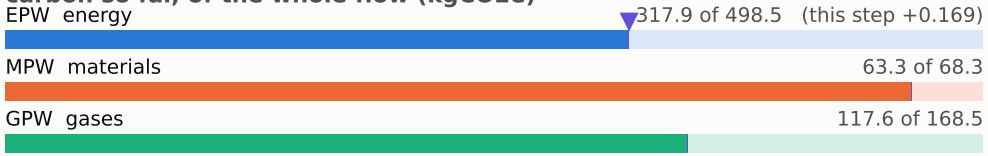


this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

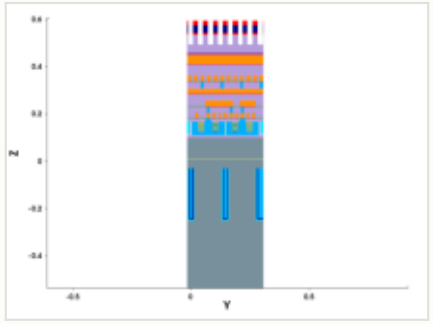
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

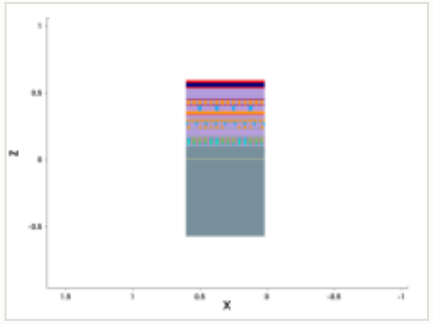


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



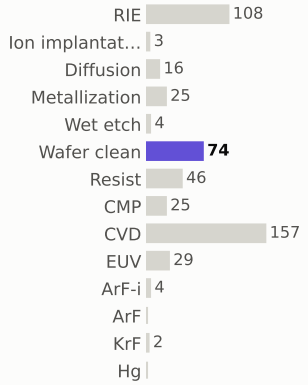
y cut at 0.184 µm



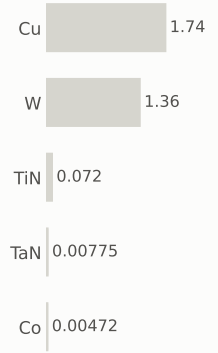
mask for this step

no mask for this step

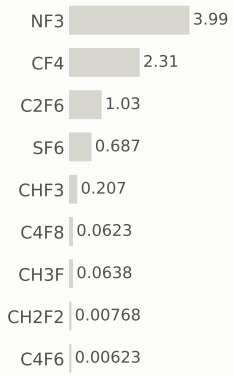
steps by process area



materials deposited (g)



gases emitted (g)



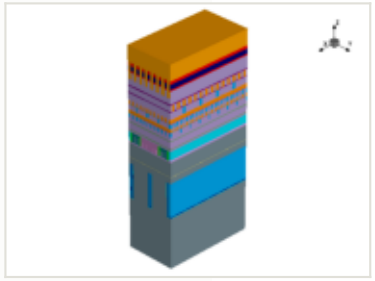
Step #5180: Resist

depo_fill · Resist

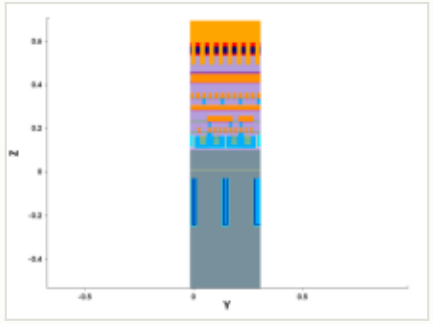
0.169	499.0	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

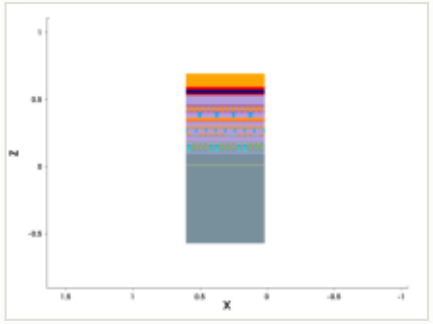
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

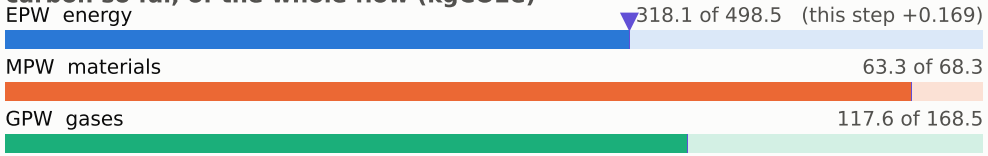
no mask for this step

this step

process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

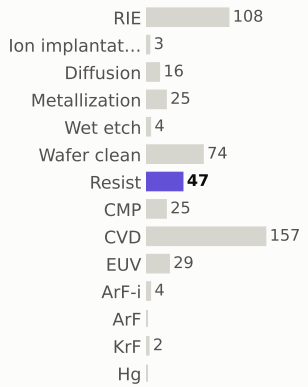
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

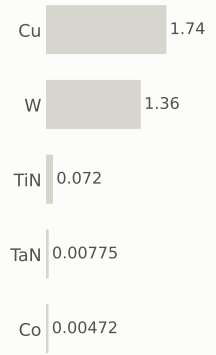


embodied carbon up to each step (kgCO2e per wafer)

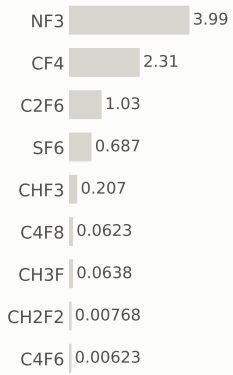
steps by process area



materials deposited (g)



gases emitted (g)



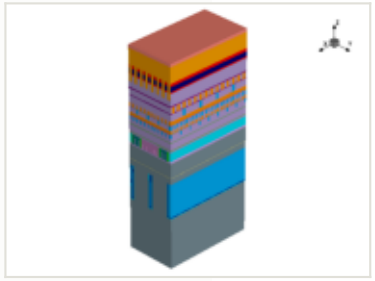
Step #5190: CVD

depo_fill · CVD

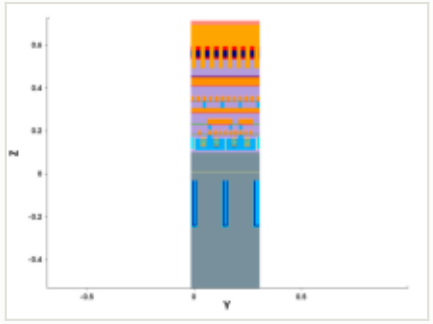
1.31	500.3	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

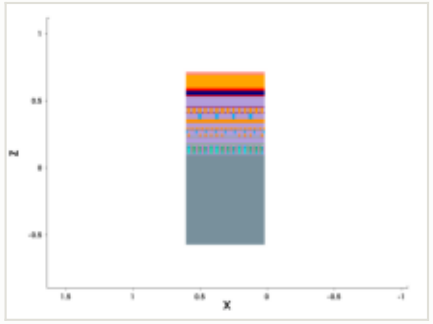
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

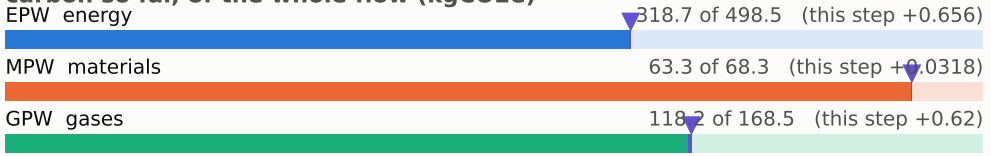
no mask for this step

this step

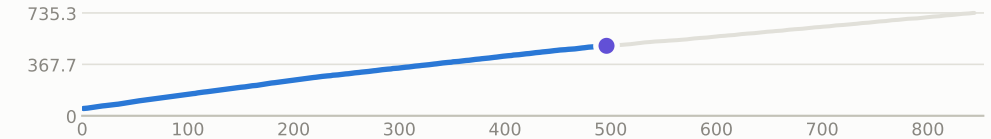
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

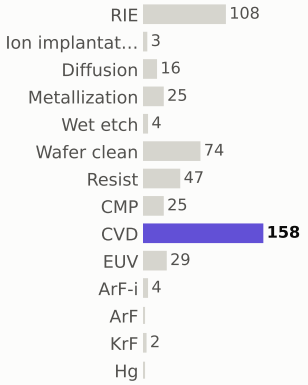
carbon so far, of the whole flow (kgCO2e)



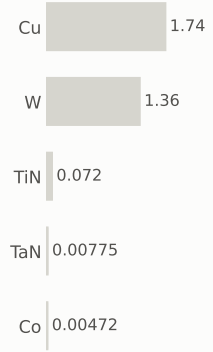
embodied carbon up to each step (kgCO2e per wafer)



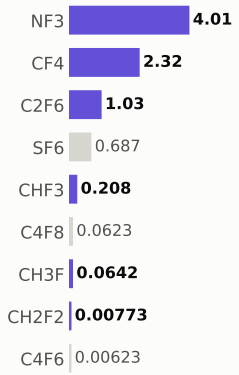
steps by process area



materials deposited (g)



gases emitted (g)



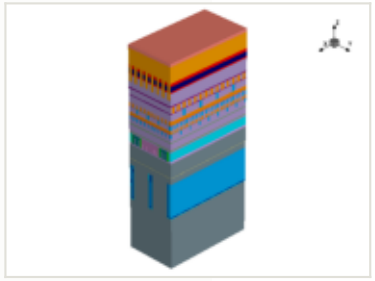
Step #5200: Wafer_clean

clean · Wafer_clean

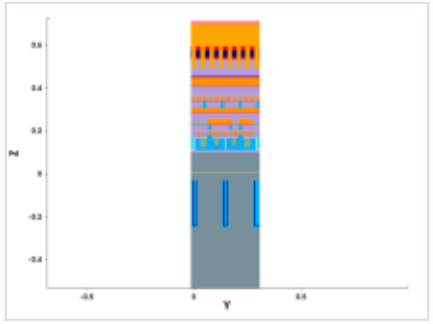
0.169	500.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

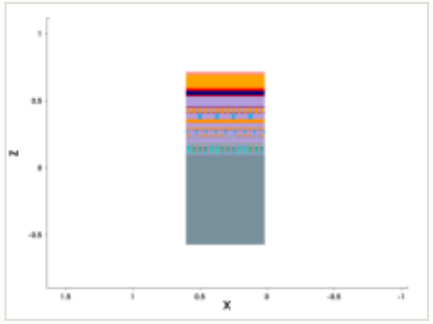
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

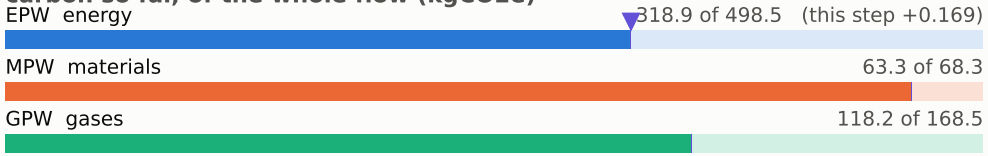
no mask for this step

this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

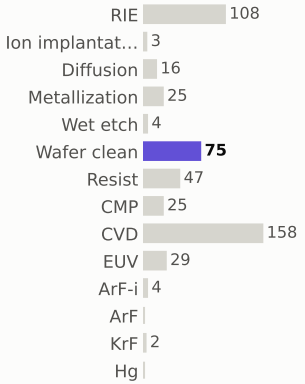
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

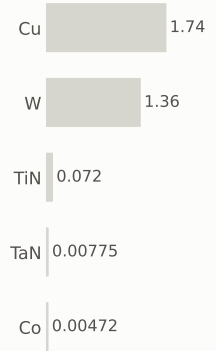


embodied carbon up to each step (kgCO2e per wafer)

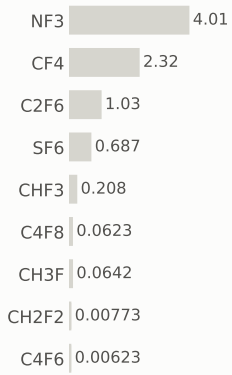
steps by process area



materials deposited (g)



gases emitted (g)



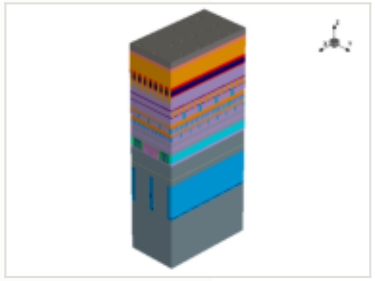
Step #5210: ArF-i

litho_patterning · ArF-i

0.217	500.7	0.029%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

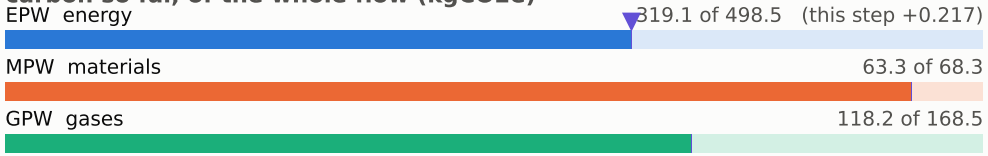


this step

process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

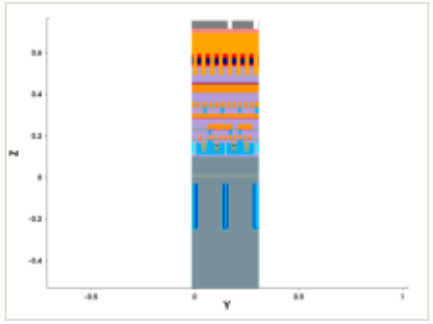
EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

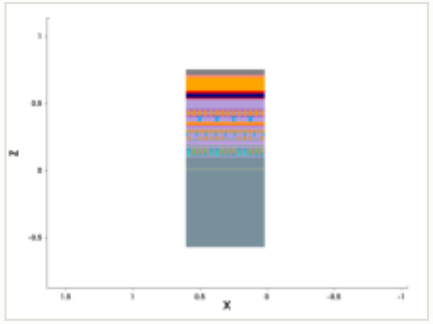


embodied carbon up to each step (kgCO2e per wafer)

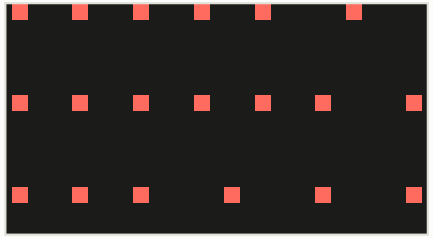
x cut at 0.294 µm



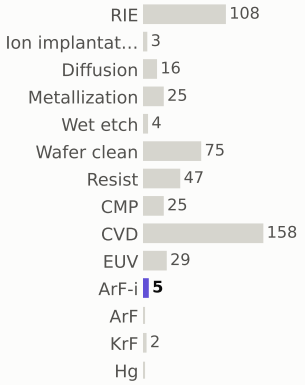
y cut at 0.184 µm



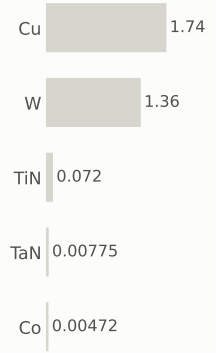
mask: layer 10421:0 (GDSII)



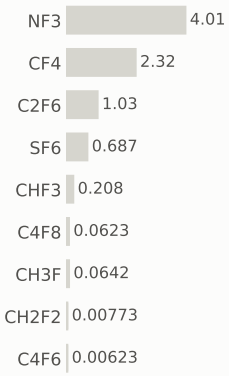
steps by process area



materials deposited (g)



gases emitted (g)



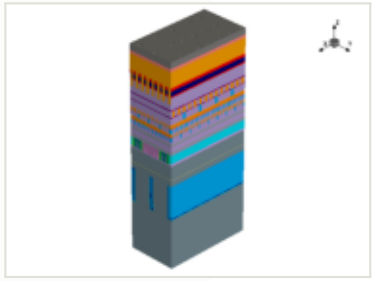
Step #5220: RIE

etch_anisotropic · RIE

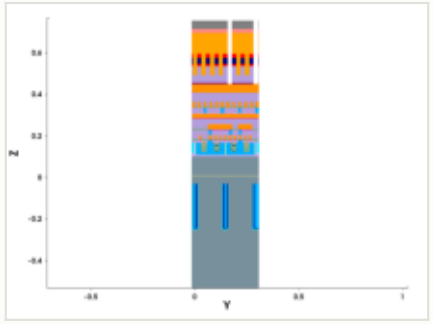
1.04	501.7	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

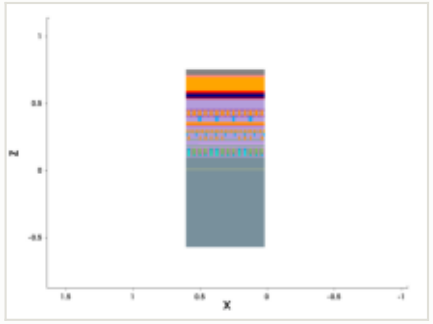
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

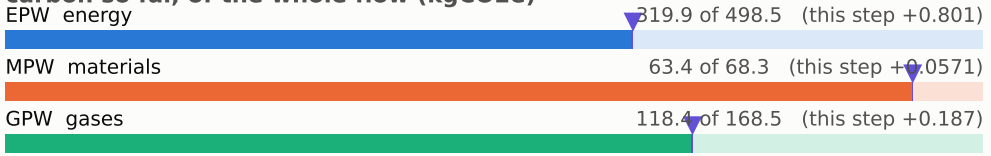
no mask for this step

this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

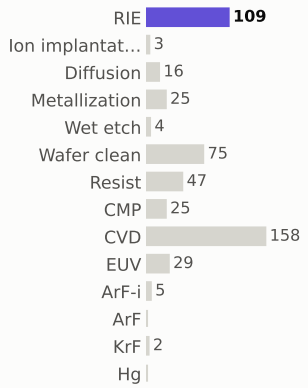
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

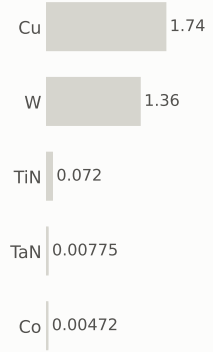


embodied carbon up to each step (kgCO2e per wafer)

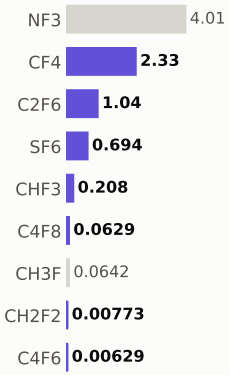
steps by process area



materials deposited (g)



gases emitted (g)



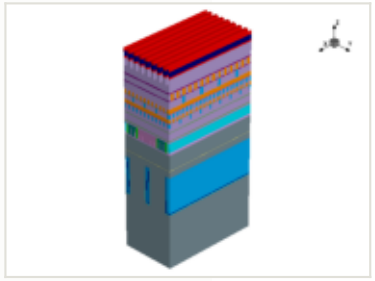
Step #5230: Wafer_clean

etch_strip · Wafer_clean

0.169	501.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

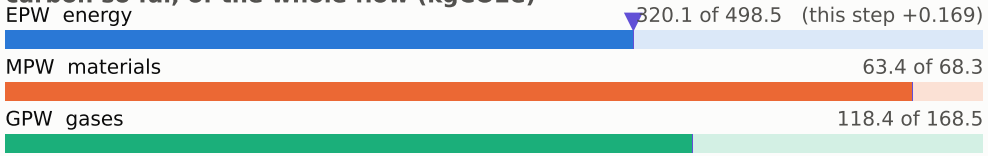


this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

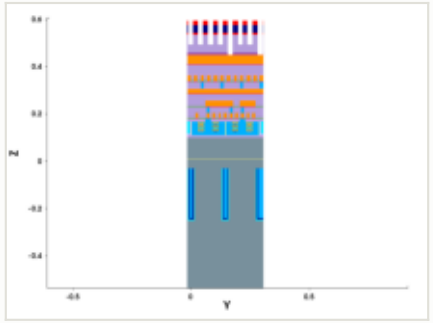
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

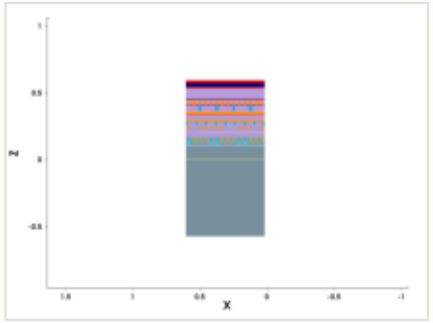


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



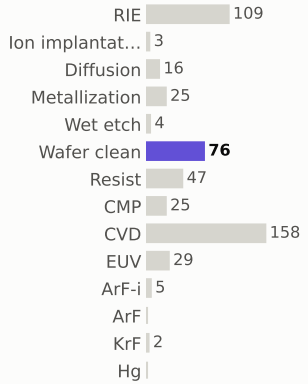
y cut at 0.184 µm



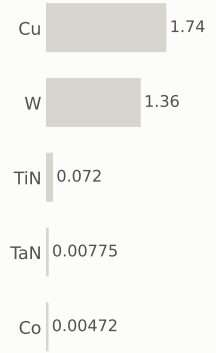
mask for this step

no mask for this step

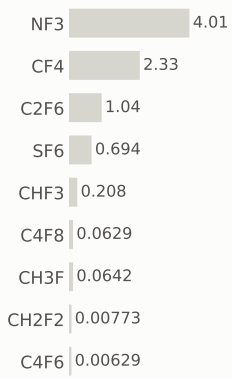
steps by process area



materials deposited (g)



gases emitted (g)



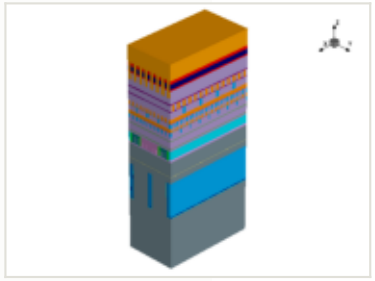
Step #5240: Resist

depo_fill · Resist

0.169	502.0	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

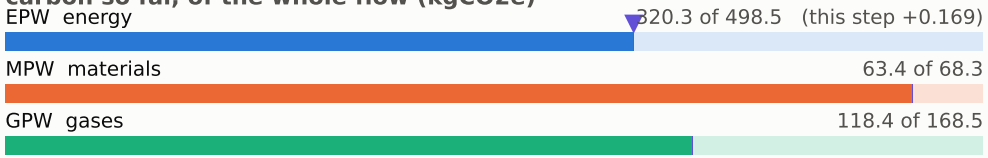


this step

process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

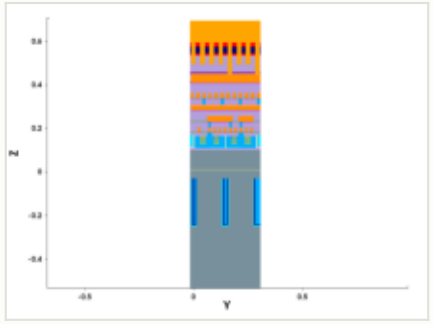
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

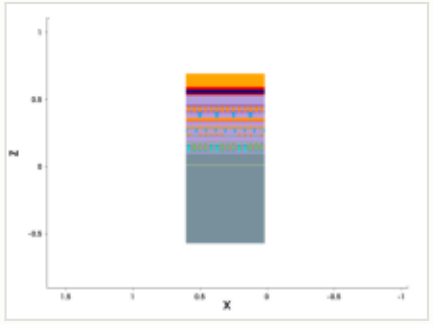


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



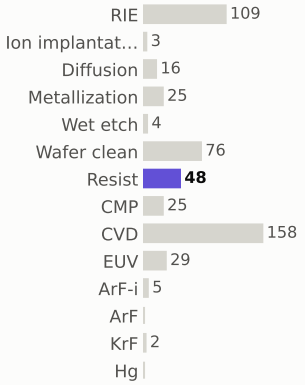
y cut at 0.184 µm



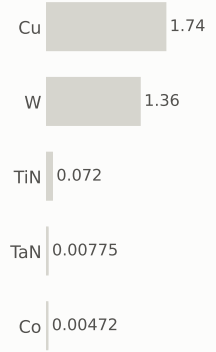
mask for this step

no mask for this step

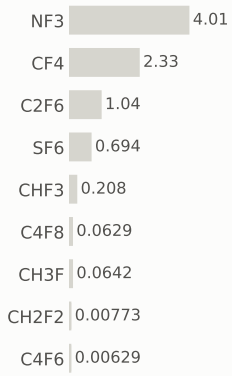
steps by process area



materials deposited (g)



gases emitted (g)



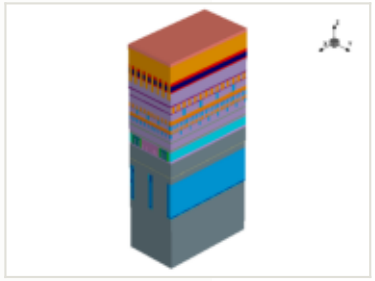
Step #5250: CVD

depo_fill · CVD

1.31	503.4	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

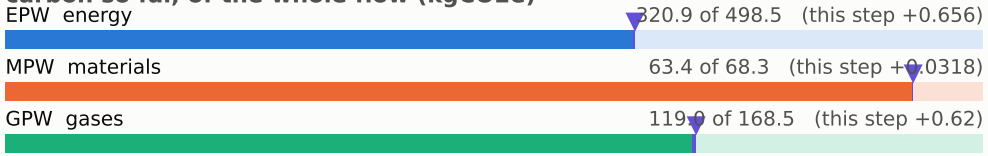


this step

process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

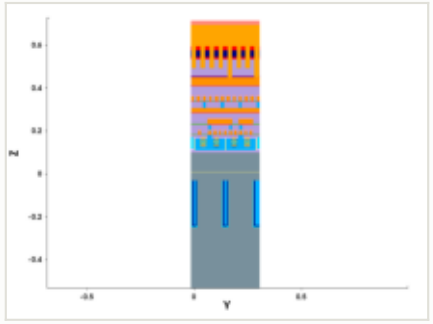
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

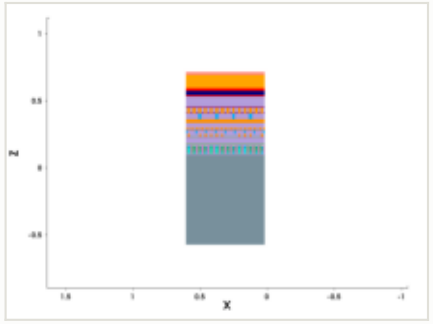


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



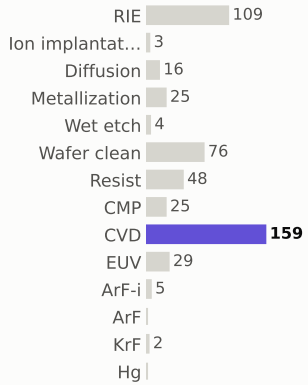
y cut at 0.184 µm



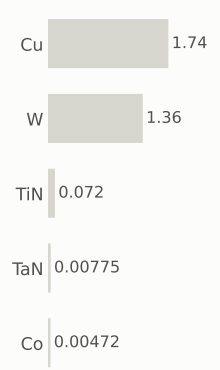
mask for this step

no mask for this step

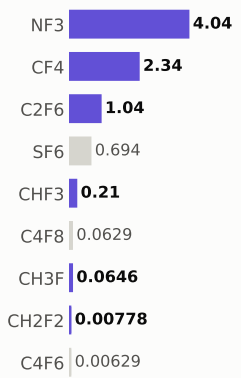
steps by process area



materials deposited (g)



gases emitted (g)



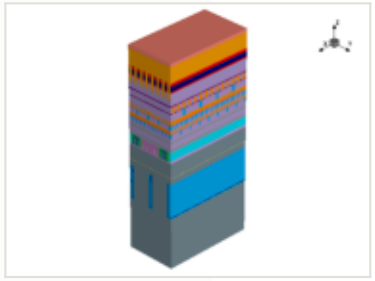
Step #5260: Wafer_clean

clean · Wafer_clean

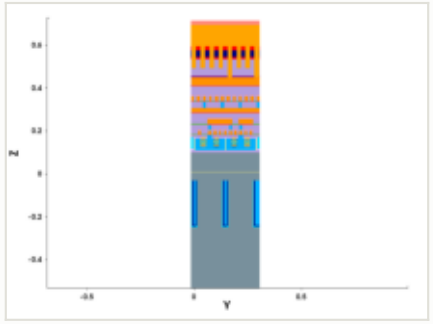
0.169	503.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

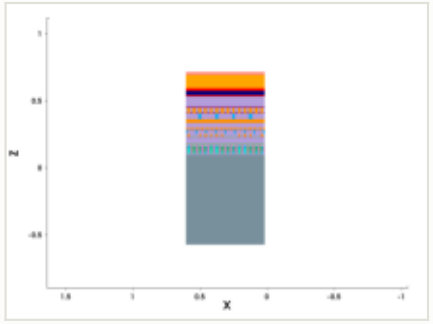
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

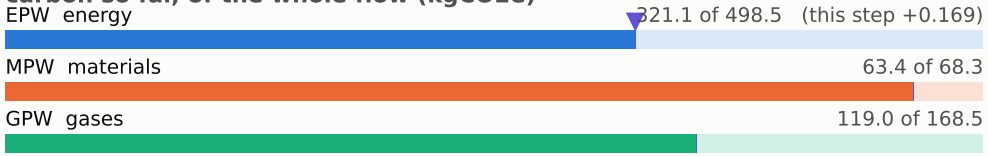
no mask for this step

this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

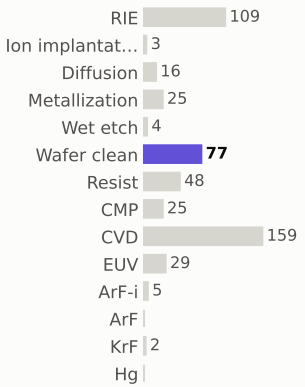
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

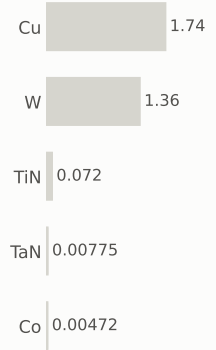


embodied carbon up to each step (kgCO2e per wafer)

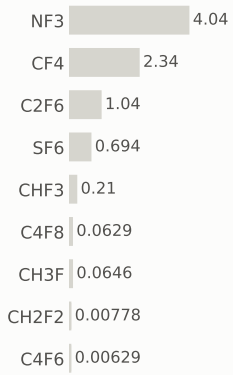
steps by process area



materials deposited (g)



gases emitted (g)



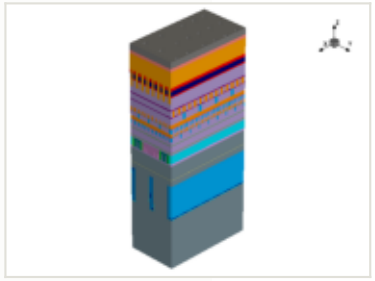
Step #5270: ArF-i

litho_patterning · ArF-i

0.217	503.7	0.029%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

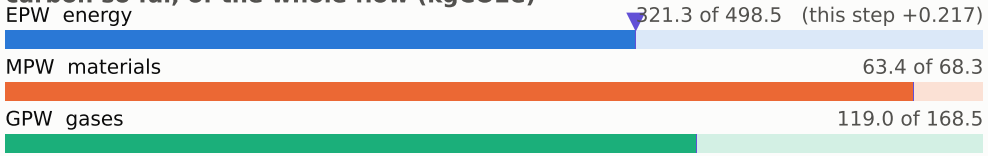


this step

process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

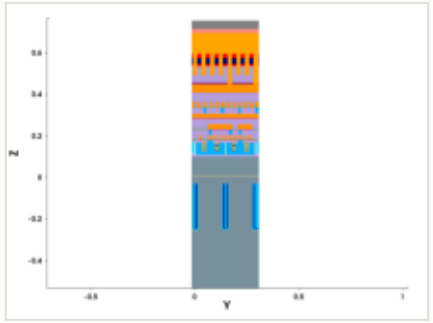
EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

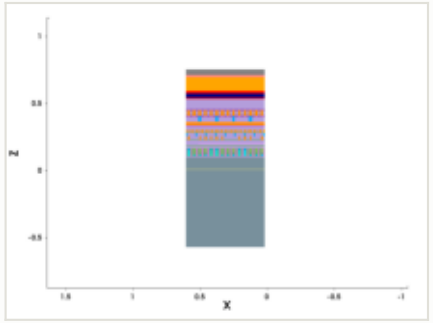


embodied carbon up to each step (kgCO2e per wafer)

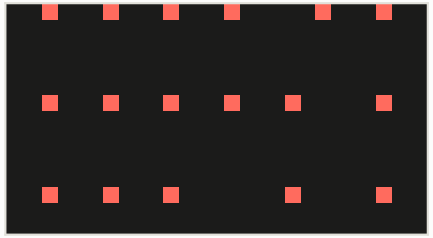
x cut at 0.294 µm



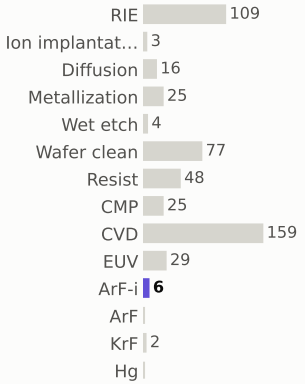
y cut at 0.184 µm



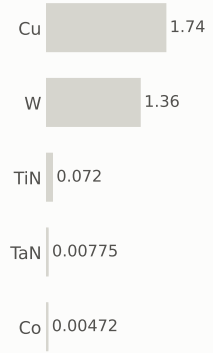
mask: layer 10422:0 (GDSII)



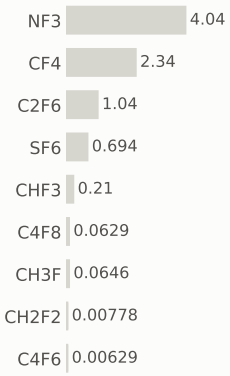
steps by process area



materials deposited (g)



gases emitted (g)



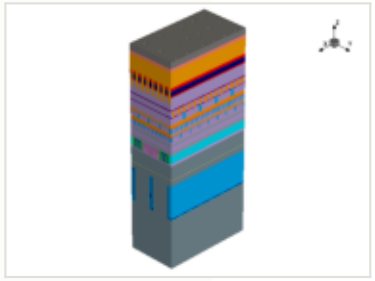
Step #5280: RIE

etch_anisotropic · RIE

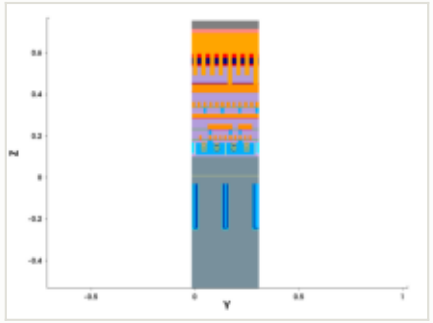
1.04	504.8	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

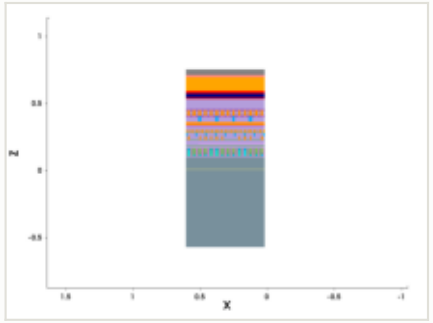
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

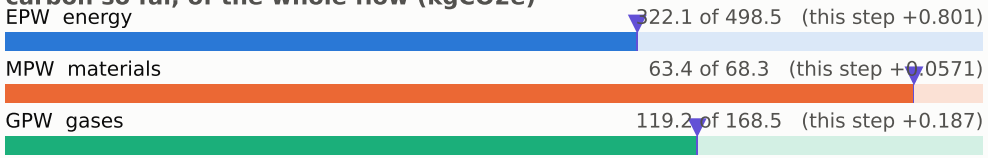
no mask for this step

this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

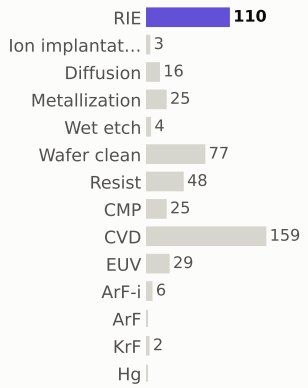
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

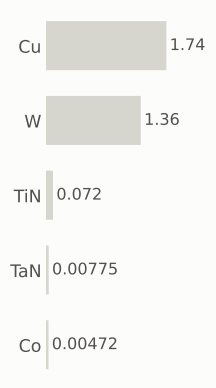


embodied carbon up to each step (kgCO2e per wafer)

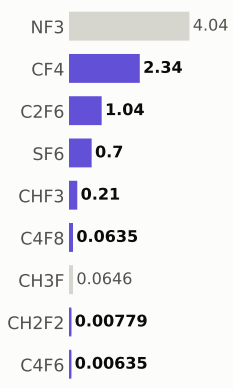
steps by process area



materials deposited (g)



gases emitted (g)



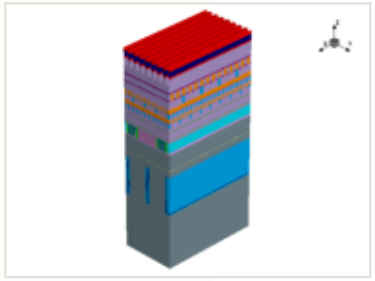
Step #5290: Wafer_clean

etch_strip · Wafer_clean

0.169	505.0	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

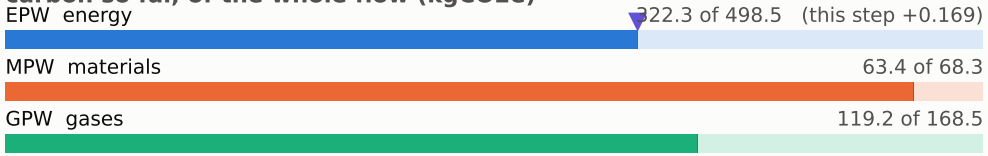


this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

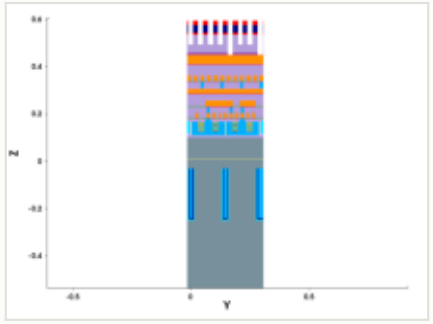
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

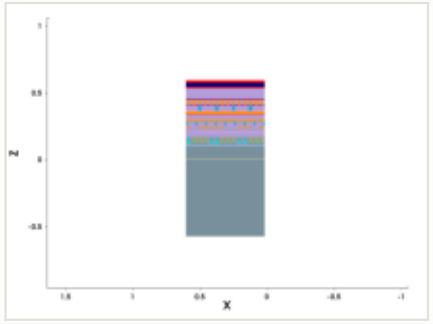


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



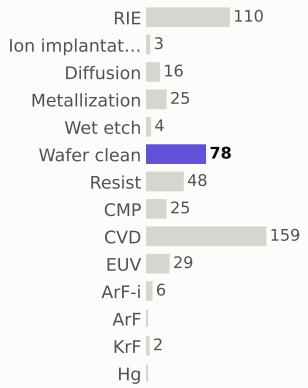
y cut at 0.184 µm



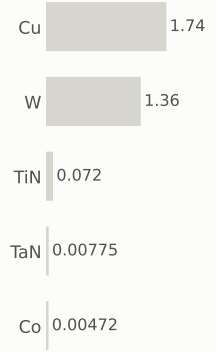
mask for this step

no mask for this step

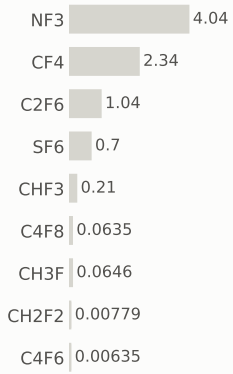
steps by process area



materials deposited (g)



gases emitted (g)



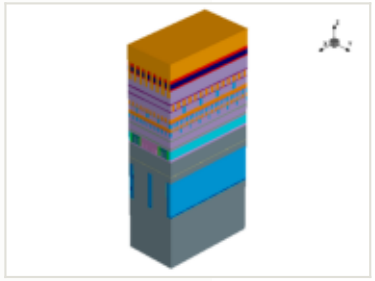
Step #5300: Resist

depo_fill · Resist

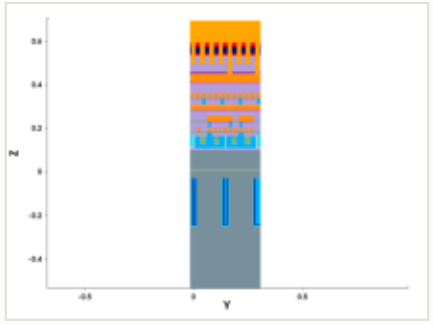
0.169	505.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

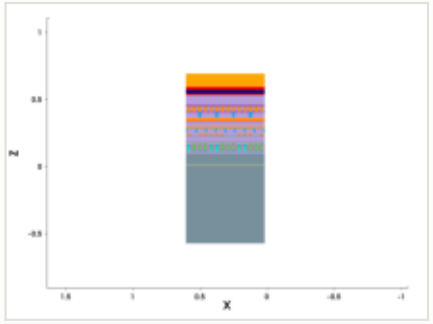
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

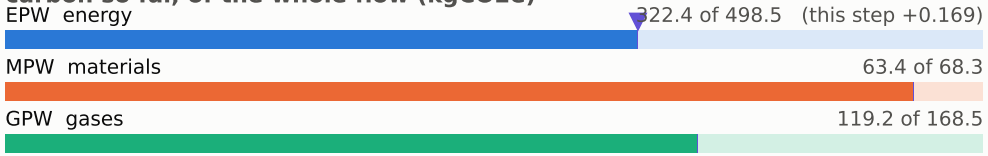
no mask for this step

this step

process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

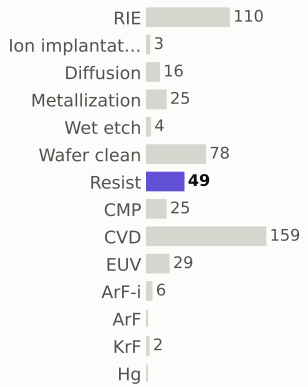
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

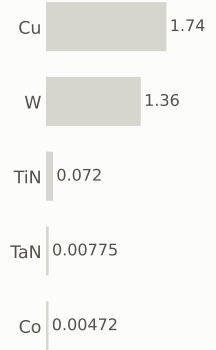


embodied carbon up to each step (kgCO2e per wafer)

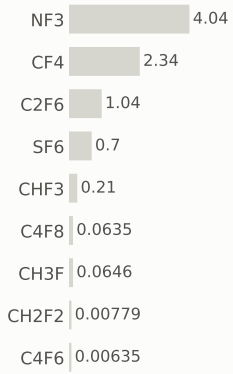
steps by process area



materials deposited (g)



gases emitted (g)



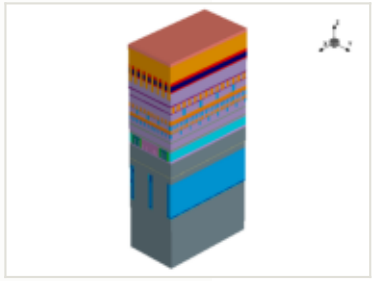
Step #5310: CVD

depo_fill · CVD

1.31	506.4	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

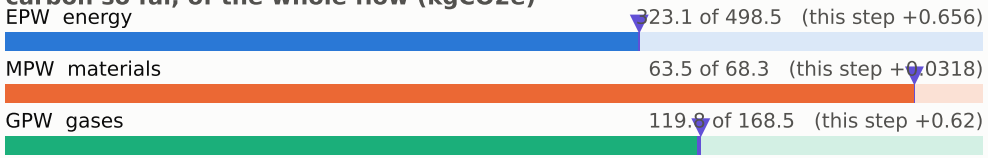


this step

process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

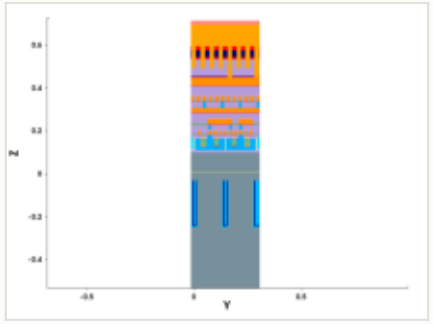
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

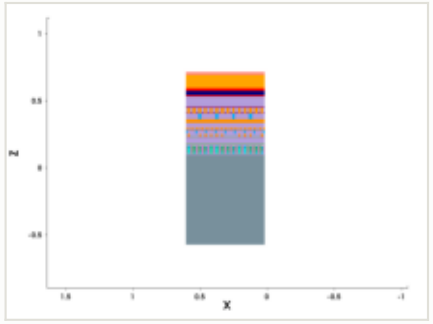


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



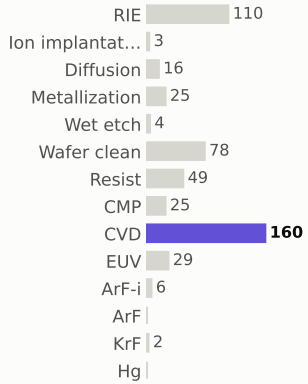
y cut at 0.184 µm



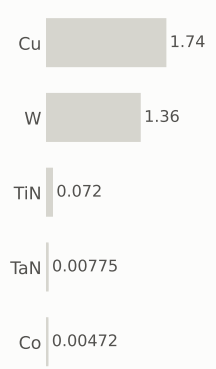
mask for this step

no mask for this step

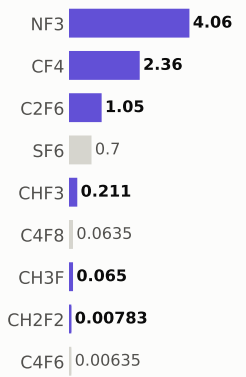
steps by process area



materials deposited (g)



gases emitted (g)



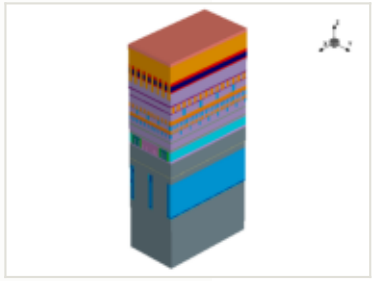
Step #5320: Wafer_clean

clean · Wafer_clean

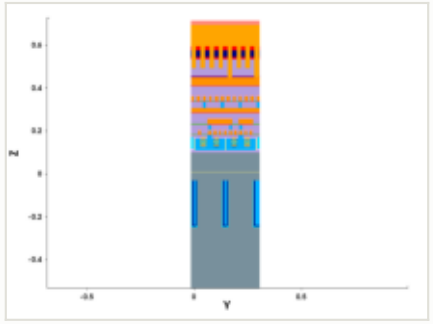
0.169	506.6	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

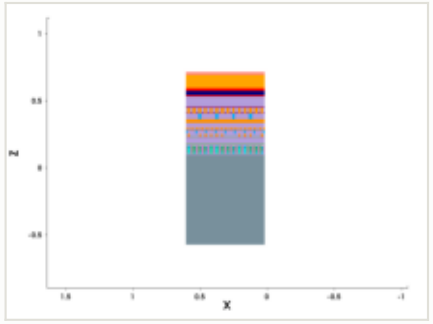
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

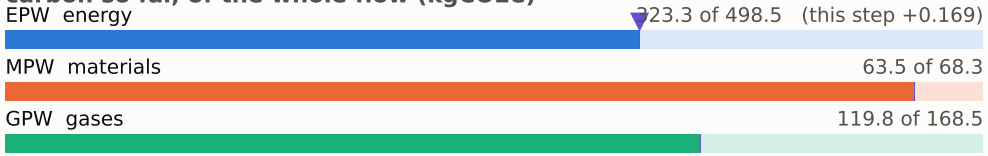
no mask for this step

this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

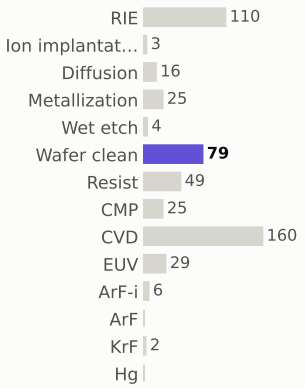
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

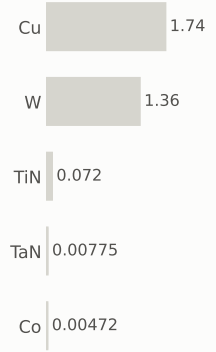


embodied carbon up to each step (kgCO2e per wafer)

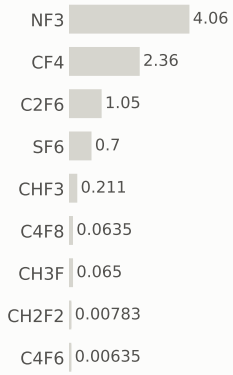
steps by process area



materials deposited (g)



gases emitted (g)



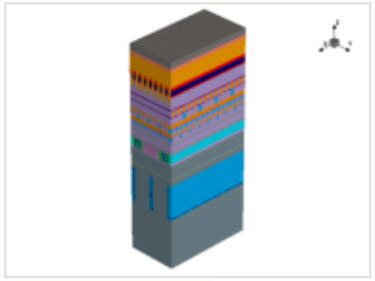
Step #5330: ArF-i

litho_patterning · ArF-i

0.217	506.8	0.029%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

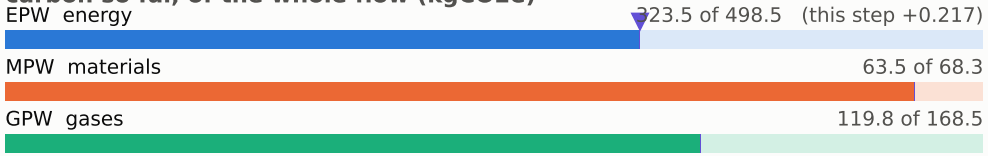


this step

process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

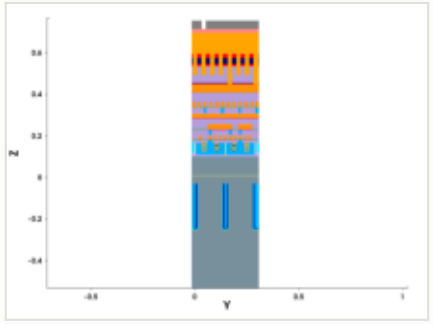
EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

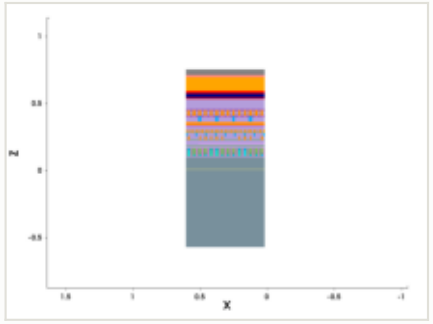


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm

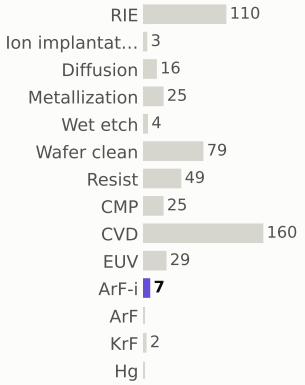


y cut at 0.184 µm

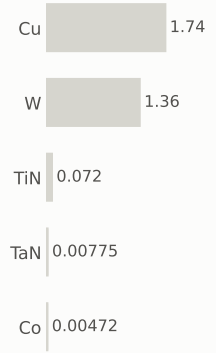


mask: layer 10423:0 (GDSII)

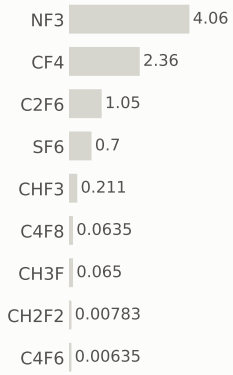
steps by process area



materials deposited (g)



gases emitted (g)



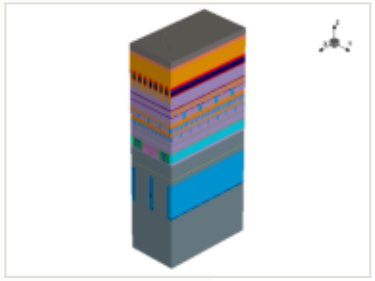
Step #5340: RIE

etch_anisotropic · RIE

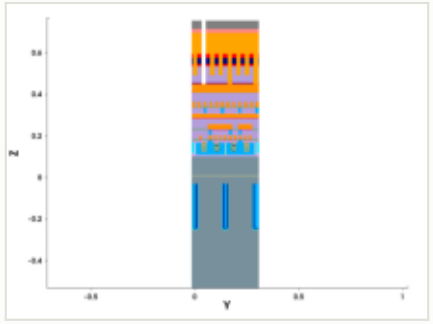
1.04	507.9	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

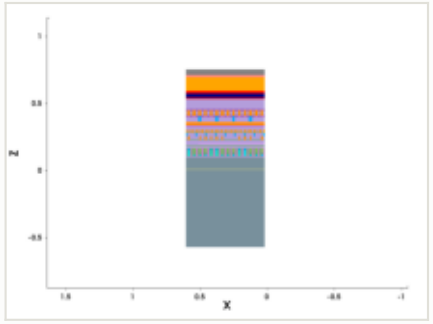
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

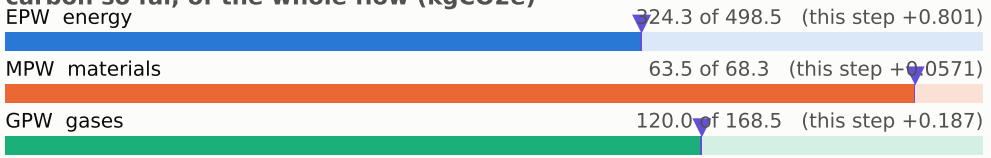
no mask for this step

this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

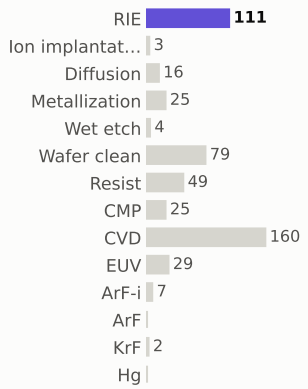
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

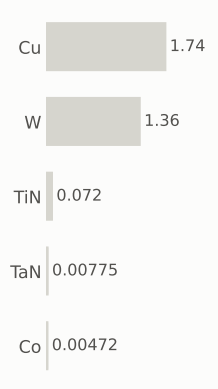


embodied carbon up to each step (kgCO2e per wafer)

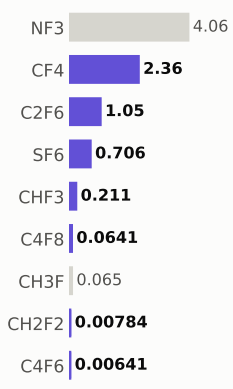
steps by process area



materials deposited (g)



gases emitted (g)



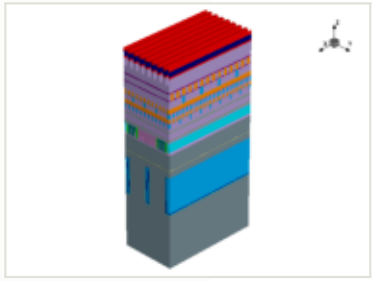
Step #5350: Wafer_clean

etch_strip · Wafer_clean

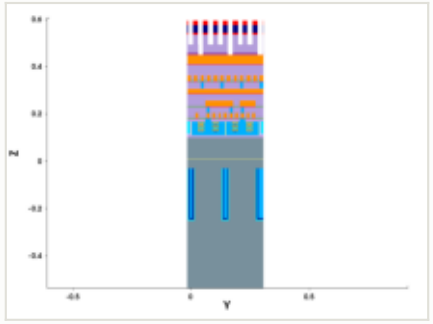
0.169	508.0	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

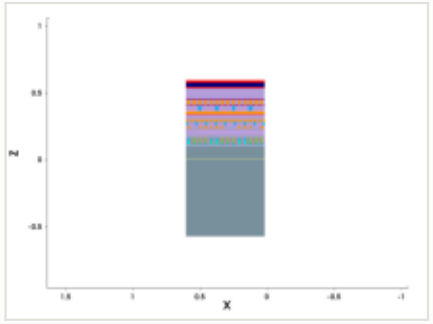
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



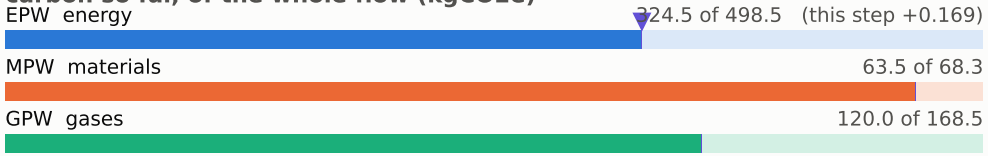
mask for this step

no mask for this step

this step

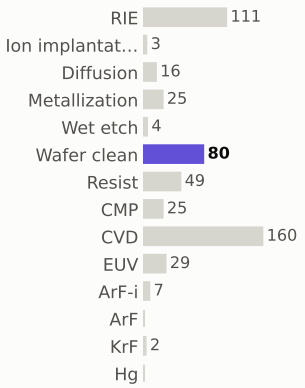
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

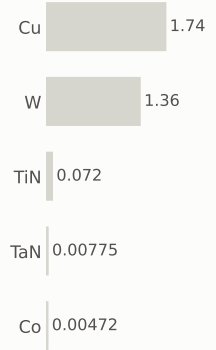


embodied carbon up to each step (kgCO2e per wafer)

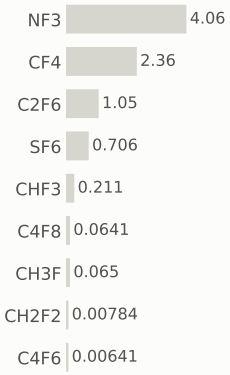
steps by process area



materials deposited (g)



gases emitted (g)



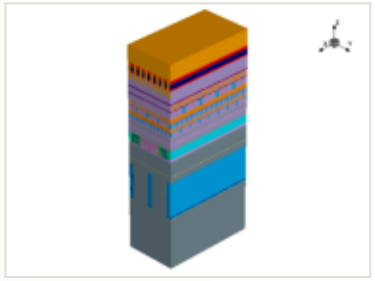
Step #5360: Resist

depo_fill · Resist

0.169	508.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

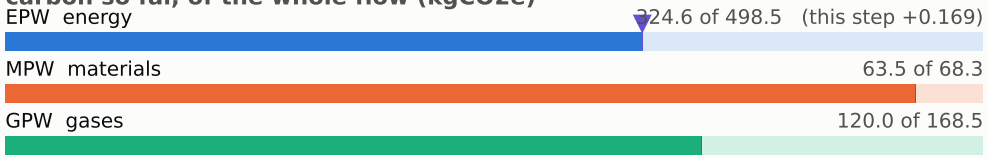


this step

process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

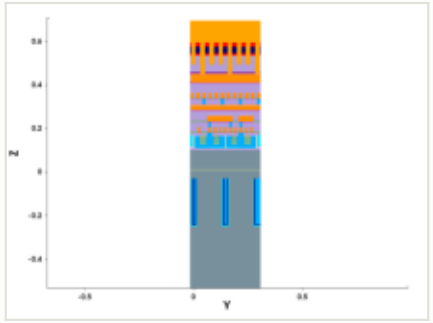
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

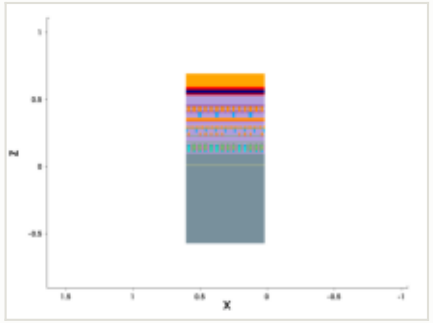


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



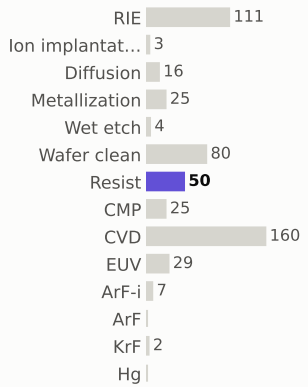
y cut at 0.184 µm



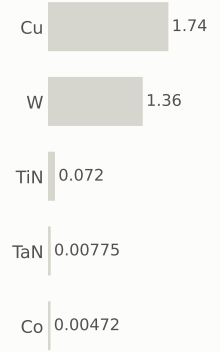
mask for this step

no mask for this step

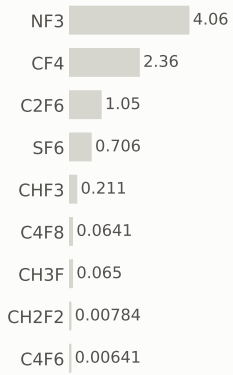
steps by process area



materials deposited (g)



gases emitted (g)



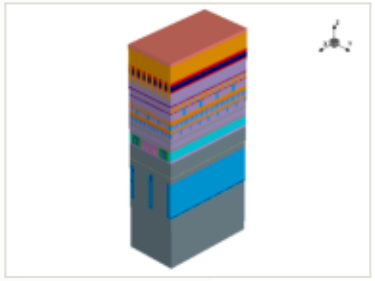
Step #5370: CVD

depo_fill · CVD

1.31	509.5	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

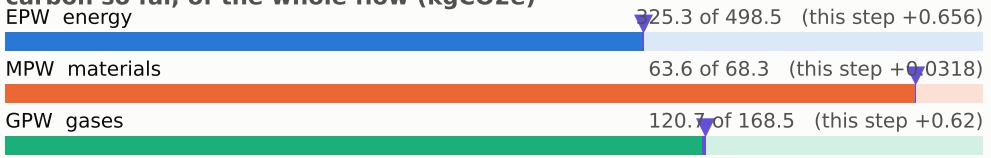


this step

process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

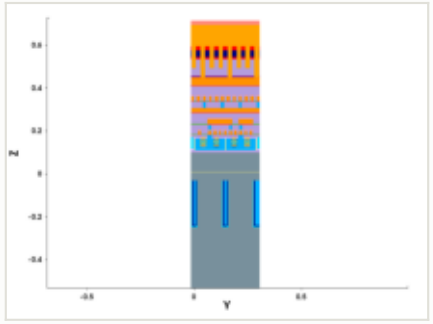
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

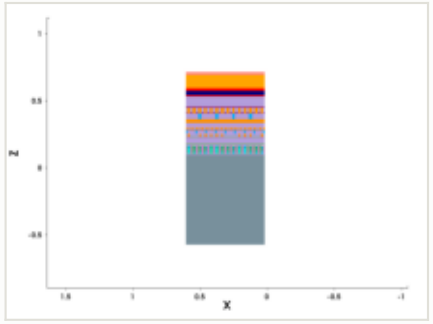


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



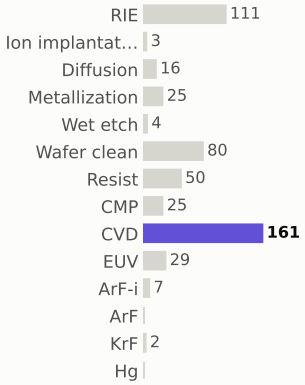
y cut at 0.184 µm



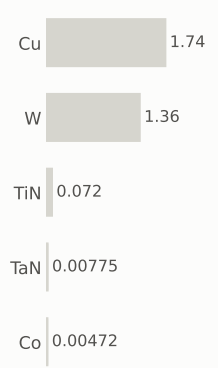
mask for this step

no mask for this step

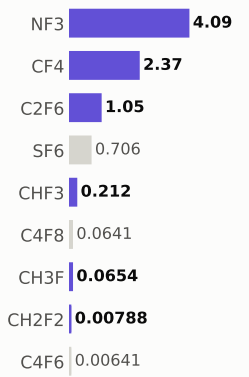
steps by process area



materials deposited (g)



gases emitted (g)



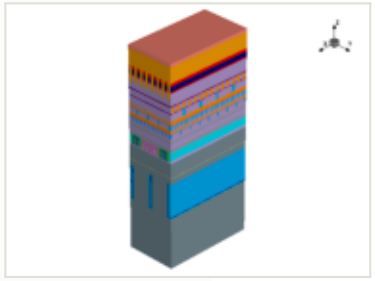
Step #5380: Wafer_clean

clean · Wafer_clean

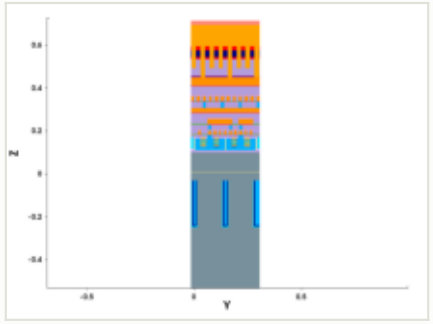
0.169	509.7	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

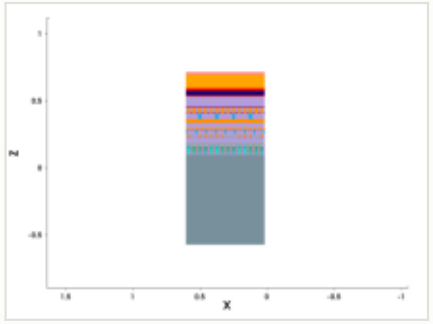
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

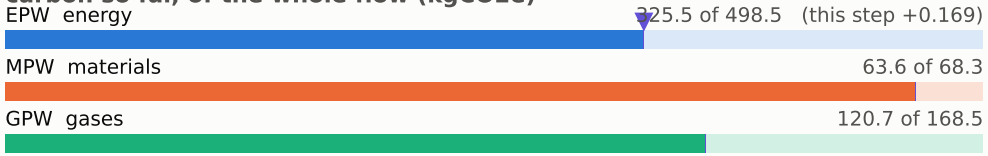
no mask for this step

this step

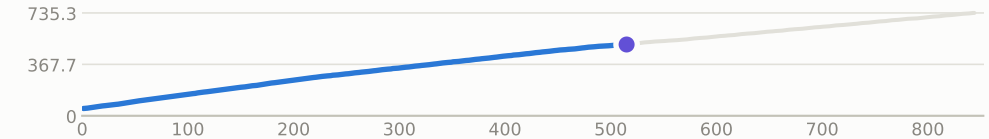
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

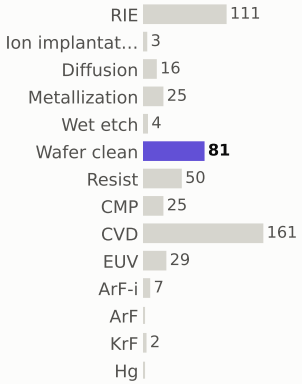
carbon so far, of the whole flow (kgCO2e)



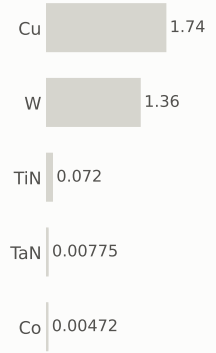
embodied carbon up to each step (kgCO2e per wafer)



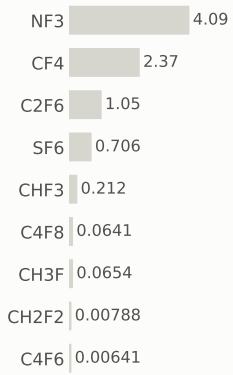
steps by process area



materials deposited (g)



gases emitted (g)



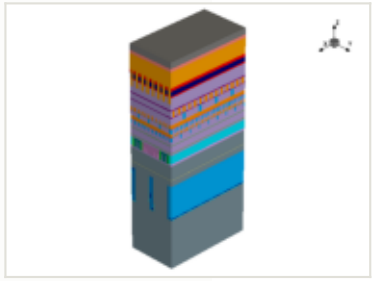
Step #5390: ArF-i

litho_patterning · ArF-i

0.217	509.9	0.029%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

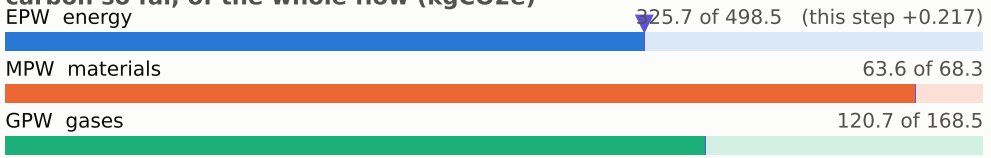


this step

process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

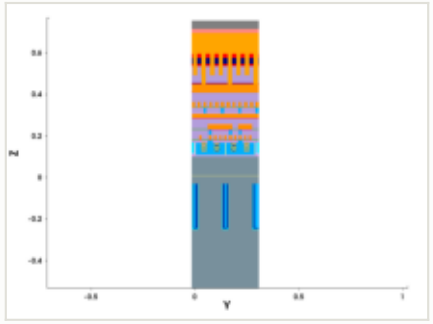
EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

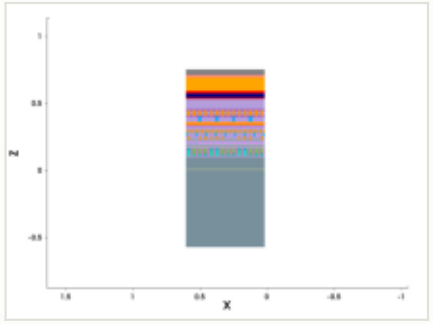


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



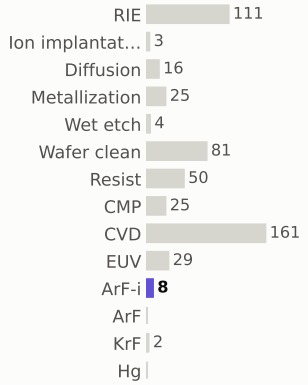
y cut at 0.184 µm



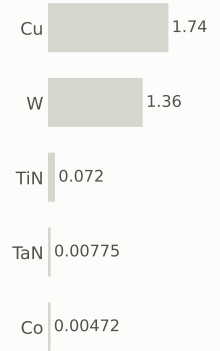
mask: layer 10424:0 (GDSII)

nothing in the clip window

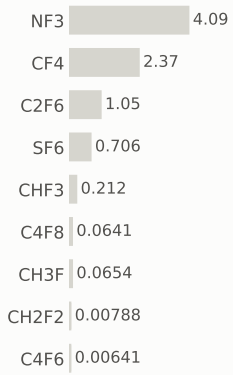
steps by process area



materials deposited (g)



gases emitted (g)



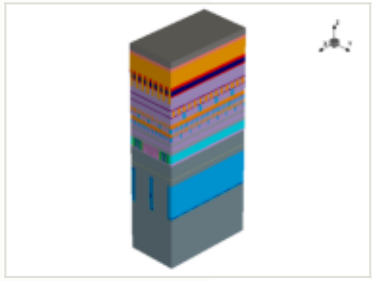
Step #5400: RIE

etch_anisotropic · RIE

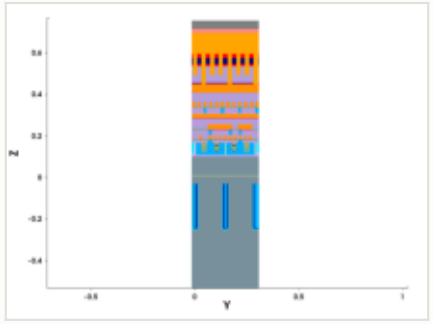
1.04	510.9	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

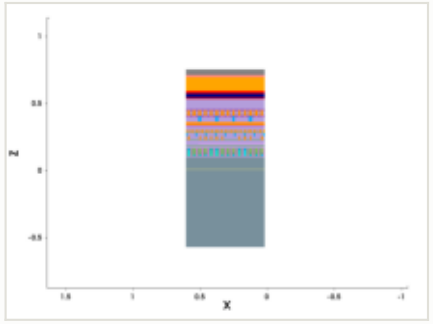
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

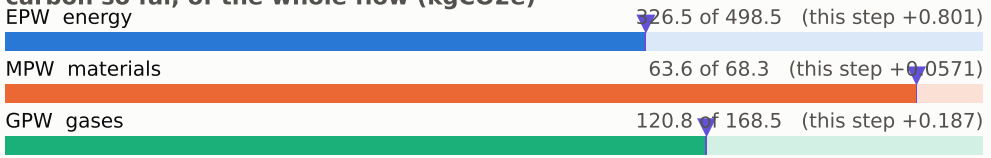
no mask for this step

this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

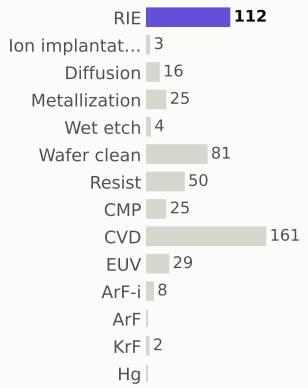
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

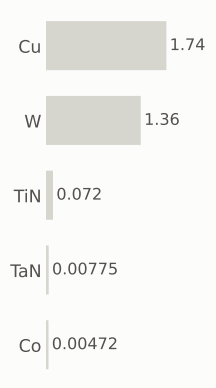


embodied carbon up to each step (kgCO2e per wafer)

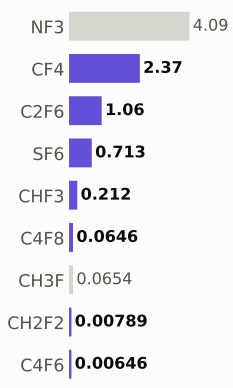
steps by process area



materials deposited (g)



gases emitted (g)



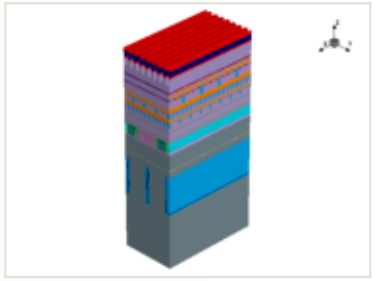
Step #5410: Wafer_clean

etch_strip · Wafer_clean

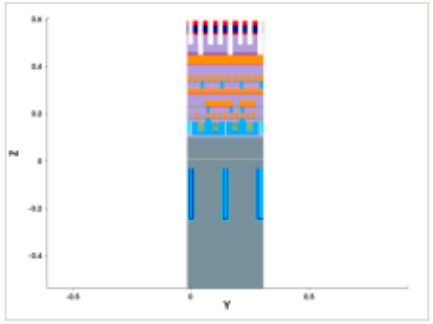
0.169	511.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

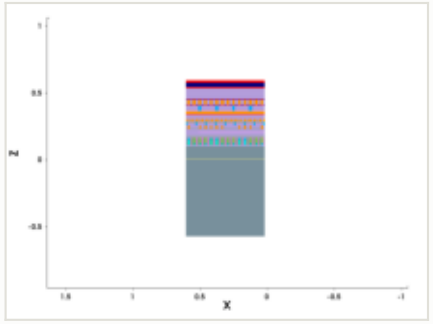
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



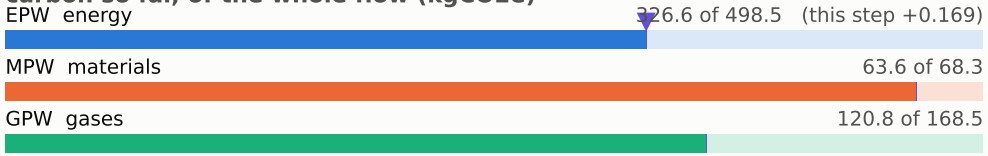
mask for this step

no mask for this step

this step

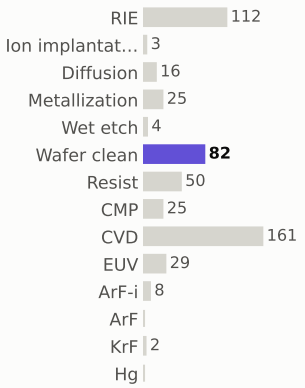
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

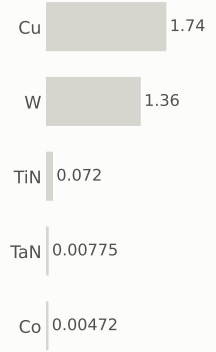


embodied carbon up to each step (kgCO2e per wafer)

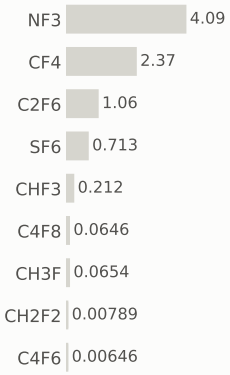
steps by process area



materials deposited (g)



gases emitted (g)



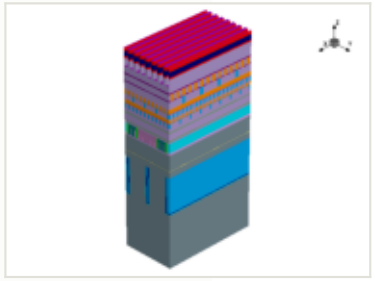
Step #5420: Metallization

depo_ellipsoidal · Metallization

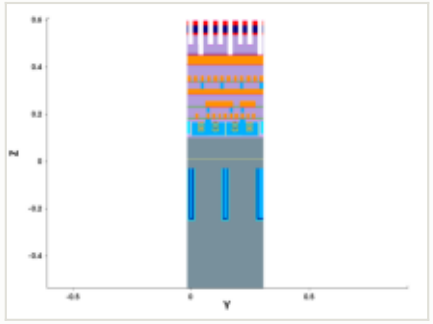
1.2	512.3	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

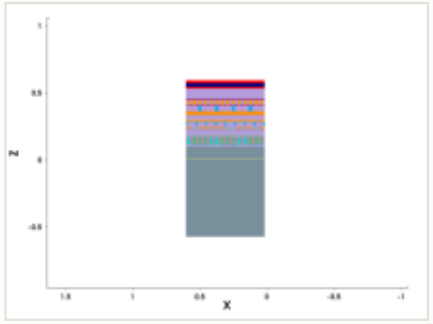
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

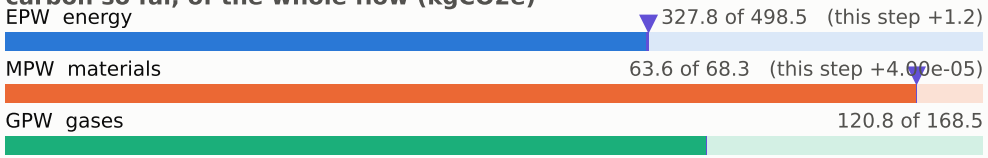
no mask for this step

this step

process area	Metallization
material	TaN
thickness	0.0016 µm
material GWP	25.8159 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

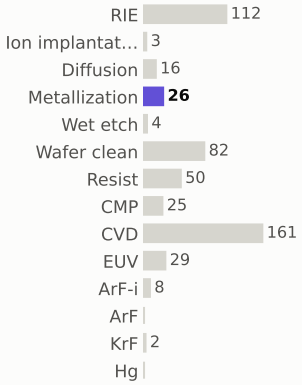
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.00e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

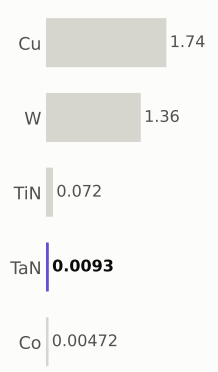


embodied carbon up to each step (kgCO2e per wafer)

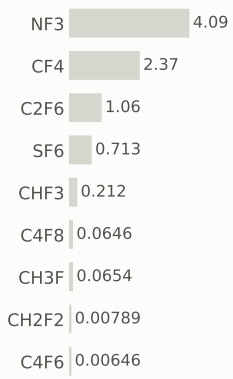
steps by process area



materials deposited (g)



gases emitted (g)



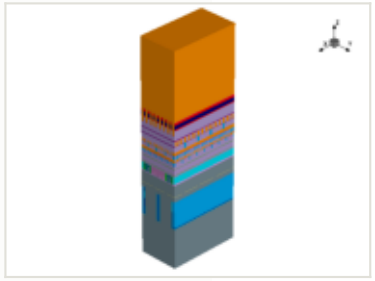
Step #5430: Metallization

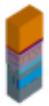
depo_fill · Metallization

1.21	513.5	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

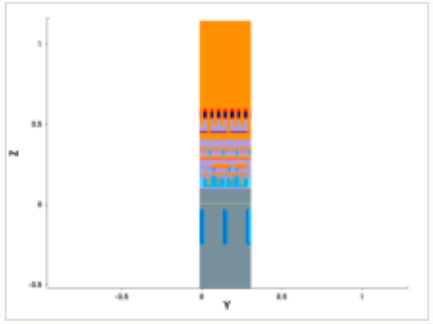
3D structure after this step

zoomed to fit

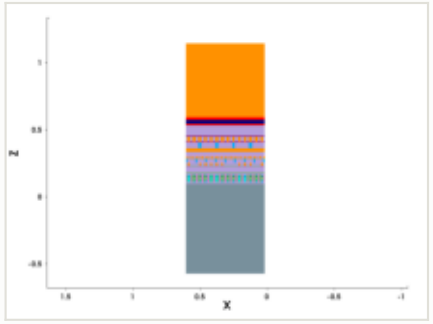




x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

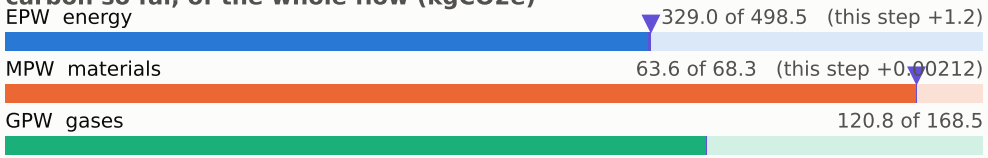
no mask for this step

this step

process area	Metallization
material	Cu
thickness	0.55 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

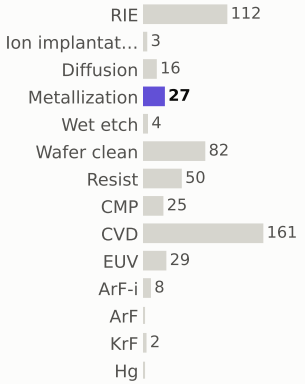
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00212 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

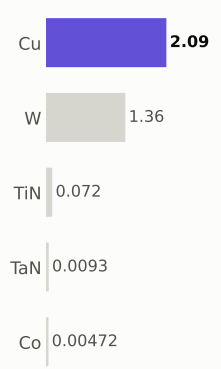


embodied carbon up to each step (kgCO2e per wafer)

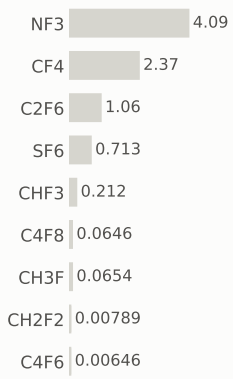
steps by process area



materials deposited (g)



gases emitted (g)



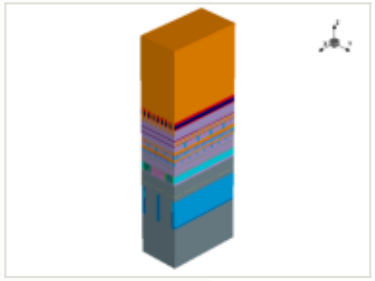
Step #5440: Diffusion

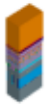
treatment · Diffusion

0.801	514.3	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

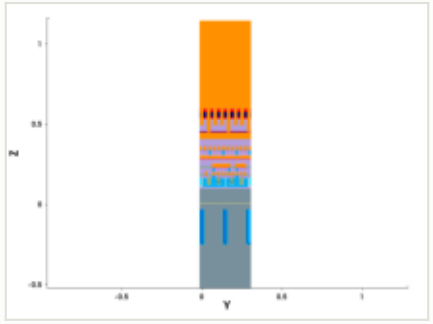
3D structure after this step

zoomed to fit

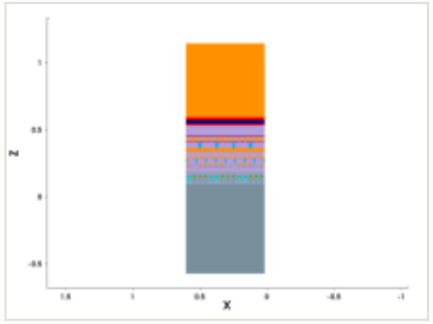




x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

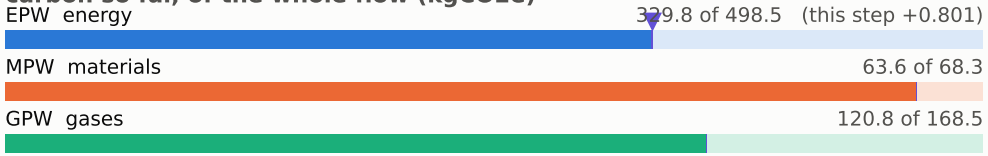
no mask for this step

this step

process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

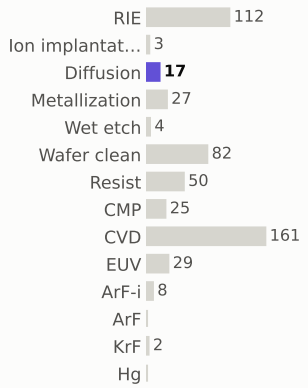
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

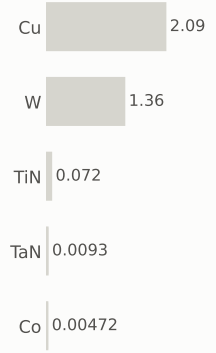


embodied carbon up to each step (kgCO2e per wafer)

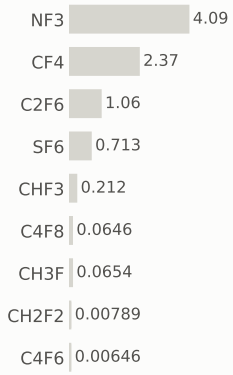
steps by process area



materials deposited (g)



gases emitted (g)



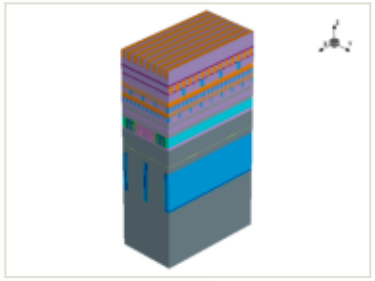
Step #5450: CMP

etch_cmp · CMP

0.169	514.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

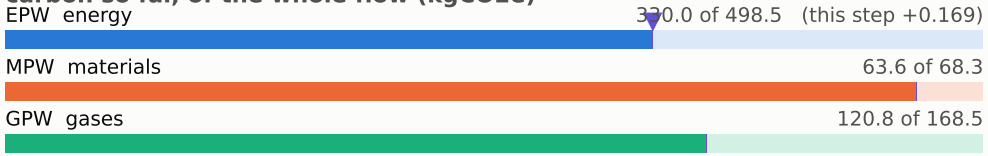


this step

process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

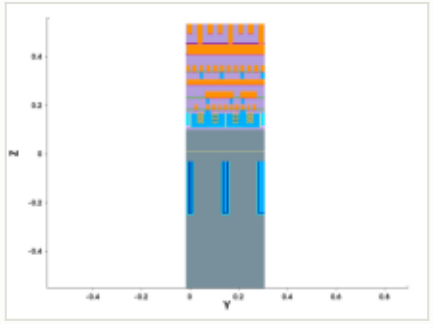
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

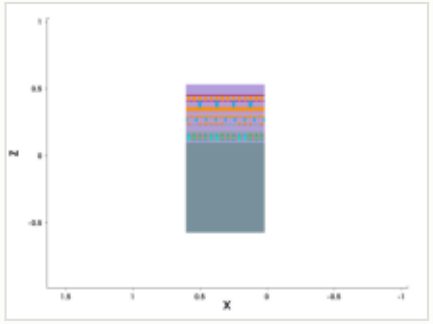


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



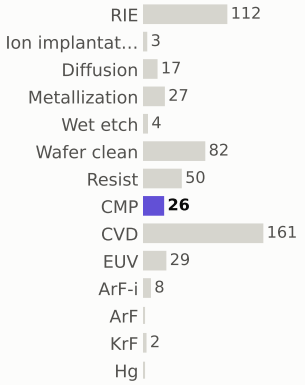
y cut at 0.184 µm



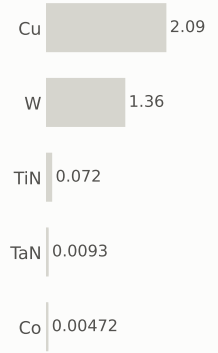
mask for this step

no mask for this step

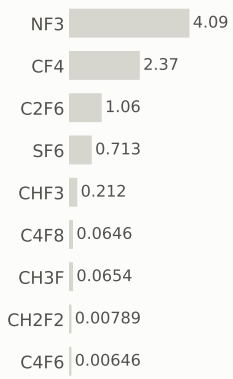
steps by process area



materials deposited (g)



gases emitted (g)



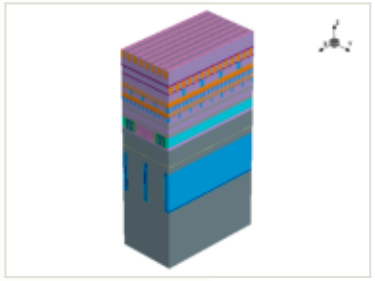
Step #5460: Metallization

depo_isotropic · Metallization

1.2	515.7	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

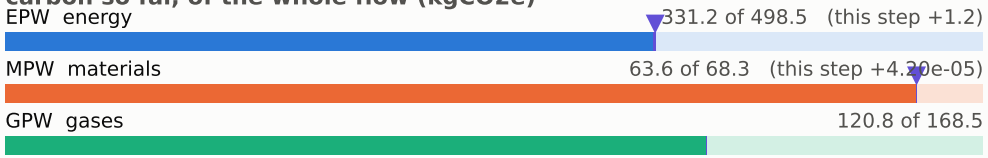


this step

process area	Metallization
material	Co
thickness	0.0015 µm
material GWP	44.5 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

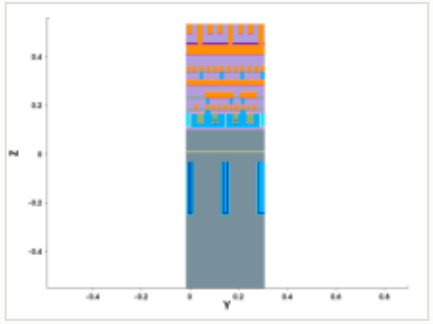
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.20e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

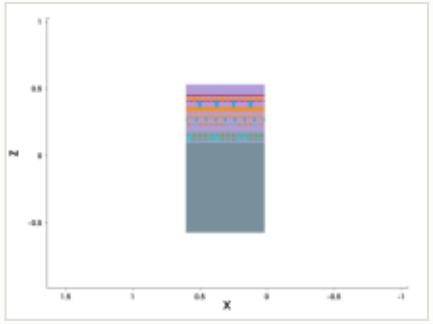


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



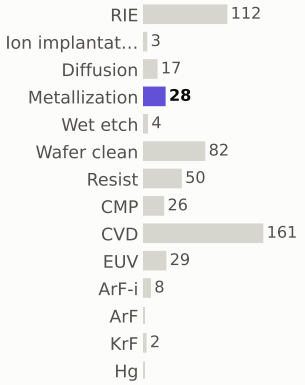
y cut at 0.184 µm



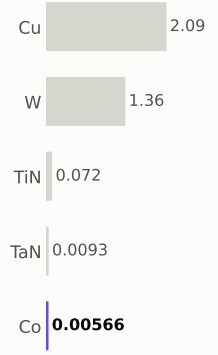
mask for this step

no mask for this step

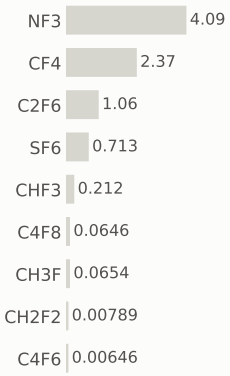
steps by process area



materials deposited (g)



gases emitted (g)



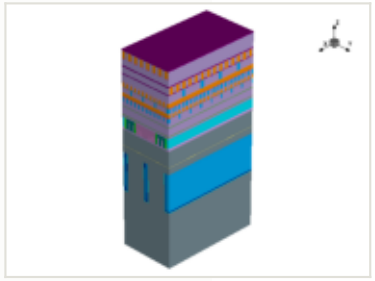
Step #5480: CVD

depo_fill · CVD

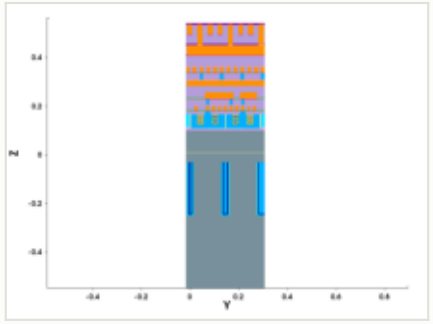
1.31	517.0	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

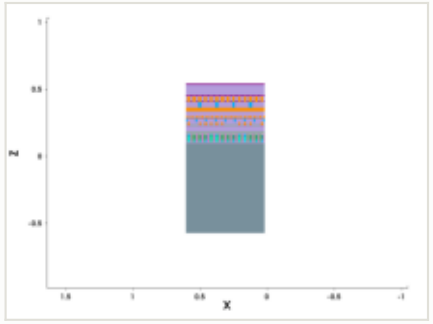
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

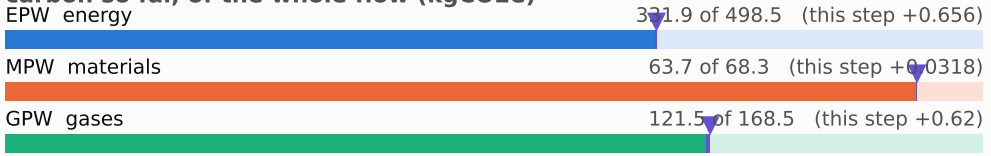
no mask for this step

this step

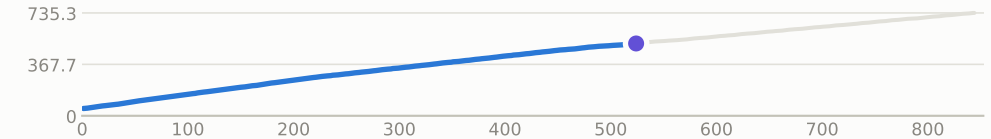
process area	CVD
material	—
thickness	0.006 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

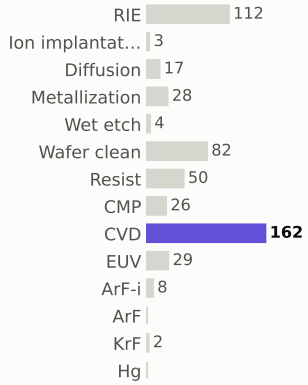
carbon so far, of the whole flow (kgCO2e)



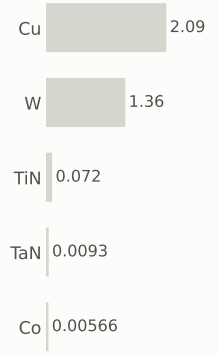
embodied carbon up to each step (kgCO2e per wafer)



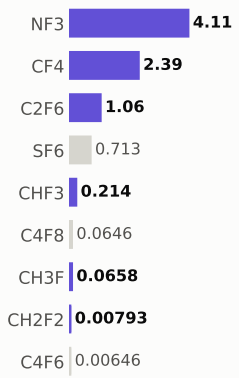
steps by process area



materials deposited (g)



gases emitted (g)



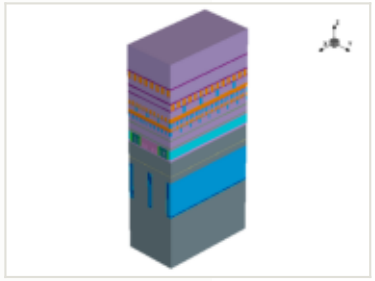
Step #5490: CVD

depo_fill · CVD

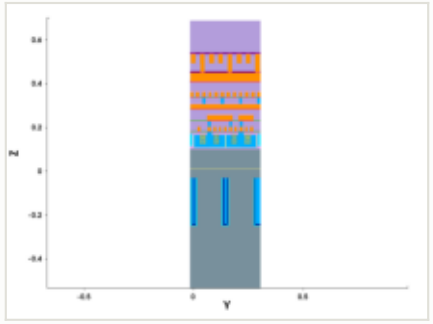
1.31	518.3	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

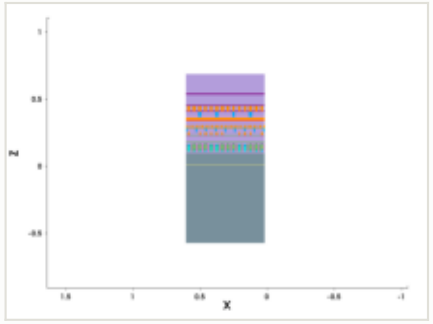
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

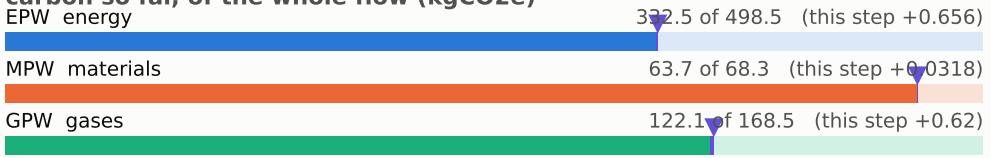
no mask for this step

this step

process area	CVD
material	—
thickness	0.146 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

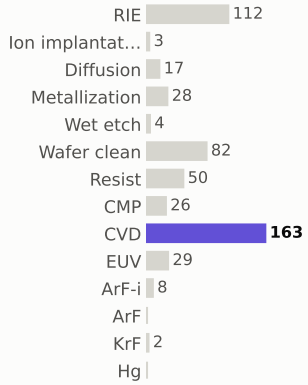
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

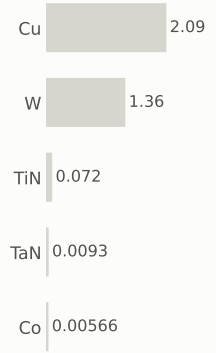


embodied carbon up to each step (kgCO2e per wafer)

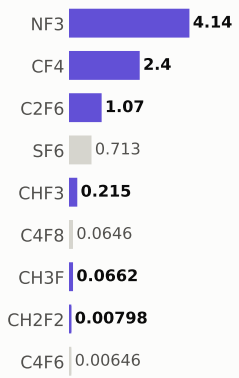
steps by process area



materials deposited (g)



gases emitted (g)



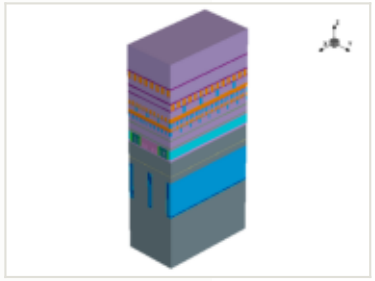
Step #5500: Diffusion

treatment · Diffusion

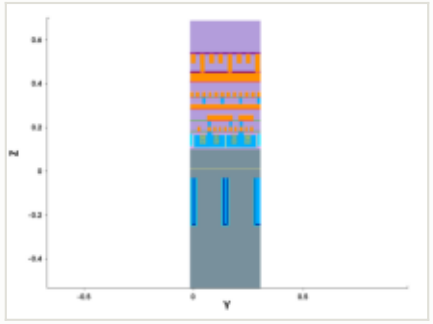
0.801	519.1	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

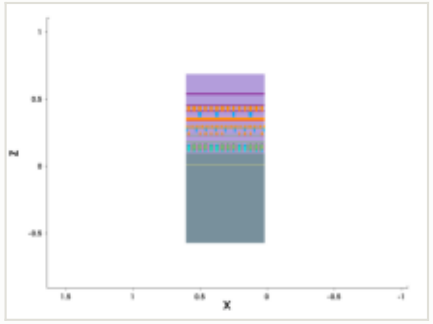
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

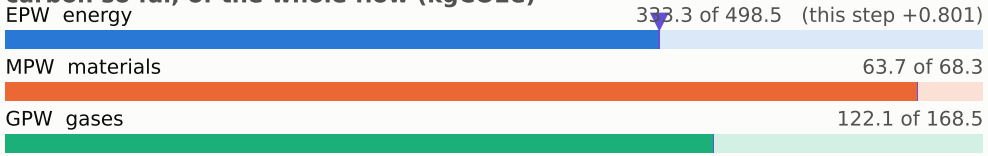
no mask for this step

this step

process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

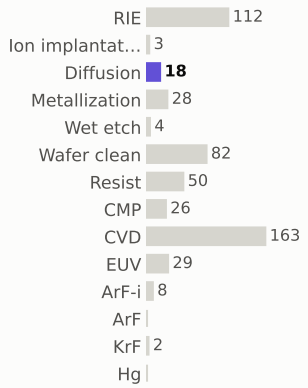
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

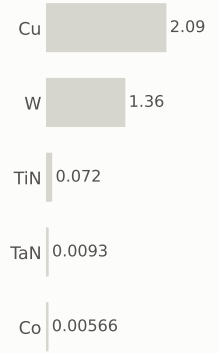


embodied carbon up to each step (kgCO2e per wafer)

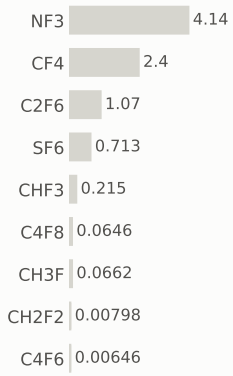
steps by process area



materials deposited (g)



gases emitted (g)



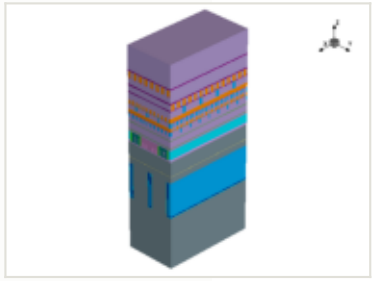
Step #5510: Wafer_clean

clean · Wafer_clean

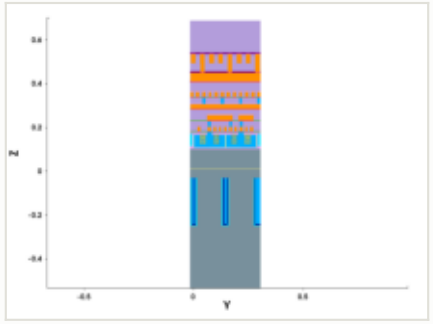
0.169	519.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

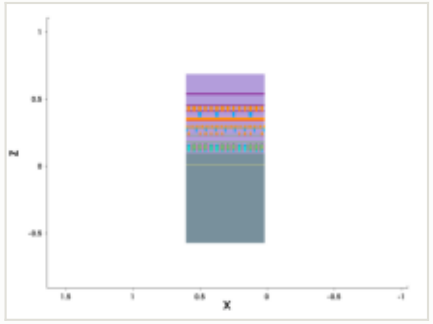
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

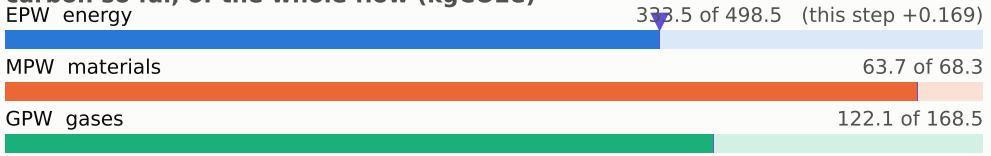
no mask for this step

this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

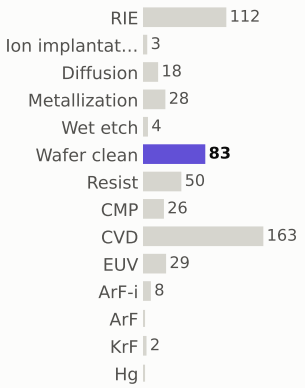
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

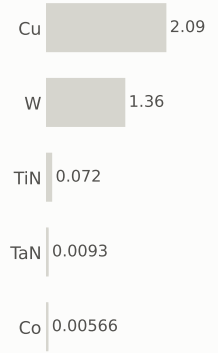


embodied carbon up to each step (kgCO2e per wafer)

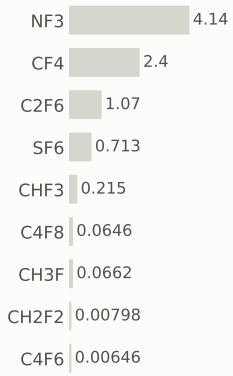
steps by process area



materials deposited (g)



gases emitted (g)



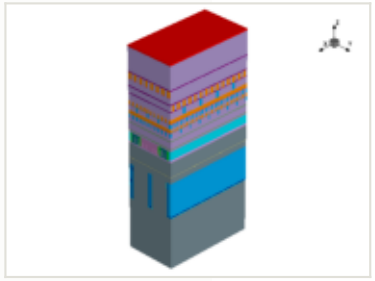
Step #5520: CVD

depo_fill · CVD

1.31	520.6	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

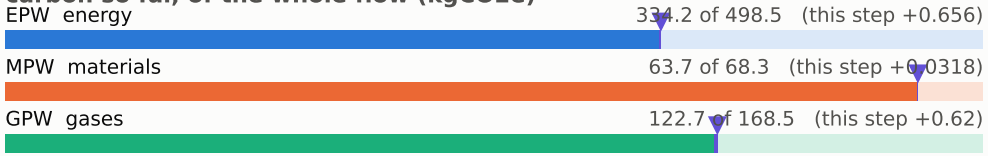


this step

process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

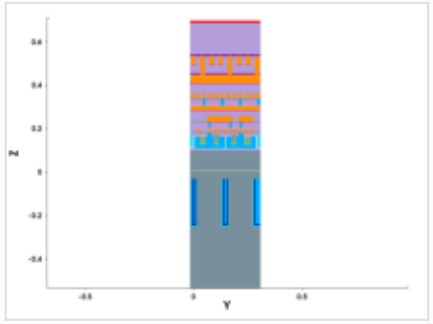
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

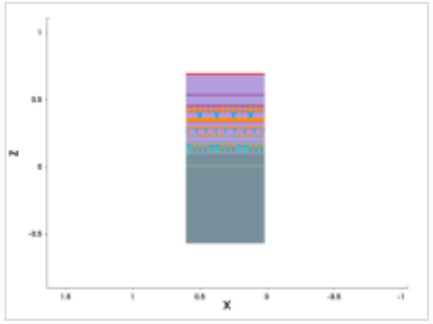


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



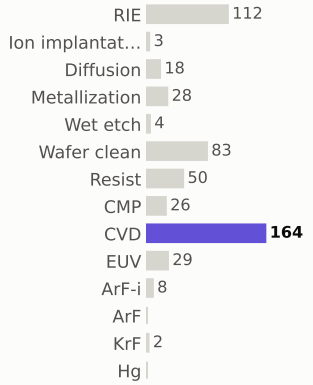
y cut at 0.184 µm



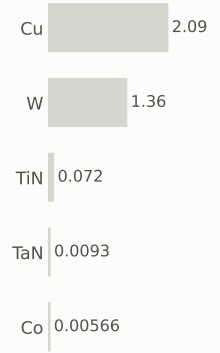
mask for this step

no mask for this step

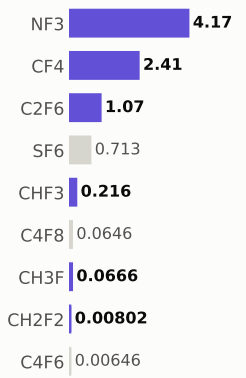
steps by process area



materials deposited (g)



gases emitted (g)



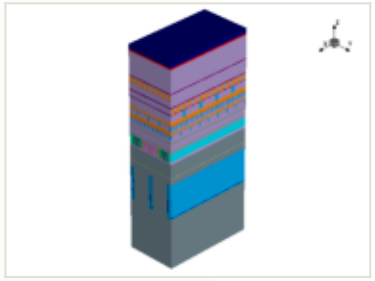
Step #5530: CVD

depo_fill · CVD

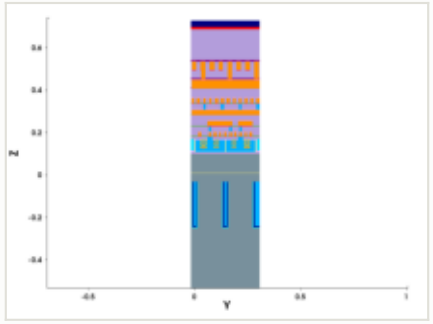
1.31	521.9	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

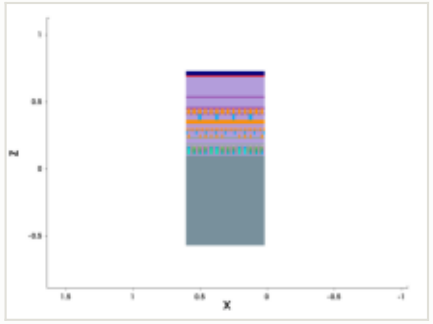
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

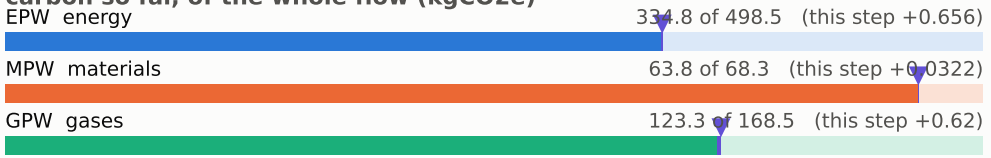
no mask for this step

this step

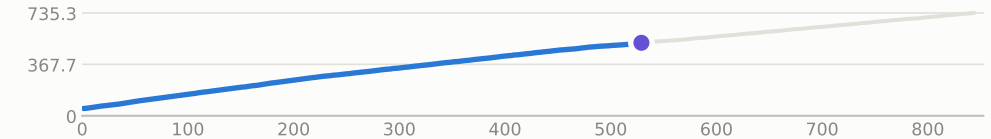
process area	CVD
material	TiN
thickness	0.03 µm
material GWP	43.9778 gCO2e/g
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0322 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

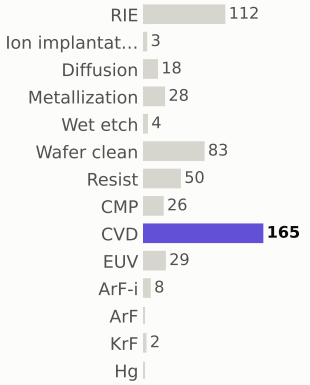
carbon so far, of the whole flow (kgCO2e)



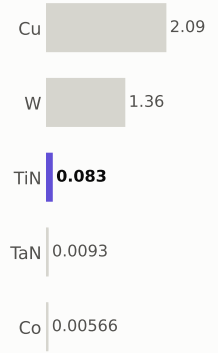
embodied carbon up to each step (kgCO2e per wafer)



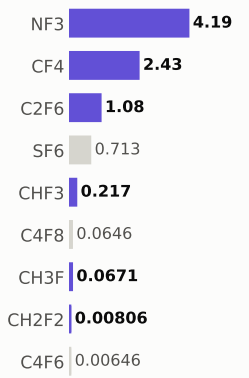
steps by process area



materials deposited (g)



gases emitted (g)



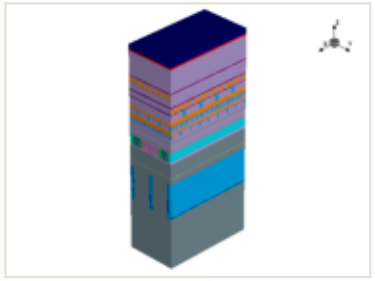
Step #5540: Wafer_clean

clean · Wafer_clean

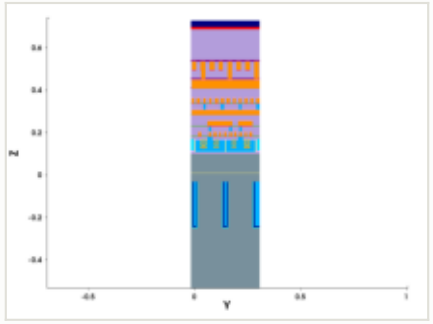
0.169	522.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

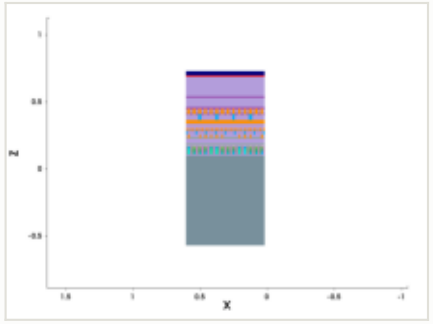
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

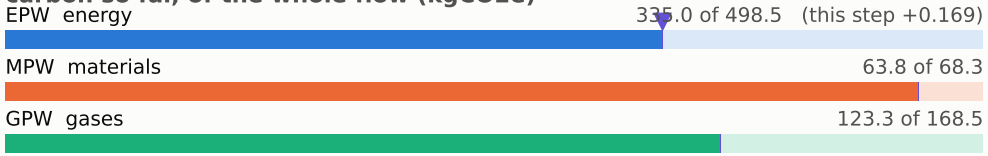
no mask for this step

this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

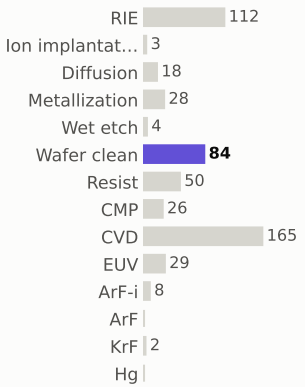
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

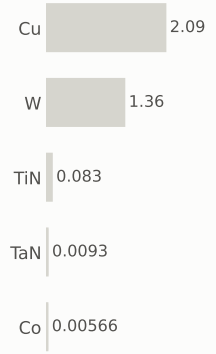


embodied carbon up to each step (kgCO2e per wafer)

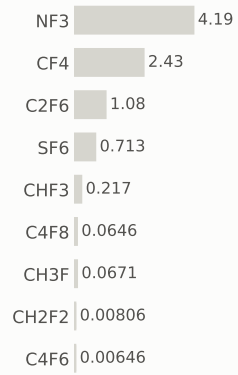
steps by process area



materials deposited (g)



gases emitted (g)



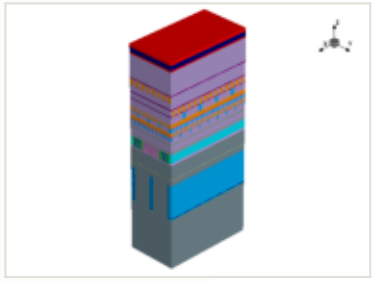
Step #5550: CVD

depo_fill · CVD

1.31	523.4	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

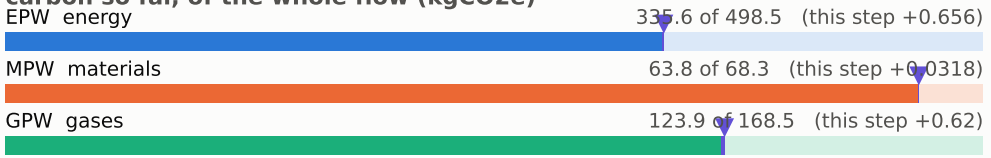


this step

process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

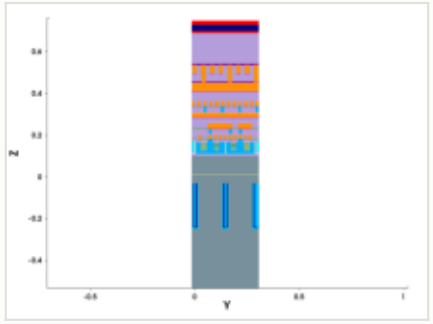
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

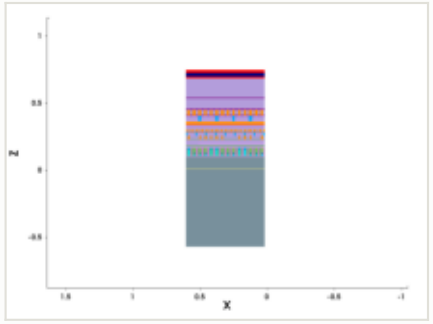


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



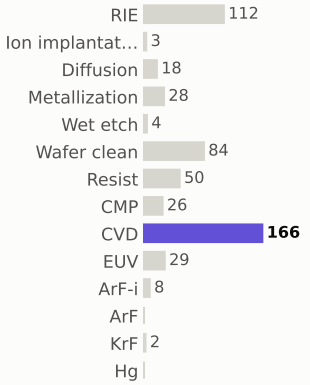
y cut at 0.184 µm



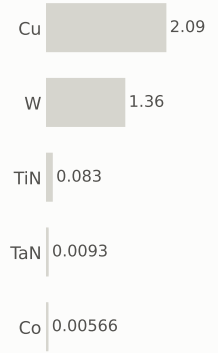
mask for this step

no mask for this step

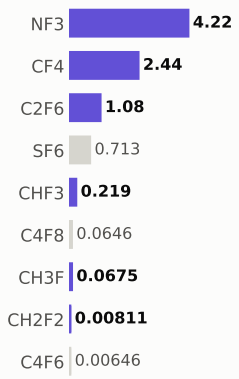
steps by process area



materials deposited (g)



gases emitted (g)



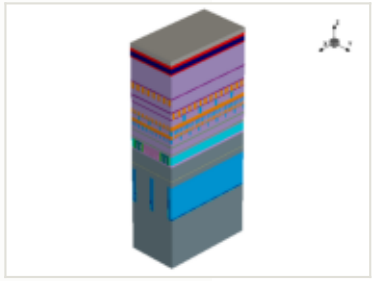
Step #5560: CVD

depo_fill · CVD

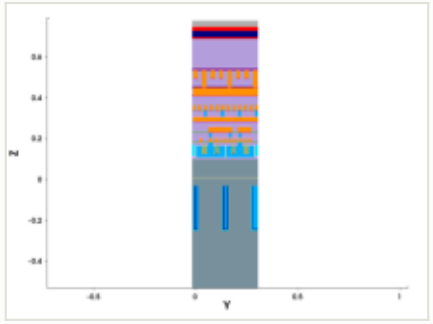
1.31	524.7	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

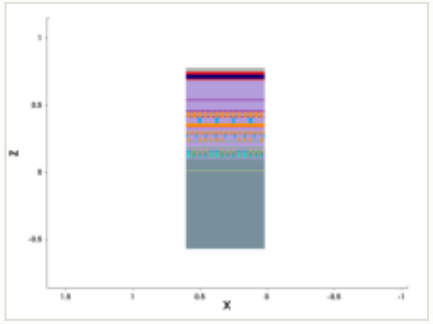
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

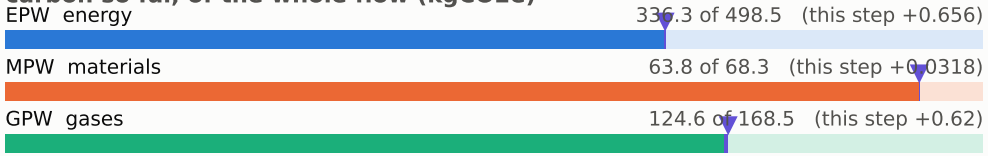
no mask for this step

this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

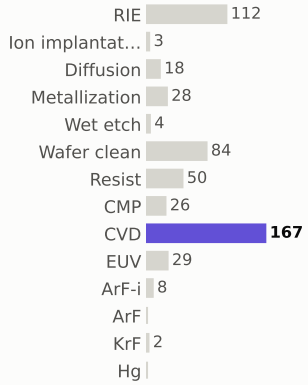
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

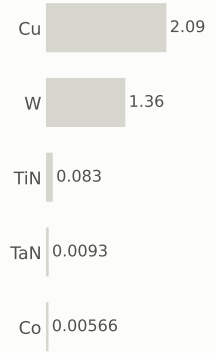


embodied carbon up to each step (kgCO2e per wafer)

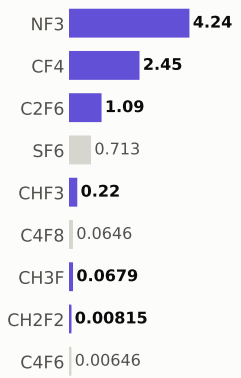
steps by process area



materials deposited (g)



gases emitted (g)



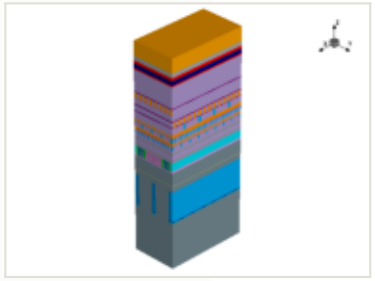
Step #5570: Resist

depo_fill · Resist

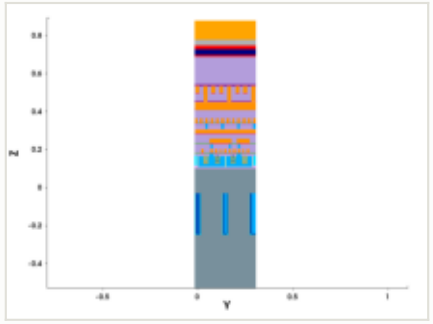
0.169	524.8	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

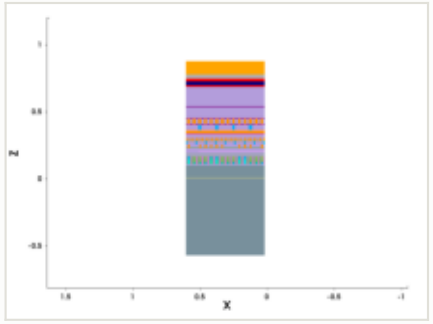
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



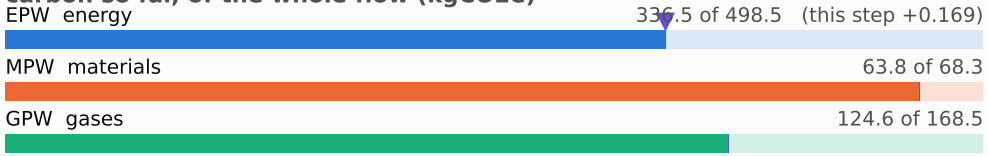
mask for this step

no mask for this step

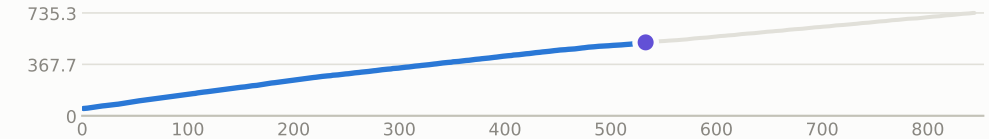
this step

process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	0.1 µm	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

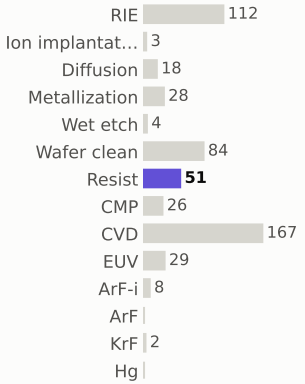
carbon so far, of the whole flow (kgCO2e)



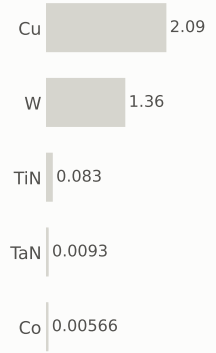
embodied carbon up to each step (kgCO2e per wafer)



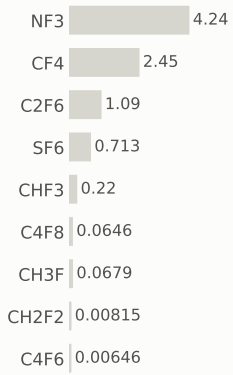
steps by process area



materials deposited (g)



gases emitted (g)



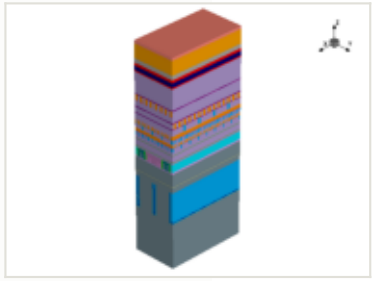
Step #5580: CVD

depo_fill · CVD

1.31	526.2	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

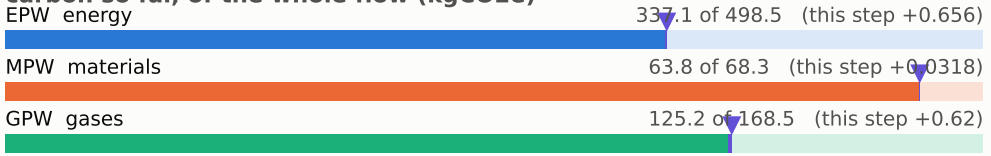


this step

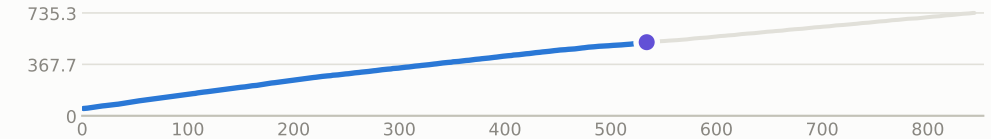
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

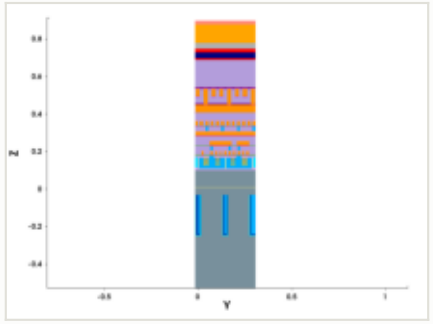
carbon so far, of the whole flow (kgCO2e)



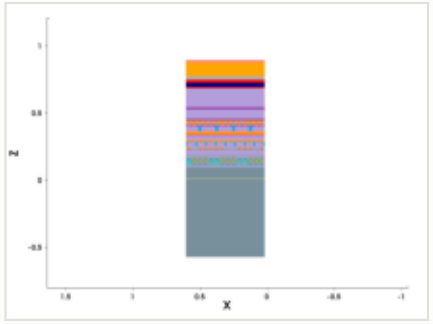
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



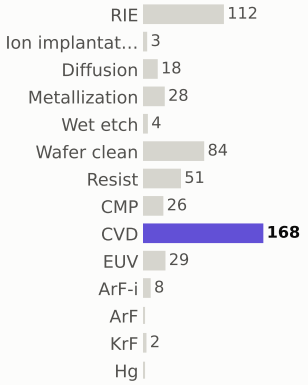
y cut at 0.184 µm



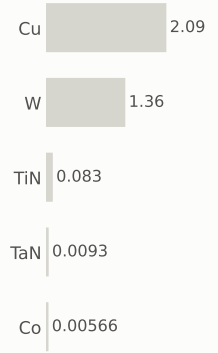
mask for this step

no mask for this step

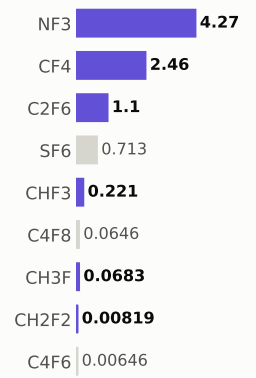
steps by process area



materials deposited (g)



gases emitted (g)



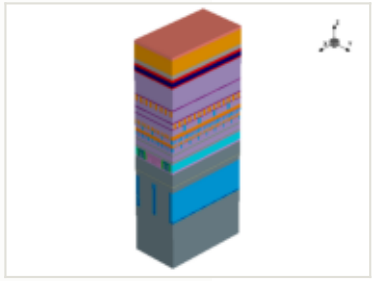
Step #5590: Wafer_clean

clean · Wafer_clean

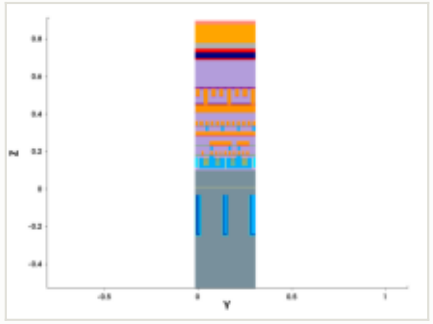
0.169	526.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

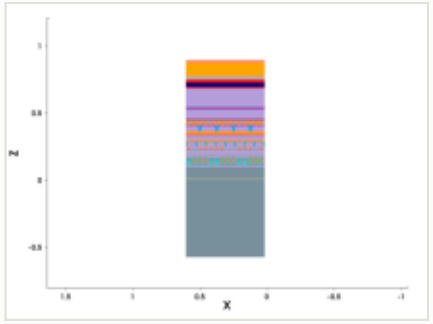
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

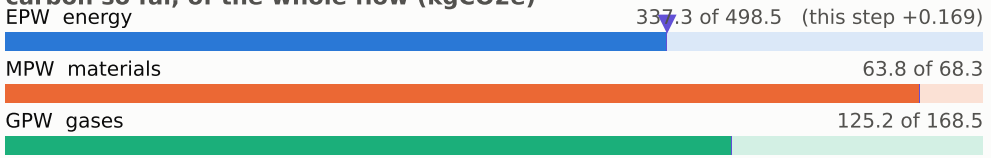
no mask for this step

this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

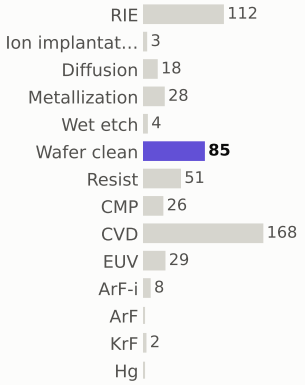
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

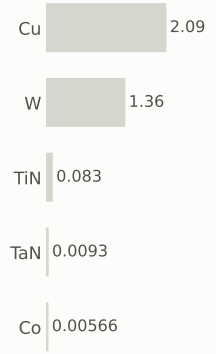


embodied carbon up to each step (kgCO2e per wafer)

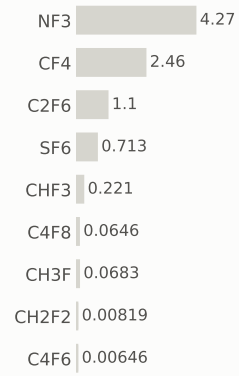
steps by process area



materials deposited (g)



gases emitted (g)



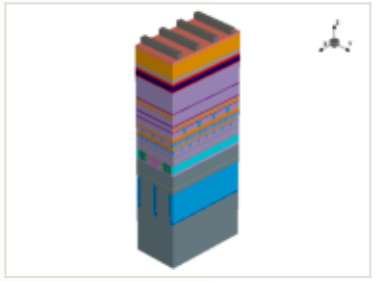
Step #5600: ArF-i

litho_patterning · ArF-i

0.217	526.5	0.029%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

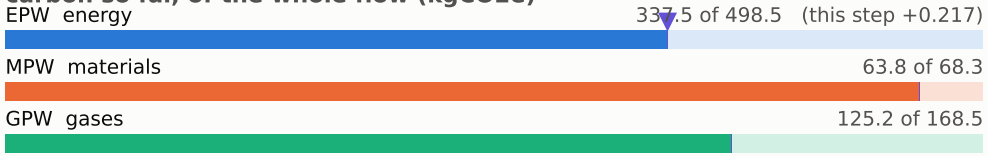


this step

process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

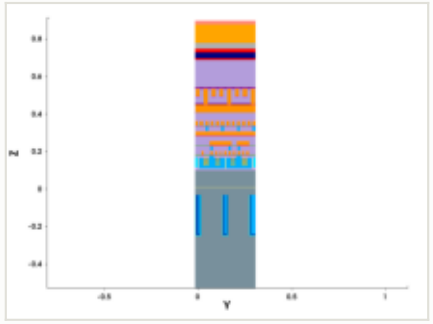
EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

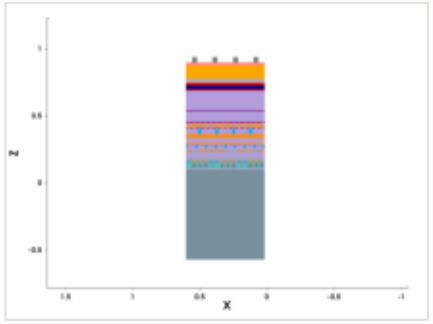


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm

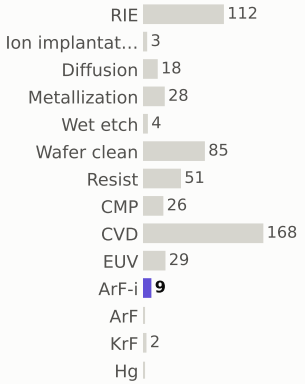


y cut at 0.184 µm

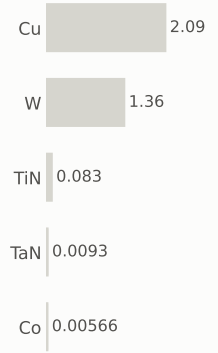


mask: layer 10501:0, positive (GDSII)

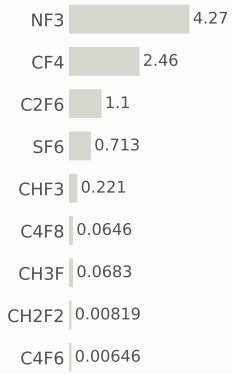
steps by process area



materials deposited (g)



gases emitted (g)



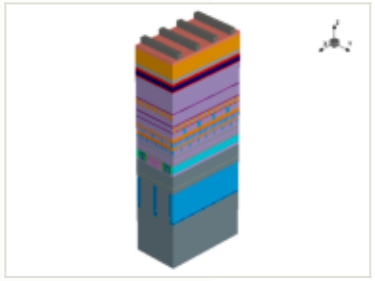
Step #5610: Ion_implantation

treatment · Ion_implantation

0.801	527.3	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



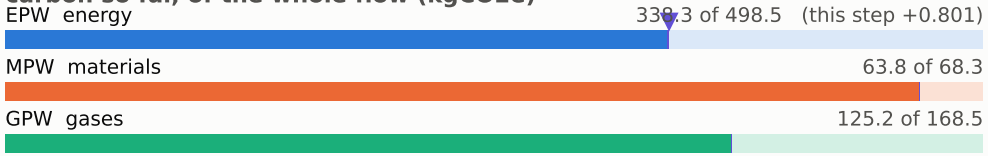
this step

process area	Ion_implantation
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

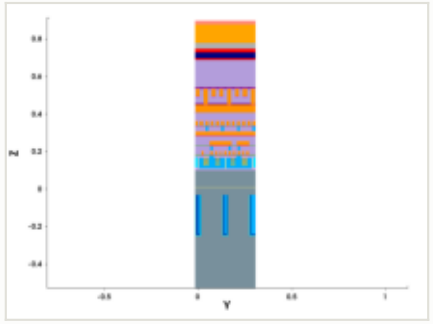
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

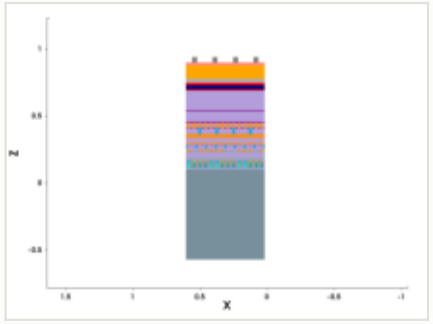


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



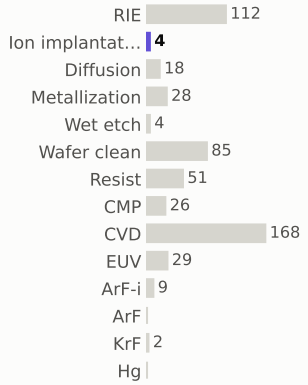
y cut at 0.184 µm



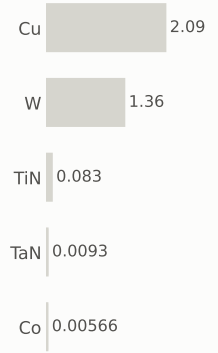
mask for this step

no mask for this step

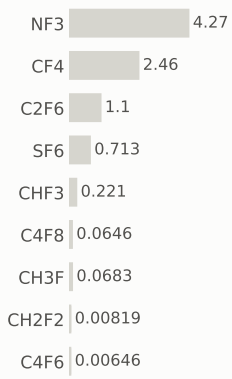
steps by process area



materials deposited (g)



gases emitted (g)



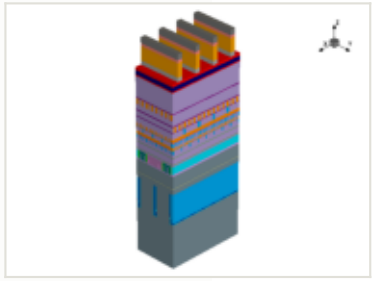
Step #5620: RIE

etch_anisotropic · RIE

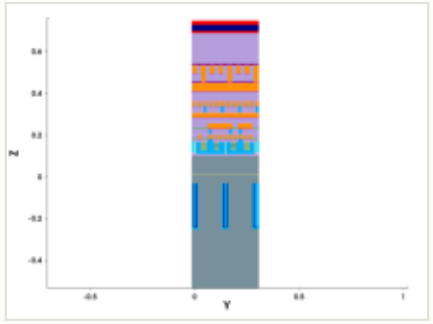
1.04	528.4	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

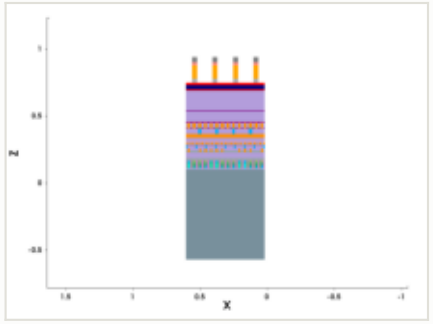
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

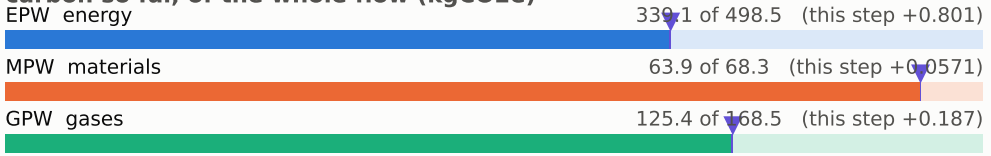
no mask for this step

this step

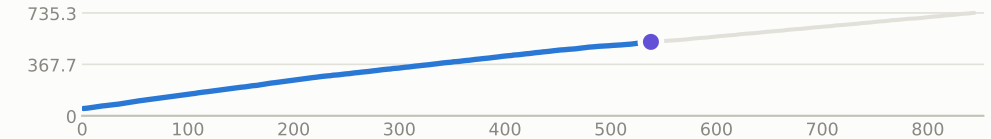
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

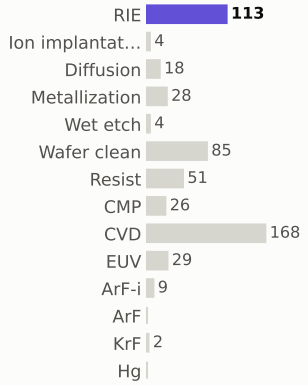
carbon so far, of the whole flow (kgCO2e)



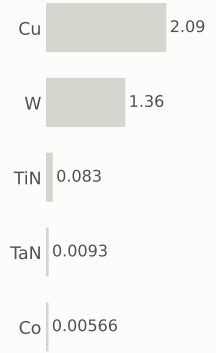
embodied carbon up to each step (kgCO2e per wafer)



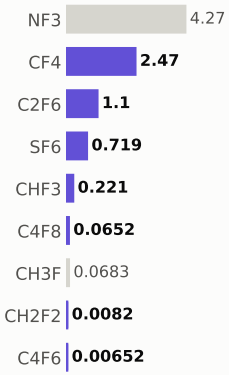
steps by process area



materials deposited (g)



gases emitted (g)



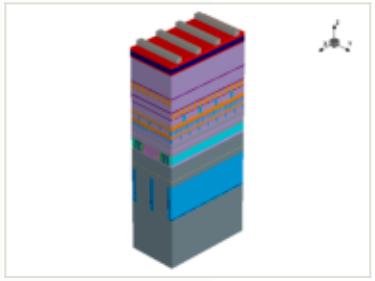
Step #5630: Wafer_clean

etch_strip · Wafer_clean

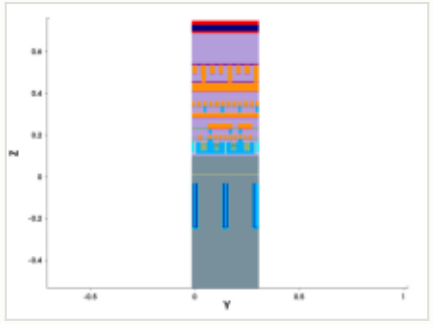
0.169	528.6	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

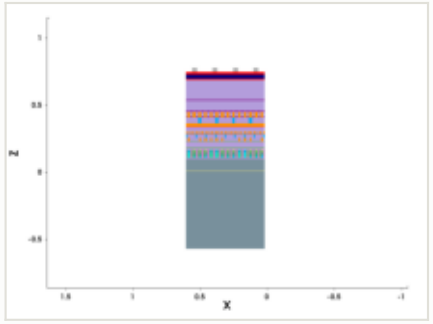
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



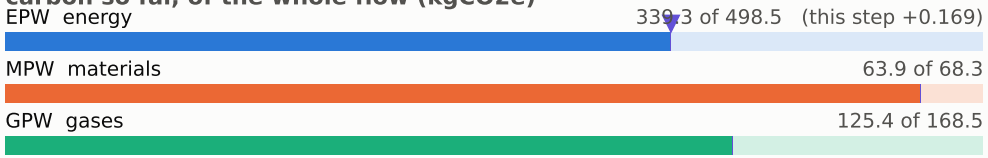
mask for this step

no mask for this step

this step

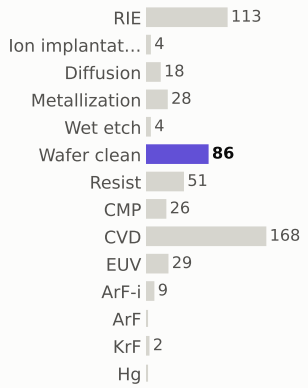
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

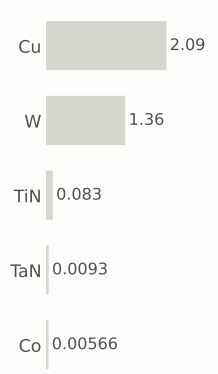


embodied carbon up to each step (kgCO2e per wafer)

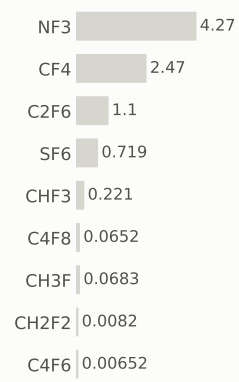
steps by process area



materials deposited (g)



gases emitted (g)



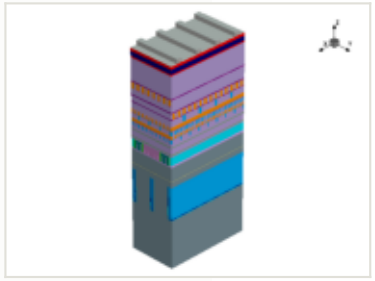
Step #5640: CVD

depo_isotropic · CVD

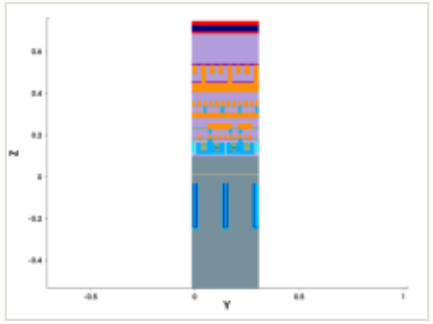
1.31	529.9	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

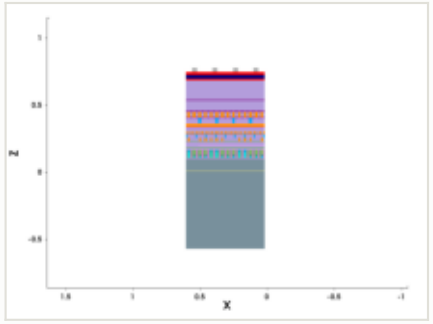
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

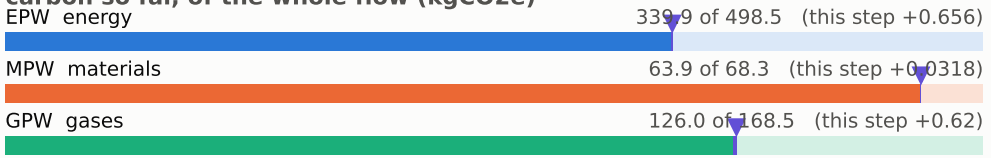
no mask for this step

this step

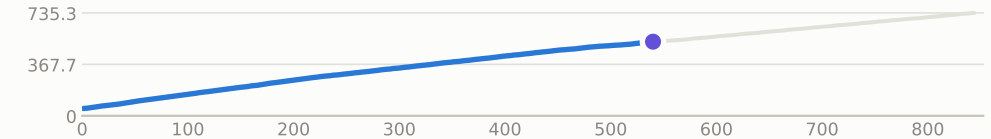
process area	CVD
material	—
thickness	0.001 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

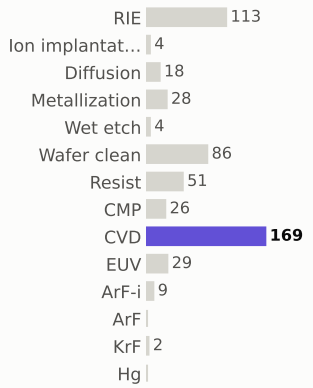
carbon so far, of the whole flow (kgCO2e)



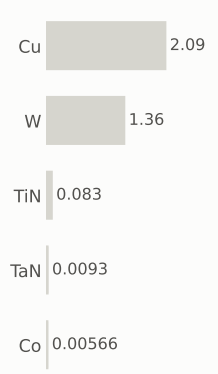
embodied carbon up to each step (kgCO2e per wafer)



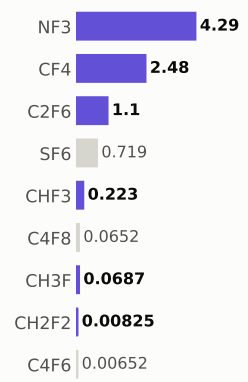
steps by process area



materials deposited (g)



gases emitted (g)



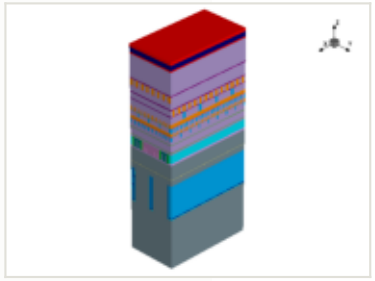
Step #5650: RIE

etch_anisotropic · RIE

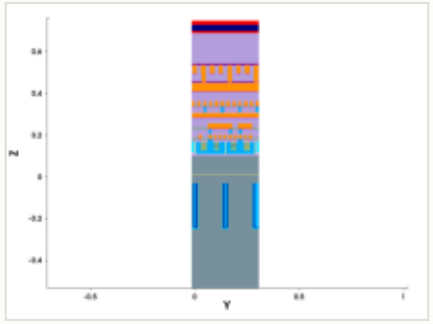
1.04	530.9	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

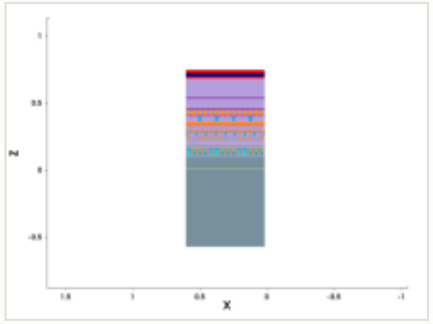
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

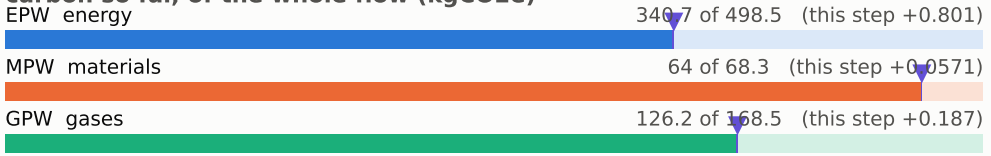
no mask for this step

this step

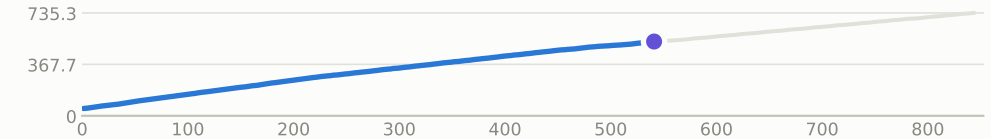
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

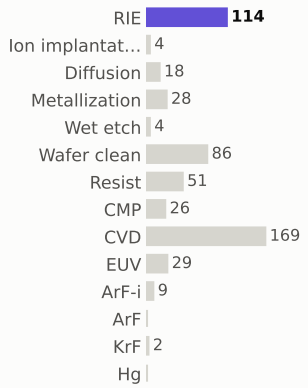
carbon so far, of the whole flow (kgCO2e)



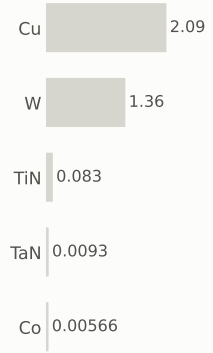
embodied carbon up to each step (kgCO2e per wafer)



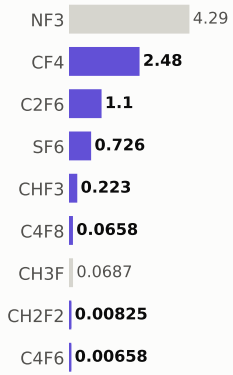
steps by process area



materials deposited (g)



gases emitted (g)



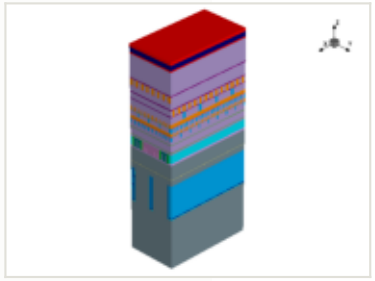
Step #5660: Wafer_clean

etch_strip · Wafer_clean

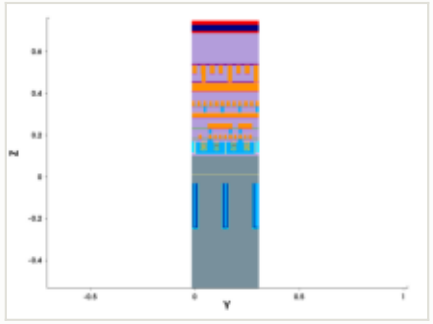
0.169	531.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

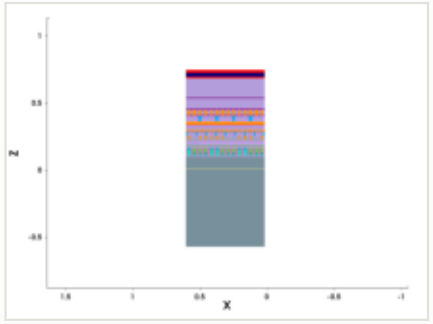
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



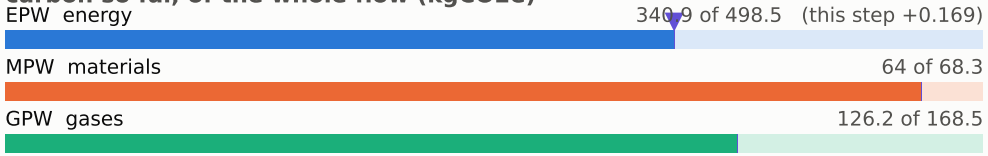
mask for this step

no mask for this step

this step

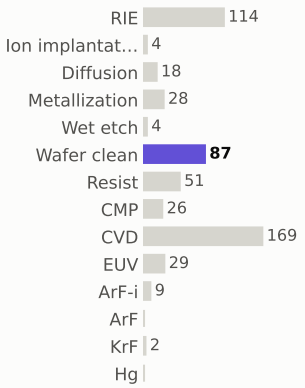
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

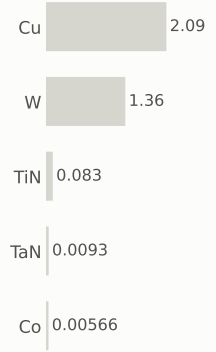


embodied carbon up to each step (kgCO2e per wafer)

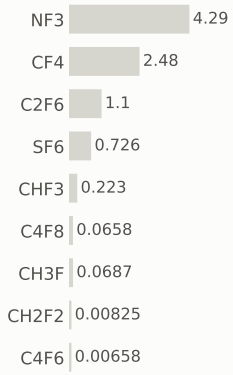
steps by process area



materials deposited (g)



gases emitted (g)



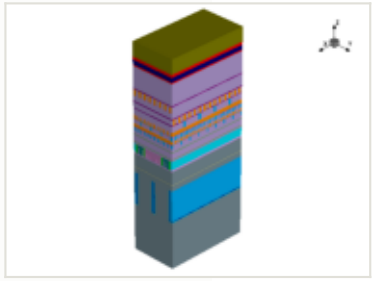
Step #5670: Resist

depo_fill · Resist

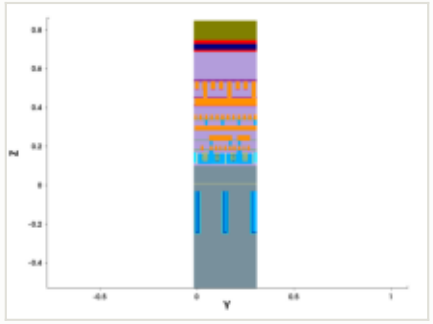
0.169	531.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

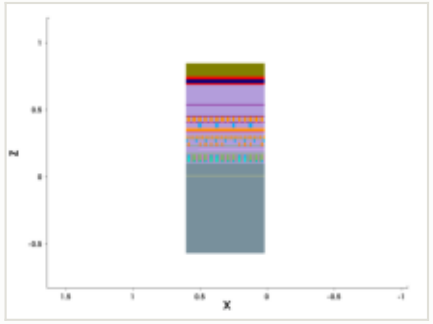
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



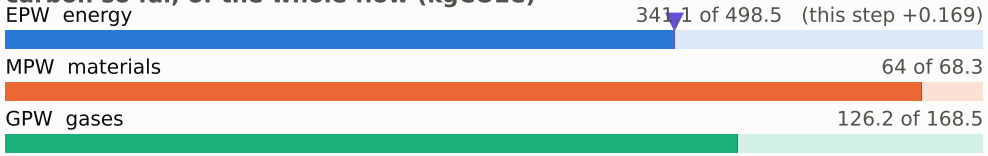
mask for this step

no mask for this step

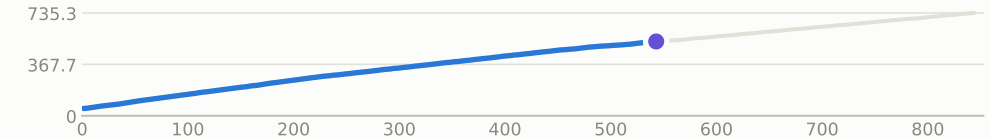
this step

process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	0.1 µm	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

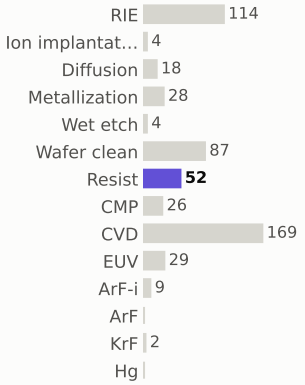
carbon so far, of the whole flow (kgCO2e)



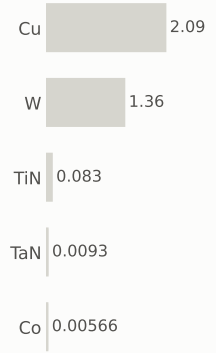
embodied carbon up to each step (kgCO2e per wafer)



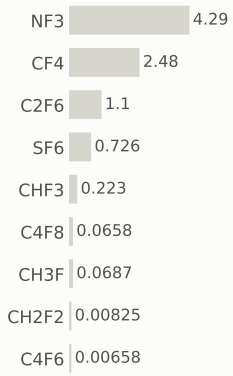
steps by process area



materials deposited (g)



gases emitted (g)



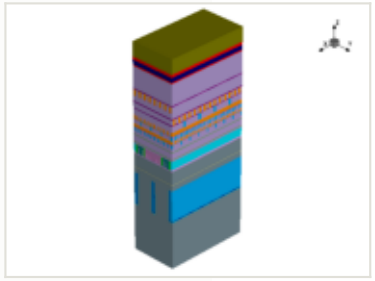
Step #5680: Wafer_clean

clean · Wafer_clean

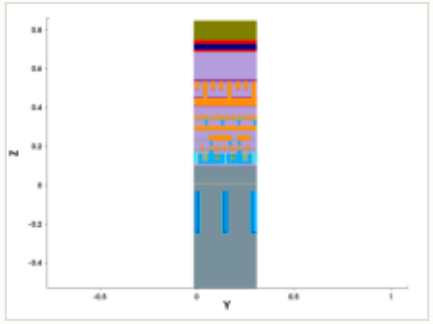
0.169	531.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

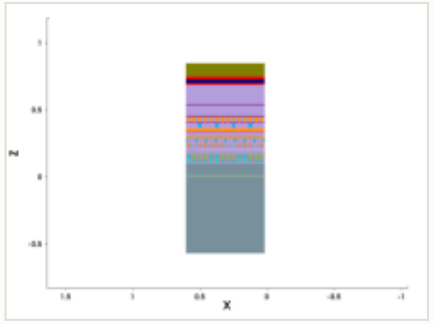
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

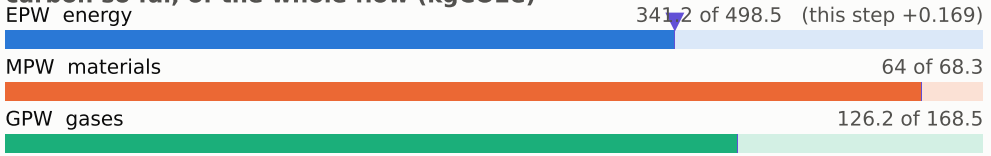
no mask for this step

this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

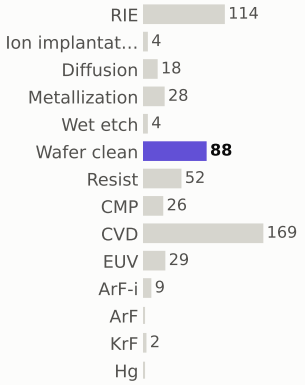
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

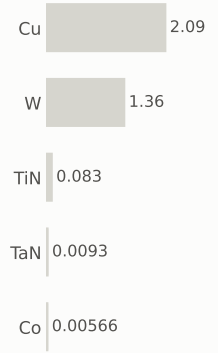


embodied carbon up to each step (kgCO2e per wafer)

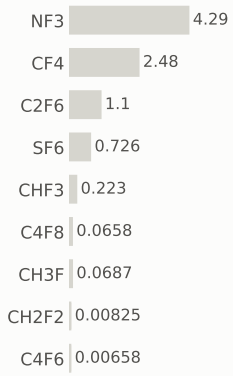
steps by process area



materials deposited (g)



gases emitted (g)



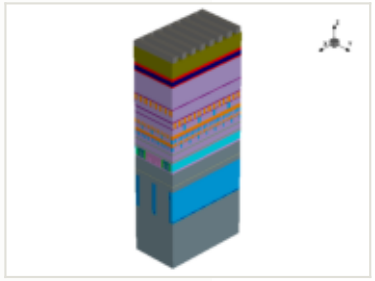
Step #5690: ArF-i

litho_patterning · ArF-i

0.217	531.6	0.029%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

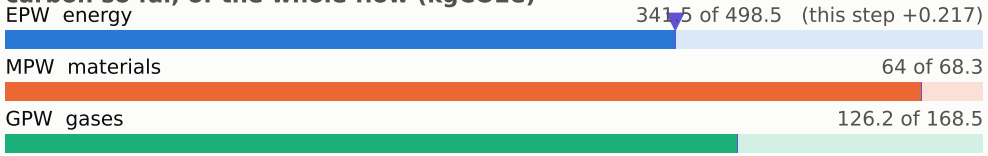


this step

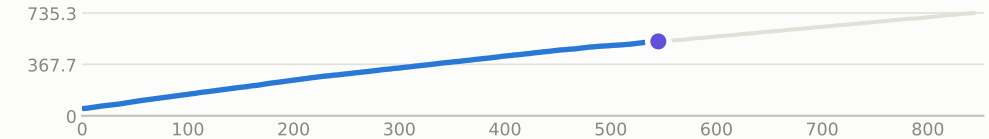
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

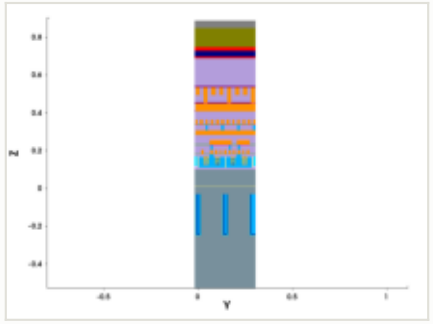
carbon so far, of the whole flow (kgCO2e)



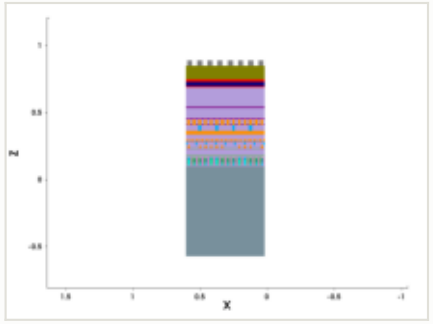
embodied carbon up to each step (kgCO2e per wafer)



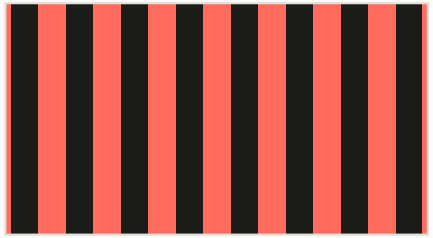
x cut at 0.294 µm



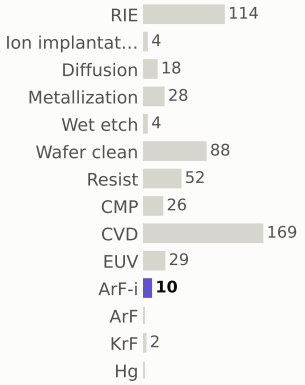
y cut at 0.184 µm



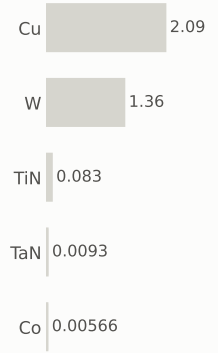
mask: layer 50:0 (GDSII)



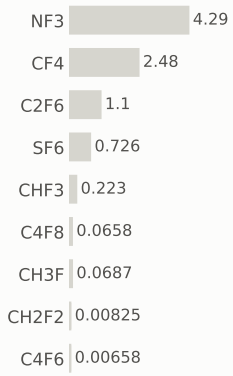
steps by process area



materials deposited (g)



gases emitted (g)



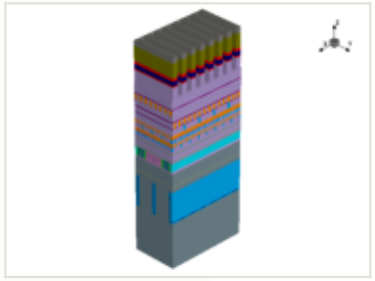
Step #5700: RIE

etch_anisotropic · RIE

1.04	532.7	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

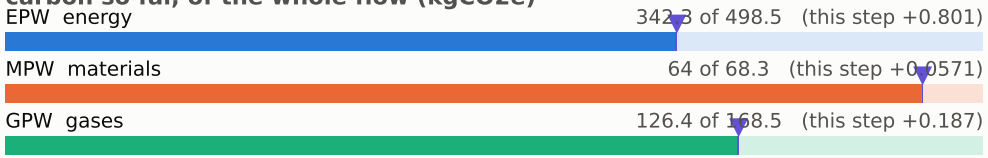


this step

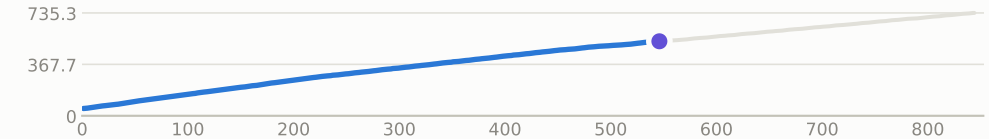
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

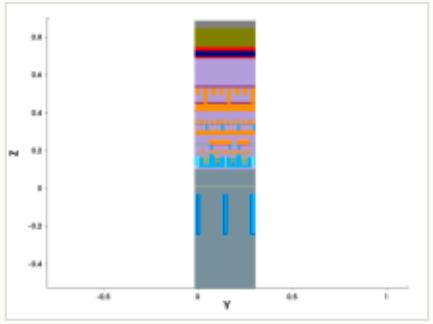
carbon so far, of the whole flow (kgCO2e)



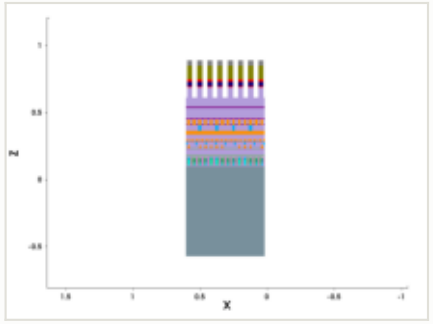
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



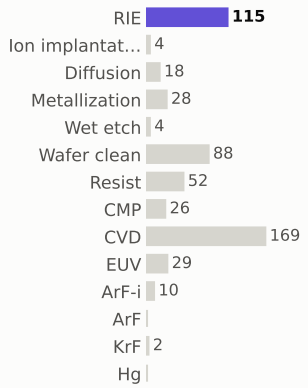
y cut at 0.184 µm



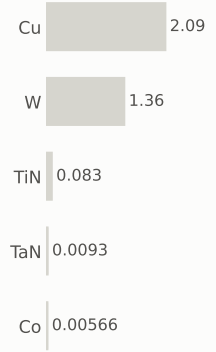
mask for this step

no mask for this step

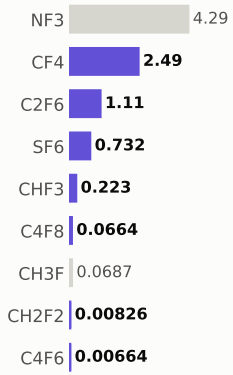
steps by process area



materials deposited (g)



gases emitted (g)



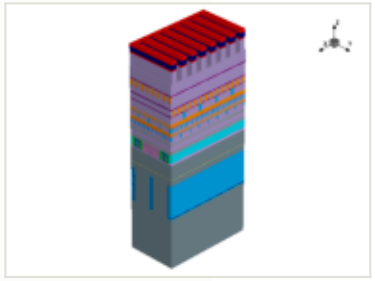
Step #5710: Wafer_clean

etch_strip · Wafer_clean

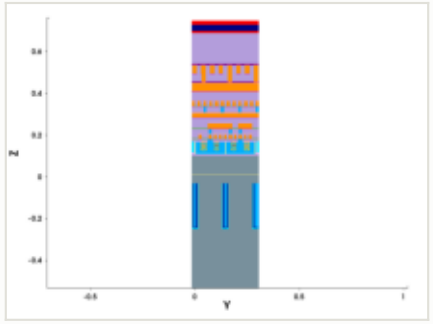
0.169	532.8	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

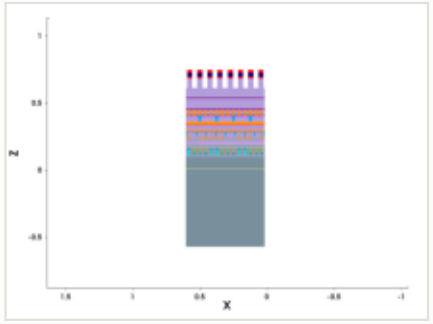
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



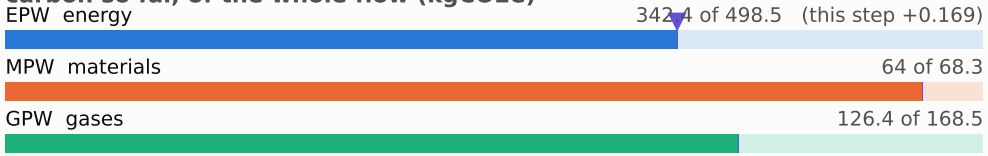
mask for this step

no mask for this step

this step

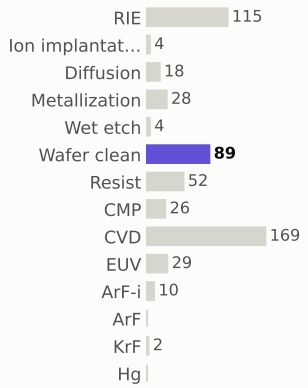
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

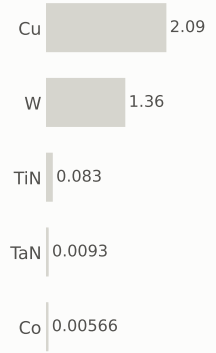


embodied carbon up to each step (kgCO2e per wafer)

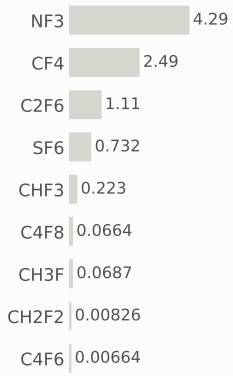
steps by process area



materials deposited (g)



gases emitted (g)



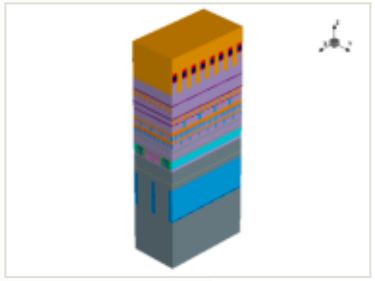
Step #5720: Resist

depo_fill · Resist

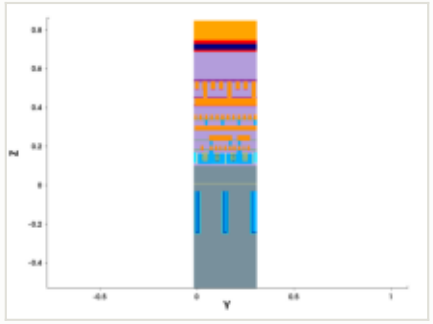
0.169	533.0	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

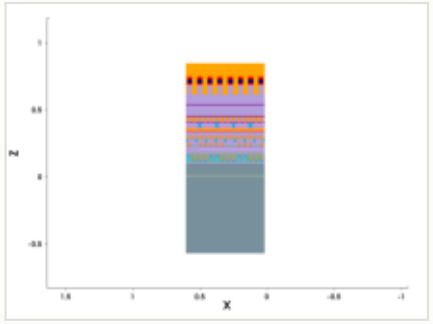
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

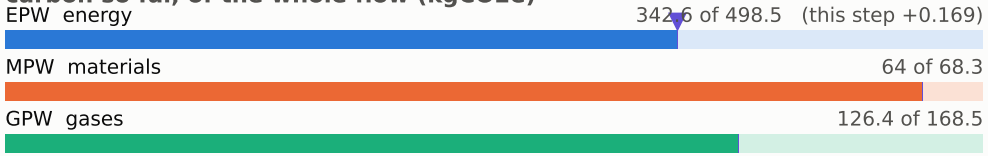
no mask for this step

this step

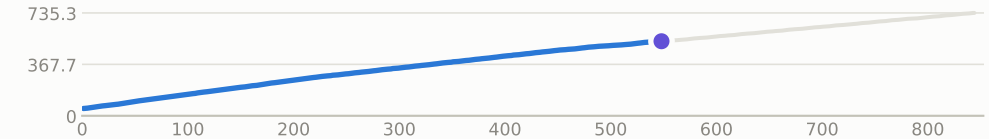
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

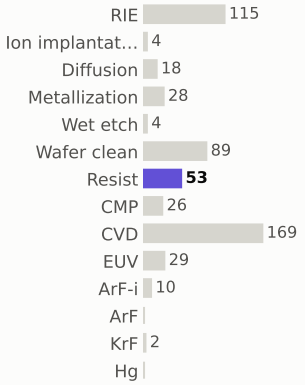
carbon so far, of the whole flow (kgCO2e)



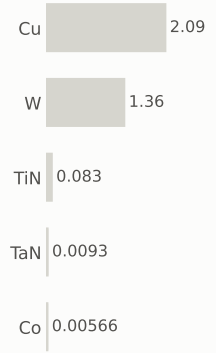
embodied carbon up to each step (kgCO2e per wafer)



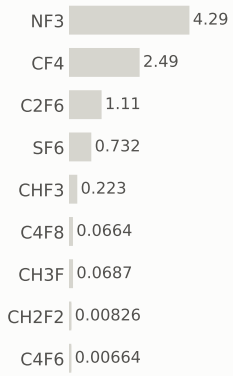
steps by process area



materials deposited (g)



gases emitted (g)



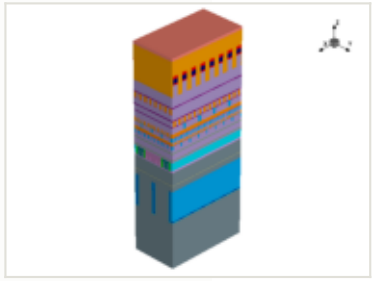
Step #5730: CVD

depo_fill · CVD

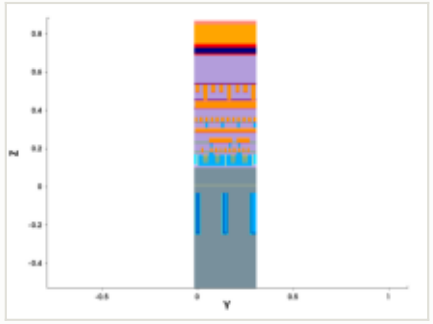
1.31	534.3	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

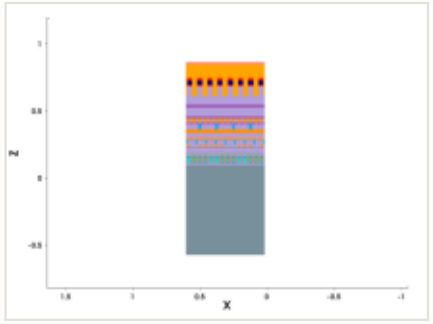
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

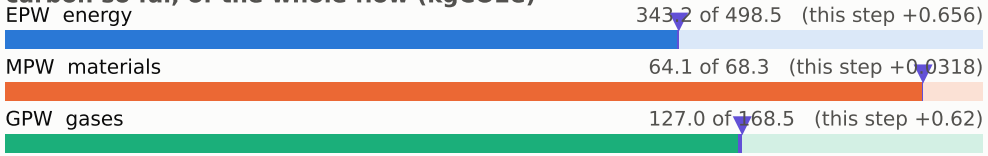
no mask for this step

this step

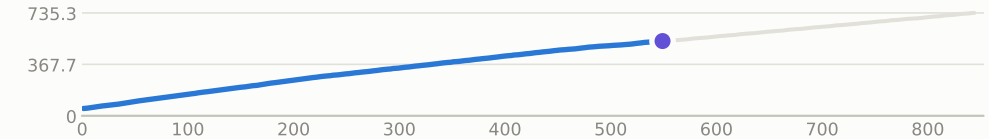
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

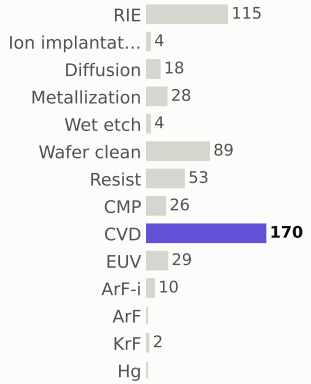
carbon so far, of the whole flow (kgCO2e)



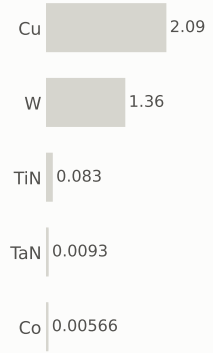
embodied carbon up to each step (kgCO2e per wafer)



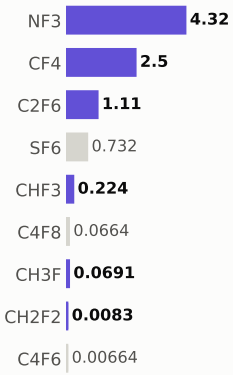
steps by process area



materials deposited (g)



gases emitted (g)



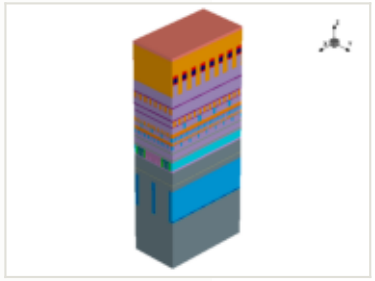
Step #5740: Wafer_clean

clean · Wafer_clean

0.169	534.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



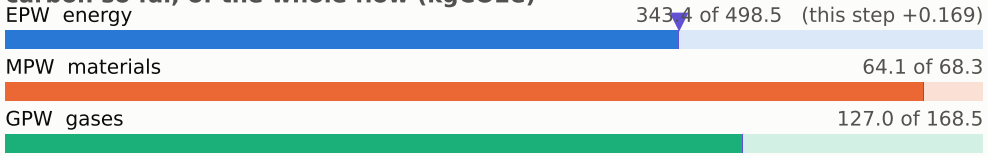
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

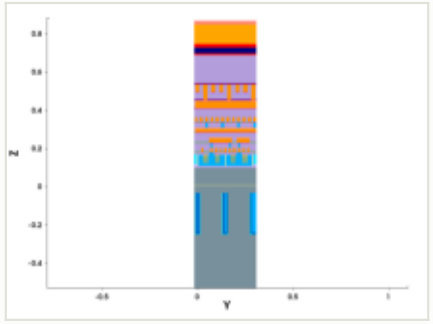
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

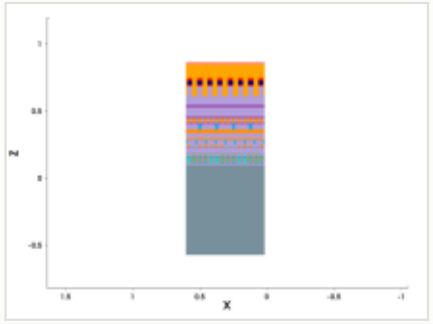


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



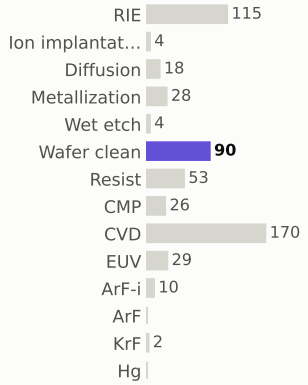
y cut at 0.184 µm



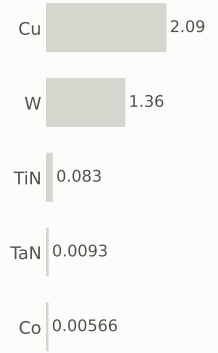
mask for this step

no mask for this step

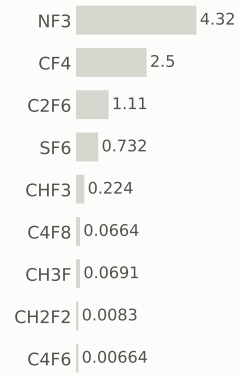
steps by process area



materials deposited (g)



gases emitted (g)



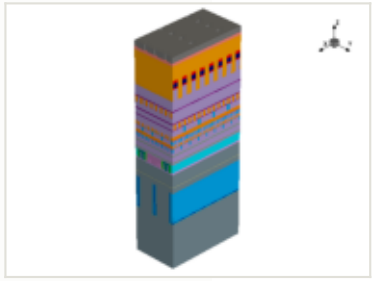
Step #5750: ArF-i

litho_patterning · ArF-i

0.217	534.7	0.029%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

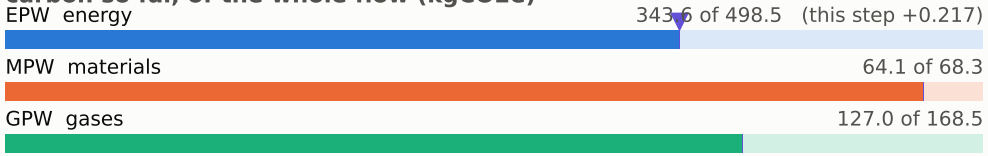


this step

process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

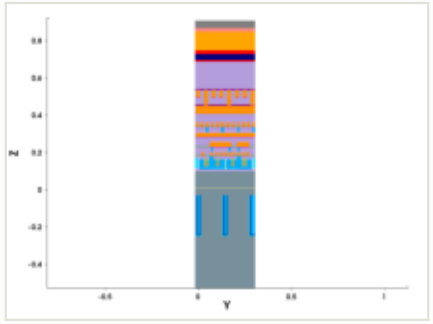
EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

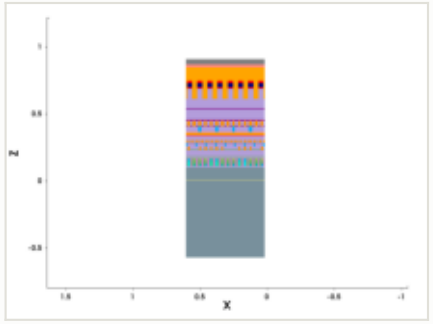


embodied carbon up to each step (kgCO2e per wafer)

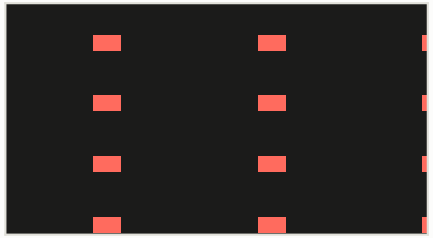
x cut at 0.294 µm



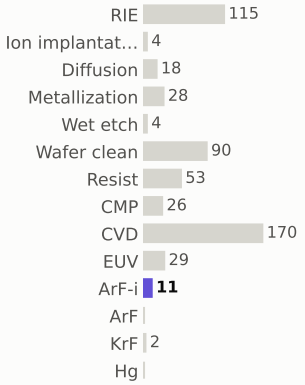
y cut at 0.184 µm



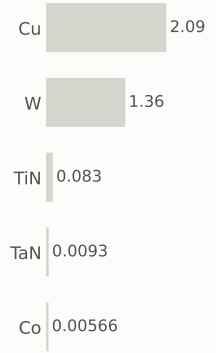
mask: layer 10471:0 (GDSII)



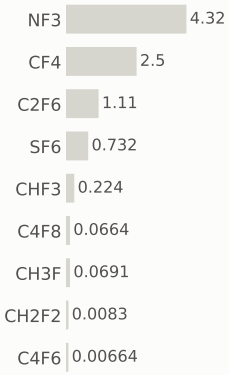
steps by process area



materials deposited (g)



gases emitted (g)



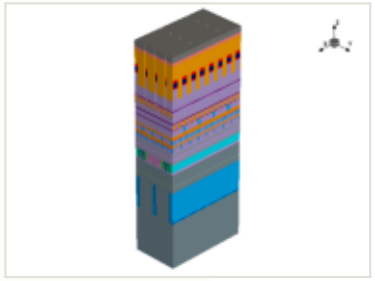
Step #5760: RIE

etch_anisotropic · RIE

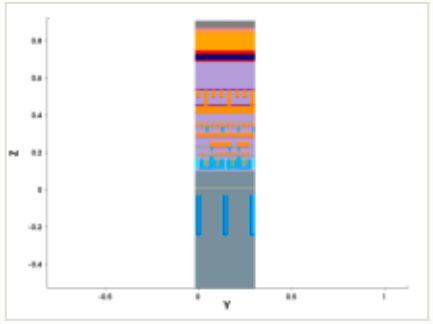
1.04	535.7	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

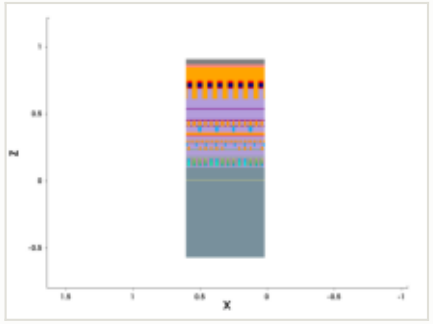
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

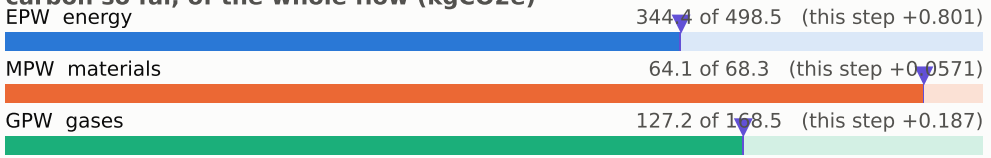
no mask for this step

this step

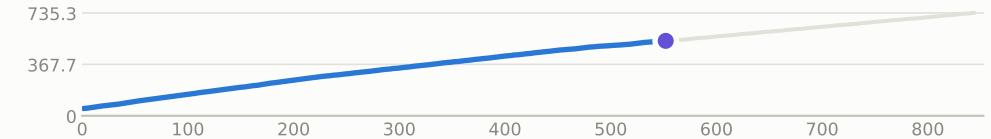
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

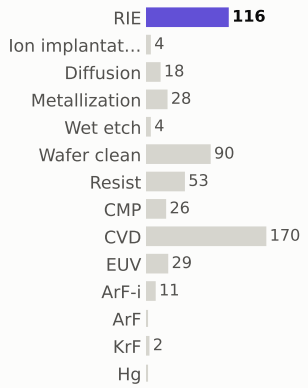
carbon so far, of the whole flow (kgCO2e)



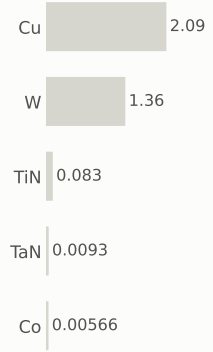
embodied carbon up to each step (kgCO2e per wafer)



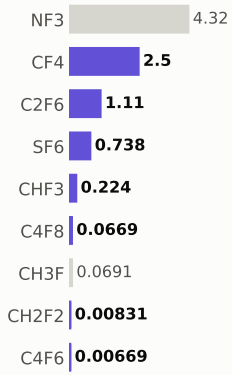
steps by process area



materials deposited (g)



gases emitted (g)



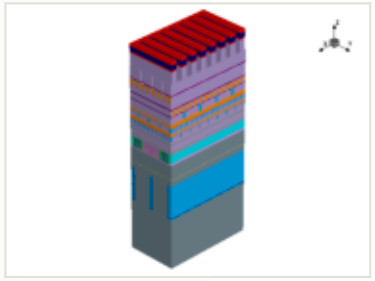
Step #5770: Wafer_clean

etch_strip · Wafer_clean

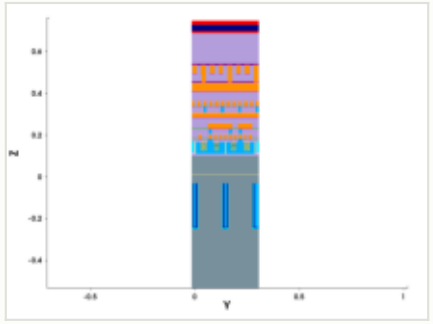
0.169	535.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

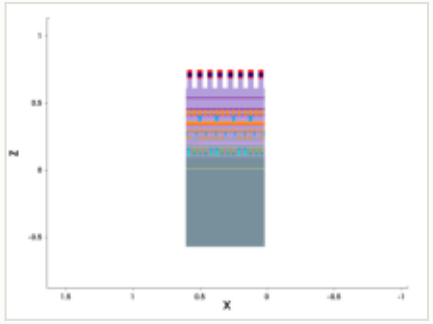
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



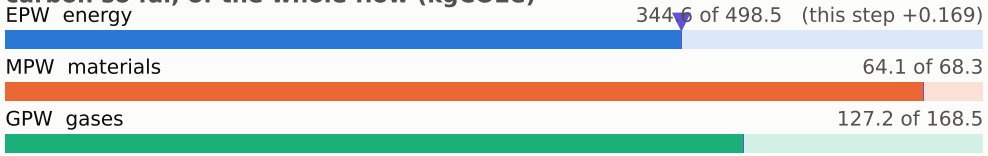
mask for this step

no mask for this step

this step

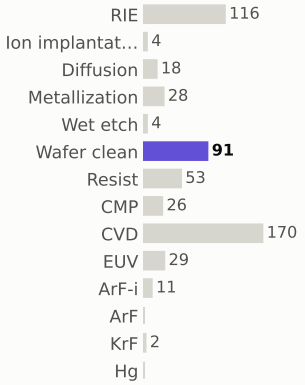
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

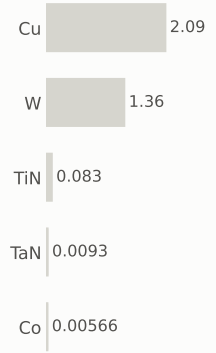


embodied carbon up to each step (kgCO2e per wafer)

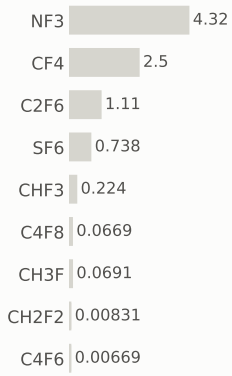
steps by process area



materials deposited (g)



gases emitted (g)



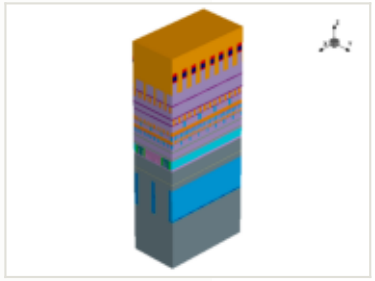
Step #5780: Resist

depo_fill · Resist

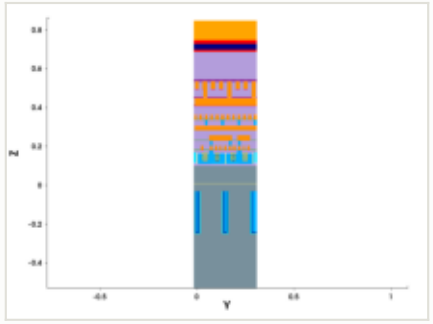
0.169	536.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

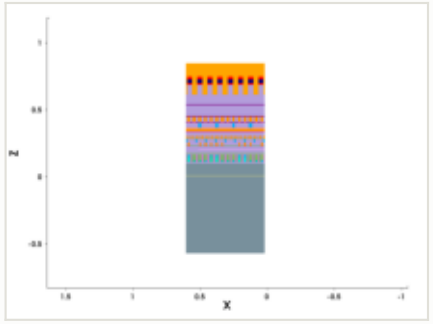
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

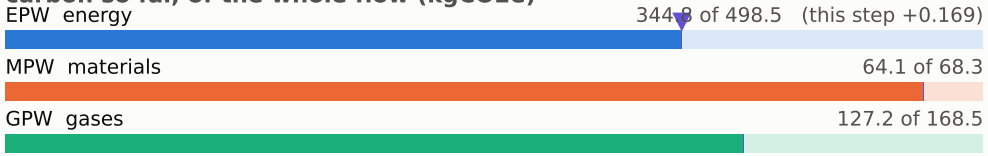
no mask for this step

this step

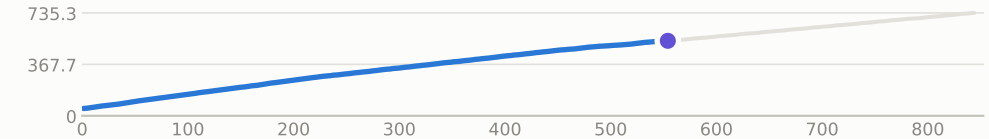
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

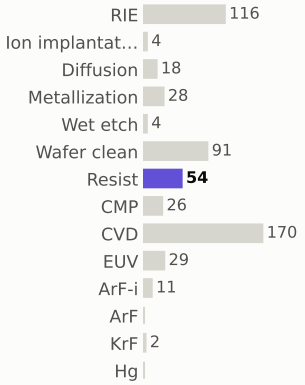
carbon so far, of the whole flow (kgCO2e)



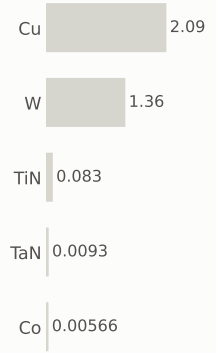
embodied carbon up to each step (kgCO2e per wafer)



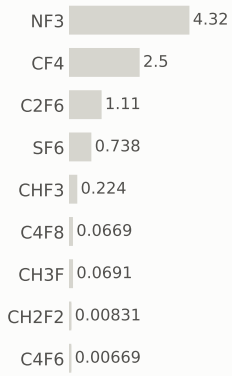
steps by process area



materials deposited (g)



gases emitted (g)



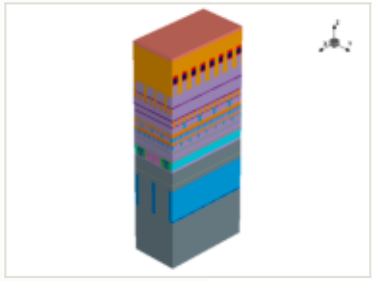
Step #5790: CVD

depo_fill · CVD

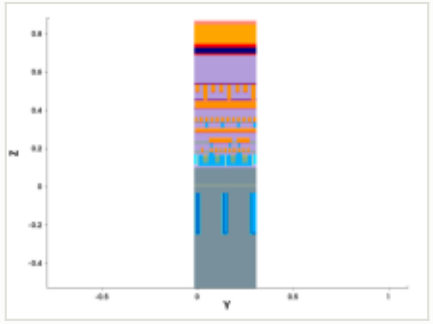
1.31	537.4	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

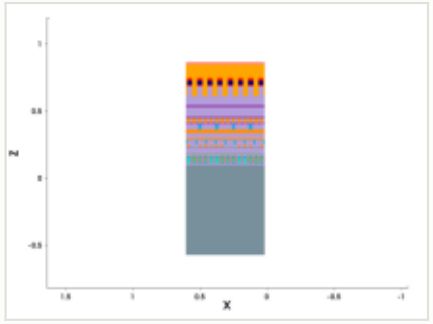
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

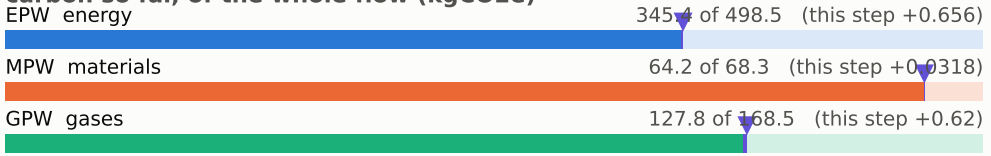
no mask for this step

this step

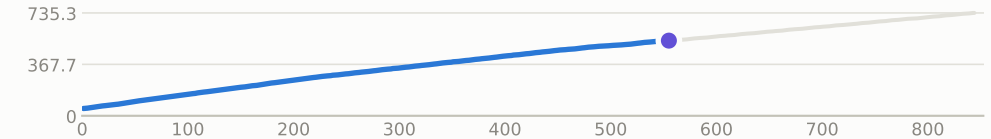
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

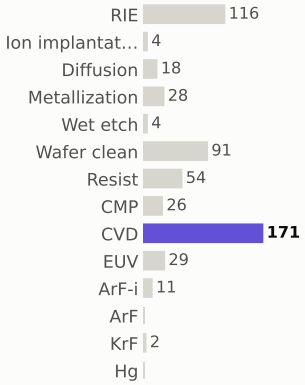
carbon so far, of the whole flow (kgCO2e)



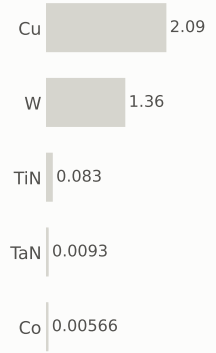
embodied carbon up to each step (kgCO2e per wafer)



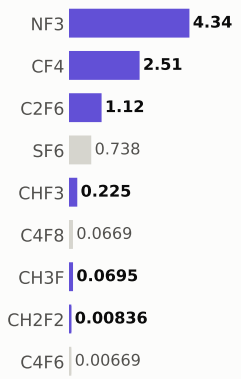
steps by process area



materials deposited (g)



gases emitted (g)



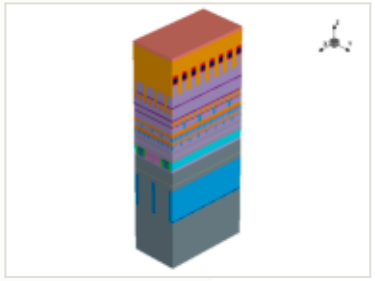
Step #5800: Wafer_clean

clean · Wafer_clean

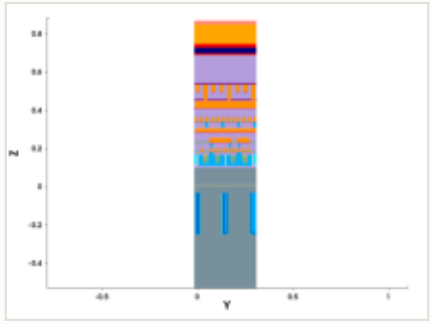
0.169	537.6	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

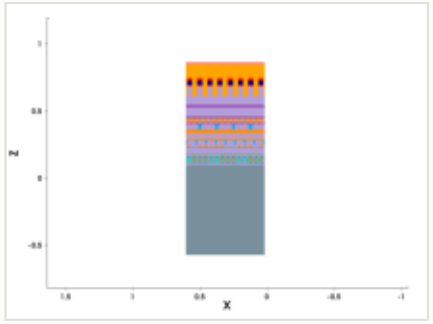
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

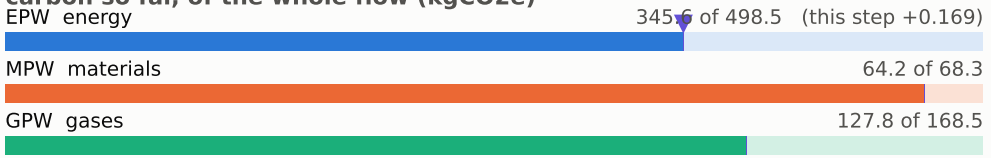
no mask for this step

this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

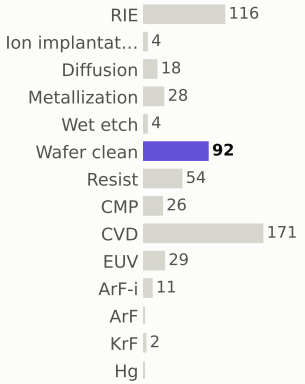
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

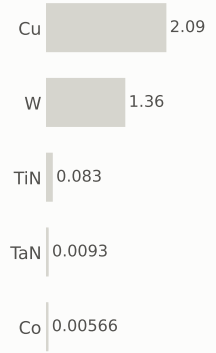


embodied carbon up to each step (kgCO2e per wafer)

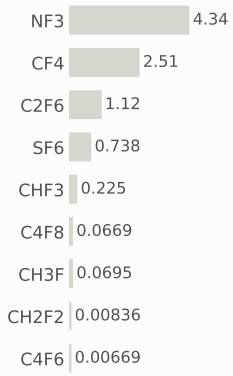
steps by process area



materials deposited (g)



gases emitted (g)



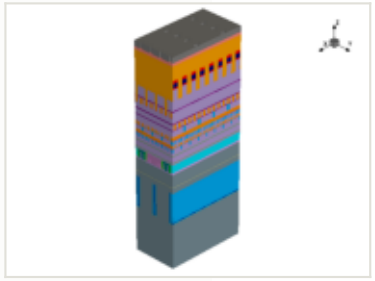
Step #5810: ArF-i

litho_patterning · ArF-i

0.217	537.8	0.029%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

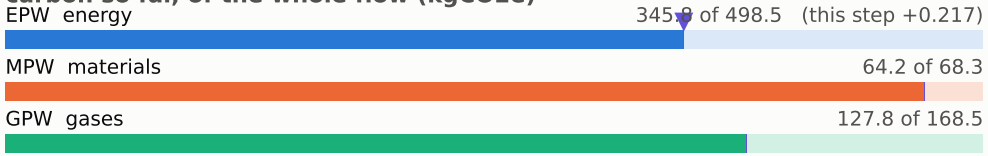


this step

process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

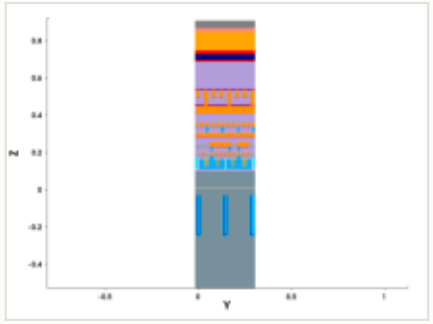
EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

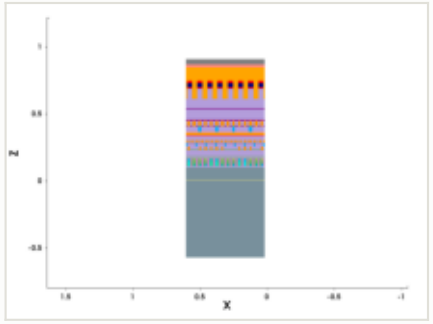


embodied carbon up to each step (kgCO2e per wafer)

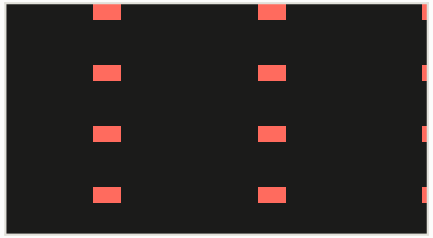
x cut at 0.294 µm



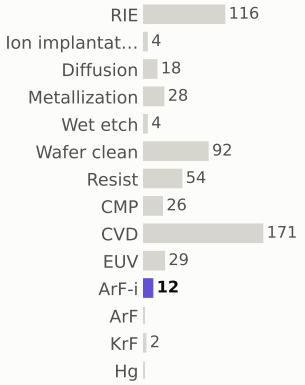
y cut at 0.184 µm



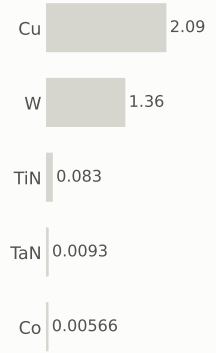
mask: layer 10472:0 (GDSII)



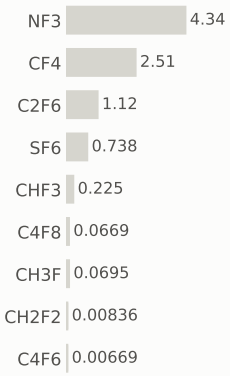
steps by process area



materials deposited (g)



gases emitted (g)



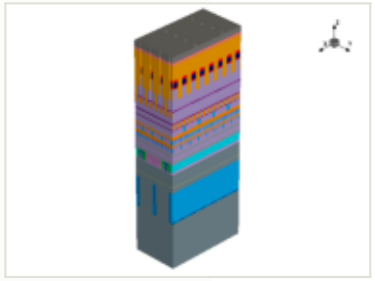
Step #5820: RIE

etch_anisotropic · RIE

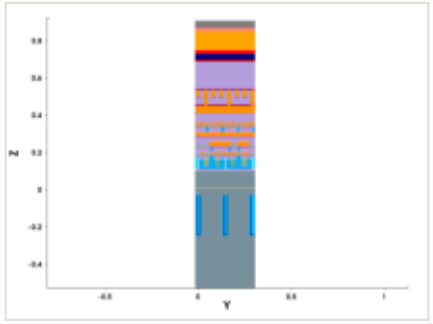
1.04	538.8	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

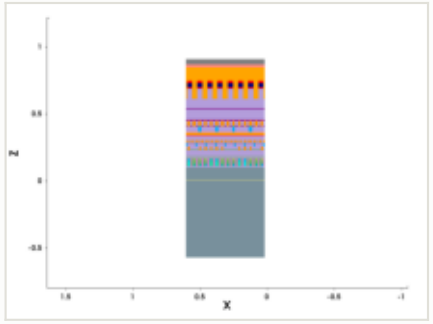
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

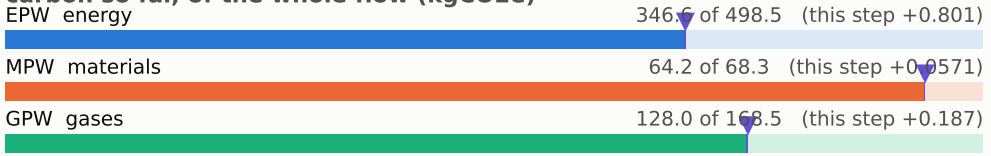
no mask for this step

this step

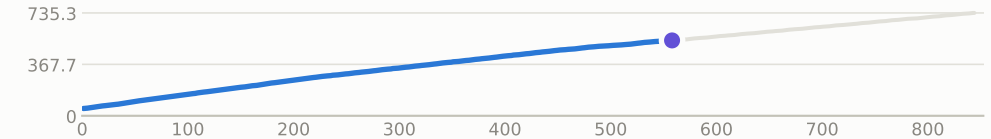
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

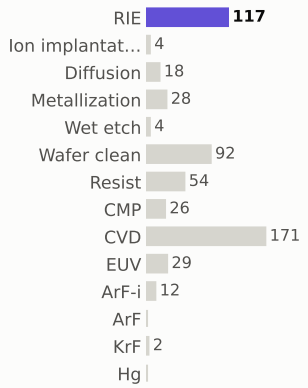
carbon so far, of the whole flow (kgCO2e)



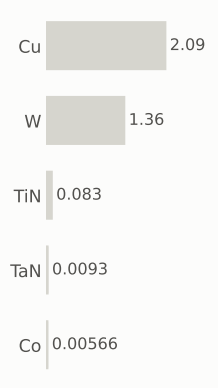
embodied carbon up to each step (kgCO2e per wafer)



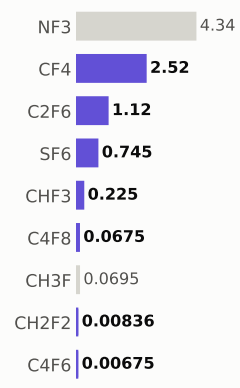
steps by process area



materials deposited (g)



gases emitted (g)



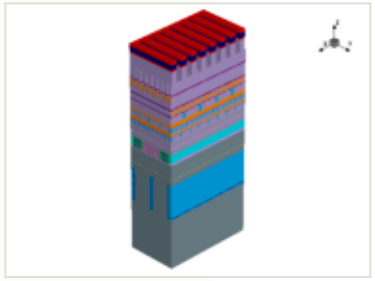
Step #5830: Wafer_clean

etch_strip · Wafer_clean

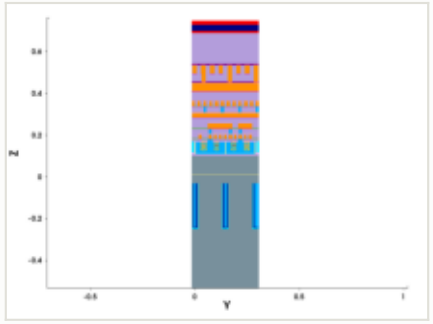
0.169	539.0	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

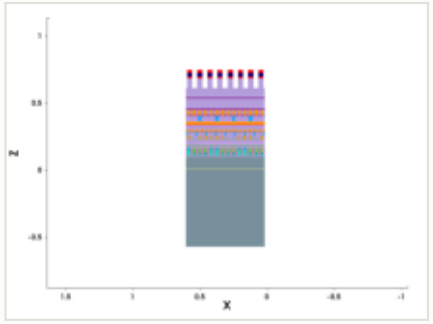
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



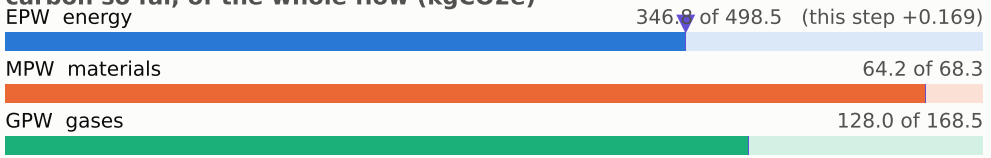
mask for this step

no mask for this step

this step

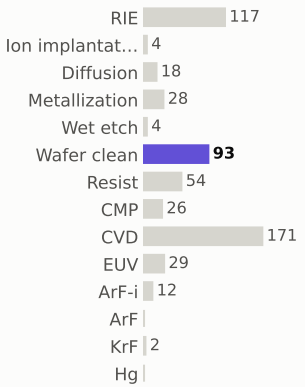
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

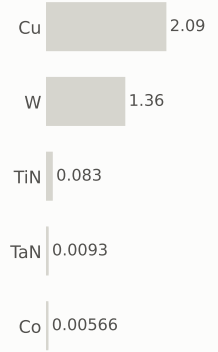


embodied carbon up to each step (kgCO2e per wafer)

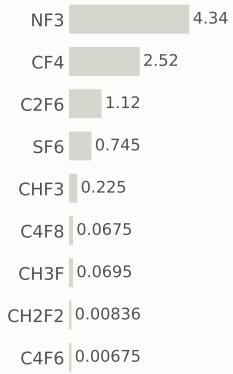
steps by process area



materials deposited (g)



gases emitted (g)



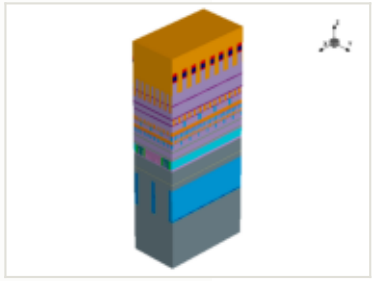
Step #5840: Resist

depo_fill · Resist

0.169	539.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

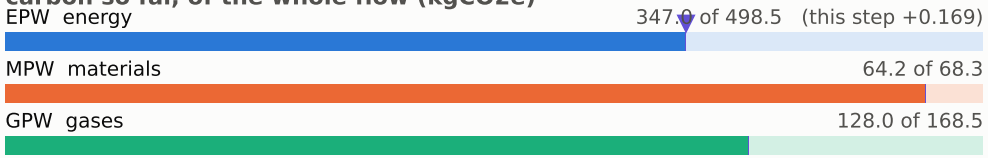


this step

process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

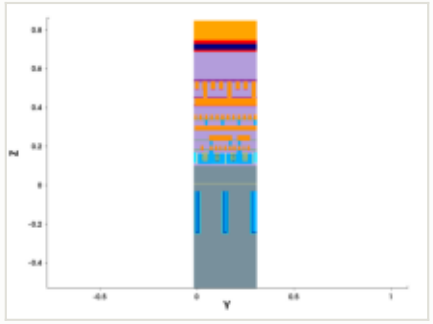
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

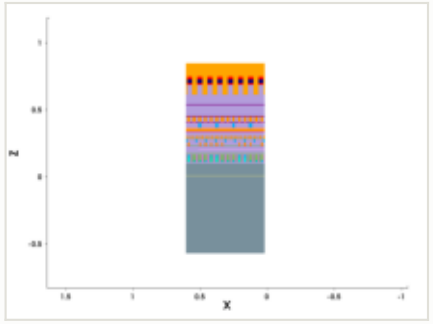


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



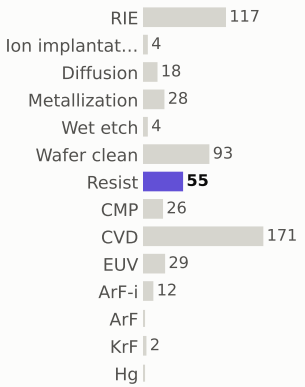
y cut at 0.184 µm



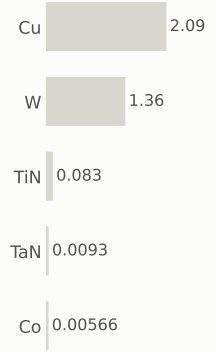
mask for this step

no mask for this step

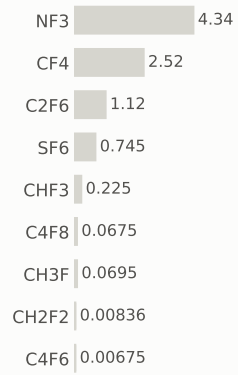
steps by process area



materials deposited (g)



gases emitted (g)



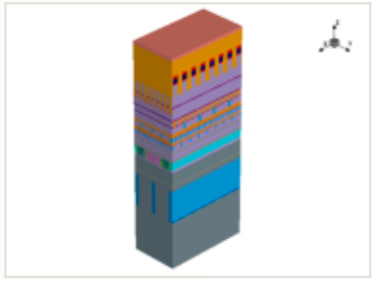
Step #5850: CVD

depo_fill · CVD

1.31	540.5	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

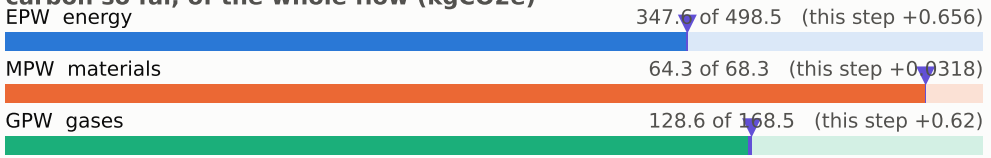


this step

process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

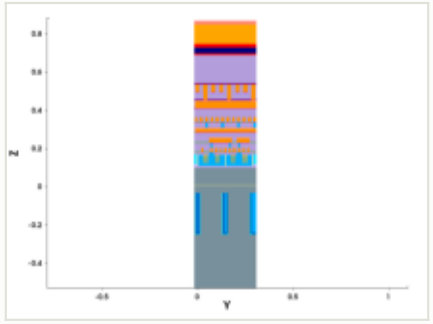
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

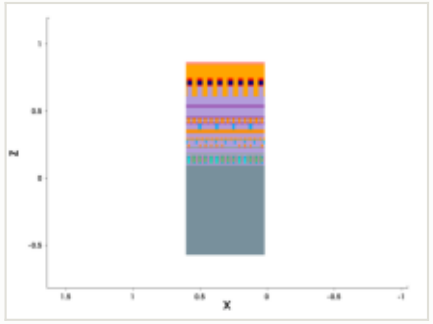


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



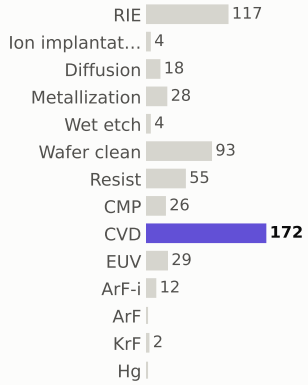
y cut at 0.184 µm



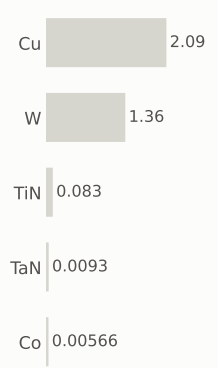
mask for this step

no mask for this step

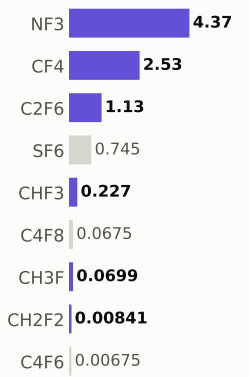
steps by process area



materials deposited (g)



gases emitted (g)



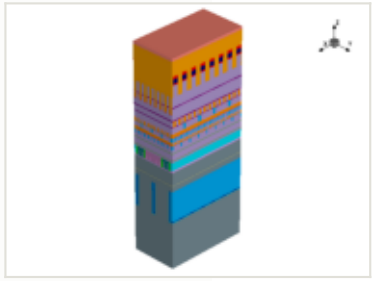
Step #5860: Wafer_clean

clean · Wafer_clean

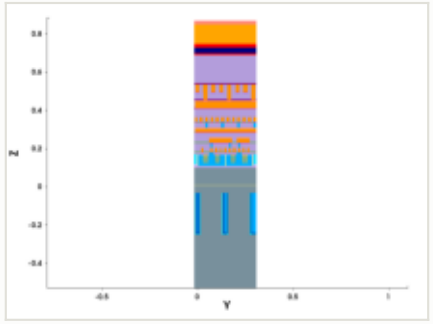
0.169	540.6	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

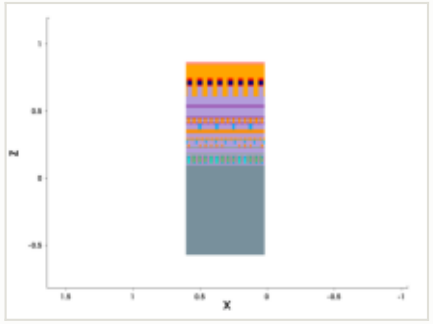
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

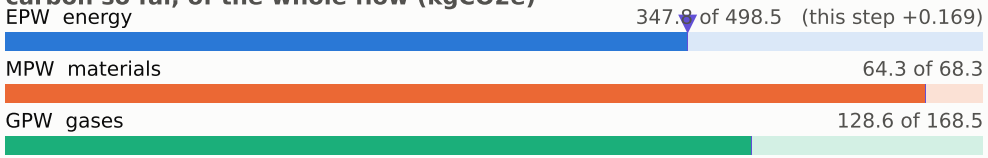
no mask for this step

this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

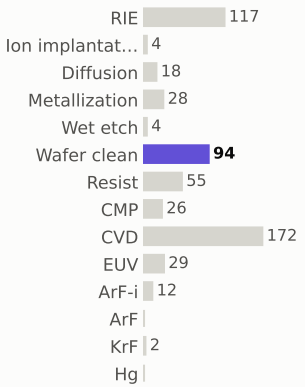
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

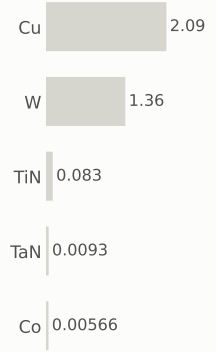


embodied carbon up to each step (kgCO2e per wafer)

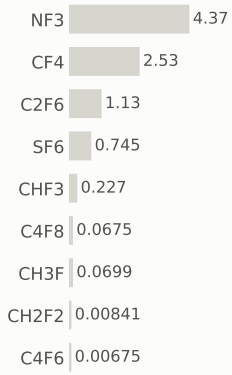
steps by process area



materials deposited (g)



gases emitted (g)



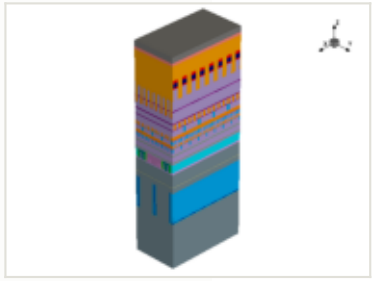
Step #5870: ArF-i

litho_patterning · ArF-i

0.217	540.9	0.029%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

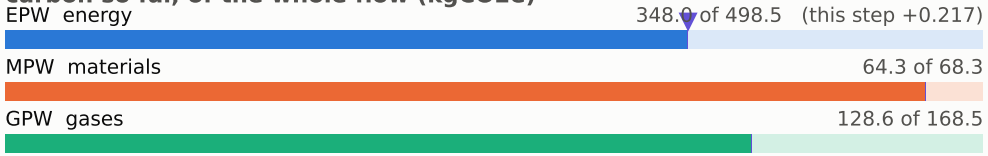


this step

process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

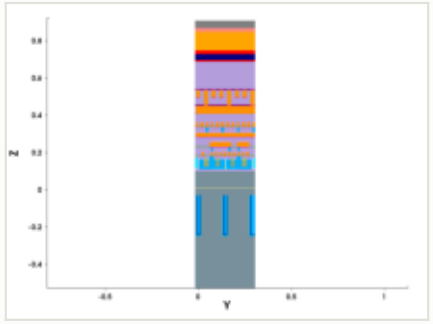
EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

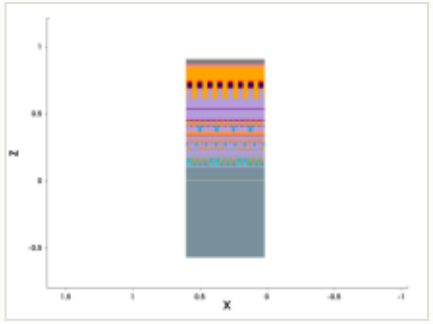


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



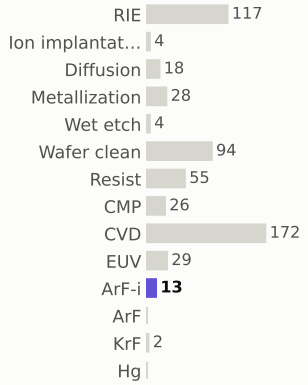
y cut at 0.184 µm



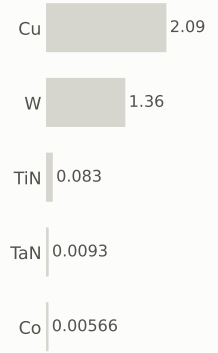
mask: layer 10473:0 (GDSII)

nothing in the clip window

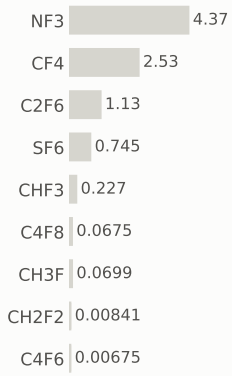
steps by process area



materials deposited (g)



gases emitted (g)



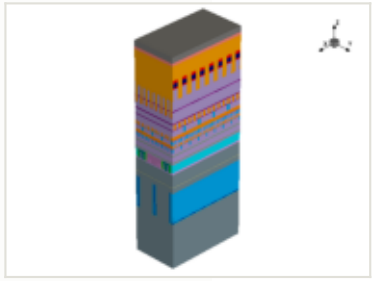
Step #5880: RIE

etch_anisotropic · RIE

1.04	541.9	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

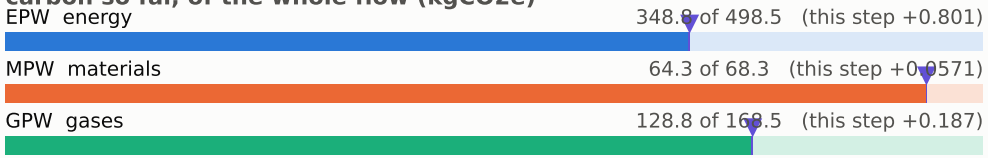


this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

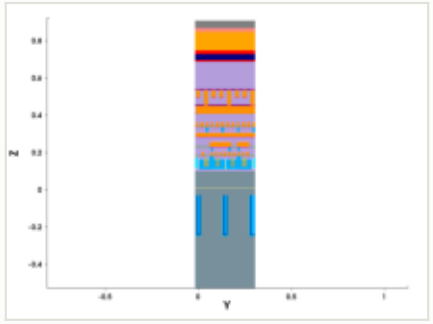
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

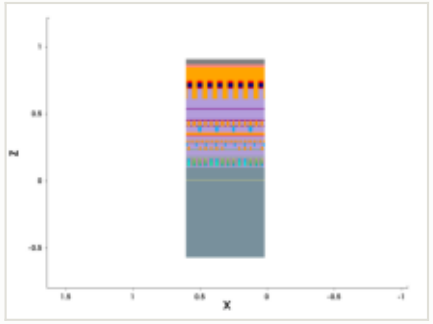


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



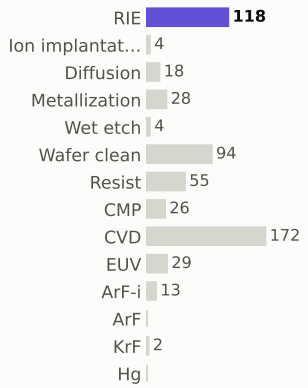
y cut at 0.184 µm



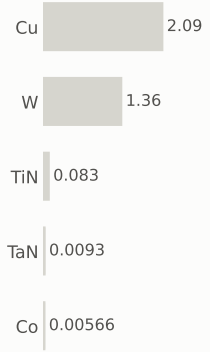
mask for this step

no mask for this step

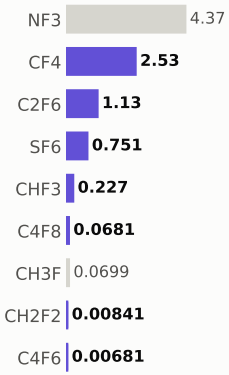
steps by process area



materials deposited (g)



gases emitted (g)



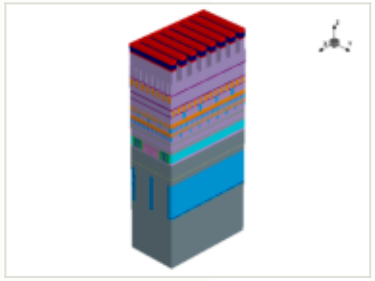
Step #5890: Wafer_clean

etch_strip · Wafer_clean

0.169	542.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

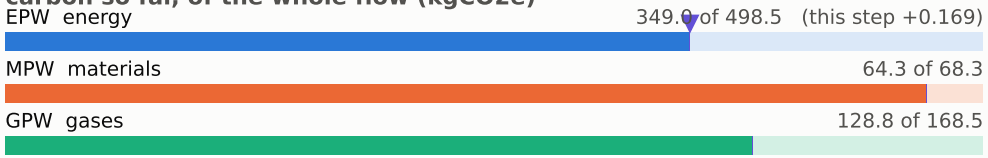


this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

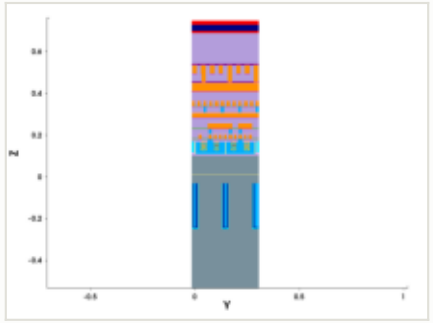
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

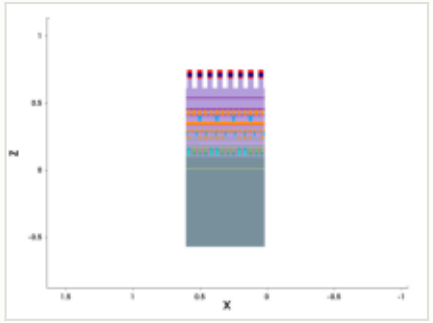


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



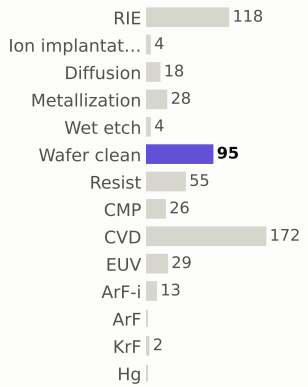
y cut at 0.184 µm



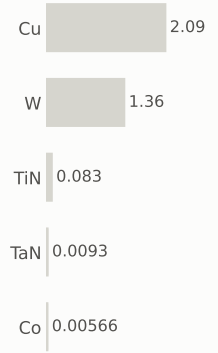
mask for this step

no mask for this step

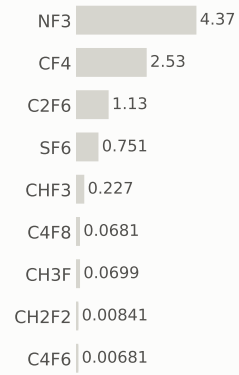
steps by process area



materials deposited (g)



gases emitted (g)



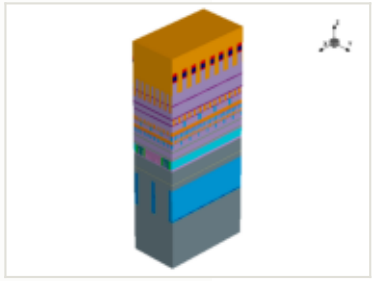
Step #5900: Resist

depo_fill · Resist

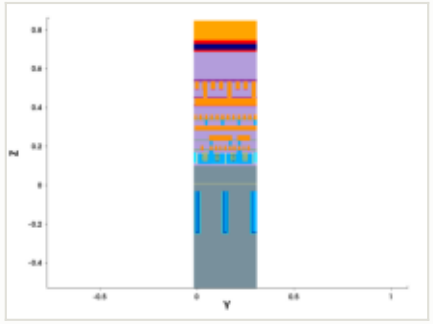
0.169	542.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

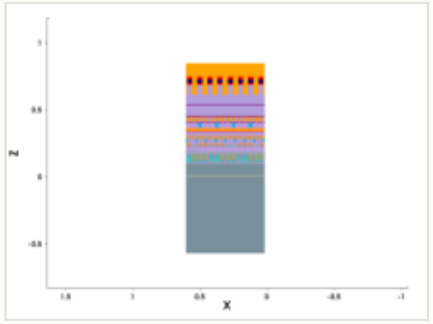
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



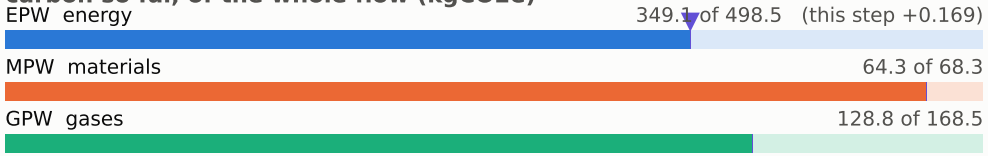
mask for this step

no mask for this step

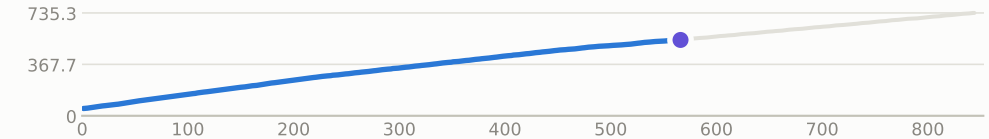
this step

process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	0.1 µm	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

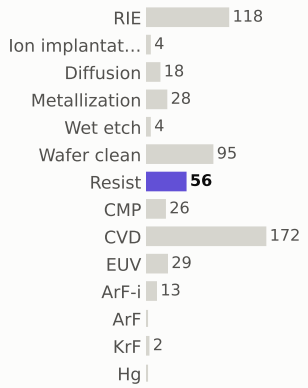
carbon so far, of the whole flow (kgCO2e)



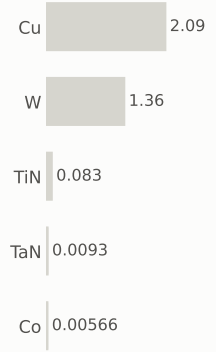
embodied carbon up to each step (kgCO2e per wafer)



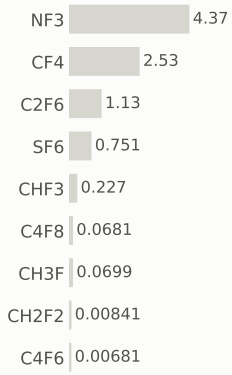
steps by process area



materials deposited (g)



gases emitted (g)



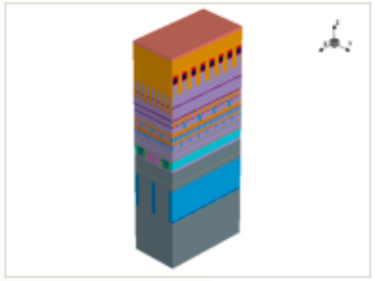
Step #5910: CVD

depo_fill · CVD

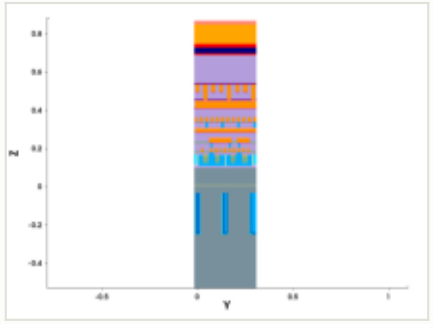
1.31	543.5	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

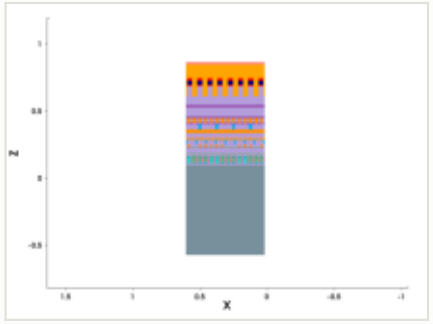
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

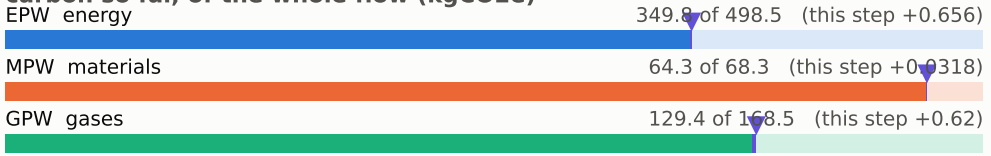
no mask for this step

this step

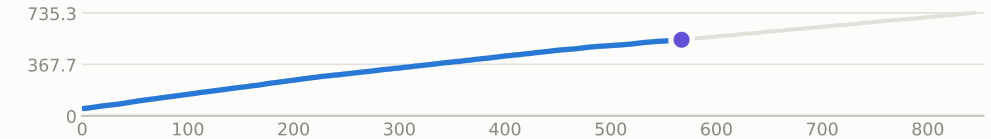
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

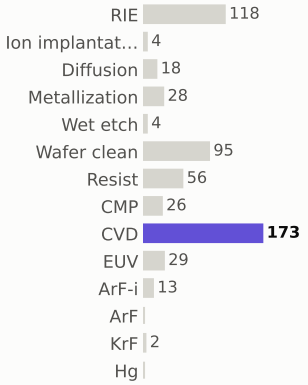
carbon so far, of the whole flow (kgCO2e)



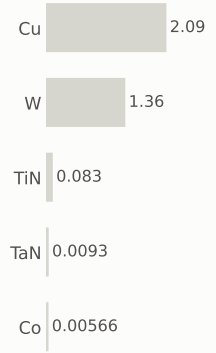
embodied carbon up to each step (kgCO2e per wafer)



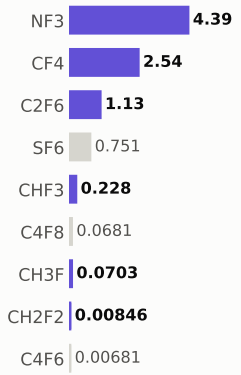
steps by process area



materials deposited (g)



gases emitted (g)



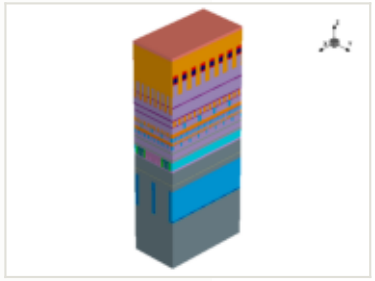
Step #5920: Wafer_clean

clean · Wafer_clean

0.169	543.7	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



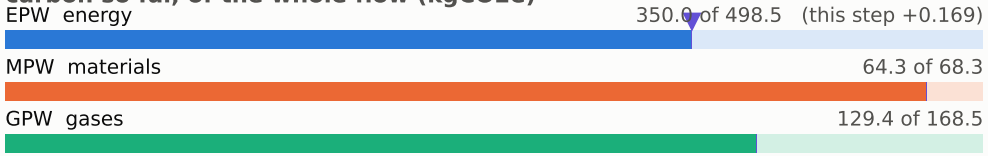
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

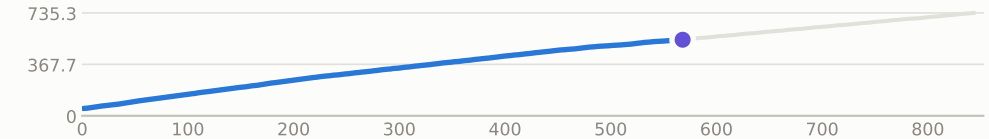
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

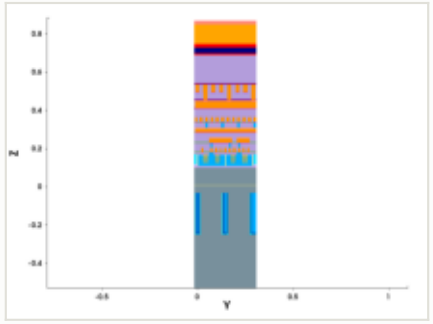
carbon so far, of the whole flow (kgCO2e)



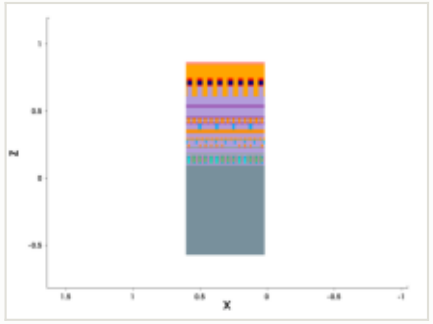
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



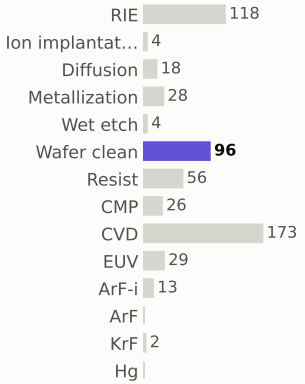
y cut at 0.184 µm



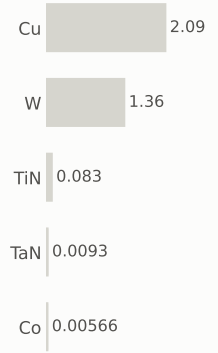
mask for this step

no mask for this step

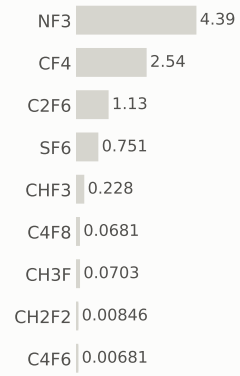
steps by process area



materials deposited (g)



gases emitted (g)



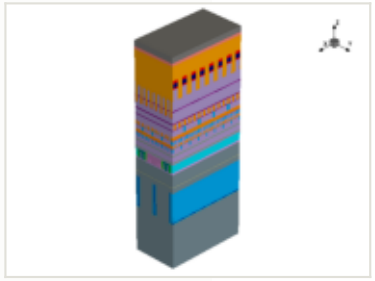
Step #5930: ArF-i

litho_patterning · ArF-i

0.217	543.9	0.029%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

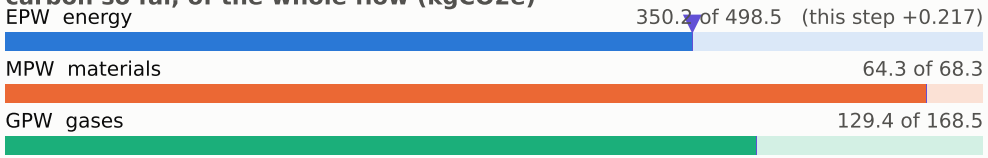


this step

process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

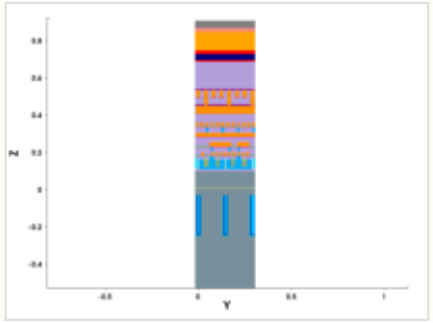
EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

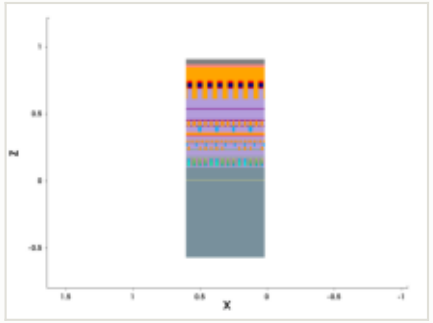


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



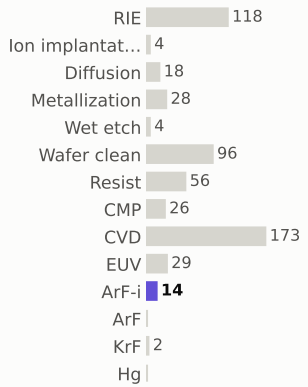
y cut at 0.184 µm



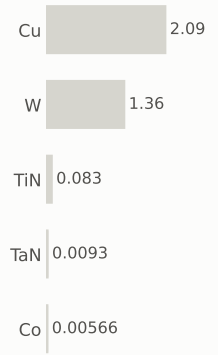
mask: layer 10474:0 (GDSII)

nothing in the clip window

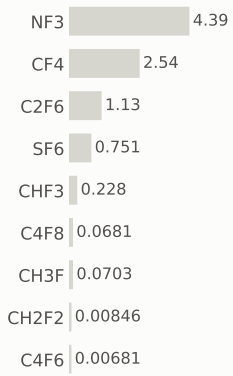
steps by process area



materials deposited (g)



gases emitted (g)



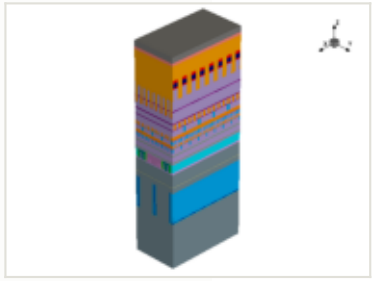
Step #5940: RIE

etch_anisotropic · RIE

1.04	545.0	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

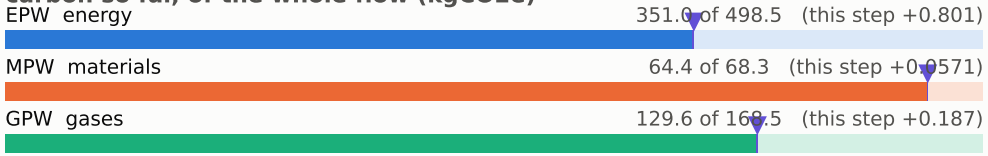


this step

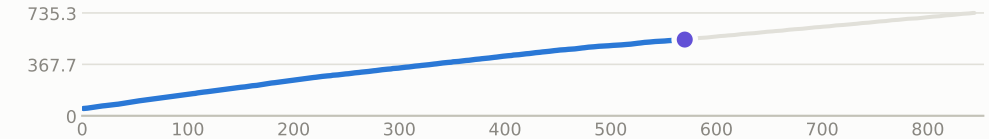
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

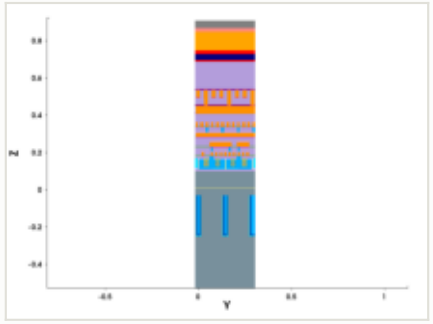
carbon so far, of the whole flow (kgCO2e)



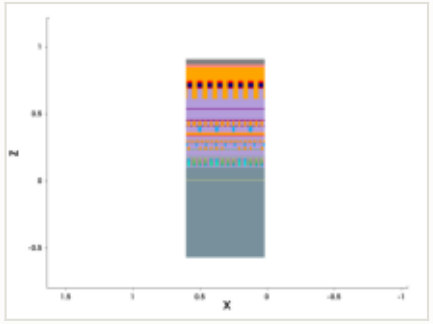
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



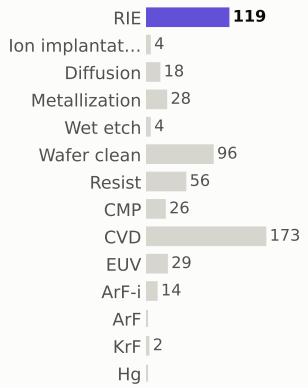
y cut at 0.184 µm



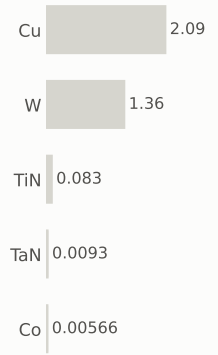
mask for this step

no mask for this step

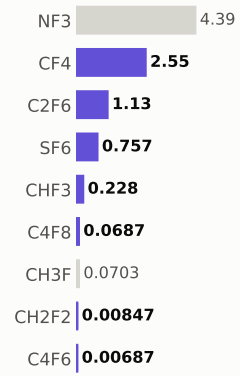
steps by process area



materials deposited (g)



gases emitted (g)



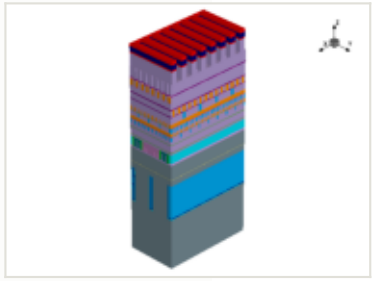
Step #5950: Wafer_clean

etch_strip · Wafer_clean

0.169	545.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

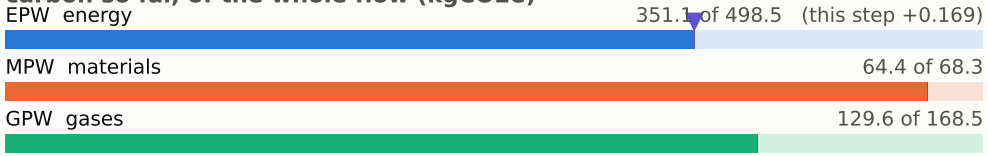


this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

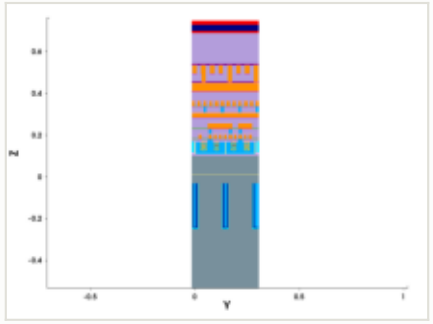
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

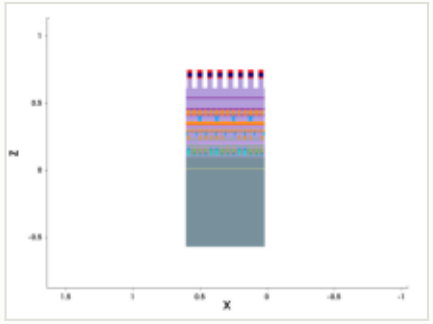


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



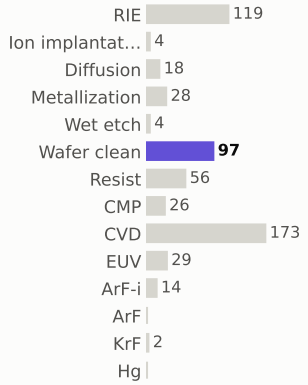
y cut at 0.184 µm



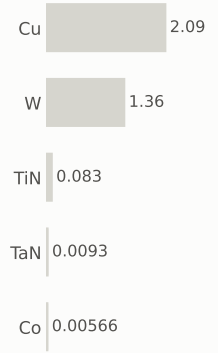
mask for this step

no mask for this step

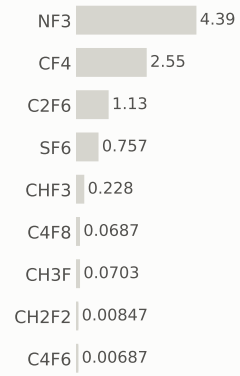
steps by process area



materials deposited (g)



gases emitted (g)



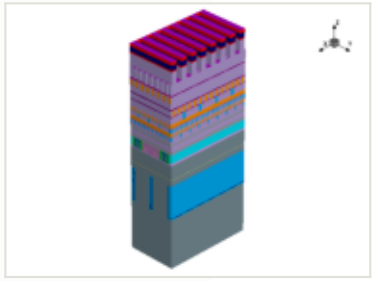
Step #5960: Metallization

depo_ellipsoidal · Metallization

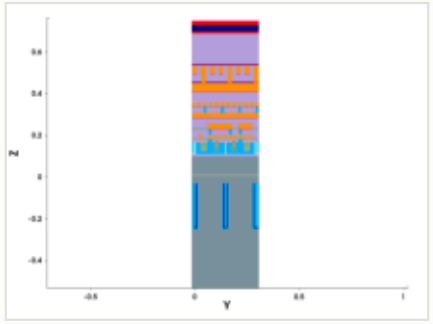
1.2	546.3	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

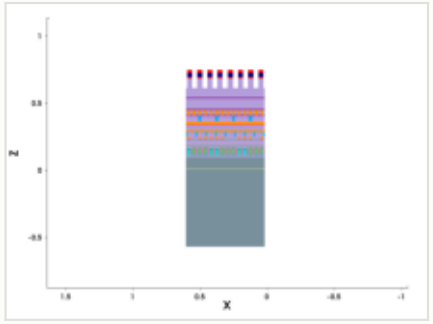
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

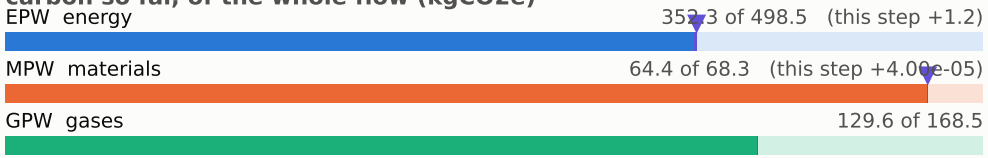
no mask for this step

this step

process area	Metallization
material	TaN
thickness	0.0016 µm
material GWP	25.8159 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

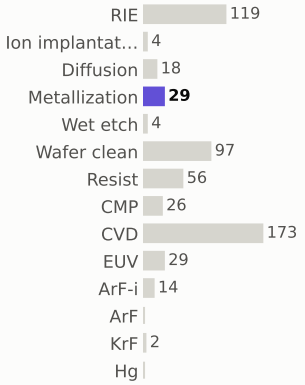
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.00e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

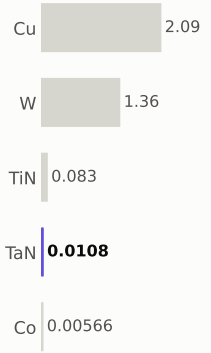


embodied carbon up to each step (kgCO2e per wafer)

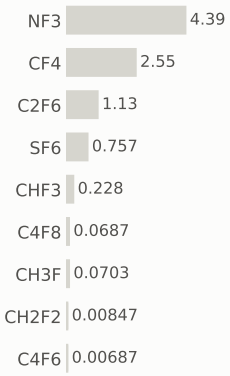
steps by process area



materials deposited (g)



gases emitted (g)



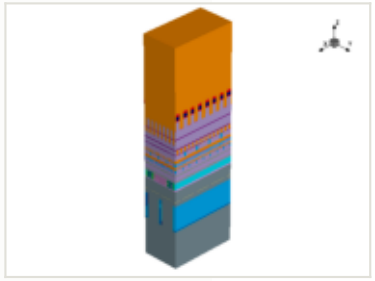
Step #5970: Metallization

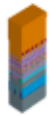
depo_fill · Metallization

1.21	547.6	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



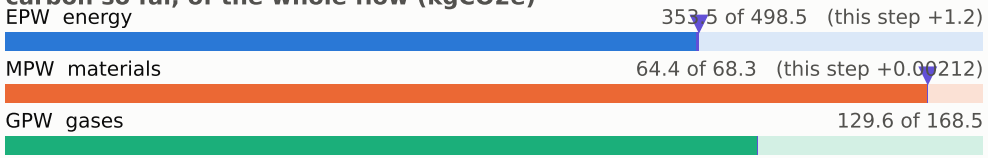


this step

process area	Metallization
material	Cu
thickness	0.55 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

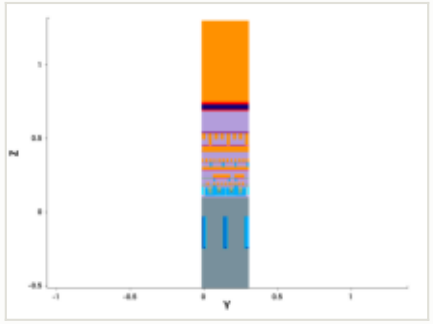
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00212 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

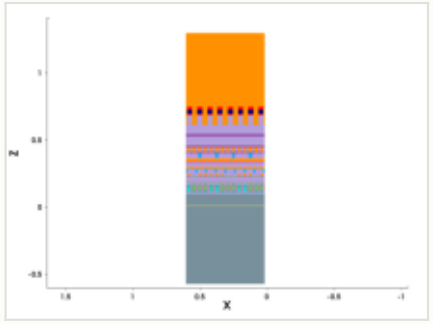


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



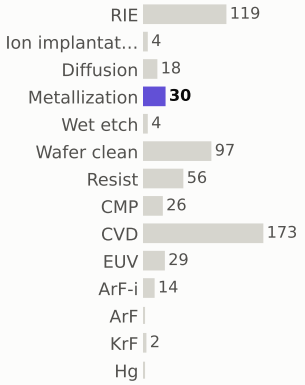
y cut at 0.184 µm



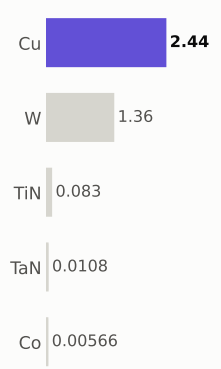
mask for this step

no mask for this step

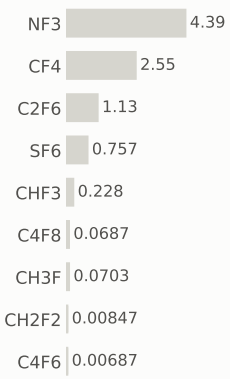
steps by process area



materials deposited (g)



gases emitted (g)



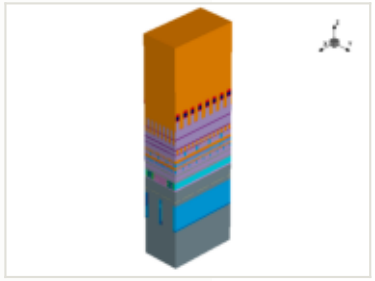
Step #5980: Diffusion

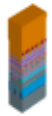
treatment · Diffusion

0.801	548.4	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

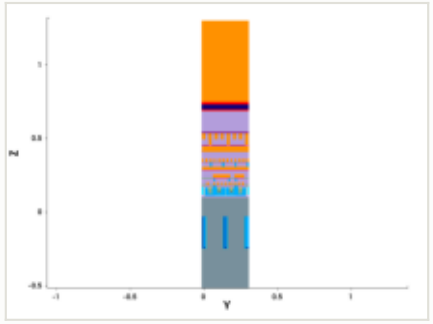
3D structure after this step

zoomed to fit

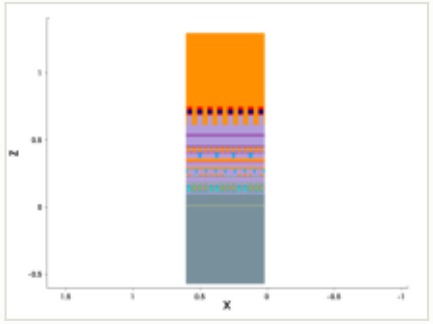




x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

no mask for this step

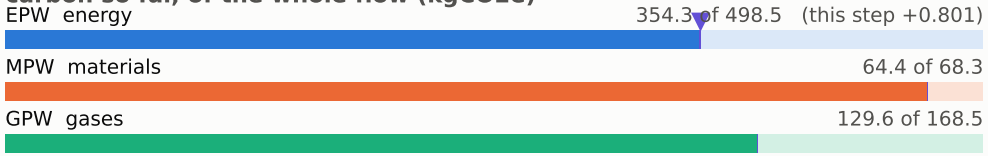
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

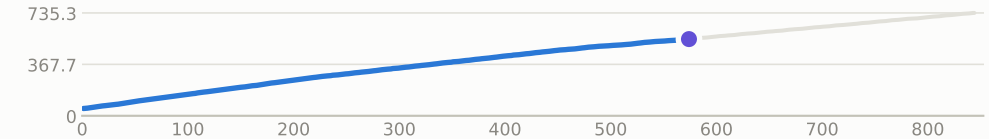
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

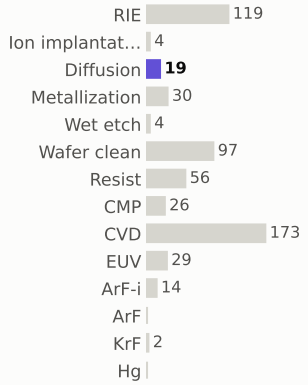
carbon so far, of the whole flow (kgCO2e)



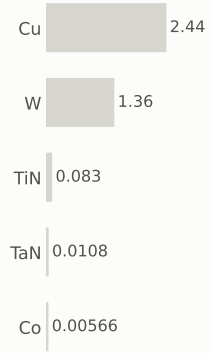
embodied carbon up to each step (kgCO2e per wafer)



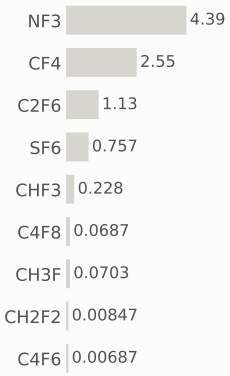
steps by process area



materials deposited (g)



gases emitted (g)



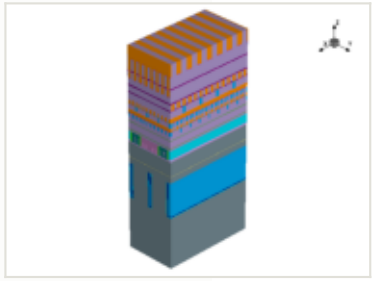
Step #5990: CMP

etch_cmp · CMP

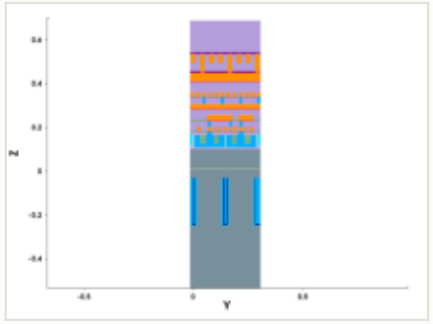
0.169	548.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

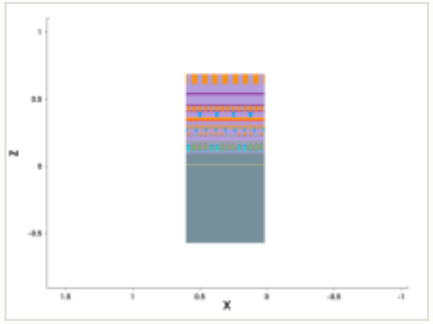
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

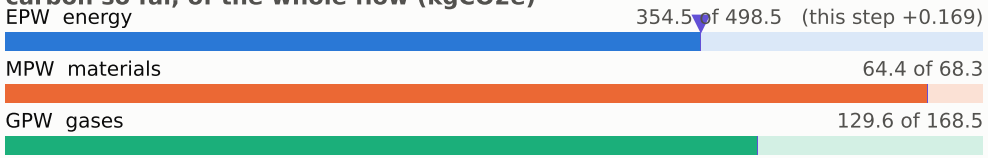
no mask for this step

this step

process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

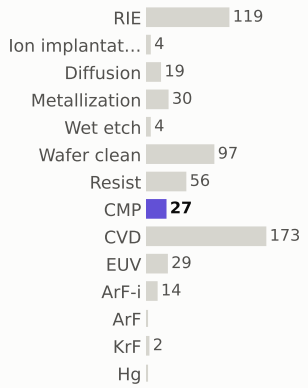
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

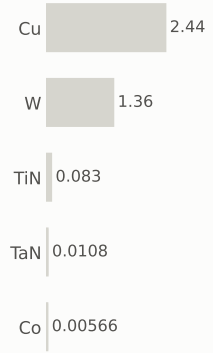


embodied carbon up to each step (kgCO2e per wafer)

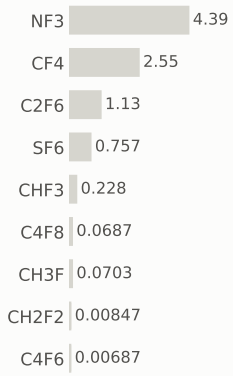
steps by process area



materials deposited (g)



gases emitted (g)



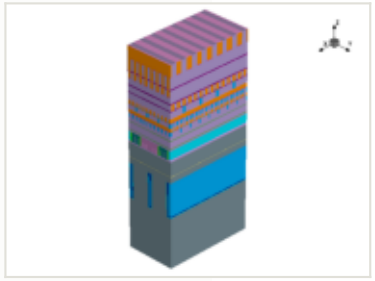
Step #6000: Metallization

depo_isotropic · Metallization

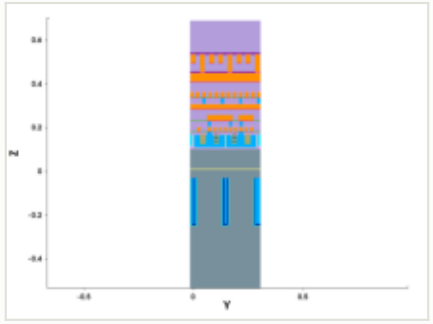
1.2	549.7	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

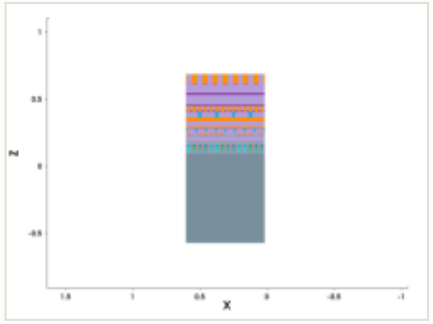
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

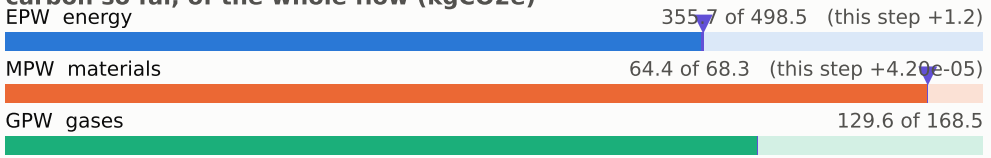
no mask for this step

this step

process area	Metallization
material	Co
thickness	0.0015 µm
material GWP	44.5 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

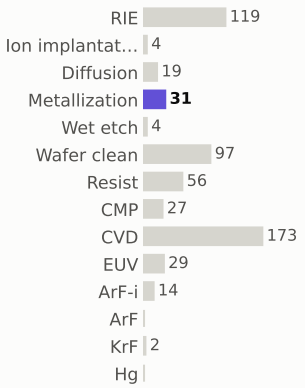
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.20e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

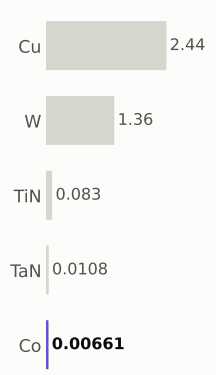


embodied carbon up to each step (kgCO2e per wafer)

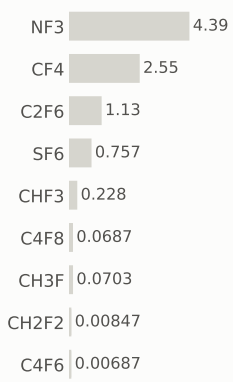
steps by process area



materials deposited (g)



gases emitted (g)



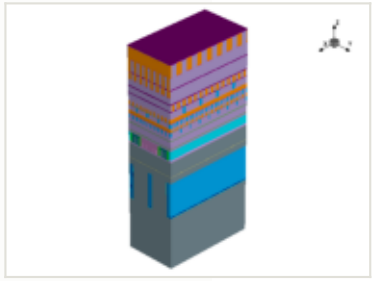
Step #6020: CVD

depo_fill · CVD

1.31	551.0	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

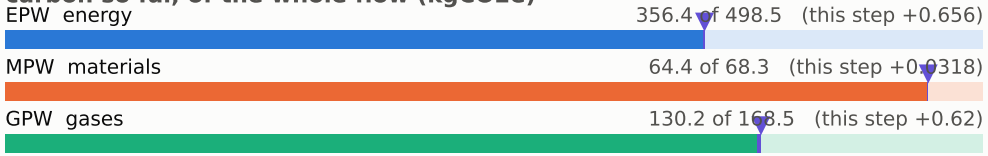


this step

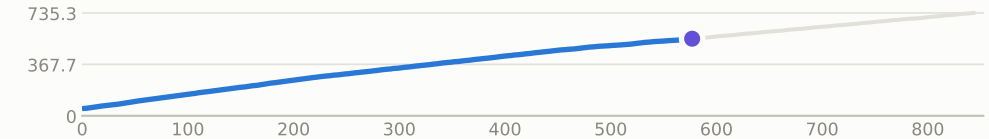
process area	CVD
material	—
thickness	0.006 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

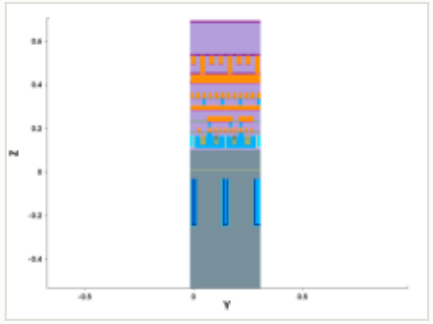
carbon so far, of the whole flow (kgCO2e)



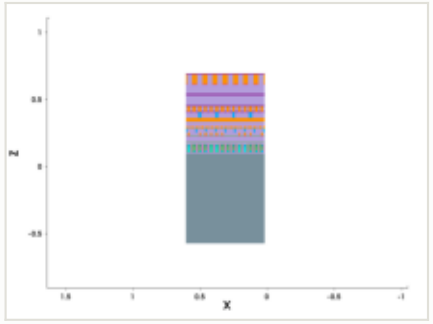
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



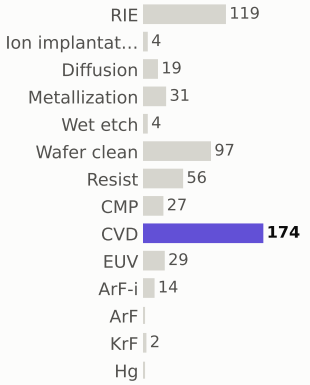
y cut at 0.184 µm



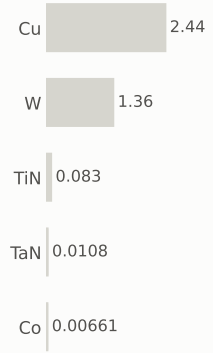
mask for this step

no mask for this step

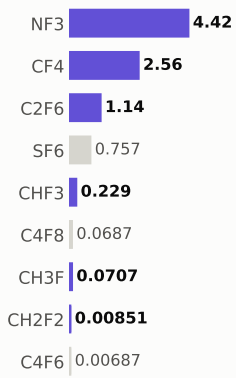
steps by process area



materials deposited (g)



gases emitted (g)



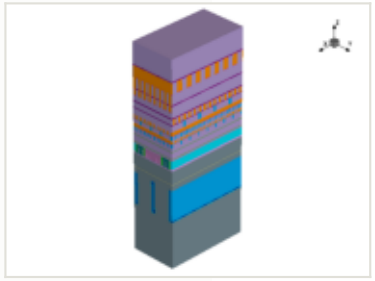
Step #6030: CVD

depo_fill · CVD

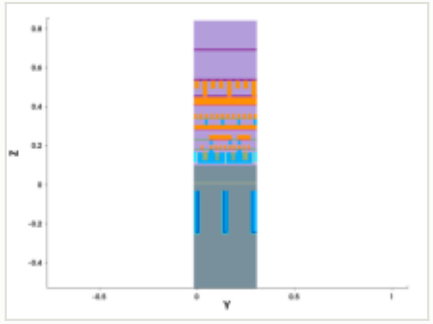
1.31	552.3	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

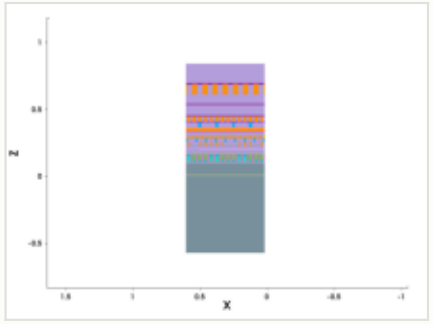
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

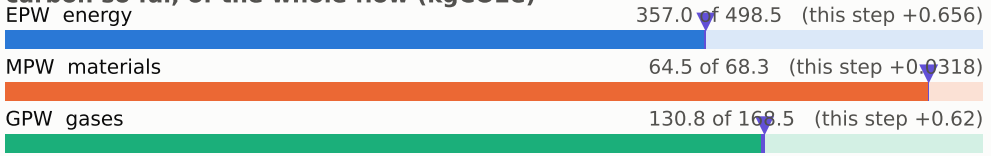
no mask for this step

this step

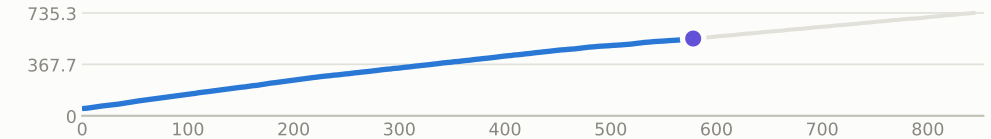
process area	CVD
material	—
thickness	0.146 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

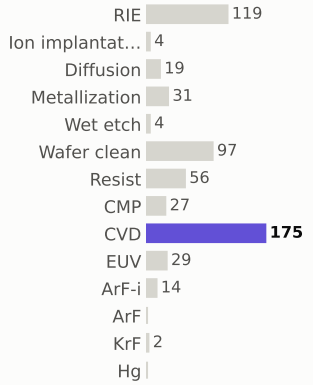
carbon so far, of the whole flow (kgCO2e)



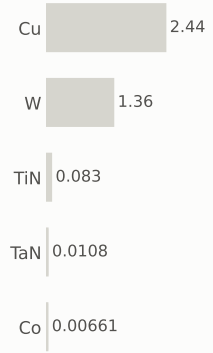
embodied carbon up to each step (kgCO2e per wafer)



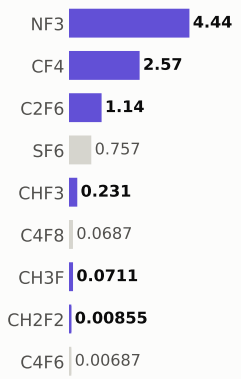
steps by process area



materials deposited (g)



gases emitted (g)



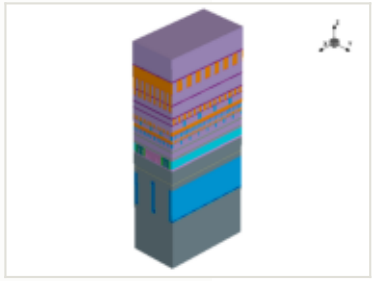
Step #6040: Diffusion

treatment · Diffusion

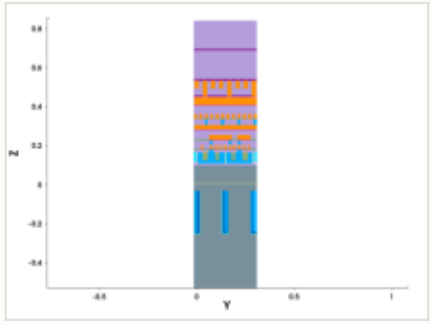
0.801	553.1	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

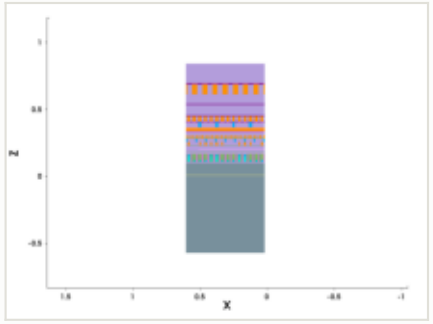
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

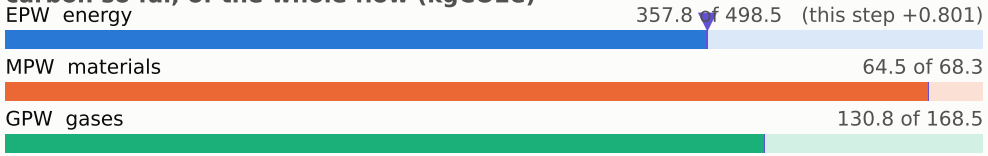
no mask for this step

this step

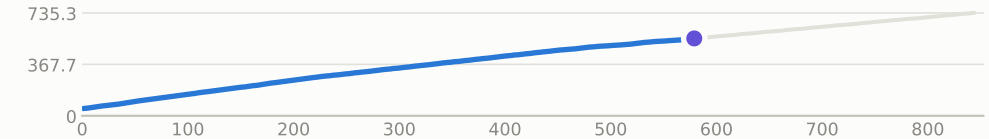
process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

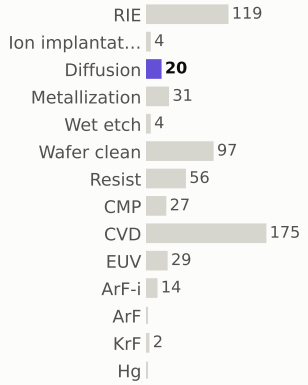
carbon so far, of the whole flow (kgCO2e)



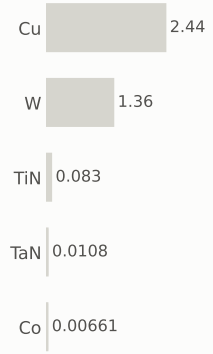
embodied carbon up to each step (kgCO2e per wafer)



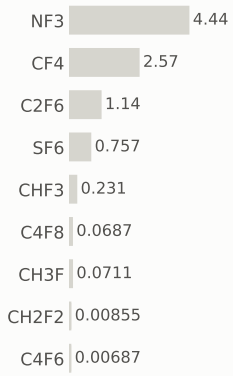
steps by process area



materials deposited (g)



gases emitted (g)



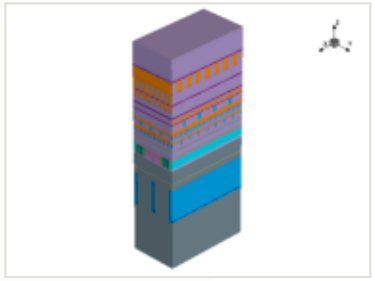
Step #6050: Wafer_clean

clean · Wafer_clean

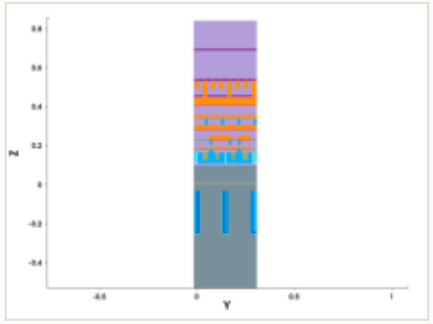
0.169	553.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

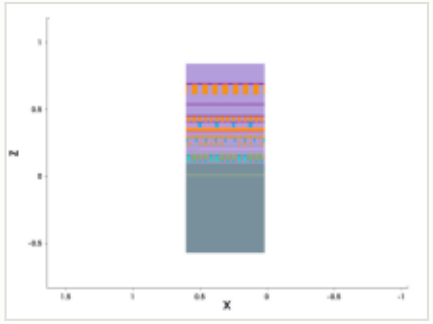
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

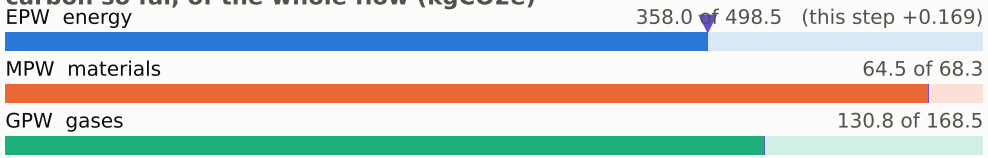
no mask for this step

this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

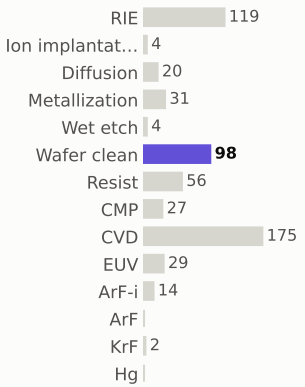
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

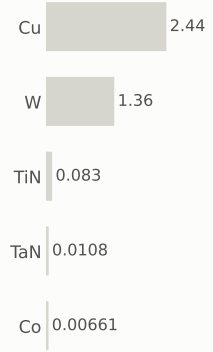


embodied carbon up to each step (kgCO2e per wafer)

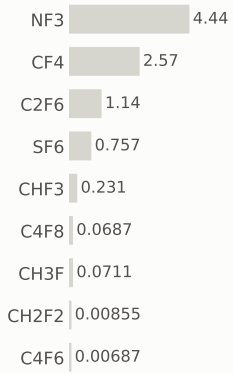
steps by process area



materials deposited (g)



gases emitted (g)



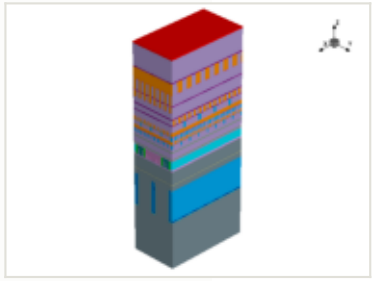
Step #6060: CVD

depo_fill · CVD

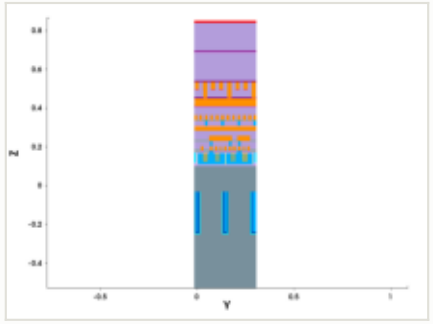
1.31	554.6	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

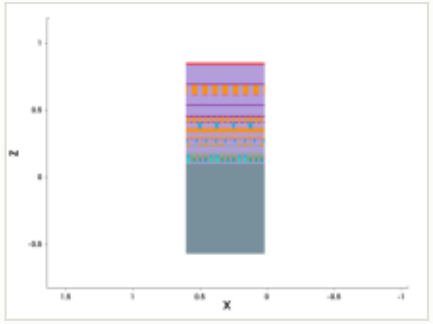
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

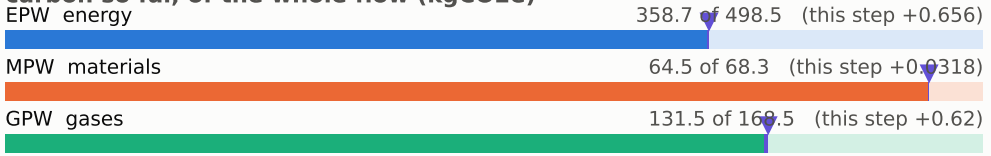
no mask for this step

this step

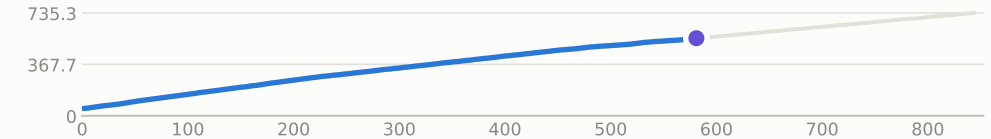
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

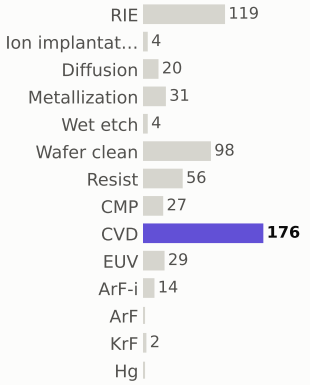
carbon so far, of the whole flow (kgCO2e)



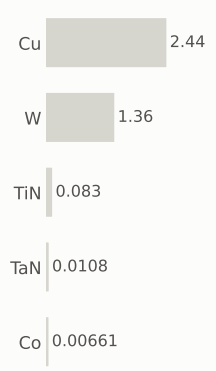
embodied carbon up to each step (kgCO2e per wafer)



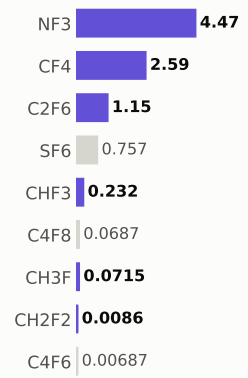
steps by process area



materials deposited (g)



gases emitted (g)



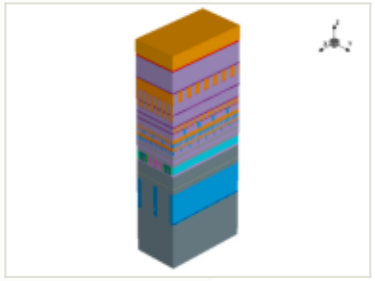
Step #6070: Resist

depo_fill · Resist

0.169	554.8	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

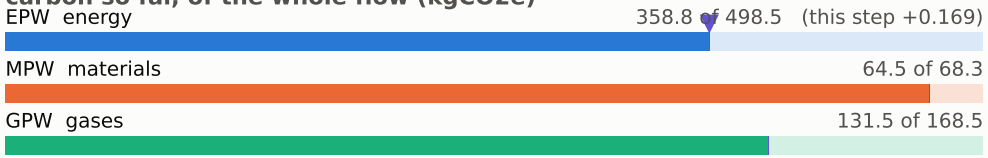


this step

process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

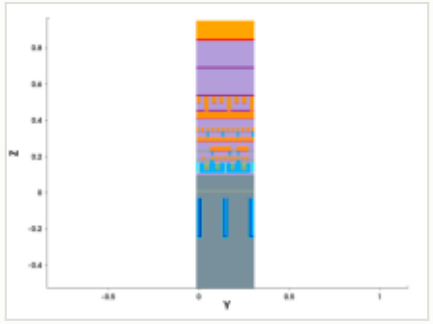
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

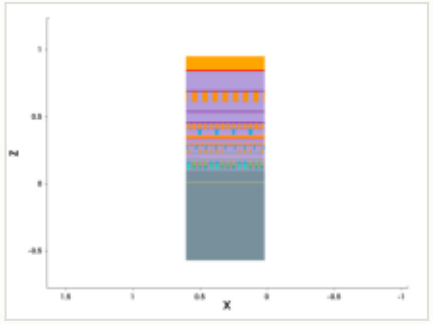


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



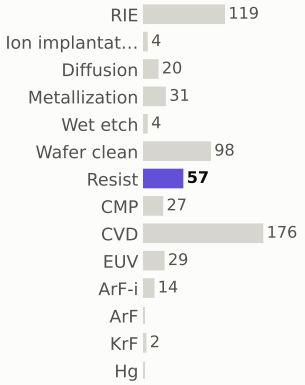
y cut at 0.184 µm



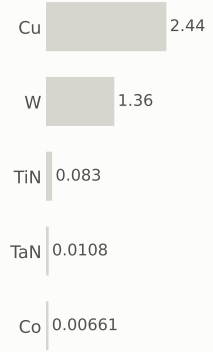
mask for this step

no mask for this step

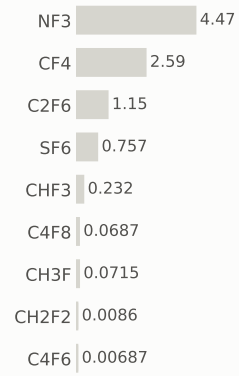
steps by process area



materials deposited (g)



gases emitted (g)



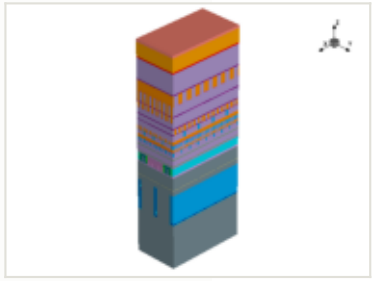
Step #6080: CVD

depo_fill · CVD

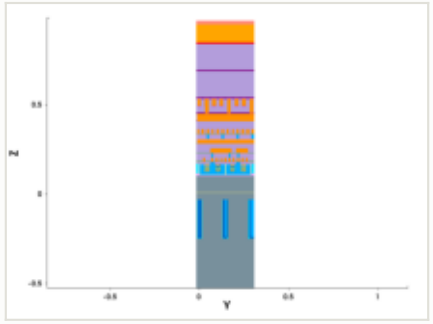
1.31	556.1	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

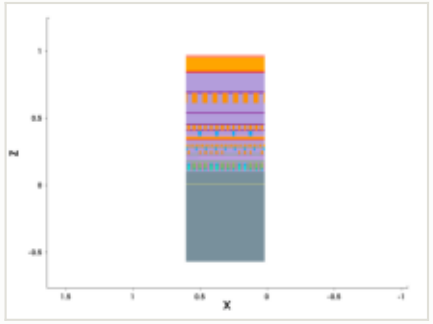
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

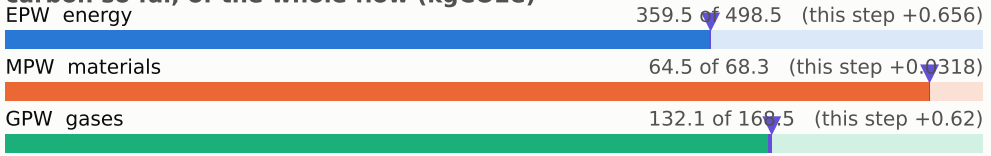
no mask for this step

this step

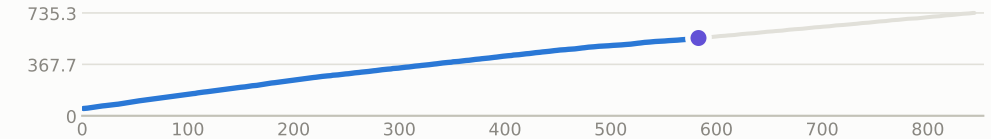
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

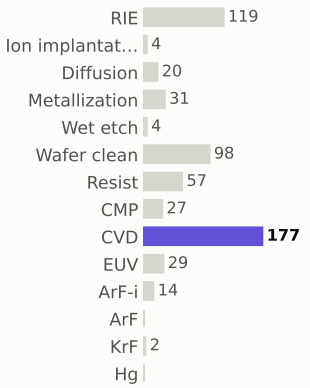
carbon so far, of the whole flow (kgCO2e)



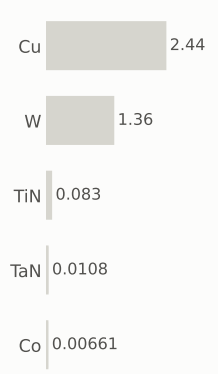
embodied carbon up to each step (kgCO2e per wafer)



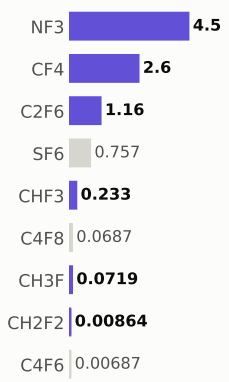
steps by process area



materials deposited (g)



gases emitted (g)



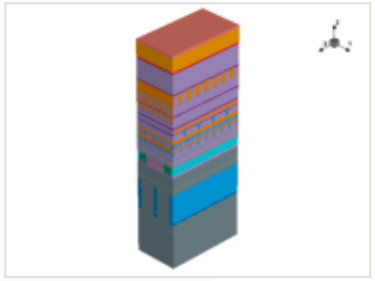
Step #6090: Wafer_clean

clean · Wafer_clean

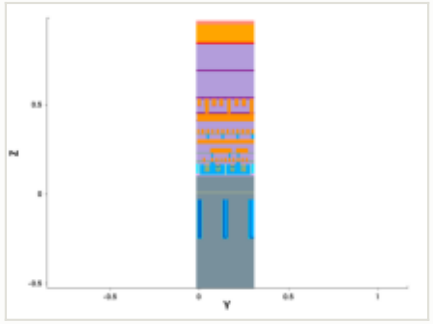
0.169	556.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

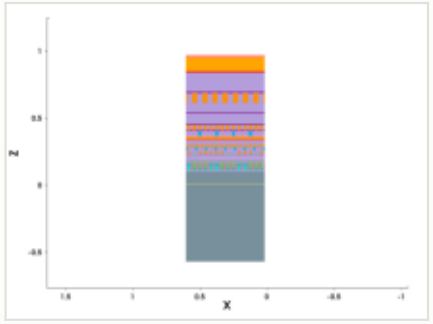
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

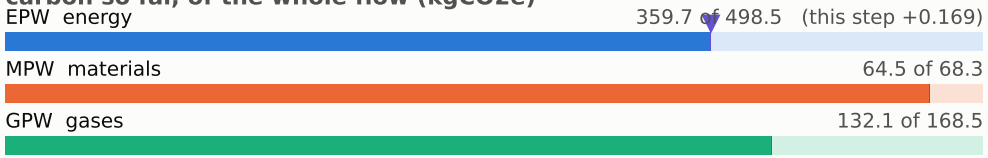
no mask for this step

this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

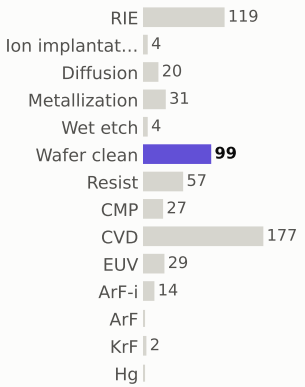
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

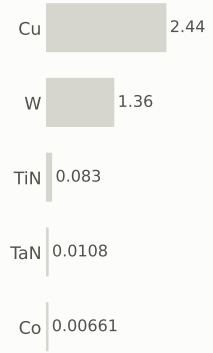


embodied carbon up to each step (kgCO2e per wafer)

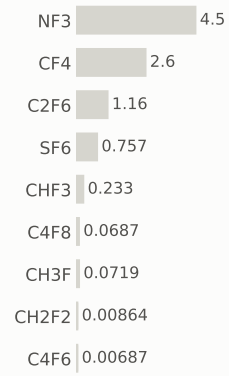
steps by process area



materials deposited (g)



gases emitted (g)



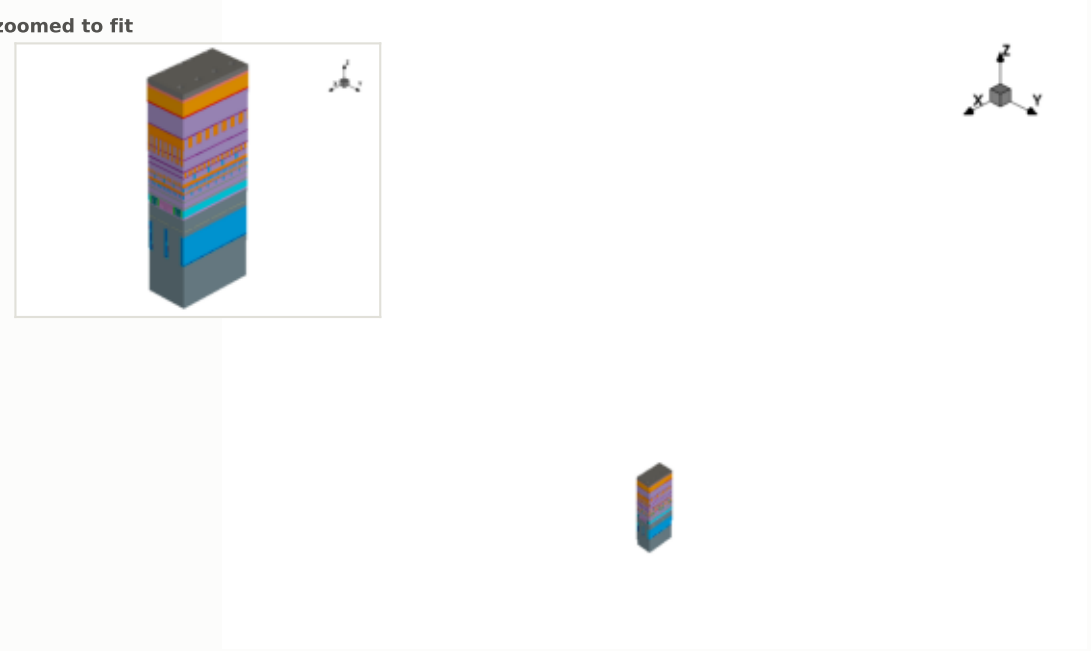
Step #6100: ArF-i

litho_patterning · ArF-i

0.217	556.5	0.029%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

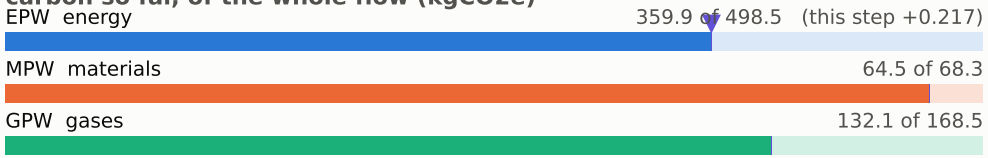
zoomed to fit



this step

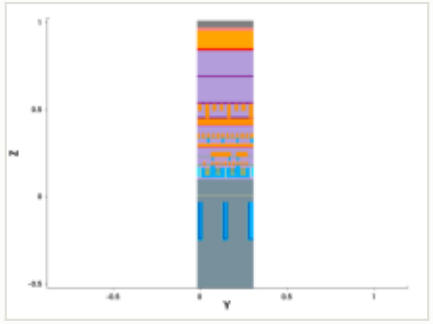
process area	ArF-i	EPW	0.217 kgCO2e
material	—	EPW tool	0.155 kgCO2e
thickness	—	EPW overhead	0.0619 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.36 kWh + 0.144 overhead	embodied carbon	**0.217 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

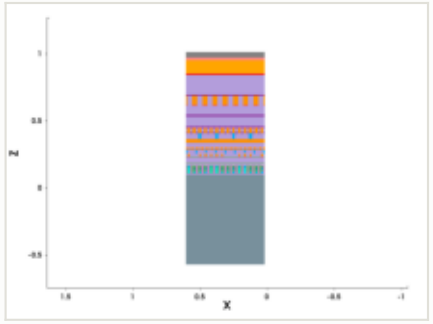


embodied carbon up to each step (kgCO2e per wafer)

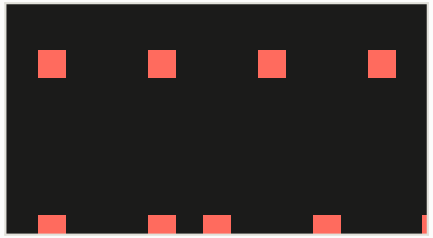
x cut at 0.294 µm



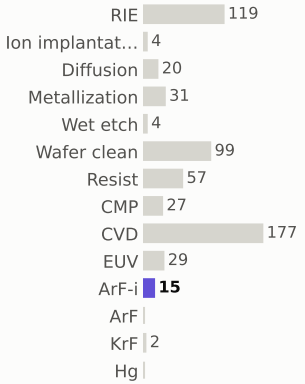
y cut at 0.184 µm



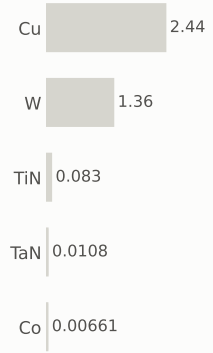
mask: layer 10521:0 (GDSII)



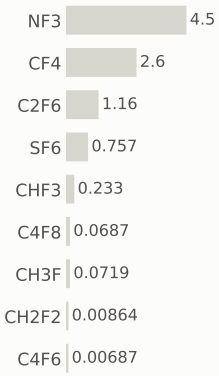
steps by process area



materials deposited (g)



gases emitted (g)



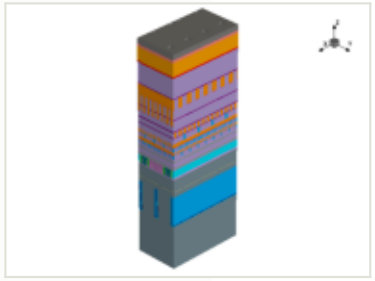
Step #6110: RIE

etch_anisotropic · RIE

1.04	557.5	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

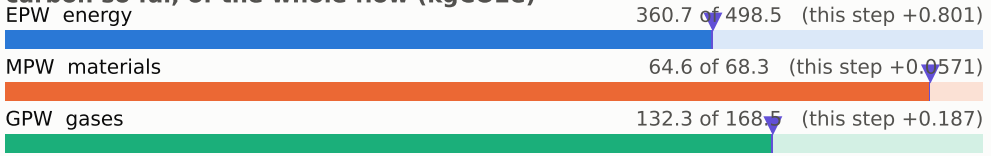


this step

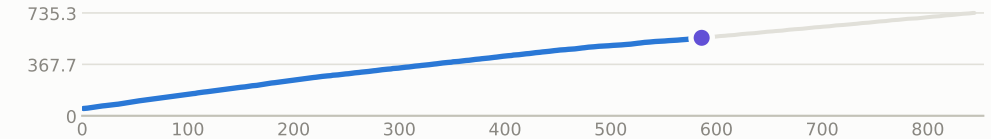
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

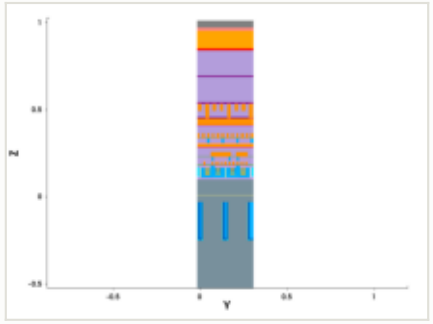
carbon so far, of the whole flow (kgCO2e)



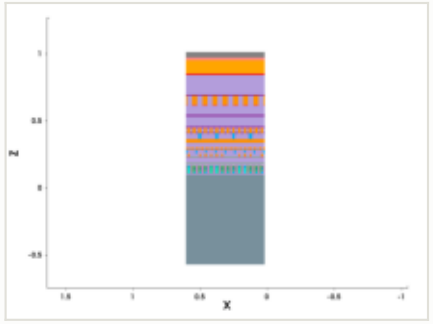
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



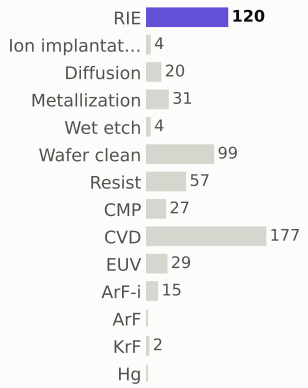
y cut at 0.184 µm



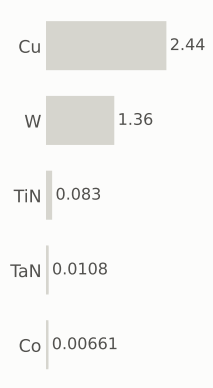
mask for this step

no mask for this step

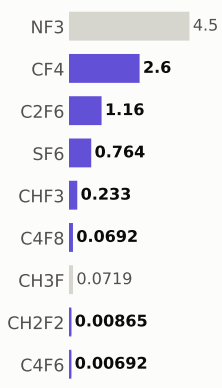
steps by process area



materials deposited (g)



gases emitted (g)



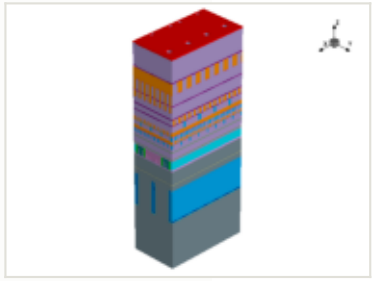
Step #6120: Wafer_clean

etch_strip · Wafer_clean

0.169	557.7	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

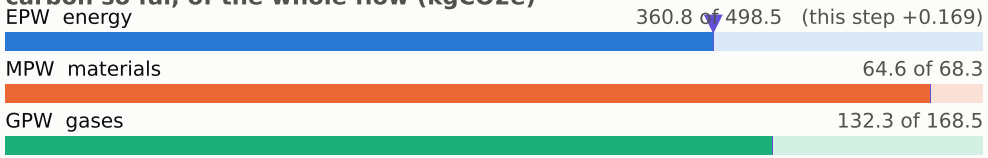


this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

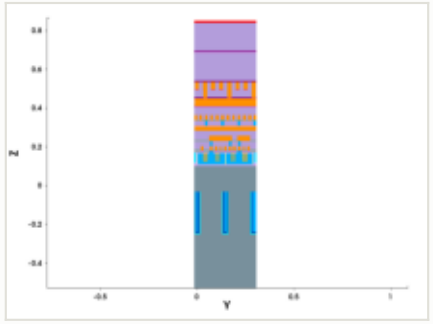
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

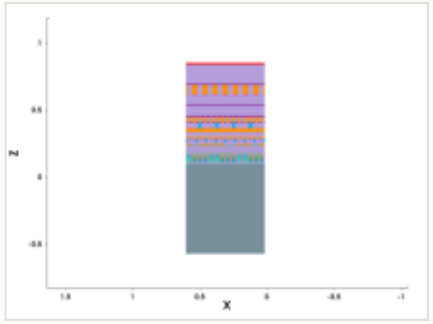


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



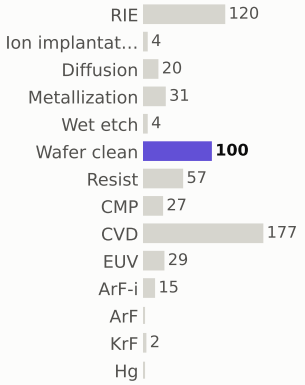
y cut at 0.184 µm



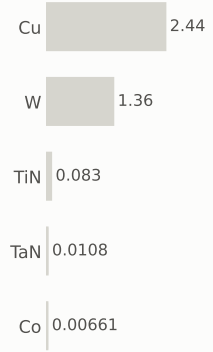
mask for this step

no mask for this step

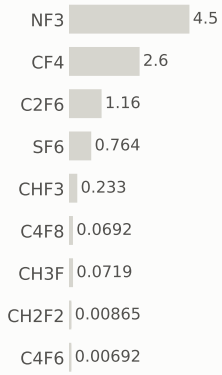
steps by process area



materials deposited (g)



gases emitted (g)



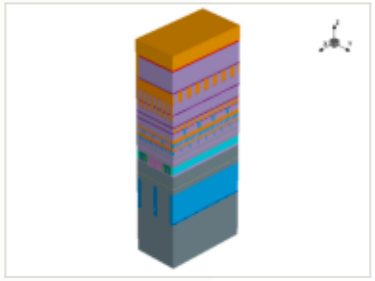
Step #6130: Resist

depo_fill · Resist

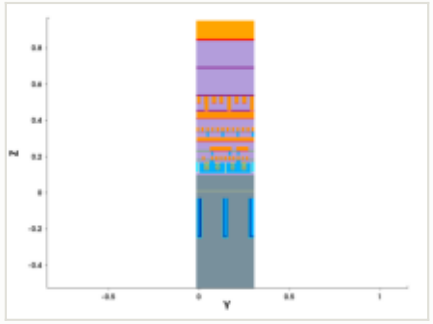
0.169	557.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

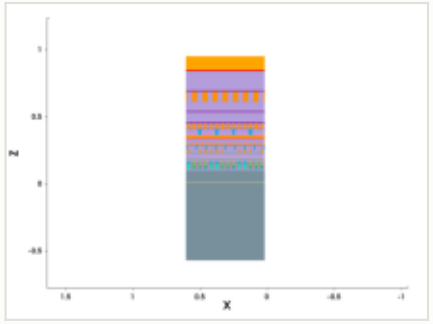
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



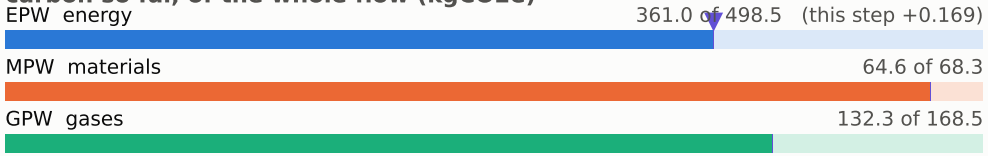
mask for this step

no mask for this step

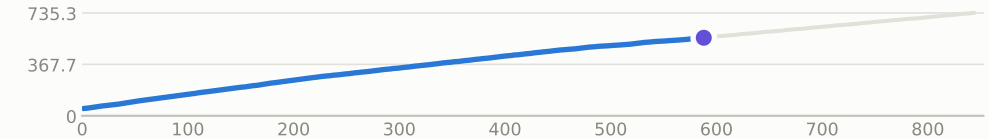
this step

process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	0.1 µm	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

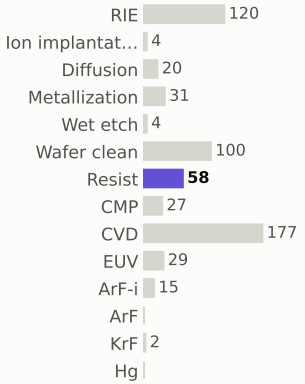
carbon so far, of the whole flow (kgCO2e)



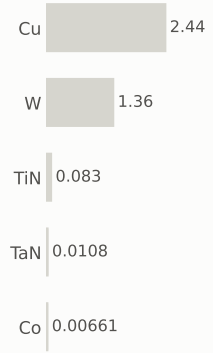
embodied carbon up to each step (kgCO2e per wafer)



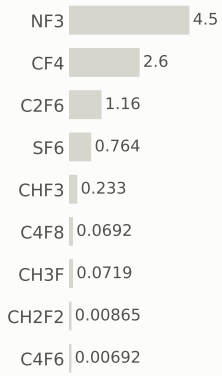
steps by process area



materials deposited (g)



gases emitted (g)



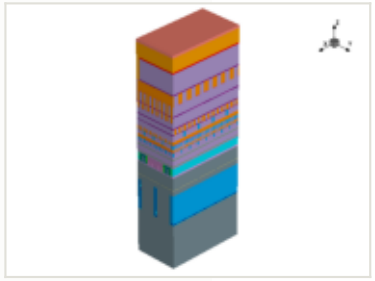
Step #6140: CVD

depo_fill · CVD

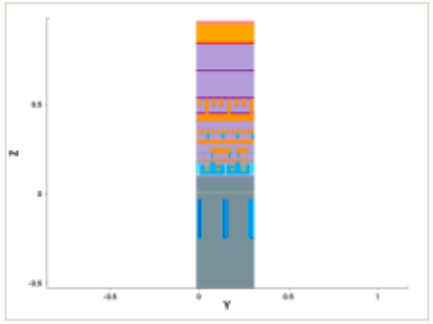
1.31	559.2	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

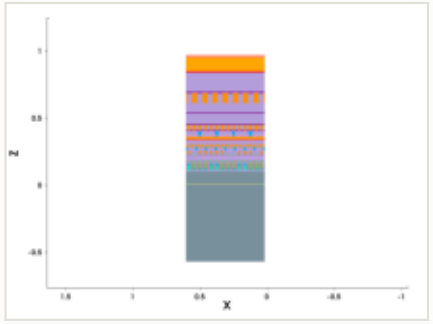
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

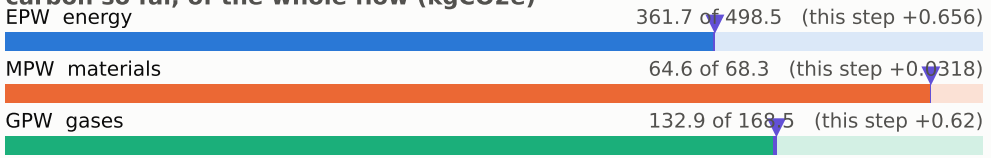
no mask for this step

this step

process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

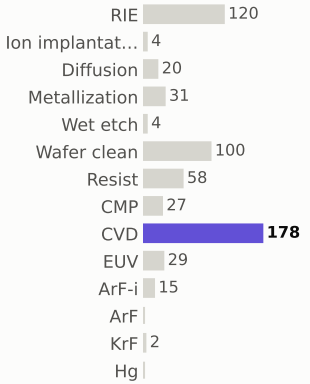
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

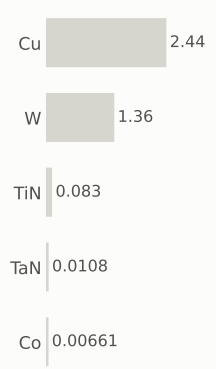


embodied carbon up to each step (kgCO2e per wafer)

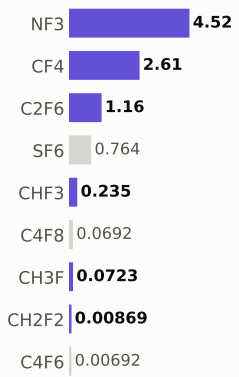
steps by process area



materials deposited (g)



gases emitted (g)



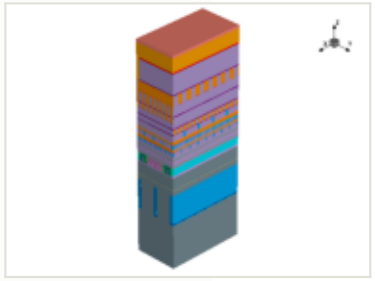
Step #6150: Wafer_clean

clean · Wafer_clean

0.169	559.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



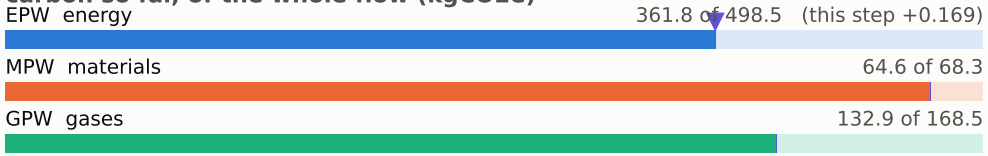
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

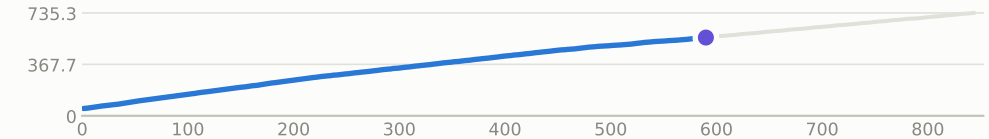
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

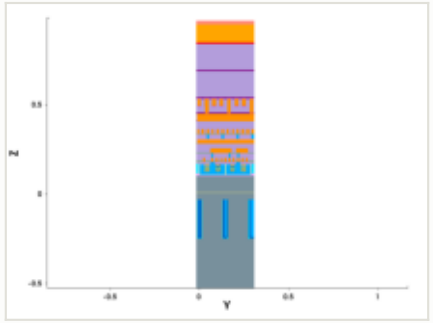
carbon so far, of the whole flow (kgCO2e)



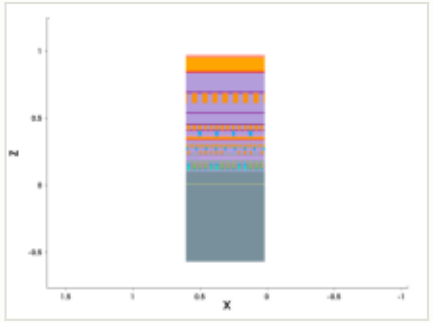
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



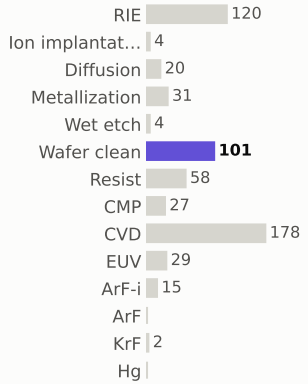
y cut at 0.184 µm



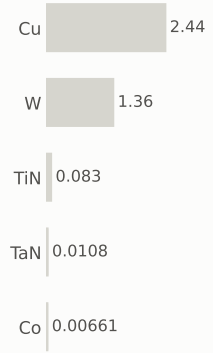
mask for this step

no mask for this step

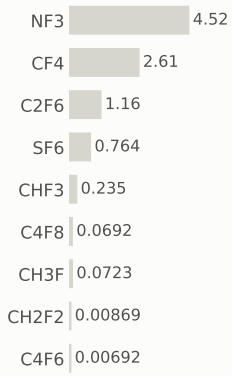
steps by process area



materials deposited (g)



gases emitted (g)



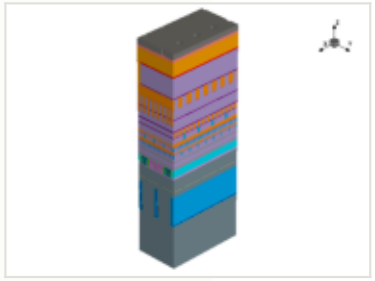
Step #6160: ArF-i

litho_patterning · ArF-i

0.217	559.6	0.029%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

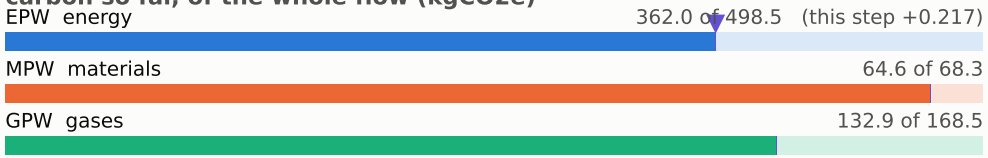


this step

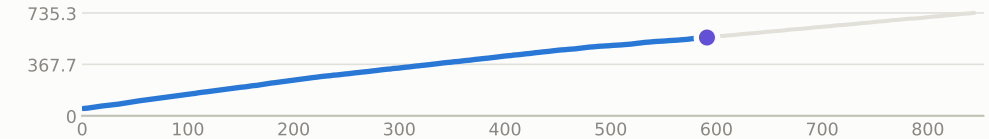
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

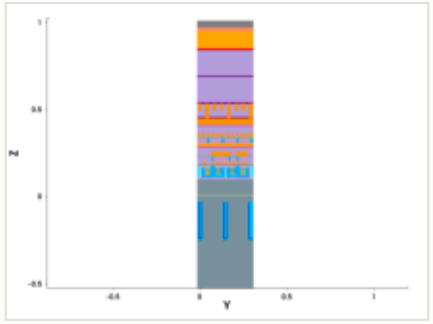
carbon so far, of the whole flow (kgCO2e)



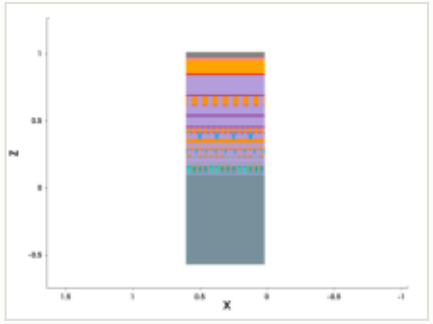
embodied carbon up to each step (kgCO2e per wafer)



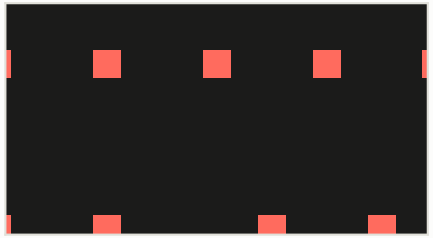
x cut at 0.294 µm



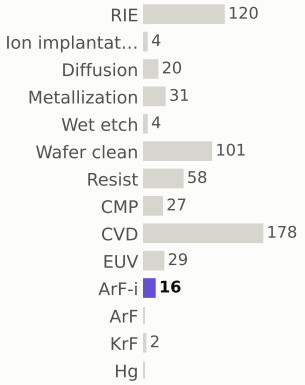
y cut at 0.184 µm



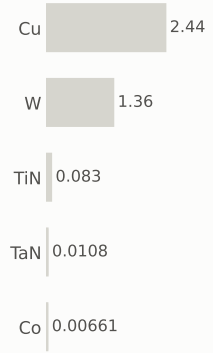
mask: layer 10522:0 (GDSII)



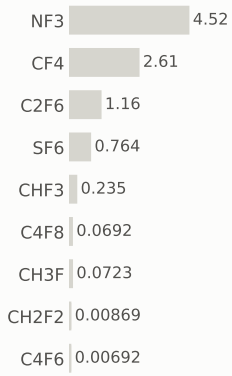
steps by process area



materials deposited (g)



gases emitted (g)



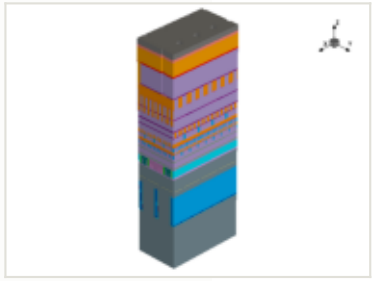
Step #6170: RIE

etch_anisotropic · RIE

1.04	560.6	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

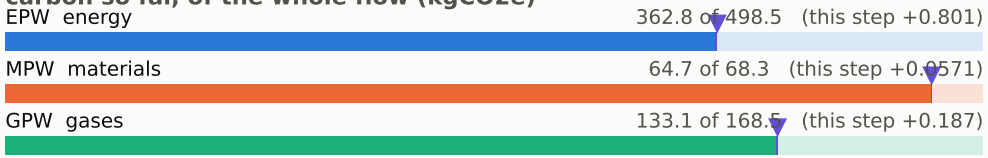


this step

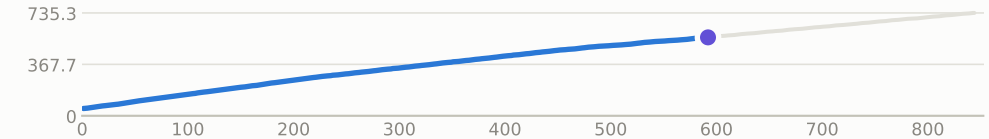
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

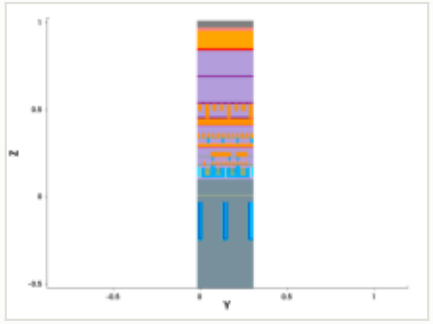
carbon so far, of the whole flow (kgCO2e)



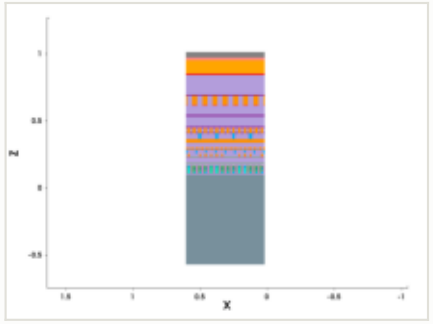
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



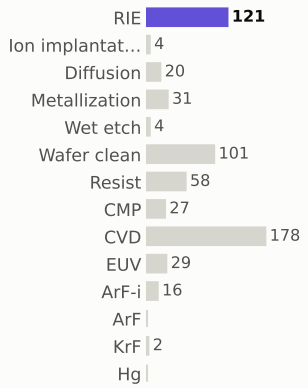
y cut at 0.184 µm



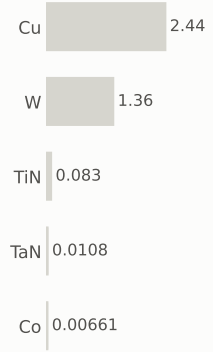
mask for this step

no mask for this step

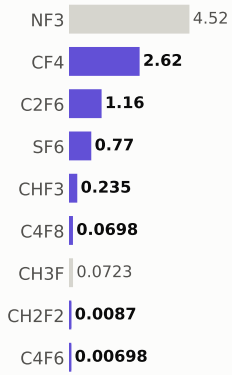
steps by process area



materials deposited (g)



gases emitted (g)



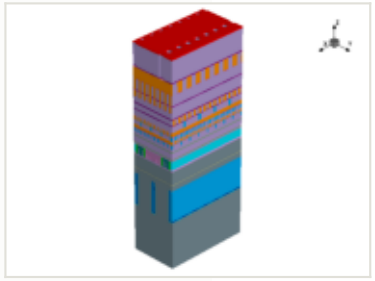
Step #6180: Wafer_clean

etch_strip · Wafer_clean

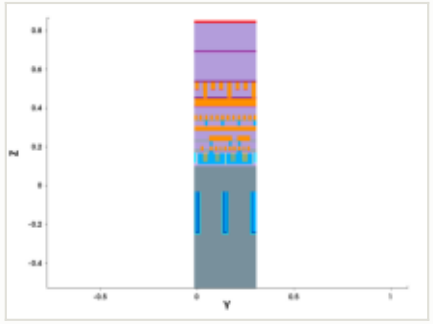
0.169	560.8	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

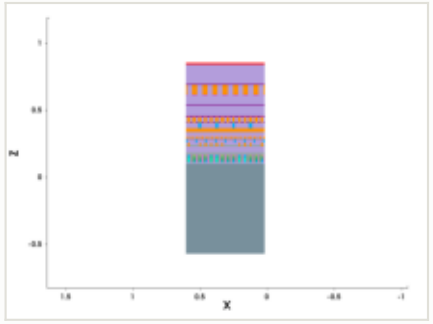
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



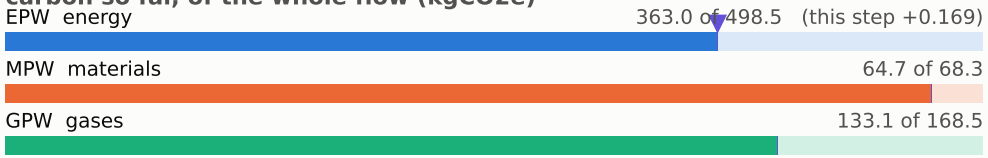
mask for this step

no mask for this step

this step

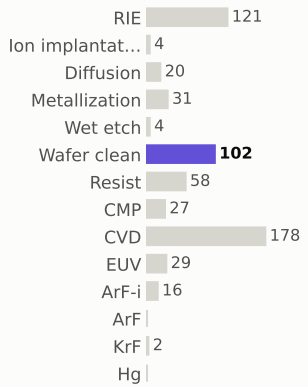
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

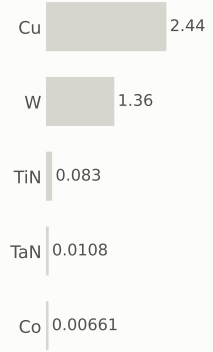


embodied carbon up to each step (kgCO2e per wafer)

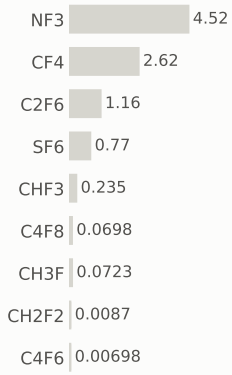
steps by process area



materials deposited (g)



gases emitted (g)



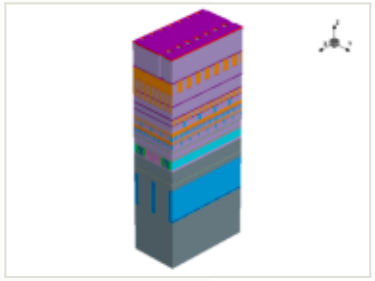
Step #6190: Metallization

depo_ellipsoidal · Metallization

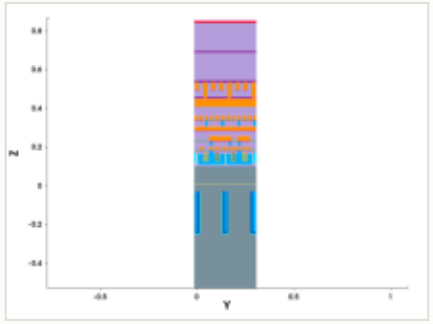
1.2	562.0	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

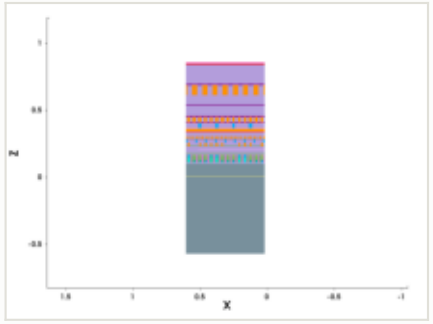
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

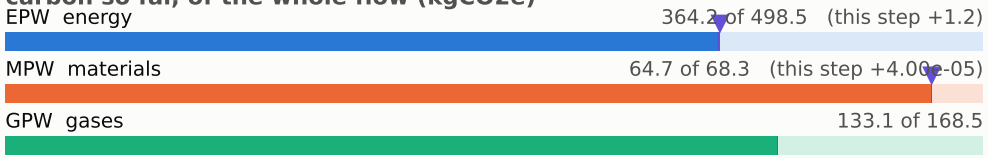
no mask for this step

this step

process area	Metallization
material	TaN
thickness	0.0016 µm
material GWP	25.8159 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

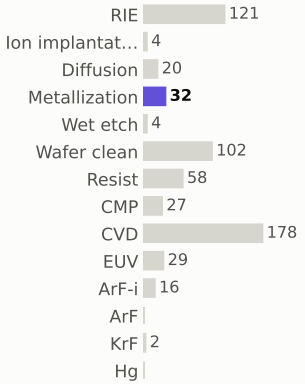
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.00e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

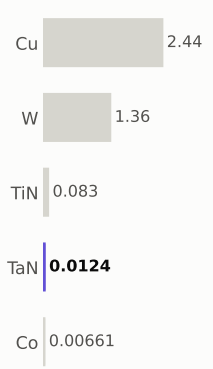


embodied carbon up to each step (kgCO2e per wafer)

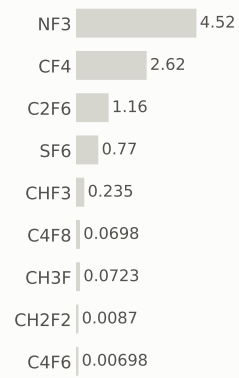
steps by process area



materials deposited (g)



gases emitted (g)



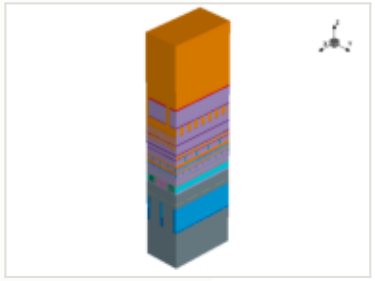
Step #6200: Metallization

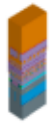
depo_fill · Metallization

1.21	563.2	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

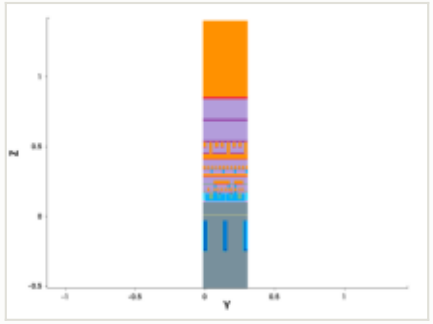
3D structure after this step

zoomed to fit

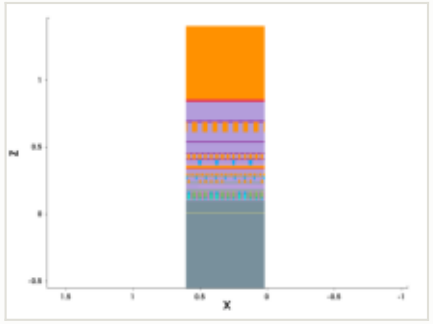




x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

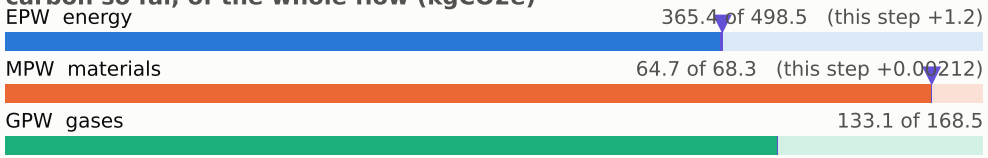
no mask for this step

this step

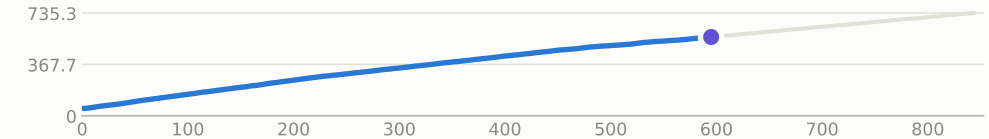
process area	Metallization
material	Cu
thickness	0.55 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00212 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

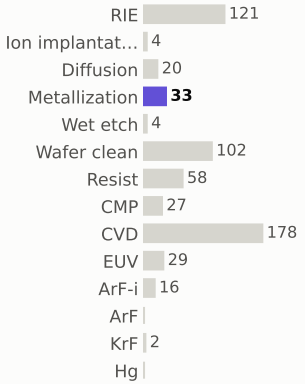
carbon so far, of the whole flow (kgCO2e)



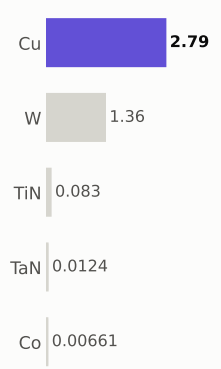
embodied carbon up to each step (kgCO2e per wafer)



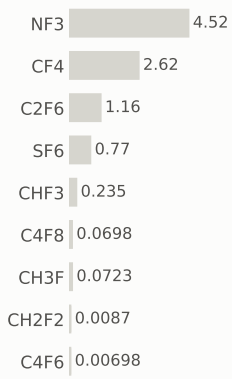
steps by process area



materials deposited (g)



gases emitted (g)



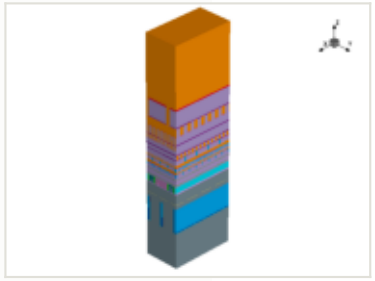
Step #6210: Diffusion

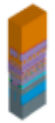
treatment · Diffusion

0.801	564.0	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





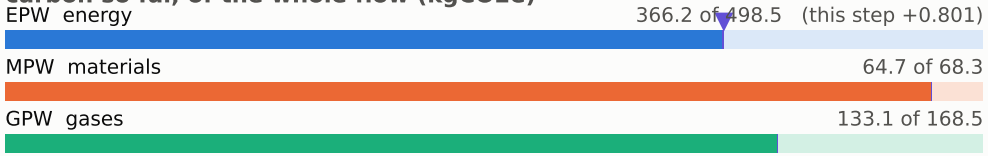
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

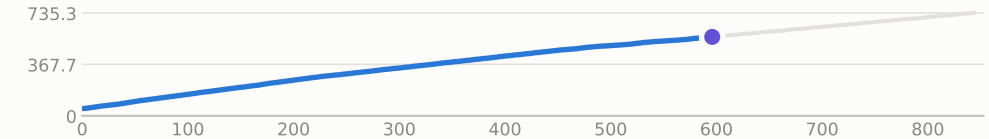
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

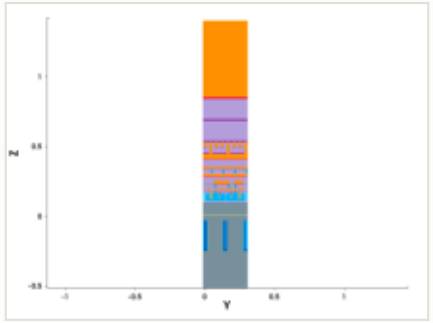
carbon so far, of the whole flow (kgCO2e)



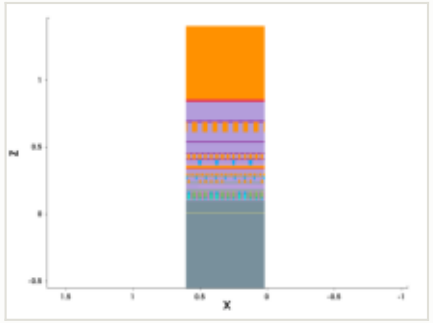
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



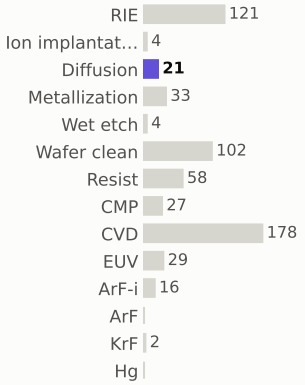
y cut at 0.184 µm



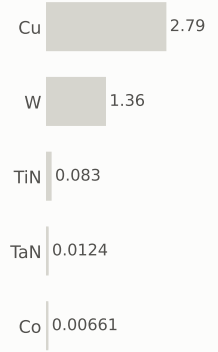
mask for this step

no mask for this step

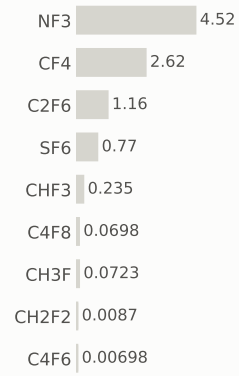
steps by process area



materials deposited (g)



gases emitted (g)



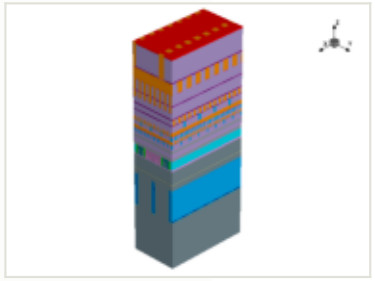
Step #6220: CMP

etch_cmp · CMP

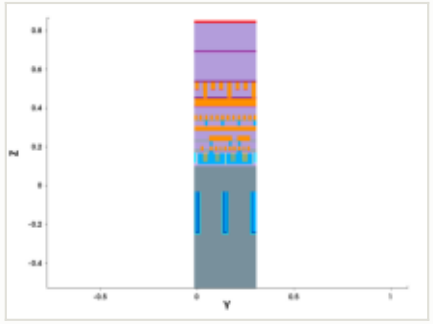
0.169	564.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

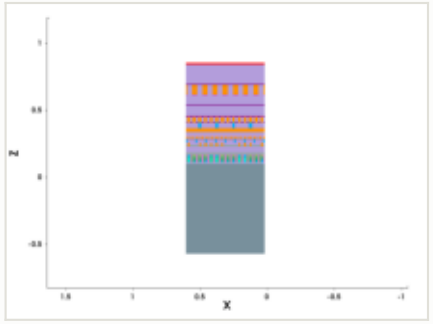
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

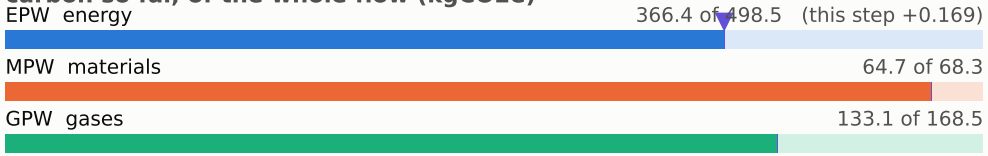
no mask for this step

this step

process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

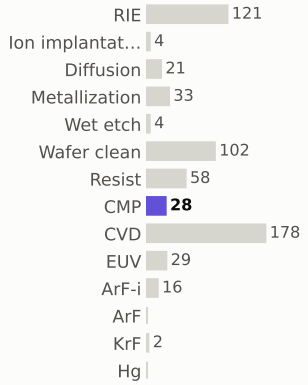
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

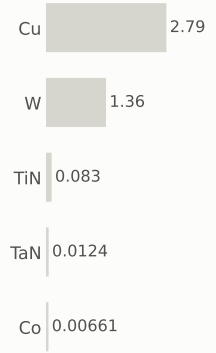


embodied carbon up to each step (kgCO2e per wafer)

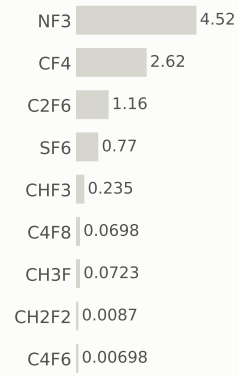
steps by process area



materials deposited (g)



gases emitted (g)



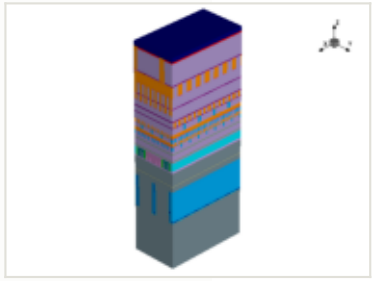
Step #6230: CVD

depo_fill · CVD

1.31	565.5	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

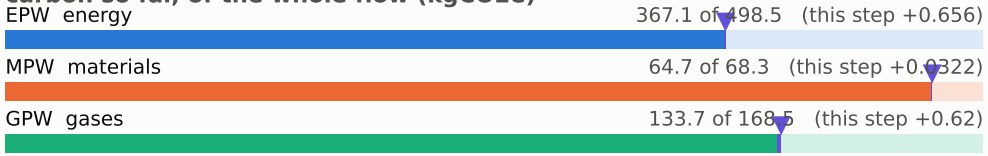


this step

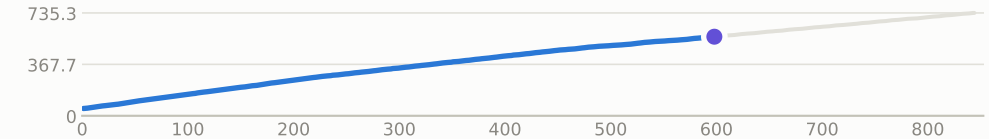
process area	CVD
material	TiN
thickness	0.03 µm
material GWP	43.9778 gCO2e/g
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0322 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

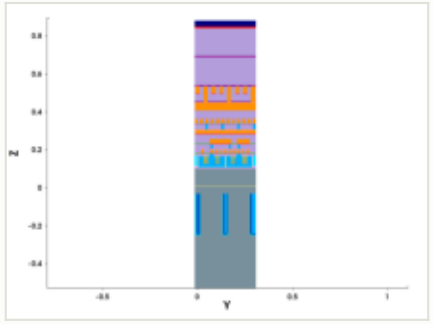
carbon so far, of the whole flow (kgCO2e)



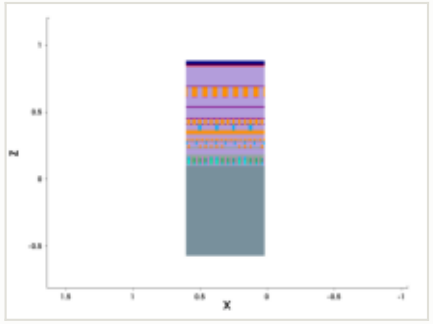
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



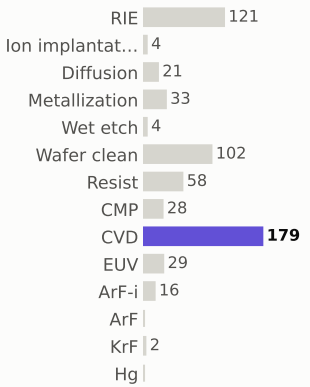
y cut at 0.184 µm



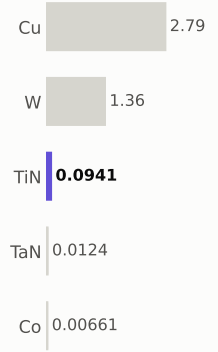
mask for this step

no mask for this step

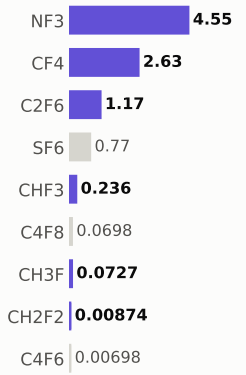
steps by process area



materials deposited (g)



gases emitted (g)



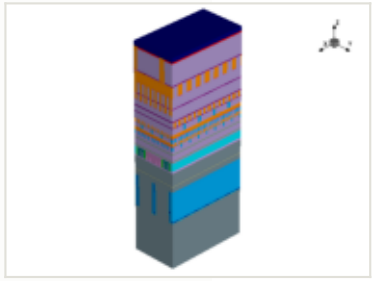
Step #6240: Wafer_clean

clean · Wafer_clean

0.169	565.6	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



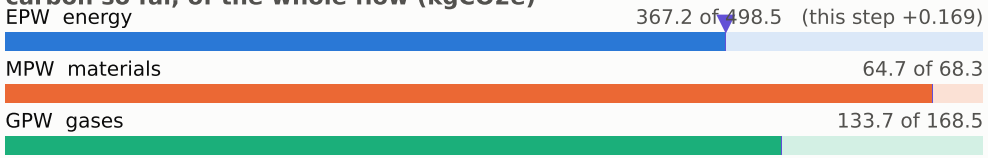
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

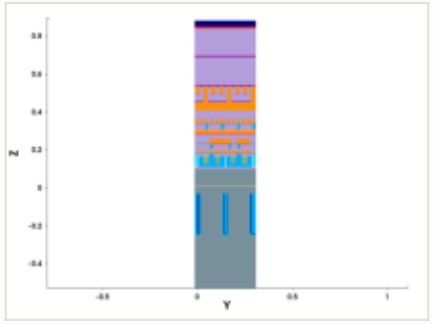
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

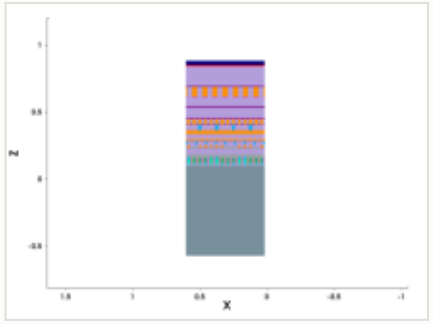


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



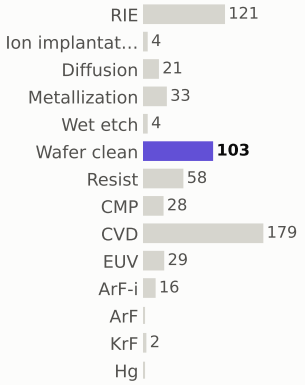
y cut at 0.184 µm



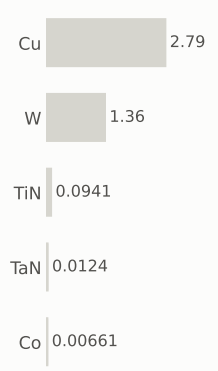
mask for this step

no mask for this step

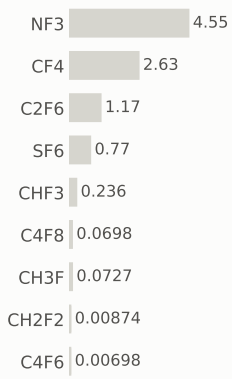
steps by process area



materials deposited (g)



gases emitted (g)



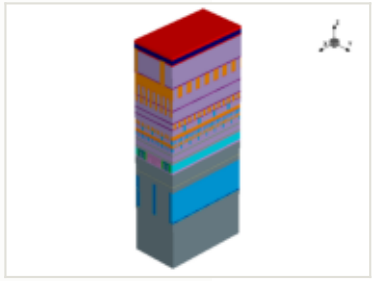
Step #6250: CVD

depo_fill · CVD

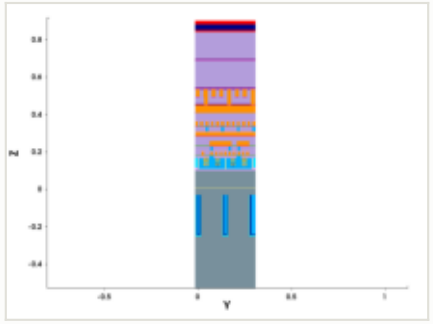
1.31	566.9	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

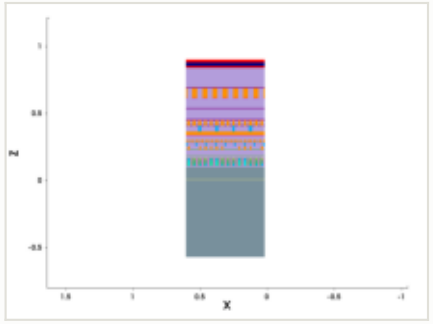
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

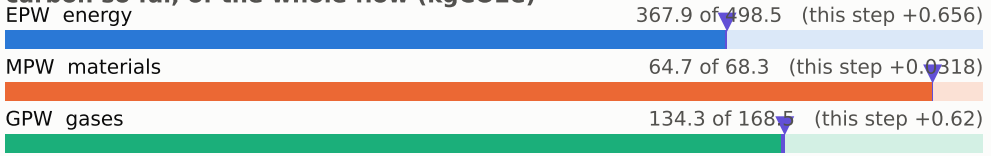
no mask for this step

this step

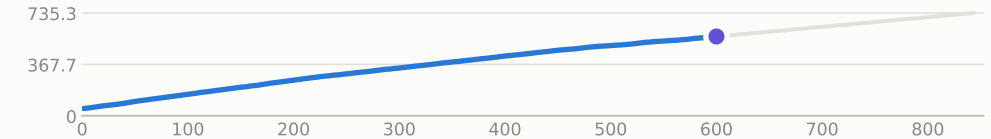
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

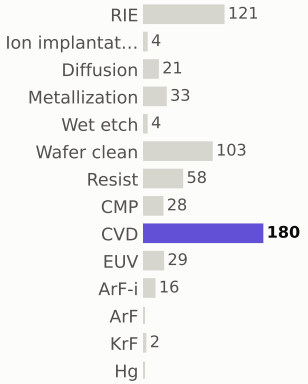
carbon so far, of the whole flow (kgCO2e)



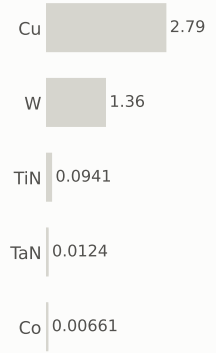
embodied carbon up to each step (kgCO2e per wafer)



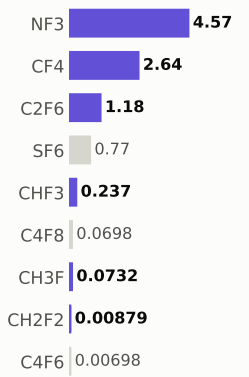
steps by process area



materials deposited (g)



gases emitted (g)



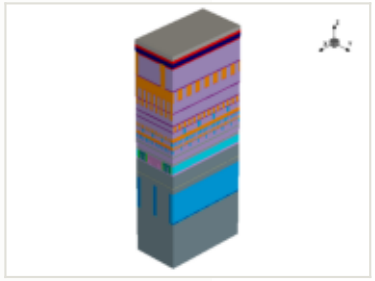
Step #6260: CVD

depo_fill · CVD

1.31	568.2	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

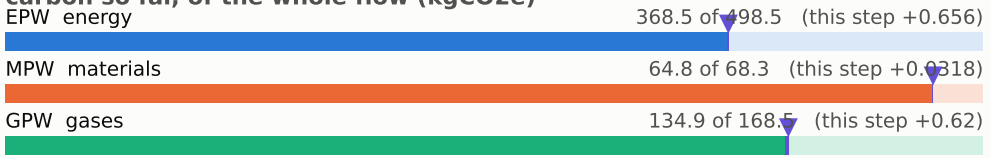


this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

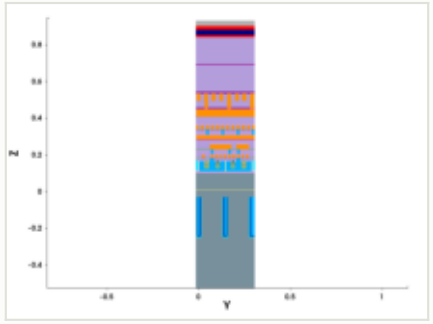
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

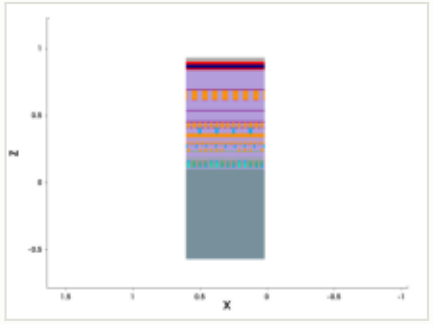


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



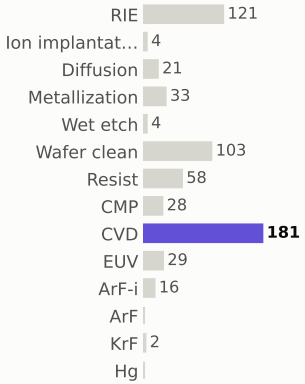
y cut at 0.184 µm



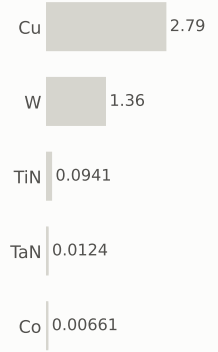
mask for this step

no mask for this step

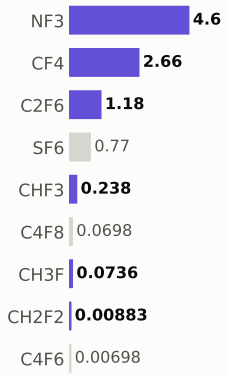
steps by process area



materials deposited (g)



gases emitted (g)



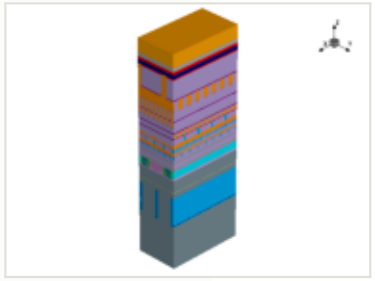
Step #6270: Resist

depo_fill · Resist

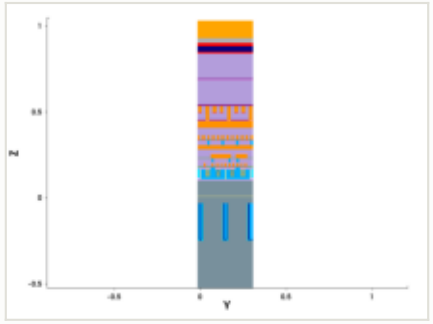
0.169	568.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

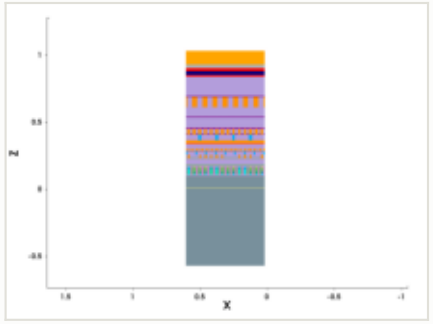
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



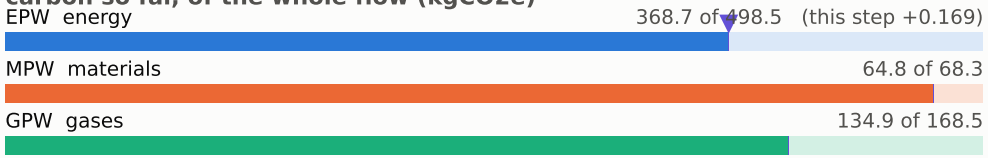
mask for this step

no mask for this step

this step

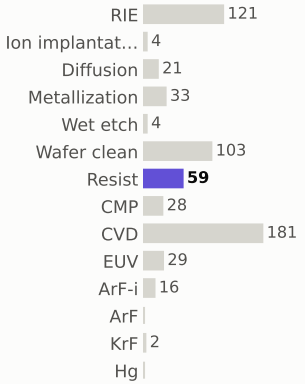
process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	0.1 µm	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

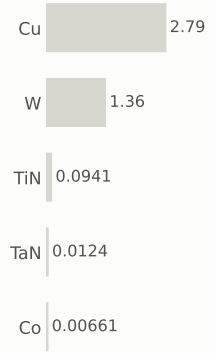


embodied carbon up to each step (kgCO2e per wafer)

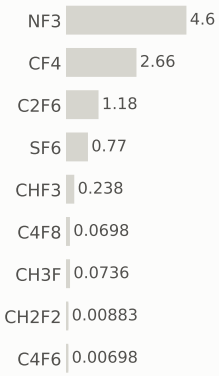
steps by process area



materials deposited (g)



gases emitted (g)



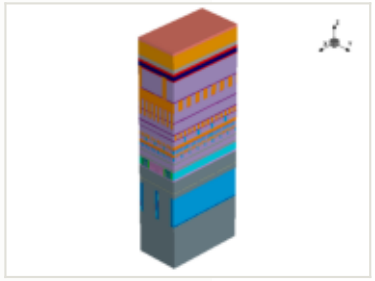
Step #6280: CVD

depo_fill · CVD

1.31	569.7	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

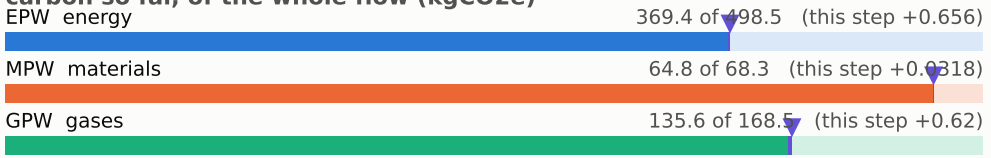


this step

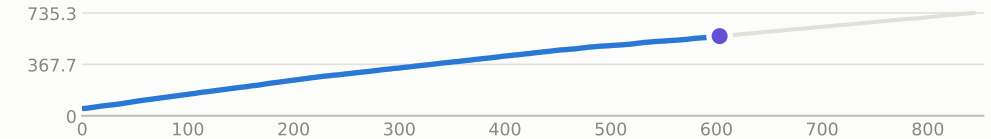
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

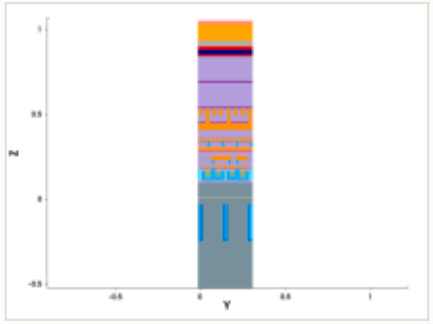
carbon so far, of the whole flow (kgCO2e)



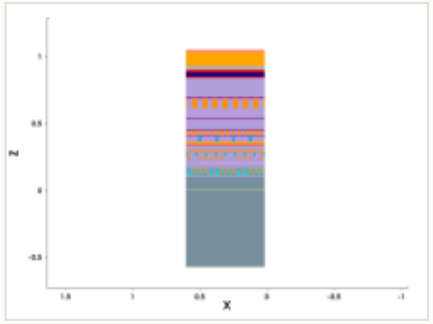
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



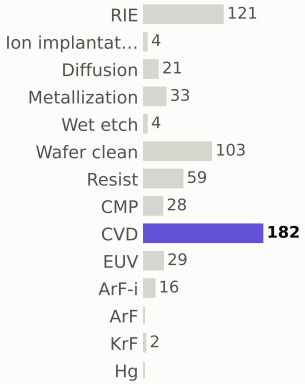
y cut at 0.184 µm



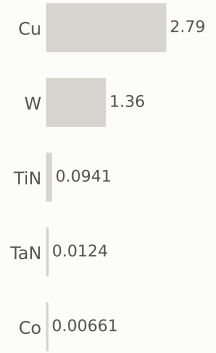
mask for this step

no mask for this step

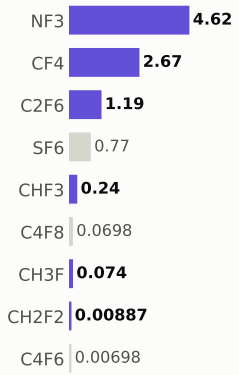
steps by process area



materials deposited (g)



gases emitted (g)



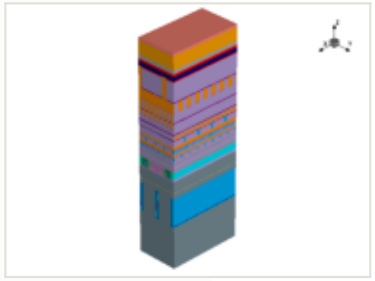
Step #6290: Wafer_clean

clean · Wafer_clean

0.169	569.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



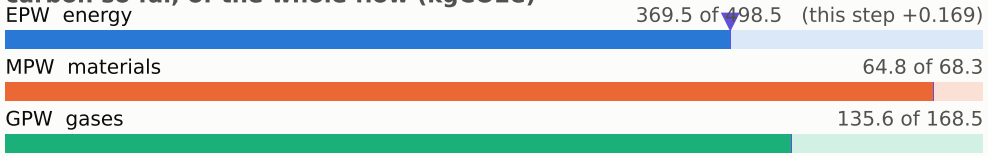
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

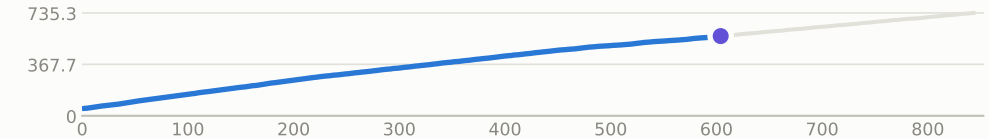
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

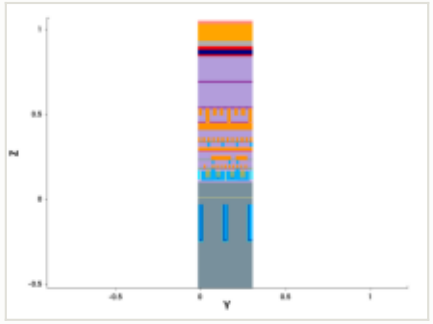
carbon so far, of the whole flow (kgCO2e)



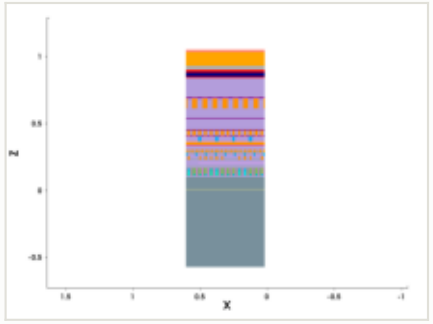
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



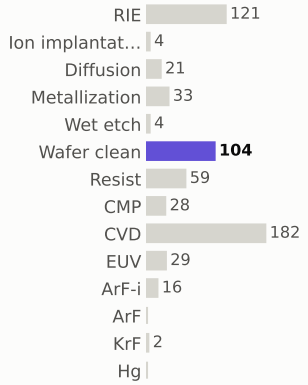
y cut at 0.184 µm



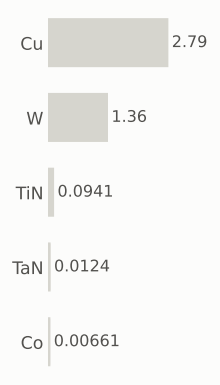
mask for this step

no mask for this step

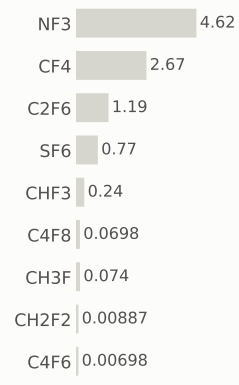
steps by process area



materials deposited (g)



gases emitted (g)



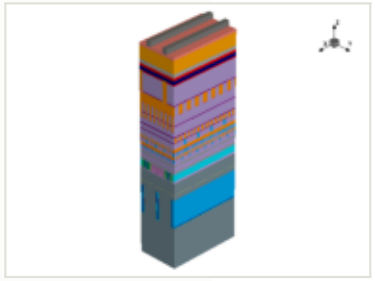
Step #6300: ArF-i

litho_patterning · ArF-i

0.217	570.1	0.029%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

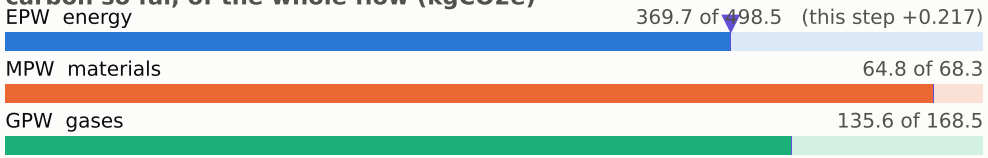


this step

process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

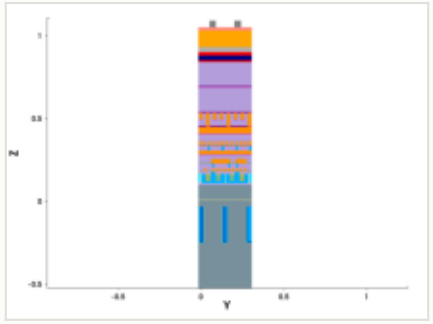
EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

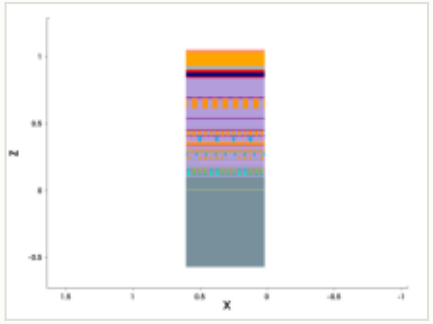


embodied carbon up to each step (kgCO2e per wafer)

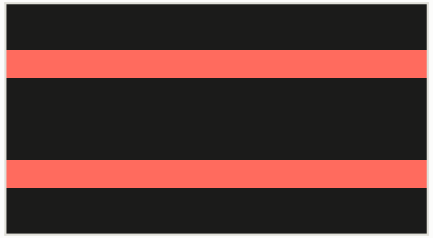
x cut at 0.294 µm



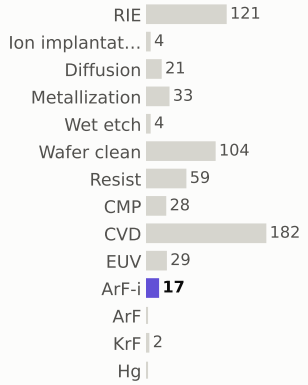
y cut at 0.184 µm



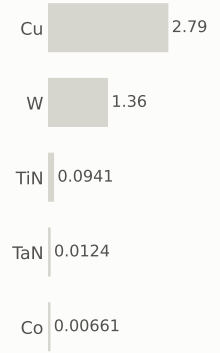
mask: layer 10551:0, positive (GDSII)



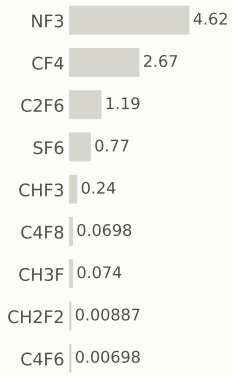
steps by process area



materials deposited (g)



gases emitted (g)



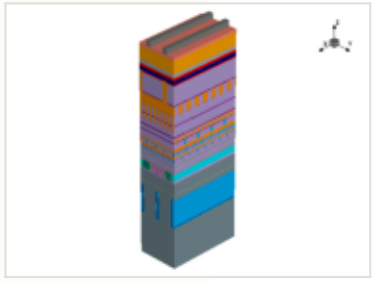
Step #6310: Ion_implantation

treatment · Ion_implantation

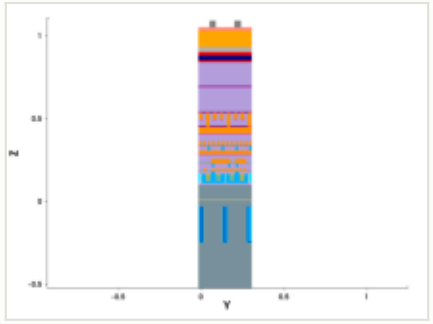
0.801	570.9	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

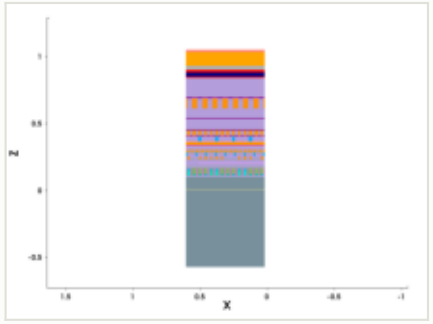
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

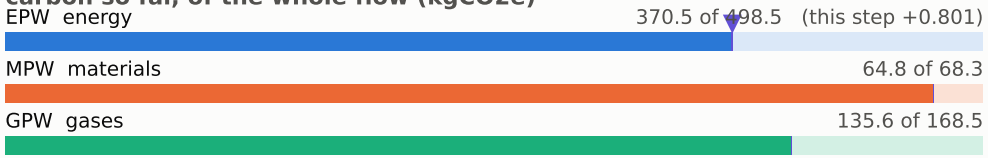
no mask for this step

this step

process area	Ion_implantation	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

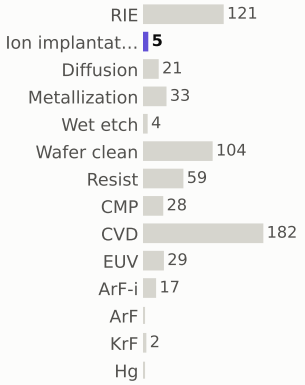
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

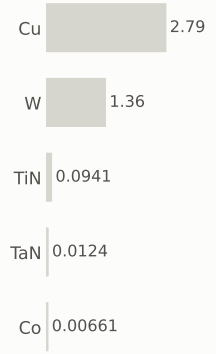


embodied carbon up to each step (kgCO2e per wafer)

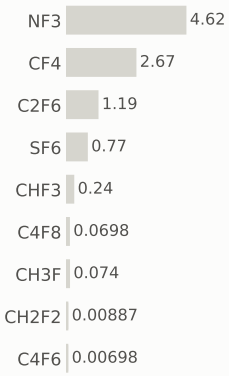
steps by process area



materials deposited (g)



gases emitted (g)



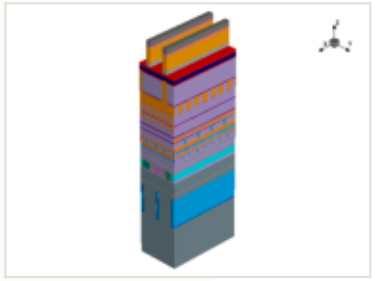
Step #6320: RIE

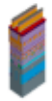
etch_anisotropic · RIE

1.04	572.0	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

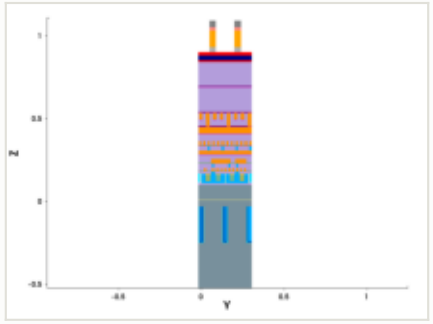
3D structure after this step

zoomed to fit

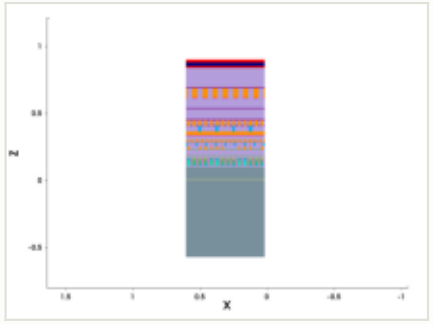




x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

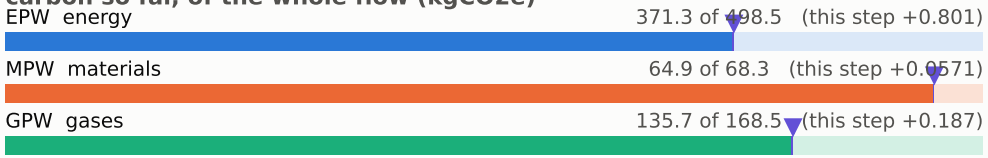
no mask for this step

this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

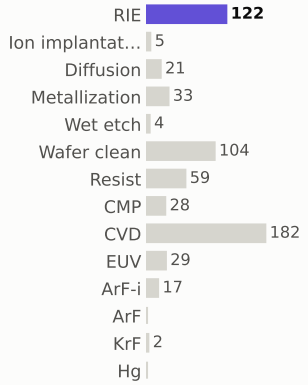
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

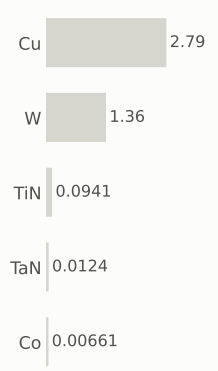


embodied carbon up to each step (kgCO2e per wafer)

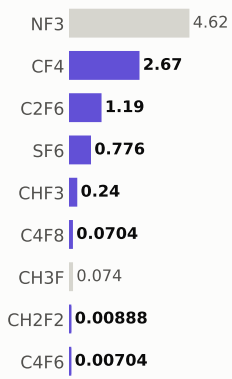
steps by process area



materials deposited (g)



gases emitted (g)



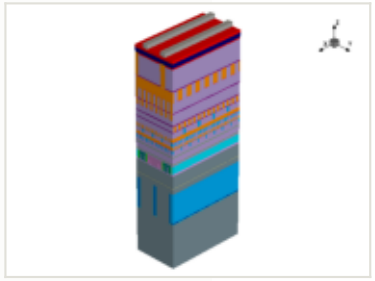
Step #6330: Wafer_clean

etch_strip · Wafer_clean

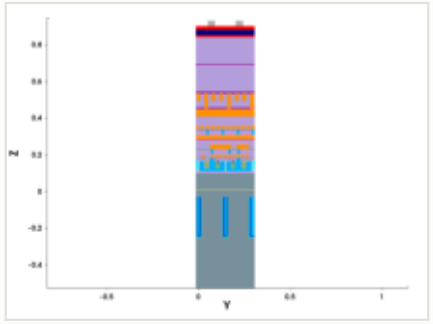
0.169	572.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

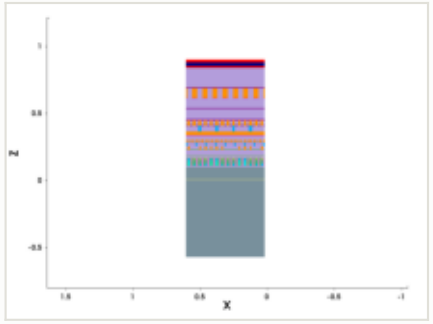
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



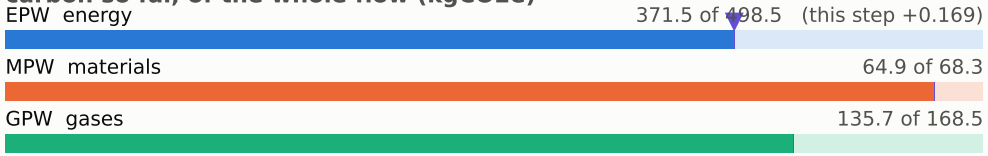
mask for this step

no mask for this step

this step

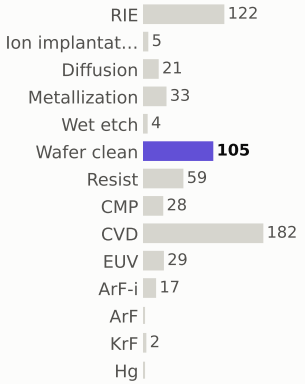
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

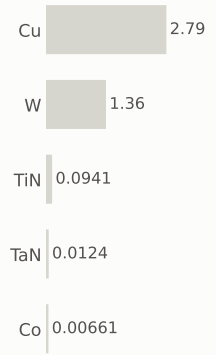


embodied carbon up to each step (kgCO2e per wafer)

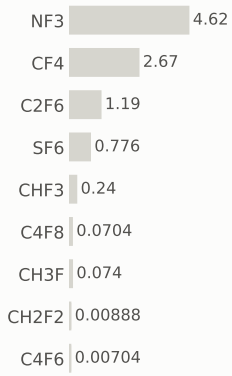
steps by process area



materials deposited (g)



gases emitted (g)



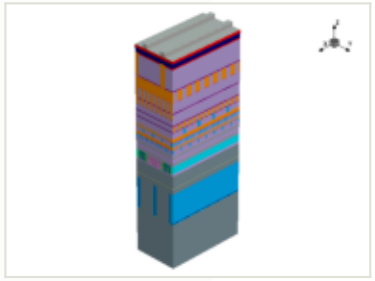
Step #6340: CVD

depo_isotropic · CVD

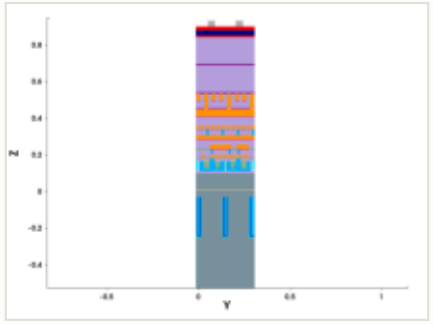
1.31	573.4	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

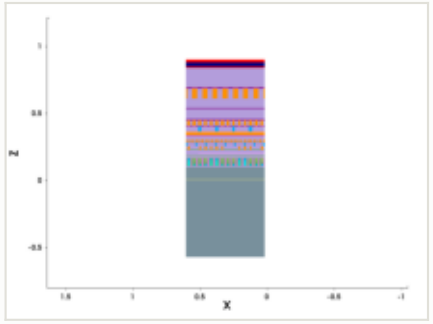
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

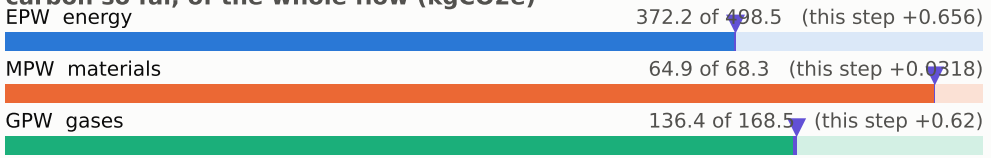
no mask for this step

this step

process area	CVD
material	—
thickness	0.001 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

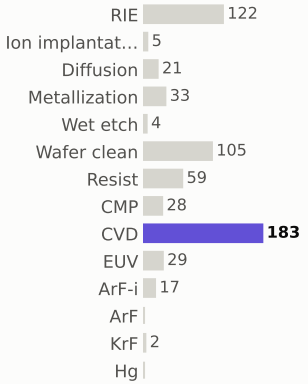
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

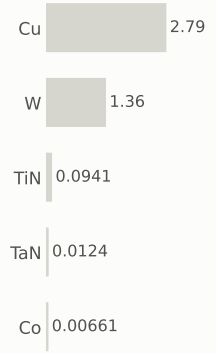


embodied carbon up to each step (kgCO2e per wafer)

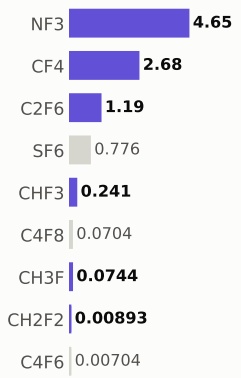
steps by process area



materials deposited (g)



gases emitted (g)



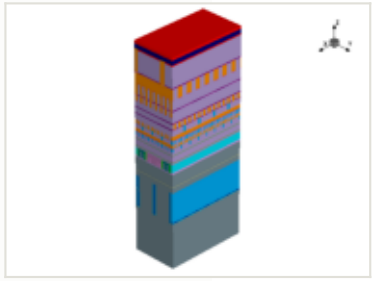
Step #6350: RIE

etch_anisotropic · RIE

1.04	574.5	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

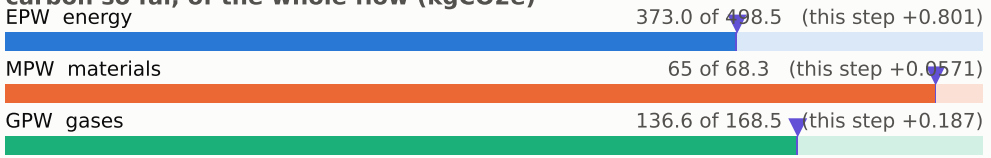


this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

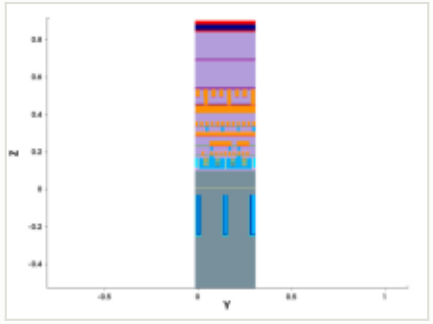
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

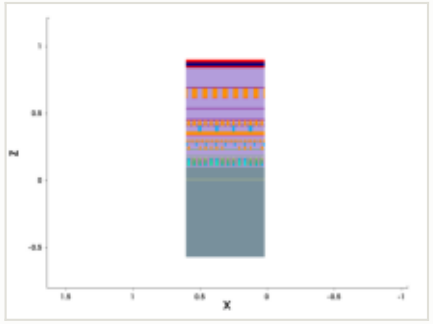


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



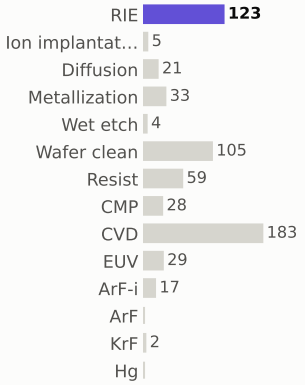
y cut at 0.184 µm



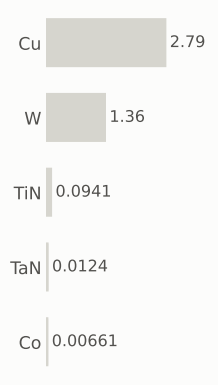
mask for this step

no mask for this step

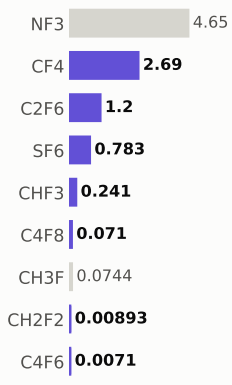
steps by process area



materials deposited (g)



gases emitted (g)



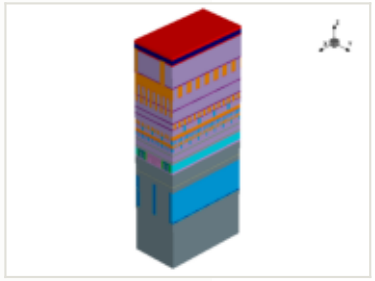
Step #6360: Wafer_clean

etch_strip · Wafer_clean

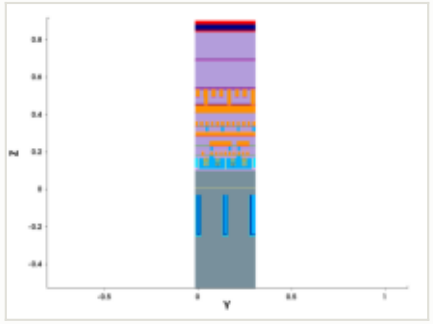
0.169	574.6	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

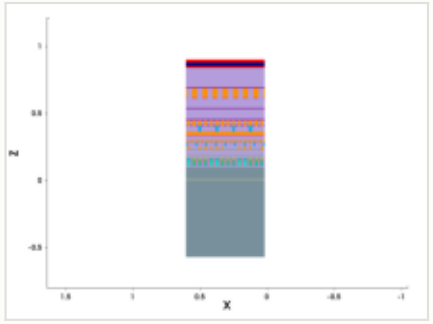
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

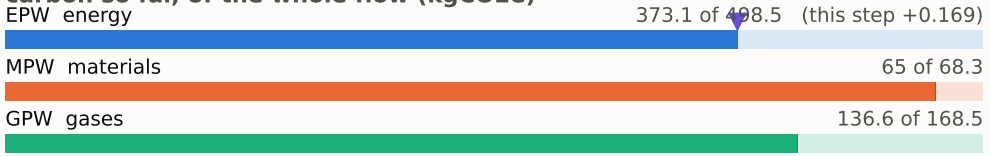
no mask for this step

this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

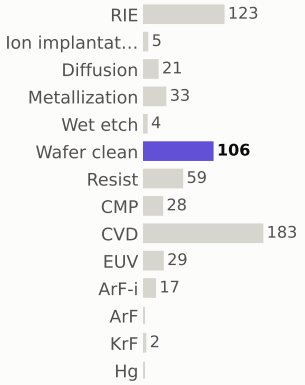
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

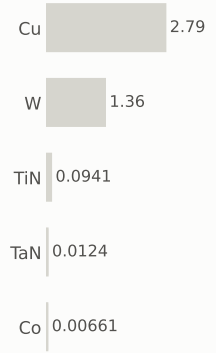


embodied carbon up to each step (kgCO2e per wafer)

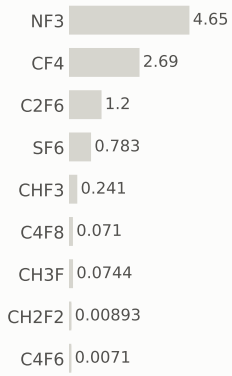
steps by process area



materials deposited (g)



gases emitted (g)



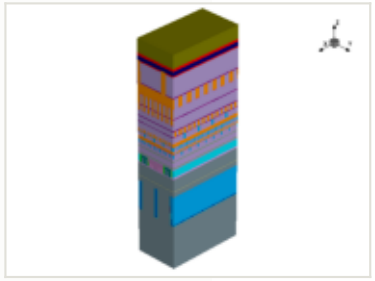
Step #6370: Resist

depo_fill · Resist

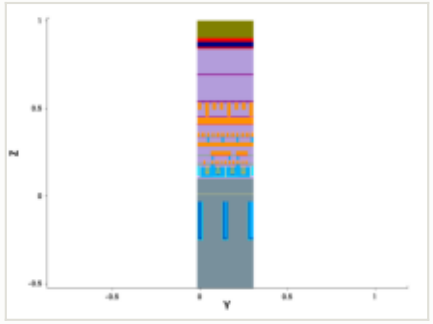
0.169	574.8	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

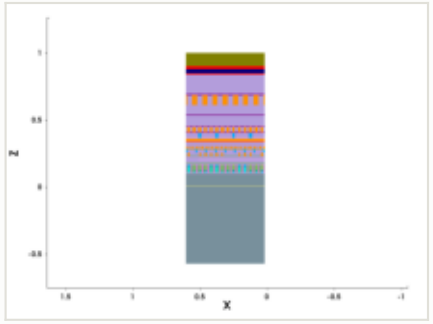
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



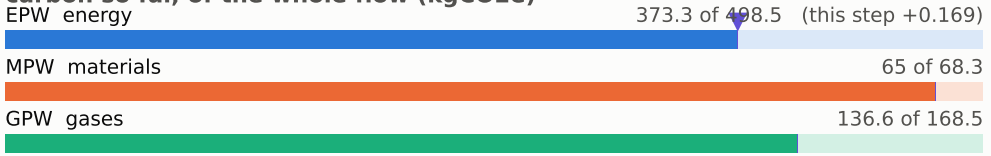
mask for this step

no mask for this step

this step

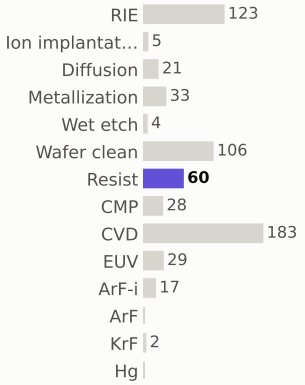
process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	0.1 µm	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

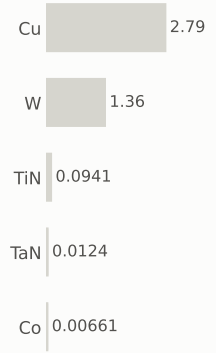


embodied carbon up to each step (kgCO2e per wafer)

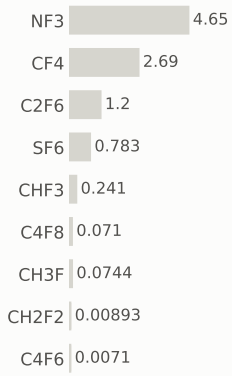
steps by process area



materials deposited (g)



gases emitted (g)



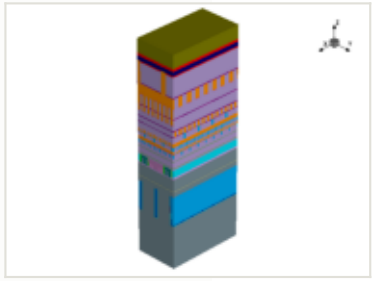
Step #6380: Wafer_clean

clean · Wafer_clean

0.169	575.0	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



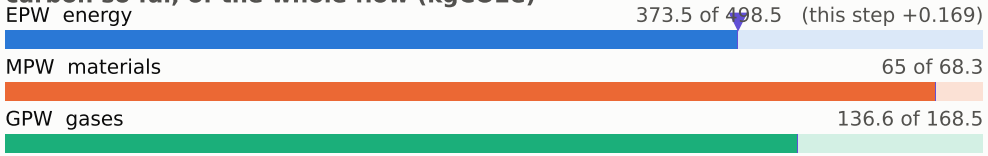
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

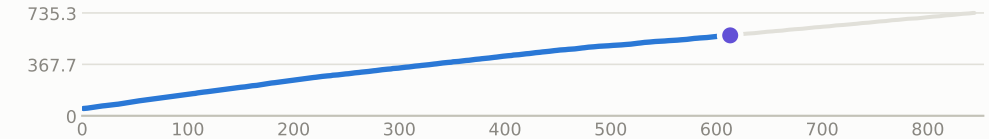
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

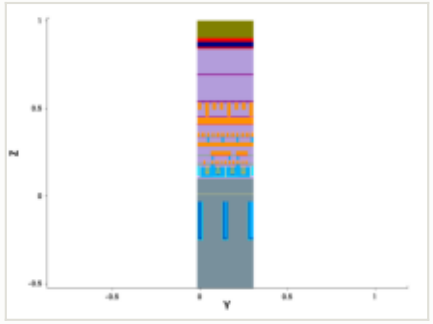
carbon so far, of the whole flow (kgCO2e)



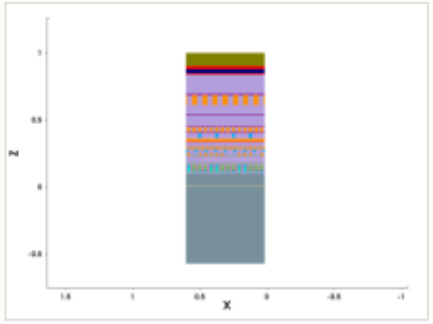
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



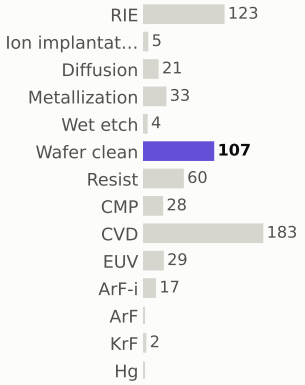
y cut at 0.184 µm



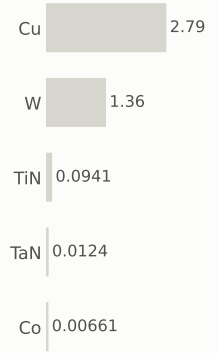
mask for this step

no mask for this step

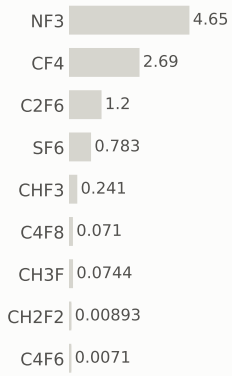
steps by process area



materials deposited (g)



gases emitted (g)



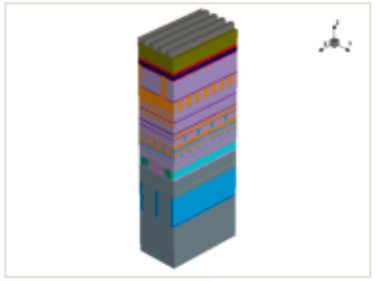
Step #6390: ArF-i

litho_patterning · ArF-i

0.217	575.2	0.029%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

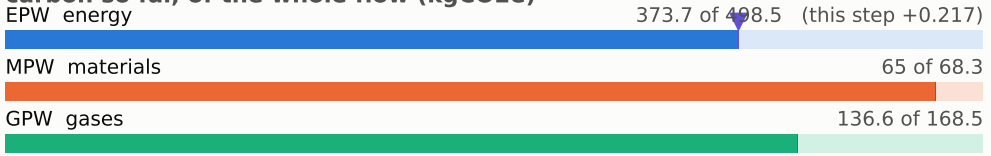


this step

process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

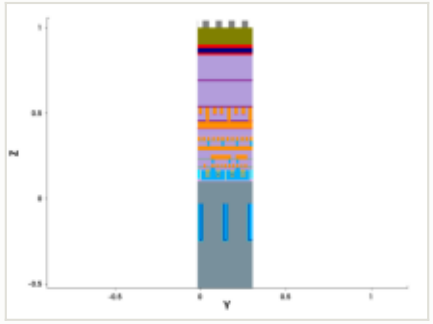
EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

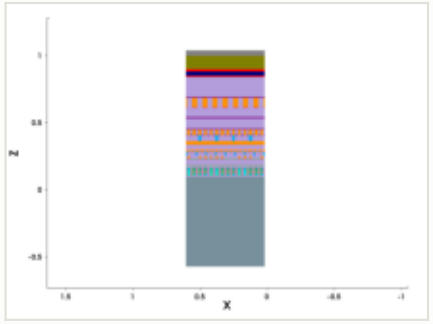


embodied carbon up to each step (kgCO2e per wafer)

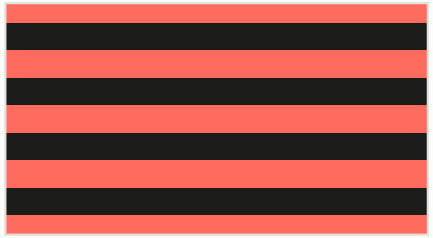
x cut at 0.294 µm



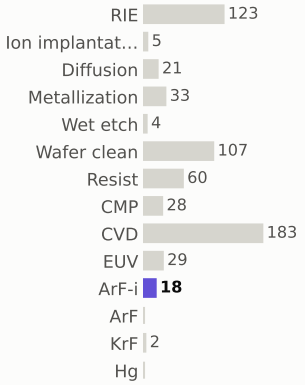
y cut at 0.184 µm



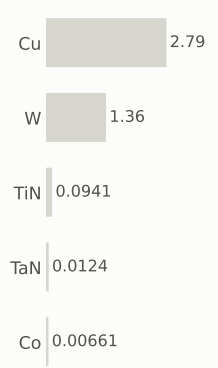
mask: layer 55:0 (GDSII)



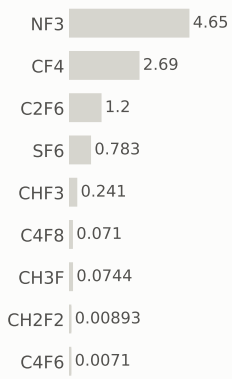
steps by process area



materials deposited (g)



gases emitted (g)



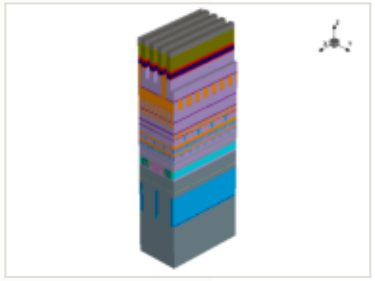
Step #6400: RIE

etch_anisotropic · RIE

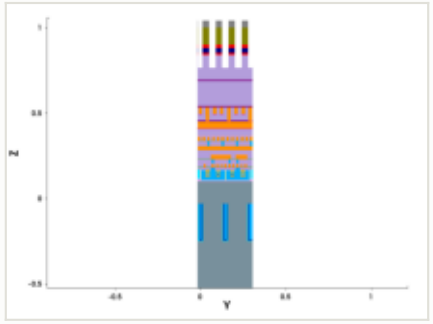
1.04	576.2	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

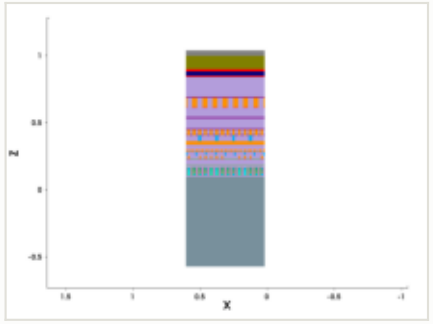
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

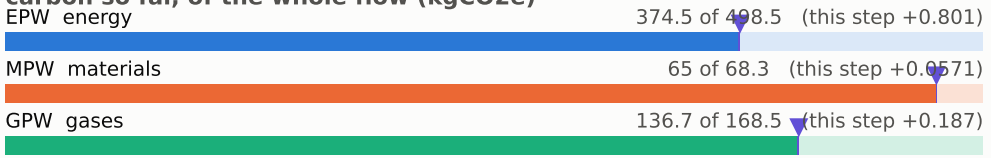
no mask for this step

this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

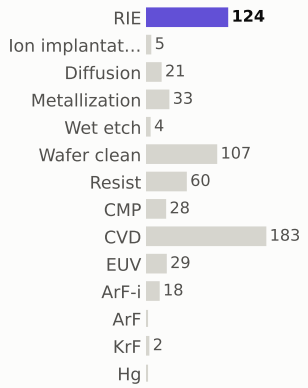
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

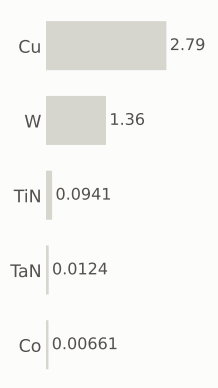


embodied carbon up to each step (kgCO2e per wafer)

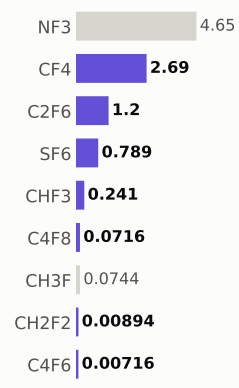
steps by process area



materials deposited (g)



gases emitted (g)



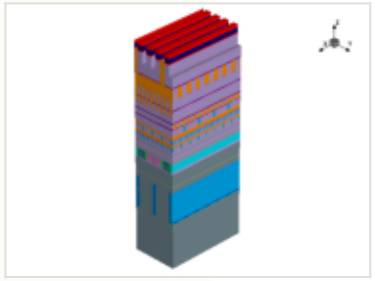
Step #6410: Wafer_clean

etch_strip · Wafer_clean

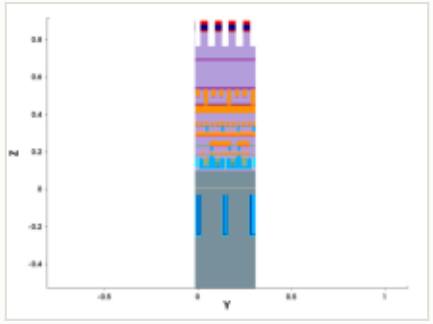
0.169	576.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

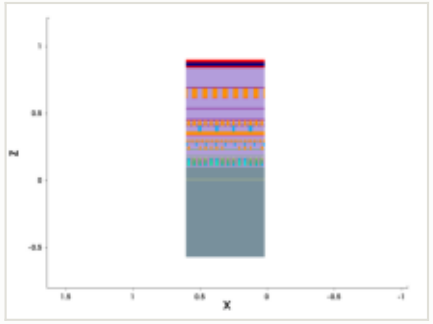
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



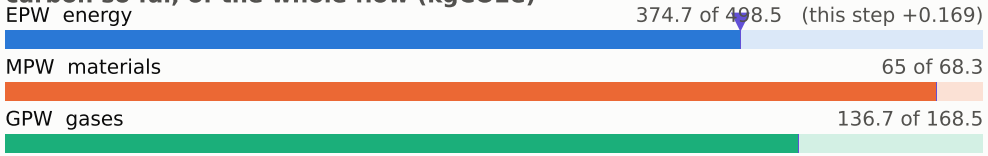
mask for this step

no mask for this step

this step

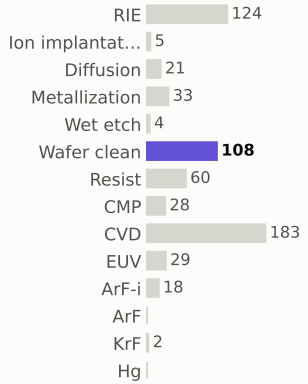
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

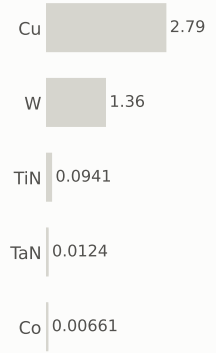


embodied carbon up to each step (kgCO2e per wafer)

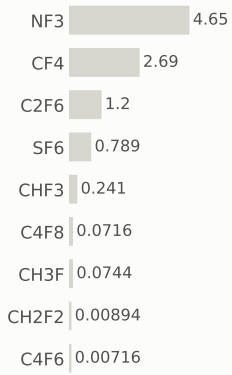
steps by process area



materials deposited (g)



gases emitted (g)



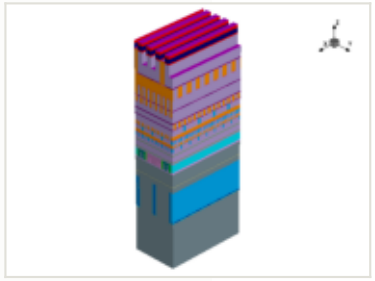
Step #6420: Metallization

depo_ellipsoidal · Metallization

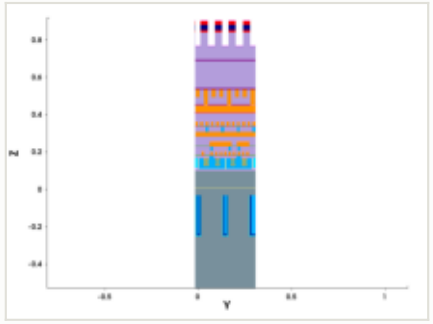
1.2	577.6	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

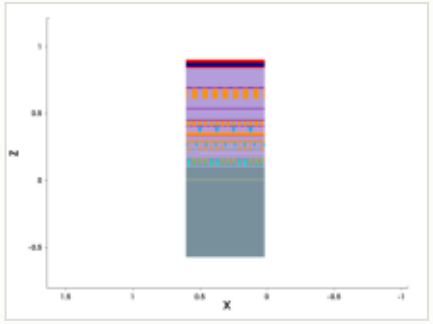
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

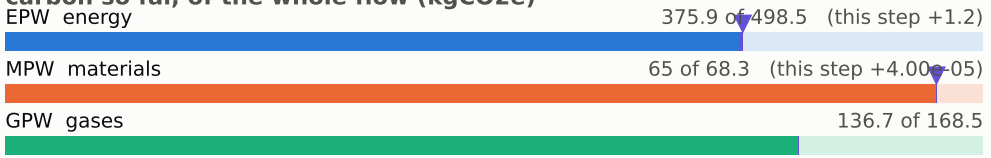
no mask for this step

this step

process area	Metallization
material	TaN
thickness	0.0016 µm
material GWP	25.8159 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

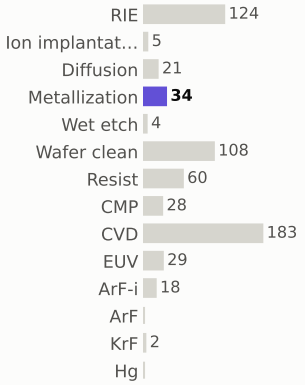
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.00e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

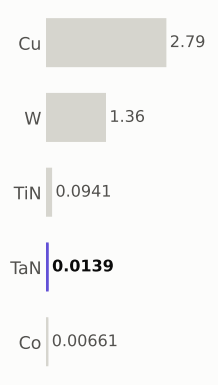


embodied carbon up to each step (kgCO2e per wafer)

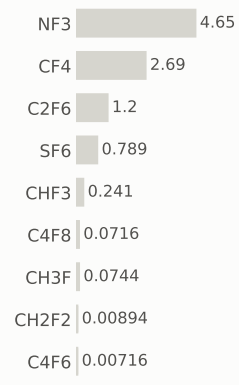
steps by process area



materials deposited (g)



gases emitted (g)



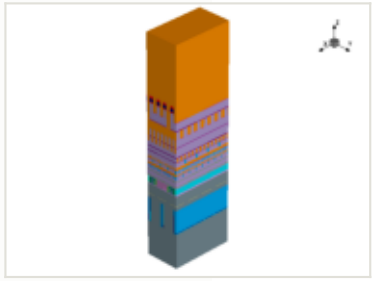
Step #6430: Metallization

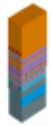
depo_fill · Metallization

1.21	578.8	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

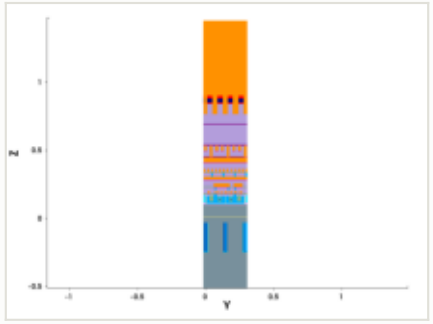
3D structure after this step

zoomed to fit

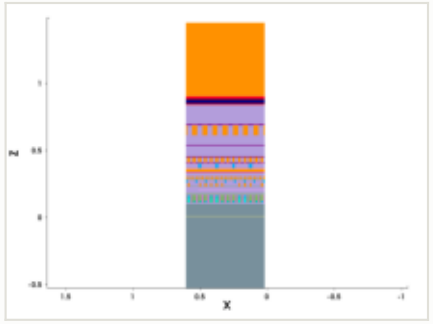




x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

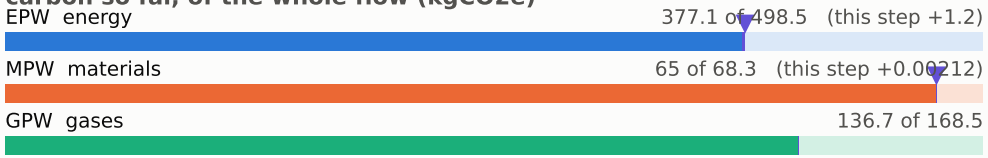
no mask for this step

this step

process area	Metallization
material	Cu
thickness	0.55 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

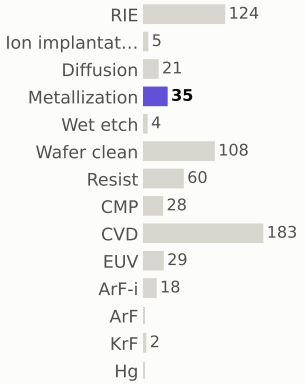
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00212 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

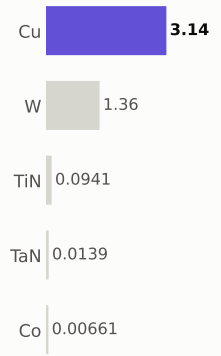


embodied carbon up to each step (kgCO2e per wafer)

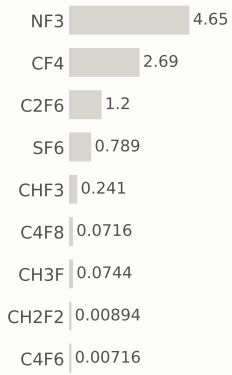
steps by process area



materials deposited (g)



gases emitted (g)



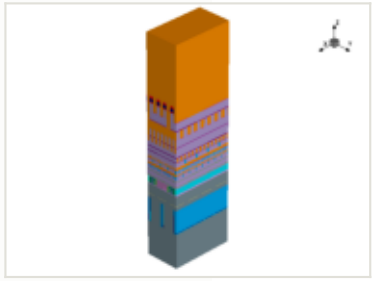
Step #6440: Diffusion

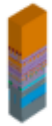
treatment · Diffusion

0.801	579.6	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

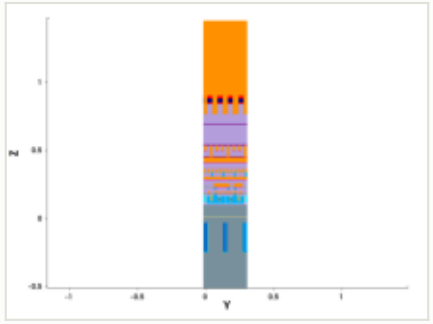
3D structure after this step

zoomed to fit

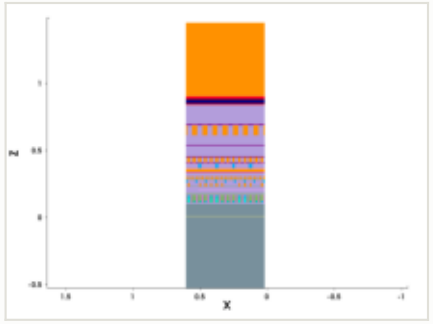




x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

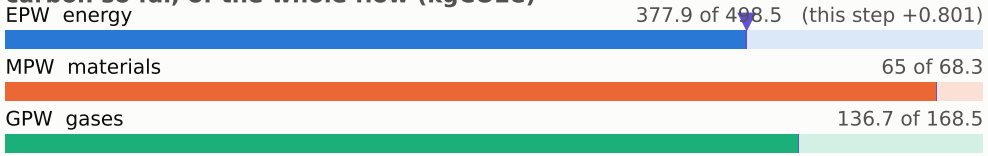
no mask for this step

this step

process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

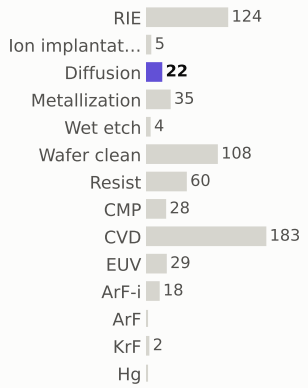
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

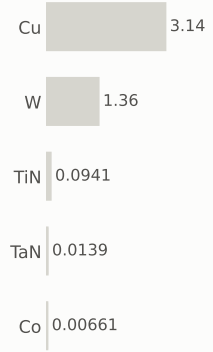


embodied carbon up to each step (kgCO2e per wafer)

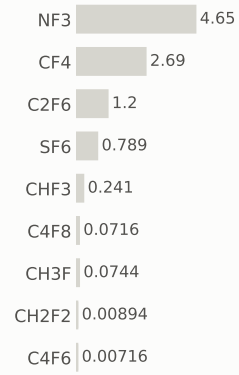
steps by process area



materials deposited (g)



gases emitted (g)



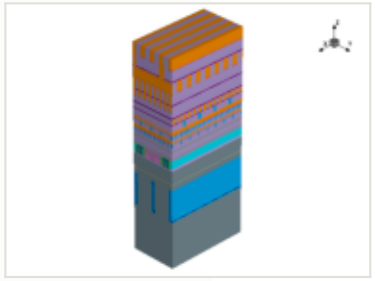
Step #6450: CMP

etch_cmp · CMP

0.169	579.8	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

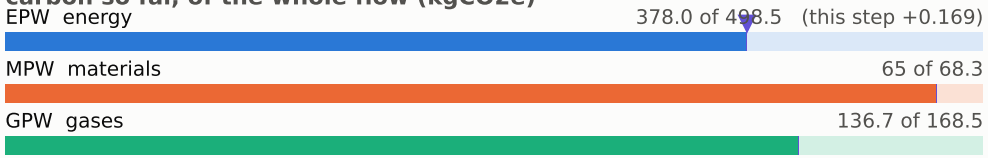


this step

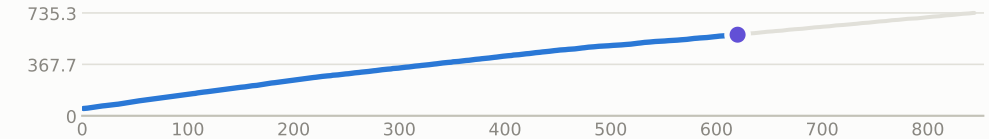
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

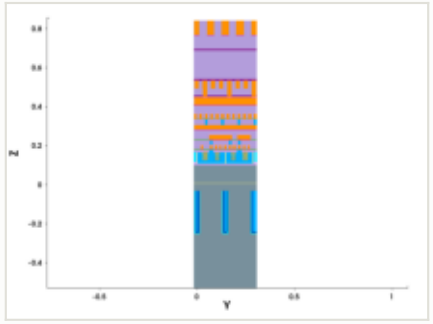
carbon so far, of the whole flow (kgCO2e)



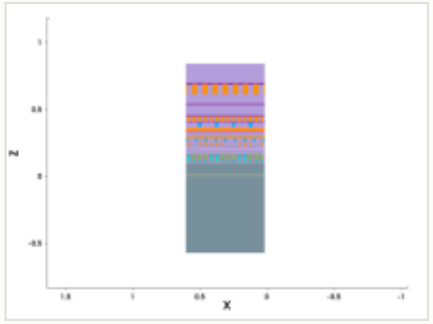
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



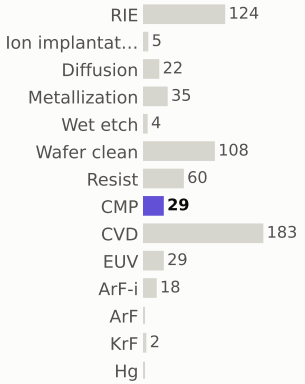
y cut at 0.184 µm



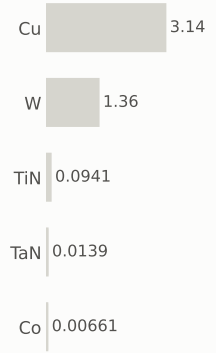
mask for this step

no mask for this step

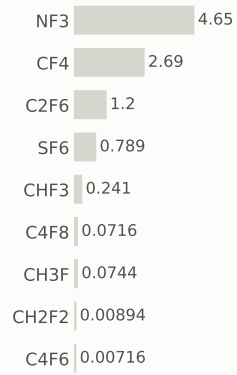
steps by process area



materials deposited (g)



gases emitted (g)



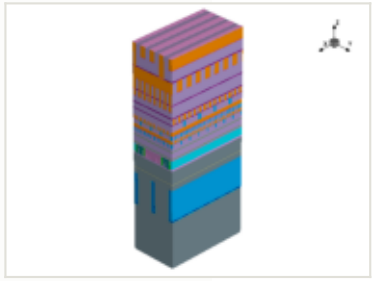
Step #6460: Metallization

depo_isotropic · Metallization

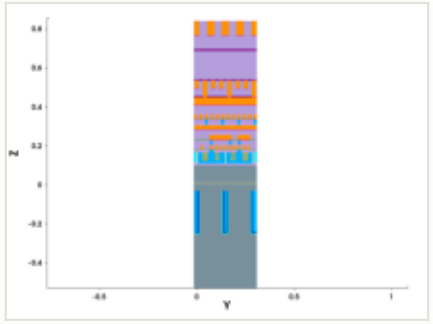
1.2	581.0	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

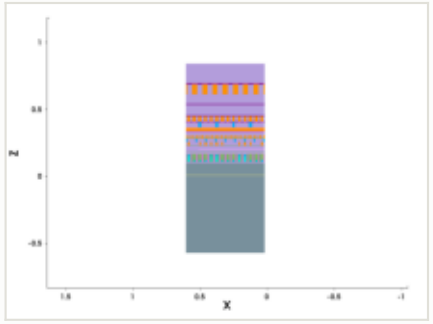
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

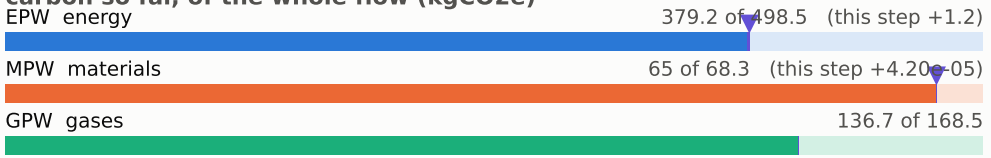
no mask for this step

this step

process area	Metallization
material	Co
thickness	0.0015 µm
material GWP	44.5 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

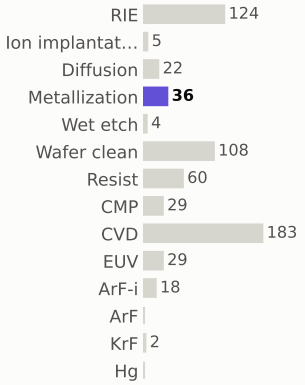
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.20e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

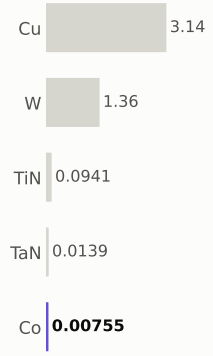


embodied carbon up to each step (kgCO2e per wafer)

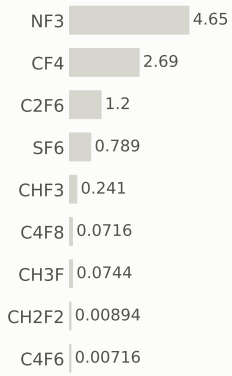
steps by process area



materials deposited (g)



gases emitted (g)



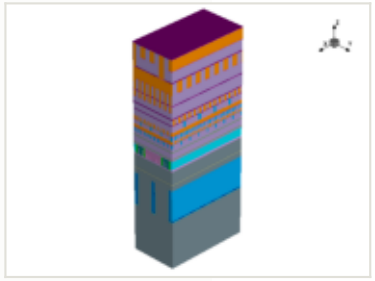
Step #6480: CVD

depo_fill · CVD

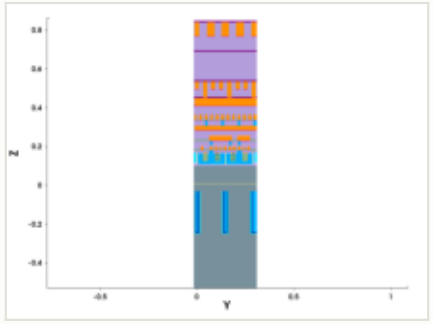
1.31	582.3	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

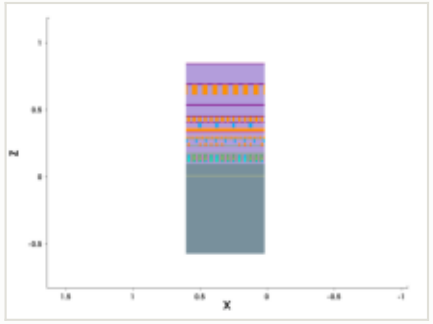
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

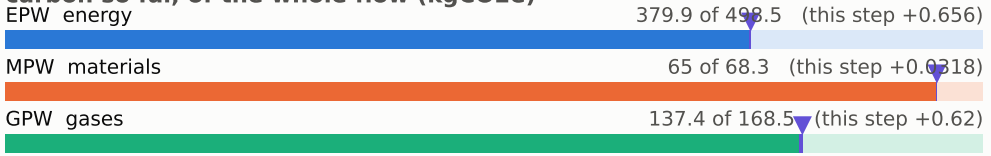
no mask for this step

this step

process area	CVD
material	—
thickness	0.006 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

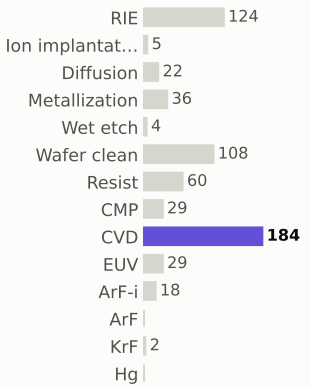
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

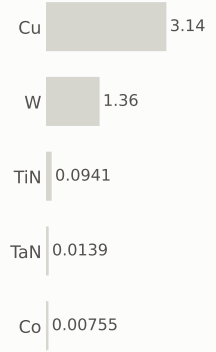


embodied carbon up to each step (kgCO2e per wafer)

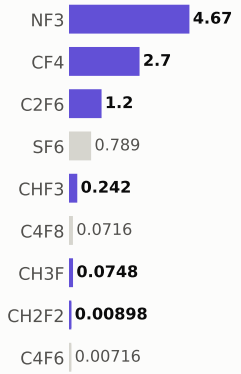
steps by process area



materials deposited (g)



gases emitted (g)



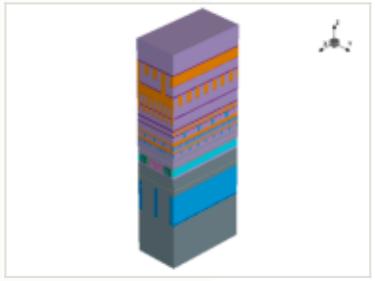
Step #6490: CVD

depo_fill · CVD

1.31	583.6	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

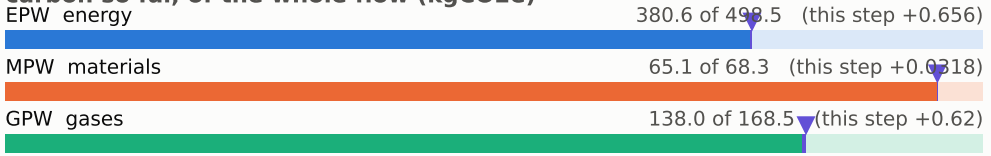


this step

process area	CVD
material	—
thickness	0.146 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

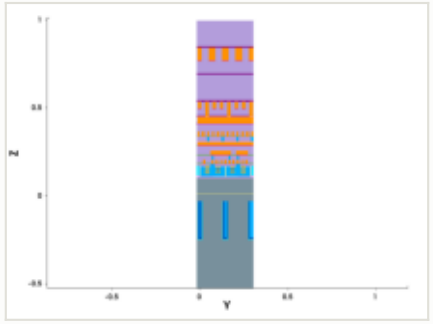
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

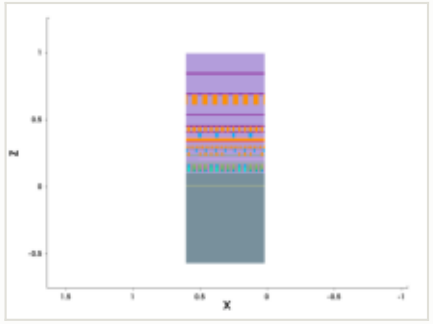


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



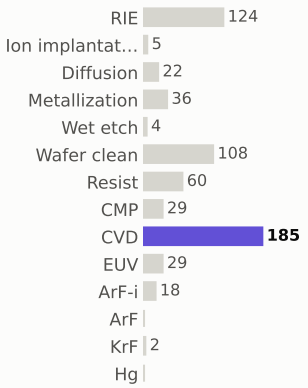
y cut at 0.184 µm



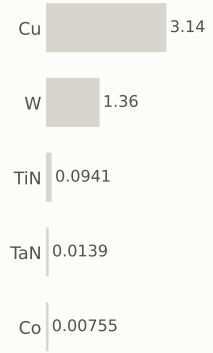
mask for this step

no mask for this step

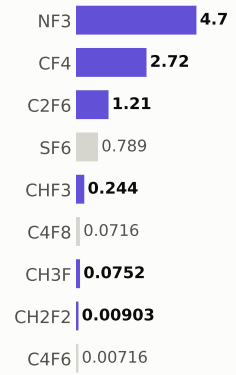
steps by process area



materials deposited (g)



gases emitted (g)



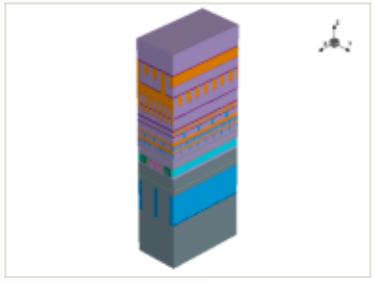
Step #6500: Diffusion

treatment · Diffusion

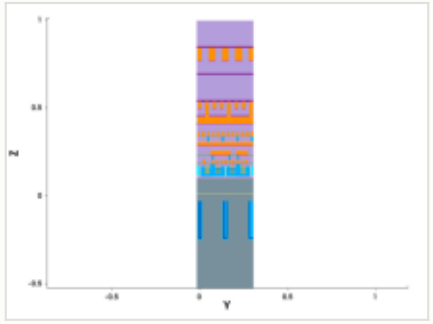
0.801	584.4	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

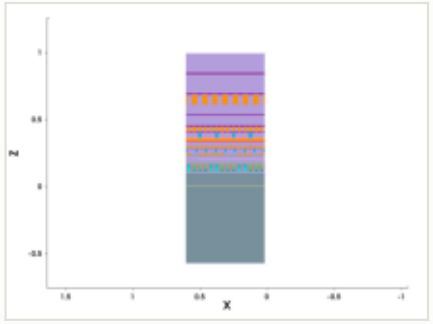
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

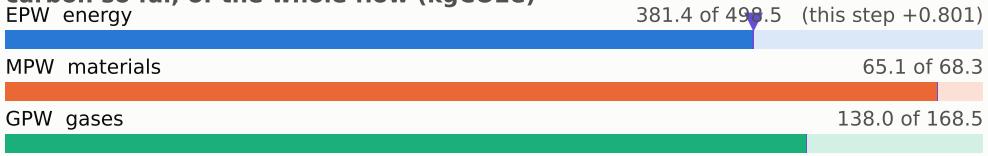
no mask for this step

this step

process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

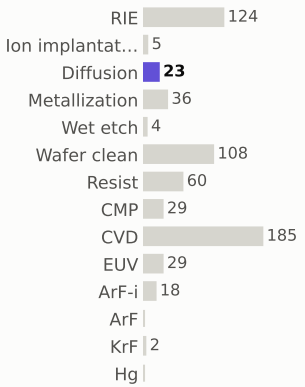
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

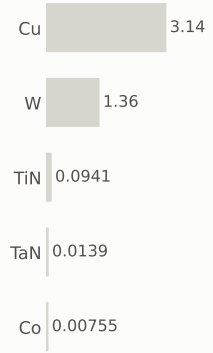


embodied carbon up to each step (kgCO2e per wafer)

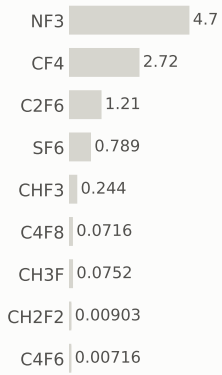
steps by process area



materials deposited (g)



gases emitted (g)



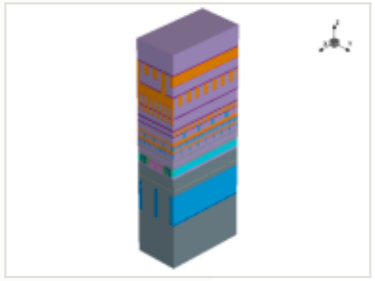
Step #6510: Wafer_clean

clean · Wafer_clean

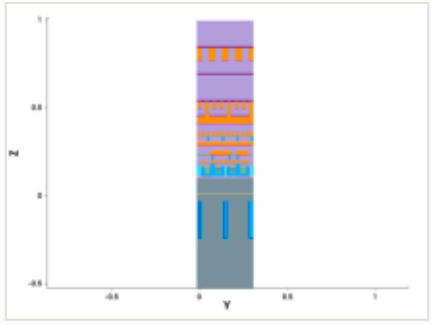
0.169	584.6	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

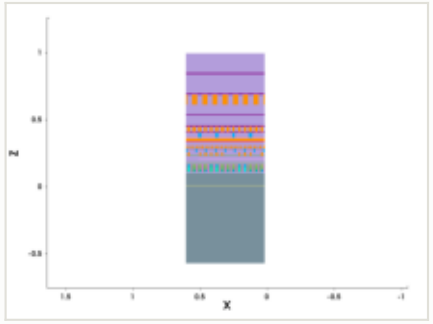
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

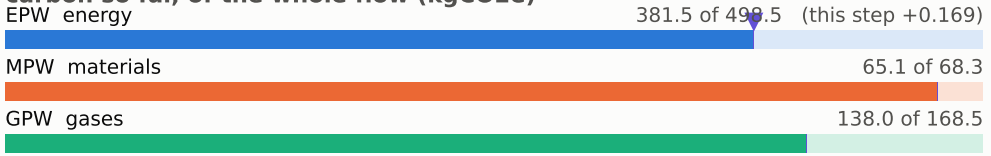
no mask for this step

this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

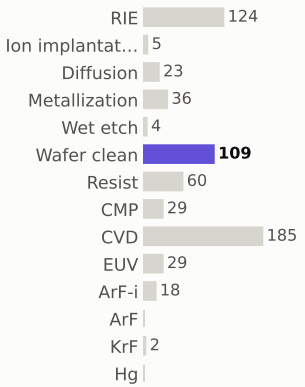
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

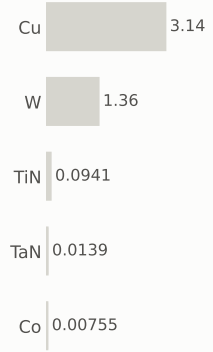


embodied carbon up to each step (kgCO2e per wafer)

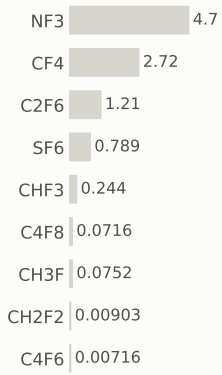
steps by process area



materials deposited (g)



gases emitted (g)



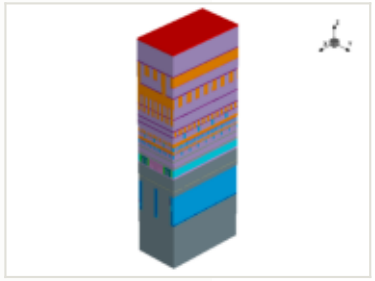
Step #6520: CVD

depo_fill · CVD

1.31	585.9	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

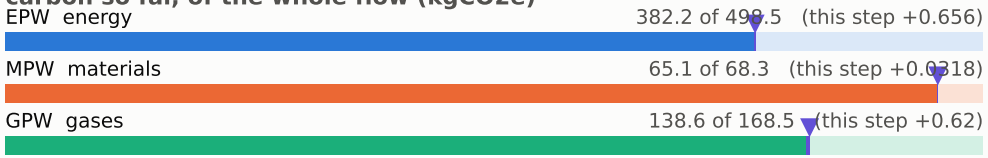


this step

process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

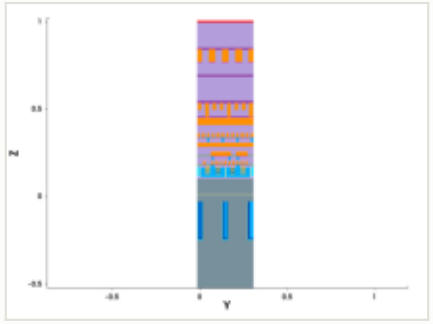
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

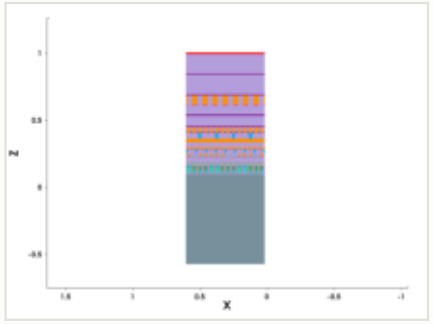


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



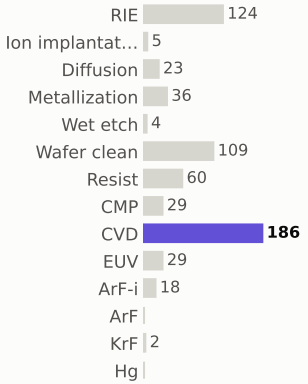
y cut at 0.184 µm



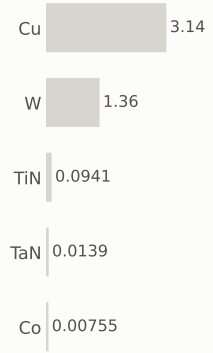
mask for this step

no mask for this step

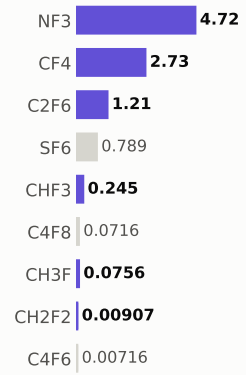
steps by process area



materials deposited (g)



gases emitted (g)



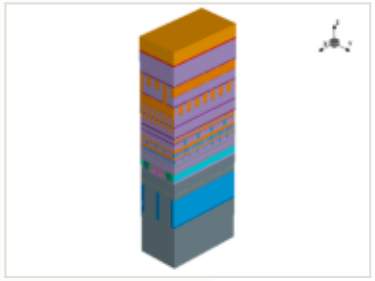
Step #6530: Resist

depo_fill · Resist

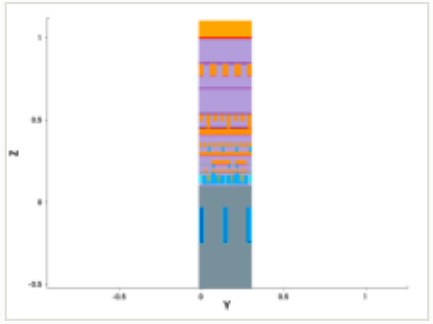
0.169	586.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

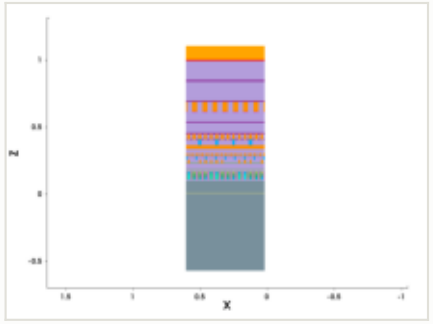
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



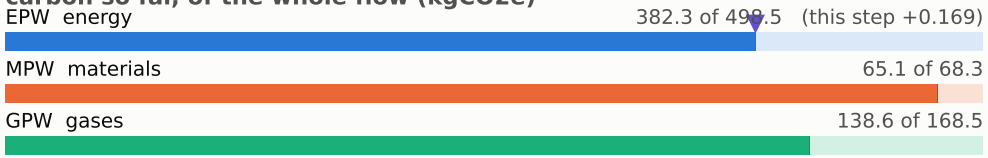
mask for this step

no mask for this step

this step

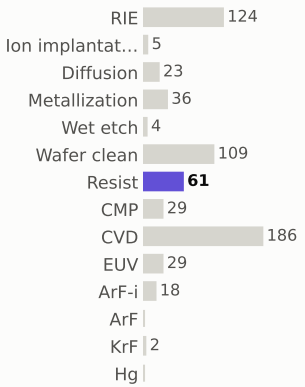
process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	0.1 µm	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

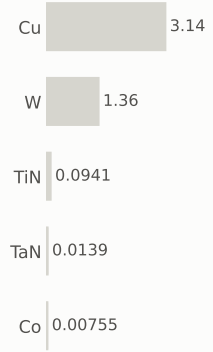


embodied carbon up to each step (kgCO2e per wafer)

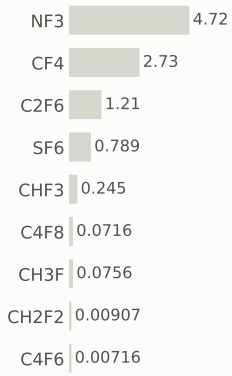
steps by process area



materials deposited (g)



gases emitted (g)



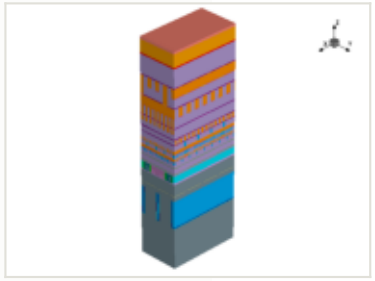
Step #6540: CVD

depo_fill · CVD

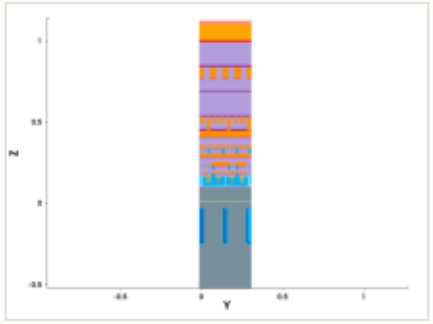
1.31	587.4	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

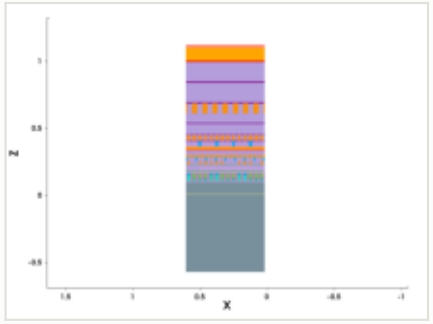
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

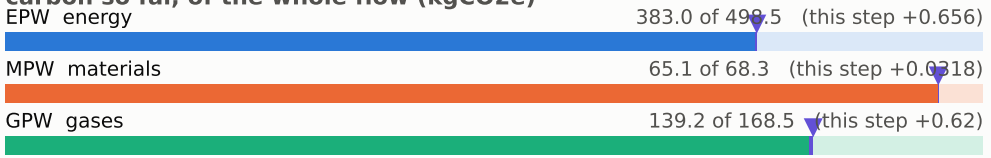
no mask for this step

this step

process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

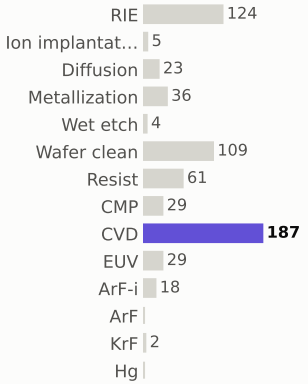
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

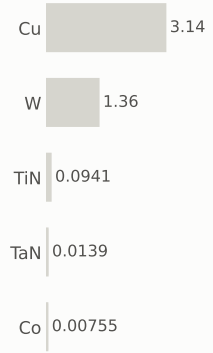


embodied carbon up to each step (kgCO2e per wafer)

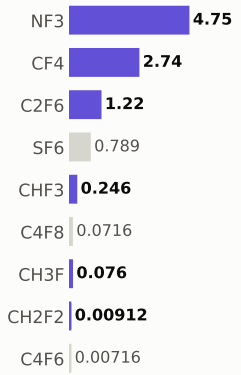
steps by process area



materials deposited (g)



gases emitted (g)



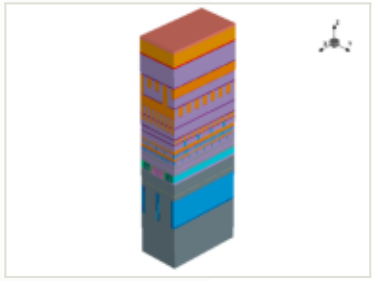
Step #6550: Wafer_clean

clean · Wafer_clean

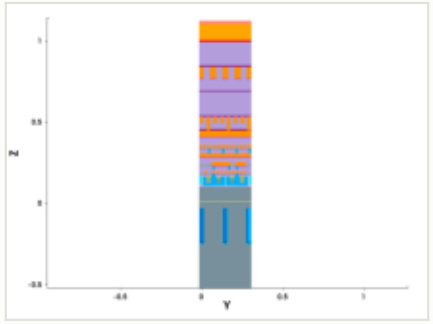
0.169	587.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

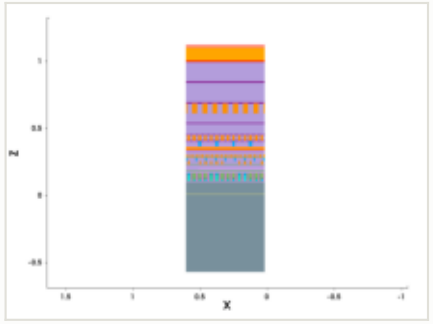
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

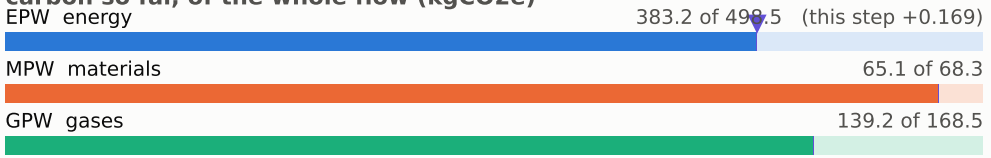
no mask for this step

this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

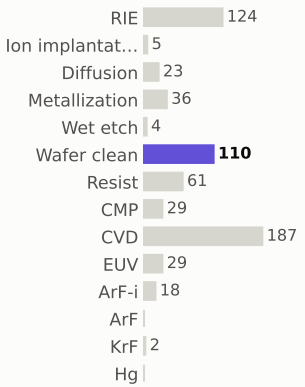
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

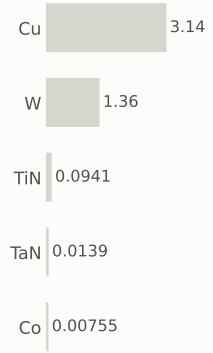


embodied carbon up to each step (kgCO2e per wafer)

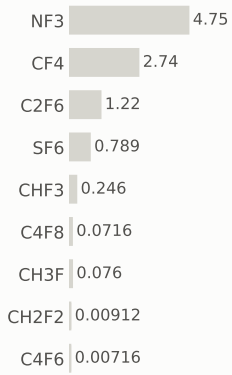
steps by process area



materials deposited (g)



gases emitted (g)



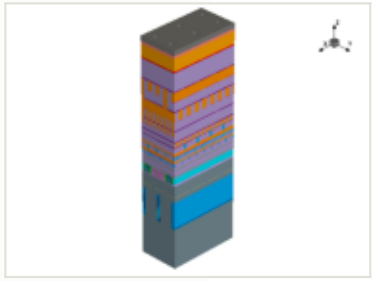
Step #6560: ArF-i

litho_patterning · ArF-i

0.217	587.7	0.029%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

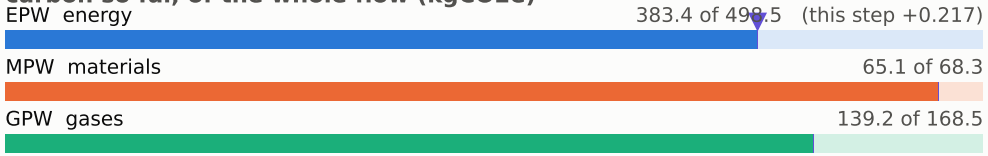


this step

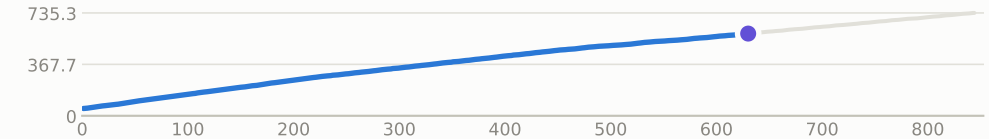
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

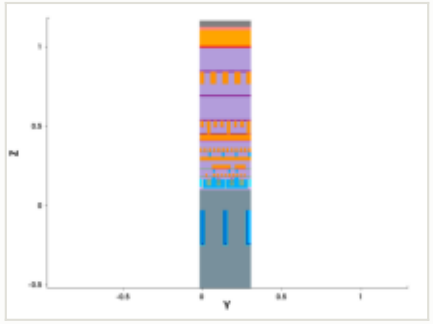
carbon so far, of the whole flow (kgCO2e)



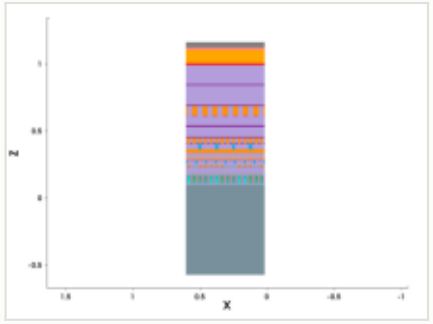
embodied carbon up to each step (kgCO2e per wafer)



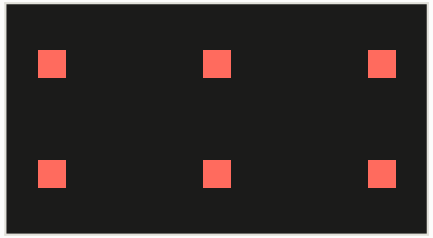
x cut at 0.294 µm



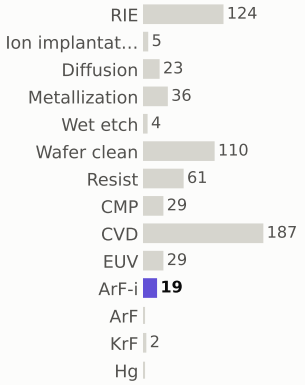
y cut at 0.184 µm



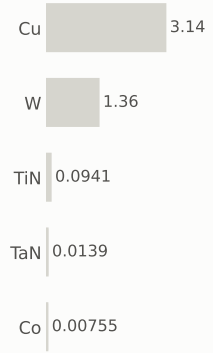
mask: layer 10571:0 (GDSII)



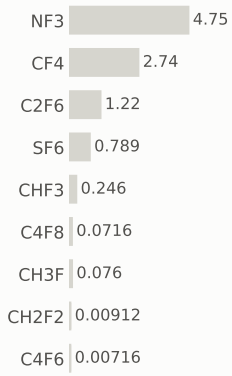
steps by process area



materials deposited (g)



gases emitted (g)



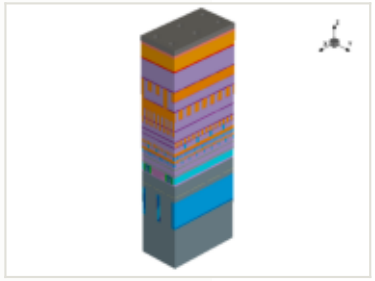
Step #6570: RIE

etch_anisotropic · RIE

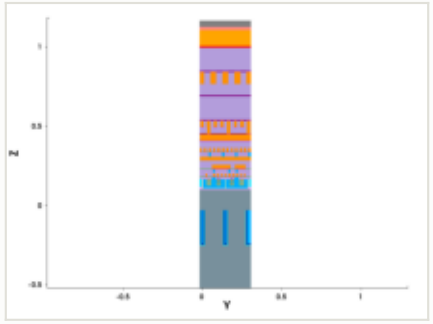
1.04	588.8	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

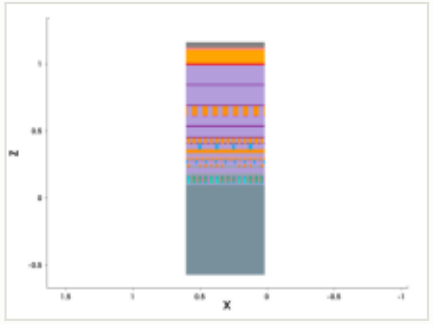
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

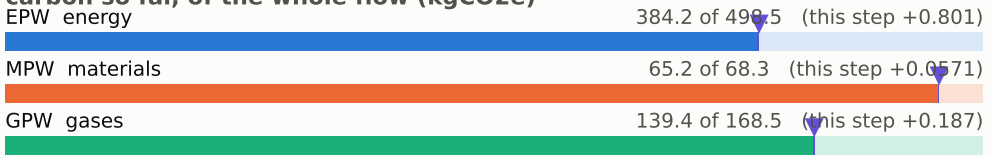
no mask for this step

this step

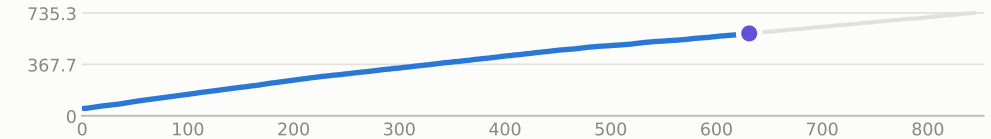
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

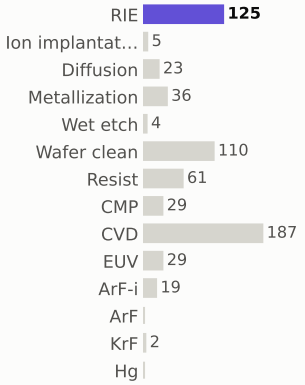
carbon so far, of the whole flow (kgCO2e)



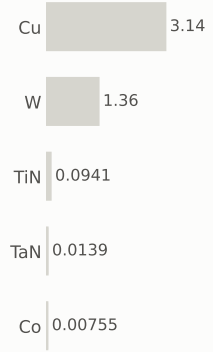
embodied carbon up to each step (kgCO2e per wafer)



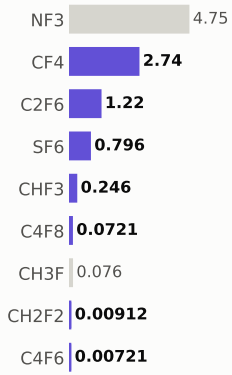
steps by process area



materials deposited (g)



gases emitted (g)



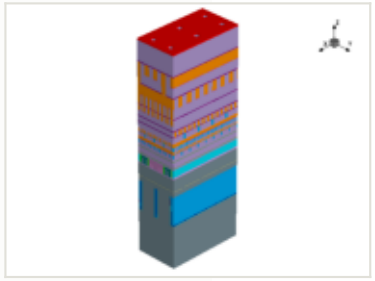
Step #6580: Wafer_clean

etch_strip · Wafer_clean

0.169	589.0	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

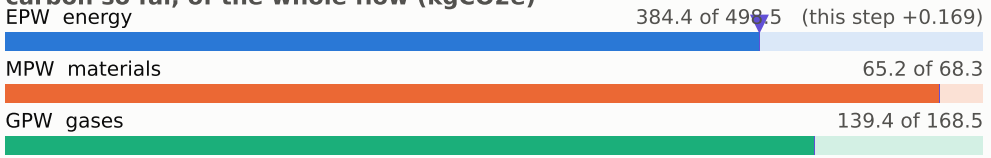


this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

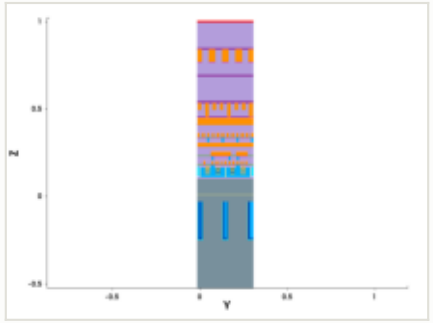
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

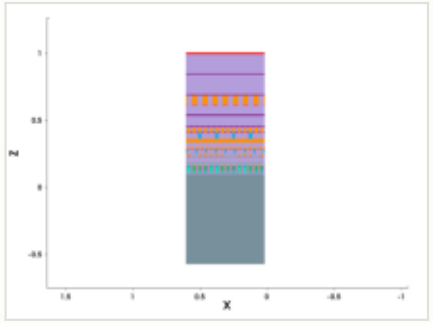


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



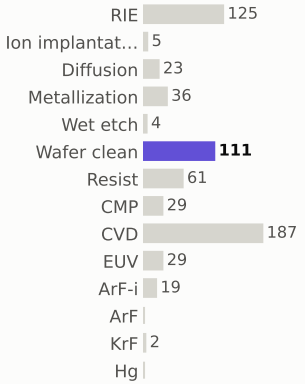
y cut at 0.184 µm



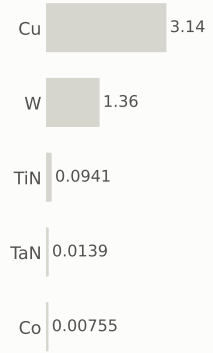
mask for this step

no mask for this step

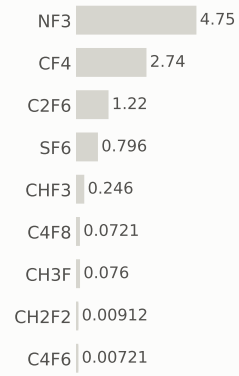
steps by process area



materials deposited (g)



gases emitted (g)



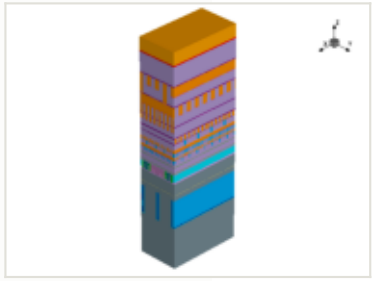
Step #6590: Resist

depo_fill · Resist

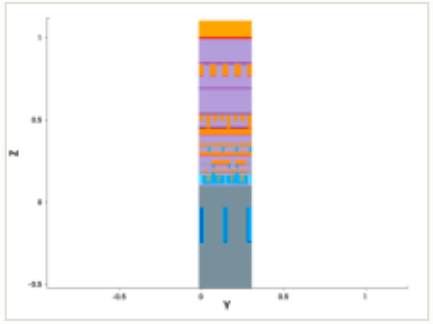
0.169	589.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

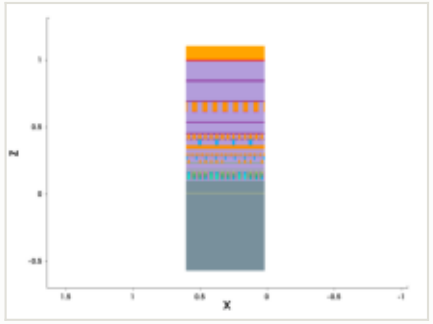
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



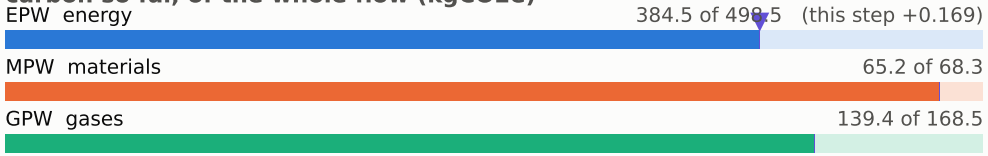
mask for this step

no mask for this step

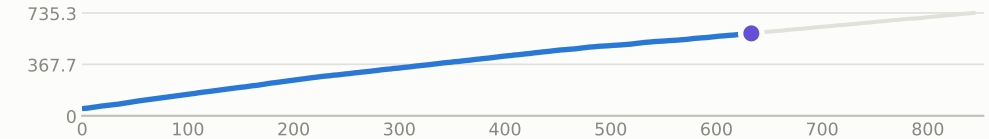
this step

process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	0.1 µm	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

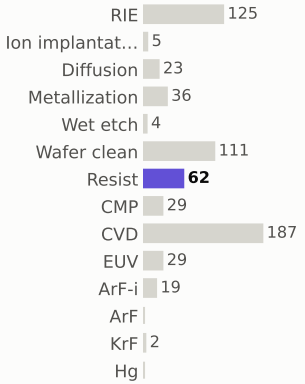
carbon so far, of the whole flow (kgCO2e)



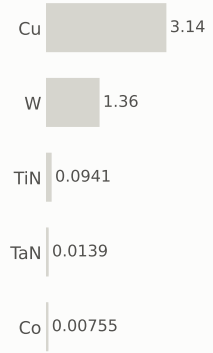
embodied carbon up to each step (kgCO2e per wafer)



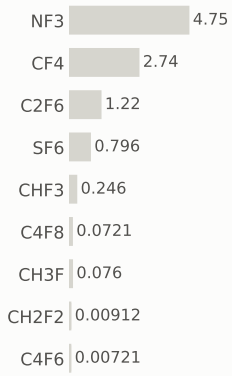
steps by process area



materials deposited (g)



gases emitted (g)



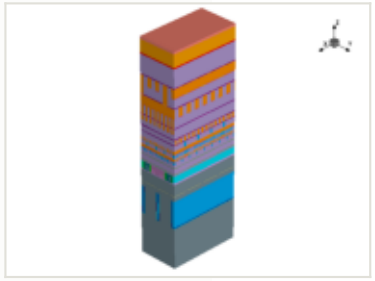
Step #6600: CVD

depo_fill · CVD

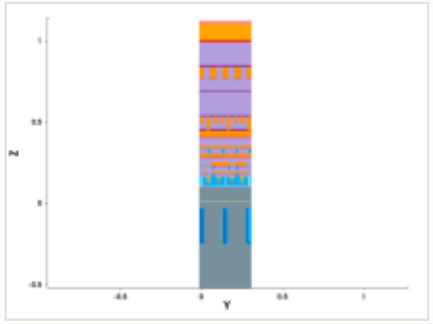
1.31	590.4	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

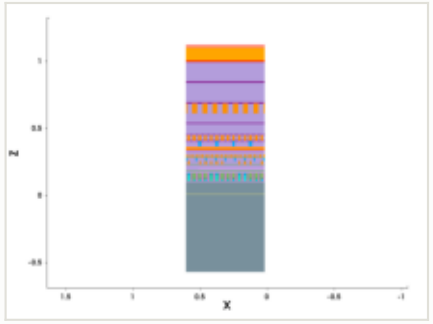
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

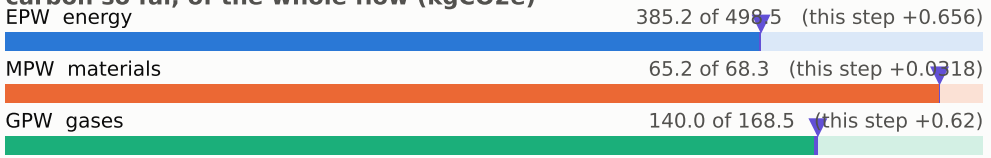
no mask for this step

this step

process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

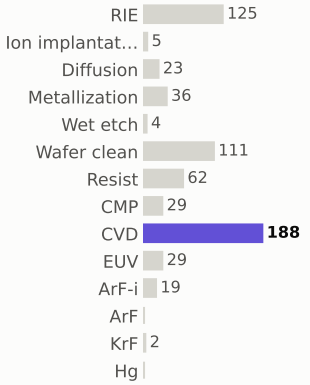
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

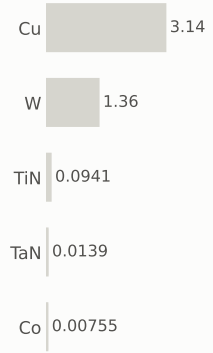


embodied carbon up to each step (kgCO2e per wafer)

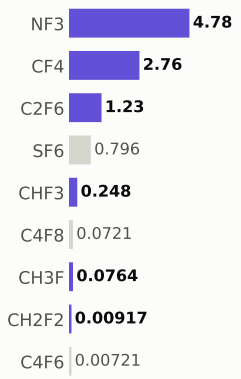
steps by process area



materials deposited (g)



gases emitted (g)



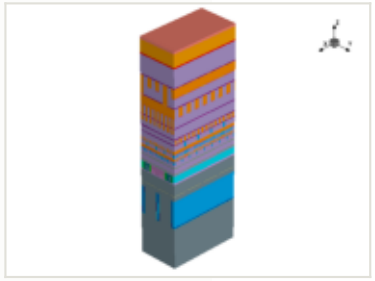
Step #6610: Wafer_clean

clean · Wafer_clean

0.169	590.6	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



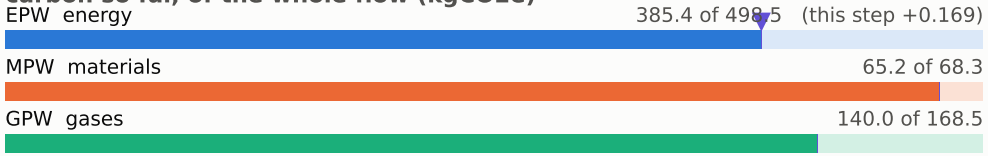
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

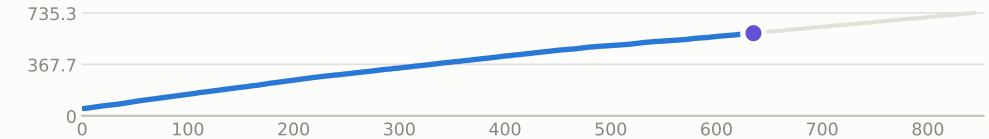
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

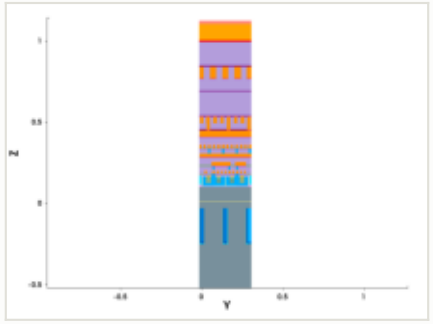
carbon so far, of the whole flow (kgCO2e)



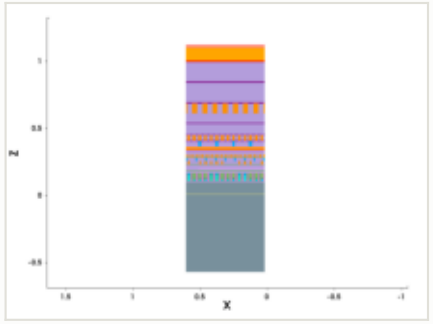
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



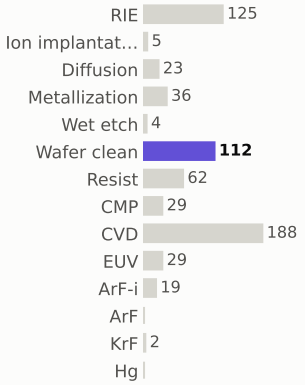
y cut at 0.184 µm



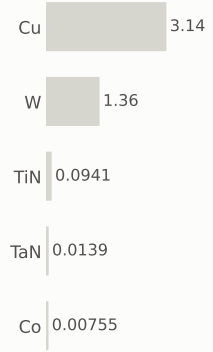
mask for this step

no mask for this step

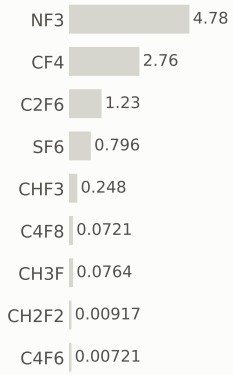
steps by process area



materials deposited (g)



gases emitted (g)



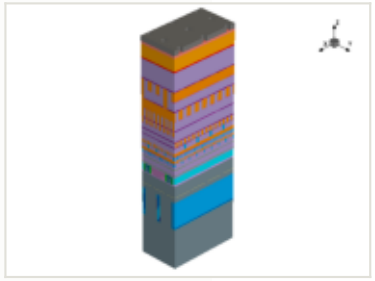
Step #6620: ArF-i

litho_patterning · ArF-i

0.217	590.8	0.029%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

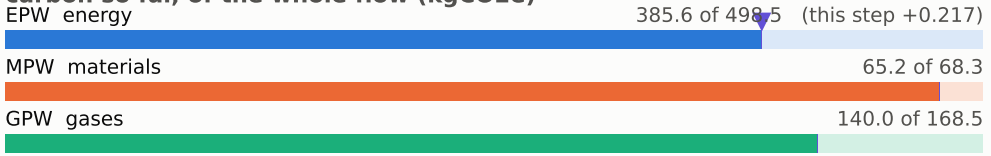


this step

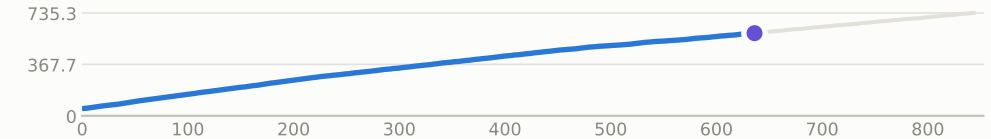
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

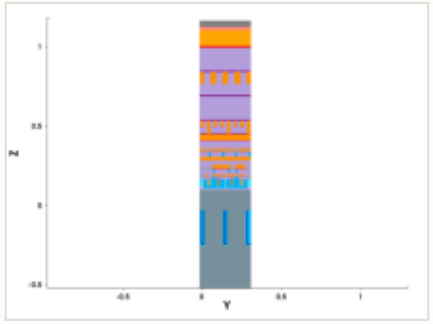
carbon so far, of the whole flow (kgCO2e)



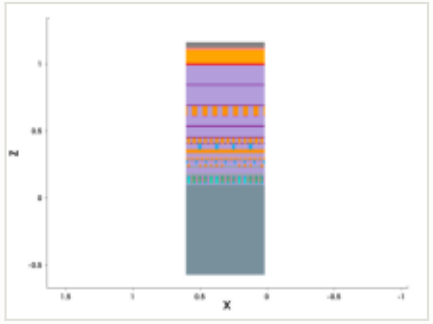
embodied carbon up to each step (kgCO2e per wafer)



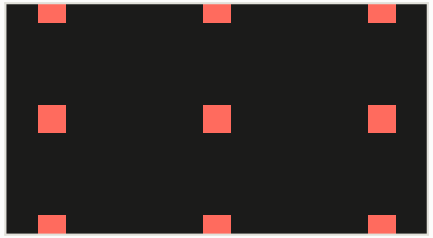
x cut at 0.294 µm



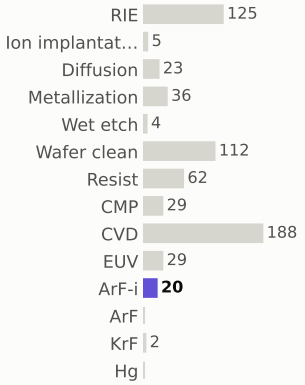
y cut at 0.184 µm



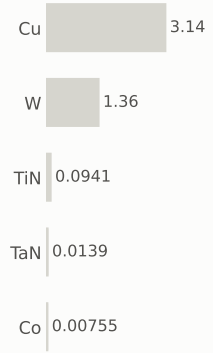
mask: layer 10572:0 (GDSII)



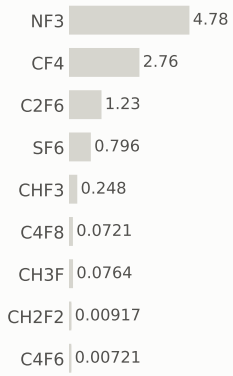
steps by process area



materials deposited (g)



gases emitted (g)



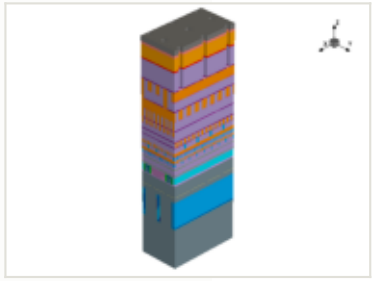
Step #6630: RIE

etch_anisotropic · RIE

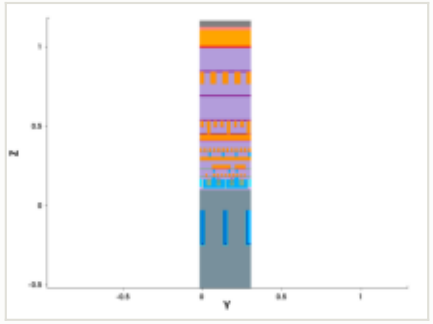
1.04	591.9	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

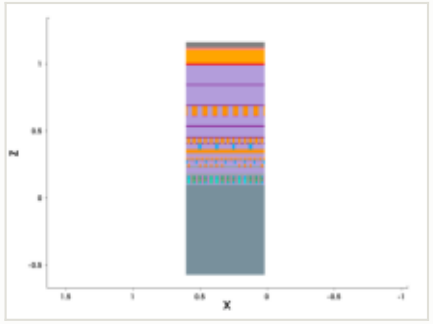
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

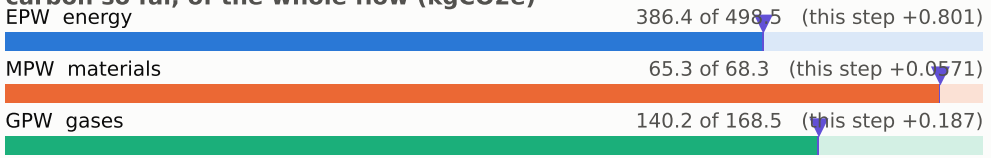
no mask for this step

this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

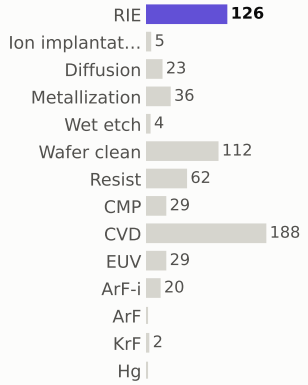
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

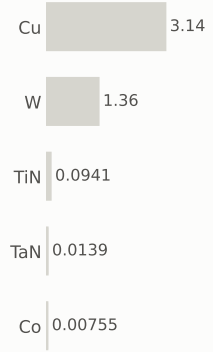


embodied carbon up to each step (kgCO2e per wafer)

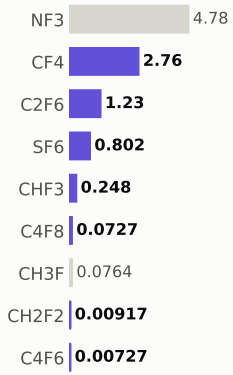
steps by process area



materials deposited (g)



gases emitted (g)



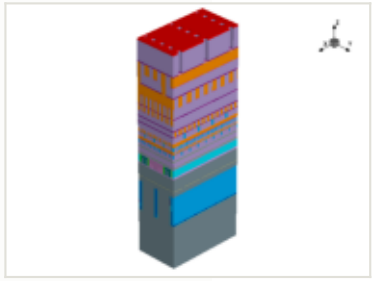
Step #6640: Wafer_clean

etch_strip · Wafer_clean

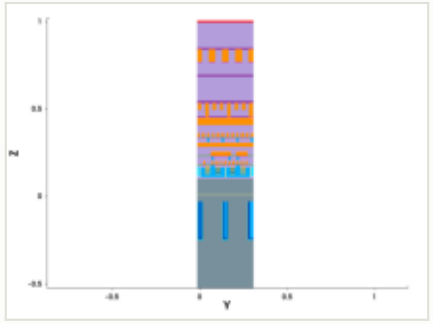
0.169	592.0	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

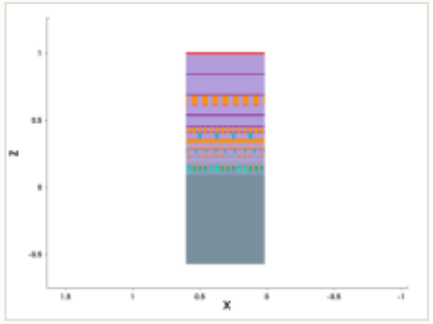
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



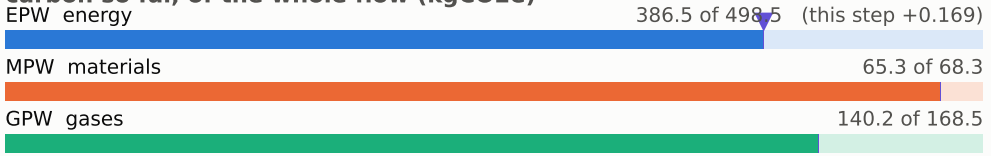
mask for this step

no mask for this step

this step

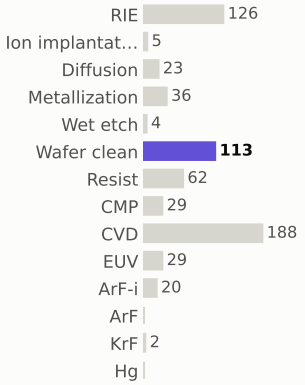
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

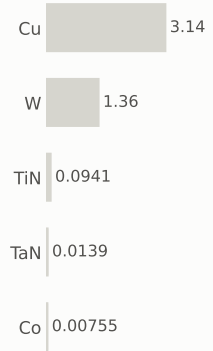


embodied carbon up to each step (kgCO2e per wafer)

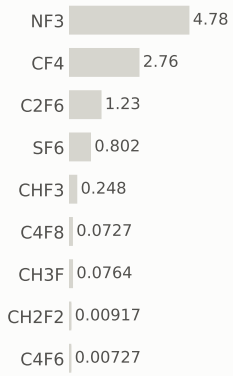
steps by process area



materials deposited (g)



gases emitted (g)



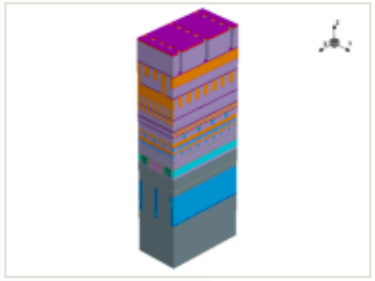
Step #6650: Metallization

depo_ellipsoidal · Metallization

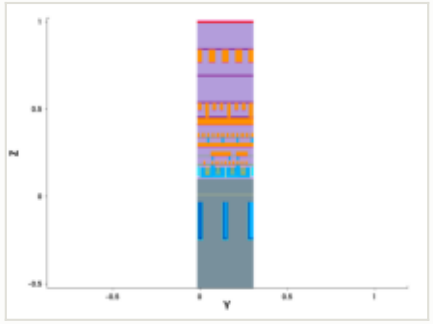
1.2	593.2	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

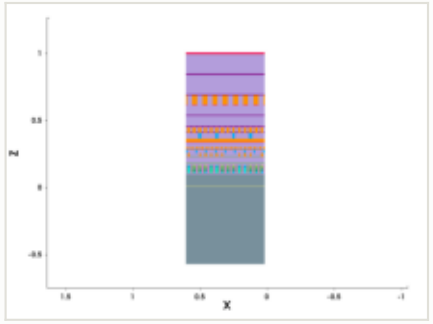
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

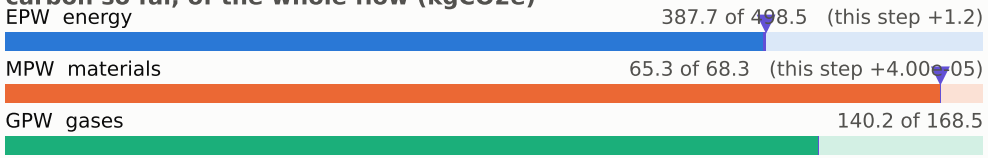
no mask for this step

this step

process area	Metallization
material	TaN
thickness	0.0016 µm
material GWP	25.8159 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

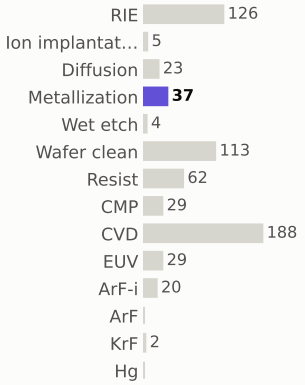
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.00e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

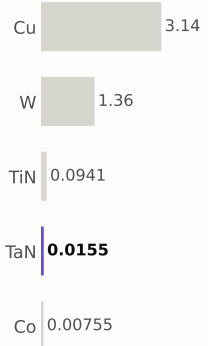


embodied carbon up to each step (kgCO2e per wafer)

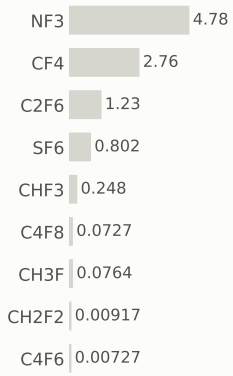
steps by process area



materials deposited (g)



gases emitted (g)



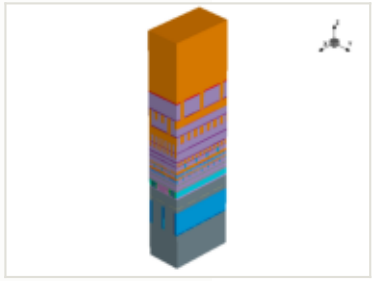
Step #6660: Metallization

depo_fill · Metallization

1.21	594.4	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

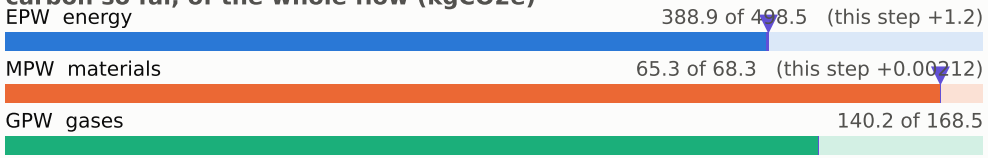


this step

process area	Metallization
material	Cu
thickness	0.55 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

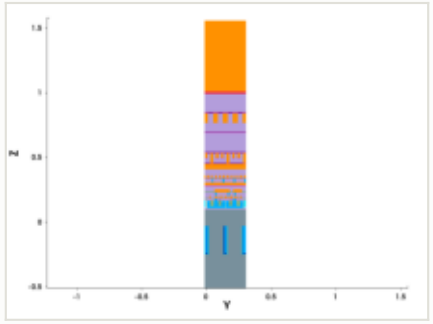
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00212 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

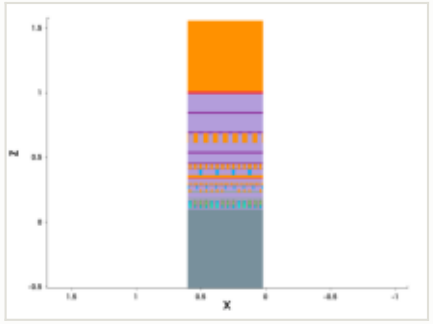


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



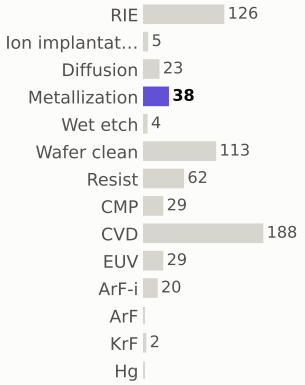
y cut at 0.184 µm



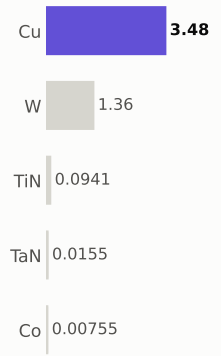
mask for this step

no mask for this step

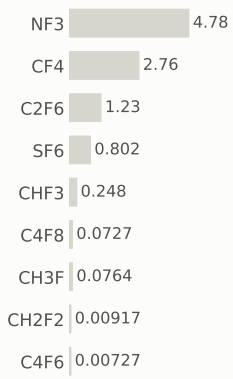
steps by process area



materials deposited (g)



gases emitted (g)



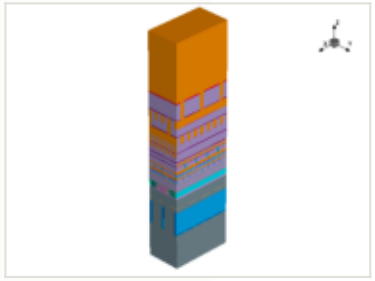
Step #6670: Diffusion

treatment · Diffusion

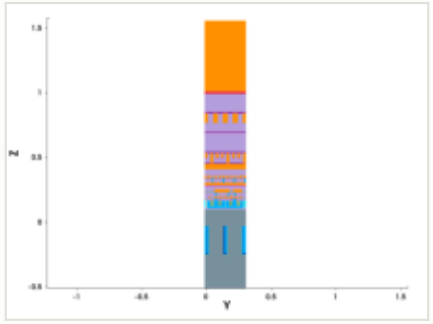
0.801	595.2	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

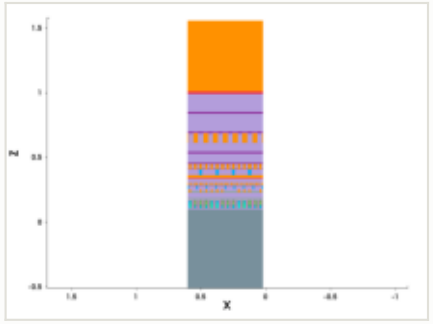
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

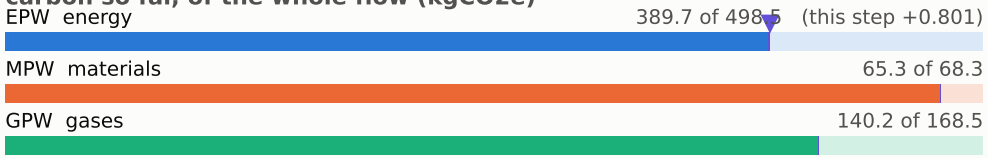
no mask for this step

this step

process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

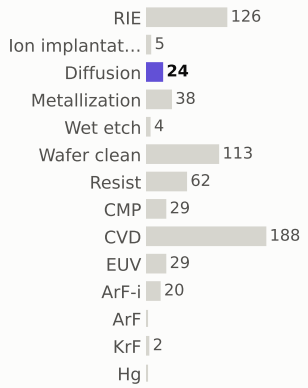
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

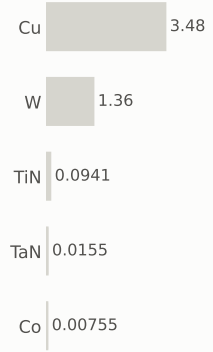


embodied carbon up to each step (kgCO2e per wafer)

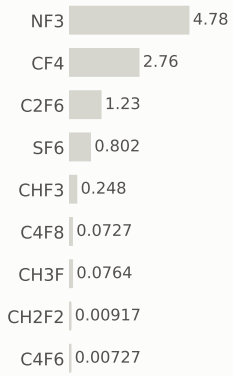
steps by process area



materials deposited (g)



gases emitted (g)



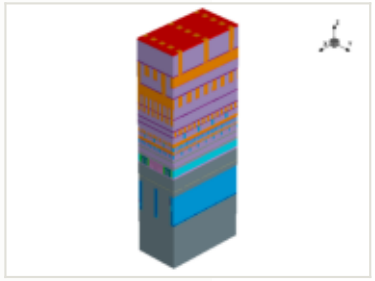
Step #6680: CMP

etch_cmp · CMP

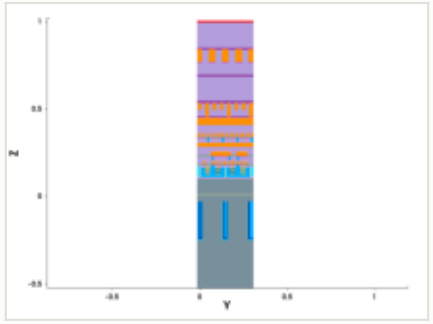
0.169	595.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

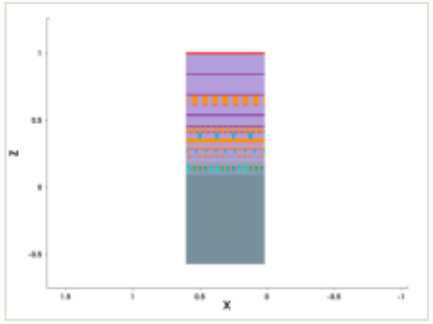
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

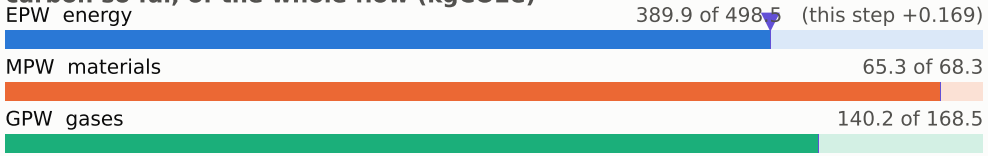
no mask for this step

this step

process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

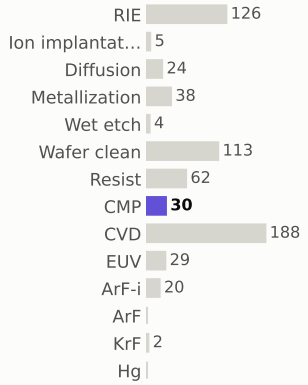
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

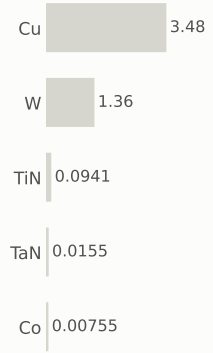


embodied carbon up to each step (kgCO2e per wafer)

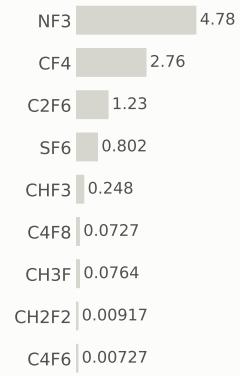
steps by process area



materials deposited (g)



gases emitted (g)



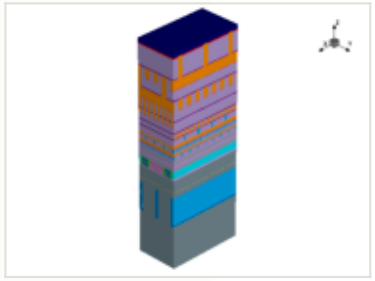
Step #6690: CVD

depo_fill · CVD

1.31	596.7	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

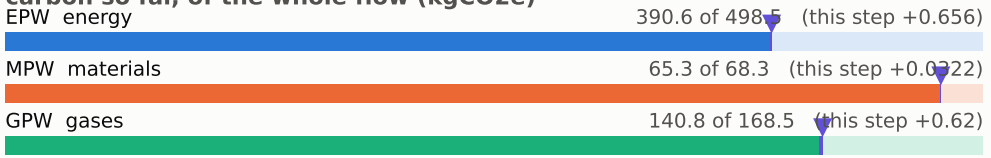


this step

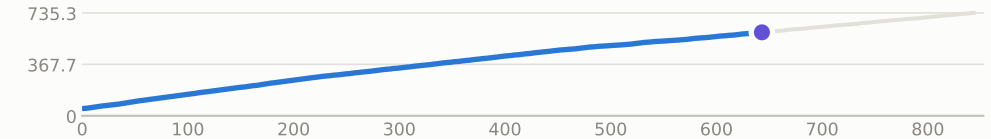
process area	CVD
material	TiN
thickness	0.03 µm
material GWP	43.9778 gCO2e/g
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0322 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

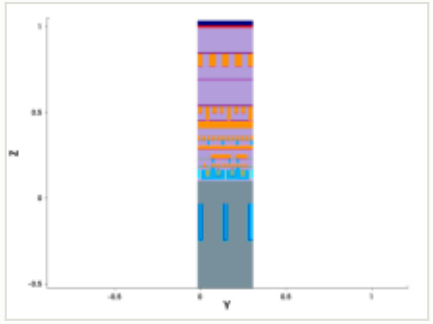
carbon so far, of the whole flow (kgCO2e)



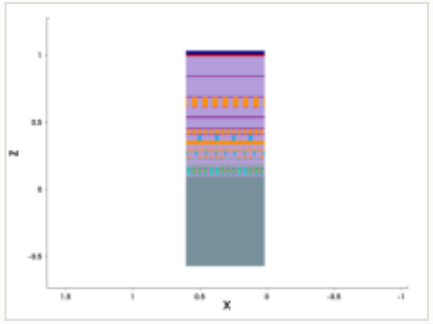
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



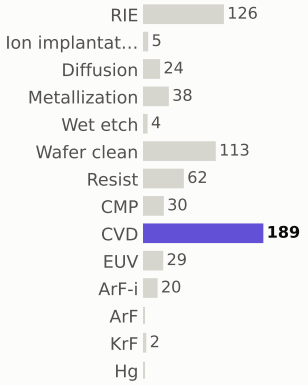
y cut at 0.184 µm



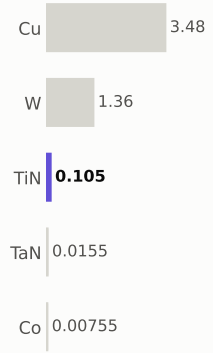
mask for this step

no mask for this step

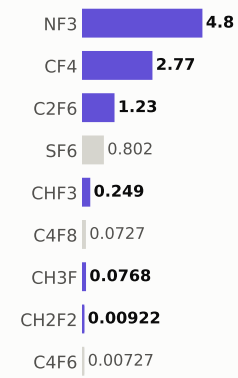
steps by process area



materials deposited (g)



gases emitted (g)



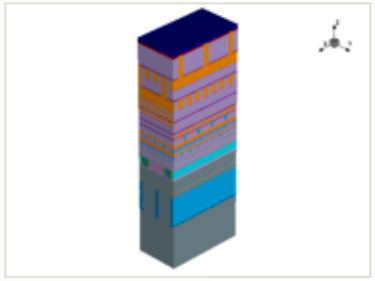
Step #6700: Wafer_clean

clean · Wafer_clean

0.169	596.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



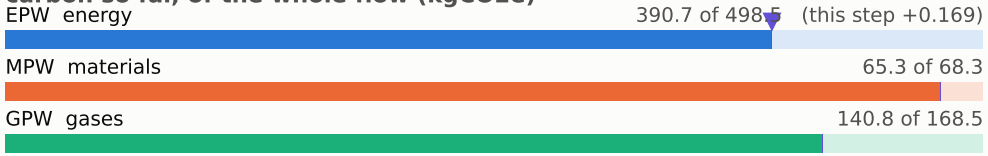
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

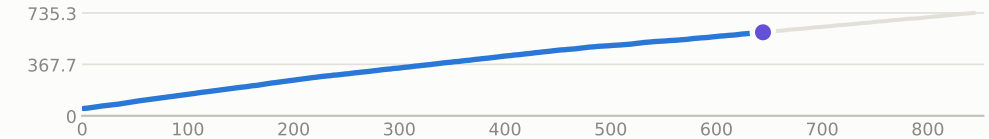
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

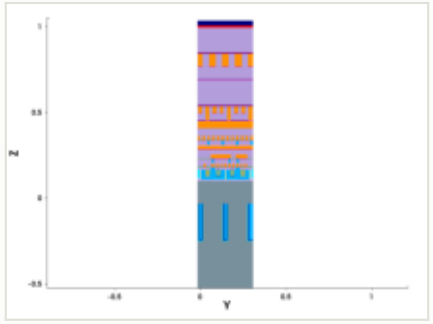
carbon so far, of the whole flow (kgCO2e)



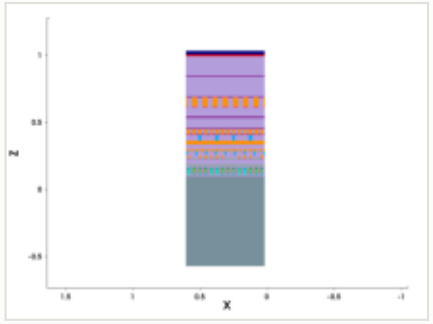
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



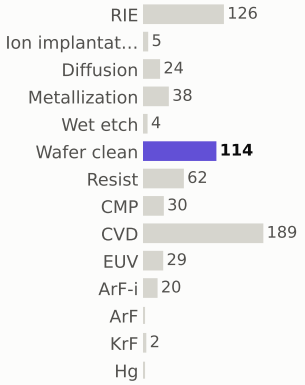
y cut at 0.184 µm



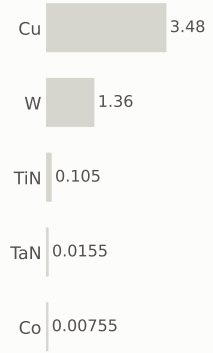
mask for this step

no mask for this step

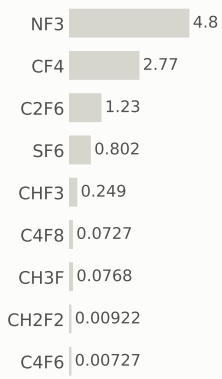
steps by process area



materials deposited (g)



gases emitted (g)



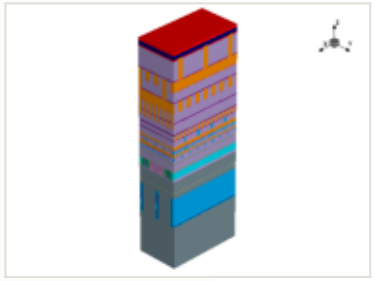
Step #6710: CVD

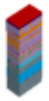
depo_fill · CVD

1.31	598.2	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



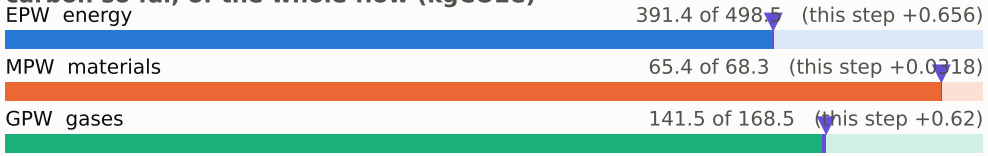


this step

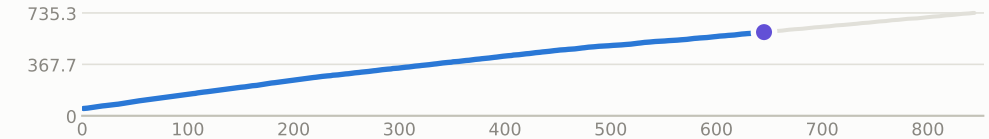
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

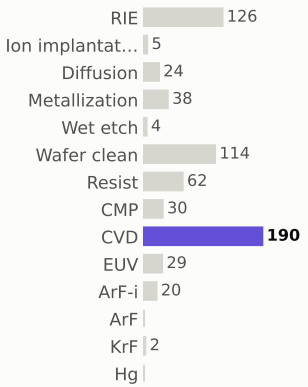
carbon so far, of the whole flow (kgCO2e)



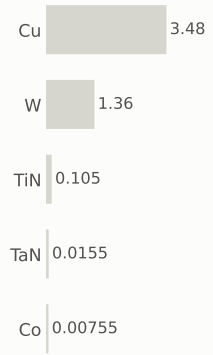
embodied carbon up to each step (kgCO2e per wafer)



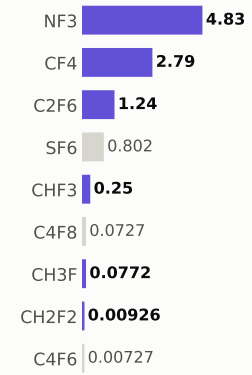
steps by process area



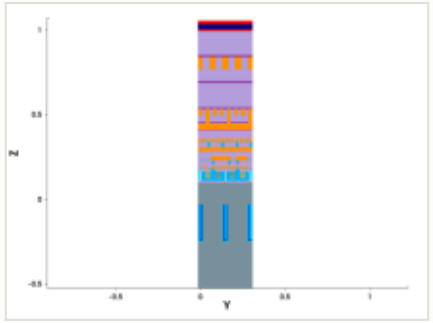
materials deposited (g)



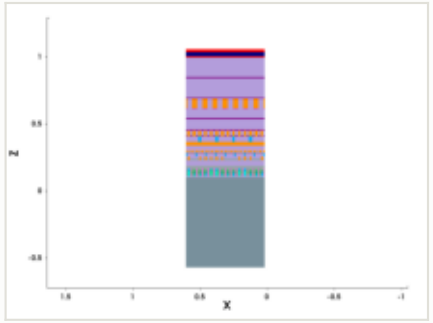
gases emitted (g)



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

no mask for this step

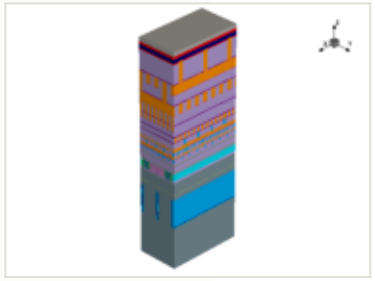
Step #6720: CVD

depo_fill · CVD

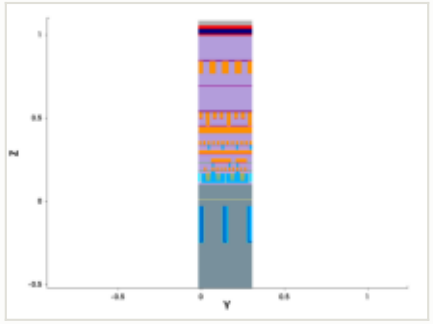
1.31	599.5	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

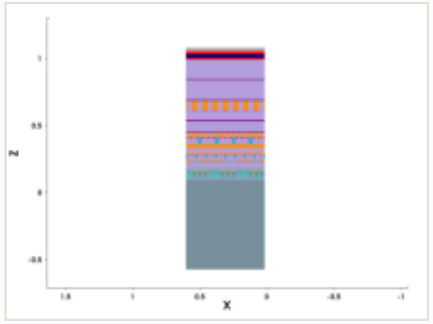
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

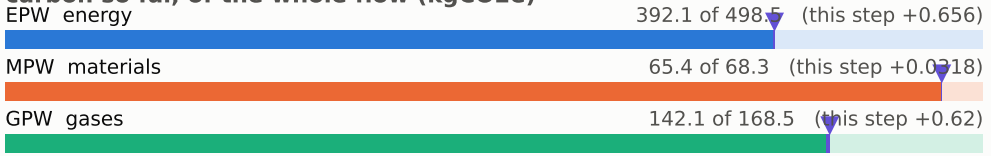
no mask for this step

this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

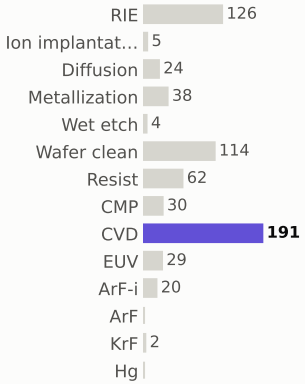
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

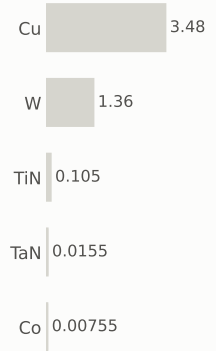


embodied carbon up to each step (kgCO2e per wafer)

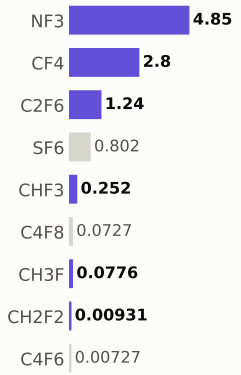
steps by process area



materials deposited (g)



gases emitted (g)



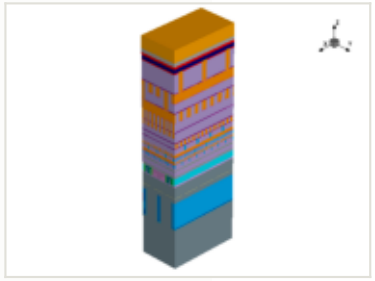
Step #6730: Resist

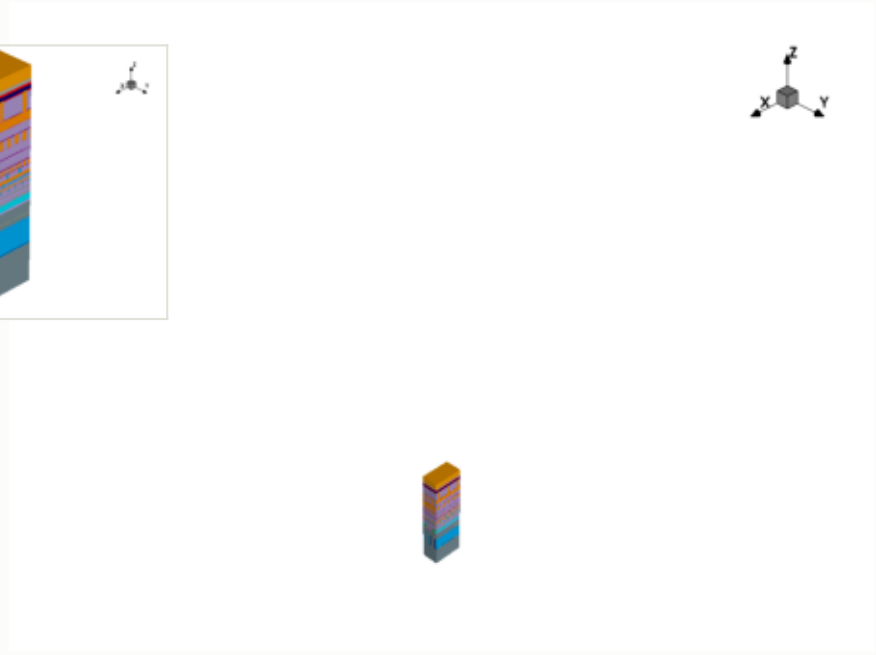
depo_fill · Resist

0.169	599.7	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



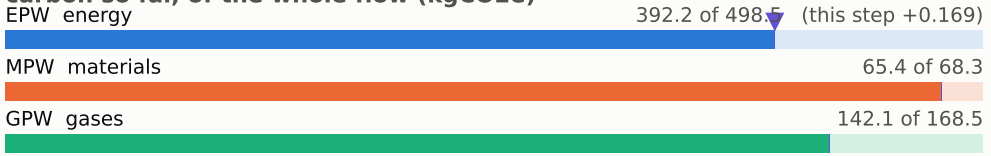


this step

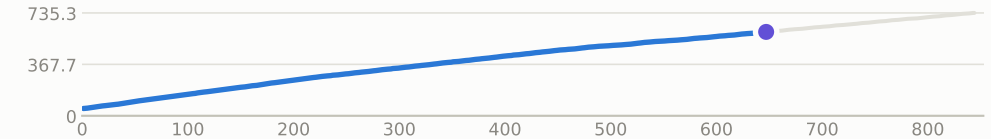
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

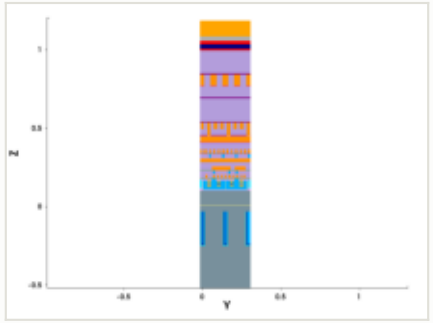
carbon so far, of the whole flow (kgCO2e)



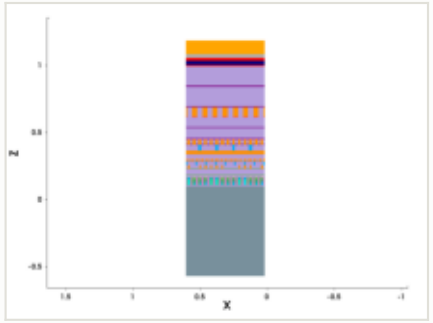
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



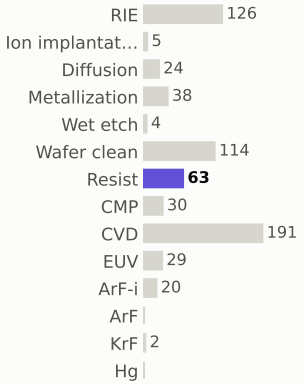
y cut at 0.184 µm



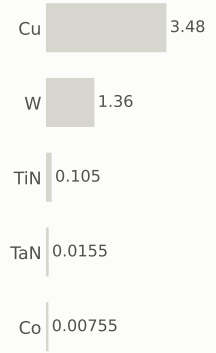
mask for this step

no mask for this step

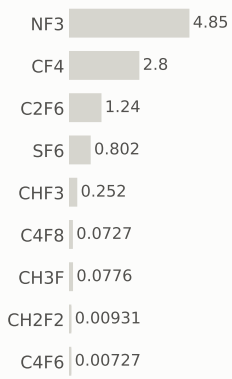
steps by process area



materials deposited (g)



gases emitted (g)



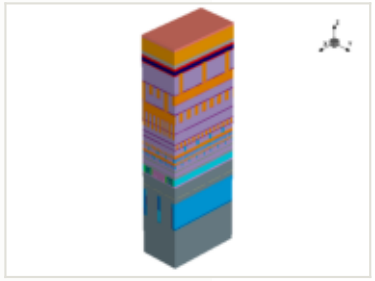
Step #6740: CVD

depo_fill · CVD

1.31	601.0	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

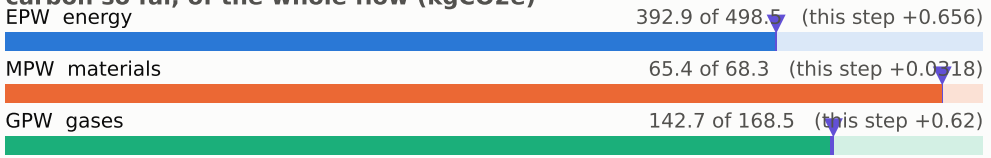


this step

process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

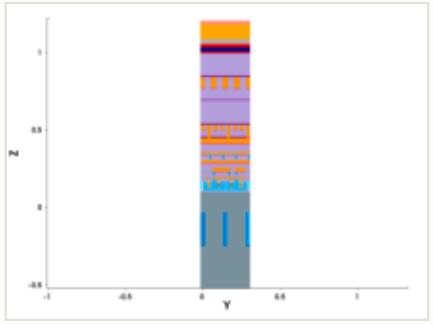
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

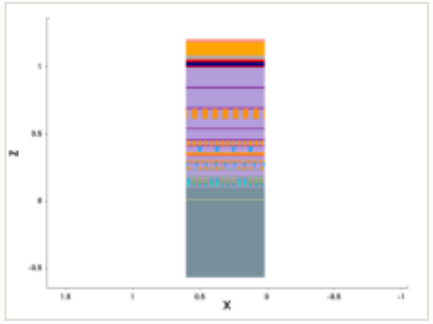


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



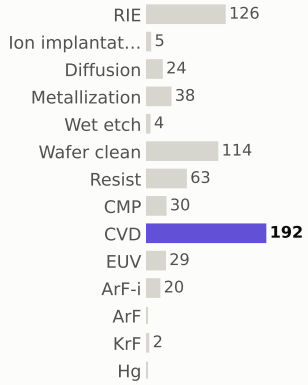
y cut at 0.184 µm



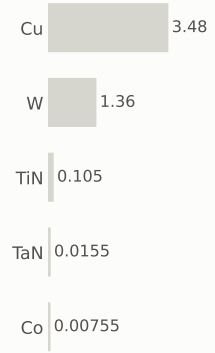
mask for this step

no mask for this step

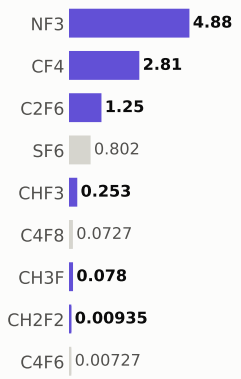
steps by process area



materials deposited (g)



gases emitted (g)



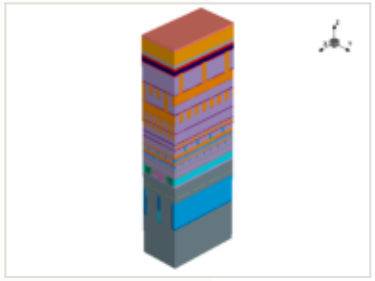
Step #6750: Wafer_clean

clean · Wafer_clean

0.169	601.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



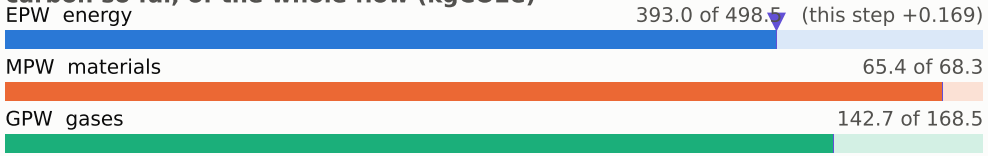
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

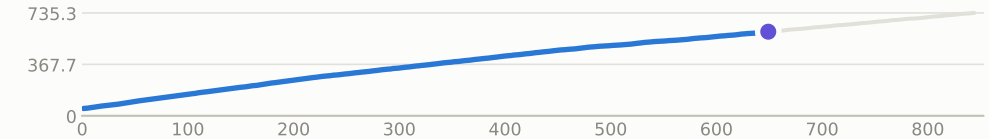
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

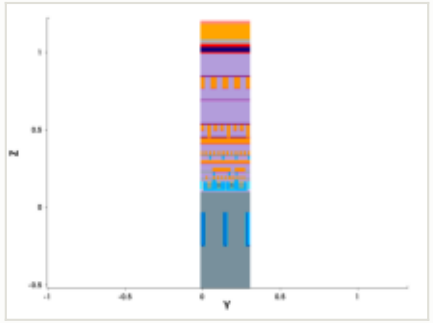
carbon so far, of the whole flow (kgCO2e)



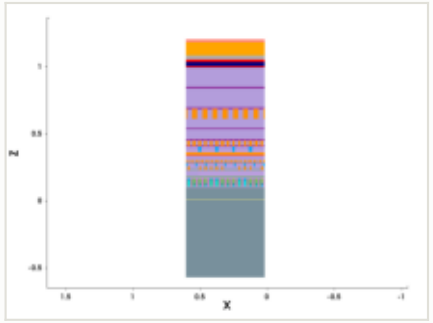
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



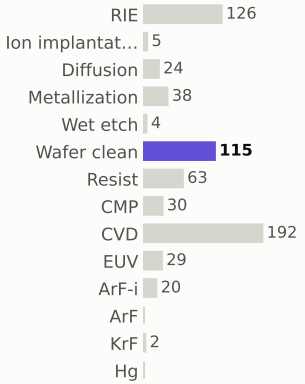
y cut at 0.184 µm



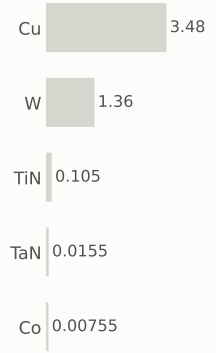
mask for this step

no mask for this step

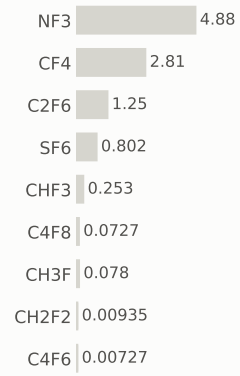
steps by process area



materials deposited (g)



gases emitted (g)



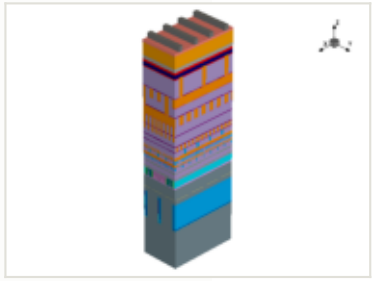
Step #6760: ArF-i

litho_patterning · ArF-i

0.217	601.4	0.029%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

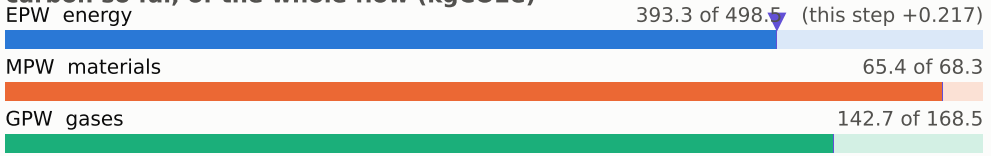


this step

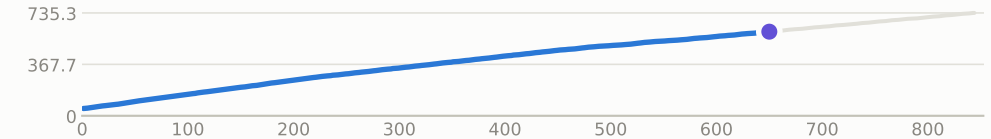
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

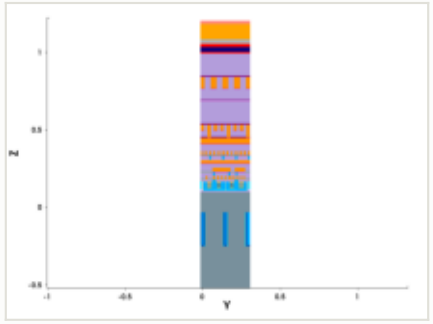
carbon so far, of the whole flow (kgCO2e)



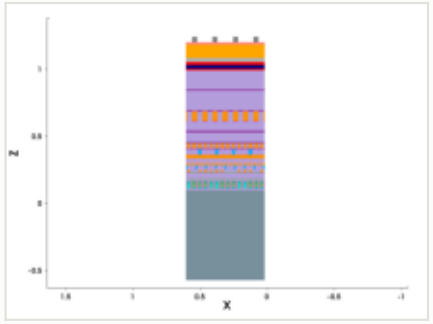
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm

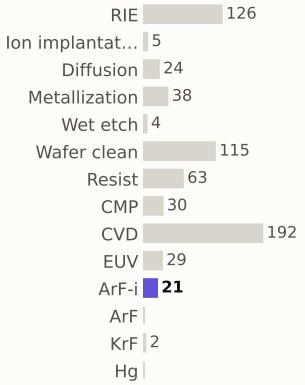


y cut at 0.184 µm

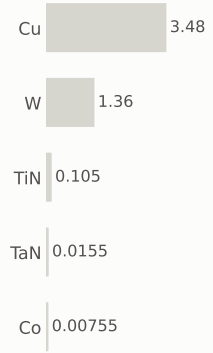


mask: layer 10601:0, positive (GDSII)

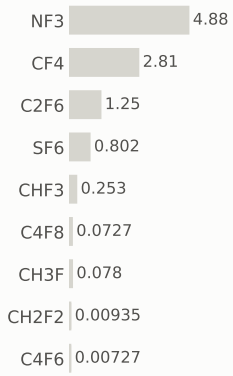
steps by process area



materials deposited (g)



gases emitted (g)



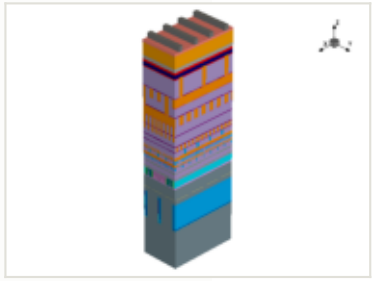
Step #6770: Ion_implantation

treatment · Ion_implantation

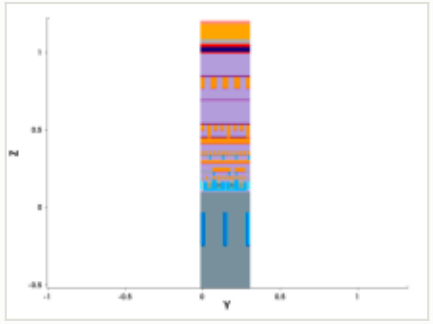
0.801	602.2	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

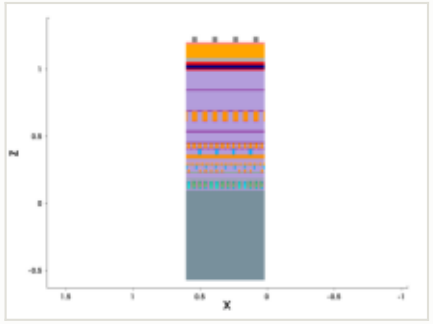
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

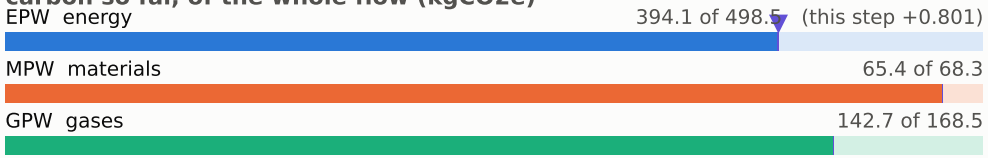
no mask for this step

this step

process area	Ion_implantation	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

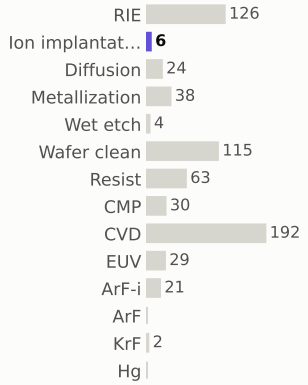
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

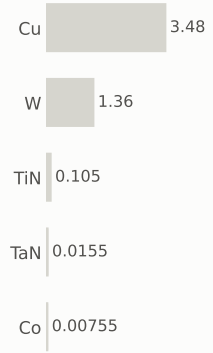


embodied carbon up to each step (kgCO2e per wafer)

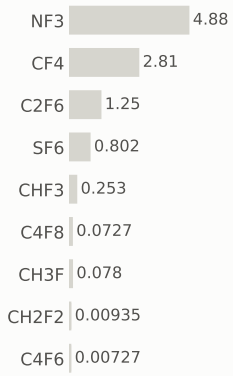
steps by process area



materials deposited (g)



gases emitted (g)



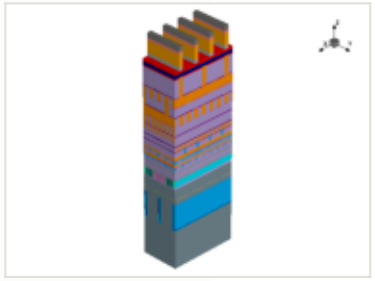
Step #6780: RIE

etch_anisotropic · RIE

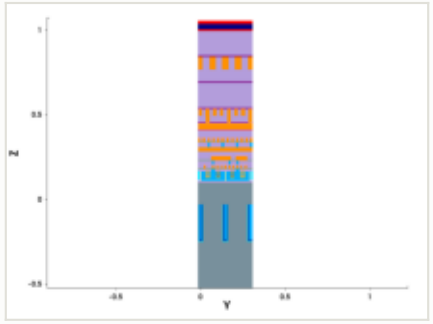
1.04	603.2	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

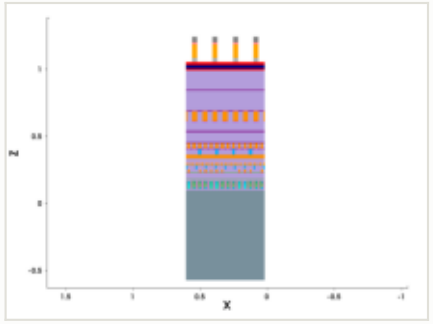
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

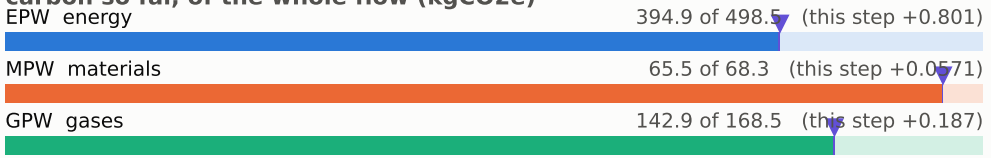
no mask for this step

this step

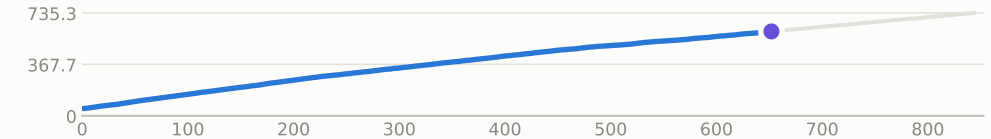
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

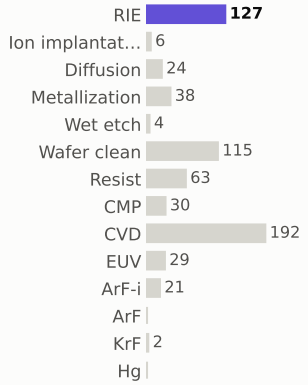
carbon so far, of the whole flow (kgCO2e)



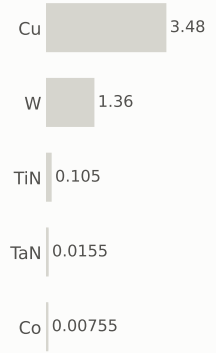
embodied carbon up to each step (kgCO2e per wafer)



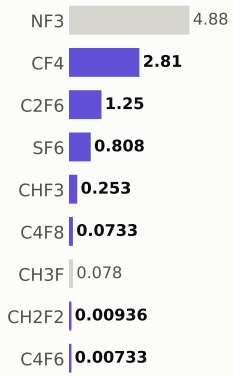
steps by process area



materials deposited (g)



gases emitted (g)



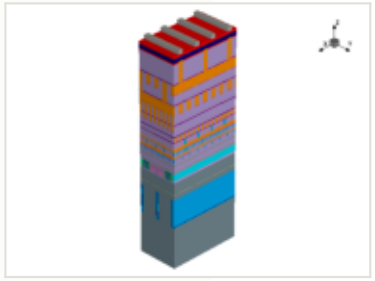
Step #6790: Wafer_clean

etch_strip · Wafer_clean

0.169	603.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

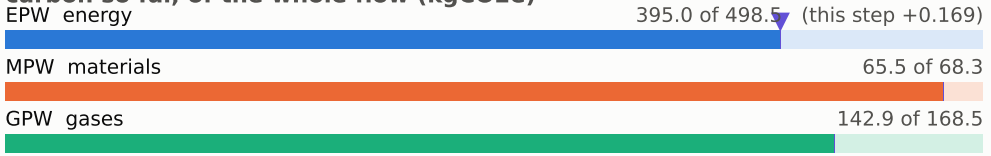


this step

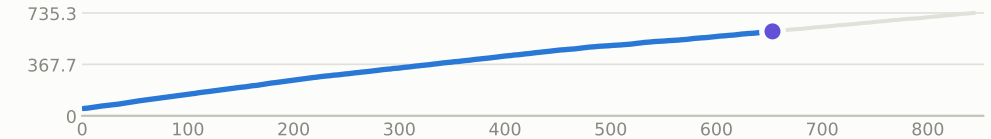
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

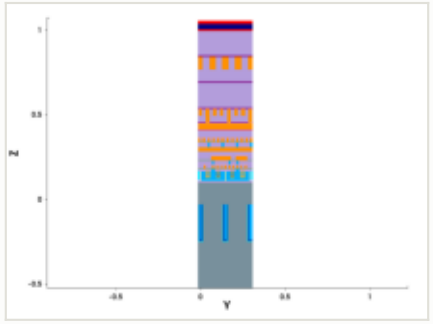
carbon so far, of the whole flow (kgCO2e)



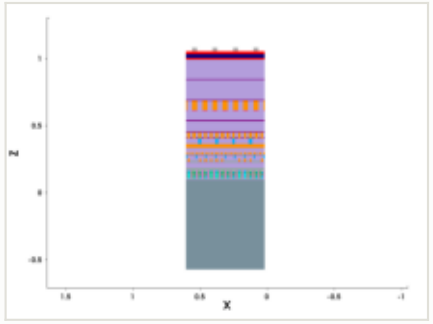
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



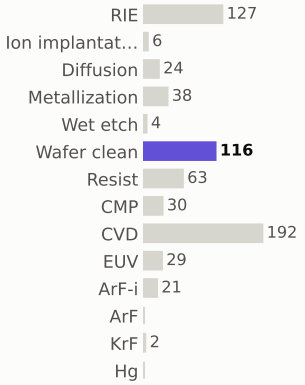
y cut at 0.184 µm



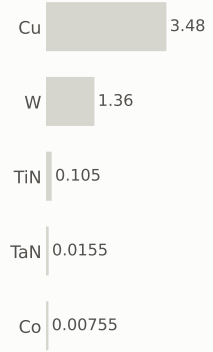
mask for this step

no mask for this step

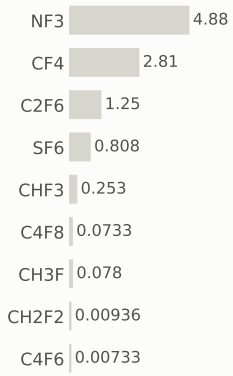
steps by process area



materials deposited (g)



gases emitted (g)



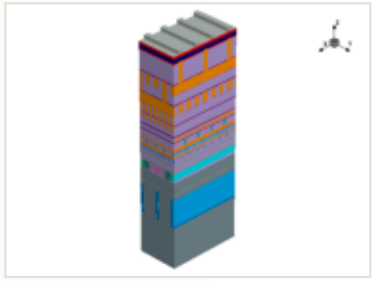
Step #6800: CVD

depo_isotropic · CVD

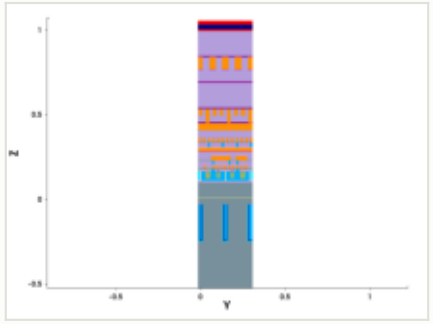
1.31	604.7	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

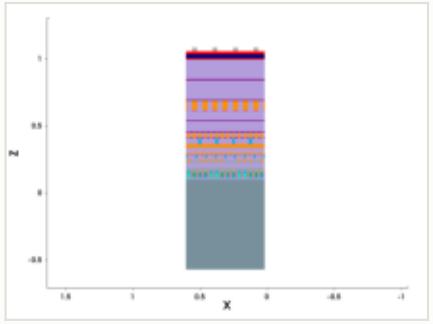
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

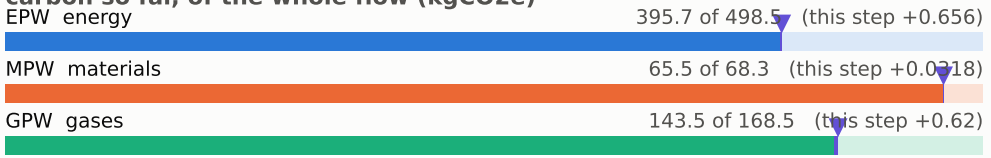
no mask for this step

this step

process area	CVD
material	—
thickness	0.001 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

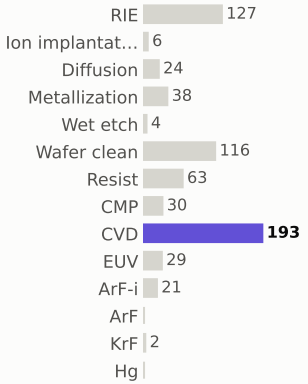
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

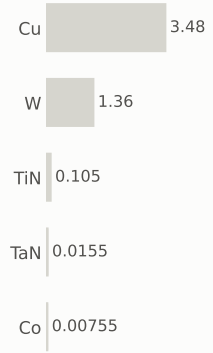


embodied carbon up to each step (kgCO2e per wafer)

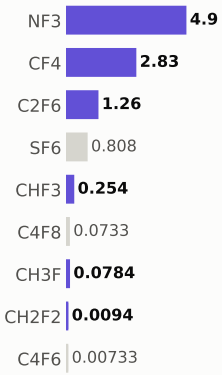
steps by process area



materials deposited (g)



gases emitted (g)



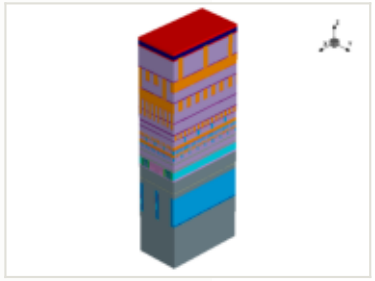
Step #6810: RIE

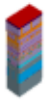
etch_anisotropic · RIE

1.04	605.7	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

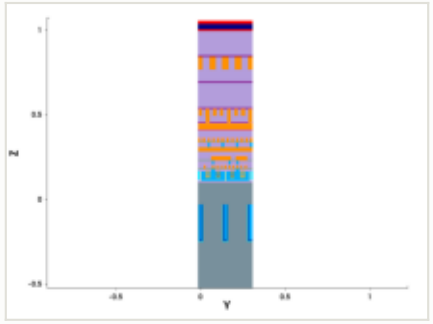
3D structure after this step

zoomed to fit

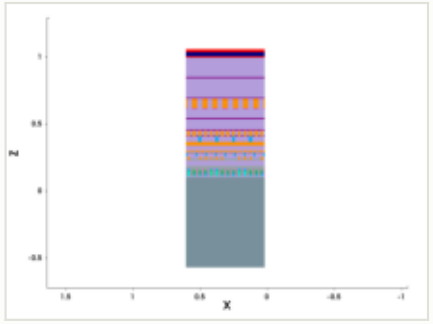




x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

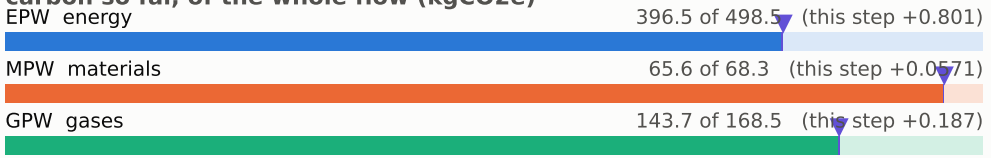
no mask for this step

this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

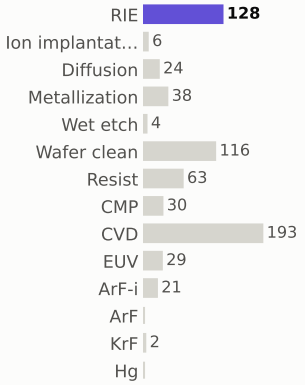
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

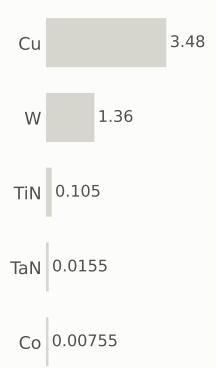


embodied carbon up to each step (kgCO2e per wafer)

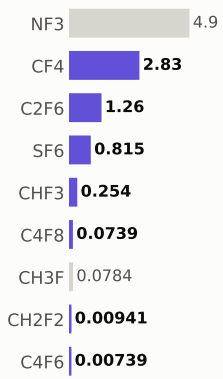
steps by process area



materials deposited (g)



gases emitted (g)



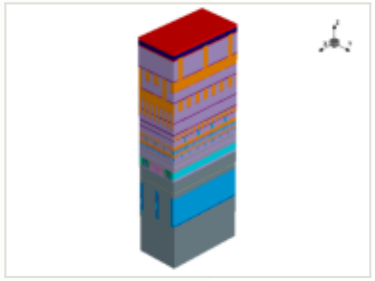
Step #6820: Wafer_clean

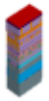
etch_strip · Wafer_clean

0.169	605.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

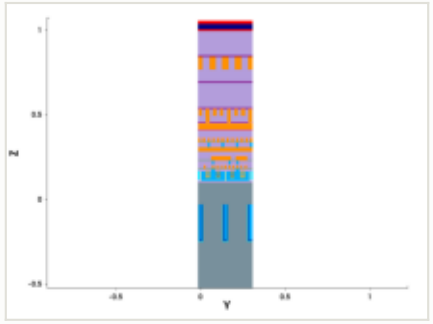
3D structure after this step

zoomed to fit

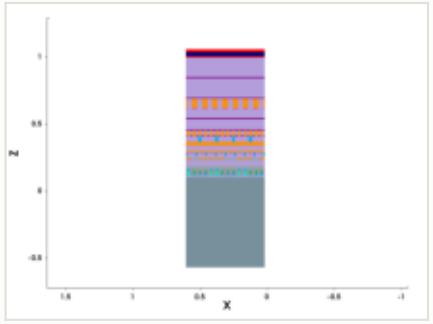




x cut at 0.294 µm



y cut at 0.184 µm



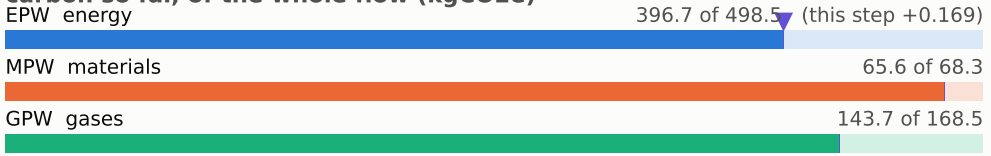
mask for this step

no mask for this step

this step

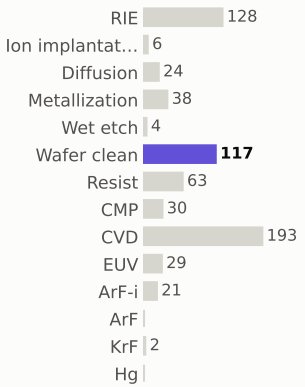
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

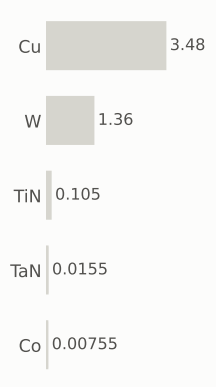


embodied carbon up to each step (kgCO2e per wafer)

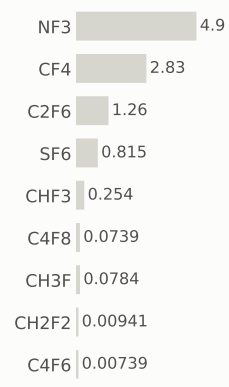
steps by process area



materials deposited (g)



gases emitted (g)



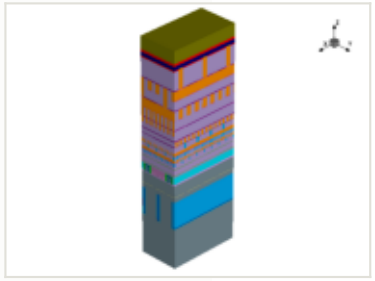
Step #6830: Resist

depo_fill · Resist

0.169	606.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

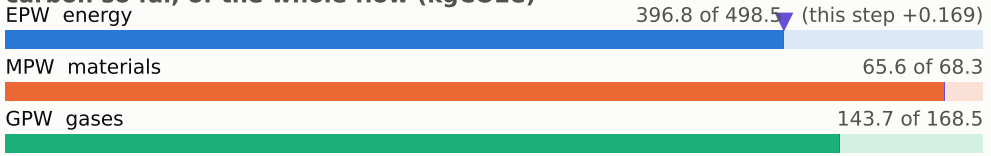


this step

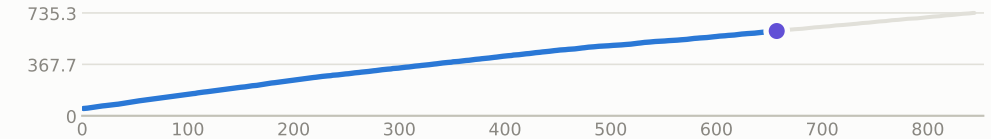
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

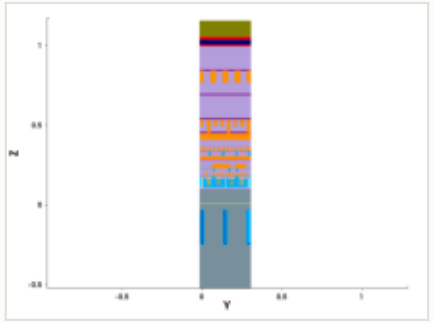
carbon so far, of the whole flow (kgCO2e)



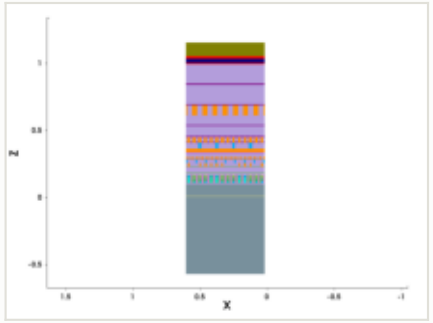
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



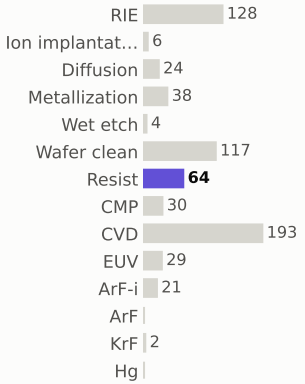
y cut at 0.184 µm



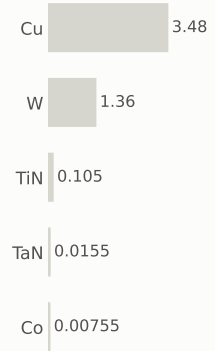
mask for this step

no mask for this step

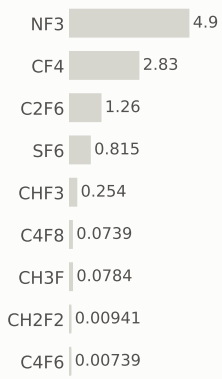
steps by process area



materials deposited (g)



gases emitted (g)



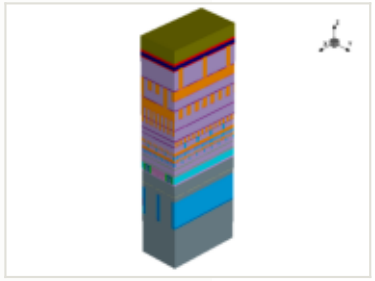
Step #6840: Wafer_clean

clean · Wafer_clean

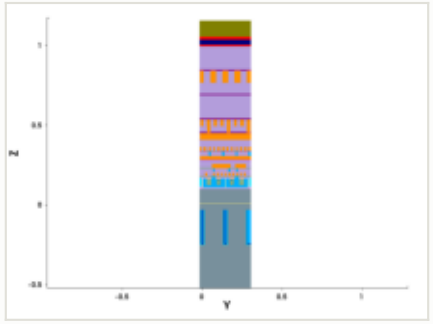
0.169	606.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

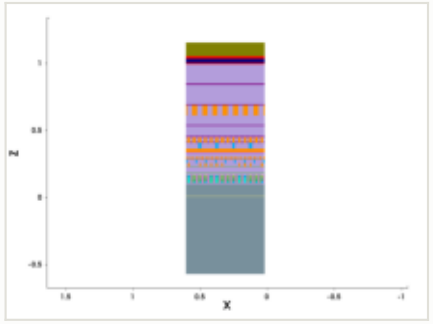
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

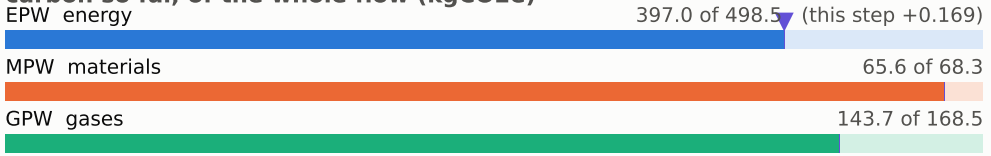
no mask for this step

this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

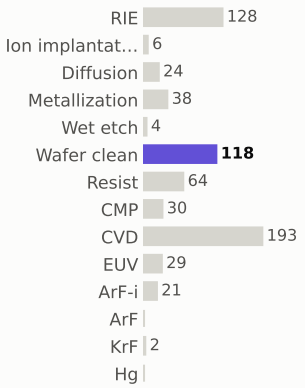
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

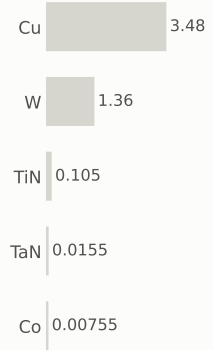


embodied carbon up to each step (kgCO2e per wafer)

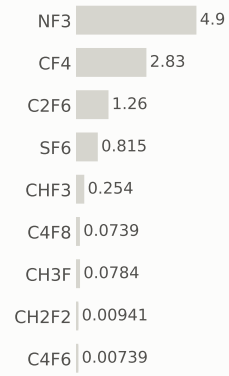
steps by process area



materials deposited (g)



gases emitted (g)



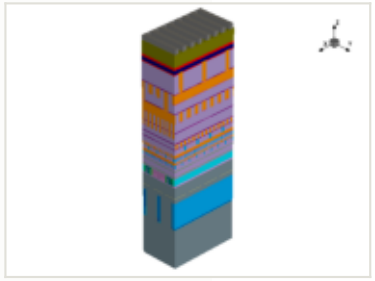
Step #6850: ArF-i

litho_patterning · ArF-i

0.217	606.5	0.029%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

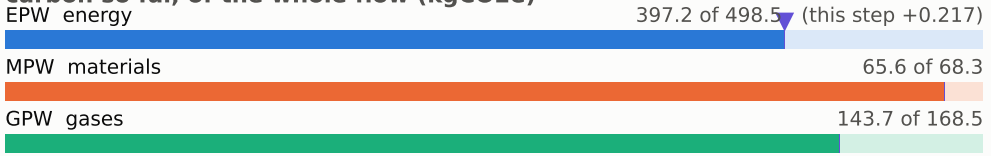


this step

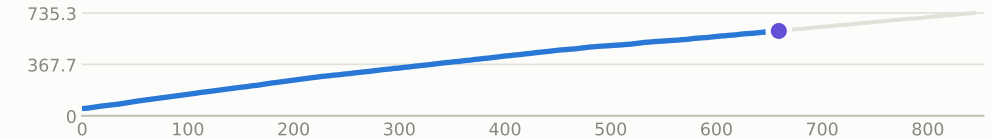
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

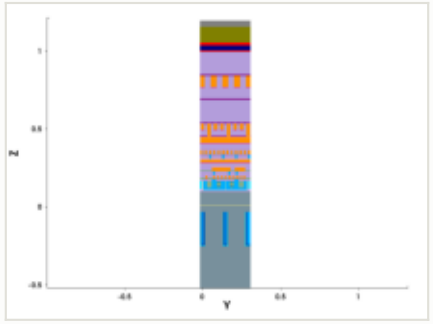
carbon so far, of the whole flow (kgCO2e)



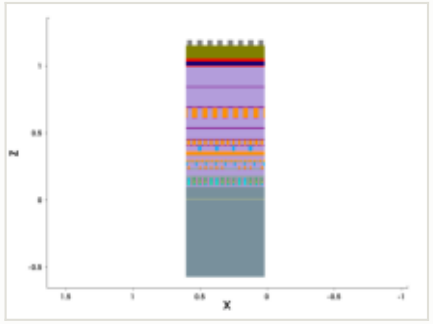
embodied carbon up to each step (kgCO2e per wafer)



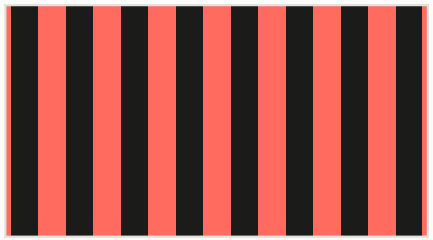
x cut at 0.294 µm



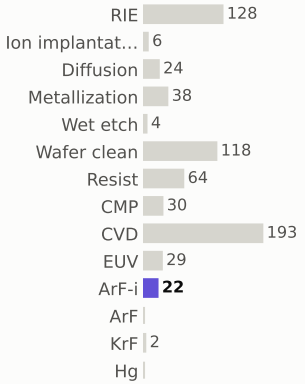
y cut at 0.184 µm



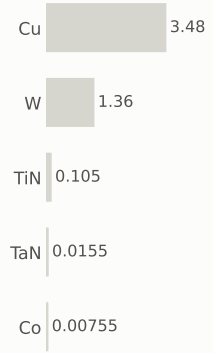
mask: layer 60:0 (GDSII)



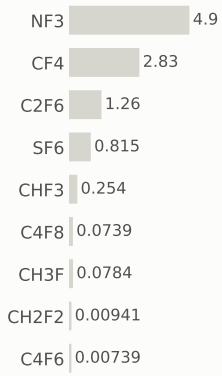
steps by process area



materials deposited (g)



gases emitted (g)



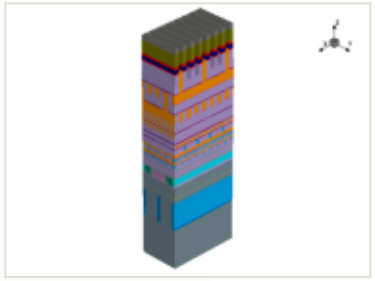
Step #6860: RIE

etch_anisotropic · RIE

1.04	607.5	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

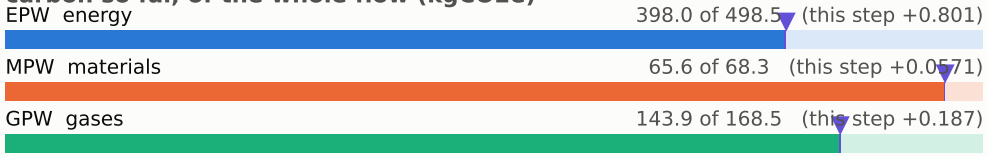


this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

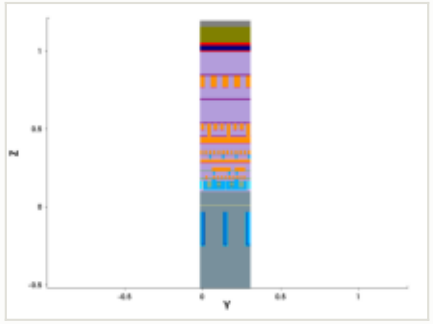
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

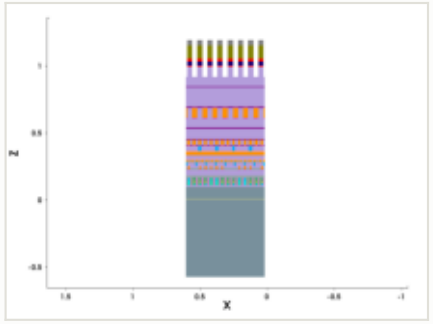


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



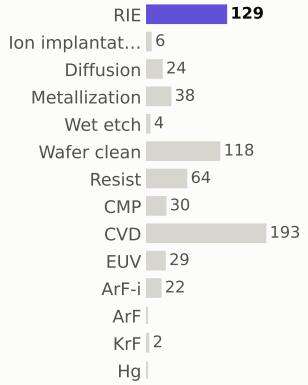
y cut at 0.184 µm



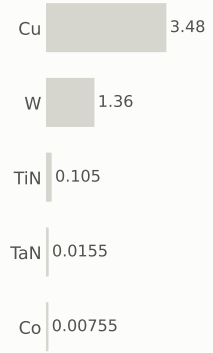
mask for this step

no mask for this step

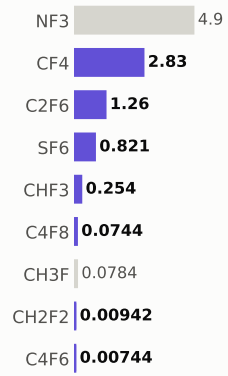
steps by process area



materials deposited (g)



gases emitted (g)



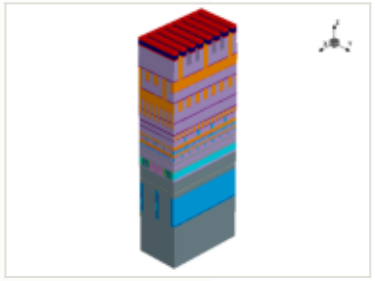
Step #6870: Wafer_clean

etch_strip · Wafer_clean

0.169	607.7	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

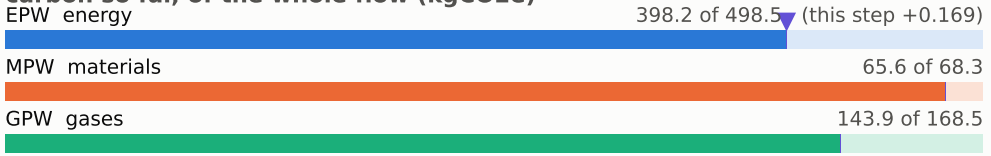


this step

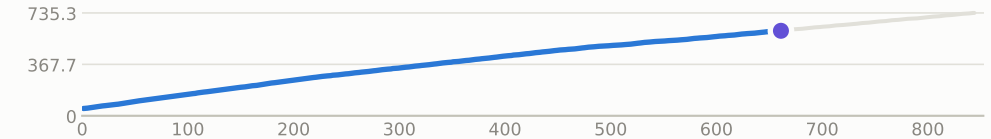
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

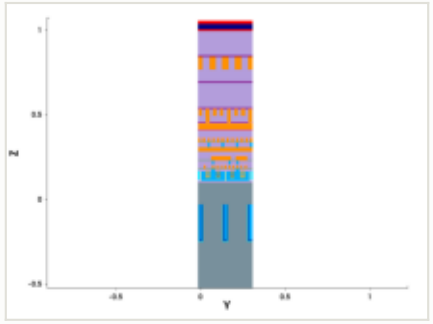
carbon so far, of the whole flow (kgCO2e)



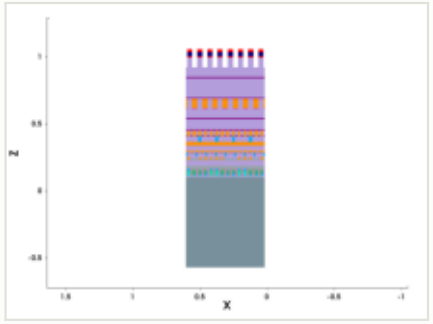
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



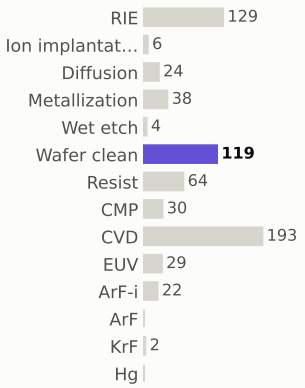
y cut at 0.184 µm



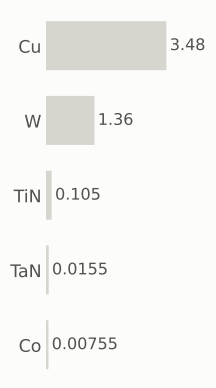
mask for this step

no mask for this step

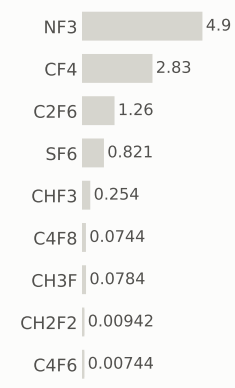
steps by process area



materials deposited (g)



gases emitted (g)



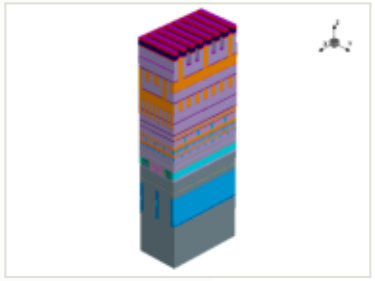
Step #6880: Metallization

depo_ellipsoidal · Metallization

1.2	608.9	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

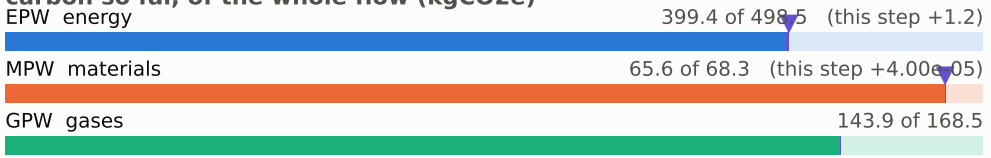


this step

process area	Metallization
material	TaN
thickness	0.0016 µm
material GWP	25.8159 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

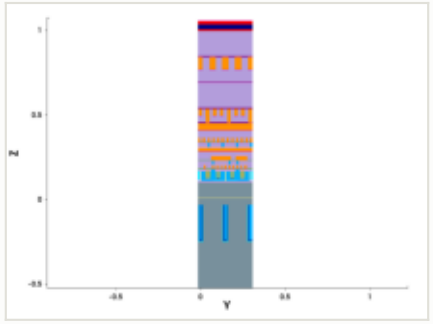
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.00e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

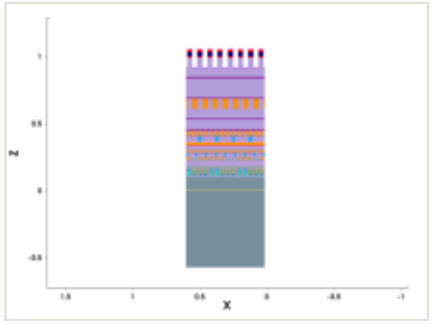


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



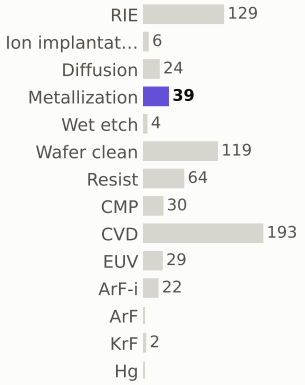
y cut at 0.184 µm



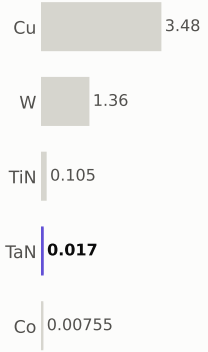
mask for this step

no mask for this step

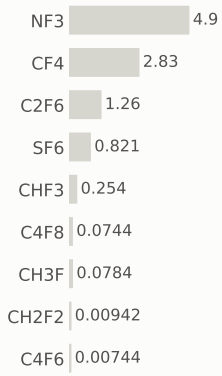
steps by process area



materials deposited (g)



gases emitted (g)



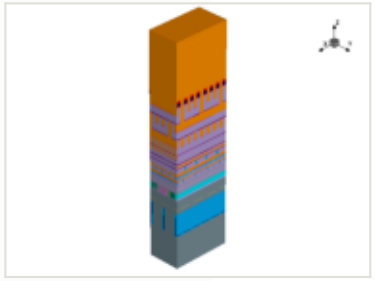
Step #6890: Metallization

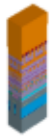
depo_fill · Metallization

1.21	610.1	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

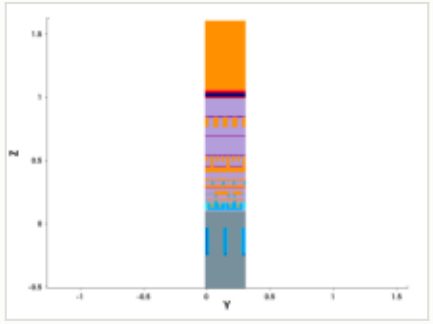
3D structure after this step

zoomed to fit

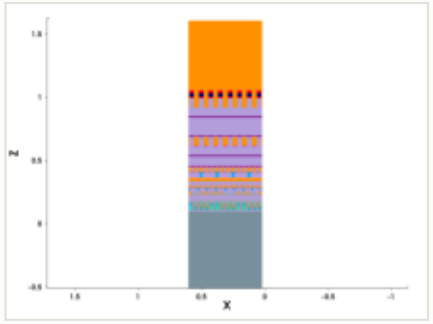




x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

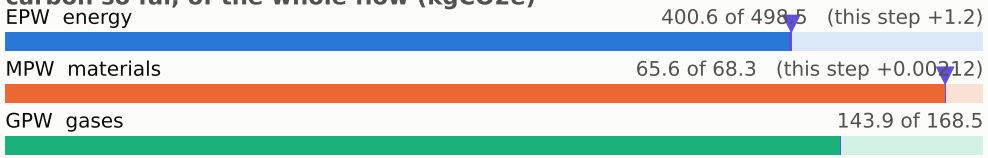
no mask for this step

this step

process area	Metallization
material	Cu
thickness	0.55 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

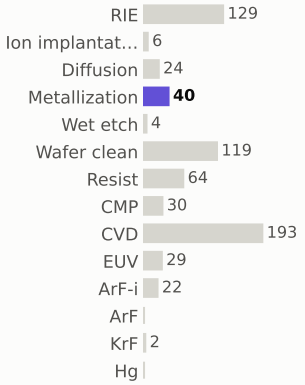
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00212 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

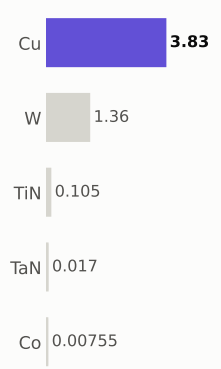


embodied carbon up to each step (kgCO2e per wafer)

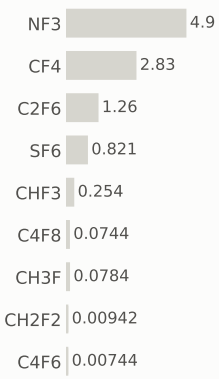
steps by process area



materials deposited (g)



gases emitted (g)



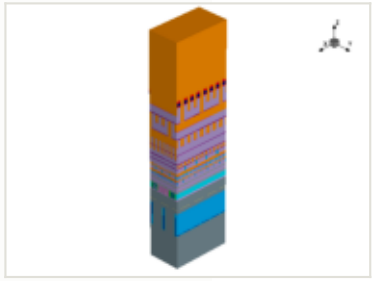
Step #6900: Diffusion

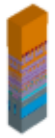
treatment · Diffusion

0.801	610.9	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

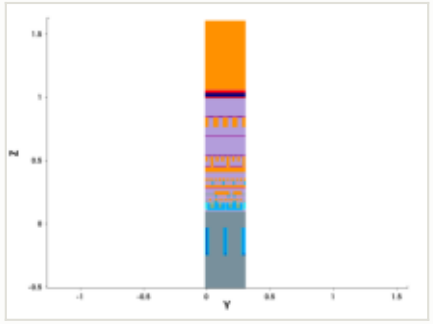
3D structure after this step

zoomed to fit

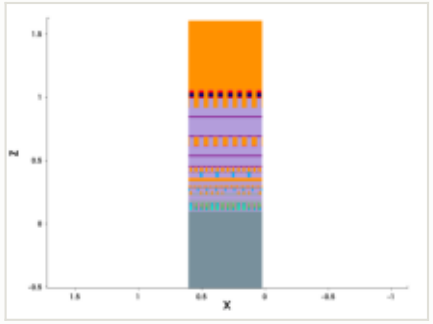




x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

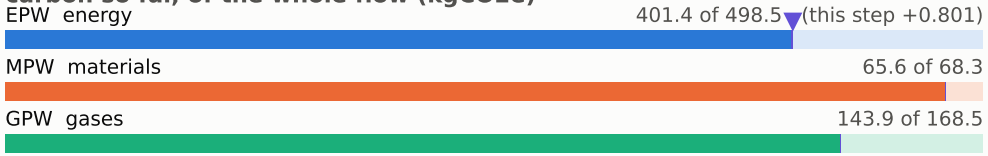
no mask for this step

this step

process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

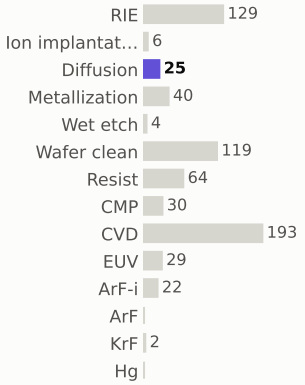
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

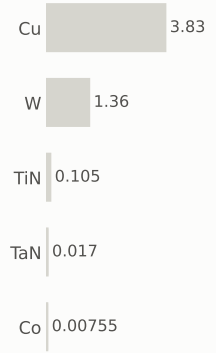


embodied carbon up to each step (kgCO2e per wafer)

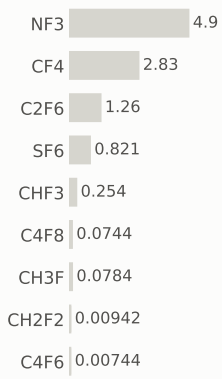
steps by process area



materials deposited (g)



gases emitted (g)



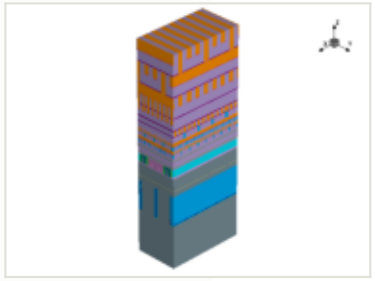
Step #6910: CMP

etch_cmp · CMP

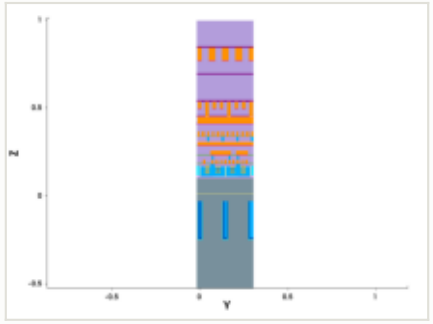
0.169	611.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

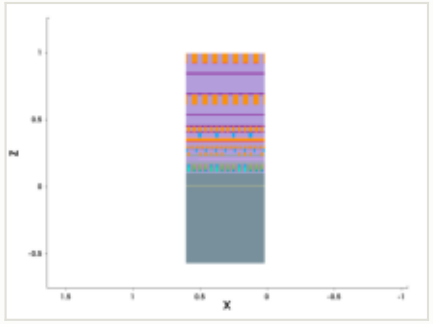
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

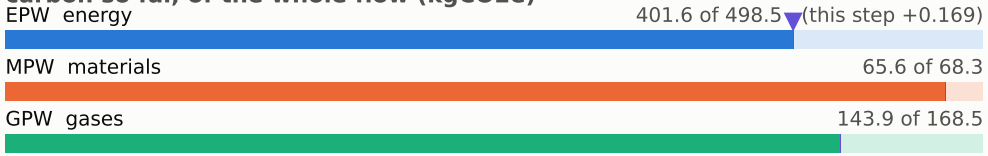
no mask for this step

this step

process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

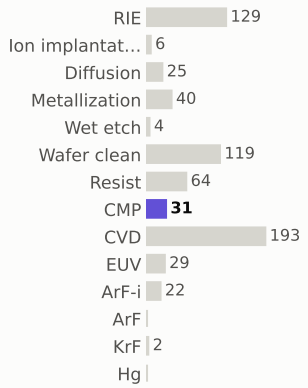
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

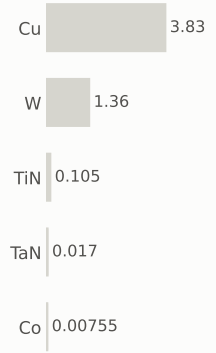


embodied carbon up to each step (kgCO2e per wafer)

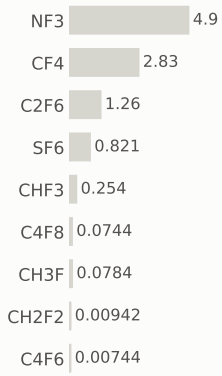
steps by process area



materials deposited (g)



gases emitted (g)



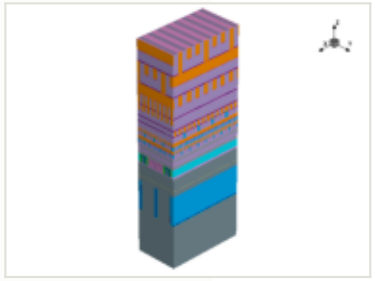
Step #6920: Metallization

depo_isotropic · Metallization

1.2	612.3	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

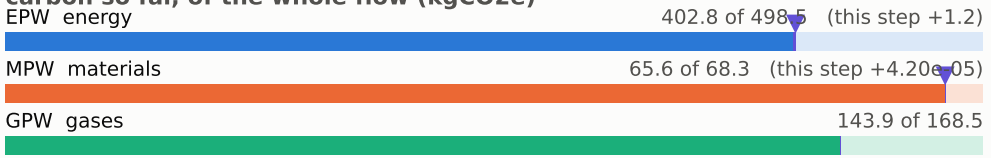


this step

process area	Metallization
material	Co
thickness	0.0015 µm
material GWP	44.5 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

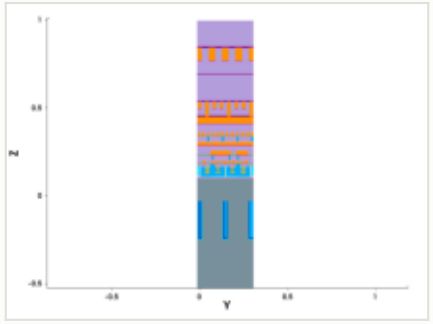
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.20e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

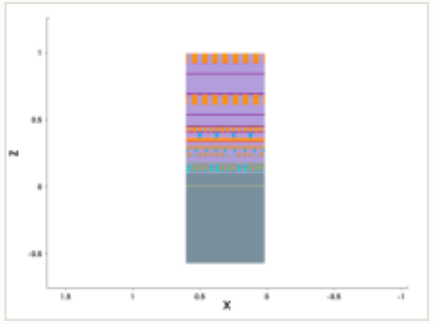


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



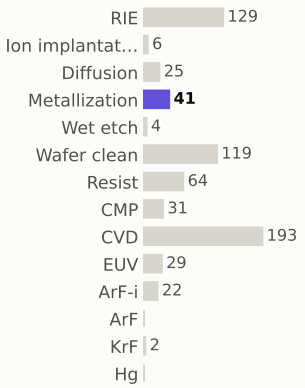
y cut at 0.184 µm



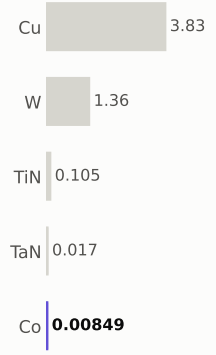
mask for this step

no mask for this step

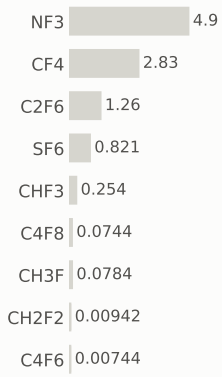
steps by process area



materials deposited (g)



gases emitted (g)



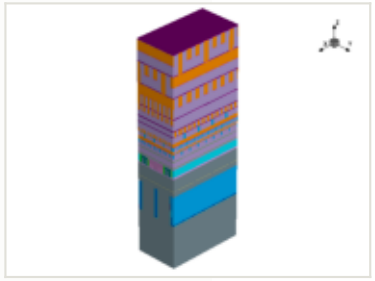
Step #6940: CVD

depo_fill · CVD

1.31	613.6	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

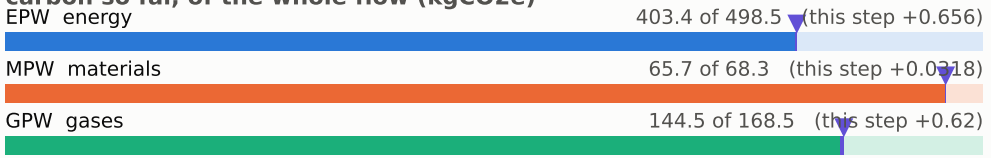


this step

process area	CVD
material	—
thickness	0.006 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

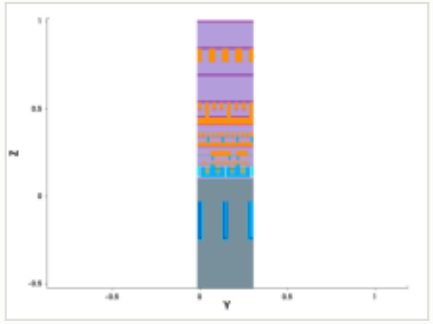
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

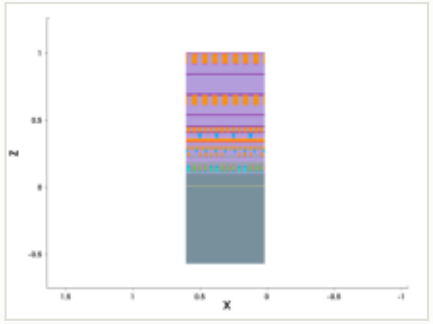


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



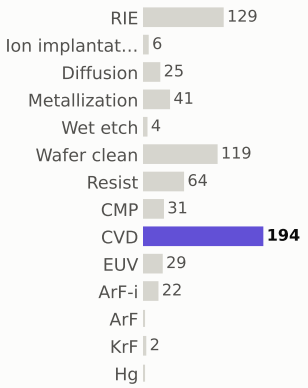
y cut at 0.184 µm



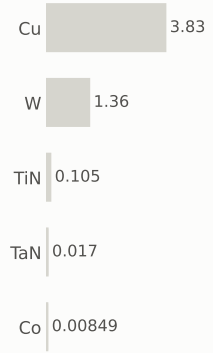
mask for this step

no mask for this step

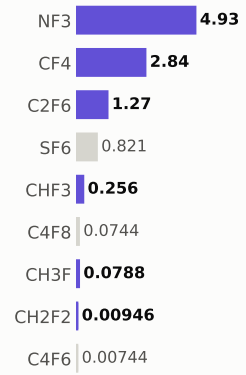
steps by process area



materials deposited (g)



gases emitted (g)



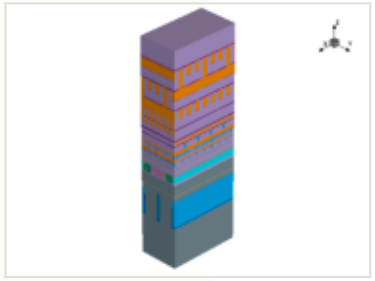
Step #6950: CVD

depo_fill · CVD

1.31	614.9	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

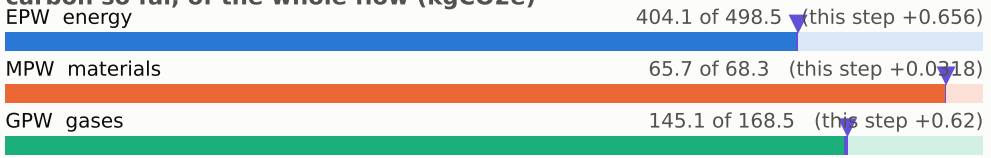


this step

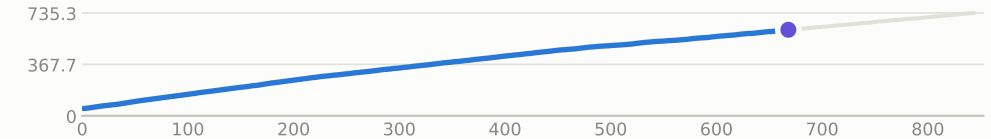
process area	CVD
material	—
thickness	0.146 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

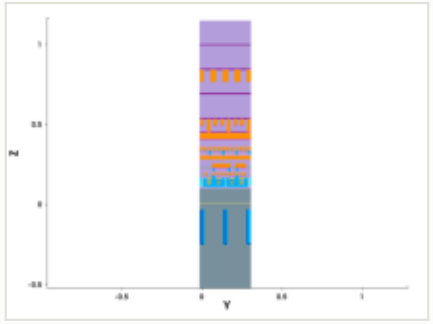
carbon so far, of the whole flow (kgCO2e)



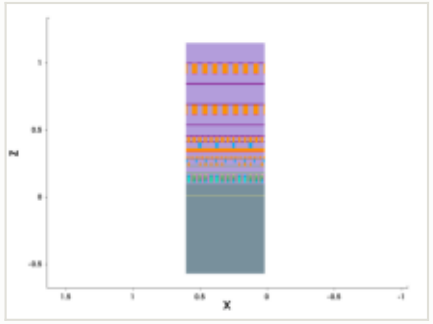
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



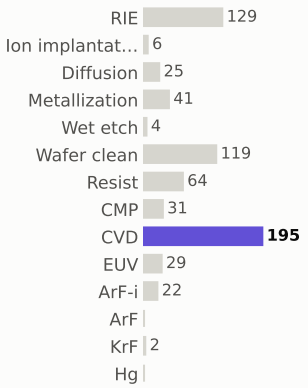
y cut at 0.184 µm



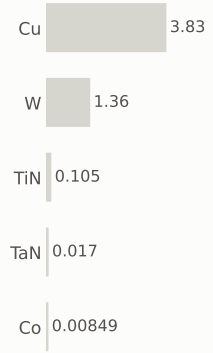
mask for this step

no mask for this step

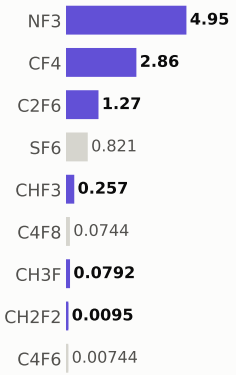
steps by process area



materials deposited (g)



gases emitted (g)



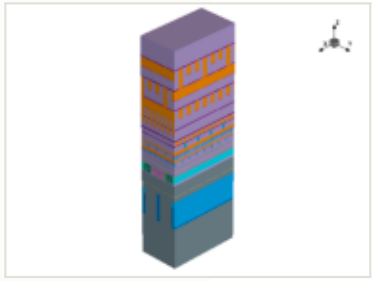
Step #6960: Diffusion

treatment · Diffusion

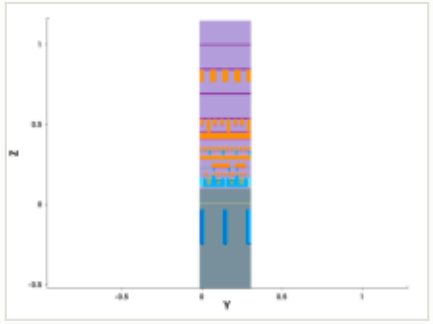
0.801	615.7	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

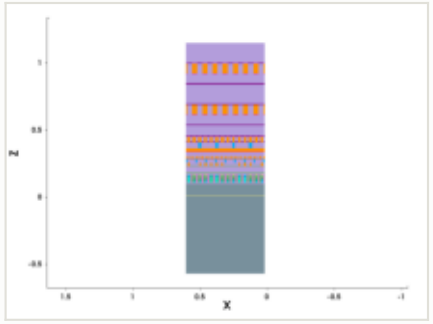
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

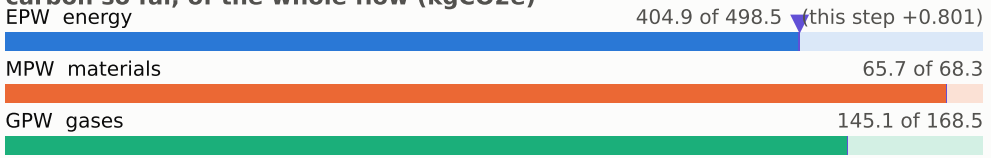
no mask for this step

this step

process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

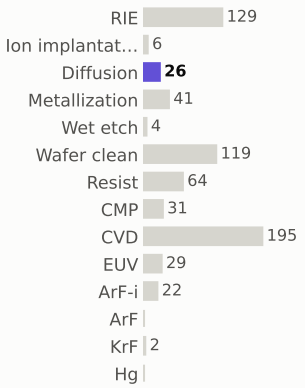
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

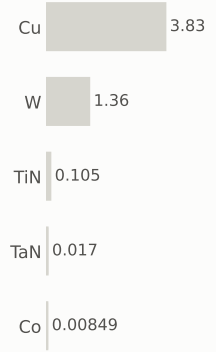


embodied carbon up to each step (kgCO2e per wafer)

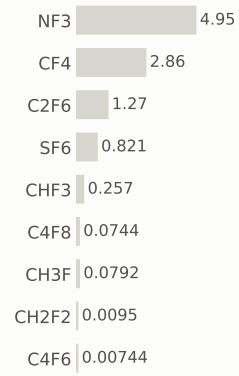
steps by process area



materials deposited (g)



gases emitted (g)



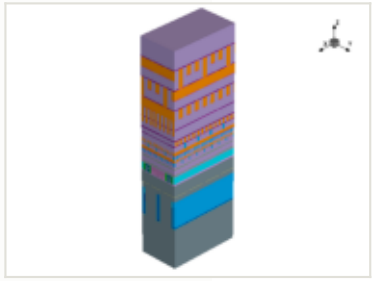
Step #6970: Wafer_clean

clean · Wafer_clean

0.169	615.8	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



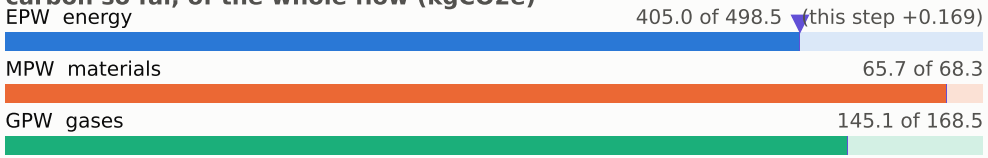
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

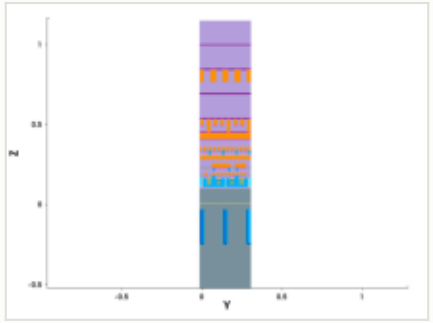
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

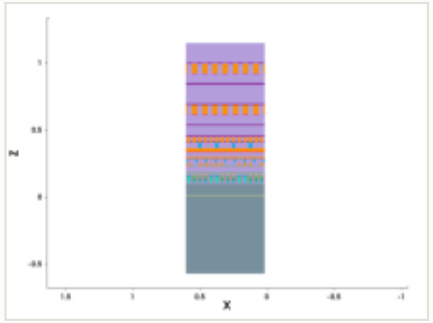


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



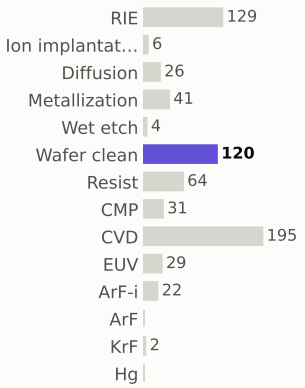
y cut at 0.184 µm



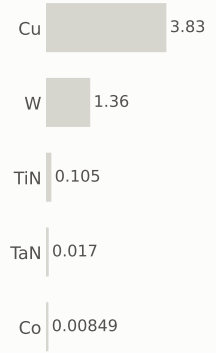
mask for this step

no mask for this step

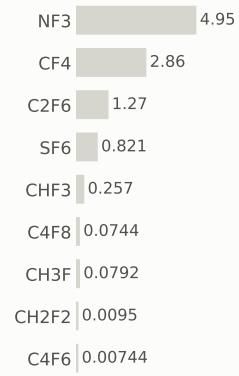
steps by process area



materials deposited (g)



gases emitted (g)



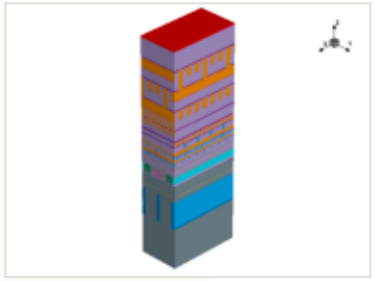
Step #6980: CVD

depo_fill · CVD

1.31	617.2	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

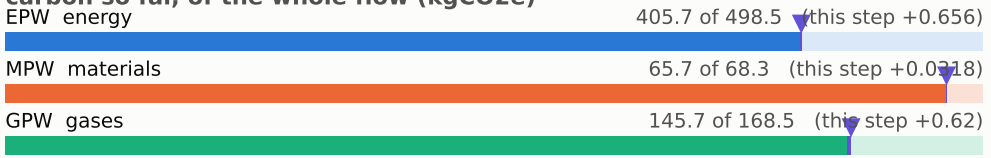


this step

process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

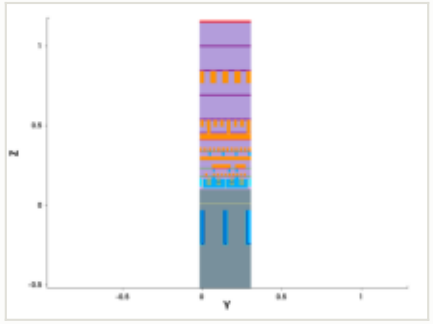
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

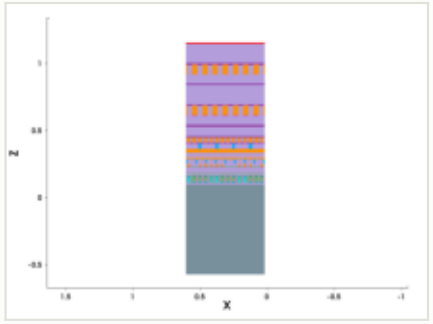


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



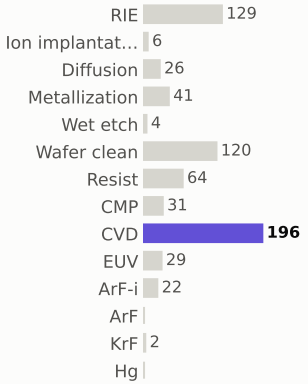
y cut at 0.184 µm



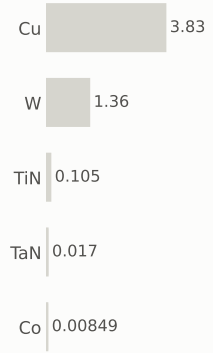
mask for this step

no mask for this step

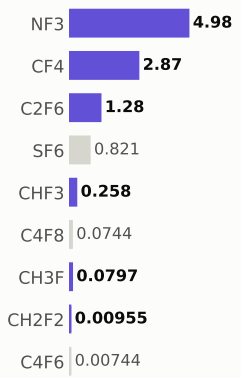
steps by process area



materials deposited (g)



gases emitted (g)



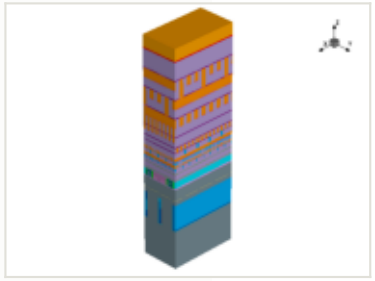
Step #6990: Resist

depo_fill · Resist

0.169	617.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

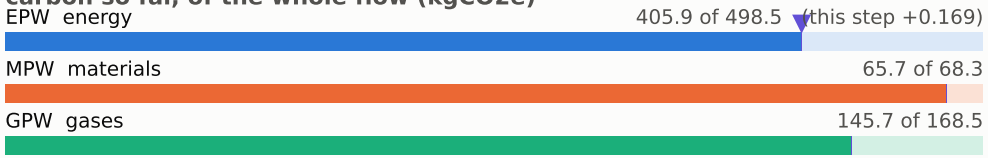


this step

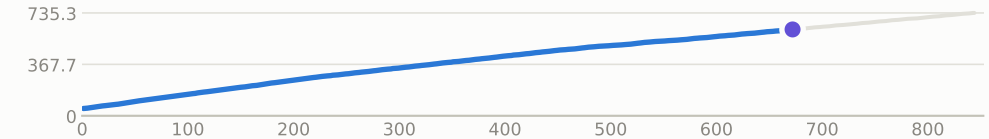
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

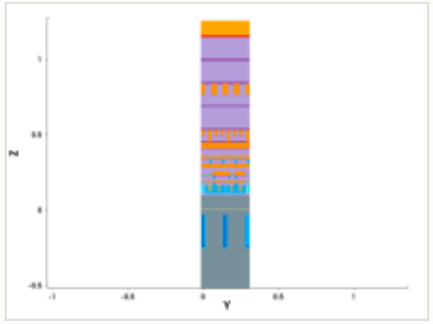
carbon so far, of the whole flow (kgCO2e)



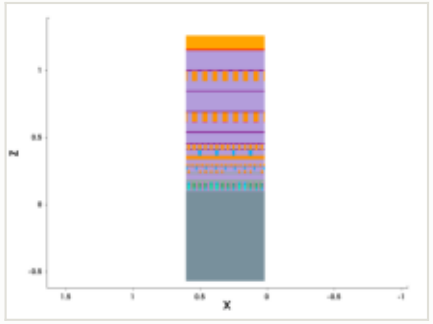
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



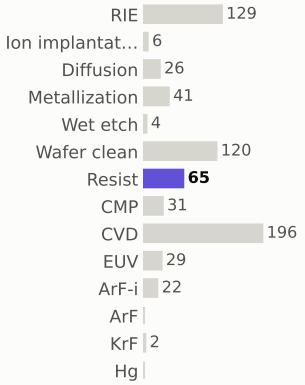
y cut at 0.184 µm



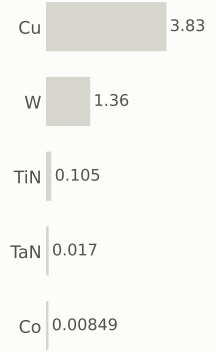
mask for this step

no mask for this step

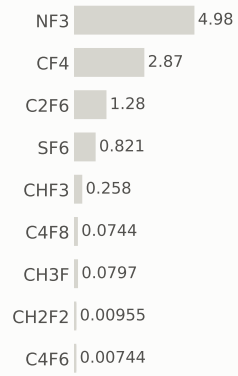
steps by process area



materials deposited (g)



gases emitted (g)



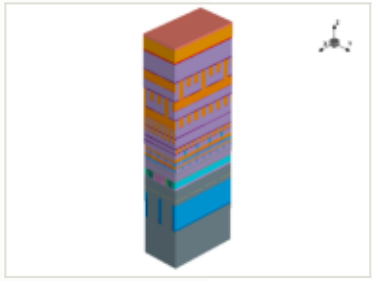
Step #7000: CVD

depo_fill · CVD

1.31	618.6	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

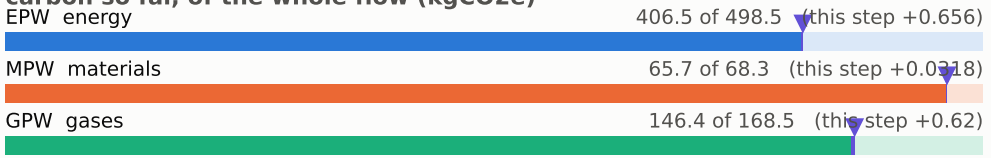


this step

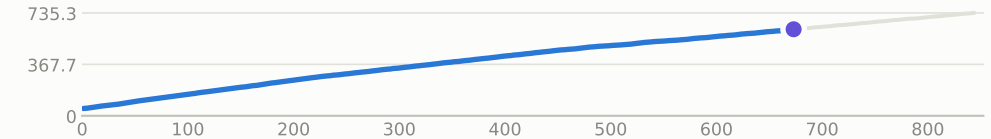
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

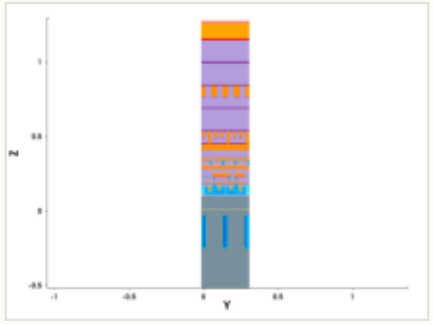
carbon so far, of the whole flow (kgCO2e)



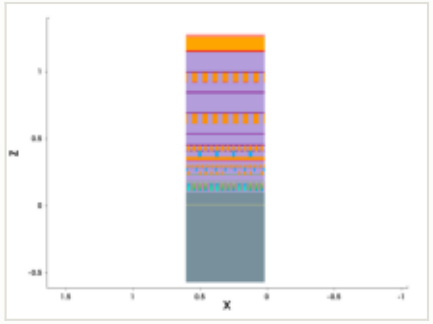
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



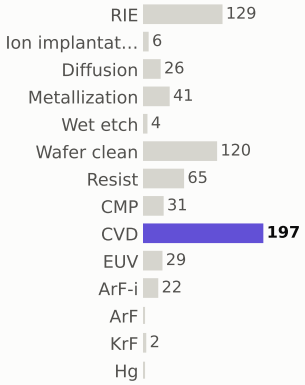
y cut at 0.184 µm



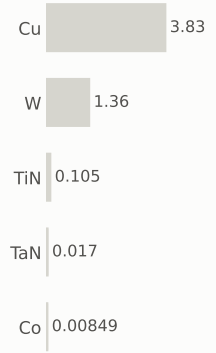
mask for this step

no mask for this step

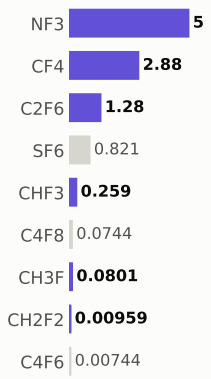
steps by process area



materials deposited (g)



gases emitted (g)



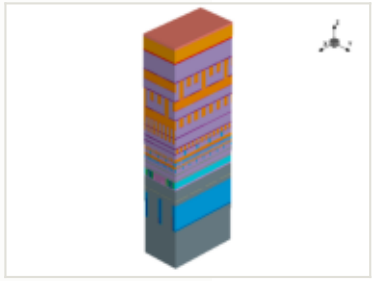
Step #7010: Wafer_clean

clean · Wafer_clean

0.169	618.8	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



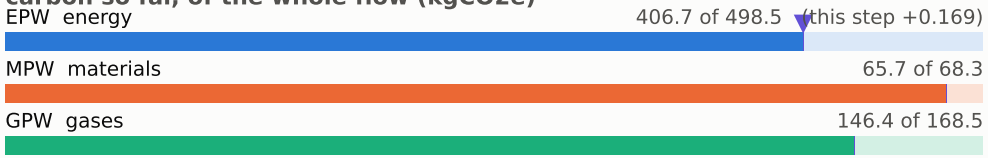
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

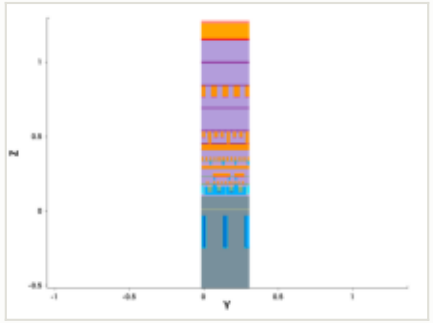
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

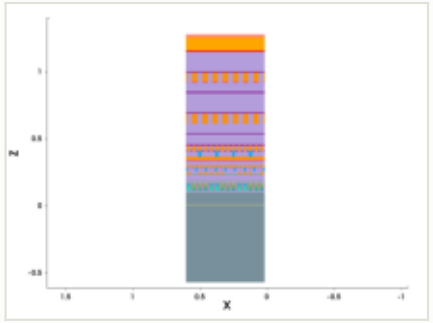


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



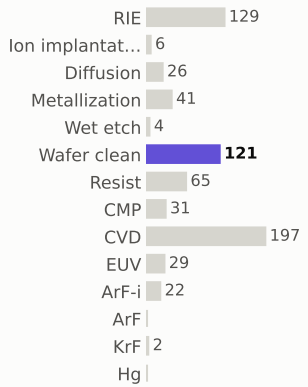
y cut at 0.184 µm



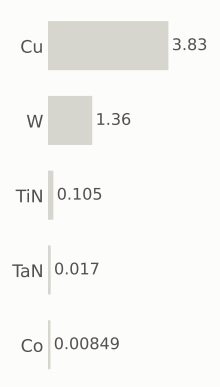
mask for this step

no mask for this step

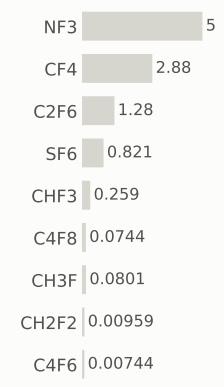
steps by process area



materials deposited (g)



gases emitted (g)



Step #7020: ArF-i

litho_patterning · ArF-i

0.217	619.0	0.029%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

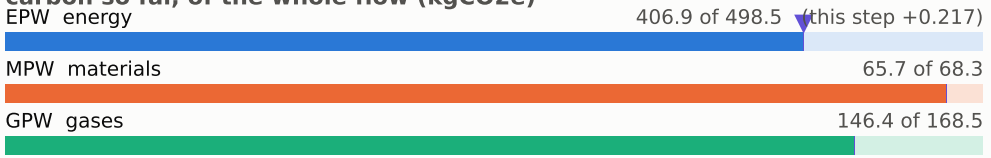
3D structure after this step

zoomed to fit

this step

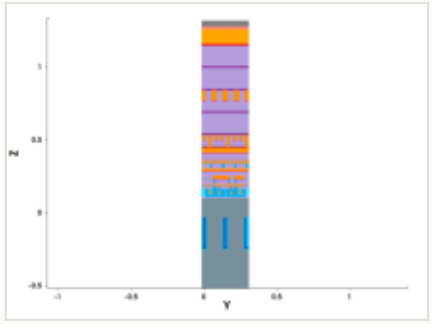
process area	ArF-i	EPW	0.217 kgCO2e
material	—	EPW tool	0.155 kgCO2e
thickness	—	EPW overhead	0.0619 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.36 kWh + 0.144 overhead	embodied carbon	**0.217 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

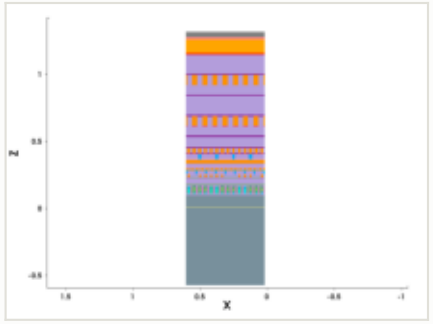


embodied carbon up to each step (kgCO2e per wafer)

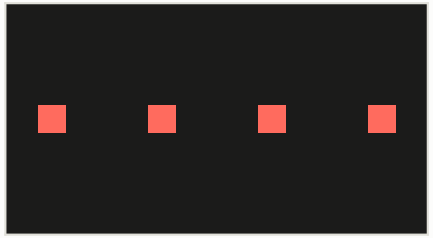
x cut at 0.294 µm



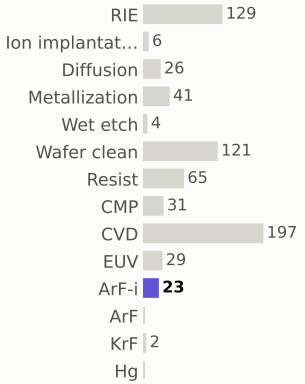
y cut at 0.184 µm



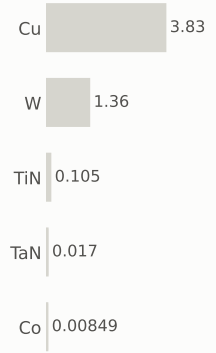
mask: layer 10621:0 (GDSII)



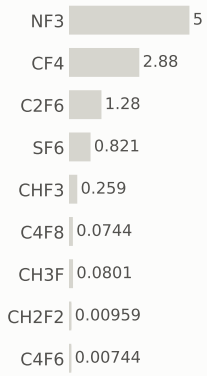
steps by process area



materials deposited (g)



gases emitted (g)



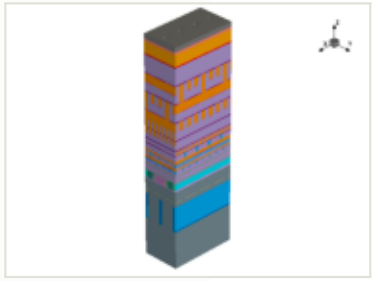
Step #7030: RIE

etch_anisotropic · RIE

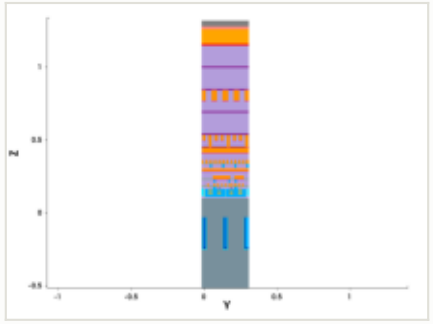
1.04	620.1	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

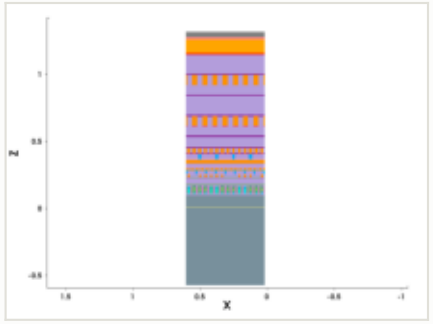
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

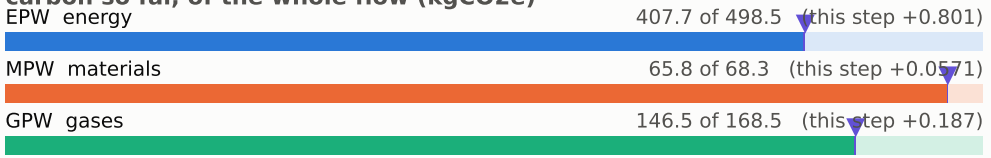
no mask for this step

this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

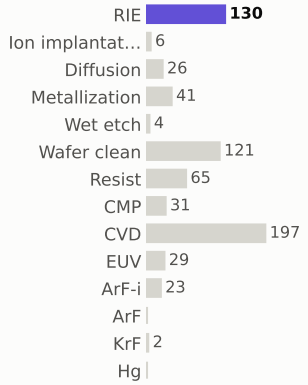
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

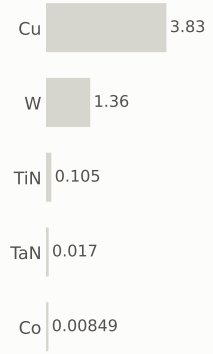


embodied carbon up to each step (kgCO2e per wafer)

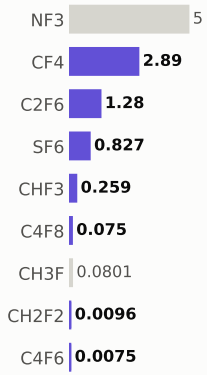
steps by process area



materials deposited (g)



gases emitted (g)



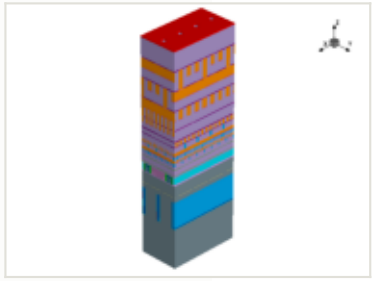
Step #7040: Wafer_clean

etch_strip · Wafer_clean

0.169	620.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

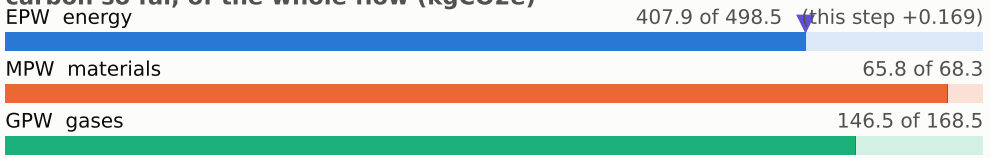


this step

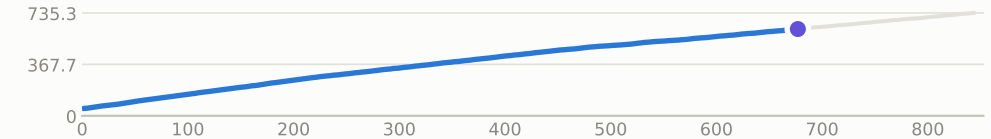
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

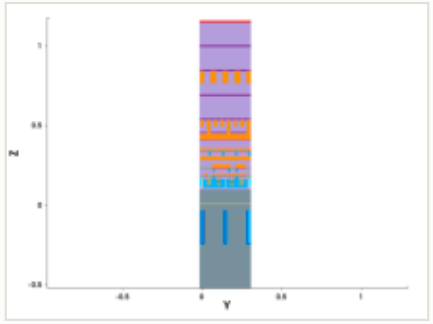
carbon so far, of the whole flow (kgCO2e)



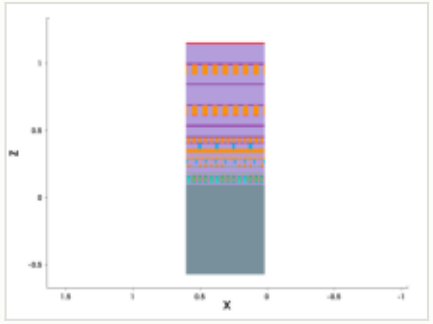
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



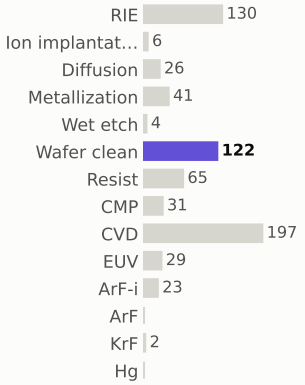
y cut at 0.184 µm



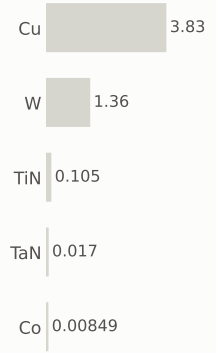
mask for this step

no mask for this step

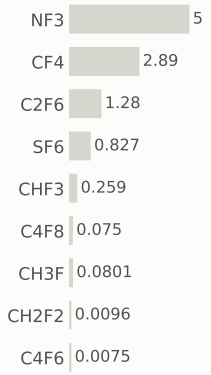
steps by process area



materials deposited (g)



gases emitted (g)



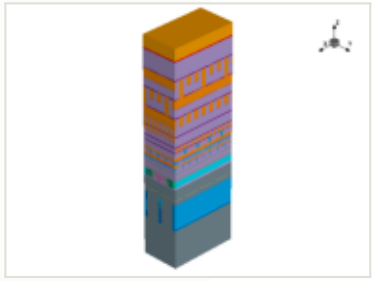
Step #7050: Resist

depo_fill · Resist

0.169	620.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

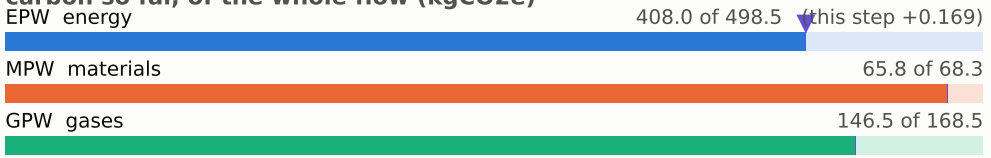


this step

process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

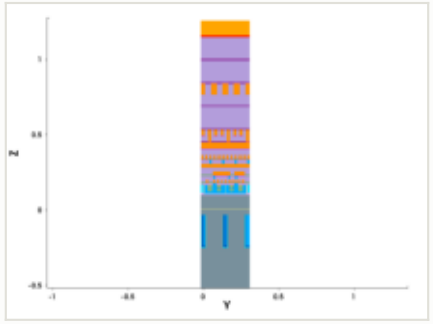
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

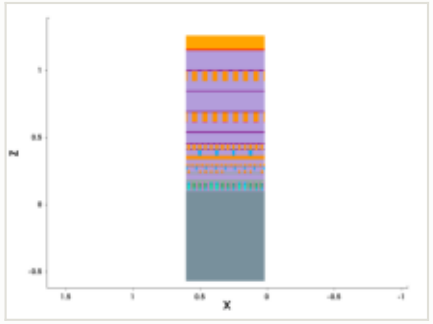


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



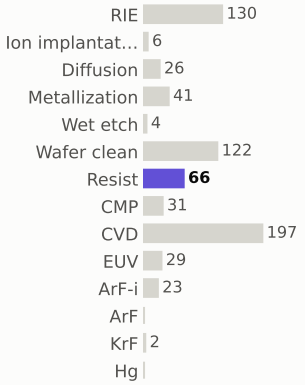
y cut at 0.184 µm



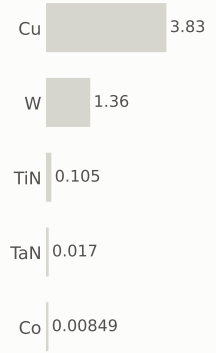
mask for this step

no mask for this step

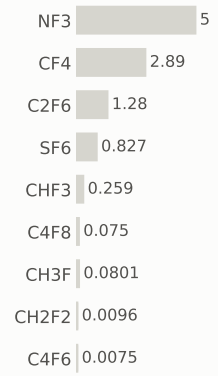
steps by process area



materials deposited (g)



gases emitted (g)



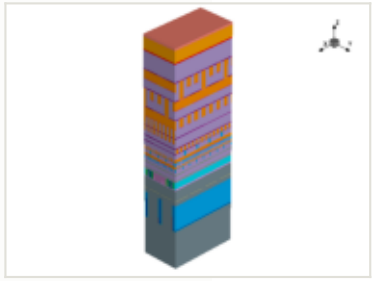
Step #7060: CVD

depo_fill · CVD

1.31	621.7	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

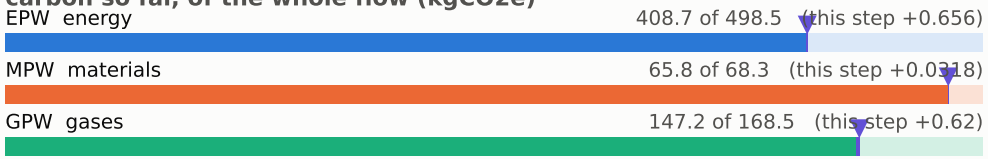


this step

process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

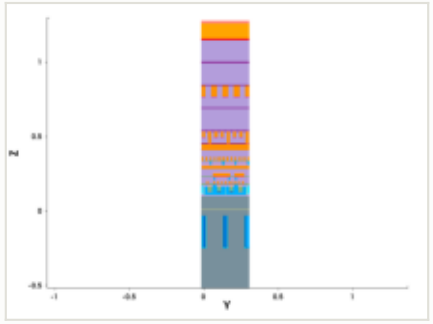
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

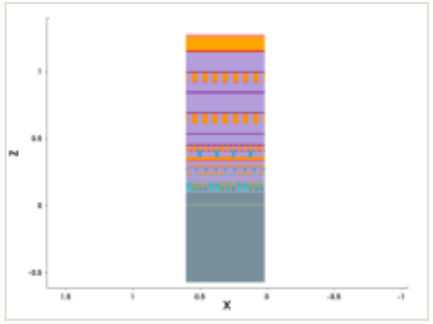


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



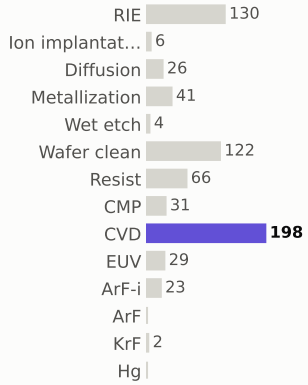
y cut at 0.184 µm



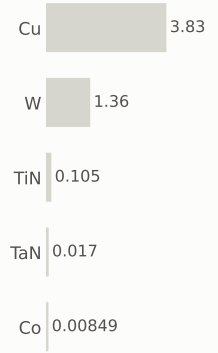
mask for this step

no mask for this step

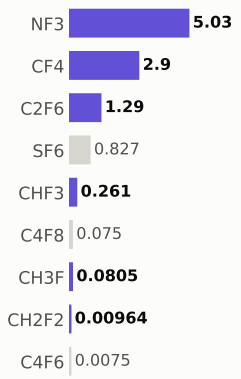
steps by process area



materials deposited (g)



gases emitted (g)



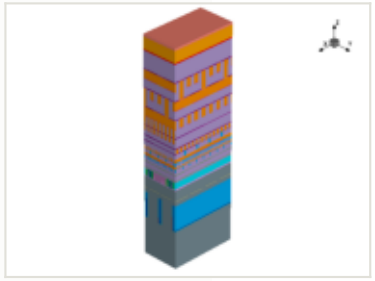
Step #7070: Wafer_clean

clean · Wafer_clean

0.169	621.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



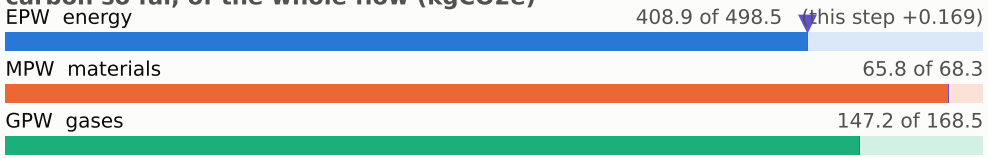
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

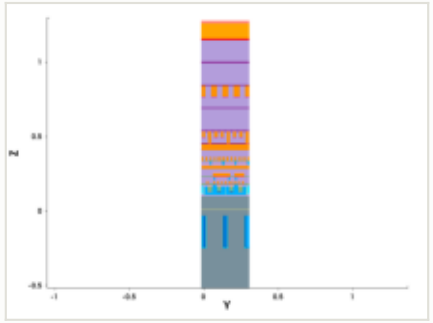
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

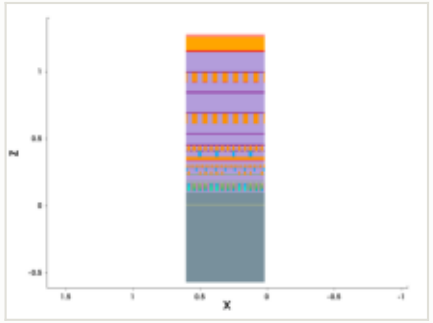


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



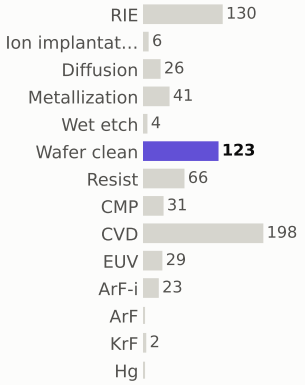
y cut at 0.184 µm



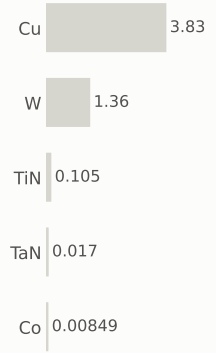
mask for this step

no mask for this step

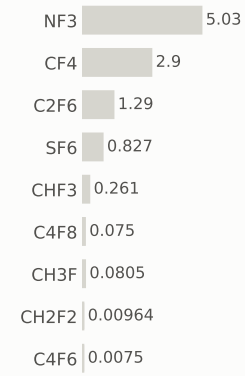
steps by process area



materials deposited (g)



gases emitted (g)



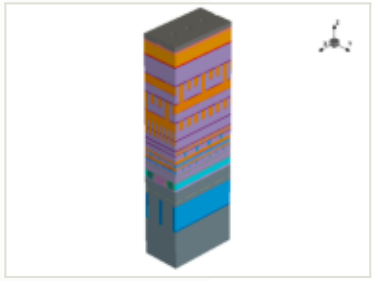
Step #7080: ArF-i

litho_patterning · ArF-i

0.217	622.1	0.029%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

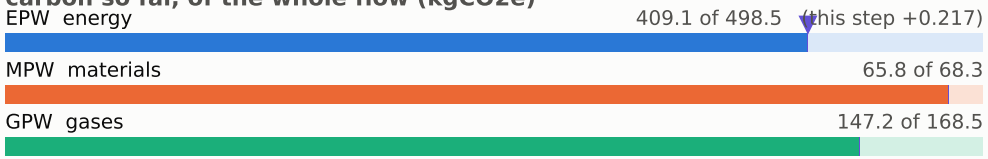


this step

process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

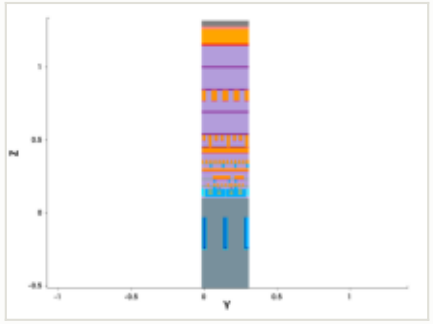
EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

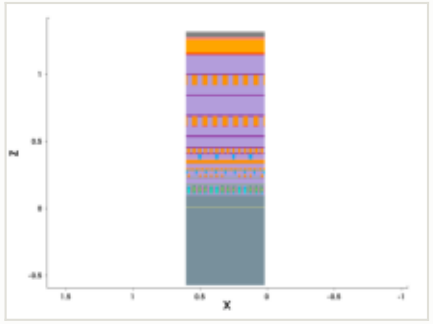


embodied carbon up to each step (kgCO2e per wafer)

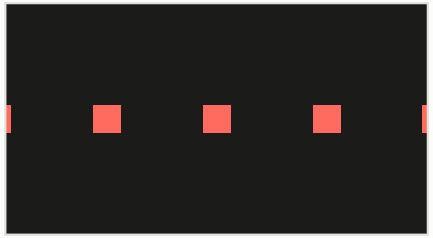
x cut at 0.294 µm



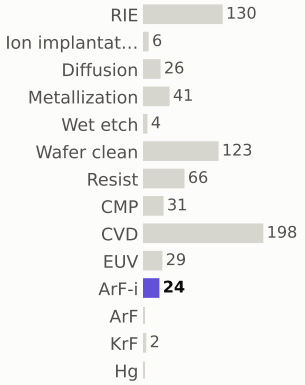
y cut at 0.184 µm



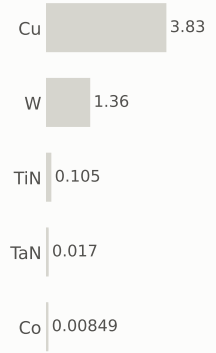
mask: layer 10622:0 (GDSII)



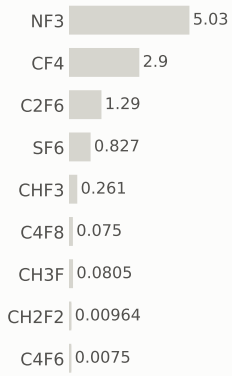
steps by process area



materials deposited (g)



gases emitted (g)



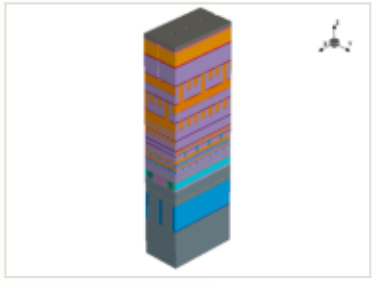
Step #7090: RIE

etch_anisotropic · RIE

1.04	623.1	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

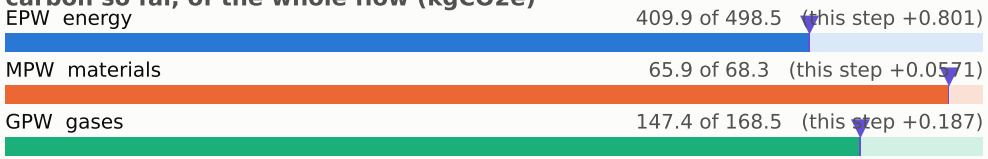


this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

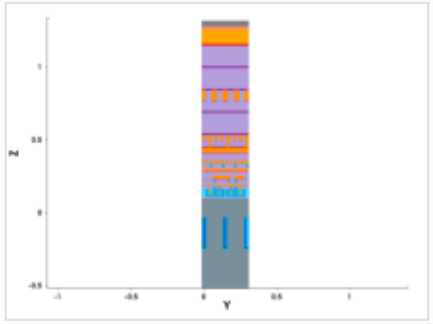
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

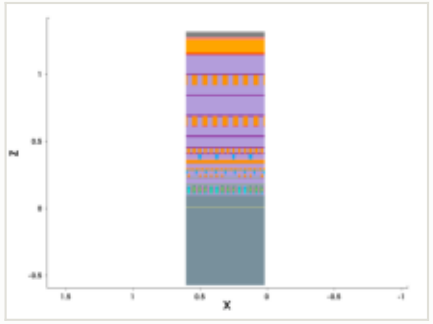


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



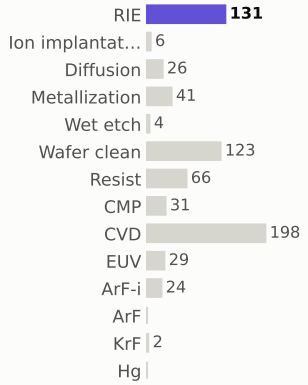
y cut at 0.184 µm



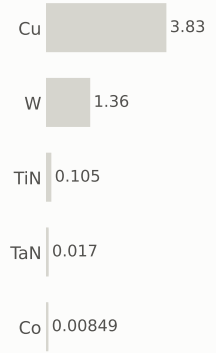
mask for this step

no mask for this step

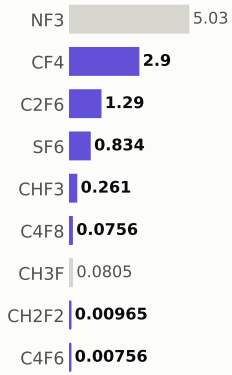
steps by process area



materials deposited (g)



gases emitted (g)



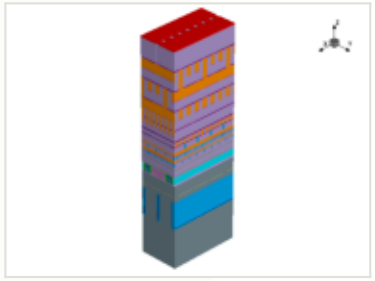
Step #7100: Wafer_clean

etch_strip · Wafer_clean

0.169	623.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

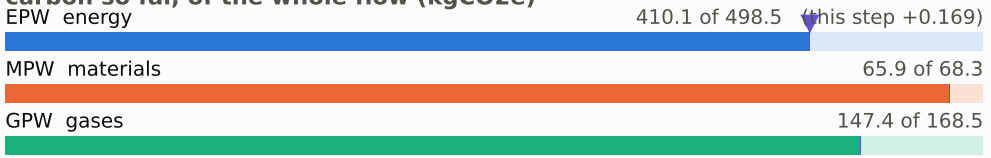


this step

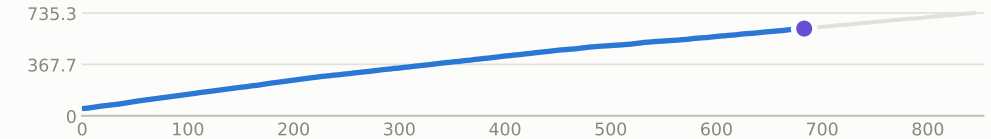
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

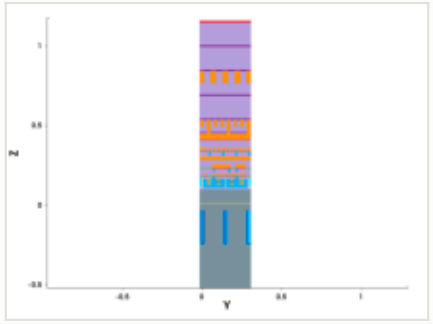
carbon so far, of the whole flow (kgCO2e)



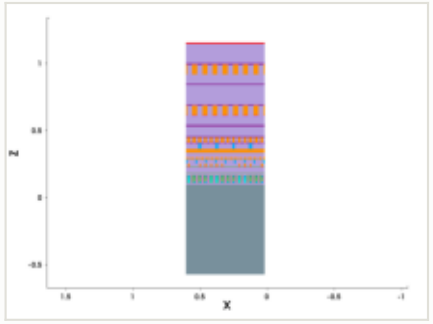
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



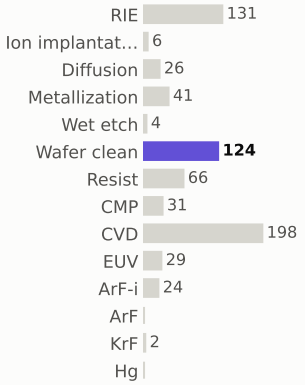
y cut at 0.184 µm



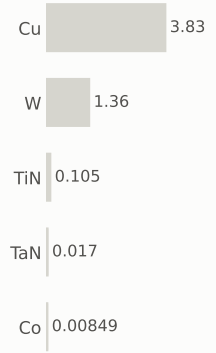
mask for this step

no mask for this step

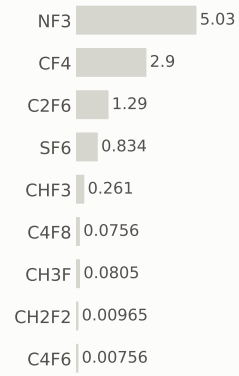
steps by process area



materials deposited (g)



gases emitted (g)



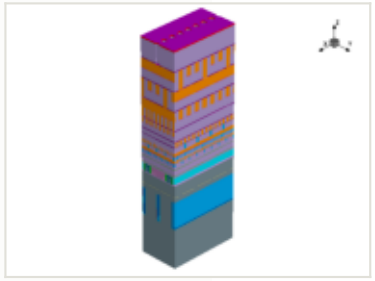
Step #7110: Metallization

depo_ellipsoidal · Metallization

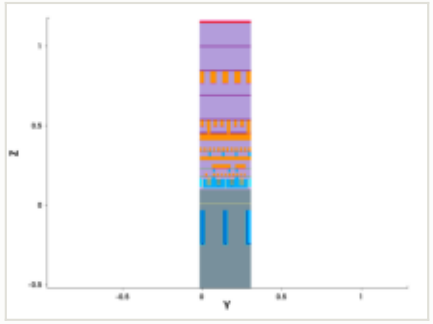
1.2	624.5	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

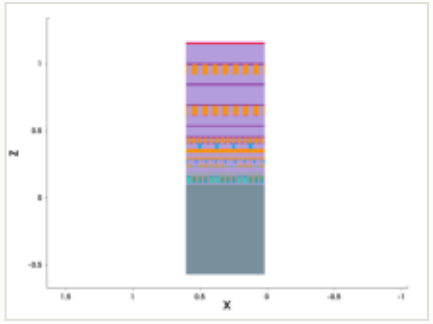
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

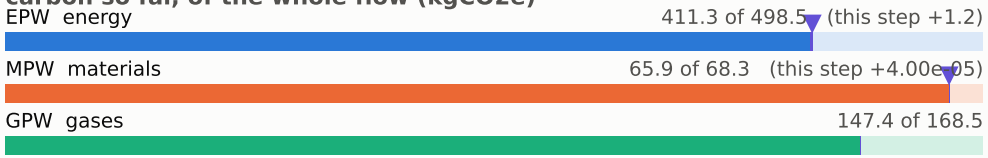
no mask for this step

this step

process area	Metallization
material	TaN
thickness	0.0016 µm
material GWP	25.8159 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

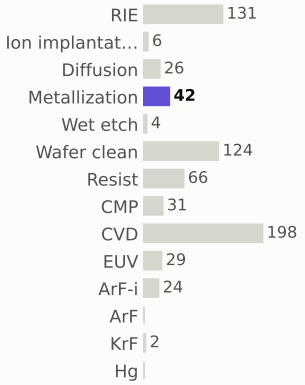
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.00e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

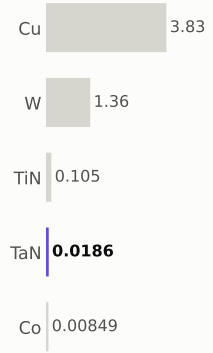


embodied carbon up to each step (kgCO2e per wafer)

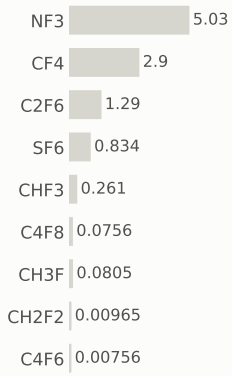
steps by process area



materials deposited (g)



gases emitted (g)



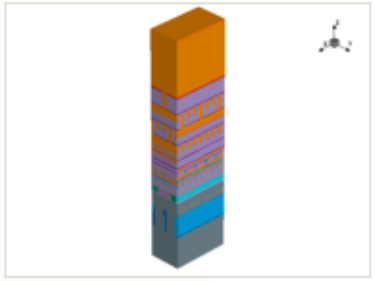
Step #7120: Metallization

depo_fill · Metallization

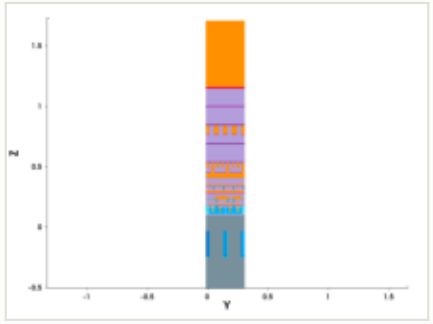
1.21	625.7	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

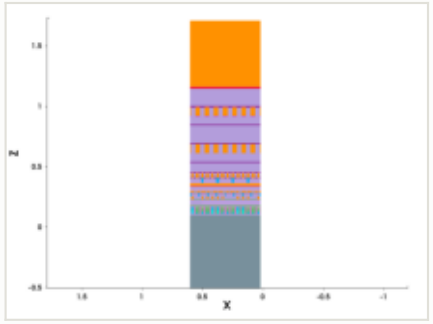
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

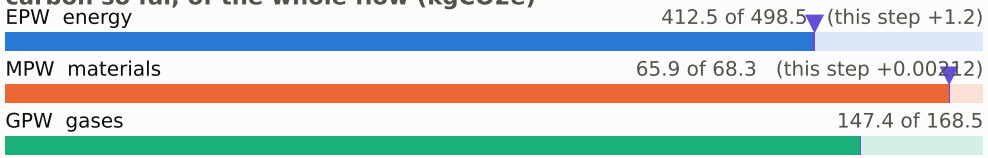
no mask for this step

this step

process area	Metallization
material	Cu
thickness	0.55 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

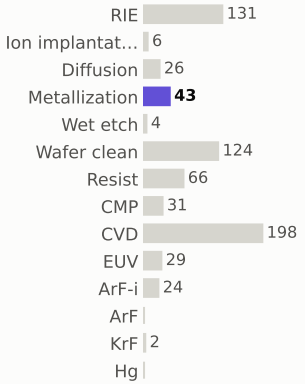
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00212 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

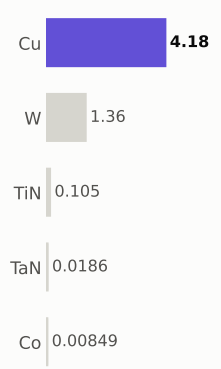


embodied carbon up to each step (kgCO2e per wafer)

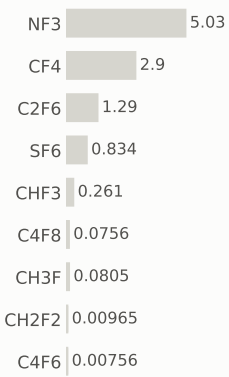
steps by process area



materials deposited (g)



gases emitted (g)



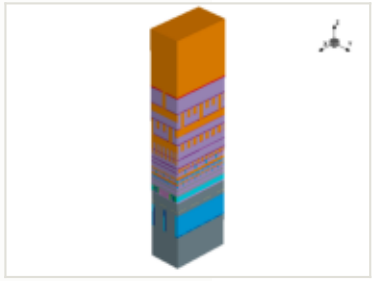
Step #7130: Diffusion

treatment · Diffusion

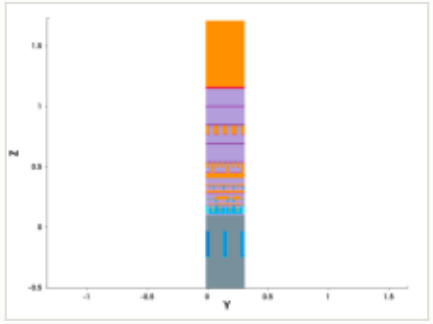
0.801	626.5	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

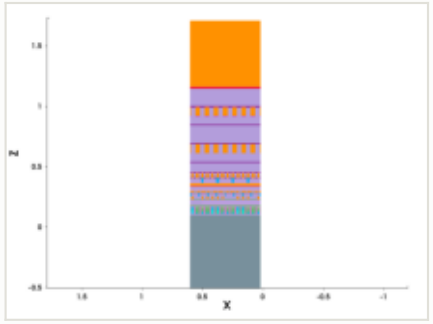
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

no mask for this step

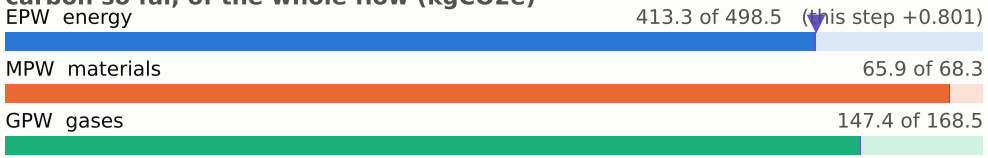
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

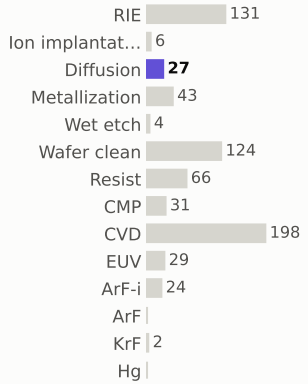
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

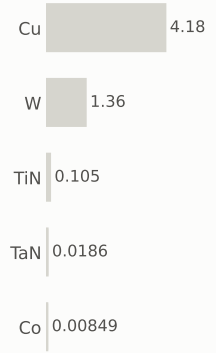


embodied carbon up to each step (kgCO2e per wafer)

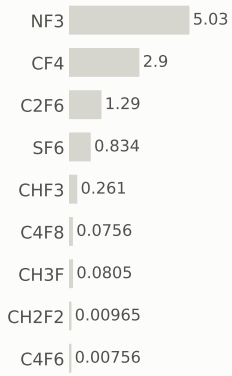
steps by process area



materials deposited (g)



gases emitted (g)



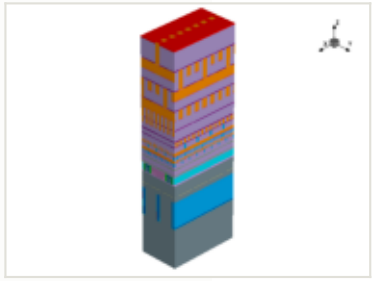
Step #7140: CMP

etch_cmp · CMP

0.169	626.7	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

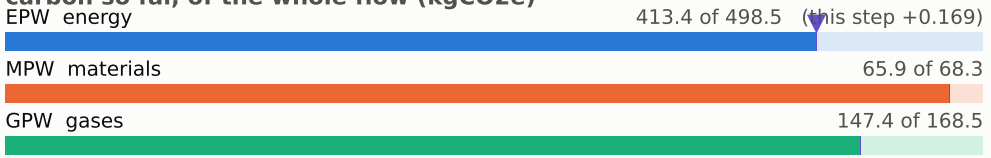


this step

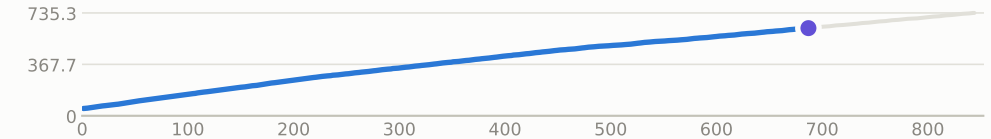
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

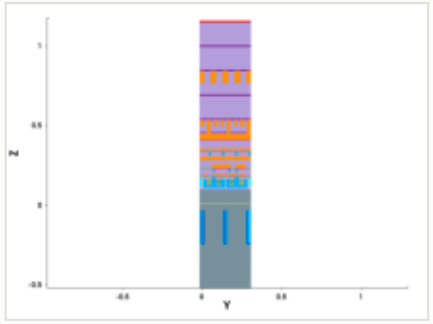
carbon so far, of the whole flow (kgCO2e)



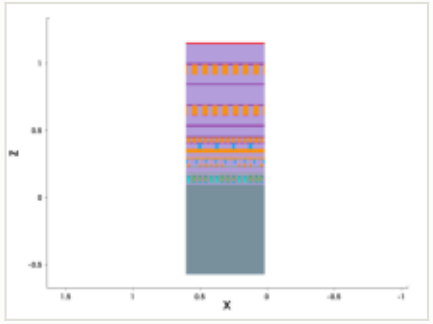
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



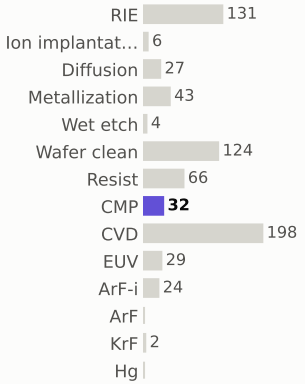
y cut at 0.184 µm



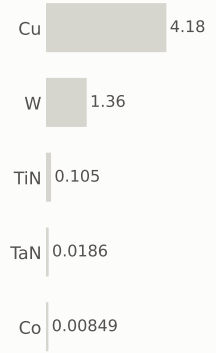
mask for this step

no mask for this step

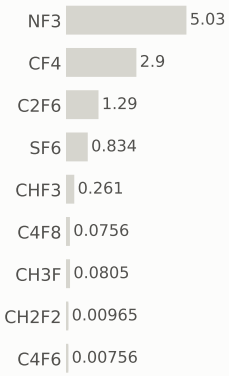
steps by process area



materials deposited (g)



gases emitted (g)



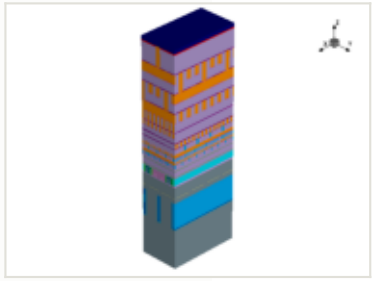
Step #7150: CVD

depo_fill · CVD

1.31	628.0	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

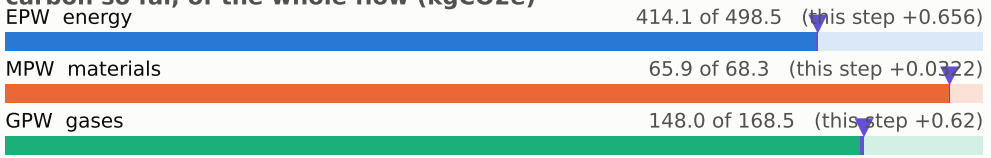


this step

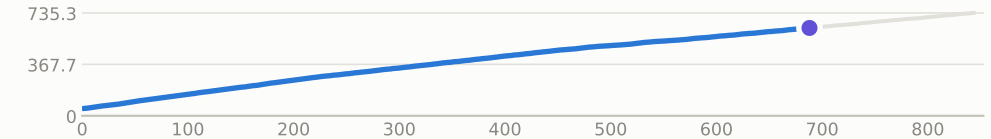
process area	CVD
material	TiN
thickness	0.03 µm
material GWP	43.9778 gCO2e/g
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0322 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

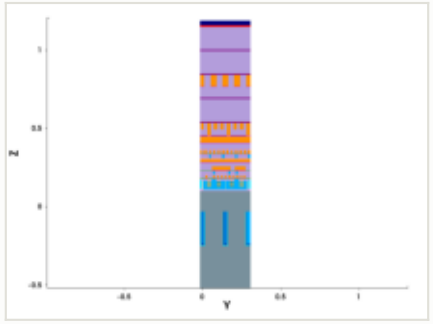
carbon so far, of the whole flow (kgCO2e)



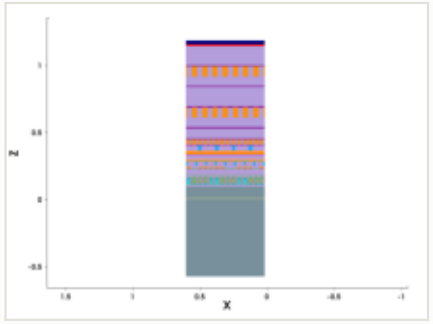
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



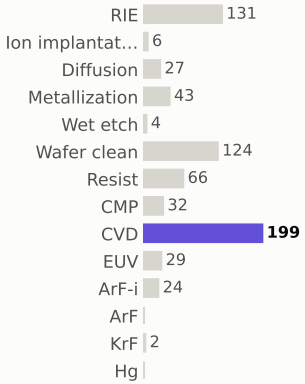
y cut at 0.184 µm



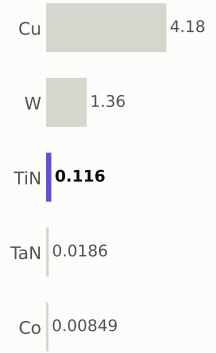
mask for this step

no mask for this step

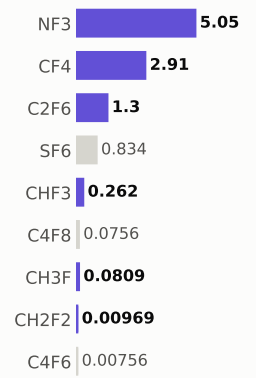
steps by process area



materials deposited (g)



gases emitted (g)



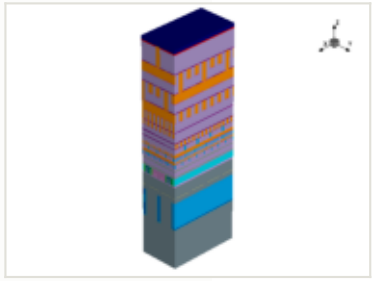
Step #7160: Wafer_clean

clean · Wafer_clean

0.169	628.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



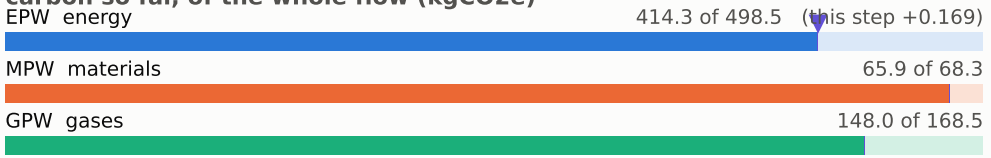
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

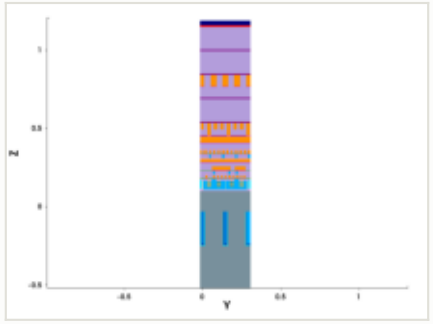
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

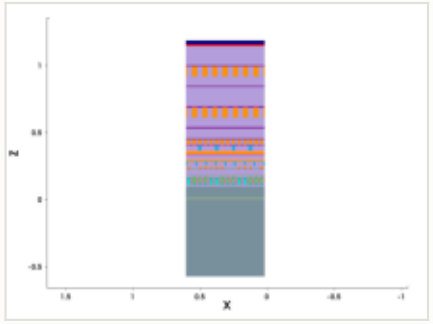


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



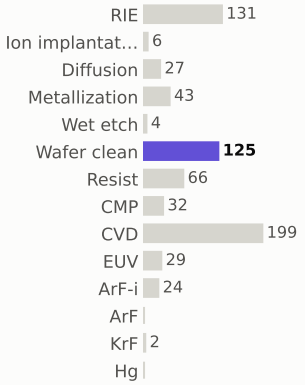
y cut at 0.184 µm



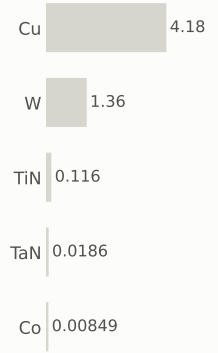
mask for this step

no mask for this step

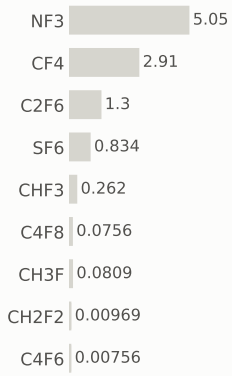
steps by process area



materials deposited (g)



gases emitted (g)



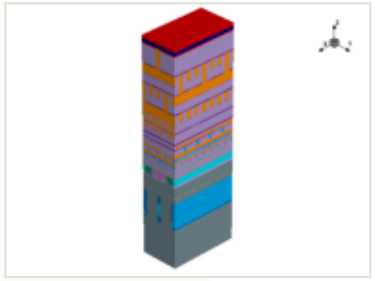
Step #7170: CVD

depo_fill · CVD

1.31	629.5	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

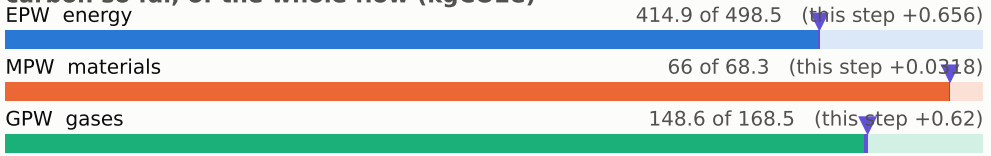


this step

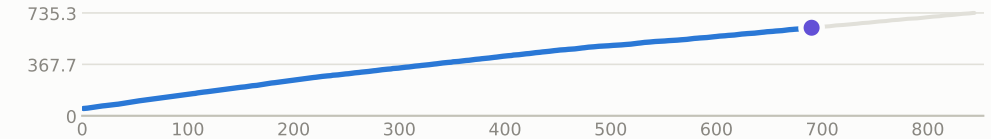
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

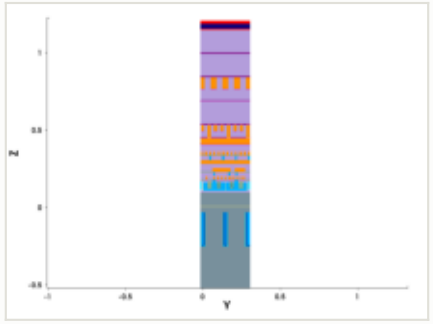
carbon so far, of the whole flow (kgCO2e)



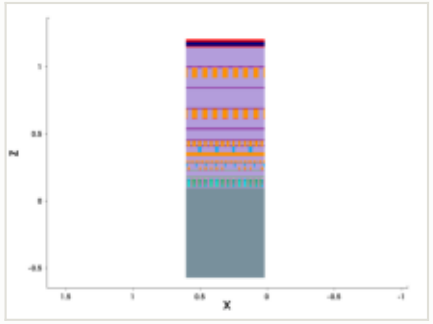
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



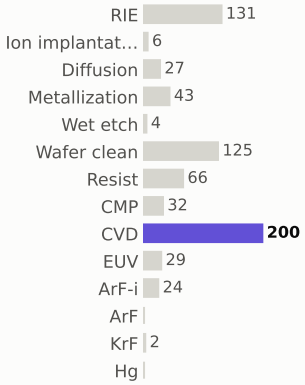
y cut at 0.184 µm



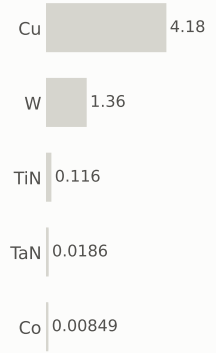
mask for this step

no mask for this step

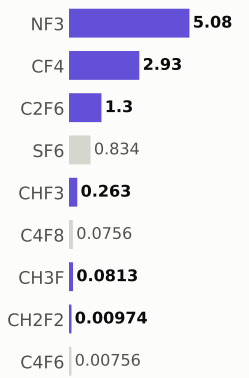
steps by process area



materials deposited (g)



gases emitted (g)



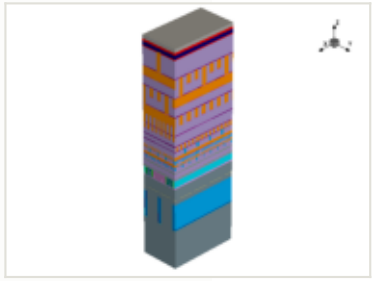
Step #7180: CVD

depo_fill · CVD

1.31	630.8	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

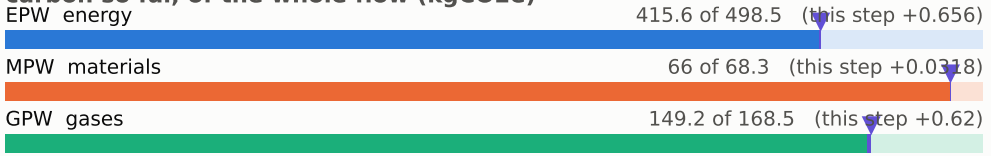


this step

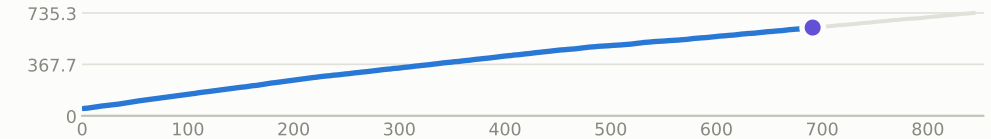
process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

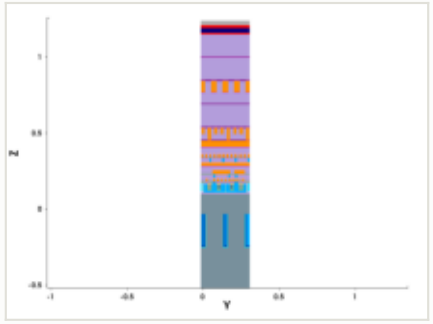
carbon so far, of the whole flow (kgCO2e)



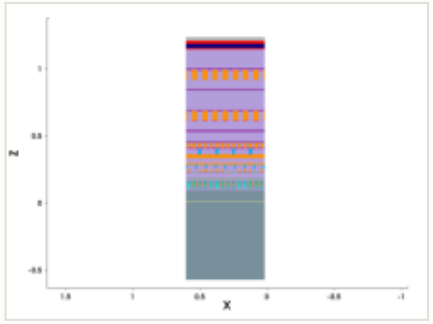
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



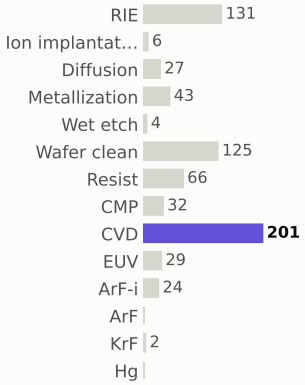
y cut at 0.184 µm



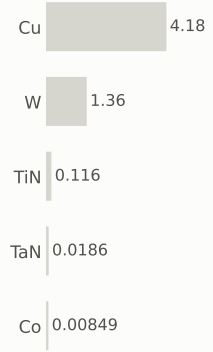
mask for this step

no mask for this step

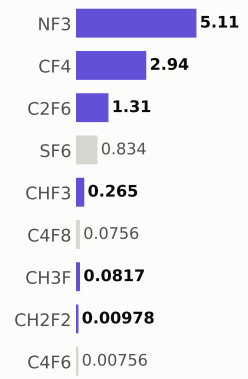
steps by process area



materials deposited (g)



gases emitted (g)



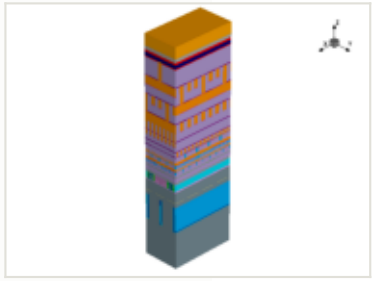
Step #7190: Resist

depo_fill · Resist

0.169	630.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

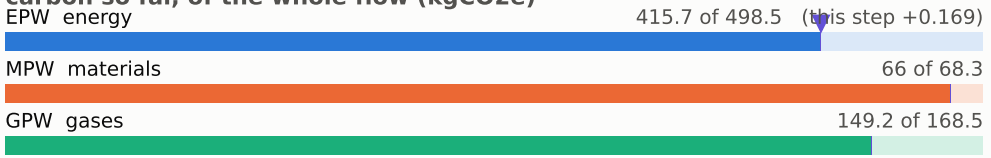


this step

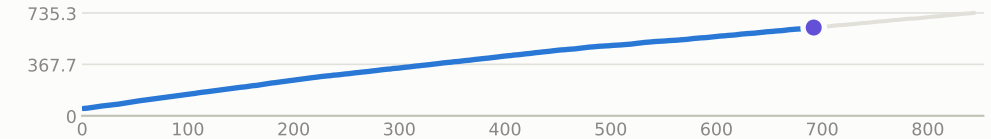
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

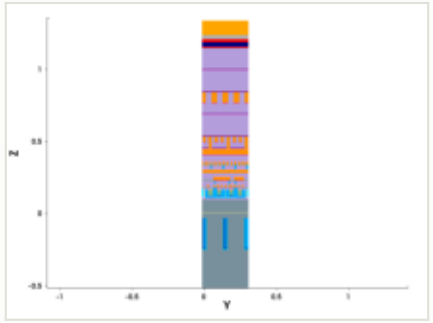
carbon so far, of the whole flow (kgCO2e)



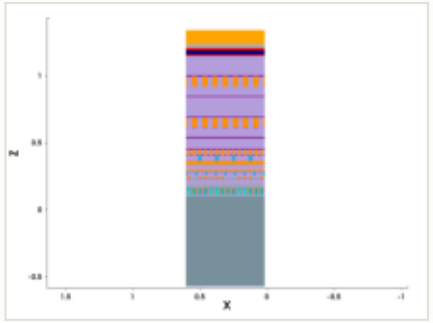
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



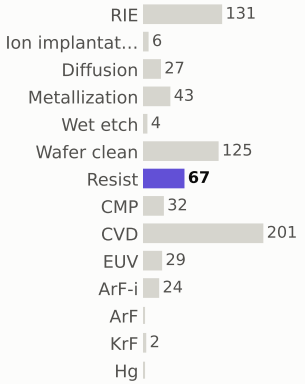
y cut at 0.184 µm



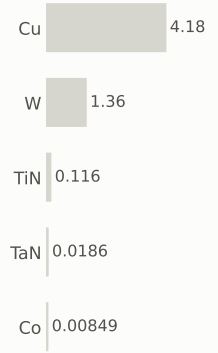
mask for this step

no mask for this step

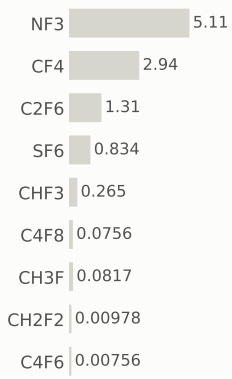
steps by process area



materials deposited (g)



gases emitted (g)



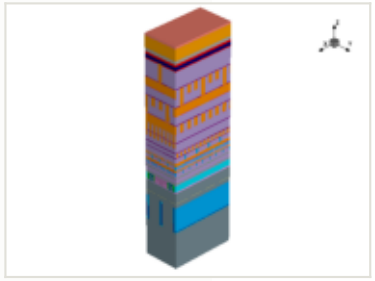
Step #7200: CVD

depo_fill · CVD

1.31	632.3	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

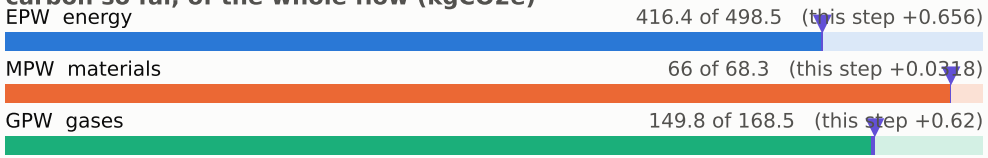


this step

process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

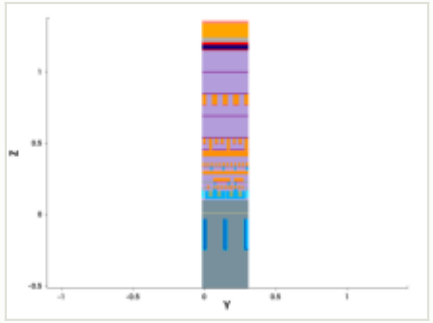
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

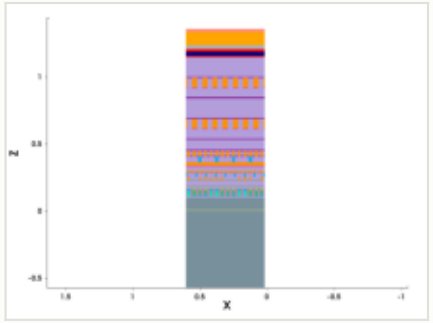


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



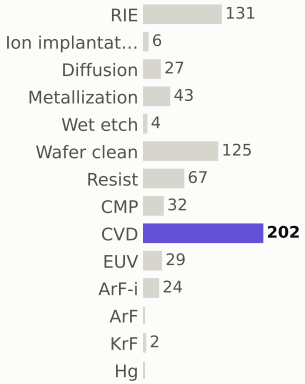
y cut at 0.184 µm



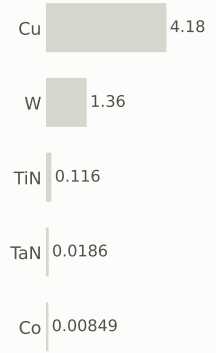
mask for this step

no mask for this step

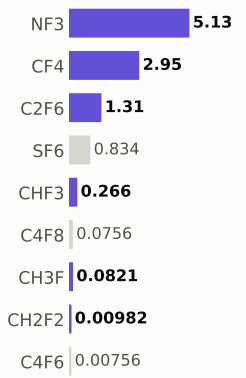
steps by process area



materials deposited (g)



gases emitted (g)



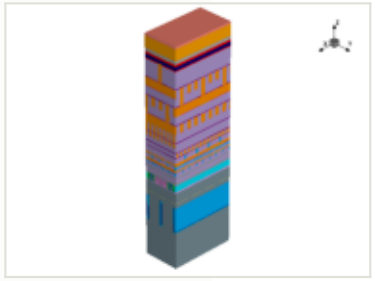
Step #7210: Wafer_clean

clean · Wafer_clean

0.169	632.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



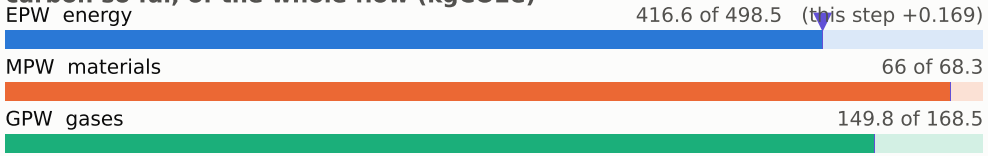
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

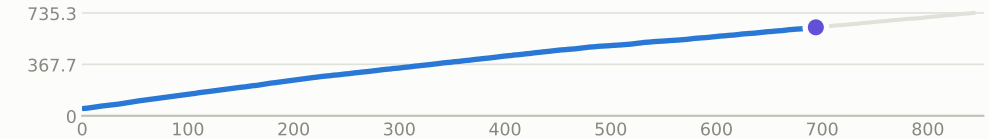
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

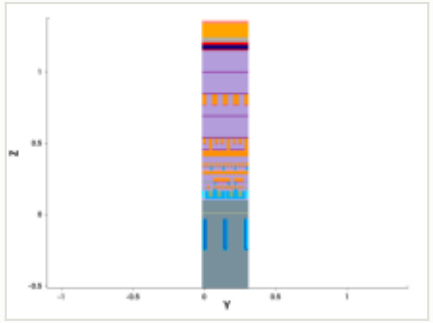
carbon so far, of the whole flow (kgCO2e)



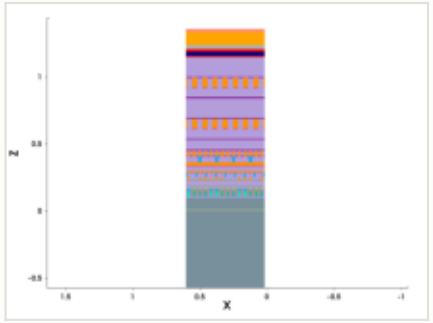
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



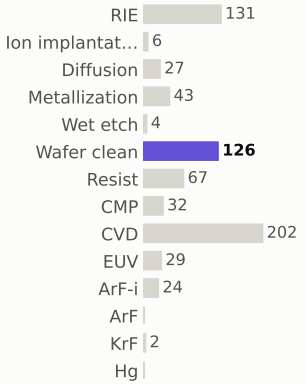
y cut at 0.184 µm



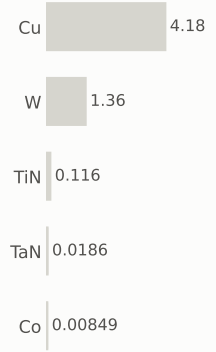
mask for this step

no mask for this step

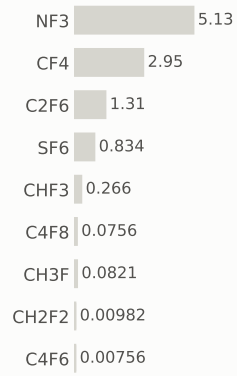
steps by process area



materials deposited (g)



gases emitted (g)



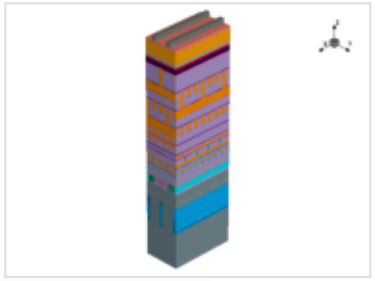
Step #7220: ArF-i

litho_patterning · ArF-i

0.217	632.6	0.029%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

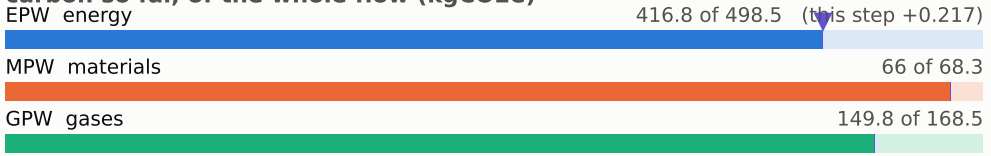


this step

process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

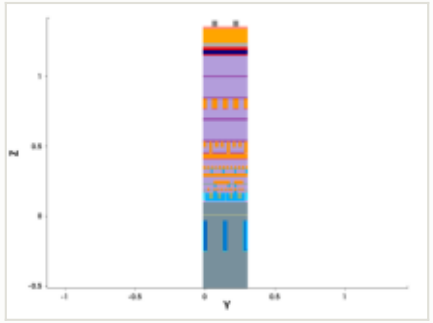
EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

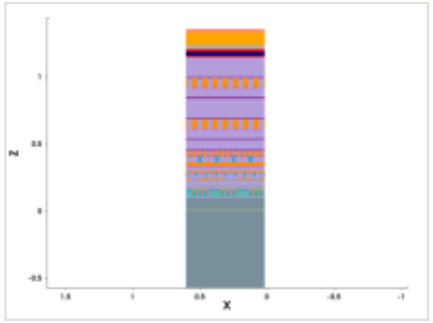


embodied carbon up to each step (kgCO2e per wafer)

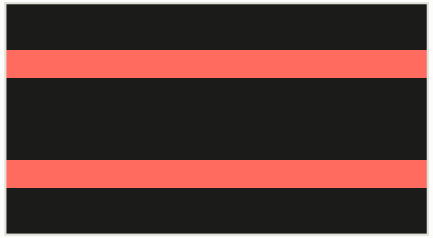
x cut at 0.294 µm



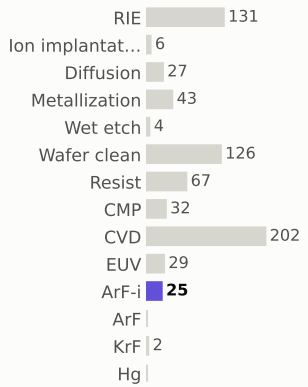
y cut at 0.184 µm



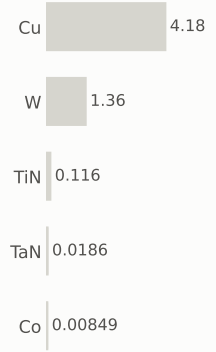
mask: layer 10651:0, positive (GDSII)



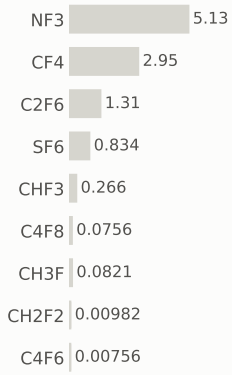
steps by process area



materials deposited (g)



gases emitted (g)



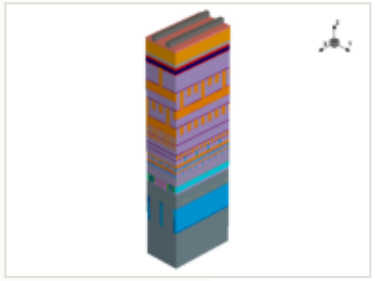
Step #7230: Ion_implantation

treatment · Ion_implantation

0.801	633.4	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



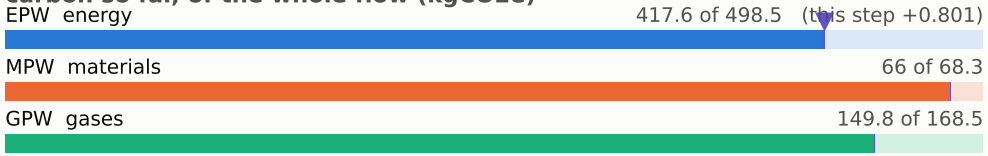
this step

process area	Ion_implantation
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

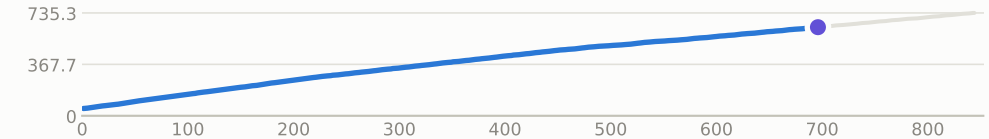
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

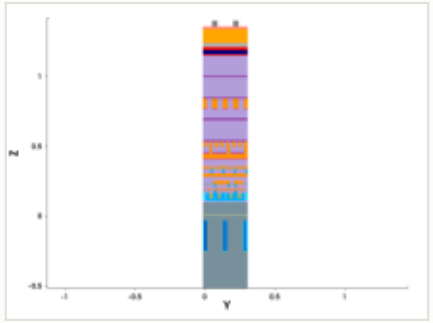
carbon so far, of the whole flow (kgCO2e)



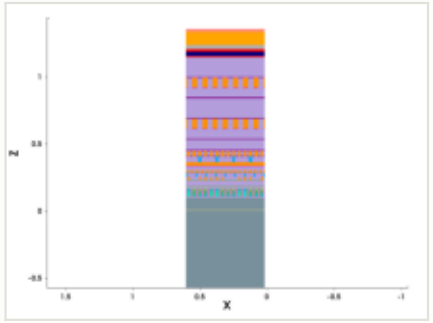
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



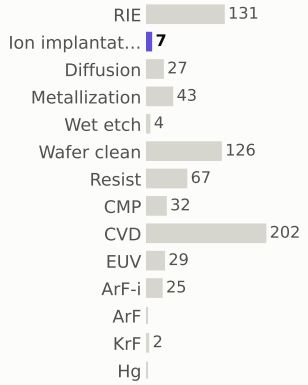
y cut at 0.184 µm



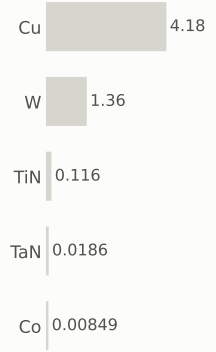
mask for this step

no mask for this step

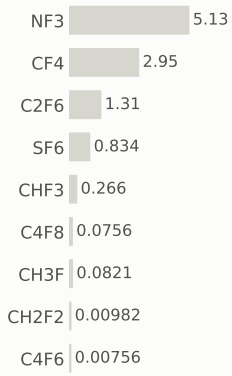
steps by process area



materials deposited (g)



gases emitted (g)



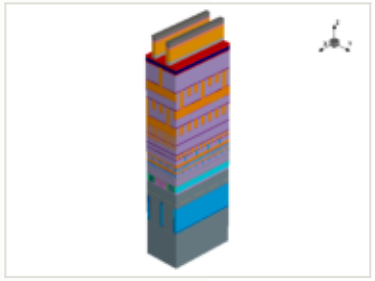
Step #7240: RIE

etch_anisotropic · RIE

1.04	634.5	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

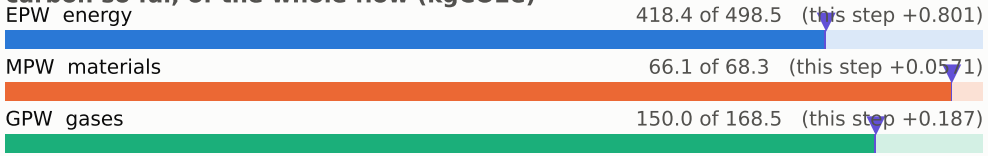


this step

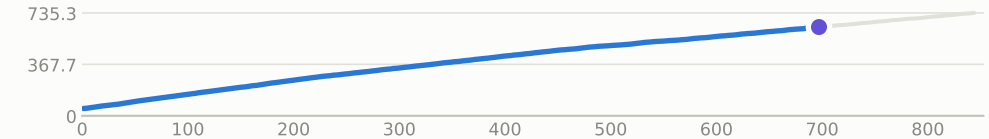
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

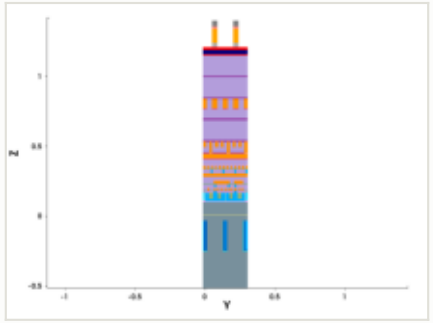
carbon so far, of the whole flow (kgCO2e)



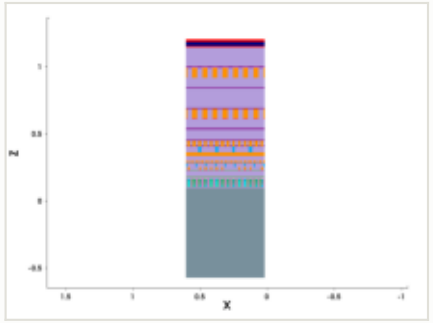
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



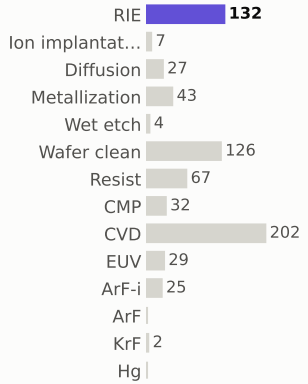
y cut at 0.184 µm



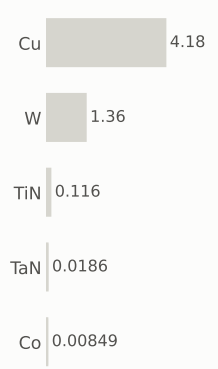
mask for this step

no mask for this step

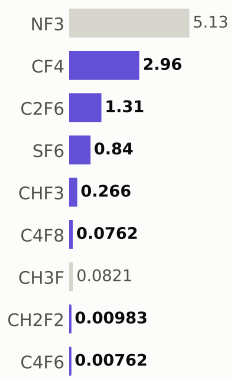
steps by process area



materials deposited (g)



gases emitted (g)



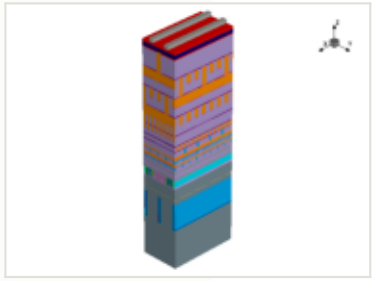
Step #7250: Wafer_clean

etch_strip · Wafer_clean

0.169	634.7	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

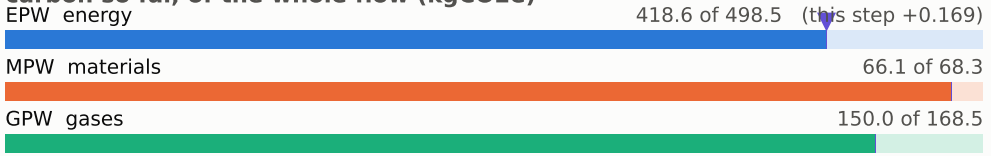


this step

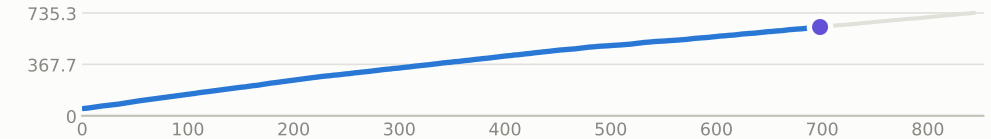
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

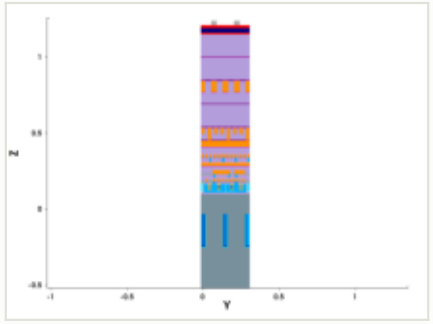
carbon so far, of the whole flow (kgCO2e)



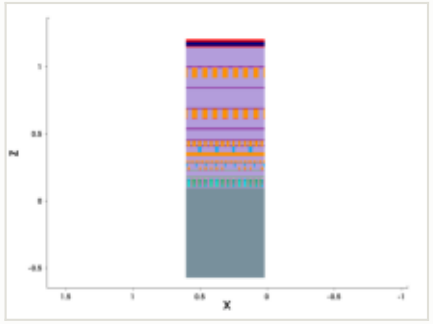
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



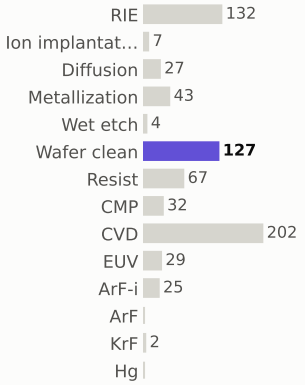
y cut at 0.184 µm



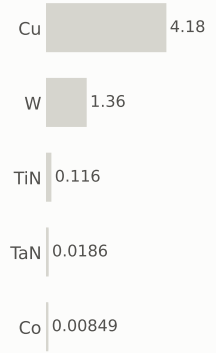
mask for this step

no mask for this step

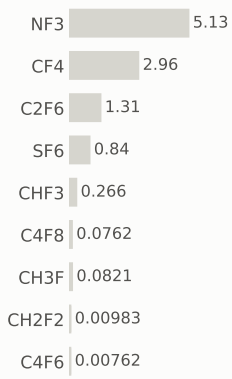
steps by process area



materials deposited (g)



gases emitted (g)



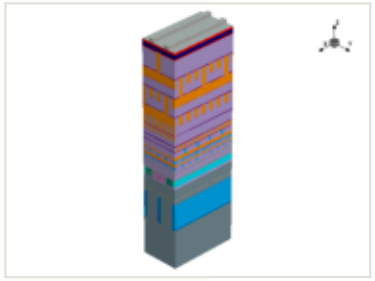
Step #7260: CVD

depo_isotropic · CVD

1.31	636.0	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

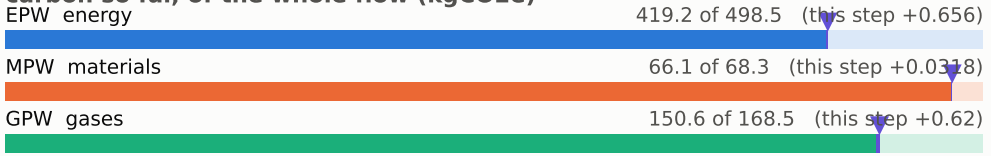


this step

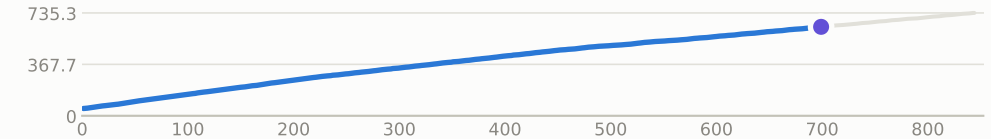
process area	CVD
material	—
thickness	0.001 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

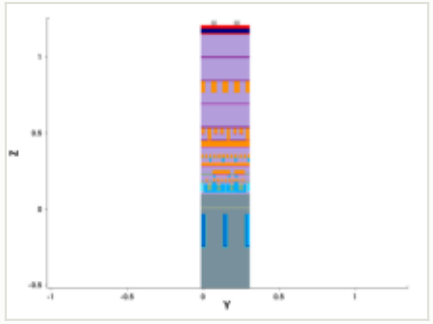
carbon so far, of the whole flow (kgCO2e)



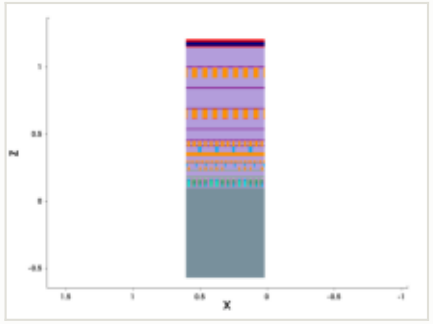
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



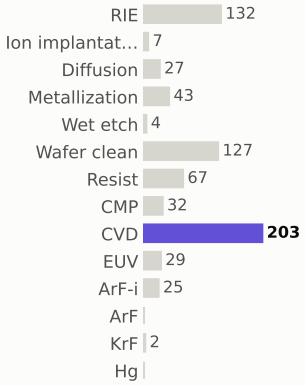
y cut at 0.184 µm



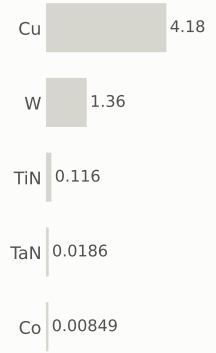
mask for this step

no mask for this step

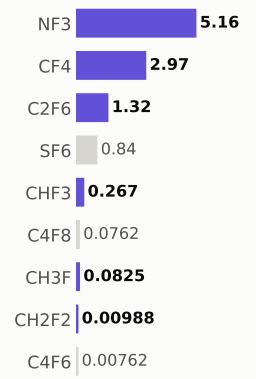
steps by process area



materials deposited (g)



gases emitted (g)



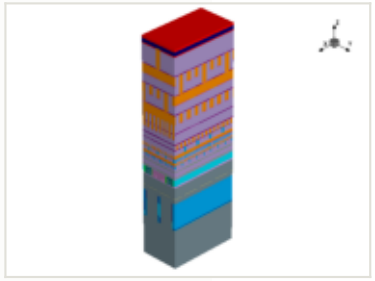
Step #7270: RIE

etch_anisotropic · RIE

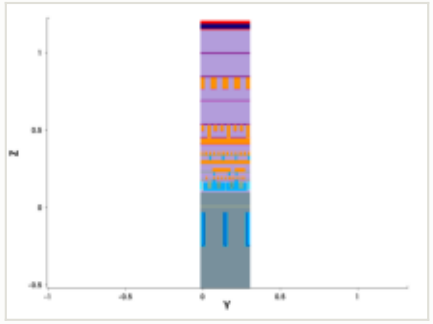
1.04	637.0	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

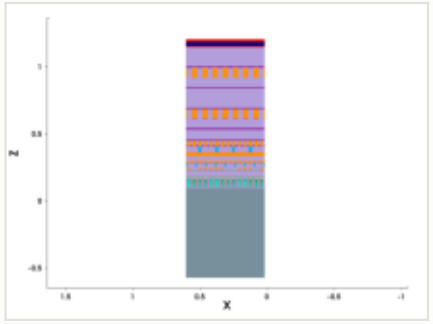
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

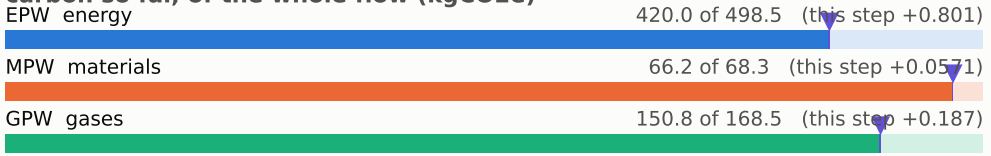
no mask for this step

this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

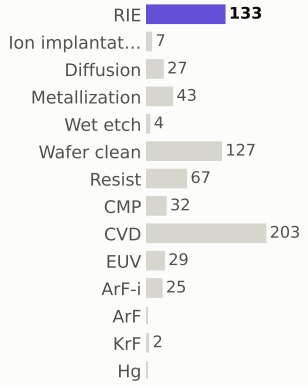
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

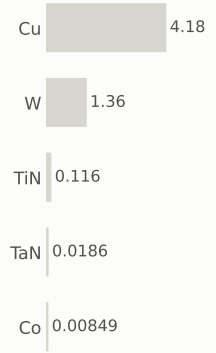


embodied carbon up to each step (kgCO2e per wafer)

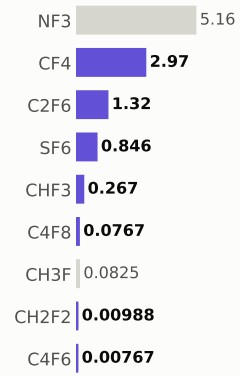
steps by process area



materials deposited (g)



gases emitted (g)



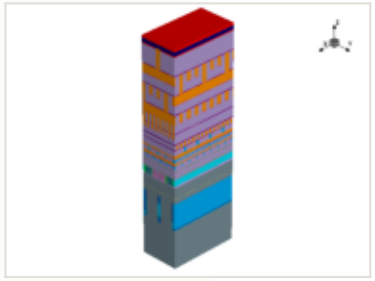
Step #7280: Wafer_clean

etch_strip · Wafer_clean

0.169	637.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

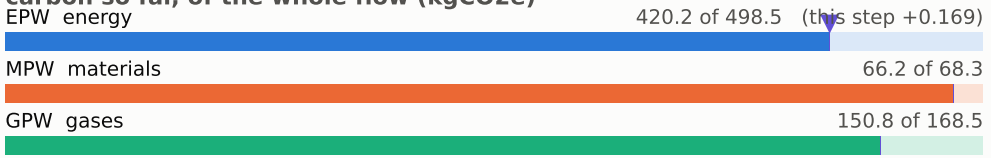


this step

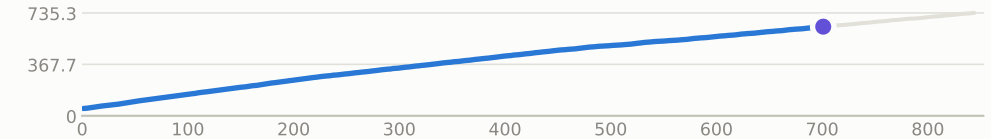
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

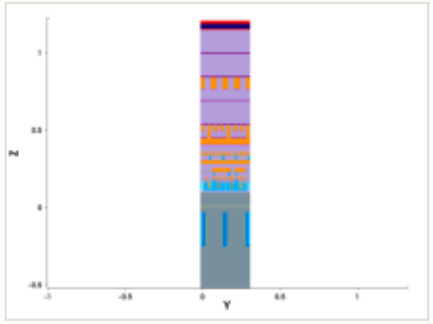
carbon so far, of the whole flow (kgCO2e)



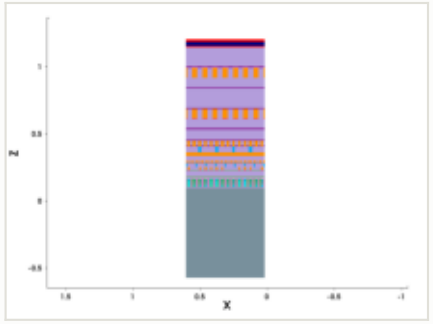
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



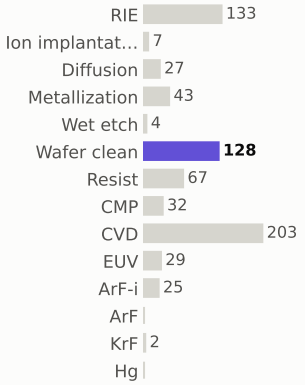
y cut at 0.184 µm



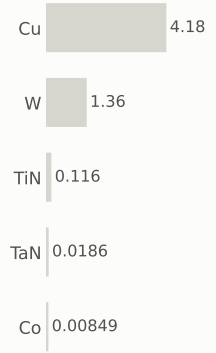
mask for this step

no mask for this step

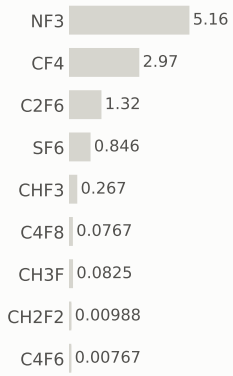
steps by process area



materials deposited (g)



gases emitted (g)



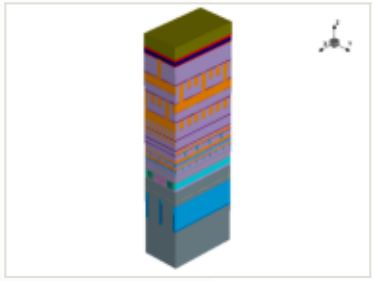
Step #7290: Resist

depo_fill · Resist

0.169	637.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

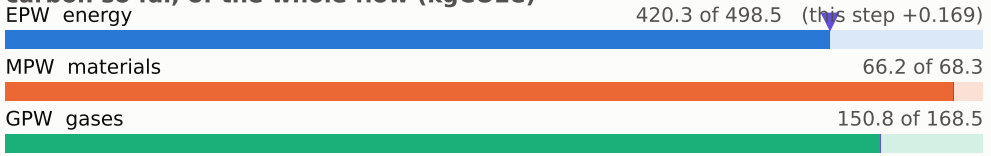


this step

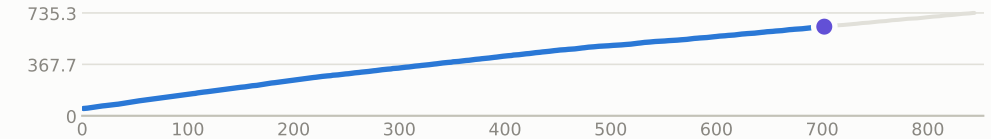
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

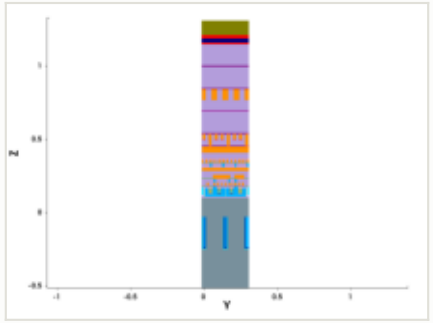
carbon so far, of the whole flow (kgCO2e)



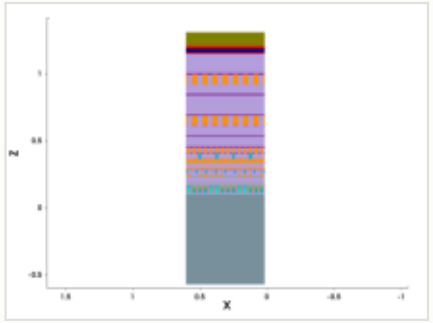
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



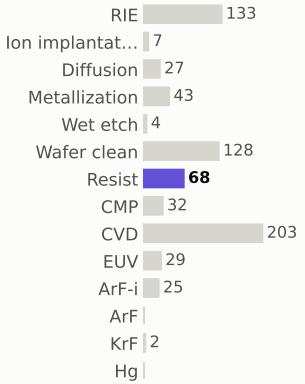
y cut at 0.184 µm



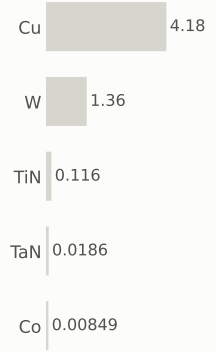
mask for this step

no mask for this step

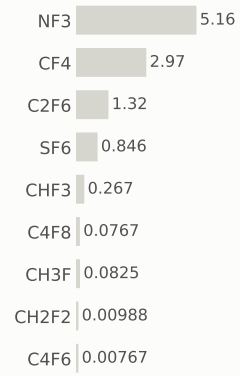
steps by process area



materials deposited (g)



gases emitted (g)



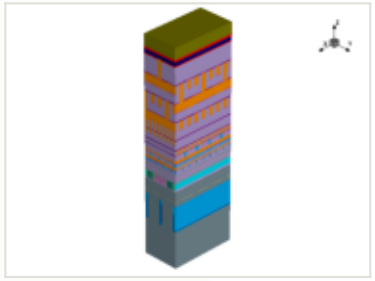
Step #7300: Wafer_clean

clean · Wafer_clean

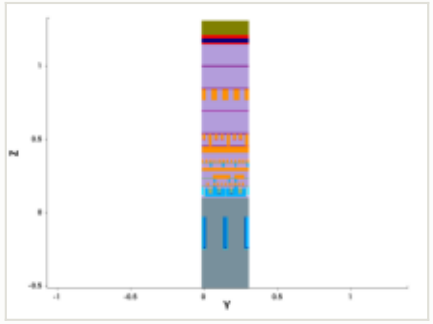
0.169	637.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

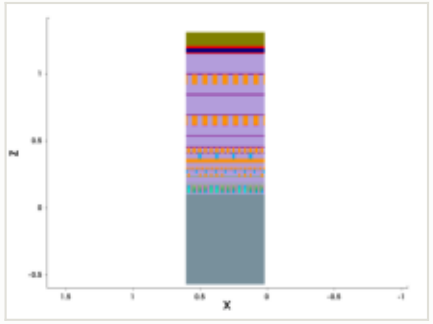
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

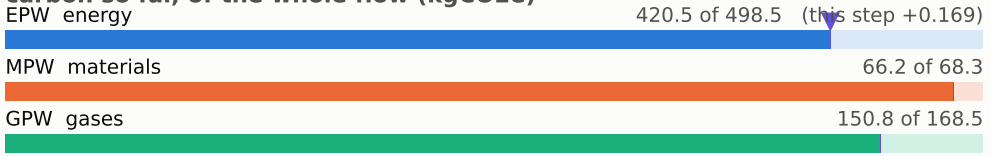
no mask for this step

this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

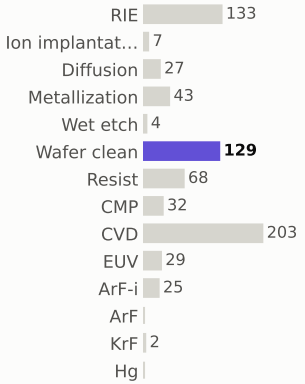
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

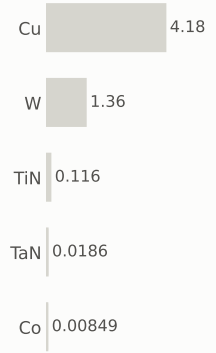


embodied carbon up to each step (kgCO2e per wafer)

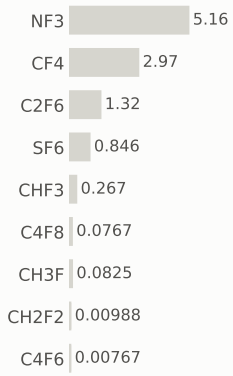
steps by process area



materials deposited (g)



gases emitted (g)



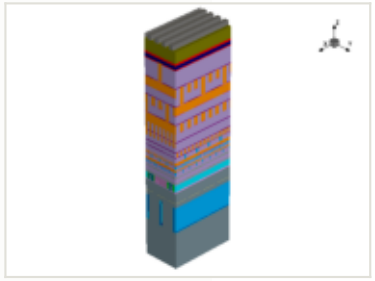
Step #7310: ArF-i

litho_patterning · ArF-i

0.217	637.7	0.029%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

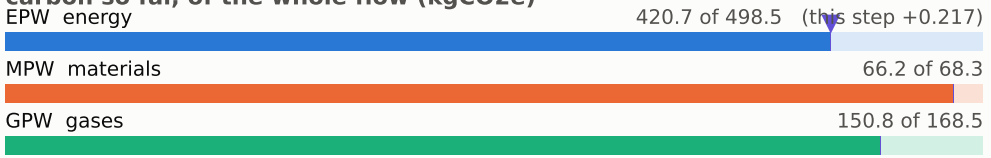


this step

process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

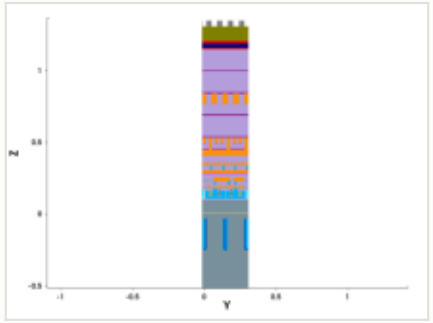
EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

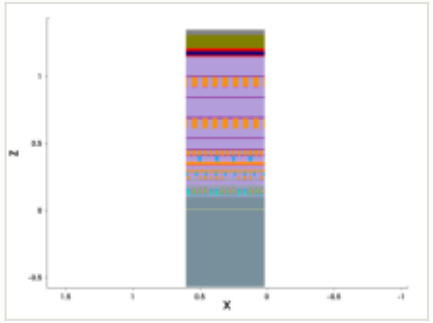


embodied carbon up to each step (kgCO2e per wafer)

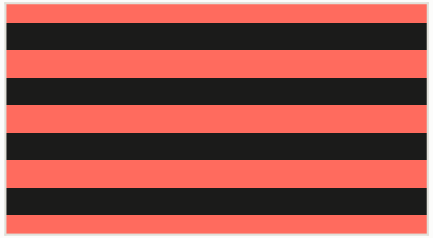
x cut at 0.294 µm



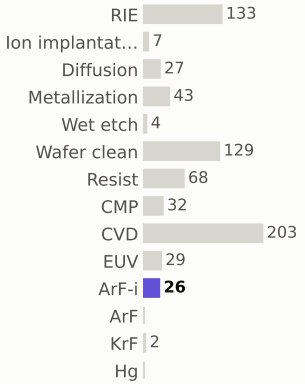
y cut at 0.184 µm



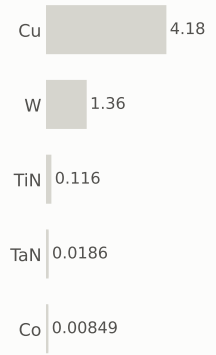
mask: layer 65:0 (GDSII)



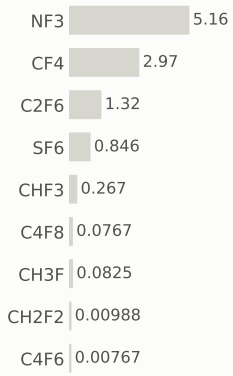
steps by process area



materials deposited (g)



gases emitted (g)



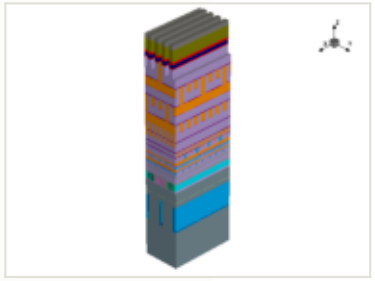
Step #7320: RIE

etch_anisotropic · RIE

1.04	638.8	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

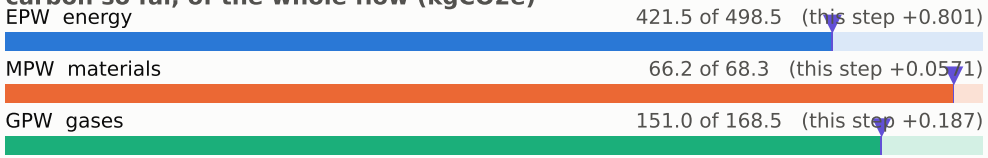


this step

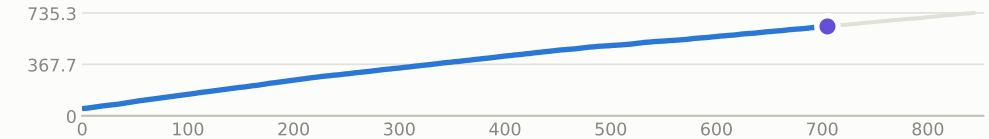
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

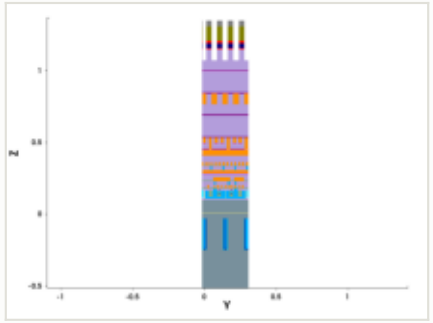
carbon so far, of the whole flow (kgCO2e)



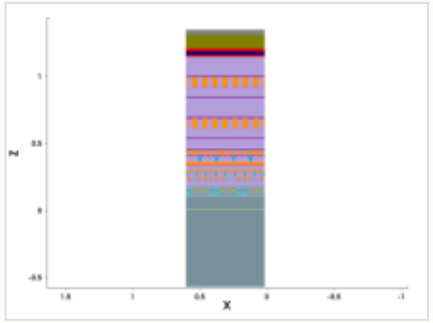
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



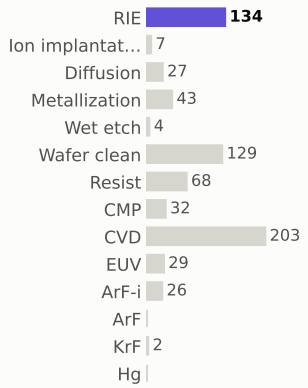
y cut at 0.184 µm



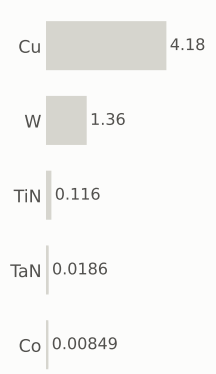
mask for this step

no mask for this step

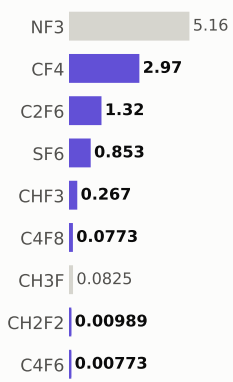
steps by process area



materials deposited (g)



gases emitted (g)



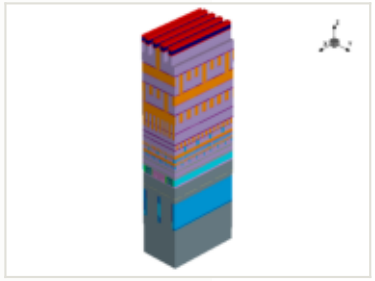
Step #7330: Wafer_clean

etch_strip · Wafer_clean

0.169	638.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

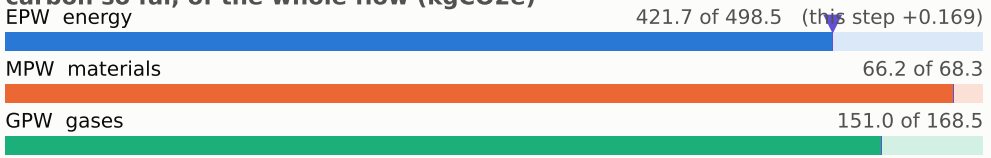


this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

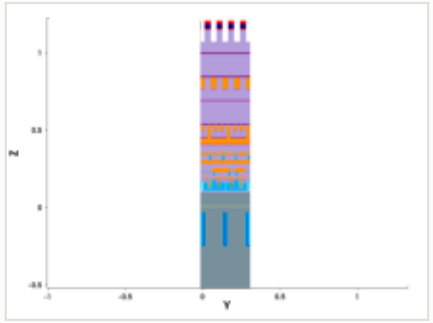
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

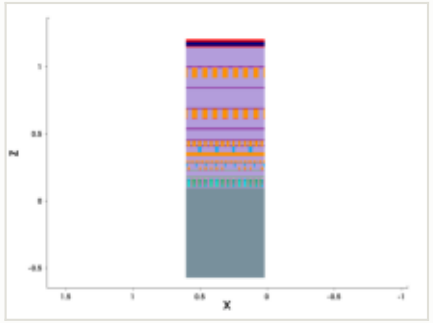


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



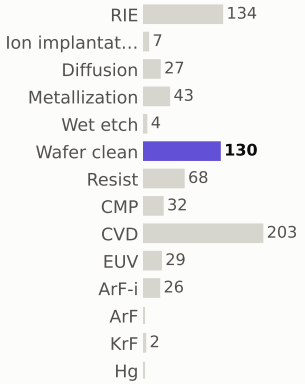
y cut at 0.184 µm



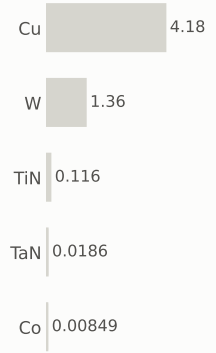
mask for this step

no mask for this step

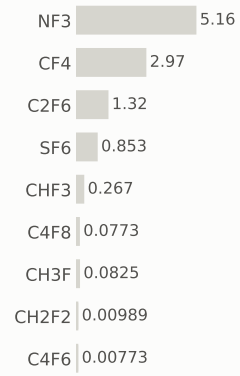
steps by process area



materials deposited (g)



gases emitted (g)



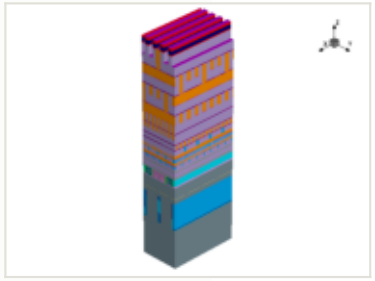
Step #7340: Metallization

depo_ellipsoidal · Metallization

1.2	640.1	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

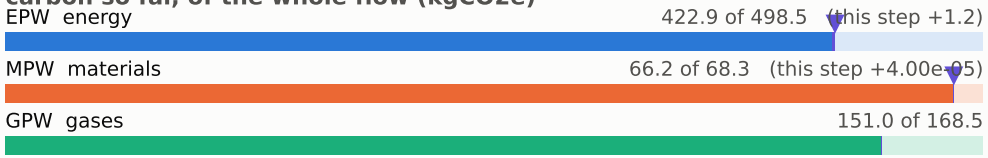


this step

process area	Metallization
material	TaN
thickness	0.0016 µm
material GWP	25.8159 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

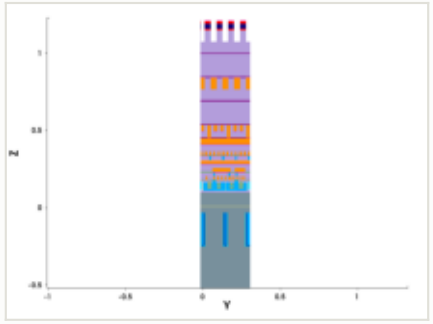
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.00e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

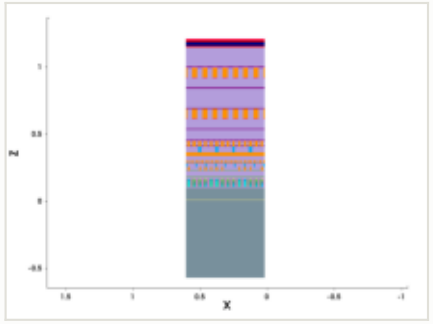


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



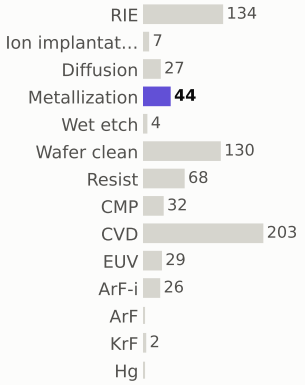
y cut at 0.184 µm



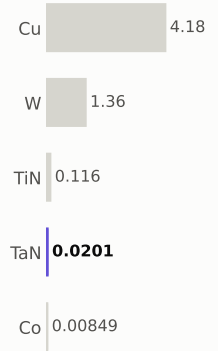
mask for this step

no mask for this step

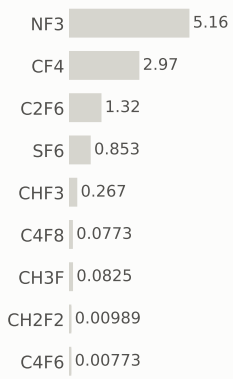
steps by process area



materials deposited (g)



gases emitted (g)



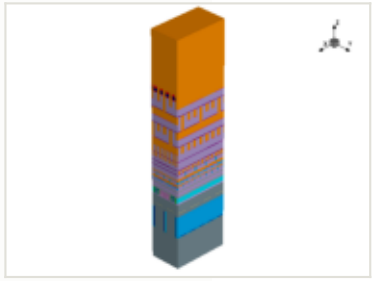
Step #7350: Metallization

depo_fill · Metallization

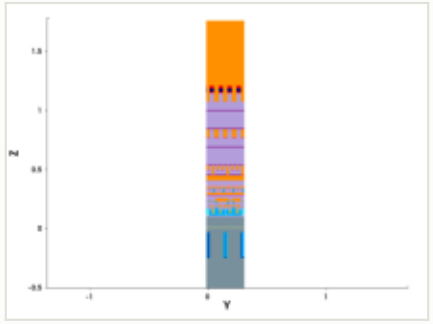
1.21	641.4	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

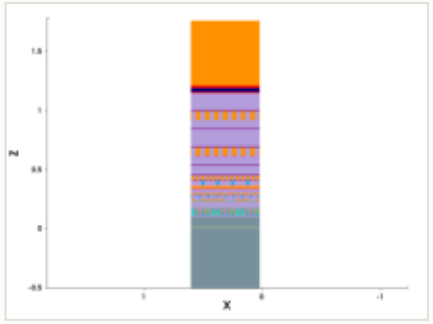
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

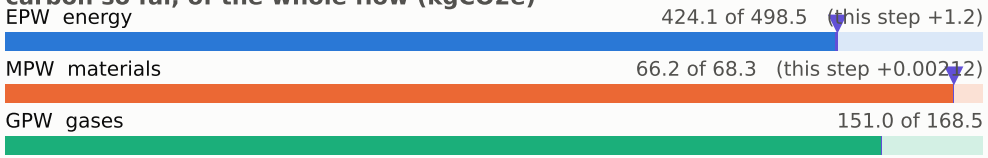
no mask for this step

this step

process area	Metallization
material	Cu
thickness	0.55 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

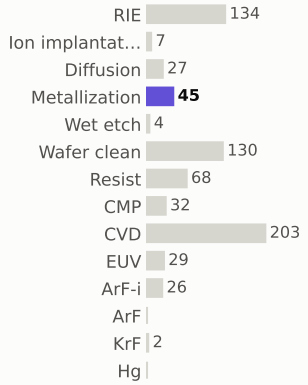
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00212 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

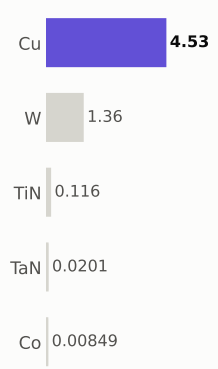


embodied carbon up to each step (kgCO2e per wafer)

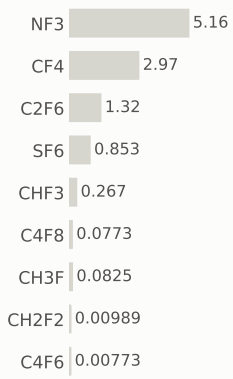
steps by process area



materials deposited (g)



gases emitted (g)



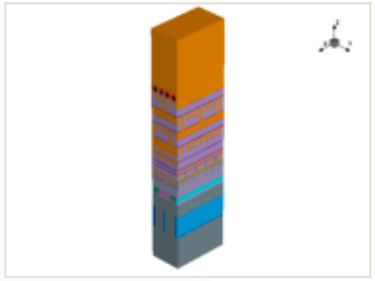
Step #7360: Diffusion

treatment · Diffusion

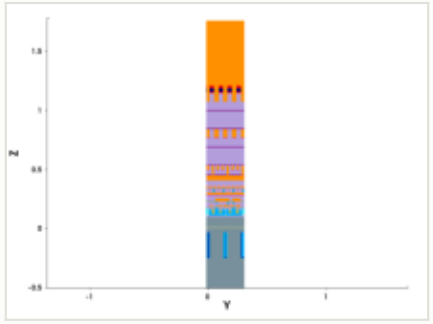
0.801	642.2	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

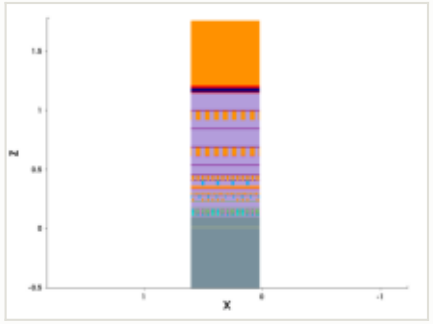
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

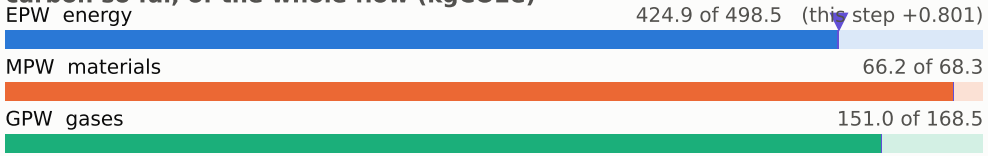
no mask for this step

this step

process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

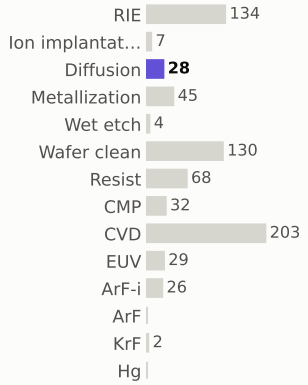
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

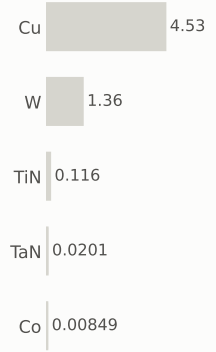


embodied carbon up to each step (kgCO2e per wafer)

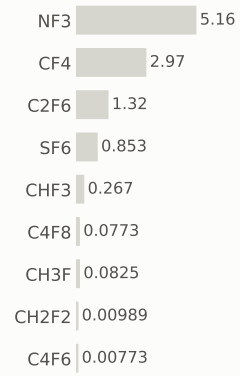
steps by process area



materials deposited (g)



gases emitted (g)



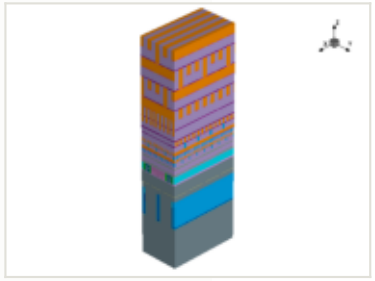
Step #7370: CMP

etch_cmp · CMP

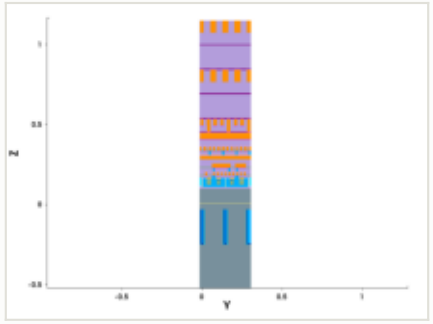
0.169	642.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

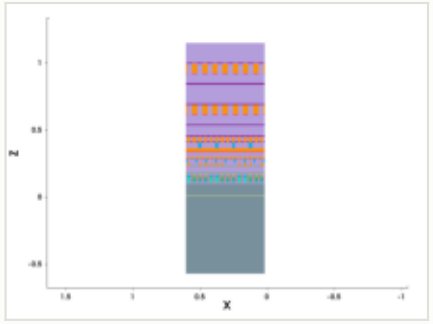
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

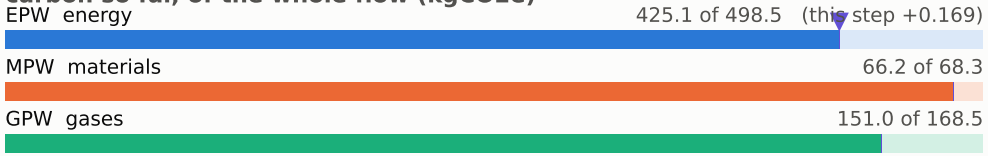
no mask for this step

this step

process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

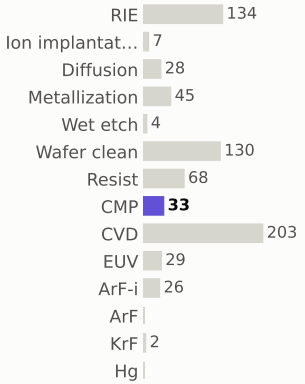
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

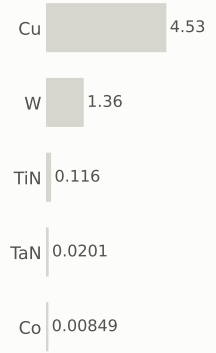


embodied carbon up to each step (kgCO2e per wafer)

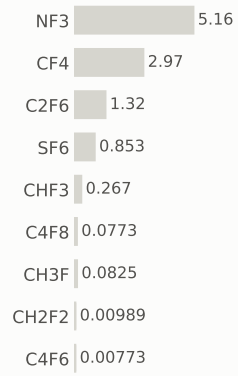
steps by process area



materials deposited (g)



gases emitted (g)



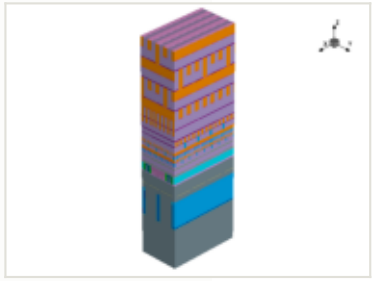
Step #7380: Metallization

depo_isotropic · Metallization

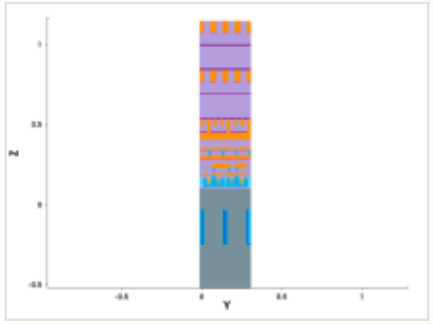
1.2	643.5	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

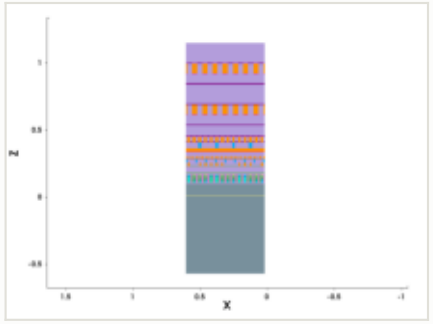
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



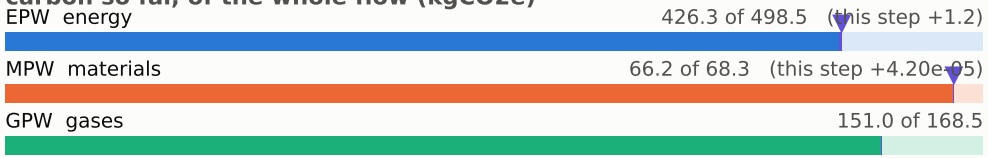
mask for this step

no mask for this step

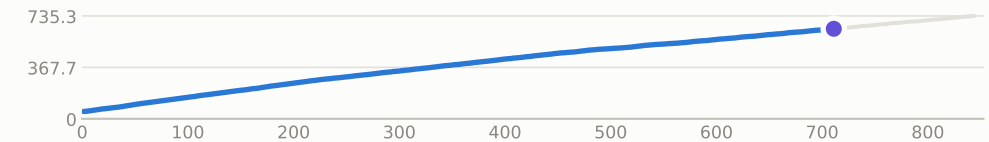
this step

process area	Metallization	EPW	1.2 kgCO2e
material	Co	EPW tool	0.86 kgCO2e
thickness	0.0015 µm	EPW overhead	0.344 kgCO2e
material GWP	44.5 gCO2e/g	MPW	4.20e-05 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	2 kWh + 0.8 overhead	embodied carbon	**1.2 kgCO2e**

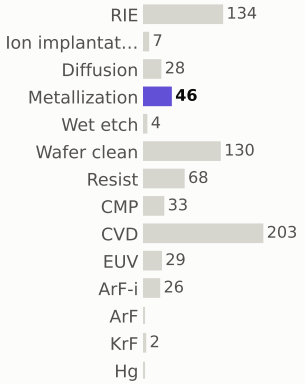
carbon so far, of the whole flow (kgCO2e)



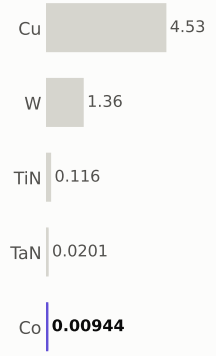
embodied carbon up to each step (kgCO2e per wafer)



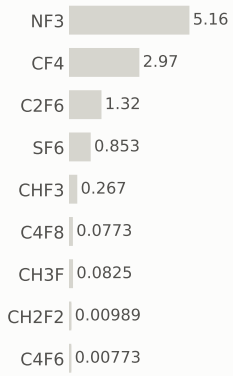
steps by process area



materials deposited (g)



gases emitted (g)



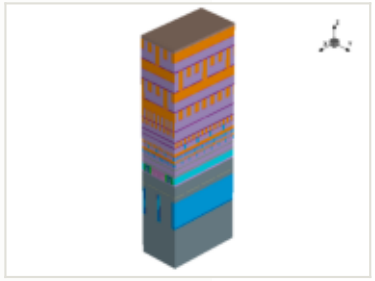
Step #7400: CVD

depo_fill · CVD

1.31	644.8	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

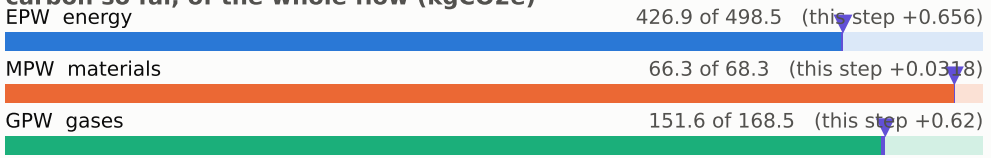


this step

process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

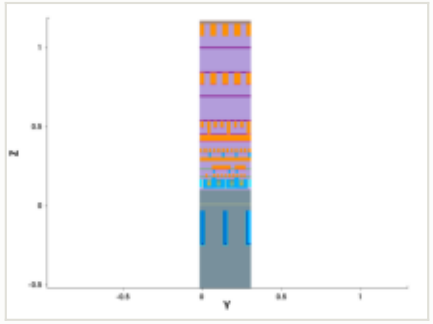
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

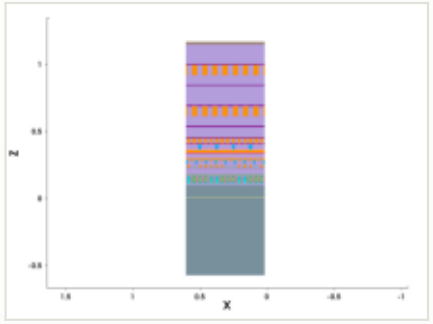


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



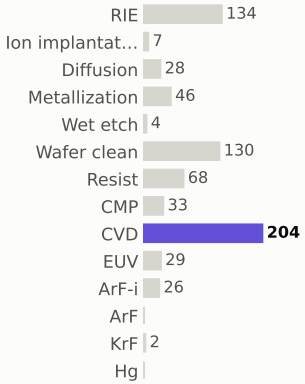
y cut at 0.184 µm



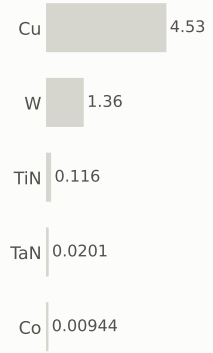
mask for this step

no mask for this step

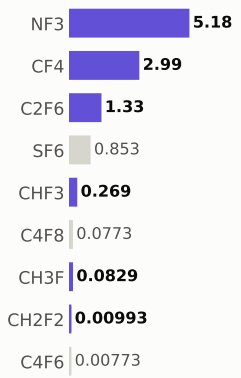
steps by process area



materials deposited (g)



gases emitted (g)



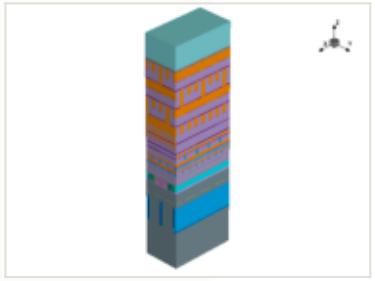
Step #7410: CVD

depo_fill · CVD

1.31	646.1	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

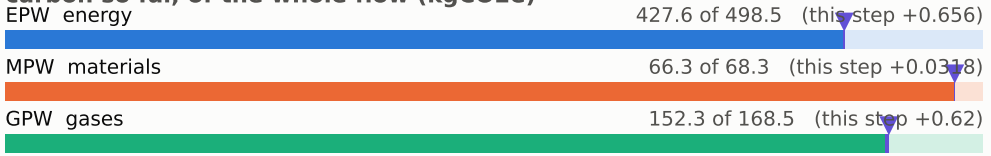


this step

process area	CVD
material	—
thickness	0.204 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

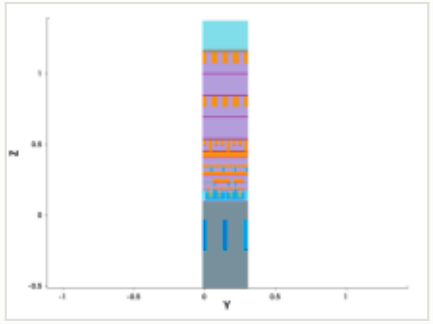
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

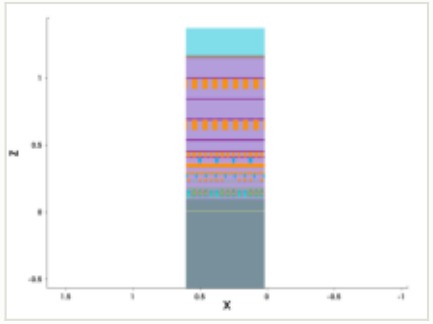


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



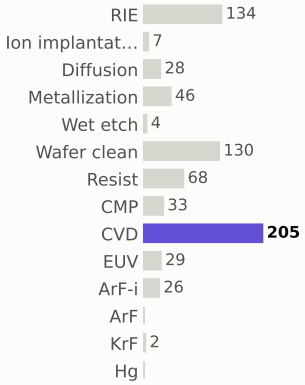
y cut at 0.184 µm



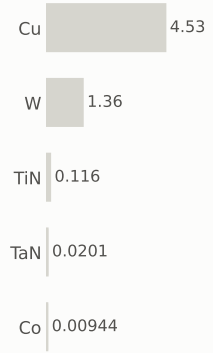
mask for this step

no mask for this step

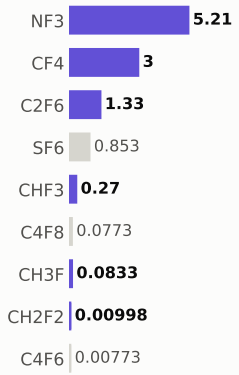
steps by process area



materials deposited (g)



gases emitted (g)



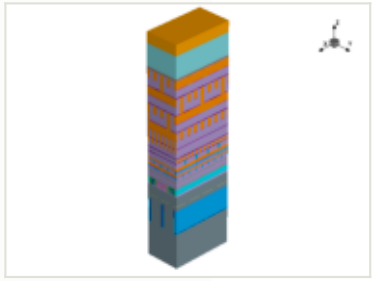
Step #7420: Resist

depo_fill · Resist

0.169	646.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

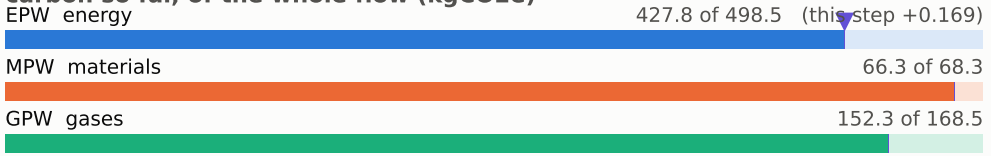


this step

process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

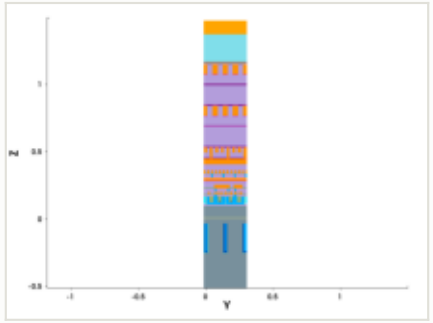
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

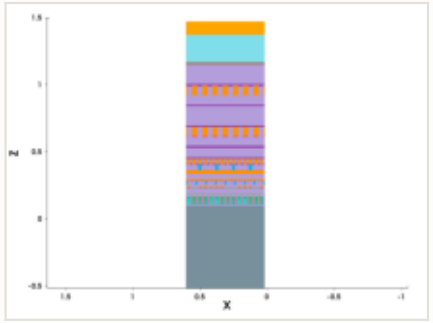


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



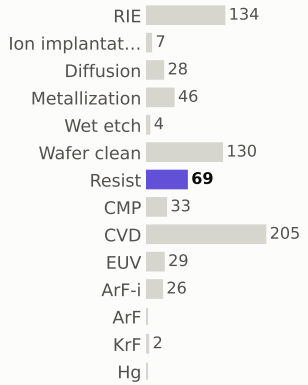
y cut at 0.184 µm



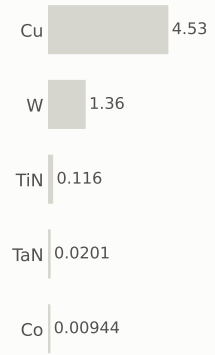
mask for this step

no mask for this step

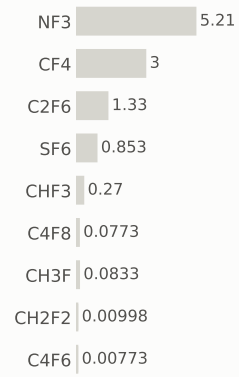
steps by process area



materials deposited (g)



gases emitted (g)



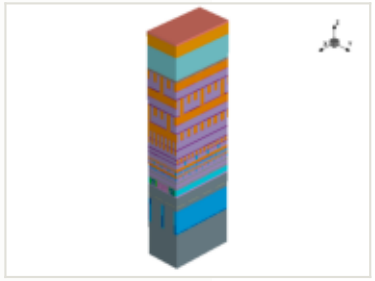
Step #7430: CVD

depo_fill · CVD

1.31	647.6	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

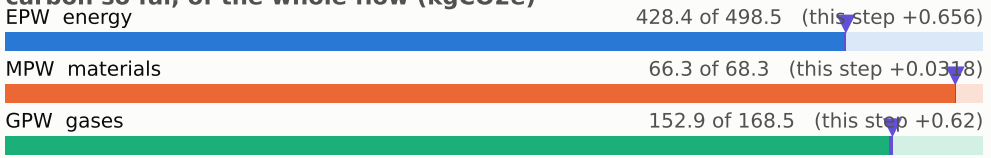


this step

process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

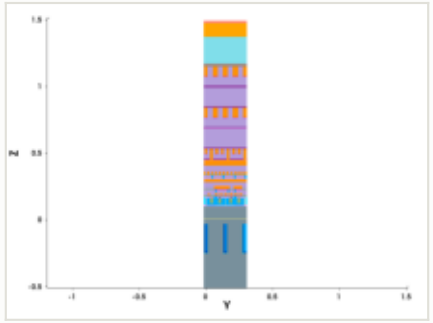
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

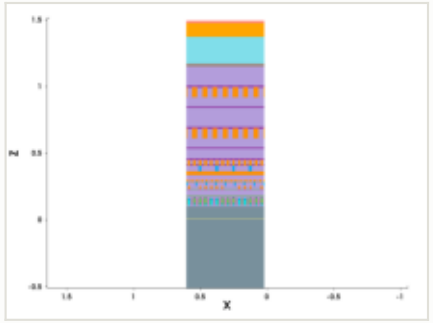


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



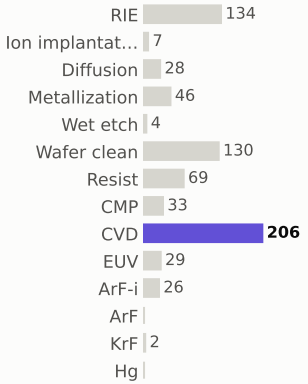
y cut at 0.184 µm



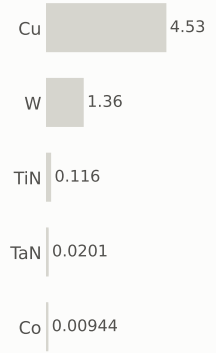
mask for this step

no mask for this step

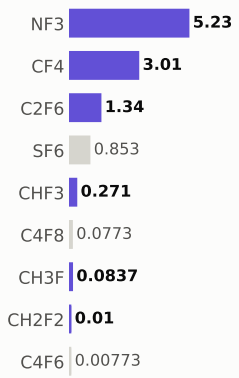
steps by process area



materials deposited (g)



gases emitted (g)



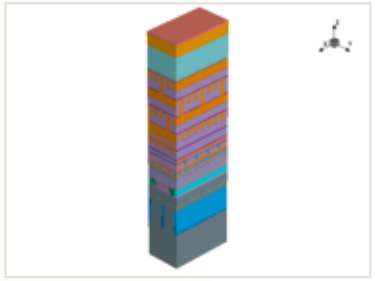
Step #7440: Wafer_clean

clean · Wafer_clean

0.169	647.8	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



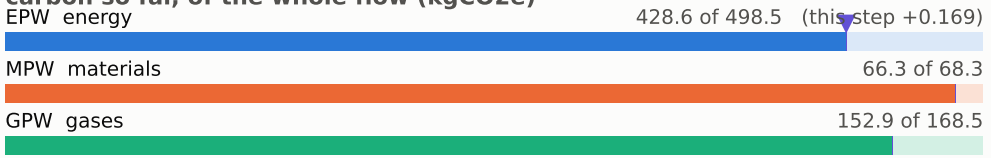
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

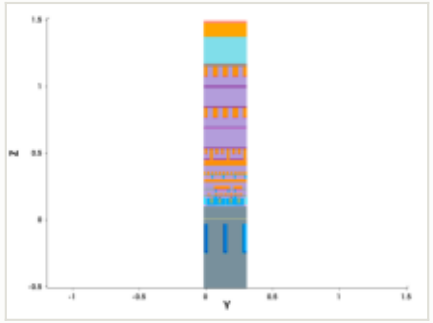
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

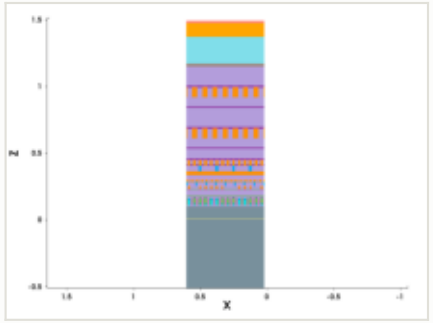


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



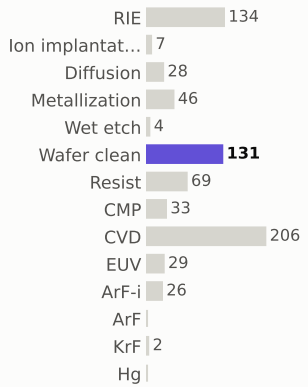
y cut at 0.184 µm



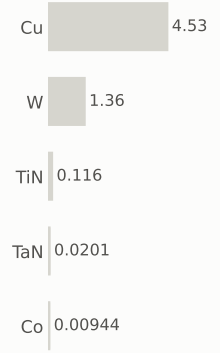
mask for this step

no mask for this step

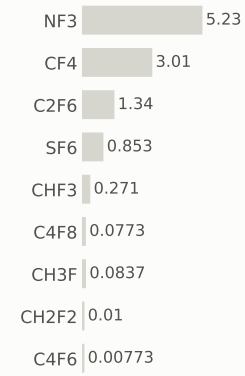
steps by process area



materials deposited (g)



gases emitted (g)



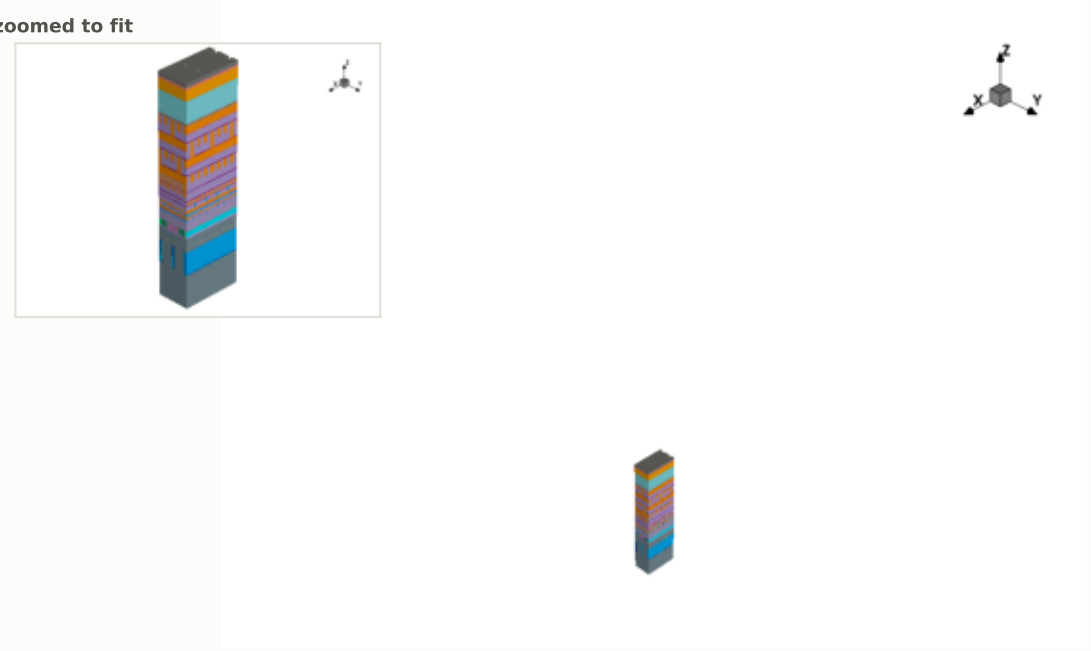
Step #7450: ArF-i

litho_patterning · ArF-i

0.217	648.0	0.029%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

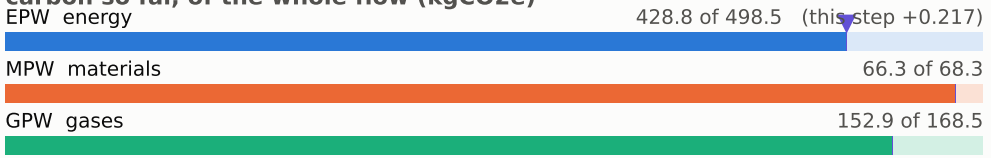
zoomed to fit



this step

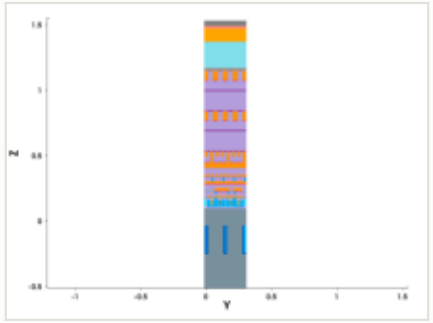
process area	ArF-i	EPW	0.217 kgCO2e
material	—	EPW tool	0.155 kgCO2e
thickness	—	EPW overhead	0.0619 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.36 kWh + 0.144 overhead	embodied carbon	**0.217 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

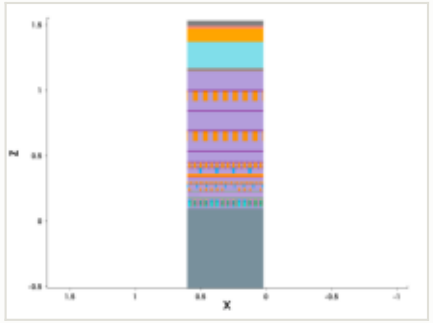


embodied carbon up to each step (kgCO2e per wafer)

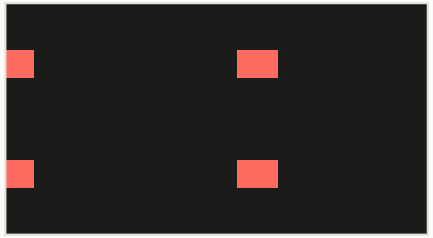
x cut at 0.294 µm



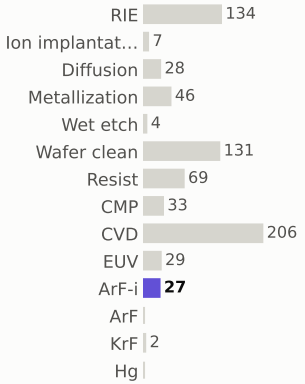
y cut at 0.184 µm



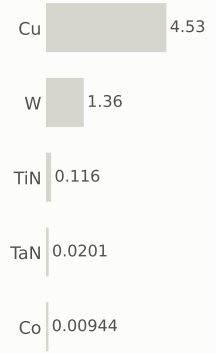
mask: layer 10671:0 (GDSII)



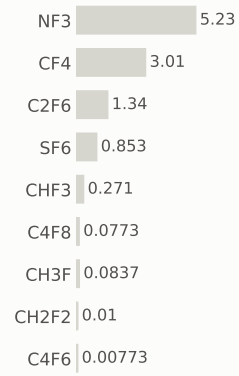
steps by process area



materials deposited (g)



gases emitted (g)



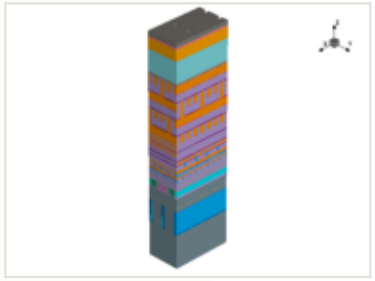
Step #7460: RIE

etch_anisotropic · RIE

1.04	649.1	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

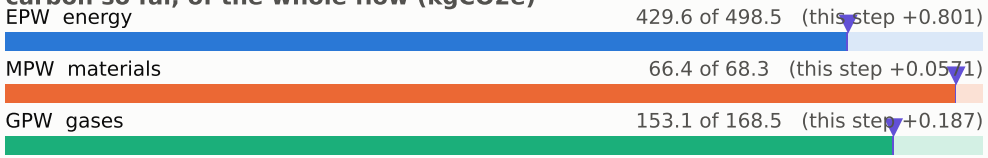


this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

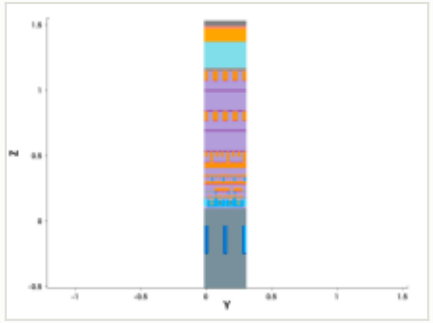
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

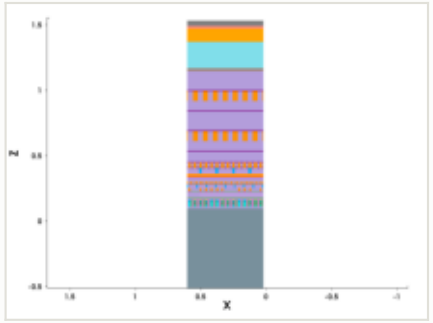


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



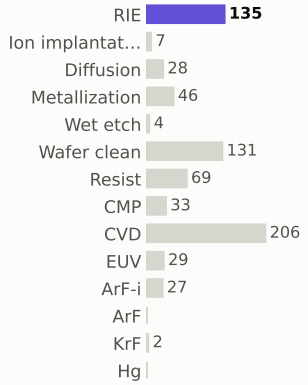
y cut at 0.184 µm



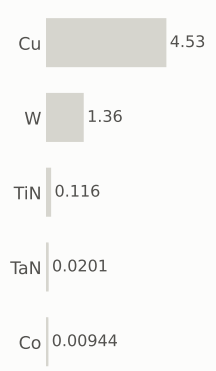
mask for this step

no mask for this step

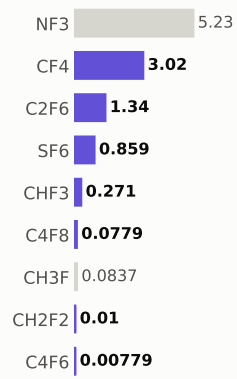
steps by process area



materials deposited (g)



gases emitted (g)



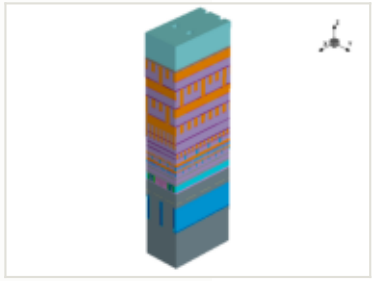
Step #7470: Wafer_clean

etch_strip · Wafer_clean

0.169	649.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

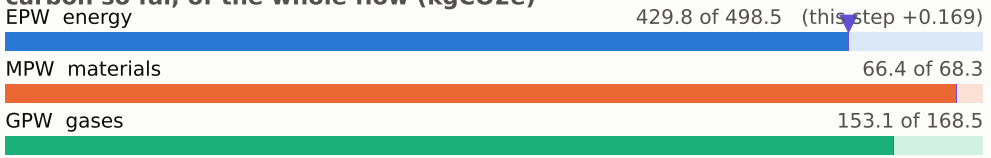


this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

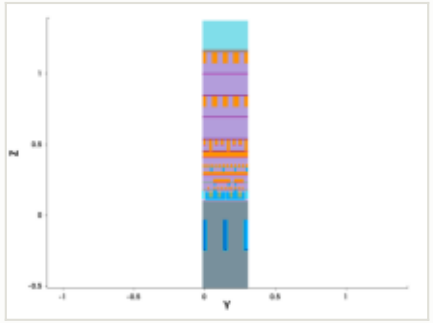
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

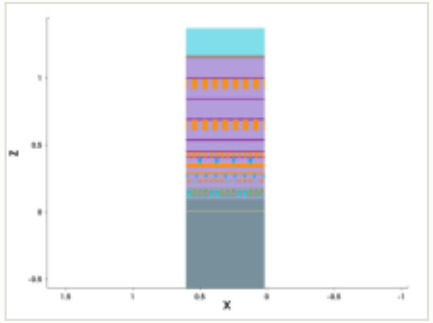


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



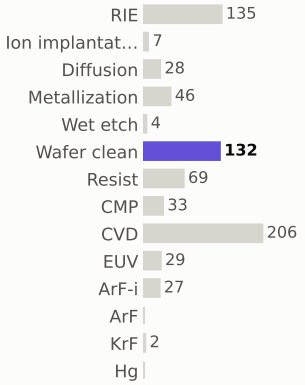
y cut at 0.184 µm



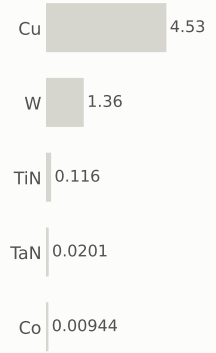
mask for this step

no mask for this step

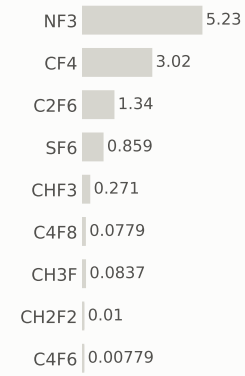
steps by process area



materials deposited (g)



gases emitted (g)



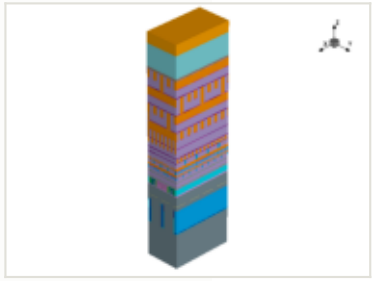
Step #7480: Resist

depo_fill · Resist

0.169	649.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

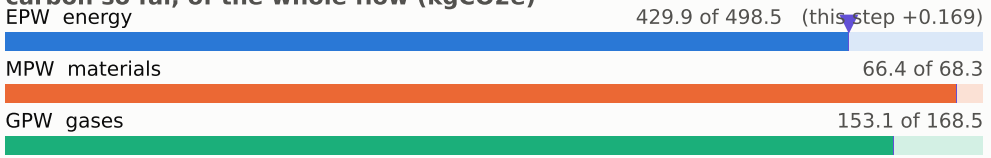


this step

process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

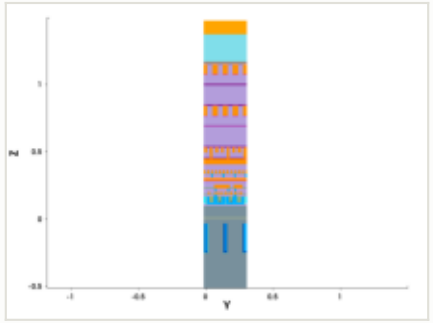
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

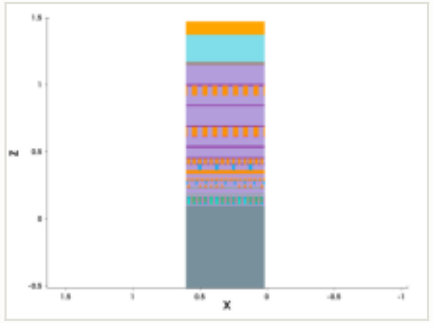


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



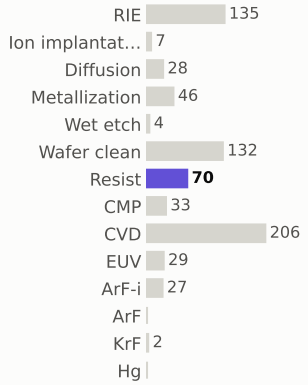
y cut at 0.184 µm



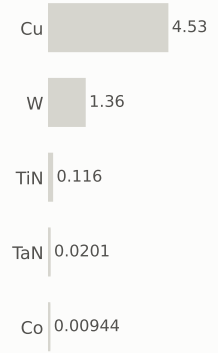
mask for this step

no mask for this step

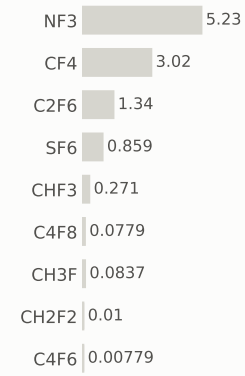
steps by process area



materials deposited (g)



gases emitted (g)



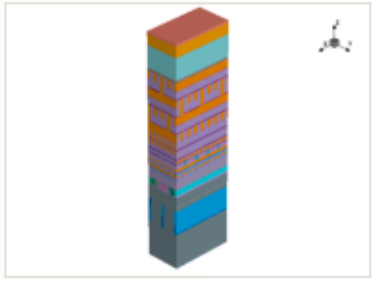
Step #7490: CVD

depo_fill · CVD

1.31	650.7	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

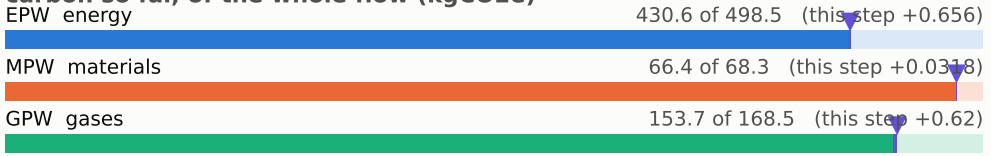


this step

process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

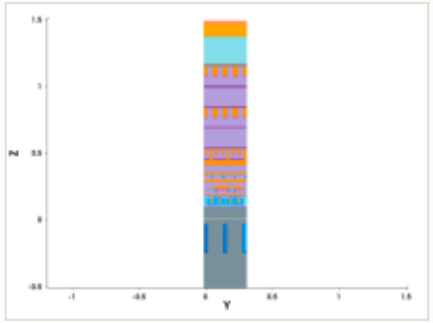
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

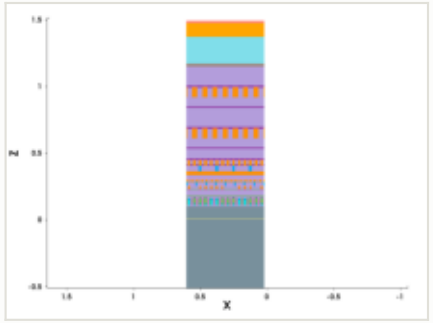


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



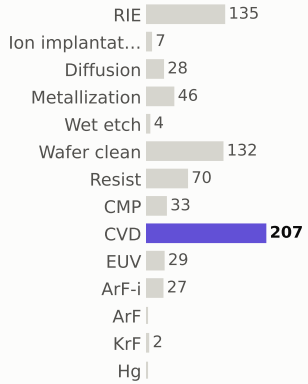
y cut at 0.184 µm



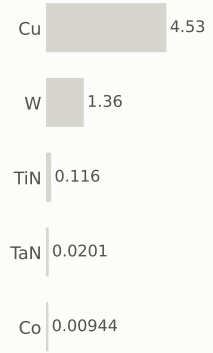
mask for this step

no mask for this step

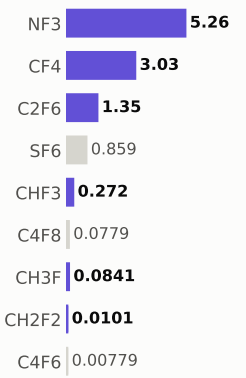
steps by process area



materials deposited (g)



gases emitted (g)



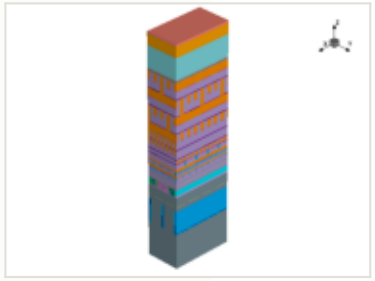
Step #7500: Wafer_clean

clean · Wafer_clean

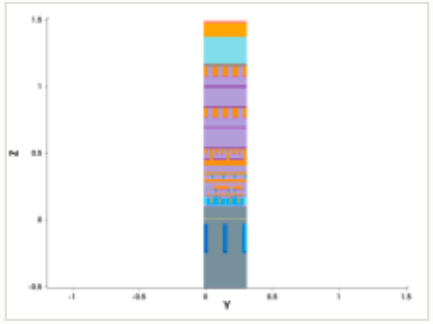
0.169	650.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

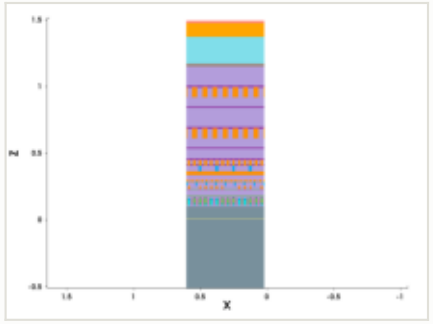
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

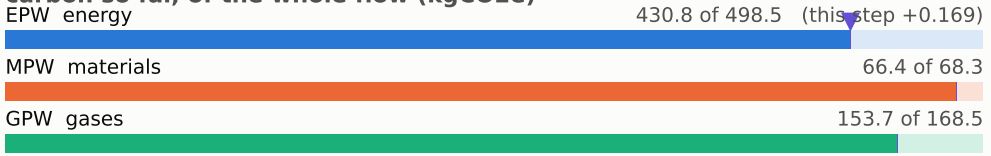
no mask for this step

this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

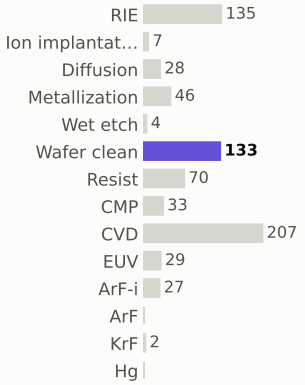
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

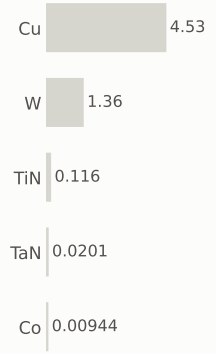


embodied carbon up to each step (kgCO2e per wafer)

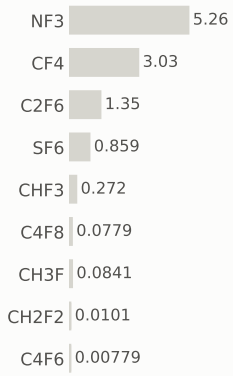
steps by process area



materials deposited (g)



gases emitted (g)



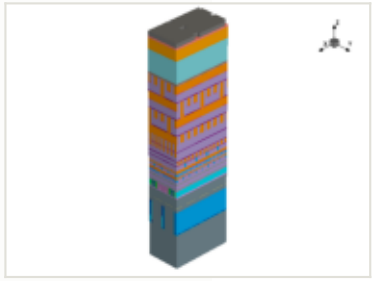
Step #7510: ArF-i

litho_patterning · ArF-i

0.217	651.1	0.029%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

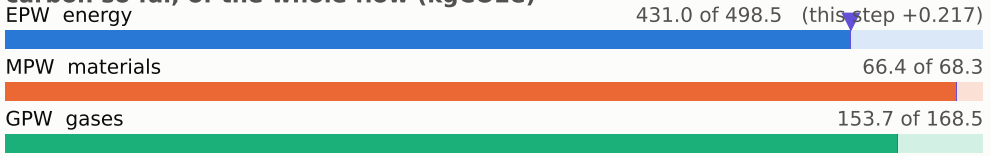


this step

process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

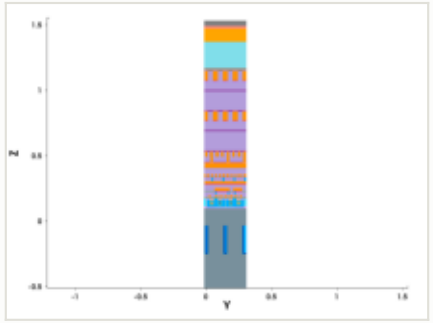
EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

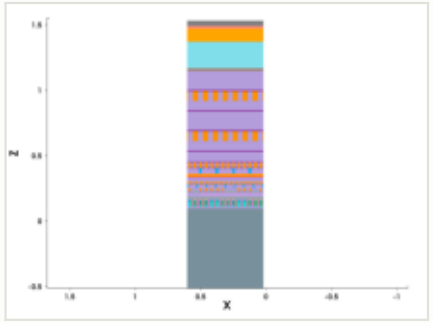


embodied carbon up to each step (kgCO2e per wafer)

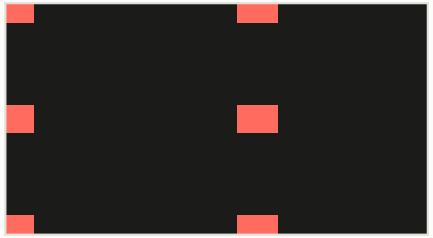
x cut at 0.294 µm



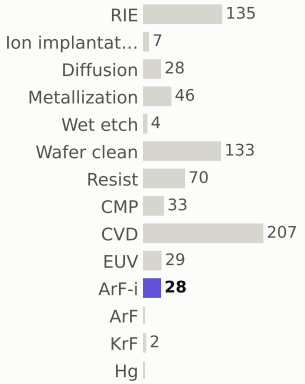
y cut at 0.184 µm



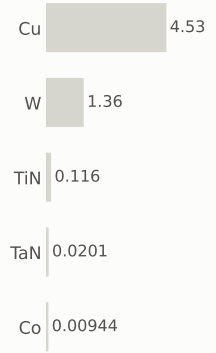
mask: layer 10672:0 (GDSII)



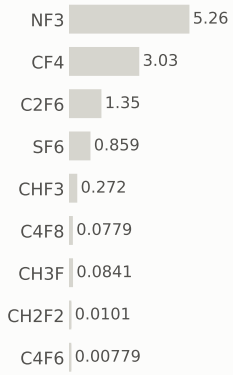
steps by process area



materials deposited (g)



gases emitted (g)



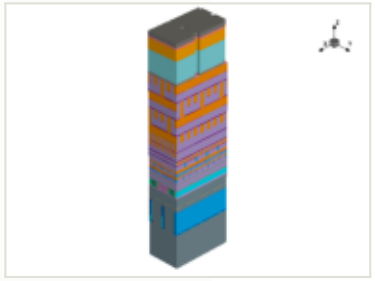
Step #7520: RIE

etch_anisotropic · RIE

1.04	652.1	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

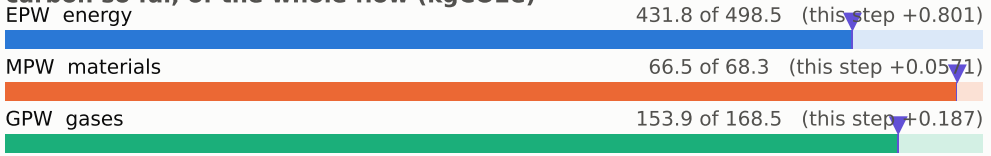


this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

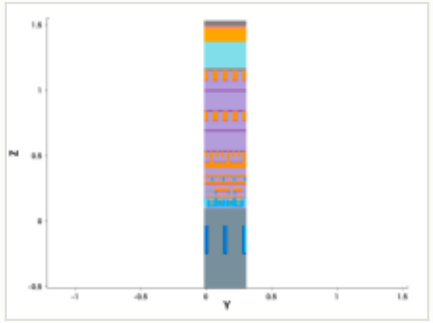
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

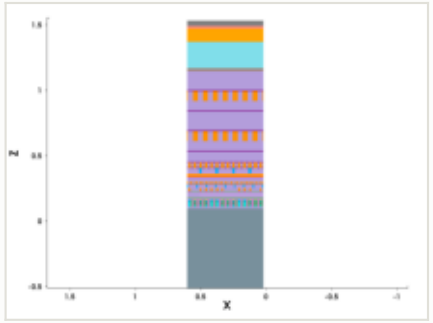


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



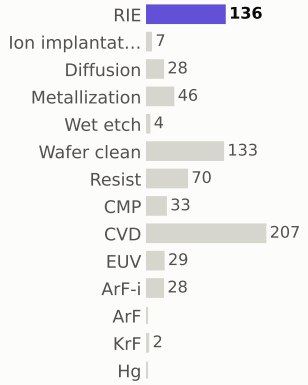
y cut at 0.184 µm



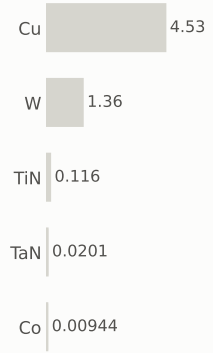
mask for this step

no mask for this step

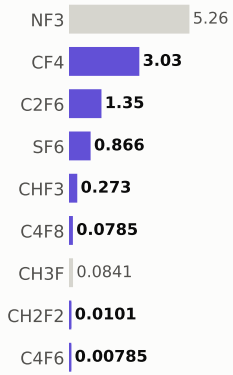
steps by process area



materials deposited (g)



gases emitted (g)



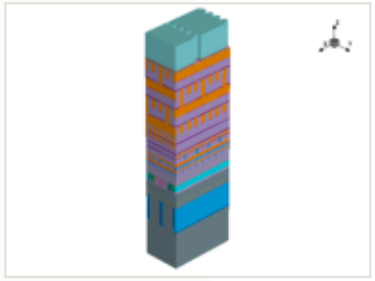
Step #7530: Wafer_clean

etch_strip · Wafer_clean

0.169	652.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

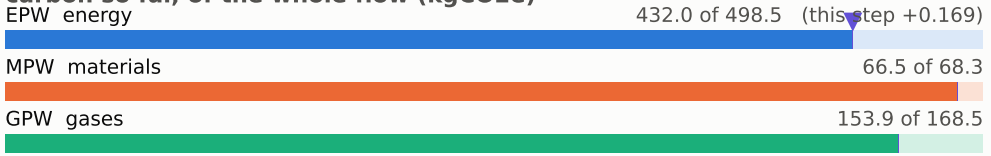


this step

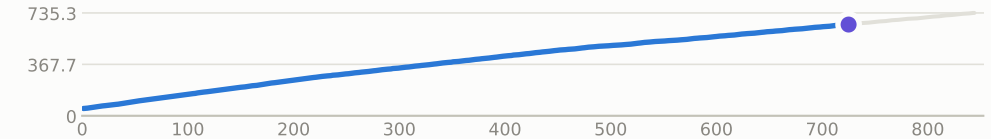
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

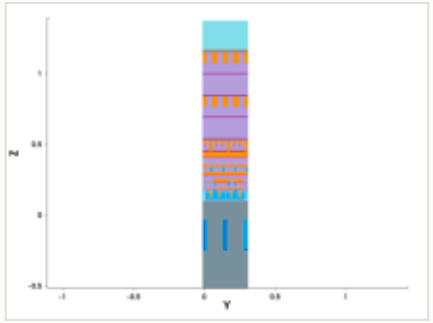
carbon so far, of the whole flow (kgCO2e)



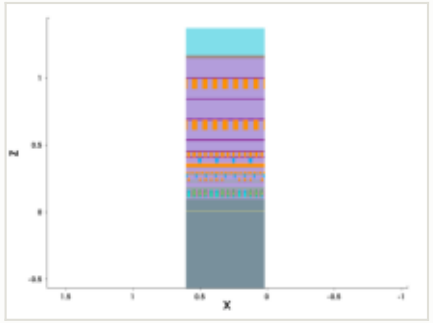
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



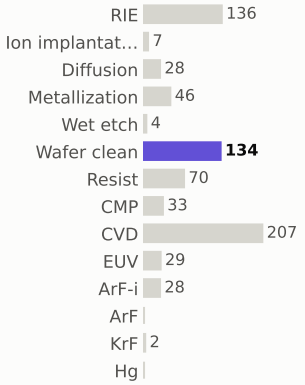
y cut at 0.184 µm



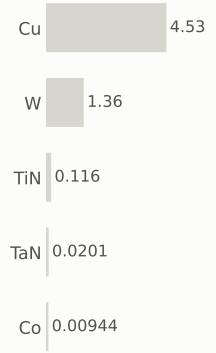
mask for this step

no mask for this step

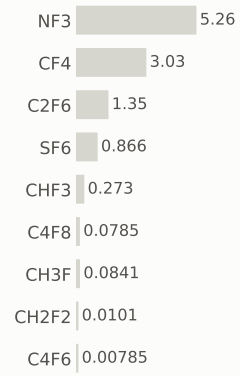
steps by process area



materials deposited (g)



gases emitted (g)



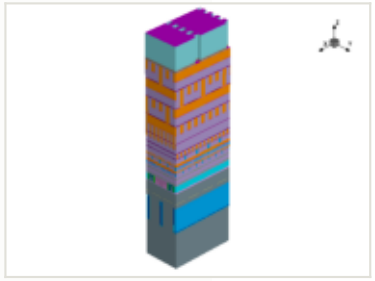
Step #7540: Metallization

depo_ellipsoidal · Metallization

1.2	653.5	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

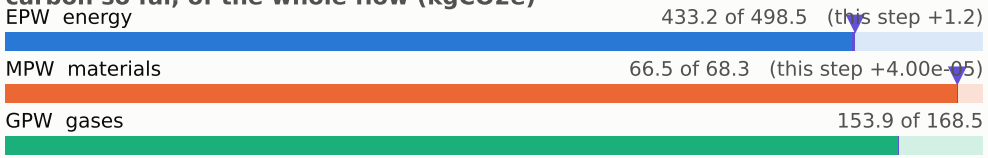


this step

process area	Metallization
material	TaN
thickness	0.0016 µm
material GWP	25.8159 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

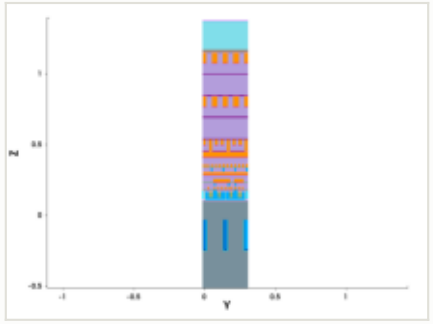
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.00e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

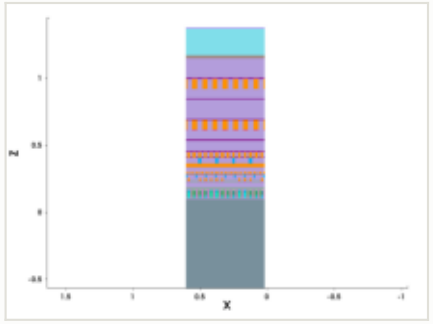


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



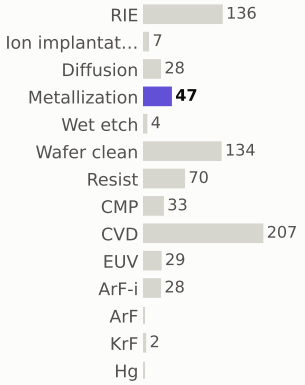
y cut at 0.184 µm



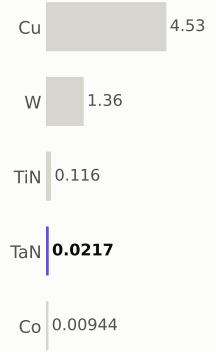
mask for this step

no mask for this step

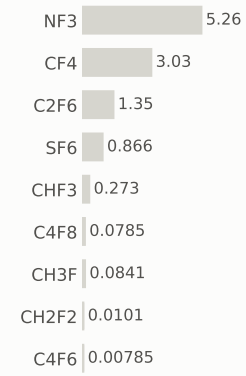
steps by process area



materials deposited (g)



gases emitted (g)



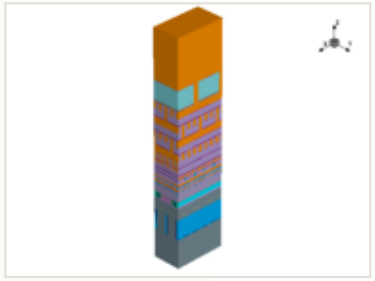
Step #7550: Metallization

depo_fill · Metallization

1.21	654.7	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

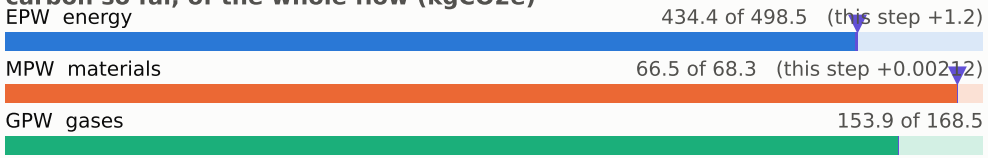


this step

process area	Metallization
material	Cu
thickness	0.55 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

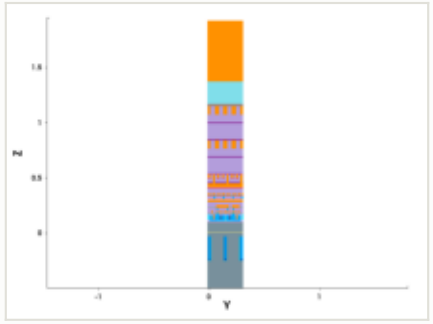
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00212 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

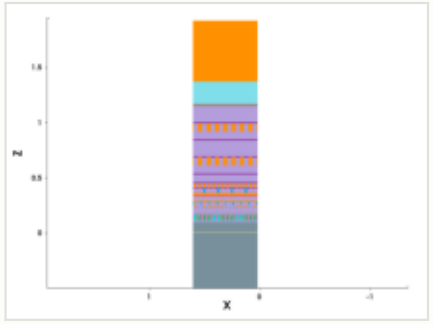


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



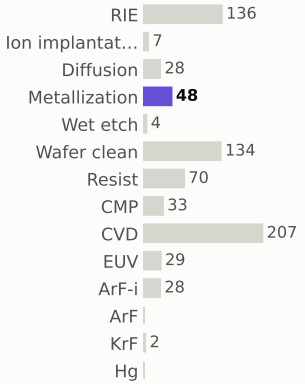
y cut at 0.184 µm



mask for this step

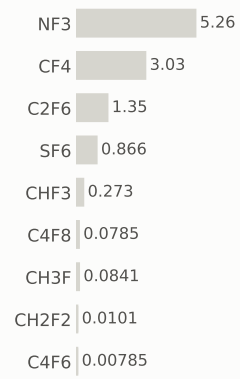
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



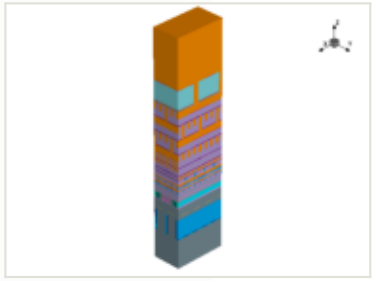
Step #7560: Diffusion

treatment · Diffusion

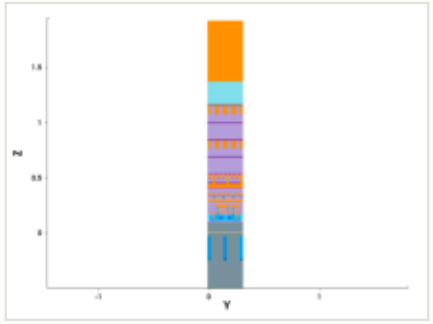
0.801	655.5	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

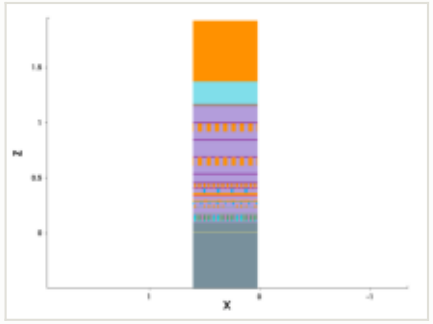
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

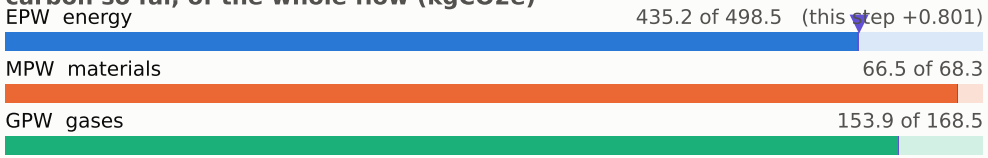
no mask for this step

this step

process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

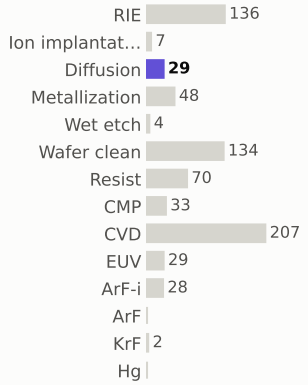
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

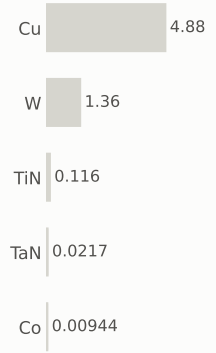


embodied carbon up to each step (kgCO2e per wafer)

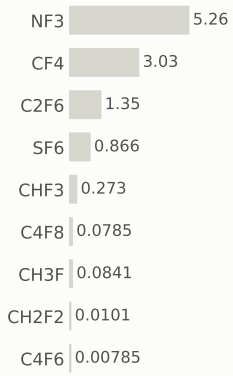
steps by process area



materials deposited (g)



gases emitted (g)



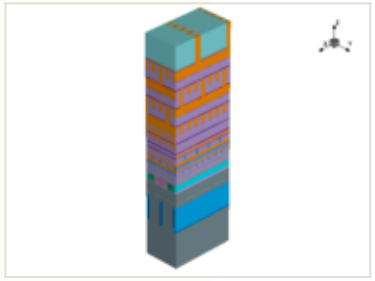
Step #7570: CMP

etch_cmp · CMP

0.169	655.7	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

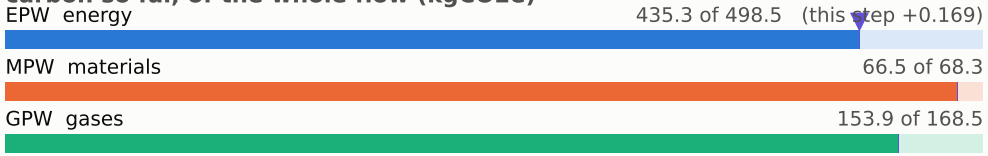


this step

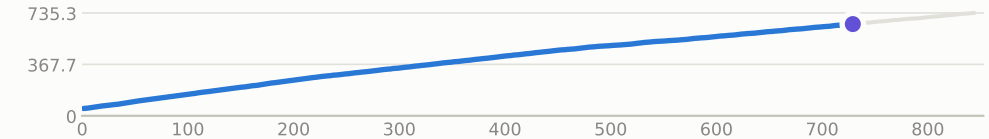
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

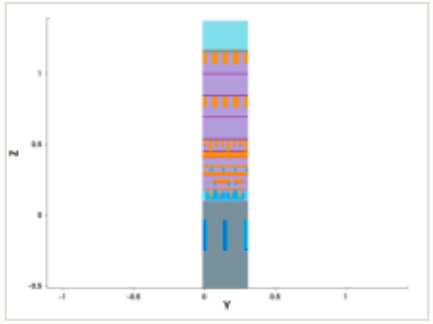
carbon so far, of the whole flow (kgCO2e)



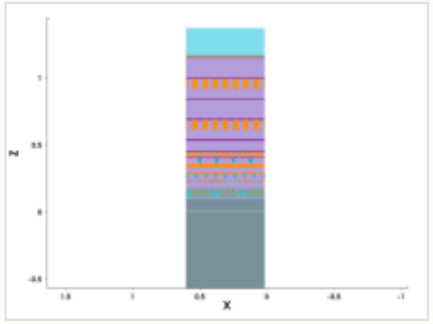
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



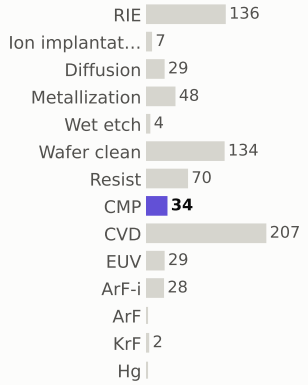
y cut at 0.184 µm



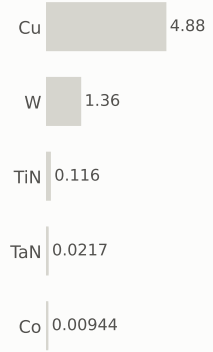
mask for this step

no mask for this step

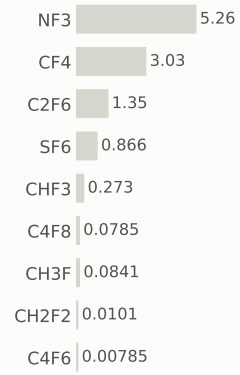
steps by process area



materials deposited (g)



gases emitted (g)



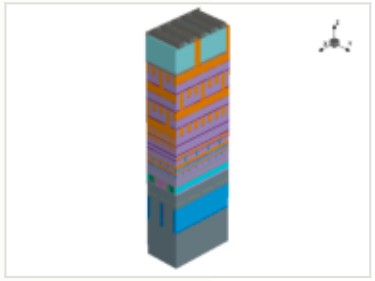
Step #7580: ArF

litho_patterning · ArF

0.199	655.9	0.027%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

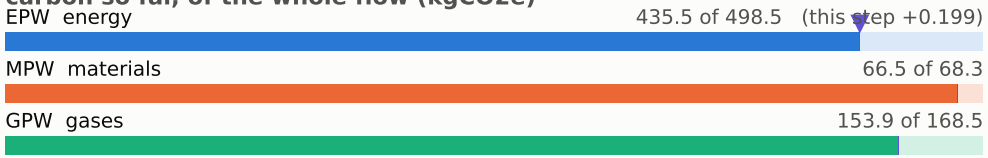


this step

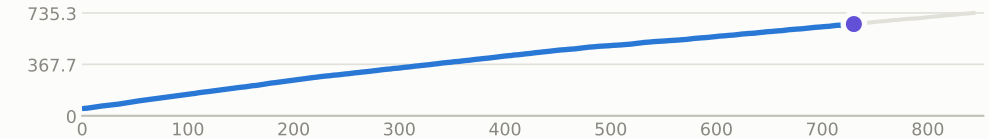
process area	ArF
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

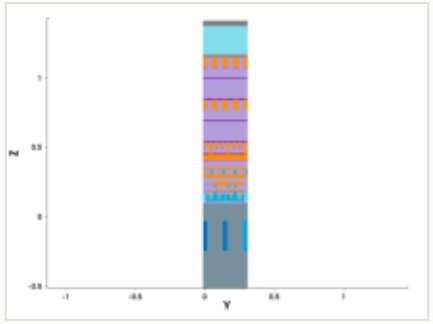
carbon so far, of the whole flow (kgCO2e)



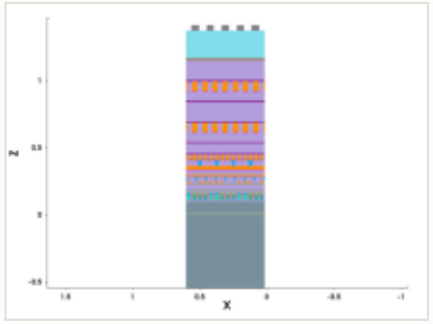
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm

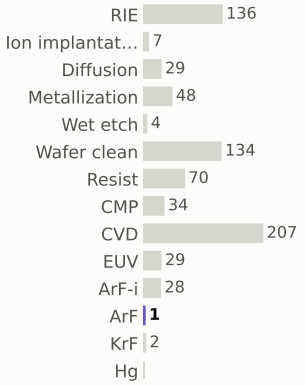


y cut at 0.184 µm

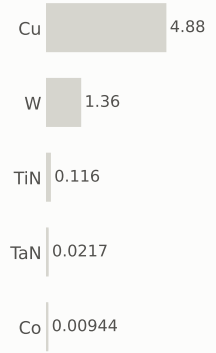


mask: layer 70:0 (GDSII)

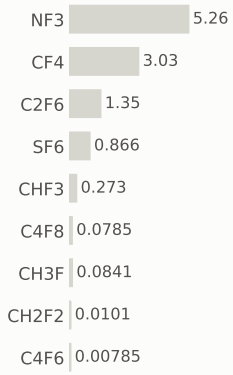
steps by process area



materials deposited (g)



gases emitted (g)



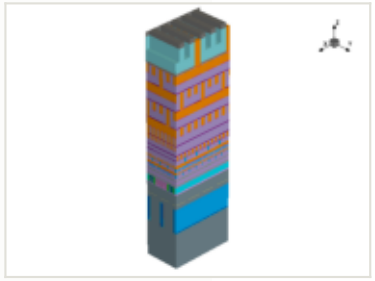
Step #7590: RIE

etch_anisotropic · RIE

1.04	656.9	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

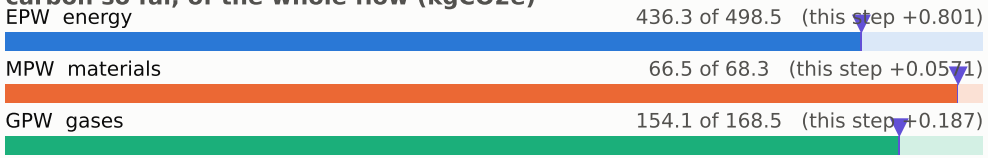


this step

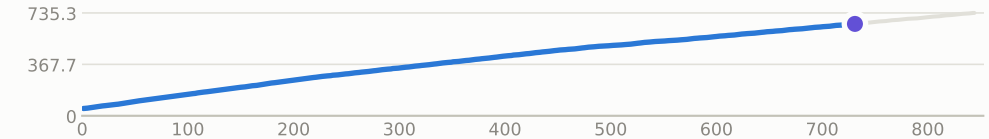
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

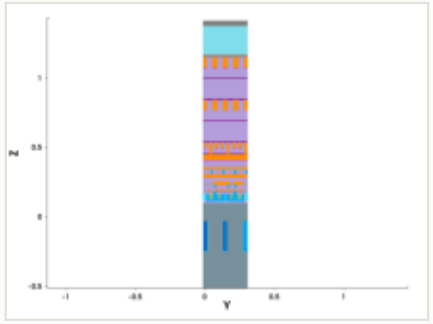
carbon so far, of the whole flow (kgCO2e)



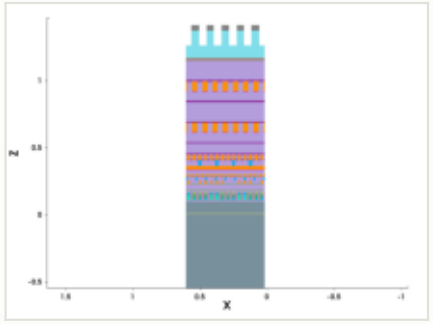
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



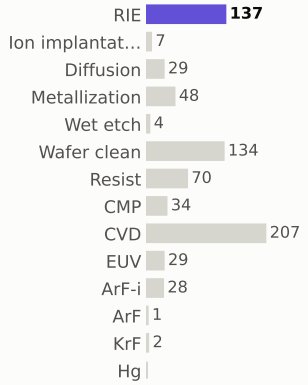
y cut at 0.184 µm



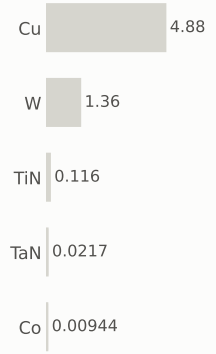
mask for this step

no mask for this step

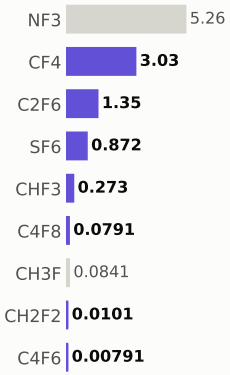
steps by process area



materials deposited (g)



gases emitted (g)



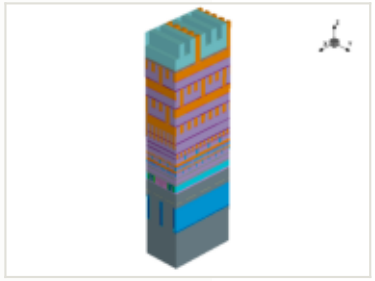
Step #7600: Wafer_clean

etch_strip · Wafer_clean

0.169	657.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

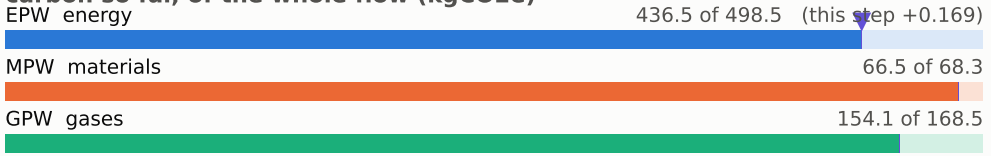


this step

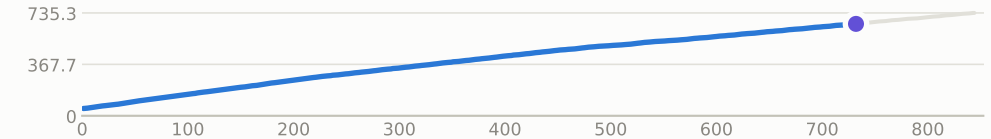
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

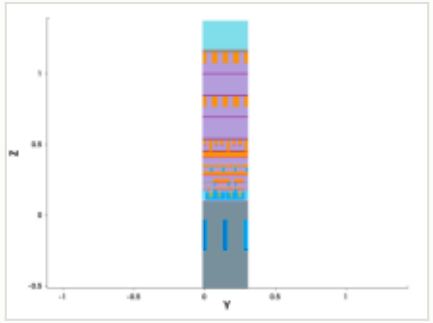
carbon so far, of the whole flow (kgCO2e)



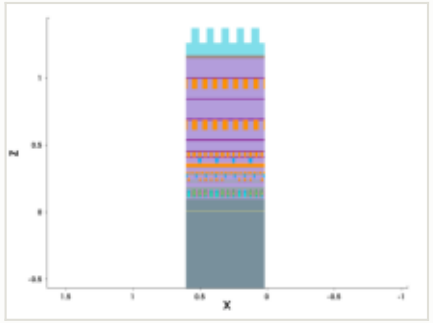
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



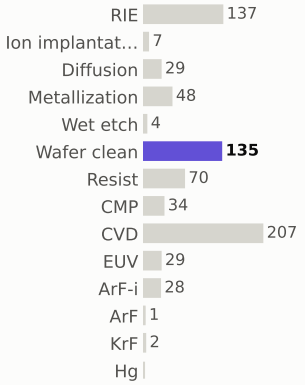
y cut at 0.184 µm



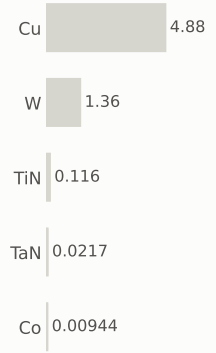
mask for this step

no mask for this step

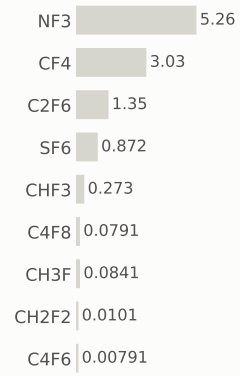
steps by process area



materials deposited (g)



gases emitted (g)



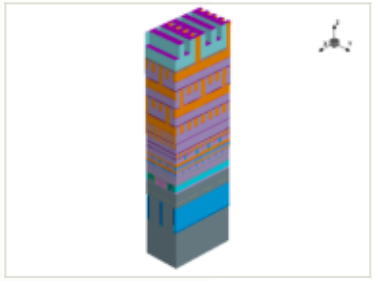
Step #7610: Metallization

depo_ellipsoidal · Metallization

1.2	658.3	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

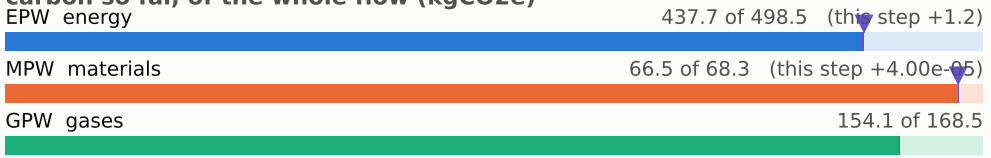


this step

process area	Metallization
material	TaN
thickness	0.0016 µm
material GWP	25.8159 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

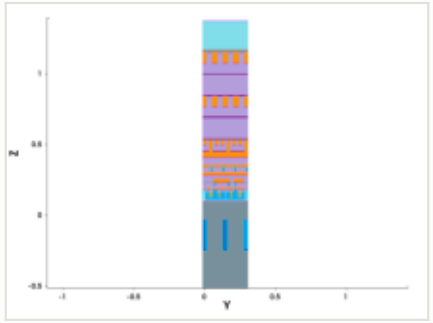
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.00e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

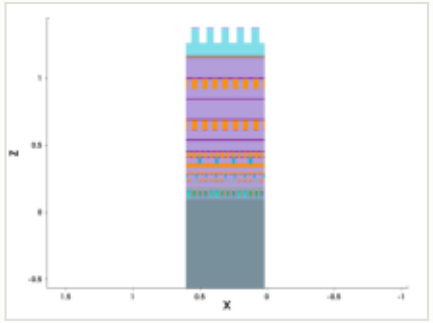


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



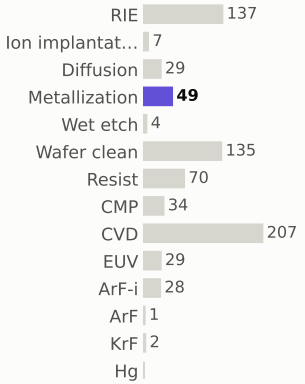
y cut at 0.184 µm



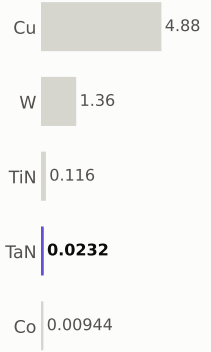
mask for this step

no mask for this step

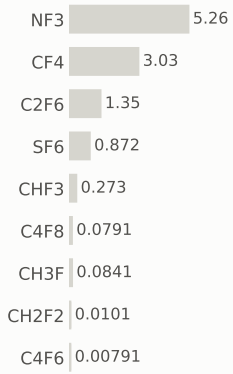
steps by process area



materials deposited (g)



gases emitted (g)



Step #7620: Metallization

depo_fill · Metallization

1.21	659.5	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

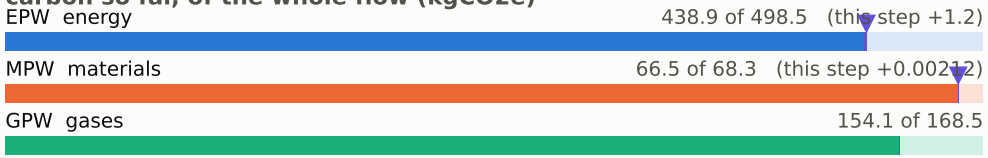
zoomed to fit

this step

process area	Metallization
material	Cu
thickness	0.55 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

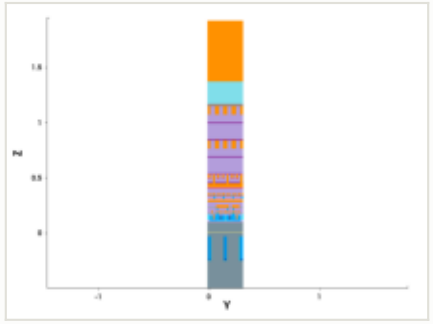
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00212 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

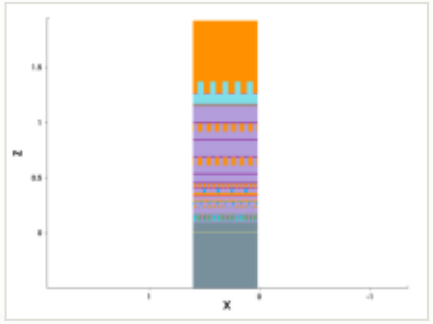


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



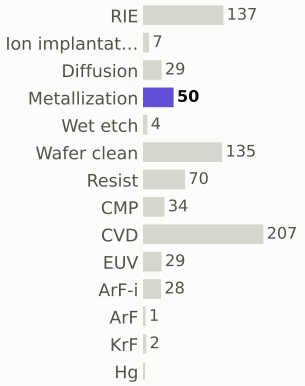
y cut at 0.184 µm



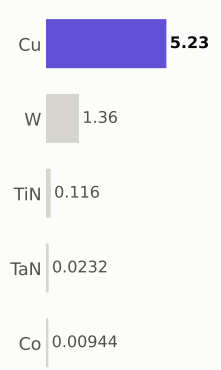
mask for this step

no mask for this step

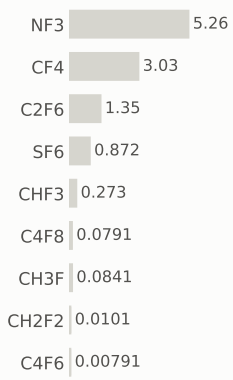
steps by process area



materials deposited (g)



gases emitted (g)



Step #7630: Diffusion

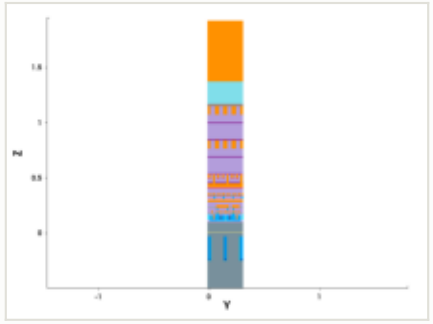
treatment · Diffusion

0.801	660.3	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

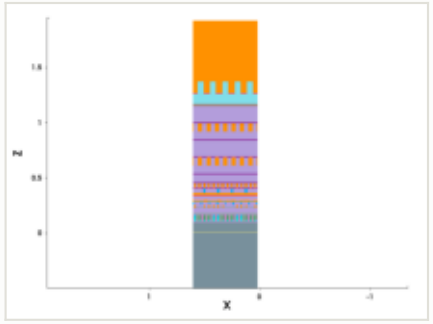
3D structure after this step

zoomed to fit

x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

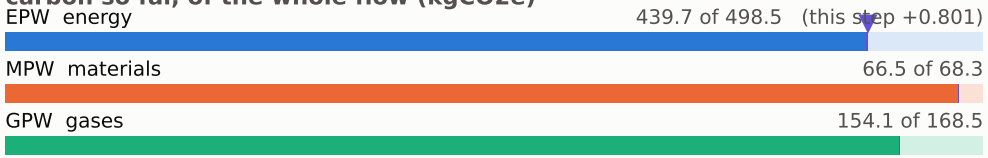
no mask for this step

this step

process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

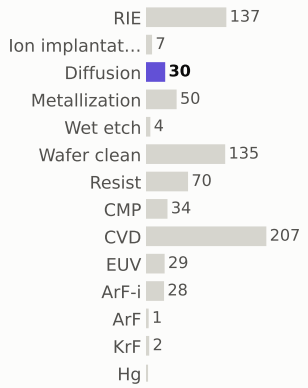
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

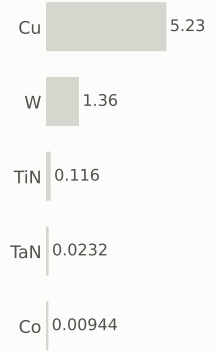


embodied carbon up to each step (kgCO2e per wafer)

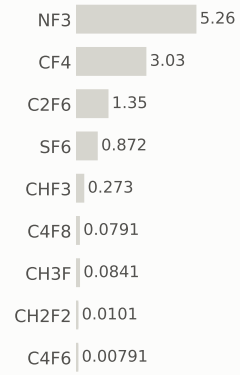
steps by process area



materials deposited (g)



gases emitted (g)



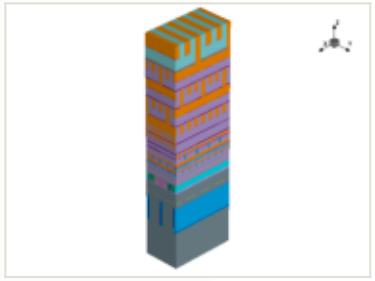
Step #7640: CMP

etch_cmp · CMP

0.169	660.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

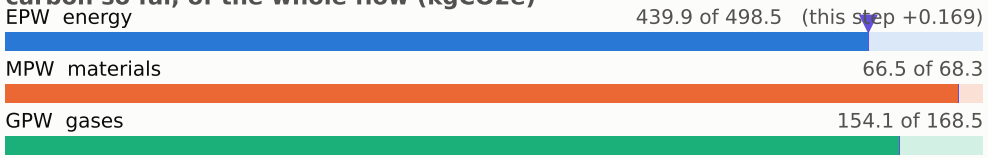


this step

process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

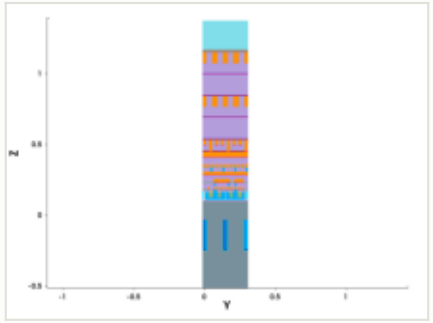
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

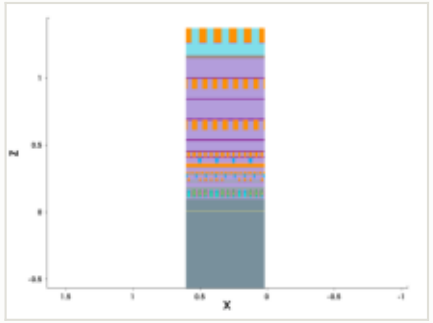


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



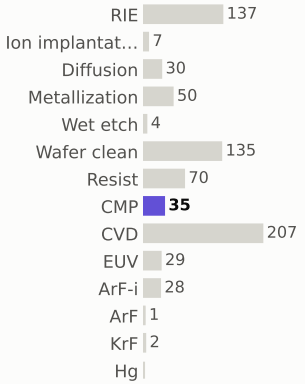
y cut at 0.184 µm



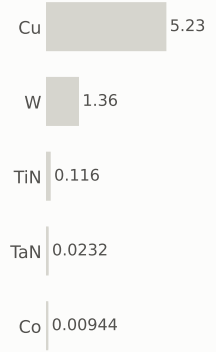
mask for this step

no mask for this step

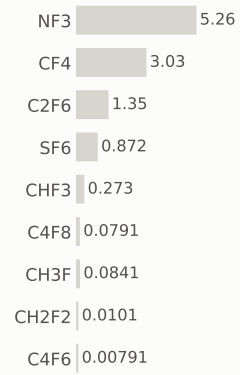
steps by process area



materials deposited (g)



gases emitted (g)



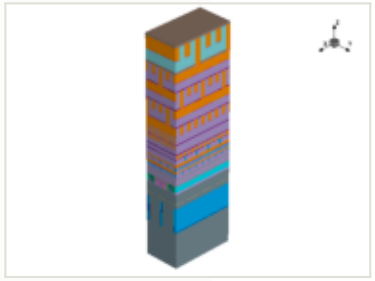
Step #7660: CVD

depo_fill · CVD

1.31	661.8	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

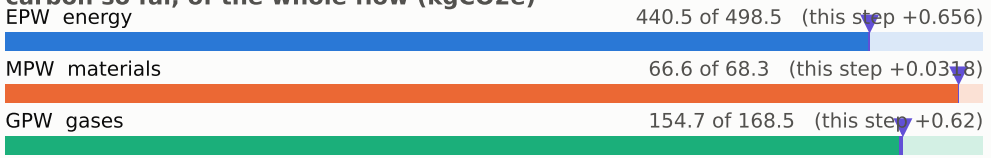


this step

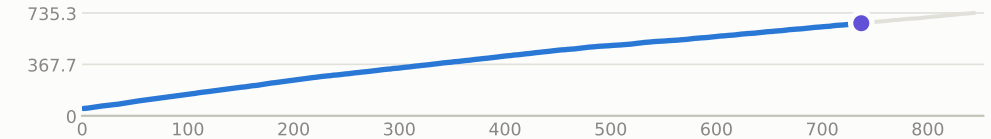
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

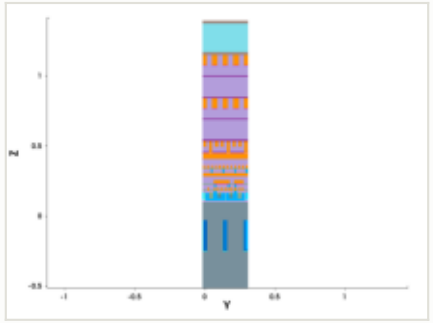
carbon so far, of the whole flow (kgCO2e)



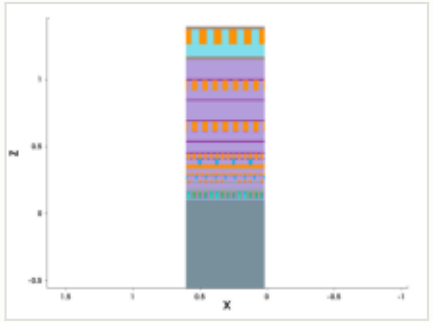
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



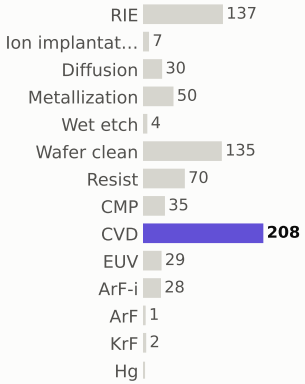
y cut at 0.184 µm



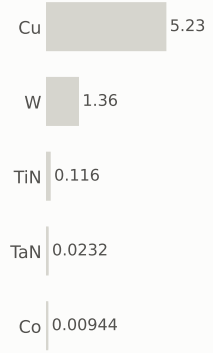
mask for this step

no mask for this step

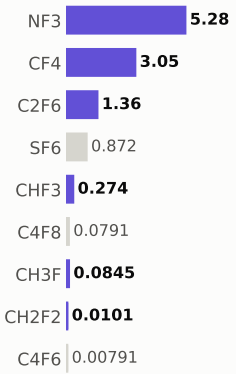
steps by process area



materials deposited (g)



gases emitted (g)



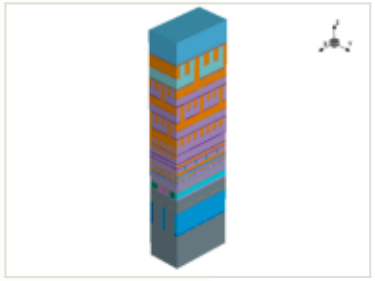
Step #7670: CVD

depo_fill · CVD

1.31	663.1	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

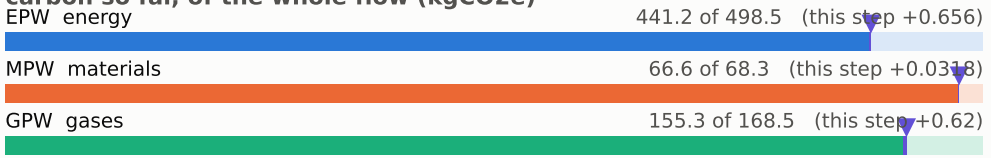


this step

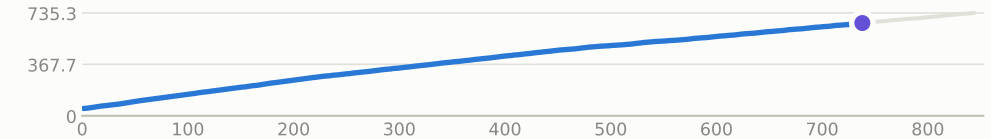
process area	CVD
material	—
thickness	0.204 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

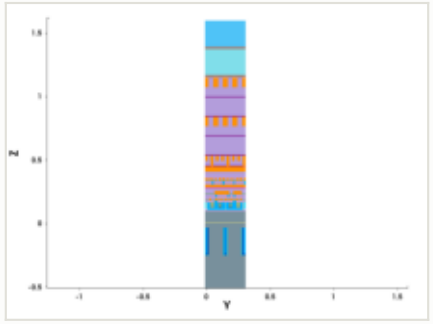
carbon so far, of the whole flow (kgCO2e)



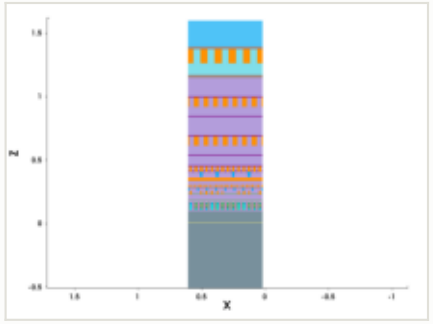
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



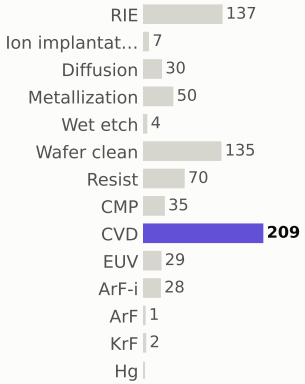
y cut at 0.184 µm



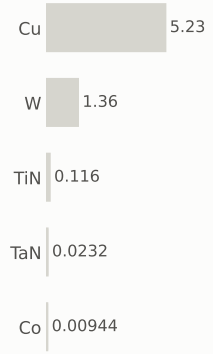
mask for this step

no mask for this step

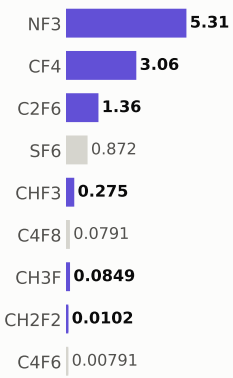
steps by process area



materials deposited (g)



gases emitted (g)



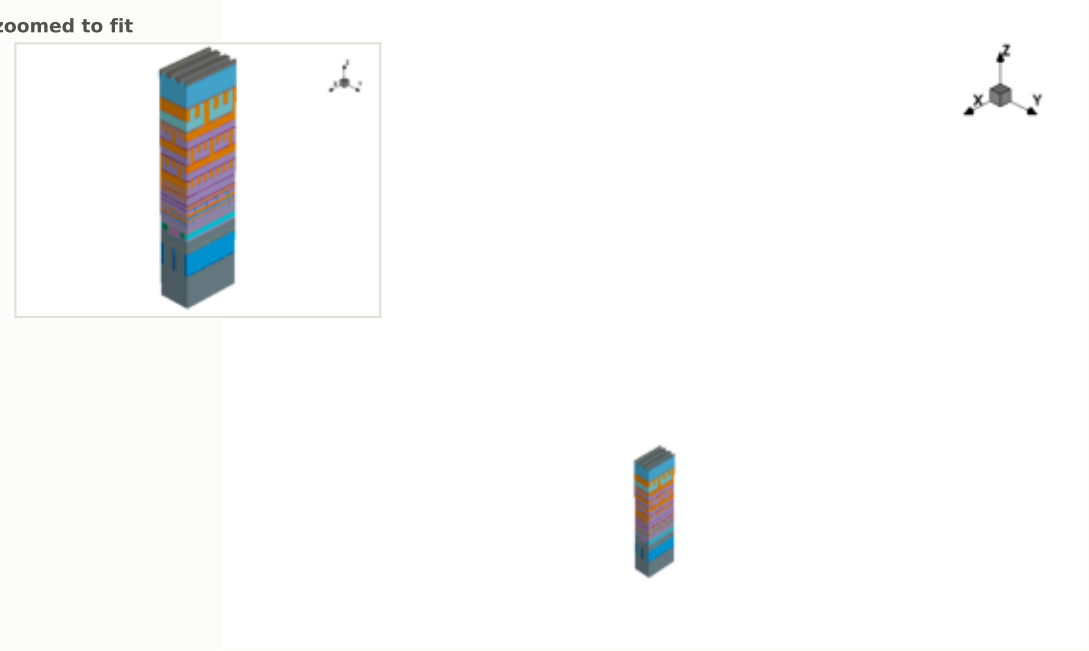
Step #7680: ArF

litho_patterning · ArF

0.199	663.3	0.027%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

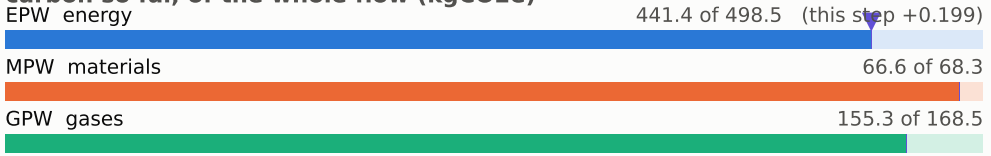
zoomed to fit



this step

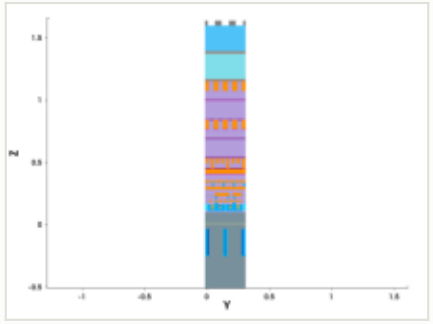
process area	ArF	EPW	0.199 kgCO2e
material	—	EPW tool	0.142 kgCO2e
thickness	—	EPW overhead	0.0568 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.33 kWh + 0.132 overhead	embodied carbon	**0.199 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

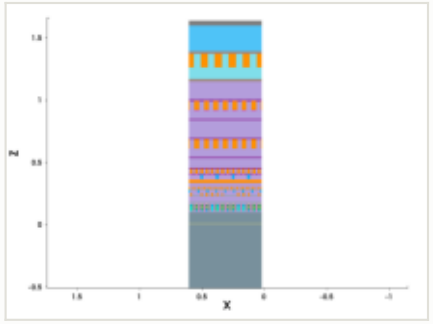


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm

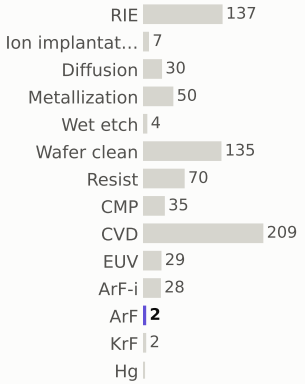


y cut at 0.184 µm

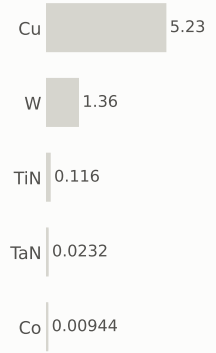


mask: layer 75:0 (GDSII)

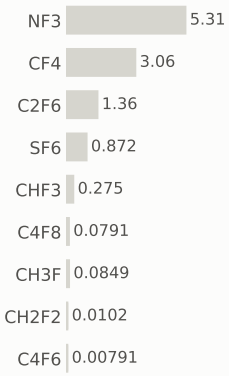
steps by process area



materials deposited (g)



gases emitted (g)



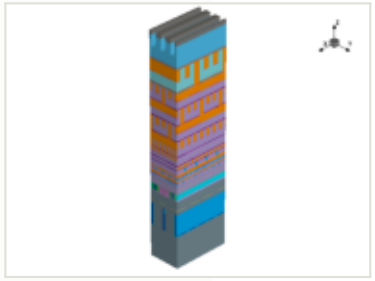
Step #7690: RIE

etch_anisotropic · RIE

1.04	664.3	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

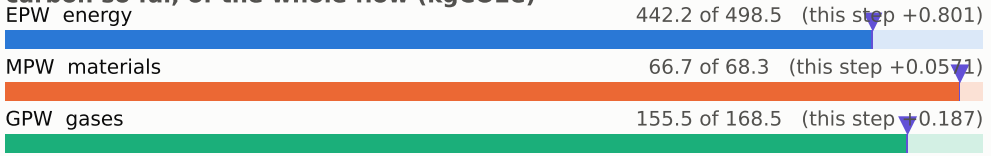


this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

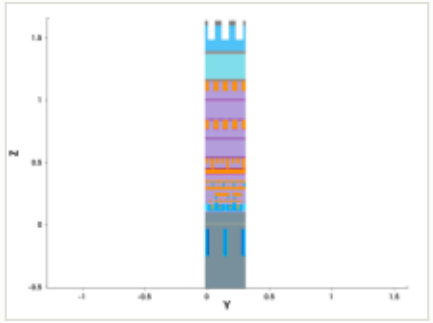
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

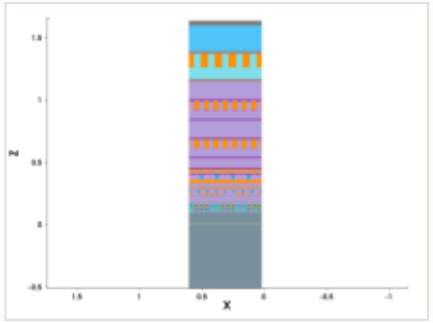


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



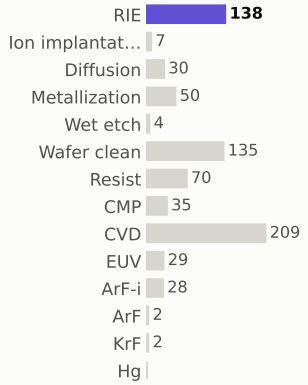
y cut at 0.184 µm



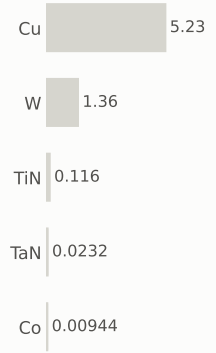
mask for this step

no mask for this step

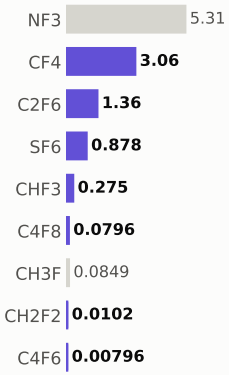
steps by process area



materials deposited (g)



gases emitted (g)



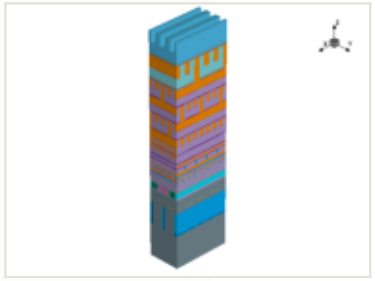
Step #7700: Wafer_clean

etch_strip · Wafer_clean

0.169	664.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

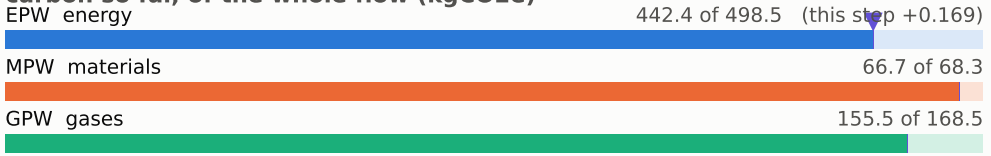


this step

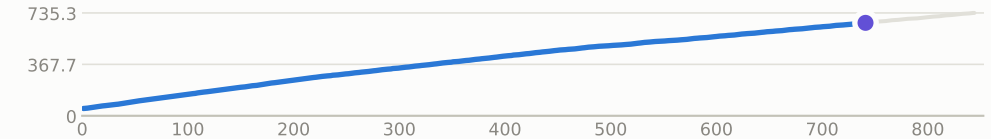
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

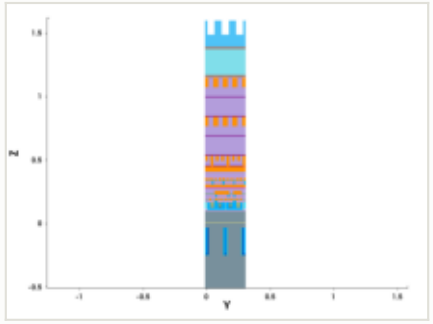
carbon so far, of the whole flow (kgCO2e)



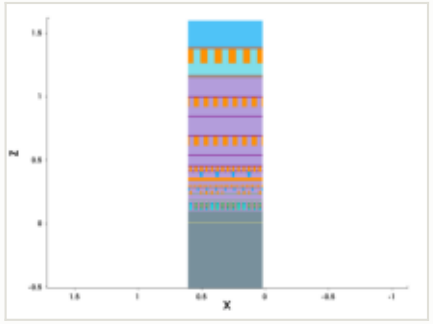
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



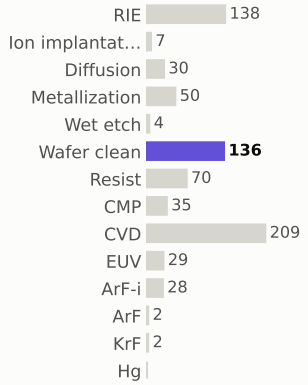
y cut at 0.184 µm



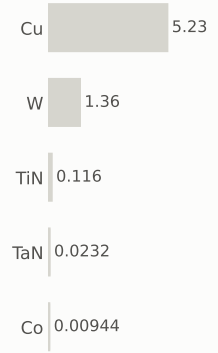
mask for this step

no mask for this step

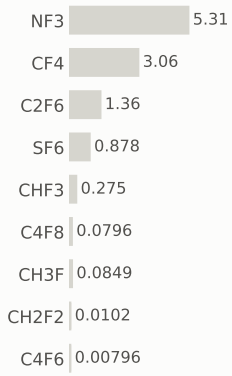
steps by process area



materials deposited (g)



gases emitted (g)



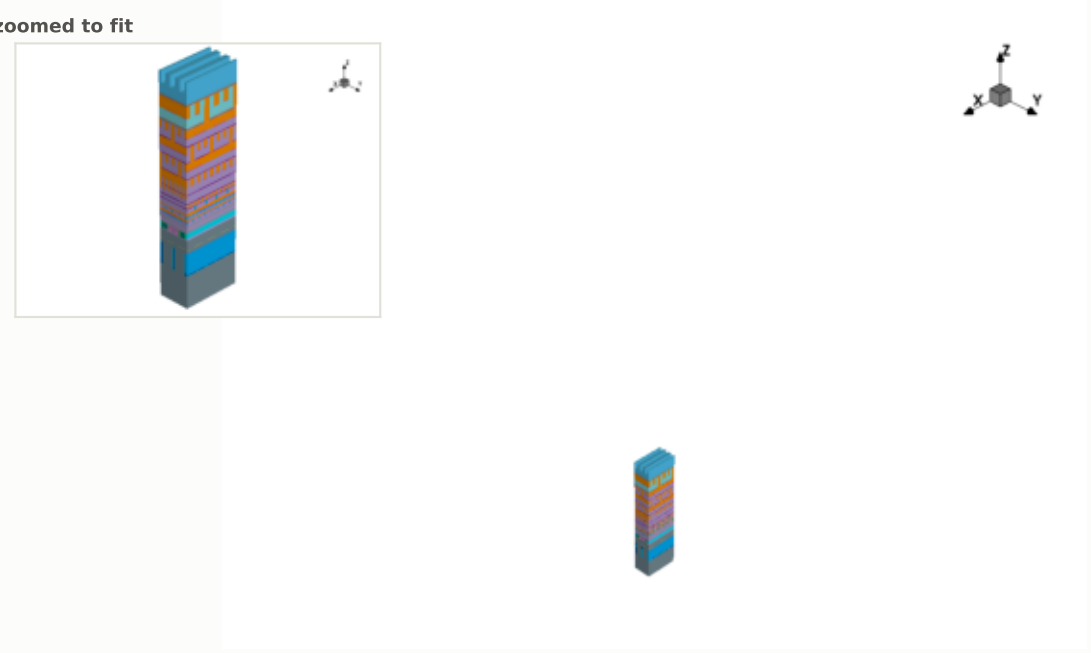
Step #7710: RIE

clean · RIE

1.04	665.5	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

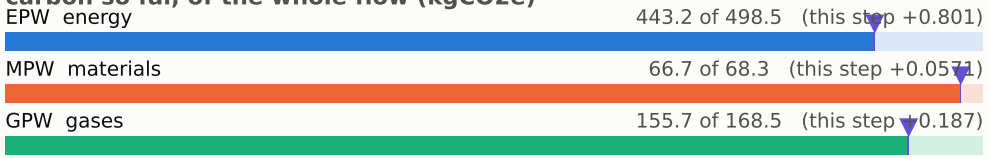


this step

process area	RIE	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

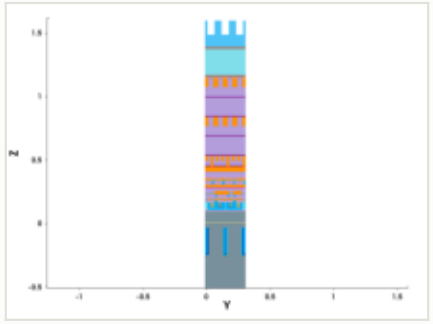
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

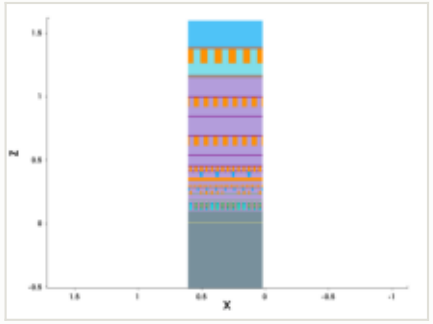


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



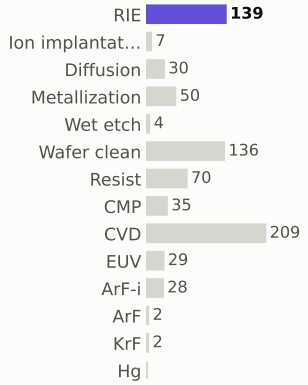
y cut at 0.184 µm



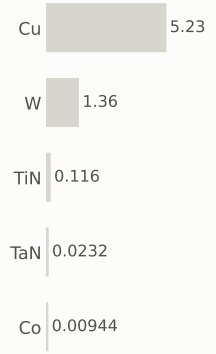
mask for this step

no mask for this step

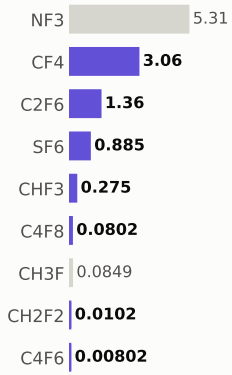
steps by process area



materials deposited (g)



gases emitted (g)



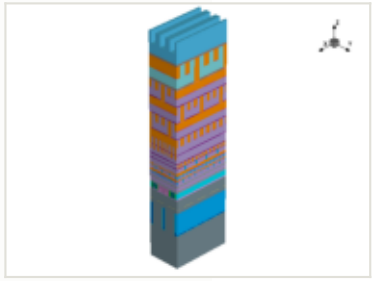
Step #7720: Wafer_clean

clean · Wafer_clean

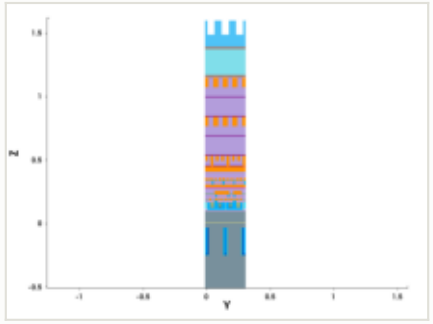
0.169	665.7	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

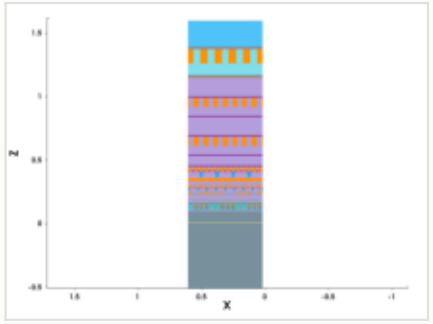
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

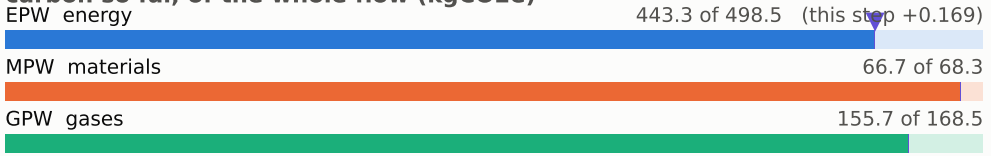
no mask for this step

this step

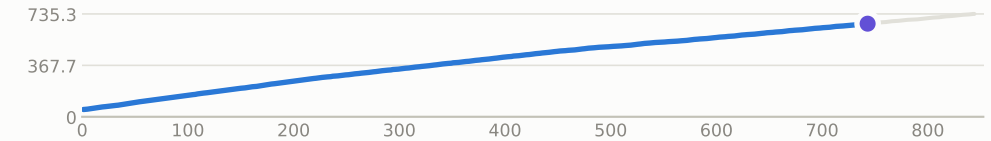
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

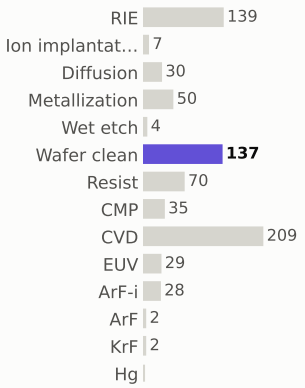
carbon so far, of the whole flow (kgCO2e)



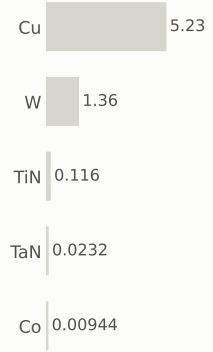
embodied carbon up to each step (kgCO2e per wafer)



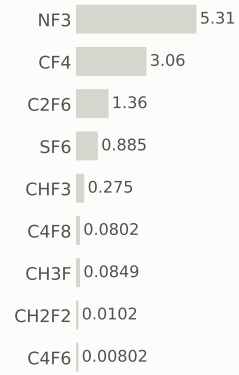
steps by process area



materials deposited (g)



gases emitted (g)



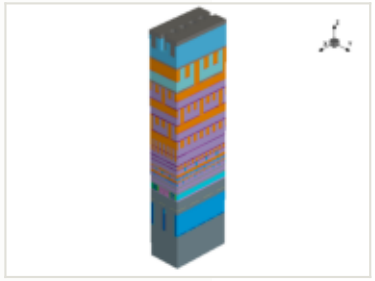
Step #7730: ArF

litho_patterning · ArF

0.199	665.9	0.027%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

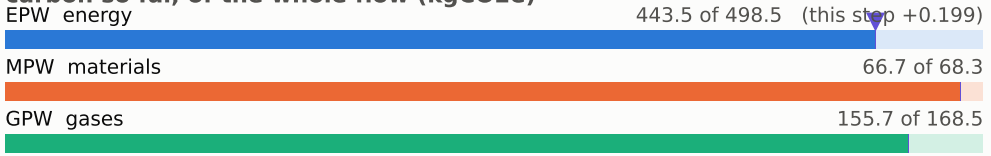


this step

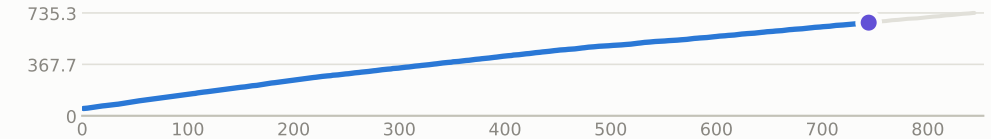
process area	ArF
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

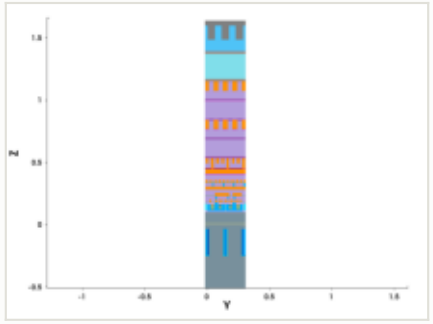
carbon so far, of the whole flow (kgCO2e)



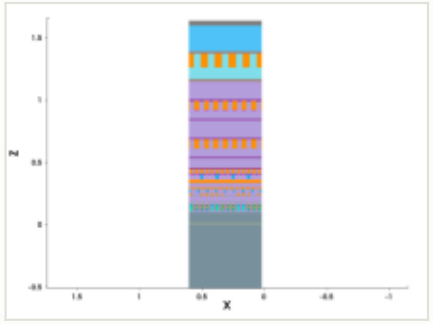
embodied carbon up to each step (kgCO2e per wafer)



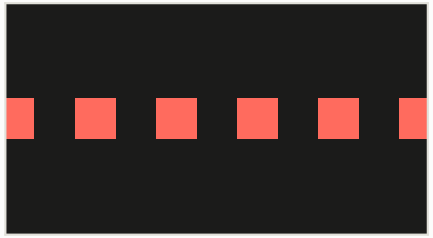
x cut at 0.294 µm



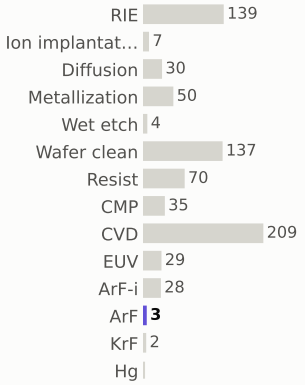
y cut at 0.184 µm



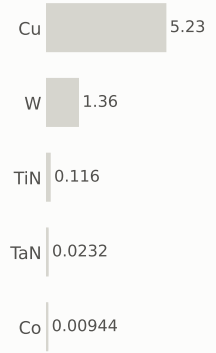
mask: layer 72:0 (GDSII)



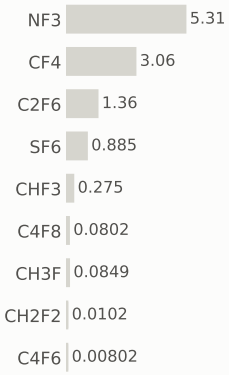
steps by process area



materials deposited (g)



gases emitted (g)



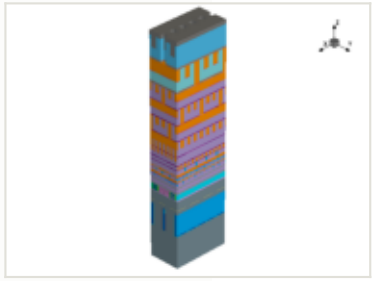
Step #7740: RIE

etch_anisotropic · RIE

1.04	667.0	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

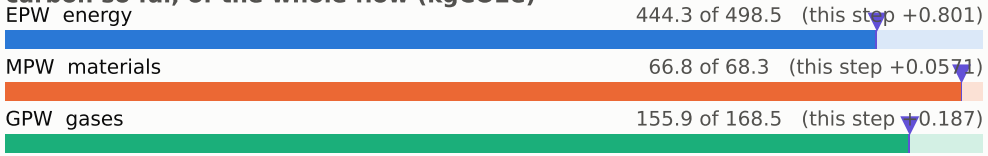


this step

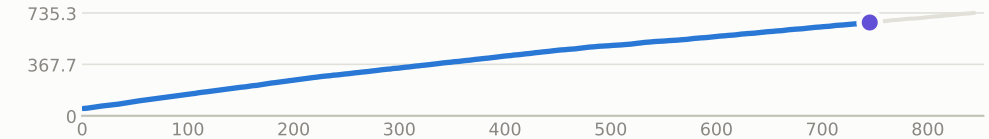
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

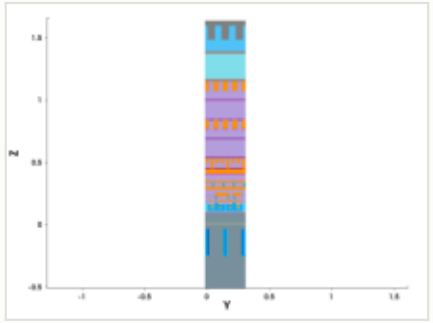
carbon so far, of the whole flow (kgCO2e)



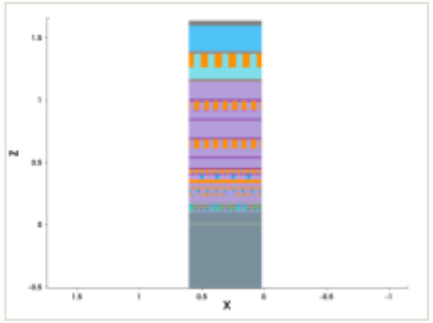
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



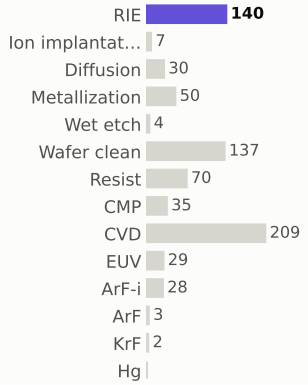
y cut at 0.184 µm



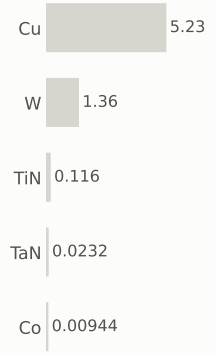
mask for this step

no mask for this step

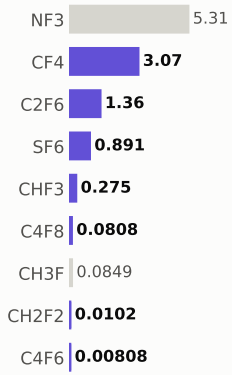
steps by process area



materials deposited (g)



gases emitted (g)



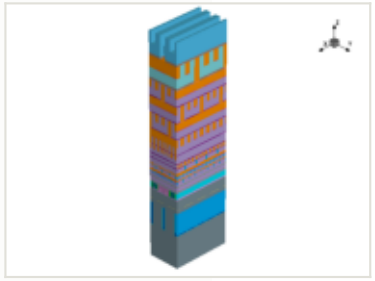
Step #7750: Wafer_clean

etch_strip · Wafer_clean

0.169	667.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

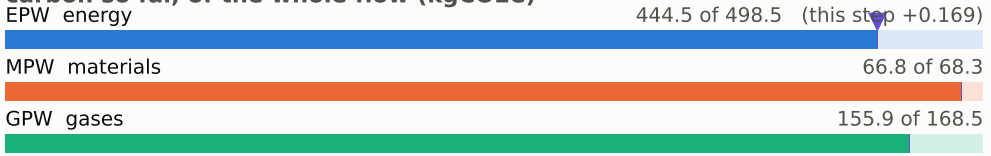


this step

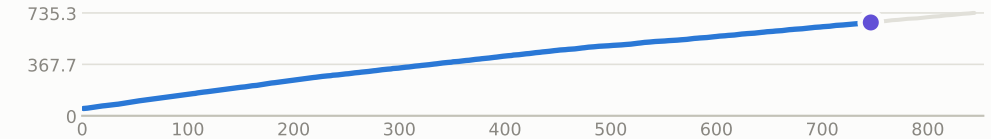
process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

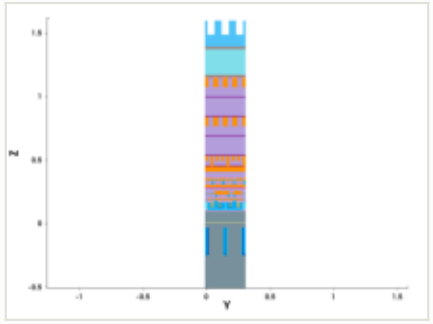
carbon so far, of the whole flow (kgCO2e)



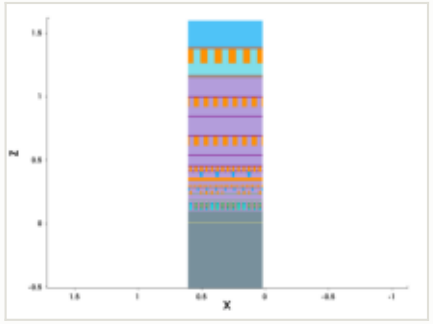
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



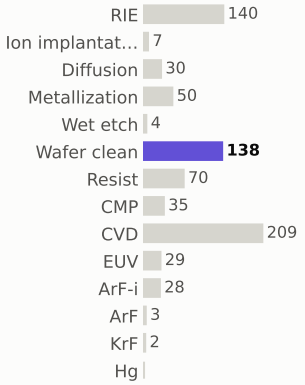
y cut at 0.184 µm



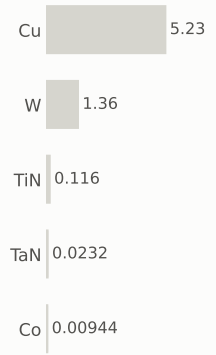
mask for this step

no mask for this step

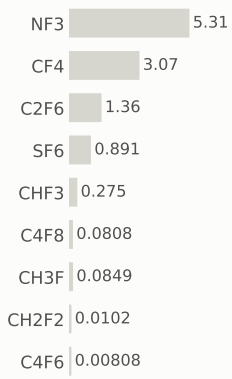
steps by process area



materials deposited (g)



gases emitted (g)



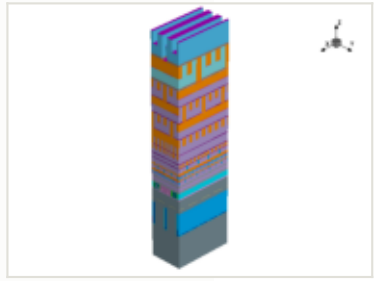
Step #7760: Metallization

depo_ellipsoidal · Metallization

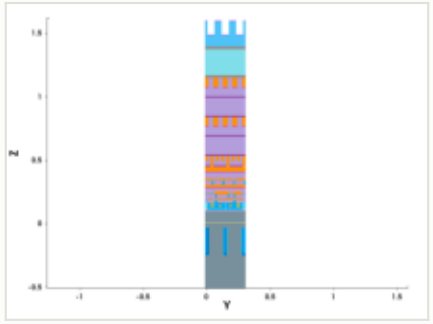
1.2	668.3	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

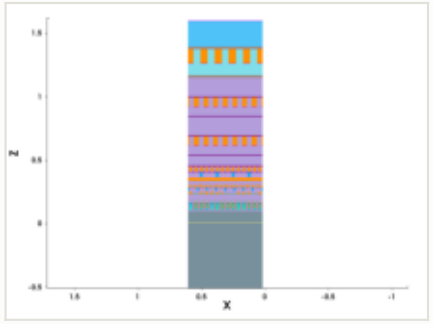
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

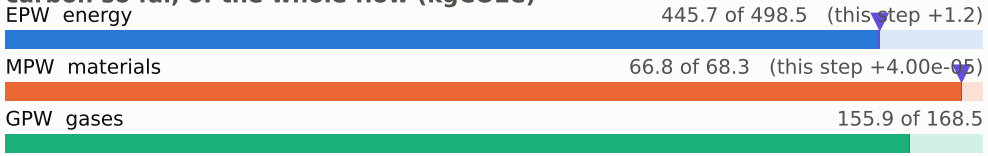
no mask for this step

this step

process area	Metallization
material	TaN
thickness	0.0016 µm
material GWP	25.8159 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

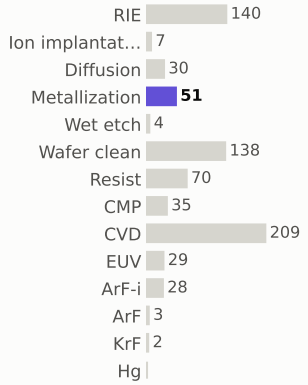
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.00e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

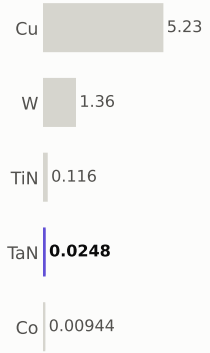


embodied carbon up to each step (kgCO2e per wafer)

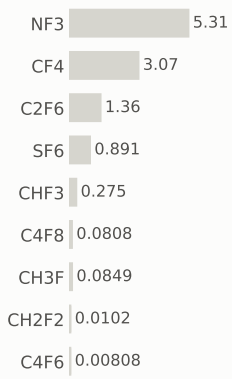
steps by process area



materials deposited (g)



gases emitted (g)



Step #7770: Metallization

depo_fill · Metallization

1.21	669.5	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

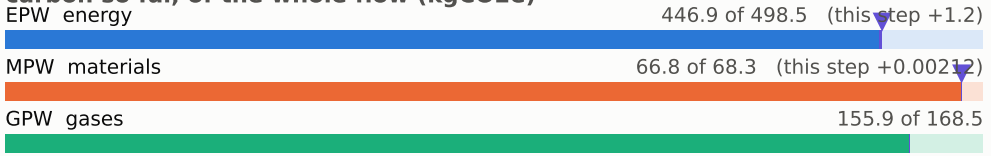
zoomed to fit

this step

process area	Metallization
material	Cu
thickness	0.55 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

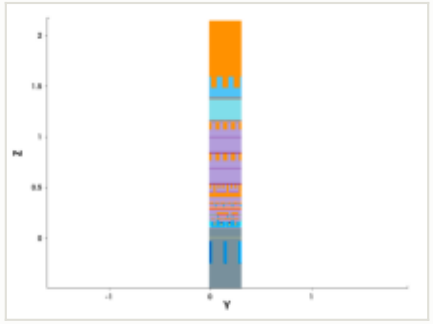
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00212 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

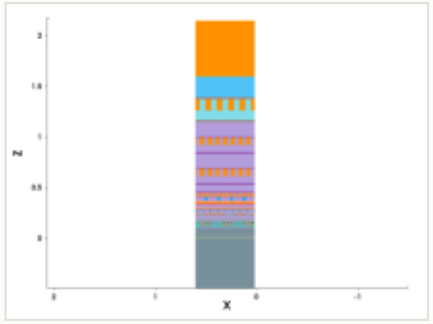


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



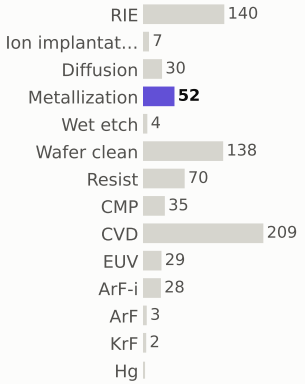
y cut at 0.184 µm



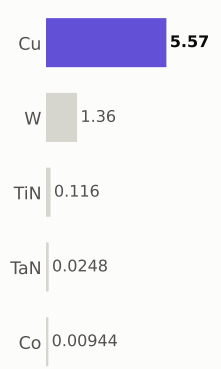
mask for this step

no mask for this step

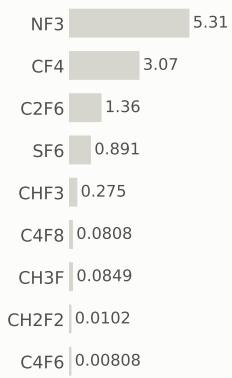
steps by process area



materials deposited (g)



gases emitted (g)



Step #7780: Diffusion

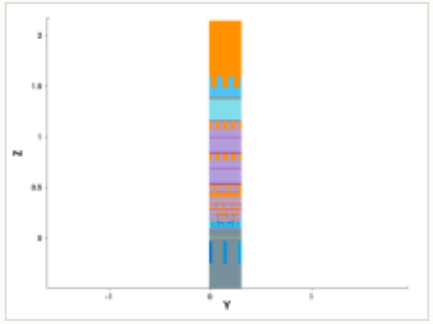
treatment · Diffusion

0.801	670.3	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

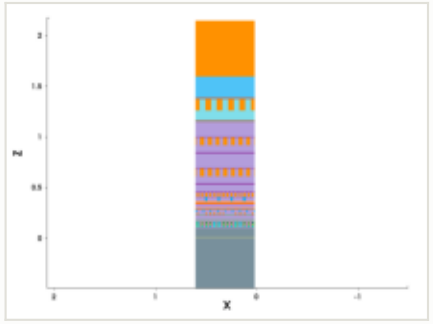
3D structure after this step

zoomed to fit

x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

no mask for this step

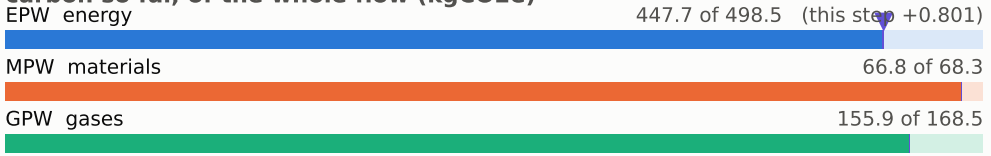
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

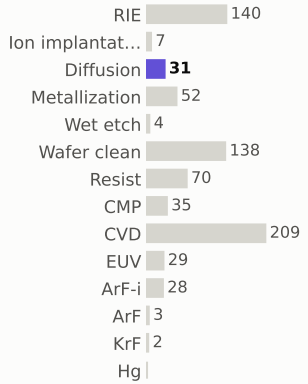
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

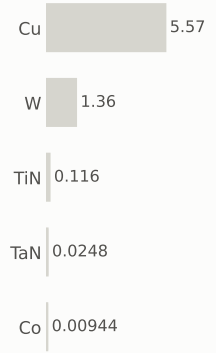


embodied carbon up to each step (kgCO2e per wafer)

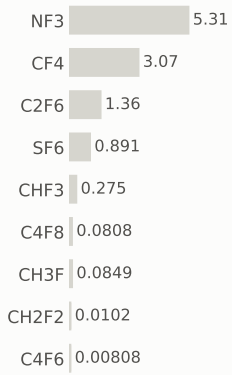
steps by process area



materials deposited (g)



gases emitted (g)



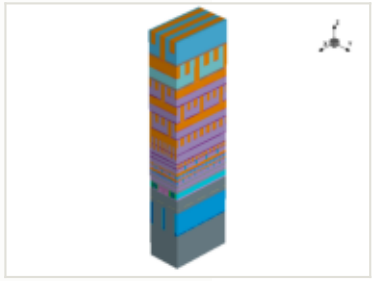
Step #7790: CMP

etch_cmp · CMP

0.169	670.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

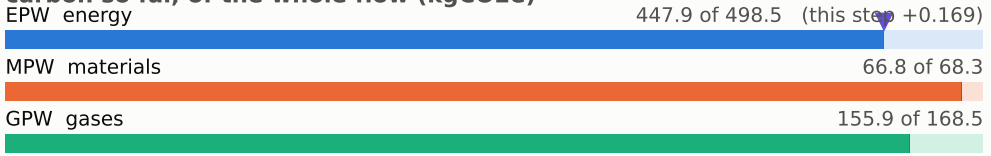


this step

process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

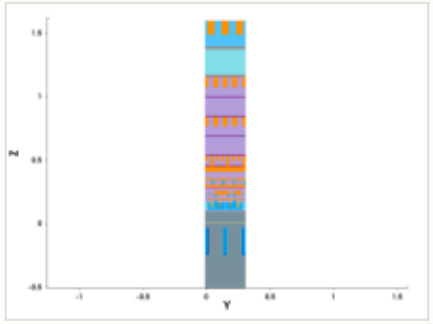
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

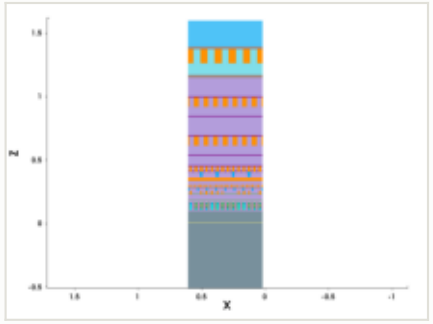


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



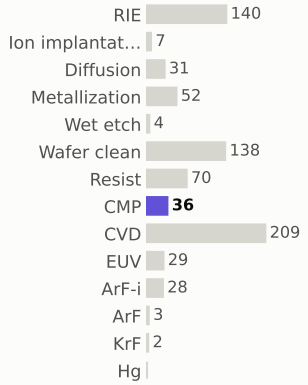
y cut at 0.184 µm



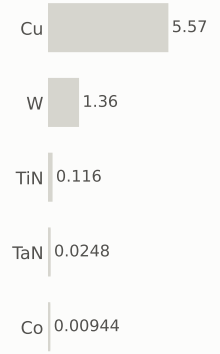
mask for this step

no mask for this step

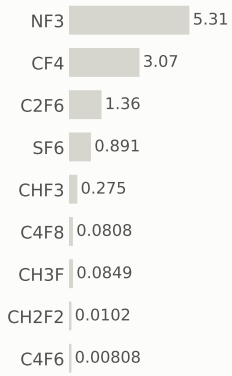
steps by process area



materials deposited (g)



gases emitted (g)



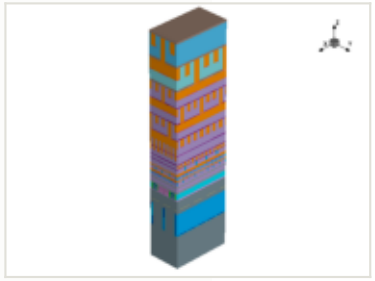
Step #7810: CVD

depo_fill · CVD

1.31	671.8	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

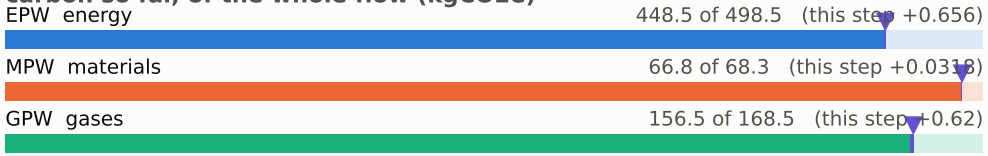


this step

process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

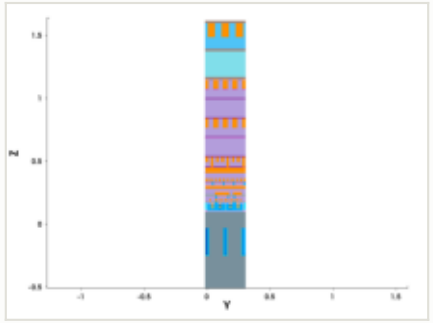
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

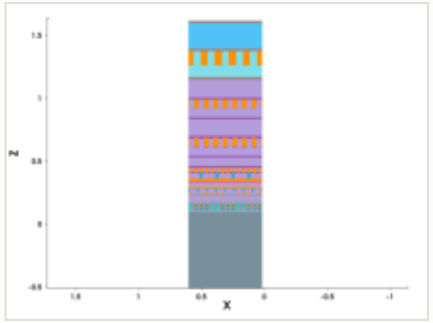


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



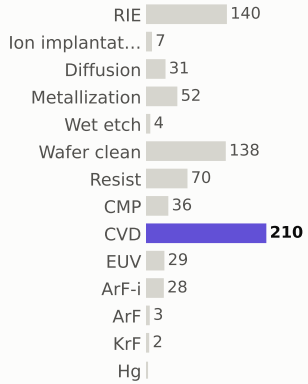
y cut at 0.184 µm



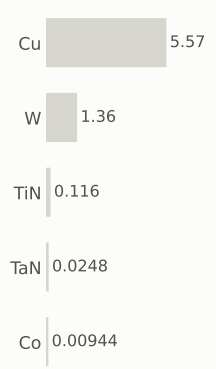
mask for this step

no mask for this step

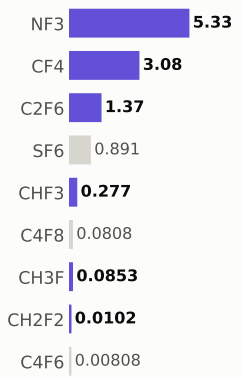
steps by process area



materials deposited (g)



gases emitted (g)



Step #7820: CVD

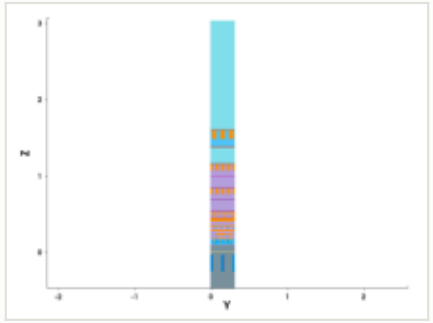
depo_fill · CVD

1.31	673.1	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

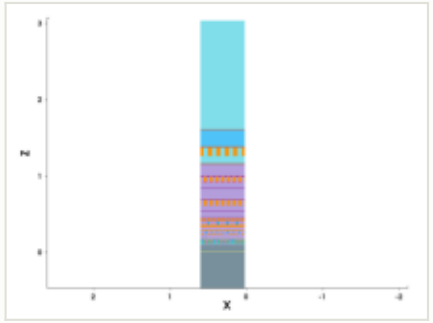
3D structure after this step

zoomed to fit

x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

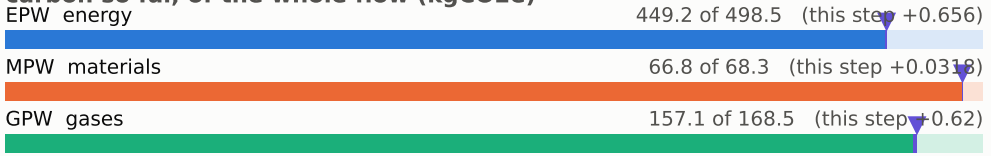
no mask for this step

this step

process area	CVD
material	—
thickness	1.42 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

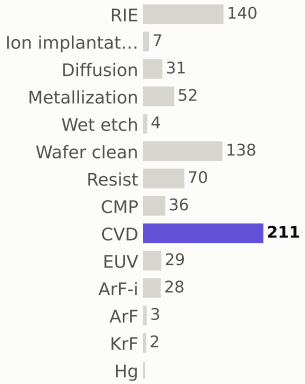
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

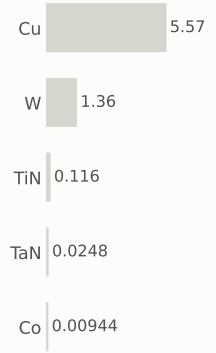


embodied carbon up to each step (kgCO2e per wafer)

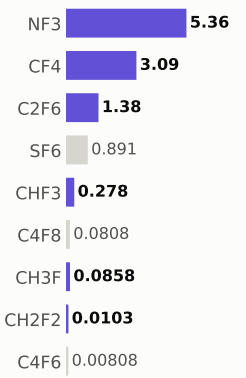
steps by process area



materials deposited (g)



gases emitted (g)



Step #7830: Hg

litho_patterning · Hg

0.199	673.3	0.027%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

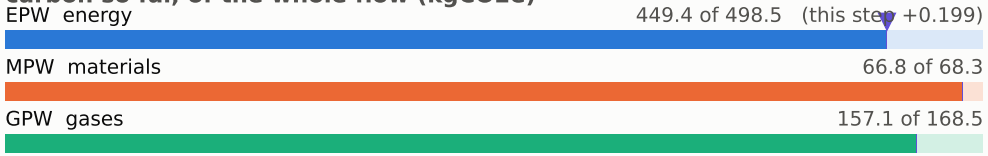
zoomed to fit

this step

process area	Hg
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

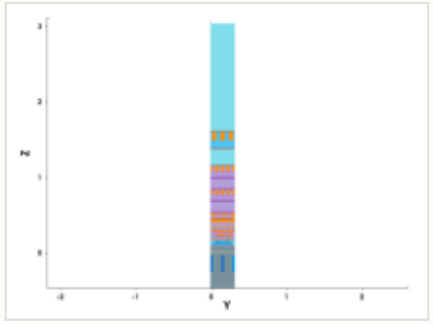
EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

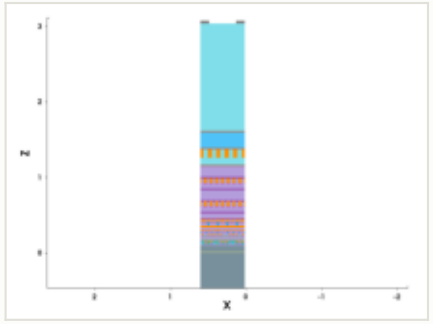


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm

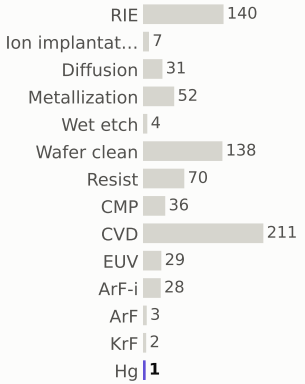


y cut at 0.184 µm

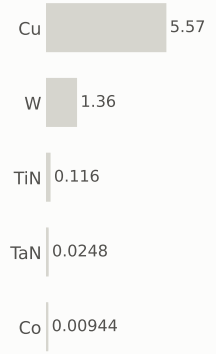


mask: layer 80:0 (GDSII)

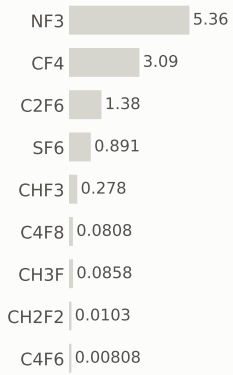
steps by process area



materials deposited (g)



gases emitted (g)



Step #7840: RIE

etch_anisotropic · RIE

1.04	674.4	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

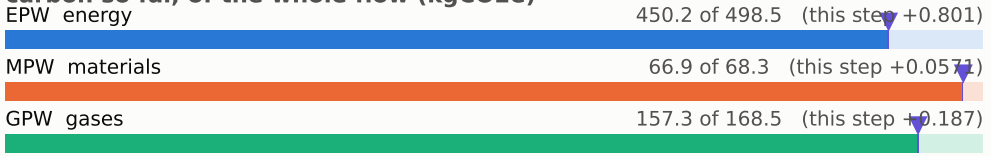
zoomed to fit

this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

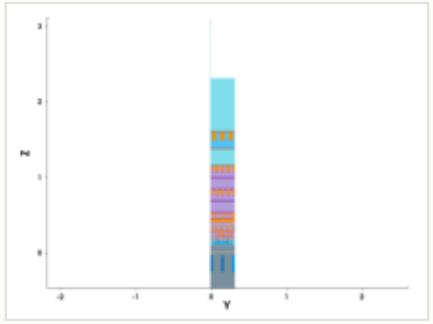
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

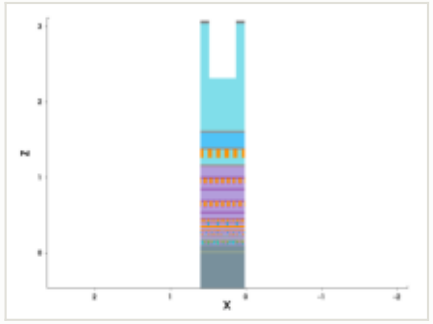


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



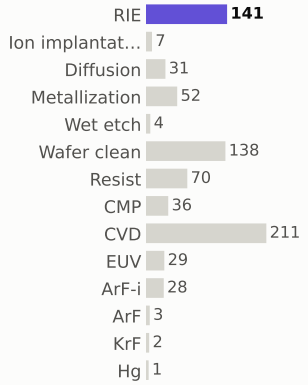
y cut at 0.184 µm



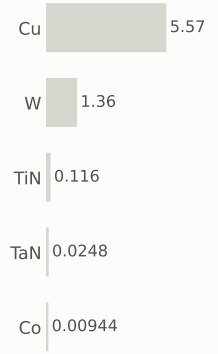
mask for this step

no mask for this step

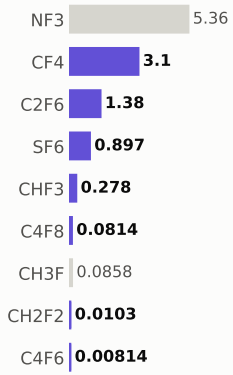
steps by process area



materials deposited (g)



gases emitted (g)



Step #7850: Wafer_clean

etch_strip · Wafer_clean

0.169	674.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

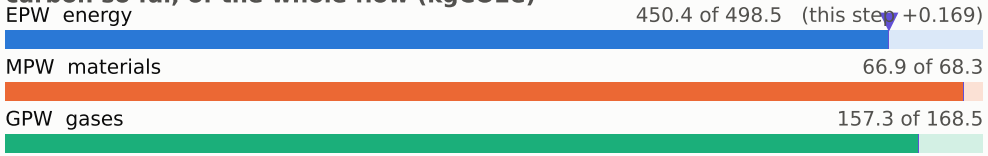
3D structure after this step

zoomed to fit

this step

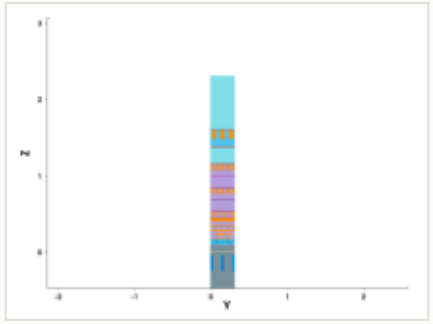
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

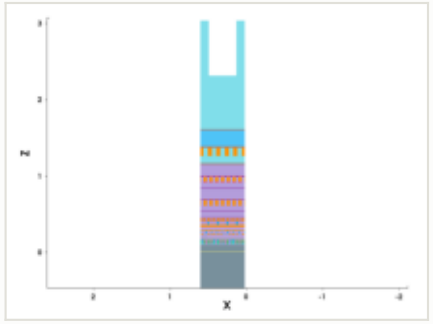


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



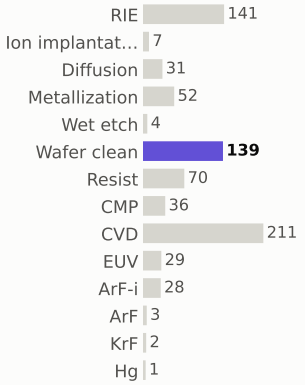
y cut at 0.184 µm



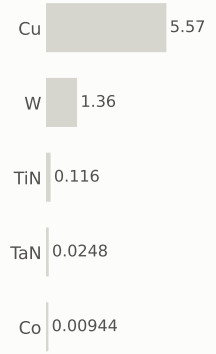
mask for this step

no mask for this step

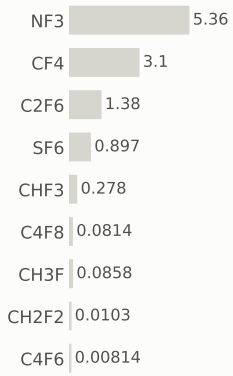
steps by process area



materials deposited (g)



gases emitted (g)



Step #7860: RIE

clean · RIE

1.04	675.6	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

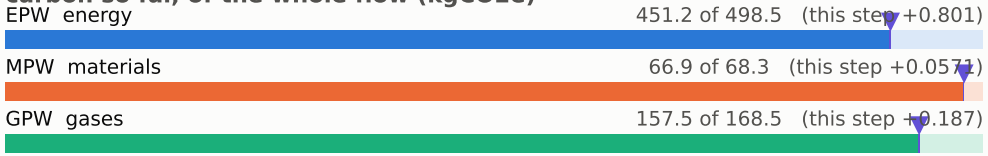
zoomed to fit

this step

process area	RIE	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

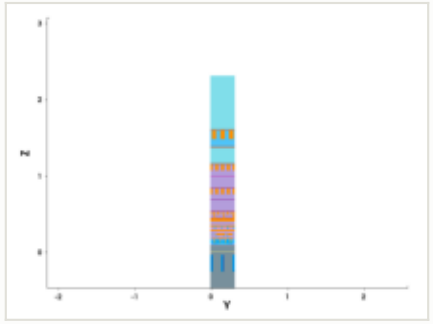
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

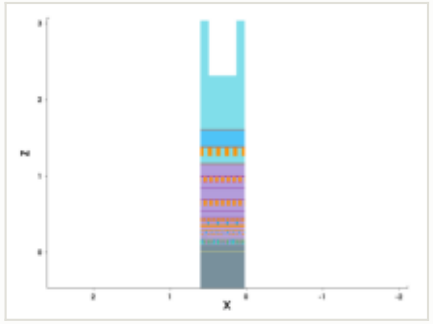


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



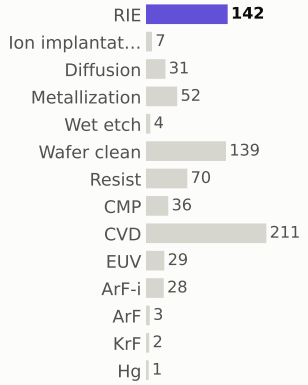
y cut at 0.184 µm



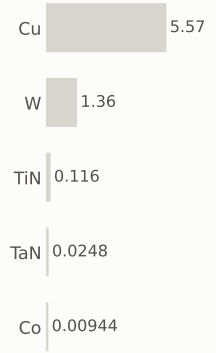
mask for this step

no mask for this step

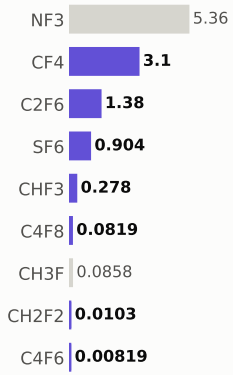
steps by process area



materials deposited (g)



gases emitted (g)



Step #7870: Wafer_clean

clean · Wafer_clean

0.169	675.7	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

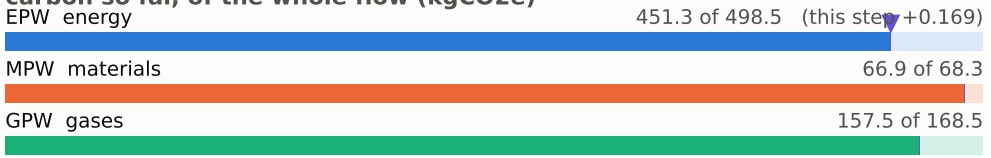
zoomed to fit

this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

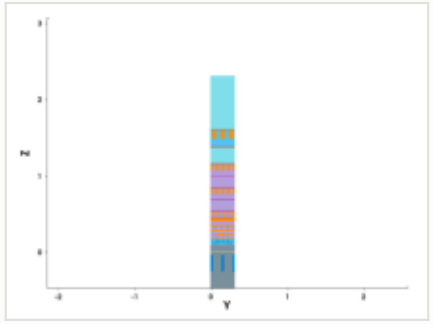
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

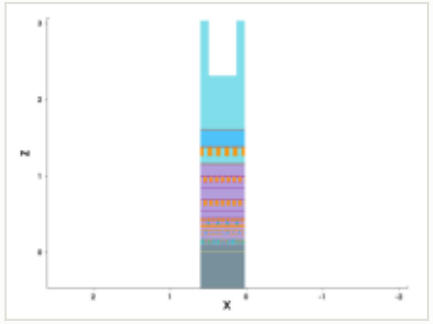


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



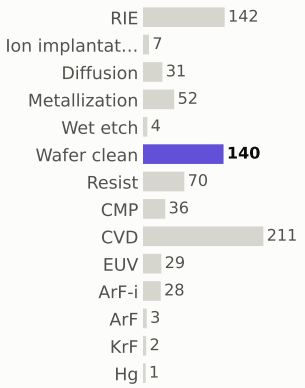
y cut at 0.184 µm



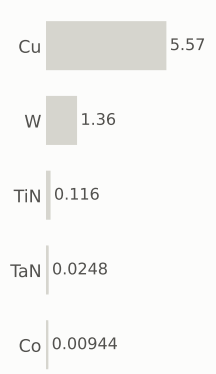
mask for this step

no mask for this step

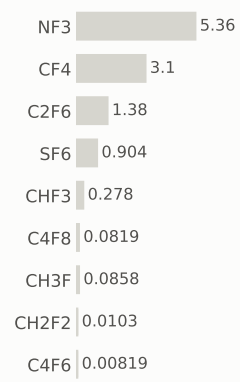
steps by process area



materials deposited (g)



gases emitted (g)



Step #7880: ArF

litho_patterning · ArF

0.199	675.9	0.027%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

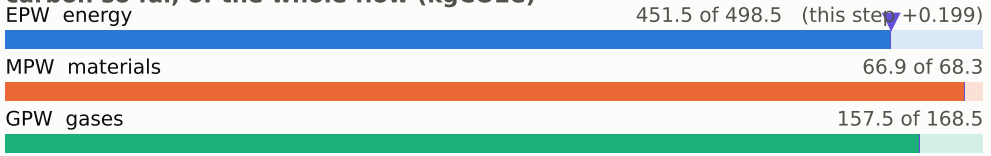
3D structure after this step

zoomed to fit

this step

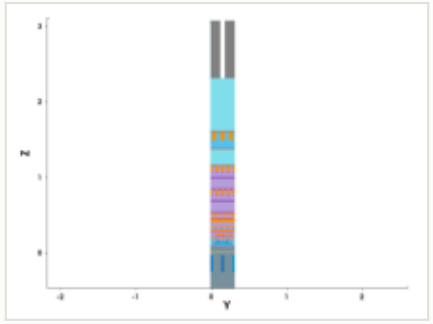
process area	ArF	EPW	0.199 kgCO2e
material	—	EPW tool	0.142 kgCO2e
thickness	—	EPW overhead	0.0568 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.33 kWh + 0.132 overhead	embodied carbon	**0.199 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

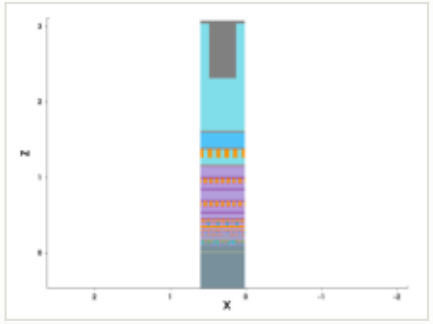


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm

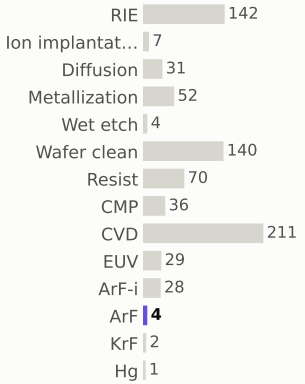


y cut at 0.184 µm

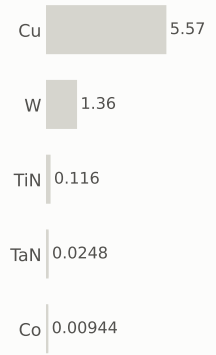


mask: layer 77:0 (GDSII)

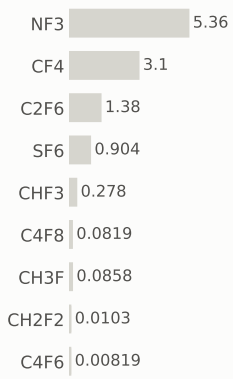
steps by process area



materials deposited (g)



gases emitted (g)



Step #7890: RIE

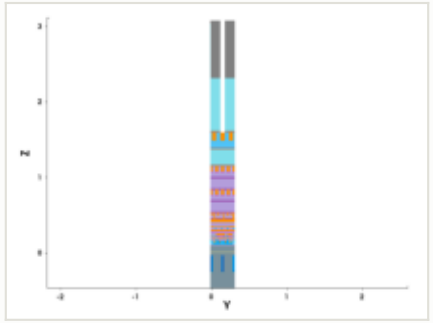
etch_anisotropic · RIE

1.04	677.0	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

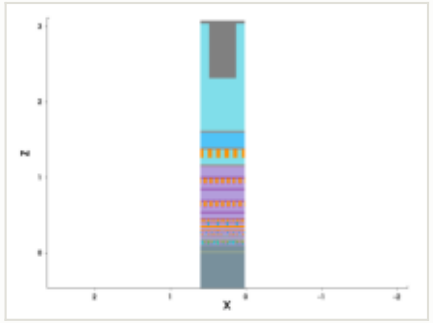
3D structure after this step

zoomed to fit

x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

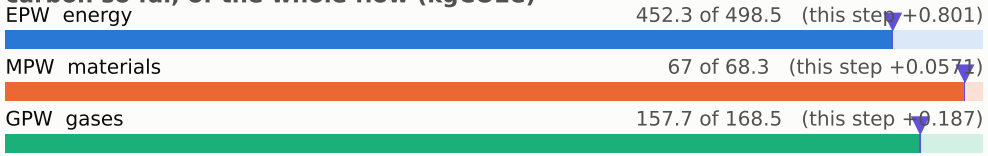
no mask for this step

this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

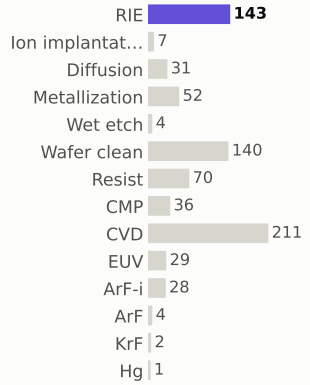
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

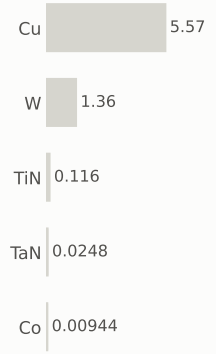


embodied carbon up to each step (kgCO2e per wafer)

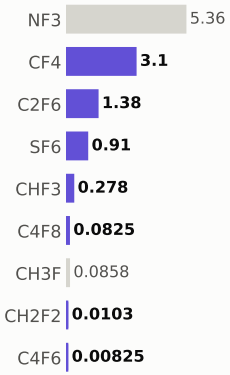
steps by process area



materials deposited (g)



gases emitted (g)



Step #7900: Wafer_clean

etch_strip · Wafer_clean

0.169	677.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

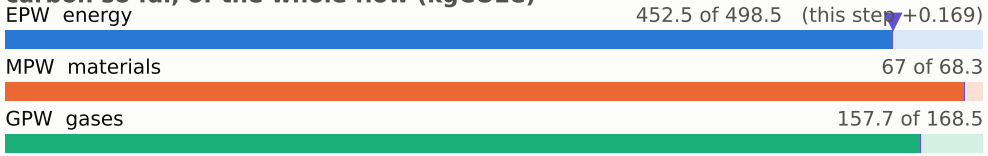
zoomed to fit

this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

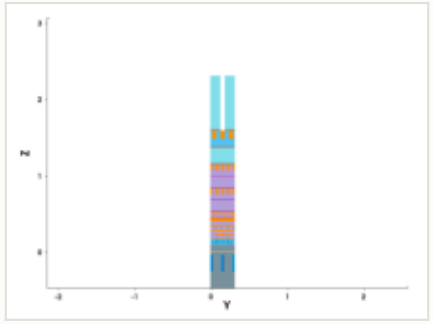
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

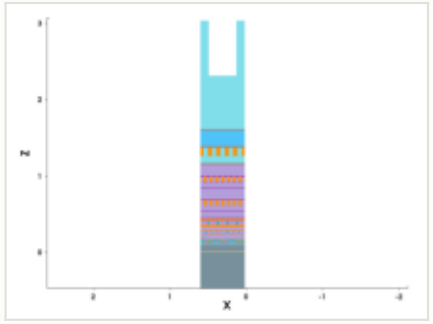


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



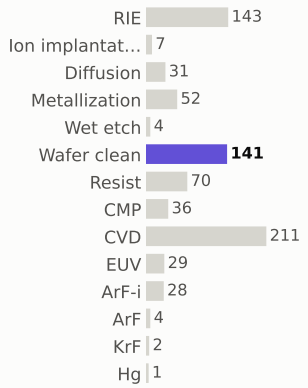
y cut at 0.184 µm



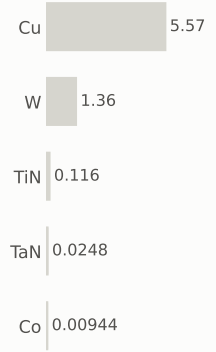
mask for this step

no mask for this step

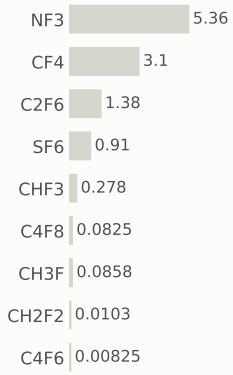
steps by process area



materials deposited (g)



gases emitted (g)



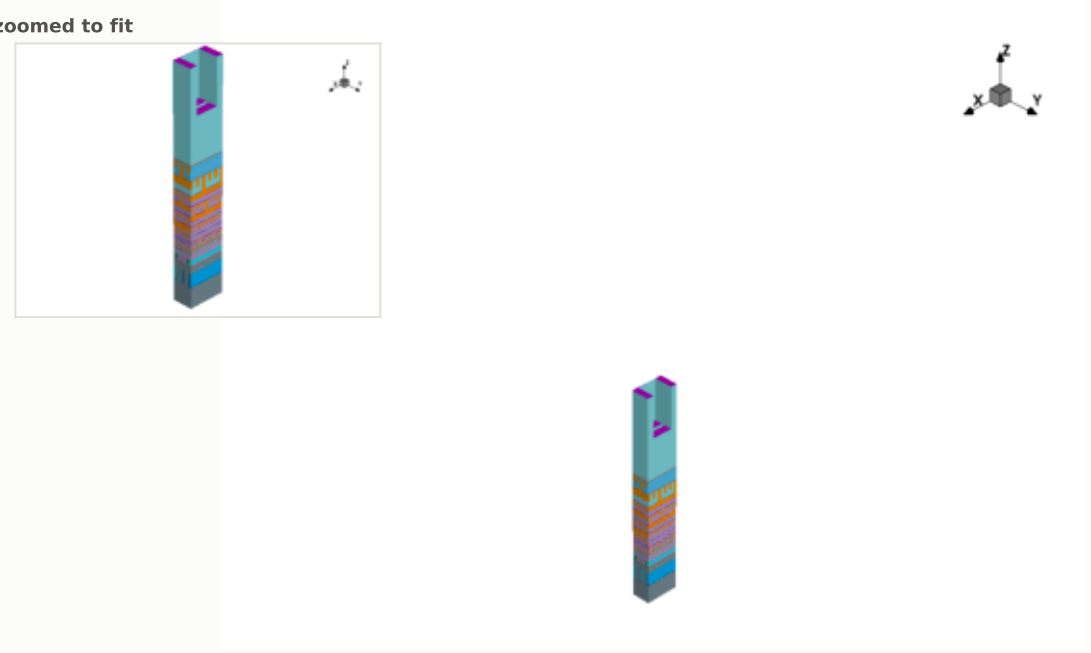
Step #7910: Metallization

depo_ellipsoidal · Metallization

1.2	678.4	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

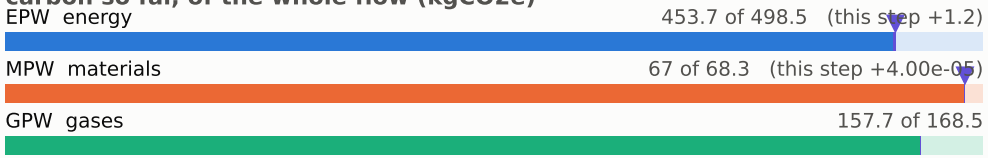
zoomed to fit



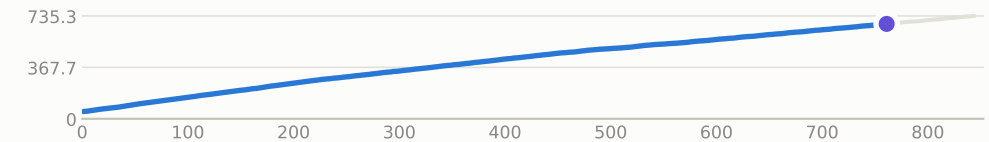
this step

process area	Metallization	EPW	1.2 kgCO2e
material	TaN	EPW tool	0.86 kgCO2e
thickness	0.0016 µm	EPW overhead	0.344 kgCO2e
material GWP	25.8159 gCO2e/g	MPW	4.00e-05 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	2 kWh + 0.8 overhead	embodied carbon	**1.2 kgCO2e**

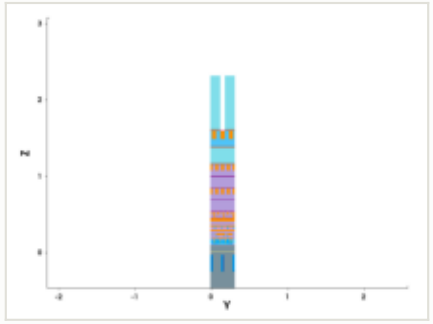
carbon so far, of the whole flow (kgCO2e)



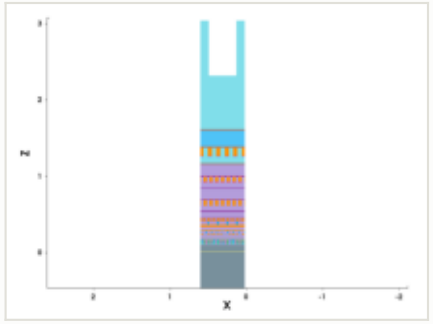
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



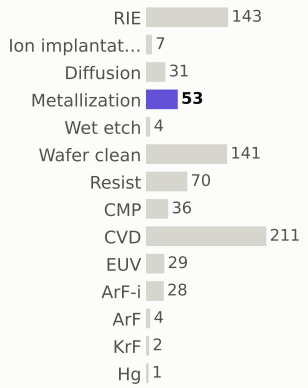
y cut at 0.184 µm



mask for this step

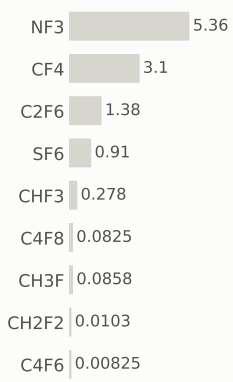
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



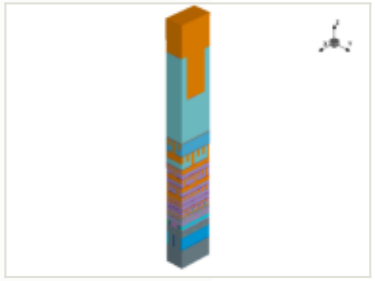
Step #7920: Metallization

depo_fill · Metallization

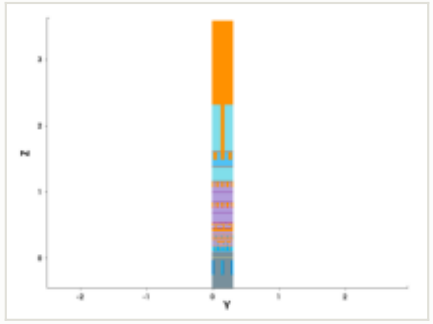
1.21	679.6	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

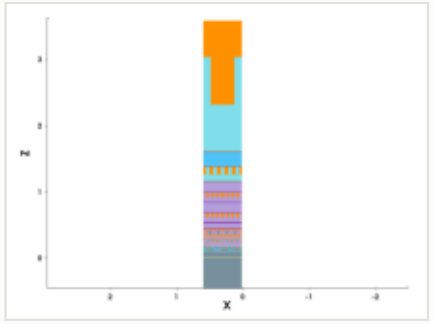
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

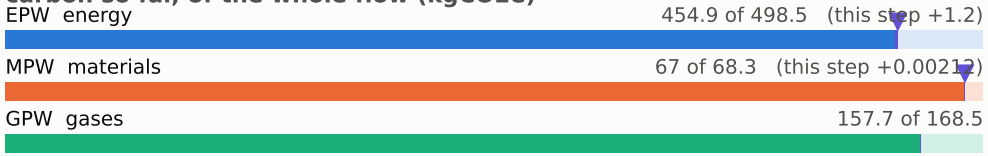
no mask for this step

this step

process area	Metallization
material	Cu
thickness	0.55 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

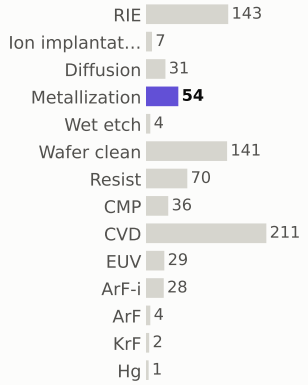
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00212 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

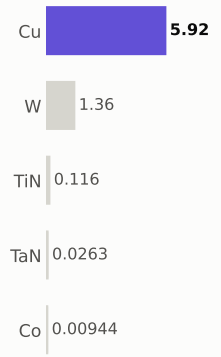


embodied carbon up to each step (kgCO2e per wafer)

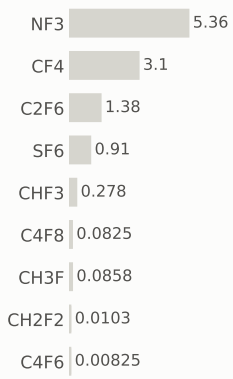
steps by process area



materials deposited (g)



gases emitted (g)



Step #7930: Diffusion

treatment · Diffusion

0.801	680.4	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

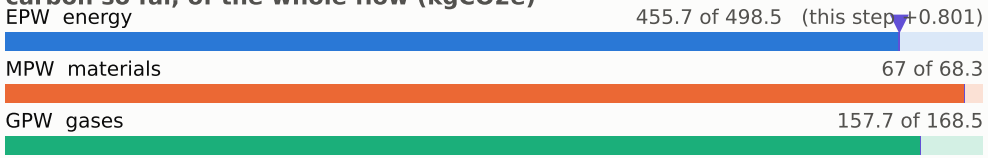
zoomed to fit

this step

process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

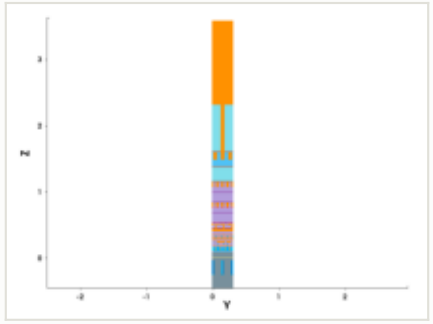
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

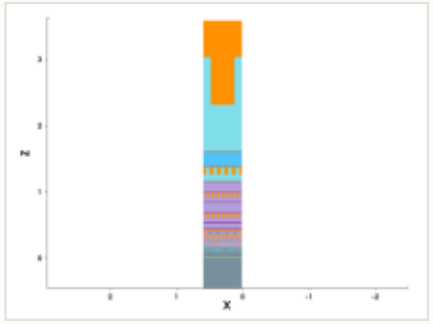


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



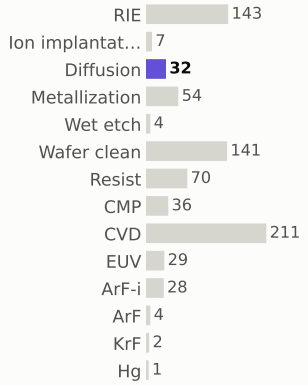
y cut at 0.184 µm



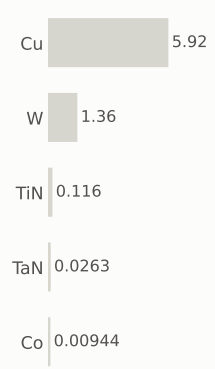
mask for this step

no mask for this step

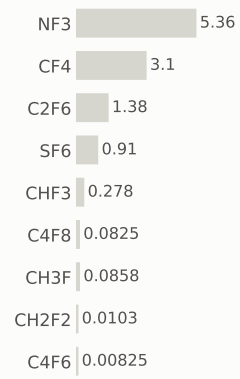
steps by process area



materials deposited (g)



gases emitted (g)



Step #7940: CMP

etch_cmp · CMP

0.169	680.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

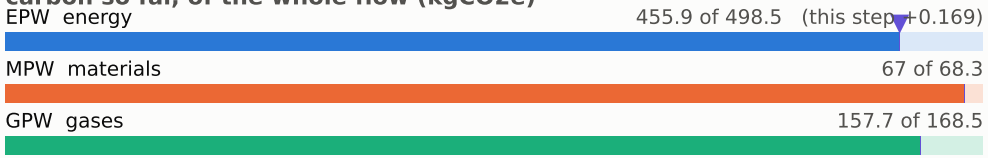
3D structure after this step

zoomed to fit

this step

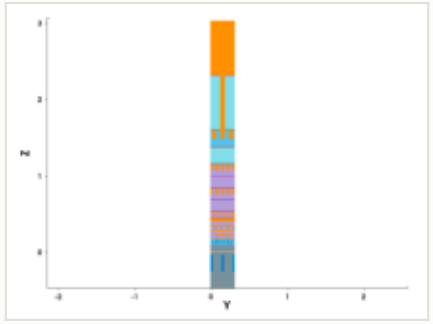
process area	CMP	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

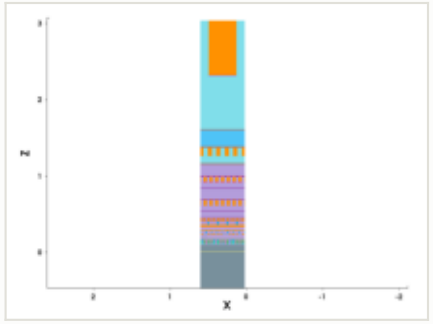


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



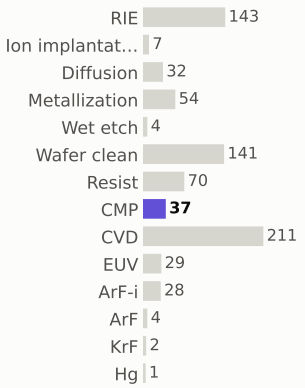
y cut at 0.184 µm



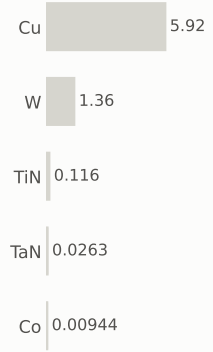
mask for this step

no mask for this step

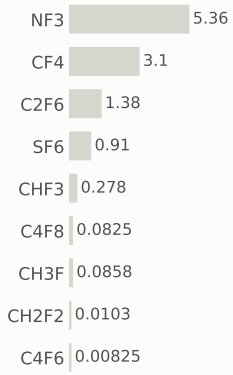
steps by process area



materials deposited (g)



gases emitted (g)



Step #7960: CVD

depo_fill · CVD

1.31	681.8	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

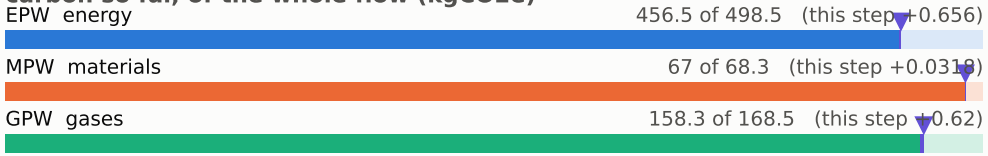
3D structure after this step

zoomed to fit

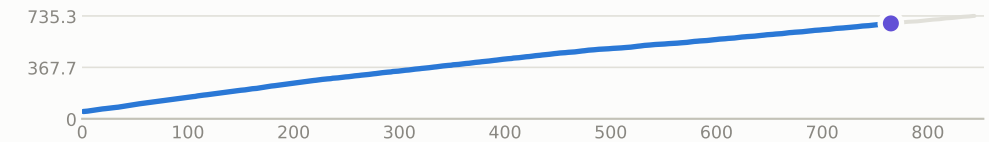
this step

process area	CVD	EPW	0.656 kgCO2e
material	—	EPW tool	0.469 kgCO2e
thickness	0.02 µm	EPW overhead	0.187 kgCO2e
material GWP	—	MPW	0.0318 kgCO2e (gas 0.0318)
input gases	NF3 2.54 g	GPW	0.62 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**1.31 kgCO2e**

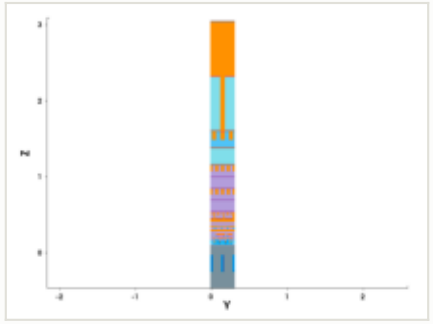
carbon so far, of the whole flow (kgCO2e)



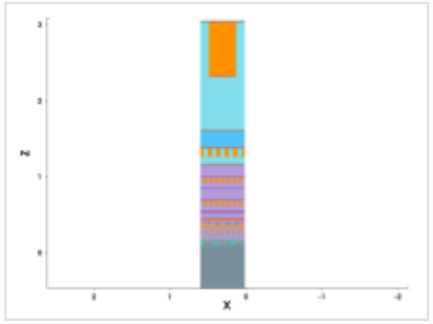
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



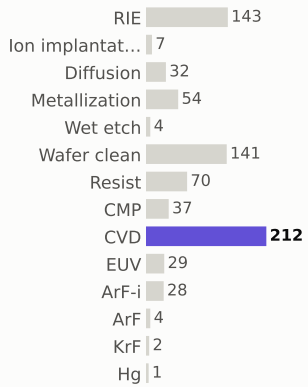
y cut at 0.184 µm



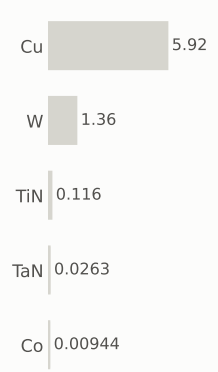
mask for this step

no mask for this step

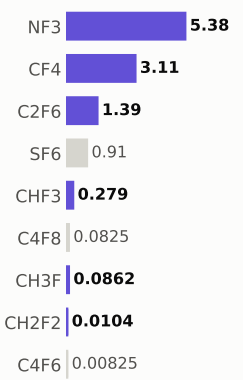
steps by process area



materials deposited (g)



gases emitted (g)



Step #7970: CVD

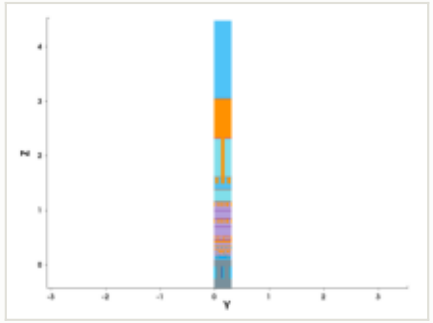
depo_fill · CVD

1.31	683.2	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

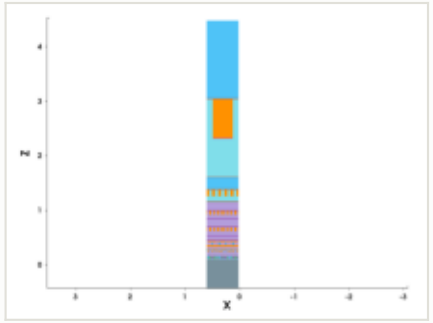
3D structure after this step

zoomed to fit

x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

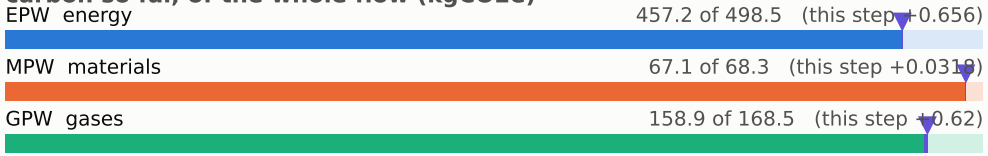
no mask for this step

this step

process area	CVD
material	—
thickness	1.42 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

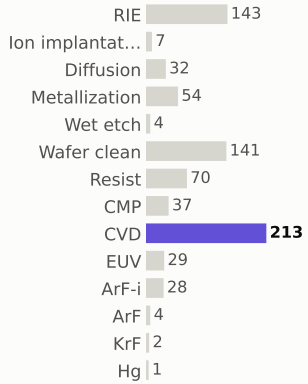
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

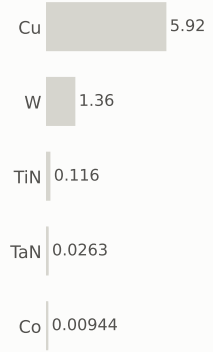


embodied carbon up to each step (kgCO2e per wafer)

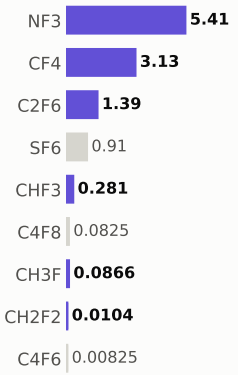
steps by process area



materials deposited (g)



gases emitted (g)



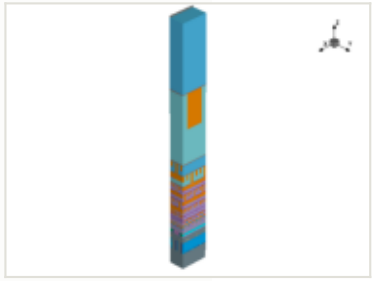
Step #7980: Hg

litho_patterning · Hg

0.199	683.3	0.027%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

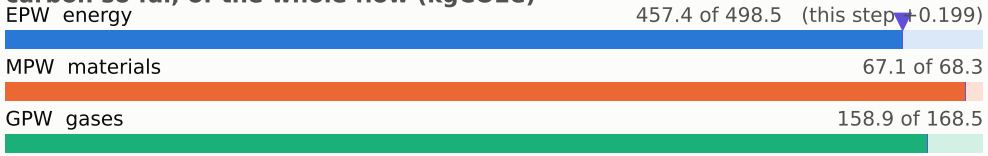


this step

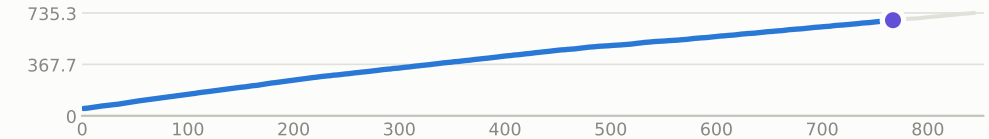
process area	Hg
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

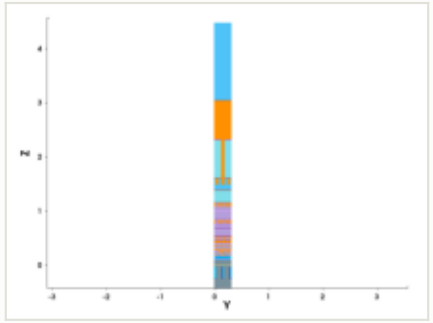
carbon so far, of the whole flow (kgCO2e)



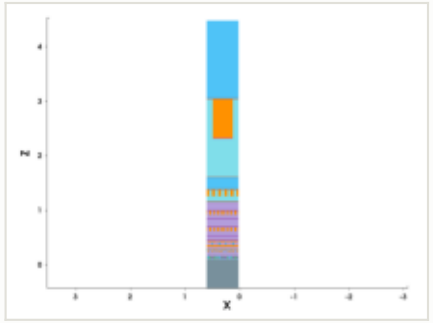
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm

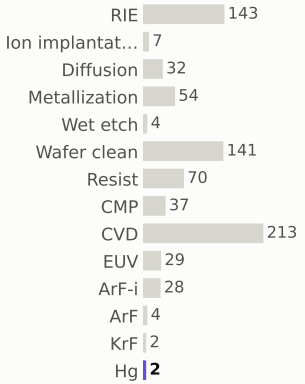


y cut at 0.184 µm

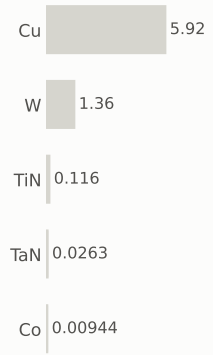


mask: layer 85:0 (GDSII)

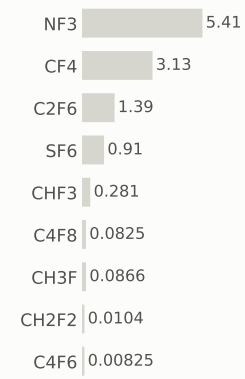
steps by process area



materials deposited (g)



gases emitted (g)



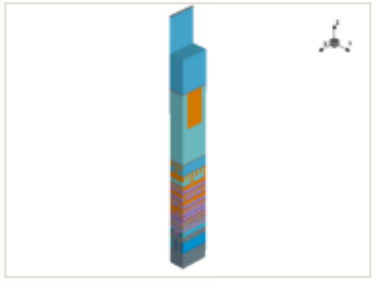
Step #7990: RIE

etch_anisotropic · RIE

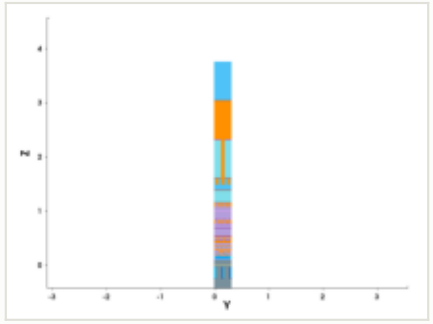
1.04	684.4	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

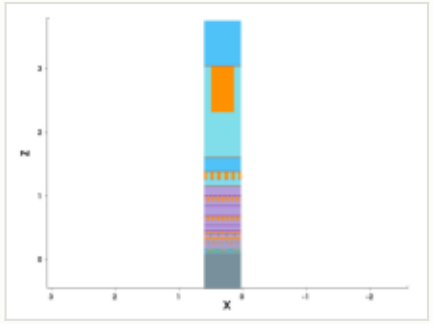
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

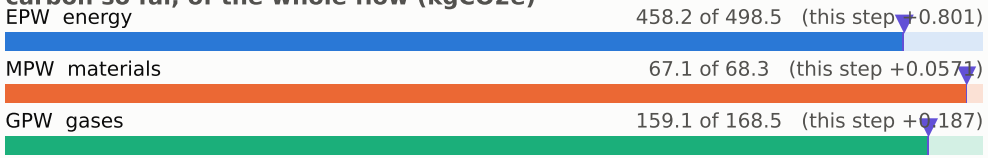
no mask for this step

this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

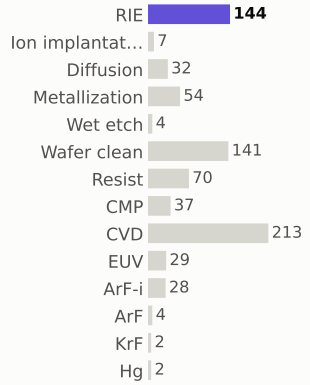
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

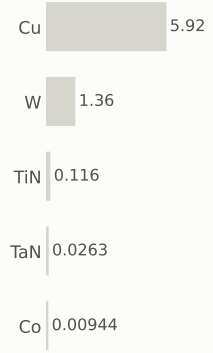


embodied carbon up to each step (kgCO2e per wafer)

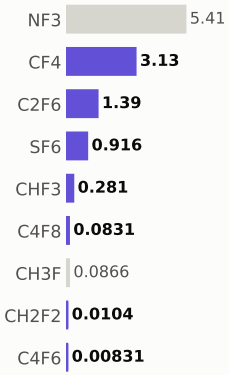
steps by process area



materials deposited (g)



gases emitted (g)



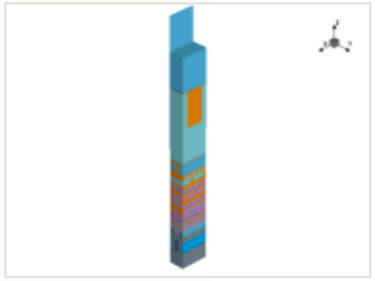
Step #8000: Wafer_clean

etch_strip · Wafer_clean

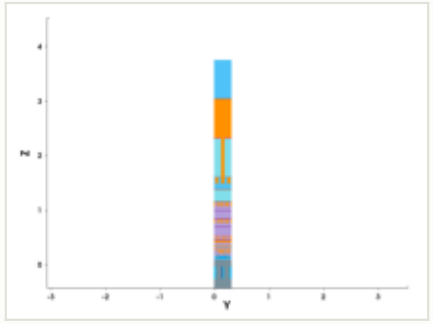
0.169	684.6	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

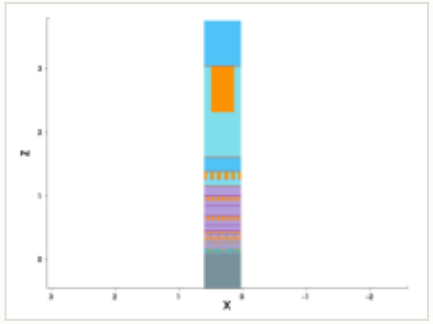
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



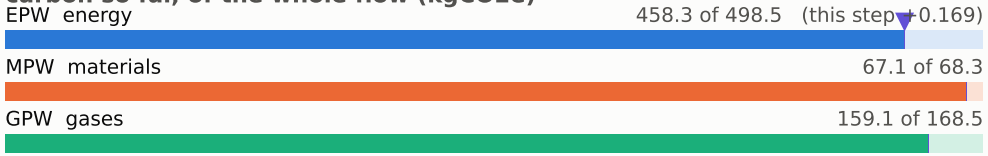
mask for this step

no mask for this step

this step

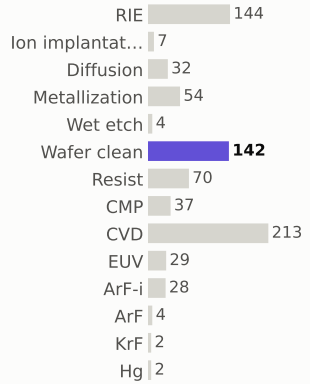
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

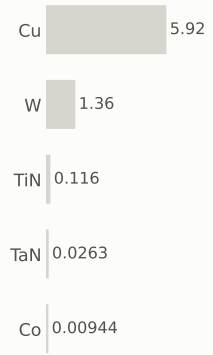


embodied carbon up to each step (kgCO2e per wafer)

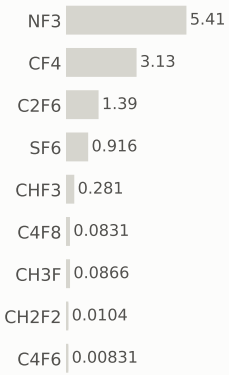
steps by process area



materials deposited (g)



gases emitted (g)



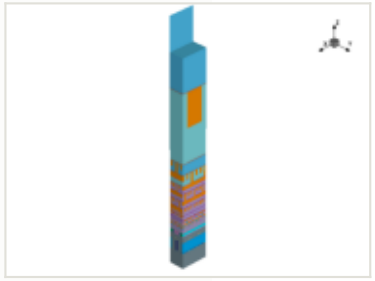
Step #8010: RIE

clean · RIE

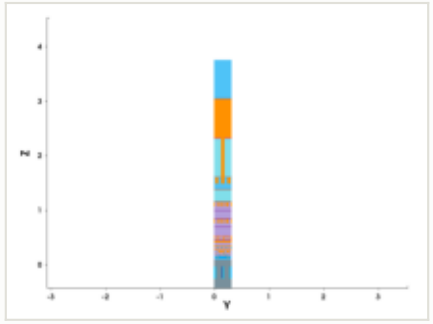
1.04	685.6	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

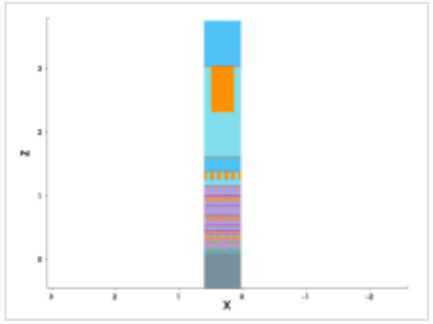
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

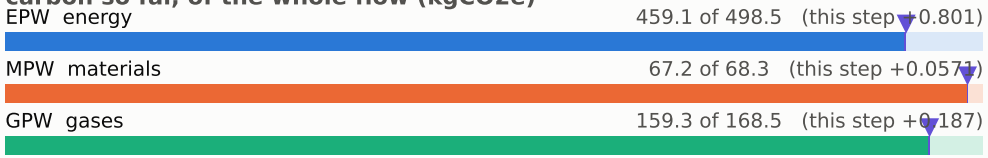
no mask for this step

this step

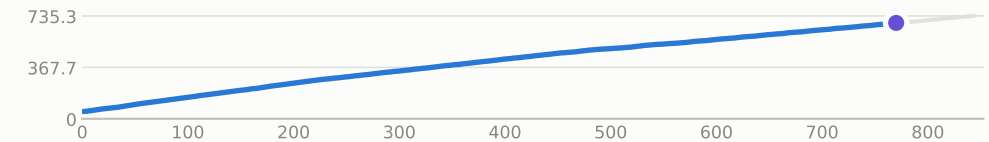
process area	RIE	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

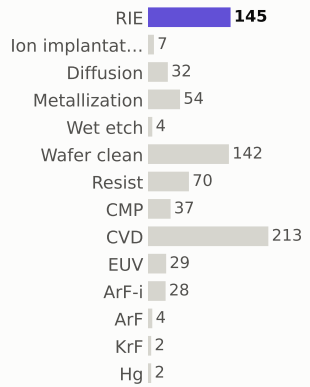
carbon so far, of the whole flow (kgCO2e)



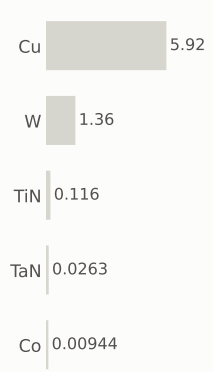
embodied carbon up to each step (kgCO2e per wafer)



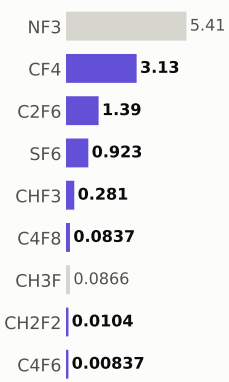
steps by process area



materials deposited (g)



gases emitted (g)



Step #8020: Wafer_clean

clean · Wafer_clean

0.169	685.8	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

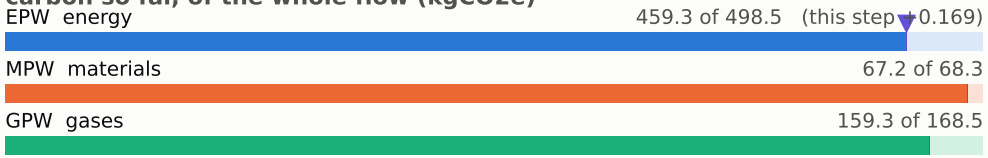
zoomed to fit

this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

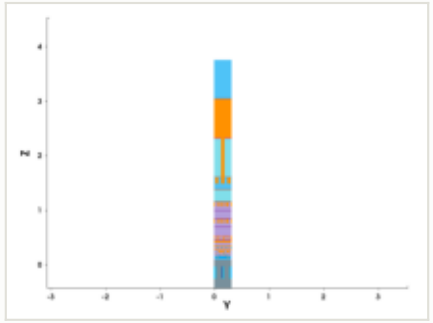
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

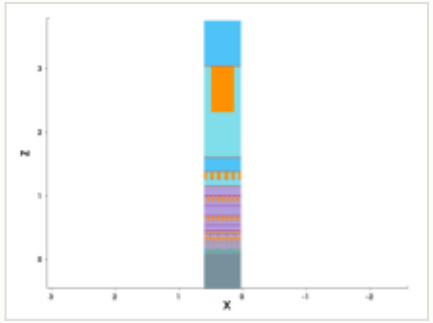


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



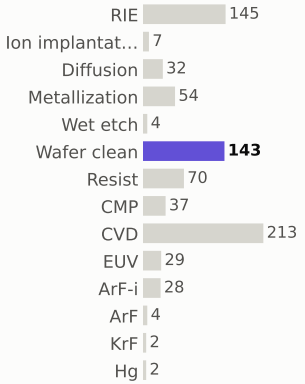
y cut at 0.184 µm



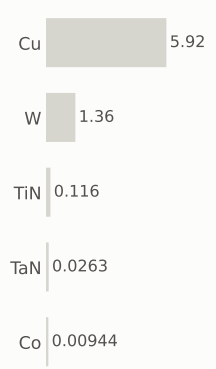
mask for this step

no mask for this step

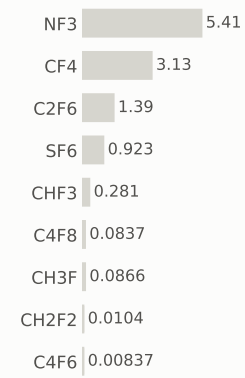
steps by process area



materials deposited (g)



gases emitted (g)



Step #8030: Hg

litho_patterning · Hg

0.199	686.0	0.027%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

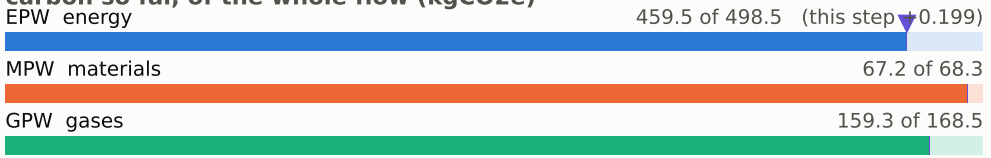
3D structure after this step

zoomed to fit

this step

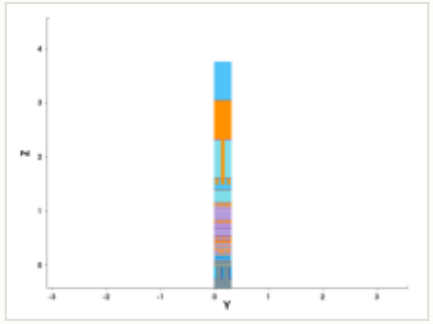
process area	Hg	EPW	0.199 kgCO2e
material	—	EPW tool	0.142 kgCO2e
thickness	—	EPW overhead	0.0568 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.33 kWh + 0.132 overhead	embodied carbon	**0.199 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

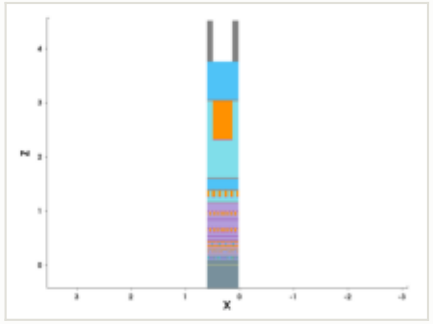


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm

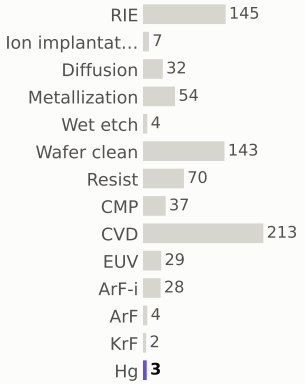


y cut at 0.184 µm

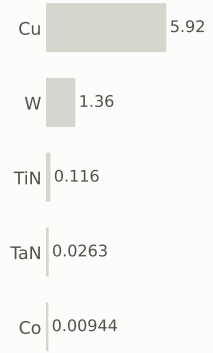


mask: layer 82:0 (GDSII)

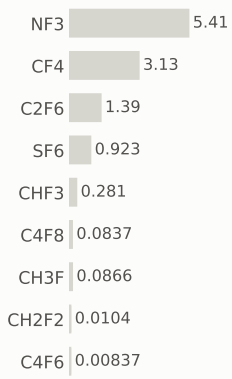
steps by process area



materials deposited (g)



gases emitted (g)



Step #8040: RIE

etch_anisotropic · RIE

1.04	687.0	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

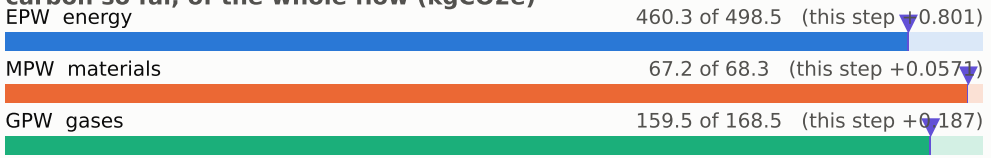
zoomed to fit

this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

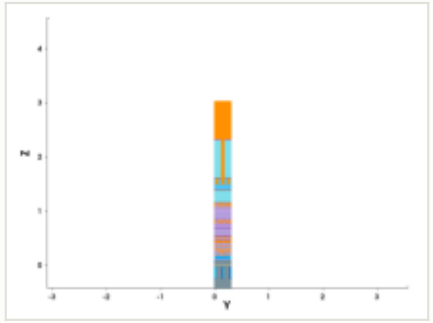
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

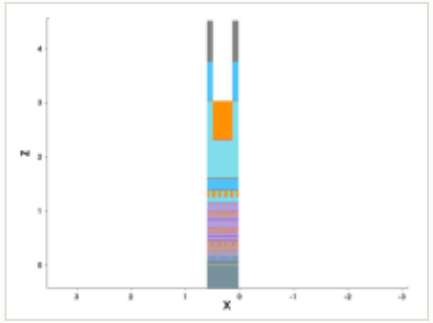


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



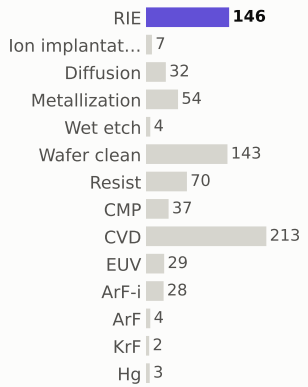
y cut at 0.184 µm



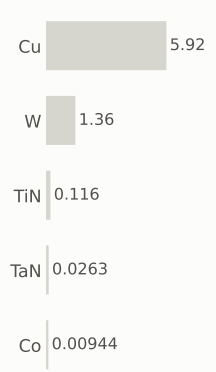
mask for this step

no mask for this step

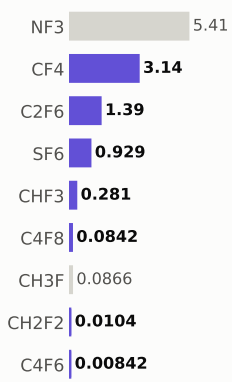
steps by process area



materials deposited (g)



gases emitted (g)



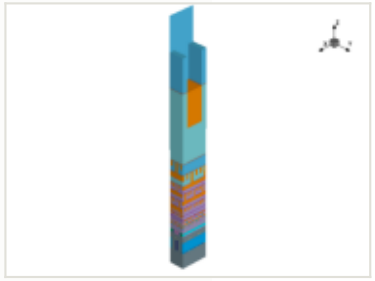
Step #8050: Wafer_clean

etch_strip · Wafer_clean

0.169	687.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

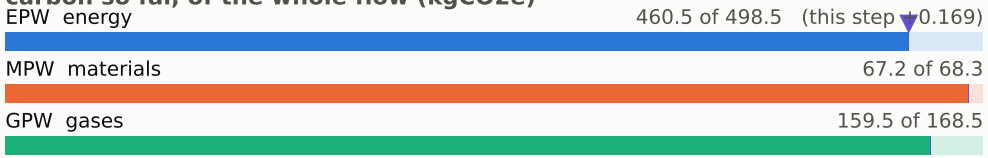


this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

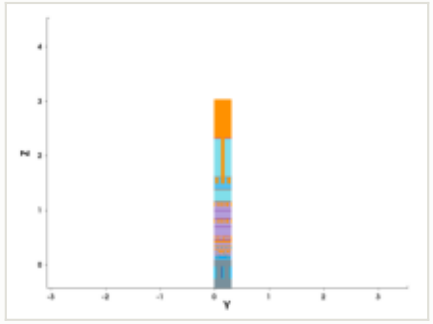
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

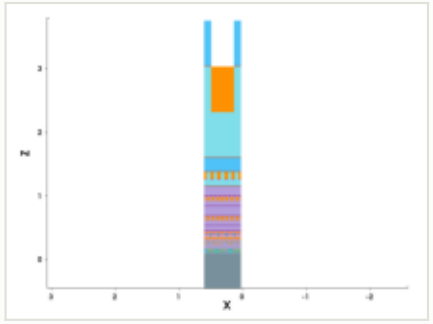


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



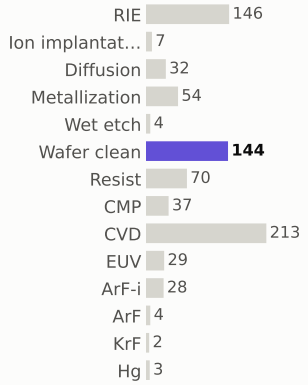
y cut at 0.184 µm



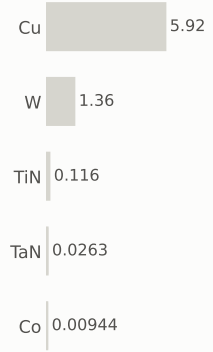
mask for this step

no mask for this step

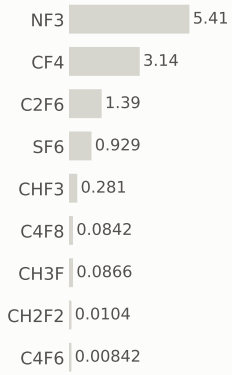
steps by process area



materials deposited (g)



gases emitted (g)



Step #8060: Metallization

depo_ellipsoidal · Metallization

1.2	688.4	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

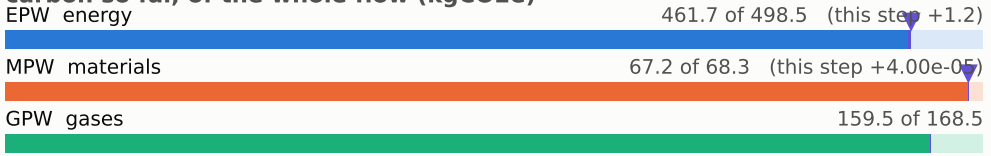
zoomed to fit

this step

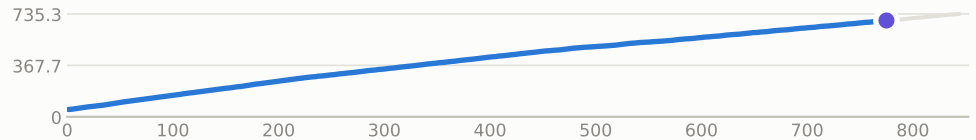
process area	Metallization
material	TaN
thickness	0.0016 µm
material GWP	25.8159 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.00e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

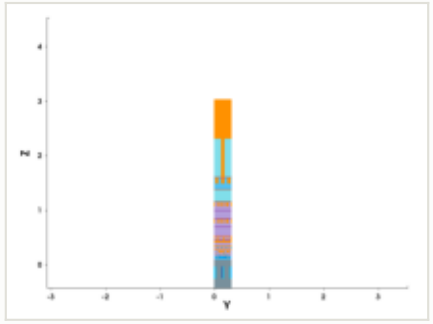
carbon so far, of the whole flow (kgCO2e)



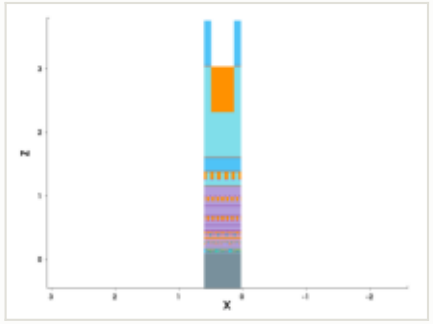
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



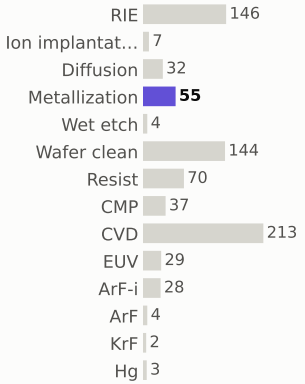
y cut at 0.184 µm



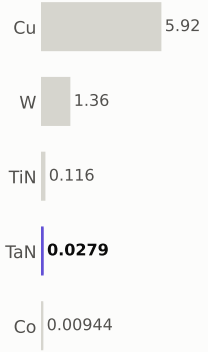
mask for this step

no mask for this step

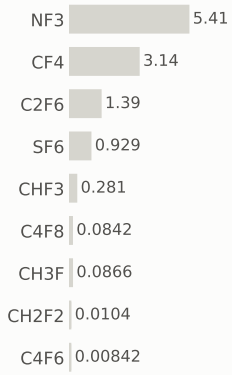
steps by process area



materials deposited (g)



gases emitted (g)



Step #8070: Metallization

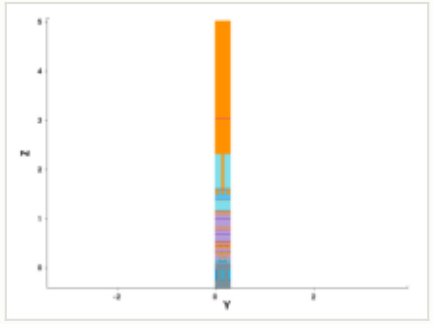
depo_fill · Metallization

1.21	689.6	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

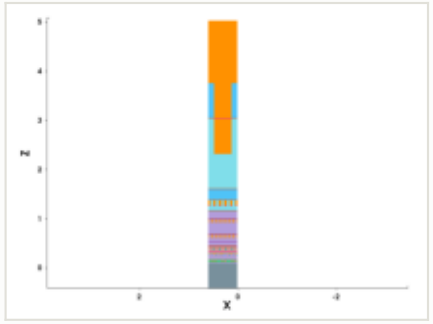
3D structure after this step

zoomed to fit

x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

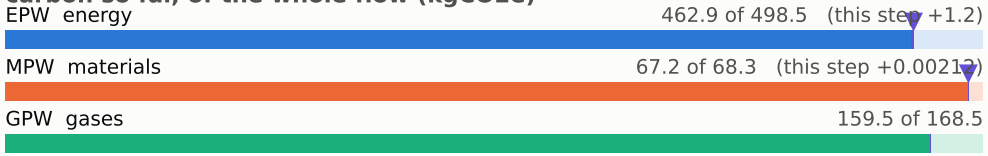
no mask for this step

this step

process area	Metallization
material	Cu
thickness	0.55 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

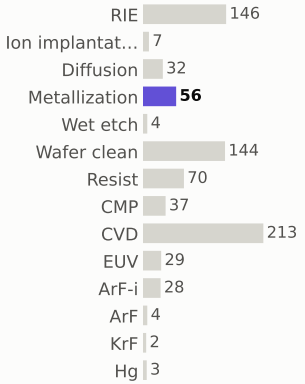
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00212 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

carbon so far, of the whole flow (kgCO2e)



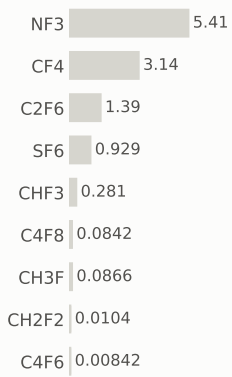
embodied carbon up to each step (kgCO2e per wafer)

steps by process area



materials deposited (g)

gases emitted (g)



Step #8080: Diffusion

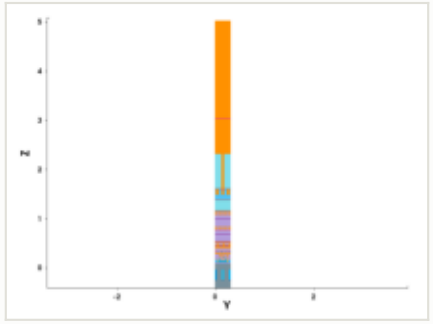
treatment · Diffusion

0.801	690.4	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

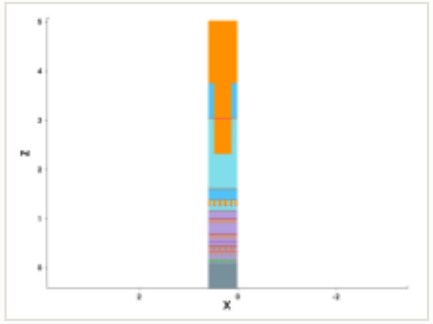
3D structure after this step

zoomed to fit

x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

no mask for this step

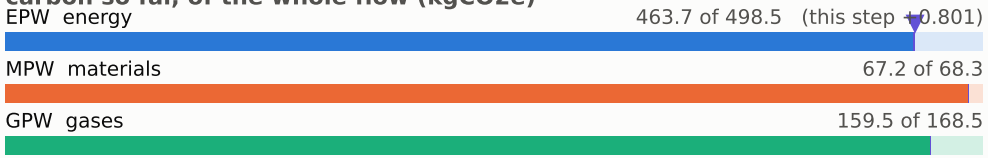
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

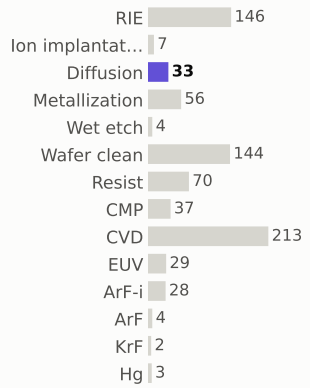
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

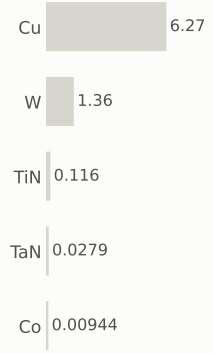


embodied carbon up to each step (kgCO2e per wafer)

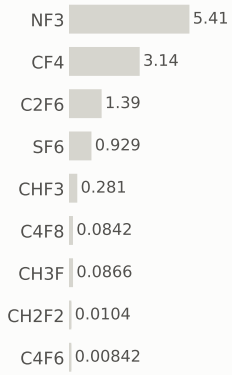
steps by process area



materials deposited (g)



gases emitted (g)



Step #8090: CMP

etch_cmp · CMP

0.169	690.6	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

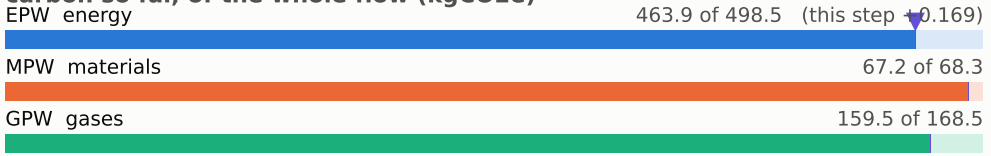
3D structure after this step

zoomed to fit

this step

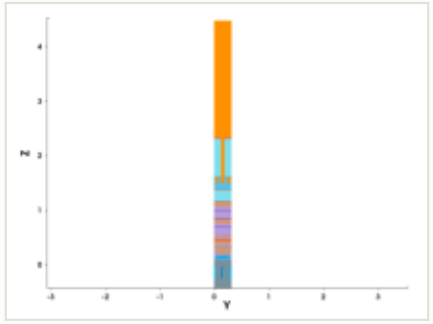
process area	CMP	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

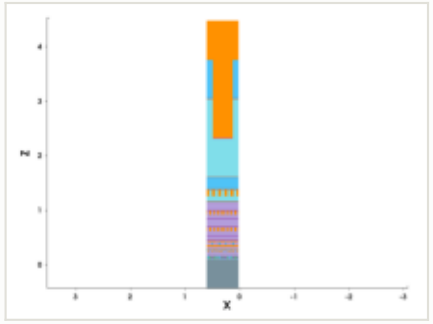


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



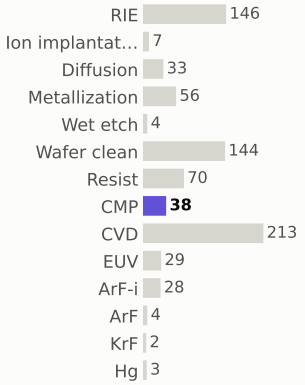
y cut at 0.184 µm



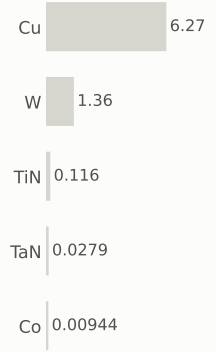
mask for this step

no mask for this step

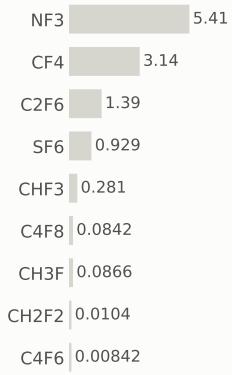
steps by process area



materials deposited (g)



gases emitted (g)



Step #8110: CVD

depo_anisotropic · CVD

1.31	691.9	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

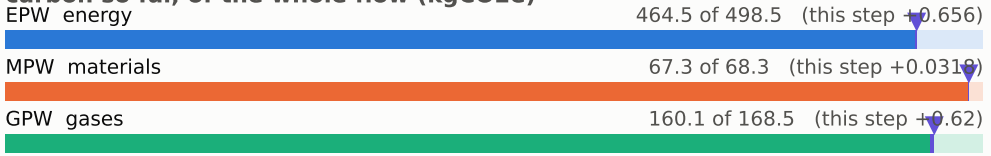
3D structure after this step

zoomed to fit

this step

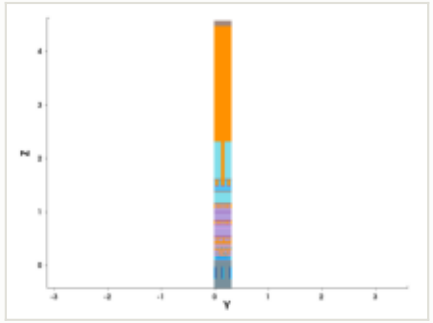
process area	CVD	EPW	0.656 kgCO2e
material	—	EPW tool	0.469 kgCO2e
thickness	0.1 µm	EPW overhead	0.187 kgCO2e
material GWP	—	MPW	0.0318 kgCO2e (gas 0.0318)
input gases	NF3 2.54 g	GPW	0.62 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

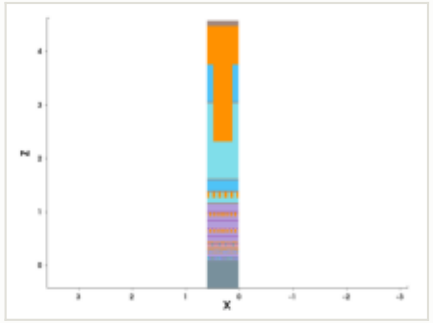


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



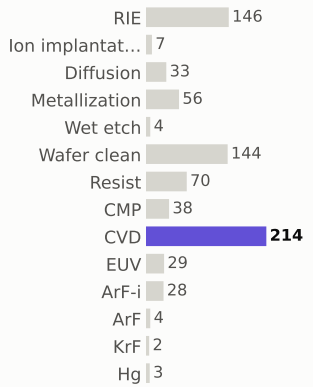
y cut at 0.184 µm



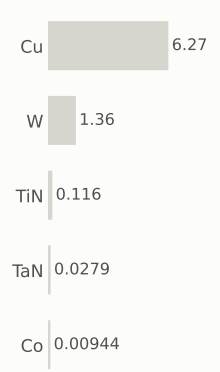
mask for this step

no mask for this step

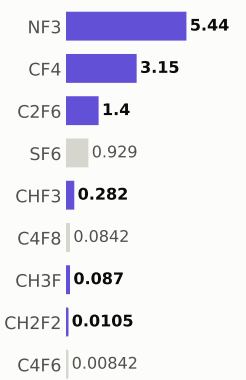
steps by process area



materials deposited (g)



gases emitted (g)



Step #8120: CVD

depo_anisotropic · CVD

1.31	693.2	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

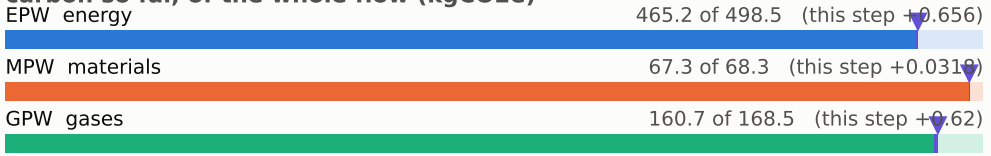
3D structure after this step

zoomed to fit

this step

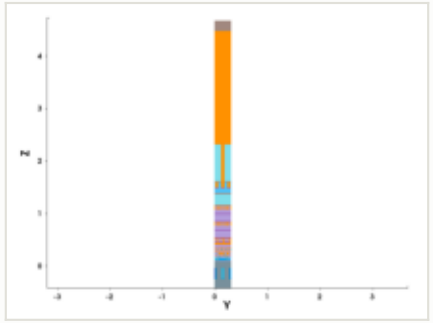
process area	CVD	EPW	0.656 kgCO2e
material	—	EPW tool	0.469 kgCO2e
thickness	0.1 µm	EPW overhead	0.187 kgCO2e
material GWP	—	MPW	0.0318 kgCO2e (gas 0.0318)
input gases	NF3 2.54 g	GPW	0.62 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

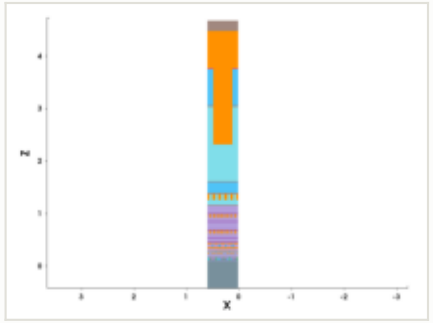


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



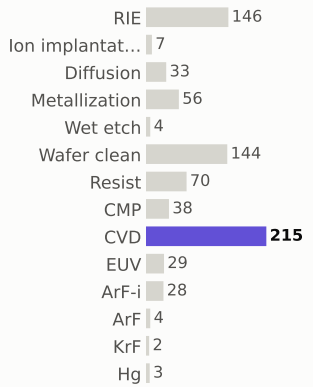
y cut at 0.184 µm



mask for this step

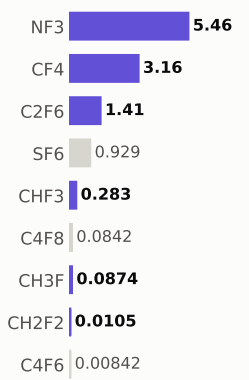
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



Step #8130: CMP

etch_cmp · CMP

0.169	693.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

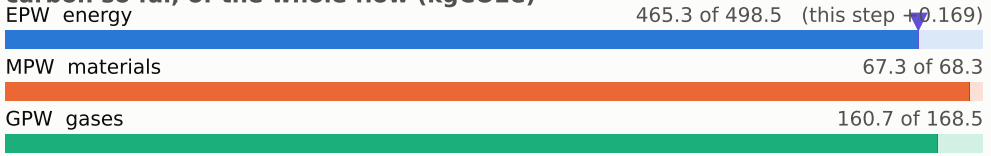
3D structure after this step

zoomed to fit

this step

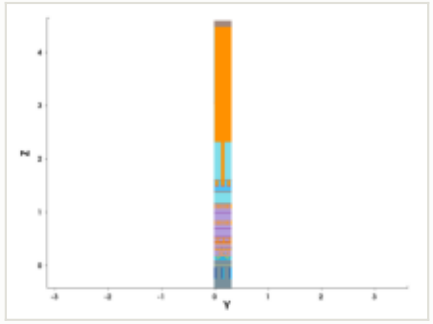
process area	CMP	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

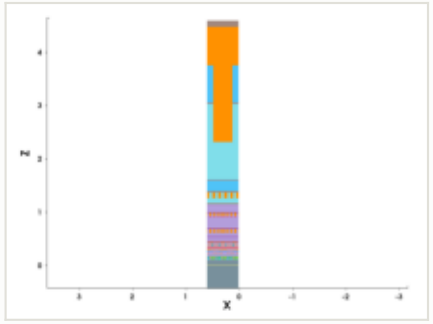


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



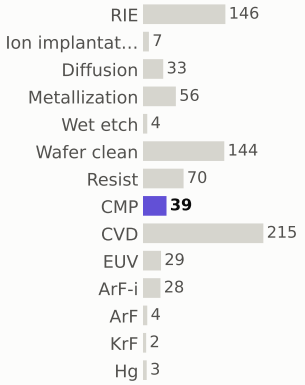
y cut at 0.184 µm



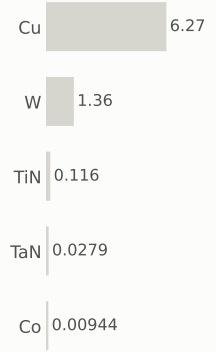
mask for this step

no mask for this step

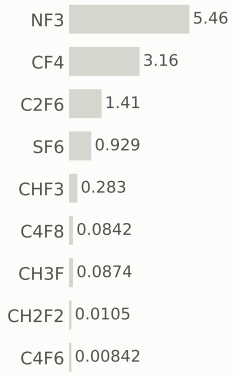
steps by process area



materials deposited (g)



gases emitted (g)



Step #8140: CMP

etch_cmp · CMP

0.169	693.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

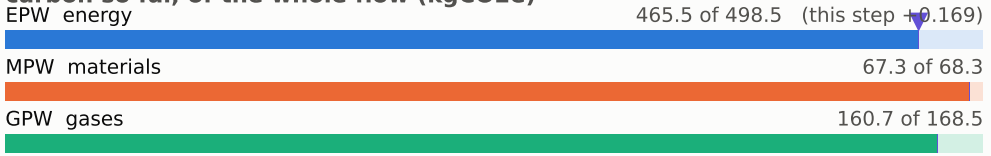
3D structure after this step

zoomed to fit

this step

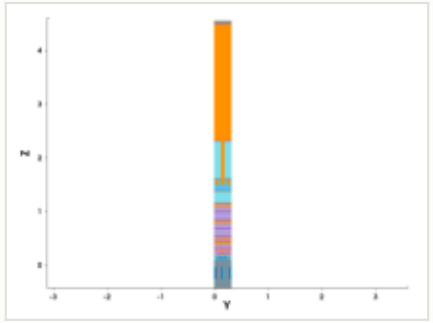
process area	CMP	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

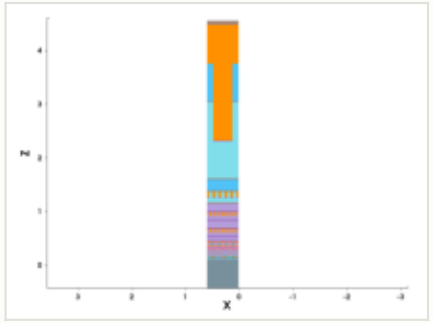


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



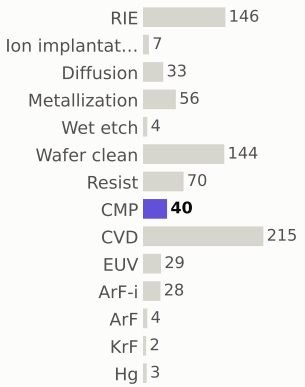
y cut at 0.184 µm



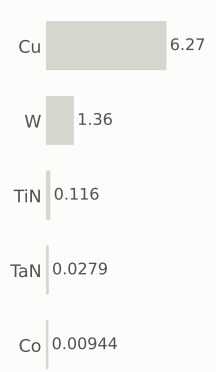
mask for this step

no mask for this step

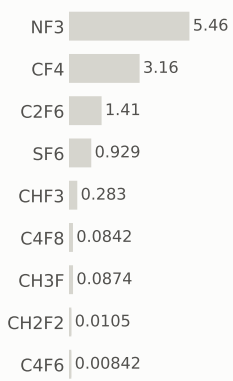
steps by process area



materials deposited (g)



gases emitted (g)



Step #8150: Wafer

bond_structure · Wafer

0	693.5	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

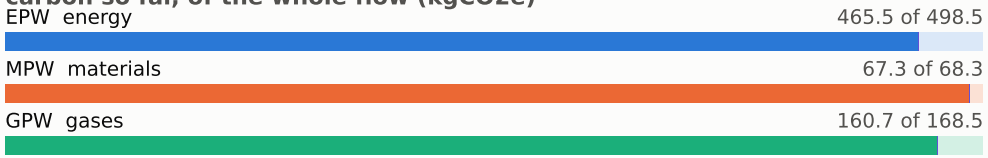
zoomed to fit

this step

process area	Wafer	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

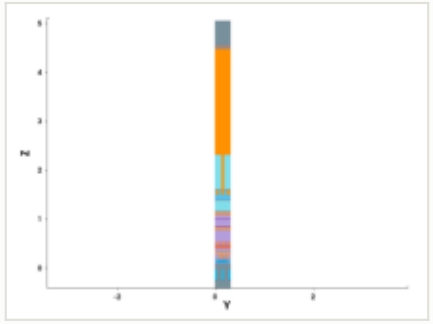
Wafer step: not counted as a process step (uncounted_process_areas)

carbon so far, of the whole flow (kgCO2e)

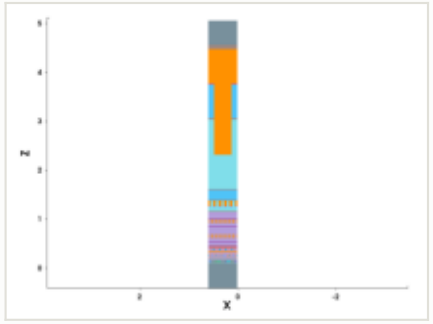


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



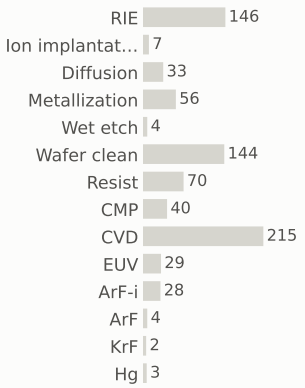
y cut at 0.184 µm



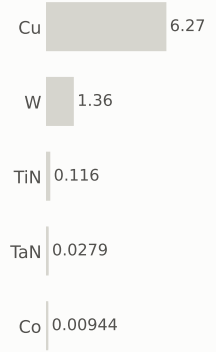
mask for this step

no mask for this step

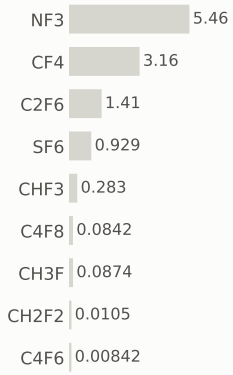
steps by process area



materials deposited (g)



gases emitted (g)



Step #8170: Wafer

flip_structure · Wafer

0	693.5	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

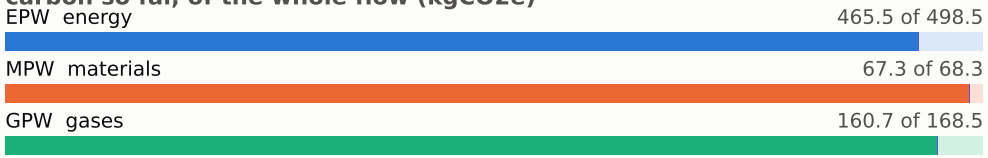
zoomed to fit

this step

process area	Wafer	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

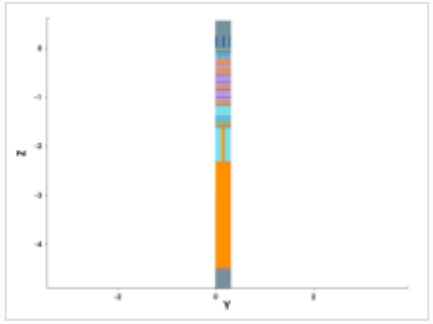
Wafer step: not counted as a process step (uncounted_process_areas)

carbon so far, of the whole flow (kgCO2e)

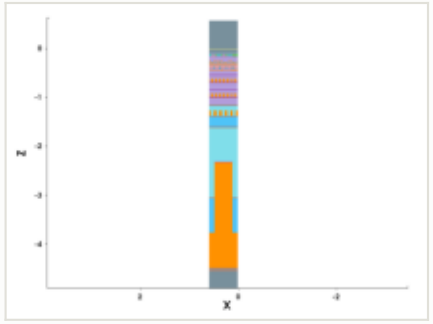


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



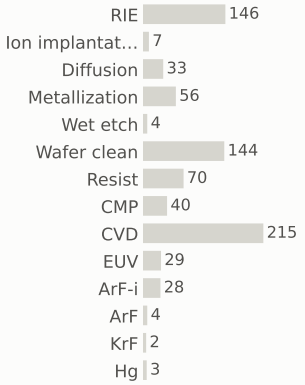
y cut at 0.184 µm



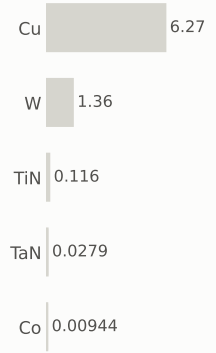
mask for this step

no mask for this step

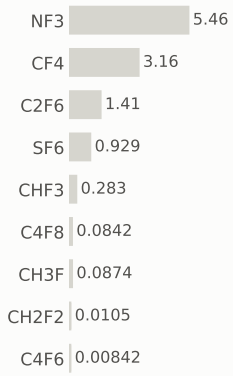
steps by process area



materials deposited (g)



gases emitted (g)



Step #8180: CMP

etch_cmp · CMP

0.169	693.7	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

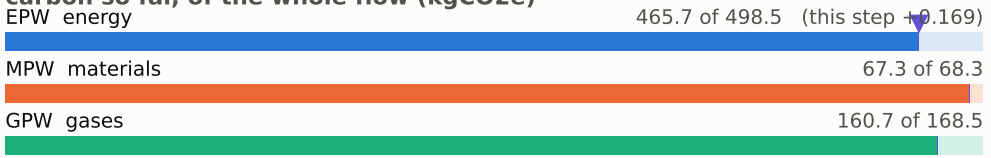
zoomed to fit

this step

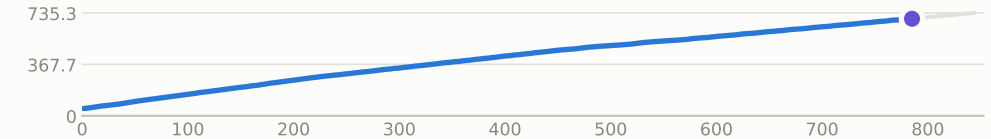
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

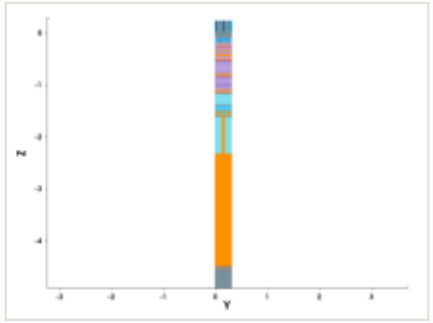
carbon so far, of the whole flow (kgCO2e)



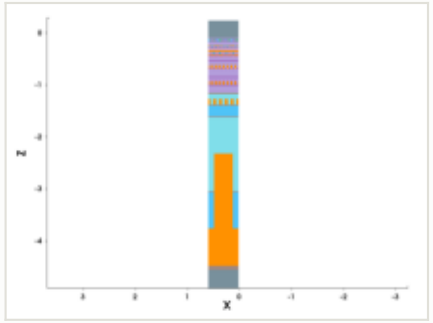
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



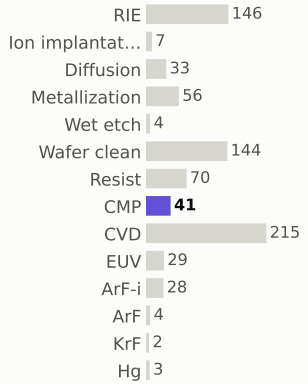
y cut at 0.184 µm



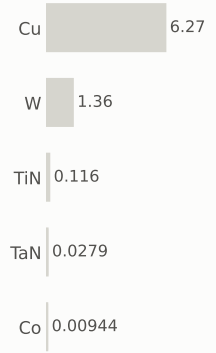
mask for this step

no mask for this step

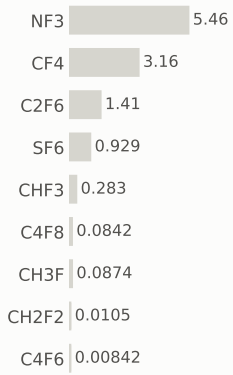
steps by process area



materials deposited (g)



gases emitted (g)



Step #8190: CMP

etch_cmp · CMP

0.169	693.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

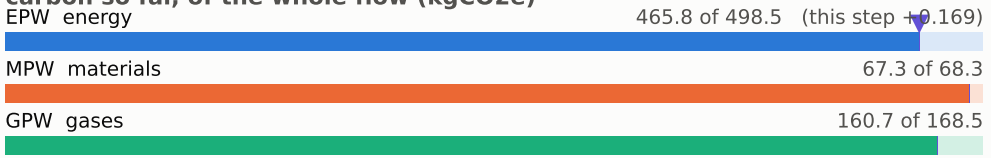
zoomed to fit

this step

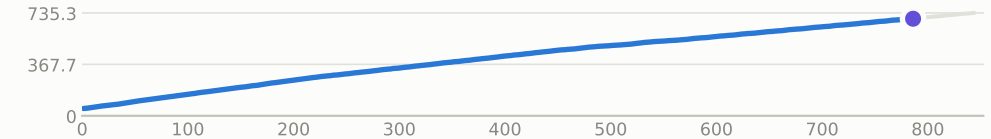
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

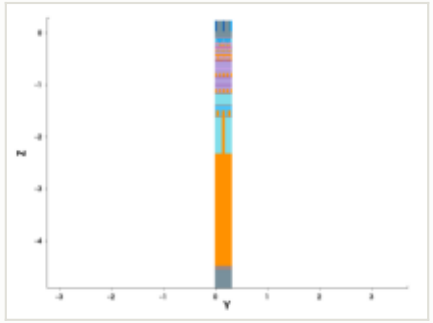
carbon so far, of the whole flow (kgCO2e)



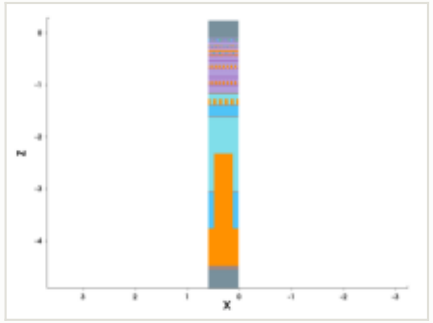
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



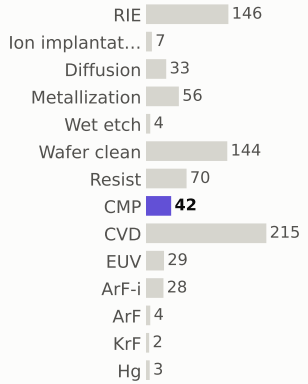
y cut at 0.184 µm



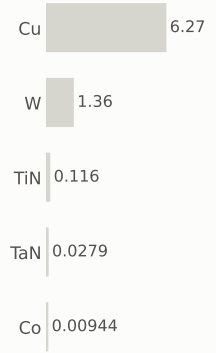
mask for this step

no mask for this step

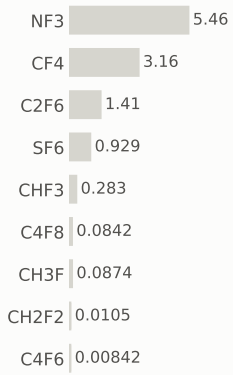
steps by process area



materials deposited (g)



gases emitted (g)



Step #8200: CVD

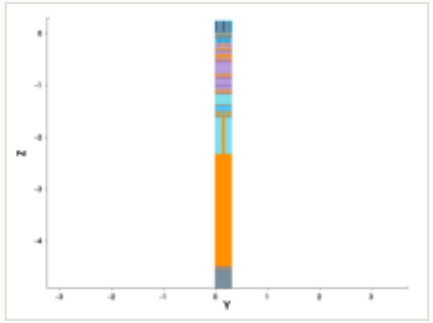
depo_anisotropic · CVD

1.31	695.2	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

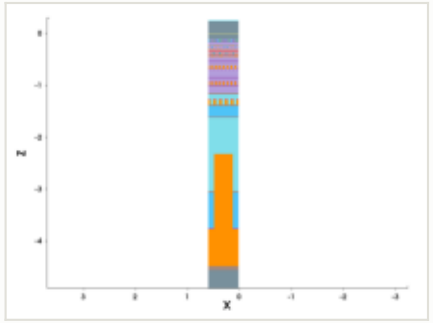
3D structure after this step

zoomed to fit

x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

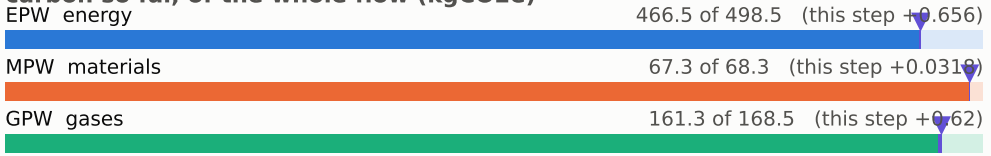
no mask for this step

this step

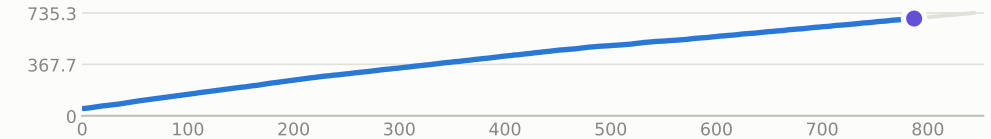
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

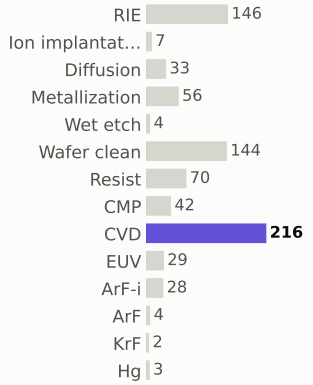
carbon so far, of the whole flow (kgCO2e)



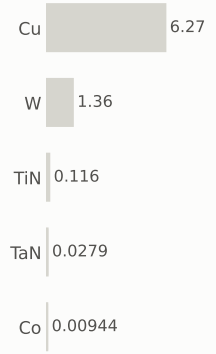
embodied carbon up to each step (kgCO2e per wafer)



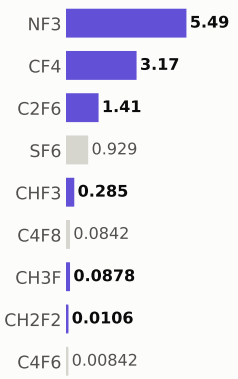
steps by process area



materials deposited (g)



gases emitted (g)



Step #8220: RIE

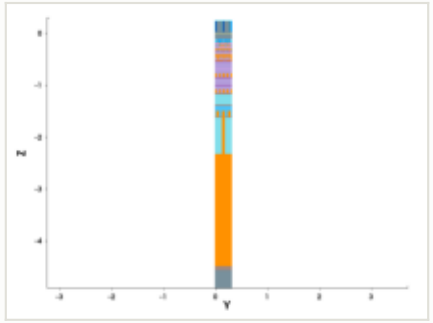
clean · RIE

1.04	696.2	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

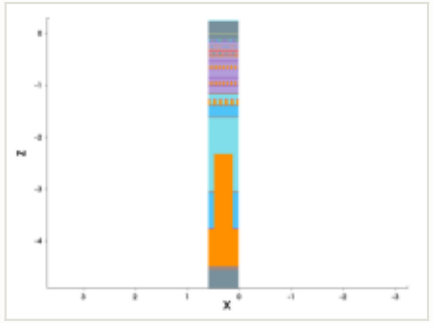
3D structure after this step

zoomed to fit

x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

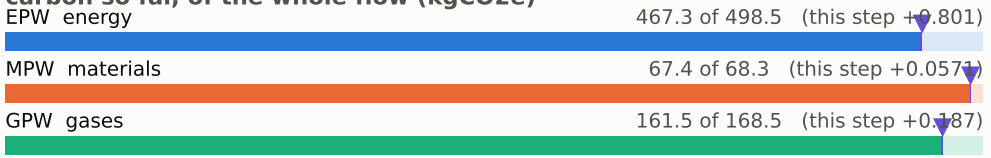
no mask for this step

this step

process area	RIE	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

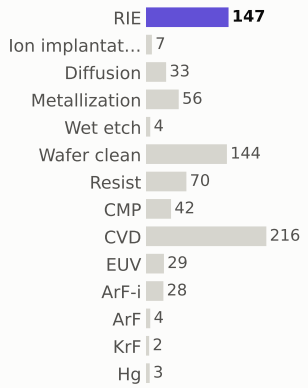
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

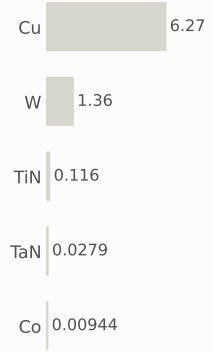


embodied carbon up to each step (kgCO2e per wafer)

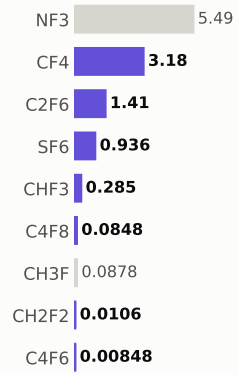
steps by process area



materials deposited (g)



gases emitted (g)



Step #8230: Wafer_clean

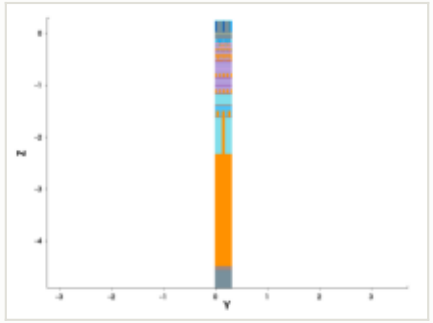
clean · Wafer_clean

0.169	696.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

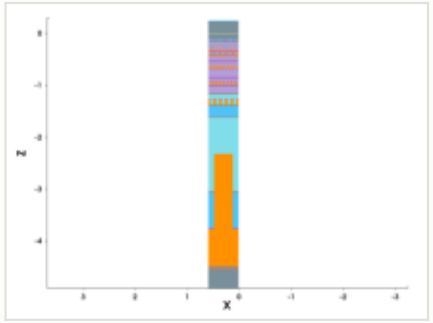
3D structure after this step

zoomed to fit

x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

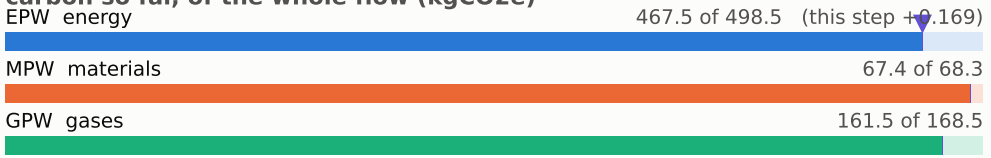
no mask for this step

this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

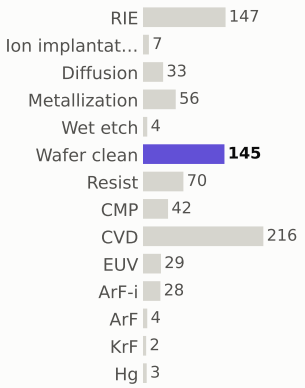
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

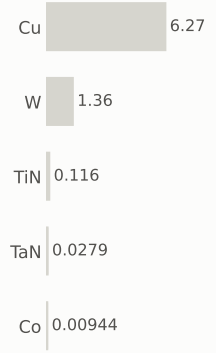


embodied carbon up to each step (kgCO2e per wafer)

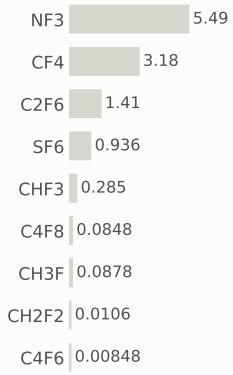
steps by process area



materials deposited (g)



gases emitted (g)



Step #8240: KrF

litho_patterning · KrF

0.199	696.6	0.027%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

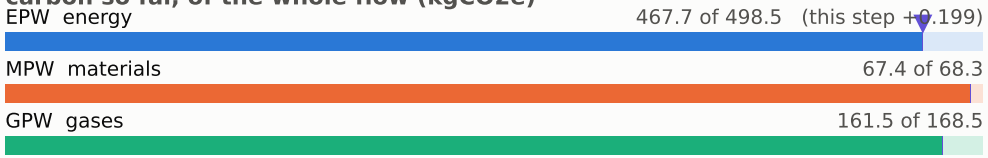
zoomed to fit

this step

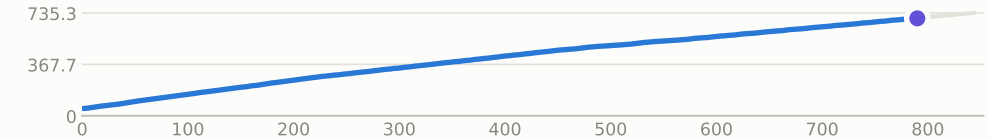
process area	KrF
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

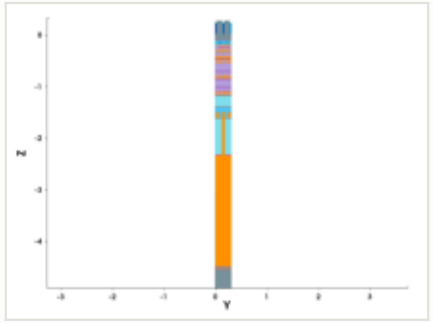
carbon so far, of the whole flow (kgCO2e)



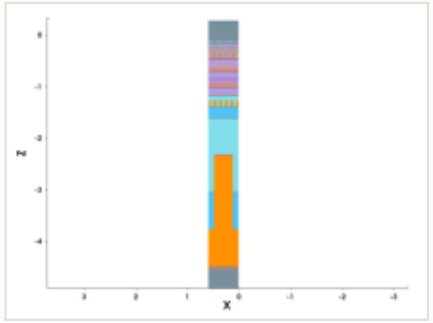
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm

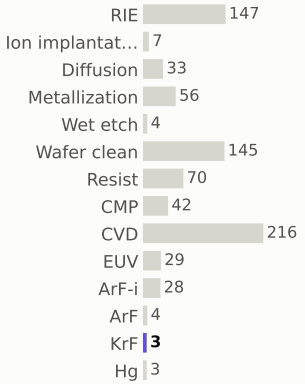


y cut at 0.184 µm

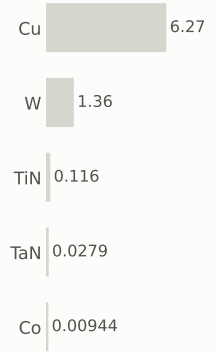


mask: layer 8:0 (GDSII)

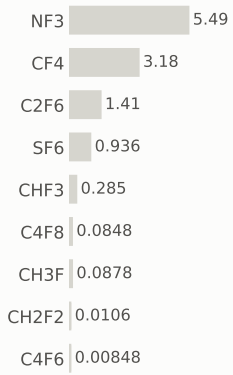
steps by process area



materials deposited (g)



gases emitted (g)



Step #8250: RIE

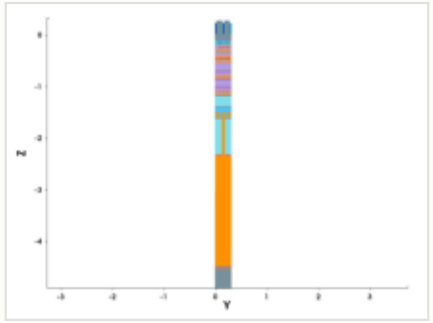
etch_anisotropic · RIE

1.04	697.6	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

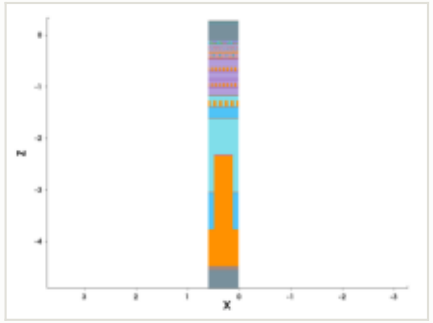
3D structure after this step

zoomed to fit

x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

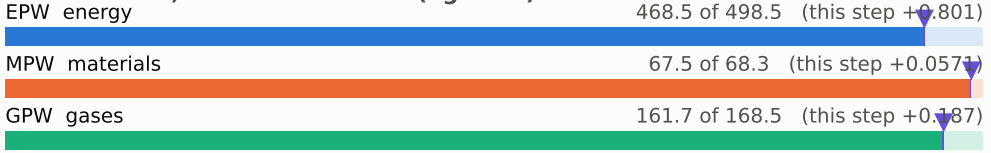
no mask for this step

this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

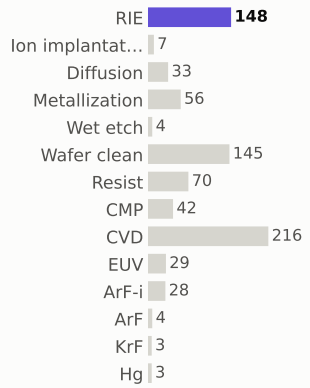
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

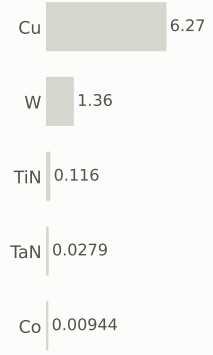


embodied carbon up to each step (kgCO2e per wafer)

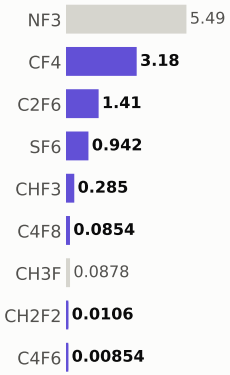
steps by process area



materials deposited (g)



gases emitted (g)



Step #8260: Wafer_clean

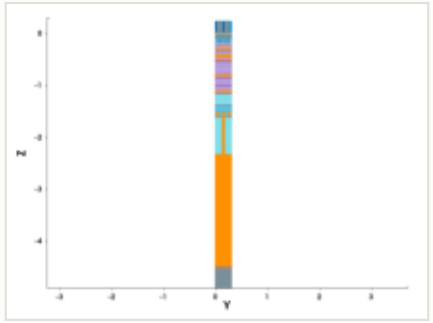
etch_strip · Wafer_clean

0.169	697.8	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

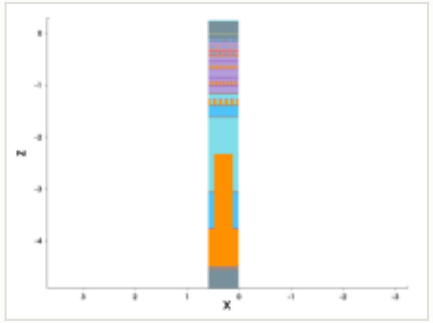
3D structure after this step

zoomed to fit

x cut at 0.294 µm



y cut at 0.184 µm



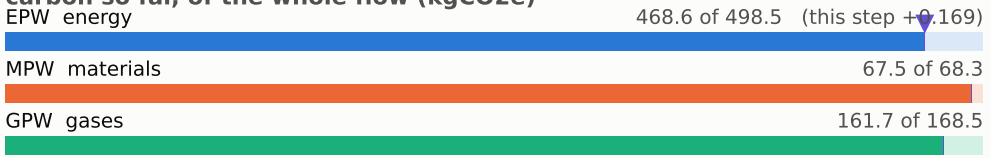
mask for this step

no mask for this step

this step

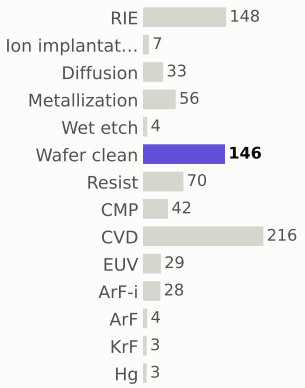
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

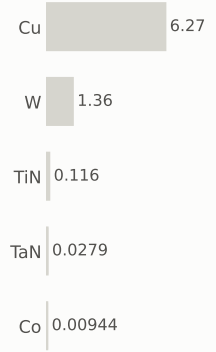


embodied carbon up to each step (kgCO2e per wafer)

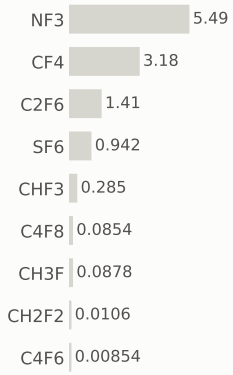
steps by process area



materials deposited (g)



gases emitted (g)



Step #8280: CVD

depo_fill · CVD

1.31	699.1	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

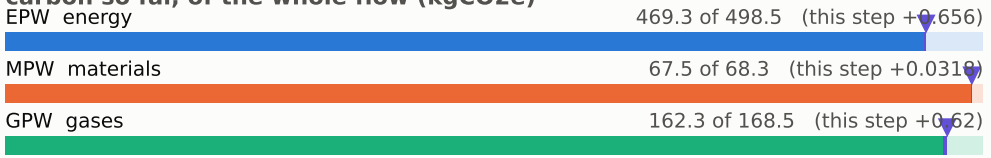
zoomed to fit

this step

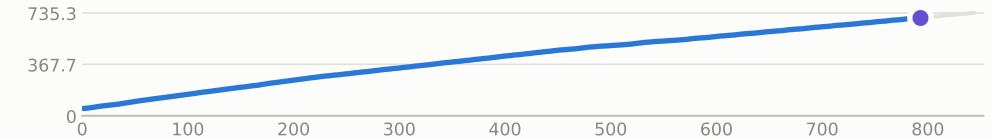
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

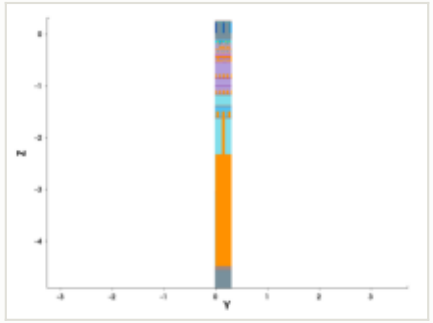
carbon so far, of the whole flow (kgCO2e)



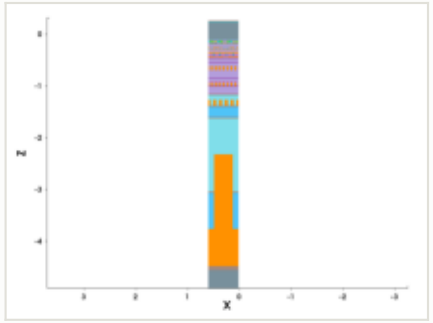
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



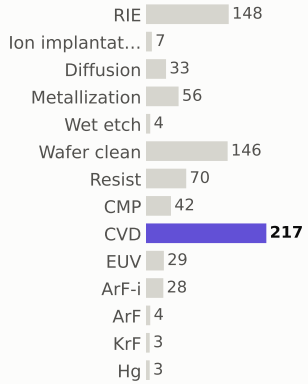
y cut at 0.184 µm



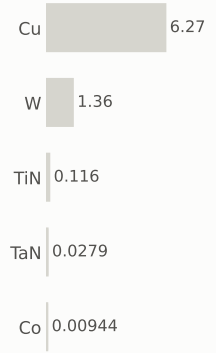
mask for this step

no mask for this step

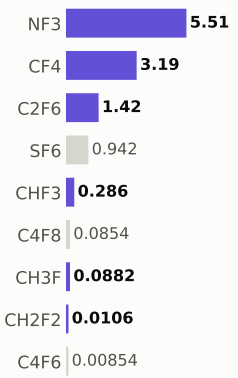
steps by process area



materials deposited (g)



gases emitted (g)



Step #8290: CVD

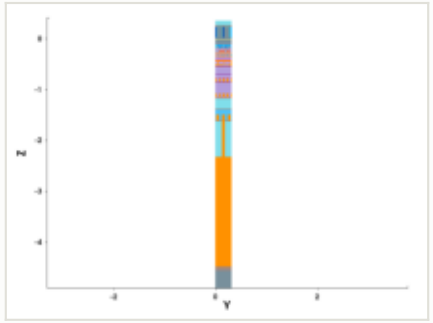
depo_fill · CVD

1.31	700.4	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

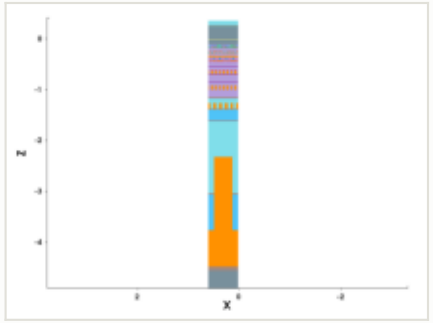
3D structure after this step

zoomed to fit

x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

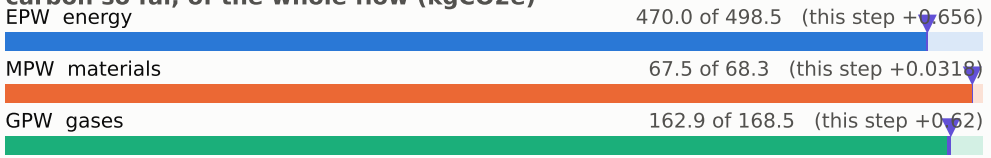
no mask for this step

this step

process area	CVD
material	—
thickness	0.092 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

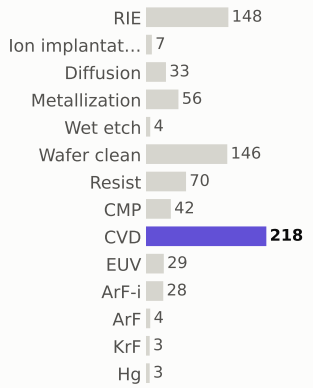
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

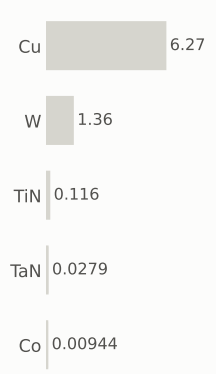


embodied carbon up to each step (kgCO2e per wafer)

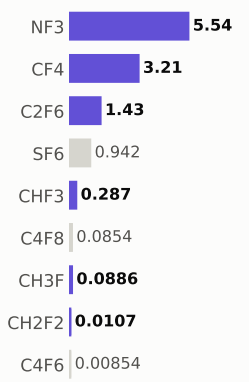
steps by process area



materials deposited (g)



gases emitted (g)



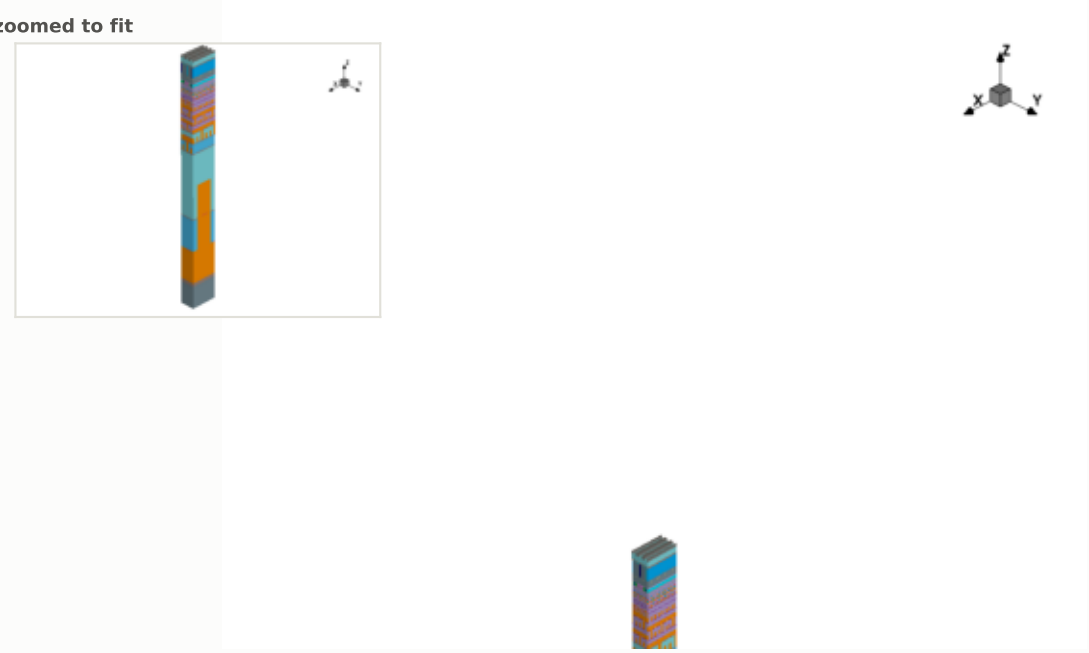
Step #8300: ArF

litho_patterning · ArF

0.199	700.6	0.027%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

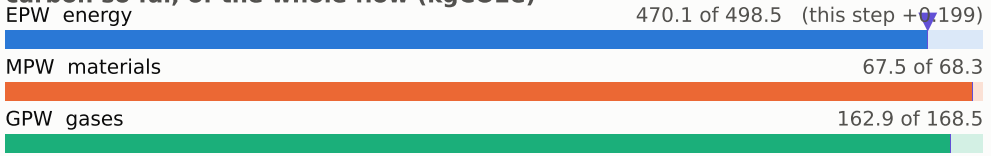
zoomed to fit



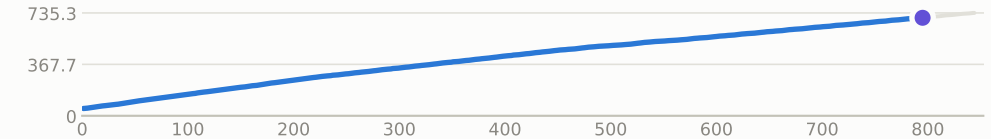
this step

process area	ArF	EPW	0.199 kgCO2e
material	—	EPW tool	0.142 kgCO2e
thickness	—	EPW overhead	0.0568 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.33 kWh + 0.132 overhead	embodied carbon	**0.199 kgCO2e**

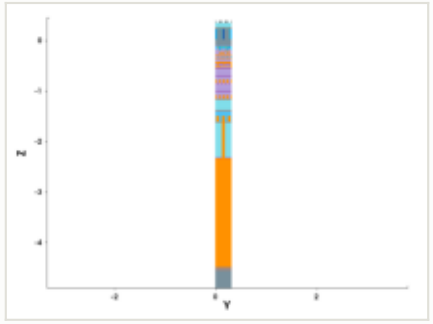
carbon so far, of the whole flow (kgCO2e)



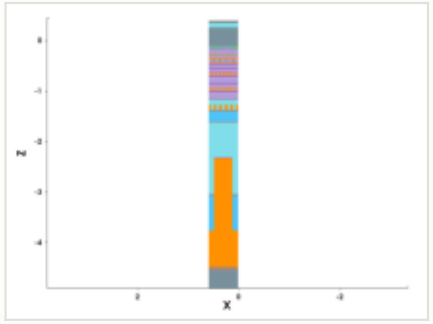
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm

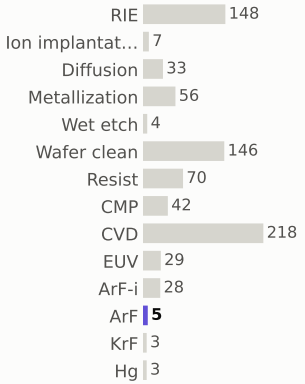


y cut at 0.184 µm

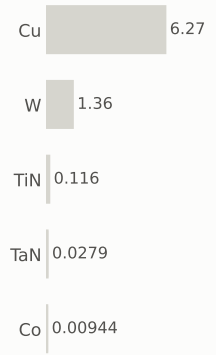


mask: layer 920:0 (GDSII)

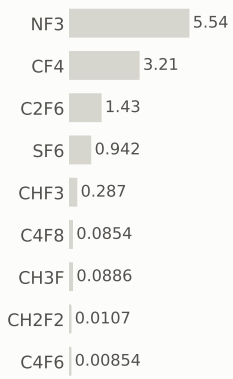
steps by process area



materials deposited (g)



gases emitted (g)



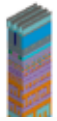
Step #8310: RIE

etch_anisotropic · RIE

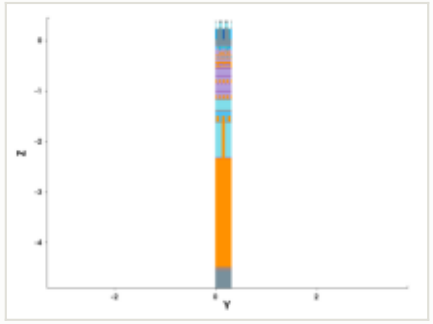
1.04	701.7	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

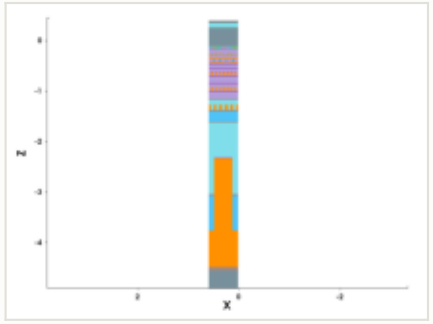
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

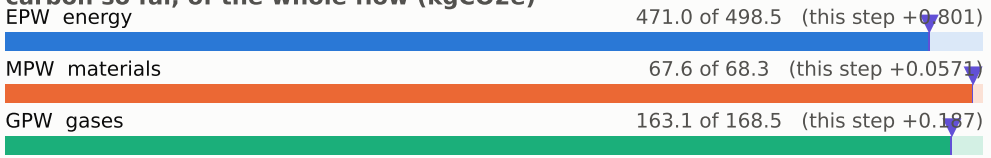
no mask for this step

this step

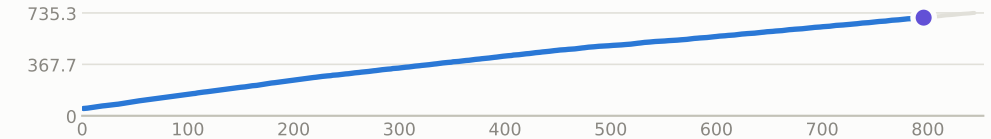
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

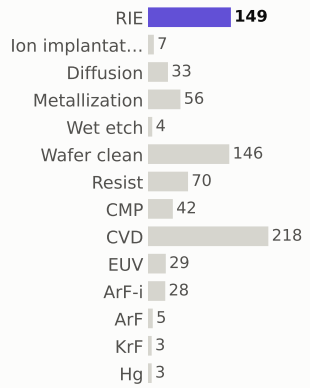
carbon so far, of the whole flow (kgCO2e)



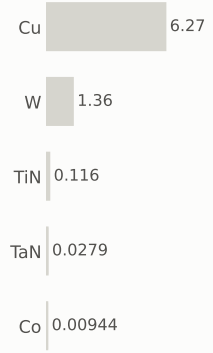
embodied carbon up to each step (kgCO2e per wafer)



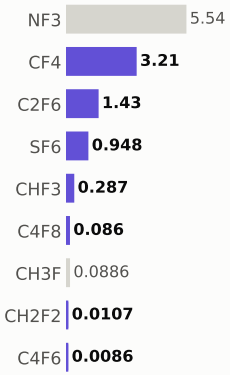
steps by process area



materials deposited (g)



gases emitted (g)



Step #8320: Wafer_clean

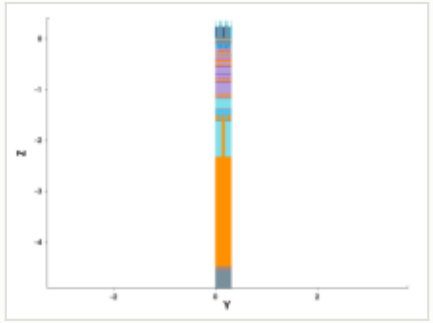
etch_strip · Wafer_clean

0.169	701.8	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

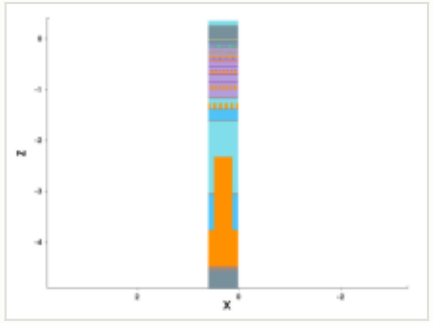
3D structure after this step

zoomed to fit

x cut at 0.294 µm



y cut at 0.184 µm



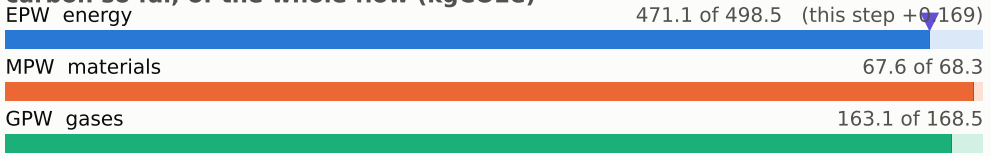
mask for this step

no mask for this step

this step

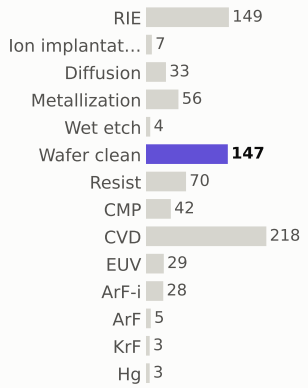
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

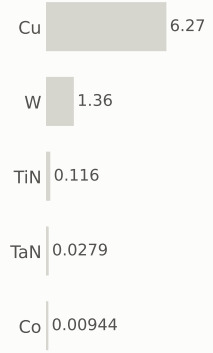


embodied carbon up to each step (kgCO2e per wafer)

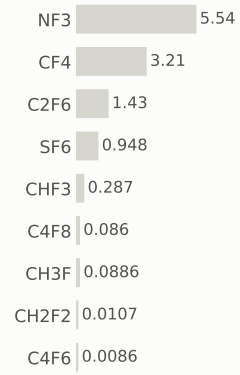
steps by process area



materials deposited (g)



gases emitted (g)



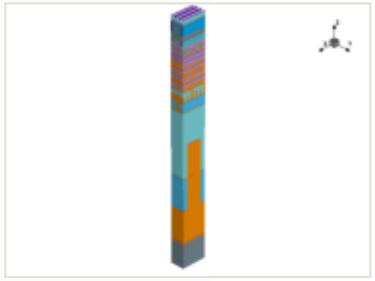
Step #8330: Metallization

depo_ellipsoidal · Metallization

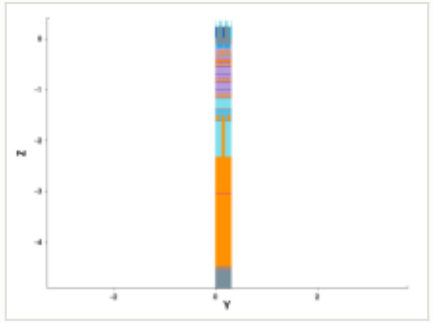
1.2	703.0	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

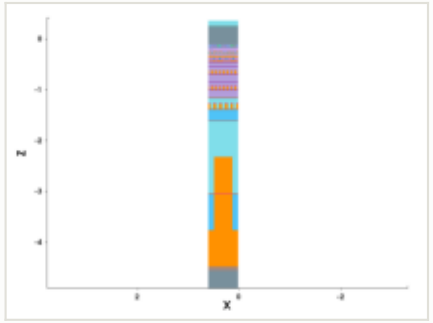
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

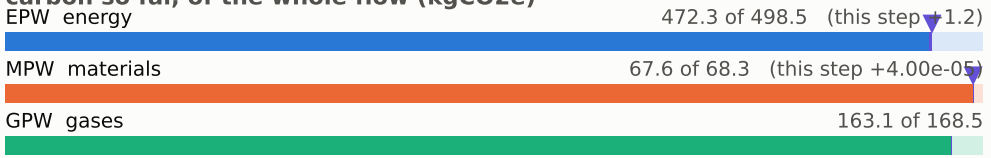
no mask for this step

this step

process area	Metallization
material	TaN
thickness	0.0016 µm
material GWP	25.8159 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

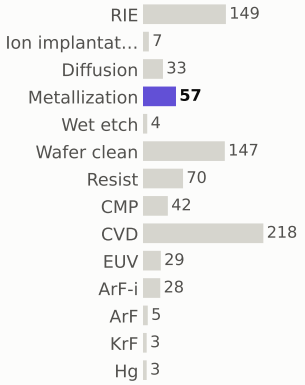
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.00e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

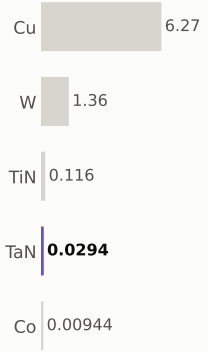


embodied carbon up to each step (kgCO2e per wafer)

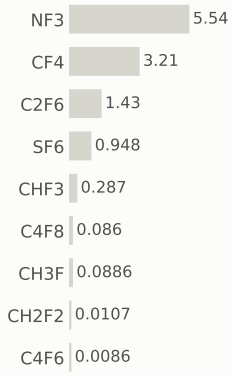
steps by process area



materials deposited (g)



gases emitted (g)



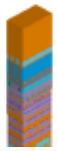
Step #8340: Metallization

depo_fill · Metallization

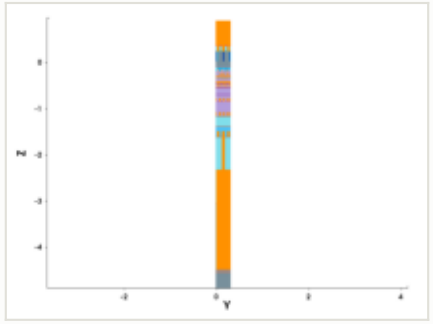
1.21	704.2	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

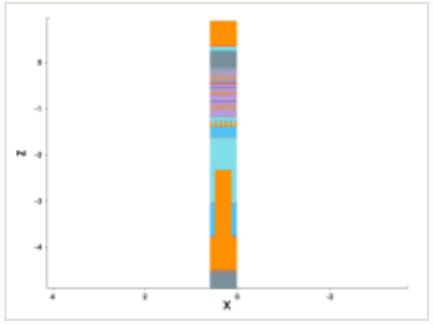
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

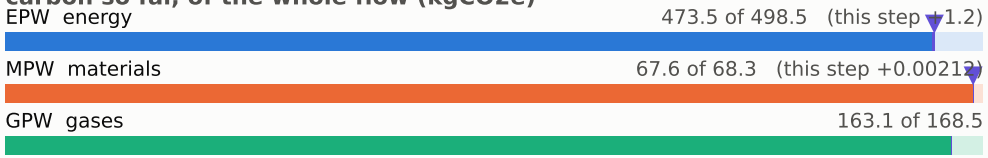
no mask for this step

this step

process area	Metallization
material	Cu
thickness	0.55 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

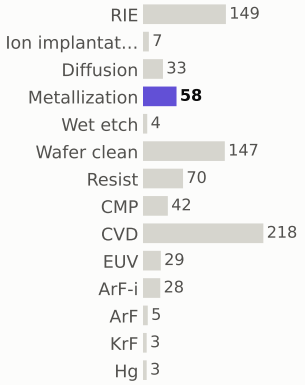
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00212 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

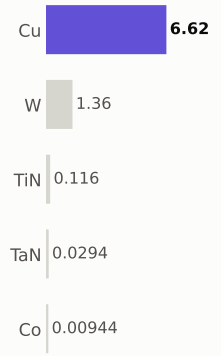


embodied carbon up to each step (kgCO2e per wafer)

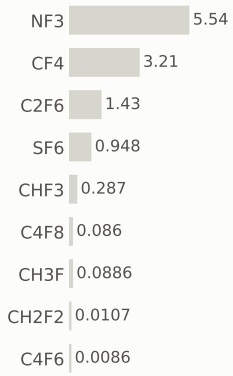
steps by process area



materials deposited (g)



gases emitted (g)



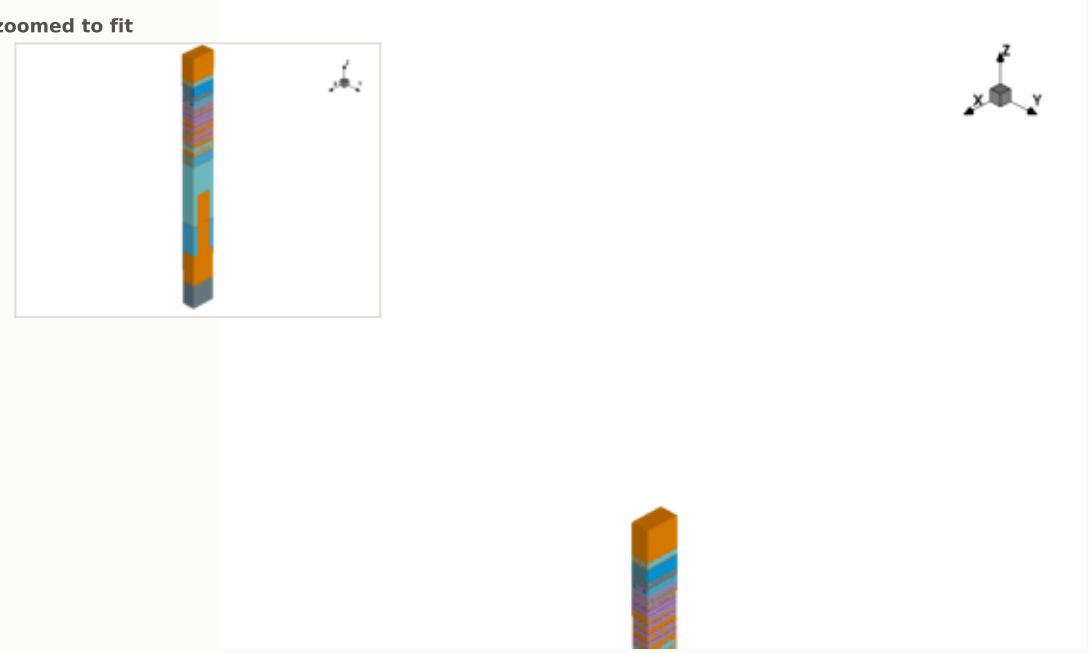
Step #8350: Diffusion

treatment · Diffusion

0.801	705.0	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

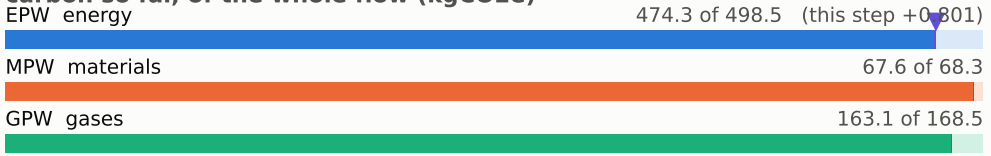


this step

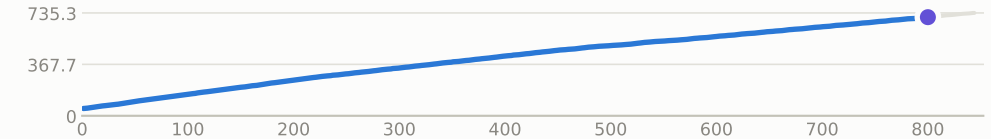
process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

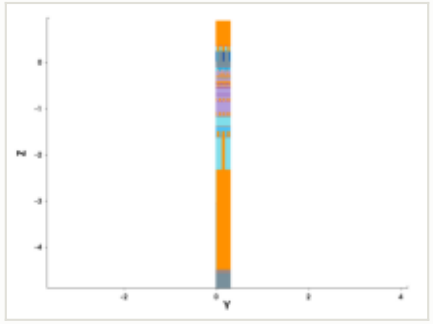
carbon so far, of the whole flow (kgCO2e)



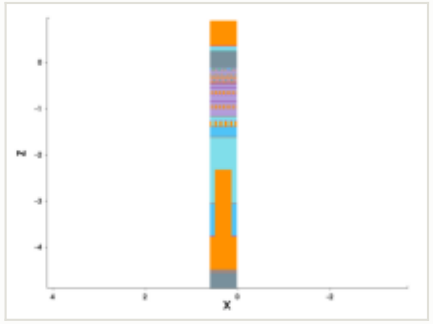
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



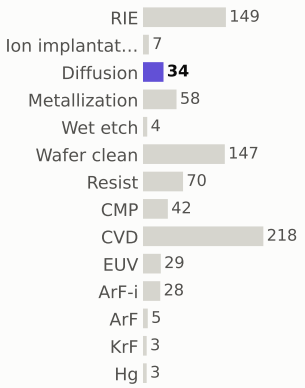
y cut at 0.184 µm



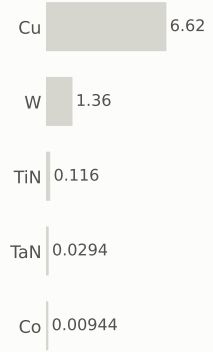
mask for this step

no mask for this step

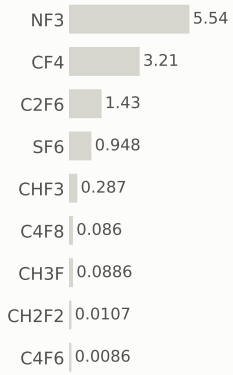
steps by process area



materials deposited (g)



gases emitted (g)



Step #8360: CMP

etch_cmp · CMP

0.169	705.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

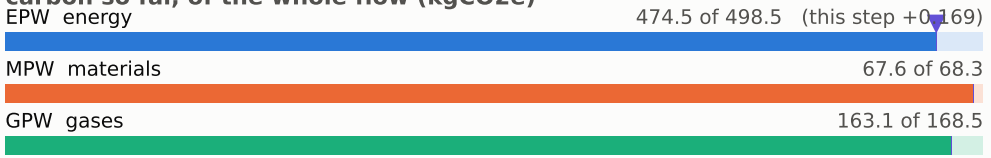
zoomed to fit

this step

process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

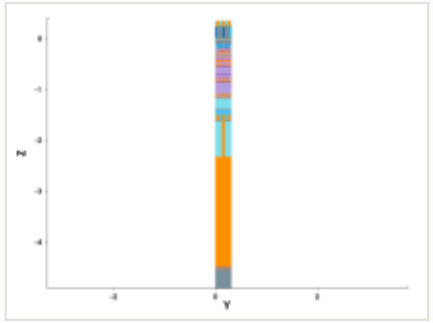
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

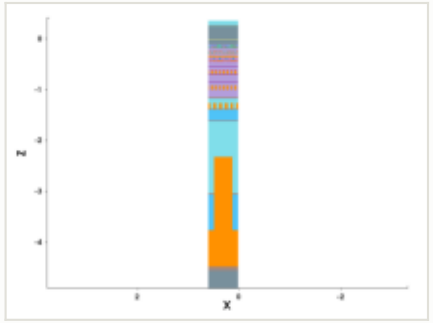


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



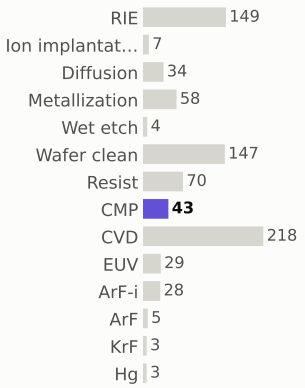
y cut at 0.184 µm



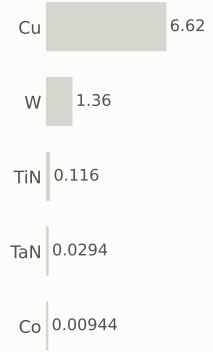
mask for this step

no mask for this step

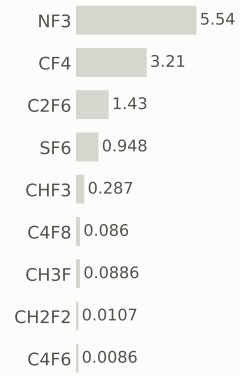
steps by process area



materials deposited (g)



gases emitted (g)



Step #8380: CVD

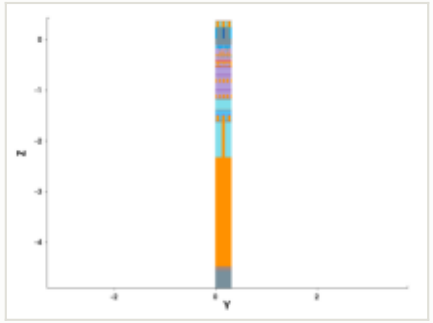
depo_fill · CVD

1.31	706.5	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

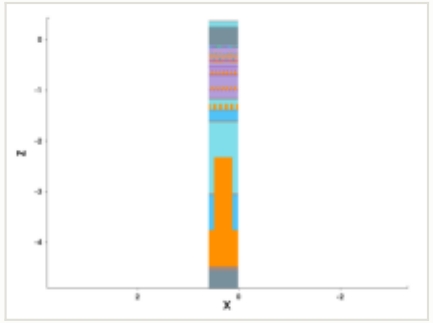
3D structure after this step

zoomed to fit

x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

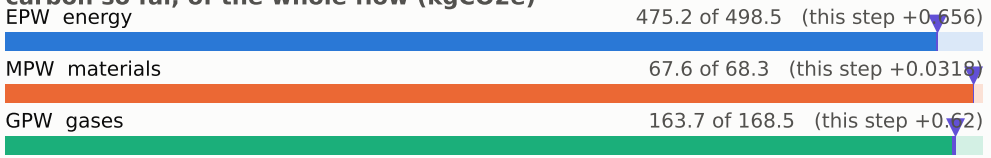
no mask for this step

this step

process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

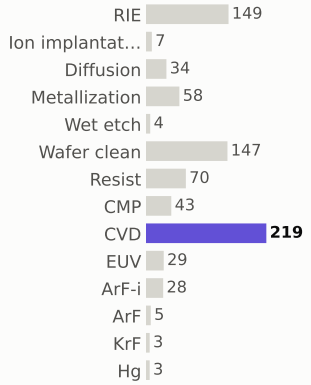
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

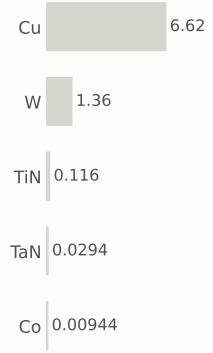


embodied carbon up to each step (kgCO2e per wafer)

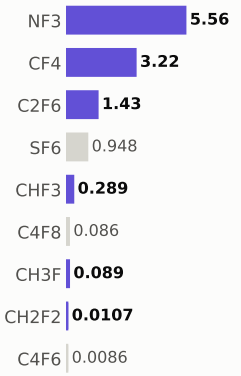
steps by process area



materials deposited (g)



gases emitted (g)



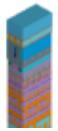
Step #8390: CVD

depo_fill · CVD

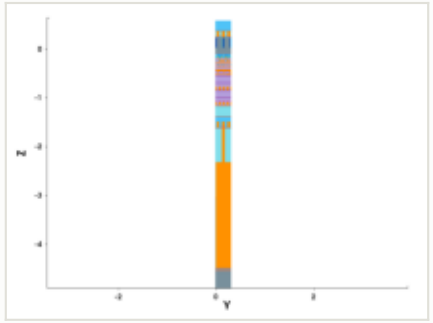
1.31	707.8	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

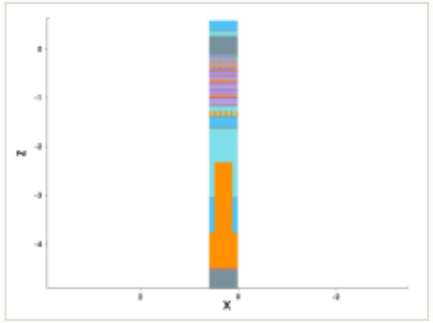
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

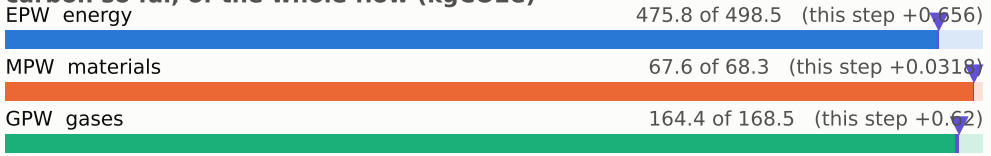
no mask for this step

this step

process area	CVD
material	—
thickness	0.204 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

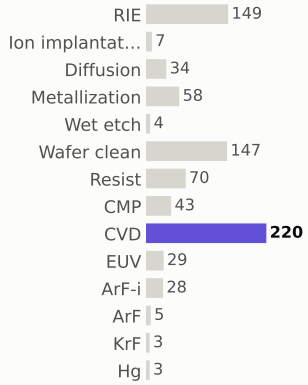
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

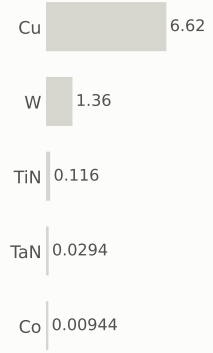


embodied carbon up to each step (kgCO2e per wafer)

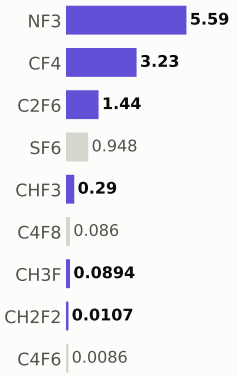
steps by process area



materials deposited (g)



gases emitted (g)



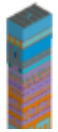
Step #8400: ArF

litho_patterning · ArF

0.199	708.0	0.027%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

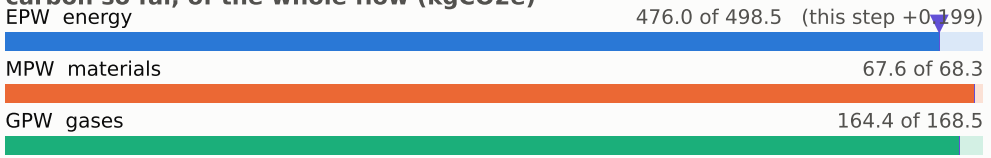


this step

process area	ArF
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

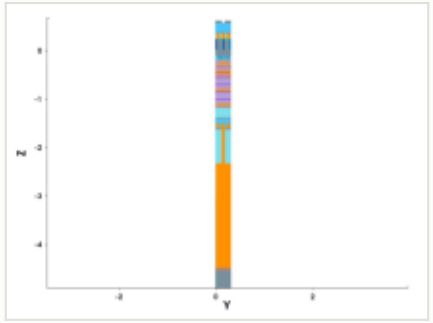
EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

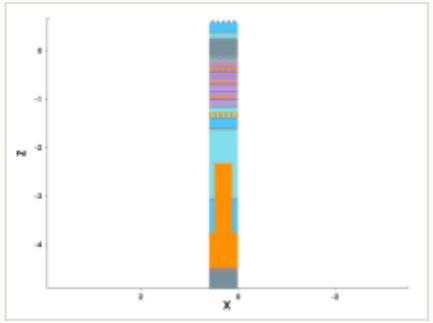


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm

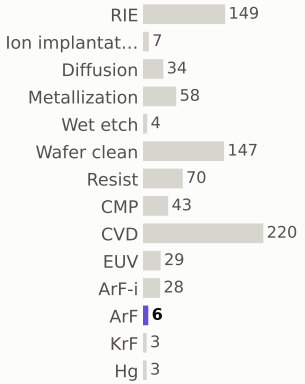


y cut at 0.184 µm

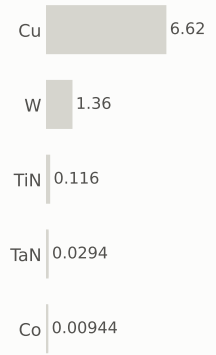


mask: layer 940:0 (GDSII)

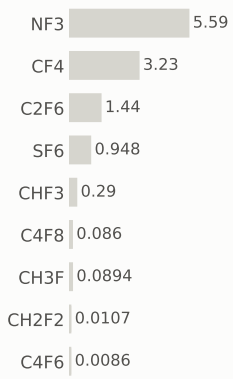
steps by process area



materials deposited (g)



gases emitted (g)



Step #8410: RIE

etch_anisotropic · RIE

1.04	709.1	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

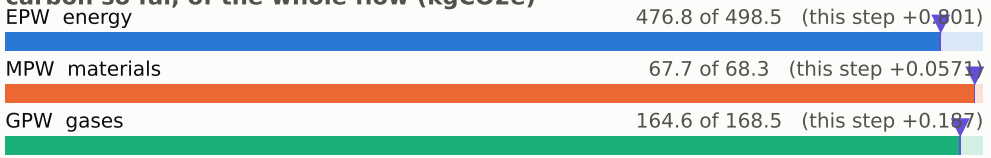
3D structure after this step

zoomed to fit

this step

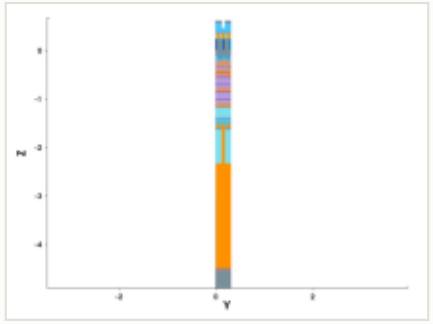
process area	RIE	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

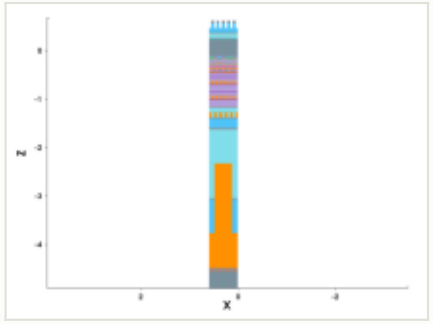


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



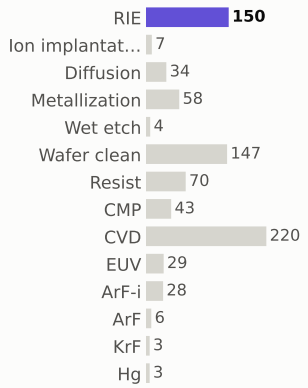
y cut at 0.184 µm



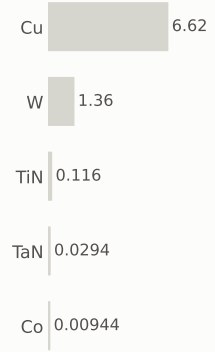
mask for this step

no mask for this step

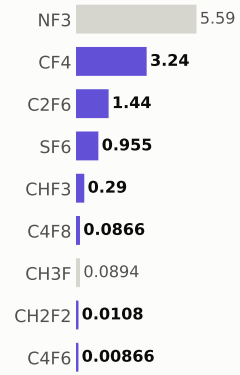
steps by process area



materials deposited (g)



gases emitted (g)



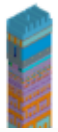
Step #8420: Wafer_clean

etch_strip · Wafer_clean

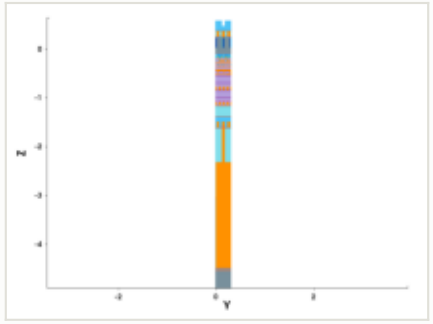
0.169	709.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

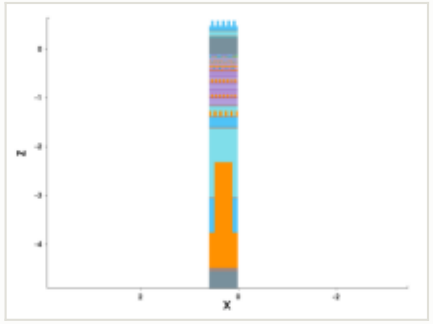
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



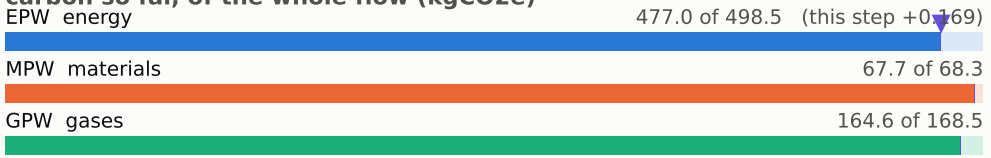
mask for this step

no mask for this step

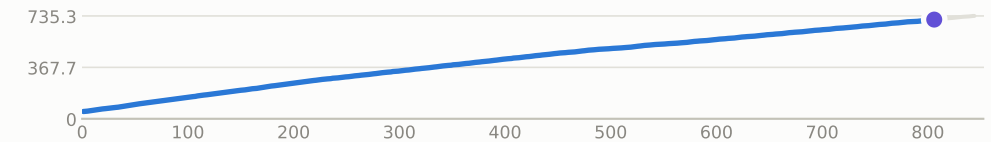
this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

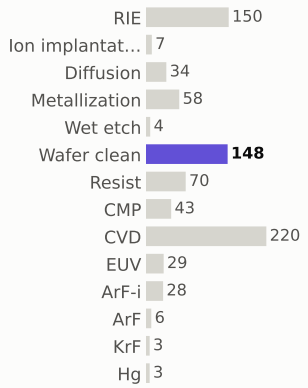
carbon so far, of the whole flow (kgCO2e)



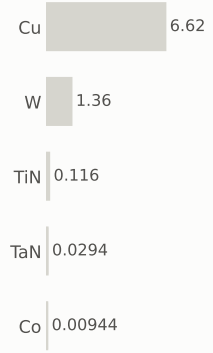
embodied carbon up to each step (kgCO2e per wafer)



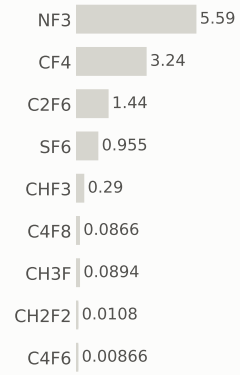
steps by process area



materials deposited (g)



gases emitted (g)



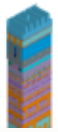
Step #8430: RIE

clean · RIE

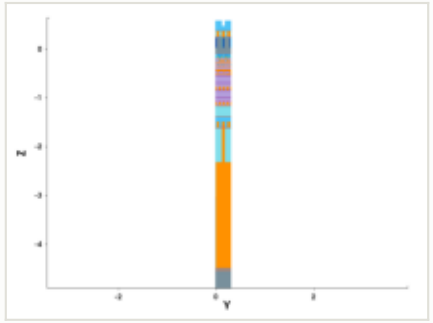
1.04	710.3	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

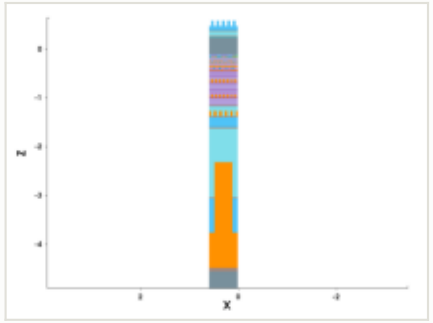
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

no mask for this step

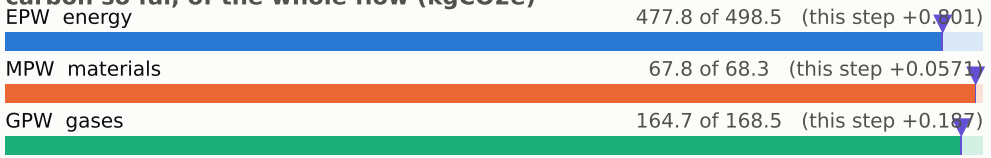
this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

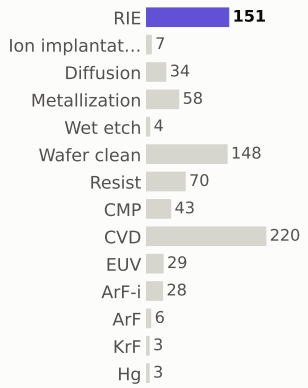
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

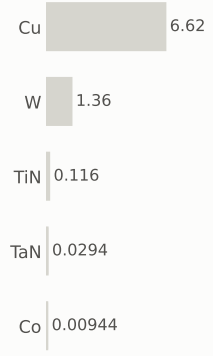


embodied carbon up to each step (kgCO2e per wafer)

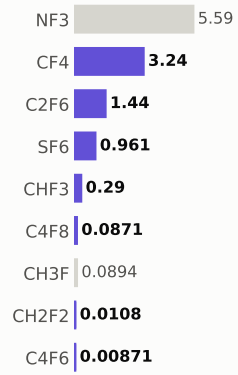
steps by process area



materials deposited (g)



gases emitted (g)



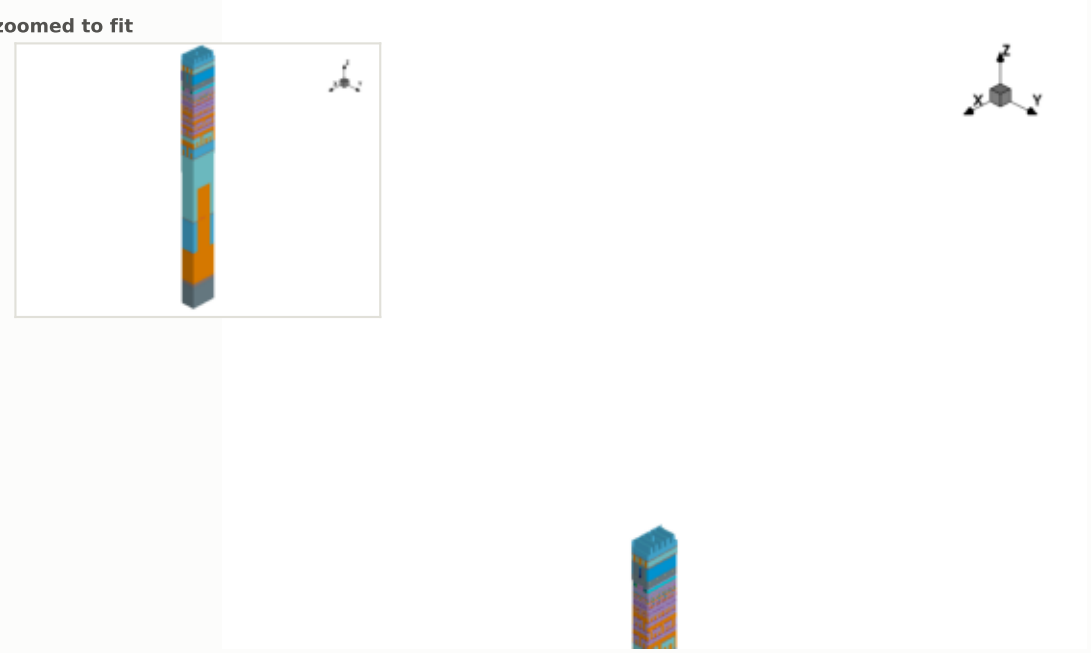
Step #8440: Wafer_clean

clean · Wafer_clean

0.169	710.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

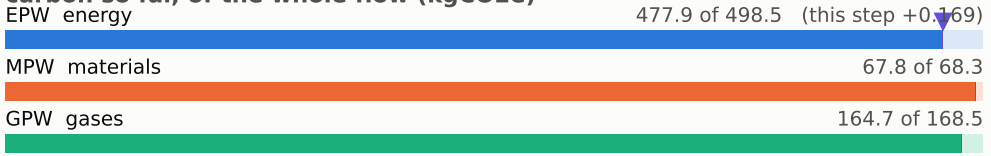


this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

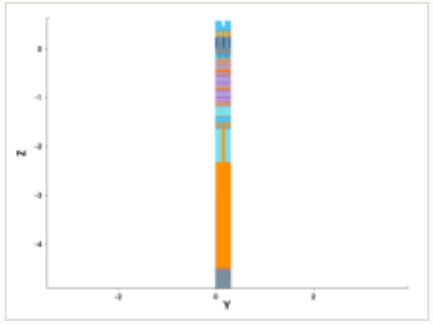
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

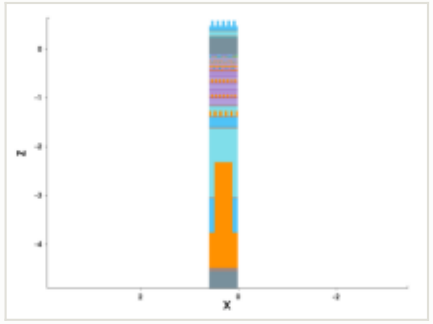


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



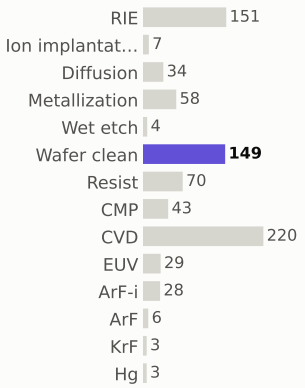
y cut at 0.184 µm



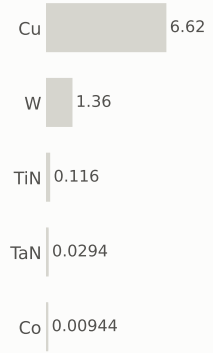
mask for this step

no mask for this step

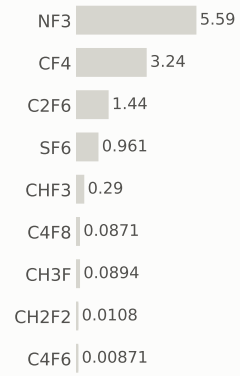
steps by process area



materials deposited (g)



gases emitted (g)



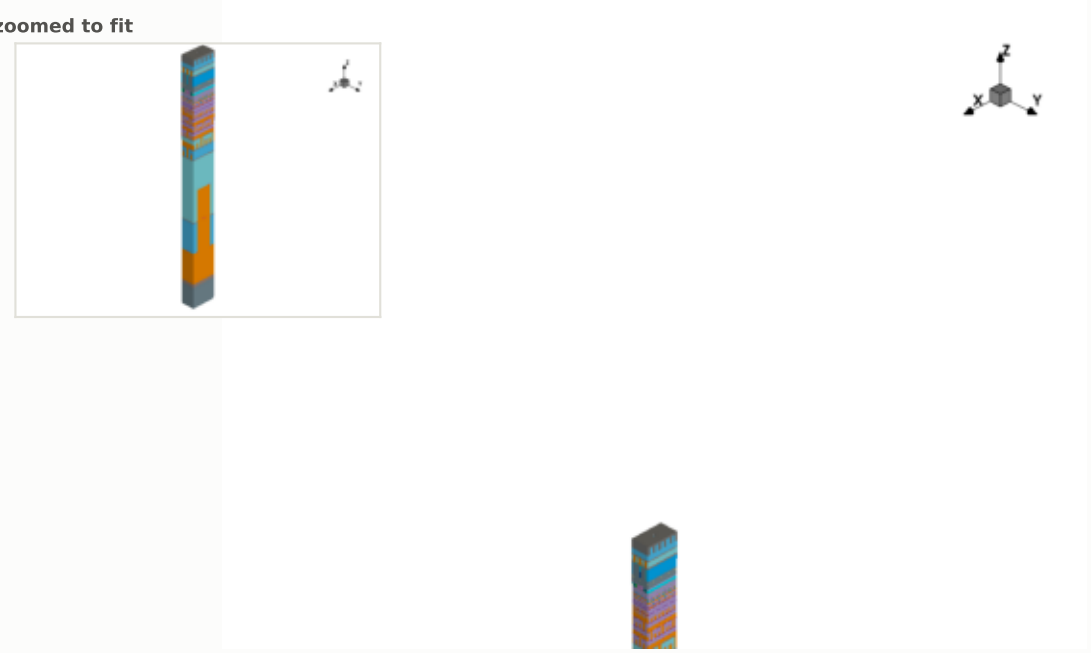
Step #8450: ArF

litho_patterning · ArF

0.199	710.6	0.027%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

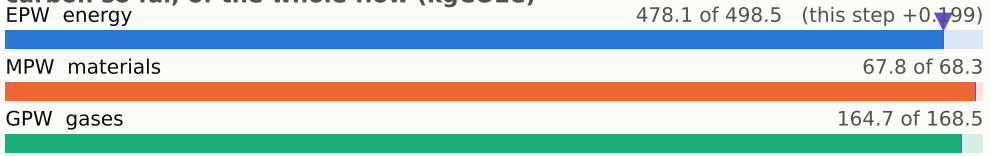


this step

process area	ArF
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

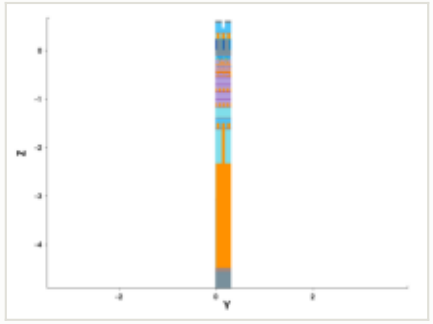
EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

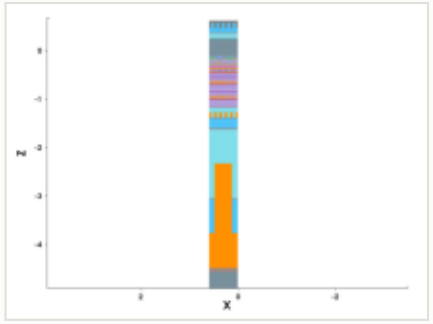


embodied carbon up to each step (kgCO2e per wafer)

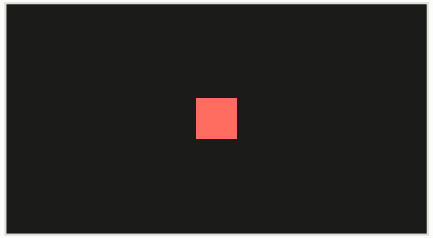
x cut at 0.294 µm



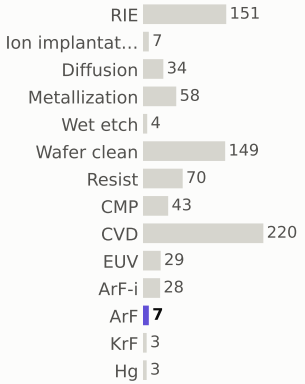
y cut at 0.184 µm



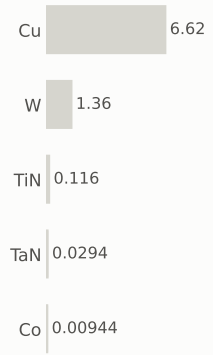
mask: layer 930:0 (GDSII)



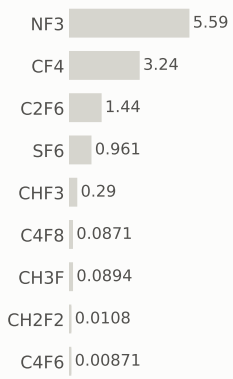
steps by process area



materials deposited (g)



gases emitted (g)



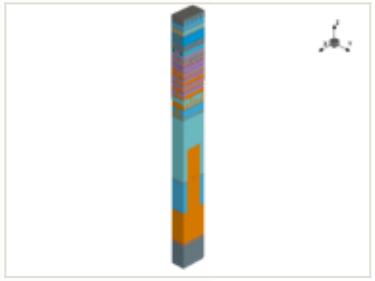
Step #8460: RIE

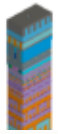
etch_anisotropic · RIE

1.04	711.7	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



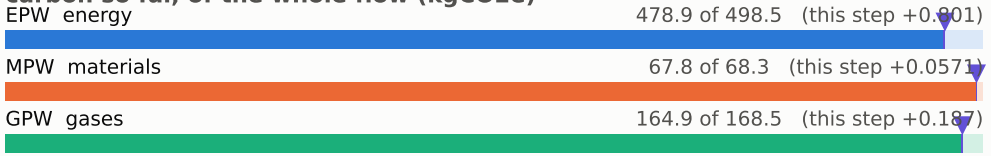


this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

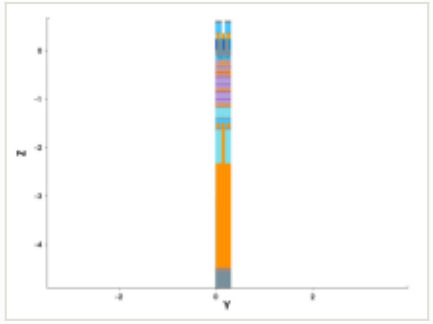
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

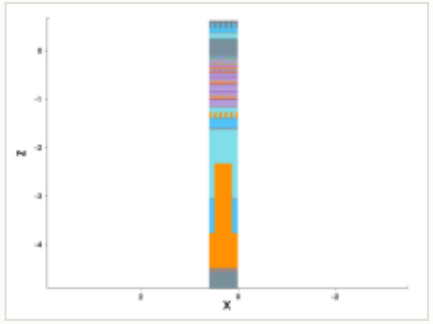


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



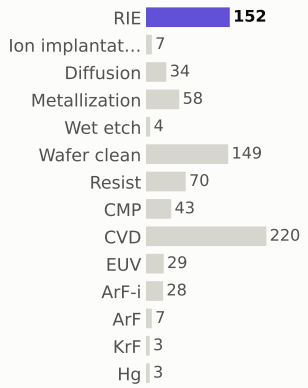
y cut at 0.184 µm



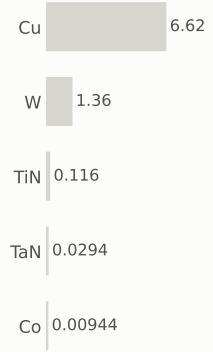
mask for this step

no mask for this step

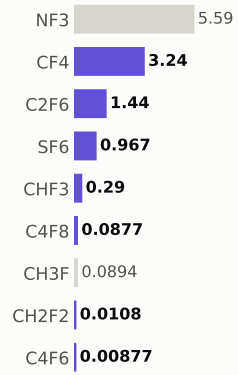
steps by process area



materials deposited (g)



gases emitted (g)



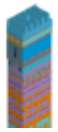
Step #8470: Wafer_clean

etch_strip · Wafer_clean

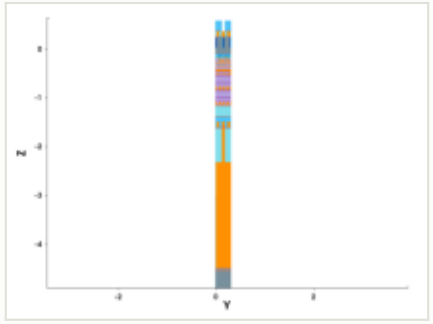
0.169	711.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

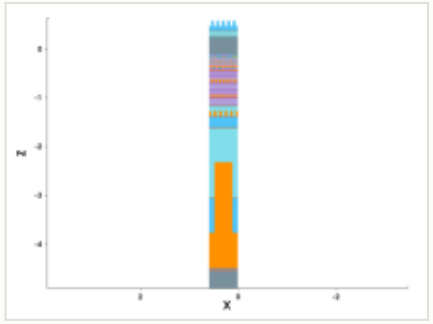
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



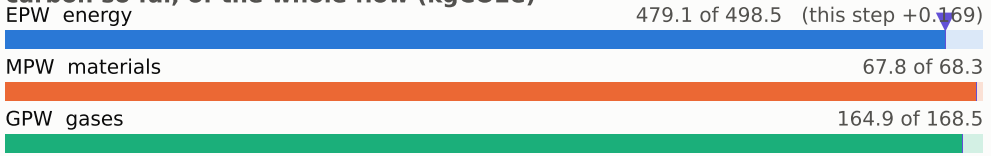
mask for this step

no mask for this step

this step

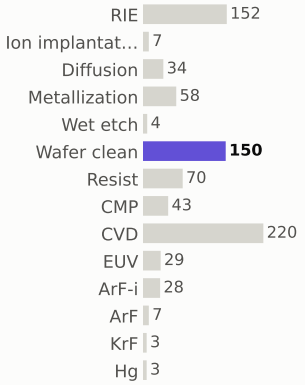
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

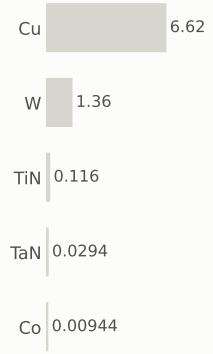


embodied carbon up to each step (kgCO2e per wafer)

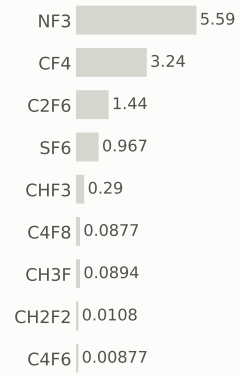
steps by process area



materials deposited (g)



gases emitted (g)



Step #8480: Metallization

depo_ellipsoidal · Metallization

1.2	713.1	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

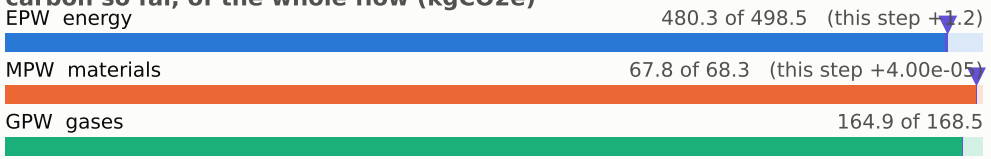
3D structure after this step

zoomed to fit

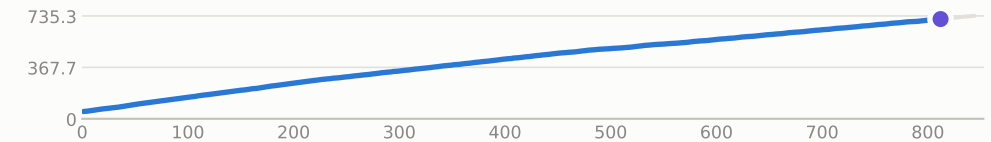
this step

process area	Metallization	EPW	1.2 kgCO2e
material	TaN	EPW tool	0.86 kgCO2e
thickness	0.0016 µm	EPW overhead	0.344 kgCO2e
material GWP	25.8159 gCO2e/g	MPW	4.00e-05 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	2 kWh + 0.8 overhead	embodied carbon	**1.2 kgCO2e**

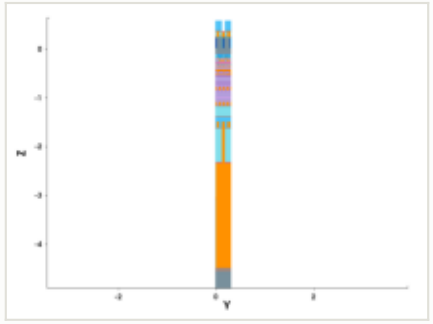
carbon so far, of the whole flow (kgCO2e)



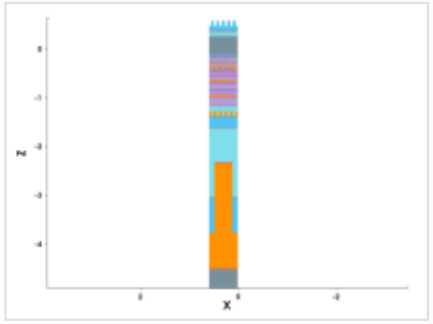
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



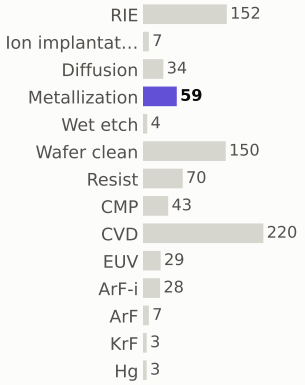
y cut at 0.184 µm



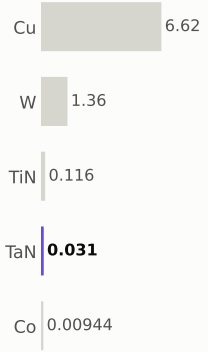
mask for this step

no mask for this step

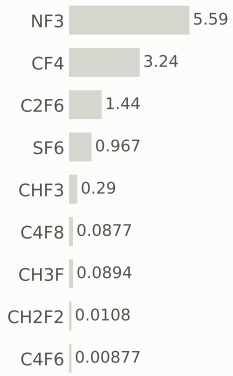
steps by process area



materials deposited (g)



gases emitted (g)



Step #8490: Metallization

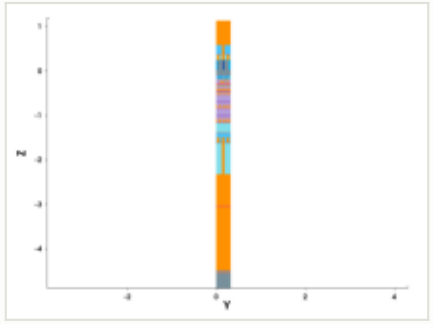
depo_fill · Metallization

1.21	714.3	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

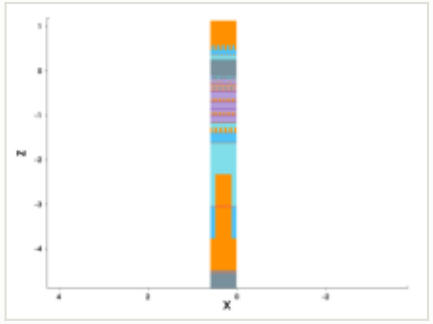
3D structure after this step

zoomed to fit

x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

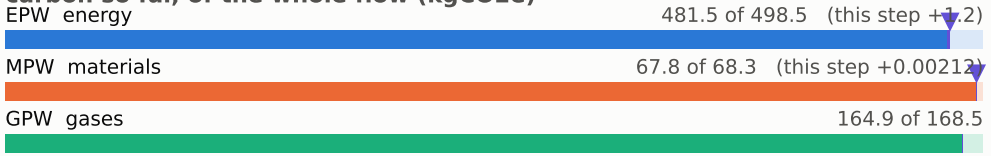
no mask for this step

this step

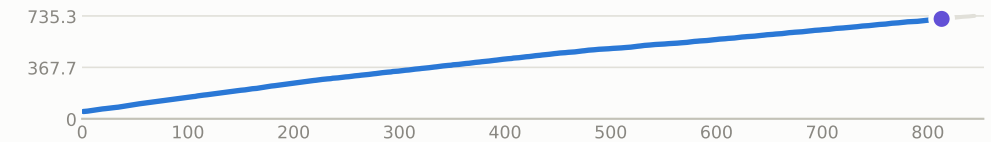
process area	Metallization
material	Cu
thickness	0.55 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00212 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

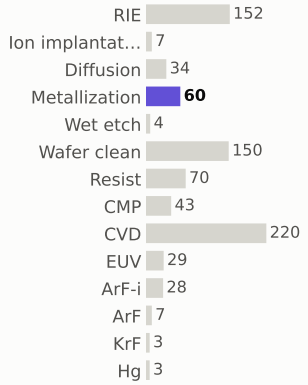
carbon so far, of the whole flow (kgCO2e)



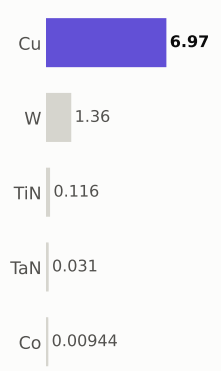
embodied carbon up to each step (kgCO2e per wafer)



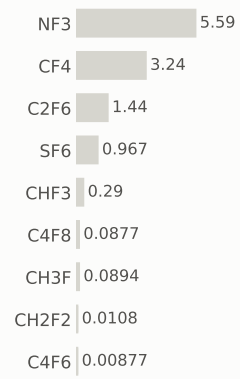
steps by process area



materials deposited (g)



gases emitted (g)



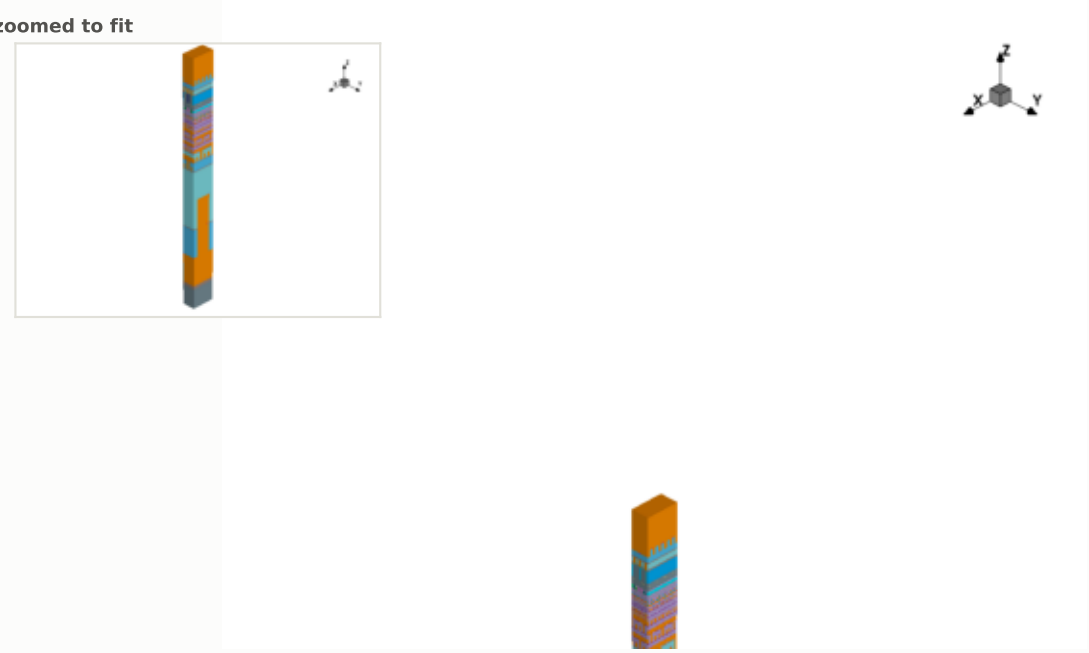
Step #8500: Diffusion

treatment · Diffusion

0.801	715.1	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

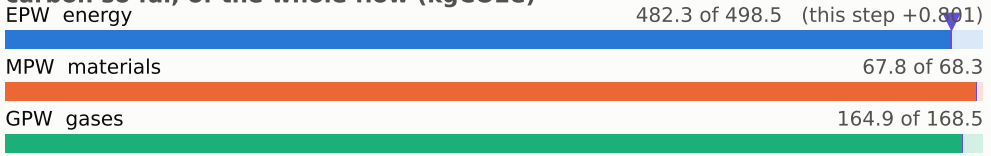


this step

process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

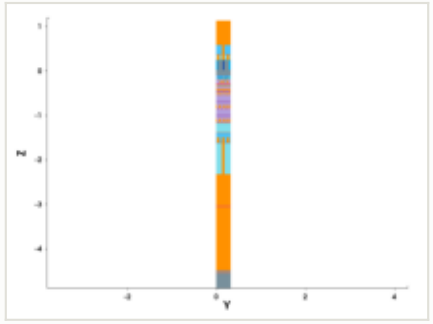
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

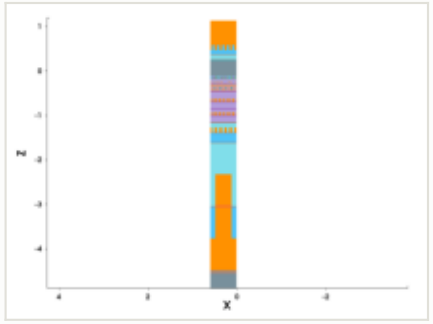


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



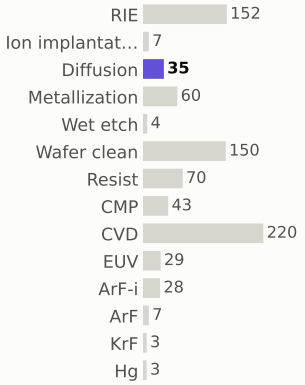
y cut at 0.184 µm



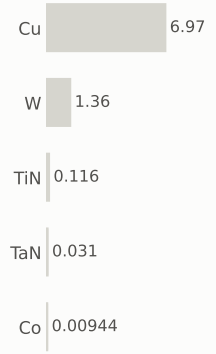
mask for this step

no mask for this step

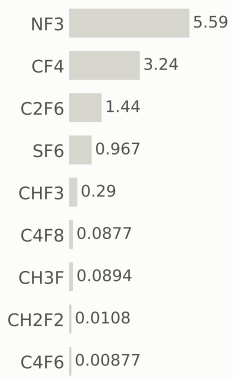
steps by process area



materials deposited (g)



gases emitted (g)



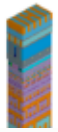
Step #8510: CMP

etch_cmp · CMP

0.169	715.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

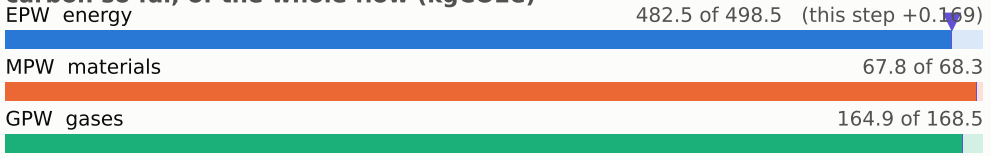


this step

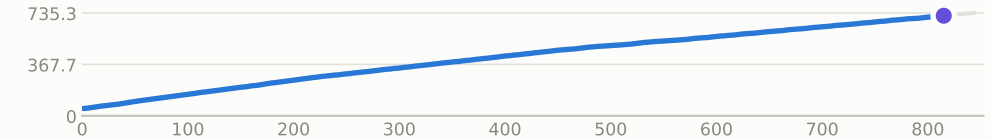
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

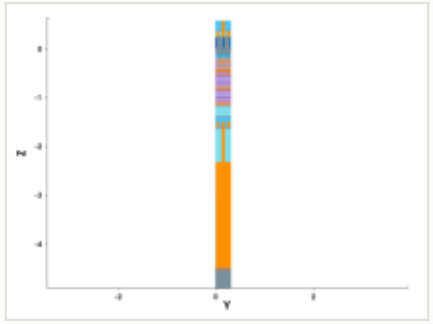
carbon so far, of the whole flow (kgCO2e)



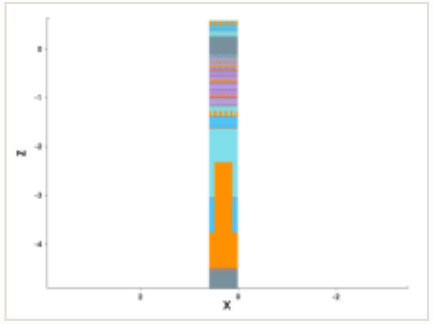
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



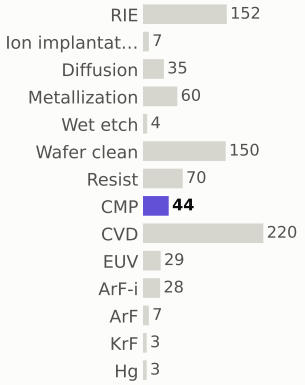
y cut at 0.184 µm



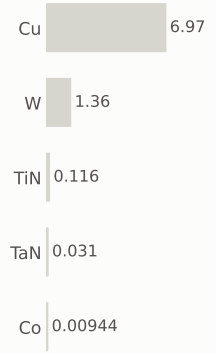
mask for this step

no mask for this step

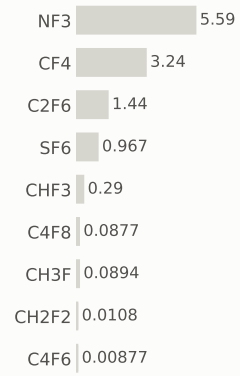
steps by process area



materials deposited (g)



gases emitted (g)



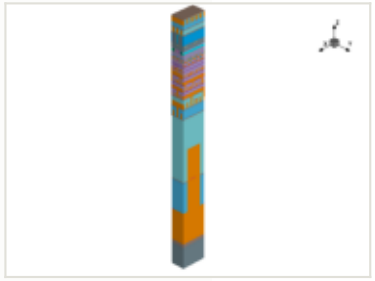
Step #8530: CVD

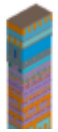
depo_fill · CVD

1.31	716.5	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

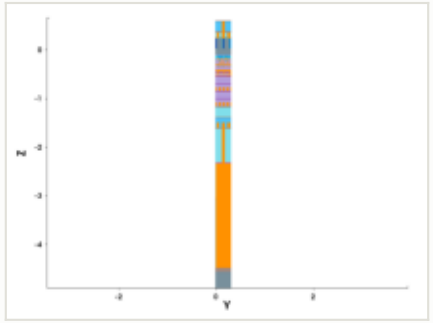
3D structure after this step

zoomed to fit

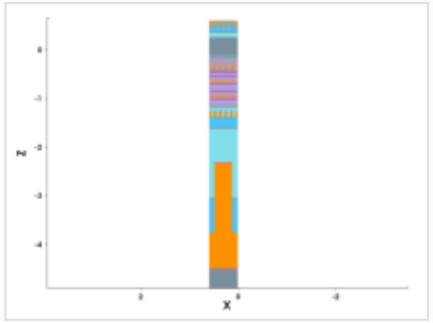




x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

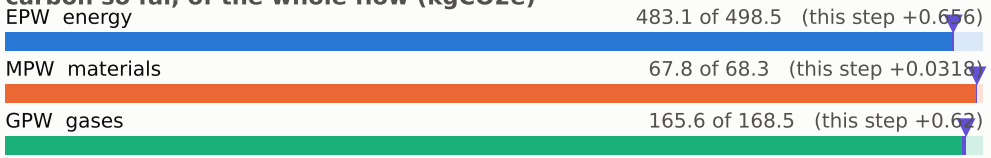
no mask for this step

this step

process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

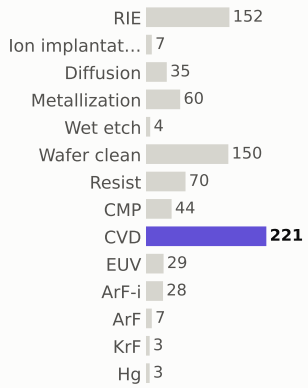
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

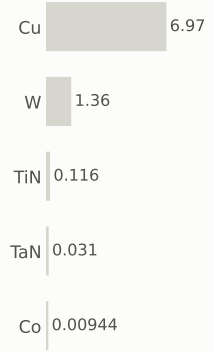


embodied carbon up to each step (kgCO2e per wafer)

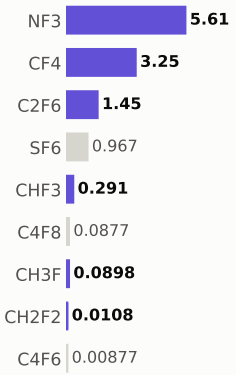
steps by process area



materials deposited (g)



gases emitted (g)



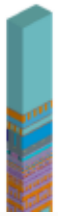
Step #8540: CVD

depo_fill · CVD

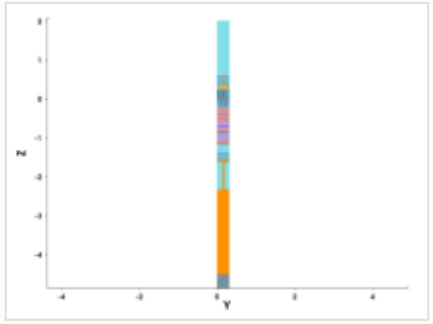
1.31	717.9	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

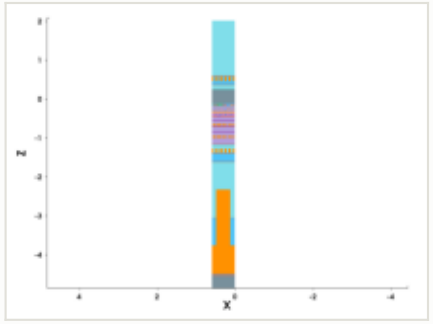
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

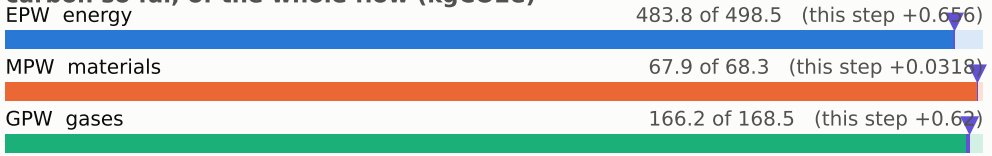
no mask for this step

this step

process area	CVD
material	—
thickness	1.42 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

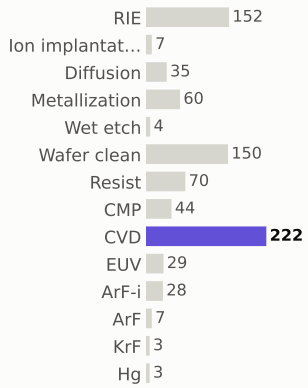
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

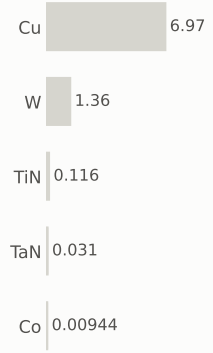


embodied carbon up to each step (kgCO2e per wafer)

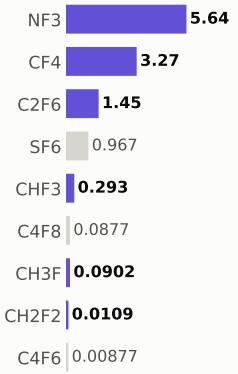
steps by process area



materials deposited (g)



gases emitted (g)



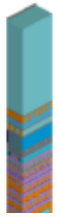
Step #8550: Hg

litho_patterning · Hg

0.199	718.0	0.027%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

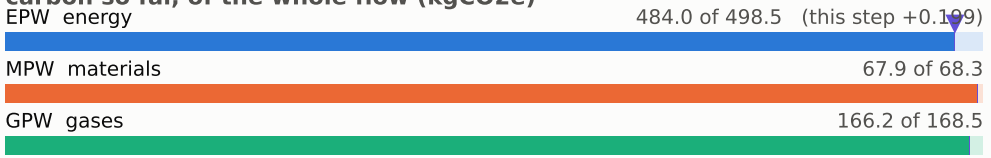


this step

process area	Hg
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

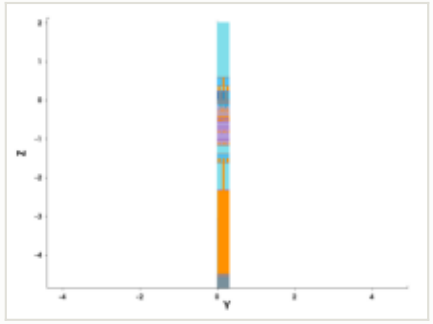
EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

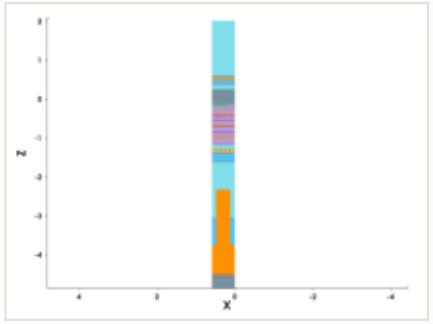


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm

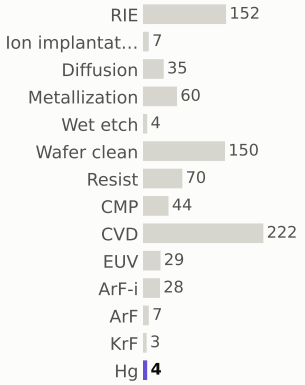


y cut at 0.184 µm

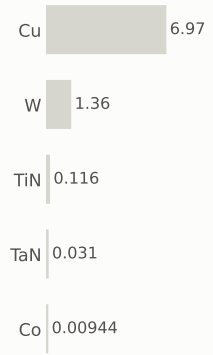


mask: layer 960:0 (GDSII)

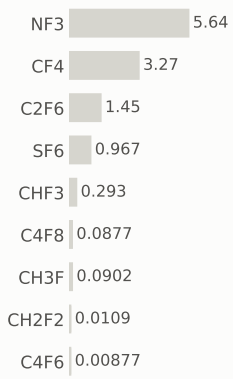
steps by process area



materials deposited (g)



gases emitted (g)



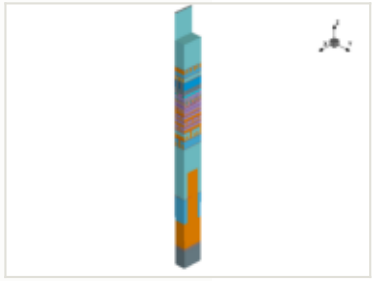
Step #8560: RIE

etch_anisotropic · RIE

1.04	719.1	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

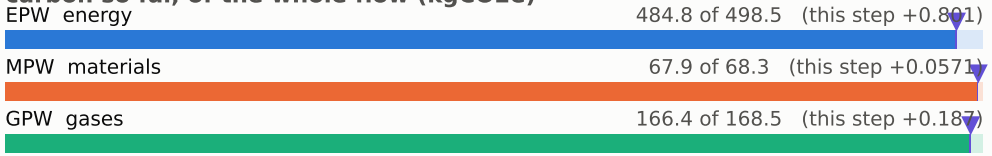


this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

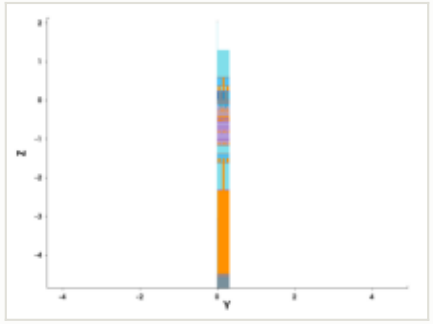
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

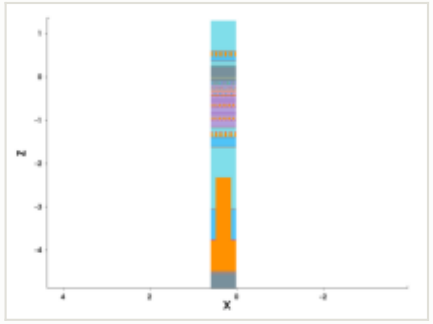


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



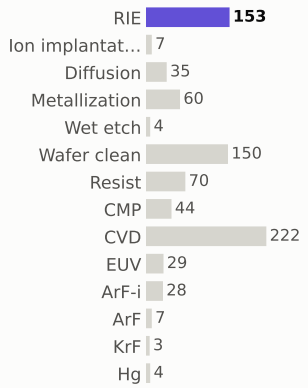
y cut at 0.184 µm



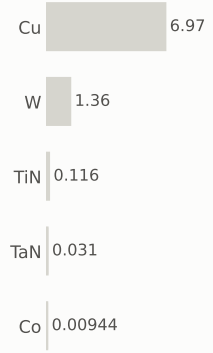
mask for this step

no mask for this step

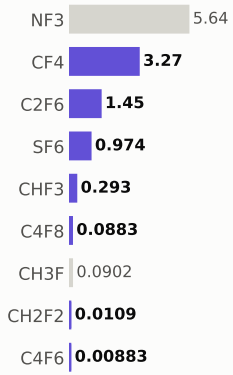
steps by process area



materials deposited (g)



gases emitted (g)



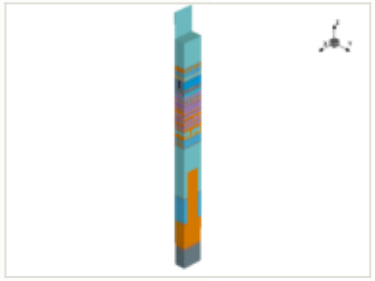
Step #8570: Wafer_clean

etch_strip · Wafer_clean

0.169	719.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

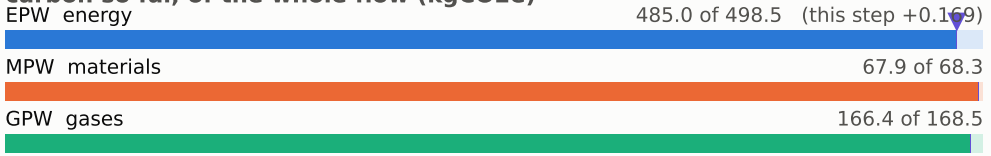
zoomed to fit



this step

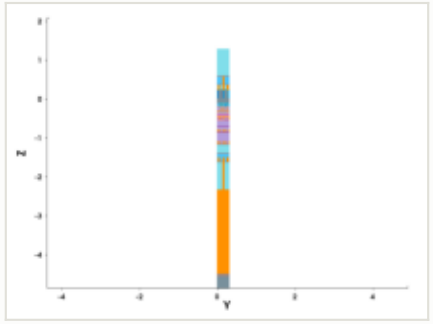
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

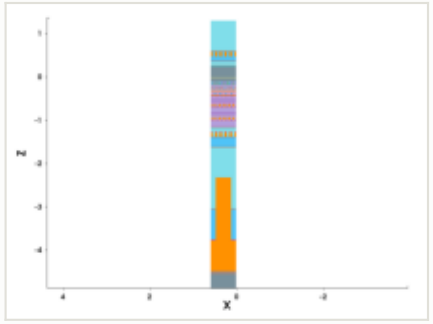


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



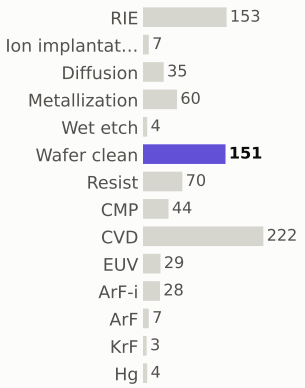
y cut at 0.184 µm



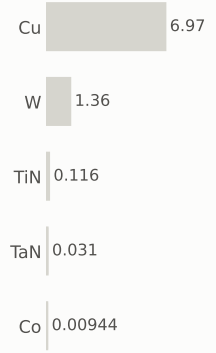
mask for this step

no mask for this step

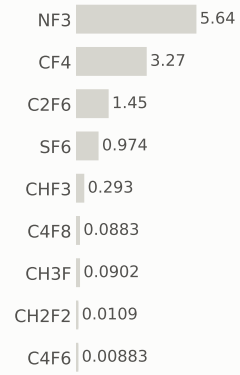
steps by process area



materials deposited (g)



gases emitted (g)



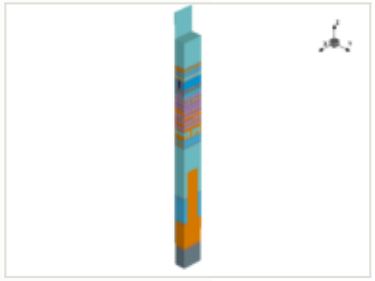
Step #8580: RIE

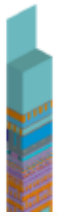
clean · RIE

1.04	720.3	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

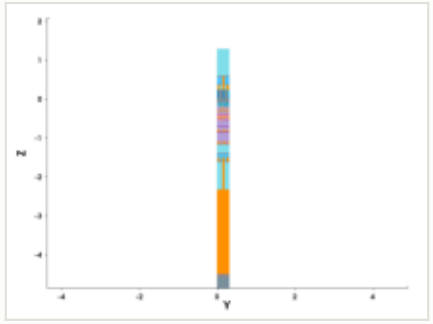
3D structure after this step

zoomed to fit

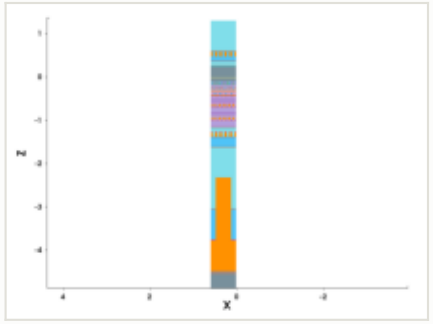




x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

no mask for this step

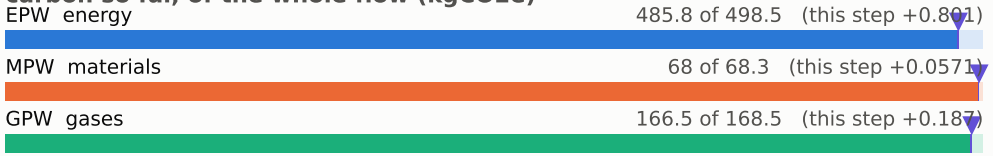
this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

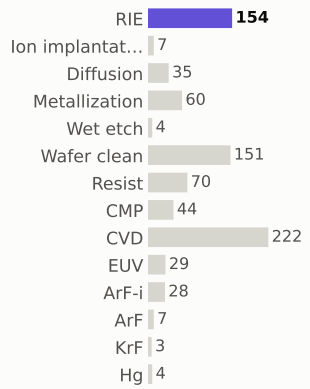
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

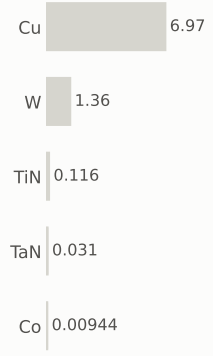


embodied carbon up to each step (kgCO2e per wafer)

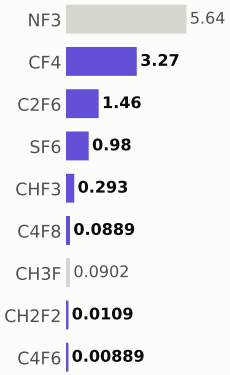
steps by process area



materials deposited (g)



gases emitted (g)



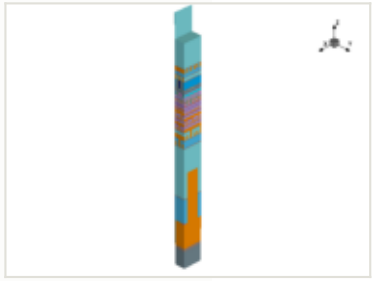
Step #8590: Wafer_clean

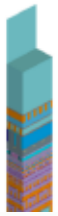
clean · Wafer_clean

0.169	720.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

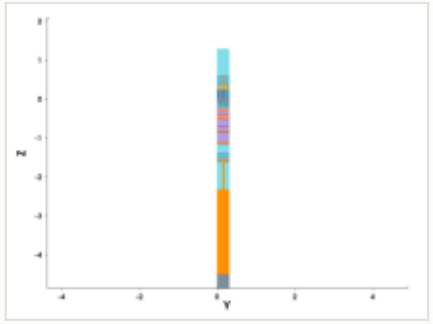
3D structure after this step

zoomed to fit

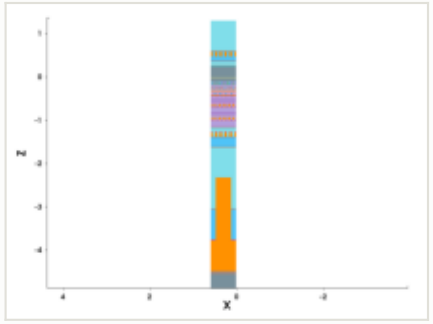




x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

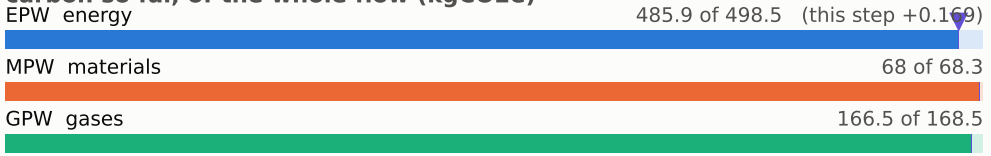
no mask for this step

this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

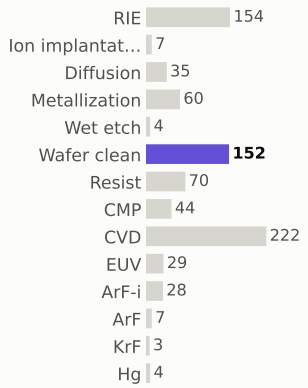
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

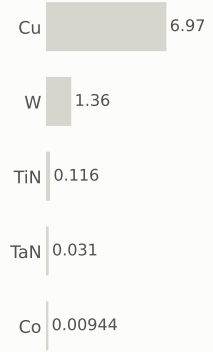


embodied carbon up to each step (kgCO2e per wafer)

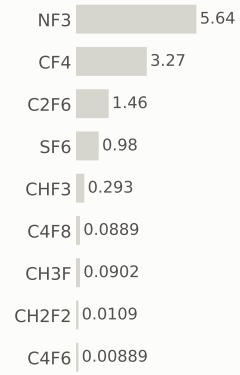
steps by process area



materials deposited (g)



gases emitted (g)



Step #8600: ArF

litho_patterning · ArF

0.199	720.7	0.027%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

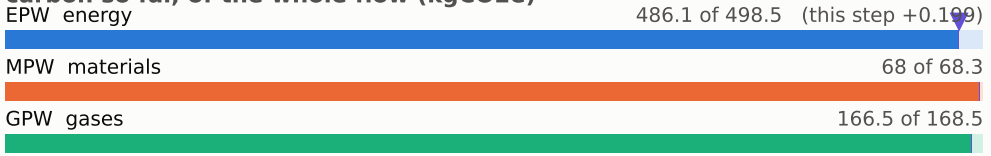
3D structure after this step

zoomed to fit

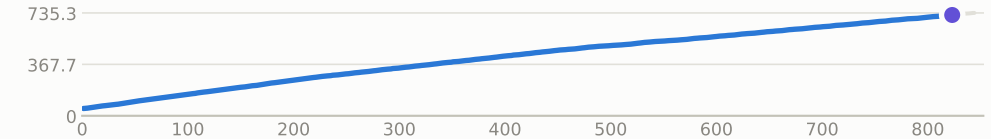
this step

process area	ArF	EPW	0.199 kgCO2e
material	—	EPW tool	0.142 kgCO2e
thickness	—	EPW overhead	0.0568 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.33 kWh + 0.132 overhead	embodied carbon	**0.199 kgCO2e**

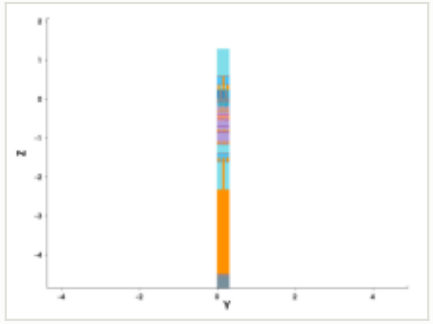
carbon so far, of the whole flow (kgCO2e)



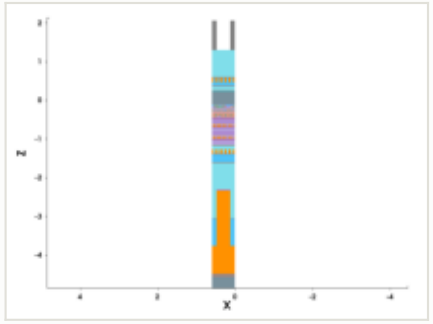
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm

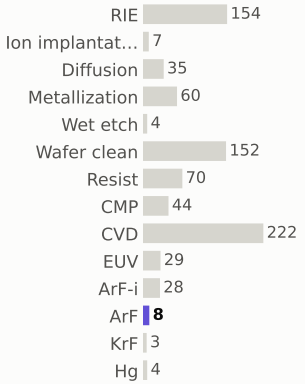


y cut at 0.184 µm

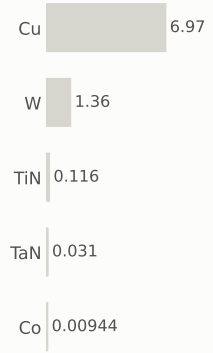


mask: layer 950:0 (GDSII)

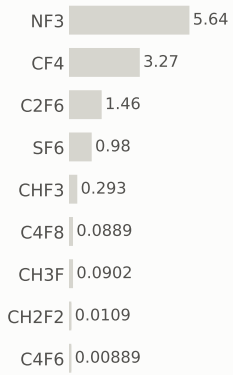
steps by process area



materials deposited (g)



gases emitted (g)



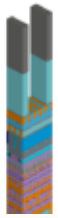
Step #8610: RIE

etch_anisotropic · RIE

1.04	721.7	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

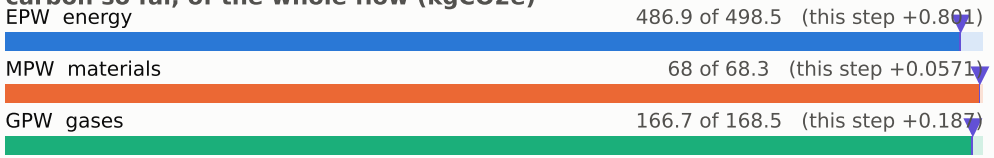


this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

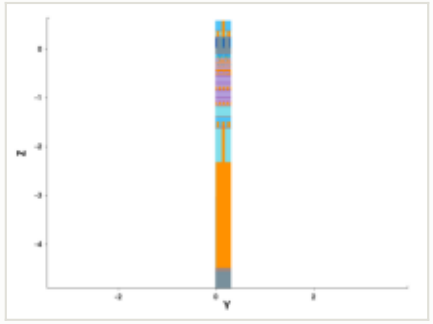
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

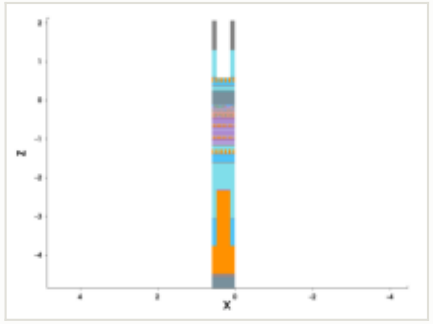


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



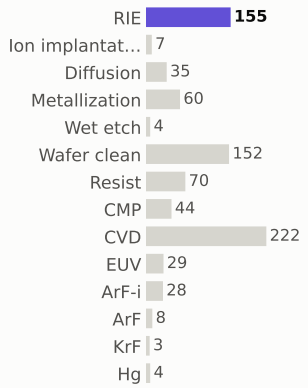
y cut at 0.184 µm



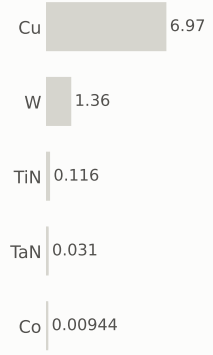
mask for this step

no mask for this step

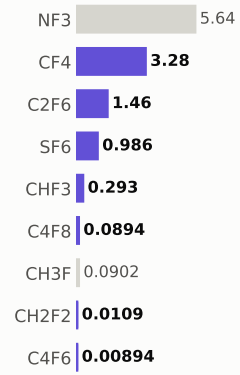
steps by process area



materials deposited (g)



gases emitted (g)



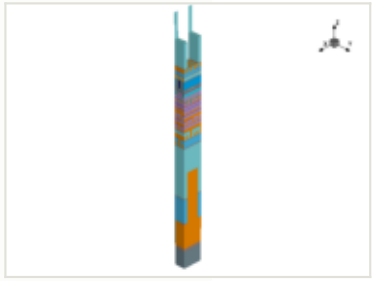
Step #8620: Wafer_clean

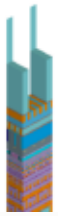
etch_strip · Wafer_clean

0.169	721.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

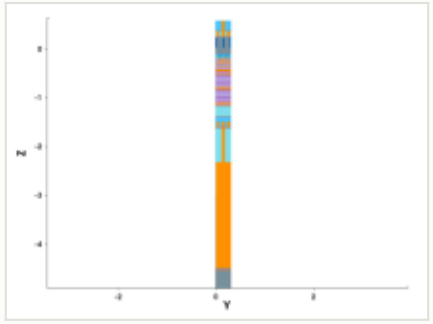
3D structure after this step

zoomed to fit

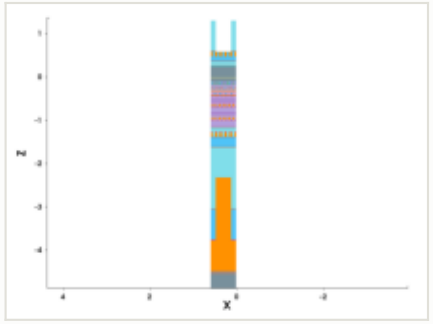




x cut at 0.294 µm



y cut at 0.184 µm



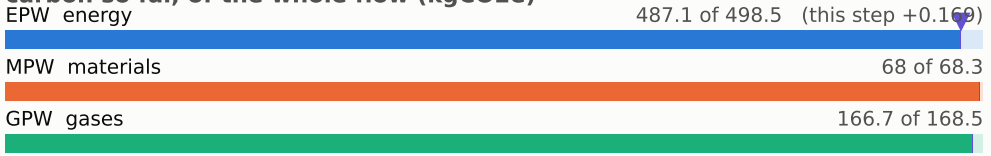
mask for this step

no mask for this step

this step

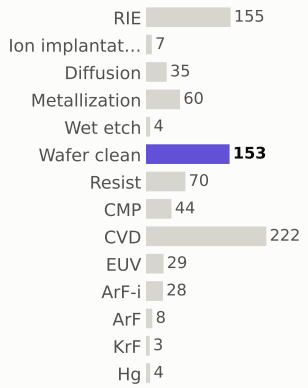
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

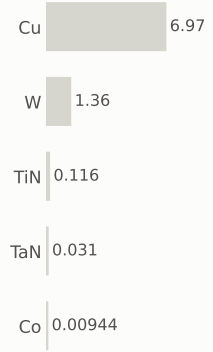


embodied carbon up to each step (kgCO2e per wafer)

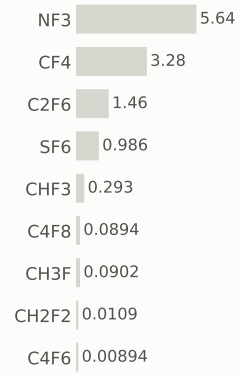
steps by process area



materials deposited (g)



gases emitted (g)



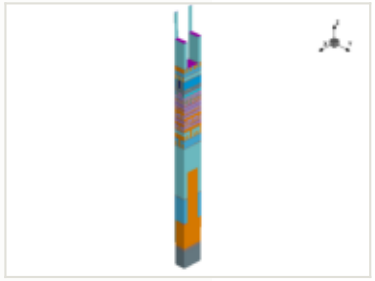
Step #8630: Metallization

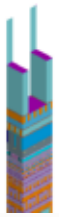
depo_ellipsoidal · Metallization

1.2	723.1	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

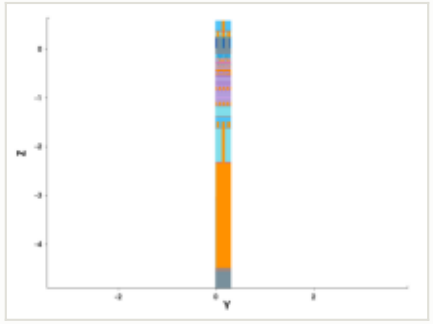
3D structure after this step

zoomed to fit

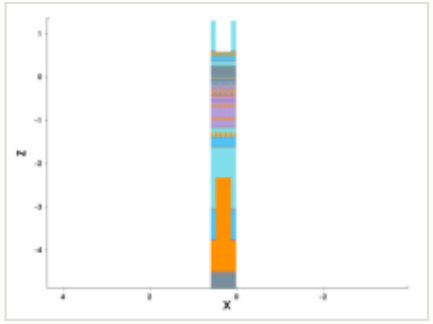




x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

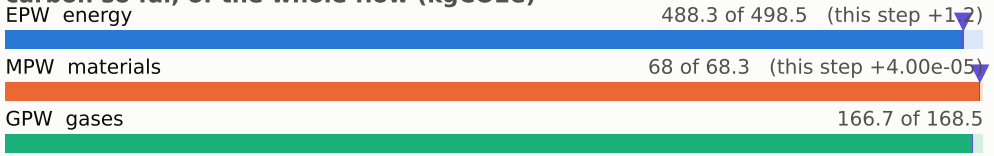
no mask for this step

this step

process area	Metallization
material	TaN
thickness	0.0016 µm
material GWP	25.8159 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

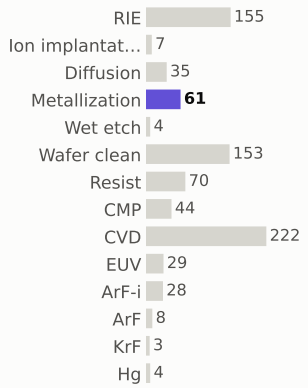
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.00e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

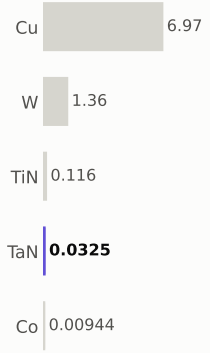


embodied carbon up to each step (kgCO2e per wafer)

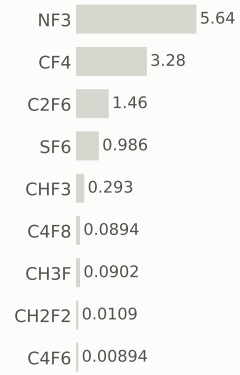
steps by process area



materials deposited (g)



gases emitted (g)



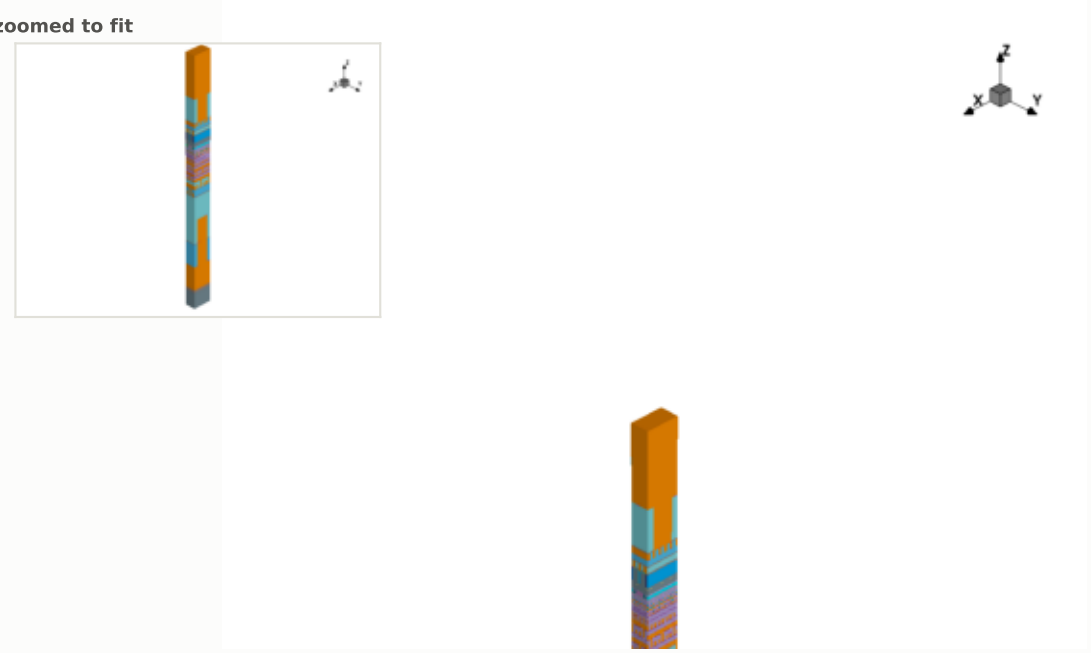
Step #8640: Metallization

depo_fill · Metallization

1.21	724.3	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

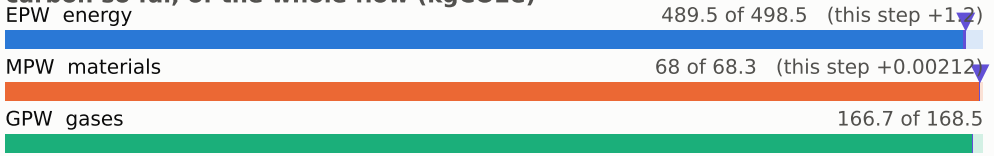


this step

process area	Metallization
material	Cu
thickness	0.55 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

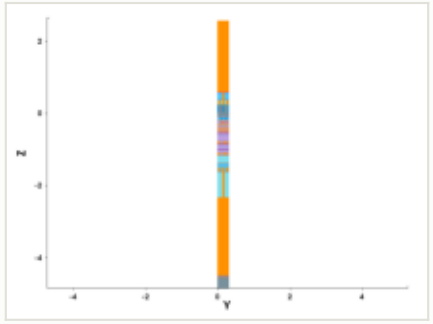
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00212 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

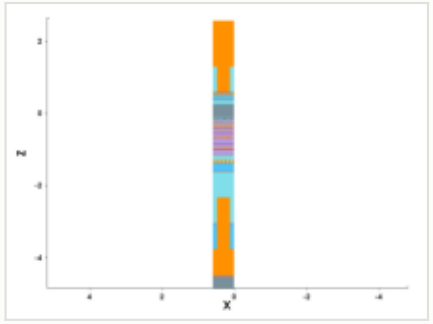


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



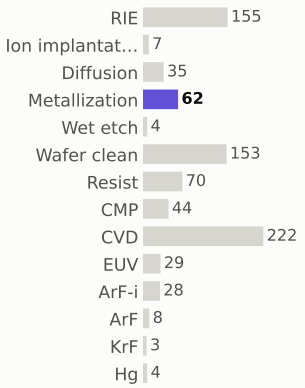
y cut at 0.184 µm



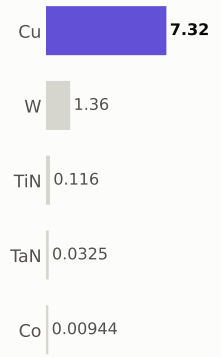
mask for this step

no mask for this step

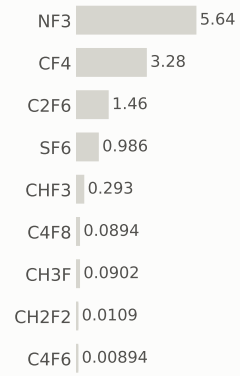
steps by process area



materials deposited (g)



gases emitted (g)



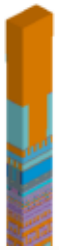
Step #8650: Diffusion

treatment · Diffusion

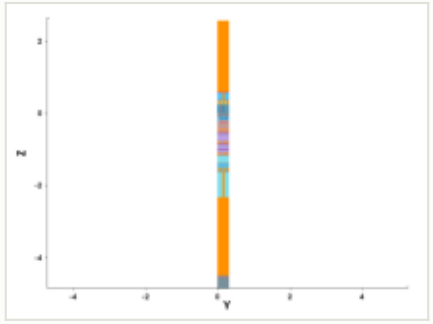
0.801	725.1	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

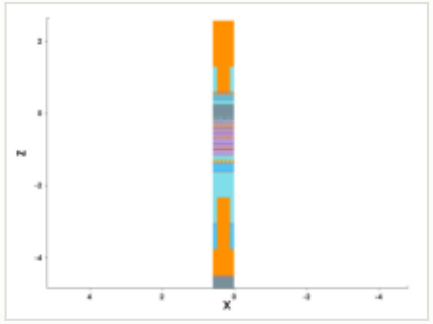
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

no mask for this step

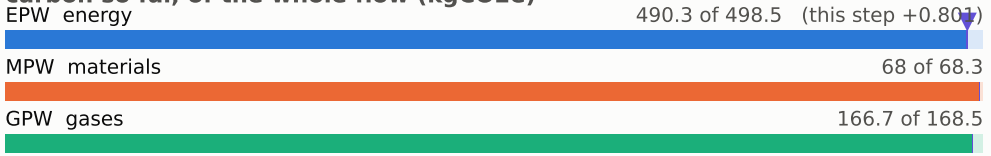
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

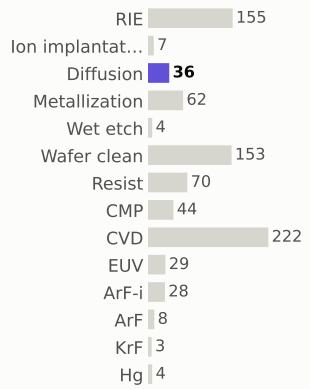
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

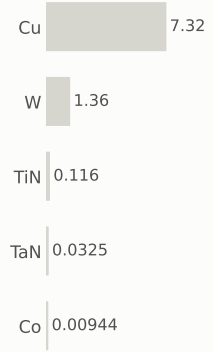


embodied carbon up to each step (kgCO2e per wafer)

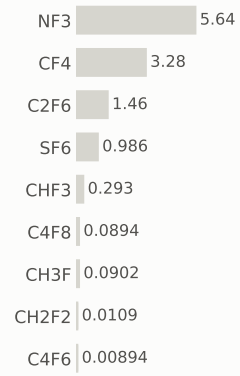
steps by process area



materials deposited (g)



gases emitted (g)



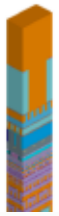
Step #8660: CMP

etch_cmp · CMP

0.169	725.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

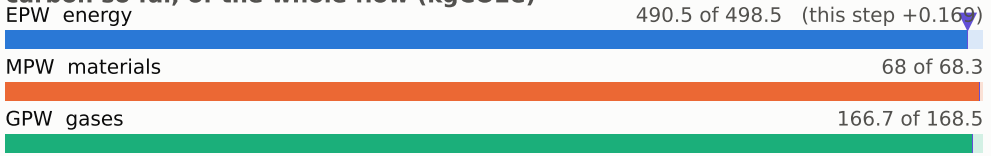


this step

process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

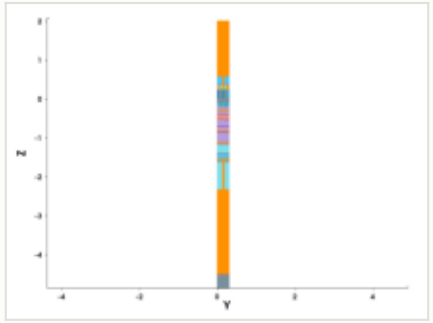
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

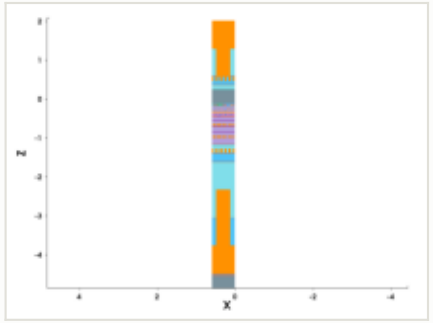


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



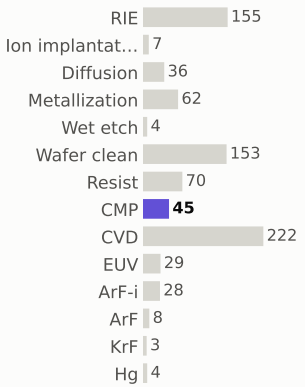
y cut at 0.184 µm



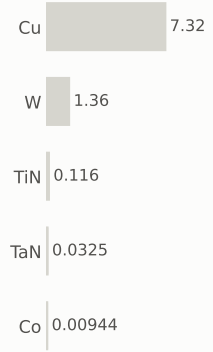
mask for this step

no mask for this step

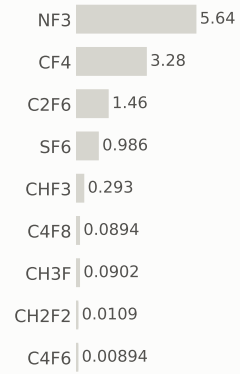
steps by process area



materials deposited (g)



gases emitted (g)



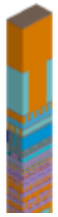
Step #8680: CVD

depo_fill · CVD

1.31	726.6	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

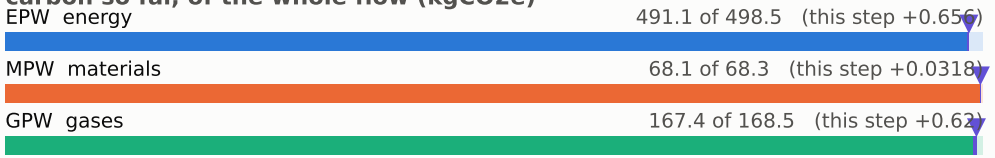


this step

process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

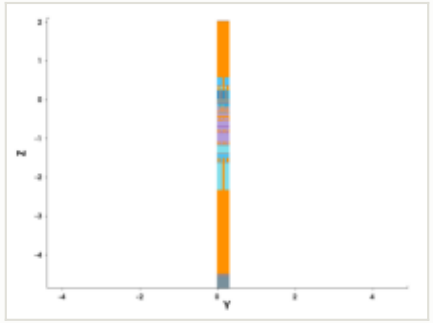
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

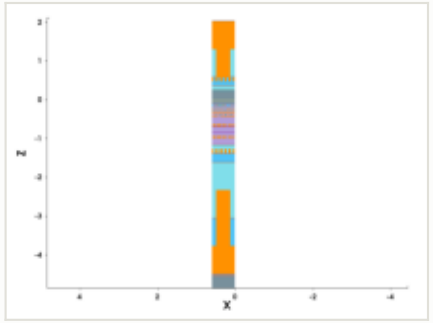


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



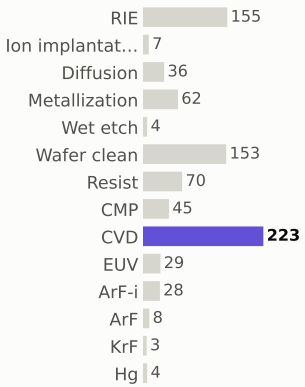
y cut at 0.184 µm



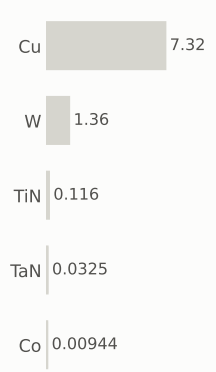
mask for this step

no mask for this step

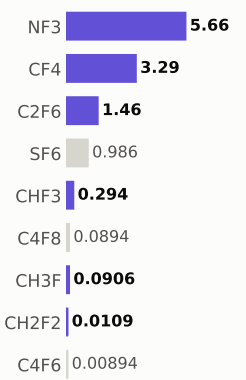
steps by process area



materials deposited (g)



gases emitted (g)



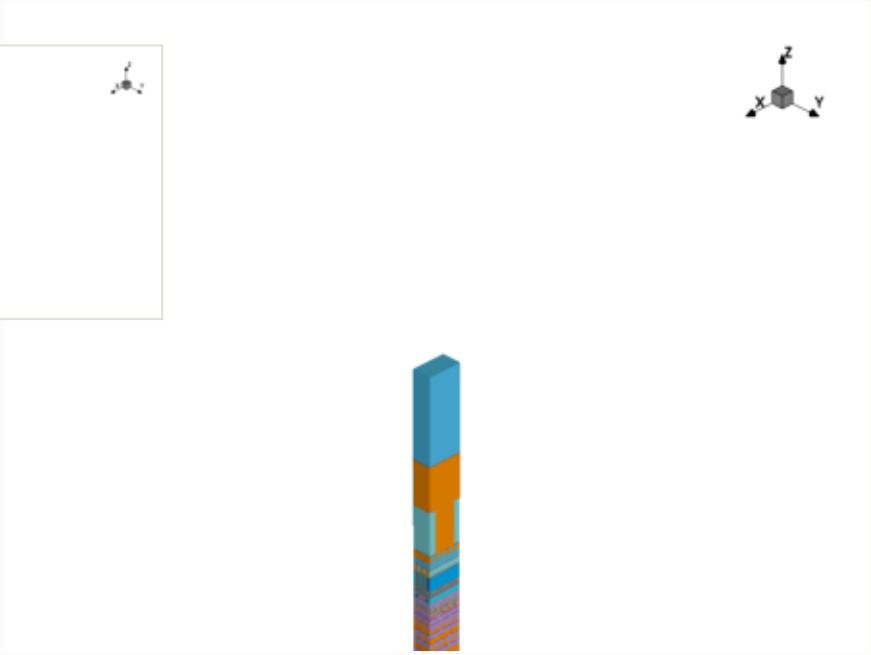
Step #8690: CVD

depo_fill · CVD

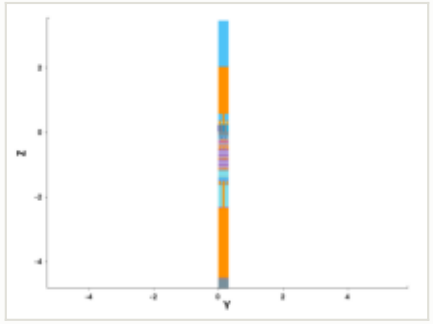
1.31	727.9	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

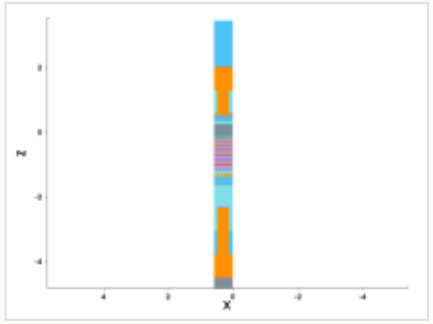
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

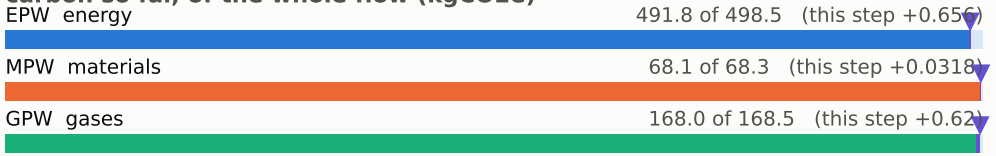
no mask for this step

this step

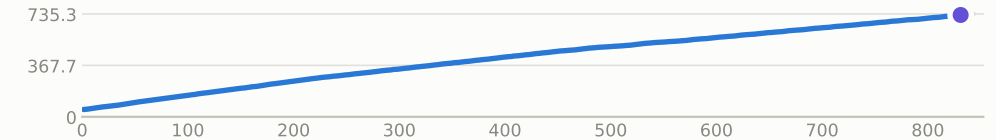
process area	CVD
material	—
thickness	1.42 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

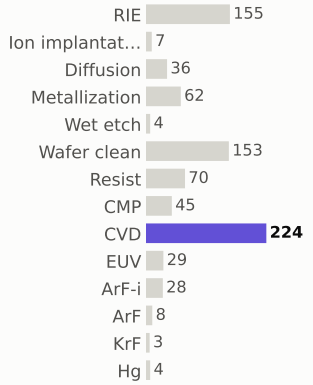
carbon so far, of the whole flow (kgCO2e)



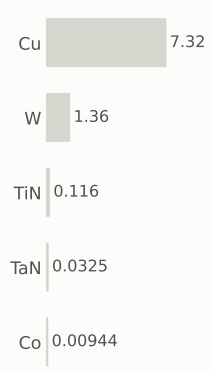
embodied carbon up to each step (kgCO2e per wafer)



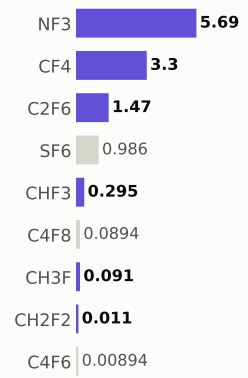
steps by process area



materials deposited (g)



gases emitted (g)



Step #8700: Hg

litho_patterning · Hg

0.199	728.1	0.027%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

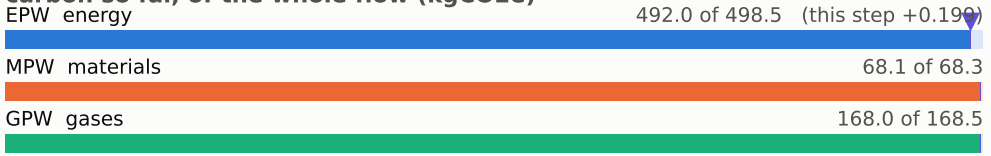
3D structure after this step

zoomed to fit

this step

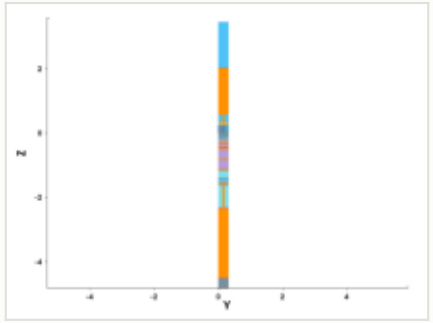
process area	Hg	EPW	0.199 kgCO2e
material	—	EPW tool	0.142 kgCO2e
thickness	—	EPW overhead	0.0568 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.33 kWh + 0.132 overhead	embodied carbon	**0.199 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

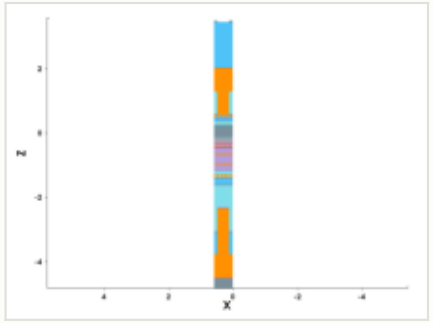


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm

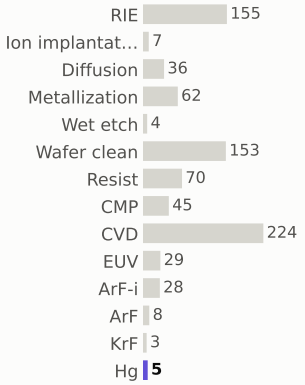


y cut at 0.184 µm

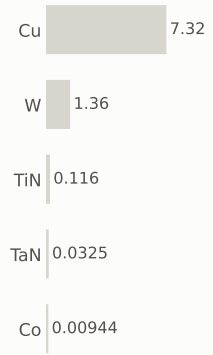


mask: layer 980:0 (GDSII)

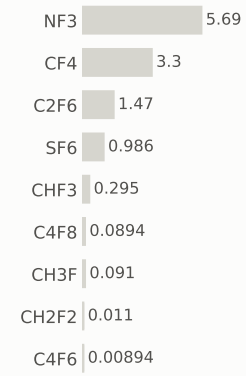
steps by process area



materials deposited (g)



gases emitted (g)



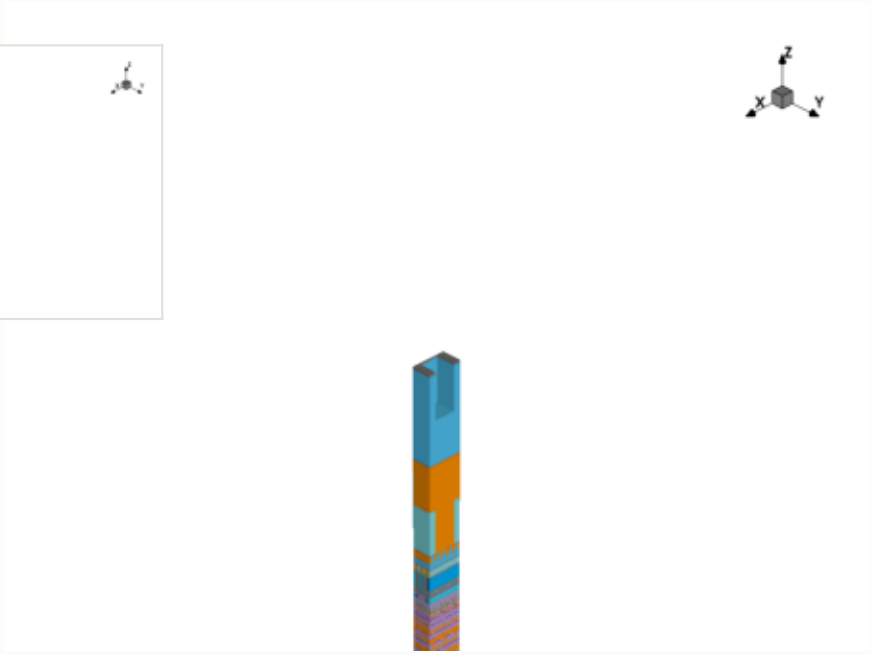
Step #8710: RIE

etch_anisotropic · RIE

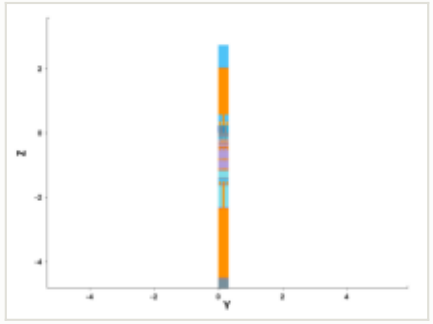
1.04	729.1	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

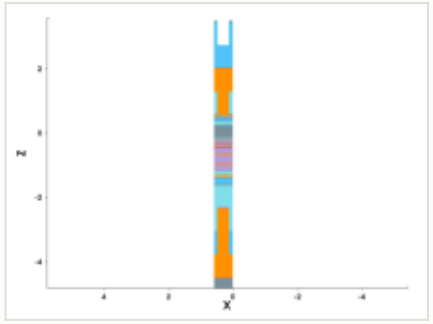
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

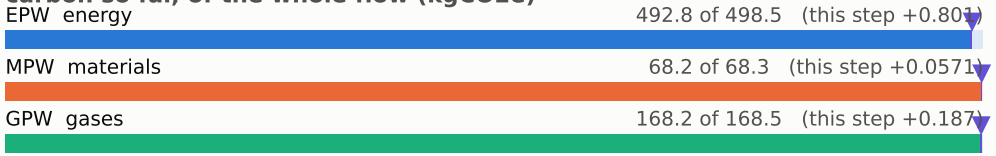
no mask for this step

this step

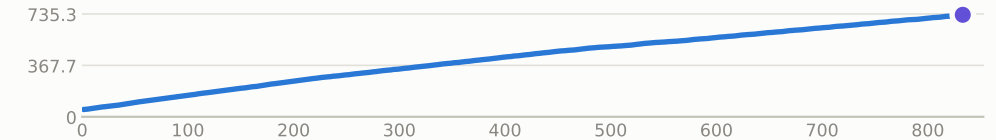
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

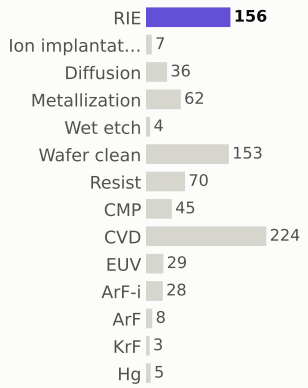
carbon so far, of the whole flow (kgCO2e)



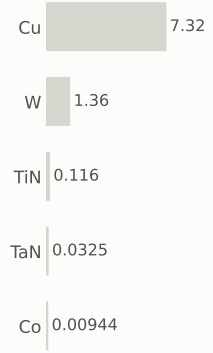
embodied carbon up to each step (kgCO2e per wafer)



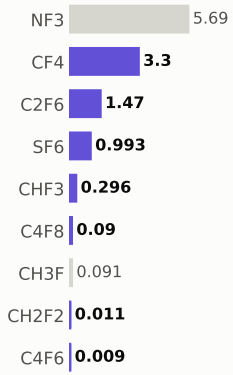
steps by process area



materials deposited (g)



gases emitted (g)



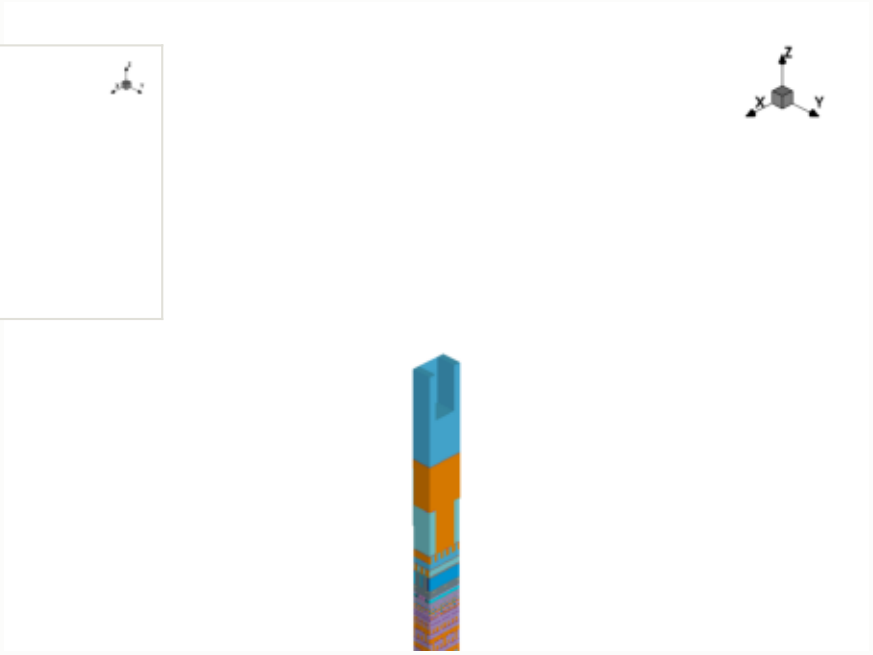
Step #8720: Wafer_clean

etch_strip · Wafer_clean

0.169	729.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

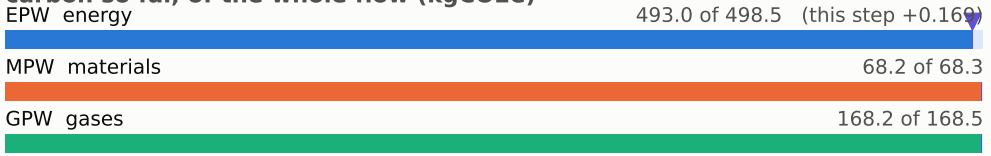
zoomed to fit



this step

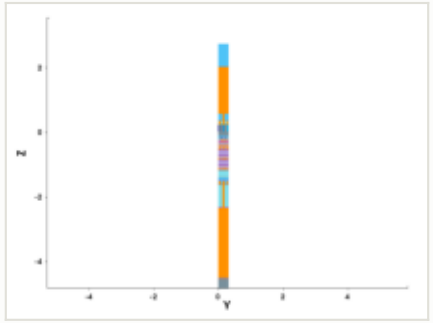
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

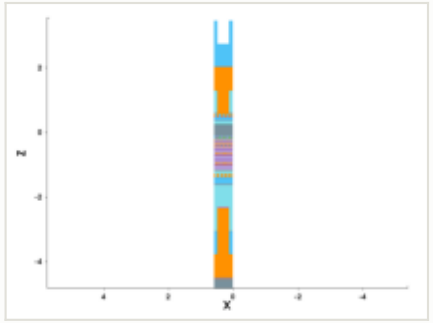


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



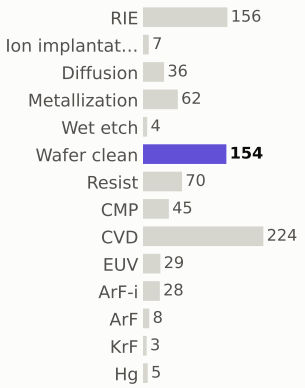
y cut at 0.184 µm



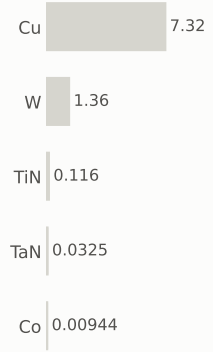
mask for this step

no mask for this step

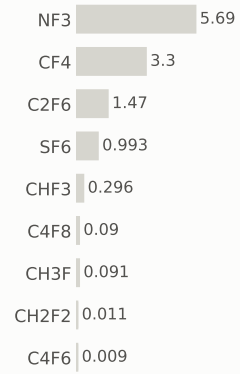
steps by process area



materials deposited (g)



gases emitted (g)



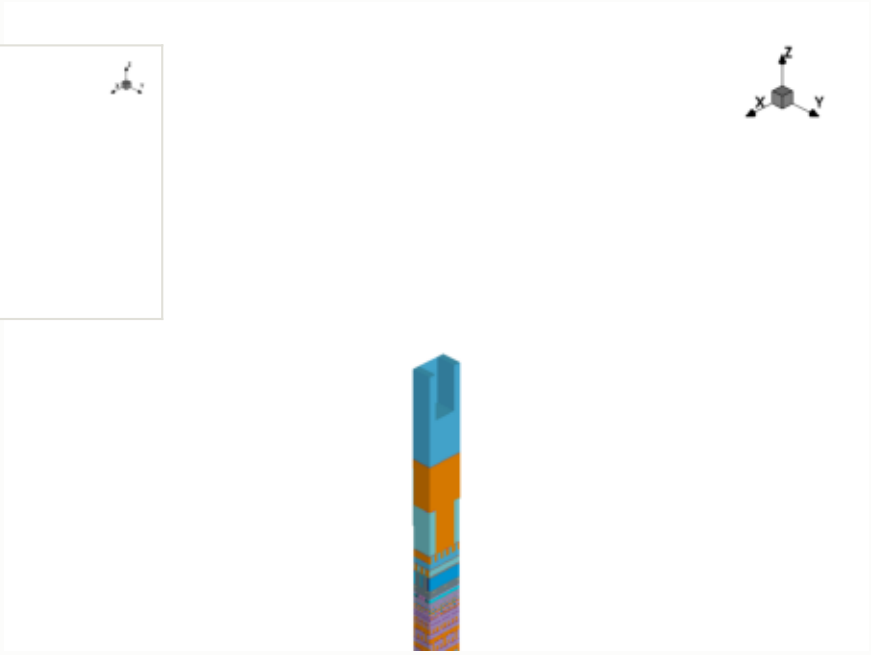
Step #8730: RIE

clean · RIE

1.04	730.3	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



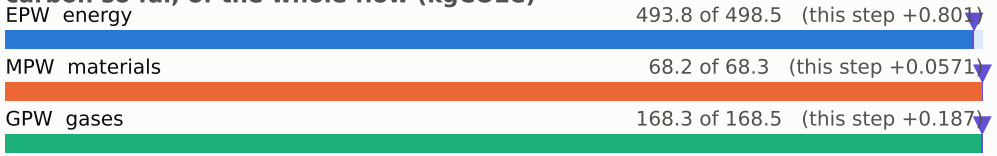
this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

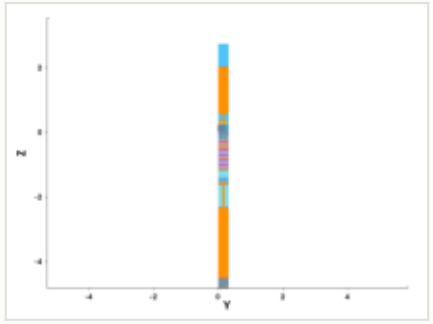
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

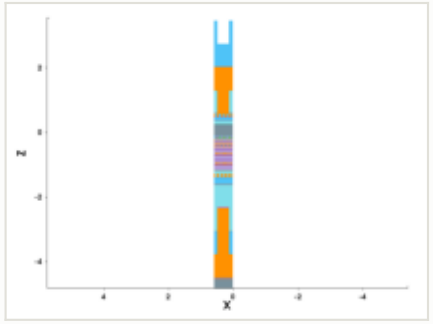


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



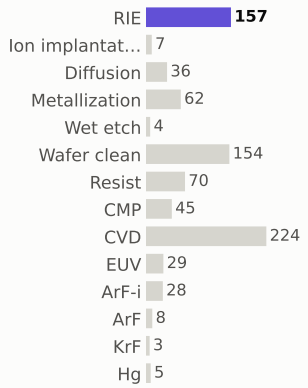
y cut at 0.184 µm



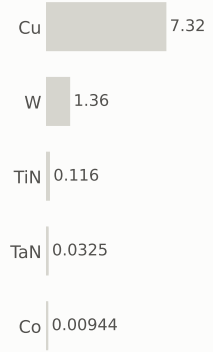
mask for this step

no mask for this step

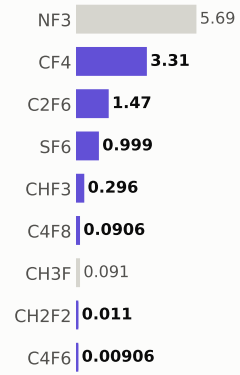
steps by process area



materials deposited (g)



gases emitted (g)



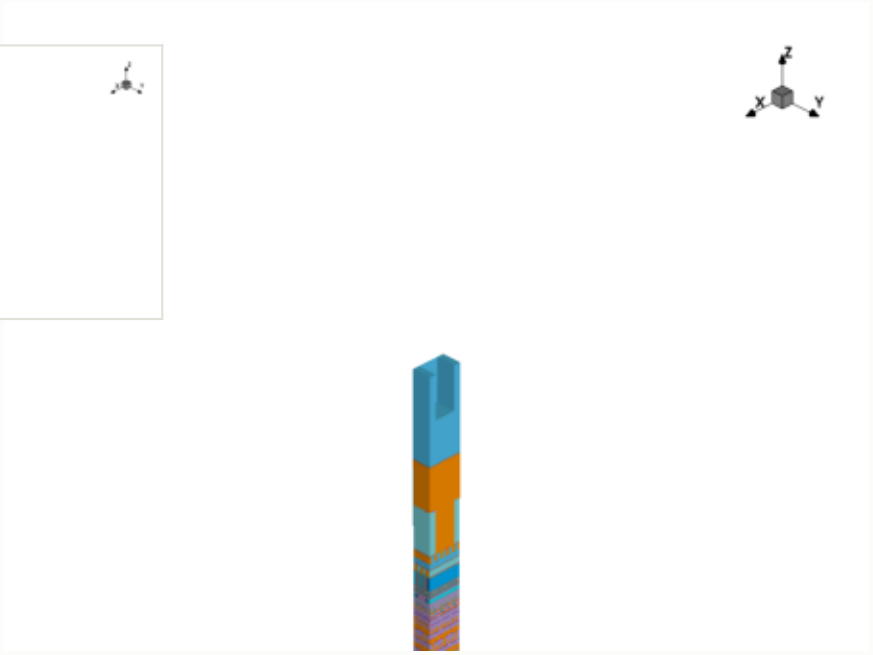
Step #8740: Wafer_clean

clean · Wafer_clean

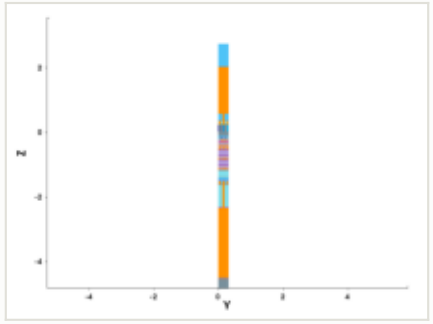
0.169	730.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

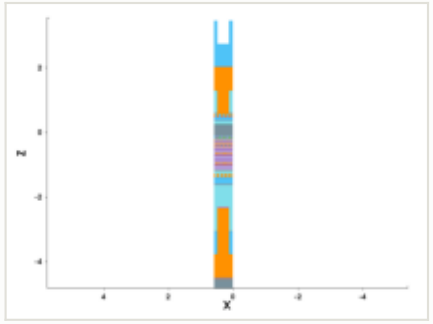
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

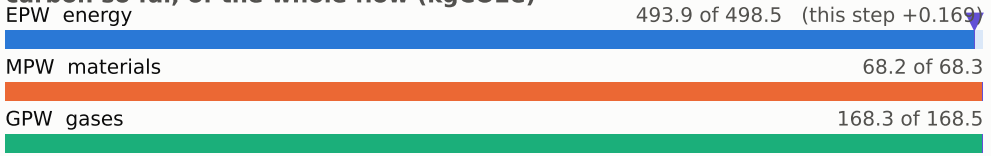
no mask for this step

this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

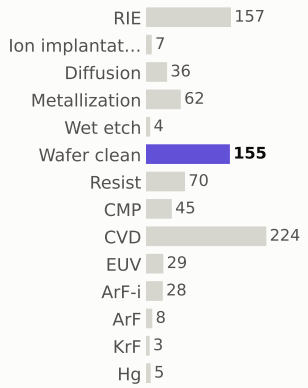
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

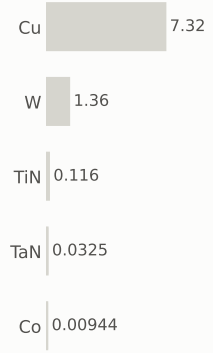


embodied carbon up to each step (kgCO2e per wafer)

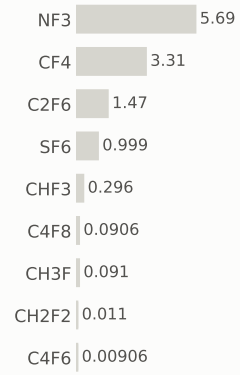
steps by process area



materials deposited (g)



gases emitted (g)



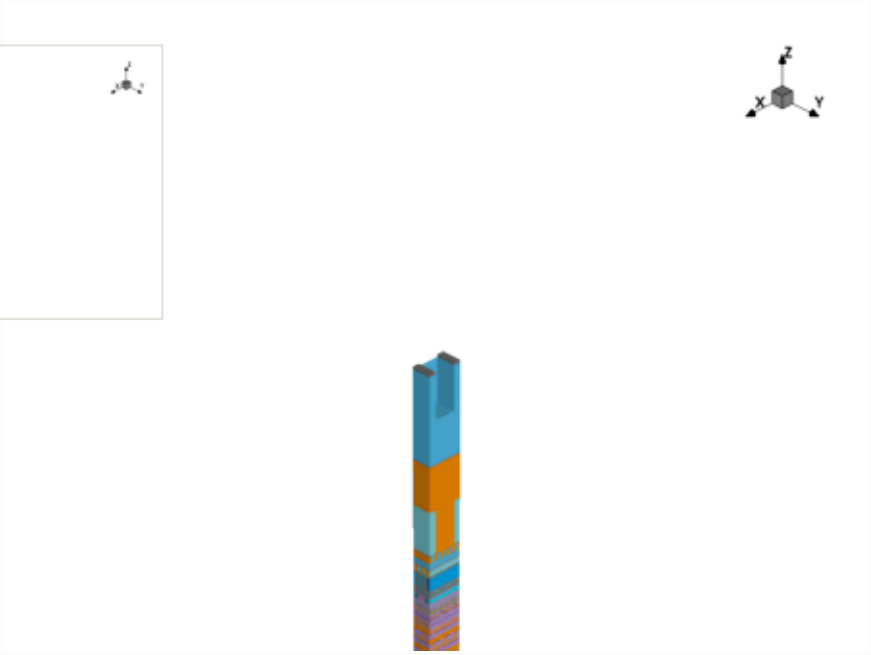
Step #8750: Hg

litho_patterning · Hg

0.199	730.7	0.027%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

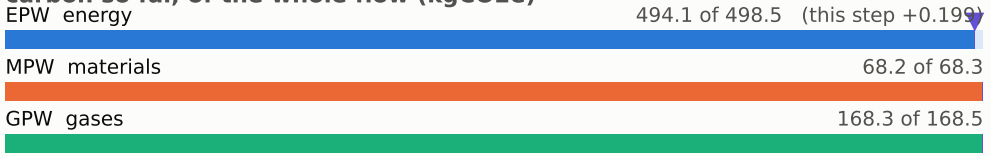


this step

process area	Hg
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

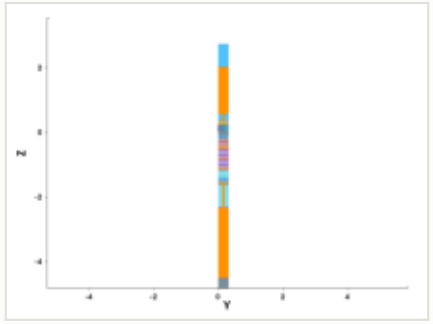
EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

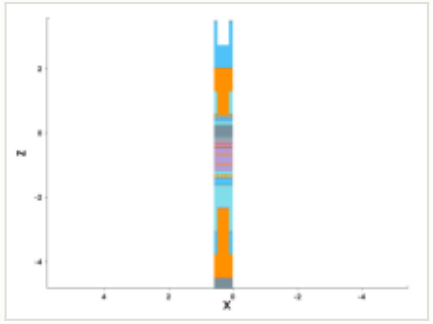


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm

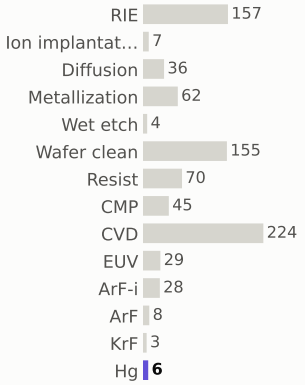


y cut at 0.184 µm

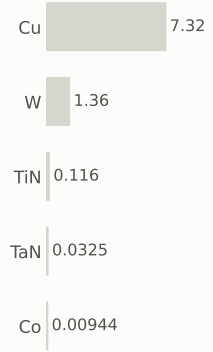


mask: layer 970:0 (GDSII)

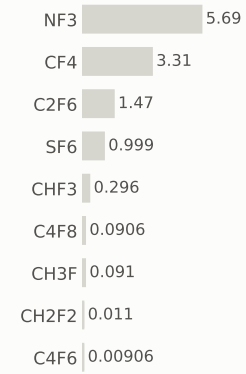
steps by process area



materials deposited (g)



gases emitted (g)



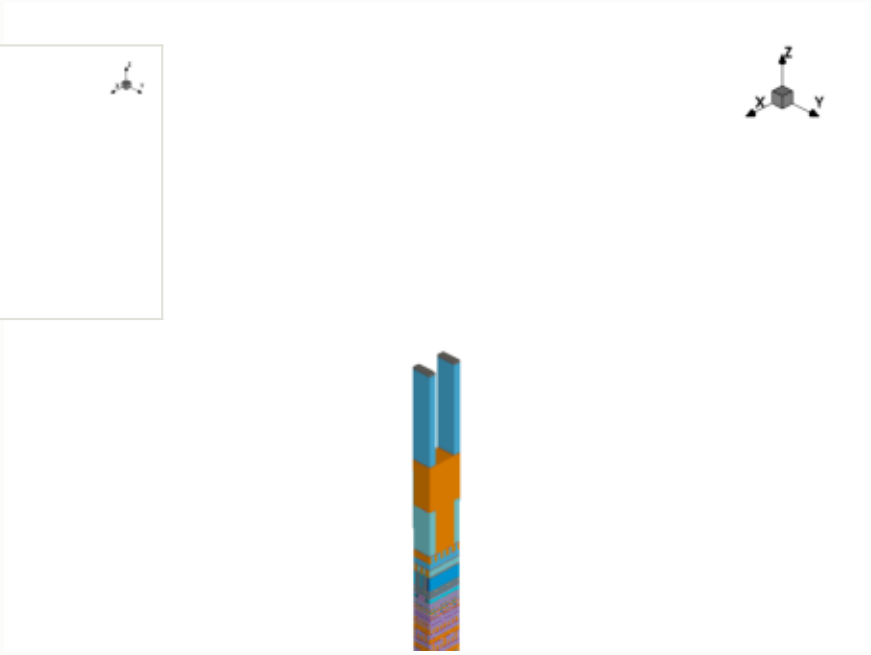
Step #8760: RIE

etch_anisotropic · RIE

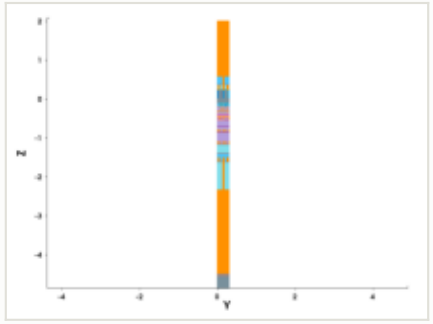
1.04	731.8	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

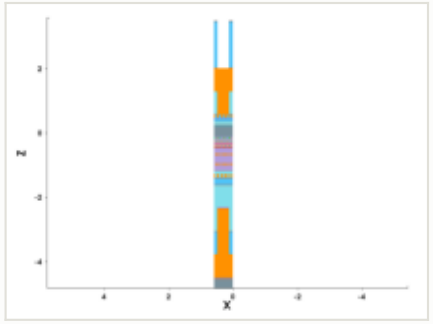
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

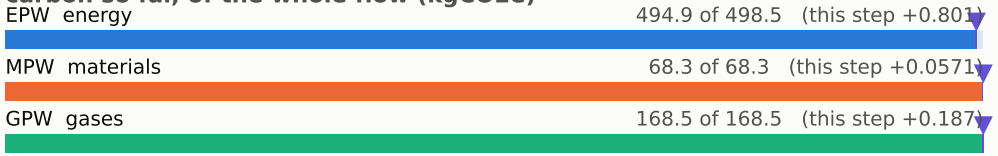
no mask for this step

this step

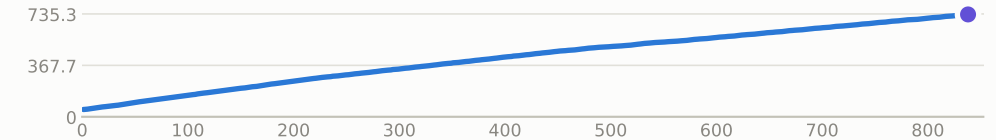
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

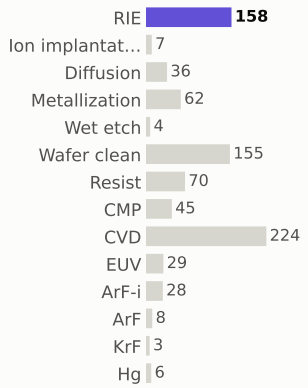
carbon so far, of the whole flow (kgCO2e)



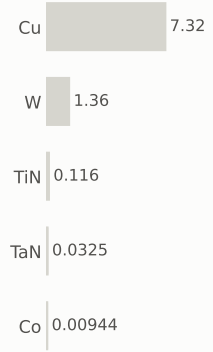
embodied carbon up to each step (kgCO2e per wafer)



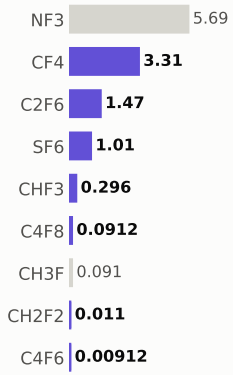
steps by process area



materials deposited (g)



gases emitted (g)



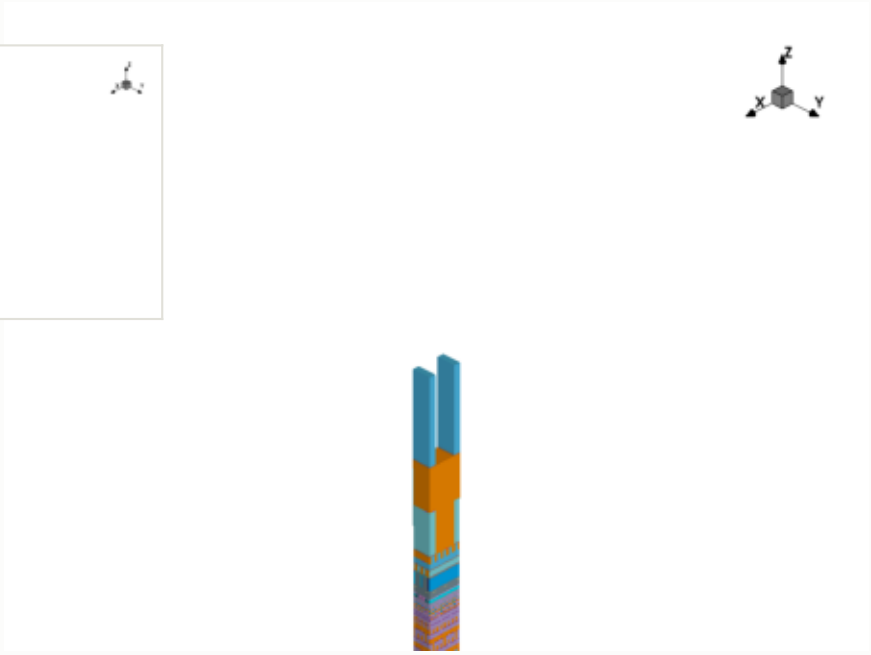
Step #8770: Wafer_clean

etch_strip · Wafer_clean

0.169	731.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

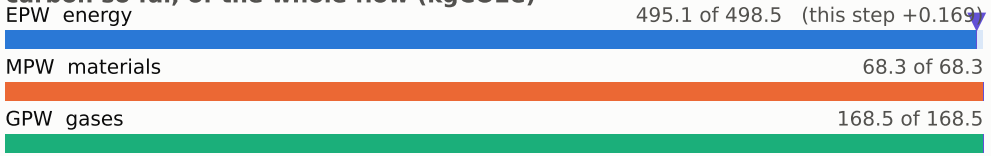
zoomed to fit



this step

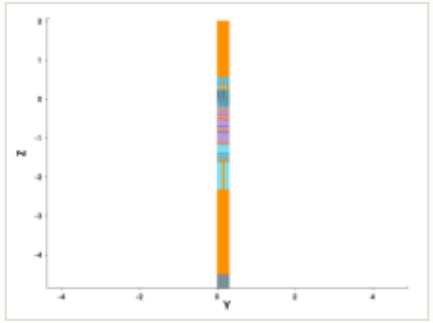
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

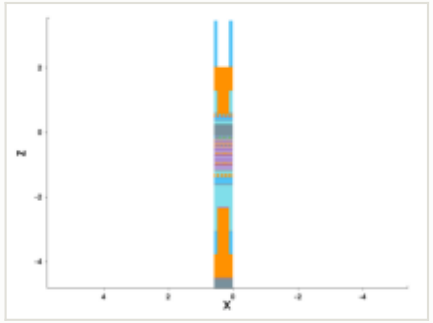


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



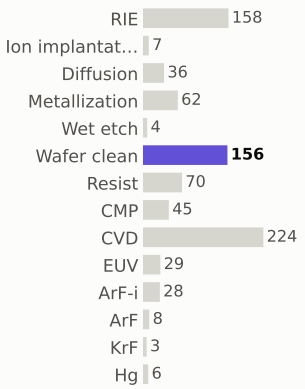
y cut at 0.184 µm



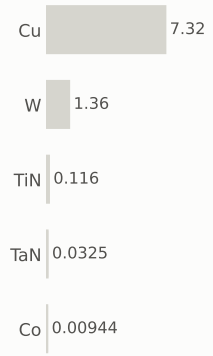
mask for this step

no mask for this step

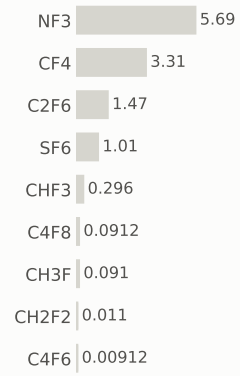
steps by process area



materials deposited (g)



gases emitted (g)



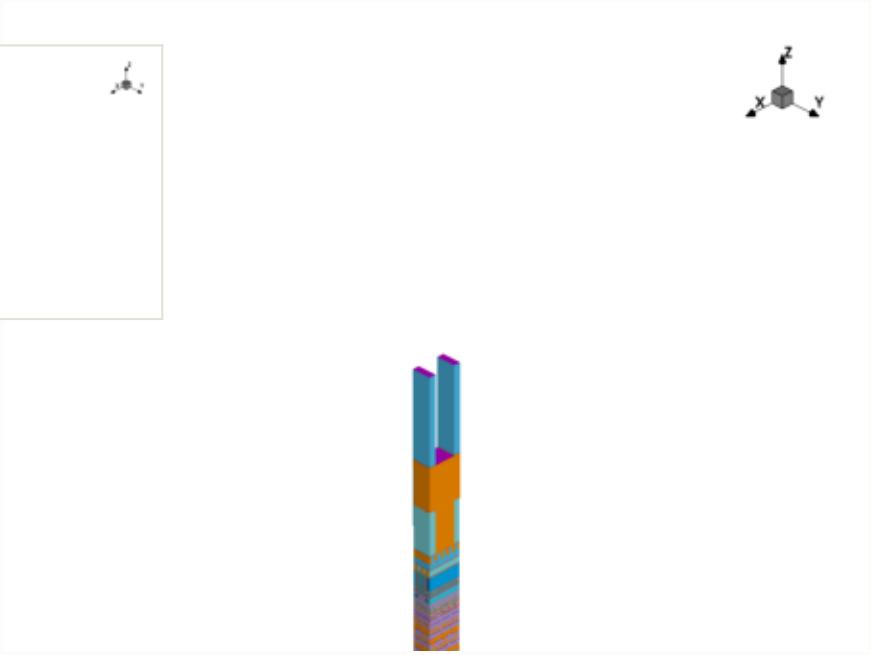
Step #8780: Metallization

depo_ellipsoidal · Metallization

1.2	733.1	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

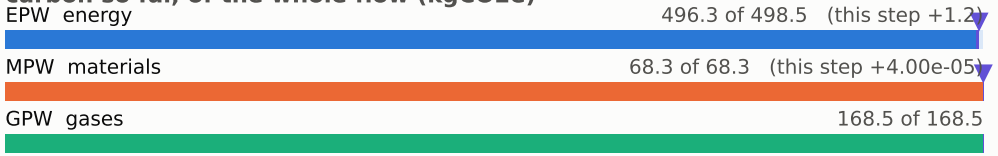


this step

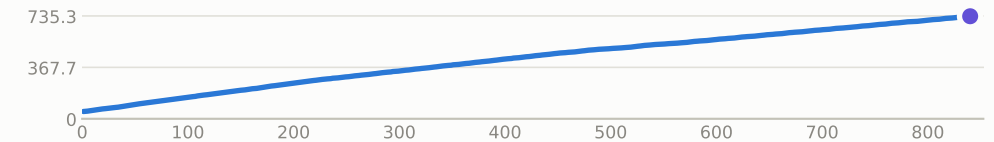
process area	Metallization
material	TaN
thickness	0.0016 µm
material GWP	25.8159 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.00e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

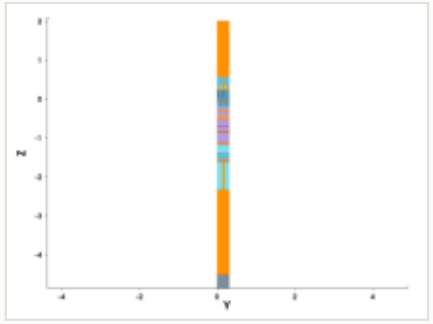
carbon so far, of the whole flow (kgCO2e)



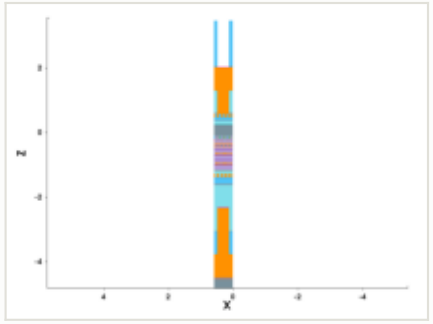
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.294 µm



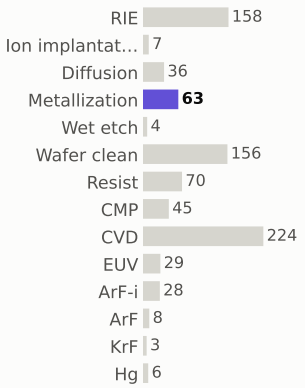
y cut at 0.184 µm



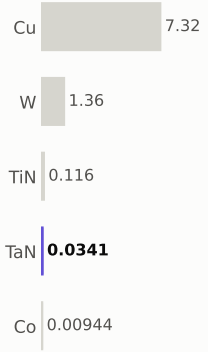
mask for this step

no mask for this step

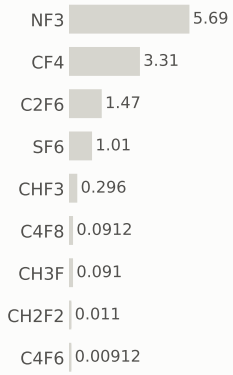
steps by process area



materials deposited (g)



gases emitted (g)



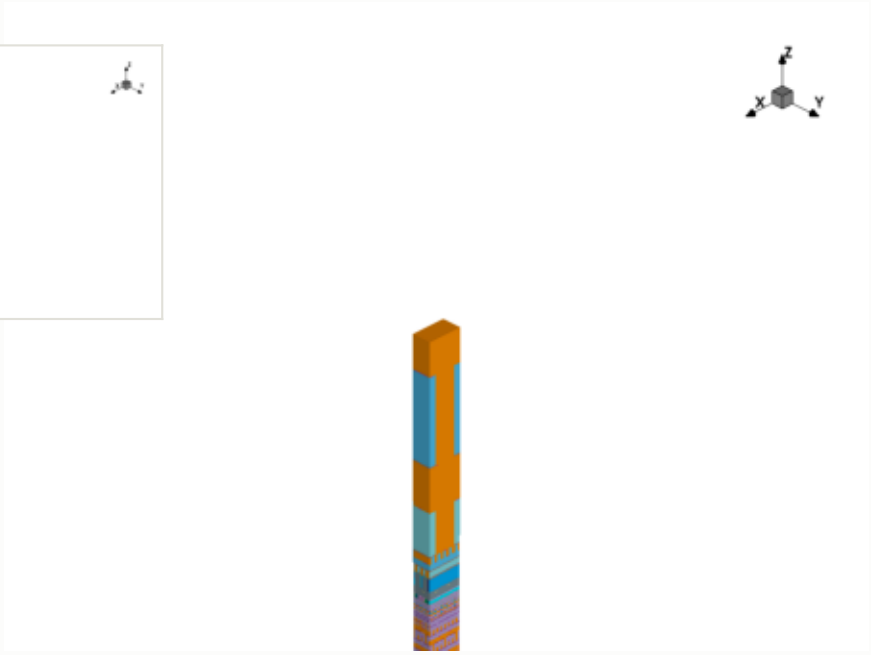
Step #8790: Metallization

depo_fill · Metallization

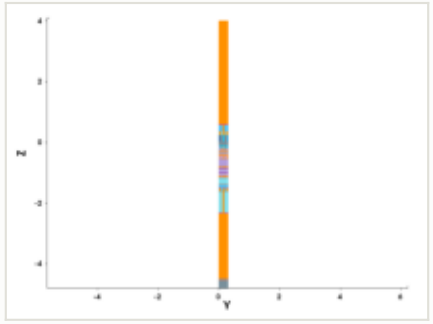
1.21	734.3	0.16%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

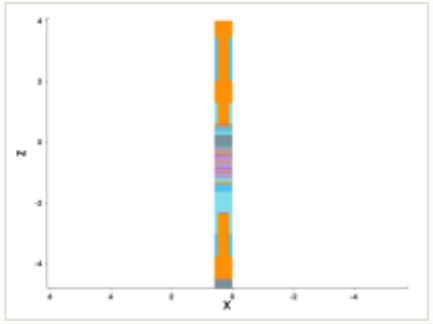
zoomed to fit



x cut at 0.294 µm



y cut at 0.184 µm



mask for this step

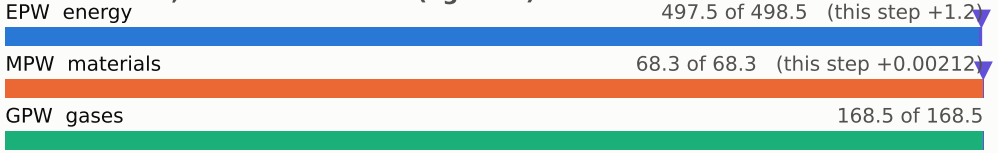
no mask for this step

this step

process area	Metallization
material	Cu
thickness	0.55 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

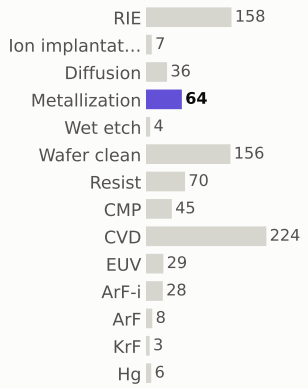
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00212 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

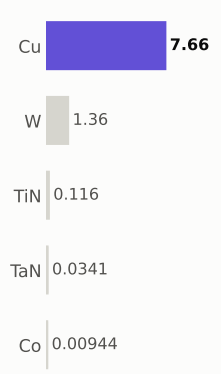


embodied carbon up to each step (kgCO2e per wafer)

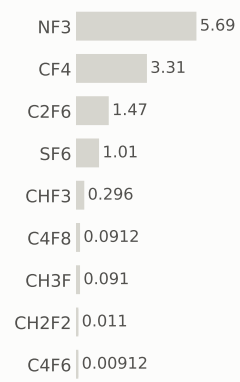
steps by process area



materials deposited (g)



gases emitted (g)



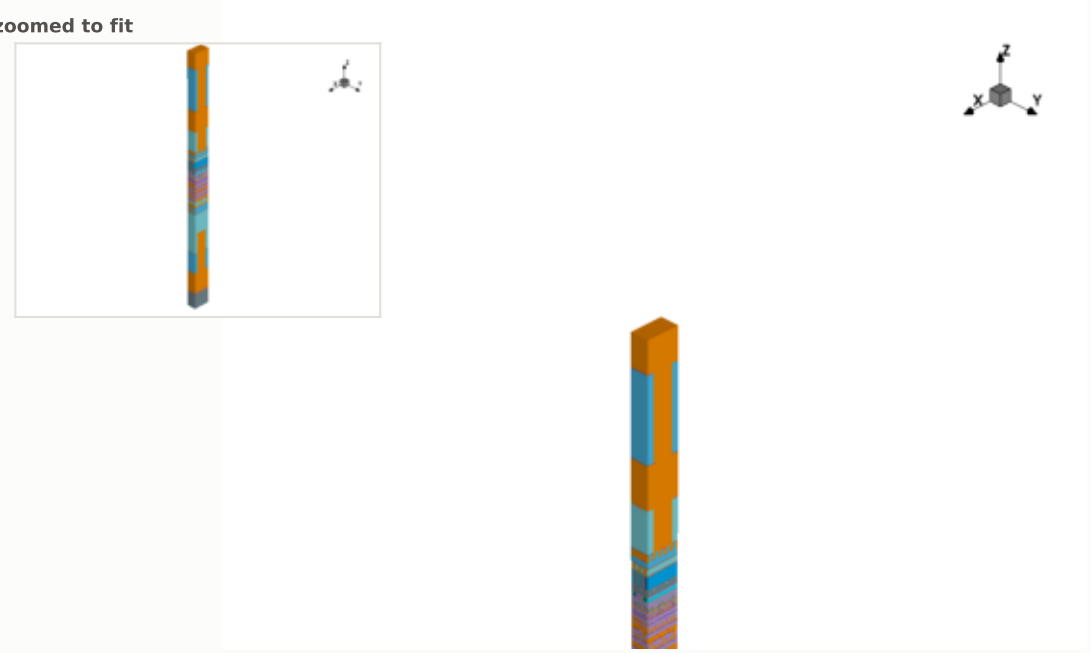
Step #8800: Diffusion

treatment · Diffusion

0.801	735.1	0.11%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

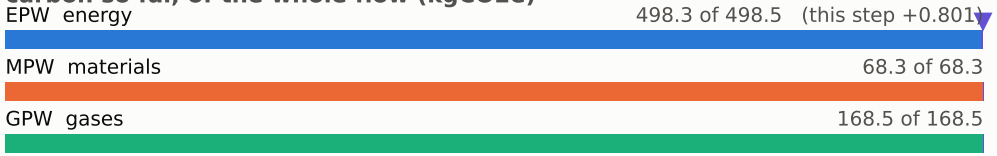


this step

process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

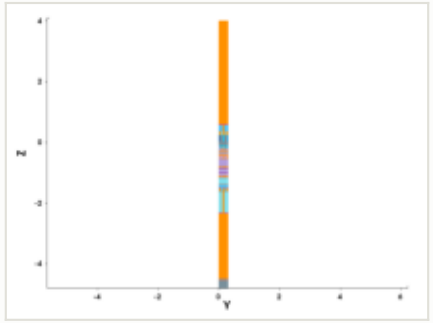
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

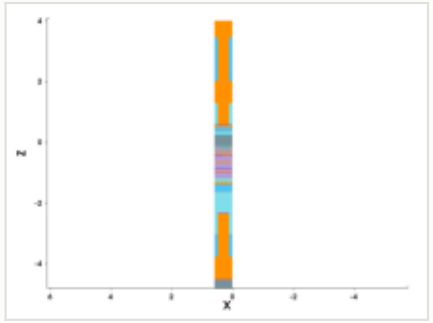


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



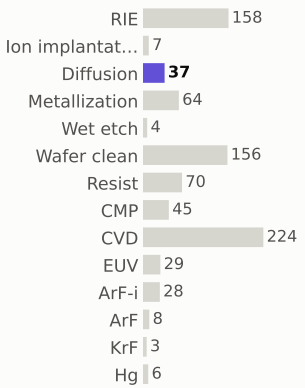
y cut at 0.184 µm



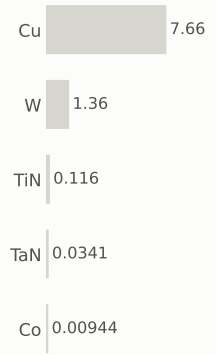
mask for this step

no mask for this step

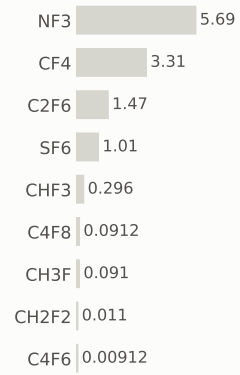
steps by process area



materials deposited (g)



gases emitted (g)



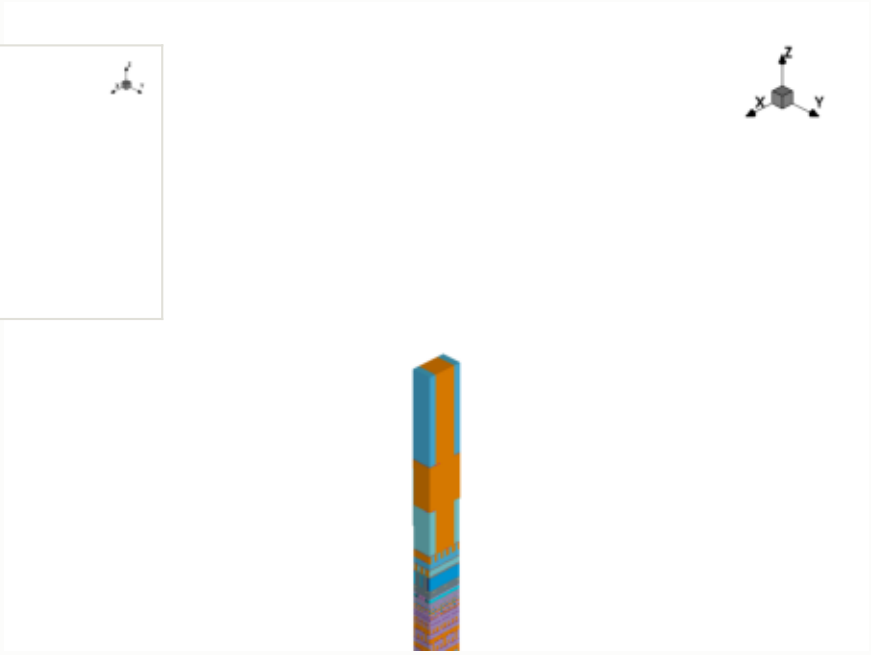
Step #8810: CMP

etch_cmp · CMP

0.169	735.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

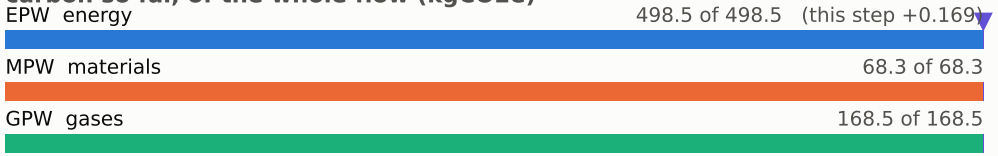
zoomed to fit



this step

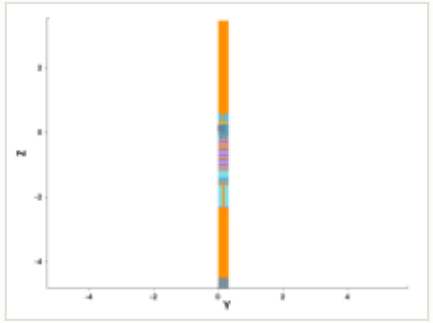
process area	CMP	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

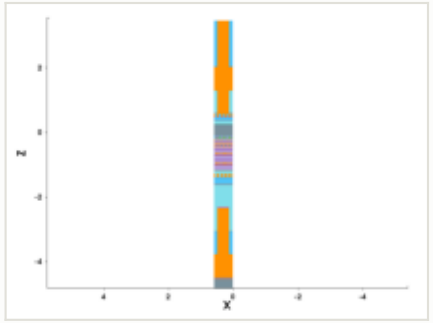


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



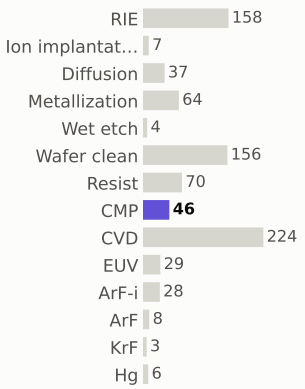
y cut at 0.184 µm



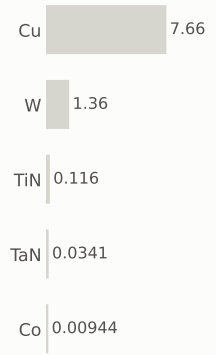
mask for this step

no mask for this step

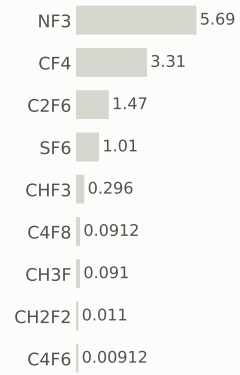
steps by process area



materials deposited (g)



gases emitted (g)



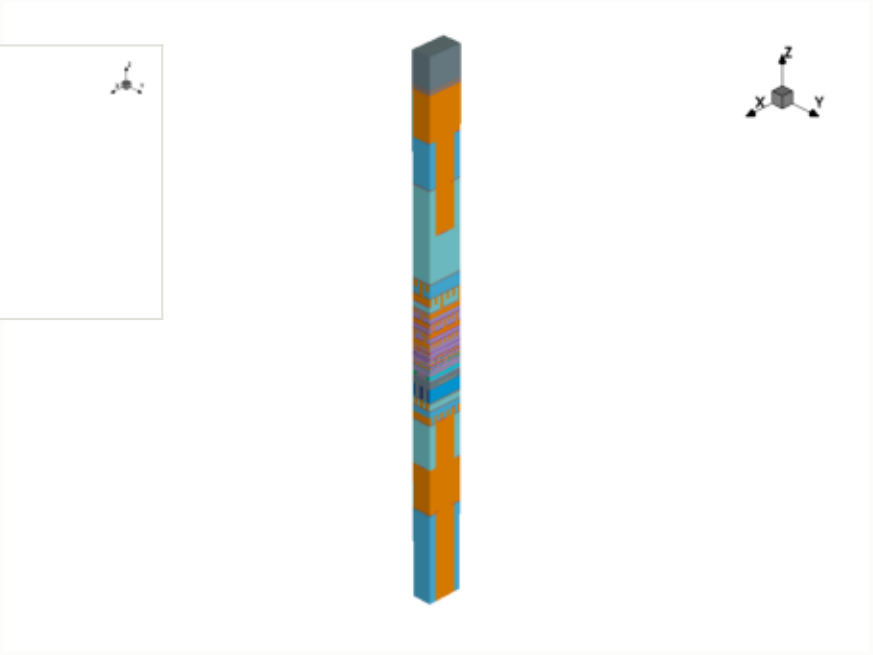
Step #8830: Wafer

flip_structure · Wafer

0	735.3	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

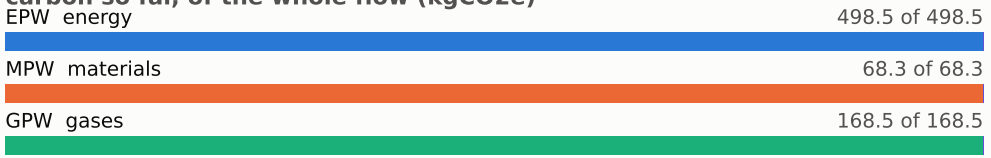


this step

process area	Wafer	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

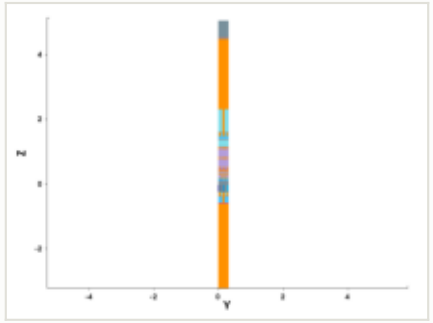
Wafer step: not counted as a process step (uncounted_process_areas)

carbon so far, of the whole flow (kgCO2e)

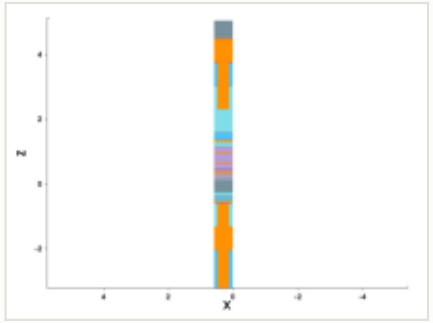


embodied carbon up to each step (kgCO2e per wafer)

x cut at 0.294 µm



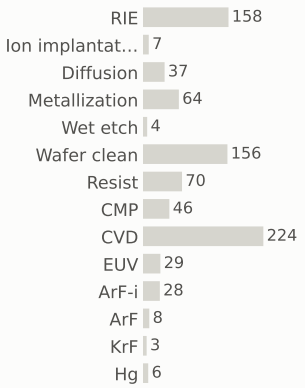
y cut at 0.184 µm



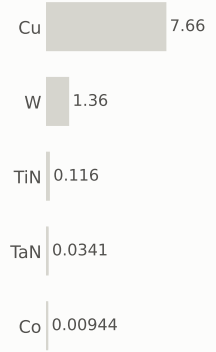
mask for this step

no mask for this step

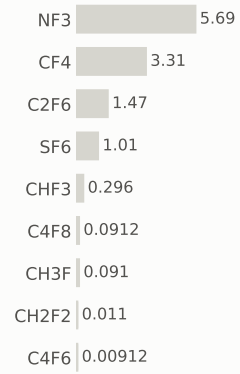
steps by process area



materials deposited (g)

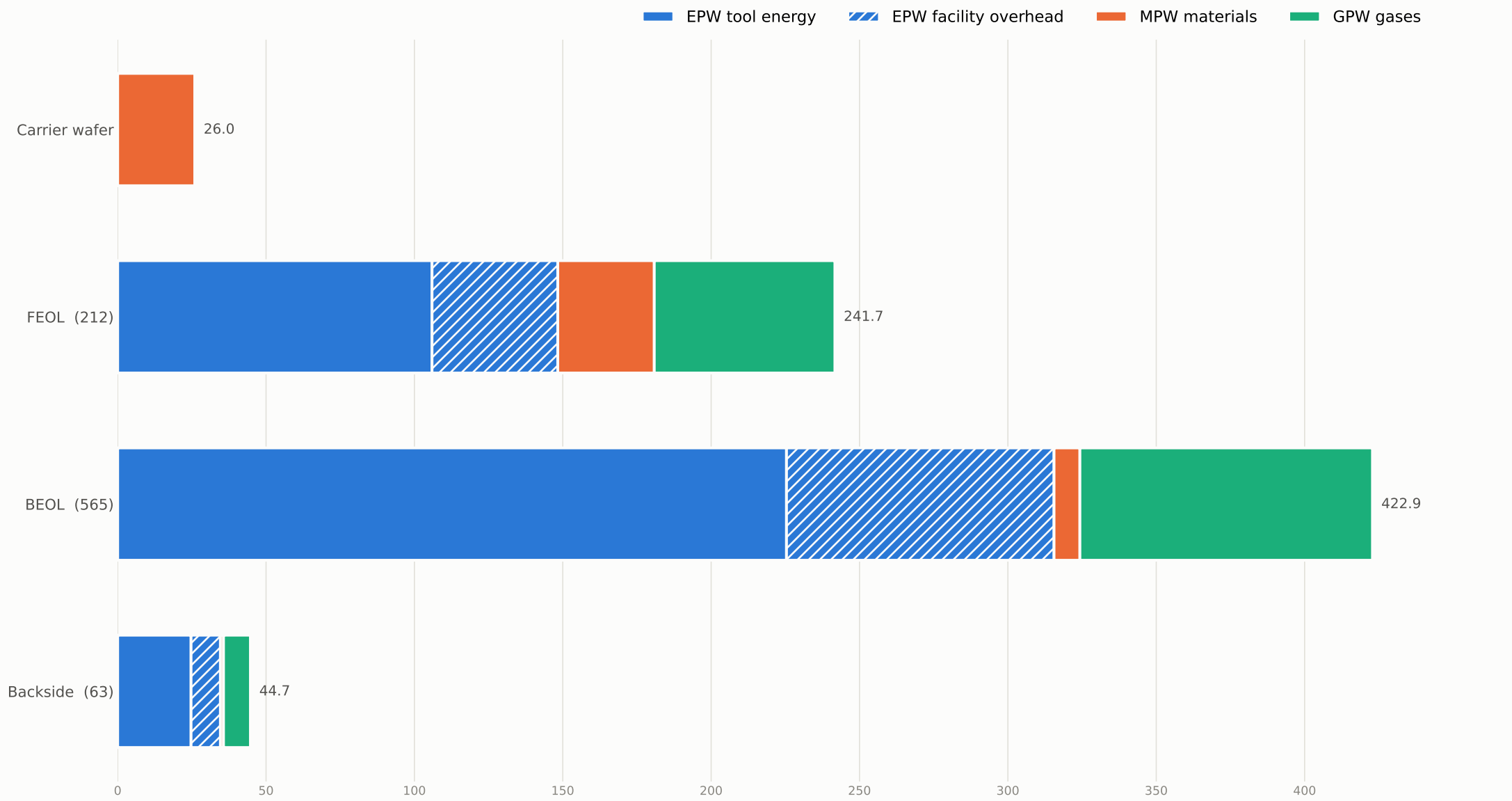


gases emitted (g)



Carbon by module

kgCO2e per wafer, in flow order (steps in brackets)



Totals by process area

kgCO2e per wafer; EPW = tool + facility overhead (overhead energy = tool energy x 0.4)

process area	steps	tool kWh	overhead kWh	EPW tool	EPW overhead	EPW	MPW	GPW	total
CVD	224	244.16	97.66	104.99	42.00	146.98	7.12	138.98	293.08
RIE	158	210.14	84.06	90.36	36.14	126.50	9.02	29.55	165.07
Metallization	64	128.00	51.20	55.04	22.02	77.06	0.15	0.00	77.21
EUV	29	94.25	37.70	40.53	16.21	56.74	0.00	0.00	56.74
Diffusion	37	49.21	19.68	21.16	8.46	29.62	0.00	0.00	29.62
Wafer_clean	156	43.68	17.47	18.78	7.51	26.30	0.00	0.00	26.30
Resist	70	19.60	7.84	8.43	3.37	11.80	0.00	0.00	11.80
CMP	46	12.88	5.15	5.54	2.22	7.75	0.00	0.00	7.75
ArF-i	28	10.08	4.03	4.33	1.73	6.07	0.00	0.00	6.07
Ion_implantation	7	9.31	3.72	4.00	1.60	5.60	0.00	0.00	5.60
ArF	8	2.64	1.06	1.14	0.45	1.59	0.00	0.00	1.59
Hg	6	1.98	0.79	0.85	0.34	1.19	0.00	0.00	1.19
Wet_etch	4	1.12	0.45	0.48	0.19	0.67	0.00	0.00	0.67
KrF	3	0.99	0.40	0.43	0.17	0.60	0.00	0.00	0.60
Wafer		0.00	0.00	0.00	0.00	0.00	52.00	0.00	52.00
all	840	828.04	331.22	356.06	142.42	498.48	68.29	168.54	735.30